

RiskAgility FM U.S. Library

User Guide

Version 4.0

WillisTowersWatson 

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User Guide

General books

- G01 Introduction
- G02 Code management
- G03 Input management
- G04 Run management
- G05 Output management
- G06 Customization

Company books

- C01 Profit calculations
- C02 Economic scenarios
- C03 Asset modeling
- C04 GAAP accounting
- C05 PBA
- C06 Hedging

Liability books

- L01 Term life
- L02 Traditional life
- L03 Universal life
- L04 Variable universal life
- L05 Indexed universal life
- L06 Fixed annuity
- L07 Variable annuity
- L08 Fixed indexed annuity
- L09 Payout annuity
- L10 Long term care
- L12 Externally projected liabilities
- L13 Riders
- L14 Actuarial math library

Asset books

- A01 Bonds
- A02 Mortgages
- A03 Securitized assets
- A04 Interest rate derivatives
- A05 Real estate
- A06 Equities
- A07 Equity index options
- A08 Externally projected assets

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G01 INTRODUCTION

This book provides an overview of the RiskAgility FM U.S. Library.

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1 USER GUIDE OVERVIEW

This User guide documents the features of the RiskAgility FM U.S. Library. This guide consists of the books listed below. Each book is organized using headings like this:

- Chapter – The first level (such as 1)
- Section – The second level (such as 1.1)
- Topic – The third level (such as 1.1.1)

1.1 Books index

1.1.1 General books

G01 Introduction (this book)
G02 Code management
G03 Input management
G04 Run management
G05 Output management
G06 Customization

1.1.2 Company books

C01 Profit calculations
C02 Economic scenarios
C03 Asset modeling
C04 GAAP accounting
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C06 Hedging

1.1.3 Liability books

L01 Term life
L02 Traditional life
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1.1.4 Asset books

A01 Bonds

A02 Mortgages

A03 Securitized assets

A04 Interest rate derivatives

A05 Real estate

A06 Equities

A07 Equity index options

A08 Externally projected assets

1.2 Using this guide

Each book begins with its table of contents.

You can also search for a phrase in the entire user guide (all books together). Alternative names for various concepts are included, to improve the chance that your search will be successful.

Information discussed in another place are referenced in the text, such as: 'See the Point-in-time valuation chapter of the PBA book.'

2 U.S. LIBRARY COMPONENTS

2.1 Solution

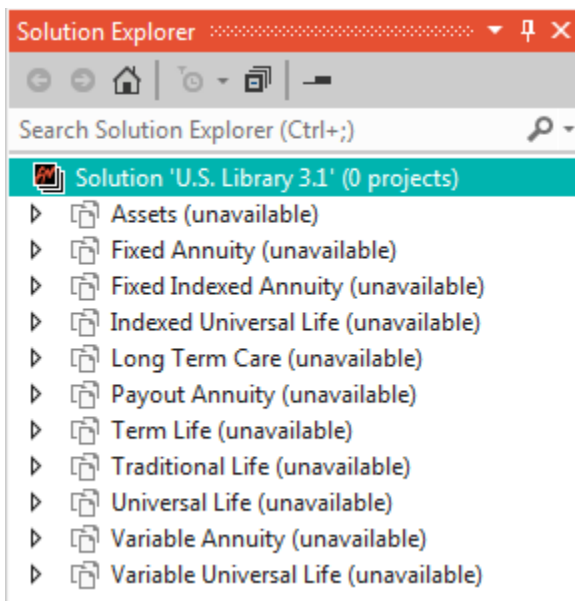
The RiskAgility FM U.S. Library x.y solution, where x represents the major version number and y represents a minor version number, provides the basic structure and functionality you need to create and manage the individual **Projects** (discussed below) which are contained within the RiskAgility FM U.S. Library x.y.

The solution files, represented by a .sln file and .suo file, contain definition and configuration information relating to the U.S. Library Solution and its associated Projects. The definition and configuration information includes the following:

- General information about the Projects and how they are structured in the Solution
- The RiskAgility FM Setup Options you define, such as the System Paths
- Customization you apply to the RiskAgility FM work area

A single solution, named 'U.S. Library x.y', is delivered with the library. The solution files are stored in a 'RiskAgility FM U.S. Library x.y' child folder of the folder selected during the installation process.

Load the 'U.S. Library x.y' solution file into RiskAgility FM by selecting File-Open Project/Solution, navigating to '..\RiskAgility FM U.S. Library x.y', and selecting the U.S. Library x.y.RAFM_.sln file. The solution is loaded into the Solution Explorer as follows:



Although a solution file is provided with the delivery of the RiskAgility FM U.S. Library, you may select to create your own solution for the U.S. Library by using the File-New-Project menu selection and specifying to create a new RiskAgility FM Solution.

Select Tools-Options-Startup and select 'Load last loaded solution', if you desire RiskAgility FM to reload the last loaded solution each time you open RiskAgility FM.

2.2 Projects

As illustrated above, the delivered RiskAgility FM U.S. Library solution contains references to the RiskAgility FM U.S. Library projects.

A RiskAgility FM U.S. Library project is a collection of files which contain the code definitions, input management definitions, run management definitions, and output definitions required to model U.S life insurance products and assets.

Separate projects are provided for each product type, plus an assets project containing the individual assets. This approach is used because projects are easier to work with and manage when they are smaller in size. For example, smaller projects load, unload, respond to user actions, and compile faster. They are also smaller in size when backed up.

The delivered 'U.S. Library x.y' solution file contains all projects of the library, and contains **fixed paths** expecting the project files to be stored in '.\Projects\[Project Name]' folders under the '[Install Path]\RiskAgility FM U.S. Library x.y' folder. **The paths to the projects cannot be altered in the delivered solution file.** Each RiskAgility FM U.S. Library project is referenced through a .msproj file, which is stored in the root folder for that project.

Projects are referenced in an unloaded state within the delivered solution. This reduces the time required to load the solution and initial memory requirements of the solution. Projects are loaded into the solutions by right clicking on the project and selecting Reload Project. However, only those projects for which you are licensed can be loaded into the solution. Best practice is to unload the projects before closing RiskAgility FM by right clicking on the project name and selecting Unload Project. This will ensure that the loading time and initial memory requirements of the solution are minimized.

If you select to create your own solution file for the RiskAgility FM U.S. Library x.y, you can then add the projects to the solution by right clicking on the solution name, selecting Add-Existing Project, navigating to the project folder, and selecting the .msproj file. Again, only those projects for which you are licensed can be added to your custom solution.

Finally, each RiskAgility FM U.S. Library project may also be loaded into RiskAgility FM as its own solution. This is accomplished through the File-Open-Project/Solution, navigating to the folder of the project, and selecting the [Project Name].RAFM_sln file. For example, the Fixed Annuity project may be loaded into RiskAgility FM by navigating to the 'RiskAgility FM U.S. Library 1.0\Projects\Fixed Annuity' folder and selecting Fixed Annuity.RAFM_sln.

2.3 Global data

The RiskAgility FM U.S. Library is delivered with global files and data that is referenced or ready to be used in the projects. The global data includes:

Item	Description
ActuarialMath.dll/lib	The Towers Watson actuarial math library which calculates formulaic cash values and reserves for term, traditional life, and long term care products. Both 32bit and 64 bit versions are provided with the 64bit versions contained in the x64 folder
Adppmdl.dll/lib	The Andrew Davidson prepayment model library Both 32bit and 64 bit versions are provided with the 64bit versions contained in the x64 folder

Item	Description
Cmosub32.dll and Cmosub64.dll	The Intex solutions library Both 32bit and 64 bit versions are provided with the 64bit versions contained in the x64 folder
Economy.dll/lib	The Towers Watson economy library which provides for retrieval of scenario rates and is used to generate new scenario files for inner loop valuations
Rafmsqlite.lib	The Towers Watson RiskAgility FM SQLite library which is used for dependent task processing to retrieve results from the output files into the projection
AG38 Lapse Rates.xlsm	Actuarial guideline 38 lapse rates
AG48 Actuarial Method NPR Percentages.xlsm	Actuarial guideline 48 percentages applicable to the NPR in determination of the actuarial method
Guideline Premium Test Corridor.xlsm	IRC section 7702 guideline premium test corridor percentages by attained age
IMR Amortization Rates.xlsm	Amortization percentages for IMR realized capital gains varying by years to maturity and years after sale. Separate schedules are included for each calendar year and by bond, mortgage, or other
Mortality Improvement.xlsm	The 2012 projection scale G mortality improvement rates, varying by gender, calendar year, and attained age
Mortality Select Multiples.xlsm	Mortality select multiple for: 1980 CSO 10 year, 1980 CSO 15 year, 1980 CSO 20 year, 1980 CSO 10 year aggregate, 1980 CSO 15 year aggregate, 1980 CSO 20 year aggregate, 2001 CSO 25 year age nearest birthday 2001 CSO 25 year aggregate age nearest birthday 2001 CSO 25 year age last birthday 2001 CSO 25 year aggregate age last birthday 2001 CSO 25 year preferred class age nearest birthday 2001 CSO 25 year preferred class age last birthday
VA Standard Scenario Mortality Multiples.xlsm	Mortality multiples for variable annuity standard scenario reserve and capital calculations. Multiples vary by reserves versus capital calculations and attained age
VA Standard Scenario Mortality.xlsm	Mortality rates for the 1994 US MGDB tables. Rates vary by gender and attained age. Both age nearest birthday and age last birthday are included

Item	Description
Valuation Drop and Recovery Rates.xlsm	Valuation drop and recovery rates for actuarial guideline 34 and standard scenario calculations. Drop and recovery rates vary by asset class and projection year. For standard scenario calculations, rates vary by reserves versus capital.
Valuation Interest Rates.xlsm	Statutory, Tax, and Nonforfeiture rates for life insurance, payout annuity, and deferred annuity. For life insurance and payout annuities, valuation rates vary by duration and issue year. For deferred annuities, valuation rates are included for all SVL variations of CARVM definitions, cash settlement definitions, interest guarantee definitions, guaranteed duration ranges, valuation plan types, and vary by issue year.
VM20 NPR Assumptions.xlsm	VM 20 net premium reserve assumptions for expense allowance, lapse rates, and interest rates

2.4 Utilities

The RiskAgility FM U.S. Library is delivered with Excel based utilities which are used to assist with the creation of model point files, mortality table external sources, standard reports, and scenarios.

The utilities provided include:

Item	Description
Liability Model Point CSV Builder Tool.xlsm	Tool for creating the CSV files for liability model points
Supplemental Model Point CSV Builder Tool.xlsm	Tool for creating the CSV files for reinsurance and rider liability model points
Asset Model Point CSV Builder Tool.xlsm	Tool for creating the CSV files for asset model points
Global Mortality Table External Source Creator.xlsm	Tool for creating Excel workbooks of global mortality rates drawing from the SOA Excel database
Reporting Tool.xlsm	Tool for creating workbooks of standard reports from the output created by the RiskAgility FM U.S. Library projects
AAA Scenarios CSV Translator.xlsm	Tool for translating the CSV files created by the Academy of Actuaries generator into a scenario file in the format required by the RiskAgility FM U.S. Library projects
TAS Scenario File Translator.xlsm	Tool for translating a TAS scenario file into the format required by the RiskAgility FM U.S. Library projects

Each utility is discussed in detail in other user guide books.

2.5 Templates

Each RiskAgility FM U.S. Library project contains templates which provide a structure for file management, input management, run management, and output management.

The structure and use of the templates are discussed in other user guide books.

3 FILE MANAGEMENT

3.1 Standard folders

RiskAgility FM U.S. Library x.y is delivered with the following standard directory structure:

Folder	Description
[Install Path]\RiskAgility FM U.S. Library x.y	Root folder for the RiskAgility FM U.S. Library solution. Referred to as [Root] below
[Root]\Projects	The root folder for the RiskAgility FM U.S. Library projects. Referred to as [Projects] below
[Projects]\Assets	The root folder for the Assets project. Referred to as [Project] below.
[Projects]\Fixed Annuity	The root folder for the Fixed Annuity project. Referred to as [Project] below.
[Projects]\Fixed Indexed Annuity	The root folder for the Fixed Indexed Annuity project. Referred to as [Project] below.
[Projects]\Indexed Universal Life	The root folder for the Indexed Universal Life project. Referred to as [Project] below.
[Projects]\Long Term Care	The root folder for the Long Term Care project. Referred to as [Project] below.
[Projects]\Payout Annuity	The root folder for the Payout Annuity project. Referred to as [Project] below.
[Projects]\Term Life	The root folder for the Term Life project. Referred to as [Project] below.
[Projects]\Traditional Life	The root folder for the Traditional Life project. Referred to as [Project] below.
[Projects]\Universal Life	The root folder for the Universal Life project. Referred to as [Project] below.
[Projects]\Variable Annuity	The root folder for the Variable Annuity project. Referred to as [Project] below.
[Projects]\Variable Universal Life	The root folder for the Variable Universal Life project. Referred to as [Project] below.

Folder	Description
[Project]\ProjectData	Root folder for project data for the project. Referred to as [ProjectData] below.
[ProjectData]\Global	Global data for the project. Includes required DLLs/LIBs and Excel based standard industry data
[ProjectData]\Template	Template data for the project. Includes template assumption workbooks and model point files.
[Root]\Utilities	The root folder for the RiskAgility FM U.S. Library utilities. Referred to as [Utilities] below
[Utilities]\Model Point Tools	Folder for the Excel based model point CSV creator tools
[Utilities]\Mortality Table Tools	Folder for the Excel based mortality table tool
[Utilities]\Reporting Tools	Folder for the Excel based reporting tool
[Utilities]\Scenario Tools	Folder for the Excel based scenario tools

Best practice is to store all project data and capture all project output in children folders of the [Project]\ProjectData folder.

3.2 Modeling project folders

In addition to the standard folders, RiskAgility FM U.S. Library projects are designed to manage all project data and output for a modeling project in children folders of the [Project]\ProjectData folders.

For example, use a 2013YearEnd folder under the ProjectData folder to capture the assumptions, data, scenario, and output for the 2013 year end modeling project.

RiskAgility FM U.S. Library projects utilize the following modeling project folders:

Folder	Description
..\Assumptions	Folder for the external source files, such as Excel workbooks
..\Data	Folder for the model point files
..\Scenarios	Folder for the scenario files
..\Output	Folder for the output from the projections

This structure demonstrates a best practice for setting up a new modeling project, such as for a new reporting period. For example, copy the project data folder for year-end 2013 modeling project into the new project folder for first quarter 2014 values.

3.3 Wildcards

The RiskAgility U.S. Library projects use the following standard wildcards:

Wildcard keyword	Default replacement string	Description
proj_data	<*project_directory*>ProjectData\ [Modeling Project Name]	Path for the modeling project data. [Modeling Project Name] is a folder such as '2013YearEnd'.
global	<*project_directory*>ProjectData\ Global\	Global external sources
assump	<*proj_data*>Assumptions\	Folder for external source files
data	<*proj_data*>Data\	Folder for model point files; EPA and EPL files
scen	<*proj_data*>Scenarios\	Folder for specified scenario files
scen_gen	<*scen*>Generated\	Folder for generated scenario files
output	<*proj_data*>Output\	Output directory

In addition, these grid specific wildcards are included to facilitate distribution of the data to the grid

Wildcard keyword	Default replacement string	Description
grid_input_global	<*global*>	Required for referencing the global folder for distributing the projections to the grid
grid_input_scen	<*scen*>	Required for referencing the specified scenario files for distributing the projections to the grid
grid_output_data	<*data*>	Required for referencing the data files for distributing the projections to the grid
grid_output_scen	<*scen_gen*>	Required for referencing the generated scenario files for distributing the projections to the grid

The RiskAgility FM U.S. Library projects wildcards utilize the <*project_directory*> system wildcard in order to provide full relativity of paths from the project folder.

Since the <*project_directory*> wildcard is reconciled to the folder where the .msproj file resides, the project can be stored in any location and, as long as the relative folder structure remain unaltered, the wildcard keywords referencing will reconcile to the appropriate path.

Best practice is to use wildcards for all references, never hard-coded directory paths.

This use of the <*proj_data*> wildcard coincide with the best practice for setting up and managing modeling projects. For each modeling project, it is best practice to create a wildcard set and adjust the [Modeling Project Name] element of the <*proj_data*> wildcard.

For example, a '2014Q1' wildcard set contains the wildcards where the 'proj_data' wildcard references the correct 'ProjectData\2014Q1\' path, whereas the '2013Q4' wildcards set would have <*proj_data*> referencing the 'ProjectData\2013Q4\' path. Then, the appropriate wildcard set is selected in all the projection tasks.

4 STANDARD FEATURES

4.1 Calculation types

4.1.1 Rate generation

Historical rates are updated and new economic scenarios are generated branching from the valuation date. See the Economic scenario generation chapter in the Economic scenario book.

4.1.2 Factors

Life product projects calculate specified values per unit in force such as for cash values and statutory reserves.

4.1.3 Valuation

Valuation results are calculated for point in time valuations or for use in experience projections. New business model points (issued after the projection date) are ignored.

- **Point-in-time valuations** produce results at the projection date ($t = 0$).
- **Inner loop valuations** produce results at each valuation date for a nested projection.

4.1.4 Pricing

Special pricing measures are calculated, such as the internal rate of return (IRR).

- **Cell pricing** produces pricing results separately for each model point. Each model point must be issued on the first day of the projection. For example, a 2014/12 projection date must have all model points issued 2015/01/01.
- **New business** produces pricing results for a portfolio. Each model point must be issued during the projection. For example, a 2014/12 projection date must have all model points issued 2015/01/01 and later. New business projections can exclude assets (liability-only new business), or can include assets (asset/liability new business).

Free surplus is zero, since distributable earnings are automatically paid out, as expected for pricing measure calculations like the IRR. The initial free surplus variable (for liability-only portfolio projections) is ignored.

4.1.5 Projection

Other projection tasks, i.e., when not a rate generation, factors, valuation, or pricing type, are normal projections, such as for an experience projection. Portfolio projections can include free surplus and pay out shareholder dividends as specified for the projection; see the Capital and surplus chapter in the Profit calculations book.

4.2 Projection types

4.2.1 Portfolio versus seriatim runs

A projection or valuation can either produce portfolio or seriatim results:

Portfolio projection – A company (and business segment) projection, where investment income, taxes, target capital, distributable earnings, and free surplus are aggregate results, rather than being calculated separately for each model point. The company submodel is the top model. Shareholder dividend distributions are a specified assumption; see the Capital and surplus chapter of the Profit calculations book.

Seriatim projection – Results are calculated separately for each model point. Thus results by report group include book profit after tax, and distributable earnings. The product model is the top model. Seriatim projections are liability-only projections, since assets cannot be projected without a business segment. Free surplus is zero; distributable earnings are automatically paid out.

4.2.2 Standard versus nested projections

A projection can include 'inner loop' valuation tasks:

Standard projection – A single projection task calculating results for each stochastic projection task loop, using formulaic valuations without nested inner loop calculations.

Nested projection – The main (experience) projection is projected along 'outer' projection set loops, with 'inner' projection set sub loop valuations at various points through the projection, including the projection date ($t = 0$). The projection consists of a projection set, with a separate projection task for each type of valuation, and for advancing the outer loop experience projection. These projection tasks are repeated through each valuation date using a nested projection processing. See the time step projections chapter, below.

Both standard and nested projections can be either portfolio or seriatim projections.

4.2.3 Stochastic versus deterministic projections

Both standard and nested projections, whether portfolio or seriatim, can be either stochastic or deterministic.

Any liability-only projection, except factors calculations, can be stochastic, including pricing projections. Factors calculations, which are a special calculation type for life insurance products, are not stochastic. Projections including assets must be stochastic. Valuations can be stochastic, when appropriate.

The basis for a stochastic projection is the economic scenario file (for future scenarios) and the historic rates file (for past economic rates needed such as for calculating moving average rates during the first months of the projection). See the Economic scenarios book.

The Projection task loops property in the Run parameters set specifies which scenarios are processed. For example, specify '1-1000' to calculate results for the first 1000 sets of economic scenario rates in the scenario file. Specify '1,8,25' to run only the first, 8th, and 25th sets of rates. Specify '1' to run only the first scenario set.

Stochastic projections produce different results for each scenario when the following input variables are linked to economic rates in the scenario file:

- **Competitor rates** – Competitor rates are stochastic when the scenario rate multiple is not zero. Otherwise, the competitor rate is deterministic. See the Competitor rates chapter of the Profit calculations book.
- **Asset yield** – For liability-only projections, the asset yield is an assumption variable. When the Asset yield definition is Stochastic, define the asset yield using one of the competitor rates. Alternatively, the Asset yield definition can be Deterministic, defined as a fixed rate by calendar month.
- **Discount rates** – Discount rates that vary during the projection are stochastic when the scenario rate multiple is not zero. Otherwise the discount rate is deterministic. See the Discount rates section in the Present values chapter of the Profit calculations book.

- **Inflation rate** – The inflation rate is read from the scenario file when the Inflation definition is Stochastic. Alternatively, the Inflation definition can be Deterministic, defined as a fixed rate by calendar month. See the Inflation section in the Expenses chapter of the Profit calculations book.

A liability-only projection or valuation is deterministic when each of the above stochastic assumptions is deterministic. In this case, there is no need to specify a scenario file.

Specify the Projection task loops property in the Run parameter set as '1' to run the task only once.

Annuities and variable products

Deterministic projections are not available for annuity products, and for any product type offering variable or indexed fund values. The scenario file is required to specify those fund value growth rates.

4.2.4 Asset/liability versus liability-only

A portfolio projection or valuation (either standard or nested) can include assets.

Portfolio tasks are asset/liability projections or valuations when the Include assets? project variable (in the Run parameter set) is Yes. Otherwise, the task is a liability-only projection or valuation.

When assets are projected, the portfolio run is an asset/liability projection or valuation:

- Asset yield is the result calculated from projected assets.
- Investment income uses the projected asset cash flows.
- Capital gains are included in investment income.
- Asset values on the balance sheet, and for selling assets, are calculated from the projected assets.
- Asset based reserves (AVR and IMR) are included in the balance sheet, and thus affect profits.
- Risk based capital C1 component is based on projected asset values.
- Initial free surplus is determined by the amount of initial assets.

When assets are excluded, the portfolio task is a liability-only projection or valuation:

- Asset yield is an input assumption.
- Investment income is the asset yield applied to the asset values and cash flows.
- Capital gains are zero.
- Asset values on the balance sheet, and for selling assets, are simply the asset values and cash flows accumulated using the asset yield.
- Asset based reserves (AVR and IMR) are zero.
- Risk based capital C1 component is simply a percent of the statutory reserve.
- Initial free surplus is an input assumption.

All seriatim projections are liability-only calculations. Free surplus is zero, since distributable earnings are automatically paid out when profits are calculated separately for each model point.

4.2.5 Single business segment versus total company models

Portfolio projections can vary in scope, from a simple single business segment projection to a dynamic multiple business segment total company model projection.

Specify the business segments to include in the projection in the segments_selected variable on the Company input page under the Model Setup: Company category. Specify the segments as a comma separate list of segment ids, such as Seg1,Seg2,Seg3. Any number of business segments may be specified for multiple business segment projections.

Single business segment

The simplest type of portfolio projection has a single business segment, with a single asset portfolio. Assets backing liabilities, target capital, and free surplus earn the same investment income and capital gain rates.

When a single segment is specified in the `segments_selected` variable and 100% of target capital is invested in the business segment (the default value for the `tgt_cap_pct_inv_in_seg` variable on the Company input page in the Target Capital: Segment category), the projection is a single business segment projection. There is no corporate segment for these projections.

Single business segments and corporate segment

When a single segment is specified in the `segment_selected` variable, but you also specify that less than 100% of target capital is invested in the asset portfolio backing the business segment (i.e. the value for the `tgt_cap_pct_inv_in_seg` variable on the Company input page in the Target Capital: Segment category is less than 100%), the excess target capital, and free surplus, is invested in the 'corporate' segment. Thus assets backing capital and surplus can be invested differently than assets backing liabilities.

The corporate segment is backed by its own asset portfolio. For liability-only projections, the asset yield rates are different assumptions between the business and corporate segments. For asset/liability projections, the corporate segment has different asset model points and a different investment strategy than the business segment.

At the end of each month, the amount of business segment assets is adjusted to match the liabilities plus the amount of target capital invested with the business segment. The difference (positive or negative) from the amount of assets before this adjustment is a transfer to the corporate segment.

The corporate segment invests its starting assets and transfers from the business segment, less shareholder dividend profit distributions

Multiple business segments and corporate segment

When you more than one segment in the `segment_selected` variable, the model always includes a 'corporate' segment with the specified business segments. The excess target capital, and free surplus, is invested in the 'corporate' segment.

Each business segment and the corporate segment are backed by its own asset portfolio. For liability-only projections, the asset yield rates are different assumptions between the business and corporate segments. For asset/liability projections, each business segment and corporate segment has different asset model points and different investment strategies.

When multiple business segments and the corporate segment are included, total company taxes and target capital are determined for the company in total, with the corporate segment holding the difference between the total company amount and the sum of the amounts calculated in the business segments.

4.2.6 *Independent model points*

Use independent model point processing to reduce memory use during portfolio projections. This feature does not apply for seriatim projections, which calculate results for each model point one at a time.

Portfolio projections normally use 'period-by-period' model point processing, where all model points are held in memory and projected together through each month. This is required to calculate the asset yield for asset/liability projections, which can be used to set the crediting and dividend interest rates.

By pre-calculating results for independent model points, one at a time before the portfolio projection begins, the final portfolio projection uses less memory. A model point is independent when it can be calculated before the aggregate projection results are available; this is discussed further below.

Independent model point processing can eliminate the need for using sliding windows. Or, both features can be used together to enable larger models to fit in available memory.

It is recommended that you always use independent model point processing for long term care portfolio projections, since long term care model point calculations are never dependent on the asset yield.

Excluding assets

For liability-only projections, where the asset yield is an assumption rather than a result, every model point can be projected separately. This effectively eliminates memory constraints, so that the projection can handle a very large number of model points.

Specify to use independent model point processing by setting the Use independent model point processing? project variable (in the Run parameter set) to Yes.

The first model point is loaded into memory, and immediately calculated through the end of the projection. The model point results needed for segment and GAAP cohort calculations are loaded into a storage array, then the model point is removed from memory. Then, each successive model point is processed, adding its results to the storage array.

When the last model point has been processed, the company and segment calculations are processed to finish the projection. The calculated results are the same; independent model point processing only changes the order of the calculations.

Independent model point processing slows down the calculation speed somewhat, since the process calculates each result (needed for the storage array) in a defined order, rather than using the feature to determine the order of calculations automatically.

Including assets

For asset/liability portfolio projections, a model point is independent of the segment results only when the crediting rate (or dividend interest rate) is not based on the asset yield. The asset yield can only be calculated when all model points have been projected through each date. Thus the crediting/dividend rate cannot be based directly on the asset yield, and cannot be constrained to be within a spread from the asset yield. This determination is automatically made for each model point, based on those input variables.

When you specify independent model point processing for an asset/liability projection, the process discussed above operates for each independent projection model point. The independent model points are loaded into the storage array, for use during the portfolio projection calculations.

The portfolio projection begins after all independent model points have been processed. The results for independent model points are simply read from the storage array. The remaining (dependent) model points are now projected as usual during the portfolio projection. Company, segment, and asset results are calculated as the total of the independent and dependent model point results. Thus the projection can combine both independent and dependent products together.

To make a model point independent, define the crediting rate (or dividend interest rate) using competitor rates, without reference to the asset yield. This definition takes the view that the crediting/dividend rate is primarily driven by market forces, not whatever the company's investment portfolio happens to produce. Appropriate investment strategies must be determined in order to achieve the desired spread from the asset yield.

The projection checks each model point to determine whether its crediting/dividend rate is linked to the asset yield (dependent model point) or not (independent model point). (The result of this check is stored in the product submodel scalar, `independent_flag`.)

4.3 Product types

4.3.1 Term life

The key features for the term life product are:

- Level term, ART, Level term to ART, and mortgage decreasing term product variations
- Return of premium term plan or return of premium rider
- Net level, FPT, SNFL, or table base plan cash values with nonforfeiture exception rule testing
- AG45 or percent of premium ROP plan cash values
- ROP term plan reserve options including Unusual Cash Value method
- Mortality anti-selection using Dukes-McDonald or table
- Shock lapse varying by premium increase percentage at end of level term period
- Term conversion decrement
- Substandard multiples on experience and reserve mortality
- YRT, coinsurance, and modified coinsurance, with capabilities for multiple reinsurance treaties
- Redundant reserve securitization through letter of credit
- Accidental death benefit riders

4.3.2 Traditional life

The key features for the traditional (permanent) life product are:

- Participating or non-participating
- Single life, Joint life, or Frasierized last survivor
- Death and endowment benefits
- Active and RPU policies
- Net level, FPT, SNFL, or table base plan cash values
- Policy loans
- Experience mortality improvements
- Term conversion mortality treatment
- Specified, contribution component, or component adjusted input dividend scales
- Cash, accumulate, PUA, or OYT/PUA dividend options
- YRT, coinsurance, and modified coinsurance, with capabilities for multiple reinsurance treaties
- Accidental death benefit riders

4.3.3 Universal life

The key features for the universal life product are:

- Sum assured, Plus Fund, Plus ROP base death benefit
- Extended maturity option
- Single life, First-to-die joint life, or Frasierized last survivor
- Cumulative premium or shadow fund based secondary guarantees
- Multiple shadow fund structure

- Partial withdrawals
- Policy loans
- Experience mortality improvements
- Term conversion mortality treatment
- Base and competitive driven excess lapses
- YRT, coinsurance, and modified coinsurance, with capabilities for multiple reinsurance treaties
- Accidental death benefit riders

4.3.4 *Variable universal life*

The key features of the variable universal life product are:

- Sum assured, Plus Fund, Plus ROP base death benefit
- Extended maturity option
- Single life, First-to-die joint life, or Frasierized last survivor
- GMDB based on cumulative premiums test
- Partial withdrawals
- Policy loans, with loaned fixed fund tracking and transfers to/from variable funds
- Fixed fund and up to 6 variable funds (can customize for more)
- Dollar cost averaging for user specified months from issue
- Account value rebalancing
- Asset yield and competitive based crediting strategies on fixed fund
- Experience mortality improvements
- Term conversion mortality treatment
- Base and competitive driven excess lapses
- YRT and modified coinsurance, with capabilities for multiple reinsurance treaties
- Accidental death benefit riders

4.3.5 *Indexed universal life*

The key features for the indexed universal life product are:

- Sum assured, Plus Fund, Plus ROP base death benefit
- Extended maturity option
- Single life, First-to-die joint life, or Frasierized last survivor
- Cumulative premium secondary guarantees
- Point to point, Moving average, Monthly sum cap, crediting designs with premium payment bucket structure
- Dynamic indexing cap calculations
- Partial withdrawals
- Policy loans
- Experience mortality improvements
- Term conversion mortality treatment

- Base and competitive cap driven excess lapses
- YRT, coinsurance, and modified coinsurance, with capabilities for multiple reinsurance treaties
- Accidental death benefit riders

4.3.6 Fixed annuity

The key features of the fixed annuity product are:

- Single or recurring premium
- Fund or cash value death benefit
- Penalty free withdrawals
- Market value adjustments
- Bail out provisions
- Asset yield and competitive based crediting strategies
- Experience mortality improvements
- Base and competitive driven excess lapses
- Annuitizations treated as surrender
- Coinsurance

4.3.7 Fixed indexed annuity

The key features of the fixed indexed annuity product are:

- Single premium only
- Fund or cash value death benefit
- Penalty free withdrawals
- GMWB benefit rider
- Fixed rate, Point to point, Moving average, Monthly sum cap, crediting designs
- Dynamic participation and indexing cap calculations
- Experience mortality improvements
- Base and competitive driven excess lapses
- Annuitizations treated as surrender
- Coinsurance

4.3.8 Payout annuity

The key features of the payout annuity product are:

- Single premium only
- Immediate annuity, installment refund annuity, cash refund annuity, and structured settlement policy types
- Certain only, single life, and joint and last survivor
- Multiple benefit streams or 'months since issue' tables to define annuity benefits
- Solve for benefit or premium at issue, using annuity purchase rate calculation
- Actual status last survivor treatment with user specified survivor status at projection date

- User specified reduced survivor benefit percentage
- Experience and reserve mortality improvements
- Experience and reserve substandard extra deaths
- Coinsurance

4.3.9 Variable annuity

The key features of the variable annuity product are:

- Single or recurring premium
- Fund or cash value death benefit
- Penalty free withdrawals
- All four major GMxB features, with ability to mix/match on single model point
- Enhanced earnings benefit
- Fixed fund and up to 6 variable funds (can customize for more)
- Dollar cost averaging for user specified months from issue
- Account value rebalancing
- Asset yield and competitor based crediting strategies on fixed fund
- Experience mortality improvements
- Base and rider's in-the-moneyness driven excess lapses
- Annuitizations treated as surrender
- Modified coinsurance

4.3.10 Long term care

The key features of the long term care product are:

- Support for three care types out-of-box; home health care, assisted living facility, and nursing home
- Daily long term care and premium waiver benefits
- Benefit inflation
- Separate benefit utilization from specified daily benefit
- Return of premium on death, lapse, or as a maturity benefit
- Active and RPU premium status
- Premium refunds and rate increases
- Mortality improvement and anti-selection
- Base and shock lapses
- Claim projection using full first principles, a first principles approximation method, or with specified claim costs
- Option for exposure to apply to policies not on claim for full first principles projections
- Recovery from and transfer between care types for full first principles projections
- YRT, coinsurance, and modified coinsurance, with capabilities for multiple reinsurance treaties

4.4 Asset types

4.4.1 *Bond*

The key features of the bond asset type are:

- Fixed or adjustable coupon rate
- Zero coupon and original issue discounts
- Sinking fund provision
- Call and put options
- Option price schedules and make-whole provisions
- Municipal bonds
- Deterministic and stochastic defaults
- Financial based option exercise

4.4.2 *Mortgage*

The key features of the mortgage asset type are:

- Fixed or adjustable payment rate
- Servicing fees
- Delayed amortization
- Balloon payment on specific date/period or per a schedule
- Prepayments
- Prepayment penalties and make-whole provisions
- Deterministic and stochastic defaults
- Refinance, restructure, foreclosure decrements
- Nonfinancial and financial based prepayment exercise, including link to Andrew Davidson prepayment model
- Link to Intex Solutions library

4.4.3 *Securitized asset*

The key features of the securitized asset type are:

- A, B, C, Z, and PAC 2A owned tranche structure
- Supporting tranches and PACs
- Collateral asset characteristics similar to mortgage asset type
- Deterministic and stochastic defaults
- Link to Intex Solutions library

4.4.4 *Interest rate derivative*

The key features of the interest rate derivative asset type are:

- Swap, cap, or floor structure
- Fixed and floating rates
- Rate specific interest rate term, multiple, spread, and payment mode

- Delayed payment start
- Deterministic and stochastic defaults

4.4.5 Real estate (property)

The key features of the real estate (property) asset type are:

- Linked to scenario index
- Delayed payment start
- Market value growth adjustments
- Rental growth
- Depreciation

4.4.6 Equity

The key features of the equity asset type are:

- Linked to scenario index
- Market value growth adjustments
- Dividend yield adjustments

4.4.7 Equity index option

The key features of the equity index option asset type are:

- Linked to scenario index, with multiple and addition adjustments
- Call or Put option, or Futures contract
- Point to point, Highest gain, Asian lookback
- Flexible lookback years and sampling interval
- Maximum payout index value

4.5 Reserve and capital calculations

4.5.1 Formulaic based reserves

The key features for formulaic reserves are:

- Net level, FPT, and CRVM reserves for term life, traditional life, and long term care
- Regulation XXX, New York Regulation 147, and ROP term plan reserves for term life
- UL CRVM, Regulation XXX, New York Regulation 147, AG 38-8A, AG 38-8B, AG 38-8C, AG 38-8D, and AG 38-8E reserving for universal life
- UL CRVM and AG 37 for variable universal life
- CARVM AG 33 for fixed annuity, fixed indexed annuity, and payout annuity
- CARVM AG 34 and AG 39 for variable annuity
- CARVM AG 35 for fixed indexed annuities

4.5.2 Formulaic based capital

The key features for formulaic capital are:

- RBC C1 percent of statutory liabilities (liability model), and asset based for ALM

- RBC C2 percent of mortality and morbidity
- RBC C3 percent life insurance net reserves
- RBC C3 percent annuity net reserves, split by low risk, medium risk, and high risk annuities
- RBC C4 percent annualized premiums (excluding single premium) and percent separate account liabilities

4.5.3 Principles based reserves

The key features for the principles based reserves are:

- AG 43 for variable annuity product reserves
- VM 20 for life insurance products reserves
- VM 22 exposure for fixed annuities
- AG 48 required level of primary security for letter of credit reserve financing

4.5.4 Principles based capital

The key features for the principles based capital are:

- RBC C3 Phase II for variable annuities
- RBC C3 Phase III for life insurance and fixed annuities

4.5.5 U.S. GAAP

The key features for the U.S. GAAP calculations are:

- SFAS 60 factors for term life, traditional life , and long term care products
- SFAS 91 schedules for payout annuity
- SFAS 97 limited pay factors for term life, traditional life, payout annuity, and long term care products
- SFAS 97 retrospective deposit schedules for universal life, indexed universal life, variable universal life, fixed annuity, fixed indexed annuity, and variable annuity, incorporating SFAS 133/157 and SOP 03-1
- SFAS 120 schedules for traditional life
- SFAS 115 asset sales class treatment

5 STANDARDS

5.1 Naming standards

Variables and formulas follow the same naming convention, so that formulas have names consistent with the input variables which affect them. Names typically begin with the main concept, followed by types, ending in variations. Common abbreviations and acronyms are used to shorten the names to make them easier to understand in formulas.

The description of each formula spells out the full words which are abbreviated in the name. For variables, the descriptions are intended to make the 'input screens' (input pages) easier to understand.

5.2 Abbreviations

The following are the standard abbreviations used in constructing names:

Abbreviation	Replaces
acctg	accounting
accr	accrued
accum	accumulated, accumulation (typically with interest)
acq	acquisition
addn	addition
addl	additional
adj	adjusted, adjustment
admin	administration
aft	after
alloc	allocation
allow	allowance
alt	alternate
amortzn	amortization
amt	amount
anniv	anniversary
annuitzn	annuitization
ann	annuity
anticip	anticipated
appl	applicability, applicable
approx	approximation
assess	assessments
class	asset class
assign	assignment
avail	available
avg	average
bal	balance
bef	before
beg	beginning
ben	benefit
bon	bonus
bk	book
borrow	borrowing, borrowed
break	breakpoint
bus	business
calc	calculation
cal	calendar
cap	capital
capzn	capitalization
capzd	capitalized
chg	charge
chgs	charges
coins	coinsurance

Abbreviation	Replaces
coll	collected
comb	combination, combined
comm	commission
comp	competitor
concurr	concurrent
contribn	contribution
cumul	cumulative (typically without interest, i.e., not accumulated)
curr	current
dth	death
dths	deaths
decr	decrease
dflt	default
def	deficiency
defn	definition
denom	denominator
dep	deposit
diff	difference, differential
disab	disability income
disc	discount
distrib	distributable
distribn	distribution
div	dividend
dur	duration
dyn	dynamic
earn	earned
eff	effective
endow	endowment
eqt	equity
exceptn	exception
excl	excluding
exp	expense
exper	experience
expon	exponent
face	face amount
freq	frequency
func	function
grp	group
guar	guarantee, guaranteed
guide	guideline
illus	illustration
imm	immediate
improve	improvement
incid	incidence
inc	income
incr	increase

Abbreviation	Replaces
incur	incurred
indem	indemnity
info	information
inf	in force (following the subject), or in-force (preceding the subject)
infl	inflation
init	initial
ins	insurance
int	interest
interp	interpolation
inv	investment
iter	iteration
lapse	lapse
liab	liability
life1	life 1 (usually excluded from names)
life2	life 2
maint	maintenance
match	matching
mgmt	management
mkt	market
max	maximum
min	minimum
modco	modified coinsurance
mth	month
mothly	monthly
mths	months
morb	morbidity
mort	mortality
mtg	mortgage
mult	multiple
neg	negative
nonf	nonforfeiture
num	number
numer	numerator
obj	objective
opt	option
occ	occupation
par	par value
wthdrl	partial withdrawal
pmt	payment
pct	percent, percentages
persist	persistency
pol	policy
port	portfolio
prelim	preliminary
prem	premium

Abbreviation	Replaces
waiver	premium waiver
prepm	prepayment
prev	previous
prin	principal
phldr	policyholder
prod	product
proj	projection
purch	purchase
purp	purpose
qtr	quarter
qtrly	quarterly
qtrs	quarters
rebal	rebalancing
ref	refund
reins	reinsurance
reinv	reinvestment
rpt	report
req	required
res	reserve
restrict	restricted
rev	revenue
scen	scenario
sch	schedule
sched	scheduled
seg	segment
settle	settlement
shldr	shareholder
sprd	spread
std	standard
stat	statutory
stoch	stochastic
strat	strategy
struct	structured
surp	surplus
surr	surrender
tbl	table
tgt	target
termin	termination
thresh	threshold
trad	traditional
ult	ultimate
unadj	unadjusted
undistr	undistributed
unrestrict	unrestricted
valn	valuation

Abbreviation	Replaces
val	value
wthdrl	withdrawal
wo	without
yr	year
yrs	years

5.3 Acronyms

The following are the standard acronyms used in constructing names:

Acronym	Replaces
abs	asset based security
alr	active life reserve
ag	actuarial guideline (followed by cardinal number, like ag43)
arb	adjustable rate bond
arm	adjustable rate mortgage
art	annually renewable term
alm	asset liability model
avr	asset valuation reserve
air	assumed investment rate
bocy	beginning of calendar year
bom	beginning of month
bopy	beginning of policy year
boq	beginning of quarter
bs	Black Scholes
qtd	calendar quarter to date
cm	calendar month (when part of a name; cal_mth when by itself)
cy	calendar year (when part of a name; cal_yr when by itself)
ytd	calendar year to date
cp	collateral pool
coi	cost of insurance
carvm	commissioners annuity reserve valuation method
crvm	commissioners reserve valuation method
cusip	committee on uniform security identification procedures
da	deferred annuity
dac	deferred acquisition cost
dlr	disabled life reserve
eocy	end of calendar year
emo	extended maturity option
eom	end of month
eopy	end of policy year
eoq	end of quarter
eio	equity index option
egm	estimated gross margin
egp	estimated gross profit
eti	extended term insurance
epa	externally projected assets

Acronym	Replaces
epl	externally projected liabilities
fia	fixed indexed annuity
fy	first year
ga	general account
gt	greater than
gaap	generally accepted accounting principles
gmp	guaranteed maturity premium
gmdb	guaranteed minimum death benefit
gmab	guaranteed minimum accumulation benefit
gmib	guaranteed minimum income benefit
gmwb	guaranteed minimum withdrawal benefit
glp	guideline level premium
gsp	guideline single premium
hel	home equity loan
id	identification
ia	index asset
imr	interest maintenance reserve
io	interest only
ird	interest rate derivative
irr	internal rate of return
iul	Indexed universal life
lt	less than
loc	letter of credit
ltc	long term care
mva	market value adjustment
mvrn	market value reserve method
mp	model point
mec	modified endowment contract
nar	net amount at risk
np	net premium
nlp	net level premium
nsp	net single premium
ny	New York (followed by regulation number, like ny147)
oyt	one year term rider
opt	option
pa	payout annuity
pu	paid up
pua	paid up addition
pfw	penalty free withdrawal
py	policy year (when part of a name; pol_yr when by itself)
pytd	policy year to date
pp	premium paying
pv	present value
po	principal only
pba	principle-based approach

Acronym	Replaces
pdf	probability distribution function
pad	provision for adverse deviation
rng	random number generator
re	real estate (property)
rpu	reduced paid up
ry	renewal year
rop	return of premium
roe	return on equity
rbc	risk based capital
sg	secondary guarantee
sa	separate account
sec	securitized asset
sop	standard of practice (followed by # without space, like sop031)
snfl	standard nonforfeiture law
ul	universal life
ulcrvm	universal life commissioners reserve valuation method
url	unearned revenue liability
vpt	vanishing premium test
vagb	variable annuity guarantee benefits
vaglb	variable annuity guarantee living benefits
vul	variable universal life
wac	weighted average coupon
wal	weighted average life
wl	whole life
yrt	yearly renewable term

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G02 CODE MANAGEMENT

This book discusses code management of RiskAgility FM U.S. Library projects.

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1 CODE MANAGER STRUCTURE

1.1 Principles

1.1.1 Common structure

Although each project can be used by itself, it is important to view the projects as a collection which is developed, maintained, and delivered as a single solution.

Each project has a common structure, such as definitions for project variables, DLLs/LIBs, Externs, model classes, submodel structures, and calculation structure. Functionality is encapsulated into reusable components as much as practical. These components are consistent across product types. This promotes consistency of definitions and approaches both within and between the projects.

1.1.2 Common functions

All projects include reusable components for functions that are common across the projects. The term 'component' represents a DLL/LIB, Extern, model class, or set of definitions for a particular business domain. For example, the rates_economy model class is a reusable component, so is the seg_comp model class. Using common functions reduces development, maintenance, and customization efforts.

In order to ensure reusability, development and maintenance of the shared components must consider the wide range of uses and variations which are required for the component. Careful management of the reusable components must also be considered.

1.1.3 Encapsulation

Many of the project components are also developed to provide some level of encapsulation. In generic programming terms, encapsulation is an approach by which objects hide their complexities from other objects and communicate through a well-defined interface.

While RiskAgility FM itself does not provide a high degree of encapsulation (since every variable and formula is exposed to all other model classes), several model classes have been developed which are intended to have a high degree of encapsulation. Generally, the approach taken for these model classes is through an approach of setting up variables as parameters, using 'get' functions or columns to retrieve results, and then using reset() or resetValues() to clear out the results before the next use.

1.1.4 Consistency

Maintaining consistency of the components across the projects is critical to ensure integrity of the code. A manual process must be employed which ensures that all components are developed and maintained properly, since RiskAgility FM does not automatically maintain consistency of the components across projects.

The first key to ensuring consistency is agreeing where the single source for the definitions of that component reside. Once this is agreed, a procedure must be put in place to ensure that changes to the 'master' definitions for that component are made only in that project. For example, it may be decided that a component be declared as having its source definitions in the variable annuity project. Then all 'master' changes to the component should then be made only in the variable annuity project.

The second key is ensuring that comprehensive coverage and variation support are maintained. This includes understanding the current, and future, use of the component to ensure that changes made will not adversely impact other parts of the library.

The third key is agreeing on an 'integration' strategy and developing integration procedures. Changes in consistent components must be deployed to the projects which use them in a structured way and at appropriate times, using utilities of the RiskAgility platform such as model class compare. Without a successful integration strategy, the components will quickly lose consistency across the projects.

1.2 Project variables

All projects include project variables for run parameters which are specific to the calculations of the library, such as the projection date, whether to include assets, etc.

The following project variables are included:

Project variable	Description
Projection date [proj_date]	In-force values, and present value calculations, are as of the end of the month specified as yyyy/mm, such as 2014/12 for December 31, 2014
Pricing projection? [pricing_defn]	Yes for projections calculating special pricing results like IRR (all model points must be issued at the start of a seriatim projection [cell pricing], or during a portfolio projection [new business pricing]); No otherwise
Valuation years [valn_yrs]	Valuation tasks (target column is valn_tgt_...) project this many years from the projection date; Except that GAAP valuations ignore this variable since they project to the DAC amortization years; Has no effect on projection tasks since they project to the 'Projection period to' property
Project GAAP results? [gaap_defn]	Yes for projection tasks to include GAAP calculations, reading GAAP factors/schedules from the output specified from previous valuation tasks; No otherwise;
Include assets? [assets_defn]	Yes for portfolio projection and valuation tasks to project asset/liability results; No for liability-only task;
Include target capital? [tgt_cap_defn]	Yes for projection tasks to include target capital results; No to treat target capital as zero; No otherwise;
Use independent model point processing? [independent_mp_defn]	Yes for portfolio projection and valuation tasks to project 'independent' model points using seriatim calculations before starting the portfolio calculations; No otherwise;
Validate data definitions? [validate_data_defn]	Yes to include optional input checking to produce warning messages in the run log; No otherwise
Include decrements? [decrements_defn]	No to treat mortality, lapse, and annuitization decrement assumptions as zero; Yes for normal calculations
Discount rates selected [disc_rates_selected]	Specify stochastic discount rates using a comma separated list with no spaces; these discount rates are specified on the Company input page
Portfolio yield definition [disc_rate_port_yld_defn]	Before Tax to discount using the asset yield; After Tax to discount using the asset yield times (1 – tax rate)
Valuation periods selected [valn_periods_selected]	Valuation period for each projection set sub loop. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month 24, and so on. The first valuation period must be 0.

Project variable	Description
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs which generate large scenario files.

1.3 DLLs/LIBs

Links to external libraries are used in the RiskAgility FM U.S. Library to provide functionality to the project which has been developed and packaged externally from RiskAgility FM into dynamic link libraries (DLLs) and/or associated library files (LIB). These external libraries are referenced using the DLLs/LIBs container of the Code Manager of the project.

The following external libraries are included:

DLLs/LIBs name	Description
ActuarialMathLibrary DLL 32bit	32bit version DLL of the Towers Watson actuarial math library which calculates formulaic cash values and base plan reserves for term, traditional life, and long term care products as well as reserves for riders.
ActuarialMathLibrary DLL 64bit	64bit version DLL of the Towers Watson actuarial math library which calculates formulaic cash values and base plan reserves for term, traditional life, and long term care products as well as reserves for riders.
ActuarialMathLibrary LIB 32bit	32bit version LIB of the Towers Watson actuarial math library which calculates formulaic cash values and base plan reserves for term, traditional life, and long term care products as well as reserves for riders.
ActuarialMathLibrary LIB 64bit	64bit version LIB of the Towers Watson actuarial math library which calculates formulaic cash values and base plan reserves for term, traditional life, and long term care products as well as reserves for riders.
ADCo DLL 32bit	32bit DLL of the Andrew Davidson prepayment model library
ADCo DLL 64bit	64bit DLL of the Andrew Davidson prepayment model library
ADCo LIB 32bit	32bit LIB of the Andrew Davidson prepayment model library
ADCo LIB 64bit	64bit LIB of the Andrew Davidson prepayment model library
Economy DLL 32bit	32bit DLL of the Towers Watson economy library which provides for retrieval of scenario rates and is used to generate new scenario files for inner loop valuations

DLLs/LIBs name	Description
Economy DLL 64bit	64bit DLL of the Towers Watson economy library which provides for retrieval of scenario rates and is used to generate new scenario files for inner loop valuations
Economy LIB 32bit	32bit LIB of the Towers Watson economy library which provides for retrieval of scenario rates and is used to generate new scenario files for inner loop valuations
Economy LIB 64bit	64bit LIB of the Towers Watson economy library which provides for retrieval of scenario rates and is used to generate new scenario files for inner loop valuations
RAFM SQLite LIB 32bit	32bit LIB of the Towers Watson RiskAgility FM SQLite library which is used for dependent task processing to retrieve results from the output files into the projection
RAFM SQLite LIB 64bit	64bit LIB of the Towers Watson RiskAgility FM SQLite library which is used for dependent task processing to retrieve results from the output files into the projection

1.4 Externs

All projects include global values and functions which are applicable across all projects of the solution. These items are defined using the Externs container of the Code Manager of the project.

There are two types of global values:

- Global constants are named with all capital letters, and are assigned specific values as part of their definitions. All references to these values use the global value name as opposed to the assigned value.
- Global calculation flags and variables are named using lower case letters, and are assigned values as the projections are run. The global flags (variables which include '_flag' in the name) are set to either true or false, based on the values of model parameters during the model startup. The global variables are filled with values as required during initialization or during a specific calculation.

Functions which perform generic tasks, such as rounding a result or retrieving a month from a date character variable, are declared and implemented within the Externs container so that they are accessible across the model classes of the project. These functions are mostly generic C++ routines, and use parameters for all data. They do not reference any specific submodels or complex special constructs.

The following Extern elements are included:

Extern name	Description
All Model Classes	'Externed' global variables and function declarations referenced by all model classes. This provides for the appropriate visibility of the global values and functions across all model classes of the project.
Intex	Declarations of data structures for linking to the library from Intex Solutions, Inc.

Extern name	Description
Rates Only	Declarations for the global values and functions referenced by the rates_economy model class. The rates_economy model class is used since this provides a single model class to declare the variables and functions which is used in all projects

1.5 Model classes

All projects include model classes required to support the calculation results, assets, and products included in the solution. The model classes contain the variables and formulas required to model the calculations of the business objects.

Since business objects generally fall into Company, Economy, GAAP, PBA, Asset, and Liability groups, the model classes have been organized into the group by including the group as part of the model class name. The model classes are organized into the groups as follows:

Model class group	Model classes
Company	comp_comp, hdg_grp_comp, seg_comp
Economy	rates_economy
GAAP	sfas60_gaap, sfas91_gaap, sfas97lp_gaap, sfas97rd_gaap, sfas120_gaap, sfas133_gaap
PBA	ann_pba_pba, life_pba_pba, va_pba_pba
Asset	a_subport_asset, adco_asset, asset_asset, bond_asset, bond_cf_asset, eio_asset, epa_asset, eqt_asset, intex_asset, invstrat_asset, ird_asset, mtg_asset, mtg_cf_asset, re_asset, sec_asset, sec_cf_asset, sec_cp_asset

Model class group	Model classes
Liability	liab_liab, epl_liab, fa_liab, facarvm_liab, fia_liab, fiaafund_liab, fiacarvm_liab iul_liab, iul_ifund_liab, iulafund_liab, iul_py_liab, iulcvres_liab ltc_liab, ltc_cc_liab, ltc_ccct_liab, ltc_fp_liab, ltc_fpct_liab, ltc_py_liab pa_liab, paben_liab, pavaln_liab term_liab, term_py_liab trad_liab, trad_div_liab, trad_py_liab ul_liab, ulcvres_liab, ul_ifund_liab, ul_py_liab va_liab, va_afund_liab, vacarvm_liab vul_liab, vulafund_liab, vulcvres_liab, vulifund_liab, vul_py_liab
Reinsurance	treaty_reins
Riders	rider_liab rider_py_liab

More details regarding the calculations of the various model classes are included in the Submodel structures sections of other books in this user guide.

2 MODEL CLASS STRUCTURE

2.1 Principles

2.1.1 Comprehensive coverage

All model classes are developed to provide coverage for a range of uses. In order to be comprehensive, a model class may be required to include definitions which don't apply for all projects.

For example, segment level separate account functionality is only required for variable product projects. It is not required for the universal life project. However, since basically all other segment level functionality is common across variable annuity and universal life, a comprehensive seg_comp model class must contain the separate account items and then, since the seg_comp model class is included consistently in variable annuity and universal life, these columns reside in both projects. Including these extra definitions in the universal life project is a better approach than have two separate seg_comp model classes.

2.1.2 Variations support

All model classes are also developed to provide the ability to support variations across the range of uses. For example, a model class may be required to support product-specific variations and calculation result variations, while still being easily included in any project of the solution and in projections which process a portfolio.

For example, SFAS97 GAAP functionality is required for both the variable annuity and universal life projects, and also is required for portfolio and cell pricing projections. Sharing one common model class for this functionality is better than implementing four distinct SFAS97 GAAP model classes.

The key to variations support is the use of compiler directives and unique model pointer conditions. Consider this sample snippet of code for a column of the sfas97rd_gaap model class:

```

acq_exp(sfas97rd_gaap) -> X
1  if (t <= start_period || t > final_period)
2      return NO_AVG;
3
4  if (this == sfas97rd)
5      return sum_over_gaap("acq_exp", t);
6
7  #if defined(__UL_LIAB_H_)
8  if (this == ul_sfas97rd)
9  {
10     if (!reins_flag) //Gross
11         return ul->acq_exp(t);
12
13     //Acquisition expenses are not reinsured.
14     return 0.0;
15 }
16 #endif
17
18 #if defined(__VUL_LIAB_H_)
19 if (this == vul_sfas97rd)
20 {
21     if (!reins_flag) //Gross
22         return vul->acq_exp(t);
23
24     //Acquisition expenses are not reinsured.
25     return 0.0;
26 }
27 #endif
--

```

Variation support across projects is provided through the compiler directive conditions of “`#if defined(__UL_LIAB_H_)`” and “`#if defined(__VUL_LIAB_H_)`”. The global constants “`__UL_LIAB_H_`” and “`__VUL_LIAB_H_`” are defined automatically when the `ul_liab` or `vul_liab` model class exists in a project. Of course, since there are separate projects for universal life and variable universal life, the “`__UL_LIAB_H_`” global constant will not be defined in the variable universal life project, nor will the “`__VUL_LIAB_H_`” be defined in the universal life project. At compile time, the compiler directive conditions are evaluated by the C++ compiler. If the compiler directive condition is true, the code within the `#if` and `#endif` delimiters is included in the compiled code. Thus, the compiled code for the variable annuity project will not even include the code for the universal life case.

Variation support for calculation results within a project is provided through the unique model pointer conditions using the ‘`this`’ pointer. Every object has access to its own address through the ‘`this`’ pointer. When there are multiple instances of submodels, the ‘`this`’ pointer can be compared to the pointer addresses created based on the unique submodel names. If the condition is true, then the code for that context is executed. By using this approach, the `sfas97rd_gaap` model class interfaces with the product model classes for cell pricing as well as the portfolio model classes, and functions in multiple model structures.

2.2 Categories

All model classes are designed with a similar category tree structure, which facilitates organizing variables and formulas together into logical groupings.

The top level categories separate the variables and formulas by major categories, such as Run Control, Company, Economy, Investment, and product level groupings. These major categories correspond directly to the input management structure, discussed in the Input Management book.

The second level categories collect the variables and formulas for specific features within the major categories. For example, the Competitor Rates category of the Company category collects the variables and formulas for parameterizing and determining competitor rates.

When a feature has several options, there may be a third level category included. However, third level categories are not common.

2.3 Externs

Variables whose scope must cover across the objects of a model classes are defined in the START_EXTERNS/END_EXTERNS section of the startup column for the model class. This includes special data containers, such as maps, and constants.

When the variable's scope is required across objects of other model classes, such as to the children of a parent model class, the variables are 'externed' in the START_EXTERNS/END_EXTERNS section of the startup of the other model class.

2.4 Class variables

Class variables not expected to be set by the user, which cannot be represented as scalars, are created as class variables in the START_CLASS/END_CLASS startup column for the model class. This is commonly used for special data containers, such as maps and SmartArrays and is also used for initial variables for the model classes whose initial values may be set within the code.

All class variables defined in START_CLASS/END_CLASS are initialized at object creation by specifying initialization code within the START_CONSTRUCTOR/END_CONSTRUCTOR section of the startup column. As, when required, there is also termination code within START_TERMINATOR/END_TERMINATOR.

2.5 Variables

Variables which are expected to be set by the user or are required for data retrieval processing are defined as RiskAgility FM model class variables.

The most common variable types used in the model classes are floating point number, integer number, and character. Floating point arrays and integer arrays are rarely used and the variable type of Table is never used. Floating point number and Integer number variables are used for all numeric data and Character variables are used for variables which represent standard character values or choice lists.

- Variables are categorized and ordered logically within the category
- Variable names adhere to the naming and abbreviation standards.
- Variables include appropriate default values and, for numeric variables, ranges and decimals. Six is typically specified as the number of decimals for the floating point number variables.
- Variables include a description which is in 'sentence case'. That is, the first letter of the description is capitalized, but other words are not capitalized except as needed for acronyms. The descriptions typically end with a colon (":") since this is an input prompt but may end in a "?" if the description asks for a question to be answered. The descriptions are not help messages. They are short and do not contain instructions. Instructions are included in the related help message.
- Variables include a help message which complements the information provided in this user guide. The help messages provide detailed information regarding the treatment of the variable in the calculations and, for choice lists, the effect of each of the choices.
- When a character variable is a choice list, the Character Type of 'Enumerated, Error if Undefined' is used, which provides for improved performance of comparisons since the comparison value is an enumeration and not a string, and produces an error if the data is invalid.
- Variables which are expected to be set by the user have the 'Set Value in Input Manager' property defined as 'All'. However, variables which are set in the code and used for data retrieval through the input manager, referred to as a 'Hidden lookup key' will have this property set to User Defined with no submodels selected through the Set Value in Input Manager dialog.

2.6 Scalars

Scalars are used for results which are not dependent on time, or for parameters that are calculated at run-time:

- For a value set at the start of the calculation depending on other input variables.
- For business object characteristics, such as gender, in submodel classes instead of duplicating the variables to the submodel classes.
- For calling columns at set periods, such as at issue, the projection date, and maturity, to reduce sliding windows sizes.

For calculation results which are unaffected by time, such as for a monthly accumulation factor of a fixed rate.

Since RiskAgility FM does not force a scalar to calculate just by naming it in a formula line, code must be added to force scalar calculations within a formula. This is done using this standard expression that makes it clear what the formula is doing:

```
double force_scalar_calc = [scalar name];
```

2.7 Columns

Columns are used for a calculation result which is dependent on time or are desired to be included in the output.

- Columns are categorized but are in alphabetical order within the category
- Column names adhere to the naming and abbreviation standards.
- Column headers are set equal to the column names
- Column 'Combine Groups By' properties are set as required for group reporting where Average is used for columns which cannot be summed properly across model points, such as ages and rates
- Column 'Discount Use' and 'Discount Timing' properties are set to allow for columns to calculate discounted values. The discount use of Yes is used for cash flow columns and No for balance sheet columns. The discount timing is used to specify the timing in a projection period the cash flow is assumed to occur. The timing choices of Beginning and End are used appropriately based on the timing of the treatment for the column calculation. A choice of Middle is used when the column is determined using a formula containing results from a mix of cash flow timings.
- Column 'Rate Use' is set as required for discounting. A rate use of 'Stream' is used for those columns which represent discount rates, such as disc_rate_1, disc_rate_2, disc_rate_3, and disc_rate_port_yld which apply as a stream discount rate to the cash flow columns. The 'Spot' is not used.
- Column 'Rebase Type' is set as required for rebasing of asset and liability calculation, such as for present value calculations for bonds and reserving calculations for universal life.

2.8 Temporary tables

Temporary tables are used in the following cases:

- When a 'two dimensional' column is required, for example, one which varies by time and a second dimension
- When results need to be held across data points without recalculation.

Examples of temporary tables include the IMR amortization tables of the seg_comp model class and the capitalized premium tax amortization schedules. These are temporary tables since the values vary by projection month and relative calendar year.

Generally, temporary tables in model classes which will be arrayed have been avoided in favor of SmartArrays.

2.9 External functions

External functions are used in the projects to:

- Hold a subroutine that is called many times
- Collect calculations that use multiple parameters and create results which are used but do not need to be stored or included in output results
- Implement a procedure

Extensive use of external functions exists in the rates_economy model class to retrieve rates from the scenario files and in the invstrat_asset model class to drive the purchase and sales process for the assets. External functions are also implemented in a number of model classes to sum over elements of the arrays of other model classes.

2.10 Equality comparisons

Equality comparison thresholds are used since checking equality of a floating point values to a constant value, such as 0.0, will likely never succeed. So, instead all projects use a fabs comparison against the threshold amount.

The following threshold amounts are defined as global values:

- rate_ratio_threshold (10^{-6}) for rates and similar values
- model_point_amount_threshold (10^{-4}) for results for model point results
- aggregate_amount_threshold (10^{-2}) for business segment and company results.

Compare to direct values are used when the test is not for equality, such as “cash_val > 0.0” and “pol_loan < cash_val”.

3 SUBMODEL STRUCTURE

3.1 Standalone submodels

All projects use standalone submodels to represent a single instance of a business object. Standalone submodels are used for company, rates, asset, liab, intex, and adco.

Individual asset type and product type submodels, such as bond and ul, are treated as standalone when running with the asset or product as the target model object.

3.2 Array submodels

All projects use data driven, dependent data driven arrays, and user defined arrays for all arrayed submodels. Data driven arrays are used for all base liability and asset model points. Dependent data driven arrays are used for payout annuity benefit streams, reinsurance records, and riders. User defined arrays are used for arrays with multiple elements, such as the subaccount elements for variable annuity model points. Fixed Arrays are not used, due to their significant memory overhead.

Sizing of the data driven and dependent data arrayed submodels are determined based on the number of model points to be processed in the projection. The sizing of the data driven array is handled automatically through the RiskAgility FM core.

When asset purchases occur, the data driven arrays for the asset model points, such as bond, are increased in size to add the newly purchased model points.

Sizing of the user defined arrayed submodels is determined based on entries for specific variables, such as segments_selected of the company submodel or the gaap_cohorts_selected_sfas97rd variables of the segment submodel. Code is included in startup to resize the arrays based on variable values.

3.3 Clone submodels

Clone submodels are only used for present value calculations for bonds, mortgages, and securitized assets.

Since clones require significant memory, special coding is included to reuse the clone elements across the asset model points. This improves the memory performance but requires very complex logic to shrink and grow the clones for the model points before they can be used in the present value calculation process.

Examples of clone submodels are bond_pv, mtg_pv, and sec_pv

3.4 'Working' submodels

Some submodels are treated as 'working' submodels, which are shared across a number of business objects. These submodels are used to calculate a single, or collection, of results in a controlled fashion. Once these calculations are complete, the working submodel may be 'shrunk' to clear the memory it used.

Examples of working submodels are life_pba, va_pba, and ul_cvres.

G03 INPUT MANAGEMENT

This book discusses the input management of RiskAgility FM U.S. Library projects.

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1 INPUT MANAGER STRUCTURE

RiskAgility FM presents wide flexibility for managing input assumptions for projections. The key principle is to link the projection to data available within the company, rather than having to transform into special formats.

This user guide presents instructions and options for managing input following the templates included with the projects. This is only one option, but is used consistently here in order to explain input manager operations. Of course, you can use the full flexibility of the RiskAgility FM platform, appropriate to your situation.

1.1 Input managers

1.1.1 *Input manager display*

The input manager (IM) of a project displays the input pages, data pages, and external sources. It is through the input and data pages that you specify the links to external source information. As stated above, this user guide takes the approach that all data and assumptions are specified in Excel workbooks, so that the IM contains only links and not actual input values.

1.1.2 *Template input manager*

Each RiskAgility FM U.S. Library project provides a template input manager which is linked to the Company target model object. This input manager contains a standard structure for managing data externally using Excel workbooks for assumptions and comma separated value (CSV) files for model point files.

Create an input manager for each modeling project, such as one for '2014 embedded value' and one for '2014 product pricing', starting from either the Template input manager or an existing input manager.

You can copy to a new input manager, such as creating the '2015 embedded value' IM starting from the work as of the previous date. Then, the changes you make to the new input manager do not affect the input structures for previous modeling projects.

1.2 Input pages

1.2.1 *General structure*

Input for variables which are not managed as model point variables is managed through 'input pages'. Each input page is a collection of variables that belong together, and which often change between alternative projection tasks.

A common structure of input pages is used across all projects, derived from the code manager category structure of the project. This approach separates input into company, economy, investment, and product assumptions. The product assumptions are then further broken down into product specifications, funds, expenses, mortality, policyholder, and reinsurance. This structure provides a level of granularity of assumptions so that assumptions can be managed by different groups and be organized together based on the modeling project requirements.

The names of the input pages are Pascal case names. For the product input pages, the product abbreviation is combined with the input page name. For example, ULProduct, TradExpenses, etc.

Within each input page, headers are used to further group the input into logical groupings, where the group headers are based on the second and third levels of the code manager category structure of the project.

The input variables (discussed below) are displayed on the input page in an order appropriate for the features of the project and consistent with the order of the variables in the code manager categories. For example, the variables which are used to define the general structure of the secondary guarantee feature for universal life are ordered within the Secondary Guarantee : General category on the ULProduct input page

1.2.2 *Company input page*

The Company input page is required for all tasks. It specifies the variables in the company, segment, PBA, and GAAP submodels. It supplies these same values to the counterpart variables in the product submodels. For example, the `tax_rate` variable in the company submodel applies for portfolio projections, while the `tax_rate` variable in the product submodel applies for seriatim projections.

The Company input page is similar between projects for each product type. The differences are:

- Variables with counterparts in the product submodels are associated to the product type for each project. For example, the `tax_rate` variable is associated to `trad->tax_rate` in the Traditional life project, and to `fa->tax_rate` in the Fixed annuity project.
- GAAP, PBA, and LOC categories (and the variables they collect) are included only for the methods appropriate for each product type. For example, the Traditional life project includes variables for SFAS 60 GAAP, while the Fixed annuity project includes variables for SFAS 97 retrospective deposit GAAP.

Categories

The variables on this page are organized using these major categories:

- Model setup – The segments, GAAP cohorts, LOC cohorts (life only), and hedge groups (FIA, VA, and IUL only) selected
- Discount rates – The stochastic discount rates (which are listed in the Discount rates selected project variable specified in the run parameter set); see the Present values chapter of the Profit calculations book
- Competitor rates – See the Competitor rates chapter of the Profit calculations book
- Asset reserves – See the AVR and IMR chapters of the Asset processing book
- Risk based capital – See the Capital and surplus chapter of the Profit calculations book
- Target capital – See the Capital and surplus chapter of the Profit calculations book
- Income tax – See the Taxes chapter of the Profit calculations book
- Profits – See the Profit distributions section in the Capital and surplus chapter of the Profit calculations book
- PBA – See the PBA book
- Letter of Credit – See the PBA book
- GAAP – See the GAAP accounting book.

1.2.3 *Economy input page*

The Economy input page is typically used for all tasks. The exception is for deterministic projections which do not use scenario rates; see the Stochastic versus deterministic projections section in the Projection types chapter of the Introduction book. Factors calculation (life products only) is a deterministic projection.

This input page specifies the variables in the rates submodel. It is the same for all product types. The Economic scenarios book describes these functions.

Categories

The variables on this page are organized using two major categories:

- Scenarios – Using scenario rates, including asset default rates and credit spreads
- Generation – These variables only apply for scenario generation tasks, which have rates as the top model.

1.2.4 Investment input page

The Investment input page is used only for asset/liability tasks. It is the same for all product types. The Asset modeling, and the book for each asset type, describe these functions.

This input page specifies the variables in the investment strategy submodel, and in the asset submodels (including asset group and the submodels for each asset type), with the exception of variables on asset model point records.

Categories

The variables on this page are organized using these major categories:

- Asset run control – Options governing asset calculations
- Cash – Cash investments
- Decrements – Adjustments to asset default rates, assumptions for exercising call, put, and prepayment options
- Balance sheet – Asset accounting definitions
- Asset groups – Asset group definitions and assumptions
- Strategy – Investment strategy definitions, including sales
- Purchase asset definition – Variables defining assets to be purchased during the projection, separate for each asset type.

1.2.5 Hedging input page

The Hedging input page is used only for hedging tasks. It is the same for all product types. The Hedging book describe these functions.

This input page specifies the variables in the hedging submodel.

Categories

The variables on this page are organized using these major categories:

- Pregenerated hedges – Pregenerated hedging results files
- Hedge groups – Hedge group definitions and assumptions
- Hedge valuation – Variable annuity guaranteed benefit hedge valuation definitions and assumptions
- Hedge strategy – Variable annuity guaranteed benefit hedge strategies

1.2.6 Product input pages

Each product type has input pages containing the variables in the product submodels that are not specified on the Company input page. See the Input organization chapter of the book for each product type.

1.3 Data pages

1.3.1 General structure

Input variables managed as model point files are collected into 'data pages'. Each data page only contains the model point variables of a single model object.

A common structure of data pages is used across all projects. This structure separates model point input into a page for each product type, a page for each asset type which is included in the project, and pages for externally projected liabilities and externally projected assets.

Similarly to the input page names, the names of the data pages use Pascal case. Example names are ULData, TradData, TermReinsData, TermRiderData, PADData, PABenData, BondData, and EpaData.

1.3.2 Product type data pages

The Model point records chapter in the book for each product type discusses the model point variables.

1.3.3 Reinsurance record type data pages

Dependent reinsurance model point records allow the user to link data points contained in one model point file to data points in another model point file. The reinsurance model point file is a dependent data file which links to the product type model point file via the variable `mp_id`. The Reinsurance model point records chapter in the book for each product type discusses the model point variables.

1.3.4 Rider record type data pages

Dependent rider model point records allow the user to link data points contained in one model point file to data points in another model point file. The rider model point file is a dependent data file which links to the product type model point file via the variable `mp_id`. The Riders book discusses the rider model point variables.

1.3.5 Asset type data pages

The Model point records chapter in the book for each asset type discusses the model point variables.

1.4 Input variables

1.4.1 Input variable types

Input variables define the information which can be entered for an input or data page.

The most common variable types used for input variables are floating point number, integer number, and character. Floating point arrays and integer arrays are rarely used and the variable type of Table is never used. Floating point number and Integer number variables are used for all numeric data and Character variables are used for variables which represent standard character values or choice lists.

Instead of entering variable values directly on the input pages, the value for each input variable is typically a reference to an external source containing the values.

1.4.2 Code variable associations

In order to provide input to the model objects, input variables must be associated with one or more code variables of the same variable type as the input variable. Variables on an input page may be associated with code variables from any submodel which exists within the model tree of the target model object for the Input Manager.

A single input variable may be associated with many code variables, thus providing a mechanism for a single input to populate code variables for more than one submodel of the model tree. For example, the `tax_rate` input variable on the Company input page is associated with a code variable in the company submodel and a code variable in the product submodel.

1.4.3 Table style indexing

Input variables may require 'table style' variations beyond what can be provided through a single instance of the input variable. A 'table style' is defined as a structure of tabular data with specific column and row structures, such as issue age by policy year, and attained age. When 'table style' variations are required, indexing of the input variable by table style can be used to provide for variations handled within the input manager while simplifying the management of the data externally.

Here are two examples of variables that require table style variations:

- Some life products have premiums varying by: (1) issue age (level premiums), (2) issue age and policy year, (3) attained age (ART premiums), (4) issue age and benefit term (level term plans), and so on.

- Some life products have statutory mortality varying by: (1) issue age and policy year (select mortality) or (2) attained age (ultimate mortality).

Specifying all these values in a single tabular grid would produce unwieldy tables that are hard to manage. Instead, the preferred approach is to provide 'table style' variations of the input variable so that each variation can reference a different external source range with the appropriate table style.

Using the preferred approach requires the following:

- A [variable_name]_tbl_id variable which gives a name to each set of values, such as the name of a set of premium rates
- A [variable_name]_tbl_style variable which specifies the table style indexing for the input variable, such as IssueAge, IssueAgeByPolYr, AttainedAge, and IssueAgeByBenTerm. Each '[variable_name]_tbl_style' variable includes a unique set of table styles. For example, a particular variable might include the IssueAgeByPolYr or PolYrByIssueAge table style, depending on which format best suits the table values, but both formats should not be used together.
- Indexing the [variable_name] input variable by the [variable_name]_tbl_style input variable, providing the index variations required.

The template input manager limits the use of indexing by table styles to variables where there is a substantial amount of input that would be much larger without this indexing. Follow this procedure to introduce table style indexing for an input variable:

- Create a '[variable_name]_tbl_id' input variable for the name of each table.
- Create a '[variable_name]_tbl_style' input variable for indexing the [variable_name] input variable. Specify in the table styles in the character list property of the input variable. For example, the character list might be AttainedAge, IssueAge, and IssueAgeByPolYr.
- Create '[variable_name]_tbl_id' and '[variable_name]_tbl_style' code variables and associate them with the corresponding input variables. These are only for the purpose of linking the input variables so they know which model point is currently being calculated. The code variables have default value of NONE, with no choice list. The Help message is "DO NOT REMOVE. This variable is associated with an input variable used for linking to external source tables."
- Index the [variable_name] input variable using the '[variable_name]_tbl_style' input variable as the index. This creates a row for each defined table style. Note the index is the input variable, not the code variable, so that a user can add a new table style without having to open the code explorer.

The following illustrates the structure of the input manager with table style indexing for the annual_prem_per_unit input variable:

annual_prem_tbl_id		NONE
annual_prem_tbl_style		IssueAge
annual_prem_per_unit	Annual premium per unit	Indexed
AttainedAge		0
IssueAgeByPolYr		0
IssueAge		0

where the annual_prem_tbl_style input variable contains the following properties:

Input Variable Details			
Input Variable annual_prem_tbl_style			
Variable Type		Character	
Code Variable Associations			
Name	Model Object	Model Class	Type
annual_prem_tbl_style	term_py	term_py_liab	Character
Validation			
Character List		AttainedAge, IssueAge, IssueAgeByPolYr	
Values			
AttainedAge			
IssueAge			
IssueAgeByPolYr			

and the annual_prem_per_unit input variable is indexed as follows:

Input Variable Details			
Input Variable annual_prem_per_unit			
Variable Type		Floating Point	
Code Variable Associations			
Name	Model Object	Model Class	Type
annual_prem_per_unit	term_py	term_py_liab	Floating Point
Validation			
Range From		0	
Range To		9999999999	
Runtime Failure		Error	
Index Variable			
 annual_prem_tbl_style (TermProduct)			

1.4.4 Table ID only indexing

An advantage of indexing by table style is that the tables vary by tbl_id instead of by plan code, valuation code, etc. This helps reduce the size of the tabular ranges in cases where the same large table applies for several plan codes, for example.

Thus this technique can be used even when the values do not vary by table style. For example, the XXX reserve select mortality multiples all use the PolYrByIssueAge table style, but a certain table applies for different variables and for multiple valuation codes.

In this case, only the [variable_name]_tbl_id input and code variables are included. There is no indexing by table style, but the table names can still be used.

1.5 External sources

1.5.1 Why external sources?

Instead of entering variable values directly on the input pages, the value for each input variable is typically a reference to an external source containing the values. Advantages include:

- After an initial setup process, all assumptions can be managed without using the input manager. For example, a new plan code can be added simply by extending the ranges in the external source Excel workbooks.
- In many cases, input can be retrieved directly from current company information, without having to recreate using a new format.
- Tables can vary using whatever configuration is appropriate.
- Once the external source references have been set up, input for a different modeling project can be set simply by changing wildcards and copying the external source files. For example, changing from last year's embedded values to this year's.

1.5.2 Excel external sources

The template input manager includes Excel individual external sources required for managing all non-model point input externally. The external sources are structured as follows:

- Referencing the appropriate assumption workbooks (discussed below) using the appropriate <*assump*>, or <*global*> wildcards,
- Use the Named Range selection, referencing the named ranges from the reference assumption workbook

The use of user defined names has not been used in the template external sources so that the input manager input page displays the full external source reference, including the wildcard, workbook, and range.

1.5.3 Plain text external sources

The template input manager includes plain text individual external sources required for managing the model point input. The external sources are structured as follows:

- Referencing the appropriate comma separated value (CSV) file using the appropriate <*data*> wildcard,
- Selecting 'Comma' as the delimiter

Again, the use of user defined names has not been used in the template external sources so that the data page displays the full external source reference

1.5.4 Composite external sources

For several U.S. Library projects, the template input manager includes composite external sources which provide for additional lookup references to the individual external sources. For example, the mortality tables used for universal life model points may vary by whether the model point is the result of a normal issue or from a term conversion. The composite external sources are structured as follows:

- A composite key choice list variable is specified for each composite external source,
- The required excel external source range is then selected for each selection of the choice list variable,

Composite external sources have been defined for base experience mortality tables, mortality multiples, mortality additions, and YRT rates. The use of user defined names, using a Pascal case naming convention, has been used for the template composite external sources so that the composite external sources can be easily identified in the input manager.

1.6 Assumption sets

You can create multiple assumption sets for each input page. For example, the ULMortality input page could include 'Baseline' and 'High' assumption sets.

Each assumption set can have different values for the input variables collected on the input page. For example, the 'High' Mortality assumption set could link to a different external source table of Mortality adjustment multiples.

1.6.1 Default assumption set

The template input manager is populated with a default assumption set, where all associated input variables are set to display the code default value.

The default assumption set is intended only to provide a way to examine the default values of the associated code variables through the input manager. This assumption set is not intended to be used in any projection tasks.

1.6.2 Baseline assumption set

The template input manager is also populated with a baseline assumption set, where all associated input variables are linked to external sources with standard external source lookup definitions.

1.6.3 Linked assumption sets

Best practice is to link the alternative assumption sets to the Baseline assumption set. Then the external source links on the Baseline set also apply for each linked assumption set. Right-click to unlink whatever variables you want to change, then specify the alternative external source links. Then the assumption sets clearly show which specific input variables have different links from the Baseline assumptions.

1.6.4 Assumption set ID input variables

The template input managers include an important method for efficient handling of variations between assumption sets of the Company, Economy, and Investment input, without the need for adjusting the external link definitions of the input variables.

At the top of the input page, an 'Assumption Setup' group of input variables is included. Input variables are included within that group, are named [xxx]_assump_set_id, and are associated with similarly named code variables. The only use of these input variables is to be a lookup key for the associated input variables on this input page (see the Indexing by assumption set ID topic in the External lookups chapter below).

The big advantage with this approach is that there are no changes in the external source links between the alternative assumption sets. Each additional assumption set is created simply by linking to the Baseline set. The entire variation in values occurs in the external source ranges, where [xxx]_assump_set_id is a column in each range.

Using this approach, the only external source links to specify and check are in the Baseline assumption set. This method works when the variations can be accomplished using an [xxx]_assump_set_id column within each single range.

Company assumption set IDs

The Company input page includes the following disassociated input variables:

- company_assump_set_id is for referencing general company and segment variables
- pba_assump_set_id is for referencing variables under the PBA category
- loc_gpr_assump_set_id is for referencing variables under the Letter of credit GPR category (life only)
- gaap_assump_set_id is for referencing variables under the GAAP category.

With these assumption set id variables, the various Company assumption sets can include alternatives using the same general company and segment variables but having different assumptions, such as for different types of PBA and GAAP calculations.

Economy assumption set IDs

The Economy input page includes the following disassociated input variables:

- scen_assump_set_id is for referencing variables under the Scenarios category, and
- gen_assump_set_id is for referencing under the Generation category.

With these assumption set id variables, the various Economy assumption sets can include alternatives using the same initial scenario rates but having different scenario generation assumptions, such as for different types of PBA and GAAP valuations.

Investment assumption set ID

The Investment input page includes the following disassociated input variables:

- inv_assump_set_id is for referencing variables on the Investment input page which vary by assumption set

2 EXTERNAL SOURCE FILES

2.1 Modeling project files

2.1.1 Assumption workbooks

Each modeling project uses the following assumption workbooks:

- Company.xlsm – Input variables on the Company input page. Includes worksheets for Company, Segment, Life PBA, Annuity PBA, VA PBA, LOC GPR, SFAS60 GAAP, SFAS91 GAAP, SFAS97LP GAAP, SFAS97RD GAAP, SFAS120 GAAP
- Economy.xlsm – Input variables on the Economy input page. Includes worksheets for Scenarios and Generation
- Hedging.xlsm – Input variables on the Hedging input page. Includes worksheets for Hedges and HedgeStrategies
- Investment.xlsm – Input variables on the Investment input page. Includes worksheets for Asset Run Control, Cash, Decrements, Sec Cash Flows, Balance Sheet, Asset Groups, RefinanceForeclosureRestructure, Strategy, Purch Bond Defn, Purch Mtg Defn, Purch Sec Defn, Purch Ird Defn, Purch Re Defn, Purch Eqt Defn, and Purch Eio Defn
- Reinsurance.xlsm – Input variables related to various reinsurance treaty specifications and assumptions. Includes a single Reinsurance worksheet.
- [product type] – Product.xlsm – Input variables on each of the input pages related to product definitions and assumptions. Includes worksheets, as appropriate, for Product, Fund, Dividends, Expenses, Commissions, Mortality, Policyholder
- [product type] – Valuation.xlsm – Input variables related to valuation definitions and assumptions. Includes a single Valuation worksheet.
- Riders.xlsm – Input variables related to various rider specification and assumptions. Includes a single Riders worksheet
- Riders – Valuation.xlsm – Input variables related to rider valuation definitions and assumptions. Includes a single Valuation worksheet.

For example, the Traditional Life – Product.xlsm has this name to distinguish it from, say, the Term Life – Product.xlsm workbook when these two products are run together.

2.1.2 **Mortality table workbooks**

Each modeling project typically uses the following mortality table workbooks:

- [product type] – Mortality Tables SelectAndUltimate.xlsm – Select and ultimate mortality tables (life products)
- [product type] – Mortality Tables Attained Age.xlsm – Ultimate mortality tables.
- [product type] – Mortality Tables Adjustments.xlsm – Multiples, additions, and improvement percentages applied to the base table rates.

For example, the Traditional Life – Mortality Tables SelectAndUltimate.xlsm has this name to distinguish it from, say, the Term Life – Mortality Tables SelectAndUltimate.xlsm workbook when these two products are run together.

If riders are included, the modeling project may use the additional mortality table workbooks:

- Riders – Mortality Tables SelectAndUltimate.xlsm – Select and ultimate mortality tables (life products)
- Riders – Mortality Tables Attained Age.xlsm – Ultimate mortality tables.
- Riders – Mortality Tables Adjustments.xlsm – Multiples, additions, and improvement percentages applied to the base table rates.

2.1.3 **Model point files**

Each modeling project typically uses the following model point files:

- [product type]MP-[usage].csv – Liability model point records for the product type and usage. For example, TradMP-Proj.csv for traditional life model points
- [product type]ReinsMP-[usage].csv – Dependent reinsurance model point records for the product type and usage. For example, TradReinsMP-Proj.csv for traditional life reinsurance model points
- [product type]RiderMP-[usage].csv – Dependent rider model point records for the product type and usage. For example, TradRidersMP-proj.csv for traditional life riders model points.
- [asset type]MP-[usage].csv – Asset model point records for the asset type and usage. For example, BondMP-Proj.csv for existing bond model points

2.1.4 **Scenario files**

Each modeling project typically uses historical rates and projection scenario rates files:

- [YYYYMM]MonthlyHistoricalRates.csv – Monthly historical rates through YYYYMM. For example, 201312MonthlyHistoricalRates.csv
- [YYYYMM]MonthlyProjectedScenarios.csv – Projected scenarios starting YYYYMM. For example, 201312MonthlyProjectedScenarios.csv

2.2 **Assumption workbooks**

The template input manager references template assumption workbooks, located in the ProjectData\Template\Assumptions child folder of the project folder.

2.2.1 **General structure**

The general structure of the template assumption workbooks is as follows:

- Macro enabled with macros that allow for adding columns and rows to defined ranges
- Contains a table of contents worksheet with hyperlinks to allow you to jump to the range for a selected subject. Then each range includes a 'Back to TOC' hyperlink to return to the table of contents.
- Gridlines are not displayed

- Has a different worksheet for each major collection of variables.
- Includes hyperlinks to the left of each range that allows the user to return either to the table of contents or back to the main range at the top of the worksheet.
- Uses Pascal case and standard abbreviations for range names, using the variable name when an individual range is required
- Displays the range names above the range using a 'GetNameRange' user defined function.
- Provides for shading to distinguish the lookup terms and keys from the values
- Uses bold text to distinguish the column and row lookup terms
- Displays the default values for each variable in column A, or, if required, as a row of values above the range. The default values are consistent with the default values of the variables in the code manager.
- Uses left alignment for column A and the row lookup key names and keys. The column lookup key names and keys, and variable values are right aligned
- Uses cell number formatting appropriate for the variable value, consistent with the expected interpretation of the variable input and number of decimal places for the variable values
- Uses data validation lists for choice list variables
- Uses conditional formatting with blue text highlighting to identify variables changed from default
- Uses a main range, individual ranges, and alternative ranges to handle variable value variations

2.2.2 **Main range**

The main range, located at the top of each worksheet, is designed to replicate the structure of the input pages as much as is practical, including the group headers. Within the main range, each input variable is included as a row. The columns are appropriate variations in the variable values, such as assumption set id, plan code, valn_code, treaty_code, or rider_code.

When the variable values vary only by the variable name and a single column lookup key, this main range contains the values. When the values vary by other or additional lookup keys so that it cannot be set in the main range, a row for the variable is still included in the main range with a hyperlink to a range that specifies the values.

2.2.3 **Individual ranges**

When a variable values vary by more than the variable name, those variables require a range individual for that variable, set out in a range different from the main range. For example, the deterministic asset yield and fixed maintenance expense vary by date. Values for each of these variables are included in an individual range for that variable, below the main range.

Each range is typically designed to use rows for the lookup key that can have more values. For the two examples above, 'date' is the row lookup key such as for the possibility that the table values might vary monthly for, say, 30 years.

2.2.4 **Alternative ranges**

When other projections use an alternative table of values, that alternative table is another range located directly below the first version of it.

Each version must have its own range name. For example, a range name of *MortMult110Pct* is a range name which could be used as an alternative range name for a *MortMult* range.

2.2.5 **Table index ranges**

When an input variable is indexed by table style, special ranges are included in the assumption workbook to manage the values of the [xxx]_tbl_id and [xxx]_tbl_style variables. Three ranges are included;

- TblIndexByPlanCode – Values for [xxx]_tbl_id and [xxx]_tbl_style which vary by plan code, such as annual_premium_tbl_id and annual_prem_tbl_style

A value of NONE is used in the range for [xxx]_tbl_id when a table does not apply, such as for a particular plan code.

2.3 Mortality table workbooks

2.3.1 General structure

Rates for mortality tables are collected into separate workbooks with several worksheets and external source ranges. The general structure of the workbooks are as follows:

- Contains a table of contents worksheet with hyperlinks to allow you to jump to the range for a selected subject.
- Includes documentation of the external source format and date created
- Includes reference to issue ages included and SOA table reference
- Has a different worksheet for each mortality table.
- Uses Pascal case and standard abbreviations for range names
- Sets the external source range name for each individual table as the name of the table, with a leading underscore (since Excel does not allow a range name to begin with a number). For example, _2001CsoAlb.
- The column lookup key names are gender, smoking_class, rate_class, and issue_age (S&U tables) as appropriate for the table variations and structure.
- The row lookup key is either policy_yr (S&U tables) or attained_age (ultimate tables).

2.3.2 Mortality table creator tool

A special Excel tool provided with the RiskAgility FM U.S. Library is used to create workbooks of the mortality rates, drawing the mortality rates from the industry mortality tables published by the Society of Actuaries. Thus you can use these publically available values instead of creating separate versions of these standard tables.

The 'Global Mortality Table External Source Creator.xlsm' utility is located in the Utilities\Mortality Table Tools folder of the RiskAgility FM U.S. Library x.y environment. Follow the instructions provided on the 'Welcome' worksheet to create the desired mortality table workbooks.

The tool creates a worksheet and range for each selected table, plus a worksheet which combines all the tables into a single external source range. It is this combined range that is the external source range for looking up mortality variable values.

For life products, create two workbooks using this tool -- one for select and ultimate tables ([Product type] – Mortality.xlsx) and one for attained age tables ([Product type] – MortalityAttainedAge.xlsx). The select and ultimate tables are translated into a policy year by issue age structure.

2.4 Model point files

2.4.1 General structure

Model point records are specified into separate files by product and asset type. The general structure of the model point files are as follows:

- Are comma separated value (CSV) files
- Includes the input variable names in row 1

- Generally, the first column is a character field for the model point group, using the pipe ('|') character to create a combination group, such as Plan A|2010 Issues|Male|NonSmoker. However, for payout annuity benefit streams, reinsurance, and riders, the first column is the 'keyfield' used to link the dependent data to the base model point file.

2.4.2 Liability model point files

Refer to the model point records chapter in each of the product type user guide books for a discussion of the model point records for the liabilities.

2.4.3 Reinsurance model point files

Refer to the reinsurance model point records chapter in each of the product type user guide books for a discussion of the model point records for the liabilities.

2.4.4 Rider model point files

Refer to the model point records chapter in the Riders user guide book for a discussion of the model point records for the riders.

2.4.5 Asset model point files

Refer to the model point records chapter in each of the asset type user guide books for a discussion of the model point records for the assets.

2.4.6 Model point creator tools

Three special Excel tools provided with the RiskAgility FM U.S. Library can be used to create the liability, reinsurance, and asset model point files.

The 'Liability Model Point CSV Builder Tool.xlsm', 'Supplemental Model Point CSV Builder Tool.xlsm', and 'Asset Model Point CSV Builder Tool.xlsm' workbooks are located in the Utilities\Model Point Tools folder of the RiskAgility FM U.S. Library x.y environment. Follow the instructions provided on the 'Welcome' worksheet to create the desired model point files.

The tool creates the required CSV file for the selected liability or asset type.

2.5 Scenario files

2.5.1 General structure

Historical and projected economic scenario rates are collected into separate files. The general structure of the scenario files are as follows:

- Stored in comma separated value (CSV) format
- Includes data structures for interest rates, interest volatilities, index rates, index volatilities, and miscellaneous rates.

Please refer to the Economic scenarios user guide book for a detailed discussion of the scenario files

2.5.2 Scenario tools

To assist in the creating and viewing of the scenario files, several scenario tools are provided with the RiskAgility FM U.S. Library. These tools can assist with the translation of scenario from one format to the required CSV format.

Please refer to the Economic scenarios user guide books for a detailed discussion of the scenario tools.

3 EXTERNAL LOOKUPS

3.1 External lookup definitions

The lookup definitions for an input variable describe how the columns and rows of an external source will be referenced to draw a value from the external source for an input variable.

3.1.1 General structure

The general structure of the lookup definitions in the template input manager is as follows:

- Key name – Not used
- Lookup Terms – Selected from constant, input variables, code variables, code scalars, and columns as required. System variables are not used.
- Lookup Model Object – Specified as Absolute or Relative as required. Relative is required when an input variable is linked to code variables from more than one model object and the lookup term for the lookup should be referenced from the model object which is performing the lookup.
- Type – Specified as Character or Numeric as required. Constant lookup terms are always Character type and column lookup terms are always numeric. Input variable, code variable, and code scalar lookup terms must be set appropriate for the type of the variable. For example, a code variable lookup term using `rate_term` must be specified as a numeric.
- Before First – Only applicable when the Type property is Numeric. Most commonly set to First.
- Missing – Commonly set to Error when the Type property is Character. When the Type property is Numeric, is set to Previous or Linear per the desired treatment for the data. For example, Linear is used as the Missing property for the `rate_term` lookup key for retrieving deterministic credit spreads.
- After Last – Only applicable when the Type property is Numeric. Most commonly set to Last
- Transformation properties – Not used

3.1.2 Indexing by variable name

When variable values vary by the variable name and perhaps one other variation in the external source, such as a plan code, a constant lookup key is specified where the constant value is the text name of the input variable.

For example, on the Company input page, the row lookup term for the `init_free_surp` variable is the constant text 'init_free_surp'.

For variable name lookup keys, the Lookup Model Object property does not apply, the Type property is set to Character, the Before First and After Last properties do not apply and Error is used as the Missing property.

3.1.3 Indexing by static text

When a variable value varies by a constant text value in the external source, a constant lookup key is specified where the constant value is the specified text.

For example, on the Investment input page, a column lookup term for the `call_price_pct` variable is the constant character value 'CallPricePct'.

For static text lookup keys, the Lookup Model Object property does not apply, the Type property is set to Character, the Before First and After Last properties do not apply, and Error is used as the Missing property.

3.1.4 Indexing by assumption set ID

When a variable value varies by an assumption set ID, an input variable lookup key is specified where the input variable lookup term is the appropriate assumption set ID.

For example, on the Economy input page, a column lookup term for the scen_id variable is the scen_assump_set_id input variable.

For assumption set ID lookup keys, the Lookup Model Object property does not apply, the Type property is set to Character, the Before First and After Last properties do not apply and Error is used as the Missing property.

3.1.5 Indexing by submodel ID

When a variable value varies by a submodel identifier, a code variable lookup key is specified where the code variable is the appropriate submodel ID variable.

For example, on the Company input page, a code variable lookup term for the asset_yld_defn variable is the seg_id code variable.

For submodel ID lookup keys, the Lookup Model Object property is set to Relative for those variables which have multiple associations, the Type is set to Character, the Before First and After last properties do not apply and Error is used as the Missing property.

3.1.6 Indexing by model point characteristics

When a variable value varies by model point characteristic, a code variable or scalar lookup key is specified where the code variable is the appropriate model point characteristic variable or scalar.

For example, on the TradPolicyholder input page, a code variable lookup term for the lapse_rate_base variable is the issue_age code variable.

Also, on the ULFund input page, a scalar lookup term for the load_per_unit_sf variable is the ins_issued_per_pol code scalar.

For model point characteristic lookup keys, the Lookup Model Object property is set to Absolute since all model point characteristics are submodel specific, the Type is set to Character or Numeric as appropriate, and the Before First, After Last, and Missing properties are set as appropriate for character or numeric values.

3.1.7 Indexing by calculated variable keys

Some variable values vary by values of keys which change during the projection. For example, the competitor rate variables must vary by the lookup key comp_rate_id. Each time the function get_comp_rate is called, this formula sets the current competitor rate name as the value for the comp_rate_id variable. Since each of the competitor rate variables are linked to an external source that varies by comp_rate_id, the lookup functions return the definitions for that rate.

Other examples include:

- Variables specifying stochastic discount rates which vary by disc_rate_id.
- Scenario credit spread assumptions which vary by quality_id and rate_term.
- Shock lapse rates which vary by prem_incr_pct (the percent increase in the premium) and yrs_since_incr (years since the time of the premium increase).
- Purchase asset variables which vary by purch_asset_id and, when appropriate, mths_since_issue of the inv_strategy submodel

Code variables which represent calculated variable keys are also referred to as **hidden lookup keys**. These code variables have the 'Set Value in Input Manager' property set to 'User Defined' with no submodels selected. They are not associated with any input variables and are thus hidden from the user.

For calculated variable lookup keys, the Lookup Model Object property is set to Relative for those variables which have multiple associations, the Type is set to Character or Numeric as appropriate, and the Before First, After Last, and Missing properties are set as appropriate for character or numeric values.

Warning! The external source range must use the exact names of the hidden variable lookup keys set by the formulas.

3.1.8 Indexing by mortality selectors

Mortality variables are special since they must be read twice, once for life 1 and again for life 2, for joint life products and for blended mortality annuities. The variables classified as 'mortality variables', and thus treated specially, include cash value and reserve mortality, XXX reserve mortality select multiples, YRT reinsurance rates, and COI rates.

Mortality selectors are calculated variable keys as follows:

- `mort_issue_age` – Set equal to `issue_age` for life 1 and `issue_age_life2` for life 2
- `mort_pol_yr` – Set equal to the appropriate policy year value. Equals 't' in the policy year submodels, or calculated from the `commencement_period`.
- `mort_attained_age` – Set equal to `mort_issue_age` + `mort_pol_yr` – 1
- `mort_gender` – Set equal to `gender` for life 1 and `gender_life2` for life 2
- `mort_smoking_class` – Set equal to `smoking_class` for life 1 and `smoking_class_life2` for life 2
- `mort_rate_class` – Set equal to `rate_class` for life 1 and `rate_class_life2` for life 2
- `mort_rate_class_res` – Set equal to `rate_class_res` for life 1 and `rate_class_res_life2` for life 2

Warning! The links to the external source range must use these special lookup keys, otherwise using gender (for example) as a lookup key would return the life 1 rate also for life 2.

For the mortality selector lookup keys, the Lookup Model Object property is set to Absolute, the Type is set to Character or Numeric as appropriate, and the Before First, After Last, and Missing properties are set as appropriate for character or numeric values.

Note that attained age must use the `mort_attained_age` mortality selector lookup key, not the `attained_age` column, since issue age can vary between life 1 and life 2. Also, calculation of mortality improvement loops through calendar years since the mortality table's base year, while holding `mort_attained_age` constant.

Product types that do not include joint life or blended mortality features still use this system for consistency.

3.1.9 Indexing by time period value

When a value varies by time period value, such as by calendar year, policy year, attained age, date, etc., a column lookup key is specified where the column is the time period calculation.

For example, on the Company input page, a column lookup term for the `asset_yld_deterministic` variable is the date column.

Common time period value column lookup keys are:

- `cal_yr(t)` – Calculates the calendar year value, i.e. 2015, at period 't'
- `cal_mth(t)` – Calculates the calendar month (1 – 12) at period 't'
- `pol_yr(t)` – Calculates the policy year (1, 2, 3, ...) of the liability model point at period 't'
- `pol_mth(t)` – Calculates the policy month (1 – 12) of the liability model point at period 't'
- `attained_age(t)` – Calculates the attained age (0, 1, 2, ...) based on the issue age of life 1 of the liability model point at period 't'
- `mths_since_issue(t)` – Calculates the months since issue of the model point at period 't'
- `date(t)` – Calculates the number of days since 1899 for the first day of period 't'. This matches with the date function value in Excel.

For time period value lookup keys, the Lookup Model Object property is set to Relative for those variables which have multiple associations, the Type is set to Numeric, and the Before First, After Last, and Missing properties are set as appropriate for numeric values.

3.1.10 Indexing by table style

When an input variable is indexed by table style, a combination of constant value, input variable, code variable, and time period values are required to obtain the proper value from the external source.

Table ID input variable lookup definition

Table ID input variables link to the *TblIndexByPlanCode* external source ranges using a code variable column lookup key of either `plan_code` or `treaty_code` and a constant row lookup key of the variable name.

For example, on the TradProduct input page, the `annual_prem_tbl_id` variable is linked to the *TblIndexByPlanCode* range of the Traditional Life – Product.xlsm workbook and uses the code variable `trad->plan_code` as the column lookup key and the constant lookup key of `'annual_prem_tbl_id'` as the row lookup key.

Table style input variable lookup definition

Table style input variables also link to the *TblIndexByPlanCode* external source ranges using a code variable column lookup key of either `plan_code` or `treaty_code` and a constant row lookup key of the variable name.

For example, on the TradProduct input page, the `annual_prem_tbl_style` variable is linked to the *TblIndexByPlanCode* range of the Traditional Life – Product.xlsm workbook and uses the code variable `trad->plan_code` as the column lookup key and the constant lookup key of `"annual_prem_tbl_style"` as the row lookup key.

Indexed variable lookup definition

Each variation of the indexed variable is linked to the appropriate external source range for the variable values using the `[xxx]_tbl_id` as an input variable lookup key, plus other lookup keys as required to retrieve the values.

For example, on the TradProduct input page, the `annual_prem_per_unit` variable for the IssueAgeByPolYr index is linked to the *PremCurrTblsIssueAge* range of the Traditional Life – Product.xlsm workbook and uses the `annual_prem_tbl_id` input variable, code variables of `gender`, `smoking_class`, `rate_class`, and the `pol_yr` column as column lookup keys and the code variable of `issue_age` as the row lookup key.

3.2 External lookup processing

3.2.1 Static lookup key value

When the lookup keys for an input variable do not change during the projection, the external source lookup is performed a single time and the value is stored in the associated code variable of the model object. This is typically the case for lookup keys based on constant values, input variables, code variables, and scalars.

3.2.2 Change in lookup key value

When the lookup keys for an input variable change during the projection, the external source lookup is performed each time the code variables change. This is typically the case for lookup keys based on calculated variable keys.

3.2.3 Implicit indexing

Input management through external sources is very powerful. It provides for defining variables which may represent tabular data as integer or floating point variables and then developing formulas which reference the code variables simply, without any lookup parameters.

Of course, when the variable values do vary by a time period, the external lookup processing must return the value for the appropriate time period. This is accomplished through **implicit indexing** which reconciles all columns as lookup keys based on the referencing submodel 't' value for the period the lookup is being performed.

3.2.4 Offset lookups

When the value is needed as of a time different from the current projection period, implicit indexing does not work. Instead, the formula must pass that appropriate 't' value to the external lookup process. This is accomplished through special syntax in the formulas that adds ".AtTime(...)" to the variable name. The value within the parentheses is the referencing submodel 't' period which will be used in the lookups for the column lookup keys. For example (using life product formulas):

- $t - 1$ or $t + 1$ for the value of the variable during the previous or next period
- commencement_period + 1 for the value of the variable during the first policy year, when the referencing is performed from the monthly submodels
- prem_term_period for the value of the variable during the policy year when premiums end
- maturity_period for the value of the variable during in the last policy year.

Note that the lookup definitions are the same for looking up the current value versus the value for some other time. It is the formula using the input value that must specify the time period for which to retrieve the value.

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G04 RUN MANAGEMENT

This book discusses the run management of RiskAgility FM U.S. Library projects.

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1 RUN MANAGER STRUCTURE

RiskAgility FM presents wide flexibility for managing projections.

This user guide presents instructions and options for managing projections following the templates included with the projects. This is only one option, but is used consistently here in order to explain run manager operations. Of course, you can use the full flexibility of the RiskAgility FM platform, appropriate to your situation.

1.1 Run managers

1.1.1 Run manager display

The run manager (RM) of a project displays the run pages, and run controller of the run manager. It is through the run pages and the run controller that you specify the projection tasks and projections sets for the project.

1.1.2 Template run manager

Each RiskAgility FM U.S. Library project provides a template run manager which is linked to the Template input manager. This run manager contains a standard structure for managing projection tasks and projection sets.

Create a run manager for each modeling project, such as one for '2014 embedded value' and one for '2014 product pricing'. Then, create the run pages required for the modeling project linking to the input manager of the same modeling project.

Important! Since the links of the run pages of the Template run manager to the Template input manager cannot be altered, you cannot copy the template run manager to a new run manager for the new modeling project.

1.2 Run pages

1.2.1 General structure

A 'run page' provides an interface to define projection tasks. Each 'run page' is linked to an input manager and target model object when the page is created. Once the input manager and target model object is selected, they cannot be changed.

A common structure of run pages is as follows:

- Linked to the input manager of the same name as the run manager
- First level separation into 'Standard', 'PBA', 'GAAP', 'LOC GPR', and 'Hedging'
- Second level separation into 'Rate Gen', 'Seriatim', and 'Portfolio'

The first level separation is suggested as best practice for run management since PBA, GAAP, and Hedging projection tasks are commonly organized together into a nested projection, while Standard projections typically produce full statutory financial statement results independently from other projections.

The second level separation is required since different target model objects are required for the run pages for rate generation, versus seriatim, versus portfolio.

1.2.2 Standard seriatim run page

Use the 'Standard Seriatim' run page to define projection tasks which use the product or asset type as the top model, and are not used for PBA, GAAP, or Hedging calculations. For example, a cell pricing projection which uses formulaic reserves and capital would be defined on the 'Standard Seriatim' run page.

1.2.3 *Standard portfolio run page*

Use the 'Standard Portfolio' run page to define projection tasks which use company as the top model, and are not used for PBA, GAAP, or Hedging calculations. For example, an asset liability projection of in force business which uses formulaic reserves and capital would be defined on the 'Standard Portfolio' run page.

1.2.4 *PBA rate gen run page*

Use the 'PBA Rate Gen' run page to define projection tasks which use the rates submodel as the top model and generate inner loop scenarios for PBA valuations. For example, the stochastic exclusion test scenarios generation for use in an inner loop PBA deterministic reserve valuation would be defined on the 'PBA Rate Gen' run page.

1.2.5 *PBA seriatim run page*

Use the 'PBA Seriatim' run page to define projection tasks which use the product submodel as the top model and determine PBA seriatim results. For example, the point in time standard scenario amount valuation for variable annuities would be defined on the 'PBA Seriatim' run page.

1.2.6 *PBA portfolio run page*

Use the 'PBA Portfolio' run page to define projection tasks which use the company submodel as the top model and determine PBA portfolio level results. For example, the point in time scenario amount valuation for a block of Term life insurance would be defined on the 'PBA Portfolio' run page.

1.2.7 *GAAP rate gen run page*

Use the 'GAAP Rate Gen' run page to define projection tasks which use the rates submodel as the top model and generate inner loop scenarios for GAAP valuations. For example, the mean reversion scenarios generation for use in an SFAS 97 retrospective deposit unlocking would be defined on the 'GAAP Rate Gen' run page.

1.2.8 *GAAP seriatim run page*

Use the 'GAAP Seriatim' run page to define projection tasks which use the product submodel as the top model and determine GAAP seriatim results. For example, the SFAS 60 GAAP factors valuation for traditional life would be defined on the 'GAAP Seriatim' run page.

1.2.9 *GAAP portfolio run page*

Use the 'GAAP Portfolio' run page to define projection tasks which use the company submodel as the top model and determine GAAP portfolio results. For example, the point in time SFAS 97 retrospective deposit DAC valuation for universal life would be defined on the 'GAAP Portfolio' run page.

1.2.10 *Hedging rate gen run page*

Use the 'Hedging Rate Gen' run page to define projection tasks which use the rates submodel as the top model and generate baseline and shocked scenarios for Hedging valuations. For example, the risk neutral interest rate and equity volatility shocked scenarios generation for use in a hedging valuation would be defined on the 'Hedging Rate Gen' run page.

1.2.11 *Hedging portfolio run page*

Use the 'Hedging Portfolio' run page to define projection tasks which use the company submodel as the top model and determine Hedging portfolio results. For example, the point in time hedging valuation for variable annuities would be defined on the 'Hedging Portfolio' run page.

1.3 Run parameter sets

1.3.1 General structure

A 'run parameter set' provides an interface to define projection task run parameters. Best practice is to create run parameter sets which can be reused across a number of projection tasks. Run parameter sets typically vary based on characteristics as follows:

- Calculation type
- Seriatim versus portfolio
- Based on selections for choice list project variable values
- Selected target
- Projection period to value
- Projection date project variable
- Projection task loops

Using the above structure, a naming convention is established so that it is not necessary to open the run parameter set when determining if a run parameter set should be selected.

The name is developed using values for each of the characteristic, each separated by a space:

#	Element value	Conditions and notes
1	RateGen, Factors, Valn, Pricing, Proj	Target model object = rates Target column = factors_tgt (applies to life products only) Target column = valn_tgt_xxx pricing_defn = Yes, and Target column != valn_tgt_xxx pricing_defn = No; Target column != valn_tgt_xxx
2	Seriatim, Portfolio	Target model object = [product] Target model object = company; Leave this element out of the name for Factors and RateGen
3a	GAAP	Only when gaap_defn = Yes and not Valn; Leave this element out of the name otherwise
3b	ALM	Only when assets_defn = Yes Leave this element out of the name otherwise
3c	IMP	Only when independent_mp_defn = Yes and not Seriatim; Leave this element out of the name otherwise
3d	Hedge	Only when the target columns are valn_tgt_hedge_xxx or proj_tgt_hedge_stmt

#	Element value	Conditions and notes
4	[target]	Example: [target column name], all, irr For targeting output definitions, include the name of the output definitions For Valn use target name without 'valn_tgt_' part of name Leave this element out of the name for Factors (only one possible target)
5	[Proj Period To]	For Proj and Pricing, the number of periods such as 1440 or 1600 When value is 0, leave this element out of the name
6	[Projection date]	The proj_date value, such as 2014_12 (cannot use '/' character)
7	[Projection task loops]	Identify the projection task loops only when they create multiple run parameter objects that would otherwise have the same names. For example: 1to5, 1_2_5_10 (cannot use ',' character)

The following are some example run parameter object names:

- Pricing Seriatim all 1600 2014_12 1to10
- Proj Portfolio ALM GAAP IMP distrib_earnings 60 2014_06
- Proj Portfolio Hedge 360 2015_12
- Valn Seriatim pba_det_res 2014_06
- Valn Portfolio IMP pba_pol_res 2014_06.
- Valn Portfolio ALM pba_scen_amt 2014_12 1to1000
- Valn Portfolio Hedge shock 2015_12

The following are examples of names that cannot occur:

- Pricing Seriatim IMP all 1600 2014_12 (IMP cannot apply for Seriatim task)
- Valn Seriatim pba_det_res 0 2014_06 (Proj Period To element is blank for Valn task)
- Valn Seriatim GAAP gaap_sfas97rd 2014_06 (GAAP project variable does not affect Valn task).

Note there is no designation for nested projections, since that is not specified by the run parameter set.

1.3.2 Run parameters

Run parameters are properties required by the RiskAgility FM platform. Parameterization of the run parameters is as follows:

Property	Values
Target object	Typically selected as Target Column Target Output Definitions may be selected for projections which are not valuation or factor projections

Property	Values
Target column	Only submodels of 'company', '[product]', and 'rates' may be selected as the model object Can be the key result column from the top model, like distrib_earnings, or irr (for pricing/new business projections). Valuation tasks must name the valn_tgt_xxx column which specifies the results to be valued.
Projection period from	Typically 0
Projection period to	For projection tasks, the number of periods to include in present values. For valuation tasks (target column is valn_tgt_xxx), 0 since the valuation projection is specified by the Valuation years project variable.
Projection period min	0 for liability only portfolio tasks, since there should be no calculations before the projection date. -xxx for seriatim tasks, where xxx is before the earliest issue period to allow for external source lookup keys calculating before the projection date (such as pol_yr) -xxx for ALM portfolio tasks, where xxx is before the earliest issue period of existing assets Asset type top models may need to reach back to the earliest issue date for the existing assets.
Projection period max	At least as long as the calculations for each submodel. For example, UL individual fund calculations must calculate to maturity when used for solving, even when the projection period is shorter. For ALM projections, the asset submodels must calculate 360 months (30 years) longer than the projection period, in order to calculate the market value of bonds (etc.) purchased at the end of the projection. Setting this value longer than needed allocates (and thus uses) more memory for the projection; however submodels for model point calculations automatically reduce memory use to the maturity period for each model point.
Projection task loops	Typically, the scenario paths to process, for example 1-1000, or 1,8,25, 1-10,500. However, for hedging valuations, this specifies the combination of shocks and scenarios to process. The values specified here determine the values of • The proj_task_loop_num, first_proj_task_loop, is_first_proj_task_loop, last_proj_task_loop, is_last_proj_task_loop, and num_of_proj_task_loops system variables
Wildcard set	Select the wildcard set which contains the paths to the appropriate external data for the projection task.
Assumption set restore	Yes for seriatim tasks, and when independent model point processing applies for portfolio tasks; No otherwise

Property	Values
Extract model points	Yes to enable the Extract model points feature specified for the model point file in the Input manager; has no effect when this feature is not used for the model point file
Sliding windows	Do Not Slide for seriatim runs and when small models or independent model point processing enables portfolio runs to fit within available memory; Slide to reduce memory requirements during a projection task which does not otherwise fit into available memory
Distribution	For portfolio tasks, use Split by Projection Loop to distribute the projection task loops to the cores on the grid. For seriatim tasks, could be more efficient to use Split by Group when number of model points exceed number of projection task loops Specify a Target Split for Distribution value for the minimum number of Projection Loops or Groups that will be allocated to a Grid Task in a Distributed Processing environment. The ideal setting for this value requires thorough analysis of the model and grid configuration.

1.3.3 Project variables

Project variables are included for those run parameters which are specific to the calculations of the library. Parameterization of the project variables is as follows:

Project variable	Description
Projection date [proj_date]	In-force values, and present value calculations, are as of the end of the month specified as yyyy/mm, such as 2014/12 for December 31, 2014
Pricing projection? [pricing_defn]	Yes for projections calculating special pricing results like IRR (all model points must be issued at the start of a seriatim projection [cell pricing], or during a portfolio projection [new business pricing]); No otherwise
Valuation years [valn_yrs]	Valuation tasks (target column is valn_tgt_XXX) project this many years from the projection date; Except that GAAP valuations ignore this variable since they project to the DAC amortization years; Has no effect on projection tasks since they project to the 'Projection period to' property
Project GAAP results? [gaap_defn]	Yes for projection tasks to include GAAP calculations, reading GAAP factors/schedules from the output specified from previous valuation tasks; No otherwise;
Include assets? [assets_defn]	Yes for portfolio projection and valuation tasks to project asset/liability results; No for liability-only task;

Project variable	Description
Include target capital? [tgt_cap_defn]	Yes for projection tasks to include target capital results; No to treat target capital as zero; No otherwise;
Use independent model point processing? [independent_mp_defn]	Yes for portfolio projection and valuation tasks to project 'independent' model points using seriatim calculations before starting the portfolio calculations; No otherwise;
Validate data definitions? [validate_data_defn]	Yes to include optional input checking to produce warning messages in the run log; No otherwise
Include decrements? [decrements_defn]	No to treat mortality, lapse, and annuitization decrement assumptions as zero; Yes for normal calculations
Discount rates selected [disc_rates_selected]	Specify stochastic discount rates using a comma separated list with no spaces; these discount rates are specified on the Company input page
Portfolio yield definition [disc_rate_port_yld_defn]	Before Tax to discount using the asset yield; After Tax to discount using the asset yield times (1 – tax rate)
Valuation periods selected [valn_periods_selected]	For projection tasks which are included in nested projections, must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces; has no effect for standard projections. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0.

1.4 Output parameter set

1.4.1 General structure

An 'output parameter set' provides an interface to define projection task output parameters. Best practice is to create output parameter sets which can be reused across a number of projection tasks. Output parameter sets typically vary based on characteristics as follows:

- Calculation type
- Seriatim versus portfolio
- All, Statutory, or GAAP output
- Column output to value

Using the above structure, a naming convention is established so that it is not necessary to open the output parameter set when determining if a run parameter set should be selected.

The name is developed using values for each of the characteristic, each separated by a space:

#	Element value	Conditions and notes
1	Factors, Valn, Proj	Target column = factors_tgt (applies to life products only) Target column = valn_tgt_xxx Used for both pricing and projections, Target column != valn_tgt_xxx
2	Seriatim, Portfolio	Target model object = [product] Target model object = company; Leave this element out of the name for Factors
3	All Stat GAAP Hedge	Produces output for all columns in the top model Produces output for all statutory reporting columns in the top model and PBA submodels Produces output for all GAAP reporting columns in the top model and GAAP submodels Produces output for the hedge reporting columns in the top model and the hedge submodel Leave this element out of the name for Factors
4	[Column Output To]	The number of periods in the output, such as 240 When value is 0, leave this element out of the name

The following are some example output parameter object names:

- Factors 1440
- Proj Seriatim All 360
- Valn Portfolio GAAP
- Valn Portfolio Hedge

1.4.2 Output parameters

Output parameters are properties required by the RiskAgility FM platform. Parameterization of the output parameters is as follows:

Property	Values
Column output from	Typically 0
Column output to	The number of periods required in the output, such as 240 May be set to 0 for PBA valuations
Fixed discount rate list	Fixed discount rates, such as '8,10,12' for 8%, 10%, and 12% Leave this blank for Valn output parameter sets

Property	Values
Use column discount rates	Select to calculate present values using stochastic discount rates, and the asset yield Do not select for Valn output parameter sets
IRR wanted	Not used, since the RiskAgility FM system IRR does not cater for losses after profits
Output manager	Select the appropriate output manager for the projection in order to limit the output captured by the projection. For example, select the Valn GAAP output manager for a 'Valn Portfolio GAAP 240' output parameter set See the discussion of output managers in the Output management user guide book
Model object output	Select All Columns, No Columns, or Output Definitions for the model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions
Model object period	Not commonly used. Is only applicable when an output manager is selected and period definitions have been defined
Model object individual output	Not included as standard feature. Refer to the RiskAgility FM help for a discussion on defining and using individual output
Model object group expression	Not commonly used. Specify a group expression to override the group field of the model point files

1.5 Goal seek parameter set

1.5.1 General structure

A 'goal seek parameter set' provides an interface to define properties for running a projection in goal seek mode. Best practice is to create goal seek parameter sets which can be reused across a number of projection tasks. Goal seek parameter sets typically vary based on characteristics as follows:

- Variable submodel
- Variable name
- Target column submodel
- Target column name
- Targeted result
- Targeted result period

Using the above structure, a naming convention is established so that it is not necessary to open the goal seek parameter set when determining if a goal seek parameter set should be selected.

The name is developed using values for each of the characteristic, each separated by a space:

#	Element value	Conditions and notes
1	Variable submodel	The submodel of the dependent variable name, such as Company, UL, Trad, etc.
2	Variable name	The name of the dependent variable name, such as tgt_cap_rbc_mult, maint_exp_per_unit, comm_pct_prem_input, etc.
3	Target column submodel	The submodel of the target column, such as Company, UL, Trad, etc.
4	Target column name	The name of the target column, such as irr
5	Targeted result	The targeted result of the target column. For example 15pct for a 15% irr target
6	Targeted result period	The period for the target result Leave this out if the target period is 0

The following are some examples of goal seek parameter object names:

- Company maint_exp Company free_surp 5000000 360
- UL comm_pct_prem_input UL irr 15pct

1.5.2 Goalseek parameters

Goal seek parameters are properties required by the RiskAgility FM platform. Parameterization of the goal seek parameters is as follows:

Property	Values
Variable name	Select the submodel and variable which represents the variable for the goal seek process
Target value	Specify the targeted result for the goal seek process
Initial variable value	Specify the initial value of the goal seek variable
Goal seek iterations	Specify the maximum number of goal seek iterations.
Sensitivity	Specify the proximity of the target column value to the specified target value required for goal seeking. If the value of the target column falls within range of the target value within the maximum number of iterations specified, goal seeking is considered successful.

1.6 Projection tasks

1.6.1 General structure

A 'projection task' provides a structure to select assumption sets, run parameter sets, output parameter sets, and if applicable, goal seek parameter sets required to perform required model calculations.

Projection tasks are created on the run pages and the structure of the projection task parameters is dependent on the run page properties for the input manager and target model object. Thus, it is critical to select the appropriate run page when creating the projection tasks.

For example, a projection task to perform a 2014 embedded value seriatim projection for universal life model points, where a formulaic approach to reserves and capital is utilized, would be defined on the Standard Seriatim run page within the '2014 embedded value' run manager, since that run page is linked to the '2014 embedded value' input manager and has 'ul' as the target model object.

Best practice for projection task names is as follows:

- Use short but descriptive names
- Use standard abbreviations and acronyms
- Do not include elements in the name which are already specified in the run page name. For example, do not include 'Portfolio' in a projection task that is defined on the 'Standard Portfolio' run page. Also, do not include 'PBA' or 'GAAP' in the name of projection tasks which are defined on those pages
- Use baseline in the name for baseline projections
- Include information regarding the sensitivity test in the name of a sensitivity run
- Include ALM in the name for projections including assets
- Include IMP in the name for projections using independent model point processing

1.6.2 Projection task parameters

Projection task properties include the selection of the assumption sets, run parameter sets, output parameter sets, and if applicable, the goal seek parameter set. Parameterization of the projection task properties are as follows:

Parameter	Values
Company assumption set	Select the company input page assumption set from the linked input manager Not applicable for 'Rate Gen' projection tasks*
Economy assumption set	Select the economy input page assumption set from the linked input manager Not applicable for 'Rate Gen' projection tasks*
Investment assumption set	Select the investment input page assumption set from the linked input manager Not applicable for 'Seriatim' or 'Rate Gen' projection tasks*
Hedging assumption set	Select the hedging input page assumption set from the linked input manager. Not applicable for 'Seriatim' or 'Rate Gen' projection tasks*

Parameter	Values
Reinsurance assumption set	Select the reinsurance input page assumption set from the linked input manager. Not applicable for 'Rate Gen' projection tasks*
[Prod]xxx assumption sets	Select the product input page assumption sets from the linked input manager Not applicable for 'Rate Gen' projection tasks*
Riders assumption set	Select the rider input page assumption set from the linked input manager. Not applicable for 'Rate Gen' projection tasks*
[Prod]Data assumption sets	Select the product data page assumption sets from the linked input manager Not applicable for 'Rate Gen' projection tasks*
[Prod][xxx]Data assumption sets	Select the dependent data page assumption sets from the linked input manager, where xxx is Ben for payout annuity benefits, Reins for reinsurance, and Rider for riders. Not applicable for 'Rate Gen' projection tasks*
[Asset]Data assumption sets	Select the asset type data page assumption set from the linked input manager Not applicable for 'Seriatim' or 'Rate Gen' projection tasks*
Run parameter set	Select the appropriate run parameter set for the projection task For example, select the 'Valn Portfolio pba_scen_amt 2014_12 1to1000' run parameter set for a projection task defined on the 'PBA Portfolio' run page that calculates the PBA scenario amount over 1000 scenarios as of December 2014
Output parameter set	Select the appropriate output parameter set for the projection task For example, select the 'Proj Seriatim Stat' output parameter set for a projection task defined on the 'Standard Seriatim' run page that calculates statutory results for individual model points
Goal seek parameter set	Select the appropriate goal seek parameter set for the projection task For example, select the 'UL comm_pct_prem_input UL irr 15pct' goal seek parameter set for a projection task defined on the 'Standard Seriatim' run page that calculates the irr for individual universal life model points. Warning! Goal seeking will only work correctly when the selected run parameter set and goal seek parameter set contain consistent information.

*Input and data pages which are not applicable for projection tasks of a particular run page are shown in *italics* and are not selectable.

1.7 Run controller

1.7.1 General structure

A 'projection set' provides a structure for collecting projection tasks and other tasks together and running them through a single job submission. The tasks are ordered and dependencies are specified. Tasks which are independent are run in parallel.

Projection set loops and sub loops are included as required for nested projections (discussed in detail in the Nested projections chapter below).

Best practice for projection set names is as follows:

- Use short but descriptive names
- Use standard abbreviations and acronyms
- Use separate projection sets for 'Standard', 'PBA', 'GAAP', and 'Hedging' projection tasks, including 'Standard', 'PBA', 'GAAP', and 'Hedging' in the names
- Use separate projection sets for 'Valn', 'Pricing', and 'Proj' projection tasks, including 'Valn', 'Pricing', and 'Proj' in the names. An exception to this rule is that inner loop valuations for nested projections are included inside a 'Pricing' or 'Proj' nested projection. (See the Nested projections chapter below.)

1.7.2 Projection set properties

Projection set properties include projection set loops, independent loops, include sub loops, projection set sub loops, projection tasks, and task dependencies. Parameterization of the projection set properties are as follows:

Property	Values
Projection set loops	The loops to run for the projection set. For example, 1-100, 1,5,8,10, 1-5,20 The values specified here determine the values of ● The <*proj_set_loop_num*> wildcard ● The proj_set_loop_num, overall_first_proj_set_loop, overall_last_proj_set_loop, overall_is_first_proj_set_loop, overall_is_last_proj_set_loop, and overall_num_of_proj_set_loops system variables
Independent loops	Yes, when each projection set loop can be calculated at the same time
Include sub loops	Yes, for nested projections (See the Nested projections chapter below)
Projections before sub loop	Typically none
Projection set sub loops	The sub loops to run for the projection set. For example, 1-5, 12. The values specified here determine the values of ● The <*proj_set_sub_loop_num*> wildcard System variables proj_set_sub_loop_num, overall_first_proj_set_sub_loop, overall_last_proj_set_sub_loop, overall_is_first_proj_set_sub_loop, overall_is_last_proj_set_sub_loop, and overall_num_of_proj_set_sub_loops

Property	Values
Sub loop projections	The projection or other task included in the projection set. When projection set sub loops are included, all projection tasks are typically included within the Sub Loop section. Important Note: After a projection task is selected, the run page identifier of the projection task is not displayed within the projection set grid. Left click on the projection task to review the full properties of the projection task, including the run page. Also, specify the task dependencies. Tasks may be dependent on any number of tasks above the selected task. Reorder the tasks as required to develop appropriate task dependencies
Projections after sub loop	Typically none

2 STANDARD PROJECTIONS

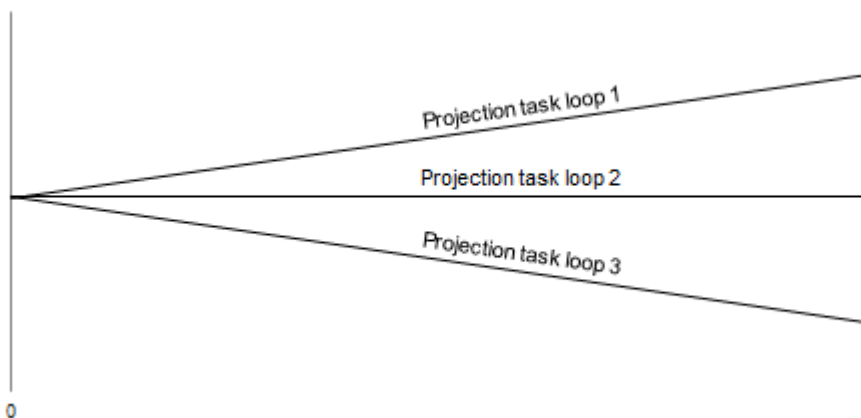
2.1 Standard projection overview

2.1.1 Use cases

Standard projections are applicable when a projection does not include inner loop valuations. That is, standard projections use only projection task loops; they do not use projection set loops and projection set sub loops.

2.1.2 Projection structure

Standard projections and valuations calculate results over scenario paths branching from the projection date.



2.1.3 Standard projection scenarios

For standard tasks, the scenarios to be calculated are the projection task loops specified in the Run parameter set. For example, specify "1-5,10,12" to project using scenarios 1 through 5, 10, and 12.

Standard task scenario rates must be specified in the scen_id variable (on the Economy page). Standard features of the U.S. Library do not include on-the-fly generation of these 'outer loop' scenario rates. See the Scenario tools chapter of the Economic scenarios book for information about creating scenario files.

When the task is deterministic, there is no scenario file and there should be only one projection task loop. See the Stochastic versus deterministic projections topic in the Standard Features chapter of the Introduction book.

Inner loop scenarios are not used by standard projection and valuation tasks; they only apply for tasks within a nested projection. See the Nested projections chapter, below.

2.2 Standard portfolio projections

This section discusses standard portfolio projections, which project results for the company and business segment, either including or excluding assets.

2.2.1 Company top model

Specify a portfolio projection by creating a projection task on a Run page that uses the company submodel as the target model object, such as the 'Standard Portfolio' run page. The following shows an example portfolio projection model tree, using the universal life project:



2.2.2 Including assets

Portfolio projections and valuations can include assets in the projection, since the projection uses aggregate results.

2.2.3 New business pricing

When the Pricing projection? project variable in the Run parameters set is Yes, special pricing measures are calculated for the new business portfolio.

2.2.4 Portfolio profit results

Portfolio projections calculate aggregate results in the company and segment submodels for investment income, taxes, target capital, distributable earnings, and free surplus. Thus profit reports for the entire projection show book profit before tax, book profit after tax, the effect of target capital, and distributable earnings.

Full profit results are available for report groups in seriatim projections (discussed below), where all calculations are performed for model points since there is no segment submodel.

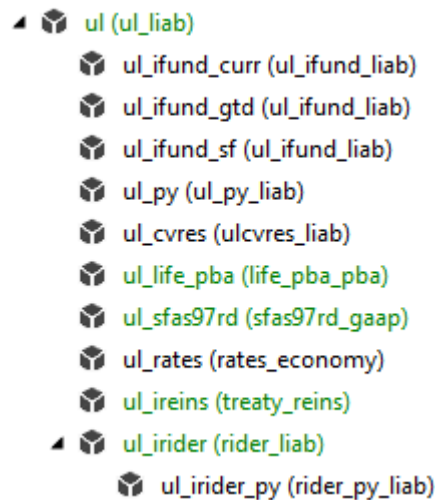
See the Profit definitions chapter of the Company and segments book for definitions of results such as book profit, distributable earnings, and undistributed earnings.

2.3 Standard seriatim projections

This section discusses standard seriatim projections, which project liability-only results separately for each model point.

2.3.1 *Product top model*

Specify a seriatim projection by selecting the product type submodel to be the top model. The following shows an example seriatim projection model tree, using the universal life project:



Assets cannot be included in seriatim projections, which process each model point separately without calculating aggregate results.

Independent model point processing (see the Independent model points topic in the Standard features chapter of the Introduction book) does not apply for seriatim projections, since all model points are automatically calculated separately without this special processing.

2.3.2 *Surplus*

There is no free surplus for seriatim projections. Initial free surplus is automatically zero, and distributions automatically pay out the full amount of distributable earnings. Thus the product submodels have no counterpart for these input variables and columns which are in the segment submodel.

2.3.3 *Cell pricing*

When the Pricing projection? project variable in the Run parameter set is Yes, special pricing measures are calculated for each pricing cell. See the Calculation types section in the Standard features chapter of the Introduction book.

2.3.4 *Seriatim profit results*

For seriatim projections, profits are calculated by model point, including investment income, target capital, and tax calculated for each model point. Thus report groups (which combine model point results) include full reporting of book profit before tax, book profit after tax, the effect of target capital, and distributable earnings.

Profit calculations for seriatim projections include:

- GAAP results are calculated using a GAAP schedule for each model point separately.
- PBA reserves and capital are calculated for each model point separately.
- Tax columns in the product type submodel are for the purpose of calculating profits in seriatim projections. These columns are not called during portfolio projections, where taxes are calculated only using aggregate results.
- Investment income is allocated between profit components the same way as for portfolio projections. Just as for the tax columns, these investment income columns in the product type submodel are called only during seriatim projections.

2.4 Point-in-time valuations

This section discusses point-in-time valuations, which calculate values at $t = 0$ for each projection task loop. Inner loop valuations are discussed in the Nested projections chapter, below.

A projection task produces valuation results when its target column is one of the 'valn_tgt_xxx' columns provided with the project. See the GAAP accounting and PBA books for the target column appropriate for each type of valuation.

2.4.1 Portfolio valuations

Portfolio valuations have company as the top model, calculating results for the business segment (for PBA), GAAP cohorts (for GAAP valuations), or Hedge groups (for Hedge valuations).

Specify the target column as appropriate for the valuation to be performed. For example:

- GAAP valuation (valn_tgt_gaap_sfas97rd or valn_tgt_gaap_sfas120)
- PBA valuation (valn_tgt_pba_det_res or valn_tgt_pba_scen_amt)
- Hedging valuation (valn_tgt_hedge_shock or valn_tgt_hedge_final).

For PBA and GAAP stochastic valuations, each projection task loop produces the $t = 0$ result for that scenario path. Then, the reporting tool combines these results, such as making the Conditional Tail Expectation (CTE) calculation for stochastic PBA reserves and capital.

For Hedging shock valuations, each projection task loop produces the $t = 0$ present value of claims and charges for the combination of a shock type and economic scenario path. A Hedging final valuation then calculates the $t = 0$ economic reserve (present value claims – present value charges) for each shock type.

2.4.2 Seriatim valuations

Seriatim valuations have the product type submodel as the top model, calculating results separately for each model point.

Specify the target column as appropriate for the valuation to be performed. For example:

- GAAP factors valuation (valn_tgt_gaap_sfas60_sfas97lp_sfas120_nlp for Trad).
- Gross premium reserve (GPR) valuation (valn_tgt_gpr). This is always a seriatim projection, even though the GPR is collected and used within cohorts defined for this purpose.

When the valuation results will be used in a seriatim projection, a valuation task that is normally a portfolio valuation will instead be a seriatim valuation. This applies for GAAP and PBA valuations, such as for use in cell pricing and other seriatim projections. Of course, this means that GAAP results are calculated separately for each model point, rather than grouping results for the block of business or within each GAAP cohort.

2.4.3 Including assets

Portfolio valuations can include assets in the projection, since the projection uses aggregate results. See the asset/liability versus liability-only topic in the Standard features chapter of the Introduction book.

Seriatim valuations cannot include assets since each model point is projected separately.

2.4.4 Valuation years

The 'Projection period from' and 'Projection period to' properties in the Run parameters set should both be 0 for valuation tasks. Specifying a different 'from' value produces an error. Specifying a different 'to' value is ignored.

Instead, specify the time for projecting each valuation using the Valuation years project variable in the Run parameters set. These are full years from the valuation date. For example, specifying 30 years for a 2015/06 (June 30, 2015) valuation date means that values are projected through June 2045.

2.4.5 Column output

The 'Column output from' property in the Output parameters set should be 0. The 'Column output to' property is 0 when the valuation result is only the present value calculations at the valuation date. Specify a higher 'Column output to' period when you want to see the projected results used to calculate the present values, such as the liability cash flows.

An exception is for a projection task which is producing a GAAP schedule for use by a subsequent projection, rather than only valuing the DAC amount. Specify the number of months for GAAP schedule output.

Another exception is for GPR valuations, which produce GPR results for each future date. Specify the number of months for GPR valuation results output.

2.5 Standard projection setup

2.5.1 Run page

Standard projections are specified on run pages as follows:

Standard projection type	Run page
Rate generation	Target model object = 'rates'. For example, use the • 'PBA Rate Gen' run page for generating for PBA valuations • 'GAAP Rate Gen' run page for generating for GAAP valuations • 'Hedging Rate Gen' run page for generating for Hedging valuations
Standard seriatim	Target model object = '[prod]', where [prod] = 'ul', 'fa', etc. For example, use the 'Standard Seriatim' run page
Standard portfolio	Target model object = 'company' For example, use the 'Standard Portfolio' run page

2.5.2 Input and data page assumptions

Select the appropriate assumption sets for standard projections from the linked input manager for each of the required input and data pages.

2.5.3 Run parameter set

Create run parameter sets for standard projections which include the following settings:

Property	Values
Target object	For projection tasks, select the key result column from the top model, like distrib_earnings, or irr (for pricing/new business projections). Valuation tasks must name the valn_tgt_xxx column which specifies the results to be valued. Target Output Definitions may be selected for projections which are not valuation or factor projections
Target column	Only submodels of from 'company', '[product]', and 'rates' may be selected as the model object Can be the key result column from the top model, like distrib_earnings, or irr (for pricing/new business projections). Valuation tasks must name the valn_tgt_xxx column which specifies the results to be valued.
Projection period from	Typically 0
Projection period to	For projection tasks, the number of periods to include in present values. For valuation tasks (target column is valn_tgt_xxx), 0 since the valuation projection is specified by the Valuation years project variable.
Projection period min	0 for liability only portfolio tasks, since there should be no calculations before the projection date. -xxx for seriatim tasks, where xxx is before the earliest issue period to allow for external source lookup keys calculating before the projection date (such as pol_yr) -xxx for ALM portfolio tasks, where xxx is before the earliest issue period of existing assets Asset type top models may need to reach back to the earliest issue date for the existing assets.
Projection period max	At least as long as the calculations for each submodel. For example, UL individual fund calculations must calculate to maturity when used for solving, even when the projection period is shorter. For ALM projections, the asset submodels must calculate 360 months (30 years) longer than the projection period, in order to calculate the market value of bonds (etc.) purchased at the end of the projection. Setting this value longer than needed allocates (and thus uses) more memory for the projection; however submodels for model point calculations automatically reduce memory use to the maturity period for each model point.

Property	Values
Projection task loops	Typically, the scenario paths to process, for example 1-1000, or 1,8,25, or 1-10,500. However, for variable annuity hedging valuations, this represents the combination of shocks and scenarios to process,
Wildcard set	Select the wildcard set which contains the paths to the appropriate external data for the projection task.
Assumption set restore	Yes for seriatim tasks, and when independent model point processing applies for portfolio tasks; No otherwise
Extract model points	Yes to enable the Extract model points feature specified for the model point file in the Input manager; has no effect when this feature is not used for the model point file
Sliding windows	Do Not Slide for seriatim runs and when small models or independent model point processing enables portfolio runs to fit within available memory; Slide to reduce memory requirements during a projection task which does not otherwise fit into available memory
Distribution	For portfolio tasks, use Split by Projection Loop to distribute the projection task loops to the cores on the grid. For seriatim tasks, could be more efficient to use Split by Group when number of model points exceed number of projection task loops Specify a Target Split for Distribution value for the minimum number of Projection Loops or Groups that will be allocated to a Grid Task in a Distributed Processing environment. The ideal setting for this value requires thorough analysis of the model and grid configuration.

2.5.4 Project variables

Additionally, the run parameter set should include the following project variable settings.

Property	Values
Projection date [proj_date]	In-force values, and present value calculations, are as of the end of the month specified as yyyy/mm, such as 2014/12 for December 31, 2014
Pricing projection? [pricing_defn]	Yes for projections calculating special pricing results like IRR (all model points must be issued at the start of a seriatim projection [cell pricing], or during a portfolio projection [new business pricing]); No otherwise
Valuation years [valn_yrs]	Valuation tasks (target column is valn_tgt_XXX) project this many years from the projection date; Except that GAAP valuations ignore this variable since they project to the DAC amortization years; Has no effect on projection tasks since they project to the 'Projection period to' property

Property	Values
Project GAAP results? [gaap_defn]	Yes for projection tasks to include GAAP calculations, reading GAAP factors/schedules from the output specified from previous valuation tasks; No otherwise; Has no effect on valuation tasks
Include assets? [assets_defn]	Yes for portfolio projection and valuation tasks to project asset/liability results; No for liability-only task; Has no effect on seriatim tasks
Include target capital? [tgt_cap_defn]	Yes for projection tasks to include target capital results; No to treat target capital as zero; No otherwise; Has no effect on valuation tasks
Use independent model point processing? [independent_mp_defn]	Yes for portfolio projection and valuation tasks to project 'independent' model points using seriatim calculations before starting the portfolio calculations; No otherwise; Has no effect on seriatim tasks
Validate data definitions? [validate_data_defn]	Yes to include optional input checking to produce warning messages in the run log; No otherwise
Include decrements? [decrements_defn]	No to treat mortality, lapse, and annuitization decrement assumptions as zero; Yes for normal calculations
Discount rates selected [disc_rates_selected]	Specify stochastic discount rates using a comma separated list with no spaces; these discount rates are specified on the Company input page
Portfolio yield definition [disc_rate_port_yld_defn]	Before Tax to discount using the asset yield; After Tax to discount using the asset yield times (1 – tax rate)
Valuation periods selected [valn_periods_selected]	Does not affect standard projections
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Does not affect standard projections

2.5.5 Output parameter set

Create an output parameter set for standard projections which include the following settings.

Property	Values
Column output from	Typically 0
Column output to	The number of periods required in the output, such as 240 May be set to 0 for PBA valuations

Property	Values
Fixed discount rate list	Fixed discount rates, such as '8,10,12' for 8%, 10%, and 12% Leave blank for valuation projections
Use column discount rates	Selected to calculate present values using stochastic discount rates, and the asset yield Do not select for valuation projections
IRR wanted	Not used, since the RiskAgility FM system IRR does not cater for losses after profits
Output manager	Select the appropriate output manager for the projection in order to limit the output captured by the projection.
Model object output	Select All Columns, No Columns, or Output Definitions for the model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions

2.5.6 Run controller projection set

Standard projections may be organized into projection sets on the Run controller and run together by submitting the run controller. For example, baseline standard projections would be organized into a 'Standard Baseline' projection set, with mortality sensitivities organized into a 'Standard Mortality Sensitivities' projection set.

Standard projections do not require projection set loop or sub loops and may or may not have dependencies. Thus, organizing standard projections into a projection set is not required.

2.6 Goal seeking

2.6.1 Overview

Standard projections can use the goal seeking feature to find the value of a specified input variable to achieve a target result. For example, a universal life projection can find the crediting spread (from the asset yield) to achieve a specified internal rate of return (IRR).

Goal seeking finds the input variable value which achieves the target result for all the model points included in the projection. When multiple model points are included in seriatim pricing projection, the goal seek is performed across all model points, not individually for each model point. A special process using a RiskAgility FM 'other' task can provide for goal seeking on each model point individually.

2.6.2 Goal seeking variable setup

RiskAgility FM does not allow for goal seeking on a variable which is linked to an external source. For example in the Traditional Life application, you cannot select the `annualzd_prem_per_unit` variable as a goal seeking variable if it is linked to an external source range.

One approach to overcome the RiskAgility FM limitation is to introduce a custom goal seeking multiple variable that can be applied to values of the variable you wanted to goal seek on.

Follow these steps:

1. Add a new floating point number code variable in the appropriate model class of your application, with a default value of 1.0.
2. Update the model class code to apply this multiple against the value of the variable you wanted to goal seek on.
3. Associate the new code variable with an input variable on the appropriate input page of the input manager of your application.
4. Set the Value Type of the new variable as Assumption for each applicable assumption set, with an assumption set value of 1.0.

With this minor customization, you now have a variable which can be used in a goal seeking process to determine a multiple of the base external assumptions to achieve a targeted result.

2.6.3 Goal seeking parameter set setup

The goal seeking process is managed through a Goal Seek Parameter set.

Given the limitation mentioned above and the proposed approach, the variable selected in the goal seek parameter set must be the new goal seek multiple variable and the initial value set as an appropriate starting value of the multiple, which is typically the default value of 1.0.

Other parameters of the goal seek parameter set are specified as desired, such as for the target value and sensitivity for the target column, and the number of goal seek iterations.

2.6.4 Projection task setup

Set a standard projection to be a goal seeking projection by selecting the tick box for the Goal Seek Parameter Set. Then, select the appropriate goal seek parameter set for the projection task.

2.7 Running and debugging

2.7.1 Running

The most common way to run independent standard projections is by submitting the job from the run page. You can also select the single independent standard projection from the run controller projection set and submit the job.

You may specify any name you desire in the 'Job Name' field of the job submission.

Typical practice is to (1) specify the <*output*> wildcard for the 'Output Location' and, (2) manage all output in the single <*output*> folder by not selecting the 'Create subfolder for Job Name', and (3) specify to overwrite existing files.

Warning! The 'Create subfolder for Job Name' must NOT be selected when submitting a standard projection which is producing dependent output for a task within a nested projection. Selecting this option would lead to the output from the task being stored in a location which is not recognizable by the nested projection task.

Multiple independent standard projections can be run together from the run page by selecting the projections and submitting the job.

Alternatively, multiple independent standard projections can be run together from the run controller projection set. To run, submit the entire projection set or individually select the desired projection tasks and then submit the job.

If a standard projection is dependent on another projection, running that projection from the run page requires the task it is dependent on has been run and the required results are available. Selecting to submit a dependent standard projection from the run controller without also selecting the task it is dependent on will cause RiskAgility FM to raise an error.

2.7.2 *Job monitor*

Review the state and run logs of the standard projections through the job monitor. Drill down from the job to review the state of the job and run logs from the standard projections.

2.7.3 *Debugging*

Debugging of standard projections is accomplished through the RiskAgility FM integrated debugger. Follow these steps to debug a standard projection.

1. Use the code explorer to set a breakpoint in the model code
2. Select the standard projection from the run page
3. Right click and select Debug
4. Continue past any system exceptions until you arrive at your breakpoint
5. Use the debugger interface features to evaluate the calculations of the standard projection
6. Stop the debugging to return to the standard RiskAgility FM interface

Warning! RiskAgility FM does not always end the run when selecting to stop the debugging. You may need to cancel the job through the job monitor to end the execution of the job.

3 NESTED PROJECTIONS

3.1 Nested projection overview

3.1.1 *Use cases*

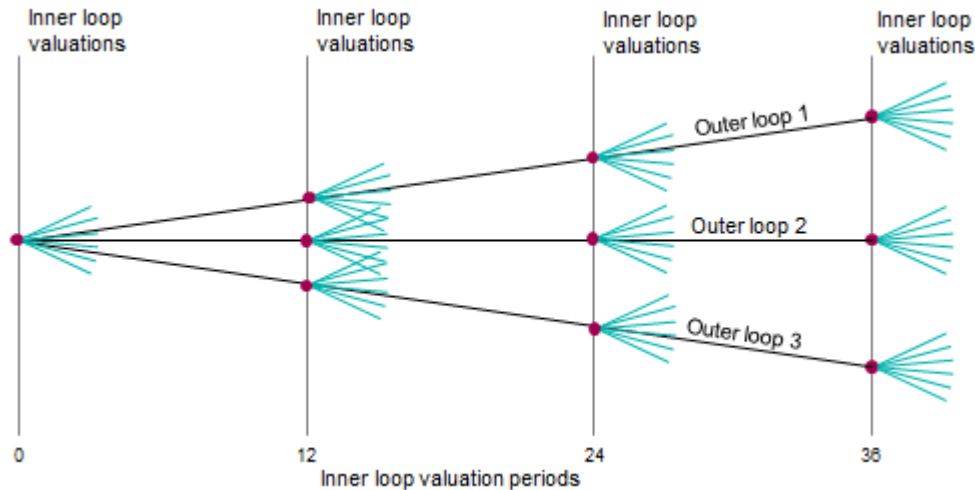
Nested projection processing is needed when special, typically stochastic, valuations must be performed along the projection scenario paths. Financial results are calculated through time, stopping at specified projection periods to perform valuation calculations.

For example, a nested projection would be appropriate if, let's say, you have been asked to produce a three-year planning projection showing how PBA reserves and capital, and dynamic GAAP unlocking, will affect your company's financial results using selected economic scenario paths.

3.1.2 *Projection structure*

To meet the above objective, a projection must process multiple outer loops along the projection horizon, with results from, typically stochastic, inner loop valuations feeding the experience projection at specified points along the projection horizon.

The diagram below shows an example projection that performs special valuation calculations at the projection date and each 12 months during the projection.



In a nested projection, the projection must progress in stages:

- 1 First, the valuations at the projection date are calculated. The experience projection uses those first valuation calculations to begin the projection for each outer loop scenario path.
- 2 The experience projection progresses through period 12, producing financial results.
- 3 The valuation tasks run using the inforce information at the end of period 12.
- 4 The experience projection uses the new valuation results to progress from period 13 to 24.
- 5 And so on until the end of the projection.

Thus each outer loop projection is broken up into 'time steps' that advance the calculations between the inner loop valuations.

3.1.3 *Dependent standard projections*

Nested projections are completed using a set of dependent standard projections. For example, standard projections which define a scenario generation, a valuation, and an experience projection are combined together with dependencies to form a nested projection.

The experience projection is dependent on the inner loop valuations, since the valuation results are used to determine the financial statements.

The valuations are dependent on scenario generation projections because the inner loop valuation calculations are projected using 'inner loop' scenario paths which branch from the 'outer loop' scenario paths at each valuation date.

The scenario generations are independent.

3.1.4 *Seriatim versus portfolio projections*

Nested projections may be either seriatim or portfolio projections. Whether a nested projection is seriatim or portfolio is based on the structure of the setup of the experience projection. A nested projection is seriatim when the experience projection is seriatim and is portfolio when the experience projection is portfolio.

As an example, a projection that is set up with cell pricing model points and 'ul' as the top model, and includes revaluations of PBA reserves and capital, is a seriatim nested projection. However, a projection that is set up with in force model points and company as the top model, and includes SFAS 97 retrospective deposit GAAP unlocking, is a portfolio nested projection.

A seriatim nested projection must have all dependent standard projections which are also seriatim. However, a portfolio nested projection may have some dependent standard projections which are seriatim and some which are portfolio.

3.2 Projection set structure

The following shows an example from the Run controller display:

C2. PBA Cell Pricing Last Run: **Job 208**

Projection Set Loops: ☒ Independent Loops ☒ Include Sub Loops

Projections before Sub Loop:

Projection	Dependencies
Enter Projection name	

Sub Loop: Projection Set Sub Loops:

Projection	Dependencies
PBA Gen SET	PBA Gen SET
PBA Det Res Cell Pricing	PBA Gen SET
PBA Gen Stoch	PBA Gen Stoch
PBA Scen Amt Res and Cap Cell Pricing	PBA Det Res Cell Pricing, PBA Scen Amt Res and...
PBA Cell Pricing	
Enter Projection name	

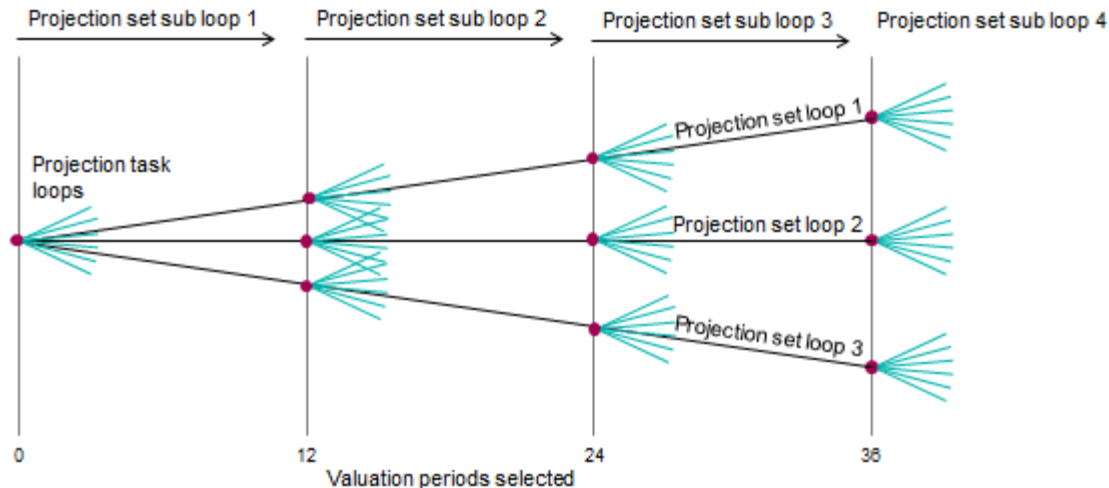
Projections after Sub Loop:

Projection	Dependencies
Enter Projection name	

- This is a PBA projection over outer loop scenarios 1, 5, and 10. (For a standard projection, these would have been the projection task loops in the Run parameter set.)
- The valuation tasks, and the experience projection task, are run within the four projection set sub loops. The Valuation periods selected project variable (in the Run parameter set for each of these tasks) defines these sub loops to begin at $t = 0, 12, 24$, and 36 .
- This projection set includes tasks to generate the inner loop scenario rates for the two valuation tasks.
- The Dependencies for each task restrict that the valuation tasks cannot begin until the scenario generation tasks have completed. The experience projection cannot start until the valuation tasks have finished.

3.2.1 Projection set loops and sub loops

Nested projections are executed in an efficient manner by taking advantage of the projection set loop and sub loop capabilities of the RiskAgility FM platform. The following diagram shows how the projection set loops and sub loops are defined for the tasks within the projection set:



- The outer loop scenarios are the 'projection set loops'. The projection set loop numbers are the paths to project from the specified scenarios file.
- The time steps are the 'projection set sub loops'. These are sequential steps, with sub loop 1 advancing from the projection date to the first inner loop valuation after the projection date, sub loop 2 advancing to the next valuation date, and so on.
- The valuation dates are specified in the 'valuation periods selected' project variable in the Run parameter set. For example "0,12,24,36" specifies that projection set sub loop 1 ends at $t = 12$, sub loop 2 ends at $t = 24$, sub loop 3 ends at $t = 36$, and the final sub loop 4 completes the projection calculations at $t = 36$.
- The inner loop scenarios for each valuation task are the 'projection task loops' specified in the Run parameter set for each task.
- The experience projection task must leave the projection task loops property blank; this task advances along the projection set loops; it has no inner loop valuation calculations.

3.2.2 Distributed processing

The nested projection structure of the projects provides for efficient distribution of compute-intensive calculations:

- In the diagram above, the three outer loops are independent of each other. Thus the calculations for each outer loop can be distributed so they calculate at the same time.
- Further, the tasks within each time step can process at the same time unless there are dependencies. In this case, the two scenario generation tasks can start at the same time. The two valuation tasks can also start at the same time (as long as their scenario generation tasks have finished.)
- Finally, the inner loop valuation scenario calculations within a valuation task can be distributed. When the PBA scenario amount task uses 1000 scenarios, those calculations can use up to 1000 calculation cores on the grid.

3.2.3 Assumption sets

Each type of valuation, and the experience projection, is performed using a different task which has assumptions appropriate for that task.

For example, the PBA valuation tasks in the above example use different Company assumption sets to specify the PBA definitions, and different Economy assumption sets to specify the appropriate scenario rates for each type of valuation.

3.3 Nested projection scenarios

3.3.1 Scenario rates

For nested projections, the `scen_id` variable (in the Economy input page) specifies the outer loop scenario paths. The projection set loops (specified for the projection set on the Run controller) selects which of these outer loops to process. These rates cannot be generated on-the-fly. See the Scenario tools chapter of the Economic scenarios book for information about creating scenario files.

Historical scenario rates must be specified in the `historic_rates_id` variable, which contains rates before the projection date such as for calculation of moving average competitor rates in the early projection periods.

The `inner_loop_valn_scen_id` variable specifies the inner loop scenario paths, which apply only for valuation tasks. The projection task loops (in the Run parameter set) selects which of these inner loops to process.

Inner loop scenarios can be generated on-the-fly for time step valuations after the projection date ($t = 0$). See the Economic scenarios generation chapter of the Economic scenarios book.

3.3.2 Outer loop scenarios

For nested projections, the outer loop scenarios to be calculated are the projection set loops specified in the Run controller. For example, specify "1-5,10,12" to project using scenarios 1 through 5, 10, and 12.

Outer loop scenario rates must be specified in the `scen_id` variable (on the Economy page). Standard features do not include on-the-fly generation of these scenario rates. See the Scenario tools chapter of the Economic scenarios book for information about creating scenario files.

3.3.3 Inner loop scenarios

Inner loop scenarios only apply for projection tasks within a nested projection. Specify the initial (at the projection date) inner loop scenarios in the `inner_loop_valn_scen_id` variable on the Economy input page.

The inner loop scenarios for valuations during the projection must be consistent with the outer loop economic rates at the time of that valuation. Thus the inner loop scenario rates should begin from a different starting point for each inner loop valuation after the projection date.

Typically, the inner loop scenarios are generated at the time of each valuation. This is done by including in the projection set a task for each set of inner loop scenarios needed by the valuation tasks.

3.4 State_of_world model point files

3.4.1 Model point file names

Since the projection tasks run at each valuation date along each outer loop path, a different set of model point records is needed for each time step. The experience projection task creates these model points at the end of each time step.

The model point records file names (on the [product][xxx]Data input pages) must include the outer loop and time step numbers, such as `mpfilename_<proj_set_loop_num>_<proj_set_sub_loop_num>.csv`. You must provide the initial file, `mpfilename_x_1.csv`, where `x` is the outer loop number. The experience projection creates the model point files for the other time steps, as discussed below.

Asset model points

This also applies for the asset model point files. When the experience projection includes assets, the experience projection creates model point files for each product and asset type included in the projection. Each new file is created at the end of each time step as the projection progresses.

The initial (at the projection date) asset model point files specify the existing assets at $t = 0$. For a pricing projection, where there are no initial existing assets, you must specify the `_x_1` version of the file even though it contains no model points. This file name is needed for the experience projection to create asset model point files for future time steps, even though it is not needed for starting the projection.

3.4.2 *At inner loop valuation dates*

The experience projection task creates new model point files which will be used for the tasks operating at each successive valuation date within each outer loop scenario path. For example, a new set of model point files is created for time step 2 (valuation at $t = 12$), time step 3 (valuation at $t = 24$), and time step 4 (valuation at $t = 36$) for each outer loop in the example projection shown above.

This is accomplished by the `write_state_of_world` function defined for each product type and asset type submodel. This function within the product type submodel creates a new model point file containing the same model point identification, insured characteristics, and policy specifications variable values, but with updated initial (as of the projection date) values. For example, the file for use at $t = 12$ contains the initial policies and other model point values as of the end of projection month 12, as calculated by the experience projection for the applicable outer loop scenario path.

This same process operates similarly for each asset type. For example, a new set of existing bond model points is created containing a record for each bond existing in the experience projection at the end of that month for the appropriate outer loop scenario path. The initial par value, book value, etc., is the value for each bond at that date. This model point file includes records for bonds purchased during the projection. Bonds which have matured or have been sold are no longer contained in the file.

These new files are .csv files which are created in the same directories where the initial model point files are located. Each of these files is named with `_<proj_set_loop_num*>_<proj_set_sub_loop_num*>` appended to the base file name.

3.4.3 *Rerunning a time step*

After the nested projection set has run completely, the model point files for each time step along each outer loop have been created. You can then re-run selected time steps without having to rerun the entire nested projection.

For example, let's say you want to investigate the detailed calculations for a valuation task at $t = 24$ for the first outer loop. You can change the Output parameter set to produce output for all periods, rather than using the normal Column output to = 0 that applies to minimize output file size during the production run. You can similarly select to output All Columns results for selected submodels. Now change the projection set to run only projection set loop 1 and projection set sub loop 3. Since `mpfilename_1_3.csv` already exists, you can run this one valuation task with full output, so you can understand the detailed calculations that produce the final valuation result for that time step.

3.4.4 *Special model point*

Each time step model point file includes a special model point which is added to the model point file. The model point ID is 'mp_time_step', issued after the final inner loop valuation date, with 0.0001 initial sum assured, policies, and units so that the effect on results is trivial.

The purpose of this special model point is to make sure that there is at least one model point that is still in force through the last inner loop valuation date. This assures that the projection calculations will continue through this time even when there is no business remaining in force. This is necessary for the runoff calculations such as for DAC tax amortization and GAAP DAC amortization.

Warning! The Extract model points feature cannot be used for nested projections. Since this special model point is added to the beginning of the model point file, the model point numbering is changed after the first time step.

3.5 *State_of_world financial results*

A special function in the top model of the experience projection, `write_financial_state_of_world`, temporarily stores financial results needed when the experience projection resumes at the start of the next time step. This transfer of results between time steps occurs 'behind the scenes', and is not included in output results. For example, year-to-date tax results are needed to continue calculations in the next time step.

This is similar to the `state_of_world` function for model point files, which is discussed below.

3.6 Nested projection output

3.6.1 Nested projection output files

Output results from nested projections are split into separate output for each time step along each outer loop. Each file has the values of `<proj_set_loop_num>` and `<proj_set_sub_loop_num>` within the naming convention structure of `@<proj_set_loop_num>_<proj_set_sub_loop_num>` appended to the normal output file base name. For example, a file including “@1_2”, is the output for projection set loop 1, projections set sub loop 2.

3.6.2 Valuation output

Valuation results used by subsequent tasks must be specified using the `_<proj_set_loop_num>_<proj_set_sub_loop_num>` addition to the base output file name. For example, the experience projection using PBA deterministic reserve valuation results specifies these results such as `<output>PBA Seriatim_PBA Det Res Cell Pricing@<proj_set_loop_num>_<proj_set_sub_loop_num>~company~liab~ul~ul_life_pba.csv`

3.6.3 Standard reporting

Individual time step output for nested projections is combined to form a single set of output files for the nested projection for standard reporting.

3.6.4 Differences from standard projections

As explained above, nested projections pass information between each time step using the ‘`write_state_of_world`’ functions for model point and financial results. This enables the nested projection to produce the same results as for the equivalent standard projection.

There may be instances, however, where the standard projection does not pass along all information needed for the nested projection to exactly reproduce the standard projection results. Such exceptions are documented in the book for each product type.

An example is the universal life feature to guarantee the crediting rate at issue for the first months of the policy. Since the model point record includes only the initial (at the projection date) crediting rate, and not also the crediting rate at issue, the temporary guarantee rate can vary between time steps. See the Interest crediting section in the Product benefits and values chapter of the Universal life book.

3.7 Nested projection setup

The following explains differences in the setup for nested projections, compared to standard projections (which are discussed in the preceding chapter).

3.7.1 Run page

Create the projections required to make up the nested projection on Run pages similarly to the setup of projection tasks on a Run pages for standard projections.

Dependent standard projection type	Run page
Rate generation	Target model object = 'rates'. For example, use the • 'PBA Rate Gen' run page for generating for PBA valuations • 'GAAP Rate Gen' run page for generating for GAAP valuations • 'Hedging Rate Gen' run page for generating for Hedging valuations
Seriatim	Target model object = '[prod]', where [prod] = 'ul', 'fa', etc. For example, use the • 'PBA Seriatim' for PBA seriatim inner loop valuations and experience projections • 'GAAP Seriatim' for GAAP seriatim inner loop valuations and experience projections
Portfolio	Target model object = 'company' For example, use the • 'PBA Portfolio' for PBA portfolio inner loop valuations and experience projections • 'GAAP Portfolio' for GAAP portfolio inner loop valuations and experience projections • 'Hedging Portfolio' for Hedging portfolio inner loop valuations

3.7.2 Run parameter set

For nested projections, the settings for the Run parameter set selected for each projection are similar to the settings for standard projections with the following additional considerations.

Property	Values
Projection task loops	For valuation tasks, the inner loop scenarios to be processed for the valuation. For experience projection tasks, this property must be left blank since the outer loop scenarios are specified for the projection set rather than for this individual task.
Extract model points	This feature cannot be used for nested projections, which add a 'dummy' model point to the file to make sure there is at least one model point that remains in force through the entire projection

3.7.3 Project variables

For nested projections, the settings for the project variables of the Run parameter set selected for each projection are similar to the settings for standard projections with the following additional considerations.

Property	Values
Projection date [proj_date]	Must be the same for all tasks within the nested projection set
Valuation periods selected [valn_periods_selected]	Must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces; has no effect for standard projections. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0.
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Must be the same for all tasks within the nested projection set. Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs which generate large scenario files.

3.7.4 Output parameter set

For nested projections, the settings for the output parameter set selected are similar to the settings for standard projections with the following additional considerations.

Property	Values
Fixed discount rate list	Not defined for the valuation or generation tasks
Use column discount rates	Not defined for the valuation or generation tasks

3.7.5 Run controller projection set

Nested projections **must** be run from a projection set from the Run controller. Run the nested projection by submitting the projection set.

Specify the following projection set properties for nested projections.

Property	Values
Projection set loops	The outer loop scenarios to project, such as "1-5,10", specified here instead of in the Run parameter set
Independent loops	Yes when each outer loop can be calculated at the same time

Property	Values
Include sub loops	Yes
Projections before sub loop	Typically none
Sub loops	The time steps to project, such as “1-4”. The Valuation periods selected property in the Run parameter set assigns a valuation period to each time step, such as “0,12,24,36”. After the full set of time steps have been run, subsequent runs can include only the time steps to be rerun (such as to analyze valuation details) such as “2”.
Sub loop projections	Each valuation task, plus the scenario generation task for each valuation (when appropriate), plus the experience projection task.
Sub loop projection dependencies	Each valuation task is dependent to start only after its scenario generation task has finished. The experience projection task is dependent to start only after the valuation tasks have finished. Some valuation tasks must wait for other valuation tasks, such as SFAS97RD GAAP that uses the SOP 03-1 valuation results for universal life products.
Projections after sub loop	Typically none

3.8 Running and debugging

3.8.1 Running

Nested projections must be run from the run controller. Right click on the projection set and submit the job.

Typical practice is to (1) specify the <*output*> wildcard for the ‘Output Location’ and, (2) manage all output in the single <*output*> folder by not selecting the ‘Create subfolder for Job Name’, and (3) specify to overwrite existing files.

Warning! The ‘Create subfolder for Job Name’ must NOT be selected when submitting a nested projection. Selecting this option would lead to the output being stored in a location which is not recognizable by all tasks within the nested projection task.

An individual projection within the nested projection may be submitted for a specific projection set loop and sub loop from the run controller when all the required information for that projection from the previous sub loop is available.

And, similar to standard projections, selecting to submit a dependent standard projection of the nested projection from the run controller without also selecting the task it is dependent on will cause RiskAgility FM to raise an error.

3.8.2 Job monitor

Review the state and run logs of the nested projections through the job monitor. Drill down from the job to review the state and run logs from each projection task within each projection set loops and sub loop.

3.8.3 Debugging

It is not possible to debug the full nested projection run through RiskAgility FM. However, it is possible to debug an individual dependent standard projection of the nested projection for a specific projections set loop and sub loop.

Follow these steps to debug one of the dependent standard projections of a nested projection for one projection set loop and sub loop.

1. Use the code explorer to set a breakpoint in the model code
2. Select the nested projection dependent standard projection from the run page
3. Right click and select Debug
4. Drill down into the Debug Properties of the Debug Job dialog
5. Set the values for the Projection Set Loops and Projection Set Sub Loops consistent with the settings for the nested projection in the projection set in the run controller
6. Set the values for the Debug Loop and Debug Sub Loop to the desired projection set loop and sub loop to debug
7. Continue past any system exceptions until you arrive at your breakpoint
8. Use the debugger interface features to evaluate the calculations of the dependent standard projection
9. Stop the debugging to return to the standard RiskAgility FM interface

Warning! RiskAgility FM does not always end the run when selecting to stop the debugging. You may need to cancel the job through the job monitor to end the execution of the job.

4 SLIDING WINDOWS

4.1 Why sliding windows?

Serial projections (where the product type submodel is the top model) have low RAM memory requirements since each model point is calculated separately, one at a time. Memory requirements can be high, however, for portfolio projections and valuations (where the company submodel is the top model) using period-by-period processing, where all model points are held in memory and projected together through each month.

The sliding windows feature of RiskAgility FM reduces the memory requirements for portfolio projections. Instead of retaining each column's values for each month (or year, for some submodels) of the projection, each column designates the 'window' of results which is retained for use in current calculations. As the projection progresses, the window 'slides', replacing the results no longer needed with new results which are needed for ongoing calculations.

The size of the window for each column is determined by the nature of the calculations. Each column has a sliding window size, which specifies the number of projection periods to be held in memory during calculations:

- A sliding window size of -1 (do not slide) indicates that all values for the column are held in memory.
- A size of 1 indicates that only the current projection period result is used in ongoing calculations.
- A size greater than 1 indicates that calculations use past values in addition to the current value.

4.2 Determining memory usage

4.2.1 Memory constraints

Sliding windows is needed when the model will not run due to memory constraints, or when run times are extremely slow. When the projection will not run, a runtime error or fatal error in the run log is produced. The memory issues arise when too much information must be held in memory.

Knowing the amount of memory used by a projection task is useful to estimate how many model points can be projected in the available RAM. Two methods are outlined below.

4.2.2 Using Windows task manager

Click on Ctrl-Alt-Delete and select Task Manager, then click on the Processes tab. The memory used by the RAFMCore.exe process is the memory used by the projection. Keep in mind that other processes are also using memory. Click on the Performance tab to see information about total memory usage.

4.2.3 Using run log information

Alternatively, the run log displays the total memory used, in the User messages area of the output window while the projection task is running. The end of the run log shows, on the 'heap' line, the memory used during the projection (the first number), plus the ending total memory used including the memory used by other applications currently running.

```
Odometer: Elapsed time = 28.15 seconds
heap    103583744    144539648    at end
Done
Whole Run: Elapsed time = 28.49 seconds
Maximum stack : 111 at record 0
```

4.3 Determining sliding window sizes

Sliding windows sizes for the columns are determined using a **Test slide** projection.

4.3.1 Test slide approaches

There are two approaches commonly used for running test slides:

- **Single run with scaled down model point file** – Run the test slide with a model point file that is a representative sample of the full set of model points where there is an expectation that the suggested sizes from the subset are likely to be adequate when running all model points together using sliding windows.
- **Recursive runs with block of model points** – Run the test slide multiple times to cover all model points in the model point file where each test slide run is executed on a block of the full model points. For example, the first test slide would use the first 500 model points, the second test slide would use the next 500, and so on. Through this approach, the sliding sizes are updated

4.3.2 Running the test slide

To run a test slide, change the Sliding Windows property in the Run parameter set for the projection task to be Test slide. Model points are processed without sliding, in order to calculate suggested sizes based on this projection.

The sliding windows values calculated during the test slide projection are stored in the folder referenced by the <*project_directory*> folder in an 'xxx_slw.txt' text file, where xxx is the name of the Run parameter set used for the test slide projection.

4.3.3 *Reviewing test slide results*

Review the sliding window results using the Sliding windows editor from within the Run parameter set. The following are considerations for reviewing the information provided in the sliding window editor:

- The old versus suggested comparison indicator provides feedback on whether the sliding size has changed since the last test sliding
- The Old value is the value of the sliding window before this test slide
- Suggested values are the sizes created during the test slide projection. Suggested size of Don't Slide means that a column was not calculated during the test slide projection. This often means the column is in a non-arrayed submodel such as the seg (segment) submodel, which never slides.
- New values are the adjusted sizes, based on a comparison between the suggested, old, and maximum sliding size values (discussed below).

Filter or compress the sections within the editor to view only the columns for a particular submodel.

4.3.4 *Maximum sliding window size*

As mention above, the New values in the sliding windows editor are adjusted values, based on a comparison between the suggested, old, and a maximum sliding size.

Through the Max sliding window size property you can control how tightly you wish to manage the memory of a projection. Generally, the larger value you select for the max sliding size the more tightly you are managing memory as more columns will have specific sliding size as opposed to being set to Do Not Slide.

Set this maximum to a high number, such as 9999, for the most tightly managed memory approach. A downside to this approach is that any variation in the model subsequent to the setup of the sliding windows values may lead to sliding windows errors. And, the sliding file may not be usable across a range of projection tasks.

Set the value to a smaller value, such as 30, when you do not require the memory to be tightly managed or when you desire to use the same sliding windows file across a range of projection tasks.

Warning! The sliding sizes in the _cvres product submodels (such as ul_cvres) should not be changed. These calculations are sensitive to issue age; the sliding sizes have been set to 121 to allow calculations for issue age 0 to run for 120 policy years.

4.3.5 *Saving the sliding window sizes*

The adjusted sliding windows values must be saved before they can be used in a projection. There are two locations for saving the sliding windows values.

Select to save the values to the specified sliding windows file by selecting the 'Update Sliding Windows File' item. Select to save the values to the column properties by selecting the 'Update Default Values'.

4.4 *Running with sliding windows*

4.4.1 *Running in slide mode*

Projections are run with sliding windows when the 'Sliding Windows' property of the run parameter set is set to Slide.

Ideally, you should first run a slide run with the reduced model point file so you can estimate the memory used for each model point, in order to estimate how many model points will fit into available memory during a large projection. If the estimated number of model points is less than the total number of model points in the full model point file, you could potentially adjust the sliding window sizes to manage memory more tightly or reduce the total number of model points included in the model point file

4.4.2 *Sliding windows errors*

When the projection attempts to call a column at a period that is outside the sliding window size, that data point is skipped and is not included in results. The projection ends with a fatal error, and the Run log includes an error message such as:

```
Sliding Error
Skip no: 1
Point skipped: Period (923) not in sliding column fund_val_aft_withdrl[924,924> in model ul_ifund/liab["main|liab|ul|ul_ifund[1]"]
called from ul_liab_stat_res_ulcrvm_per_unit(843) in model ul/liab["main|liab|ul"]for RecordNo 13 Iteration 1..
```

In the above example, the error occurred in model point 13. The error is that ul->stat_res_ulcrvm_per_unit for t = 843 is calling for ul_ifund->fund_val_aft_withdrl at t = 923, but the only values held in memory for that column are from t = 924 through 924. This indicates that the sliding size for this column must be at least 2, since the current size of 1 is not adequate.

4.4.3 *Revising the sliding windows file*

When a slide error is encountered during the projection, one of the model points does not fit the sliding sizes you have specified.

You can manually set the sliding size (for the column producing the error) using the sliding windows editor. However, you will likely find that this column uses other columns that may need their sizes altered as well.

Best practice is to make another test slide to change the sliding windows sizes, including the problem model point in the new test slide projection, as well as any others that are likely to cause problems.

Preserve the existing sliding sizes file

When you make a new test slide, the projection detects that there is already a slw.txt file present, and asks whether you want to overwrite it. Specify No, so that the existing sliding sizes are preserved. The test slide then appends new information to the existing file. Save the new sizes, and rerun the projection.

4.4.4 *Dynamic sliding test*

Use the 'Dynamic sliding test' option when you want the system to automatically adjust for sliding windows errors and restart the projection.

Set the 'Sliding Retry Count' to the desired number of retries, such as 10. Since only one error is resolved at a time, encountering 10 errors likely means that the sliding window sizes are not valid for the model points and should be adjusted through another test slide projection.

4.5 *Variation in sliding sizes*

Since run parameter sets are used across a number of projections, ideally, the sliding sizes should be set appropriately for general work, even though the sizes are higher than needed for a particular task. However, there are times when memory must be optimized for a particular projection. The following sections discuss further steps that may be needed in setting the sliding windows sizes.

4.5.1 *Variation by projection*

Required sliding sizes may vary between different projections. To preserve settings for a projection, run it as a test slide specifying an xxx_slw.txt file specific to the projection task, not the run parameter set.

4.5.2 *Variation by output settings*

Required sliding sizes may vary based on the output that is selected to be included. When you are planning to produce output for product level submodels (in addition to company or segment results), set this output to All Columns or Filter Set when running the test slides.

4.5.3 *Variation between scenario paths*

Different scenario paths may result in different sliding window size needs. Update the test slide using the scenario that is causing sliding size errors.

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G05 OUTPUT MANAGEMENT

This book discusses output management for the RiskAgility FM U.S. Library.

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1 OUTPUT MANAGER STRUCTURE

RiskAgility FM presents wide flexibility for managing output.

This user guide presents instructions and options for managing output following the templates included with the projects. This is only one option, but is used consistently here in order to explain output manager operations. Of course, you can use the full flexibility of the RiskAgility FM platform, appropriate to your situation.

1.1 Output managers

1.1.1 Output manager display

The output manager (OM) of a project displays the output definitions and period definitions of the output manager. It is through the output definition and period definitions interfaces that you specify the output definitions and period definitions for the project.

1.1.2 Template output managers

Each RiskAgility FM U.S. Library project provides template output managers. These output managers contain a standard structure for selecting output to be created from the projection tasks.

Output managers are referenced within the output parameter sets. Select the desired output manager from the set of template output managers to optimize the output captured for the various projections.

1.2 Output manager definitions

1.2.1 General structure

There is significant flexibility provided by RiskAgility FM to specify definitions for managing output within the output managers. Any number of output definitions can be defined for each submodel. Also, any number of generic period definitions can be defined.

For the RiskAgility FM U.S. Library projects, the template output managers are structured as follows:

- Separate output managers for valuations versus projections
- Output definitions for each key reporting submodel
- Separate output definitions for Statutory versus GAAP results
- Output definitions are for column output only and do not include scalars or variables, i.e. the output definitions are not Individual Output definitions
- No period definitions. Monthly results are always captured within the Column Output From and Column Output To property of the Output parameter set

1.2.2 Proj output definitions

The Proj output manager is the output manager which is used to capture the output for projection tasks which are pricing or projections.

The structure of the Proj output manager, where [prod] refers to the product submodels 'ul', 'fa', etc., is as follows:

Submodel	Output definition	Description
company	Statutory	Columns required for portfolio projection statutory reporting
company	GAAP	Columns required for portfolio projection GAAP reporting
company	Hedge	Columns required for portfolio projection Hedge reporting
rates	Statutory	Columns required for portfolio projection statutory reporting.
seg	Statutory	Columns required for portfolio projection statutory reporting.
seg	GAAP	Columns required for portfolio projection GAAP reporting
seg	Hedge	Columns required for portfolio projection Hedge reporting
ann_pba	Statutory	Columns required for portfolio projection Annuity PBA statutory reporting.
life_pba	Statutory	Columns required for portfolio projection Life PBA statutory reporting.
va_pba	Statutory	Columns required for portfolio projection VA PBA statutory reporting.
hedge_grp	Hedge	Columns required for portfolio projection Hedge reporting
sfas60	GAAP	Columns required for portfolio projection SFAS 60 GAAP cohort reporting.
sfas91	GAAP	Columns required for portfolio projection SFAS 91 GAAP cohort reporting.
sfas97lp	GAAP	Columns required for portfolio projection SFAS 97 limited pay GAAP cohort reporting.
sfas120	GAAP	Columns required for portfolio projection SFAS 120 GAAP cohort reporting.
sfas97rd	GAAP	Columns required for portfolio projection SFAS 97 retrospective deposit GAAP cohort reporting.
asset_sub_port	Statutory	Columns required for portfolio projection asset sub portfolio statutory reporting.
[prod]	Statutory	Columns required for seriatim projection statutory reporting.

Submodel	Output definition	Description
[prod]	GAAP	Columns required for seriatim projection GAAP reporting.
[prod]_ann_pba	Statutory	Columns required for cell pricing nested projection Annuity PBA statutory reporting.
[prod]_life_pba	Statutory	Columns required for cell pricing nested projection Life PBA statutory.
[prod]_va_pba	Statutory	Columns required for cell pricing nested projection VA PBA statutory.
[prod]_sfas60	GAAP	Columns required for cell pricing nested projection SFAS 60 GAAP cohort.
[prod]_sfas91	GAAP	Columns required for cell pricing nested projection SFAS 91 GAAP cohort reporting.
[prod]_sfas97lp	GAAP	Columns required for cell pricing nested projection SFAS 97 limited pay GAAP cohort reporting.
[prod]_sfas120	GAAP	Columns required for cell pricing nested projection SFAS 120 GAAP cohort reporting.
[prod]_sfas97rd	GAAP	Columns required for cell pricing nested projection SFAS 97 retrospective deposit GAAP cohort reporting.

Each output definition includes an alphabetical listing of the columns included.

1.2.3 Valn output definitions

The Valn output manager is the output manager which is used to capture the output for projection tasks which are valuations. Valuation projection tasks include both PBA and GAAP point in time and inner loop valuations.

The structure of the Valn output manager, where [prod] refers to the product submodels 'ul', 'fa', etc., is as follows:

Submodel	Output definition	Description
company	Statutory	Columns required for portfolio valuations. Includes only valn_tgt_ag38_8d and the valn_tgt_pba_xxx columns.
company	GAAP	Columns required for portfolio valuations. Includes only valn_tgt_gaap_xxx columns.
company	Hedge	Columns required for portfolio hedging valuations. Includes only valn_tgt_hedge_xxx columns.
seg	Statutory	Columns required for portfolio valuations. Includes only valn_tgt_ag38_8d and the valn_tgt_pba_xxx columns.

Submodel	Output definition	Description
seg	GAAP	Columns required for portfolio valuations. Includes only valn_tgt_gaap_xxx columns.
seg	Hedge	Columns required for portfolio hedging valuations. Includes only valn_tgt_hedge_xxx columns.
ann_pba	Statutory	Columns required for portfolio Annuity PBA statutory valuation reporting. Also includes columns required in the experience projection from the inner loop valuations for nested projections.
life_pba	Statutory	Columns required for portfolio Life PBA statutory valuation reporting. Also includes the columns required in the experience projection from the inner loop valuations for nested projections.
va_pba	Statutory	Columns required for portfolio VA PBA statutory valuation reporting. Also includes the columns required in the experience projection from the inner loop valuations for nested projections.
hedge_grp	Hedge	Columns required for portfolio hedge valuations.
sfas91	GAAP	Columns required for portfolio SFAS 91 GAAP cohort schedules. Also includes the columns required in the experience projection from the inner loop valuations for nested projections.
sfas120	GAAP	Columns required for portfolio SFAS 120 GAAP cohort schedules. Also includes the columns required in the experience projection from the inner loop valuations for nested projections.
sfas133	GAAP	Columns required for portfolio SFAS 133 GAAP cohort valuations. Also includes columns required in the experience projection from the inner loop valuations for nested projections.

Submodel	Output definition	Description
sfas97rd	GAAP	Columns required for portfolio SOP 03-1 GAAP cohort valuations. These columns are required by the SFAS 97 retrospective deposit valuation. Also includes columns required for SFAS 97 retrospective deposit GAAP cohort schedules. Also includes columns required in the experience projection from the inner loop valuations for nested projections.
[prod]	Statutory	Columns required for seriatim valuations. Includes only valn_tgt_ag38_8d and the valn_tgt_pba_xxx columns
[prod]	GAAP	Columns required for seriatim valuations. Includes only valn_tgt_gaap_xxx columns
[prod]_ann_pba	Statutory	Columns required for seriatim Annuity PBA statutory valuations which are used within cell pricing projections. Also includes columns required in the experience projection from the inner loop valuations for nested projections.
[prod]_life_pba	Statutory	Columns required for seriatim Life PBA statutory valuations which are used within cell pricing projections. Also includes columns required in the experience projection from the inner loop valuations for nested projections.
[prod]_va_pba	Statutory	Columns required for seriatim VA PBA statutory valuations of the standard scenario amounts. Also includes columns required for the 2 step RBC C3 phase II valuation processing. Also includes columns required for seriatim VA PBA statutory scenario amount valuations which are used within cell pricing. Also includes columns required in the experience projection from the inner loop valuations for nested projections.
[prod]_sfas60	GAAP	Columns required for seriatim SFAS 60 GAAP factors valuations.
[prod]_sfas91	GAAP	Columns required for cell pricing nested projection SFAS 91 GAAP cohort schedules.
[prod]_sfas97lp	GAAP	Columns required for seriatim SFAS 97 limited pay GAAP factors valuations.

Submodel	Output definition	Description
[prod]_sfas120	GAAP	Columns required for cell pricing nested projection SFAS 120 GAAP cohort schedules.
[prod]_sfas133	GAAP	Columns required for cell pricing nested projection SFAS 133 GAAP cohort valuations. Also includes columns required in the experience projection from the inner loop valuations for nested projections.
[prod]_sfas97rd	GAAP	Columns required for cell pricing nested projection SOP 03-1 GAAP cohort valuations. These columns are required by the SFAS 97 retrospective deposit valuation. Also includes columns required for cell pricing nested projection SFAS 97 retrospective deposit GAAP cohort schedules. Also includes columns required in the experience projection from the inner loop valuations for nested projections.

Each output definition includes an alphabetical listing of the columns included.

2 STANDARD REPORTING

2.1 Submodel output

2.1.1 Standard projection output

Submodel output for standard projections is structured as follows:

- Base name of output files is combination of run page name and projection task name, using an underscore (_), in the structure [Run Page Name]_[Projection Task Name]. For example, a base column output file name might be “Standard Portfolio_Baseline”
- Family of CSV output files, one for each submodel which has output selected, where the individual file is identified using the model tree structure from the top model of the model tree, separating the submodel names using the tilde (~). For example, a column output file name might be named “Standard Seriatim_Baseline~company~liab~trad.csv”
- Projection task loops are captured in the task_loop column.
- Submodel groups are identified in the ‘group’ column. Group names may be organized into a tree structure using the pipe (|) character. Group names for the user defined array submodels are defined within the code using code variables and the setGroup() function.
- Output column headers are the column names.
- Present value rows are identified by ‘PV xxx’ in the time period column. The PV disc_rate_xxx rows represent the discounting at the column discount rates.
- Blank entries represent columns which have not been executed during the projection.

2.1.2 *Nested projection output*

Submodel output for nested projection projections is structured similarly to standard projections with the following additional considerations:

- Identifiers for the projection set loop and subloop are included in the file name, using the @ symbol, between the base file name and the start of the submodel references. For example, a column output file for projection set loop 2, sub loop 3 of a nested projection might be named “PBA Portfolio_Baseline@2_3~company~seg.csv”
- For the experience projection, the outer loops are NOT captured in the task_loop column
- For the experience projection of a nested projection, a mapping of the period values within each time step to the overall projection period is required to understand the raw results. For example, when the valuation periods are 0, 3, 15, 27, 39, reporting periods 0 to 3 of the @x_1 output file map to projection periods 0 to 3, reporting periods 1 to 12 of the @x_2 output file map to projection periods 4 to 15, reporting periods 1 to 12 of the @x_3 output file map to projection periods 16 to 27, reporting periods 1 to 12 of the @x_4 output file map to projection periods 28 to 39. The output from the @x_5 results file is not used.

2.2 *Standard reports*

The following standard reporting is provided with the RiskAgility FM U.S. Library.

2.2.1 *General structure*

The standard reports use the following general structure:

- Horizontal and vertical formats, but only one format for each report
- Results by calendar month and year, starting from a specified projection date
- Reports separate by submodel
- Varying frequency of years monthly, quarterly, and annual where reporting for years which are quarterly and annual are always calendar quarters and years. Policy year reports are always policy years
- Scaled results by Dollar, Thousands, or Millions
- Decimal precision as specified. Applies only to results which are not rates

2.2.2 *Summary reports*

Summary reports include:

- Pricing – Profit measures such as PV book profit before and after tax, distributable earnings, profit margin, IRR, return on assets, break-even year and surplus strain on first year at each discount rate.
- Statutory Present Value – Present value of statutory profits at each discount rate.
- Stochastic PBA – Distribution and CTE results for PBA scenario greatest present value accumulated deficiency, scenario amounts and tax adjustment ratios.

2.2.3 *Base reports*

Base reports include:

- Statutory Balance Sheet – Asset, liability, target capital, and free surplus breakdown on a U.S. statutory basis.
- Statutory Income Statement – Development of distributable earnings and accumulation of free surplus on a U.S. statutory basis.
- Statutory Profit By Source – Development of book profit before tax on a profit by source (fund) basis.

- Economic Statement – Development of hedged and unhedged economic profit and an economic balance sheet.
- GAAP Balance Sheet – Asset, liability, and equity breakdown on a U.S. GAAP basis.
- GAAP Income Statement – Development of income after tax using a consolidated financial statement approach on a U.S. GAAP basis.
- SFAS60 GAAP Statement – Development of income statement items and balance sheet items using the SFAS 60 financial statement approach on a U.S. GAAP basis. This report is available for SFAS60 GAAP cohorts only.
- SFAS91 GAAP Statement – Development of income statement and balance sheet items using the SFAS 91 financial statement approach on a U.S. GAAP basis. This report is available for SFAS91 GAAP cohorts only.
- SFAS97 Limited Pay GAAP Statement – Development of income statement and balance sheet items using the SFAS 97 Limited Pay financial statement approach on a U.S. GAAP basis. This report is available for SFAS97 Limited Pay GAAP cohorts only.
- SFAS97 Retrospective Deposit GAAP Statement – Development of income statement items and balance sheet items using the SFAS 97 Retrospective Deposit financial statement approach on a U.S. GAAP basis. This report is available for SFAS97 Retrospective Deposit GAAP cohorts only.
- SFAS120 GAAP Statement – Development of income statement items and balance sheet items using the SFAS 120 financial statement approach on a U.S. GAAP basis. This report is available for SFAS120 GAAP cohorts only.

2.2.4 Detail reports

Detail reports include:

- In-force Statistics – Model statistics including units (life only), policies, death benefit in force, fund and cash value in force, etc.
- Fund Value Roll Forward – Fund value roll-forward from period to period resulting from additional premiums, less product loads, less guarantee charges, plus interest, less claims, and less fund value released due to decrements.
- Gross Statutory Income Statement – Development of book profit before tax for gross business on a U.S. statutory basis.
- Statutory Cost of Reinsurance Statement – Development of reinsurance cost including the change in reinsurance reserve ceded on a U.S. statutory basis.
- Long Term Care Statement – Development of gross statement results for long term care insurance business.
- Investment Income and Cash Flows – Breakdown of investment income profit components and allocations and cash flows.
- Asset Portfolio – Investment income and capital gains components, investment expense, cost basis, market value, book value and unrealized capital gains, effective duration, and asset yield for an asset portfolio.
- Asset Detail – Reconciliation of asset projection results for a reporting period. This report is available for company, segment, asset sub portfolio, and individual asset types. **Important! The earned rates section at the top of this report is only valid when the reporting frequency is monthly for all periods. User modifications to the formulas are required if quarterly or annual reporting is used.**
- PBA Details – Statutory PBA capital and reserve details, including deterministic, stochastic, and excess, split by before and after reinsurance, and before and after hedging.

- Accumulated Deficiency – Development of accumulated deficiency for PBA scenario amount valuation tasks.
- Per Unit Values - For UL products seriatim pricing projections only.
- Hedge Group Data – For FIA hedging group detail information.
- Individual Fund Value Per Unit – Details for the individual fund values for UL liability groups only.
- Dynamic Rates – Details of the annualized yield rates, crediting rates, and lapse and surrender rates. This report is available for segment and each liability product when run as the top model.
- SFAS60 Roll Forward report – Period to period roll forward of DAC, benefit reserve, and maintenance expense reserve. This report is available for SFAS60 cohorts only.
- SFAS60 Source of Earnings report – Development of SFAS 60 pre-tax GAAP income using a source (margin) basis. The margins include premium, interest, benefit, and expense. This report is available for SFAS60 cohorts only.
- SFAS97 Limited Pay Roll Forward report – Period to period roll forward of DAC, benefit reserve, and maintenance expense reserve. This report is available for SFAS97 Limited Pay cohorts only.
- SFAS97 Limited Pay Source of Earnings report – Development of SFAS 60 pre-tax GAAP income using a source (margin) basis. The margins include premium, interest, benefit, and expense. This report is available for SFAS97 Limited Pay cohorts only.
- SFAS97RD Amortization Stream – Details of the development of the estimated gross profits for SFAS97 Retrospective Deposit valuations. This report is available for SFAS97 Retrospective Deposit cohorts only.
- SFAS97RD DAC Details – Details of the development of the acquisition expense, commission, premium bonus, and YRT asset DAC amounts for SFAS97 Retrospective Deposit valuations. This report is available for SFAS97 Retrospective Deposit cohorts only.

2.2.5 Policy year reports

Policy year reports include:

- Nonforfeiture Values – Cash value and other nonforfeiture factors for term and traditional life products. This report is available for term and traditional life liability groups only.
- Policy Reserves – Statutory and Tax reserve factors for term and traditional life products. This report is available for term and traditional life liability groups only.
- ROP Term Nonforfeiture Values – Return of premium nonforfeiture factors for term life products. This report is available for term life liability groups only.
- ROP Term Policy Reserves – Return of premium reserve factors for term life products. This report is available for term life liability groups only.
- Trad PUA Values – Paid up addition factors for traditional life products. This report is available for traditional life liability groups only.

3 REPORTING TOOL UTILITY

3.1 Overview

3.1.1 Key benefits

RiskAgility FM does not provide standard reporting capabilities through the interface. All output from the projections must be viewed external to the interface using tools which can read comma separated value (CSV) files.

The reporting tool is a critical element of the RiskAgility FM U.S. Library solution. It is designed to collect results into financial statement and other actuarial reports while retaining a significant amount of flexibility around working with the results. The reporting tool provides the power to process multiple output files with great flexibility. Users can easily change the time range and frequency of the reports. Users also have the power of Excel to format, calculate, display graphically and link other workbooks and worksheets into the reports.

Also, since projections of PBA and GAAP balance sheets utilize a nested projection time step approach, results from multiple output files must be treated as the single output for a projection. The reporting tool provides a way to combine these separate output files together into a single output file per submodel, which is formatted consistently with the output from standard projections.

3.1.2 Link to Microsoft's Log parser

RiskAgility FM column output results are stored in comma separated value (CSV) files. Although, RiskAgility FM does not provide any tools within the interface for reading and manipulating the column output CSV files, there are free tools in the market which do provide these capabilities. One such tool is the free utility from Microsoft called **Log parser**.

Log parser is a powerful, versatile tool that provides universal query access to text-based data such as log files, XML files and CSV files, as well as key data sources on the Windows® operating system such as the Event Log, the Registry, the file system, and Active Directory®.

Through the capabilities of log parser, the reporting tool is able to query the column output CSV files and retrieve the specific columns and periods into the reporting tool worksheets for the output required for the desired reports.

Download Microsoft's log parser version 2.2 from <http://www.microsoft.com/en-us/download/details.aspx?id=24659>

Warning! The reporting tool will not function properly until you have downloaded and installed the log parser utility.

3.1.3 Welcome worksheet

When loaded, the reporting tool displays the Welcome worksheet. This worksheet includes general setup and getting started information, as well as high level instructions on the use of the reporting tool.

Additionally, this worksheet includes the command buttons for displaying the report definitions worksheets and for resetting the reporting parameters.

3.2 Report definitions

3.2.1 Report definition worksheets

Each report included in the reporting tool is defined on a reporting definition worksheet. These worksheets are normally hidden but may be displayed by pressing the 'Customize Reports' command button on the Welcome worksheet.

The structure of the reporting definitions worksheets is as follows:

- DataSource information as columns and reporting information as rows for horizontal reports, or reporting information as columns and DataSource information as rows for vertical reports
- Group outline for reporting frequency calculations
- Rows for report title, reporting group and scaling
- An Enable calculations command button which enables the calculations on the worksheet in order to review the values being retrieved from the source worksheets. Calculations should only be enabled temporarily during the development of the reporting definitions.

- A FORMULA START indicator to define the column or row from which the formulas should be duplicated during the report processing
- START and END indicators which define the start and end of reporting definitions
- Reporting information sections structured as appropriate for the structure of the report. For example, the statutory balance sheet includes sections for Total Assets, Statutory Reserves, Net Statutory Reserve, Total Liabilities, and Total Capital and Surplus.
- DataSource information for each reporting level, which includes the submodel for the column result, the column header (which is the column name), a formula for the column index of the source data, and a reference formula to the source data. The reference formulas generally use SUMIFS with INDEX and INDIRECT offsets to retrieve and calculate the reporting results from the source worksheets taking into account appropriate income statement or balance sheet summations for the reporting frequencies. The 'ReplaceMe' references are replaced during the macro execution by the appropriate reference to the source worksheets.
- Excel Data groupings provide for outlining and breaking down the key reporting items, such as the breakdown of premium income reporting item into direct premium paid and reinsurance premium paid

3.2.2 **Report mapping worksheet**

Each report included in the reporting tool is mapped to reporting selections and report definitions worksheets on the reporting summary (RptSummary) worksheet. This worksheet is normally hidden but may be displayed by pressing the 'Customize' command button on the Welcome worksheet.

The structure of the reporting summary mapping worksheet is as follows:

- Blocks for each of the Summary, Base, Detail, and Policy Year report types
- Header information for the report selections desired (updated automatically during processing)
- Columns for each report and rows of mapping information
- Specification of report definition worksheet
- Orientation selection of 'H' for horizontal and 'V' for vertical
- Specification of order of report in report list
- Rows which calculate whether a report is available based on current reporting selections (updated automatically during processing)
- Mapping of reports to calculation types of pricing, projection, and valuation
- Mapping of reports to reporting levels of Company, Rates, Segment, Asset, GAAP Cohorts, PBA, Product Liability, Product Liability Submodel, Product PBA, and Product GAAP Submodel levels. For example, a report which is mapped for Company, Segment, and Product Liability is a report which can be selected for portfolio runs with Company as a target model object and for seriatim runs with product as the target model object.

3.2.3 **Source data worksheets**

Following a query from the projection column output files, results are collected on 'Source' worksheets. These worksheets are normally hidden but may be displayed by pressing the 'Customize' command button on the Welcome worksheet.

Separate worksheets are included for each reporting level. The source worksheets are named Source ([submodel]). For example, Source (comp) is the source worksheet for the company submodel source information.

The structure of the source data worksheets is as follows:

- Required information as columns and detail information as rows

- A reporting group column for the selected reporting group
- The task loop associated with the selected reporting group
- The submodel group
- Calendar month and relative calendar year. Required since all yearly reporting is calendar year based, not projection year based.
- Income statement periods which indicate how a result from a row is included in the reporting. For monthly frequency, the income statement periods match the periods from the output. For quarterly or annual frequency, the income statement period is repeated, indicating that the results for the periods in the output should be summed for the reporting period
- Balance sheet periods which indicate whether a results form a row is included in the reporting. For monthly frequency, the balance sheet periods match the periods from the output. For quarterly or annual frequency, the balance sheet period is either equal to the reporting period or set to -1. A -1 value indicates the row is not included in the report.
- A grid of the queried results in the format consistent with the CSV files

3.3 Report processing

3.3.1 *Combine TimeStep worksheet*

As discussed in the Nested projection output topic, nested projections create separate output files for each time step of each outer loop, and thus cannot be used directly for standard reporting. The output files must first be combined into a new output file which will have the same structure as an output file from a standard projection.

Use the functions of the 'Combine TimeStep' worksheet to create a new standard projection output file from the output files of a nested projection. Considerations for each step are as follows:

Step 1: Select Output file

First, use the "Select Output File" command button to select one of the files from the nested projection. Any one of the files from any of the submodels may be selected. The file selected is then displayed on the worksheet.

Step 2: Select Definitions file

Next, create a comma separated value file which contains the definitions of the nested projection outer loops and valuation periods. The structure of the CSV file should similar to the following:

	A	B	C	D	E
1	OuterLoops	x	y	z	
2	ValnPeriods	a	b	c	
3	ValnFlag	n			

where

- x,y,z etc. are the values of the outer loops. Items x, y, and z can be defined as individual numbers, such as 1, 5, 10 or as dash separated ranges, such as 1-5,10, 15-20.
- a,b,c, etc. are the values of the valuation periods. Items a, b, and c can be defined as individual numbers, such as 0, 3, 15 or as dash separated ranges, such as 0-3,12-24, 36-48. The value of 0 must always be included in the definition in the ValnPeriods row.
- n specifies whether the output files should be treated as results from a valuation or from a projection. Specify 'n' as 1 to indicate the output should be treated as output from nested projection valuation

projection tasks. **Note: This row is optional and may be excluded from the combine time step definitions file when the definition file is for combining results from a nested projection.**

Then, use the “Select Definitions File” command button to select the nested projection definitions CSV file. The file selected is then displayed on the worksheet.

Step 3: Combine

Finally, review the command line information which is displayed on the worksheet to ensure it is correct and then use the “Combine” command button to activate the combine time step process using the ‘Combine TimeStep.exe’ program provided with the reporting tool.

Considerations for the combine time steps process are as follows:

- The new output files are stored in a ‘CombinedTS’ sub folder of the folder for the nested projection output files selected in Step 1.
- The name of the new output files is the base name of the nested projection output files
- A new file is created for each submodel for which there is a nested projection output file
- The output by submodel group is retained
- The outer loop values, as specified in the nested projection definitions file become the task loops of the new combined files
- Excluding the first time step, time 0 results from the time-step output files are ignored unless the ValnFlag row is included with a value of 1. In that case, the time 0 values from each time-step file are retained as treated as the result for the valuation period in the combined file.
- Present values are recalculated for each column for the combined file
- The processing time required to complete the combine time step process depends on the number of outer loops, number of valuation periods, the length of each time step, how many submodels have output, and the number of columns included in the output files.

When the combine time step process is completed, standard reporting may be used on the new combined file.

3.3.2 Control worksheet

Use the functions of the ‘Control’ worksheet to specify and activate the reporting processing.

The control worksheet is structured so that a user should generally process through each step sequentially, starting with step 1 and ending with step 9 or 10. If there are no existing parameters, the control worksheet highlights each step in Red until a parameter has been specified.

Considerations for each step are as follows:

Step 1: Run Type

Select the run type of the projection to report on. Available options are Pricing, Projection, and Valuation. The run type selection controls which reports are available for selection (see step 7).

Step 2: Projection Date

Specify the projection date of the projection in MM/YYYY format. This date is used to control the calendar month and year displayed in the reports and the treatment for quarterly and annual frequencies.

Important! For proper reporting of calendar year results, this value must match the date specified for the projection in the run parameter set proj_date project variable.

Step 3: Output File

Press the “Select File” command button to navigate and select the output file to report on. Any of the submodel CSV files may be selected.

Following the selection, the path and base name of the selected file is displayed in the worksheet.

Step 4: Reporting Level

Select the reporting level for the desired reports from the drop down list which has been developed based on the collection of submodel output files which are available for the base output file projection task. The reporting level defines which reporting groups are available and which results are used in the report creation.

Consider a portfolio projection which has output for the company, seg, and fa submodels. That is, there are [Base Name]~company.csv, [Base Name]~company~seg.csv, and [Base Name]~company~liab~fa.csv files available for the [Base Name] projection. For this case, four reporting levels are available, 'All Reporting Level Options', 'Company', 'FA Liability Sub Model' and 'Segment'.

Upon selection of the reporting level, a list of available reports is the updated displayed within the grid for step 6 (discussed below).

Step 5: Select Reporting Group

Select the reporting group for the reporting using the drop down list which is based on the Reporting Level selection.

In addition to the specific reporting groups which were defined for the model, summary reporting groups are also available for selection. The summary reporting groups sum the results of the individual reporting groups which match the criteria of the summary reporting level.

For example, consider reporting for the FA Liability reporting level where the model point groups included a combination of plan code, gender, and issue age. The select reporting groups drop down list will include each combination of the projection task loop, plan, gender, and issue age group (as defined in the model point file), plus summaries for projection task loop, plan, gender, summaries for projection task loop and plan, and summaries for the projection task loop, which are identified by the grouping [proj task loop]*. There will not be a summary which combined the projection task loops since these results should never be summed for reporting.

Specify the reporting groups to process as either 'All', 'Selected', or from the list of individual reporting groups. If 'Selected' is chosen, the list of reporting groups to process is specified on the 'Reporting Groups' worksheet (discussed below). When the Reporting Level is specified as 'All Reporting Level Options', the only available selections are 'All' and 'Selected'.

When the reporting level is specified as 'All Reporting Level Options' or '[prod] Liability Sub Model' level, specify the Segment from which segment results should be captured for segment allocations using the Segment Group selection field.

Warning! For Liability Sub Model reporting for multiple segment projections, it is critical that the liability model point groups only contain model points of a single segment. Otherwise, the allocation bases from the liability groups will not be consistent with the segment amounts.

Warning! For asset sub portfolio reporting, selecting 'All' for the reporting groups can create a very lengthy report creation process since there tends to be many groups to process. A suitable alternative is to specify 'Selected' and select the reporting groups to process in smaller batches.

Step 6: Select Reports

Specify the reports to create by selecting 'Yes' for the desired reports.

Step 7: Reporting Frequency

Use the fields for reporting frequency to specify the CTE percentages for PBA summary reports, years monthly, quarterly, and annual for base and detail reports, and number of policy years for policy year reports.

The sum of the number of years monthly, quarterly, and annual represent the total number of **calendar years** for the reporting, even if this is less than the amount of output periods available in the column output files. The first calendar year is a partial calendar year when the projection month is not December.

For example, consider a projection date of 9/2013 and 237 periods of output. Specifying 10 years monthly, 5 years quarterly, and 5 years annual results in an income statement with results reported monthly for 3 months in the first calendar year, monthly for 12 months for the next 9 calendar years, quarterly results in months 3, 6, 9, and 12 for the next 5 calendar years, and annual results in months 12 for final 5 calendar years.

Step 8: Scale and Precision (amounts only)

Select the scale for the reports as Dollar, Thousands, or Millions and the number of decimals for the results.

This selection only applies to amounts. It does not apply to reporting of rates, such as interest rates, decrement rates, etc.

Step 9: Create Reports

Press the “Create Reports” command button to create the desired reports.

The reports are created using a Log parser query of the submodel CSV file for the desired reporting level. A single query is developed to retrieve results from a single reporting group of the results for the columns required for the desired reports. The query is stored on the Macro Data worksheet and the results of the query are stored on the Source worksheets.

Worksheets are created within the reporting tools for each selected report. The worksheets are displayed between the placeholder “Reports=>” and “<=Reports” worksheets. Processing updates are provided in the status bar of Excel as the reports are created and a pop up box is provided when the process is completed.

When multiple reporting groups are processed, the report creation process is run multiple times, with each loop creating the desired reports for one of the selected reporting groups. As each group is completed, the reports for that group are automatically exported into an '[base output file name] – [group name].xlsx' file, where the “|” of group names are replaced with “.”, which is stored in a 'Reports' sub folder of the folder for the selected output. For example, if the base output file name is 'Standard_Seriatim_Baseline' and the reporting group being processed is 1|FA01|Male|35, the workbook of exported results is named 'Standard_Seriatim_Baseline – 1.FA01.Male.35.xlsx'.

When the process completes, the only report worksheets included in the reporting tool are those of the last group processed. These reports can be manually exported to worksheets using the Export Finalized Reports step (see below)

Step 10: Export Finalized Reports

Step 9 processing produces worksheets for the desired reports inside the reporting tool, as well as automatically exporting the reports when multiple reporting groups are selected. Reports for the worksheets contained in the reporting tool workbook are finalized to a separate Excel workbook using the “Export Reports” command button.

Press the “Export Reports” command button and use the Windows Explorer dialog to select a folder and name for the workbook of finalized reports. The starting folder is automatically a 'Reports' folder of the folder of the output file selected in Step 3 and the recommended name for the Excel file is the base name of the output file selected in Step 3 and the group processed, using the same naming convention used when processing multiple groups (see Step 9 above).

When “Save” is pressed, finalized versions of the worksheets from the reporting tool are created in the specified workbook.

During the finalization step, values in the report worksheets of the finalized workbook which reference the 'Source' worksheets are converted to the numeric values. However, formulas of the report worksheets which sum group items, create subtotals, etc. are retained in the finalized workbook.

3.3.3 Reporting Groups worksheet

When the Select Reporting Group(s) selection is specified as 'Selected', specify the reporting groups to process by specifying 'Y' in the 'Selected' column (column J) of the 'Select Reporting Group(s)' grid.

No other changes should be made to this worksheet.

3.4 Report customization

3.4.1 Customizing an existing report

Customizing an existing report involves adding rows or columns, depending on the orientation of the report, to an existing report. Perform these steps:

Press the “Customize Reports” command button on the Welcome worksheet to display the hidden definition worksheets.

Next, navigate to the report definition worksheet for the report which you want to customize. The report definition worksheets are named [ReportName]Rpt, for example, StatIncomeRpt. Review the structure of the report definitions worksheets per the information described above.

If desired, press the “Enable Calculation” command button so that you can review results being populated on the report from the information currently captured in the source information during the customization of the report.

To include a new reported value, add a row or column where the new reported item should be inserted. For example, to add a breakdown of premium between first year and renewal, you could insert additional rows on the StatIncomeRpt worksheet for these premium amounts. If the item is a key reporting result, a new row would be added outside of any Excel data grouping. If the item is just a breakdown of a key results, the rows could be added into the existing Excel data groupings.

Update the DataSource, ColHeader, ColIndex, and Formula items as required for each reporting level section for the newly added rows. Typically, a copy and paste of an existing row or column is a good starting point for the items of a new row or column.

Be sure to review and update the Formula for rows or columns on the report which contain formulas that reference other rows or columns on the worksheet to ensure that the new reported item is being captured correctly into final reported results, sub totals, and checks.

If you have enabled calculations for the customization, be sure to press the “Disable Calculation” command button again to turn off the automatic worksheet calculation.

When you are happy with your customization, press the “Done Customizing” command button on the Welcome worksheet to hide the definition worksheets.

3.4.2 Adding a custom report

Adding a custom report involves adding a new report worksheet and updating the report mapping worksheet to allow for selecting the new report. Perform these steps:

Press the “Customize Reports” command button on the Welcome worksheet to display the hidden definition worksheets.

Next, navigate to an existing report definition worksheet which best represents the orientation and type of report that you are wanting to add, create a copy of that worksheet, and give it a new name. For example, if you are wanting to add a custom horizontal balance sheet type report, it would be best to create a copy of the ‘StatBalanceRpt’ worksheet with a new ‘[CustomName]Rpt’ name.

If desired, press the “Enable Calculation” command button so that you can review results being populated on the report from the information currently captured in the source information during the customization of the report.

If you have enabled calculations for the customization, be sure to press the “Disable Calculation” command button again to turn off the automatic worksheet calculation.

Now, navigate to the ‘RptSummary’ worksheet. Review the structure of the report definitions worksheets per the information described above.

Add a new column for the custom report in the appropriate section for Summary Report, Base Reports, Detail Reports, or Policy Year Reports. Typically copying the column of information for the report which was copied for the new report definition will provide a good starting point for the summary criteria of the new report.

Verify that the range which contains the columns for each of the Summary Report, Base Reports, Detail Reports, or Policy Year Reports also includes the new column. These ranges are named USLMSSummaryRptOption, USLMSBaseRptOption, USLMSDetailRptOption, and USLMSPolicyYearRptOption respectively.

Update the items as required in each of the Reports, Run Type, and Reporting Level sections. Be sure to update the 'Order in Report List' for all reports should you want your custom report to be shown in a particular order. Otherwise, set the 'Order in Report List' for the new report to be one value higher than the current highest value for that item.

When you are happy with the customization, press the "Done Customizing" command button on the Welcome worksheet to hide the definition worksheets.

G06 CUSTOMIZATION

This book discusses customizations of the RiskAgility FM U.S. Library.

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1 CUSTOMIZATION OVERVIEW

1.1 Customization objectives

1.1.1 *Extend standard features*

RiskAgility FM U.S. Library projects deliver robust features combined with extensive flexibility for customization.

In areas common across users, features include extensive common variations. For example, extensive features are provided for the economy, assets, investing, taxes, accounting, reserving, capital and surplus, calculation types, and profit calculations.

However, for areas of company products and practices, providing every variation out-of-the-box would result in clumsy variables and formulas, which are harder to understand and have degraded calculation speed and memory performance. Instead, basic features are provided as a basis for client customization.

Use the powerful development flexibility of the RiskAgility FM to customize the RiskAgility FM U.S. Library project for your own company products and practices.

1.1.2 *Ensure upgrade path*

To facilitate upgrades to future releases, it is important for customization to add to the variables and formulas instead of rearranging them. Changing a column's formula is acceptable; changing the structure of the submodels is not.

The following are important considerations for customizations.

Do:

- Save a copy of the Willis Towers Watson delivered solution before any customizations – This will be used as a reference when preparing for upgrades to new versions
- Enhance project model tree while maintaining common structure – Take advantage of the common structures of the library which permits combining definitions of asset and product types together
- Include a master customization control variable – Setting this variable value to No deactivates all customizations.
- Include client identifier in the names for all client specific definitions – Simplifies differentiating client definitions from Willis Towers Watson definitions
- Create client specific template input managers –Provides a client specific structure for creating input managers for modeling project work which will not be adversely impacted by future releases
- Create client specific template external source workbooks (if applicable) – Provides a client specific structure for creating assumptions for modeling project work which will not be adversely impacted by future releases
- Create client specific output manager and standard reports – Provides client specific structure for output and reporting which will not be adversely impacted by future releases

Do not:

- Change submodel structures – Do not move submodels into a new structure. Do not move variables and columns between submodels.
- Rename variables or columns – Future releases will continue to use these names, so you would have to rename them for every future release.
- Delete variables or columns – Simply leave unused variables at their default values, and ignore unused columns, so that you won't have to delete them for every future release.

- Change option list choices – Add new choices rather than changing existing ones, for the same reason as not renaming or deleting variables.
- Modify the template input managers – Future releases will introduce changes to the Template input manager which might conflict with changes you have made.
- Modify the template external source workbooks – Future releases will introduce changes to the Template external source workbooks which may conflict with changes you have made.
- Modify the ‘out-of-the-box’ output managers and standard reports – Future releases will introduce changes to the output managers and standard reports which may conflict with changes you have made.

For example, clients are expected to change the formula for calculating the cost of insurance (COI) charge for their specific universal life products. This will often involve replacing the standard formula with the operation consistent with company products. This can include the addition of new input variables and supporting columns, but should not include renaming, deleting, or moving of these items.

Important! Future releases may require changes

While it is our goal to facilitate upgradeability to future releases, the RiskAgility FM U.S. Library will at times undergo changes which improve performance efficiency. This can especially occur when requirements change due to new regulations and economic environments.

RiskAgility FM U.S. Library projects are not constrained to maintain legacy features and designs which may have been customized by users. While new releases will strive to maintain backward compatibility as far as practical, our goals to optimize performance and provide the best system will at times require user customization to be implemented over again. Users may sometimes be required to customize to preserve legacy features which are removed from the standard features.

1.2 Customization process

1.2.1 Overview

A well-defined customization process aims to ensure timely and high quality completion of an activity to enhance the RiskAgility FM U.S. Library project for your individual client requirements.

While one rigid customization process cannot apply to all customization activities, the process should always consist of the same key phases and activities, which are discussed in topics below. It is the level of content or emphasis within a particular phase which will vary from one customization activity to another.

And, of course, all customization activities, no matter how large or small, must be supported by appropriate project management.

1.2.2 Requirements gathering

The aim of the requirements gathering activity is to ensure an understanding of the feature before proceeding to design and implementation of the customization.

Business requirements provide an overview of the feature, and list the detailed requirements. The requirements are stated specifically, i.e., it will be clear whether the requirement is met (yes/no). The document generally tells ‘what’, not ‘how’. However, the requirements should include a detailed description of how a feature should work when this is an important practical consideration.

1.2.3 Analysis & design

The aim of analysis and design is to fully research the customization before advancing to implementation work. Both high level and detail design step ensures a full understanding of how the customization will be implemented and what parts of the project are affected.

For example, a detail design document may be produced which describes:

- Model structure changes required
- Model class changes required
- Input manager changes required
- External source structure changes required
- Output manager changes required
- Reporting changes required

1.2.4 Prototyping

The aim of prototyping is to extend the understanding of the proposed customization beyond what is provided by the analysis and design.

Prototypes, or proof of concepts, provide a physical representation of what the customization might look like, what the inputs, or outputs, what the processes might be, and how the user would interact with the interface to achieve the desired results.

The objectives of the prototype are developed prior to proceeding with the development of the prototype. Once the prototype is complete, it is reviewed to ensure the objectives were met and then presented to the development team and end users for feedback.

All prototyping efforts are accompanied by prototype brief and prototype review documents. The prototype brief document is produced prior to proceeding with the prototyping effort and the prototype review document is produced as the final step of the prototyping process.

1.2.5 Off-line implementation

The aim of off-line implementation activity is to fully implement the customization in a way that does not affect the production environment or other customizations currently underway. Off-line implementation is an iterative process of development, testing, peer review, and show & tells feedback.

Development

Development includes full implementation of changes required to code definitions and formulas, input definitions, and output definitions. All standards should be followed during off-line implementation.

Best practice is that the implementation of customization changes should follow the development standards and adhere to the structures discussed in this user guide. This includes:

- Understand the structures and processes explained in the Code management, Input management, Run management, and Output management user guide books.
- New variables (both code and input) and columns should adhere to the naming convention explained in Introduction user guide book. You should also consider including an abbreviation of your company name as a prefix so that definitions added by you can be easily identified.
- Use a comment marker to designate customization changes in formulas. Then the Find and replace formula utility can easily show a list of all customized formulas, and you can navigate to the places where changes have been made.
- Document changes, such as maintaining good Help messages for every variable, and explaining formula changes by including comments. This may also include developing a client specific addendum to the U.S. Library user guide.

Backups

Best practice is to backup the project before and after making changes. For example, name each backup file including the date and time of the backup. Using the yyyyymmdd hhmm [am/pm] date format causes the files to sort in chronological order.

Testing

Unit testing is completed as a part of the off-line implementation. This testing is completed by the actuarial modeler using a small set of test cases to assure the implementation is correct.

Peer review

Frequent peer reviews during the off-line implementation phase are critical to ensure the customization is being implemented properly, particularly when the customization is significant. The peer review should include a review of the technical implementation and for standards.

Show & tells feedback

As parts of the customization are completed, business stakeholders and subject matter experts are provided the opportunity to react to the functionality and implementation through a show & tell activity. This activity may highlight requirement changes and areas for improvement around the implementation which leads to returning to activities of the elaboration phase and then proceeding again into construction. It can even lead to a complete abandonment of the currently implemented solution.

1.2.6 Acceptance testing

The aim of acceptance testing is for someone other than the actuarial modeler to assure that there are no errors. This step should not find errors if the previous steps have been followed rigorously.

Full testing should be performed using the off-line implementation project if practical. If so, the test cases are included in the regression testing done by the actuarial modeler after on-line implementation. This testing is done following on-line implementation when the feature depends significantly on other features also being implemented.

1.2.7 Regression testing

The aim of regression testing is to make sure that your projections produce the same results after a customization when the customized feature is not activated. This can be done by comparing a key result, such as a present value of distributable earnings, for a series of projections.

When the results following the customization do not match the previous projections, research the differences by successively reducing the projection to isolate the cause of the difference. For example, run the projection using only the liability model points for one plan code at a time.

When you find a plan code with different results, test individual model points to isolate a single model point which produces a difference. Using that plan code, use a present value result to compare each cash flow and profit component to find the first column which has different results.

Now with that column isolated, and running a single model point, you can trace the results through debugger or introduce logging to further isolate the cause of the difference and introduce the correction.

1.2.8 On-line implementation

The aim of on-line implementation is to install the feature quickly (it has already been fully developed and tested) to minimize delays on other development and releases. Adding this feature must not introduce errors so that the development version remains an appropriate basis for ongoing development and releases.

1.2.9 Change tracking

The aim of change tracking is to document the changes made since the last deployment of the library into production. The change tracking information should include (at a minimum) the following items:

- Date of change – The date of the change
- Changed By – Who made the change
- Component changed – [Project Name], Utilities, etc.

- Type of change – Development or Maintenance
- Description of change – A sufficient detailed description of the change made. For significant customizations, this might be a reference to another document which more details
- Code updates – Model classes, variables and formulas changed
- Input updates – Input managers, input pages, input variables, external links, external sources workbooks changed
- Output updates – Output managers, output definitions, reporting tool definitions changed

1.3 Customization management

Typically, actuarial model development and maintenance is completed by a team of individuals. No matter the amount of customizations required, governance, security, and audit capabilities of the customization are critical. This section describes approaches which can be used to manage customizations of RiskAgility FM U.S. Library projects.

1.3.1 RiskAgility FM Team

Customizations to RiskAgility FM U.S. Library projects can be managed using Willis Towers Watson's RiskAgility FM Team.

RiskAgility FM Team was built with collaboration in mind and interfaces directly with RiskAgility FM so there is no separate system to learn. It enables multiple users to effectively work on a U.S. Library project at the same time through the use of source control for versioning and built-in compare and merge utilities. It also provides role based security for additional control of your models. Users may be given access to all parts of the project or just certain parts to ensure only authorized users may make changes.

RiskAgility FM Team features manage the customization process through an integration with Microsoft Team Foundation Server (TFS). Once a project has been checked in to the TFS source control through the RiskAgility FM Team features, users do not work directly with source controlled versions but instead check out a copy for development or testing work. Once the changes are complete, they may be checked back in to source control and saved as a new version.

Each version is saved with information about which parts of the project have been modified, who made the changes, when it was made and user comments describing the changes. This provides a complete version history that may be accessed at any time. The version history may be used to analyze differences in versions of the model from a high level down to a very fine level of detail. For example, the comparison utility may indicate a high level change such as adding a new projection or something more granular such as differences in specific lines of code in a formula.

Versioning extends beyond the code to also include each of the input managers, output manager, run managers, run and output parameter sets, and wildcards (references to data locations). This preserves your model setup, such as the projections and input setup, in addition to the code. Values stored as inputs directly to the system are included in the versioning. Versioning does not apply to external data files and output since these files typically are replaced when the development model is promoted to a production version. External data, DLL and LIB files, or output could be added separately in TFS and versioned alongside the project if desired.

Since various users may be making updates to different local copies simultaneously and checking in their changes, Team edition provides Get Latest Version functionality to easily merge updates between versions. If a newer version has not been added since check out, a user's local copy is just added as a new version to source control. If a newer version is available, the local version is updated as follows:

- New component files (Input Managers, Run Managers, etc.) in the latest version are added to the local copy,
- Component files deleted from the latest version are deleted from the local copy if they have not been modified since checkout,

- Unmodified component files in the local version are replaced by the server version,
- Component files modified in the local copy are preserved and users may resolve any conflicts between the two versions through a Resolve Conflicts dialog.

Where conflicts exist users may choose to take the server version, keep the local version, use the compare utility to show differences or use the merge utility to combine changes from each. The merge utility may be used to manage each conflict separately to replace the local copy with the latest version information, leave the local copy unchanged or to remove the item in the local version.

If you would like more information on RiskAgility FM Team, please contact

RiskAgility FM Team requires licensing in addition to RiskAgility FM and RiskAgility FM U.S. Library. Please contact Willis Towers Watson if you are interested in learning more about how RiskAgility FM Team can help your company manage your customizations.

1.3.2 Manual procedures

As discussed above, RiskAgility FM Team provides very comprehensive features to help you manage your customizations. When RiskAgility FM Team is not used, customizations should only be performed with strict adherence to a set of manual standard procedures.

These procedures would ideally include procedures around the setup and management of:

- Local environments for the customization
- Approach to off-line implementation and testing
- Change tracking, and
- Merging changes to the master environment

Please contact Willis Towers Watson if you are interested in having us work with you to develop a set of manual procedures for managing your customizations to the RiskAgility FM U.S. Library.

2 CUSTOMIZATION APPROACHES

2.1 Adding/modifying features

2.1.1 Overview

A customization for adding/modifying features is appropriate when the code, input, and output definitions of the out of the box projects do not contain the definitions required for the modeling.

2.1.2 Code customization

Code customizations for adding/modifying features are typically model class definitions updates. Considerations for code customizations are as follows:

- If not already included, add a code variable at the company and product level to control the inclusion of the customizations
- Add new client model classes or submodels as required, following model structure standards, and including a prefix to identify client model classes and submodels from the base model classes and submodels. Of course, DO NOT remove the base model classes or restructure the model tree.
- Add code variables and formulas as required, following the code definition standards, and including a prefix to identify client variables from the base variables
- Structure code updates of base formulas to separate client code modifications from the base code, such that the client code is only executed when the control variable is set to Yes. This approach simplifies merging changes during future upgrades

- Ensure all updates to base model classes are within the principles of comprehensive coverage and variations support (see the Code management user guide book) so that the customizations will function properly for both seriatim and portfolio projects

2.1.3 Input customization

Input customizations for adding/modifying features typically require updates to input managers and the external source workbooks for new variables.

Considerations for input manager customizations are as follows:

- If not already completed, copy the Template IM input manager of the product type project to a new client template input manager
- Create input variables which associate to the new code variables on the appropriate input page
- Create external source links and definitions for the new input variables

Considerations for external source workbooks are as follows:

- Update existing ranges or add new ranges within the template workbooks for the new variables as appropriate

2.1.4 Output customization

Output customizations for adding/modifying features are only required when the customization introduces a formula which is required for nested projection dependent output or is a reported item for the standard reporting.

Considerations for output manager customizations are as follows:

- Update the output manager output definitions for formulas added to the company, segment, PBA, GAAP, and product level submodels which are required for nested projection dependent output or for standard reporting.

Considerations for reporting tool customizations are as follows:

- If not already completed, create a client specific reporting tool workbook. This approach ensures that the master reporting tool is saved for reference during upgrades to newer versions
- Update the desired reports following the setup and structure of the base report

2.2 Adding asset types

2.2.1 Overview

A customization for adding asset types is appropriate when the code, input, and output definitions of the product type project do not contain the asset definitions required for the modeling.

2.2.2 Code customizations

Since all product type projects utilize the common company, segment, economy, and investment structures, which already handle all the asset types of the assets project, code customizations to add additional asset types to a product type project is a model tree structure update.

Perform these steps:

- Load the assets and product type projects into the solution
- Copy/Paste the desired asset model classes, for example `ird_asset`, from the asset project to the product type project. A top model for the asset type submodel is added to the product type project.
- Rename the asset type submodel in the model tree to match the name of the submodel in the assets project

- Build the required model tree in the product type project for the added asset types. Use the Move To... command to move the top model asset type submodel into the existing model tree so that Company is the only top model in the product type project
- Modify the properties of the added asset type submodel to match the values from the submodels in the assets project
- Update the Model References properties of the model classes as required to provide for interfacing between existing and new submodels
- Modify the 'Set Value in Input Manager' property for each variable which should have a setting other than 'All'*
- Save the customized product type project
- Build the customized project

***RiskAgility FM does NOT copy the 'Set Value in Input Manager' property value of the variables when copying model classes from one project to another and that property is reset to the 'All' selection. This will lead to compile errors if the variable is set in the model code. To resolve the compile error, you must manually update the property of the code variable to have the correct setting.**

2.2.3 Input customizations

Since the Template input manager for each product type project contains input variables on the Investment input page which are associated with the code variables of the included asset types and a data page for the model points of each asset type, input manager customizations are required to add an asset type to a product type project.

Perform these steps:

- Load the assets and product type projects into the solution
- If not already completed, copy the Template IM input manager of the product type project to a new client template input manager
- Load the new client template input manager and set the scope to only view the Dissoc. Code Variables, filtering on the variables from the added asset types
- Update the Investment input page of the new client template input manager to associate the code variables from the new asset type submodels to the input variables in the same manner as they are associated in the Investment input page in the assets project. For example, the ird submodel mkt_val_calc_freq code variable is associated with the mkt_val_calc_freq input variable within the Asset Run Control group.
- Update the Investment input page of the new client template input manager to have input variables and external links for the new asset type submodels similarly to the input variables on the Investment input page in the assets project. For example, a new input variable for ird_bk_val_defn is added into the Balance Sheet group, associated with the bk_val_defn ird code variable, and populated with an appropriate external source link to the Investment external source workbook.
- Load the External Sources of the new client template input manager and add external sources for the new asset type model point files. For example, add a plain text external source referencing <*data*>IrdMP.csv when interest rate derivatives are being added into the product type project
- Copy/Paste the 'xxxData' pages for the new asset types from the Template IM of the assets project to the new client template input manager. For example, copy the IrdData input page from the Template IM of the assets project to the new client template input manager. With this step, all variables defined on the data pages are removed from this list of dissociated variables for the input manager.
- Select the new 'xxxData' page in the new client template input manager, highlight the 'Default' assumption set and reselect the '<*data*>xxxMP.csv external source. This step populates the 'Default' assumption set with the variable links.

No customizations are required to the Template Investment.xlsm external source workbook since it already contains all variables for all asset types

2.2.4 Output customizations

Output customizations are usually not required when adding asset types to a product type project since there are no asset type specific output manager definitions and standard reports.

2.3 Combining product types

2.3.1 Overview

A customization to combine product type is appropriate when there is a desire for a single project to have the capabilities to model multiple product types.

2.3.2 Create new project

Depending upon the expected usage of the project with combined product types, it may be appropriate to create a new project to contain the combined product types.

If desired, perform these steps:

- Select one of the product type projects which would be included in the combined project in the Solution Explorer
- Right click and select 'Copy Project...', specify the name of the new project, and browse for the location of the new project. Best practice is to select the 'RiskAgility FM U.S. Library x.y\Projects' folder as the location for the new project.
- Copy the 'ProjectData' folder from the copied project to the folder of the new project. For example, copy the 'ProjectData' folder from 'Projects\Universal Life' to 'Projects\Life Insurance'. This step populates the new project with the same structure as contained in the copied project.

2.3.3 Code customization

Since all product type projects utilize the common company, segment, economy, and investment structures, which already handle all the product types, code customizations to combine product types is a model tree structure update.

Perform these steps:

- Load the product type projects into the solution. This may include the new combined product type project created from one of the desired product type projects.
- Copy/Paste the model classes required for the product types from the source product type projects to the destination combined product type project. This would include all the product type submodels, the required PBA submodels, and the required GAAP submodels. A top model for each of the model classes copied is added to the destination project.
- Build the required model tree in the combined product type project for the added model classes. Use the Move To... command to move the top model submodels into the existing model tree so that Company is the only top model in the project
- Modify the properties of the added submodels to match the values from the submodels in the source project
- Update the Model References properties of the model classes as required to provide for interfacing between existing and new submodels
- Modify the 'Set Value in Input Manager' property for each variable which should have a setting other than 'All'*

- Save the customized product type project
- Build the customized project

***RiskAgility FM does NOT copy the 'Set Value in Input Manager' property value of the variables when copying model classes from one project to another and that property is reset to the 'All' selection. This will lead to compile errors if the variable is set in the model code. To resolve the compile error, you must manually update the property of the code variable to have the correct setting.**

2.3.4 Input customization

Since the Template IM input manager for each product type project contains input variables on the Company and Economy input page which are associated with the code variables of the included product types, distinct input pages for each product type and a data page for the model points of each product type, input manager customizations are required to combine product types.

Perform these steps:

- Load the product type projects into the solution
- If not already completed, copy the Template IM input manager of the product type project to a new client template input manager
- Load the new client template input manager and set the scope to only view the Dissoc. Code Variables, filtering on the variables from the added product types
- Update the Company input page of the new client template input manager to associate the code variables from the new product type submodels to the input variables in the same manner as they are associated in the Company input page in the source project. For example, associate the comp_rate_type code variable with the comp_rate_type input variable in the Competitor Rates group.
- Update the Economy input page of the new client template input manager to associate the code variables from the new product type submodels to the input variables in the same manner as they are associated in the Economy input page in the source project. For example, associate the scen_id code variable with the scen_id input variable in the Scenarios : Rates group.
- Copy the [Product Type] – Product.xlsm, [Product Type] – Mortality Tables.xlsx and [Product Type] – Mortality Tables Attained Ages.xlsx external source workbooks from the Template\Assumptions folder of the source product type project environment to the combined product type project environment
- Load the External Sources of the new client template input manager and add external sources for the new product type external source workbooks. For example, add an Excel external source referencing the xxProd' range in the <*assump*>[Product Type] – Product.xlsm external source
- Copy/Paste the input pages for the new product type from the Template IM of the source project to the new client template input manager. All input pages for the product type should be copied. For example, copy the xxxProduct input page from the Template IM of the source product type project to the new client template input manager. With this step, all variables for the product defined on those input pages are removed from this list of dissociated variables for the input manager.
- Copy the xxMP.csv, external source workbook, where xxx is the abbreviation for the product type, from the Template\Data folder of the source product type project environment to the combined product type project environment
- Load the External Sources of the new client template input manager and add external sources for the new product type model point files. For example, add a plain text external source referencing <*data*>xxxMP.csv
- Copy/Paste the 'xxxData' pages for the new product type from the Template IM of the source product type project to the new client template input manager. With this step, all variables defined on the data pages are removed from this list of dissociated variables for the input manager.

- Select the new 'xxxData' page in the new client template input manager, highlight the 'Default' assumption set and reselect the '<*data*>xxxMP.csv' external source. This step populates the 'Default' assumption set with the variable links.

No customizations are required to the Template Company.xlsm and Economy.xlsm external source workbooks since they already contains all variables for all product types

2.3.5 *Output customization*

Since the output managers contain output definitions separate for each product submodel, customizations are required for the output managers.

Perform these steps:

- Load the product type projects into the solution
- Edit the output managers of the combined product type project and add output definitions for the product type submodels which were added. This includes adding output definitions for the product type submodel, the product level GAAP submodels, and the product level PBA submodels.

Customizations are usually not required to the standard reporting when creating a project with combined product types since the reporting tool is already setup to handle all product types.

C01 PROFIT CALCULATIONS

This book discusses profit calculations. These calculations occur within the product submodel when running a seriatim projection.

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1 OVERVIEW

1.1 Calculation level

Profit calculations occur within the company and segment submodel for portfolio projections and within the product submodel for seriatim projections.

1.2 Seriatim projections

Seriatim projection tasks calculate investment income, taxes, target capital, free surplus, and distributable earnings individually for each model point. Profits are calculated using formulas in the product submodels.

A projection task is a seriatim projection when the product submodel is the top model.

1.3 Business segment projections

A portfolio projection calculates aggregate profits for a business segment and the corporate segment:

- A 'business segment' is typically a block of business which is backed by its own investment portfolio, and therefore has a different investment strategy from other segments of the company's business.
- The 'corporate segment' invests free surplus, undistributed earnings, and any target capital that is not invested in the business segment asset portfolio.
- The company submodel aggregates the segment results, and makes final profit calculations.

Portfolio projection tasks calculate investment income, taxes, target capital, free surplus, and distributable earnings using aggregate results, rather than calculating these results separately for each model point. Profits are calculated using formulas in the company and segment submodels. The product submodel calculates model point cash flows and values, which are aggregated by the segment submodel.

A projection task is a portfolio projection or valuation when the company submodel is the top model. This is different from seriatim projections, which have the product submodel as the top model, and calculate profit results separately for each model point.

1.4 Portfolio projection differences from seriatim

1.4.1 Aggregate results

Portfolio projections calculate aggregate results for a business segment, while seriatim projections calculate results for each model point separately. Due to the different nature of these calculations, portfolio projection profit results can be different from the sum of model point profits from a seriatim projection.

Portfolio projections calculate aggregate results in the company and segment submodels for investment income, taxes, target capital, distributable earnings, and free surplus. Thus, profit reports for the entire projection show book profit before tax, book profit after tax, the effect of target capital, and distributable earnings.

1.4.2 Projecting assets

Only portfolio projections can project assets. When the Include assets? project variable (in the Run parameters object) is Yes, this is an asset/liability projection. Otherwise, this is a liability-only projection, which has the asset yield specified as an assumption.

Seriatim projections are always liability-only projections, since the profits are calculated for each model point separately.

1.4.3 Free surplus

Only portfolio projections can include free surplus. Portfolio projections specify the amount of profits to pay out as shareholder dividends, and can include an amount (positive or negative) of initial (at the projection date) free surplus. See the Capital and surplus chapter, below.

Note, however, that free surplus is zero for new business projections, which are calculating special pricing measures that require free surplus to be excluded. A portfolio projection is a new business projection when the Pricing projection? project variable (in the Run parameters object) is Yes.

Seriatim projections exclude free surplus, which exists for the company, not for each model point.

1.4.4 Segment calculation differences

Book profit before tax can be different between a portfolio projection and a seriatim projection using the same model points. For example:

- Portfolio projections can include fixed expenses in addition to the model point expenses.
- Risk based capital, which is the basis for target capital, is different using aggregate calculations than summing individual model point calculations. When shareholder dividends are paid less frequently than monthly, investment income allocated to undistributed earnings is included in book profit. Since undistributed earnings are determined using target capital, this difference in target capital can affect book profits.
- Portfolio projections can include assets, so that liabilities include asset based reserves.

2 DATES AND TIMING

2.1 Projection year

2.1.1 Projection date

The projection date is the end of the calendar month and year specified as yyyy/mm in the Run parameters object. The projection starts on the first day of the next month. Present values are discounted to the projection date.

For example, a projection date of 2014/12 begins from initial values as of December 31, 2014, and has the first new business model points issued on January 1, 2015.

2.1.2 Projection period

Specify the projection period in the Run parameters object. The Projection period to property is the number of months from the projection date. For example, a 2014/06 projection date projecting for 126 periods ends at December 2020.

The exception is for time-step projections, which are controlled by the inner loop valuation periods. The projection task for the experience (outer loop) projection must have the Projection period to property at least long enough to reach between each valuation date. Any value longer than that is ignored, since the outer loop projection stops in order to perform valuation tasks at the next valuation date.

The Projection period from property should be 0, indicating that calculations begin from the projection date.

Valuation iteration projection years

The projection period property does not apply for valuation tasks. Valuation results are calculated through the valuation years project variable specified in the Run parameters object. [valn_yrs]

2.2 Calendar year

The projection date is the end of the specified calendar year and month. Calendar year and month are calculated through the projection, such as for looking up values by date.

Profit distributions occur on a calendar year cycle. For example, distributing annually in a projection starting 2014/06 results in distributions occurring at the end of projection months 6, 18, 20, and so on.

Taxes are calculated on a calendar year cycle. Year-to-date taxes are calculated from the projection date, and from the beginning of each calendar year. The monthly tax amount is the change in the year-to-date taxes.

2.3 Assumed timing of events

Calculations assume that premiums, model point issues, and related acquisition expenses occur at the beginning of each month. Decrements, policyholder transactions, and maintenance expenses are assumed to occur at the end of each month. This simplifies formulas and reduces calculation time since interest crediting (and separate account fund values growth) are calculated only at the end of the month, when they are also needed to produce values for balance sheet and profit calculations.

Economic scenario rates change monthly. There is no need to determine mid-month rates, since decrements and transactions occur only at the end of each month.

3 COMPETITOR RATES

'Competitor rates' are external financial rates which may influence policyholder behavior and affect crediting rates. These are sometimes called 'benchmark rates', 'market rates', or 'smart rates'.

3.1 Stochastic model point calculations

Competitor rates are linked to scenario rates, and are thus stochastic. The value for each rate varies over time and between each scenario path. You can construct a deterministic competitor rate by setting the scenario rate multiple to zero.

Each competitor rate is constant for all dates within a particular month, since scenario rates only vary by month. For example, a competitor rate calculated for June 1 (for determining the crediting rate), and again for June 30 (for excess lapses), has the same value at both dates.

3.2 Input organization

The competitor rate input must lookup definitions using the `comp_rate_id` hidden code variable, which is set by formulas. Select the lookup term as the `comp_rate_id` code variable from either the company (for portfolio projections) or product (for seriatim projections) submodel. Change the lookup type to be Relative, so that the input manager uses the appropriate submodel for the current `comp_rate_id` value.

The `comp_rate_type` variable specifies whether a particular competitor rate is a Single, Weighted, or Compared type of calculation (see descriptions below). The formulas then use the variables related to each type of rate.

The following shows an example external source defining competitor rates:

VarID	comp_rate_id	5Year	10Year	5_10_Avg	10_Avg_Max	Max_5_10
comp_rate_type		Single	Single	Weighted	Weighted	Compared
comp_rate_single_scen_yr		5.00	10.00			
comp_rate_single_avg_mths		1	1			
comp_rate_single_mult		1.00000	1.00000			
comp_rate_single_addn		0.00%	0.00%			
comp_rate_weighted_rate1_id				5Year	10Year	
comp_rate_weighted_rate1_weight				0.50000	0.25000	
comp_rate_weighted_rate2_id				10Year	5_10_Avg	
comp_rate_weighted_rate2_weight				0.50000	0.50000	
comp_rate_weighted_rate3_id				NONE	Max_5_10	
comp_rate_weighted_rate3_weight				0.00000	0.25000	
comp_rate_compared_defn						Maximum Rate
comp_rate_compared_rate1_id						5Year
comp_rate_compared_rate2_id						10Year
comp_rate_compared_rate3_id						NONE

Competitor rates are used in calculations when the name (comp_rate_id) is specified for some input variable. For example, the indexed policy loan rate, crediting rate definition or constraints, excess lapse rate parameters, and so on, depending on the features varying by product type.

3.3 Single rates

Any number of 'single' competitor rates can be specified as a monthly geometric average of a scenario yield rate, times a multiple then plus an addition. Each 'single' competitor rate is defined by specifying:

- Scenario years – The effective yield for a point on the risk free yield curve
- Moving average months
- Multiple
- Addition.

The moving average is a geometric average, calculated using effective annual rates adjusted by the multiple, then the addition:

$$(1+i)^m = (1+i_1) \times (1+i_2) \times \dots \times (1+i_m)$$

3.4 Weighted combination rates

'Weighted' combination competitor rates can be based on other single or combination rates. Each rate is the weighted sum of other competitor rates. The rates must not create a circular definition.

Each 'weighted' competitor rate is defined by specifying the name and weight for up to three other competitor rates. (You can customize to extend to more than three competitor rates.) The weights are not normalized, i.e., they are not forced to add to 100%. A weight can be negative.

3.5 Compared combination rates

'Compared' combination competitor rates are the maximum or minimum of other competitor rates. The rates must not create a circular definition.

Each 'weighted' competitor rate is defined by specifying the comparison definition (either Maximum or Minimum), plus the names for up to three competitor rates. (You can customize to extend to more than three competitor rates.)

4 EXPENSES

4.1 Investment expenses

Investment expenses reduce investment income. Investment expenses are a percent of each asset amount. See the book for each asset type.

4.2 Product expenses

Product expenses are calculated for each model point. These expenses vary by the amount of business being processed. Acquisition expenses (related to issuing the policy) are reported separately from maintenance (ongoing) expenses. Reinsurance administration expenses are included with product maintenance expenses.

Premium taxes are included with maintenance expenses on reports.

Policy loan expenses are treated as investment expenses on reports. That is, policy loan interest (which is part of investment income) is net of policy loan expenses. However, the principle-based (PBA) deterministic reserve for life products includes policy loan expenses in the present value of expenses.

4.3 Fixed expenses

For portfolio projections, fixed expenses can be added for the business segment. These are in addition to the product expenses calculated for the business included in each model point. Reports include fixed expenses with maintenance expenses.

4.4 Inflation

The inflation rate is included in the scenarios file; see the Economic scenarios book. Deterministic projections can assume fixed inflation rates, specified on the Economy input page.

Product expenses not expressed as a percent are adjusted by inflation, starting from the inflation start date specified on the Product input page.

Fixed expenses (which apply for the business segment in portfolio projections) are also adjusted by inflation, starting from the inflation start date specified on the Company input page.

Investment expense rates are not subject to inflation.

5 INVESTMENT INCOME

5.1 Investment income on assets

Asset/liability projections calculate investment income by projecting the assets for the corporate and business segments, then adjusting capital gains for asset sales, defaults, and prepayments. Cash flows and amounts not invested earn interim cash interest.

Investment income on invested assets includes:

- Ordinary investment income – Interest (including interest on cash and interim cash), dividends, and rent, plus the increase in cost basis (amortized cost basis growth or decrease), less investment expenses.
- Realized capital gains – Realized capital gains are: (1) the amounts received on sale of assets less the cost basis of those assets, plus (2) prepayment amounts received less the related reduction in the cost basis, less (3) the cost basis plus accrued income for asset defaults.

- Increase in unrealized capital gains – Unrealized capital gains are recognized in statutory accounting when assets are valued at market value. (Note that U.S. taxes exclude unrealized capital gains, while statutory income includes it.)

Total investment income adds:

- Separate account investment income – Separate account growth amounts are aggregated from liability model points.
- Policy loan interest – Policy loan interest is aggregated from liability model points. The interest applies regardless of whether it is paid by the policyholder or whether the accrued interest increases the outstanding policy loan balance. Policy loan expenses then reduce policy loan interest. (PBA calculations include policy loan expenses in the gross premium reserve calculation of the present value of expenses.)

Investment income on assets only includes the investment income on assets that have a portfolio definition of 'Investment'. The investment income of 'Hedging' assets is not included in this amount.

5.2 Liability-only projections

For projections without assets, the asset yield is an assumption. The asset yield can be either:

- Stochastic – one of the competitor rates, which are calculated based on economic scenario rates.
- Deterministic – a fixed rate varying by calendar month.

When index credit hedging is included in liability only projections, the asset yield assumption is treated as the asset yield on non-hedge assets.

5.3 Allocation to profit components

Ordinary investment income for each month is allocated by profit component:

Profit Component	Allocation Basis
Book profit	Interest on direct cash flows, plus investment income on reserves and undistributed earnings, less investment income on reinsurance reserves
Reserves	Investment income on invested assets, less investment income on reinsurance reserves, less investment income on target capital, undistributed earnings, and free surplus
Reinsurance reserves	Previous EOM reinsurance reserves
Target capital	Previous EOM target capital
Undistributed earnings	Previous EOM undistributed earnings
Free surplus	Previous EOM free surplus
Fixed fund values (for SFAS 97 GAAP EGP)	Previous EOM fixed account fund value

The net investment income allocated to undistributed earnings is included in book profit investment income. Thus, the frequency for distributing earnings affects book profit. Since target capital determines undistributed earnings, target capital has some effect on book profit when shareholder dividends are paid less frequently than monthly.

Realized capital gains, and the increase in unrealized capital gains, are allocated using these same bases, except that cash flows are ignored.

Policy loan investment income is included in book profit.

Separate account investment income is excluded from profits since it is exactly offset by the increase in the separate account liabilities.

6 CAPITAL AND SURPLUS

6.1 Target capital

6.1.1 Specified capital

Target capital (sometimes called target surplus, required surplus, required capital, or solvency margin) is the level of capital that the insurer regards as necessary to meet risk, regulatory, and rating agency requirements.

Target capital is a specified multiple of risk based capital (see this topic, below). The AVR and IMR are considered to be additions (can be negative) to this amount.

Target capital is calculated differently between portfolio and seriatim projections:

- Portfolio projections calculate the risk based capital and target capital for each business segment, and for the total company. There is no allocation to model point report groups for portfolio projections.
- Seriatim projections calculate the risk based capital and target capital separately for each model point. The sum of risk based capital and target capital across the seriatim model points will be different than the aggregate calculation for those same model points within a portfolio projection, due to the difference between individual and aggregate risk based capital results.

6.1.2 AVR

The asset valuation reserve (AVR) is considered to be part of target capital. This has some effect on book profits, since investment income is allocated to the AVR.

See the Asset valuation reserve chapter of the Asset modeling book.

6.1.3 IMR

The interest maintenance reserve (IMR) is considered to be part of target capital. Investment income allocated to the IMR is part of the investment income on target capital. The amortization of the IMR is allocated to reserves, target capital, and free surplus, thus directly affecting book profits.

See the Interest maintenance reserve chapter of the Asset modeling book.

6.2 Investment of target capital

6.2.1 Business segment target capital

For portfolio projections, a portion of target capital can be invested in the corporate segment instead of keeping all assets backing target capital in the business segment.

When part (or all) of the business segment's target capital is invested in the corporate segment, 'distributable earnings' for the business segment includes the effect of only the target capital invested in the business segment. The total distributable earnings, including the effect of total target capital, is calculated in the company submodel.

For example, when target capital is invested in the corporate segment, 'distributable earnings' for the business segment is the same as book profit after tax. The corporate segment calculates the 'distributable earnings' from the target capital.

Warning! When the projection is a total company asset/liability projection, such that free surplus does not apply in the business segments, the initial target capital invested in the business segment may be adjusted from the specified amount so that initial assets equal initial liabilities plus the target capital, and free surplus is zero.

6.2.2 Corporate segment target capital

Risk based capital and target capital is calculated for the total company. Any difference (can be negative) from the target capital calculated for business segments is invested in the corporate segment.

6.3 Risk based capital

6.3.1 RBC amount

U.S. risk based capital (RBC) is:

$$\sqrt{\left((C1 \text{ other assets component} + C3 \text{ component})^2 + (C1 \text{ equity assets component})^2 + (C2 \text{ component})^2 \right)} + C4 \text{ component}$$

- C1 is the risk of asset default,
- C2 is the risk of inadequate product pricing,
- C3 is the risk of changes in interest and other market rates, and
- C4 is the general business risk.

RBC component percentages should be specified using after tax rates.

6.3.2 PBA capital

When PBA capital is calculated, this amount replaces the C3 component of RBC. The other components, and total RBC, are calculated as usual.

6.3.3 C1 component

The RBC C1 equity risk component is:

$$\left(\sum_{\text{assets}} \text{book value for equity assets} \times \text{RBC C1 risk percent} \right) \times \text{RBC asset concentration adjustment}$$

The RBC C1 other assets risk component is:

$$\begin{aligned} & \left(\sum_{\text{assets}} \text{book value for other assets} \times \text{RBC C1 risk percent} \right) \\ & \times \text{RBC portfolio size adjustment} \\ & \times \text{RBC asset concentration adjustment} \\ & + \text{cash and interim cash less temporary borrowing} \times \text{RBC C1 risk percent} \\ & + \text{reinsured statutory reserve} \times \text{RBC C1 reinsurance reserve percent} \end{aligned}$$

The RBC C1 risk percentages are asset model point input variables. Specify these rates by mapping out the percent for each individual asset using the specified rules. For example, bond rates vary by NAIC Securities Valuation Office (SVO) reserve class, mortgage rates vary by the type of mortgage (such as residential, commercial, and farm), and equities vary between affiliate etc. classifications.

For liability-only projections, specify the C1 component as a specified average percent of statutory reserves including the separate account, net of reinsurance reserves.

6.3.4 C2 component

RBC C2 mortality risk

For individual life insurance products, the RBC C2 mortality risk is the net amount at risk times the RBC C2 percent. (The prescribed percent is 0.15% up to the first \$500 million, 0.10% up to \$5 billion, 0.08% up to \$25 billion, and 0.0585% for the excess.)

There is no contribution from annuity products.

RBC C2 morbidity risk

For long term care (LTC) products, the RBC C2 morbidity risk is the sum of: (1) a percent of statutory net claim reserves, plus (2) a percent of the previous 12 months of statutory net earned premiums, plus (3) a percent of the previous 12 months of statutory net incurred claims.

6.3.5 C3 component

The C3 component is the PBA capital result. There is no split between C3 'interest' and 'equity' components.

When PBA capital is not calculated, the RBC C3 risk component is a percent of:

- Life insurance and LTC reserves net of policy loans and modified coinsurance reserves (The prescribed percent is 0.5% or 1.15%*.)
- Annuity general account reserves net of coinsurance reserves. Specify an average percent which approximates the mix of: (1) 'low risk' annuities with market value adjusted (MVA) cash values or not withdrawable (0.5% or 0.75%*), (2) 'medium risk' annuities with a surrender charge of 5% or more (1.0% or 1.5%*), and (3) 'high risk' annuities without a significant surrender charge (2% or 3%*).

*The lower percentages apply when the company submits an unqualified actuarial opinion based on asset adequacy testing. The percentages can be adjusted by a multiple which approximates the C3 risk amount determined by cash flow testing analysis. This multiple is limited by a maximum 200% and minimum 50%.

Each annuity product is classified as being either low risk, medium risk, or high risk for the purpose of aggregating reserves for calculating the RBC C3 component.

6.3.6 C4 component

The RBC C4 risk component is a percent of:

- Annualized premiums (2%), excluding single premiums.
- Separate account liabilities (0.05%).

6.4 Transfers to the corporate segment

At the end of each month, portfolio projections transfer an amount to the corporate segment so that business segment assets match the reserves and target capital invested in the business segment. Of course, this transfer can be positive or negative.

The corporate segment invests assets backing target capital not invested in the business segment, undistributed earnings, and free surplus.

This monthly distribution from the business segment is the 'distributable earnings' calculated for the business segment. This 'distributable earnings' for the business segment includes the effect of only the target capital invested in the business segment. The total distributable earnings, including the effect of total target capital, is calculated in the company submodel.

Single segment projections

When all liability and asset model points are assigned to a single business segment and 100% of target capital is invested in the business segment, the projection is a single business segment projection. Then `tcm_flag = false` and there is no corporate segment. Columns in the segment submodel are the total projection results; the company submodel retrieves these values for producing reporting results. Assets backing target capital and free surplus earn the same investment rates as the assets backing liabilities.

6.5 Shareholder dividends

Shareholder dividends (also called profit distributions) pay out cash from the corporate segment, thus reducing invested assets. When the 'dividend' is negative, this represents amounts paid in to the company, increasing invested assets.

Profits are distributed from the projection either at the end of each month, calendar quarter, or calendar year. This profit distribution mode determines the frequency for discounting the present value of profits; see the Present values chapter, below.

Seriatim and pricing projections automatically distribute the full amount of distributable earnings. There is no free surplus, and the initial free surplus variable is ignored.

Portfolio projections* distribute a percent of distributable earnings, plus a percent of free surplus as of the end of the previous month. *The exception is that new business pricing projections automatically pay out the full distributable earnings, as explained above.

6.6 Undistributed earnings

Undistributed earnings (also called undistributed surplus) occur when shareholder dividends are paid less frequently than monthly. For months between distributions, distributable earnings are added to undistributed earnings on the balance sheet. Undistributed earnings can be negative, particularly in the first year of pricing projections.

Investment income allocated to undistributed earnings is included in book profit. Target capital thus affects book profits due to this effect.

6.7 Free surplus

6.7.1 Portfolio non-pricing projections

Free surplus (also called unallocated surplus) is: (1) total assets, less (2) liabilities, (3) target capital, and (4) undistributed earnings. Free surplus can be negative. Assets backing free surplus are invested in the corporate segment.

Free surplus does not apply for the following cases, which automatically pay out distributable earnings:

- Seriatim projections, which calculate profits separately for each model point.
- New business pricing projections, which calculate special pricing measures that assume all profits are distributed. This applies when the Pricing projection? project variable in the Run parameters object is Yes for a portfolio projection, which is then a 'New business pricing' projection. (Cell pricing projections also exclude free surplus, since they are both pricing and seriatim projections.)

6.7.2 Initial free surplus

For asset/liability projections, initial free surplus is the result of the initial (at the projection date) book value of assets.

For liability-only projections, specify the initial free surplus for the projection.

7 TAXES

7.1 Corporate segment

7.1.1 *Portfolio projections*

For portfolio projections and valuations, taxes are calculated for the total company, and for each business segment. There is no allocation to model point report groups.

The tax amounts calculated for each segment are distributed to the corporate segment. Total company tax payments are then paid from the corporate segment.

7.1.2 *Seriatim projections*

For seriatim projections, taxes are calculated separately for each model point since there are no aggregate calculations.

7.2 Tax payments

7.2.1 *Tax cash flows*

Taxes are cash flows (can be negative) from the corporate and business segments:

- The tax calculated for the business segment is a cash flow paid from the business segment.
- The tax cash flow from the corporate segment is the tax for the total company, less the tax cash flows from the business segment. This includes the tax on corporate segment investment income, plus the difference between total company taxes and the sum of the taxes calculated for each segment.

7.2.2 *Year-to-date calculation*

Taxes are calculated within each calendar year. The tax cash flow each month is the current less the previous year-to-date tax amount.

Projections starting during the tax year ignore taxes before the projection date. That is, initial year-to-date taxes are zero.

7.2.3 *Immediate loss recognition*

Negative tax amounts increase profits immediately. For example, this might assume that the loss is offset by positive taxes from other business segments.

Pricing options

For cell pricing and new business (portfolio) projections, however, you can also choose more conservative options:

- Tax loss carry forward – The tax is zero. The loss amount is carried forward until it is offset against future positive taxes. There is no limit to the number of carry forward years.
- Zero tax – The tax is zero, with no effect on future taxes.

Zero tax means that the tax on book profit is not less than the tax on investment income on target capital, so that the total tax on distributable earnings is zero. It is only the remainder of the negative tax on book profit which is carried forward to offset against future positive amounts.

7.3 Tax on income

7.3.1 Taxable income

Life insurance company taxable income (LICTI) is split into three pieces for calculating book profit and distributable earnings:

- book profit taxable income
- + investment income on target capital
- + investment income on free surplus

7.3.2 Book profit taxable income

The book profit component of taxable income is:

- premium income
- + tax basis book profit investment income
- benefits
- tax basis reserve increase
- commissions
- expenses less claim expense reserve increase
- + capitalized premiums less amortization
- tax basis dividend benefits
- reinsurance cost
- net transfers to separate account
- amortization of prior capitalized premiums
- tax exempt income for book profit

See the Profit definitions chapter, below, for a detailed discussion of various profit components. The following describes special aspects for tax calculations:

Tax basis investment income is statutory investment income, less the increase in unrealized capital gains. The total amount for the business segment is allocated to the profit components for book profit, target capital, and free surplus (see the Investment income chapter, above).

Tax basis reserve increase is calculated using tax reserves instead of statutory reserves. The reserve is net of separate account liabilities. This includes the tax basis loading increase, which operates the same as the statutory loading increase discussed in the Profit definitions chapter, below.

Expenses exclude the increase in health claim expense reserves [future release], which are not tax deductible.

Capitalized premiums less amortization is the increase in taxable income due to the DAC tax (discussed separately below).

Tax basis dividend benefits are the dividends paid, plus a percent (such as 50%) of the dividend liability increase.

Taxable income adjustments can be included in the amortization of prior capitalized premiums. See the Taxable income adjustments section, below.

Tax exempt income for book profit includes the portion of the tax exempt income on municipal bonds less the investment expenses attributed to those bonds allocated to book profits.

7.3.3 *Investment income on target capital*

The investment income on target capital is the tax basis investment income allocated to target capital, which includes AVR and IMR, less the tax exempt income allocated to target capital.

7.3.4 *Investment income on free surplus*

Similarly, the investment income on free surplus is the tax basis investment income allocated to free surplus, less the tax exempt income allocated to free surplus.

7.3.5 *Tax calculation*

The tax rate on realized capital gains can be different from the tax rate on the rest of LICTI. Thus the tax on each of the above income amounts is:

$$\text{taxable income} \times \text{corporate tax rate} + \\ \text{realized capital gains} \times (\text{capital gains tax rate} - \text{corporate tax rate})$$

7.4 Capitalized premium tax (DAC tax)

7.4.1 *Projected premiums*

Tax on book profit includes the capitalized premium tax, which is also called the DAC tax. A percent of taxable premium income is capitalized and added to taxable income. Then this capitalized amount is amortized over future years as subtractions from taxable income. This is a proxy for amortizing acquisition expenses in profit results, rather than as occurring at issue.

Specify the capitalized premium percent on the Product input page, in the Premiums category. (The prescribed rate is 1.75% for deferred annuity products, 7.70% for individual life insurance products, and 2.05% for group life contracts.) Qualified pension plans are exempt from this tax. For deferred annuity products, specify the percent of premiums that is subject to this tax.

Taxable premium income is paid premium, minus net reinsurance premiums, plus reinsurance benefits (death benefits and non-death benefits). Net reinsurance premium is reinsurance premiums paid, less (1) reimbursed premium taxes, (2) expense allowances net of any chargebacks, and (3) coinsurance experience refunds.

The capitalized premium income for each tax year is the taxable premium income times the percent of premium capitalized.

The capitalized amounts are amortized linearly over 10 years. The amortization starts in the middle of the tax year in which the premium income occurs. Thus the amortization is 5% in the first year, 10% in years 2 through 10, and 5% in the 11th year. During the year of the premium payment, 5% of the amount is amortized at the same time it is capitalized. In subsequent years, one twelfth of the annual amount is amortized in each month.

7.4.2 *Past premiums*

Specify on the Company input page the unamortized capitalized premium income for premiums received before the projection date. The unamortized amount must vary by calendar date, showing the runoff of this initial amount over time. [tax_capzed_prem_unamortzd]

7.5 Taxable income adjustments

Specify on the Company input page the unamortized additions to taxable income. This is treated as an increase in book profit taxable income. The unamortized amount must vary by calendar date, showing the runoff of this initial amount over time. [taxable_inc_addn_unamortzd]

8 LETTER OF CREDIT FINANCING

8.1 Overview

Letter of credit (LOC) financing creates a non-invested LOC asset that offsets the 'redundant' part of the statutory reserve. The LOC asset operates similarly to the deferred gross premium asset for mean reserves.

Entering into a LOC financing agreement offsets part of the reserve, thus increasing free surplus. Book profit is increased by the LOC asset increase, offsetting part of the statutory reserve increase. Book profit is also decrease by the LOC cost, which is the financing interest paid on the LOC asset.

Please refer to the LOC economic reserve and LOC projections setup chapters of the C05 PBA book for information regarding the setting up valuation and projections which include Letter of Credit reserve financing.

8.2 LOC cohorts

Any number of financing transactions can be included as LOC cohorts. Each cohort defines the financing rate for the transaction and discount rates used when the economic reserve uses a gross premium reserve methodology (discussed below).

Model points are assigned to the LOC cohorts using plan code and issue year criteria.

8.3 LOC asset

The balance sheet includes the LOC asset. This is a non-invested asset, which operates as an offset to the statutory reserves.

The LOC asset is:

$$\max[0, \text{statutory reserves net of reinsurance} - \text{loc economic reserve net of reinsurance}]$$

The LOC economic reserve can be determined using a gross premium reserve methodology or the required level of primary security determined using the actuarial method described in Actuarial Guideline 48.

Specify the selection for the LOC economic reserve using the 'loc_econ_res_defn' variable in the Letter of Credit: General section of the Company input page.

When the LOC economic reserve definition is 'Gross Premium Reserve, the economic reserve is determined using a gross premium reserve methodology, where the gross premium reserve is the present value of model point cash flows, calculated using the discount rate specified for each financing transaction.

When the LOC economic reserve definition is 'AG 48', the economic reserve is determined using the actuarial method of Actuarial Guideline 48. The actuarial method is the greater of the deterministic reserve, stochastic reserve (when included), and the applicable percentage of the NPR determined using the methodologies of VM-20.

8.4 LOC cost

The LOC cost is the interest cost associated with holding the LOC asset. This cost is determined monthly and is calculated as the monthly financing rate times the LOC asset at the end of the previous period. Each LOC cohort can have a different financing rate.

As simple interest methodology is used to compute the LOC cost. That is, the monthly financing rate is the specified annual rate divided by 12.

Book profits are reduced by the LOC financing cost.

9 SEPARATE ACCOUNT

Separate accounts are included for Variable Annuities and Variable Universal Life products where initial fund values and a percentage of premium are allocated to any number of variable accounts linked to scenario indexes.

9.1 Separate account assets

Investment processing invests the company's assets excluding the separate account fund values, which are invested as directed by policyholders. The separate account asset value is always equal to the separate account fund values.

9.2 Separate account reserve

Specify the separate account reserve through the sa_res-defn variable on the Separate Account category of the Company Input page as either:

- Separate account fund value – The separate account reserve for a policy equals the separate account fund value.
- Minimum of the separate account fund value and the basic reserve – The separate account reserve for a policy equals the smaller of the separate account fund value and the policy's basic reserve.
- Minimum of the separate account fund value and the allocation basic reserve – The separate account reserve for a policy equals the smaller of the separate account fund value and an allocated portion of the policy's basic reserve.

The basic reserve for a policy is the statutory reserve excluding any reserve for guaranteed death or living benefits and, when used, the allocation factor equals the ratio of the separate account fund value to the total policy fund value.

When the separate account reserve does not equal the separate account fund value, a separate account reserve expense allowance liability is established. This amount equals the negative of the difference between the separate account fund value and the separate account reserve. The separate account fund value and this amount are reported as separate items on the balance sheet.

9.3 Separate account transfers

The effect of the separate account on cash flows and profits are collected into separate account transfer cash flows and a net transfer to separate account amount.

The transfer cash flows include the premiums, transfers of separate account fund values to fixed fund values, loads and charges taken from separate account fund values, and separate account fund values released on claim payments. The transfer cash flow amounts are calculated as the amounts received from the separate account, which are added to direct cash flows.

The net transfer to separate account amount includes the transfer cash flows plus the transfer required to maintain the separate account surplus, which equals the change in the separate account reserve expense allowance. The net transfer to separate account amount is calculated as the amount paid to the separate account, which is a subtraction within the profit calculation.

10 PROFIT DEFINITIONS

10.1 Book profit before tax

10.1.1 Book profit

The statutory book profit is the profit with the effects of capital and surplus removed. Thus it does not include investment income on assets allocated to capital and surplus, taxes on that investment income, or the changes in capital and surplus. However, book profit includes the effect of undistributed earnings, which is affected by target capital.

Book profit before tax is:

```

premium income
+ book profit investment income
– IMR impact on book profit
– benefits
– reserve increase
– commissions
– expenses
– reinsurance cost
– hedge cost
– dividend benefits
– net transfers to separate account

```

10.1.2 Premium income

Premium income includes:

```

premium income
= gross premiums paid
+ dividend option income
+ annuitization income
+ deferred gross premium increase

```

Gross premiums paid are net of premiums paid by applying policyholder dividends.

Dividend option income is the amount of policyholder dividends applied to purchase paid-up additions and one year term insurance, to accumulate at interest, or to reduce premiums.

Annuitization income is the amount of policy proceeds applied to purchase payout annuities.

Deferred gross premium increase applies when the Mean or Interpolated mean method is used for calculating mid-year statutory reserves for term and traditional life products. The mid-year reserve assumes the annual premium is paid at the start of the policy year. For premium modes other than annual, this premium has not actually been paid at the beginning of the year. The following income statement transactions account for this balance sheet treatment: (1) The reserve increase (based on the annual

premium assumption) has been included in benefits. (2) The increase in the deferred gross premiums is included in premium income. (3) The increase in loadings on deferred premiums is also included in benefits. (4) The increase in loadings on deferred premiums is the increase in the deferred gross premiums, minus the increase in the deferred valuation-basis premiums. Substandard premiums are excluded from this calculation. See the Deferred premium asset section in the Statutory and tax reserves section of the Term life and Traditional life books.

10.1.3 Book profit investment income

The investment income allocated to book profit is discussed in the Investment income chapter, above. Investment income is net of investment expenses.

10.1.4 IMR impact on book profit

The IMR impact on book profit is the net increase in the IMR which is allocated to book profit, based on the statutory reserve and undistributed earnings at the end of the previous month. This is the same basis as for allocating realized capital gains. The IMR impact accounts for book profit realized capital gains which are added to the IMR and then subsequently amortized back into income.

10.1.5 Benefits

Benefits include:

- death benefits
- + maturity and endowment benefits
- + annuity benefits
- + health claims paid
- + surrender benefits
- + conversion costs

Death benefits are the basic policy death benefits, the death claims from policyholder dividend options, and terminal dividends paid on death.

Maturity and endowment benefits are amounts paid to surviving policyholders on the policy maturity date, plus pure endowments paid in any year. Maturity benefits include policyholder dividend option amounts paid upon expiration of the policy.

Annuity benefits are the benefits paid from payout annuities.

Health claims paid are the cash claim payments including premium waiver benefits, such as for long term care (LTC) products. This is not health claims incurred, which are the benefits paid (included here), plus the increase in the claim reserves (included in the statutory reserve increase), not including the increase in health claim expense reserves (included in maintenance expenses).

Surrender benefits include payment of cash values, policyholder dividend option cash values, and terminal dividends paid on surrender. Surrender benefits include partial withdrawal benefits paid when policyholders withdraw part of the cash value or dividend option amounts without terminating the policy. Partial withdrawal benefits include fees that reduce the part of the withdrawal that is paid to the policyholder.

Conversion costs occur when term insurance policies are replaced by permanent insurance policies issued without underwriting. This cost can apply for traditional life products. Term conversion costs generally represent the present value of profits on regularly issued permanent products, less the present value of profits on policies issued from a term conversion.

10.1.6 Reserve increase

The reserve increase is:

- base policy reserve increase
- + dividend option reserve increase
- + health claim reserve increase
- + statutory loading increase
- LOC asset increase

Base policy reserve increase is net of the separate account increase, which is included in the net transfer to the separate account adjustment to book profits.

Policyholder dividend options reserve increase is the increase in reserves for dividend option benefits purchased from policyholder dividends. This does not include the increase in the policyholder dividend liability (for dividends not yet paid or applied), which is included in the reporting of policyholder dividends incurred.

Health claim reserve increase does not include the increase in claim expense reserves, which is included in maintenance expenses.

Statutory loading increase is the counterpart to the deferred gross premium increase discussed above. See the Deferred premium asset section in the Statutory and tax reserves section of the Term life and Traditional life books.

LOC asset increase is the increase in the letter of credit financing asset, which offsets part of the increase in the statutory reserve. This amount is already net of any reinsurance.

10.1.7 Commissions

Commissions include commissions paid, less commission chargebacks.

10.1.8 Expenses

Expenses include:

- acquisition expenses
- + maintenance expenses
- + claim expense reserve increase
- + premium taxes
- + LOC costs

Acquisition expenses are the expenses related to the sale of new business.

Maintenance expenses are the expenses related to the administration of in force business. These include variable maintenance expenses (expenses varying by the amount of business in force), plus fixed expenses (additional business segment expenses in portfolio projections).

Claim expense reserve increases, for disability and indemnity benefits, are included with expenses. Thus they are not included in the statutory reserve increase.

Premium taxes are considered to be expenses.

Letter of credit (LOC) costs reduce the book profit before taxes.

Investment expenses and policy loan expenses are not included in the reporting of expenses. Instead, they reduce the amount of investment income.

10.1.9 Dividend benefits

Policyholder dividend benefits incurred are:

dividends paid
 + terminal dividends
 + dividend liability increase

Dividends paid include dividends paid in cash, used to reduce premiums, and applied to purchase paid-up additions and one year term insurance or to accumulate at interest.

Terminal dividends are the dividends paid on the termination of the policy, such as paid on lapse and death.

The dividend liability includes dividends declared but not yet paid. Dividends are declared at the end of the calendar year, for payment on the policy anniversaries in the next calendar year.

10.1.10 Reinsurance cost

The reinsurance cost is:

reinsurance premiums
 + increase in reinsurance gross deferred premiums
 + investment income on reinsurance reserves and cash flows
 + capital gains on reinsurance reserves
 – reinsurance benefits
 – increase in reinsurance reserves
 – increase in statutory loading on reinsurance premiums
 + reinsurance expenses
 – reinsurance allowances
 – increase in reinsured dividend liability

Reinsurance premiums include the premiums paid for ceded reinsurance, plus the related increase in gross deferred premiums. The amount is net of reinsurance premium refunds.

Investment income and capital gains allocated to ceded reinsurance reserves is reduced by the IMR impact on capital gains on reinsurance reserves. Reinsurance investment income includes interest on reinsurance cash flow.

The statutory reinsurance reserve increase includes reinsurance statutory loading increases.

Reinsurance dividend benefits are the increase in the reinsurance dividend liability.

10.1.11 Hedge cost

The hedge cost is the negative of the addition of the hedge cash flows and the increase in the hedge market values.

10.1.12 Separate account net transfers

Profit reports show the full amount of premiums paid by policyholders, and the full amount of claim payments. The effect of the separate account is collected into an adjustment for the 'net transfers to the separate account'

10.2 Book profit after tax

Book profit after taxes is:

- book profit before tax
- tax on book profit income
- tax on book profit realized capital gains

10.3 Distributable earnings

Distributable earnings are the company's statutory profit adjusted to remove the effect of free surplus, calculated on each profit distribution date. Distributable earnings are:

- undistributed earnings
- + book profit after tax
- + target capital effect

10.3.1 Undistributed earnings

Distributable earnings include the undistributed earnings as of the end of the previous month. Undistributed earnings are the distribution amounts not distributed in previous months. Investment income allocated to undistributed earnings is included in book profit.

10.3.2 Target capital effect

The effect of target capital is:

- net investment income on target capital
- target capital increase
- tax on target capital investment income
- tax on target capital realized capital gains
- IMR impact on target capital

Investment income allocated to target capital includes realized capital gains, plus the increase in unrealized capital gains that is recognized in statutory profit. Investment income is net of investment expenses. Investment income allocated to AVR and IMR is included in the target capital effect.

Taxes on target capital include tax on the investment income allocated to target capital, plus the tax on target capital. See the Taxes chapter above.

The IMR impact on target capital is the portion of the net increase in the interest maintenance reserve (IMR) allocated to target capital, based on the target capital, AVR, and IMR at the end of the previous month. This is the same basis as for allocating realized capital gains.

11 PRESENT VALUES

Present values are discounted to the projection date.

11.1 Profit discount frequency

11.1.1 Profit calculation interval

Present values of profit results are calculated using the profit distribution mode (annual, quarterly, or monthly), using the standard convention that profits occur at the end of each period. For example, when distributions are made annually, profits are discounted from the end of each calendar year. Profits for

calculating present values are zero in months without distributions, then the sum of the previous amounts in the distribution month.

11.1.2 Profit components

Present values of cash flows are calculated by discounting each cash flow from the date it occurs in the projection. Cash flows like premium payments are assumed to occur at the beginning of the month. Cash flows like payments to policyholders are assumed to occur at the end of the month.

Present values of non-cash flow profit components are calculated by discounting from the end of each month. For example, this applies for the present value of increases in reserves and target capital.

Present values of profits are discounted from the end of each profit calculation interval, as explained above.

11.1.3 Balancing adjustments

Since the present values of profits and cash flows use different timing conventions, reporting of the present value of profit components (premiums, investment income, benefits, etc.) includes the following adjustments:

- Cash flow timing – The balancing item for the PV of book profit components to add to the PV of book profits before taxes.
- Tax timing – The balancing item for the PV of book profits before taxes, less the PV of taxes on book profit, to equal the PV of book profits after taxes.
- Target capital effect – The balancing item for the PVs of book profits after taxes, target capital increases, investment income on target capital, and tax on that investment income, to equal the PV of distributable earnings.

11.2 Discount rates

11.2.1 Fixed discount rates

Specify fixed discount rates in the Output parameter set. This is the Fixed Discount Rate property. For example, specify '8, 10, 12' to calculate present values at 8%, 10%, and 12%. (Note that this property works differently than for specifying assumptions, where you enter '.08' to specify a rate of 8%.)

11.2.2 Column discount rates

Discount rates varying during the projection are 'column' discount rates. The standard features provide for up to three of these stochastic discount rates. (You can customize to extend to provide additional rates.)

Stochastic discount rates

Activate column discount rates by:

- In the Output parameter set, check the Use Column Discount Rates property.
- In the Run parameter set, specify the discount rate names in the `disc_rates_selected` project variable. Each name is separated by a comma. For example, '90Day,10Year'.
- On the Company input page, define the discount rate variables, looking up definitions using the `disc_rate_id` hidden code variable, which is set by formulas. Select the lookup term as the `disc_rate_id` code variable from either the company (for portfolio projections) or product (for seriatim projections) submodel. Change the lookup type to be Relative, so that the input manager uses the appropriate submodel for the current `disc_rate_id` value.

The following shows an example external source defining discount rates defining 90-day and 10-year risk free scenario rates:

	disc_rate_id	90Day	10Year
VarID			
disc_rate_scen_yr		0.25	10.00
disc_rate_mult		1.00000	1.00000
disc_rate_addn		0.00%	0.00%

Portfolio yield discount rate

Column discount rates include discounting using either the before or after tax asset yield. Specify in the Run parameter set project variable whether this discount rate is before or after tax. [disc_rate_port_yld_defn]

11.3 Internal rate of return

The internal rate of return (IRR), also called the return on investment (ROI), is the discount rate at which the present value of profits is zero. IRR is calculated in columns irr and irr_without_tgt_cap.

The standard IRR calculation would produce misleading results when the profit stream includes large book losses later in the projection. For example, consider the following profit stream: -40 in year 1, 33 in each of years 2 through 4, then -60 in year 5. The IRR calculation makes no sense in this case. The present value is zero at two discount rates: 2.87% and 24.03%. The maximum present value is reached at a discount rate of 11.52%.

To solve this problem, reports show the adjusted internal rate of return instead of the standard IRR. The adjusted internal rate of return is the same as the IRR under the classic pattern of initial losses followed by gains.

The adjusted IRR is determined by working backwards from the end of the projection. When the present value of future profit plus the current period's profit is negative, the sum is discounted using the after tax asset yield. This can be viewed as setting up a reserve to cover the present value of those future losses. The original profit stream is replaced by the adjusted stream, then the standard IRR is calculated.

For example, consider the above example with an after-tax asset yield of 6.0%. Since the final year's profits are negative, they are discounted to the fourth year. Since the discounted final profits plus the fourth year profits are still negative, this total is also discounted back to the third year. By the third year, the discounted future profits plus the current year profits are positive, and the adjusted IRR can be calculated. The adjusted IRR in this case is 7.47%, which is the IRR for the profit stream: -40 in year 1, 33 in year 2, 10.73 in year 3, and 0 in years 4 and 5.

C02 ECONOMIC SCENARIOS

This book explains the basic features for using and generating economic scenarios.

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1 INTRODUCTION

1.1 Terminology

- **Economic condition** – An element which is used to describe the economy, such as a yield curve, yield curve volatility, index, index volatility, or miscellaneous rate.
- **Economic environment** – A collection of economic conditions for a specified date (year, month, day), and country.
- **Historic rate** – A rate of an economic condition for a period prior to the projection date.
- **Scenario rate** – A rate of an economic condition for a period after the projection date.
- **Historic rates** – The collection of economic environments for periods through the projection date. These economic environment are collected together into an historic rates file.
- **Scenario** – The collection of economic environments for periods after the projection date. A scenario identifier collects these economic environments.
- **Scenarios** – The collection of the scenario elements where each scenario is identified by a scenario number. The scenarios are combined in a scenarios file, which includes the economic environment at the projection date.
- **Outer loop historic rates and scenarios** – Historic rates and scenarios required for calculating the projected results branching from the projection date. Outer loop historic rates include economic environments for periods up to the projection date. The scenarios file includes multiple scenarios of economic environments for periods after the projection date. Other terms used for the outer loop projection scenarios are 'experience' or 'planning' scenarios.
- **Inner loop scenarios** – Scenarios required for valuation projections at a valuation date along the outer loop paths, including the projection date. The inner loop scenarios are commonly referred to as 'nested' scenarios, which are typically generated 'on-the-fly' using information from the outer loop historic rates and scenario.

1.2 Usage

1.2.1 Valuation

Outer loop historic rates and scenarios are used for point-in-time statutory, GAAP, and hedging valuations of in force business at the projection date.

Inner loop scenarios, and the inner loop historic rates (a hidden variable which is updated at the end of each time step), are used for statutory, GAAP, and hedging valuation projections of in force business such as are required to provide for the nested projections with principle-based approaches reserves and capital, GAAP SFAS 133/157 and SOP 03-1, and greek hedge strategies.

1.2.2 Pricing

Outer loop historic rates and scenarios are used for scenario based pricing of new business, both for a single cell and a portfolio, and for market consistent pricing of charges for guarantees in variable products.

When future statutory, GAAP, or hedging valuations require stochastic projections, inner loop scenarios are also used. This creates a nested stochastic-on-stochastic projection structure.

1.2.3 Projections

Outer loop historic rates and scenarios are used for business planning projections (liability-only, and asset/liability) of cash flows, income statements, and balance sheets.

As is the case with pricing, when future statutory, GAAP, or hedging valuations require stochastic projection, then inner loop scenarios are also used. This creates a nested stochastic-on-stochastic projection structure.

2 ECONOMIC SCENARIOS

2.1 Scenario input

2.1.1 Economy input page

Specify assumptions for scenarios under the Scenarios: Rates category of the Economy input page

2.1.2 Historic rates

For standard projections and point-in-time valuations, specify the historic rates in the `historic_rates_id` variable.

Historic rates are required for all projections using scenario rates. This file must contain all the yield curves, volatilities, indexes, and miscellaneous rates contained in the scenario file. Historic rates are specified through economic environments referenced by calendar year and month. Elements may be included in the historic rates files which are not included in the projected files, but the converse is not the case. Yield curves, volatilities, indexes, and miscellaneous rates included in the scenario files but not in the historic rates are ignored.

The historic rates variables mentioned above must reference a file with an extension of “.csv”. Best practice is to collect the files together into the folder represented by the `<*scen*>` wildcard.

2.1.3 Scenarios

For standard projections and point-in-time valuations, specify the scenarios in the `scen_id` variable. For nested projections, also specify the `t = 0` inner loop scenarios in the `inner_loop_valn_scen_id` variable respectively, and how the inner loop file is reused at each subsequent inner loop valuation date.

Scenarios are required for all stochastic projections.

The scenario id variables mentioned above must reference a file with an extension of “.csv”. Both of these files must be in the comma separated value (CSV) format acceptable to the U.S. Library. Best practice is to collect the files together into the folder represented by the `<*scen*>` wildcard. There is no need to include the `<*proj_set_loop_num*>` and `<*proj_set_sub_loop_num*>` wildcard as part of the file name for the inner loop files, since this referencing is handled automatically.

2.1.4 Other parameters

Specify other parameters as follows:

Missing values treatment

When a scenario file does not contain economic environments for each month, specify using the `'missing_val_defn'` variable if the missing values are (1) linearly interpolated, or (2) held constant at the last provided rates.

The missing values treatment applies to yield curve rates, yield curve volatilities, and index volatilities only. For index rates, inflation rates, and default rates, the next specified value is always used as the rate for the missing economic environments.

Open file retry attempts

Projections which are distributed to a compute cluster typically share data, including the economic scenarios files. In rare occurrences, file contention issues may occur when two workers attempt to open the file at the same time. These issues are typically resolved by waiting a short period and then attempting again to open the file.

Specify the number of attempts to open the economic scenario files before an error is issued, using the 'open_file_retry_attempts' variable. Increasing this value typically eliminates the file contention issues which may be experienced on a compute cluster when multiple workers are using the same economic scenario files.

Yield curve memory treatment

When yield curves are accessed, a full yield curve is created in memory with values for every maturity term 0 to 600. These can be retained in memory in order to speed retrieval of values on repeated calls to the same yield curve or released from memory after each reference.

When the projection does not have enough memory to run, you can specify to recalculate the yield curve (by releasing it from memory), using the 'yld_curve_mem_defn' variable.

2.2 Scenario types

2.2.1 Deterministic scenarios

Deterministic scenarios are simply economic conditions with user specified values. These scenarios are typically created manually.

A common set of deterministic scenarios used for U.S. cash flow testing is specified by New York Regulation 126. The following is the Willis Towers Watson interpretation for our provision of scenario files to users for cash flow testing:

- Level – The initial yield curve stays in effect.
- Increasing – The yield curve increases linearly to 5% higher rates 10 years after the projection date, then stays level.
- Up/down – The yield curve increases linearly to 5% higher rates in 5 years, then decreases linearly to the current yield curve 10 years after the projection date, and stays level.
- Pop-up – The yield curve immediately increases by 3%, then stays level.
- Decreasing – The yield curve decreases linearly to 5% lower rates 10 years after the projection date, then stays level.
- Down/up – The yield curve decreases linearly by 5% over 5 years, then increases to the current yield curve the next 5 years, then stays level.
- Pop-down – The yield curve immediately decreases by 3%, then stays level.
- Inverted – The 10-year rate increases linearly to 4% higher rates over 5 years, then decreases linearly to the starting rate over the next 5 years, then stays level. The 90-day rate increases linearly to a rate that is 2% higher than the 10-year rate at the end of 5 years, then linearly to the starting rate over the next 5 years, then stays level. This scenario is not explicitly part of the New York Regulation 126 scenarios, but may be used as a sensitivity test.

The decreasing interest scenarios are subject to a minimum interest rate of half of the starting nominal semi-annual bond equivalent yields.

2.2.2 *Stochastic scenarios*

Stochastic scenarios are economic conditions created through time evolving from an initial condition, using an appropriate n-state evolution process, calibrated model parameters, and a stochastic process. There can be many stochastic scenarios, such as 1,000 or more.

One common set of stochastic scenarios are those used for principle-based approaches (PBA) reserves and capital calculations as prescribed by the American Academy of Actuaries (AAA). Regulations permit use of either the academy generated scenarios (selected from a set of 10,000) or scenarios generated by another model which meets specified academy calibration criteria.

3 ECONOMIC ENVIRONMENTS

3.1 Dates and timing

3.1.1 *Scenario dates*

Economic environments apply as of fixed calendar dates for historic rates, and relative month from the projection (or valuation) date for scenarios.

For example, an historic economic environment may apply for the month of 2009/12, while an economic environment in a scenario applies for month 1, 2, 3, etc., following the projection (or valuation) date.

3.1.2 *Monthly or daily historical rates*

The economic environments of the historical rates apply to either all dates in a specific calendar month or for a specific calendar day, depending on the structure of the date information for the economic environments in the file.

Daily environments apply when the file contains a 'Day' column in addition to a 'Year' and 'Month' column for each economic environment. The missing values treatment applies for dates which are not included in the file.

3.1.3 *Monthly scenario rates*

The economic environments of the projected scenario file apply for all dates during a particular month. This is because economic environments for scenario files only vary by month since the projection (or valuation) date.

3.2 Currency exchange rates

All projections are for a single country and single currency. All economic conditions must belong to the "USA" country code.

3.3 Interest rates

3.3.1 *Yield curves*

Interest rate economic conditions for an economic environment are organized into yield curves elements, and then further organized by months to maturity.

An economic environment must include information for the government risk free yield curve. An unlimited number of credit yield curves can also be included. Yield curves are defined using a quality grade identifier, such as 'Govt' (name required) for risk free rates, 'AAA', 'BB', etc. Each yield curve must be specified using at least two terms (months to maturity).

Yield curve rates may be specified as yield rates or spot rates and as either effective annual or as nominal semi-annual rates. Conversion processes are used to convert from the specified rate basis and return type to the form requested.

3.3.2 Interpolation

Linear or cubic spline interpolation may be used to determine the rates which are not specifically provided on the yield curve. For example, the yield curve structure may only contain two specified points on the yield curve; yet other points on the curve may be required. The interpolation occurs before the yield curve rates are converted to another form.

3.3.3 Yield rates

Bond (yield) rates are the par rates to each maturity term for a specified payment mode. Semiannual rates are used in the U.S. These rates are called 'bond equivalent yields' (BEY) elsewhere in the documents.

When spot rates are specified in the economic conditions, yield rates are derived.

A 'boot strapping' algorithm, based on effective annual rates, is used to calculate yield rates consistent with the specified spot rates for each successive month, until a full spot rate curve of 600 months to maturity is developed. For months before the first payment term, the 6 month term, the yield rate is assumed equal to the spot rate.

The specific algorithm used to calculate yield rates from spot rates is included in the `calc_yld_from_spot_rate` external function of the rates submodel.

3.3.4 Spot rates

Spot rates are the yields on the zero coupon bonds to each maturity term.

When yield rates are included in the economic environments, spot rates are derived.

A 'boot strapping' algorithm, based on effective annual rates, is used to calculate spot rates consistent with the specified bond equivalent yield rates for each successive month, until a full spot rate curve of 360 months to maturity is developed. For months before the first payment term, the 6 month term, the spot rates is assumed equal to the bond equivalent yield rate.

The specific algorithm used to calculate spot rates from BEY rates is included in the `calc_spot_from_yld_rate` external function of the rates submodel.

3.3.5 Forward rates

When BEY rates are included in the economic environments, forward rates are derived. A forward curve is the series of forward rates for the same maturity, but with different forward periods. An ' $n \times (n + m)$ ' forward rate is an interest rate payable on a forward loan that commences n months from the current date. It matures m months after that, and it accumulates interest to maturity.

Forward rates are frequently used to assure that there is no arbitrage in the pricing of interest rate derivatives. Forward rates can be used to hedge by locking the current value of a security into a forward contract.

Forward rates for each term on the curve are calculated from the spot rates, using the spot rate for the requested term and a forward term.

The specific algorithm used to calculate forward rates from spot rate is included in the `GET_FORWARD_RATE` section of the `get_int_rate` external function of the rates submodel.

3.3.6 Credit spreads

Yield curves for quality grades which are not specifically provided in the generated scenarios, but are required for asset projections, are created automatically using credit spread definitions specified for the projection.

Specify the additional yield curve definitions using the Scenarios: Assumptions category on the Economy input page.

Specified credit spreads are a spread to either risk free spot rates or risk free yield rates. The credit risk curve for the quality grade is the risk free rate plus the spread for the months to maturity.

If the credit spread is a spot spread, a credit risk spot curve is determined as the risk free spot curve plus the spot spread and the credit risk yield curve is derived from that curve. If the credit spread is a yield spread, a credit yield curve is determined as the risk free yield curve plus the yield spread from which a credit risk spot curve can be derived.

Deterministic credit spread

Deterministic credit spreads are specified by term to maturity, and may further vary by calendar date.

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. This must be an external source varying by hidden variable lookup keys `quality_id` and `rate_term`; and can also vary by the date column. Credit spreads typically vary by date when spreads that are consistent with current market values are different from those assumed to apply for normal market conditions going forward.

The missing rules of the external source lookup definitions define how to determine credit spreads for rate terms to maturity and dates which are not specified. Grading of spreads must be accomplished by creating separate credit spread definitions for each calendar date.

The following example illustrates a deterministic credit spread grid for AA and B quality grades:

quality_id	AA	AA	AA	AA	B	B	B	B
rate_term	0.25	1	5	10	0.25	1	5	10
date								
1/1/1900	0.10%	0.20%	0.60%	0.80%	0.80%	1.00%	1.20%	1.50%

Stochastic credit spread

Stochastic credit yield curves for a quality grade are a combination of two credit yield curves which are specifically provided. For example, when the economic environment includes credit yield curves for AA and B grade bonds, the spot curve for BB-grade bonds can be specified as x% of the AA grade yield curve, plus (1 – x%) of the B grade yield curve. The credit risk curve for lower grade bonds can be specified as a multiple of the B grade yield curve.

Specify the stochastic credit spread as a multiple of the credit spreads for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup key `yld_curve_id` and hidden variable lookup key `quality_id`.

In the following example, the stochastic credit spread for A-rated securities is 80% of the spread for AA-rated securities plus 20% of the spread for B-rated securities.

quality_id	A
yld_curve_id	
Govt	0.0000%
AA	80.0000%
B	20.0000%

Both additional credit yield curve derivation methods may be employed to define a credit risk curve. The final credit risk curve is then equal to the risk free curve plus the sum of the deterministic credit risk spread and the stochastic credit risk spread.

3.3.7 Shifted yield curves

The discussion above focuses on 'base' yield curves. In addition to those yield curves, a second set of yield curves support effective duration calculations for fixed income assets.

These 'shifted' yield curves apply a 1 basis point (0.01%) parallel shift to the spot rates of each yield curve, creating a shifted spot curve from which a shifted yield curve is derived. For credit risk curves, the spot shift is applied to the risk free spot curve, to create a shifted risk free spot curve, from which a shifted yield curve is derived.

If the credit risk spread is a spot spread, a shifted credit risk spot curve is calculated from the shifted risk free spot curve and a shifted credit risk yield curve is derived from that curve. However, if the credit risk spread is a yield spread, a shifted risk free yield curve is determined first, to which the yield spread is added to arrive at a shifted credit risk yield curve. Then, finally the shifted credit risk spot curve is derived from the shifted credit risk yield curve.

The shifted yield curve approach discussed here is different from a shocked scenario process which is represented by a separate scenario file that has been created through a generation process based on shocking of certain initial conditions, such as a specific key rate or index volatility. Therefore, it is possible to have shifted yield curves based on a yield curve of a shocked scenario file.

3.4 Index rates

Index rates economic conditions are organized into index rate elements. An economic environment may include an unlimited number of indexes. Each index is described by an index ID, which is specified on each asset or liability linking to the index.

While there is yet no independent standard for the identifier for an index, best practice is to utilize the identifiers for the indexes proposed by the American Academy of Actuaries.

3.4.1 Usage

Index rates determine the performance for equity, real estate assets, as well as for equity 'index assets'.

Index rates are referenced as the growth rates for separate account fund values.

For indexed annuity and life products, the indexed fund value is linked to an index.

3.4.2 Index rates

Income return, dividend yield, index growth, and total returns are available for the index. The index rates may be specified as effective annual or as actual monthly rates.

The scenario index must contain at least one 'income' rate and one 'growth' rate. For example, the scenario index may contain the dividend yield and the growth rate, or it could contain the dividend yield and the total return rate. If the scenario index only contains the dividend yield and income return, an error is raised.

When the scenario index does not contain all four rates, the missing rates are determined using the following relationship formulas:

$$\text{total return} = \text{income return} + \text{growth rate}$$

$$\text{income return} = (1 + \text{growth rate}) \times \text{dividend yield}$$

$$\text{growth rate} = (1 + \text{total return}) / (1 + \text{dividend yield}) - 1$$

When the rates specified as actual monthly rates, the specified rates are converted to effective annual before deriving the other rates.

3.4.3 Index values

Index values are also available for the index. The index value may be specified as part of the scenario index data for the historic rates. **Index values cannot be specified as part of the scenario index in the scenario file.**

When the index value is not specified, the index is assumed to have a value of 1.0 at the projection date. Then, index values for historic environments before the projection date are determined from that initial index value using the index growth rates.

Index values for the projected periods are calculated from the index value at the projection date (either the value specified or a value of 1.0) and the growth rates.

3.5 Volatility rates

Interest rate and index volatility economic conditions are organized into interest and index volatility elements respectively, and then further organized into moneyiness levels (for index volatilities) and terms (tenors) to maturity. An economic environment may include interest rate volatilities for calculating interest rate derivative market values, plus the option price for bond call and put options. Similarly, an economic environment may include volatility of asset prices for calculating equity index option assets.

For each yield curve, volatilities are specified for **at least two terms**. For each index, volatilities may be specified for any number of moneyiness levels and for **at least two tenors**. Volatility rates for moneyiness levels not included are linearly interpolated between the moneyiness levels included. Each volatility rate may be specified as an effective annual or as an actual monthly rate.

Additional volatilities

Specify the additional volatility definitions using the Scenarios: Assumptions category on the Economy input page. Deterministic volatilities for a quality grade or index are effective annual rates by term (tenor) and moneyiness.

	index_id	US	US	US	US	US	US
	moneyiness	1.00000	1.00000	1.00000	1.05000	1.05000	1.05000
	rate_term	1.00000	5.00000	10.00000	1.00000	5.00000	10.00000
date							
1/1/1900		25.0000%	18.0000%	14.0000%	26.0000%	20.0000%	16.0000%

This is an external source varying by hidden variable lookup keys index_id, moneyiness, and rate_term; and can also vary by the date column. The missing rules of the external source lookup definitions define how to determine volatilities for moneyiness levels, rate terms to maturity, and dates which are not specified.

Stochastic volatilities for a quality grade (index) are a combination of two yield curves (indexes) which are specifically provided. For example, when the economic environment includes volatilities for AA and B grade bonds, the volatility for A-grade bonds can be specified as x% of the AA grade volatility, plus (1 – x%) of the B grade volatility. The volatility for lower grade bonds can be specified as a multiple of the B grade volatility.

Specify the stochastic volatility as a multiple of the volatilities for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup key yld_curve_id and hidden variable lookup key quality_id.

In the following example, the stochastic volatility for A-rated securities is 80% of the volatility for AA-rated securities plus 20% of the volatility for B-rated securities.

quality_id	A
yld_curve_id	
Govt	0.0000%
AA	80.0000%
B	20.0000%

3.6 Inflation rate

Inflation economic conditions are organized into miscellaneous rate elements. An economic environment may include one inflation rate. This rate is automatically applied to expense rates. The inflation index may be specified as effective annual or as an actual monthly rate.

The application of inflation is discussed in the Inflation topic in the Liability calculations section of the projection structure book.

Deterministic inflation rate

Pricing actuaries commonly desire to utilize a single deterministic path of inflation rates during the pricing process. This can be accommodated through use of a deterministic scenario file that contains the deterministic inflation rate. An alternative is to select the deterministic inflation definition and specify a table of deterministic inflation rates.

Specify the inflation definitions using the Scenarios: Assumptions category on the Economy input page. Vary the inflation by date using an external source varying by the date column.

3.7 Default rate

Default rates economic conditions, based on quality grades, are organized into miscellaneous rate elements. Defaults are not specified by asset category or by individual asset.

An economic environment may include default rates for an unlimited number of quality grades. Each default index is referenced using a quality grade ID. Default rates may be specified as effective annual or as an actual monthly rate.

Credit yield curve and default rate consistency

Defaults rates for quality grades can only be explicitly specified in the economic environment when a credit yield curve is also explicitly specified. Thus stochastic default rates are consistent with the stochastic credit spreads.

Additional credit default rates derivation

Default rates for quality grades which are not specifically provided in the generated scenarios, but are required for asset projections, are created automatically using credit default definitions specified for the projection.

Specify the additional yield curve definitions under the Scenarios: Assumptions category on the Economy input page.

Deterministic default rates vary by quality id, and can also vary by date. Specify deterministic default rates on the Economy input page under the Scenarios: Assumptions category. These definitions must be an external source varying by the hidden variable lookup key quality_id; they can also vary by the date column.

quality_id	AA	B
date		
1/1/1900	0.1000%	0.1500%

The missing rules of the external source lookup definitions define how to determine default rates for dates which are not specified.

Stochastic credit defaults for a quality grade are a combination of two default rates which are specifically provided. For example, when the economic environment includes default rates for AA and B grade bonds, the default rate for A grade bonds can be specified as x% of the AA grade default rate, plus (1 – x%) of the B grade default rate. The default rate for lower grade bonds can be specified as a multiple of the B grade rate.

Specify the stochastic default rate as a multiple of the default rates for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup key `yld_curve_id` and hidden variable lookup key `quality_id`.

In the following example, the stochastic default rate for A-rated securities is 80% of the scenario default rate for AA-rated securities plus 20% of the scenario default rate for B-rated securities.

quality_id		A
yld_curve_id		
Govt		0.0000%
AA		80.0000%
B		20.0000%

Both credit default definition methods may be employed to define a default rate for a quality grade. The final default rate is then equal to sum of the default rates resulting from combining the two methods.

3.8 Available economic conditions

The economic conditions for interest rates, interest volatilities, index rates, and index volatilities which are defined in the historical rates files are the conditions which are available in the projection. Economic conditions which only exist in the scenarios file will not be available during the projection.

Best practice is to ensure that the economic conditions structure of the historical rates file and scenarios files are consistent.

4 ECONOMIC SCENARIOS FILES

4.1 Scenario file considerations

4.1.1 Naming convention

Economic scenario files may be named using any valid combination of characters and numbers per the Windows file naming conventions. Best practice is to create names which distinguish between the following characteristics of the scenario files:

- Scenarios start date (yyyymm, interpreted as EOM) – To distinguish between historic and scenario rates generated at differing scenarios start dates.
- Real world scenarios versus risk neutral scenarios
- Frequency of economic conditions, such as Monthly, Quarterly, Annual
- Historic rates versus scenario rates
- Baseline (unshocked) – Such as for risk neutral rates before applying shocks.
- Shock method (parallel or specific), shock basis (interest rate, index volatility), a shock direction, and a shock amount. For example, risk neutral rates with a yield curve rate parallel shift of 100 basis points.

Several examples of economic scenario file names using the above conventions are:

201512MonthlyHistoricRates.csv

201512RWQuarterlyProjectedScenarios.csv

201512RNAnnualProjectedScenarios_Baseline.csv

201512RNAnnualProjectedScenarios_ParallelIntUp100.csv

4.1.2 Economic environment frequency

Economic scenario files include flexibility around the frequency for the economic environments, allowing the user to specify any number of economic environments to define the historical rates and projected scenarios.

As discussed, the missing values treatment method is used to determine the yield and volatility rates for economic environments requested which are not included in the file. For the indexes, inflation, and default economic conditions, the rates specified are interpreted as the accumulated amounts over the sparse period.

4.2 Pre-packaged economic scenarios files

Willis Towers Watson provides historic rates, and scenario files commonly used for U.S. projections. These files are provided in our supported formats.

4.2.1 Historic rates

Historic rates include economic conditions from 1982 forward the U.S. risk free bond (yield) curve (10 terms, treasury constant maturities), the S&P 500 equity index, and the consumer price inflation rate.

4.2.2 Deterministic scenarios

Deterministic scenarios include those used for U.S. NY Regulation 126, NAIC Model Actuarial Opinion and Memorandum, and PBA stochastic exclusion test scenario projections.

In addition to a base set of NY Regulation 126 scenarios, a set is provided that reflects a normalized yield curve after three years. The projected rates include monthly economic conditions for fifty years from the Scenarios start date for the U.S. risk free BEY curve (10 terms, treasury constant maturities).

4.2.3 Stochastic scenarios

Stochastic scenario files include, for RBC C3 Phase I projections, sets of 12, 50, and 200 scenarios. The projected rates include monthly economic conditions for thirty years from the Scenarios start date for the U.S. risk free bond (yield) curve (10 terms, treasury constant maturities).

4.3 Historic rates and scenario file format

Historic rates and scenario files must adhere to the comma separated value (CSV) format required by the U.S. Library. Examples of the format for the scenario files are included in the 'Template\Scenarios' folder included with the U.S. Library projects.

Files or scenario data in other formats must be translated into the required format through the use of scenario translator tools.

4.3.1 Basic structure

All historic rates and scenario files are structured with a header section and a detail information section.

The header section is the 'Meta data' of the file and contains rows to describe general treatment of the information in the file and the economic conditions included in the file. Each row contains the appropriate number of items to describe the economic conditions including the position in the detail section contain where the relevant information for each economic condition is found.

The detail information section of the file contains rows for each economic environment included in the file, organized by year, month, and day (optional) for the historic rates file and scenario and month for scenario rates file. Each row contains the number of items required to fully describe the economic environment, and must be aligned properly with the information provided in the header.

The specific details of the header and detail sections are discussed further below.

4.3.2 File header structure

Historic rates and scenario files must contain a header section which describes the economic conditions contained in the file.

The header section of the file begins at the top row of the file. Each row for the header section is declared by a 'Header' identifier as the first entry in the row. The header section ends when a row with an entry of Header,End is detected.

Descriptive information

A description for the scenario file may be included in the header of the file. The structure of the description information header row is **Header,Description,[DescripInfo]**.

where:

[DescripInfo] is any series of text and include spaces and other characters; but must not contain a comma.

Interpolation

The interpolation treatment for the file must be included in the header of the file. The structure of the interpolation information is as follows:

Header,Interpolation,[InterpType].

where:

[InterpType] can be CubicInterp or LinearInterp.

Currency

The currency basis for the file must be included in the header of the file. The structure of the currency basis is as follows:

Header,Currency,[CurrencyBasis].

where:

[Currency] basis is USD, since, currently all economic scenario files used in the U.S. Library are U.S. dollar based.

Yield curves

Any number of yield curves may be included in the file by specifying yield curve rows in the header. However, a Govt yield curve **must** be specified.

The structure of the yield curve rows are as follows:

Header,YieldCurve,[YieldCurveName],Terms:[Terms],[YieldBasis],[RateBasis],[StartCol],[EndCol]

where:

[YieldCurveName] is the name of the yield curve, such as Govt for the treasury yield curve,

[Terms] is a dash separated list of maturity term months, such as 3-6-12-24-36-60-84-120-240-360,

[YieldBasis] is either Yield Rates or Spot Rates,

[RateBasis] is either NomSa or Annual,

[StartCol] is the position in the detail section where the yield curve information begins, and

[EndCol] is the position in the detail section where the yield curve information ends.

Important! Yield curves must contain at least two terms. Otherwise, an error will be encountered when attempting to access the yield curve elements in the projection.

Yield curve volatilities

Any number of yield curve volatilities may be included in the file by specifying yield curve volatility rows in the header.

The structure of the yield curve volatility rows are as follows:

Header,YieldCurveVol,[YieldCurveVolName],Terms:[Terms],[RateBasis],[StartCol],[EndCol]

where:

[YieldCurveVolName] is the name of the yield curve, such as Govt for the treasury yield curve,

[Terms] is a dash separated list of maturity term months, such as 3-6-12-24-36-60-84-120-240-360,

[RateBasis] is either Annual or Actual Monthly,

[StartCol] is the position in the detail section where the yield curve volatility information begins, and

[EndCol] is the position in the detail section where the yield curve volatility information ends.

Important! Yield curve volatilities must contain at least two terms. Otherwise, an error will be encountered when attempting to access the yield curve volatility elements in the projection.

Index rates

Any number of index rates may be included in the file by specifying index rate rows in the header.

The structure of the index rates rows are as follows:

Header,Index,[IndexName],[RateTypes],[RateBasis],[StartCol],[EndCol]

where:

[IndexName] is the name of the index, such as US,

[RateTypes] is a dash separated list of rate types included, where the rate type options are Dividend Yield, Income Return, Index Growth, Total Return, and for indexes included in the historic rate file, Index Value. The minimum number of items in the rate type is 2, and there must be at least one of Dividend Yield and Income Return included and at least one of Index Growth and Total Return included. For example, a rate type of Income Return-Index Growth is acceptable, but a rate type of Dividend Yield-Income Return is not acceptable.

[RateBasis] is either Annual or Actual Monthly,

[StartCol] is the position in the detail section where the index rate information begins, and

[EndCol] is the position in the detail section where the index rate information ends.

The equity types element contains a list of equity type (referred to as indexes in this document) names contained in the scenario collection. Each equity type is identified by an equity type ID.

Index volatilities

Any number of index volatilities may be included in the file by specifying index volatility rows in the header.

The structure of the index volatility rows are as follows:

Header,IndexVol,[IndexName],[MoneynessLevel],[Tenors],[RateBasis],[StartCol],[EndCol]

where:

[IndexName] is the name of the index, such as US,

[MoneynessLevel] is the moneyness level for the volatility structure, such as 1.0 for an at-the-moneyness volatility term structure,

[Tenors] is a dash separated list of maturity tenor months, such as 12-60-120,

[RateBasis] is either Annual or Actual Monthly,

[StartCol] is the position in the detail section where the index volatility information begins, and

[EndCol] is the position in the detail section where the index volatility information ends.

Important! Index volatilities must contain at least two tenors. Otherwise, an error will be encountered when attempting to access the index volatility elements in the projection.

Misc rates

Any number of miscellaneous rates may be included in the file by specifying miscellaneous rate rows in the header.

The structure of the miscellaneous rate rows are as follows:

Header,MiscRate,[MiscRateName],[RateBasis],[StartCol],[EndCol]

where:

[IndexName] is the name of the miscellaneous rate, such as Infl,

[RateBasis] is either Annual or Actual Monthly,

[StartCol] is the position in the detail section where the miscellaneous rate information begins, and

[EndCol] is the position in the detail section where the miscellaneous rate information ends.

Since miscellaneous rates are used for inflation and defaults, and all that identifies a miscellaneous rate is a single ID, a naming convention is employed to ensure that the ID clearly identifies the rate. The inflation rate is identified as 'Infl' and a miscellaneous rate representing a default rate by quality grade must contain an identifier such as 'AA_Dflt'.

Of course, user defined miscellaneous rates may be included which do not represent either the inflation rate or a default rate. When user defined rates are included, it is up to the user to properly interpret the miscellaneous rates retrieved.

Sample file header

A sample file header section of an historic rates file (as displayed in Notepad) is as follows:

```
Header,Description,Template Historical Rates File
Header,Interpolation,CubicInterp
Header,Currency,USD
Header,YieldCurve,Govt,Tenors:3-6-12-24-36-60-84-120-240-360,Yield Rates,NomSA,3,12
Header,YieldCurveVol,Govt,Tenors:12-60-120,Annual,13,15
Header,IndexVol,US,1,Tenors:12-60-120,Annual,16,18
Header,IndexVol,US,0.95,Tenors:12-60-120,Annual,19,21
Header,IndexVol,US,1.05,Tenors:12-60-120,Annual,22,24
Header,Index,US,Income Return-Index Growth-Index Value,Annual,25,27
Header,MiscRate,Infl,Annual,28,28
Header,End
```

4.3.3 Detail information structure

Historic rates and scenario files must contain a detail information section which captures the economic environments contained in the file.

The detail information section of the file begins in the row immediately following the Header,End row. The detail information section ends at the end of the file.

Detail descriptor

The first row in the detail information section is a detail descriptor row. This row provides titles for each item in the detail information section.

The number of items in this row must equal the last position specified for the items included in the file header structure. For example, when the last position is 27 in the file header, then 27 elements must be included in the detail descriptor row.

Historic economic environments

After the detail descriptor row are all the economic environments for the file. For the historic file, the economic environment are organized by calendar year, month, as if desired day.

Similarly to the detail descriptor row, the number of items in each economic environment row must equal the last position specified for the items included in the file header structure. For example, when the last position is 27 in the file header, then 27 elements must be included in each economic environment row.

The first element of each economic environment row is the calendar year (YYYY).

The second element of the economic environment row is the calendar month (MM).

The third element of the economic environment row might be the calendar day (DD), as this is optional.

The third (or fourth, depending on whether the historic file contains daily environments) and subsequent elements of the economic environment rows are the rates for the economic conditions specified in the file header. The positions in the economic environment row for each item must align with the start column and end column values. For example, if the start column and end column for the US index are 25 and 26 respectively, and the rates included are Income Return and Index Growth, then the US index income return must be in element 25 and the index growth must be in element 26 of all economic environment rows for all scenarios.

Scenario economic environments

After the detail descriptor row are all the economic environments for the file. For the scenario file, the economic environment are organized by scenario and period.

Similarly to the detail descriptor row, the number of items in each economic environment row must equal the last position specified for the items included in the file header structure. For example, when the last position is 27 in the file header, then 27 elements must be included in each economic environment row.

The first element of each economic environment row is the scenario number. This number is repeated for each row for all economic environments for the periods covering the scenario.

The second element of the economic environment row is the calendar month (MM). For the scenario file For the scenario file, the second element of the economic environment row is the period, starting at 0 and increasing through the final period of the scenario.

The third and subsequent elements of the economic environment rows are the rates for the economic conditions specified in the file header. The positions in the economic environment row for each item must align with the start column and end column values. For example, if the start column and end column for the US index are 25 and 26 respectively, and the rates included are Income Return and Index Growth, then the US index income return must be in element 25 and the index growth must be in element 26 of all economic environment rows for all scenarios.

of the TAS scenario file specifies the information included, which can include multiple yield curves with flexible yield terms, multiple indexes, and multiple miscellaneous rates.

Yield curves translation

For yield curves, a 'Govt' yield curve is always translated, along with a set of volatilities. When additional yield curves are included in the monthly format files, they are also translated. If the additional yield curves are spreads, the spreads are assumed to be spreads to the 'Govt' yield rates, and the translation creates a term structure of bond yield rates for the additional yield curve.

When translating a TAS quarterly file, the yield term rates are stored as bond equivalent rates and the volatilities are stored as effective annual rates. When translating a TAS monthly file, the yield term rates are stored using the basis in the TAS file and volatilities are stored as effective annual rates.

For the TAS quarterly format files, the following effective annual volatilities are assumed:

Yield curve term	Effective annual volatility
d03	0.2637
d06	0.2511
d12	0.2323
d24	0.2086
d36	0.1879
d60	0.1617
d84	0.1450
d120	0.1297
d240	0.1228
d360	0.1196

For the TAS monthly format files, the volatilities are translated from the TAS scenario file.

Index translation

For indexes, the translation depends on the format of the TAS scenario file. For the quarterly format scenario file, two indexes are translated; a 'US' index and an 'RE' index. For the monthly scenario file, all the indexes included in the scenario file are translated.

When translating a TAS quarterly file, the index rates are stored as effective annual dividend yields and growth rates. When translating a TAS monthly file, the same index rates are stored in the CSV as are present in the TAS file, using the same rate basis.

For the TAS quarterly format file, the following volatilities are assumed for the 'US_100' volatility term structure.

Tenor	Effective annual volatility
d03	0.28157
d06	0.3175
d12	0.4575
d24	0.2836
d36	0.2378
d60	0.1624

For the TAS monthly format files, the volatilities are translated from the TAS scenario file.

Miscellaneous rates translation

For miscellaneous rates, the translation also depends on the format of the TAS scenario file. For the quarterly format scenario file, only an inflation index and a single generic default rate are translated. For the monthly scenario file, the miscellaneous rates translated can include any number of inflation and default rates.

When translating a TAS quarterly file, the miscellaneous rates are stored as effective annual rates. For the monthly scenario file, the miscellaneous rates are stored using the same rate basis as the TAS file.

For the TAS quarterly format file, the single default rate item is stored using a 'Generic_Dflt' ID.

5.2 AAA scenarios CSV translator tool

The AAA scenarios CSV translator tool translates AAA prepackaged scenarios, or AAA ESG generated scenarios, specified in the comma separated value (CSV) format into the U.S. Library CSV format. This tool eases the conversion from the scenario format used in AAA CSV files into the required format.

The expected source of CSV files from the AAA should be those produced by the 'Interest_Rate_Generator_AIRG_Version 7_1_201406.xls' Excel workbook or the prepackaged scenarios from the AAA website, which include CSV files for each index and separate term to maturity for the government yield curve.

Once opened, instructions for working the tool are provided on the 'Welcome' worksheet.

5.2.1 Control worksheet

Specify the translation process using the fields on the 'Control' worksheet.

Specify the path and file name for the CSV file to create and the number of years for the economic environments to be included in the CSV file.

The 'Translate Academy CSV Files' button activates the translation process and the progress is displayed in the status bar at the bottom of the Excel workbook. A confirmation button is displayed when the translation is complete.

5.2.2 YldCurveData worksheet

Specify the translation criteria for the AAA yield curves through the fields on the YldCurveData worksheet. This worksheet must be completed as desired before the translation process is activated.

Specify to include one or many yield terms from one or many yield curves in the translation. Expand the rows in the 'green' area as desired ensuring that no blank rows are included between items. Rows may be included in this worksheet even if the yield curve terms are not desired in any particular translation, since the include column selection can be set to Yes or No.

The yield curve name column represents the yield curve name which will be used in the CSV file. The yield term ID must represent a month and follow the naming convention such as '3', '6', '12', '120', etc. A static effective annual volatility for an 'at the money' option may be specified in volatility column. This rate will be assumed for all scenarios and all periods. Set the volatility amount to zero if you do not desire a volatility to be included for a specific term. When all volatilities are specified as zero, no volatilities for the yield curve will be included in the CSV file. Finally, double click on the cells in the CSV file column to browse to select the AAA CSV file which contains the rates for the yield curve terms.

5.2.3 IndexData worksheet

Specify the translation criteria for the AAA indexes and combination indexes through the fields on the IndexData worksheet. This worksheet must be completed as desired before the translation process is activated.

Specify to include one or more indexes in the translation. Expand the rows in the 'green' areas as desired ensuring that no blank rows are included between items. Rows may be included in this worksheet even if the indexes are not desired in any particular translation, since the include column selection can be set to Yes or No.

For the AAA indexes, the index name column represents the index name which will be used in the USLMS ESML file. Specify a static effective annual income return in the income return column may and a static

effective annual volatilities for an 'at the money' option in the two volatility columns. These rates will be assumed for all scenarios and all periods. Set the volatilities to zero if you do not desire a volatility to be included for a specific index. Double click on the cells in the CSV file column to browse to select the AAA CSV file which contains the total return accumulation factors for the index. Finally, select whether the CSV file contains cumulative accumulation factors or monthly accumulation factors. The AAA prepackaged scenario files use monthly returns; while the Interest_Rate_Generator_AIRG_Version 7_1_201406.xls (from <http://www.soa.org/research/software-tools/research-scenario.aspx>) has returns cumulative from the beginning of the projection period. Please be certain to validate what type of return is include before proceeding to perform the translation, since the AAA may change how the factors in their source files are represented.

For the combination indexes, the index name column represents the index name which will be used in the CSV file. Use the drop down lists to select the two indexes which are used to derive the combination index. Specify the weight of index 1 in the index 1's weight column and the correlation of the two indexes. Carefully consider the setting for correlation, since using a default value of zero for the correlation could underestimate the volatility of the combination index.

5.2.4 Translation process

When the translation process is activated, the translator first validates many of the input selections. An error message is produced if input errors are encountered.

The translation process then proceeds to translate all the desired yield curve terms and indexes, reading in the information from specified CSV files.

Yield terms translation

For yield terms, the rates are read directly from the specified CSV files to develop the yield curves. The volatility curves are created from the volatilities specified in the YldCurveData worksheet.

Index translation

For the AAA indexes, total returns are read from the specified CSV files and income returns and volatilities are read from the data specified in the IndexData worksheet. Monthly accumulation factors are developed from the total returns of the CSV file which are then converted to monthly total returns, which, when adjusted by monthly income returns, allow for determining a monthly growth rate. The monthly growth rate is converted to an effective annual rate and included in the CSV file along with the effective annual income return and effective annual volatility curve.

For the combination indexes, the monthly income return and growth rate are combined using the weighting factor. These rates are converted to effective annual rate and included in the CSV file. The combination index volatility is developed using the correlation. The volatility for the combination index equals the square root of the sum of 1) the square of the weight for index 1 times the volatility for index 1 plus 2) two times the weight for index 1 times the volatility for index 1 time the weight for index 2 times the volatility for index 2 times the correlation plus 3) the square of the weight for index 2 times the volatility for index 2.

6 ECONOMIC SCENARIOS GENERATION

6.1 Generation uses

6.1.1 PBA considerations

Principle-based approaches to reserves and capital included in projections require inner loop economic scenarios beginning from the branching date. This section discusses the application of the various scenario generation processes to PBA projections.

VA standard scenario calculations

VA standard scenario revaluations for reserves do not require inner loop economic scenarios. All projection specifications are prescribed in the regulations.

VA standard scenario revaluations for capital do not specifically require inner loop economic scenarios per the RBC capital requirement. However, since the 10 year risk free rate at the branching date is required for the discount rate, inner loop economic scenario files must be used. Therefore, a generation process is needed to create the new historic file for branching dates.

Deterministic life reserve calculations

Deterministic life reserve revaluations require projection of cash flows using scenario 12 of the stochastic exclusion test scenarios. The stochastic exclusion test scenarios are created using the real world stochastic exclusion test generation approach.

Scenario amount calculations

Scenario amount revaluations require stochastic real world economic scenarios beginning from the branching date. These are created using a real world stochastic generation approach and either the 'Random Scenarios' or 'Academy Selected Scenarios' real world stochastic generation definition. When using the "Random Scenarios" definition, you can specify calibration criteria for yield curve rates and spreads and index rates wealth factors and then perform a calibration to ensure the generated scenarios pass a set of calibration tests, such as those specified by the American Academy of Actuaries.

Inner loop economic scenarios generation for variable annuity projections typically specify to reuse the specified inner loop scenarios. If assets are included in the projection or if the general account is significant for a liability-only projection, selecting to reuse all rates may not be appropriate. In that case, selecting to reuse index rates only may produce a more desirable result.

6.1.2 GAAP considerations

GAAP revaluations and unlocking included in projections require inner loop economic scenarios beginning from the branching date. Risk neutral and real world stochastic scenarios, along with a mean reverting deterministic scenario may be necessary.

This section discusses the application of the various scenario generation processes to GAAP projections.

SFAS 133

SFAS 133 revaluations for embedded derivative fair values require stochastic risk neutral economic scenarios beginning from the branching date. These economic scenarios are created using the risk neutral stochastic generation approach.

SOP 03-1

SOP 03-1 revaluations of benefit ratios require stochastic real world economic scenarios beginning from the branching date. The economic scenarios are created using the real world stochastic generation approach. SOP 03-1. For variable annuity projections, the index rates from the specified inner loop scenario file are commonly reused.

DAC Unlocking

Because GAAP best estimate assumptions have no foreknowledge of the future, it is not appropriate to continue to use outer loop (experience projection) scenario rates during DAC valuations from the branching date. Instead, a single inner loop deterministic scenario for DAC unlocking is created using the real world mean reversion generation approach.

The period to grade may be set such that no grading occurs during the mean reversion generation. That is, the GAAP schedule calculation assumes the starting rates remain level through the inner loop valuation. Note – This option should be used with care, since this could result in excess lapses continuing indefinitely into the future when the GAAP valuation occurs when excess lapse formulas are active.

6.1.3 Hedging considerations

Point-in-time valuations and inner loop valuations within a projection for the embedded derivatives of the guaranteed benefits of variable annuities require the generation of baseline and a set of shocked risk neutral economic scenarios beginning from the branching date. These economic scenarios are created using the risk neutral stochastic generation approach with a generation definition of 'Shocked Scenarios'.

6.2 Generation projection tasks

6.2.1 Overview

The rates submodel, along with the companion Economy DLL, provide the ability to derive historic rates and to generate risk neutral and real world scenarios for use in point-in-time hedging valuation as well as statutory and GAAP valuations occurring at valuation dates along the path of an outer loop scenario. This capability is very important within a nested projection.

Scenarios can be generated stochastically, using random numbers generated from the Mersenne-Twister random number generator (for risk neutral stochastic scenario generation) or the Academy of Actuaries random number generator (for real world stochastic generation), derived using a special set of random numbers (stochastic exclusion test generation), or derived through a mean reversion process. Additionally, the generated projection scenario file may contain rates reused from the specified inner loop scenario file. When rates are reused, the rates starting at the projection date apply starting at the valuation date. This would be appropriate, for example, for index rates when the assumption that future growth of indexes is not tied to past performance.

Parameters used for the generation (for risk neutral stochastic scenario generation) and the generated scenarios (for real world stochastic generation) may be calibrated. For risk neutral stochastic generation, a Levenberg-Marquardt optimization process is used to arrive at the calibrated parameters. For real world stochastic generation, the American Academy of Actuaries' calibration tests are used to ensure the generated interest and index rates adhere to the calibration criteria.

Since the initial valuation ($t = 0$) of a nested projection can be thought of as a first valuation date, an inner loop generation may be utilized to generate inner loop valuation rates branching from the projection date. For generation of normal scenarios, unless specified, no inner loop generation will occur at that time period. Instead, the initial valuation uses the specified inner scenario file, plus the inner loop historic rates (which initially contains a copy of the historic rates file).

6.2.2 Scenario rates

Specify the outer loop, historic rates, and inner loop scenarios rates required for the generation under the Scenarios: Rates category of the Economy input page.

For a generation, the outer loop scenario file contains the rates which become historic rates for periods before the valuation period and the initial conditions, the historic rates contain the rates which are the historic rates for the periods before the projection date, and the inner loop scenario file contains the rates which are reused should the `inner_loop_valn_scen_reuse_defn` be set to All Rates or Index Rates Only.

6.2.3 Generation parameters

Generation parameters provide the scenario generator with the information necessary to setup and execute the appropriate generation process. Specify the scenario generation parameters under the Generation: Control category of the Economy input page.

The appropriate selections for each of these parameters are discussed in the individual generation sections of this document.

6.2.4 Economy data

Economy data provide the scenario generator with the information necessary to determine which economic conditions should be included in the generation, the interest, index, and miscellaneous rates initial economic conditions and generation assumptions. Specify scenario generation economy data under the Generation: Economy Data category of the Economy input page.

The amount and structure of economy data required differs based on whether the generation is real world or risk neutral, and for real world, whether the generation is using a stochastic, stochastic exclusion test, or mean reversion generation approach.

The appropriate selections for each of the economy data variables are discussed in the individual generation sections of this document.

6.2.5 Shock control

Shock control parameters provide the scenario generator with the information necessary to execute a risk neutral shocked scenario generation process. Specify shock control parameters under the Generation: Shock Control category of the Economy input page.

6.2.6 Calibration data

Calibration data provides the scenario generator with the information necessary to execute a calibration process as part of the scenario generation. Specify calibration data under the RN Generation: Calibration Data and RW Generation: Calibration Data categories of the Economy input page.

The structure and required setup of the calibration data is significantly different between a risk neutral generation basis, which calibrates a set of interest and index parameters which feed the generation, and the real world generation basis, which runs a loop until a set of generated scenarios is accepted or to the number of calibration trials.

The appropriate selections for each of the calibration data variables are discussed in the individual generation sections of this document.

6.2.7 Projection task

All scenario generation projection tasks are defined on run pages which uses 'rates_economy' as the target model object. The diagram below shows an example generation projection task from the 'PBA Rate Gen' run page of the Variable Annuity application:

Name	RW Proj Scen Gen
- Input Pages - Company - Economy - Investment - Hedging - VAProduct - VAValuation - VAFund - VAExpenses - VAMortality - VAPolicyholder - VAData - EplData - BondData - MtgData - IrdData - EioData - EpaData - Parameter Sets - Run Parameter Set - Output Parameter Set - Goal Seek Parameter Set	STAT RW Inflation ▼ RateGen generate 2013_12 ▼ Default ▼ ☐ ▼

The only assumption set applicable for the generation tasks is the Economy assumption set. The selection for the Run parameter set and Output parameter sets are discussed in section below.

6.2.8 Run parameter set

Specify the following information in the Run parameter set for the scenario generation task.

Property	Value
Target column	generate
Projection period from	0
Projection period to	0
Projection period min	0
Projection period max	0
Projection task loops	Typically 1. However, for projection tasks which are generating shocked risk neutral scenarios, set the projection task loops to the number of shocked scenarios to generate. For example, if you are generating a baseline, parallel interest shocks up and down, and equity volatility shocks for each of 4 indexes, set the projection task loops equal to 1-7.
Wildcard set	Select the appropriate wildcard set for accessing the external assumptions
Distribution	Typically not used. However, for projection tasks which are generating are generating shocked risk neutral scenarios, use Split by Projection Loop to distribute the projection task loops to the cores on the grid. Specify a Target Split for Distribution value for the minimum number of Projection Loops that will be allocated to a Grid Task in a Distributed Processing environment. Since typically the number of cores on the grid will exceed the number of shocked scenarios to generate, this value should typically be set to 1.

6.2.9 Project variables

Also specify the project variables in the Run parameter set.

Property	Value
Projection date [proj_date]	The projection date for the nested projection

Property	Value
Valuation periods selected [valn_periods_selected]	For generation projection tasks which are included in nested projections, must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces; has no effect for standard projections. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0.
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs which generate large scenario files.

6.2.10 Output parameter set

No output is created by the scenario generation projection tasks since the only columns executed are the startup, initialize, generate, and finalize column of the rates submodel.

6.2.11 Nested projection setup

Nested projections perform scenario generations at the projection date ($t = 0$), if desired, and at each successive valuation date. For example, scenarios might be generated at the end of each year from the projection date ($t = 12, 24, 36$, and so on).

Nested projections must be run within a projection set from the Run controller. Run the nested projection by submitting the projection set.

For nested projections, the scenario paths to be processed are specified for the projection set on the Run controller, instead of in the Run parameter set.

Property	Values
Projection set loops	The outer loop scenarios to project, such as "1-5,10"
Independent loops	Yes when each outer loop can be calculated at the same time
Include sub loops	Yes; this is the driver for the nested projection
Projections before sub loop	Typically none

Property	Values
Sub loops	The time steps to project, such as “1-4”. The Valuation periods selected property in the Run parameter set assigns a valuation period to each time step, such as “0,12,24,36”. After the full set of time steps have been run, subsequent runs can include only the time steps to be rerun (such as to analyze valuation details) such as “2”.
Sub loop projections	Includes the scenario generation task, plus each valuation (when appropriate), plus the experience projection task.
Sub loop projection dependencies	Specify the dependencies of the projection tasks. Typically the valuation tasks are dependent on the scenario generation tasks. The experience projection task is dependent to start only after the valuation tasks have finished.
Projections after sub loop	Typically none

The diagram below shows an example from the Run controller display:

The screenshot shows the 'UL Business Planning' interface with the following configuration:

- Projection Set Loops:** 1,5
- Independent Loops:** ☒
- Include Sub Loops:** ☒
- Projections before Sub Loop:** (Empty table)
- Sub Loop:** (Empty field)
- Projection Set Sub Loops:** 1-4
- Sub Loop Projections:**

Projection	Dependencies
PBA Gen SET	
Valn Det Res Proj	PBA Gen SET
PBA Gen Stoch	Valn Det Res Proj
Valn Scen Amt Res Cap Proj	PBA Gen Stoch
Valn AG38 8D Proj	PBA Gen Stoch
Experience Proj	Valn Det Res Proj, Valn Scen Amt Res Cap Proj, Va...
- Projections after Sub Loop:** (Empty table)

Note: In the original image, 'PBA Gen SET' and 'PBA Gen Stoch' are circled in red.

6.3 Risk neutral stochastic generation

6.3.1 Overview

Risk neutral stochastic generation is the process of generating risk neutral scenarios for use in market consistent valuations and projections.

6.3.2 Scenario rates

Specify the outer loop, historic rates, and inner loop scenarios rates required for the generation under the Scenarios: Rates category of the Economy input page.

For a generation, the outer loop scenario file contains the rates which become historic rates for periods before the valuation period and the initial conditions, the historic rates contain the rates which are the historic rates for the periods before the projection date, and the inner loop scenario file contains the rates which are reused should the inner_loop_valn_scen_reuse_defn be set to All Rates or Index Rates Only.

6.3.3 Generation parameters

Specify scenario generation parameters as follows for the risk neutral generation process.

Variable Name	Value
gen_defn	'Normal Scenarios' if you wish to generate a single set of risk neutral scenarios. 'Shocked Scenarios' if you wish to generate a set of shocked risk neutral scenarios.
gen_approach_defn	'Stochastic'.
gen_basis_defn	'Risk Neutral'.
gen_scenarios	Number of scenarios to generate.
gen_mths	Number of months of rates to generate from the branching date.
gen_rates_freq	'Monthly' to capture monthly economic environments into the generated file. 'Annual' to capture annual economic environment into the generated scenario file.
gen_initial_valn_period	'Yes' if you wish to generate the risk neutral rates branching from the projection date. 'No' otherwise. This variable has no effect when shocked risk neutral scenarios are generated. A set of shocked scenarios are always generated at the projection date.
gen_random_seed	Seed for the random number generator.

Variable Name	Value
gen_rn_int_proj_defn	'Constant' if you wish to assume a constant volatility for the interest rates during the calibration and generation. 'Term Structure' if you wish to assume a term structure of volatilities for the interest rates during the calibration and generation. Important: When calibration is included and the interest calibration data source is 'Swap Volatility', the interest projection definition is automatically set to 'Term Structure'. And, when calibration is included and the interest calibration data source is 'Bond Volatility', the interest projection definition is automatically set to 'Constant'.
gen_rn_bond_terms	An array of bond terms to be used in the risk neutral quick generation approach. They array may contain 4 elements where each element represents a term in years for a bond rate.
gen_detail_defn	'Yes' if you wish to create the generation audit report. 'No' otherwise.
gen_detail_scen	An integer number specifying the generated scenario to include in the generation audit report
gen_calib_defn	'Yes' if you want to perform the calibration process for interest rates and indexes as part of the scenario generation process. 'No' otherwise
gen_calib_retry_attempts	An integer number specifying the number of trials for the calibration process.
gen_calib_detail_defn	'Yes' if you wish to create the calibration audit report. 'No' otherwise

Note: The generation process does not allow missing rates within the historic rates. The last date within the historical data file must be greater than or equal to the projection date.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.3.4 Economy data

Specify economy data as follows for the risk neutral generation process.

Variable Name	Value
econ_data_yld_curve_selected	Specify Govt for this variable and the Govt yield curve must exist in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file. No other yield curves should be specified since the risk neutral generator cannot generate corporate bond returns.
econ_data_yld_curve_term_selected	A comma separated list of terms which describe the initial yield curve. For example, to describe the fill 10 point yield curve, specify 0.25, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 20.0, 30.0 for this variable.
econ_data_yld_curve_vol_mr_term_selected	A comma separated list of terms to use to describe the mean reversion rate and volatility parameters for the generation. For example, 0.25, 1.0, 5.0, 10.0. The short rate term of 0.25 must be included in the list since the risk neutral generation requires a short rate parameter.
econ_data_yld_curve_mr_rates	Mean reversion rates required for the risk neutral generation process. These rates can either be calibrated rates or initial values which feed the calibration process. The mean reversion rates typically vary by yield curve id ('Govt'), term to maturity and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of yld_curve_id and rate_term as column lookup keys and the 'rates' submodel column date as the row lookup key. Interpolation is typically used for missing rate terms.

Variable Name	Value
econ_data_yld_curve_vol	Yield curve volatilities required for the risk neutral generation process. These rates can either be calibrated rates or initial values which feed the calibration process. The yield curve volatilities typically vary by yield curve id ('Govt'), term to maturity (5.0) and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of yld_curve_id and rate_term as column lookup keys and the 'rates' submodel column date as the row lookup key. Interpolation is typically used for missing rate terms.
econ_data_index_rate_selected	A comma separated list of equity indexes to include in the generation process. For example, to include the US and Small indexes, specify US, Small for this variable. The indexes must exist in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file.
econ_data_index_rate_moneyness_selected	You must specify 1.0 for this variable since the risk neutral generation process required 'at-the-money' volatilities. No other moneyness levels are required.
econ_data_index_rate_tenor_selected	A comma separated list of year to expiry tenors which describe the index volatility curves. For example, you could specify 1.0, 5.0, 10.0 for volatilities for a 1 year, 5 year, and 10 year expiry options.
econ_data_index_rate_vol	Implied volatilities of the underlying index required for the risk neutral generation process. These rates can either be calibrated rates or initial values which feed the calibration process. The index rate volatilities typically vary by index id ('US'), moneyness level (1.0), tenor (5.0), and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of index_id, moneyness, and rate_term as column lookup keys and the 'rates' submodel column date as the row lookup key. Previous is typically used for missing moneyness levels and interpolation is typically used for missing rate terms.

Variable Name	Value
econ_data_correlation	Correlations for the short rate of the 'Govt' yield curve and the index rates included in the generation process. The 'Govt' yield curve and index rates specified must exist in the economic scenario files specified for both the outer and inner loop. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variable of corrid1 as the column lookup key and corrid2 as the row lookup key, thereby allowing for the matrix of correlations across the short rate and index rates to be specified.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.3.5 Shock control

Specify shock control parameters as follows for the risk neutral generation process.

Variable Name	Value
shock_freq	An integer value representing the frequency for the shock generation within a nested projection. For example, 1 is specified for the generation to occur monthly within the nested projection horizon, 3 for the generation to occur quarterly, etc.
eqt_price_shock_defn	'Yes' if an equity asset price shock is included in the combination shock element. 'No' otherwise.
int_rate_shock_defn	'Yes' if an interest rate shock is included in the combination shock element. 'No' otherwise.
int_rate_shock_id	Text representing the interest rate shock to apply for the combination shock element.
int_rate_shock	A floating point value representing the shock amounts for the interest rate shock. The shock amounts may be either positive, representing an up shock, or negative, representing a down shock, and are expressed as decimal values. (0.01 = 1%).

Variable Name	Value
eqt_vol_shock_defn	'Yes' if an equity volatility shock is included in the combination shock element. 'No' otherwise.
eqt_vol_shock_id	Text representing the equity volatility shock to apply for the combination shock element.
eqt_vol_shock	A floating point value representing the shock amounts for the equity volatility shock. The shock amounts may be either positive, representing an up shock, or negative, representing a down shock, and are expressed as decimal values. (0.01 = 1%).
phldr_shock_defn	'Yes' if a policyholder shock is included in the combination shock element. 'No' otherwise.
scen_file_name_addn	Text representing the addition to the scenario name for the shock in the scen_file_name_addn variable. This text will be appended to the base name of the scenario file, historical rates file, or the inner loop scenario file to produce the name of the shock file. Best practice is to specify a name addition that will create a shocked scenario file which adheres to the naming convention discussed in scenario file considerations section above. For example, 'Baseline' would be an appropriate value for the scen_file_name_addn variable for the shock which represents the baseline scenarios, and 'ParallelIntRateUp' would be an appropriate value for the scen_file_name_addn for the shock which represent a parallel up 1 for the interest rates.

All the variables discussed above are typically linked to external source ranges of the Hedging.xlsm workbook using appropriate column and row lookup keys. Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.3.6 Calibration criteria

Specify calibration criteria as follows for the risk neutral generation process.

Variable Name	Value
calib_yld_curve_selected	To calibration the nominal interest rates, specify Govt for this variable and the Govt yield curve must exist in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file.
calib_yld_curve_data_source_defn	'Swaption Volatility' if the nominal interest rates calibration is to swaption prices. 'Bond Volatility' if the nominal interest rates calibration is to nominal bond prices.
calib_yld_curve_normalize_wgths_defn	'Yes' if the calibration should normalize the weights for each of the swaption or bond when determining the calibrated values. 'No' otherwise
calib_yld_curve_swap_maturities_selected	A comma separated list of maturities (expressed in years) for the swaptions to include in the calibration process. For example, specify 1.0, 5.0, 10.0 for this variable to include swaptions maturing in 1 year, 5, years, and 10 years respectively in the calibration process.
calib_yld_curve_swap_tenors_selected	A comma separated list of tenors (expressed in years) for the swaptions to include in the calibration process. For example, specify 1.0, 5.0, 10.0 for this variable to include swaptions using 1 year, 5, year, and 10 year tenors.
calib_yld_curve_swaption_vol	Swaption volatilities required for the risk neutral calibration process. The swaption volatilities typically vary by yield curve id ('Govt'), maturity (5.0) and tenor (10.0). This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of yld_curve_id, swap_maturity, and swap_tenor as column lookup keys. Previous is typically used for missing maturities and tenors.
calib_yld_curve_swaption_wght	Swaption weights required for the risk neutral calibration process. The swaption weights typically vary by yield curve id ('Govt'), maturity (5.0) and tenor (10.0). This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of yld_curve_id, swap_maturity, and swap_tenor as column

Variable Name	Value
	lookup keys. Previous is typically used for missing maturities and tenors.
calib_yld_curve_bond_terms_selected	A comma separated list of terms (expressed in years) for the bonds to include in the calibration process. For example, specify 1.0, 5.0, 10.0 for this variable to include bonds using 1 year, 5, year, and 10 year terms.
calib_yld_curve_bond_vol	Bond volatilities required for the risk neutral calibration process. The bond volatilities typically vary by yield curve id ('Govt') and maturity (5.0). This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of yld_curve_id and rate_term as column lookup keys. Previous is typically used for missing rate terms.
calib_yld_curve_bond_wght	Bond weights required for the risk neutral calibration process. The bond weights typically vary by yield curve id ('Govt') and maturity (5.0). This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of yld_curve_id and rate_term as column lookup keys. Previous is typically used for missing rate terms.
calib_index_rate_selected	A comma separated list of equity indexes to include in the calibration process. For example, to include the US and Small indexes, specify US, Small for this variable. The indexes included here must be listed in the econ_data_index_rate_selected variable and must exist in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file.
calib_index_rate_normalize_wgths_defn	'Yes' if the calibration should normalize the weights for each of the indexes when determining the calibrated values. 'No' otherwise The normalize weights definition typically varies by index id ('US'). This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variable of index_id.

Variable Name	Value
calib_index_rate_div_yld	A floating point value representing the dividend yield for the index used in the calibration. This dividend yield is needed for the index option pricing formula. The dividend yield typically varies by index id ('US'). This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variable of index_id.
calib_index_rate_opt_defn	'Call' if the index option used in the calibration is a European call option. 'Put' if the index option used in the calibration is a European put option. The index rate option definition typically varies by index id ('US'). This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variable of index_id.
calib_index_rate_moneyness_selected	A comma separated list of moneyness levels (expressed in decimal value) for the index options used in the calibration process. For example, specify 1.0, 1.05 for this variable to include at-the-money and in-the-money index options in the calibration.
calib_index_rate_terms_selected	A comma separated list of terms (expressed in years) for the index options to include in the calibration process. For example, specify 1.0, 5.0, 10.0 for this variable to include index options expiring in 1 year, 5, years, and 10 years.
calib_index_rate_opt_vol	Index option volatilities required for the risk neutral calibration process. The index option volatilities typically vary by index id ('US'), moneyness level (1.0), and term (5.0). This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of index_id, moneyness, and rate_term as column lookup keys. Previous is typically used for missing moneyness and rate terms.
calib_index_rate_opt_wght	Index option weights required for the risk neutral calibration process. The index option weights typically vary by index id ('US'), moneyness level (1.0), and term (5.0). This variable is commonly linked to

Variable Name	Value
	an external source which uses the hidden 'rates' submodel code variables of index_id, moneyness, and rate_term as column lookup keys. Previous is typically used for missing moneyness and rate terms.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.3.7 Generation process

The generation processing for risk neutral scenarios first creates new historical rates, then, if requested, calibrates the generation parameters to the specified calibration criteria, then generates the new risk neutral scenarios.

Historic rate derivation

The economic conditions included in the generated historic rates file are the economic conditions which are included in the outer loop scenario file. Any economic conditions which are only included in the specified historic rates file are excluded, since there are no values for those economic conditions included in the outer loop scenario file.

Economic conditions for the generated historic rates are derived at a valuation date by taking economic conditions from the historic rates for periods prior to the projection date and combining those with economic conditions from the current scenario for periods from the projection date to the valuation date.

The economic environments of historic rates are stored monthly for the user specified number of months subsequent to the valuation date. The interpolation method for the generated historic rate file is the interpolation method of the specified historic rate file.

Parameter calibration

The parameters specified for the various economy data variables for volatilities and mean reversion rates may be already calibrated parameters or initial guess values for a parameter calibration process.

When the gen_calib_defn is 'Yes', the specified values are considered to be initial guess for a calibration process which determines the appropriate values for the parameters by comparing the model values with market data, utilizing the Leveberg Marquardt non-linear optimization routine.

The calibration process is initialized based on the yield curves and indexes to calibrate, the various options specified for the calibration, and the selection and structure of the market data. The initial guesses for the calibration parameters are captured from the economy data. Values for the market data are determined, such as prices for swaptions and index call options. For indexes, the calibration is always to the at-the-money option volatilities.

The market data and weights are provided to the Leveberg Marquardt optimization routine which evaluates a sum of squares error on the initial parameters. The routine then adjusts the parameters, calling the appropriate routine to redetermine the values for the market data, in an attempt to reduce the sum of squares error. The whole process is repeated until the internal acceptance criteria is met or the number of trials exceeds 100.

Once determined, the calibrated parameters for the yield curve and indexes are provided to the scenario generator.

Scenario generation

Risk neutral interest rates are generated using a Gaussian Heath Jarrow Morton (HJM) framework. Within the Gaussian HJM framework, a one factor Hull White (HW) interest model is utilized, allowing for either a constant volatility or term structure of volatilities approach. The initial economic conditions for the risk free rates are captured from the current scenario in the outer loop at the valuation period. When the generation is for interest rate shock scenarios, the effective annual shocks are added to the effective annual risk free yield rates, floored at zero. The forward rate curve is then determined utilizing the appropriate boot strapping of yield rates to spot rates to forward rates. The forward rates are then developed using the Hull White model from which zero coupon bond prices and yield rates are determined. The mean reversion and volatility parameters used in the generation are either the parameters specified or the calibrated parameters

A 'quick generation' approach is used for risk neutral scenario interest rate generation which limits the amount of items generated to several user specified bond terms of the yield curves. The quick generation process involves creating only the zero coupon bond prices and returns, plus the yield curve cash account accumulation (beta) which are created as part of the nominal interest rate generation process, thus reducing the interest rate generation time.

Risk neutral index growth rates are generated using a term structure of volatility generation model approach where the value of the index is modeled as geometric Brownian motion with a stochastic interest rate and piecewise constant volatility parameters. The index return is based on the scenario specific beta forward rate from the Hull White process, the index volatility, and a dependent random number.

The economic environments of scenario rates are stored at the user defined frequency. The interpolation method for the generated scenario file is the interpolation method of the specified outer loop scenario file.

6.3.8 Auditing

The calibration and generation of risk neutral scenarios can be audited through the risk neutral calibration and generation audit reports.

Calibration audit report

When the `gen_calib_detail_defn` is 'Yes', a comma separated value (CSV) file is created in the 'generated' folder that contains the details of the calibration process. The name of the calibration audit report is the name of the generated scenario file appended with '_CalibAuditReport'. For example, if the scenario file being generated is `YYYYMMMonthlyRNScenarios_1_1.csv`, the calibration audit report will be named `YYYYMMMonthlyRNScenarios_1_1_CalibAuditReport.csv`.

The calibration audit report is structured with initial sections for the main calibration parameters, economy data structures, initial assumptions and calibration definitions and criteria. Then, for each item being calibrated, sections are included for the calibration setup, the calibration targets, and the Levenberg-Marquardt setup and trials. The final calibration results are displayed at the end of the file.

Generation audit report

When the `gen_detail_defn` is 'Yes', a comma separated value (CSV) file is created in the 'generated' folder that contains the details of the generation process. The name of the generation audit report is the name of the generated scenario file appended with '_GenAuditReport'. For example, if the scenario file being generated is `YYYYMMMonthlyRNScenarios_1_1.csv`, the generation audit report will be named `YYYYMMMonthlyRNScenarios_1_1_GenAuditReport.csv`.

The generation audit report is structured with initial sections for the main generation parameters, economy data structures, and initial assumptions and values, including covariance assumptions, Cholesky decomposition matrices, and independent random numbers. Then, for the scenario number specified in the `gen_detail_scen`, detail is shown for the projection of the nominal process and for each index being generated. The final results for the scenario file are displayed at the end of the file, taking into account whether the economic environments are captured on a monthly or annual frequency.

6.4 Real world stochastic generation

6.4.1 Overview

Real world stochastic scenario generation is the process of generating stochastic scenarios using the same generation parameters and methodology used within the American Academy of Actuaries' stochastic scenario generator (AIRG).

6.4.2 Scenario rates

Specify the outer loop, historic rates, and inner loop scenarios rates required for the generation under the Scenarios: Rates category of the Economy input page.

For a generation, the outer loop scenario file contains the rates which become historic rates for periods before the valuation period and the initial conditions, the historic rates contain the rates which are the historic rates for the periods before the projection date, and the inner loop scenario file contains the rates which are reused should the inner_loop_valn_scen_reuse_defn be set to All Rates or Index Rates Only.

6.4.3 Generation parameters

Specify scenario generation parameters as follows for the real world stochastic generation process.

Variable Name	Value
gen_defn	'Normal Scenarios'.
gen_approach_defn	'Stochastic'.
gen_basis_defn	'Real World'.
gen_rw_stoch_defn	'Random Scenarios', when the scenarios are generated using random draws from a single random number generator. 'Academy Selected Scenarios', when the scenarios generated are derived by applying the Academy scenario picking process to 10,000 scenarios generated using the Academy generation process.
gen_scenarios	Number of scenarios to generate. When selecting to reuse the inner loop scenario rates, the number of scenarios defined in the inner loop scenario file must exceed the value specified in the 'gen_scenarios' variable. Otherwise, the scenario generator will issue a fatal error. Also, when the reuse selection is 'All Rates' the value of gen_scenarios is ignored and the inner loop scenario simply refers to the original inner loop scenario file.
gen_mths	Number of months of rates to generate from the branching date.

Variable Name	Value
gen_rates_freq	'Monthly' to capture monthly economic environments into the generated file. 'Annual' to capture annual economic environment into the generated scenario file.
gen_initial_valn_period	'Yes' if you wish to generate the real world rates branching from the projection date. 'No' otherwise.
gen_random_seed	Seed for the random number generator.
gen_detail_defn	'Yes' if you wish to create the generation audit report. 'No' otherwise.
gen_detail_scen	An integer number specifying the generated scenario to include in the generation audit report
gen_rw_mrp_mean_yrs_1	Years lookback for determining the mean result for the mean reversion parameter. For both AAA and NAIC methodology treatment, set this variable to 3.
gen_rw_mrp_mean_yrs_2	Years lookback for determining the second mean result for the mean reversion parameter. For NAIC methodology, set this value to 10. For AAA methodology, leave this value at 0.
gen_rw_mrp_median_yrs	Years lookback for determining the median result for the mean reversion parameter. For both NAIC and AAA methodology, set this value to 50. When the gen_rw_mrp_mean_yrs_2 value is zero, the AAA methodology of subtracting 0.25% from the median rate is applied.
gen_calib_defn	'Yes' if you wish to perform the generation calibration tests. 'No' otherwise.
gen_calib_retry_attempts	Number of attempts to generate scenarios which pass the calibration tests.
gen_calib_detail	'Yes' if you wish to create the calibration audit report. 'No' otherwise.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.4.4 Economy data

Specify economy data as follows for the real world stochastic exclusion test generation process.

Variable Name	Value
econ_data_yld_curve_selected	Specify Govt for this variable and the Govt yield curve must exist in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file. No other yield curves should be specified.
econ_data_yld_curve_term_selected	Specify the full 10 point terms of the 'Govt' yield curve as 0.25, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 20.0, 30.0 for this variable.
econ_data_index_rate_selected	A comma separated list of indexes to include in the generated scenario file. For example, to only include the US and Small indexes, specify US, Small for this variable. The index identifiers in this comma separated list must match the identifiers of the indexes included in the AAA scenarios generation. The allowable identifiers, which are case sensitive are: AGGR, Balanced, Bond, INTL, Money, Small, US, ITGVT, and LTCORP. The indexes must exist in in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file.
econ_data_misc_rates_selected	While not specifically required, best practice is to specify 'Infl' for this variable in order to align the economy data with the internal setup of an inflation index for the generation.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.4.5 Shock control

Shock control parameters do not apply for real world stochastic exclusion test generation.

6.4.6 Calibration criteria

In order to apply the scenario calibration methodology and requirements outlined in the ESWG June 2008 report, *Report from the American Academy of Actuaries' Economic Scenario Work Group* and Appendix 2 of the LCAS's June 2005 report, *Recommended Approach for Setting Regulatory Risk-Based Capital Requirements for Variable Annuities and Similar Products* to the generated real world stochastic scenarios, specify calibration criteria as follows.

Variable Name	Value
calib_yld_curve_rates_rate_term	The risk free yield term in years used for the calibration rates test. Terms specified in the Academy methodology are 1 and 20.
calib_yld_curve_rates_tgt_horizon	The time horizon in years for the calibration rates test. Time horizons specified in the Academy methodology are 1, 5, 10, and 30
calib_yld_curve_rates_percentile	The percentile value for the calibration rates test. Percentile values specified in the Academy methodology are 0.05 (5%) and 0.95 (95%).
calib_yld_curve_rates_percentile_tgt	The percentile targets for the calibration rates test. The values specified in the Academy methodology differ for each of the 16 tests.
calib_yld_curve_rates_rate_adj_a	The rate adjustment 'A' allowable in the Academy methodology. For the 1 year horizon test, the value of 'A' is 0.01 (1%). For all other horizon, the value is (0.005) (0.5%).
calib_yld_curve_rates_rate_adj_b	The rate adjustment 'B' allowable in the Academy methodology. For the 1 year horizon test, the value of 'B' is 0.2 (20%). For all other horizon, the value is (0.1) (10%).
calib_yld_curve_rates_pass_pct	The percentage of the yield curve rate tests which must pass for the yield curve calibration set to pass. Specify 0.9 for 90%.
calib_yld_curve_sprds_tgt_horizon	The time horizon in years for the calibration spread test. Specify 30 for the Academy methodology.
calib_yld_curve_sprds_percentile	The percentile value for the calibration spreads test. Percentile values specified in the Academy methodology are 0.05 (5%) and 0.95 (95%).
calib_yld_curve_sprds_percentile_tgt	The percentile target value for the calibration spreads test. The values specified in the Academy methodology differ for each of the 2 tests.

Variable Name	Value
calib_yld_curve_sprds_sprd_adj	The adjustment to the Academy spread for the calibration spread test. The value in the Academy methodology is 0.005 (0.5%). The adjustment is added for the low spread and subtracted for the high spread.
calib_yld_curve_sprds_pass_pct	The percentage of the yield curve spread tests which must pass for the yield curve calibration set to pass. Specify 0.9 for 90%.
calib_index_rates_tgt_horizon	The time horizon in years for the calibration index rates wealth accumulation test. The Academy methodology provides for horizons of 1, 5, 10, and 20.
calib_index_rates_percentile	The percentile value for the calibration index rates wealth accumulation test. The Academy methodology provides for percentiles of 0.025 (2.5%), 0.05 (5%), 0.1 (10%), 0.9 (90%), 0.95 (95%) and 0.975 (97.5%).
calib_index_rates_percentile_tgt	The index rate wealth accumulation percent of a 1 dollar investment at the time horizon for the calibration index rates wealth accumulation test. The values specified in the Academy methodology differ for each of the 22 tests.
calib_index_rates_pass_pct	The percentage of the index rate wealth accumulation tests which must pass for the yield curve calibration set to pass. Specify 0.9 for 90%.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.4.7 Generation process

The generation processing for real world stochastic scenarios first creates new historical rates, then, generates the new real world scenarios, running a loop, if requested, until the generated scenarios pass the calibration tests or the number of calibration trials are completed.

Historic rate derivation

The economic conditions included in the generated historic rates file are the economic conditions which are included in the outer loop scenario file. Any economic conditions which are only included in the specified historic rates file are excluded, since there are no values for those economic conditions included in the outer loop scenario file.

Economic conditions for the generated historic rates are derived at a valuation date by taking economic conditions from the historic rates for periods prior to the projection date and combining those with economic conditions from the current scenario for periods from the projection date to the valuation date.

The economic environments of historic rates are stored monthly for the greater of the `gen_rw_mrp_mths_max` or the user specified number of months subsequent to the valuation date. The greater of these two values is used because the correct mean reversion point for the stochastic generation can only be calculated in the generation when the new history file contains the sufficient history values. The interpolation method for the generated historic rate file is the interpolation method of the specified historic rate file.

Scenario generation

The economic conditions included in the generated scenario file are determined using the parameters and methodologies of the Academy of Actuaries' stochastic scenario generator.

The parameters for the generation are an inherent part of the generator and are not required to be set by the user through the Economy data. The initial conditions for the risk free rates and inflation rate are captured from the current scenario in the outer loop at the valuation date and the mean reversion parameter is calculated from the rates captured into the newly generated historic rates file.

The economic conditions for all 10 terms of the 'Govt' yield curve are generated. For the indexes, the economic conditions for all 9 Academy indexes are generated. However, only the indexes listed in the `econ_data_index_rates_selected` will be included in the generated file. And, finally an inflation index is generated.

When the `'gen_rw_stoch_defn'` is 'Random Scenarios', the scenarios are generated using draws from a single instance of the Academy random number generator and the specified, or calibration adjusted (see below), random number seed. The number of scenarios generated is the number specified in the `gen_scenarios` variable.

When the `'gen_rw_stoch_defn'` is 'Academy Selected Scenarios', the scenarios are generated using separate random number generators for interest, indexes, and inflation where scenario specific random number seeds are applied. Two generation loops are executed. The first loop generates the long term rate and the scenario significance measures. Then, the scenarios are ranked and the subset, as specified in the `gen_scenarios` value, are selected using the Academy of Actuaries scenario picking process. Then, the second generation loop is executed, and generates the interest rates, index growth rates, and inflation rate for the final scenario file, using the appropriate scenario specific random numbers and separate generators.

When the `inner_loop_valn_scen_reuse_defn` variable is 'All Rates' or 'Index Rates Only', the economic conditions for each scenario of the generated scenario file yield curve and/or indexes are captured from the corresponding scenario of the inner loop scenario file. When rates are reused, the stochastic generation for the reused economic conditions are ignored. The specified inner loop scenario file must have at least as many scenarios and economic environments in each scenario as are required based on the values specified for the `gen_scenarios` and `gen_mths` variables.

The economic environments of scenario rates are stored at the user defined frequency. The interpolation method for the generated scenario file is the interpolation method of the specified outer loop scenario file.

Scenario calibration

When the `'gen_rw_stoch_defn'` is 'Random Scenarios' the scenarios generated can be accepted or rejected using the specified calibration tests.

When calibration is selected, the calibration process described in the ESWG June 2008 report, *Report from the American Academy of Actuaries' Economic Scenario Work Group* and Appendix 2 of the LCAS's June 2005 report, *Recommended Approach for Setting Regulatory Risk-Based Capital Requirements for Variable Annuities and Similar Products* is executed to ensure that the generated scenarios pass the user specified percentage of calibration tests.

The calibration process involves generating a scenario file on which the calibration tests are performed. If the scenario file meets the user specified passing percent passes for all three calibration test sets (yield rates, yield spreads, and index rates), the scenario file is accepted. However, if any one of the calibration test sets fail, the scenario file is rejected. The random seed is altered and a new scenario file is generated. The

process is repeated until a scenario file is accepted or the number of calibration trials exceed the value specified for the `gen_calib_retry_attempts` variable.

6.4.8 Auditing

The calibration and generation of real world stochastic scenarios can be audited through the real world calibration and generation audit reports.

Calibration audit report

When the `gen_calib_detail_defn` is 'Yes', a comma separated value (CSV) file is created in the 'generated' folder that contains the details of the calibration process. The name of the calibration audit report is the name of the generated scenario file appended with '_CalibAuditReport'. For example, if the scenario file being generated is `YYYYMMMonthlyRWScenarios_1_1.csv`, the calibration audit report will be named `YYYYMMMonthlyRWScenarios_1_1_CalibAuditReport.csv`.

The calibration audit report is structured with initial sections for the main calibration parameters, and when included, yield rates, yield spreads, and index wealth factor calibration tests and criteria. Then, for each calibration trial, a section for generated rates and values needed for the calibration is included, and when included, basic statistics and calibration test values and results are included for each of yield rates, yield spreads, and index wealth factor calibration. The final calibration pass/fail result is provided at the end of the file.

Generation audit report

When the `gen_detail_defn` is 'Yes', a comma separated value (CSV) file is created in the 'generated' folder that contains the details of the generation process. The name of the generation audit report is the name of the generated scenario file appended with '_GenAuditReport'. For example, if the scenario file being generated is `YYYYMMMonthlyRWScenarios_1_1.csv`, the generation audit report will be named `YYYYMMMonthlyRWScenarios_1_1_GenAuditReport.csv`.

When `gen_rw_stoch_defn` is 'Academy Selected Scenarios', the generation detail is only captured for the second loop. That is, the generation detail is after the application of the scenario picking process to reduce the 10,000 scenarios down to the user specified subset.

The generation audit report is structured with initial sections for the main generation parameters, economy data structures, and initial assumptions and values, including the mean reversion point, and generation and correlation parameters. When the `gen_rw_stoch_defn` is 'Academy Selected Scenarios', sections are included for the scenario significance measure and ranking process, showing the scenarios selected for the user specified subset. Then, for the scenario number specified in the `gen_detail_scen`, detail is shown for the projection of the interest rates, for each index, and for the inflation rate. The final results for the scenario file are displayed at the end of the file, taking into account which indexes are to be included, and whether the economic environments are captured on a monthly or annual frequency.

6.5 Real world stochastic exclusion test generation

6.5.1 Overview

Stochastic exclusion test scenario rates generation is the process of creating the 16 scenarios as specified for the stochastic exclusion test (SET) in VM-20.

6.5.2 Scenario rates

Specify the outer loop, historic rates, and inner loop scenarios rates required for the generation under the Scenarios: Rates category of the Economy input page.

For a real world stochastic exclusion test generation, the outer loop scenario file contains the rates which become historic rates for periods before the valuation period and the initial conditions and the historic rates contain the rates which are the historic rates for the periods before the projection date.

6.5.3 Generation parameters

Specify scenario generation parameters as follows for the real world stochastic exclusion test generation process.

Variable Name	Value
gen_defn	'Normal Scenarios'.
gen_approach_defn	'Stochastic Exclusion Test'
gen_basis_defn	'Real World'.
gen_mths	Number of months of rates to generate from the branching date.
gen_rates_freq	'Monthly' to capture monthly economic environments into the generated file. 'Annual' to capture annual economic environment into the generated scenario file.
gen_detail_defn	'Yes' if you wish to create the generation audit report. 'No' otherwise.
gen_detail_scen	An integer number specifying the generated scenario to include in the generation audit report. This value should always be 1 when generating the real world mean reversion scenario.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.5.4 Economy data

Specify economy data as follows for the real world stochastic exclusion test generation process.

Variable Name	Value
econ_data_yld_curve_selected	Specify Govt for this variable and the Govt yield curve must exist in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file. No other yield curves should be specified.
econ_data_yld_curve_term_selected	Specify the full 10 point terms of the 'Govt' yield curve as 0.25, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 20.0, 30.0 for this variable.

Variable Name	Value
econ_data_index_rate_selected	A comma separated list of indexes to include in the generated scenario file. For example, to only include the US and Small indexes, specify US, Small for this variable. The index identifiers in this comma separated list must match the identifiers of the indexes included in the AAA scenarios generation. The allowable identifiers, which are case sensitive are: AGGR, Balanced, Bond, INTL, Money, Small, US, ITGVT, and LTCORP. The indexes must exist in in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file.
econ_data_misc_rates_selected	While not specifically required, best practice is to specify 'Infl' for this variable in order to align the economy data with the internal setup of an inflation index for the generation.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.5.5 Shock control

Shock control parameters do not apply for real world stochastic exclusion test generation.

6.5.6 Calibration criteria

Calibration criteria does not apply for real world stochastic exclusion test generation.

6.5.7 Generation process

The generation processing for real world stochastic exclusion test scenarios first creates new historical rates, then generates the new real world stochastic exclusion test scenarios.

Historic rate derivation

The economic conditions included in the generated historic rates file are the economic conditions which are included in the outer loop scenario file. Any economic conditions which are only included in the specified historic rates file are excluded, since there are no values for those economic conditions included in the outer loop scenario file.

Economic conditions for the generated historic rates are derived at a valuation date by taking economic conditions from the historic rates for periods prior to the projection date and combining those with economic conditions from the current scenario for periods from the projection date to the valuation date.

The economic environments of historic rates are stored monthly for the user specified number of months subsequent to the valuation date. The interpolation method for the generated historic rate file is the interpolation method of the specified historic rate file.

Scenario generation

The economic conditions included in the generated scenario file are the economic condition information specified in the `econ_data_yield_curve_selected`, `econ_data_yield_curve_term_selected`, and `econ_data_index_rate_selected` variables which also exist in the specified historic rates file and outer loop scenario file.

The economic conditions for the generated scenario file are determined using the Academy of Actuaries' stochastic scenario generator parameters and methodology and the stochastic exclusion test methodology described in Appendix 1 of the *VM-20 Requirements For Principle-Based Reserves For Life Products* section of the Valuation Manual.

Similarly to the real world stochastic generation process, you may select to reuse the inner loop rates by setting the `inner_loop_valn_scen_reuse_defn` variable to 'All Rates' or 'Index Rates Only'. However, this is not typically done since the stochastic exclusion test scenario generation is focused on creating scenarios with various shock levels of interest rates and must align properly with the current scenario of the outer loop scenario file at the valuation date.

The economic environments of scenario rates are stored at the user defined frequency. The interpolation method for the generated scenario file is the interpolation method of the specified outer loop scenario file.

6.5.8 Auditing

The generation of real world stochastic exclusion test scenarios can be audited through the real world generation audit reports.

When the `gen_detail_defn` is 'Yes', a comma separated value (CSV) file is created in the 'generated' folder that contains the details of the generation process. The name of the generation audit report is the name of the generated scenario file appended with '_GenAuditReport'. For example, if the scenario file being generated is `YYYYMMMonthlyRWSETScenarios_1_1.csv`, the generation audit report will be named `YYYYMMMonthlyRWSETScenarios_1_1_GenAuditReport.csv`.

The generation audit report is structured with initial sections for the main generation parameters, economy data structures, and initial assumptions and values, including the mean reversion point, and generation and correlation parameters. Then, for the scenario number specified in the `gen_detail_scen`, detail is shown for the projection of the interest rates, for each index, and for the inflation rate. The final results for the scenario file are displayed at the end of the file, taking into account which indexes are to be included, and whether the economic environments are captured on a monthly or annual frequency.

6.6 Real world mean reversion generation

6.6.1 Overview

Mean reversion scenario rates generation is the process of creating a single scenario with rates that grade from the economic conditions as defined by the current outer loop scenario at the branching date to rates specified by the user.

6.6.2 Scenario rates

Specify the outer loop, historic rates, inner loop scenarios rates, and the inner loop reuse definition for the generation under the Scenarios: Rates category of the Economy input page.

For a real world mean reversion generation, the outer loop scenario file contains the rates which become historic rates for periods before the valuation period, the historic rates contain the rates which are the historic rates for the periods before the projection date, and the inner loop scenario file contains the index rates which are reused should the `inner_loop_valn_scen_reuse_defn` be set to 'Index Rates Only'.

A selection of 'All Rates' for the `inner_loop_valn_reuse_defn` is not appropriate for this generation approach and is treated the same as a selection of 'None'.

6.6.3 Generation parameters

Specify scenario generation parameters as follows for the real world mean reversion generation process.

Variable Name	Value
gen_defn	'Normal Scenarios'.
gen_approach_defn	'Mean Reversion' if the index income return and growth rates should revert to user specified values. 'Mean Reversion With Index Spread' if the index growth rate should revert to a spread from a user specified 'Govt' yield rate.
gen_basis_defn	'Real World'.
gen_mths	Number of months of rates to generate from the branching date.
gen_rates_freq	'Monthly' to capture monthly economic environments into the generated file. 'Annual' to capture annual economic environment into the generated scenario file.
gen_mean_reversion_mths	Number of months to grade from the economic conditions at the branching date to the user specified mean reversion rates. May be: '0' for immediate reversion to specified rates. A value larger than 'gen_mths' for no mean reversion.
gen_detail_defn	'Yes' if you wish to create the generation audit report. 'No' otherwise.
gen_detail_scen	An integer number specifying the generated scenario to include in the generation audit report. This value should always be 1 when generating the real world mean reversion scenario.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.6.4 Economy data

Specify economy data as follows for the real world mean reversion generation process.

Variable Name	Value
econ_data_yld_curve_selected	Specify Govt for this variable and the Govt yield curve must exist in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file. No other yield curves should be specified.
econ_data_yld_curve_term_selected	Specify the full 10 point terms of the 'Govt' yield curve as 0.25, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 20.0, 30.0 for this variable.
econ_data_yld_curve_vol_mr_term_selected	Specify the full 10 point terms of the 'Govt' yield curve as 0.25, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 20.0, 30.0 for this variable.
econ_data_yld_curve_mr_rates	Mean reversion rates required for the real world mean reversion generation process. The mean reversion rates typically vary by yield curve id ('Govt'), term to maturity and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of yld_curve_id and rate_term as column lookup keys and the 'rates' submodel column date as the row lookup key. Interpolation is typically used for missing rate terms.
econ_data_index_rate_selected	A comma separated list of indexes to include in the generation process. For example, to include the US and Small indexes, specify US, Small for this variable. The indexes must exist in in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file.
econ_data_index_rate_mr_inc_ret	Mean reversion index income return rates required for the real world mean reversion generation process. The mean reversion rates typically vary by index id ('US'), and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of index_id as the column lookup key and the 'rates' submodel column date as the row lookup key.

Variable Name	Value
econ_data_index_rate_mr_grwth	Mean reversion index growth rates required for the real world mean reversion generation process. The mean reversion rates typically vary by index id ('US'), and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of index_id as the column lookup key and the 'rates' submodel column date as the row lookup key.
econ_data_index_rate_mr_term	Mean reversion index rate risk free terms required for the real world mean reversion with index spread generation process. The index mean reversion risk free terms typically vary by index id ('US'), and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of index_id as the column lookup key and the 'rates' submodel column date as the row lookup key.
econ_data_index_rate_mr_sprd	Mean reversion index rate spreads to risk free rates required for the real world mean reversion with index spread generation process. The index mean reversion spreads to risk free rates typically vary by index id ('US'), and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of index_id as the column lookup key and the 'rates' submodel column date as the row lookup key.
econ_data_misc_rates_selected	A comma separated list of miscellaneous rates to include in the generation process. For example, to include the Infl miscellaneous rate, specify Infl for this variable. The miscellaneous rates must exist in the economic scenario files specified for both the outer and inner loop as well as in the historic rate file.

Variable Name	Value
econ_data_misc_rates	Mean reversion rates for the miscellaneous rates required for the real world mean reversion generation process. The mean reversion rates typically vary by the miscellaneous rates id ('Infl') and perhaps by date. This variable is commonly linked to an external source which uses the hidden 'rates' submodel code variables of index_id as the column lookup key and the 'rates' submodel column date as the row lookup key.

All the variables discussed above are typically linked to external source ranges of the Economy.xlsm workbook using appropriate column and row lookup keys. Please refer to the Economy.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Economy input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

6.6.5 Shock control

Shock control parameters do not apply for real world mean reversion generation.

6.6.6 Calibration criteria

Calibration criteria does not apply for real world mean reversion generation.

6.6.7 Generation process

The generation processing for real world mean reversion scenarios first creates new historical rates, then generates the new real world mean reversion scenarios.

Historic rate derivation

The economic conditions included in the generated historic rates file are the economic conditions which are included in the outer loop scenario file. Any economic conditions which are only included in the specified historic rates file are excluded, since there are no values for those economic conditions included in the outer loop scenario file.

Economic conditions for the generated historic rates are derived at a valuation date by taking economic conditions from the historic rates for periods prior to the projection date and combining those with economic conditions from the current scenario for periods from the projection date to the valuation date.

The economic environments of historic rates are stored monthly for the user specified number of months subsequent to the valuation date. The interpolation method for the generated historic rate file is the interpolation method of the specified historic rate file.

Scenario generation

The economic conditions included in the generated scenario file are the economic condition information specified in the econ_data_yield_curve_selected, econ_data_yield_curve_term_selected, econ_data_index_rate_selected, and econ_data_misc_rates_selected variables which also exist in the specified historic rates file and outer loop scenario file.

For both generation approaches, economic conditions for the generated scenario file 'Govt' yield curve are determined using a linear interpolation between the rates for the current scenario in the outer loop at the valuation date and the user specified mean reversion rates.

When the generation approach is mean reversion, the economic conditions for the generated scenario file indexes are determined using linear interpolation between the rate for the current scenario in the outer loop at

the valuation date and the user specified mean reversion rates. Both an income return and index growth rate are determined by the mean reversion process.

When the generation approach is mean reversion with index spread, the economic conditions for the generated scenario file indexes are determined as a user specified spread from the user specified term of the risk free yield rate. Under this approach, only an index growth rate is determined by the mean reversion process.

However, if the `inner_loop_valn_scen_reuse_defn` variable is 'Index Rates Only', then the economic conditions for generated scenario file indexes are captured from scenario 1 of the inner loop scenario file and the mean reversion parameters are ignored.

For both generation approaches, economic conditions for the generated scenario file miscellaneous rates are determined using a linear interpolation between the rates for the current scenario in the outer loop at the valuation date and the user specified mean reversion rates.

The economic environments of scenario rates are stored at the user defined frequency. The interpolation method for the generated scenario file is the interpolation method of the specified outer loop scenario file.

6.6.8 Auditing

The generation of real world mean reversion scenarios can be audited through the real world generation audit reports.

When the `gen_detail_defn` is 'Yes', a comma separated value (CSV) file is created in the 'generated' folder that contains the details of the generation process. The name of the generation audit report is the name of the generated scenario file appended with '_GenAuditReport'. For example, if the scenario file being generated is `YYYYMMMonthlyRWSETScenarios_1_1.csv`, the generation audit report will be named `YYYYMMMonthlyRWSETScenarios_1_1_GenAuditReport.csv`.

The generation audit report is structured with initial sections for the main generation parameters, economy data structures, and initial assumptions and mean reversion values. Then, for the single mean reversion scenario, detail is shown for the projection of the interest rates, for each index, and for the inflation rate. The final results for the scenario file are displayed at the end of the file, taking into account whether the economic environments are captured on a monthly or annual frequency.

7 SUBMODELS STRUCTURE

7.1 Rates submodels

The 'rates' submodel is a standalone instance of the `rates_economy` model class and contains all variables and formulas for supporting economic scenarios functionality. This submodel is a shared component of the U.S Library and is included consistently in all U.S. Library projects.

The rates submodel is a child of the top model for all projections. For portfolio projections (which include business segment calculations), the rates submodel is a child of the company submodel.

For cell pricing or seriatim valuation projections, the rates submodel is a child of the individual liability submodel (such as `ul`) or of the individual asset submodel (such as `bond`).

When the rates submodel is included as a child of an individual liability or individual asset submodel, the 'rates' submodel is identified by the unique name, such as `ul_rates`. Even when these unique names are included, the formulas in the code still reference the 'rates' pointer reference. This is accomplished by setting the generic rates pointer equal to the unique `ul_rates` pointer during the startup processing of the projection.

7.1.1 Variables

Code variables for specifying economic scenarios parameters are contained in the rates submodel.

In addition to the variables defined as code variables, 'rates' class variables are also included. These variables are used to reference the object of the Economy DLL (discussed more below), to collect the names of the economic elements included, and to hold economic rates retrieved from the economic scenario files. The class variables which collect the names of the economic elements included and hold the economic rates retrieved provide efficiency in rate retrieval and reuse.

7.1.2 Formulas

Formulas for managing economic scenarios are contained in the rates submodel. A combination of columns, scalars, and external functions are used to provide the required functionality, with only a very limited number of columns included.

The formulas in the rates/economy model class have been written to provide functionality encapsulation and reuse across both portfolio projections and seriatim projection, as well as for all assets and product types.

Encapsulation and reuse has been accomplished through use of specific interface formulas for the 'rates' submodel. All other submodels interact with economic scenarios through this interface, passing in the parameters required based on the context of the usage.

Key formulas in the rates submodel include:

Startup

The 'startup' column is automatically executed during model initialization. This column does the work to set up the data structures for the economic scenario rates including: (1) instantiation of the Economy object, and (2) manages the loading of historic rates and the initial information of the scenario file by executing the `scen_set_info_load` function.

Initialize

The 'initialize' column is executed when referenced from the top model of the projection. This formula manages the loading of the required scenario by executing the `scen_rates_load` function.

get_xxx

The 'get_xxx' functions provide the interface of the rates submodel to all other components of the projects. These functions are executed to retrieve economic rates from the scenarios.

Distinct functions are used for the different types of economic rates and the parameters of the functions generally specify the time period (t), identifier, rate type, and return basis for the rates to be retrieved. The specific details of the parameters of the functions are provided as comments in the functions.

Generate

The 'generate' column controls the inner loop economic scenario generation process including: (1) setup of Economy object generation parameters by executing 'setup_generation_parameters', (2) setup of Economy object economy data by executing 'setup_economy_data', (3) setup of Economy object calibration criteria by executing 'setup_calibration_criteria', and (4) execution of the generation process by executing the 'Generate' method of the Economy object.

7.2 Economy library

A majority of the economic scenario functionality required for the U.S. Library projects is provided through functionality of the Willis Towers Watson Economy library. This external library provide for additional functionality encapsulation and reuse, simplifying the overall complexity of the solution.

Using an external library for economic scenario functionality is beneficial because economic scenario functionality is considered standard functionality and does not typically require client customization. Thus only the limited amount of data and methods required to interface with the scenario functionality is exposed to the user and to other parts of the projects.

The 'rates' submodel contains a link to the Willis Towers Watson Economy DLL, which is a C++ library for managing economic scenarios data contained in economic scenario files and generating inner loop economic scenario rates.

The economy library provides a limited application programmer interface (API) which includes a stub LIB (Economy.lib), and a DLL (Economy.DLL). The economy library is linked to the rates_economy model class by including the API definitions within the START_HEADER END_HEADER section of the startup. A reference to <global*>Economy.lib, <global*>Economy.dll are also included in the 'DLLs/LIBs' properties of the project.

The rates_economy model class includes references to the classes of the economy library. The economy library objects are instantiated during the startup processing of the model and properties of the economy classes are populated with the required information. The 'Econ' object is the main interfacing object into the economy library.

Economic scenario files are opened using the 'ScenSetInfoLoad' method of the economy class and projected economic scenario files are loaded using the 'ScenLoadProjRates'. Economic rates are retrieved using 'Getxxx' methods of the economy class and inner loop economic scenarios are generated using the 'Generate' method of the economy class.

C03 ASSET MODELING

This book discusses asset modeling for investment assets, and the asset, inv_strategy, and asset_sub_port submodels. This book also discusses asset based reserves, which are calculated in the company and segment submodels.

This book does not discuss modeling of hedging assets which is discussed in the Hedging book.

The book for each asset type discusses individual asset calculations.

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1 INPUT ORGANIZATION

1.1 Terminology

The following terms are used to distinguish between different assets:

- Investment asset – Assets held for the general purpose of generating investment income.
- Hedging asset – Assets held for a specific purpose, such as for index credit or variable annuity GMxB greek hedging.
- Existing asset – Initial (as of the projection date) assets.
- Purchase asset – Assets purchased during the projection.

Policy loans and deferred premium assets are not investment assets. These assets are calculated by each liability model point, and aggregated within each business segment.

1.2 Investment input page

The Investment input page specifies the variables defining asset sub portfolios and investment strategies. This input page is organized using categories:

- Asset Run Control – Asset sub portfolios included in the projection, asset market value calculation frequencies, proxy bond definition (see the Asset processing chapter of the PBA book), asset detail report controls, Andrew Davidson DLL variables (see the Mortgages book), and Intex DLL variables (see the Securitized Assets book)
- Cash – Cash and borrowing interest rates
- Decrements – Assumptions for mortgage defaults, bond sinking fund payments, and exercising of call, put, and prepayment options
- Balance Sheet – Definitions for asset book values
- Asset Sub Portfolios – Assignment of individual assets into asset sub portfolios used for reporting and investment strategy processing
- Strategy – Investment frequency, purchase assets included in the projection, investment strategy definitions, planned investment strategies, strategy changes based on economic conditions, and asset sales strategies
- Purchase Asset Definitions – The definition of the assets purchased during the projection. See the book for each asset type.

1.3 Company input page

Asset based reserve variables are associated on the Company input page. These are AVR and IMR definitions.

1.4 Asset model points

Individual asset variables are defined on the model point records for each asset type. See the book for each asset type.

2 CASH AND BORROWING

2.1 Cash account

2.1.1 *Invested cash*

Each investment strategy specifies a percent of invested assets (excludes hedge assets) to be held in the cash account. The cash account is the first use of investable cash. Available cash flows are first used to restore the cash account to the specified percent. Investment assets are purchased only when there is investable cash remaining.

Specify the initial (as of the projection date) cash balance for the corporate segment and each business segment.

2.1.2 *Cash interest rate*

The cash interest rate is the annual effective yield for a specified term of a specified quality grade, times a multiple, then plus an addition. The cash rate definitions are specified in the Cash category of the Investment input page. The monthly cash rate is

$$(1 + \text{annual rate})^{1/12} - 1$$

2.1.3 *Interim cash*

The cash account is split between invested cash and 'interim cash', which consists of amounts not yet invested due to investment processing thresholds and timing. Interim cash is added to the amount to be invested at the next investment date.

Interim cash earns either the cash interest rate, or the segment asset yield rate so that amounts not yet invested do not reduce investment earnings.

2.2 Borrowing

2.2.1 *Temporary borrowing*

When negative cash flows result in borrowing, invested cash is reduced to zero, and the borrowed balance is projected as negative invested cash. Future positive cash flows are first used to repay borrowing, then to rebuild the cash account, before buying assets.

2.2.2 *Borrowing interest rate*

The borrowing interest rate is the annual effective yield for a specified term of a specified quality grade, time a multiple, then plus an addition. The borrowing rate definitions are specified in the Cash category of the Investment input page. The monthly borrowing rate is:

$$(1 + \text{annual rate})^{1/12} - 1$$

When the initial (as of the projection date) cash account is negative, this is the initial borrowed balance.

3 ASSET SUB PORTFOLIOS

Asset sub portfolios are the foundation for investment strategies. Each investment strategy directs the distribution of investments between asset sub portfolios. Asset sales occur by asset sub portfolio priority. Asset sub portfolio reports show the investments and transactions for each asset sub portfolio.

3.1 Asset sub portfolios selected

Specify the names of the asset sub portfolios in the `asset_sub_ports_selected` variable under the Run Control category on the Investment input page. Formulas set the hidden variable `asset_sub_port_id` using the name of the current asset sub portfolio from this list. Each asset sub portfolio definition variable must use this variable as the lookup key linking to the external source values.

Specify the names of the asset sub portfolios as a comma separated list with no spaces between the names. Spaces are permitted in the sub portfolio names.

The asset sub portfolios must be listed in the hierarchical order for assigning assets to the sub portfolios. See the Asset sub portfolio assignments section, below.

3.2 Asset sub portfolio definitions

Variables under the Asset Sub Portfolios category on the Investment input page define each asset sub portfolio:

- Asset type – Each asset sub portfolio may only contain only one asset type, either: Bond, Mortgage, Securitized Asset, Interest Rate Derivative, Equity, Real Estate, or Index Option.
- Portfolio – Each asset sub portfolio may only contain only Investment or Hedging assets.
- Asset categories – Specify the asset categories assigned to the asset sub portfolio as a comma separated list with no spaces between the names. For example, the value "Govt,AAA,AA,A" specifies that these four asset categories are included in the asset sub portfolio. Specify the default value of ALL to assign all asset categories related to the specified asset type.
- Existing or purchase assets – The asset sub portfolio can be restricted to contain only existing or purchase assets.
- Minimum duration – An asset transfers out of this asset sub portfolio when its duration is less than this minimum. (See the Asset sub portfolio duration measure topic, below.)
- Maximum duration – An asset is not assigned to this asset sub portfolio when its duration exceeds this maximum. This maximum must be the minimum for another asset sub portfolio, so that there are no gaps or overlaps between the duration ranges between asset sub portfolios.
- Minimum size – Individual assets are sold when their statutory book value becomes less than this amount, such as due to defaults, partial sales, and partial prepayments. This assumption avoids calculation time spent processing insignificant assets.
- Include in duration matching? – When this option is No, assets in this sub portfolio are not included in calculating the duration of invested assets, which is used in duration matching investment strategies. This option only affects bonds, mortgages, collateralized assets, and interest rate derivatives. There is no duration calculation for equities, equity index options, and real estate.
- Sales priority – Within each sales class, assets are first sold from the asset sub portfolio with the first priority, such as 1, then progressing to the next priority number, such as 2, and so on. (See the Investment asset sales chapter, below.) When multiple asset sub portfolios have the same priority, assets within these sub portfolios are considered together during sales processing. A priority of 0 means that assets will not be sold from this sub portfolio.

- Rebalancing tolerance percent – When asset sales occur due to rebalancing (see the Investment asset sales chapter, below), assets are sold from asset sub portfolios whose statutory book value of assets exceeds the target percent of total investment assets by this percent. For example, when the target percent is .30 (30%), a tolerance of .02 (2%) causes rebalancing sales when this asset sub portfolio has over 32% of investment assets.
- Maximum monthly turnover rate – Rebalancing asset sales are restricted to be not more than this annual percent of the statutory book value of each asset sub portfolio.
- Minimum monthly turnover rate – Rebalancing processing sells at least this annual percent of the statutory book value of each asset sub portfolio. For example, a portion of equities might be sold regularly in order to realize capital gains.
- Investment expense rate – The investment expense at the end of each month is the statutory book value (plus accrued interest) at the beginning of the month, times this annual expense rate, divided by 12. Investment expenses thus vary by asset sub portfolio. For example, investment expenses might be higher for lower grade bonds.
- Proxy bond asset category – The proxy bond created to replace the individual assets in this asset sub portfolio has this asset category, which must match one of the bond asset categories. Proxy bonds can apply during PBA valuation inner loops. See the Asset processing chapter of the PBA book.
- Proxy bond sales class – The proxy bond created to replace the individual assets in this asset sub portfolio has this sales class, which must be Available for Sale, Trading, Held to Maturity, or Not Assigned.

3.3 Asset sub portfolio assignments

Assets are assigned to the asset sub portfolios by processing through the sub portfolio definitions in the order they appear in the `asset_grps_selected` variable. When an asset meets the definitions of the first asset sub portfolio, it is assigned to that asset sub portfolio. Then processing continues to assign the next asset. When an asset does not meet the definitions of the first asset sub portfolio, the second asset sub portfolio record is processed, and so on until reaching an asset sub portfolio with definitions matching the asset.

An error message is produced when an existing asset cannot be assigned to a sub portfolio at the projection date. Warning messages are produced, and the asset sub portfolio remains unchanged, when a purchase asset is unable to be assigned to an asset sub portfolio during the projection.

Thus it is important to list the asset sub portfolio definitions in hierarchical order. For example, the 'high grade bonds' asset sub portfolio might define that it includes asset categories Govt, AAA, AA, and A. Then the 'low grade bonds' sub portfolio could define that it includes 'ALL' asset categories. The high grade bonds sub portfolio must be tested first, so that it contains the bonds with the defined asset categories, then all other bonds are assigned to the low grade bonds sub portfolio. If these asset sub portfolios were listed in the wrong order, all bonds would be assigned to the low grade bonds asset sub portfolio, with no assets assigned to the high grade bonds sub portfolio.

3.4 Asset sub portfolio duration measure

Asset sub portfolio assignments are dynamic. The individual assets are assigned to the asset sub portfolios each month. When an asset is reassigned to a different sub portfolio, because its duration has changed, its values transfer out of its former sub portfolio and into the new one.

The duration measure is the same for all asset sub portfolios:

- Effective duration – The change in market value when the scenario yield curve is shifted, divided by the 0.0001 shift in the yield curve. This duration measure applies only in months when asset market values are calculated; assets remain in their old sub portfolios in other months.

- **Weighted average life** – The weighted years to each scheduled principal payment. This duration measure applies only in months when asset market values are calculated.
- **Years to maturity** – This duration measure applies in all months, even when asset market values are not calculated.

4 INVESTMENT STRATEGIES

4.1 Investment strategy definitions

4.1.1 Strategy names

Investment strategy definition variables must use the hidden variable lookup key `inv_strat_id`, which is set by the formulas processing the strategy:

- For planned investment strategies, this is the strategy specified for the date currently being projected. See the Planned investment strategies section, below.
- For strategies varying by economic condition, this is the strategy specified for the test number currently being projected. See the Economic condition investment strategies section, below.

4.1.2 Strategy variables

Each of the following investment strategy variables must use hidden variable `inv_strat_id`, which is set by the formulas, as the lookup key:

- **Cash percent** – The targeted cash balance is this percent of the statutory book value of investment assets. The amount needed to restore the cash balance to this level is subtracted from the amount available to invest. When the current cash balance is greater than this level, the excess is added to the amount available to invest.
- **Asset sub portfolios distribution method** – The target percent for each asset sub portfolio applies according to this option. The Cash option specifies that the amount allocated for purchases in each asset sub portfolio is the amount to invest times the percent for that sub portfolio. The Book option specifies for investments to be allocated to result in the statutory book value of assets to have this asset mix between sub portfolios.
- **Asset sub portfolio target percentages** – Specify the target percent (0.2 = 20%) for each asset sub portfolio, using the hidden variable `asset_sub_port_id` as the lookup key link to the external source. The total of the percentages for all asset sub portfolios should add to 1.00 (100%), otherwise the specified values are normalized so they do add to that result.
- **Purchase asset sub portfolio identification and weight** – Specify the sub portfolio identification and weight (0.2 = 20%) within the sub portfolio for each purchase asset, using the hidden variable `purch_asset_id` and a constant value of `SubPortID` and `WghtWithinSubPort` as the lookup key links to the external source. The total of the percentages for all purchase assets within each sub portfolio should add to 1.0 (100%), otherwise the specified values are normalized so they do add to that result.

4.2 Purchase asset definitions

The allocation percentages determine the part of the available cash flow that is used to purchase assets for each asset sub portfolio. You define investment assets to be created by these purchases.

Investment assets differ from existing assets in that they are hypothetical assets, to be created in each potential economic condition in the future. For example, a 10-year bond investment asset may be defined to have a coupon rate based on the current 10-year risk free bond equivalent yield at the time the bond is created. This bond will have a high coupon rate when interest rates are high, and a low coupon rate when interest rates are low.

Assets model point data specifies all the characteristics of the assets already owned, since those characteristics are known with certainty. Investment assets, however, have characteristics based on rules defined in the purchase asset definitions. For example, purchase asset definitions specify that the new bond has a maturity period of 10 years. Then the maturity date of the bond is determined when this bond is purchased. The coupon rate, call and put options, and other characteristics are defined based on economic conditions at that time.

The yield rate for investment bonds, mortgages, and collateralized assets is always based on the asset category definition at time of purchase. You can define the asset's coupon rate to be different from this yield rate. For example, when a bond is to be purchased at a discount, you can specify a positive differential between the investment asset yield and the coupon rate. When a bond is to be purchased at a premium, you can specify a negative differential between the investment asset yield and the coupon rate. To purchase a zero coupon bond, you can specify a zero coupon rate. When the interest rate on a mortgage is specified as a multiple of a risk-free rate plus a spread (e.g. the five year risk-free rate plus 120 basis points), the mortgage is purchased at a premium or discount so that the mortgage's net yield at purchase equals the asset category yield rate at purchase.

For securitized assets, purchase asset definitions are defined for both the securitized asset and its collateral pool. For example, an investment securitized asset defines the tranche owned in terms of percentages of the total collateral principal allocated to sequential and planned/targeted amortization class tranches with payment priority higher, the same, and lower than the tranche owned. Then the purchase asset designated as the mortgage pool backing the securitized asset is defined in terms such as the maturity period and mortgage interest rates. The collateral pool portfolio definition designates that a purchase asset is used to define a collateralized asset, and is not part of the investment portfolio.

It is important to make sure that each purchase asset has characteristics representing the asset sub portfolio it is purchased for. For example, assume the long-term bond asset sub portfolio is defined as bonds having effective duration greater than 10. When a purchase asset for this sub portfolio has an initial duration less than 10, this bond transfers to the intermediate-term or short-term bonds asset sub portfolio at time of purchase. This may or may not be a problem, depending on the interest rate environment at time of purchase. You can check for these transfers by reviewing the Asset Sub Portfolio reports or by reviewing the individual asset detail results.

4.3 Planned strategies

The 'planned' investment strategy applies when the current economic conditions do not warrant use of a special strategy. The planned investment strategy variables specify the name of the strategy to use, plus the names of alternative strategies to use when changes are needed for duration matching and the duration targets and tolerances (see the Duration matching chapter, below). These items use the hidden variable `inv_strat_id` lookup key for reading the investment strategy definition variables.

The planned investment strategy can change over time by using the date column (in the investment strategy submodel) as the lookup key linking to the external source.

4.4 Economic condition strategies

4.4.1 Special strategies

The planned investment strategy can be replaced when the current scenario rates are extreme enough to meet one of the specified 'economic condition' tests. Specify the number of tests in variable `inv_strat_economic_tests`. When the number of tests is the zero default value, this feature does not apply.

The special strategy variables specify the name of the planned strategy to use, plus the names of alternative strategies to use when changes are need for duration matching and duration target and tolerance increases (see the Duration matching chapter, below).

The formulas step through each of the tests by setting the hidden variable `inv_strat_economic_test_num`. This must be used as the lookup key linking to the external source.

For example, when the number of tests is 3, the formulas first evaluate the conditions for test 1, using the specified strategy when these conditions are met. When test 1 fails, the formulas evaluate test 2, using that strategy when the test is passed. When test 2 fails, the formulas evaluate test 3. When test 3 also fails, the planned investment strategy applies.

4.4.2 *Economic condition tests*

Each economic condition threshold is defined using:

- **Test months value** – The value for each rate is either the Average, Minimum, or Maximum of the values for the months in the test period.
- **Number of test months** – For example, the rates could be the average over the past six months.
- **Rates calculation** – The test is either Subtraction (rate 1 – rate 2) or Division (rate 1 / rate 2).
- **Rate 1 term** – The first test rate is the scenario bond equivalent yield for this specified maturity years.
- **Rate 2 term** – The second test rate is the scenario bond equivalent yield for this specified maturity years.
- **Rate 2 lag months** – A value of 0 specifies that rate 2 uses the current month's scenario rates. A higher value specifies to use scenario rates for a previous month. For example, defining rate 1 as 0.25 years, and rate 2 as 0.25 years with a lag of 3, means that the test measures the change in the 90-day risk free scenario rate over the past three months.
- **Comparison condition** – The special economic condition can exist when the comparison calculation is Greater Than or Less Than the fixed test value.
- **Fixed test value** – The comparison condition is tested using this value. Using the above example with a test of Greater Than 0.0050 means the economic condition exists when the 90-day risk free scenario rate has increased by more than 50 basis points (0.50%) over the past three months.

The tests must be processed in the correct hierarchical order when they are not mutually exclusive. For example, test 1 might test for a steeply inverted yield curve (say 10-year rate / 90-day rate < 50%), then test 2 might test for a lesser inverted yield curve (say 10-year rate / 90-day rate < 0%). Obviously, test 2 cannot be tested before test 1 since its conditions are met by both tests.

4.4.3 *Replacement strategy*

When one of the economic conditions is found to be true, the planned investment strategy is replaced by the strategy specified for this test. The alternative strategies for use in duration matching are also replaced.

In addition, the duration matching target and tolerance values can also be changed. The values are increased by the specified amounts; the 'increase' can be negative.

4.5 *Asset purchases*

4.5.1 *Amount to invest*

The amount to be used to purchase investment assets consists of (1) interim cash (cash flows and amounts not previously invested), less hedging asset purchases plus hedging sales. That is, hedging strategies are performed first, then the investment strategy processes the remaining amount to invest.

The cash for asset purchases is the cash available for investment less the amount required to increase the cash account to the target percent of estimated invested assets.

Policy loans and deferred premium assets are automatically handled during the calculations for liability model points. They are not considered to be part of investment assets, and so are not affected by investment strategies. The amount of these assets is determined before determining the amount available to invest.

4.5.2 Sub portfolio allocations

The cash for asset purchases is allocated to the asset subportfolios based on the asset subportfolio distribution method.

When the distribution method is 'Cash', the amount allocated for purchases in each asset sub portfolio is the amount to invest times the percent for that sub portfolio.

When the distribution method is 'Book', the amount allocated for purchases in each asset sub portfolio is the amount necessary to result in the statutory book value of assets to have the desired mix between sub portfolios. A sub portfolio which is already above the desired mix will not have any amount allocated to it.

4.5.3 Individual asset allocations

The cash for asset purchases within each sub portfolio is further allocated to the purchase assets specified for the asset sub portfolio.

The amount allocated to each purchase asset within a sub portfolio is always the amount allocated to the sub portfolio times the purchase asset weight within the sub portfolio.

4.5.4 Size thresholds

To avoid trivial asset purchases, a minimum standard investment amount, specified in the `inv_amt_min` variable on the Investment input page, applies to asset purchases. When the amount of assets that would be purchased, is below this minimum, the amount accumulates as interim cash. Interim cash is added to the cash flow to be invested at the next investment opportunity.

4.5.5 Investing frequency

You can process investment strategies at the end of each month for the first years of the projection, then investing at the end of each calendar quarter for a specified number of years, and then investing at the end of each calendar year for the remainder of the projection. When investing is not performed at the end of a month, the amount to invest is added to interim cash for investing at the next opportunity.

5 NEGATIVE CASH FLOW STRATEGIES

5.1 Negative cash flow strategy definitions

The following characteristics describe the negative cash flow strategy:

- **Negative cash flow strategy definition** – The negative cash flows strategy can be to borrow, purchase negative assets, or sell assets to cover the negative cash flow.
- **Cash balance definition and threshold** – The cash balance can be reduced to a specified threshold, adjusted for borrowing to a threshold, or targeted to equal the cash account balance at the threshold of the investment strategy.
- **Negative assets strategy** – When the negative cash flow strategy is to purchase negative assets, negative assets can be purchased based on any investment strategy which uses an asset sub portfolio distribution method of 'Cash'.
- **Sales selection definition and priorities** – When the negative cash flow strategy is to sell assets, sales can be processed on a pro rata basis or to target certain assets to maximum capital gains, reduce the duration, or increase the duration.

5.2 Planned strategies

The 'planned' negative cash flow strategy applies when the current economic conditions do not warrant use of a special strategy.

The planned negative cash flow strategy variables specify the strategy to use and can change over time by using the date column (in the company submodel) as the lookup key linking to the external source.

5.3 Economic condition strategies

5.3.1 *Special strategies*

The planned negative cash flow strategy can be replaced when the current scenario rates are extreme enough to meet one of the specified 'economic condition' tests. Specify the number of tests in variable `neg_cash_flow_strat_economic_tests`. When the number of tests is the zero default value, this feature does not apply.

The formulas step through each of the tests by setting the hidden variable `neg_cash_flow_strat_economic_test_num`. This must be used as the lookup key linking to the external source.

For example, when the number of tests is 3, the formulas first evaluate the conditions for test 1, using the specified strategy when these conditions are met. When test 1 fails, the formulas evaluate test 2, using that strategy when the test is passed. When test 2 fails, the formulas evaluate test 3. When test 3 also fails, the planned investment strategy applies.

5.3.2 *Economic condition tests*

Each economic condition threshold is defined using:

- **Test months value** – The value for each rate is either the Average, Minimum, or Maximum of the values for the months in the test period.
- **Number of test months** – For example, the rates could be the average over the past six months.
- **Rates calculation** – The test is either Subtraction (rate 1 – rate 2) or Division (rate 1 / rate 2).
- **Rate 1 term** – The first test rate is the scenario bond equivalent yield for this specified maturity years.
- **Rate 2 term** – The second test rate is the scenario bond equivalent yield for this specified maturity years.
- **Rate 2 lag months** – A value of 0 specifies that rate 2 uses the current month's scenario rates. A higher value specifies to use scenario rates for a previous month. For example, defining rate 1 as 0.25 years, and rate 2 as 0.25 years with a lag of 3, means that the test measures the change in the 90-day risk free scenario rate over the past three months.
- **Comparison condition** – The special economic condition can exist when the comparison calculation is Greater Than or Less Than the fixed test value.
- **Fixed test value** – The comparison condition is tested using this value. Using the above example with a test of Greater Than 0.0050 means the economic condition exists when the 90-day risk free scenario rate has increased by more than 50 basis points (0.50%) over the past three months.

The tests must be processed in the correct hierarchical order when they are not mutually exclusive. For example, test 1 might test for a steeply inverted yield curve (say 10-year rate / 90-day rate < 50%), then test 2 might test for a lesser inverted yield curve (say 10-year rate / 90-day rate < 0%). Obviously, test 2 cannot be tested before test 1 since its conditions are met by both tests.

5.3.3 *Replacement strategy*

When one of the economic conditions is found to be true, the planned negative cash flow strategy is replaced by the strategy specified for this test.

5.4 *Borrowing*

When borrowing is used to cover negative cash flow, the cash account is first reduced to zero, and if necessary, a borrowed balance amount is calculated to cover the negative cash flow.

5.5 *Negative asset purchases*

5.5.1 *Strategy selection*

When negative assets are purchased to cover negative cash flow, the specified negative asset strategy is selected.

5.5.2 *Processing sequence*

The following process applies when negative assets are being purchased to cover the negative cash flow:

1. Reduce the cash account – The cash account is first reduced to a minimum amount plus a minimum percent of estimated investment assets.
2. Borrow – When there is still a remaining negative amount, funds are borrowed up to a maximum amount plus a maximum percent of estimated investment assets. (Specify the threshold using the option to Borrow to Threshold rather than only to Reduce Cash Balance to Threshold.)
3. Purchase negative assets – Finally, negative assets are purchased to cover any remaining negative amount using the same approach as investment assets are purchased.

5.5.3 *Size threshold*

The same size threshold used for investment asset purchases applies to negative asset purchases.

5.5.4 *Investing frequency*

The above process normally occurs at the end of any month when cash flow is negative. When investments are only being purchased quarterly or annually, the above process applies at the end of each calendar quarter or calendar year. When investments are purchased only quarterly or annually, the entire cash flow (positive or negative) accumulates as interim cash until the end of the quarter or year.

5.5.5 *Future treatment*

The negative assets purchased through the negative assets investment process are treated as held to maturity assets and are not included in any future negative cash flow sales, rebalancing, or duration matching.

5.6 *Negative cash flow sales*

5.6.1 *Processing sequence*

The following process applies when assets are sold to cover the negative cash flow:

1. Reduce the cash account – The cash account is first reduced to a minimum amount plus a minimum percent of estimated investment assets.

2. Borrow – When there is still a remaining negative amount, funds are borrowed up to a maximum amount plus a maximum percent of estimated investment assets. (Specify the threshold using the option to Borrow to Threshold rather than only to Reduce Cash Balance to Threshold.)
3. Sell assets – Finally, assets are sold to cover any remaining negative amount.

The above process normally occurs at the end of any month when cash flow is negative. When investments are only being purchased quarterly or annually, the above process applies at the end of each calendar quarter or calendar year. When investments are purchased only quarterly or annually, the entire cash flow (positive or negative) accumulates as interim cash until the end of the quarter or year.

5.6.2 Asset sales process

When assets are sold to cover negative cash flow, they are sold in the following priority order:

1. Every asset has a sales classification, either Available for Sale, Trading, Held to Maturity, or Not Assigned. You can assign a priority for each classification. When assets are sold, assets with a higher selling priority ('1' is the highest) are sold first. For example, setting a priority of '1' for sales classifications of Available for Sale, Trading, and Not Assigned, and a priority of '2' for Held to Maturity results in assets Held to Maturity being sold only as a last resort, after all other assets have been sold.
2. You can also set sales priorities for asset sub portfolios. These are used to differentiate between assets when it is not necessary to sell all assets with the same asset sales classification. For example, when the asset sub portfolios for short-term, medium-term and long-term bonds all have an asset sales priority of '1', these sub portfolios are all considered for sale together. When, however, the sales priority is 1 for short- and medium-term bonds and 2 for long-term bonds, then long-term bonds are only sold after all short- and medium-term bonds have been sold. These priorities between asset sub portfolios refer only to sales of assets with the current highest sales classification priority.

A sales priority of zero indicates that the assets cannot be sold.

5.6.3 Partial sales

When some, but not all, assets within the same sales priority level are sold, individual assets are selected using a method such as:

- Pro rata – the same proportion of each asset is sold
- Longest duration – individual assets are sold in decreasing order of effective duration
- Shortest duration – the assets are sold in increasing order of effective duration
- Maximize capital gains – the assets are sold in decreasing order of capital gains that would be realized

When the amount of any asset remaining after a sale is below the minimum asset size (specified for each asset sub portfolio), the remainder of the asset is sold also. This applies regardless of the sales classification priority. The sale of small assets reduces calculation time by eliminating the processing of trivially small assets.

5.6.4 Size thresholds

To avoid trivial asset sales, the minimum standard investment amount applies for negative cash flows, in addition to asset purchases. When the amount of assets that would be sold, or the amount of negative assets that would be purchased, is below this minimum, the negative amount accumulates as interim cash. Interim cash is added to the cash flow to be invested at the next investment opportunity.

When a partial sale of an asset would result in its falling below the minimum standard investment amount, the entire asset is sold.

5.6.5 Asset sales timing

Before assets can be sold, asset market values must be calculated. To reduce calculation time, you can choose to sell assets only at certain times:

- **At asset frequency** – Assets are sold only when market values are calculated for all assets as specified in the Market value calculation frequency option. When assets cannot be sold, the cash account is first reduced. Then, when the cash account is reduced to zero, borrowing is used.
- **Whenever needed** – Asset market values are calculated whenever asset sales are required to cover negative cash flow. In this case, market values are calculated only for assets in each priority level considered for asset sales.

You can specify in existing assets data that certain assets are to be sold at designated times during the projection. Planned sales are always made at the planned sales date, even when there is positive cash flow.

5.6.6 Profit and tax effect

Since investment asset sales affect investment income, taxes, and distributions for the month, the sales process includes the calculation of a 'capital gains cash factor from sales'. This factor takes into account the amount of gain from the sale that is expected to be paid in taxes or distributions and thus will increase the amount of assets to be sold for a negative cash flow strategy sale over the original cash flow deficit. Using this approach provides for the amount of assets sold to take into account the financial impacts.

Further, investment asset sales lead to a recalculation of the income statement and balance sheet for the month. This is required because the investment income, taxes, and distributions calculated before the sales will be impacted by the sales. As discussed in the submodel structure section below, this special processing requires that 'before sale' amounts be included for a number of financial statement results.

6 REBALANCING

Suppose, for example, that you wish 20% of the book value of assets be maintained in an asset sub portfolio, with a tolerance of 2%. If the actual percent of assets in that sub portfolio becomes higher than 22%, the sub portfolio is over funded and asset should be removed from that sub portfolio. Similarly, if the actual percentage of assets in that sub portfolio becomes lower than 18%, the sub portfolio is underfunded and asset should be added to that sub portfolio.

Rebalancing is the process used to retain the desired mix of assets across asset sub portfolios.

6.1 Rebalance definitions

6.1.1 Investment strategy

Rebalancing can only be performed when the selected investment strategies uses a book value distribution method. This is because, under the book distribution method, the percentages specified for the asset sub portfolio represent the percentage of estimated investment assets to maintain in each asset sub portfolio.

Rebalancing does not apply when the investment strategy uses the cash distribution method.

6.1.2 Sales definitions

While it is true that purchases alone could potentially rebalance the asset sub portfolios to the desired mix, rebalancing is most effective when both sales and investments are used. The following characteristics describe the rebalancing sales features:

- **Rebalance frequency** – The frequency for rebalancing sales.
- **Rebalance sales selection definition and priorities** – Rebalance sales can be processed on a pro rata basis or to target certain assets to maximum capital gains, reduce the duration, or increase the duration.

- **Turnover rates** – Specified as part of the asset sub portfolio, the maximum and minimum turnover rates set a cap and floor on the amount of asset sales for rebalancing.

6.2 Rebalance sales

6.2.1 Sub portfolio selection

Rebalancing sells assets from asset sub portfolios when the proportion of asset book values in the sub portfolio exceeds its allocation percent, plus a tolerance percent. For example, you might specify that 20% of the book value of assets be maintained in an asset sub portfolio, with a tolerance of 2%. When the actual percent of assets exceeds 22%, assets are sold to return the percent to 20%. These proceeds are added to the amount for asset purchases.

When a special investment strategy applies because of the current economic condition, rebalancing can incorporate asset sales to move the asset mix toward the new asset sub portfolio targets.

To reduce calculation time, you can restrict the frequency of rebalancing sales, such as processing rebalancing only annually at the end of each calendar year.

6.2.2 Turnover restrictions

You can restrict the rate of asset sales for rebalancing by setting a maximum turnover rate. The rebalancing sales from each asset sub portfolio are limited to the book value (plus accrued interest and rents) of the asset sub portfolio, times the maximum turnover rate. Thus the movement to the desired asset distribution can be restricted to gradual changes at each opportunity.

You can force the sale of assets from selected asset sub portfolios by setting a minimum turnover rate. Each time that assets can be sold, they are automatically sold within the sub portfolio at an amount equal to the book value (plus accrued interest and rents) of the asset sub portfolio, times the minimum turnover rate. This can be used to realize capital gains gradually by forcing the sale of equities and property (real estate).

6.2.3 Asset sales process

When assets are sold for rebalancing, they are sold in the following priority order:

1. Every asset has a sales classification, either Available for Sale, Trading, Held to Maturity, or Not Assigned. You can assign a priority for each classification. When assets are sold, assets with a higher selling priority ('1' is the highest) are sold first. For example, setting a priority of '1' for sales classifications of Available for Sale, Trading, and Not Assigned, and a priority of '2' for Held to Maturity results in assets Held to Maturity being sold only as a last resort, after all other assets have been sold.
2. You can also set sales priorities for asset sub portfolios. These are used to differentiate between assets when it is not necessary to sell all assets with the same asset sales classification. For example, when the asset sub portfolios for short-term, medium-term and long-term bonds all have an asset sales priority of '1', these sub portfolios are all considered for sale together. When, however, the sales priority is 1 for short- and medium-term bonds and 2 for long-term bonds, then long-term bonds are only sold after all short- and medium-term bonds have been sold. These priorities between asset sub portfolios refer only to sales of assets with the current highest sales classification priority.

A sales priority of zero indicates that the assets cannot be sold.

6.2.4 Partial sales

When some, but not all, assets within the same sales priority level are sold, individual assets are selected using a method such as:

- Pro rata – the same proportion of each asset is sold
- Longest duration – individual assets are sold in decreasing order of effective duration

- Shortest duration – the assets are sold in increasing order of effective duration
- Maximize capital gains – the assets are sold in decreasing order of capital gains that would be realized

When the amount of any asset remaining after a sale is below the minimum asset size (specified for each asset sub portfolio), the remainder of the asset is sold also. This applies regardless of the sales classification priority. The sale of small assets reduces calculation time by eliminating the processing of trivially small assets.

6.2.5 Asset sales timing

Before assets can be sold, asset market values must be calculated. To reduce calculation time, you can choose to sell assets only at certain times:

- **At asset frequency** – Assets are sold only when market values are calculated for all assets as specified in the Market value calculation frequency option. When assets cannot be sold, the cash account is first reduced. Then, when the cash account is reduced to zero, borrowing is used.
- **Whenever needed** – Asset market values are calculated whenever asset sales are required to cover negative cash flow. In this case, market values are calculated only for assets in each priority level considered for asset sales.

You can specify in existing assets data that certain assets are to be sold at designated times during the projection. Planned sales are always made at the planned sales date, even when there is positive cash flow.

6.2.6 Profit and tax effect

Since investment asset sales affect investment income, taxes, and distributions for the month, the sales process includes the calculation of a 'capital gains cash factor from sales'. This factor takes into account the amount of gain from the sale that is expected to be paid in taxes or distributions and thus will increase the amount of assets to be sold over the original cash flow deficit. Using this approach provides for the amount of assets sold to take into account the financial impacts.

Further, investment asset sales lead to a recalculation of the income statement and balance sheet for the month. This is required because the investment income, taxes, and distributions calculated before the sales will be impacted by the sales. As discussed in the submodel structure section below, this special processing requires that 'before sale' amounts be included for a number of financial statement results.

6.3 Rebalance investments

6.3.1 Sub portfolio selection

Rebalancing investments occur in asset sub portfolios when the proportion of asset book values in the sub portfolio is below its allocation percent, plus a tolerance percent. For example, you might specify that 20% of the book value of assets be maintained in an asset sub portfolio, with a tolerance of 2%. When the actual percent of assets is below 18%, assets are purchased in the sub portfolio sold to return the percent to 20%.

When a special investment strategy applies because of the current economic condition, rebalancing can incorporate asset purchases to move the asset mix toward the new asset sub portfolio targets.

6.3.2 Sub portfolio allocations

Since the distribution method must be 'Book' for rebalance investments, the amount allocated for purchases in each asset sub portfolio is the amount necessary to result in the statutory book value of assets to have the desired mix between sub portfolios. A sub portfolio which is already within the tolerance of the desired mix will not have any amount allocated to it.

For rebalancing, the cash for asset investments typically includes the cash for rebalance sales.

6.3.3 *Individual asset allocations*

The cash for asset purchases within each sub portfolio is further allocated to the purchase assets specified for the asset sub portfolio.

The amount allocated to each purchase asset within a sub portfolio is always the amount allocated to the sub portfolio times the purchase asset weight within the sub portfolio.

6.3.4 *Size thresholds*

To avoid trivial asset purchases, a minimum standard investment amount, specified in the `inv_amt_min` variable on the Investment input page, applies to asset purchases. When the amount of assets that would be purchased, is below this minimum, the amount accumulates as interim cash. Interim cash is added to the cash flow to be invested at the next investment opportunity.

6.3.5 *Investing frequency*

You can process rebalance investment strategies at the end of each month for the first years of the projection, then investing at the end of each calendar quarter for a specified number of years, and then investing at the end of each calendar year for the remainder of the projection. When investing is not performed at the end of a month, the amount to invest is added to interim cash for investing at the next opportunity.

7 DURATION MATCHING

7.1 Duration matching definitions

7.1.1 *Alternative investment strategies*

Duration matching applies when alternative investment strategies, specified as part of the planned and economic condition strategies, are different from the specified planned or economic strategies.

The alternate investment strategies are:

- **Shorter** – The first strategy to be used to shorten the duration of assets. The allocation percentages should be higher for asset sub portfolios with purchase assets that have lower effective durations. For example, the allocation percent for the short term bonds asset sub portfolio is typically higher than in the normal strategy. Similarly, the allocation to the long term bonds asset sub portfolio is typically lower than in the normal strategy.
- **Shortest** – A second strategy to shorten the duration of assets in cases where the shorter investment strategy cannot reach the asset duration target
- **Longer** – The first strategy to be used to lengthen the duration of assets
- **Longest** – A second strategy to lengthen the duration of assets in cases where the longer investment strategy cannot reach the asset duration target

The alternative investment strategies may use either the 'Cash' or 'Book' asset sub portfolio distribution method. It is expected, however, that all alternative strategies use the same distribution method and that the method is the same as the planned strategy. When the distribution strategies do not match, and a merge strategy is required, the distribution method of the planned strategy is used.

7.1.2 *Durations targets and tolerances*

When duration matching applies, duration targets and tolerances including economic condition increases to those amounts, specified as part of the planned and economic condition strategies, are used to determine the effective duration target for the asset portfolio.

The duration targets and tolerances for planned strategy definitions are:

- **Duration matching target** – The target effective duration for investment assets for the calendar date.
- **Duration matching tolerance** – The maximum variation from the target effective duration allowable for the calendar date.

The duration target and tolerance increases, for economic condition strategy definitions are:

- **Duration matching target increase** – The increase to apply to the calendar date duration matching target when the economic condition strategy is selected.
- **Duration matching tolerance increase** – The increase to apply to the calendar date duration matching tolerance when the economic condition strategy is selected.

The duration matching targets can be held constant over the projection, when the block of liabilities includes future premiums and future sales of new policies. The targets can be reduced over time when the block of business is closed, and thus aging to shorter durations over time. The special duration matching target and tolerance adjustments can be applied for the special strategies assigned by economic condition, when those conditions would significantly affect the duration of the liabilities.

7.1.3 *Rebalance sales treatment*

When duration matching applies and the planned and alternate strategies use the 'Book' asset sub portfolio distribution method, rebalance sales may be included to make the movement of the asset duration to the target more powerful. Assets are sold from the asset sub portfolios most over the proportion specified by the target percentages. Then, the proceeds of those sales increase asset purchases using the alternative investment strategy.

Duration matching sales definitions are:

- **Duration matching includes making rebalance sales** – Specify 'Yes' to include rebalance sales as part of the duration matching strategy evaluation and final processing. When 'No' is specified, rebalance sales occur before any duration matching strategy evaluation and final processing such that the duration strategy evaluations and final processing will only include asset purchases.

7.2 Duration matching process

7.2.1 *Asset applicability*

Duration matching applies only to positive bonds, mortgages, securitized assets, and interest rate derivatives. All other asset types, including cash, are excluded from duration matching calculations. You can also specify for certain asset sub portfolios to be excluded from duration matching calculations. The calculation excludes asset sub portfolios where the Include in duration matching? option is No.

7.2.2 *Using only new purchases*

When duration matching applies, and the planned and alternative strategies use the 'Cash' asset sub portfolio distribution method such that rebalance sales are not included, the distribution of purchases between asset sub portfolios uses the linear combination of the planned strategy and the alternative strategy that best reaches the target.

For example, assume that the planned strategy would produce an effective duration higher than the duration target plus the tolerance amount. Instead, the allocation percent for each asset sub portfolio is a linear combination of the planned investment strategy and the shorter investment strategy. The linear combination, such as 40% of the planned strategy assets and 60% of the shorter strategy assets, is determined so that the resulting effective duration of assets matches the target.

When buying 100% of the assets using the shorter strategy still does not reach the target, the linear combination uses the shortest strategy and the planned strategy. When buying 100% of the assets using the shortest strategy still does not reach the duration target, the shortest strategy is still used, and the target is not obtained. Duration matching may apply to reach the target at the next opportunity.

7.2.3 *Using rebalance sales and new purchases*

When duration matching applies, and the planned and alternative strategies use the 'Book' asset sub portfolio distribution method such that rebalance sales are included, and the rebalance sales are included as part of the duration matching strategy evaluation and final processing, the following process is used:

1. Rebalancing sales and purchases are estimated using the planned strategy. When this results in the effective duration of assets being within the tolerance range of the target, those assets are sold and purchased.
2. When that asset portfolio would be outside of the tolerance range, the estimation process is repeated, but using the shorter or longer alternative strategy for both the rebalancing sales and purchases. When the target would be reached, the projection solves for a linear combination of the planned and the shorter or longer investment strategies that approximates the duration matching target.
3. When the target would still not be reached, the estimation process is repeated, but using the shortest or longest investment strategy. When the target would be reached, the projection solves for a linear combination of the planned and the shortest or longest investment strategies that approximates the duration matching target.
4. When the target would still not be reached, the projection uses the strategy (shortest or longest) to move as far towards the duration matching target as possible.

Including rebalancing sales in duration matching increases calculation times, since a number of 'virtual' sales and investments must be performed to evaluate the asset portfolio duration under each of the alternative strategies, and possibly a merged strategy. You can offset this by choosing a lower frequency for rebalancing sales, such as annually. This process already includes a calculation time offset since the inclusion of selling assets is so powerful. Thus the asset duration is more likely to be within the tolerance range in the future, so that duration matching processing need not be repeated as often.

7.2.4 *Portfolio duration calculation*

The portfolio effective duration is determined as the aggregate of the individual asset's market value for effective duration times the effective duration divided by the aggregate of the individual asset's market value for effective duration.

The aggregate of the individual asset's market value for effective duration times the effective duration and aggregate of the individual asset's market value for effective duration are first calculated based on the asset portfolio before considering any adjustments from duration matching.

As each alternative strategy is evaluated, a portfolio duration is calculated by temporarily adjusting the aggregate amounts for amounts from expected sales, if applicable, and purchases.

The final portfolio duration is calculated as the aggregate of the individual asset's market value for effective duration times the effective duration divided by the aggregate of the individual asset's market value for effective duration of the final portfolio.

7.2.5 *Duration matching timing*

Since the matching of asset duration targets requires that market values be calculated, asset duration targets are considered only in projection months when asset market values are calculated.

8 OTHER SALES

8.1 Planned and minimum size sales

Automatic asset sales are processed before determining the amount available to invest:

- Assets are sold on the planned sale date specified for each asset model point.
- Assets are also sold when their book values fall below the minimum asset size for their asset sub portfolio.

These sales occur each month, regardless of the market value calculation frequency.

9 ASSET VALUATION RESERVE

Asset based reserves are projected only when assets are included in the projection. The asset valuation reserve (AVR) is calculated for the company and for each segment.

For total company models, the AVR sub-components are calculated for each business segment and the total company. For each sub-component, the difference between the total company amount and the sum of the amount for the business segments is held in the corporate segment.

There is no allocation of the AVR or its effects to report groups.

9.1 Input variables

The AVR contribution, objective, and maximum percentages are model point input variables. Specify these rates by mapping out the percent for each individual asset using the specified rules. For example, bond rates vary by NAIC Securities Valuation Office (SVO) reserve class, mortgage rates vary by the type of mortgage (such as residential, commercial, and farm), and equities vary between affiliate etc. classifications.

9.2 AVR components

The AVR is an amount set aside to address the credit-related (default) and equity risks of the asset portfolio. The AVR is separated into four sub-components:

9.2.1 *Default component*

- Bonds and preferred stocks (includes securitized assets)
- Mortgages

9.2.2 *Equity component*

- Common stocks
- Real estate and other assets

9.2.3 *Negative AVR*

Each sub-component amount can be floored at zero or allowed to be negative.

For total company models, the sub-components for each segment are allowed to be negative even when the option is selected to floor the AVR sub-components at zero. The flooring of the sub-components amounts is performed within the company AVR calculations.

9.3 Initial reserves

The AVR is calculated at the end of each calendar year. Dates in between calculate the AVR for the portion of the current calendar year.

The first step in calculating each AVR sub-component is to make the following adjustments to the AVR from the previous calendar year end. This result is referred to as the initial reserve for each AVR sub-component:

- Add the realized capital gains, net of capital gains tax, for eligible asset sales and call or put options made year-to-date. Eligible assets are equities, equity options, and real estate assets. Other asset types are considered to have gains related to interest rate changes, and thus increase the IMR instead of the AVR.
- Add the amount of the increase in the amount of unrealized capital gains on equities. Note that the realized capital gains on equities from asset sales are included in the first adjustment, above.
- Add the basic contribution for the assets assigned to the sub-component. This is the book value of each asset times the AVR contribution percent on the model point record (see below).
- Subtract the amount of asset defaults.
- Subtract the write-down amounts for foreclosure and restructure of mortgage assets.

9.4 Transfers and adjustments

9.4.1 Objective reserves

The objective reserve for each sub-component is the book value of each asset times the AVR objective percent on the model point record.

9.4.2 Maximum reserves

The maximum reserve for each sub-component is the book value of each asset times the AVR maximum percent on the model point record.

9.4.3 Reserve balance before transfers

The initial reserve for each sub-component is then increased by the AVR amortization percent times the difference between the objective reserve and the initial reserve (positive or negative). This result is referred to as the reserve balance before transfers.

The AVR amortization percent can vary by calendar year. For calculations before the end of the calendar year, the percent is adjusted by the proportion of the calendar year which has currently expired.

9.4.4 Sub-component excess transfers

When the balance before transfers of any one of the four sub-components exceeds the maximum reserve for that sub-component, the excess is transferred to its sister sub-component, but only to the extent that the sister sub-component does not exceed its maximum reserve. For example, when the mortgage sub-component is below its maximum reserve, any excess on the bond and preferred stock sub-component is transferred to the mortgage sub-component to bring it up to its maximum reserve and vice versa.

9.4.5 Sub-component negative transfers

When the balance before transfers of any of the four sub-components is negative and the balance before transfers of its sister sub-component is positive, the negative amount is transferred into the sister sub-component to the extent that the transfer does not reduce the positive balance of the sister sub-component to less than 50% of its balance before the transfer.

This transfer is only processed when sub_component amounts are floored at zero.

9.4.6 Component excess transfers

When the combined reserve balance before transfers of the sub-components, in either the default component or the equity component, exceeds its respective combined maximum reserve, the excess is transferred to the other component, but only to the extent that the other component does not exceed its combined maximum reserve.

The amount transferred out of (into) each sub-component is calculated using a ratio of the excess (deficit) of the sub-component amount after the sub-component transfers.

9.4.7 Adjustments

After all transfers are completed, final adjustments are made so that each sub-component reserve is the greater of a minimum amount (a large negative number or zero when the sub-component is floored at zero) and the reserve after transfers, but not more than the maximum reserve.

9.5 AVR treatment as target capital

The asset valuation reserve (AVR) is considered to be part of target capital. That is, target capital is increased by the AVR amount. This has some effect on book profits, since investment income is allocated to the AVR.

10 INTEREST MAINTENANCE RESERVE

Asset based reserves are projected only when assets are included in the projection. The interest maintenance reserve (IMR) is calculated for each segment. There is no allocation of the IMR or its effects to report groups.

The IMR stores realized capital gains from fixed income assets, amortizing them into investment income roughly over the years when the original assets would have applied. This is a mechanism to maintain the asset yield needed to sustain product values, preventing these gains from increasing current profits at the expense of future yields supporting policyholder values.

10.1 IMR balances

10.1.1 Initial IMR

Specify the IMR balance as of the projection date, along with the projected annual amortization of this initial amount.

10.1.2 IMR increases

Realized capital gains and losses, net of capital gains tax, which are due to interest rate changes are added to the IMR. Realized capital gains and losses from bonds, mortgages, securitized assets, and interest rate derivatives are included in the IMR. The entire realized gain (loss) is considered to be due to interest rate changes, and is thus included in the IMR.

10.1.3 Negative IMR

The IMR can be negative for a particular segment, and the total IMR can be negative.

10.2 Amortization schedules

10.2.1 Initial IMR

The initial (at the projection date) IMR is amortized using the amortization schedule specified in the segment submodel. You can construct this schedule either using the grouped or seriatim amortization schedule methods.

10.2.2 IMR additions

IMR additions from sales during the projection are amortized using a calendar-year amortization schedule based on the time to maturity for the asset that was sold. Grouped amortization schedules specify the percent amortized in each calendar year, beginning with the year of the sale.

The amount amortized in the calendar year of the sale is divided evenly over the remaining months in the year of sale. The amount amortized in each month of later calendar years is the percent amortized in that year, divided by 12.

10.2.3 IMR amortization schedules

The IMR amortization schedule formulas set two hidden variable lookup keys, which must be used to link to the external source table:

- `yrs_aft_sale` – The calendar year of the sale is year '1', the next year is year '2', and so on.
- `yrs_to_maturity` – '0' when the asset has less than one year to maturity at the time of the sale, '1' when the asset has less than two years to maturity, and so on.

The Global folder contains an IMR Amortization Rates workbook, which contains standard IMR amortization schedules for sales in calendar years starting with 2005.

10.3 IMR treatment as target capital

10.3.1 Allocation of IMR impact

The IMR is treated as an addition to target capital. Thus investment income allocated to the IMR is part of investment income on target surplus.

Then IMR amortization is allocated proportionally to reserves, target capital, and free surplus. Thus the IMR functions to spread the capital gain back into investment income over time, similar to results that would have occurred if the asset had not been sold.

For example, consider one asset sale and its capital gain. At the time of the sale, the part of the gain needed to cover the capital gains tax is included in total investment income so that there is no loss from this tax. The remainder of the capital gain increases the IMR, so there is no increase in book profits, distributable earnings, and free surplus from that event. As this part of the IMR is amortized over future months (including the month of the asset sale), this reduction in the IMR increases book profits, distributable earnings, and free surplus. Thus the effect of the IMR is to spread the effect of capital gains and losses over time, instead of recognizing the effect all at the time of the sale.

10.3.2 IMR anomalies

The allocation of investment income 'blows up' when the IMR approaches or even exceeds the value of net invested assets. This can happen in **projections** running off a block of existing business. To prevent such anomalies, the IMR is reduced when it would exceed the assets invested by the segment.

11 SUBMODELS STRUCTURE

11.1 Submodels

The primary asset submodels are:

- **asset** – a standalone instance of the `asset_asset` model class which contains the formulas and variables used for asset based initialization and asset sub portfolio definitions
- **inv_strategy** – a user defined array instance of the `invstrat_asset` model class which contains the formulas and variables used for standard investment, planned and minimum size sales, rebalancing sales, duration matching investment and sales, and negative cash flow sales processing. The number of elements in the array is determined based on the number of segments in the model. There is a one to one mapping of segment elements to the `inv_strategy` elements.
- **asset_sub_port** – a three dimensional user defined array of the `a_subport_asset` model class. The number of elements in the first dimension is the number of segments in the model. The number of elements in the second dimension is determined based on the number of asset sub portfolios defined in the asset sub portfolio definitions table. The third dimension contains three elements; an element for the results of all assets in the sub portfolio, an element for results of existing assets in the sub portfolio, and an element for the results of purchase assets in the sub portfolio. The `asset_sub_port` submodel is primarily a reporting submodel of asset sub portfolio results.
- **bond, mtg, sec, ird, re, eqt, eio, epa** – data driven arrayed submodels used to record asset experience results for individual bonds, mortgages, securitized assets, interest rate derivatives, real estate (i.e. property) assets, equities, equity index options, and externally projected assets. The number of elements in these arrays is initially the number of elements in the corresponding existing asset model point data. The array sizes increase during the projection as new assets are purchased. Each of these submodels is described in the chapter for the related asset type.
- **adco** – a standalone instance of the `adco_asset` model class which contains the formulas and variables for the Andrew Davidson prepayment model. This model is used to calculating prepayments for mortgages and mortgage pools.
- **intex** – a standalone instance of the `intex_asset` model class which contains the formulas and variables for using the Intex databases to project individual mortgage (mtg) and securitized assets.

11.2 Aggregate calculations

11.2.1 Initializations

Aggregate asset initialization processing is handled through the initialize columns of the asset submodel. The initialize column of asset is called by initialize of the company submodel. Important asset initialization columns and formulas include:

- **set_asset_sub_port_map** – set up asset sub portfolio map

Investment strategy initialization processing is handled through the initialize columns of the `inv_strategy` submodel. The initialize column of asset is called by initialize of the segment submodel, through the `setup_inv_strategy` external function. Important investment strategy initialization columns and formulas include:

- **set_purch_asset_maps** – set up purchase asset maps and SmartArrays
- **set_purch_asset_defn** – validate purchase asset definitions and populate purchase asset map
- **validate_asset_data** – checks individual asset and investment strategy data

11.2.2 Segment asset processing

Aggregate asset results are calculated when formulas from the segment submodel request results from the assets by using the special `sum_over_assets` and `sum_over_asset_sub_ports` external functions. These functions loop over the individual assets of the segment for each asset type and accumulate the individual asset's contribution to the segment aggregate amount.

11.2.3 Investment processing

During each projection period, the projection can process refinance, restructure, foreclosure investments, duration matching strategies, and end of period standard investments. These investment processes are controlled by formulas in the `inv_strategy` model.

When assets are purchased, new elements are added to the individual asset type arrays. However, the new elements are not always retained. For example, during duration matching processing, the investment functions are processed in order to estimate the duration achieved by a particular strategy. These estimates are made without actually retaining the purchased assets.

The important columns for investment processing include:

- `foreclosure_cost_basis_net_write_down` – purchases foreclosure investments
- `refinance_cost_basis` – purchases refinance investments
- `restructure_cost_basis_net_write_down` – purchases restructure investments
- `cash_flow_for_inv_bef_rebal` – returns the cash flow for investment before rebalancing.
- `cash_flow_for_inv` – returns the cash flow for investment at the end of the period
- `cash_flow_for_asset_purch` – returns the cash available for asset investment at the end of the period. This amount is the cash flow after adjustment of the cash account.
- `inv_amt` – purchases standard investments at end of period and returns investment amount

Since virtual investments are considered during duration matching processing, external functions have been used for the generic investment processing algorithms. The `save_assets` parameter is used to specify if the new investment assets should be retained. The external functions for investment processing include:

- `buy_assets_inv` – controls the investment process and returns the total cash invested in investment assets
- `buy_purch_asset_inv` – purchases individual investments. The investment asset type array is increased by 1 and the new investment asset is initialized. When assets are not being saved, the asset type array is resized back to its original size to remove the temporary new investment. This function returns the investment amount in a particular investment asset.

In addition to the external functions listed above, there are external functions that control the selection and setup of the investment strategies. These external functions are as follows:

- `inv_strat_select` – selects a specific strategy and initializes variables for that strategy
- `inv_strat_set` – determines the special situation or calendar period strategy for the current projection period

11.2.4 Sales processing

Asset sales can occur due to planned sales or minimum size sales definitions, when the negative cash flow strategy specifies that assets are to be sold, and when rebalancing selling is active. These sales are controlled by formulas in the `inv_strategy` submodel.

The important columns for sales processing include:

- `cash_flow_planned_sale` – handles planned and minimum size sales and returns the estimated net cash from these sales
- `cash_flow_rebal` – handles rebalancing sales and returns the estimated net cash from these sales
- `cash_net_neg_cash_flow_sale` – handles negative cash flow sales and returns the estimated net cash from negative cash flow sales

Since virtual sales can apply while considering duration matching, external functions are used for the generic sales processing algorithms. The `sell_assets` parameter of these external functions specifies if the assets should be sold. The external functions for sales processing are as follows:

- `rebal_sale` – determines the sub portfolio adjustment factors for assets, executes the `rebal_inv_assets_sale` function, and returns the estimated net cash from these rebalancing sales
- `rebal_inv_assets_sale` – handles rebalancing sales for investment assets and returns the estimated net cash from these sales
- `sell_assets_control` – selects the sales function to use based on the type of sale and the investment asset selection criteria. These sales include planned and minimum size sales, rebalancing sales, and negative cash flow sales. The selection criteria include pro rata sales, sales to maximize capital gains, longest duration sales, and shortest duration sales. This function executes the appropriate sales function and returns the estimated net cash from these sales.
- `sell_assets_extreme_value` – sells individual unrestricted assets when assets are sold to maximize capital gains or when the assets with the longest or shortest durations are to be sold, and returns the estimated net cash from these sales
- `sell_assets_for_planned_and_min_size_sale` – sells individual planned or minimum size assets and returns the estimated net cash from these sales
- `sell_assets_pro_rata` – sells individual investment assets when assets are sold pro rata and returns the estimated net cash from these sales

Duration matching against a specified duration target involves consideration of alternative investment strategies. The appropriate investment strategy must be determined before investments are processed. Duration matching processing uses the following columns and external functions:

- `dur_bef_match` – returns the portfolio duration based on the asset portfolio before any duration matching is executed
- `dur_match_tgt` – returns the duration target based on the planned investment strategy for the period
- `dur_match_tolerance` – returns the duration tolerance based on the planned investment strategy for the period
- `dur_match_flag` – returns 1.0 when there are duration matching alternative strategies to process, 0 otherwise
- `use_dur_match_strat` – selects a duration matching strategy that comes closest to hitting the duration target. The selected strategy may be an alternative strategy or it may be created by merging an alternative strategy with the planned strategy.
- `dur_match_decr_dur` – selects a shorter duration matching strategy that comes closest to hitting the duration target when the initial portfolio duration exceeds the duration target by more than the duration tolerance. The selected strategy may be the shorter or shortest alternative strategy or it may be created by merging one of these alternative strategies with the planned strategy.
- `asset_dur_estimate` – estimates the portfolio duration obtained from a particular investment strategy
- `dur_match_incr_dur` – selects a longer duration matching strategy that comes closest to hitting the duration target when the initial portfolio duration is below the duration target by more than the duration

tolerance. The selected strategy may be the longer or longest alternative strategy or it may be created by merging one of these alternative strategies with the planned strategy

- `dur_match_solve_for_merge_strat` – creates a merged strategy by margining an alternative strategy with the planned strategy
- `cash_investment_sale_net_prelim` – estimates the net cash from investment asset sales for duration matching
- `cash_flow_port_rebal_estimate` – estimates the net cash flow from rebalancing sales for duration matching when rebalancing is based on the duration matching strategy

11.2.5 Processing order

The purchase and sale of assets at the end of the month is accomplished in the following order:

1. Minimum size sales and planned sales are processed at the end of the month.
2. Economic conditions are evaluated to determine if an investment strategy assigned by economic condition overrides the standard investment strategy assigned by calendar month.
3. Rebalancing sells assets in asset sub portfolios with too high a proportion of assets.
4. Positive cash flow and interim cash are applied to repay the borrowed balance, then to rebuild the cash account to the percentages specified in the current investment strategy.
5. When the remaining positive amount is less than the minimum standard investment amount, the amount is added to interim cash, and the end-of-month investment strategy processing is completed.
6. When the remaining positive amount is at least as large as the minimum standard investment amount, the standard investment strategy is processed. When duration matching is included, this begins with calculating the investment assets that would be purchased by the normal strategy, and then testing the resulting modified duration of assets against the tolerance range. When that test fails, further calculations are done, first to determine which assets would be purchased by alternative strategies, then by a combination of the normal and alternative strategies. When rebalancing sales are also included in the duration matching process, that testing of the effect of the normal and alternative strategies extends to include calculation of potential rebalancing sales as well.
7. When the remaining amount to invest is negative, the options for reducing the cash account, borrowing funds, selling assets, and purchasing negative assets are processed.

When rebalancing sales are also included in the duration matching process, then testing of the effect of the planned and alternative strategies extends to include the calculation of potential rebalancing sales as well. In this case, calculations loop through steps 4 to 7 described above, without actually selling or buying assets until an appropriate duration matching strategy is obtained. After a strategy has been determined, steps 4 to 7 are processed again, selling and buying assets using this strategy.

11.2.6 Detail reporting

Since many of the calculation for purchases and sales produce interim results which feed the process that adds new assets to the model or reduces the portion of assets in force, understanding the results can be challenging.

Investment, sales, and rebalancing calculation can be examined more closely by using the Create investment strategy results file? option. When this option is activated, a special text file will be created in the column output file folder that contains details on the purchases and sales performed each month from the Detail report start period to the Detail report end period.

11.3 Individual asset calculations

11.3.1 Initializations

Initializations of individual existing assets are handled through the startup and initialize columns of the individual assets. Initializations of individual investment assets are handled through the startup column and an investment asset setup function. These initializations are executed at the beginning of each iteration for all existing asset model points and when a new element is added to the asset type array for an investment asset.

Important columns and external functions for initializing individual assets include:

- initialize – saves selected pre-calculated values at the beginning of the first iteration and reloads these values on subsequent iterations. These values include such things as market values, net yields, market spreads, and anticipated yield values.
- setup_existing_asset – initializes selected existing asset variables
- setup_purchase_inv_asset – initializes selected investment assets variables

11.3.2 Individual asset projections

Individual asset results are calculated when formulas from the asset type model are referred to from formulas in seg, a_grp, inv_strategy, or asset submodels.

In most circumstances, the individual model point formulas are referred to by using the `sm_asset_type_name[mp]->Column(column_number).Value(t)` syntax inside the `sum_over_assets` external functions. However, there are cases where the formulas refer the individual asset type formulas directly using the `sm_asset_type_name[mp]->column_name(t)` syntax.

During each period, cash flow items and balance sheet results are calculated assuming the asset is held for the entire period. When trying to understand the individual asset formulas, you may find it helpful to start with the ‘_before_sale’ columns because the calculations for existing and investment assets generally begin with these values.

11.3.3 Sales processing

Individual assets may be sold in part or in total and may be sold for various sales types. In order to accommodate this, the balance sheet items must have separate columns for each possible instance of the asset. The instances are differentiated by using suffixes based on the sales types in the column name. Using an example of book value, the columns names are:

- bk_val_bef_sale – book value before any sales
- bk_val_bef_fund_match_sale – book value after planned sales but before liability hedge sales
- bk_val_bef_port_mgmt_sale – book value after planned sales and fund value hedge sales but before rebalancing sales
- bk_val_bef_neg_cash_flow_sale – book value result after planned, fund value hedge, and rebalancing sales but before negative cash flow sales
- bk_val – final book value at the end of the month after sales processing is complete

The `sold_pct_xxx` columns for individual assets do not contain formulas. Values are forced into these columns during asset sales processing in the asset submodel. For example, when an asset is sold for planned sales, the column `portion_of_asset_sold_planned_sales` is given a value of 1.0 by the appropriate sales external function in `asset/asset`.

Because of this forced processing, care must be taken when referring to instances of the balance sheet results from formulas in `asset/asset`. As an example, consider the case where the individual asset's book value is required for determining the percent of assets in a sub portfolio before rebalancing calculations. That

code cannot refer to the `bk_val` column from the individual asset as those columns are results after all sales have occurred and would result in incorrect results. The correct column to refer to is the column `bk_val_bef_sale`.

11.3.4 Intex assets

When the Use Intex databases? option is Yes, each mortgage and securitized asset record found in the Intex database is pre-projected, and the results are stored in arrays of the model point.

The process begins in the initialize column of the mortgage and securitized asset model points. When the assets are available in the Intex database, the column `intex_tgt` is executed over the projection period. The projected cash flows and balance sheet items for the asset are stored in a arrays so they can be referenced in the regular projection.

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C04 GAAP ACCOUNTING

This book discusses GAAP valuations and GAAP results in experience projections.

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1 GAAP FUNDAMENTALS

1.1 Terminology

Factors – GAAP results calculated from issue for each model point on a per unit or policy basis.

Schedules – GAAP results calculated for each cohort, which are used to roll forward cohort values between GAAP valuation dates. The first schedule is calculated at the projection date. Each schedule applies until a new schedule is created at the next unlocking date.

Cohort – A group of model points used in calculating GAAP results. Each cohort is defined as the model points issued within specified calendar years for specified plan codes.

Unlocking – GAAP schedules can be unlocked, that is, replaced by a new schedule calculated at the unlocking date. Unlocking is performed using a valuation projection. Unlocking occurs when significant changes have occurred since the current schedule was created, such as: (1) experience has emerged differently than expected, or (2) best estimate experience assumptions have changed.

DAC – Deferred Acquisition Costs (DAC) smooth GAAP profit results by deferring up front commissions and expenses by adding them to the DAC instead of including them in profit calculations. They are then included gradually in profits as they are amortized according to the GAAP schedule.

Loss recognition test – A test performed on a group or block of business using best-estimate assumptions to determine if the net liability (reserves less DAC) is sufficient to cover the minimum liability requirements. For SFAS 60, SFAS 97 limited pay method, and SFAS 120, a gross premium valuation (i.e. present value of benefits and expenses less present value of gross premiums) using current best-estimate assumptions is compared to the net GAAP liability; if the gross premium valuation is greater than the net liability, then the DAC asset must be written down and, potentially, additional reserves may be required. For Retrospective deposit SFAS 97, loss recognition limits the DAC to the present value of future gross profits, using the best estimate of the earned rate. A company may group contracts at a higher level for loss recognition testing than the level of aggregation used for recoverability testing.

Accounting statements terminology – This document references GAAP standards and guidance by the pre-codification accounting statements (e.g. SFAS 60, SFAS 97). Below is a cross-reference from the relevant pre-codification statements to the primary Accounting Standards Codification (ASC) Topics.

Pre-Codification	Primary ASC Topic
SFAS 60	944
SFAS 91	310
SFAS 97	944
SFAS 113	944
SFAS 115	320
SFAS 133	815
SOP 95-1	944
SOP 03-1	944

YRT adjustment – A YRT adjustment, referred to as a YRT asset, is established as an asset within gross cohorts when YRT reinsurance is included. Although referred to as an asset, the amount may be positive or negative.

Recoverability test – A test performed at cohort inception to determine if there are sufficient future profits to amortize the initial DAC asset from an economic perspective. If the DAC is not recoverable, the acquisition expenses are reduced until the DAC is recoverable. Recoverability tests are profitability tests of a group of insurance contracts in a given year. This test is performed only in the year of issue.

Valuation – A projection to develop GAAP factors or schedules.

Point-in-time valuation – A valuation performed at the current valuation date ($t = 0$).

Nested valuation – A valuation performed at each inner loop valuation date during a nested projection.

1.2 GAAP calculation types

1.2.1 GAAP valuations

GAAP valuations are projection tasks which are driven by a specific 'valn_tgt_gaap_xxx' target column. Each valuation task produces either:

- Factors per unit or policy (SFAS 60 method, SFAS 97 limited pay method), or
- Schedules for each cohort (SFAS 91 investment contracts method, Retrospective deposit SFAS 97 which include results from SOP 03-1 additional liability and SFAS133 valuations, and SFAS 120).

GAAP valuation projections are liability-only calculations. They can be deterministic or stochastic. They can be seriatim projections or portfolio projections.

1.2.2 Gross premium reserve valuation

Gross premium reserve (GPR) valuations used for GAAP SFAS 60, SFAS 97 limited pay method, and SFAS 120 loss recognition testing are projection tasks which are driven by a specific 'valn_tgt_gaap_gpr_xxx' target column.

GPR valuations are liability-only seriatim valuations. They calculate the GPR per unit for each future month, typically using the same assumptions as for the experience projection.

Loss recognition testing cannot be performed directly by this task, since the tests require calculations by cohort. The GPR valuation results are used by the experience projection to perform loss recognition testing.

1.2.3 Experience projection

GAAP results are calculated in experience projections when the Project GAAP results? project variable (in the Run parameter set) is Yes, and the applicable GAAP valuation results file is identified in the sfasxxx_results variable (on the Company input page under the appropriate GAAP method category).

GAAP projections include statutory results, since distributions are based on statutory distributable earnings, which are a component of GAAP equity. Thus there is no option to project only GAAP results.

Including GAAP in an experience portfolio projection requires results from GAAP valuations and gross premium reserve projections (when loss recognition is included) to be created in advance of the experience projection task. The output files from these valuations are referenced as an input into the GAAP projection.

When loss recognition is included, the testing and adjustments for loss recognition occurs in the experience projection task for each cohort

1.3 GAAP cohorts

1.3.1 Applicability

Cohorts affect the reporting of GAAP results, which are aggregated by cohort, for all GAAP methods. Each cohort produces a set of GAAP results from the projection date for the aggregate model points collected into the cohort.

There is no limit to the number of cohorts. Cohort definitions must match between the GAAP experience projection and each of the GAAP and GPR loss recognition valuations supporting it.

GAAP cohorts are required for valuation projections for the SFAS 91 investment contract method, the SFAS 97 retrospective deposit method (including the SOP 03-1 additional liability and SFAS 133 embedded derivative calculations), and for the DAC and terminal dividends calculations for SFAS 120.

For seriatim experience projections (when calculations are by model point rather than by cohort), each model point represents a separate GAAP cohort. For example, cell pricing projections calculate GAAP results for each model point separately.

1.3.2 Cohort names

The cohort IDs must be specified in the `gaap_cohorts_selected_sfasxxx` variables (one for each GAAP method) on the Company input page under the Model Setup: Segment category. Each cohort ID must be separated by a comma only, with no spaces between the cohort names. (The cohort names themselves can include spaces.)

The following example shows that the SFAS 60 cohorts are Cohort 1, Cohort 2, Cohort 3, and Cohort 4.

VarID	assump_set_id seg_id	GAAP Corp	GAAP Seg
Model Setup: Segment			
	<code>gaap_cohorts_selected_sfas60</code>		Cohort 1,Cohort 2,Cohort 3,Cohort 4

Note that there are no cohorts for the corporate segment; cohorts only apply business segments.

Formulas use these cohort names to set the hidden variable lookup key `cohort_id`, which is the link to external source tables for variables defining values for each cohort. Of course, the cohort names in the external sources must match the specified cohort IDs.

1.3.3 Cohort assignments

Model points are assigned to cohorts by issue year and plan code:

- Issue years – Specify a comma and a space between each year, such as ‘2009, 2010, 2011’. Do not use a dash to try to indicate a range of years, since ‘2009 – 2011’ would be interpreted to include only issue years 2009 and 2011, with 2010 not being assigned.
- Plan codes – Specify a comma separated list of plan codes, with no spaces between the names, such as ‘Plan A,Plan B,Plan C’.

These variables must be linked to an external source varying by hidden variable look up key `cohort_id`.

Specify the cohort definitions using the `sfasxxx_cohort_issue_yrs` and `sfasxxx_cohort_plan_codes` variables. These variables must be linked to an external source that varies by the hidden variable lookup key `cohort_id`. The `cohort_id` names must match the list specified in the corresponding `gaap_cohorts_selected_sfasxxx` variable.

Model point is assigned by testing each cohort in the order listed in the `gaap_cohorts_selected_sfasxxx` variable. When the model point matches the definition for the first cohort, it is assigned to that cohort. Otherwise the model point is tested against the definition for the second cohort, and so on until a match is found.

Every model point must be assigned to a cohort. Specify “ALL” to assign all other issue years and plan codes to a cohort.

The following example shows that:

- Plan codes Plan A, Plan B, and Plan C issued in 2009 and 2010 are assigned to Cohort 1.
- Plan codes Plan A, Plan B, and Plan C issued in 2011 and 2012 are assigned to Cohort 2.
- Plan codes Plan D, Plan E issued in 2009 and 2010 are assigned to Cohort 3.
- Plan codes Plan D, Plan E, issued in 2011 and 2012 are assigned to Cohort 4.

VarID	cohort_id	Cohort 1	Cohort 2	Cohort 3	Cohort 4
sfas97rd_cohort_issue_yrs		2009, 2010	2011, 2012	2009, 2010	2011, 2012
sfas97rd_cohort_plan_codes		Plan A,Plan B,Plan C	Plan A,Plan B,Plan C	Plan D,Plan E	Plan D,Plan E

1.3.4 Cohort variables

Some GAAP methods have variables that vary by cohort. For example, as shown above, the interest rate defined for the fixed indexed annuity valuations vary by cohort. Other examples are the initial loss recognition DAC adjustments for SFAS 60 and SFAS97 limited pay and the net reserve and benefit reserve constant yields for the SFAS 91 GAAP method.

1.3.5 Reinsurance treatment

Separate cohorts are used for gross versus ceded GAAP results in the case of coinsurance and modified coinsurance.

For YRT reinsurance, a YRT cost amount is included in the gross cohort, and a YRT reinsurance adjustment (referred to as a YRT asset which can be positive or negative) is established for future reinsurance costs.

1.4 GAAP factors

1.4.1 Applicability

GAAP factors apply for GAAP methods which develop results from issue with locked-in assumptions. These methods are SFAS 60 and SFAS 97 limited pay, plus the net level premium reserve component of SFAS 120.

1.4.2 Factors basis

GAAP factors are monthly values from issue on either a per unit or per policy basis along with the units or policies on which the factors were created.

1.4.3 Reinsurance treatment

Separate factors are developed for gross and ceded GAAP results in the case of coinsurance and modified coinsurance.

For YRT reinsurance, a YRT cost amount is included with the gross results, and a YRT reinsurance adjustment (referred to as a YRT asset which can be positive or negative) factor is established for future reinsurance costs.

1.4.4 Valuation projections

Factors are calculated **from issue** assuming one unit of insurance is issued for term and traditional life products, and one policy is issued for payout annuity and health products. Factors are produced for all model points included in the model point file.

GAAP factor valuation results are only produced when the product submodel is the top model.

1.4.5 Factors history

GAAP factors are always produced from issue. There is no mechanism to specify historical actual amounts into the calculations for the GAAP factors.

Historical DAC can be adjusted to actual levels by using the expense adjustment multiple variable (on the Expenses input page for each product type) to adjust historical expenses. See the Expenses section of the Product benefits and values chapter of the book for each product type.

1.4.6 Factors storage

Factors are stored in the output file created by the GAAP factors valuation task. The values are stored for each model point, per unit/policy in force using the GAAP valuation assumptions.

1.4.7 Experience projections

The experience projection task reads the GAAP factors when the valuation output file is named as an assumption in the appropriate GAAP submodel. The per unit/policy GAAP factors are applied to the units/policies in force, which are projected using the experience projection assumptions.

The experience GAAP amounts are the per unit/policy GAAP factor amounts, times the units/policies in force for the experience projection.

1.4.8 Loss recognition testing

Specify to include loss recognition testing in a GAAP experience projection when the `sfasxxx_loss_recognition_defn` variable is Yes, and the name of the GPR loss recognition valuation task is named in the `sfasxxx_gpr_results` variable (on the Company input page, under the category for each GAAP method using GPR valuation results).

Loss recognition events reduce the DAC amount by applying a cumulative reduction percent to the GAAP factors. The stored factors do not change.

Loss recognition testing can be performed for in force cohorts at the projection date, and can be performed at each inner loop valuation date during an experience projection using nested processing.

1.5 GAAP schedules

1.5.1 Schedule applicability

GAAP schedules apply for GAAP methods which develop results from the projection date. These methods are SFAS 91 investment contracts method, SFAS 97 retrospective deposit (including SOP 03-1 and SFAS 133) and the DAC and terminal dividend calculations for SFAS 120.

1.5.2 Schedule basis

GAAP schedules contain monthly schedule values from the projection date for the in-force model points assigned to each cohort.

1.5.3 Reinsurance treatment

Separate schedules are used for gross and ceded GAAP results in the case of coinsurance and modified coinsurance.

For YRT reinsurance, a YRT cost amount is included in the gross schedule, and a YRT reinsurance adjustment (referred to as a YRT asset which can be positive or negative) is established for future reinsurance costs.

1.5.4 Valuation projections

GAAP schedule valuations are a projection from the projection date for in force model points only.

Schedule calculations **exclude** new business model points with issue dates after the schedule valuation date. Thus, the GAAP schedule values exclude future issues. Those model points (issued in the projection after the schedule was created) will be included in the new GAAP schedule created for the cohort at the next valuation date.

For example, a portfolio pricing projection performs the first cohort calculations at the first issue date in the projection, since GAAP results must be produced starting that month. The model points (in the cohort) issued

at later dates increase the capitalized expenses, but don't affect the amortization pattern until after the first inner loop valuation date after their issue dates.

Typically GAAP projections including new business should include an inner loop valuation for December of each calendar year, so that all issues from that calendar year are included in the GAAP schedule for the cohort representing that calendar year of issue.

1.5.5 Schedule history

GAAP schedules are developed from the projection date, but may require actual amounts from the start of the cohort. When actual amounts are required from the start of the cohort, amounts for each month since inception of the cohort are read from history variables. These amounts are read into the schedule valuation for periods before the projection date, and combined with projected amounts to determine the schedule values.

Separate history is used for gross and ceded GAAP results in the case of coinsurance and modified coinsurance.

For YRT reinsurance, a YRT cost amount is included in the gross cohort, and a YRT reinsurance adjustment (referred to as a YRT asset which can be positive or negative) is established for future reinsurance costs.

1.5.6 Schedule storage

GAAP schedules are stored in the output file created by the GAAP schedules valuation task. The values are stored for each cohort.

1.5.7 Experience projections

The experience projection task reads the GAAP schedules when the valuation output file is named as an assumption in the appropriate GAAP submodel. The schedule amounts are used directly in the experience projection assumptions.

The experience GAAP amounts are the GAAP schedule amounts rolled forward using schedule amounts and actual experience.

1.5.8 Loss recognition testing

Specify to include loss recognition testing in a GAAP SFAS 97 retrospective deposit valuation or SFAS 120 experience projection when the `sfasxxx_loss_recognition_defn` variable is Yes. For SFAS 120, the output file name for the GPR loss recognition valuation task is named in the `sfasxxx_gpr_results` variable (on the Company input page, under the category for each GAAP method using GPR valuation results). For SFAS 97 retrospective deposit, the historical loss recognition amounts are included in the schedule history.

Loss recognition testing can be performed for in force cohorts at the projection date, and can be performed at each inner loop valuation date during an experience projection using nested processing.

1.5.9 Unlocking

Unlocking is automatically performed for in-force cohorts at the projection date, and can be performed at each inner loop valuation date during an experience projection using nested processing.

Unlocking is performed for each GAAP cohort.

1.6 GAAP financials

1.6.1 GAAP capitalized amounts

Amount capitalized for GAAP financials include capitalized acquisition expenses, commissions, and sales inducements, such as premium bonus.

When new business is included, capitalized amounts for the new business issued are included in the GAAP financials as the business is issued.

1.6.2 GAAP increase in liability

Depending of the GAAP method and accounting approach, the increase in GAAP liability for the financial statement is based on the GAAP factor or schedule liability amounts, produced by a GAAP valuation task. Adjustments for experience amounts, such as fund values and units or policies may be used to determine the experience liabilities.

When new business is included, the liabilities for the new business are included in the GAAP financials as the business is issued. When the liabilities are based on schedule amounts, there is no accrual of the liability for the amounts from the new business. This is because the liability accrual is based on schedule amounts which did not include the new business. The accrual of the liability will be determined in the next valuation, when the new business model points are included in the valuation as in force model points.

1.6.3 GAAP DAC amortization

The amortization of DAC is based on the GAAP factor or schedule DAC amounts, produced by a GAAP valuation task.

When new business is included and the DAC amortization is based on schedule amounts, there is no DAC amortization for the capitalized amounts from the new business. This is because the DAC amortization is based on schedule amounts which did not include the new business. The amortization of the new capitalized amounts will be determined in the next valuation, when the new business model points are included in the valuation as in force model points.

1.6.4 GAAP income before taxes

GAAP income before taxes includes investment income and realized capital gains allocated to cash flow, GAAP net liabilities, and GAAP required capital.

The Asset accounting chapter, below, discusses the value of GAAP assets for the balance sheet versus calculating income.

1.6.5 GAAP income after taxes

GAAP earnings after taxes are GAAP earnings before taxes, less actual taxes paid, and less the increase in the deferred tax liability.

1.6.6 GAAP deferred tax liability

The GAAP deferred tax liability is the current tax rate times: (1) the tax reserve net of reinsurance, less (2) future tax capitalized premium (DAC tax) amortization, less, (3) the GAAP liability net of deferred acquisition costs (DAC), and plus (4) the GAAP balance sheet value of assets less the value of assets for calculating GAAP income.

The tax reserve is reduced by the tax basis deferred net premium asset, and is increased by the tax basis dividend liability.

The Asset accounting chapter, below, discusses the value of GAAP assets for the balance sheet versus calculating income.

1.6.7 GAAP required capital

GAAP required capital is: (1) GAAP invested assets including policy loans and hedging assets, less (2) the GAAP liability net of DAC, less (3) the deferred tax liability, and less (4) GAAP free surplus.

GAAP free surplus is equal to statutory free surplus.

1.6.8 GAAP ROE

GAAP return on earnings (ROE) is calculated by the report, not during the projection. The GAAP ROE is determined as $2I / (A + B - I)$, where:

- I = GAAP income for the reporting period
- A = GAAP required capital at the beginning of the reporting period, and
- B = GAAP required capital at the end of the reporting period.

The reported GAAP ROE is constrained to be between -99.99% and 99.99%

2 SFAS 60 METHOD

2.1 Applicability

SFAS 60 GAAP accounting applies for term life, traditional life, and long term care products with premiums paid to maturity.

SFAS 60 valuations create GAAP factors using assumptions locked in at issue. Loss recognition is performed by a separate GPR valuation, which is applied for each cohort by the experience projection.

The factors are calculated by a serialtim projection task with target column `valn_tgt_gaap_sfas60_sfas97lp`. SFAS 60 GAAP factors are on-the-fly calculations at the projection date, from the issue date of each model point. That is, SFAS 60 GAAP valuations are not performed at inner loop valuation dates during nested projections.

2.2 Factors

SFAS 60 GAAP factors are calculated for each model point from issue, using the GAAP assumptions determined at issue. The exception is that the GAAP factors are replaced by the loss recognition gross premium reserve after an extreme loss recognition event that writes down all of the remaining DAC asset and still has more losses to account for. See the Loss recognition section below.

SFAS 60 GAAP factors are stored by model point. GAAP factor results are converted into results per unit or policy in force based on the units/policies projected during the GAAP valuation task. Then the GAAP results for the experience projection are these amounts per unit/policy, times the units/policies projected during the experience projection.

The GAAP factors are these amounts per unit/policy in force:

- DAC balance, both gross and ceded coinsurance
- Benefit reserve, both gross and ceded coinsurance
- Maintenance expense reserve, both gross and ceded coinsurance
- YRT reinsurance adjustment asset.

In addition to the factors, the GAAP interest rate is also captured so it will be available for use in determining the GAAP claim reserve (see the Health reserves section below).

2.3 Assumptions

2.3.1 Separate GAAP assumption variables

SFAS 60 GAAP amounts are calculated using assumptions determined at issue. Assumption variables operate the same for the GAAP factors valuation task as for the experience projection task.

For health products, benefits for SFAS 60 calculations may be specified, or experience assumptions with provisions for adverse deviation may be used.

2.3.2 Provisions for adverse deviation

SFAS 60 GAAP assumptions include provisions for adverse deviation (PAD). These are separate input variables so that the PADs are explicitly visible:

- Asset yield multiple and addition -- applies to the effective annual asset yield
- Mortality multiple and addition -- applies to the annual mortality rate, after all improvements and adjustments. These are 'mortality' variables, which operate like to base mortality table variable; see the Mortality variables section in the Input organization chapter of the book for each product type.
- Lapse multiple and addition -- applies to the annual lapse rate, after dynamic and excess lapses, but before shock lapses
- Claim cost multiple and addition (health only)
- Incidence multiple and addition (health only) -- applies to the annual morbidity rate, after all adjustments
- Termination multiple and addition (health only)
- Maintenance expense per unit multiple and addition (life only)
- Maintenance expense per policy multiple and addition
- Maintenance expense percent of premium multiple and addition
- Other maintenance expenses multiple.

PAD variables only apply during the factors valuation task. They have no effect on loss recognition and experience projection tasks.

2.4 Amortization

2.4.1 Amortization months

GAAP factors apply over the number of amortization months specified for the GAAP valuation task. Both DAC and the benefit and maintenance expense reserve calculations consider only amounts through the end of the amortization months.

When this variable is left at its zero default value, GAAP results are calculated to maturity.

2.4.2 GAAP interest rate

The GAAP interest rate is the asset yield specified for the GAAP factors valuation task, adjusted by the PAD multiple and addition. This rate is used to accumulate past values, and to discount future values.

2.5 Benefit reserve

2.5.1 Benefits

Benefits include base plan death benefits, endowment benefits, surrender benefits, maturity benefits, conversion cost, annual dividends, terminal dividends, experience incurred claims, and the statutory reserve (including the claim reserve) at the end of the amortization period.

2.5.2 Benefit reserve factor

The benefit reserve for each model point is:

$$\text{PV benefits}_t - k_{\text{benefits}} \times \text{PV gross premiums}_t$$

where:

$$k_{\text{benefits}} = \frac{[\text{PV benefits}_{\text{issue}}]}{\text{PV gross premiums}_{\text{issue}}}$$

PV = present value over amortization months, discounted at the GAAP interest rate.

The gross premiums are the current gross premiums; the guaranteed premiums have no effect.

Amounts are projected from issue using the SFAS 60 GAAP assumptions and PADs.

The benefit reserve per unit/policy GAAP factor is the model point amount, divided by the number of units/policies for each month projected by the GAAP factors valuation task.

2.5.3 After the amortization period

Statutory formulaic reserves apply after the GAAP amortization months. The benefit reserve grades into the statutory reserve at the end of the amortization period because the statutory reserve at the end of the amortization period is included in the benefits.

2.6 Maintenance expense reserve

2.6.1 Maintenance expenses

Expenses include all base plan maintenance expenses and non-deferrable commissions. Non-deferrable acquisition expenses are not considered to be policy-related expenses.

2.6.2 Maintenance expense reserve factor

The maintenance expense reserve for each model point is:

$$\text{PV maintenance expenses and non deferrable commissions}_t - k_{\text{exp}} \times \text{PV gross premiums}_t$$

where:

$$k_{\text{exp}} = \frac{[\text{PV maintenance expenses and non deferrable commissions}_{\text{issue}}]}{\text{PV gross premiums}_{\text{issue}}}$$

PV = present value over amortization months, discounted at the GAAP interest rate.

The gross premiums are the current gross premiums; the guaranteed premiums have no effect.

Amounts are projected from issue using the SFAS 60 GAAP assumptions and PADs.

The maintenance expense reserve per unit/policy GAAP factor is the model point amount, divided by the number of units/policies for each month projected by the GAAP factors projection task.

2.6.3 After the amortization period

The maintenance expense reserve is zero after the amortization period.

2.7 Other reserves

2.7.1 Unearned premium reserve

The unearned premium reserve is the gross unearned premium reserve, which is only calculated during experience projections that include SFAS 60 GAAP results.

2.7.2 *Claim reserve*

The claim reserve is the experience claims discounted at the GAAP interest rate. This amount is only calculated during experience projections that include SFAS 60 GAAP results.

2.7.3 *Rider reserve*

The rider reserve is the rider statutory reserve, and is only calculated during experience projections that include SFAS 60 GAAP results.

The rider reserve is included with the benefit reserve for the financial statement presentation.

2.8 Policyholder dividends

2.8.1 *Annual and terminal dividends*

Expected future annual and terminal dividends are included in the development of the benefit reserve.

2.8.2 *Dividend liability*

The policyholder dividend liability is zero, since the benefit reserve already includes policyholder dividends.

2.9 Deferred acquisition costs

A deferred acquisition cost (DAC) asset is established to capitalize acquisition expenses and commissions, and amortize them as a level percent of gross premiums.

2.9.1 *Capitalized expenses*

For the gross business, capitalized expenses include a specified percent of each type of acquisition expenses:

- Amount per policy
- Amount per unit (life only)
- Percent of premium.

There are no capitalized acquisition expenses for ceded business.

2.9.2 *Fixed expenses*

Capitalization does not apply for fixed expenses, which are specified for the business segment, since these are not considered to be policy-related expenses.

2.9.3 *Capitalized commissions*

For gross business, base plan capitalized commissions automatically include commission rates which are higher than the ultimate rate for the last premium payment. The capitalized percent of first year premium commission applies to the projection of annualized commissions and chargebacks in the GAAP factors projection. Thus chargebacks after the first year result in negative capitalized amounts using the capitalized commission rate for first year commissions.

For ceded business, capitalized commissions automatically include expense allowances which are higher than the ultimate rate.

2.9.4 *DAC factor*

The DAC amount for each model point is:

$$DAC_t = [DAC_{t-1} + \text{capitalized expenses and commissions}_t] \times (1 + i_t)^{1/12} - k_{DAC} \times \text{gross premium}_t$$

where:

$$k_{DAC} = \frac{PV \text{ capitalized expenses}_{issue}}{PV \text{ gross premiums}_{issue}}$$

$DAC_0 = 0$ since t represents the policy issue date.

PV = present value over amortization months, discounted at the GAAP interest rate.

Capitalized expenses are projected during the creation of the GAAP factors/schedule.

The gross premium is the current gross premium; the guaranteed premiums have no effect.

Amounts are projected from issue using the GAAP valuation assumptions.

DAC cannot be less than zero.

The DAC per unit/policy GAAP factor is the model point amount, divided by the number of units/policies for each month projected by the GAAP factors valuation task.

2.10 Reinsurance effect

2.10.1 YRT reinsurance

The YRT reinsurance effect is calculated for each model point as a 'reinsurance adjustment asset', which may be positive or negative. The reinsurance adjustment asset is:

$$\text{accumulated reinsurance cost}_t - k_{reins} \times \text{accumulated gross premiums}_t$$

where:

$$k_{reins} = \frac{PV \text{ reinsurance cost at issue}}{PV \text{ gross premiums at issue}}$$

PV = present value of projected amounts, discounted at the GAAP interest rate.

Reinsurance cost is the net cash flows of the YRT reinsurance treaties.

2.10.2 Coinsurance

When the model point includes coinsurance, the benefit reserve, maintenance expense reserve, and DAC amounts are calculated twice -- once using the gross amounts (ignoring coinsurance), then using only the ceded coinsurance amounts. The net GAAP amount is the gross result less the ceded coinsurance result.

2.10.3 Modified coinsurance

When the model point includes modified coinsurance, the benefit reserve, maintenance expense reserve, and DAC amounts are calculated twice -- once using the gross amounts (ignoring coinsurance), then using only the ceded modified coinsurance amounts.

The modified coinsurance liability is the reinsurance statutory reserve. This amount is held as a liability on the GAAP balance sheet, so the total benefit reserve for the ceding company is the benefit reserve calculated without considering reinsurance, less the portion reinsured, plus the statutory reserve held by the ceding company.

2.10.4 Multiple reinsurance treaties

When the model point has both YRT and coinsurance or modified coinsurance treaties, both of the above treatments apply.

2.11 Loss recognition

2.11.1 Loss recognition test

Loss recognition testing is a comparison, for each cohort, of the net GAAP liability against a gross premium reserve (GPR). The net GAAP liability is the GAAP benefit reserve plus the GAAP expense reserve, less the DAC. The GPR is discounted using the asset yield from the GPR valuation task.

The test is calculated during the experience projection. Specify the GAAP and GPR valuation result files (on the Company input page under the SFAS 60: GAAP: General category) for the experience projection.

The allowable DAC for the cohort is the minimum of: (1) the DAC calculated during the valuation and (2) the sum of the benefit and expenses reserves calculated from the valuation less the gross premium reserve, floored at zero. The DAC loss recognition adjustment factor is the allowable DAC divided by the DAC calculated during the valuation, capped at one and floored at zero. This ratio is multiplied against the valuation DAC factors.

If there has been loss recognition write downs in the past, specify (on the Company input page under the SFAS 60 GAAP: Cohorts category) the initial (at the projection date) DAC loss recognition adjustment factor for each cohort, since DAC cannot be written up after it has been written down.

- Gross DAC adjustment percent – the cumulative DAC write down percent for the gross DAC amount.
- Ceded DAC adjustment percent – the cumulative DAC write down percent for the ceded DAC amount.

These variables must be linked to an external source varying by hidden variable look up key cohort_id. The default value of 1.0 (100%) indicates there has been no write down.

2.11.2 GAAP results adjustment

The GAAP results for the cohort are affected by the write down of the DAC, and potentially by adjusting the amount of the GAAP benefit reserve.

When the loss recognition testing decreases the DAC to zero, and there is still a remaining excess loss, the benefit reserve is increased so that the net GAAP liability equals the gross premium reserve. The maintenance expense reserves are not adjusted.

2.12 Financial statement presentation

2.12.1 Income statement

Revenue includes the total of base plan and rider gross premium less a separate item for reinsurance premiums, reinsurance experience refund (when reporting for the segment), dividends applied to dividend options, cash flow interest, investment income and capital gains on the net GAAP liability and capital, and the modified coinsurance adjustment (when reporting for the business segment). When reporting for the business segment, reinsurance offsets are incorporated into the reported items for dividends applied to dividend options, the cash flow interest, and the investment income and capital gains on the reinsurance net GAAP liability.

Basic benefits include the total of base plan and rider direct death benefits, direct surrender benefits, direct maturity benefits, direct endowment benefits, direct health claims paid, less, when reporting for the segment, reinsurance recoveries for the basic benefits reinsured.

Other benefits include the benefit reserve increase (which includes the increase in the rider reserve), the health unearned premium reserve increase, the health claim reserve increase, the deferred profit liability increase, policyholder dividends, and the policyholder dividend liability, where, when reporting for the segment, reinsurance offsets are incorporated into the reported items. Also included is the modified coinsurance adjustment (when reporting for the ceded cohorts), and the modified coinsurance liability increase.

Expenses include the total of excess over renewal commissions, total acquisition expenses less a separate item for deferred commission and acquisition expenses, renewal commissions, maintenance expenses, premium taxes, letter of credit financing costs, conversion costs, DAC (expense DAC plus the YRT asset) amortization, and less a separate item for the YRT reinsurance cost applicable for YRT reinsurance. When reporting for the business segment, reinsurance offsets are incorporated into each reported item.

2.12.2 Balance sheet

Assets include the invested assets, other invested assets, the DAC, and the YRT reinsurance adjustment asset. When reporting for a business segment, 'reinsurance receivable' is the reinsurance net GAAP liability less the modified coinsurance liability.

Liabilities include the benefit reserve (which includes the rider reserve), the unearned premium reserve, the claim reserve, and the modified coinsurance liability.

2.12.3 Roll forward analysis

An analysis of the expense DAC and benefit reserve change is available for each cohort.

The expense DAC roll forward captures the movement of DAC from the beginning amount to the ending amount using valuation based components for additions for interest, deferred commissions, deferred acquisition costs less amounts for amortization and a final balancing adjustment. The balancing adjustment is required for the release of DAC from experience assumptions differing from valuation assumptions.

The benefit and maintenance expense reserve roll forward captures the movement of the benefit and maintenance expense reserve from the beginning amount to the ending amount using valuation based components for additions for interest and the reserve net premium less amounts for the valuation-based benefits, maintenance expenses, surrenders, dividends and a final balancing adjustment. The balancing adjustment is required for the release of the benefit reserve and maintenance expense reserve from experience assumptions differing from valuation assumptions.

2.12.4 Source of earnings

An analysis of the SFAS 60 source of earnings is available for each cohort. The source of earnings analysis breaks income into components:

- Premium margin -- the gross premiums received, less the sum of the benefit reserve net premium and the DAC net premium.
- Interest margin -- investment income on the net GAAP liability, less interest on the initial benefit reserve and the DAC asset.
- Benefit margin -- valuation-based benefits less the actual benefits paid, plus the assumed dividends in the benefits less the actual policyholder dividends paid, plus the valuation-based surrenders less the actual surrender benefits paid.
- Expense margin -- valuation-based expenses and deferrals, less the actual expenses and commissions paid.
- Other income -- adjustments to income for YRT reinsurance adjustments, unearned premium reserves, and claim reserves.

3 SFAS 91 INVESTMENT CONTRACT METHOD

3.1 Applicability

SFAS 91 includes special rules for payout annuities with 'nominal mortality risk'. These products are 'investment contracts' for US GAAP. Specify on each payout annuity model point record whether the SFAS 91 or SFAS 97 limited pay GAAP method applies.

SFAS 91 valuations create GAAP schedules for each cohort, which are created at the projection date and each inner loop valuation date. The schedules are calculated by a portfolio valuation task with target column valn_tgt_gaap_sfas91.

Loss recognition and unlocking do not apply for this GAAP method.

3.2 Schedules

The following schedule values are created for each cohort:

- Expected premiums
- Expected annuity benefits
- Expected death benefits
- Net reserve constant effective yield
- Net reserve cash flow beginning of month
- Net reserve cash flow end of month
- Net reserve at the projection date
- Benefit reserve constant effective yield
- Benefit reserve percentage of premium
- Benefit reserve cash flow beginning of month
- Benefit reserve cash flow end of month
- Benefit reserve at the projection date
- Maintenance expense reserve constant effective yield
- Maintenance expense reserve percent of premium
- Maintenance expense reserve cash flow beginning of month
- Maintenance expense reserve cash flow end of month
- Maintenance expense reserve at the projection date.

3.3 Assumptions

SFAS 91 investment contract GAAP amounts are calculated using 'best estimate' assumptions, which are set for the GAAP valuation task. Assumption variables operate the same for the GAAP valuation task as for the experience projection task.

There are no provisions for adverse deviation.

3.4 Amortization

SFAS 91 investment contract GAAP calculations apply to the end of the last contract in the cohort. There is no user specified amortization months.

3.5 Constant yields

The SFAS 91 investment contract method does not use a single GAAP interest rate in the calculations. Instead, the 'constant yield method' uses two constant yields to determine a net reserve, and then allow for the splitting of the net reserve between liabilities (the benefit reserve) and assets (the DAC)

The first constant yield is the rate which equates the net proceeds at issue to the present value of anticipated cash flows. Net proceeds are the single premium received less any deferrable commission and acquisition expenses. Anticipated cash flows are the single premium, commissions and expenses, death benefits, and annuity benefits.

The second constant yield is the rate which equates the single premium paid to the present value of non-deferred acquisition expenses and commissions, maintenance expenses, death benefits, and annuity benefits.

Specify the constant yields (on the Company input page under the SFAS 91 GAAP: Cohorts category), separate for gross and ceded, for each cohort. For in force cohorts where there is no business added to the cohort, these constant yields are used always. For in force cohorts where there is new business added to the cohort, or cohorts where there is no in force, these constant yield used are used initially but will be replaced when the constant yields are recalculated in the next valuation.

These constant yield variables must be linked to an external source varying by hidden variable look up key cohort_id.

3.6 Net reserve

3.6.1 Net reserve cash flows

Net reserve cash flows include the premium, deferrable acquisition expense and commissions, maintenance expenses, premium taxes, death benefits, and annuity benefits.

3.6.2 Net reserve constant yield

The net reserve constant yield is the rate which ensures that the present value of the anticipated reserve cash flows (premiums, less death and annuity benefits, less deferrable expenses and commissions, less maintenance expenses, less premium taxes) is zero at the beginning of the cohort. The net reserve constant yield is solved for using a Newton-Raphson method.

For the first valuation of a new cohort, experience amounts required for the net reserve calculation before the projection date are provided by the state of the world of the cohort from the experience projection.

3.6.3 Net reserve valuation

At the valuation date, the net reserve is the present value of future anticipated cash flows:

$$PV \text{ prospective net cash flows}_t$$

where:

PV = present value over the maturity of the contracts, discounted at the net reserve constant yield.

Amounts are projected using the assumptions specified for the GAAP valuation task.

3.6.4 Net reserve roll forward

In each month until the next valuation, the net reserve rolls forward using the net cash flows and the net reserve constant yield:

$$\text{Net Reserve}_t = [\text{Net Reserve}_{t-1} - \text{net reserve cash flow bom}_t] \times (1 + cy_t)^{1/12} - \text{net reserve cash flow eom}_t$$

where:

Net reserve cash flow beginning of month includes beginning of month annuity benefits, capitalized acquisition expenses, maintenance expenses, premium taxes, and capitalized commissions, less premiums paid.

Net reserve cash flow end of month includes death benefits plus end of month annuity benefits and maintenance expenses.

The net reserve constant yield rate was calculated at the projection date or the last valuation date.

The net reserve cash flows are the actual amounts each month according to the definitions used in creating the current GAAP schedule. For example, when the schedule was created on June 30 for a cohort including the current year's issues, the net reserve cash flows and the net reserve constant yield were determined only for issues through the end of June. Then the roll forward in July includes the net reserve cash flows from new issues added into the cohort in July, and so on for succeeding months.

3.7 Benefit reserve

3.7.1 Prospective benefits

Prospective benefits include death benefits and annuity benefits.

3.7.2 Benefit reserve constant yield

The benefit reserve constant yield is the rate which ensures that the present value of the anticipated benefit reserve cash flows equals zero at the beginning of the cohort. Anticipated benefit reserve cash flows are the premium, less death and annuity benefits, less non-deferrable expenses and commissions, less maintenance expenses, less premium taxes.

For the first valuation of a new cohort, experience amounts required for the benefit reserve calculation before the projection date are provided by the state of the world of the cohort from the experience projection.

The benefit reserve constant yield is solved for using a Newton-Raphson method.

3.7.3 Benefit reserve valuation

At the valuation date, the benefit reserve is the present value of future anticipated benefits

$$PV \text{ prospective benefits}_t$$

where:

PV = present value over the maturity of the contracts, discounted at the benefit reserve constant yield.

Amounts are projected using the assumptions specified for the GAAP valuation task.

3.7.4 Benefit reserve roll forward

In each month until the next valuation, the benefit reserve rolls forward using the benefit reserve cash flows and the benefit reserve constant yield.

$$\text{Benefit Reserve}_t = [\text{Benefit Reserve}_{t-1} - \text{benefit reserve cash flow bom}_t] \times (1 + cy_t)^{1/12} - \text{benefit reserve cash flow eom}_t$$

where:

Benefit reserve cash flow beginning of month includes beginning of month annuity benefits less the premium paid.

Benefit reserve cash flow end of month includes death benefits, less end of month annuity benefits.

The benefit reserve constant yield rate is the GAAP schedule amount which was calculated at the valuation date.

The benefit reserve cash flows are the actual amounts each month according to the definitions used in creating the current GAAP schedule. For example, when the schedule was created on June 30 for a cohort including the current year's issues, the benefit reserve cash flows and the benefit reserve constant yield were determined only for issues through the end of June. Then the roll forward in July includes the benefit reserve cash flows from new issues added into the cohort in July, and so on for succeeding months.

3.8 Maintenance expense reserve

3.8.1 Maintenance expenses

Maintenance expenses include non-deferrable acquisition expenses and premium taxes.

3.8.2 Maintenance expense reserve constant yield

The maintenance expense reserve constant yield is the benefit reserve constant yield

3.8.3 Maintenance expense reserve valuation

At the valuation date, the maintenance expense reserve is the present value of anticipated maintenance expenses:

$$PV \text{ prospective maintenance expenses}_t$$

where:

PV = present value over the maturity of the contracts, discounted at the maintenance expense reserve constant yield.

Amounts are projected from the projection date using the assumptions specified for the GAAP valuation task.

3.8.4 Maintenance expense reserve roll forward

In each month until the next valuation month, the maintenance expense reserve rolls forward using the maintenance expense reserve cash flows and the benefit reserve constant yield:

$$\text{Maint Exp Reserve}_t = [\text{Maint Exp Reserve}_{t-1} - \text{maint exp reserve cash flow bom}_t] \times (1 + \text{cy}_t)^{1/12} - \text{maint exp reserve cash flow eom}_t$$

where:

Maintenance expense reserve cash flow beginning of month includes the beginning of month non-deferred acquisition expenses, maintenance expenses, and premium taxes.

Maintenance expense reserve cash flow end of month includes the maintenance expenses.

The maintenance expense reserve constant yield rate is the GAAP schedule amount which was calculated at the valuation date.

The maintenance expense reserve cash flows are the actual amounts each month according to the definitions used in creating the current GAAP schedule. For example, when the schedule was created on June 30 for a cohort including the current year's issues, the maintenance expense reserve cash flows and the maintenance expense reserve constant yield were determined only for issues through the end of June. Then the roll forward in July includes the maintenance expense reserve cash flows from new issues added into the cohort in July, and so on for succeeding months.

3.9 Deferred acquisition costs

The SFAS 91 investment contract net reserve includes provision for a deferred acquisition cost (DAC) asset which incorporates capitalized acquisition expenses and commissions.

3.9.1 Capitalized expenses

For gross business, capitalized expenses include a specified percent of each type of acquisition expenses:

- Amount per policy
- Percent of premium.

There are no capitalized acquisition expenses for ceded business.

3.9.2 Fixed expenses

Capitalization does not apply for fixed expenses, which are specified for the business segment, since these are not considered to be policy-related expenses.

3.9.3 Capitalized commissions

Capitalized commissions automatically include the single premium commission.

3.9.4 DAC amount

The DAC amount for each cohort is:

$$DAC_t = \max(0.0, \text{Benefit reserve}_t + \text{Maintenance expense reserve}_t - \text{Net reserve}_t)$$

3.10 Reinsurance effect

3.10.1 YRT reinsurance

There is no YRT reinsurance for payout annuities.

3.10.2 Coinsurance

When the model point includes coinsurance, the net reserve, benefit reserve, maintenance expense reserve and DAC amounts are calculated twice, once using the gross amounts (ignoring coinsurance), then using only the ceded coinsurance amounts. The net GAAP amount is the gross result less the ceded coinsurance result.

3.10.3 Modified coinsurance

There is no modified coinsurance for payout annuities.

3.11 Financial statement presentation

3.11.1 Income statement

Revenue includes the reinsurance experience refund (when reporting for the business segment), cash flow interest, investment income and capital gains on the net GAAP liability and capital, and the modified coinsurance adjustment (when reporting for the segment). When reporting for the segment, reinsurance offsets are incorporated into the reported items the cash flow interest and the investment income and capital gains on the reinsurance net GAAP liability.

There are typically no benefits. An annuity claim cost amount may be calculated if the benefits in the reserve calculation differ from the actual benefits paid.

The required/credited interest is captured as a credited interest benefit, which is the solved for benefit reserve constant yield times the benefit reserve at the end of the previous period. When reporting for the business segment, this amount is reported net of reinsurance. When a cohort includes new business, such that the constant yields are recalculated at a valuation and lead to a change in the benefit reserve, the benefit reserve change is captured in the credited interest result in the period after the revaluation.

Expenses includes the single premium commission and total acquisition expenses less a separate item for deferred commissions and acquisition expenses, maintenance expenses, premium taxes, and DAC amortization. Since all deferrable expenses are incurred at issue, the DAC amortization is the increase in DAC, which is implied by the change in the net reserve. When reporting for the business segment, reinsurance offsets are incorporated into each reported item.

3.11.2 Balance sheet

Assets include the invested assets, other invested assets, the DAC, and (when reporting for the business segment) a reinsurance receivable equal to the reinsurance net GAAP liability adjusted to remove the modified coinsurance liability.

Liabilities include the benefit reserve, the maintenance expense reserve, and the modified coinsurance liability.

4 SFAS 97 LIMITED PAY CONTRACTS METHOD

4.1 Applicability

4.1.1 Product types

SFAS 97 includes special rules for term life, traditional life, and health products that have premium paying periods that are less than the benefit period. This method automatically applies for term life, traditional life, and health products when premiums are not paid to maturity.

These special rules also apply for payout annuities with more than a 'nominal mortality risk'. Specify on each payout annuity model point record whether the SFAS 91 or SFAS 97 limited pay GAAP method applies.

SFAS 97 limited pay valuations create GAAP factors using assumptions locked in at issue. Loss recognition is performed by a separate GPR valuation, which is applied for each cohort by the experience projection.

The factors are calculated by a seriatim projection task with target column `valn_tgt_gaap_sfas60_sfas97lp`. These GAAP factors are on-the-fly calculations at the projection date, from the issue date of each model point. That is, SFAS 97 limited pay GAAP valuations are not performed at inner loop valuation dates during nested projections.

4.2 Factors

Limited pay factors are calculated for each model point from issue, using the GAAP assumptions determined at issue. The exception is that the GAAP factors are replaced by the loss recognition gross premium reserve after an extreme loss recognition event that writes down all of the remaining DAC asset and still has more losses to account for. See the Loss recognition section below.

Limited pay factors are stored by model point. GAAP factor results are converted into results per unit/policy in force based on the units/policies projected during the GAAP valuation task. Then the GAAP results for the experience projection are these amounts per unit/policy, times the units/policies projected to be in force during the experience projection.

The GAAP factors are these amounts per unit/policy in force:

- DAC balance, both gross and ceded coinsurance
- Benefit reserve, both gross and ceded coinsurance

- Maintenance expense reserve, both gross and ceded coinsurance
- Deferred profit liability, both gross and ceded coinsurance
- YRT reinsurance adjustment asset (life insurance products only).

4.3 Assumptions

4.3.1 Separate GAAP assumption variables

Limited pay amounts are calculated using assumptions determined at issue. Assumption variables operate the same for the GAAP factors valuation task as for the experience projection task.

For health products, benefits for SFAS 97 limited pay calculations may be specified, or experience assumptions with provisions for adverse deviation may be used.

4.3.2 Provisions for adverse deviation

Limited pay GAAP assumptions include provisions for adverse deviation (PAD). These are separate input variables so that the PADs are explicitly visible:

- Asset yield multiple and addition -- applies to the effective annual asset yield
- Mortality multiple and addition -- applies to the annual mortality rate, after all improvements and adjustments. These are 'mortality' variables, which operate like to base mortality table variable; see the Mortality variables section in the Input organization chapter of the book for each product type.
- Lapse multiple and addition -- applies to the annual lapse rate, after dynamic and excess lapses, but before shock lapses
- Claim cost multiple and addition (health only)
- Incidence multiple and addition (health only) -- applies to the annual morbidity rate, after all adjustments
- Termination multiple and addition (health only)
- Maintenance expense per unit multiple and addition (life only)
- Maintenance expense per policy multiple and addition
- Maintenance expense percent of premium multiple and addition
- Other maintenance expenses multiple.

PAD variables only apply during the factors valuation task. They have no effect on loss recognition and experience projection tasks.

4.4 Amortization

4.4.1 Amortization months

For life and health products, benefit reserves, maintenance expense reserves, and the deferred profit liability apply over the number of amortization months specified for the GAAP valuation projection task. When this variable is left at its zero default value, these reserves are calculated to maturity. DAC is always calculated over the premium paying period.

For payout annuity products, factors apply to the end of the annuity contract. There is no user specified amortization months.

4.4.2 GAAP interest rate

The GAAP interest rate is the asset yield specified for the GAAP valuation task. This is the rate used to accumulate past values, and to discount future values.

4.5 Benefit reserve

4.5.1 Benefits

Benefits include death benefits, annuity benefits, endowment benefits, surrender benefits, maturity benefits, conversion costs, annual dividends, terminal dividends, experience incurred claims, and the statutory reserve (including the claim reserve) at the end of the amortization period.

4.5.2 Benefit reserve factor

The benefit reserve for each model point is:

$$PV \text{ benefits}_t - k_{\text{benefits}} \times PV \text{ gross premiums}_t$$

where:

$$k_{\text{benefits}} = \frac{PV \text{ benefits}_{\text{issue}}}{PV \text{ gross premiums}_{\text{issue}}}$$

PV = present value over amortization months, discounted at the GAAP interest rate.

The gross premium is the current gross premium; the guaranteed premiums have no effect.

Amounts are projected from issue using the limited pay GAAP assumptions and PADs.

The benefit reserve per unit/policy GAAP factor is the model point amount, divided by the number of units/policies for each month projected by the GAAP valuation projection task.

4.5.3 After the amortization period

Statutory formulaic reserves apply after the GAAP amortization months. The benefit reserve grades into the statutory reserve at the end of the amortization period because the statutory reserve at the end of the amortization period is included in the benefits.

(This does not apply to payout annuities since the amortization month is always the life of the contract.)

4.6 Maintenance expense reserve

4.6.1 Maintenance expenses

Expenses include maintenance expenses and non-deferrable commissions. Non-deferrable acquisition expenses are not considered to be policy-related expenses.

4.6.2 Maintenance expense reserve factor

For life and health products, the maintenance expense reserve for each model point is:

$$PV \text{ maintenance expenses and non - deferrable commissions}_t - k_{\text{exp}} \times PV \text{ gross premiums}_t$$

where:

$$k_{\text{exp}} = \frac{[PV \text{ maintenance expenses and non - deferrable commissions}_{\text{issue}}]}{PV \text{ gross premiums}_{\text{issue}}}$$

PV = present value over amortization months, discounted at the GAAP interest rate.

The gross premium is the current gross premium; the guaranteed premiums have no effect.

Amounts are projected from issue using the limited pay GAAP assumptions and PADs.

The maintenance expense reserve per unit/policy GAAP factor is the model point amount, divided by the number of units/policies for each month projected by the GAAP valuation projection task.

4.6.3 After the amortization period

The maintenance expense reserve is zero after the amortization period.

4.7 Other reserves

4.7.1 Unearned premium reserve

The unearned premium reserve is the gross unearned premium reserve.

4.7.2 Claim reserve

The claim reserve is the experience claims paid discounted at the GAAP interest rate.

4.7.3 Rider reserve

The rider reserve is the rider statutory reserve, and is only calculated during experience projections that include SFAS 97 limited pay GAAP results.

The rider reserve is included with the benefit reserve for the financial statement presentation.

4.8 Policyholder dividends

Limited payment participating traditional life products that do not pay dividends according to the contribution principle and are not actively managed are accounted for under SFAS 97 limited pay.

4.8.1 Annual and terminal dividends

Expected future annual and terminal dividends are included in the benefit reserve.

4.8.2 Dividend liability

The policyholder dividend liability is zero, since the benefit reserve already reflects policyholder dividends.

4.9 Deferred profit liability

The deferred profit liability (DPL) is established in addition to the benefit reserve. The DPL is sometimes referred to as an unearned profit reserve, unreleased profit reserve, or unearned revenue liability.

- For life products, the DPL is amortized so that profits emerge (when GAAP assumptions are realized) as a constant percent of the insurance amount in force.
- For health products, the DPL is amortized in proportion to the expected morbidity benefits, using the same assumptions for calculating the benefit reserve.
- For payout annuity products, the DPL is amortized in proportion to the expected annuity payments.

4.9.1 Deferred profits

Under SFAS 97 limited pay, profits for life insurance products are deferred so that they emerge in a constant relationship to the expected amount of insurance. For payout annuity contracts, profits are deferred so they emerge in a constant relationship to expected annuity benefits.

4.9.2 Deferred profit liability factor

For life insurance products, the deferred profit liability calculated for each model point is:

$$DPL_t = DPL_{t-1} \times (1+i)^{1/12} + \text{Profit \%} \times \text{gross premium}_t - k_{DPL} \times \text{insurance in force}_t$$

where:

$$\text{Profit \%} = \max(0, 1 - k_{\text{benefits}} - k_{\text{exp}} - k_{\text{DAC}})$$

$$k_{DPL} = \frac{\text{Profit \%} \times \text{PV gross premiums}_{\text{issue}}}{\text{PV insurance in force}_{\text{issue}}}$$

$DPL_0 = 0$ since t is since issue

PV = present value of projected amounts, discounted at the GAAP interest rate.

For health products, the deferred profit liability for each model point is:

$$DPL_t = DPL_{t-1} \times (1+i)^{1/12} + \text{Profit \%} \times \text{gross premium}_t - k_{DPL} \times \text{claim payment}_t$$

where:

$$\text{Profit \%} = \max(0, 1 - k_{\text{benefits}} - k_{\text{exp}} - k_{\text{DAC}})$$

$$k_{DPL} = \frac{\text{Profit \%} \times \text{PV gross premiums}_{\text{issue}}}{\text{PV claim payments}_{\text{issue}}}$$

For payout annuity products, the deferred profit liability for each model point is:

$$DPL_t = DPL_{t-1} \times (1+i)^{1/12} + \text{Profit \%} \times \text{gross premium}_t - k_{DPL} \times \text{annuity benefits}_t$$

where:

$$\text{Profit \%} = \max(0, 1 - k_{\text{benefits}} - k_{\text{exp}} - k_{\text{DAC}})$$

$$k_{DPL} = \frac{\text{Profit \%} \times \text{PV gross premiums}_{\text{issue}}}{\text{PV annuity benefits}_{\text{issue}}}$$

Since annuity benefits can be paid at either the beginning or end of the month, the annuity benefits in the DPL rollforward are the beginning of month benefits, accumulated at the GAAP interest rate, plus the end of month annuity benefits.

The deferred profit liability per policy GAAP factor is the model point amount, divided by the number of policies for each month projected by the GAAP valuation projection task.

4.9.3 After the amortization period

The deferred profit liability is zero after the amortization period.

4.10 Policyholder dividends

4.10.1 Annual and terminal dividends

Expected future annual and terminal dividends are included in the development of the benefit reserve.

4.10.2 Dividend liability

The policyholder dividend liability is zero, since the benefit reserve already reflects policyholder dividends.

4.11 Deferred acquisition costs

A deferred acquisition cost (DAC) asset is established to capitalize front-end expenses and commissions, and amortize them as a level percent of gross premiums.

For life and health products, a deferred acquisition cost asset is present during the amortization period. For payout annuity products, which are single premium contracts, no deferred acquisition cost asset is created since the amortization is instantaneous.

4.11.1 Capitalized expenses

For gross business, capitalized expenses include a specified percent of each type of acquisition expenses:

- Amount per policy
- Amount per unit (life only)
- Percent of premium.

There are no capitalized acquisition expenses for ceded business.

4.11.2 Fixed expenses

Capitalization does not apply for fixed expenses, which are specified for the business segment, since these are not considered to be policy-related expenses.

4.11.3 Capitalized commissions

For gross business, capitalized commissions automatically include commission rates which are higher than the ultimate rate for the last premium payment. The capitalized percent of first year premium commission applies to the projection of annualized commissions and chargebacks in the GAAP factors projection. Thus chargebacks after the first year result in negative capitalized amounts using the capitalized commission rate for first year commissions.

For ceded business, capitalized commissions automatically include expense allowances which are higher than the ultimate rate.

4.11.4 DAC factor

For life and health products, the DAC amount for each model point is:

$$DAC_t = [DAC_{t-1} + \text{capitalized expenses and commissions}_t] \times (1 + i_t)^{t/12} - k_{DAC} \times \text{gross premium}_t$$

where:

$$k_{DAC} = \frac{PV \text{ capitalized expenses}_{issue}}{PV \text{ gross premiums}_{issue}}$$

$DAC_0 = 0$ since t is from issue

PV = present value over amortization months, discounted at the GAAP interest rate.

Capitalized expenses are projected during the creation of the GAAP factors/schedule.

The gross premium is the current gross premium; the guaranteed premiums have no effect.

Amounts are projected from issue using the GAAP valuation assumptions.

DAC cannot be less than zero. DAC is zero after the last premium has been paid. Thus DAC is zero for single premium model points. The k_{DAC} factor is calculated for single premium model points, however, since it is used in the deferred profit liability calculation.

For payout annuity products, no DAC factor is calculated.

The DAC per unit/policy GAAP factor is the model point amount, divided by the number of units/policies for each month projected by the GAAP factors valuation task.

4.12 Reinsurance effect

4.12.1 YRT reinsurance

For life and health products, the YRT reinsurance effect for each model point is a 'reinsurance adjustment asset', which is often negative. The reinsurance adjustment asset is:

$$\text{accumulated reinsurance cost}_t - k_{\text{reins}} \times \text{accumulated gross premiums}_t$$

where:

$$k_{\text{reins}} = \frac{\text{PV reinsurance cost at issue}}{\text{PV gross premiums at issue}}$$

PV = present value of projected amounts, discounted at the GAAP interest rate.

Reinsurance cost is the net cash flows of the YRT reinsurance treaties.

4.12.2 Coinsurance

When the model point includes coinsurance, the benefit reserve and DAC amounts are calculated twice, once using the gross amounts (ignoring coinsurance), then using only the ceded coinsurance amounts. The net GAAP amount is the gross result less the ceded coinsurance result.

4.12.3 Modified coinsurance

When the model point includes modified coinsurance, the benefit reserve and DAC amounts are calculated twice, once using the gross amounts (ignoring coinsurance), then using only the ceded modified coinsurance amounts.

The modified coinsurance liability is the reinsurance statutory reserve. This amount is held as a liability on the GAAP balance sheet, so the total benefit reserve for the ceding company is the benefit reserve calculated without considering reinsurance, less the portion reinsured, plus the statutory reserve held by the ceding company.

4.12.4 Multiple reinsurance treaties

When the model point has both YRT and coinsurance or modified coinsurance treaties, both of the above treatments apply.

4.13 Loss recognition

4.13.1 Loss recognition test

Loss recognition testing is a comparison, for each cohort, of the net GAAP liability against a gross premium reserve (GPR). The net GAAP liability is the GAAP benefit reserve plus GAAP expense reserve plus the

GAAP deferred profit liability, less the DAC. The GPR is discounted using the asset yield from the GPR valuation task; see the Gross premium reserve chapter in the PBA book.

The test is calculated during the experience projection. Specify the GAAP and GPR valuation result files (on the Company input page under the SFAS 97 Limited Pay: General category) for the experience projection.

The allowable DAC for the cohort is the minimum of: (1) the DAC calculated during the valuation and (2) the sum of the benefit and expenses reserves and deferred profit liability calculated from the valuation less the gross premium reserve, floored at zero. A DAC loss recognition adjustment factor is calculated as the allowable DAC divided by the DAC calculated during the valuation, capped at one and floored at zero. This ratio is multiplied against the valuation DAC factors.

If there has been loss recognition write downs in the past, specify (on the Company input page under the SFAS 97 Limited Pay GAAP: Cohorts category) the initial (at the projection date) DAC loss recognition adjustment factor for each cohort, since DAC cannot be written up after it has been written down.

- Gross DAC adjustment percent – the cumulative DAC write down percent for the gross DAC amount.
- Ceded DAC adjustment percent – the cumulative DAC write down percent for the ceded DAC amount.

These variables must be linked to an external source varying by hidden variable look up key cohort_id. The default value of 1.0 (100%) indicates there has been no write down.

4.13.2 GAAP results adjustment

The GAAP results for the cohort are affected by the write down of the DAC, and potentially by replacing the GAAP benefit reserve, expense reserve, and deferred profit liability by the GPR.

When the loss recognition testing decreases the DAC to zero, and there is still a remaining excess loss, the benefit reserve is increased so that the net GAAP liability equals the gross premium reserve. The maintenance expenses reserve and deferred profit liability are not adjusted.

4.14 Financial statement presentation

4.14.1 Income statement

Revenue includes the total of base plan and rider gross premium less a separate item for reinsurance premiums, reinsurance experience refund (when reporting for the segment), dividends applied to dividend options (life only), cash flow interest, investment income and capital gains on the net GAAP liability and capital, and the modified coinsurance adjustment (when reporting for the segment). When reporting for the business segment, reinsurance offsets are incorporated into the reported items for dividends applied to dividend options, the cash flow interest, and the investment income and capital gains on the reinsurance net GAAP liability.

Basic benefits include the total of base plan and rider direct death benefits, direct surrender benefits, direct maturity benefits, direct endowment benefits, direct health claims paid, less, when reporting for the segment, reinsurance recoveries for the basic benefits reinsured.

Other benefits include the benefit reserve increase (which includes the increase in the rider reserve), the health unearned premium reserve increase, the health claim reserve increase, the deferred profit liability increase, policyholder dividends, and the policyholder dividend liability, where, when reporting for the segment, reinsurance offsets are incorporated into the reported items. Also included is the modified coinsurance adjustment (when reporting for the ceded cohorts), and the modified coinsurance liability increase.

Expenses includes the total of excess over renewal commissions, total acquisition expenses less a separate item for deferred commission and acquisition expenses, renewal commissions, maintenance expenses, premium taxes, letter of credit costs, conversion costs, DAC (expense DAC plus the YRT asset) amortization, and less a separate item for the YRT reinsurance cost applicable for YRT reinsurance. When reporting for the segment, reinsurance offsets are incorporated into each reported item as required.

4.14.2 Balance sheet

Assets include the invested assets, other invested assets, the DAC, and the YRT reinsurance adjustment asset. When reporting for a business segment, 'reinsurance receivable' is the reinsurance net GAAP liability less the modified coinsurance liability.

Liabilities include the benefit reserve (which includes the rider reserve), the unearned premium reserve, the claim reserve, the deferred profit liability, and the modified coinsurance liability.

4.14.3 Roll forward analysis

Because the SFAS 97 limited pay contract method is incorporated into the SFAS 60 component for life and health products, an analysis of the expense DAC and benefit reserve change is available for each cohort.

The expense DAC roll forward captures the movement of DAC from the beginning amount to the ending amount using valuation based components for additions for interest, deferred commissions, deferred acquisition costs less amounts for the DAC net premium and a final balancing adjustment. The balancing adjustment is required for the release of DAC from experience assumptions differing from valuation assumptions.

The benefit and maintenance expense reserve roll forward captures the movement of the benefit and maintenance expense reserve from the beginning amount to the ending amount using valuation based components for additions for interest and the reserve net premium less amounts for the valuation-based benefits, maintenance expenses, surrenders, dividends and a final balancing adjustment. The balancing adjustment is required for the release of the benefit reserve and maintenance expense reserve from experience assumptions differing from valuation assumptions.

The deferred profit liability roll forward captures the movement of the deferred profit liability from the beginning amount to the ending amount using valuation based components for additions for interest and the deferred profit net premium less valuation-based amortization, and a final balancing adjustment. The balancing adjustment is required for the release of the deferred profit liability from experience assumptions differing from valuation assumptions.

4.14.4 Source of earnings

Because the SFAS 97 limited pay contract method is incorporated into the SFAS 60 component for life and health products, an analysis of the source of earnings is available for each cohort. The source of earnings analysis breaks income into components.

An analysis of the SFAS 60 source of earnings is available for each cohort. The source of earnings analysis splits income into components:

- Premium margin -- the gross premiums received, less the sum of the benefit reserve net premium and the DAC net premium, less deferred profit.
- Interest margin -- investment income on the net GAAP liability, less interest on the initial benefit reserve and the DAC asset, less interest on the DPL.
- Benefit margin -- valuation-based benefits less the actual benefits paid, plus the assumed dividends in the benefits less the actual policyholder dividends paid, plus the valuation-based surrenders less the actual surrender benefits paid.
- Expense margin -- valuation-based expenses and deferrals, less the actual expenses and commissions paid.
- Other income -- adjustments to income for YRT reinsurance adjustments, unearned premium reserves, and claim reserves.

5 SFAS 97 RETROSPECTIVE DEPOSIT METHOD

5.1 Applicability

This SFAS 97 retrospective deposit method automatically applies for the universal life, variable universal life, indexed universal life, fixed deferred annuity, fixed indexed deferred annuities, and variable deferred annuity product types. These products are based on policyholder account balances, provide benefits over a long duration, and have significant sources of revenue other than investment spread income.

SFAS 97 retrospective deposit valuations create GAAP schedules for each cohort, which are created at the projection date and each inner loop valuation date. The schedules are calculated by a portfolio valuation task with target column `valn_tgt_gaap_sf97rd`.

Unlocking and loss recognition occur when each GAAP schedule is created.

5.2 Schedules

5.2.1 Monthly values

The following SFAS 97 retrospective deposit schedule values are created each month for each cohort:

- Estimated gross profits
- GAAP interest rate
- Fund values
- SOP 03-1 persistency refunds and bonuses

5.2.2 Starting values

Additionally, the following values are created at the valuation date for each cohort:

- k factor for capitalized acquisition expenses
- k factor for capitalized commissions
- k factor for premium bonuses
- k factor for YRT reinsurance asset
- k factor for SOP 03-1 persistency refunds and bonuses
- k factor for SOP 03-1 UL/IUL secondary guarantee benefits
- k factor for SOP 03-1 unearned revenue liability
- k factor for unearned revenue liability
- SOP 03-1 GMDB benefit ratio
- SOP 03-1 GMIB benefit ratio
- SOP 03-1 GMWB benefit ratio
- DAC for capitalized acquisition expenses
- DAC for capitalized commissions
- DAC for premium bonuses
- DAC YRT reinsurance asset
- Unearned revenue liability
- SOP 03-1 persistency refund and bonus liability

- Embedded derivative values for FIA SFAS 133
- DAC capitalized acquisition expenses cumulative loss recognition
- DAC capitalized commissions cumulative loss recognition
- DAC premium bonus cumulative loss recognition

5.3 Assumptions

5.3.1 Best estimate basis

SFAS 97 retrospective deposit calculations are based on 'best estimate' assumptions. Assumption variables operate the same for the GAAP valuation projection task as for the experience projection task.

5.3.2 Economic rates treatment

SFAS 97 retrospective deposit best estimate assumptions have no foreknowledge of the future. Thus GAAP schedule calculations do not use outer loop (experience projection) scenario rates at later dates along the projection path. Instead, these liability-only calculations use rates as of the valuation date, grading linearly to 'mean reversion' rates over a specified number of months. When the number of grading months is zero, the 'mean reversion' rates apply in all months.

You can specify not to grade to the mean reversion rates. The starting rates remain level through the inner loop calculations. This option should be used with care, since this could result in excess lapses continuing indefinitely into the future when the GAAP valuation occurs when excess lapse formulas are active.

Mean reversion rates are specified for:

- Risk free yield curve (for calculating competitor rates)
- Index growth rates.

Note that the asset yield is specified using a competitor rate, which are based on the scenario rates.

5.3.3 Claim and policyholder behavior rates

Non-economic assumptions continue to be calculated as they are set to apply at future dates. This is not foreknowledge of the future, but is instead using assumptions being made such as for mortality improvement rates.

Excess lapse formulas apply as usual to the economic rates used in the GAAP calculations. These formulas are the assumptions for how economic rates affect policyholder behavior in various economic conditions.

5.4 Cohort history

For each in-force cohort, specify the cohort_history_ variables (on the Company input page under the SFAS97 Retrospective Deposit GAAP: History category). These variables are located in the sfas97rd submodel. Specify the value for each month from the beginning of the cohort up to the projection date.

These variables must be linked to an external source varying by hidden variable lookup keys cohort_id and cohort_reins_defn. The cohort_reins_defn lookup key separates Gross (ignoring reinsurance) from Ceded (coinsurance and modified coinsurance treaties) values.

Some of these variables apply only for certain product types. For a projection not including those products, simply ignore (leave at zero default value) the variables that do not apply.

Variable	Value
cohort_history_agp	Actual gross profit, containing the actual profit for each month
..._capzd_prem_bonus	Premium bonus added to the fund value for annuity products
..._capzd_acq_exp	Acquisition expenses capitalized into DAC
..._capzd_comm_bom	Beginning of month commissions capitalized into DAC
..._capzd_comm_eom	End of month commissions capitalized into DAC; commission chargebacks are negative end-of-month commissions
..._deferred_load	Front end product loads deferred into the URL
..._unearn_rev_liab	Unearned revenue liability at the end of each month
..._reins_yrt_cost	Net reinsurance cash flows from YRT reinsurance
..._sop031_assess	Assessments used in calculating the SOP 03-1 additional liability: (1) COI charges, (2) non-deferred policy loads, (3) surrender gains, less (4) URL increase each month, plus (5) interest gains (interest earned less interest credited)
..._sop031_gmdb_benefits	Death benefits paid because of the GMDB benefit
..._sop031_gmib_benefits	Annuity benefits paid because of the GMIB benefit
..._sop031_gmwb_benefits	Surrender death benefits paid because of the GMWB benefit
..._sop031_sg_benefits	Death benefits paid because of the secondary guarantee for universal life products
..._sop031_coi_chg	COI charges each month. This value is used to calculate the SOP 03-1 liability for COI rates pattern
..._sop031_pbr_benefits	Persistency bonuses and refunds
..._sop031_pbr_fund_val	Fund value at the end of each month. This value is used to calculate the SOP 03-1 liability for sales inducements. Ignore this variable when this liability does not apply
..._dac_prem_bonus_loss	DAC premium bonus write-down for past loss recognition events
..._dac_acq_exp_loss	DAC expense write-down for past loss recognition events
..._dac_comm_loss	DAC commission write-down for past loss recognition events

5.5 Amortization

5.5.1 Amortization months

The amortization period ends at the specified number of months following the last issue date for the model points in the cohort calculations. For example, amortizing over 240 months for a cohort consisting of policies issued in 2014 results in an amortization period from 2014 January 1 (the first issue date) through 2034 November 30 (240 months from the last issue date on 2014 December 1).

When this variable is left at its zero default value, amortization is to the latest maturity for the model points included in the cohort.

5.5.2 GAAP interest rate

Specify (on the Company input page under the SFAS 97 Retrospective Deposit GAAP: Cohorts category), the interest rate use to accumulate past capitalized amounts and gross profits and to discount future capitalized amounts and gross profits. Separate rates can be specified for gross and ceded cohorts.

5.5.3 Loss recognition interest rate

Specify (on the Company input page under the SFAS 97 Retrospective Deposit GAAP: Cohorts category), the interest rate use to accumulate past capitalized amounts, gross profits, and loss recognition amounts and to discount future capitalized amounts and gross profits for loss recognition testing. Separate rates can be specified for gross and ceded cohorts.

5.5.4 Estimated gross profit

The DAC, DAC reinsurance YRT reinsurance adjustment, and unearned revenue liability are amortized using the estimated gross profits. EGP is:

$$\text{Investment gain} + \text{mortality gain} + \text{surrender gain} + \text{expense gain} - \text{persistency bonuses and refunds} - \text{increase in SOP 03-1 additional liability}$$

Where:

Investment gain is: (1) the net investment income and capital gains allocated to the fixed account fund value, less (2) interest credited to the fixed account fund value. Net investment income includes policy loan interest, less policy loan expenses. Investment income and credited interest on separate account fund values are excluded, since they net to zero.

Mortality gain is: (1) COI charges (including COI charges on negative fund values), less (2) death claim cost. Death claim cost is: (1) death benefits, less (2) fund value released on death. When stochastic scenarios are used to calculate SOP 03-1 liabilities, the death claim cost is the average of the cost across the stochastic scenarios used to create the current GAAP schedule. For FIA products, the death benefit base is the host contract value plus the embedded derivative.

Surrender gain is: (1) fund value released on surrender, withdrawal, and maturity, less (2) surrender, withdrawal, and maturity benefits. Annuitization is considered to be surrender for this purpose, with the benefit being the lump sum annuity present value. For FIA products, the benefit base is the host contract value plus the embedded derivative.

Expense gain is: (1) product loads net of deferred loads, less (2) non-deferred expenses and commissions. However, policy loan expenses are excluded since they are included in investment gain.

Persistency bonus and refund gain is the negative of persistency bonus and refunds.

SOP 03-1 liability change is discussed separately, below.

The **SFAS 133 liability** is ignored in the standard EGP formula. This assumes the hedging strategies are 100% effective, so that hedging investment income approximately offsets increases in hedging liabilities.

5.5.5 Actual gross profits

Actual gross profit (AGP) components the same as the EGP components, except that:

- Components reflect the actual profits that emerge during the experience projection.
- The investment gain also includes (1) the increase in the market value of hedges, plus (2) hedge cash flows. (The AGP column already accounts for this by subtracting the hedge cost.)
- SFAS 133 liability is included in the AGP calculation; the increase in the SFAS 133 liability is deducted from total AGPs.

5.5.6 Amortization stream

The DAC and the unearned revenue liability are amortized using the estimated gross profit (EGP) as the amortization stream.

The projection sets negative monthly EGP amounts to zero. This adjustment to the EGP does not apply for the preliminary EGP used to calculate the SOP 03-1 liability.

Market Value Adjustment (MVA) features do not affect the EGP stream other than the surrender gain.

5.5.7 Hedge allocation to GAAP cohorts

For FIA products, hedge cash flows and market values are projected on-the-fly, and cannot be pregenerated outside of the projection. Hedging is only modeled for point-to-point and moving average indexing methods. Monthly sum cap hedges are assumed to be 100% effective.

Hedge groups can be defined by plan code and indexing method, while GAAP cohorts can be defined by plan code and issue year. In order to determine the impact from hedging on the AGP for each cohort, hedge cash flows, and market values calculated for each hedge group, must be allocated to the GAAP cohorts.

Hedging allocation methods vary by hedge group and GAAP cohort definitions:

Hedge group defined by	GAAP cohort defined by	Mapping guideline
Indexing method	Issue Year	When GAAP cohorts are defined by issue year only, the hedge cash flows and market values for each hedge group are allocated to each GAAP cohort based on the formula below. Since multiple hedge groups can belong to the same cohort, the hedge amount for the cohort is the sum of each hedge group's allocation to that cohort.
Plan Code	Issue Year	$\text{Hedge Amt}_{\text{cohort}} = \text{Hedge Amt}_{\text{hedgroup}} \times \frac{\text{Fund Val}_{\text{cohort}}}{\text{Fund Val}_{\text{total}}}$
Indexing method	Plan Code	Hedge cash flows and market values for a particular hedge group are only allocated to a particular GAAP cohort when the plan codes for the hedge group and GAAP cohort are the same. The hedge cash flows and markets for each hedge group are allocated to the appropriate cohort based on the formula below. Since multiple hedge groups can belong to the same cohort, the hedge amount for the cohort is the sum of each hedge group's allocation to that cohort.
Plan Code	Plan Code/Issue Year	$\text{Hedge Amt}_{\text{cohort}} = \text{Hedge Amt}_{\text{hedgroup}} \times \frac{\text{Fund Val}_{\text{cohort}}}{\text{Fund Val}_{\text{hedgroup}}}$
Plan Code	Plan Code	When both the hedge group and GAAP cohort are defined by plan code, the entire hedge cash flows and market values of the hedge group are allocated to the GAAP cohort for the same plan code.
		$\text{Hedge Amt}_{\text{cohort}} = \text{Hedge Amt}_{\text{hedgroup}}$
Indexing method	Plan Code/Issue Year	Hedge cash flows and market values for a particular hedge group are allocated to the GAAP cohorts that use the same indexing method as the hedge group. The portion of the hedge group's cash flow and market value that is allocated to the cohort is:
		$\text{Hedge Amt}_{\text{cohort}} = \text{Hedge Amt}_{\text{hedgroup}} \times \frac{\text{Fund Val}_{\text{cohort}}}{\text{Fund Val}_{\text{hedgroup}}}$

Where

Hedge amount is the hedge cash flow or hedge market value for a particular hedge group or GAAP cohort.

Fund value is either the total fund value for all model points or the fund value for a particular GAAP cohort or hedge group.

5.6 Benefit reserve

The basic benefit reserve is the policy's fixed fund value. The basic benefit reserve is zero when the fixed fund value is negative. The separate account liability, which is the separate account fund value, is added to this basic benefit reserve.

For fixed indexed annuity (FIA) products, the basic benefit reserve is zero.

Market Value Adjustment (MVA) features do not affect the benefit reserve, since they define the cash value rather than the fund value.

5.7 Other liabilities

Other liabilities which may arise under this method include: (1) unearned revenue liability, (2) SOP 03-1 additional liability, and (3) SFAS 133 reserve. These additional reserves are discussed in separate sections or chapters below.

5.8 Rider reserves

The reserve for riders is the rider statutory reserve, and is only calculated during experience projections that include SFAS 97 retrospective deposit GAAP results.

The rider reserve is combined with the SOP 03-1 additional liability for financial statement presentation.

5.9 Deferred acquisition costs

A deferred acquisition cost (DAC) GAAP asset is established to capitalize up-front expenses and commissions. The DAC asset spreads capitalized expenses and commissions over the product's 'useful life', which is the specified amortization period discussed above. The DAC asset is increased by additional capitalized expenses and commissions, then by the GAAP interest rate, and is then amortized using the amortization stream.

5.9.1 Capitalized expenses

For gross business, capitalized expenses include a specified percent of each type of acquisition expenses:

- Amount per policy
- Amount per unit (life only)
- Percent of premium
- Percent of premium bonus (deferred annuity products).

There are no capitalized acquisition expenses for ceded business.

5.9.2 Fixed expenses

Capitalization does not apply for fixed expenses, which are specified for the business segment, since these are not considered to be policy-related expenses.

5.9.3 Capitalized commissions

For gross business, capitalized expenses automatically include commission rates which are higher than the ultimate rate for the last premium payment. The capitalized percent of first year premium commission applies to the projection of annualized commissions and chargebacks in the GAAP schedule projection. Thus chargebacks after the first year result in negative capitalized amounts using the capitalized commission rate for first year commissions.

For ceded business, capitalized commissions automatically include expense allowances which are higher than the ultimate rate.

5.9.4 DAC valuation

At each valuation date, the DAC asset is:

$$k_{DAC} \times PV \text{ future EGP}_0 - PV \text{ future capitalized expenses}_0$$

where:

$$k_{DAC} = \frac{\left[\text{accumulated capitalized expenses}_0 + PV \text{ future capitalized expenses}_0 \right] - \text{accumulated loss recognition}_0}{\left[\text{accumulated EGP}_0 + PV \text{ future EGP}_0 \right]}$$

PV = present value of projected amounts

Accumulated = accumulated value of history amounts

Loss recognition = adjustment made in the event of loss recognition (discussed below).

Future EGP and capitalized expense amounts are projected using the GAAP 'best estimate' assumptions (see the Unlocking section, below), using only the model points already issued by the schedule calculation date.

Experience profit and capitalized expense amounts before the projection date are specified in the cohort history variables. The projection results become the experience amounts for future valuation dates.

5.9.5 DAC roll forward

In each month until the next valuation date, the DAC balance rolls forward using the schedule amortization:

$$DAC_t = [DAC_{t-1} - + \text{capitalized expenses}_t] \times (1 + i_t)^{1/12} - k_{DAC} \times EGP_t$$

The monthly interest rate, amortization factor, and EGP are the GAAP schedule amounts which were calculated at the last valuation date.

The monthly capitalized expenses are the actual amounts capitalized each month according to the definitions used in creating the current GAAP schedule. For example, when the schedule was created on June 30 for a cohort including the current year's issues, the interest rate, amortization factor, and EGP amounts were determined only for issues through the end of June. Then the roll forward in July includes the capitalized expenses from new issues added into the cohort in July, and so on for succeeding months. At the next valuation date, the interest rate, amortization factor, and EGP amounts calculated for the new schedule will be based on the new issues up to that valuation date.

5.9.6 At the projection date

The projection date is treated as a valuation date for determining the initial DAC balance. The amortization factor is calculated using the historical EGP and capitalized expense amounts, which are the schedule history amounts specified for each cohort. The monthly schedule EGP and capitalized expense amounts are projected for use in the roll forward calculations.

5.10 Unearned revenue liability

The unearned revenue liability (URL) spreads the income from excess front-end loads over the period benefited by the loads, which is the DAC amortization period. The URL is increased by the GAAP interest rate, and by deferred loads, and is then amortized using the DAC amortization stream. The calculation process is similar to the description for DAC, above.

5.10.1 Deferred loads

Unearned revenues are amounts assessed against the policy to pay for future services. These 'excess' front end loads automatically include: (1) issue loads, and (2) other loads in excess of ultimate rates.

5.10.2 URL valuation

At each valuation date, the URL is:

$$k_{URL} \times PV EGP_0 - PV \text{ deferred loads}_0$$

where:

$$k_{URL} = \frac{[\text{accumulated deferred loads}_0 + PV \text{ deferred loads}_0]}{[\text{accumulated EGP}_0 + PV EGP_0]}$$

5.10.3 URL roll forward

In each month until the next valuation date, the URL balance rolls forward:

$$URL_t = [URL_{t-1} + \text{deferred loads}_t] \times (1 + i_t)^{1/12} - k_{URL} \times EGP_t$$

5.11 Reinsurance effect

5.11.1 YRT reinsurance

YRT reinsurance is included in the gross business cohort calculation as a 'reinsurance adjustment asset', which may be positive or negative.

At the valuation date, the reinsurance asset is:

$$k_{reins} \times PV \text{ future EGP} - PV \text{ future reinsurance cost}$$

where:

$$k_{reins} = \frac{[\text{accumulated reinsurance cost} + PV \text{ future reinsurance cost}]}{[\text{accumulated EGP} + PV \text{ future EGP}]}$$

EGP = same as for the main cohort calculation

PV = present value of projected amounts

Accumulated = accumulated value of experience amounts

Reinsurance cost is the net cash flows of the YRT reinsurance treaties

In each month until the next valuation date, the reinsurance asset rolls forward using the schedule amortization:

$$DAC YRT_t = [DAC YRT_{t-1}] \times (1 + i_t)^{1/12} + \text{reinsurance cost}_t - k_{DAC YRT} \times EGP_t$$

5.11.2 Coinsurance

When the cohort includes coinsurance, each of the GAAP results (everything discussed above) is calculated twice, once using the gross amounts (ignoring coinsurance), then using only the ceded coinsurance amounts. The net GAAP amount is the gross result less the ceded coinsurance result.

GAAP history amounts are split between gross and ceded coinsurance amounts, for the purposes of these separate calculations.

The EGP for the coinsurance calculation assumes that the reinsurer's experience mirrors the ceding company's experience. The expense gain is the same as for the gross calculation, adjusted by the expense allowances paid to the ceding company net of chargebacks.

5.11.3 Modified coinsurance

When the cohort includes modified coinsurance, each of the GAAP results (everything discussed above) is calculated twice, once using the gross amounts (ignoring coinsurance), then using only the ceded modified coinsurance amounts.

Similar to the coinsurance treatment, GAAP history amounts are split between gross and ceded modified coinsurance amounts, for the purposes of these separate calculations.

Also, the EGP for the coinsurance calculation assumes that the reinsurer's experience mirrors the ceding company's experience. The expense gain is the same as for the gross calculation, adjusted by the expense allowances paid to the ceding company net of chargebacks.

The reinsurance statutory reserve is captured as the modified coinsurance liability. This amount is held as a liability on the GAAP balance sheet which results in the total benefit reserve for the ceding company which is equal to the benefit reserve calculated without considering reinsurance, less the portion reinsured, plus the statutory reserve held by the ceding company.

5.12 Loss recognition

5.12.1 Loss recognition test

SFAS 97 retrospective deposit GAAP requires that future expected profits must be sufficient to amortize the DAC asset. A loss recognition event can occur when actual experience, or the current view of anticipated experience, is significantly worse than originally expected. The DAC amortization factor is limited to 100%, which forces an immediate write-down of the DAC asset.

Specify (on the Company input page under the GAAP: SFAS 97 Retrospective Deposit: General category) whether to include loss recognition testing. When the Loss recognition definition is Yes, the loss recognition test is performed at each valuation date. The formula results in a loss recognition amount which effectively forces the cumulative amortization factors to 100%. Note that turning off the test can result in the amortization factor exceeding 100% for some cohorts.

When Total capitalized expenses > Total EGP, then:

$$\text{loss recognition } n_t = \text{total capitalized expenses}_t - \text{previous loss recognition } n_t - \text{total EGP}_t$$

where:

Total capitalized expenses = accumulated capitalized expenses + PV capitalized expenses

Total EGP = accumulated EGP + PV EGP

Previous loss recognition = accumulated loss recognition to date

The accumulations and present values for loss recognition testing used an assumption of earned rates. Specify (on the Company input page under the GAAP: SFAS 97 Retrospective Deposit: Cohorts category), the loss recognition interest rate used to determine the accumulated and present values for the loss recognition testing. A separate rate is specified for gross and ceded cohorts.

5.12.2 GAAP results adjustment

The loss recognition amount reduces the projected DAC balances through an adjustment to the k factors for each of capitalized premium bonus, acquisition expenses, and commissions. Premium bonus DAC is written down first, followed by acquisition expense DAC, and finally commission DAC.

Once loss recognition occurs, the rate used to determine the accumulated and present values for the k factor calculations, and to determine interest on beginning DAC balances is the loss recognition rate.

5.13 Unlocking

5.13.1 Unlocking dates

The SFAS 97 retrospective deposit method calls for periodic unlocking of the DAC and amortization factors (k factors) for the URL and SOP 03-1 additional liability. Unlocking occurs when: (1) experience turns out to be different than assumed, or (2) assumptions for the future change.

The dynamic GAAP unlocking dates are the inner loop valuation dates which begin each time step. The projection date is an 'unlocking' date, since GAAP schedules are calculated on-the-fly.

5.13.2 DAC and GAAP liabilities recalculation

At each unlocking date, the amortization factors are recalculated using the 'initial calculation' formulas shown above. The changes in the DAC and GAAP liability amounts, compared to the 'roll forward' amounts, are reported as the effect of the unlocking.

There are two aspects of the unlocking calculations:

Retrospective unlocking

The accumulated EGP, capitalized expense, and deferred load results are calculated using historical amounts up to the projection date, and experience projection amounts up to the unlocking date. 'Retrospective unlocking' is the change in the DAC and URL amounts related to replacing the previous accumulation of 'schedule' amounts with the accumulation of 'experience' amounts.

Recalculated schedule amounts

The present value results, and the monthly schedule amounts for future 'roll forward' calculations, are calculated using GAAP best estimate assumptions. This is a nested projection starting from the experience projection in-force amounts as of the unlocking date.

5.14 Financial statement presentation

5.14.1 Income statement

Revenue includes product charges and fee income, rider charges, gains from surrender, release of the unearned revenue liability, which, when reporting for the segment, are reported net of reinsurance offsets. Also included is the reinsurance experience refunds (when reporting for the business segment), cash flow interest, investment income and capital gains on the net GAAP liability, and the modified coinsurance adjustment (when reporting for the business segment). When reporting for the business segment, reinsurance offsets are also incorporated into the cash flow interest and the investment income and capital gains on the reinsurance net GAAP liability.

Basic benefits include the direct excess death benefits, excess maturity benefits, excess endowment benefits, excess accumulation benefits, excess income benefits, excess withdrawal benefits, premium bonuses paid less, when reporting for the segment, reinsurance recoveries for the basic benefits reinsured.

Other benefits include the credited interest on the fixed fund, the unearned revenue liability increase, the SOP 03-1 additional liability increase (which includes the rider reserve increase), the SFAS 133 additional liability increase, where, when reporting for the segment, reinsurance offsets are incorporated into the

reported items. Also included is the modified coinsurance adjustment (when reporting for the ceded cohorts), and the modified coinsurance liability increase.

Expenses includes the total of excess over renewal commissions, total acquisition expenses less a separate item for deferred commissions and acquisition expenses, renewal commissions, maintenance expenses, premium taxes, letter of credit costs, DAC (expense DAC plus the YRT asset) amortization, the hedge (gain)/loss, and less a separate item for the YRT reinsurance cost applicable for YRT reinsurance.

5.14.2 Balance sheet

Assets include the invested assets, other invested assets, the separate account assets, hedge assets, the DAC, and the YRT reinsurance adjustment asset. When reporting for a business segment, 'reinsurance receivable' is the reinsurance net GAAP liability less the modified coinsurance liability.

Liabilities include the fixed fund value, the separate account fund value, the unearned revenue liability, the SOP 03-1 additional liability (which includes the rider reserve), the SFAS 133 additional liability, and the modified coinsurance liability.

6 SFAS 97 SFAS 133 ADDITIONAL LIABILITY

6.1 Applicability

SFAS 133, which is clarified by SFAS 157, defines the liabilities held for benefit guarantee features which are considered not to have significant mortality risks.

SFAS 133 automatically applies for fixed indexed annuity (FIA) products embedded derivatives. It can also apply for the GMWB FIA product benefits. This choice applies separately for ceded reinsurance calculations.

For variable annuity products, SFAS 133 automatically applies for GMAB benefits. It can also apply for GMWB and GMIB VA product benefits. This choice applies separately for the gross and ceded reinsurance calculation for each of these benefits.

The SFAS 133 additional liability is calculated separately for each SFAS 97 GAAP cohort. Unlocking occurs when each GAAP schedule is created.

Important Note! SFAS 133 functionality has not yet been implemented for indexed universal life (IUL) products. This will be included in a future release.

6.2 Guaranteed benefits

6.2.1 Separate valuation

SFAS 133 calculations are performed over a set of stochastic scenarios. These calculations are performed in a separate SFAS 133 valuation task. This valuation uses formulas in the sfas133 submodel.

SFAS 133 valuations create SFAS 133 liabilities for each model point, which are created at the projection date and each inner loop valuation date. The model point liabilities are calculated by a seriatim valuation task with target column valn_tgt_gaap_sfas133.

6.2.2 Risk margins

SFAS 133 calculations use assumptions which are modified to include a risk margin. Include these margins in the assumptions for the SFAS 133 valuation task.

6.2.3 Scenarios

Risk neutral scenarios must be used in SFAS 133 calculations.

GAAP valuations after the projection date use scenarios generated from the outer loop scenario risk free yield curve as of each valuation date. This can be a 'quick generation' which produces only the needed indexes, rather than the complete set of rates.

6.2.4 *Unadjusted liability calculation*

Within the separate valuation task, for each benefit, present values of expected benefits and charges are calculated across a range of risk neutral scenarios. Each cash flow is discounted at the risk free spot rate plus a provision for own credit risk. The own credit spread variable must be linked to an external source with values varying by hidden variable lookup keys `quality_id` and `rate_term`, plus the date column.

Then within the experience projection, for each benefit, the unadjusted SFAS 133 liabilities are calculated as:

$$\left[\text{PV average expected claims}(t) - \text{PV average expected charges}(t) * (1.0 - \text{required profit charge}) \right]$$

where:

The average amounts are across a range of stochastic scenarios

The expected claims are amounts paid because of the benefit, such as GMWB withdrawals after the fund value is exhausted.

The expected charges are calculated using the specified benefit charges.

The required profit charge is the amount specified on the model point record for inforce model points or solved for amount for new issue model points which ensures that the initial SFAS 133 liability is zero.

6.2.5 *Experience adjustments*

Since the assumptions used in the SFAS 133 valuation likely differ from the experience assumptions, the final SFAS 133 liabilities used in the experience projection are the unadjusted SFAS 133 liabilities times (1) a policies adjustment and (2) a fund value adjustment.

The policies adjustment is the ratio of the policies in the experience projection divided by the average policies in the valuation task.

The fund value adjustment is the ratio of the average fund value in the valuation task divided by the fund value in the experience projection.

6.3 *Embedded derivatives*

SFAS 133 automatically applies for FIA products embedded derivatives calculations, using the option budget method.

6.3.1 *Combined valuation*

Embedded derivative SFAS 133 calculations are performed using the same single economic scenario as applies for the SFAS 97 valuation. Thus, the SFAS 133 embedded derivative valuation is performed in the SFAS 97 valuation task. These calculations are performed in the `sfas97rd` submodel. There is no separate SFAS 133 valuation task, and the `sfas133` submodel (which is required for the guaranteed benefits calculations) does not apply.

The initial valuation occurs at the projection date, plus at each SFAS 97 revaluation date. The revaluation periods are the inner loop valuation dates at the beginning of each time step. SFAS 133 revaluation and SFAS 97 GAAP unlocking both occur on each valuation date.

6.3.2 *Best estimate assumptions*

SFAS 133 calculations are based on 'best estimate' assumptions. These can be the claim and policyholder behavior rates which apply for the experience projection.

6.3.3 *Scenarios*

The method used for the FIA SFAS 133 valuation is the option budget method. Specify the option budget percent on the model point record.

6.3.4 *Reserve calculation*

The SFAS 133 embedded derivative liability is calculated from a single deterministic scenario as:

Host Contract Value + Embedded Derivative

6.3.5 *Reserve valuation*

At each valuation date, the SFAS 133 liability is calculated for a single deterministic scenario. This calculation occurs in the same projection as the DAC valuation. The initial valuation occurs at the projection date and the SFAS 133 liability is revalued at each revaluation date. The revaluation periods are the inner loop valuation dates at the beginning of each time step. SFAS 133 revaluation and GAAP unlocking both occur on each inner loop valuation date.

Host contract value

The host contract value is calculated at issue as the premium paid (including the premium bonus) less the embedded derivative value. In subsequent months, the host contract value is (1) increased by the host accrual rate, (2) reduced by decrements, and (3) directly reduced by withdrawals.

The host accrual rate is calculated at issue and is the level growth rate required to accumulate the host contract value to the guaranteed fund value at the end of the amortization period. The guaranteed fund value at the end of the amortization period is a specified percent of the premium paid, excluding the premium bonus, accumulated at the guaranteed rate over the amortization period. The amortization period for each model point is the number of amortization years since the issue date of that model point.

For in-force model points, specify on the model point record the initial (at the projection date) host contract value and the host accrual rate.

Embedded derivative value

The embedded derivative value is the present value of embedded derivative cash flows over the amortization period. Embedded derivative cash flows occur when the policyholder dies, lapses, or annuitizes. The embedded derivative cash flow is the fund value in excess of the guaranteed amount. The present value of the embedded derivative cash flows is determined by discounting the cash flows by survivorship and interest. The discount rate is the risk free spot rate.

The fund value used to determine the embedded derivative: (1) includes the premium bonus, (2) is reduced by loads and charges, (3) accumulates at the GAAP interest rate, and (4) is reduced by decrements and withdrawals.

The guaranteed amount is the greater of: (1) the guaranteed minimum surrender value and (2) cumulative premiums (including the premium bonus) less cumulative withdrawals.

The guaranteed minimum surrender value: (1) is a specified percent of the premium paid (excluding the premium bonus), (2) accumulates at the minimum guaranteed rate, and (3) is reduced by decrements and withdrawals.

Embedded derivative value revaluation

At each revaluation period, the embedded derivative value is revalued prospectively from the beginning of the revaluation period. The host contract value is not revalued at each revaluation period. This is because the

initial host contract value and the host accrual rate are both calculated at issue, and the host contract value is rolled forward at the host accrual rate during the experience projection.

6.3.6 GMWB

The GMWB directly affects the embedded derivative cash flows, and therefore the SFAS 133 liability.

GMWB charges reduce both the fund value and minimum guaranteed contract value.

6.3.7 Coinsurance

When the cohort includes coinsurance, each of the GAAP results (everything discussed above) is calculated twice, once using the gross amounts (ignoring coinsurance), then using only the ceded coinsurance amounts. The net GAAP amount is the gross result less the ceded coinsurance result.

6.3.8 Between valuation dates

The FIA SFAS 133 reserve is bifurcated between the embedded derivative and the host contract value. During the experience projection, the embedded derivative is approximated between valuation periods as:

$$\text{Embedded derivative}_t = \text{Embedded derivative}_{t-1} \times \frac{\text{Fund Value}_{t-1}}{\text{Fund Value}_t}$$

A similar approximation is not needed for the host contract value, since the monthly host contract values are known at the initial valuation period and are rolled forward at a known rate of interest, the host accrual rate, in between revaluation periods.

7 SFAS 97 SOP 03-1 LIABILITY

7.1 Applicability

AICPA SOP 03-1 requires an additional reserve to be held for benefit features and charging/crediting designs which result in losses following profits in earlier policy years, and which have significant mortality risk.

The SOP 03-1 additional liability valuations create GAAP schedules for each cohort, which are created at the projection date and each inner loop valuation date. The schedules are calculated by a portfolio valuation task with target column valn_tgt_gaap_sop031.

Unlocking occurs when each GAAP schedule is created.

7.2 Calculation approach

SOP 03-1 additional liabilities are calculated using two GAAP projection tasks:

First, stochastic iterations project the amounts which will be used to calculate the benefit ratios for the guarantee benefit liabilities.

Then, a deterministic projection task calculates the benefit ratio for each of these stochastic iterations, and averages the ratios and calculates the guarantee benefit liability. This task also calculates the sales inducements, and calculates the DAC schedule.

7.3 Guaranteed benefits

7.3.1 Benefit features

For benefit features with significant mortality risk, this additional liability is for the portion of accumulated policy assessments that compensate the company for future benefits in excess of what would be paid without the feature.

A separate SOP 03-1 additional liability is calculated for each benefit, including:

- UL secondary guarantees (ULSG)
- IUL secondary guaranteed (IULSG)
- VUL and VA Guaranteed minimum death benefit (GMDB)
- VA Guaranteed minimum income benefit (GMIB)
- VA Guaranteed minimum withdrawal benefit (GMWB).
- FIA Guaranteed minimum withdrawal benefit (GMWB)

The GMIB and GMWB benefits are included when the user has selected to use an SOP 03-1 valuation approach for those benefits.

7.3.2 Benefit guarantees liability

The SOP 03-1 additional liability is calculated separately for each applicable product feature. No SOP 03-1 liability can be negative. The benefit guarantees liability is a stochastic calculation (see the Scenarios topic, below).

The additional liability for benefit guarantees is:

$$\text{benefit ratio} \times \text{accumulated actual assessments}_t - \text{accumulated actual excess benefits}_t$$

where:

$$\text{benefit ratio} = \frac{[\text{accumulated actual excess benefits}_0 + \text{PV future expected excess benefits}_0]}{[\text{accumulated actual assessments}_0 + \text{PV future expected assessments}_0]}$$

The benefit ratio is determined each time the GAAP schedule is created at the projection date, and each unlocking date.

The present value of future expected amounts are the average across the stochastic scenarios projected for this valuation. See the Scenarios topic, below.

The discount rate is the DAC amortization interest rate which would apply for each stochastic scenario.

Excess claims are benefits paid due to the benefit guarantee. That is, claims paid less the claims which would have been paid without the guarantee.

7.3.3 Assessments

The assessments for the SOP 03-1 benefit guarantee liability are the sum of:

- Expense revenue = policy loads net of deferred loads
- Charges revenue = cost of insurance charges, plus benefit guarantee charges such as for GMDB, GMAB, GMIB, and GMWB.
- Interest revenue = the spread earned on the fixed account.

7.3.4 Scenarios

The scenarios used in calculating the benefit guarantees liability are real world economic scenarios generated as of the projection date.

GAAP valuations after the projection date reuse the set of scenarios generated for this calculation at $t = 0$.

7.3.5 Approximate change in URL

The change in the URL (which is included in the assessments for the benefit ratio calculation) is approximated to remove the circularity caused by the URL being amortized using the estimated gross profit stream, which is reduced by the change in the SOP 03-1 additional liability, which is calculated using the change in the URL. For SOP 03-1 calculations, the decrease in the URL is approximated as:

$$\frac{\text{URL k factor} \times [\text{EGP stream}_t + \text{excess benefits}_t]}{[1 + \text{URL k factor} + \text{SOP 03-1 benefits k factor}]}$$

where:

EGP stream is calculated prior to the adjustment for the change in SOP 03-1 additional liability, and is not adjusted to remove negative amounts.

URL k factor is calculated using that EGP stream.

Excess benefits are the claims paid due to the guarantee, or alternatively the COI charges not collected. These were subtracted in calculating the preliminary EGP stream.

SOP 03-1 benefits k factor is calculated using the same preliminary EGP stream.

7.4 Sales inducements

An additional SOP 03-1 liability is calculated for combined persistency bonuses and refunds. This liability cannot be negative, and is zero after the last bonus/refund has been paid.

7.4.1 Sales inducements valuation

At each valuation date, the sales inducements liability is:

$$\text{PV future PBR}_0 - k_{\text{PBR}} \times \text{PV fund values}_0$$

where:

$$k_{\text{PBR}} = \frac{[\text{accumulated actual PBR}_0 + \text{PV future PBR}_0]}{[\text{accumulated actual fund values}_0 + \text{PV future fund values}_0]}$$

PBR = persistency bonuses and refunds

PV = present value of expected amounts through the last bonus/refund date, discounted using the GAAP interest rate, with the mortality decrement only (ignoring lapses and withdrawals).

Accumulated = accumulated value of history amounts

Future PBR amounts and fund values are projected using the GAAP 'best estimate' assumptions, using only the model points already issued by the schedule calculation date.

Future PBR amounts and fund values are projected assuming zero lapses.

Actual PBR amounts and fund values before the projection date are specified in the GAAP history variables. The experience projection results become the actual amounts for future valuation dates.

The projection date is treated as a valuation date for determining the initial sales inducements liability. The amortization factor is calculated using the historical fund value and persistency bonus and refund amounts,

which are GAAP history amounts specified for each cohort. The monthly schedule fund value and persistency bonus and refund amounts are projected for use in the roll forward calculations.

7.4.2 Sales inducements liability roll forward

In each month until the next valuation date, the liability amount rolls forward using the schedule amortization:

$$\text{Liability}_t = \text{Liability}_{t-1} \times (1 + i_t)^{1/12} - \text{PBR}_t + k_{\text{PBR}} \times \text{Fund value}_t$$

The monthly interest rate, amortization factor, fund value, and persistency bonuses and refunds are the GAAP schedule amounts which were calculated at the valuation date.

7.4.3 Schedule history

The cohort history variables include the fund value for each past month, since this is used for calculating the sales inducements liability.

8 SFAS 120 MUTUAL COMPANY METHOD

8.1 Applicability

SFAS 120 and SOP 95-1 GAAP accounting applies for participating traditional life products issued by mutual companies that pay dividends based on the contribution principle and actively manage those dividends. Stock companies may also elect to use these standards for dividends paid under the contribution principle.

SFAS 120 valuations create GAAP factors (for the net level premium reserve) using assumptions locked in at issue and GAAP schedules using best estimate assumptions.

Loss recognition is performed by a separate GPR valuation, which is applied for each cohort by the experience projection.

The net level premium reserve factors are calculated by a seriatim projection tasks with target column valn_tgt_gaap_sfas60_sfas97lp_sfas120_nlp. The GAAP schedules are calculated by a portfolio projection task with target column valn_tgt_gaap_sfas120.

SFAS 120 GAAP factors are on-the-fly calculations at the projection date, from the issue date of each model point. GAAP schedules for each cohort, are created at the projection date and each inner loop valuation date.

Unlocking and loss recognition occur when each GAAP schedule is created.

8.2 Factors

SFAS 120 GAAP factors for the net level premium reserve are calculated for each model point from issue, using the net level premium reserve assumptions determined at issue. The exception is that the GAAP factor is replaced by the loss recognition gross premium reserve after an extreme loss recognition event that writes down all of the remaining DAC asset and still has more losses to account for.

SFAS 120 GAAP factors are stored by model point. GAAP factor results are converted into results per unit in force based on the units projected during the GAAP valuation task. Then the GAAP results for the GAAP schedule and experience projection are these amounts per unit, times the units projected to be in force during the GAAP schedule and experience projection.

8.3 Schedules

8.3.1 Monthly values

The following SFAS 120 schedule values are created for each cohort:

- Estimated gross margins
- GAAP interest rate

8.3.2 *Starting values*

Additionally, the following values are created at the valuation date for each cohort:

- k factor for capitalized acquisition expenses
- k factor for capitalized commissions
- k factor for terminal dividend liability
- k factor for YRT reinsurance asset
- DAC for capitalized acquisition expenses
- DAC for capitalized commissions
- DAC YRT asset
- Terminal dividend liability

8.4 Assumptions

8.4.1 *Separate GAAP assumption variables*

All assumption variables operate the same for the GAAP valuation projection task as for the experience projection task.

8.4.2 *Prescribed assumptions*

The net level premium reserve is calculated using guaranteed cash value mortality and the dividend interest rate. When the dividend interest rate is unknown, specify an appropriate interest rate based on the hierarchy specified in the regulation.

8.4.3 *Best estimate basis*

GAAP schedules amounts are calculated using best estimate assumptions as of the projection date. These assumptions do not include provisions for adverse deviation (PAD).

8.5 Cohort history

For each in-force cohort, specify the cohort_history_ variables (on the Company input page under the GAAP SFAS 120: History category). These variables are located in the sfas120 submodel. Specify the value for each month from the beginning of the cohort up to the projection date.

These variables must be linked to an external source varying by hidden variable lookup keys cohort_id and cohort_reins_defn. The cohort_reins_defn lookup key separates Gross (ignoring reinsurance) from Ceded (coinsurance and modified coinsurance treaties) values.

Variable	Value
cohort_history_agm	Actual gross margins each month
..._gaap_int_rate	GAAP amortization rate
..._capzd_acq_exp	Acquisition expenses capitalized into DAC
..._capzd_comm_bom	Beginning of month commissions capitalized into DAC
..._capzd_comm_eom	End of month commissions capitalized into DAC; commission chargebacks are negative end-of-month commissions
..._terminal_div	Actual terminal dividends paid
..._reins_yrt_cost	Statutory reinsurance cost from YRT reinsurance

8.6 Amortization

8.6.1 Amortization months

DAC, the DAC YRT reinsurance asset, and the terminal dividend liability are amortized over the specified number of amortization months. When this variable is left at its zero default value, GAAP results are calculated to maturity. These calculations only consider amounts through the end of the amortization months.

The net level premium reserve considers amounts through maturity.

8.6.2 Estimated gross margins

The DAC, DAC reinsurance YRT asset, and terminal dividend liability are amortized using the estimated gross margins. EGM's are calculated as:

$$EGM_t = \text{Premiums}_t + \text{Inv Inc}_t - \text{Benefits}_t - \text{Ben Res Incr}_t - \text{Expenses}_t - \text{Dividends}_t$$

where:

Premiums are the expected gross premiums and dividends applied to purchase dividend options

Investment income is the interest earned on the NLP reserve and the beginning of month cash flows

Benefits are the death, surrender, endowment, and maturity benefits

Benefit reserve increase is the increase in the NLP reserve

Expenses include administrative expenses, maintenance expenses, non-deferrable acquisition expenses, non-deferrable commissions, the premium tax, and the reinsurance cost

Dividends are the expected annual dividends. Terminal dividends are excluded from the EGM.

The benefit reserve increase included in the EGM is based on the GAAP factor for the net level premium reserve per unit, times the units developed during the schedule projection.

8.6.3 Amortization stream

The DAC, DAC YRT reinsurance asset, and terminal dividend liability are amortized using the EGM amortization stream.

The projection sets negative monthly EGM amounts to zero.

8.6.4 GAAP interest rate

The GAAP interest rate is the asset yield for the SFAS 120 valuation task.

8.7 Net level premium reserve

8.7.1 NLP benefits

Benefits include death benefits and endowment benefits.

8.7.2 NLP reserve valuation

The NLP reserve for each model point is:

$$\text{PV benefits}_t - k_{\text{benefits}} \times \text{PV gross premiums}_t$$

where:

$$k_{\text{benefits}} = \frac{[\text{PV benefits}_{\text{issue}}]}{\text{PV gross premiums}_{\text{issue}}}$$

PV = present value over maturity, discounted at the dividend interest rate.

Amounts are projected from issue using the SFAS 120 benefit reserve assumptions, which are the guaranteed cash value mortality rate and the dividend interest rate. Expenses and other decrements do not affect the NLP reserve.

The per unit NLP reserve stored for each model point (from the issue date through maturity) is the model point amount, divided by the number of units for each month projected by the SFAS 120 valuation task.

8.8 Terminal dividend liability

The terminal dividend liability establishes a liability for the terminal dividends.

8.8.1 Terminal dividend liability valuation

At each valuation date, the terminal dividend liability for each cohort is:

$$\text{PV terminal dividends}_0 - k_{\text{TDL}} \times \text{PV EGM}_0$$

where:

$$k_{\text{TDL}} = \frac{[\text{accumulated terminal dividends}_0 + \text{PV terminal dividends}_0]}{[\text{accumulated EGM}_0 + \text{PV EGM}_0]}$$

PV = present value over maturity, discounted at the GAAP interest rate.

Accumulated = accumulated value of history amounts

EGM is the estimated gross margins.

Future EGM and terminal dividend liability amounts are projected using the GAAP 'best estimate' assumptions (see the Unlocking section, below), using only the model points already issued by the schedule calculation date.

8.8.2 Terminal dividend liability roll forward

In each month until the next valuation date, the terminal dividend liability rolls forward using the schedule amortization:

$$\text{TDL}_t = \text{TDL}_{t-1} \times (1 + i_t)^{1/12} - \text{terminal dividends}_t + k_{\text{TDL}} \times \text{EGM}_t$$

The monthly interest rate, amortization factor, and estimated gross margins (EGM) are the GAAP schedule amounts which were calculated at the valuation date.

8.9 Other reserves

8.9.1 Rider reserve

The rider reserve is the rider statutory reserve, and is only calculated during experience projections that include SFAS 120 GAAP results.

The rider reserve is included with the benefit reserve for the financial statement presentation.

8.10 Policyholder dividends

8.10.1 Annual dividends

User SFAS 120, annual dividends are not included in the benefit reserve calculation.

8.10.2 Dividend liability

The annual dividend liability is the actual accrual of dividends earned by the policyholder in the current policy year. The calculation assumes 1/12 of the annual dividend is accrued each policy month.

8.10.3 Dividend options liability

The dividend options liability is the statutory reserve for the dividend options. This amount is included with the NLP reserve when reporting the benefit reserve.

8.11 Deferred acquisition costs

A deferred acquisition cost (DAC) GAAP asset is established to capitalize up-front expenses and commissions. The DAC asset spreads capitalized expenses and commissions over the product's 'useful life', which is the specified amortization period discussed above. The DAC asset is increased by additional capitalized expenses and commissions, then by the GAAP interest rate, and is then amortized using the amortization stream.

8.11.1 Capitalized expenses

For gross business, capitalized expenses include a specified percent of each type of acquisition expenses, in excess of the ultimate rate:

- Amount per policy
- Amount per unit
- Percent of premium.

There are no capitalized acquisition expenses for ceded business.

8.11.2 Fixed expenses

Capitalization does not apply for fixed expenses, which are specified for the business segment, since these are not considered to be policy-related expenses.

8.11.3 Capitalized commissions

For gross business, capitalized commissions automatically include commission rates which are higher than the ultimate rate for the last premium payment. The capitalized percent of first year premium commission applies to the projection of annualized commissions and chargebacks in the GAAP factors projection. Thus

chargebacks after the first year result in negative capitalized amounts using the capitalized commission rate for first year commissions.

For ceded business, capitalized commissions automatically include expense allowances which are higher than the ultimate rate.

8.11.4 DAC valuation

A separate DAC asset is calculated for expenses and commissions. At each valuation date, the DAC asset for each cohort is:

$$k_{DAC} \times PV EGM_0 - PV \text{ future capitalized expenses}_0$$

where:

$$k_{DAC} = \frac{[\text{accumulated capitalized expenses}_0 + PV \text{ future capitalized expenses}_0]}{[\text{accumulated EGM}_0 + PV \text{ future EGM}_0]}$$

PV = present value over amortization months, discounted at the GAAP interest rate.

Accumulated = accumulated value of history amounts

Future EGM and capitalized expense amounts are projected using the GAAP 'best estimate' assumptions (see the Unlocking section, below), using only the model points already issued by the schedule calculation date.

Capitalized expenses are projected during the creation of the GAAP factors/schedule.

The projection date is treated as a valuation date for determining the initial DAC balance. The amortization factor is calculated using the historical EGM and capitalized expense amounts, which are GAAP history amounts specified for each cohort. The monthly schedule EGM and capitalized expense amounts are projected for use in the roll forward calculations.

Experience profit and capitalized expense amounts before the projection date are specified in the GAAP history variables. The projection results become the experience amounts for future unlocking dates.

8.11.5 DAC roll forward

In each month until the next valuation date, the DAC balance rolls forward using the schedule amortization:

$$DAC_t = [DAC_{t-1} + \text{capitalized expenses}_t] \times (1 + i_t)^{1/12} - k_{DAC} \times EGM_t$$

The monthly interest rate, amortization factor, and EGM are the GAAP schedule amounts which were calculated at the projection date or the last unlocking date.

The monthly capitalized expenses are the actual amounts capitalized each month according to the definitions used in creating the current GAAP schedule. For example, when the schedule was created on June 30 for a cohort including the current year's issues, the interest rate, amortization factor, and EGM amounts were determined only for issues through the end of June. Then the roll forward in July includes the capitalized expenses from new issues added into the cohort in July, and so on for succeeding months. At the next unlocking date, the interest rate, amortization factor, and EGM amounts calculated for the new schedule will be based on the new issues up to that unlocking date.

8.12 Reinsurance effect

8.12.1 YRT reinsurance

YRT reinsurance is calculated for each gross cohort as a 'reinsurance asset', which is often negative. At the projection date and each unlocking date, the reinsurance asset is:

$$k_{\text{reins}} \times \text{PV EGM}_0 - \text{PV reinsurance cost}_0$$

where:

$$k_{\text{reins}} = \frac{[\text{accumulated reinsurance cost}_0 + \text{PV reinsurance cost}_0]}{[\text{accumulated EGM}_0 + \text{PV EGM}_0]}$$

EGM = same as for the main cohort calculation

PV = present value of projected amounts

Accumulated = accumulated value of experience amounts

Reinsurance cost is the net cash flows of the YRT reinsurance treaties

Future EGM and reinsurance cost amounts are projected using the GAAP 'best estimate' assumptions (see the Unlocking section, below), using only the model points already issued by the schedule calculation date.

In each month until the next valuation date, the DAC reinsurance YRT asset rolls forward using the schedule amortization:

$$\text{DACYRT}_t = [\text{DACYRT}_{t-1}] \times (1 + i_t)^{1/12} + \text{reinsurance cost}_t - k_{\text{DACYRT}} \times \text{EGM}_t$$

where:

DAC YRT is the DAC reinsurance YRT asset

The monthly interest rate, amortization factor, and EGM are the GAAP schedule amounts which were calculated at the valuation date.

8.12.2 Coinsurance

When the model point includes coinsurance, the benefit reserve and DAC amounts are calculated twice, once using the gross amounts (ignoring coinsurance), then using only the ceded coinsurance amounts. The net GAAP amount is the gross result less the ceded coinsurance result.

8.12.3 Modified coinsurance

When the model point includes modified coinsurance, the benefit reserve and DAC amounts are calculated twice, once using the gross amounts (ignoring coinsurance), then using only the ceded modified coinsurance amounts.

The reinsurance statutory reserve is the modified coinsurance liability. This amount is held as a liability on the GAAP balance sheet, so the total benefit reserve for the ceding company is the benefit reserve calculated without considering reinsurance, less the portion reinsured, plus the statutory reserve held by the ceding company.

8.12.4 Multiple reinsurance treaties

When the model point has both YRT and coinsurance or modified coinsurance treaties, both of the above treatments apply.

8.13 Loss recognition

8.13.1 Loss recognition test

Loss recognition testing is a comparison, for each cohort, of the net GAAP liability against a gross premium reserve (GPR). The net GAAP liability is the GAAP benefit reserve plus GAAP terminal dividend liability, less the DAC. The GPR is discounted using the asset yield from the GPR valuation task.

The test is calculated during the experience projection. Specify the SFAS 120 NLP factors, and SFAS 120 schedule, and GPR valuation result files (on the Company input page under the GAAP: SFAS 120: General category) for the experience projection.

The allowable DAC for the cohort is the minimum of: (1) the DAC calculated during the valuation and (2) the net level premium reserve and terminal dividend liability less the gross premium reserve, floored at zero. The DAC loss recognition adjustment factor is the allowable DAC divided by the DAC calculated during the valuation, capped at one and floored at zero. This amount is applied against the valuation DAC factors.

Specify (on the Company input page under the GAAP: SFAS 120: Cohorts category) the initial (at the projection date) DAC loss recognition adjustment factor for each cohort, since DAC cannot be written up after it has been written down.

- Gross DAC adjustment percent – the cumulative DAC write down percent for the gross DAC amount.
- Ceded DAC adjustment percent – the cumulative DAC write down percent for the ceded DAC amount.

These variables must be linked to an external source varying by hidden variable look up key cohort_id. The default value of 1.0 (100%) indicates there has been no write down.

8.13.2 GAAP results adjustment

The GAAP results for the cohort are affected by the write down of the DAC, and potentially by replacing the GAAP benefit reserve and terminal dividend liability by the GPR.

When the loss recognition testing decreases the DAC to zero, and there is still a remaining excess loss, the net level premium reserve is increased so that the net GAAP liability, excluding any dividend option reserve, equals the gross premium reserve. The terminal dividend liability is not adjusted.

8.14 Unlocking

8.14.1 Unlocking dates

The SFAS 120 method calls for periodic unlocking of the DAC, terminal dividend liability, and amortization factors (k factors). Unlocking occurs when: (1) experience turns out to be different than assumed or (2) assumptions for the future change.

The dynamic GAAP unlocking dates are the inner loop valuation dates which begin each time step. The projection date is an 'unlocking' date, since GAAP schedules are calculated on-the-fly.

8.14.2 DAC and GAAP liabilities recalculation

At each unlocking date, the amortization factors are recalculated using the 'initial calculation' formulas shown above. The changes in the DAC and terminal dividend liability amounts, compared to the 'roll forward' amounts, are reported as the effect of the unlocking.

There are two aspects of the unlocking calculations:

Retrospective unlocking

The accumulated EGM and capitalized expense results are calculated using historical amounts up to the projection date, and experience projection amounts up to the unlocking date. 'Retrospective unlocking' is the change in the DAC and terminal dividend liability amounts related to replacing the previous accumulation of 'schedule' amounts with the accumulation of 'experience' amounts.

Recalculated schedule amounts

The present value results, and the monthly schedule amounts for future 'roll forward' calculations, are calculated using GAAP best estimate assumptions. This is a nested projection starting from the experience projection in-force amounts as of the unlocking date.

8.15 Financial statement presentation**8.15.1 Income statement**

Revenue includes the total of base plan and rider gross premium less a separate item for reinsurance premiums, reinsurance experience refund (when reporting for the business segment), dividends applied to dividend options, cash flow interest, investment income and capital gains on the net GAAP liability and capital, and the modified coinsurance adjustment (when reporting for the business segment). When reporting for the business segment, reinsurance offsets are incorporated into the reported items for dividends applied to dividend options, the cash flow interest, and the investment income and capital gains on the reinsurance net GAAP liability.

Basic benefits include the total of base plan and rider direct death benefits, direct surrender benefits, direct maturity benefits, direct endowment benefits, less, when reporting for the segment, reinsurance recoveries for the basic benefits reinsured.

Other benefits include the benefit reserve increase (which includes the NLP reserve increase, the rider reserve increase, the terminal dividend liability increase, and the dividend option liability increase), policyholder dividends, and the policyholder dividend liability, where, when reporting for the segment, reinsurance offsets are incorporated into the reported items. Also included is the modified coinsurance adjustment (when reporting for the ceded cohorts), and the modified coinsurance liability increase.

Expenses includes the total of excess over renewal commissions, total acquisition expenses less a separate item for deferred commission and acquisition expenses, renewal commissions, maintenance expenses, premium taxes, letter of credit costs, DAC (expense DAC plus the YRT asset) amortization, and less a separate item for the YRT reinsurance cost applicable for YRT reinsurance. When reporting for the business segment, reinsurance offsets are incorporated into each reported item as required.

8.15.2 Balance sheet

Assets include the invested assets, other invested assets, the DAC, and the YRT reinsurance asset. When reporting for a business segment, 'reinsurance receivable' is the reinsurance net GAAP liability less the modified coinsurance liability.

Liabilities include the benefit reserve (which includes the net level premium reserve, the rider reserve, the terminal dividend liability, and the dividend option reserve), the dividend liability, and the modified coinsurance liability.

9 GROSS PREMIUM RESERVES**9.1 Applicability**

Gross premium reserves (GPR) apply for loss recognition testing for the SFAS 60 method, the SFAS 97 limited pay contracts method, and the SFAS 120 mutual company method.

9.2 Factors

Gross premium reserve factors are calculated for each model point, for the projection date and each future period. The GPR amount along with the units/policies calculated in the GPR valuation can be stored by model point, or in aggregate for the model point groupings within a cohort.

When the GPR factors are stored by model point, GPR results are captured for each model point along with the units/policies determined in the GPR valuation. When used in a projection, the GPR is adjusted for the experience units/policies separately for each model point, prior to aggregating into the cohorts. This method is more accurate, since adjusting each model point accounts for the changing mix of business since the GPR valuation was performed.

When the GPR factors are stored by cohort, GPR results are aggregated to its cohort, within each model point report group, using the model point grouping capability. When used in a projection, the GPR is adjusted for the experience units/policies for the cohort. This method is more efficient, since it reduces the size of the GPR valuation output file, and adjusts valuation results for each cohort (within each reporting group).

Note – When the GPR calculations are for use in a seriatim projection, the GPR factors must be stored by model point.

9.3 Assumptions

9.3.1 *Separate GPR assumption variables*

All assumption variables operate the same for the GPR valuations as for the experience projection task.

9.3.2 *Best estimate basis*

Gross premium reserves are calculated using best estimate assumptions as of the projection date. These assumptions do not include provisions for adverse deviation (PAD).

9.4 Gross premium reserve

9.4.1 *Valuation period*

GPR factors apply over the number of periods specified in the valn_yrs project variable (in the Run parameter set) for the GPR valuation.

This value should be long enough to reach to maturity for each model point included in the GPR valuation. GPR cash flows are projected for this number of years from $t = 0$, then discounted to calculate the GPR amount at each month.

9.4.2 *GPR interest rate*

The GPR interest rate is the asset yield specified for the GPR valuation task.

The monthly discount rate = $(1 + \text{annual rate})^{1/12} - 1$

9.4.3 *Discounted cash flows*

The gross premium reserve (GPR) is the present value of liability cash flows, at the GPR interest rate. Cash flows paid out increase the reserve; cash flows received decrease the reserve. The calculation is performed at the model point level.

10 ASSET ACCOUNTING

10.1 SFAS 115

10.1.1 *Bonds, mortgages, and asset backed securities*

U.S. GAAP accounting treatment varies by the sale class of each bond, mortgage, and asset backed security asset:

Sale Class	Income Value	Balance Sheet Value
Trading	Market value	Market value
Available for sale	Amortized cost	Market value
Held to maturity	Amortized cost	Amortized cost
Not assigned	Amortized cost	Amortized cost

The value of these assets thus varies between the balance sheet and the determination of GAAP income. GAAP investment income includes market value changes only for 'trading' assets, while the GAAP balance sheet includes market value changes for 'trading' and 'available for sale' assets.

The difference between the balance sheet and income values increases the GAAP deferred tax liability. See the GAAP deferred tax liability section in the GAAP reporting chapter, above.

10.1.2 Equities and real estate

Equities and real estate assets are valued at market value for GAAP accounting. The cost basis is the actual cost.

10.2 Hedging assets

GAAP values of hedging assets for variable deferred annuity products are produced by a separate hedging projection. Projection calculations simply use these values.

11 GAAP VALUATION TASKS SETUP

This chapter gives instructions for setting up GAAP valuation projection tasks.

11.1 GAAP valuation tasks overview

11.1.1 Point-in-time valuations

GAAP valuations may be for a single point-in-time valuation, such as a valuation at December of a calendar year. Point-in-time valuations apply for all GAAP methods. The valuation tasks are performed as of the projection date (valuation date).

11.1.2 Inner loop valuations

GAAP valuations can also be included as inner loop valuations within a nested projection. For example, inner loop valuations to produce new GAAP schedules for SFAS 97 unlocking or to include new issues since the last valuation. The experience projection is dependent on the GAAP valuation tasks being completed for the current time step.

Whether a GAAP valuation should be included as an inner loop valuation depends on the GAAP treatment:

- GAAP factors, such as for SFAS 60 and SFAS 97 limited pay, are locked and thus would not be included as an inner loop valuation.
- GAAP schedules, such as for SFAS 97 retrospective deposit, are subject to unlocking, which is accomplished using an inner loop valuation.

11.1.3 Concurrent valuations

Valuations for multiple GAAP methods may be performed in a single valuation projection task for products that support multiple GAAP methods using the same calculation approach. For example, term life and traditional life GAAP valuations execute the appropriate SFAS 60 or SFAS 97 limited pay valuation based on the cohorts included and the model point characteristics. Results are captured in the appropriate GAAP submodel within the single output file name.

11.1.4 *Dependent valuations*

Some GAAP valuations may require results from other GAAP valuations. For example, the SFAS 97 retrospective deposit valuation may require results from the SFAS 133 and SOP 03-1 valuations in order to determine the final DAC and liabilities for the cohort.

SFAS 120 is another example of GAAP valuation task dependency. The net level premium reserve factors are required to produce the estimated gross margins, and are thus produced by a first valuation task. Then a second valuation task uses the factors to determine the DAC and terminal dividend liability schedules.

When there is valuation task dependency, the independent valuation task must be processed first. The results from that task are referenced as input into the dependent valuation task.

11.1.5 *Final valuations*

When GAAP valuations require more than one projection task, such as a SFAS 60 valuation where loss recognition is included, or a VA valuation where SFAS133, SOP 03-1, and DAC valuation are included, an additional projection task can be used to determine the final GAAP valuation results.

11.2 General GAAP valuation setup

The following setup instructions apply for all GAAP valuations.

11.2.1 *Input variables*

Specify GAAP cohorts to be included in the segment assumptions, on the Company input page under the Model Setup: Segment category. Specify the cohorts to include as a comma separated list of cohorts to include in the valuation.

Specify GAAP general parameters for all GAAP valuations, on the Company input page under the General category for each GAAP method.

Specify GAAP cohorts, on the Company input page under the Cohorts category for each GAAP method.

Specify the capitalized percent of each type of acquisition expenses, on the [Prod]Expenses input page under the Acquisition category.

When applicable, specify the provisions for adverse deviation (PAD) for the asset yield, on the Company input page under the Asset Yield category.

When applicable, specify the provisions for adverse deviation (PAD) for the mortality assumptions, on the [Prod]Mortality input page under the Adjustments category.

When applicable, specify the provisions for adverse deviation (PAD) for the maintenance expense assumptions, on the [Prod]Expenses input page under the Adjustments category.

When applicable, specify the provisions for adverse deviation (PAD) for the lapse assumptions, on the [Prod]Policyholder input page under the Adjustments category.

Specify, for SFAS 97 retrospective deposit and SFAS 120, GAAP history for each inforce GAAP cohort, on the Company input page under the History category for each GAAP method.

Specify results from valuations for dependent tasks such as SFAS 133, SOP 03-1, SFAS 120 NLP, and GPR valuations, on the Company input page under the General category for each GAAP method using the system <column_output_directory> wildcard as opposed to the user specified <output> wildcard. This is required so that RiskAgility FM will reference the appropriate results files when the projection is distributed to a grid.

Additionally, for nested projections with inner loop valuations, specify the <proj_set_loop_num> and <proj_set_sub_loop_num> system wildcards within the file reference so that RiskAgility FM will reference the appropriate inner loop valuation file.

An example file reference for the sfas97rd_sop031_results variable is "<column_output_directory>GAAP Portfolio_Valn SOP 03-1@<proj_set_loop_num>_<proj_set_sub_loop_num>company~seg~sfas97rd.csv"

11.2.2 Run page

Create a projection task on a Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

For cell pricing, and for seriatim based valuation tasks, create a projection task on a Run page which uses the product submodel as the target model object.

11.2.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Projection period from	0
Projection period to	0, since the valuation projection is specified by the GAAP amortization years input variable
Projection period min	-600 for seriatim tasks, to allow for external source lookup keys calculating before the projection date (such as pol_yr)
Projection period max	At least as long as the calculations for each submodel
Extract model points	No, for production tasks
Sliding windows	Do Not Slide for seriatim runs and when small models or independent model point processing enables portfolio runs to fit within available memory; Slide to reduce memory requirements during a projection task which does not otherwise fit into available memory

11.2.4 Project variables

Also specify the project variables in the Run parameter set.

Property	Values
Projection date	Valuation date is the end of the month specified as yyyy/mm
Pricing projection?	No
Valuation years	Has no effect on GAAP valuation tasks, which project over the DAC amortization months
Project GAAP results?	No, for all valuation tasks except the final valuation task
Include assets?	No, since GAAP valuations are liability-only
Include target capital?	Has no effect on valuation tasks
Use independent model point processing?	Has no effect on seriatim tasks
Validate data definitions?	No for production runs; Yes for testing

Property	Values
Include decrements?	Yes
Discount rates selected	Typically not used for valuation tasks
Portfolio yield definition	Typically not used for valuation tasks
Valuation periods selected	Has no effect on point-in-time valuations For nested projections, must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0.
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Has no effect on point-in-time valuations. For nested projections, specify how the system should manage generated scenario files selecting from: Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs that generate large scenario files.

11.2.5 Output parameter set

Property	Values
Fixed discount rate list	Typically not used for valuation tasks
Use column discount rates	No; typically not used for valuation tasks
Output manager	Select the desired output manager to limit the output captured
Model object output	Select All Columns, No Columns, or Output Definitions for the model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions

11.2.6 Run controller projection set

Point-in-time valuation tasks can be organized into a projection set on the Run controller, with appropriate task dependencies established. Then, they can be run together by submitting the projection task set from the Run controller. This is not required; standard projections can be run from the individual Run page.

Valuation tasks for GAAP projections are typically setup within a Run controller projection set and combined with an experience projection task; with appropriate task dependencies established. For GAAP projections with inner loop valuation, the valuation tasks must be combined with appropriate scenario generation tasks and an experience projection task in a Run controller projection set.

11.3 SFAS 60 setup

The following setup instructions for SFAS 60 GAAP valuations are in addition to the general instructions discussed above.

11.3.1 Applicability

SFAS 60 valuations apply for term life, traditional life, and long term care insurance products. The valuation produces SFAS 60 GAAP factors for each model point.

SFAS 60 valuations are seriatim calculations (the product submodel is the top model). A valuation can be performed for point-in-time valuation of in-force business, or for production of factors for cell pricing and portfolio projections.

11.3.2 Run page

Create a projection task on a Run page which uses the product submodel as the target model object.

11.3.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_gaap_sfas60_sfas97lp (term) valn_tgt_gaap_sfas60_sfas97lp_sfas120_nlp (trad) valn_tgt_gaap_sfas60_sfas97lp (long term care)
Projection task loops	1, since SFAS 60 is a single deterministic valuation
Assumption set restore	Yes, since this is a seriatim task

11.3.4 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	0, if only for point in time valuation At least the number of projection or time step months for inclusion in a GAAP projection

11.4 SFAS 91 setup

The following setup instructions for SFAS 91 GAAP valuations are in addition to the general instructions discussed above.

11.4.1 Applicability

SFAS 91 valuations apply to payout annuities which are classified as investment contracts. The valuation produces an SFAS 91 GAAP schedule for each cohort.

SFAS 91 valuations are cohort calculations (the company submodel is the top model). The exception is for cell pricing projections, where “pa” (payout annuity submodel) is the top model.

A valuation can be performed for point-in-time valuation of in-force business, or for production of schedules for cell pricing and portfolio projections.

11.4.2 Run page

Create a projection task on a Run page which uses the company submodel as the target model object. For cell pricing, create these projection tasks on a Run page which uses the product submodel as the target model object.

11.4.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_gaap_sfas91
Projection task loops	1, since SFAS 91 is a single deterministic valuation
Assumption set restore	Yes, when independent model point processing applies or when this is a seriatim task; No otherwise

11.4.4 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	0, for point in time valuations At least the number of projection or time step months for inclusion in a GAAP projection

11.5 SFAS 97 limited pay setup

The following setup instructions for SFAS 97 limited pay GAAP valuations are in addition to the general instructions discussed above.

11.5.1 Applicability

SFAS 97 limited pay valuations apply to term life, traditional life, long term care insurance products. This method also applies to payout annuity products that have more than a nominal mortality risk.

The valuation produces SFAS 97 limited pay GAAP factors for each model point. These calculations occur in the same task that produces SFAS 60 factors.

SFAS 97 limited pay valuations are seriatim calculations (the product submodel is the top model). A valuation can be performed for point-in-time valuation of in-force business, or for production of factors for cell pricing and portfolio projections.

11.5.2 Run page

Create a projection task on a Run page which uses the product submodel as the target model object.

11.5.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_gaap_sfas60_sfas97lp (term) valn_tgt_gaap_sfas60_sfas97lp_sfas120_nlp (trad) valn_tgt_gaap_sfas60_sfas97lp (long term care)
Projection task loops	1, since SFAS 97 limited pay is a single deterministic valuation
Assumption set restore	Yes, since this is a seriatim task

11.5.4 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	0, for point in time valuations At least the number of projection or time step months for inclusion in a GAAP projection

11.6 SFAS 97 SFAS 133 setup

The following setup instructions for SFAS 97 SFAS 133 valuations are in addition to the general instructions discussed above.

11.6.1 Applicability

Fixed indexed annuities and variable annuities use a separate SFAS 133 GAAP valuation task, which calculates the stochastic projection task loops needed for the present values of claims and charges for the SFAS 133 guaranteed benefits.

SFAS 133 valuations are seriatim calculations (the product submodel is the top model). A valuation can be performed for point-in-time valuation of in-force business, or for production of amounts for cell pricing and portfolio projections.

11.6.2 Run page

Create a projection task on a Run page which uses the product submodel as the target model object.

11.6.3 Run parameter set

Property	Values
Target column	valn_tgt_gaap_sfas133
Projection task loops	1-100 (for example); this is a stochastic valuation
Assumption set restore	Yes, when independent model point processing applies or when this is a seriatim task; No otherwise

11.6.4 Output parameter set

Property	Values
Column output from	0
Column output to	0, for point in time valuations At least the number of projection or time step months for inclusion in a GAAP projection

11.7 SFAS 97 SOP 03-1 setup

The following setup instructions for SFAS 97 SOP 03-1 valuations are in addition to the general instructions discussed above.

11.7.1 Applicability

The SOP 03-1 valuation task applies for universal life, indexed universal life, variable universal life, fixed indexed annuity, and variable annuity products.

SOP 03-1 valuations are cohort calculations (the company submodel is the top model). The exception is for cell pricing projections, where the product submodel is the top model.

A valuation can be performed for point-in-time valuation of in-force business, or for production of amounts for cell pricing and portfolio projections.

11.7.2 Run page

Create these projection tasks on a Run page which uses the company submodel as the target model object. For cell pricing, create these projection tasks on a Run page which uses the product submodel as the target model object.

11.7.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_gaap_sop031
Projection task loops	1-100 (for example); this is a stochastic valuation
Assumption set restore	Yes when independent model point processing applies or when this is a seriatim task; No otherwise

11.7.4 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	At least the number of DAC amortization months since the results of this valuation are required for the SFAS 97 retrospective deposit valuation

11.8 SFAS 97 retrospective deposit setup

The following setup instructions for SFAS 97 retrospective deposit GAAP valuations are in addition to the general instructions discussed above.

11.8.1 Applicability

SFAS 97 retrospective deposit valuations apply to universal life, indexed universal life, variable universal life, fixed annuity, fixed indexed annuity, and variable annuity products. The valuation produces an SFAS 97 retrospective deposit GAAP schedule for each cohort.

SFAS 97 retrospective deposit valuations are cohort calculations (the company submodel is the top model). The exception is for cell pricing projections, where the product submodel is the top model.

A valuation can be performed for point-in-time valuation of in-force business, or for production of schedules for cell pricing and portfolio projections.

11.8.2 Preliminary valuation task

SFAS 97 retrospective deposit valuations may require a preliminary SOP 03-1 valuation task. When applicable, that valuation tasks must be performed first. Then the SFAS 97 retrospective deposit valuation calculates DAC valuation results using those previous valuation results.

The SOP 03-1 valuation result files must be specified using the system <column_output_directory> wildcard as opposed to the user specified <output*> wildcard. This is required so that RiskAgility FM will reference the appropriate results files when the projection is distributed to a grid.

Additionally, for nested projections with inner loop valuations, the <proj_set_loop_num*> and <proj_set_sub_loop_num*> system wildcards must also be used within the file reference so that RiskAgility FM will reference the appropriate inner loop valuation file.

An example file reference for the sfas97rd_sop031_results variable is "<*column_output_directory*>GAAP Portfolio_Proj TS Valn SOP 03-1@<*proj_set_loop_num*>_<*proj_set_sub_loop_num*> company~seg~sfas97rd.csv"

11.8.3 Run page

Create these projection tasks on a Run page which uses the company submodel as the target model object. For cell pricing, create these projection tasks on a Run page which uses the product submodel as the target model object.

11.8.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_gaap_sfas97rd
Projection task loops	1, since SFAS 97 retrospective deposit is a single deterministic valuation
Assumption set restore	Yes when independent model point processing applies or when this is a seriatim task; No otherwise

11.8.5 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	0, for point in time valuations At least the number of projection or time step months for inclusion in a GAAP projection

11.9 SFAS 120 NLP setup

The following setup instructions for SFAS 120 NLP valuations are in addition to the general instructions discussed above.

11.9.1 Applicability

SFAS 120 net level premium reserve valuations apply to participating traditional life products. The valuation produces net level premium reserve factors for each model point.

A valuation can be performed for point-in-time valuation of in-force business, or for production of factors for cell pricing and portfolio projections.

11.9.2 Run page

The NLP valuation is performed by the same projection task that calculates SFAS 60 and SFAS 97 limited pay GAAP factors for traditional life model points. This task is on a Run page which uses the 'trad' submodel as the target model object.

11.9.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_gaap_sfas60_sfas97lp_sfas120_nlp
Projection task loops	1, since SFAS 120 NLP is a single deterministic valuation
Assumption set restore	Yes, since this is a seriatim task

11.9.4 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	At least the number of DAC amortization months for point in time valuations since the change in the net level premium reserve is used in the calculation of the EGM The greater of the DAC amortization months, projection, or time step months for inclusion in a GAAP projection

11.10 SFAS 120 setup

The following setup instructions for SFAS 120 GAAP valuations are in addition to the general instructions discussed above.

11.10.1 Applicability

SFAS 120 valuations apply to participating traditional life products. The valuation produces an SFAS 120 GAAP schedule for each cohort.

SFAS 120 valuations are cohort calculations (the company submodel is the top model). The exception is for cell pricing projections, where the product submodel is the top model.

A valuation can be performed for point-in-time valuation of in-force business, or for production of schedules for cell pricing and portfolio projections.

11.10.2 Preliminary valuation task

SFAS 120 valuations require a preliminary SFAS120 NLP valuation task. The SFAS 120 net level premium valuation must first produce NLP reserve factors for each model point; this is a seriatim valuation. Then the SFAS 120 valuation calculates DAC and dividend schedules using those previous valuation results.

The SFAS 120 NLP valuation result files must be specified using the system <column_output_directory> wildcard as opposed to the user specified <output*> wildcard. This is required so that RiskAgility FM will reference the appropriate results files when the projection is distributed to a grid.

An example file reference for the sfas120_nlp_results variable is "<column_output_directory>GAAP Seriatim_Valn SFAS 120 NLP~company~liab~trad~trad_sfas120.csv"

11.10.3 Run page

Create these projection tasks on a Run page which uses the company submodel as the target model object. For cell pricing, create these projection tasks on a Run page which uses the product submodel as the target model object.

11.10.4 Run parameters

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_gaap_sfas120
Projection task loops	1, since SFAS 120 is a single deterministic valuation
Assumption set restore	Yes when independent model point processing applies or when this is a seriatim task; No otherwise

11.10.5 Output parameters

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	0, for point in time valuations At least the number of projection or time step months for inclusion in a GAAP projection

11.11 Gross premium reserve setup

The following setup instructions for gross premium reserve valuations are in addition to the general instructions discussed above.

11.11.1 Applicability

SFAS 60, SFAS 97 limited pay, and SFAS 120 GAAP methods can include loss recognition testing using a preliminary GAAP GPR valuation task. This does not apply for other GAAP methods, which can test for loss recognition as part of the GAAP valuation task. The GAAP GPR valuation produces GPR values per unit in force for each model point.

GAAP GPR valuations are seriatim calculations (the product submodel is the top model). A valuation can be performed for point-in-time valuation of in-force business, or for production of factors for cell pricing and portfolio projections.

11.11.2 Input variables

Specify GAAP GPR input in the Company input page, under the 'General' section of each the applicable GAAP methods

- Store GPR valuation by cohort definition = No (valuation results stored for each model point), or Yes (results stored by cohort within report groups). When the GPR calculations are used for cell pricing, the GPR results must be stored by model point.
- GPR results; these results are used during the GAAP valuation.

The GPR valuation result files must be specified using the system <*column_output_directory*> wildcard as opposed to the user specified <*output*> wildcard. This is required so that RiskAgility FM will reference the appropriate results files when the projection is distributed to a grid.

Additionally, for nested projections with inner loop valuations, the <*proj_set_loop_num*> and <*proj_set_sub_loop_num*> system wildcards must also be used within the file reference so that RiskAgility FM will reference the appropriate inner loop valuation file.

An example file reference for the sfas60_gpr_results variable is "<*column_output_directory*>GAAP Seriatim_Valn SFAS 60 GPR@<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>company~liab~trad~trad_sfas60.csv"

11.11.3 Run page

Create a projection task on a Run page which uses the product submodel as the target model object.

11.11.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_gpr_gaap
Projection task loops	1, since GAAP GPR is a single deterministic valuation
Assumption set restore	Yes, since this is a seriatim task

11.11.5 Project variables

The Valuation years project variable must be specified, unlike the GAAP valuation tasks which are not affected by this variable.

Property	Values
Valuation years	120 years, to ensure cash flows are captured and discounted from maturity for all model points

11.11.6 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	0, for point in time valuations At least the number of projection or time step months for inclusion in a GAAP projection

11.12 GAAP final valuation setup

The following setup instructions for a final GAAP valuation are in addition to the general instructions discussed above.

11.12.1 Applicability

A final GAAP valuation task is only used for point-in-time valuations. It provides a single projection task which can bring together results of several valuation tasks to calculate the point in time GAAP balance sheet.

11.12.2 Run page

Create these projection tasks on a Run page which uses the company submodel as the target model object. For cell pricing, create these projection tasks on a Run page which uses the product submodel as the target model object.

11.12.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	gaap_liab_net_net
Projection task loops	1, since the final GAAP valuation is a single deterministic valuation
Assumption set restore	Yes when independent model point processing applies or when this is a seriatim task; No otherwise

11.12.4 Project variables

The Project GAAP results project variable must be specified, unlike the GAAP valuation tasks which are not affected by this variable.

Property	Values
Project GAAP results?	Yes, since this valuation task is calculating the initial GAAP balance sheet

11.12.5 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Column output from	0
Column output to	0

12 GAAP PROJECTION TASK SETUP**12.1 Standard versus nested projections**

When a GAAP projection is for a block of business that uses factors or for a set of inforce only business that uses schedules where there is no new business, future loss recognition, or unlocking, a standard projection may be used for the GAAP cell pricing and portfolio projection.

However, when a GAAP projection is for a model that includes new business, or when there is future loss recognition testing or unlocking, a nested projection structure must be used for the GAAP cell pricing and portfolio projection.

12.2 Valuation and experience task dependency

GAAP cell pricing or portfolio projections require valuation tasks and an experience projection. The valuation tasks must be run in advance of the experience projection and the results of the valuation tasks referenced as input into the experience projection.

When the GAAP cell pricing or portfolio projection is a standard projection, the valuation tasks can also be run as standard projections and processed in advance of the experience projection, either in the same projection set or in separate projection sets.

But, when the GAAP cell pricing or portfolio projection is a nested projection, the GAAP valuations become inner loop valuations and must be included together with the experience projection in the nested projection Run controller projection set with appropriate dependencies specified. The GAAP cell pricing or projection is then driven by the projection set loop and sub loops.

12.3 GAAP experience projection task setup

The following setup instructions apply for experience projections which include GAAP calculations.

12.3.1 Input variables

Specify GAAP cohorts the same as for the GAAP valuation tasks.

Specify GAAP cohorts for the segments the same as for the GAAP valuation tasks.

Specify the GAAP and GPR valuation results file for each applicable GAAP method. This is only the final GAAP schedule or factors; results files from preliminary valuations are ignored. For example, the projection uses only the SFAS 97 retrospective deposit results; the SOP 03-1 valuation results (which are used in the SFAS 97 valuation task) have no effect on the GAAP projections.

Specify the valuation result files using the system <*column_output_directory*> wildcard as opposed to the user specified <*output*> wildcard. This is required so that RiskAgility FM will reference the appropriate results files when the projection is distributed to a grid.

Additionally, for nested projections with inner loop valuations, specify the <*proj_set_loop_num*> and <*proj_set_sub_loop_num*> system wildcards within the file reference so that RiskAgility FM will reference the appropriate inner loop valuation file.

An example file reference for the sfas97rd_results variable is "<*column_output_directory*>GAAP Portfolio_Valn DAC@<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>company~seg~sfas97rd.csv"

12.3.2 Run page

For GAAP portfolio projections, specify the experience projection on a Run page with the Company submodel as the target model object.

For GAAP cell pricing, specify the experience projection of a Run page with the product submodel as the target model object.

12.3.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	The desired targeted amount, such as gaap_inc_aft_tax
Projection period from	0
Projection period to	For standard projections, the desired projection period For nested projections, at least as long as the maximum time between time steps
Projection period min	-600 for seriatim tasks, to allow for external source lookup keys calculating before the projection date (such as pol_yr)
Projection period max	At least as long as the calculations for each submodel
Extract model points	No, for production tasks
Sliding windows	Do Not Slide for seriatim runs and when small models or independent model point processing enables portfolio runs to fit within available memory; Slide to reduce memory requirements during a projection task which does not otherwise fit into available memory

12.3.4 Project variables

Also specify the project variables in the Run parameter set.

Property	Values
Projection date	Valuation date is the end of the month specified as yyyy/mm
Pricing projection?	Yes or No
Valuation years	Has no effect on projection tasks which are not performing valuations
Project GAAP results?	Yes; this is the switch to include GAAP results in the projection
Include assets?	Yes or No
Include target capital?	Yes typically
Use independent model point processing?	Yes or No for portfolio projections; has no effect on seriatim projections
Validate data definitions?	No for production runs; Yes for testing
Include decrements?	Yes

Property	Values
Discount rates selected	Specify the stochastic discount rates to use; these must match the disc_rate_id lookup key values used to specify these rates on the Company input page
Portfolio yield definition	Before tax (asset yield rate), or After tax (multiplied by 1 – tax rate)
Valuation periods selected	Has no effect on standard projections For nested projections, must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0. This must match the value for the related GAAP valuation tasks.
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Has no effect on standard projections. For nested projections, specify how the system should manage generated scenario files selecting from: Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs that generate large scenario files.

12.3.5 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Fixed discount rate list	For example, 8, 10, 12 for present values at 8%, 10%, and 12%
Use column discount rates	Yes to use the stochastic discount rates named in the Discount rates selected project variable
Output manager	Select the desired output manager to limit the output captured
Model object output	Select All Columns, No Columns, or Output Definitions for the model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions

12.4 Run controller projection set

GAAP standard projections should typically be set up within a projection set on the Run controller, so the entire set of tasks can be run in one step. The GAAP projection task specifies that the valuation tasks are dependent tasks, which must be completed before the projection task begins.

GAAP nested projections require the GAAP valuation tasks to be specified within the same projection set (on the Run controller) as the GAAP experience projection; with the experience projection dependent on their completion.

13 SUBMODELS STRUCTURE

GAAP valuation and projection calculations are organized in GAAP related submodels: sfas60, sfas91, sfas97lp, sfas120, sfas97rd and sfas133. They are user defined arrays containing the variables and formulas required to calculate different GAAP liabilities.

These submodels are required at two levels of the model tree. Sfasxxx submodels calculate cohort level results and are children of the segment submodel. Product version, named [prod]_sfasxxx are needed to determine individual model point results and are children of the product type submodel.

13.1 Multiple instances

The sizing of the GAAP submodels is flexible in order to support any number of cohorts, plus separation between gross and ceded results.

The cohort based GAAP submodels, sfas60, sfas91, sfas97lp, sfas120, sfas133, and sfas97rd are two dimensional arrayed submodels. The first dimension is the number of cohorts; the second dimension is '0' for the gross results and '1' for the ceded results. When the GAAP submodels are sized, the ceded element is created only when the cohort contains coinsurance or modified coinsurance model points.

The product level GAAP submodels, [prod]_sfas60, [prod]_sfas91, [prod]_sfas97lp, [prod]_sfas120, [prod]_sfas133, and [prod]_sfas97rp are single dimensional arrayed submodels, where the '0' element contains the gross results and the '1' element contains the ceded results. The ceded element is created only when the model point uses coinsurance or modified coinsurance.

The elements of the GAAP submodels are sized during the startup processing of the projection task.

13.2 Formulas

The formulas in the sfasxxx submodel class are reusable formulas across both portfolio (company is the top model) and seriatim ('product' is the top models).

This reusability has been accomplished using 'blocks' of code which apply either in the product context or in the cohort context. These blocks of code are organized using conditions on compiler directives, and unique model pointer names. The compiler directive conditions reference the compiler macros for the product type, such as '___TERM_LIAB_H_'. The unique model pointer conditions reference 'sfasxxx', and for the product instance, for example 'term_sfas60', so that they are only executed when the 'this' pointer equals the unique model pointer name.

Key formulas for GAAP submodels include:

Initialize

The 'initialize' column is executed during model initialization for each element of the sfasxxx and [prod]_sfasxxx submodel. This column is called from initialize of seg for the cohort submodels, and from initialize of 'product' for each of the product submodels.

The main use of initialize is to set up the submodel, drive the setting up of the cohorts, and to read valuation results during the experience projection.

setup_cohort

The 'setup_cohort' external function is executed during initialization of the cohort GAAP submodels when each model point is being evaluated for including in the cohort. This function evaluates the model point plan code and issue year against the cohort criteria for plan code and issue year and returns 'true' when there is a match, and 'false' otherwise.

sum_over_gaap

The 'sum_over_gaap' external function sum the results from the product level GAAP submodels into the cohort level GAAP submodel. The sum_over_gaap function is called with the column name and 't' period from the cohort GAAP submodel level.

get_xxx_valn_results

The 'get_xxx_valn_results' formulas are external functions which retrieve the factor results from the valuation task. These amounts are stored in an array for future reference.

get_gaap_schedule_items

The 'get_gaap_schedule_items' formulas are external functions which retrieve GAAP schedule results from the valuation task. These amounts are stored in an array for future reference.

write_state_of_world_financial

The 'write_state_of_world_financial' external function writes out a special file of cohort results which are required for inner loop valuations or the next experience projection. This function is executed from the finalize column of the GAAP submodel during nested projections.

read_state_of_world_financial

The 'read_state_of_world_financial' external function reads the results from the special file of cohort results that was created by the write_state_of_world_financial function. These results are required for inner loop valuations or the next experience projection. This function is executed from the initialize column of the GAAP submodel during nested projections.

get_xxx_items

The 'get_xxx_items' external function, which is executed only in the experience projection, opens the state_of_world file and retrieves information from the previous time step

Other formulas

Other formulas in the GAAP submodels are relatively straightforward. These other formulas calculate the amounts required to determine the GAAP valuation results, such as estimated gross profits and margins, accumulated values and present values of premiums, benefits, maintenance expenses, capitalized expenses and commissions, the k factors, and the resulting benefit reserve and DAC amounts. They also calculate the GAAP results during the experience projection, such as roll forwards of reserves and DAC and the income statement and balance sheet contributions of the model point and cohorts.

13.3 Seriatim versus portfolio runs**13.3.1 Seriatim projection**

When the product submodel is the top model, GAAP calculations occur in a product version GAAP submodels, such as va_sfas133 and va_sfas97rd. The product version submodels are children of the main product type submodel. In this context, the GAAP submodels reference variables and results from the product submodel, and determine single model point results.

13.3.2 Portfolio projection

For portfolio projections (which include business segment calculations), GAAP calculations occur in both the product version GAAP submodels and in the cohort GAAP submodels. In this configuration, the product

version of the GAAP submodels calculates only individual model point amounts, while the cohort GAAP submodels determine amounts for each cohort. The product version GAAP submodels reference variables and results from the product submodels, while the cohort submodels reference aggregate results from the product GAAP submodels, using a 'sum_over' external function.

13.4 Valuations versus projections

13.4.1 Valuations

During a valuation, the GAAP submodels provide for calculating the GAAP factors and schedules from $t = 0$ only.

These calculations use a cohort approach, which is driven from the segment submodel. Both the product version and cohort GAAP submodels are required for this calculation. Company is the top model but no columns except the 'valn_tgt_gaap_' columns are actually required.

13.4.2 Projections

During a projection, which could be a seriatim cell pricing or a portfolio projection, the GAAP submodels calculate the valuation amounts as discussed above. The GAAP submodels also calculate the GAAP liabilities and DAC assets during the projection. This is allowable within nested projections because the structure of the GAAP submodels provides for the results of GAAP submodels from the valuation runs to be used as input into the experience projection.

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C05 PBA

This book discusses principle-based approaches (PBA) capital and reserves. This includes:

- VM-20 reserves and RBC C3 phase III capital for life insurance products
- VM-22 reserves for fixed and indexed deferred annuity products
- AG 43 reserves and RBC C3 phase II capital for variable annuity products
- Letter of credit (LOC) reserve financing for life insurance products.

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1 PBA FUNDAMENTALS

1.1 Terminology

- **Outer loop** – Experience projection iterations, branching from the projection date.
- **Valuation iteration** – Projection to calculate a current value, rather than experience projections.
- **Inner loop** – Valuation iterations within a nested projection.
- **Nested projection** – Outer loops are projected, with inner loop valuations at various points through the projection, including the projection date ($t = 0$). The projection consists of a projection set, with a separate projection task for each type of valuation, and for advancing the outer loop experience projection. These projection tasks are repeated through each valuation date using projection set loop and sub loop processing. See the Nested projections chapter of the Run management book.
- **Point-in-time valuation** – Calculation at $t = 0$, using valuation iterations; there are no outer loops (and technically also no inner loops since this is not a projection).
- **Projection date** – The end of the specified calendar year and month which specifies the initial values on the model point records. Present values are discounted to the projection date.
- **Valuation date** – For nested projections, the points along the outer loop projection when inner loop valuations occur. For point-in-time valuations, the projection date is the valuation date.
- **PBR** – Principle-based reserve.

1.2 PBA calculation types

1.2.1 PBA valuations

PBA valuations are projection tasks which are driven by a specific 'valn_tgt_pba_xxx' target column. Each valuation task produces either:

- Standard scenario reserve or capital amounts, or
- Schedules for each cohort (SFAS 91 investment contracts method, Retrospective deposit SFAS 97 which include results from SOP 03-1 additional liability and SFAS133 valuations, and SFAS 120).

PBA valuation projections are liability-only calculations. They can be deterministic or stochastic. They can be seriatim projections or portfolio projections.

1.2.2 LOC economic reserve valuations

Gross premium reserve (GPR) valuations used for GAAP SFAS 60, SFAS 97 limited pay method, and SFAS 120 loss recognition testing are projection tasks which are driven by a specific 'valn_tgt_gaap_gpr_xxx' target column.

GPR valuations are liability-only seriatim valuations. They calculate the GPR per unit for each future month, typically using the same assumptions as for the experience projection.

Loss recognition testing cannot be performed directly by this task, since the tests require calculations by cohort. The GPR valuation results are used by the experience projection to perform loss recognition testing.

1.2.3 Experience projection

PBA results are calculated in experience projections when the applicable PBA valuation results file is identified in the xxx_pba_valn_results_xxx variable (on the Company input page under the appropriate PBA or LOC method category).

Including PBA reserves and capital and letter of credit reserve financing in an experience portfolio projection requires results from PBA and LOC economic reserve valuations to be created in advance of the experience projection task. The output files from these valuations are referenced as an input into the PBA or letter of credit reserve financing projection.

1.3 Multiple calculation together

1.3.1 Gross or net result

A PBA valuation task can calculate just gross (excluding the impact of reinsurance), just net (including the impact of reinsurance) results, or both results can be calculated together in the same valuation projection task.

1.3.2 Hedged or unhedged

A PBA valuation task can calculate just hedged (including the effect of hedging), just unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same valuation projection task.

1.3.3 Combined or separate valuations

A PBA valuation task can calculate just reserves, just capital, or reserves and capital together in same valuation projection task.

PBA results calculated together are different from the results calculated separately. The asset/liability valuation task calculating multiple PBA results develops a single asset yield, which is used for each of the PBA results calculated by that task. This is different from separate valuation tasks for each result, where assets are projected separately for each task. For example, the asset cash flows for a single PBA capital calculation includes the impact of taxes, while a single PBR calculation does not.

The asset yield affects liability cash flows when the crediting rate is based on, or constrained by, the asset yield.

The single asset yield is calculated using the reserve calculation (when it is being calculated), net of reinsurance (when included). For example, calculating PBA capital and reserves gross and net of reinsurance uses the asset cash flows projected for PBA reserves (thus before tax) net of reinsurance. Calculating only PBA capital gross and net of reinsurance uses the asset cash flows projected for capital (thus after tax) and net of reinsurance.

This use of the same asset yield is the only difference between separate and combined calculations. Even when multiple PBA results are calculated together in a valuation task, each PBA result is a separate calculation.

Of course, combining multiple PBA result calculations in the same projection task decreases calculation time. Assets are projected fewer times – just once for the standard scenario reserve and capital valuation task, and once for the stochastic reserve and capital valuation task.

When the PBA valuation task calculates only a single result, assets are projected using the cash flows for that projection. Thus each single result is based on its own asset yield. Each of these asset yield calculations is independent of the others.

For example, a separate valuation projection task can be used for each of these single PBA results:

- Gross standard scenario reserve
- Net standard scenario reserve
- Gross standard scenario capital
- Net standard scenario capital
- Gross unhedged stochastic reserve
- Gross hedged stochastic reserve

- Net unhedged stochastic reserve
- Net hedged stochastic reserve
- Gross unhedged stochastic capital
- Gross hedged stochastic capital
- Net unhedged stochastic capital
- Net hedged stochastic capital.

1.4 Asset processing

1.4.1 *Reduced asset processing*

PBA valuation calculations automatically reduce processing requirements in the following ways:

- The asset valuation reserve (AVR) is ignored.
- GAAP asset calculations do not apply.
- Risk based capital (RBC) does not apply.

1.4.2 *Investment strategy simplification*

You can specify different investment strategy assumptions for valuation calculations than for the outer loop experience projection, since each PBA valuation is performed using a separate task within the projection set. While the main investment strategy can be the same as for the outer loop, consider the following changes to streamline inner loop asset processing:

- Calculate asset market values only when needed for sales.
- Purchase assets annually or quarterly instead of monthly, particularly after the first years of the valuation projections. The amounts not invested accumulate as interim cash, which can grow at the asset yield rate.
- Set the minimum investment amount, and the minimum asset size for each asset group, high enough to avoid processing of small amounts. (This should apply for experience projections, as well.)
- Rebalance between asset groups only annually.
- Reduce or eliminate special investment strategies varying by economic condition.
- Reduce or eliminate duration matching by setting higher tolerances.

The Asset processing book explains each of these settings.

1.4.3 *Proxy bonds*

The investment strategy simplifications discussed above streamline the processing of assets purchased during inner loops. You can further increase asset processing speed by using the 'proxy bonds' feature to streamline the processing of existing assets during PBA valuations.

A special projection task creates an alternative set of bond and mortgage model point records. The individual assets within each asset group are replaced by one or two proxy bonds. A proxy bond has similar cash flow and rate characteristics as the portfolio it replaces, but requires processing only this single bond.

Specify to use proxy bonds by changing the Proxy bonds definition (in the Asset Run Control category on the Investments input page):

- Single Non Callable – A non-callable proxy bond is created for each asset group.
- Callable Separate – A callable bond is created for the assets having a call option, and a non-callable bond is created for the remaining assets within each asset group.

- Not Applicable – The valuation calculations use the same individual assets as are being projected for the experience projection.

Non-callable proxy bond

Within each asset sub portfolio, the individual assets are replaced by a proxy bond, which is created by the proxy bond creation projection:

- Maturity date – the last maturity of the individual assets.
- Coupon rate – the weighted average coupon rate, weighted by each asset's par value times its weighted average life (WAL). Including the WAL accounts for the time over which interest will be received for each asset.
- Initial par value – the sum of the individual asset par/principal values.
- Initial book value and cost basis – the sum of the individual asset values.
- Sinking fund schedule – the run-off of the individual asset par/principal values from maturities, sinking fund schedules, mortgage amortization, and balloon schedules.
- Quality rating – The proxy bond quality rating (specified in the definitions for each asset group) determines the credit spread which is used to calculate market values for sales of the proxy bond.
- Sales class – The proxy bond sales class (specified in the definitions for each asset group) determines the priority for asset sales.

Callable proxy bond

When you specify to also create a callable proxy bond within each asset sub portfolio, the above process first applies to create a non-callable proxy bond using the non-callable assets in this asset sub portfolio. Then a callable proxy bond is also created by the projection, using the callable bonds or mortgages with prepayment provisions. In addition to the above definitions, the callable proxy bond has:

- Call price schedule – the weighted average of the call price schedule for the individual assets, weighted by each asset's par/principal value. The earliest and last call dates are the earliest and latest call dates for the individual assets.

1.5 Between valuation dates

The chapters below explain the calculation of PBA reserves and risk based capital and LOC AG 48 economic reserves at each valuation date. For nested projections, where the exact PBA reserves and risk based capital and LOC AG 48 economic reserves are calculated only at each inner loop valuation date, a roll forward approach must be used to determine the amounts for the periods between the valuation dates.

The PBA reserve and capital amounts and LOC AG 48 economic reserve are rolled forward as described below for each product type. You can customize these formulas to use some other basis.

The following summarizes the roll-forward basis for each product type:

Product type	Roll-forward basis
Term life	Net premium reserve when applicable, otherwise sum assured
Traditional life	Net premium reserve when applicable, otherwise sum assured

Product type	Roll-forward basis
Universal life	Net premium reserve when applicable, otherwise max (fund value, AXXX excess premiums paid)
Variable Universal Life	Net premium reserve
Fixed annuity	Net premium reserve
Fixed indexed annuity	Net premium reserve
Variable annuity	Policies in force
Payout Annuity	Statutory CARVM reserve

2 VM-20 RESERVE

2.1 Applicability

This chapter discusses PBA reserves for life insurance products. These definitions are specified in Valuation Manual chapter 20.

2.2 Exclusion tests

You can perform the deterministic and stochastic exclusion tests by running standard projections. This test is not performed as part of the PBA calculations.

2.2.1 Deterministic exclusion test statistic

Deterministic PBA calculations are not required for blocks of business if the company demonstrates that the sum of the valuation net premiums for all future years for the block of business, determined according to exclusion test requirements, is less than the sum of the corresponding guaranteed gross premiums for the block of business.

You can perform this testing using a liability only seriatim projection and then comparing the results of the total gross premiums against the total valuation net premiums for each future year. PBA projections do not perform this test.

2.2.2 Stochastic exclusion test statistic

Stochastic PBA calculations are not required for blocks of business without a 'material tail risk', as determined by the stochastic exclusion test. You can perform this test using an asset/liability projection using a scenario file containing the 16 deterministic scenarios prescribed for this test. PBA projections do not perform this test.

For each scenario, calculate the present value of cash flows as: (1) PV benefits, plus (2) PV expenses, less (3) PV premiums, net (when applicable) of reinsurance and hedging, discounted using the asset yield rate.

The 'test ratio' numerator is the highest present value result, less the present value result for the Baseline scenario. The denominator is the Baseline scenario PV benefits net of reinsurance. The business does not have a material tail risk when the test ratio is less than 4.5%.

2.2.3 Exclusion test scenarios

The exclusion test scenarios can be generated on-the-fly, at each valuation date during nested projections. See the Economic scenarios generation chapter in the Economic scenarios book.

Note that SET scenario #12 must be generated on-the-fly in order to calculate the deterministic reserve at each inner loop valuation date during nested projections.

2.3 Net premium reserve

2.3.1 NPR effective date

PBA net premium reserves (NPR) replace the statutory reserves calculated by model point. NPR calculations are thus performed by formulas in the product type submodel, not by the life_pba submodel. See the Submodels structure chapter of the book for each product type.

NPR reserves apply when the PBA reserves and capital definition is Reserves, or Reserves and Capital, and the model point is issued on or after the PBA reserve effective date. Then each model point has its statutory reserve calculated either using VM-20 NPR (issued on or after the effective date) or CRVM (issued before the effective date).

2.3.2 Reserve basis

Specify the NPR reserve mortality, lapses, interest, expense allowances, function type (term products only), and mid-year reserve definition (term products only), on the Company input page.

2.3.3 Term life

For term life products, NPR reserves are a traditional reserve calculated using the Actuarial math library (AML). See the VM-20 NPR detail topic in the Audit reports chapter of the Actuarial math library book.

The NPR reserve between policy anniversaries is calculated using the NPR mid-year reserve definition. Each of these options operates the same as described in the Mid-year reserves section in the Statutory and tax reserves chapter of the Term life book.

2.3.4 Traditional life

For traditional life products, the PBA net premium reserve (NPR) is the statutory formulaic reserve defined for the plan code. Thus there are no special NPR calculations for this product type.

2.3.5 Universal and indexed universal life

PBA valuation date

NPR calculations are compute intensive, requiring: (1) solving calculations at issue, (2) traditional net premiums calculated at issue, and (3) monthly present values discounted from maturity starting from the current fund value, using a lapse rate calculated at the valuation date for secondary guarantee products.

Thus, you may select to determine exact NPR calculation each month or only at the end of each calendar year (see the npr_exact_calc_defn variable in the life_pba submodel). The default selection is 'Annual' with exact calculations occurring at the end of each calendar year. However, exact calculations will always be performed at the beginning of the projection (for standard projections) and on each PBA valuation date (for scenario time-step projections).

Between the exact calculation periods, the NPR is calculated using present values rolled forward from the previous exact calculation. This is the present value as of each month using the projected guaranteed fund values, guaranteed SG level premium, and SG reserve lapse rate, which were established at the last exact calculation period.

At the next exact calculation period, these values are updated based on the experience projection results, potentially causing some discontinuity in the NPR reserve from the previous month. This is the same effect as

for PBA stochastic and deterministic reserves, which are rolled forward between valuation dates. See the Between valuation dates section in the PBA overview chapter, above.

While the present value calculations are rolled forward between exact calculation periods, each month's NPR is based on the ratio of the current fund value to the guaranteed level premium fund value (see below).

Base plan reserve

The NPR is the greater of the base plan reserve, and the secondary guarantee reserve (if applicable).

The first step in calculating the base plan reserve is to solve at issue for the guaranteed level premium [solved_guar_level_prem]. This is like the GMP (for UL CRVM reserves), except that the policy matures for zero value instead of for the sum assured, and premiums are paid annually instead of modally.

The traditional valuation net premium for the base plan is then calculated using the projected guaranteed fund value [array guar_level_prem_fund] using this guaranteed level premium. The death benefit for each policy year [array dth_ben_tbl_npr_base] is the average death benefit at the end of each month using this fund value. For fund based products, traditional reserve calculations for the NPR use curtate functions.

The base plan expense allowances are specified in the life_pba submodel. The default values provide for \$2.50 per \$1000 sum assured at issue, plus 100% of the guaranteed level premium at issue, and 10% of the guaranteed level premium for the second through the fifth policy years, adjusted by the valuation net premium ratio. The expense allowance is amortized to zero at maturity.

The base plan m_{x+t} is the present value of death and maturity benefits uses the guaranteed fund value, starting from the current fund value as of the PBA valuation date, assuming that the guaranteed level premium is paid annually in future years, reduced by the present value of valuation net premiums, using the traditional net premium paid annually, and less the unamortized expense allowance.

The NPR base plan reserve is the m_{x+t} result multiplied by the ratio of the current fund value divided by the guaranteed level premium fund value.

Secondary guarantee reserve

When the model point has a secondary guarantee longer than five years from issue, the solving calculations include the guaranteed SG level premium at issue [solved_guar_sg_level_prem]. This solving uses the shadow fund or cumulative premiums test rates, assuming premiums are paid annually, resulting in the premium that keeps the secondary guarantee in force to its maturity.

The traditional net premium for the secondary guarantee is then calculated using the projected guaranteed fund value using this guaranteed SG level premium [array guar_sg_level_prem_fund]. The death benefit each policy year [array dth_ben_tbl_npr_sg] is the average death benefit at the end of each month using this fund value.

The secondary guarantee expense allowance rates and calculation methodology are the same as for the base plan. However, the expense allowance is not the same since the secondary guarantee will use a higher valuation interest rate (specified in the npr_int_rate_adj of the life_pba submodel) and the valuation net premium ratio for the secondary guarantee benefits.

The secondary guarantee net single premium is the present value of the death benefits for the remainder of the secondary guarantee period using the NPR secondary guarantee reserve interest, mortality, and lapses assumptions. The death benefits are determined using the guaranteed fund value, starting from the current fund value as of the PBA valuation date, assuming that the secondary guarantee level premium is paid annually in future years. The secondary guaranteed premium used is stored as a variable [sg_level_prem_per_unit_saved].

The secondary guarantee net single premium includes an assumption for lapses. The lapse rate is a weighted average of 'high' and 'low' lapse rates specified as part of the NPR reserve basis. The weight applied to the high lapse rate is the ratio of (1) the fully funded SG value [sg_req_single_prem_per_unit] less the current SG value on the PBA valuation date, divided by (2) the fully funded SG value less the SG value calculated from issue assuming the guaranteed SG level premium at issue has been paid annually. Since this

lapse rate is recalculated only at each exact calculation period, it is stored for use in calculations as a variable [stat_res_npr_sg_lapse_rate_saved].

The NPR secondary guarantee reserve is then determined as the net single premium times the lesser of 1 and the ratio of the actual secondary guarantee value to the fully funded secondary guarantee value, reduced by the unamortized secondary guarantee expense allowance.

2.3.6 Variable universal life

For variable universal life products, the PBA net premium reserve (NPR) is the statutory formulaic reserve defined for the plan code. Thus there are no special NPR calculations for this product type.

2.4 Starting assets

2.4.1 Circularity problem

The deterministic reserve and stochastic reserve calculations present a circularity problem. The PBR result depends on the starting amount of assets, which must be within an acceptable range of the PBR result being calculated. The amount of starting assets affects the asset yield, which can affect model point values when the crediting rate or policyholder dividend interest rate is linked to the asset yield. (This has no effect on the net premium reserve.)

Thus PBA calculations must first be run using an assumed amount of starting assets. If the resulting PBR is inconsistent with that assumption, you must adjust the assumed amount of starting assets and recalculate.

2.4.2 Initial valuation

The valuation at the projection date ($t = 0$), begins with the specified starting asset amount for PBA projections:

- Initial asset percent – The percent of existing assets specified when the valuation includes asset projections. This variable thus replaces – for PBA valuations – the Existing asset percent to use variable that normally applies for asset/liability projections.
- Initial assets – The amount of starting assets for a liability-only valuation. (This variable has no effect on an asset/liability valuation, which directly specifies the existing assets being projected.) This amount is adjusted by the initial asset percent, but that variable should typically be left at its 100% (1.00) default value for liability-only valuations.

These initial asset variables have no effect on PBA valuations after the projection date, as explained next.

2.4.3 Subsequent valuations

For PBA valuations after the projection date, the starting asset amount is calculated by the nested projection as a percent of the assets projected as of each PBA valuation date. The percent is the ratio of: (1) the NPR (formulaic statutory reserve for model points issued before the NPR effective date) net of reinsurance, to (2) the projected assets, at the end of the previous month.

2.4.4 Other starting assets

The following assets are projected separately, and are thus considered in addition to the starting assets amount:

- Initial policy loan balance – since policy loan cash flows are included in benefits for stochastic PBA calculations.
- Initial separate account asset – when variable products are included in the valuation.
- Initial market value of hedge assets – when liability projections include hedging.

2.5 Deterministic reserve

The deterministic reserve is the present value of cash flows projected for the block of business, using the deterministic scenario.

2.5.1 Deterministic scenario

Cash flows are projected using the prescribed deterministic scenario. This is a deterministic projection, using a single iteration.

Scenario definition

The deterministic scenario is scenario #12 of the stochastic exclusion test (SET) scenarios. See the Stochastic exclusion test section, below.

Scenario file

For a point-in-time valuation (only at the projection date), the stochastic scenarios are typically created using the American Academy of Actuaries (AAA) scenario tool. Select to create the Stochastic exclusion test scenarios. See the Economic scenarios book for instructions.

Of course, the historic scenario rates file is also required, to provide rates for moving averages and rate comparisons which look back to scenario rates before the projection date.

For a point-in-time valuation, specify the #12 scenario path as the Projection task loop specified in the Run parameter set. Specify the scenario file in the scen_id (Scenario file) variable, and the historic rates in the historic_rates_id (Historic rates) variable.

For nested projections, specify the initial (as of $t = 0$) scenario file (and historic scenario rates file) in the inner loop scenario file variables. The scenario file fields apply for the outer loop scenario paths.

Scenario generation

For nested projections, scenario rates for each subsequent valuation date are typically generated on-the-fly starting from the outer loop scenario yield curve at that date. The Economic scenarios book discusses the generation of inner loop economic scenario files.

2.5.2 Working reserve

Deterministic reserve calculations automatically set the working reserve to zero.

2.5.3 Taxes

Taxes are excluded from the deterministic reserve calculation.

2.5.4 Reinsurance

The deterministic reserve can calculate gross or net (including the impact of reinsurance) results, or both results can be calculated together in the same projection task.

2.5.5 Hedging

The deterministic reserve can calculate hedged (including the effect of hedging) or unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same projection task.

When calculating hedged results, the calculations incorporate the effect of hedging using information in a file projected by your separate hedging projection. This file is a comma separated value file of monthly information that contains the following columns:

Column	Description
task_loop	The task loop number for the deterministic projection
period	This represents the months from the valuation date
fund_val	Projected fund values
cash_flow	Projected hedging cash flow
mkt_val	Projected hedging market value

The hedge cash flows and market values included in the file are scaled based on the ratio of the fund values in the deterministic projection to the fund values included in the hedge result file. The hedge cash flows use the ratio as of the previous period.

2.5.6 Present value of cash flows

The deterministic reserve is:

$$\text{PV benefits} + \text{PV expenses} - \text{PV premiums} - \text{PV hedge cash flows}$$

Policy loans

The PV of benefits is increased by the initial (as of the valuation date) policy loan balance. Projected benefits are decreased by policy loan proceeds, which are the cash flows paid out when the policy loan amount is adjusted higher. Negative policy loan proceeds (loan repayments) increase projected benefits.

Expenses

The PV of expenses includes acquisition and maintenance expenses, policy loan expenses, commissions net of chargebacks, and premium taxes.

Discount rate

Present values are discounted at the asset yield for each month.

2.6 Stochastic reserve

The stochastic reserve is calculated by projecting the model over a number of stochastic scenarios, calculating the 'scenario amount' for each path, then finally calculating the 70% CTE for the 'tail' scenarios.

2.6.1 Stochastic scenarios

The model point cash flows are projected over each stochastic scenario. The following gives an overview of the scenario rates for the stochastic scenarios.

Scenario file

For a point-in-time valuation (only at the projection date), the stochastic scenarios are typically downloaded from the American Academy of Actuaries (AAA) web site, which publishes a set of 10,000 scenarios. Use the AAA tools to select a specified number of scenarios from this set. See the Economic scenarios book for instructions.

For example, when you specify 1000 scenarios, the tool produces the identification numbers for 1000 of the scenarios from this published set. Produce a scenario file containing the rates for these 1000 scenarios.

Of course, the historic scenario rates file is also required, to provide rates for moving averages and rate comparisons which look back to scenario rates before the projection date.

For a point-in-time valuation, the Projection task loops specified in the Run parameter set are the numbers of scenario paths using the scenario file specified in the scen_id (Scenario file) variable. The historic rates are specified in the historic_rates_id (Historic rates) variable.

For nested projections, specify the initial (as of $t = 0$) scenario file (and historic scenario rates file) in the inner loop scenario file variables. The scenario file fields apply for the outer loop scenario paths.

Scenario generation

For nested projections, scenario rates for each subsequent valuation date are typically generated on-the-fly starting from the outer loop scenario yield curve at that date. See the Economic scenarios book.

2.6.2 Working reserve

The stochastic reserve calculation automatically sets the working reserve to zero.

2.6.3 Target capital

Target capital is ignored for the stochastic reserve calculation.

2.6.4 Taxes

Taxes are excluded from the stochastic reserve calculation.

2.6.5 Reinsurance

The stochastic reserve can calculate gross or net (including the impact of reinsurance) results, or both results can be calculated together in the same projection task.

When both results are calculated together in the same projection, the starting assets are assumed to be on a net basis. Specify, using the life_pba_reins_gross_up_defn variable on the Company input page in the PBA: Life: General category, whether the starting assets should be adjusted up by a factor equal to $1.0 / (1.0 - \text{reinsurance percent})$ for the gross stochastic reserve calculation.

2.6.6 Hedging

The stochastic reserve can calculate hedged (including the effect of hedging) or unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same projection task.

When calculating hedged results, the calculations incorporate the effect of hedging using information in a file projected by your separate hedging projection. This file is a comma separated value file of monthly information that contains the following columns:

Column	Description
task_loop	The task loop number for the deterministic projection
period	This represents the months from the valuation date
fund_val	Projected fund values
cash_flow	Projected hedging cash flow
mkt_val	Projected hedging market value

The hedge cash flows and market values included in the file are scaled based on the ratio of the fund values in the deterministic projection to the fund values included in the hedge result file. The hedge cash flows use the ratio as of the previous period.

2.6.7 Profit distribution

There are no distribution of profits in the stochastic reserve calculations.

2.6.8 Accumulated assets

At the valuation date, the calculations begin with the starting asset amount described in the Starting assets section, above. This starting amount is projected through the valuation calculations, increased by liability cash flows and income from invested assets, and decreased by taxes and the increase in the IMR.

[assets_accum]

2.6.9 Accumulated deficiency

Since the working reserve is zero, the initial accumulated deficiency is the negative of the starting assets. For variable products, the separate account liability is also subtracted.

The accumulated deficiency is rolled forward with profits for each month. The previous deficiency is reduced by 'direct profit', then increased (when applicable) by the reinsurance cost and hedging cost. [def_accum]

Direct profit is: (1) the liability cash flow, plus (2) interest and investment income, plus (3) policy loan interest, (4) capital gains less the IMR increase, (5) increase in the separate account assets (when applicable) less the increase in the working reserve.

2.6.10 PV accumulated deficiencies

The present value at the valuation date is calculated for the accumulated deficiency at each 12 months from the valuation date, and at the end of the valuation projection. [scen_greatest_pv]

The discount rate for each month is the monthly equivalent of 1.05 (105%) times the current 1-year risk free scenario rate. This is the before tax rate, since income taxes are excluded for PBA reserve calculations.

[disc_rate_mthly]

2.6.11 Scenario amount

The 'scenario amount' for each valuation projection task loop is: (1) the greatest PV of the accumulated deficiencies, plus (2) the starting assets increased by (3) initial policy loans, (4) initial separate account assets (when applicable), and (5) initial market value of hedging assets (when applicable). [scen_amt]

2.6.12 CTE statistic

The 'stochastic amount' is the Conditional Tail Expectation (CTE) statistic for the stochastic valuation iterations. The CTE percent is 0.70 (70%) for stochastic life reserves [cte_pct].

The stochastic valuation projection task loops are ranked in order by their scenario amount. The 'tail scenarios' are the projection task loops with the highest scenario amounts. The number of tail scenarios is one minus the CTE percent times the number of stochastic scenarios, rounded down to the nearest whole number.

The stochastic amount is the average of the scenario amounts for the tail scenarios.

2.6.13 Multiple reserve calculations

When results both gross and net of reinsurance are being calculated together, the tail scenarios are selected separately for each result. That is, different scenarios may be used for calculating the gross (excluding reinsurance) PBR than are used for calculating the PBR net of reinsurance.

2.7 Final reserve

2.7.1 Principle-based reserve

The final principle-based reserve is either:

- Max of Stochastic and Deterministic and Policy Amount – The PBR is the greatest of the stochastic amount, the deterministic amount, and the net premium reserve.
- Max of Deterministic and Policy Amount – The PBR is the greater of the deterministic and net premium reserve.
- Policy Amount – The PBR is the net premium reserve.

When a deferred premium asset applies, the deferred net premium, calculated using the net premium reserve basis, is subtracted from the deterministic amount and the stochastic amounts before the comparison against the net premium reserve.

2.7.2 Excess reserve

The excess reserve is the difference between the principle-based reserve and the net premium reserve.

2.7.3 Model point reserve

The model point reserve is the sum of the model point net premium reserve and an allocated portion of the excess reserve. The allocation factor is the model point net premium reserve divided by the aggregate net premium reserve.

3 VM-22 RESERVE

3.1 Applicability

This chapter discusses PBA reserves for fixed, payout, and indexed annuity products. These definitions are specified in Valuation Manual chapter 22.

These calculations are currently as described above for the stochastic VM-20 reserve. There is no deterministic reserve, and the net premium reserve, calculated for each model point, is determined using the specified statutory formulaic reserve method.

For fixed, payout, and indexed annuities, PBA calculations are performed in the `ann_pba` submodel for portfolio projections, and in the product version of this submodel (`fa_ann_pba`, `pa_ann_pba`, and `fia_ann_pba`) for seriatim projections.

3.2 Net premium reserve

Currently, the VM-22 net premium reserve equal the statutory CARVM reserve.

3.3 Starting assets

3.3.1 Circularity problem

The stochastic reserve calculations present a circularity problem. The PBR result depends on the starting amount of assets, which must be within an acceptable range of the PBR result being calculated. The amount of starting assets affects the asset yield, which can affect model point values when the crediting rate or policyholder dividend interest rate is linked to the asset yield. (This has no effect on the net premium reserve.)

Thus PBA calculations must first be run using an assumed amount of starting assets. If the resulting PBR is inconsistent with that assumption, you must adjust the assumed amount of starting assets and recalculate.

3.3.2 Initial valuation

The valuation at the projection date ($t = 0$), begins with the specified starting asset amount for PBA projections:

- Initial asset percent – The percent of existing assets specified when the valuation includes asset projections. This variable thus replaces – for PBA valuations – the Existing asset percent to use variable that normally applies for asset/liability projections.
- Initial assets – The amount of starting assets for a liability-only valuation. (This variable has no effect on an asset/liability valuation, which directly specifies the existing assets being projected.) This amount is adjusted by the initial asset percent, but that variable should typically be left at its 100% (1.00) default value for liability-only valuations.

These initial asset variables have no effect on PBA valuations after the projection date, as explained next.

3.3.3 Subsequent valuations

For PBA valuations after the projection date, the starting asset amount is calculated by the nested projection as a percent of the assets projected as of each PBA valuation date. The percent is the ratio of: (1) the NPR net of reinsurance, to (2) the projected assets, at the end of the previous month.

3.3.4 Other starting assets

The following assets are projected separately, and are thus considered in addition to the starting assets amount:

- Initial market value of hedge assets – when liability projections include hedging.

3.4 Stochastic reserve

The stochastic reserve is calculated by projecting the model over a number of stochastic scenarios, calculating the 'scenario amount' for each path, then finally calculating the 70% CTE for the 'tail' scenarios.

3.4.1 Stochastic scenarios

The model point cash flows are projected over each stochastic scenario. The following gives an overview of the scenario rates for the stochastic scenarios.

Scenario file

For a point-in-time valuation (only at the projection date), the stochastic scenarios are typically downloaded from the American Academy of Actuaries (AAA) web site, which publishes a set of 10,000 scenarios. Use the AAA tools to select a specified number of scenarios from this set. See the Economic scenarios book for instructions.

For example, when you specify 1000 scenarios, the tool produces the identification numbers for 1000 of the scenarios from this published set. Produce a scenario file containing the rates for these 1000 scenarios.

Of course, the historic scenario rates file is also required, to provide rates for moving averages and rate comparisons which look back to scenario rates before the projection date.

For a point-in-time valuation, the Projection task loops specified in the Run parameter set are the numbers of scenario paths using the scenario file specified in the `scen_id` (Scenario file) variable. The historic rates are specified in the `historic_rates_id` (Historic rates) variable.

For nested projections, specify the initial (as of $t = 0$) scenario file (and historic scenario rates file) in the inner loop scenario file variables. The scenario file fields apply for the outer loop scenario paths.

Scenario generation

For nested projections, scenario rates for each subsequent valuation date are typically generated on-the-fly starting from the outer loop scenario yield curve at that date. See the Economic scenarios book.

3.4.2 Working reserve

The stochastic reserve calculation automatically sets the working reserve to zero.

3.4.3 Target capital

Target capital is ignored for the stochastic reserve calculation.

3.4.4 Taxes

Taxes are excluded from the stochastic reserve calculation.

3.4.5 Reinsurance

The stochastic reserve can calculate gross or net (including the impact of reinsurance) results, or both results can be calculated together in the same projection task.

When both results are calculated together in the same projection, the starting assets are assumed to be on a net basis. Specify, using the `life_pba_reins_gross_up_defn` variable on the Company input page in the PBA: Life: General category, whether the starting assets should be adjusted up by a factor equal to $1.0 / (1.0 - \text{reinsurance percent})$ for the gross stochastic reserve calculation.

3.4.6 Hedging

The stochastic reserve can calculate hedged (including the effect of hedging) or unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same projection task.

When calculating hedged results, the calculations can incorporate the effect of hedging using information projected by your separate hedging projection. A mixture of the results from a run off of existing hedges and projection of best estimate hedges can be incorporated.

The files are comma separated value files of monthly information that containing the following columns:

Column	Description
task_loop	The task loop number for the deterministic projection
period	This represents the months from the valuation date
fund_val	Projected fund values
cash_flow	Projected hedging cash flow
mkt_val	Projected hedging market value

The hedge cash flows and market values included in the file are scaled based on the ratio of the fund values in the deterministic projection to the fund values included in the hedge result file. The hedge cash flows use the ratio as of the previous period.

Specify an 'E' factor used to blend the results.

3.4.7 Profit distribution

There are no distribution of profits in the stochastic reserve calculations.

3.4.8 Accumulated assets

At the valuation date, the calculations begin with the starting asset amount described in the Starting assets section, above. This starting amount is projected through the valuation calculations, increased by liability

cash flows and income from invested assets, and decreased by taxes and the increase in the IMR. [assets_accum]

3.4.9 Accumulated deficiency

Since the working reserve is zero, the initial accumulated deficiency is the negative of the starting assets.

The accumulated deficiency is rolled forward with profits for each month. The previous deficiency is reduced by 'direct profit', then increased (when applicable) by the reinsurance cost and hedging cost. [def_accum]

Direct profit is: (1) the liability cash flow, plus (2) interest and investment income, (3) capital gains less the IMR increase, less (4) the increase in the working reserve.

3.4.10 PV accumulated deficiencies

The present value at the valuation date is calculated for the accumulated deficiency at each 12 months from the valuation date, and at the end of the valuation projection. [scen_greatest_pv]

The discount rate for each month is the monthly equivalent of 1.05 (105%) times the current 1-year risk free scenario rate. This is the before tax rate, since income taxes are excluded for PBA reserve calculations. [disc_rate_mthly]

3.4.11 Scenario amount

The 'scenario amount' for each valuation projection task loop is: (1) the greatest PV of the accumulated deficiencies, plus (2) the starting assets increased by (3) initial separate account assets (when applicable), and (4) initial market value of hedging assets (when applicable). [scen_amt]

3.4.12 CTE statistic

The 'stochastic amount' is the Conditional Tail Expectation (CTE) statistic for the stochastic valuation iterations. The CTE percent is 0.70 (70%) for stochastic life reserves [cte_pct].

The stochastic valuation projection task loops are ranked in order by their scenario amount. The 'tail scenarios' are the projection task loops with the highest scenario amounts. The number of tail scenarios is one minus the CTE percent times the number of stochastic scenarios, rounded down to the nearest whole number.

The stochastic amount is the average of the scenario amounts for the tail scenarios.

3.4.13 Multiple reserve calculations

When results both gross and net of reinsurance are being calculated together, the tail scenarios are selected separately for each result. That is, different scenarios may be used for calculating the gross (excluding reinsurance) PBR than are used for calculating the PBR net of reinsurance.

3.5 Final reserve

3.5.1 Principle-based reserve

The final principle-based reserve is the greater of (1) the stochastic amount, and (3) the aggregate net premium reserve.

3.5.2 Excess reserve

The excess reserve is the difference between the principle-based reserve and the aggregate net premium reserve.

3.5.3 *Model point reserve*

The model point reserve is the sum of the model point net premium reserve and an allocated portion of the excess reserve. The allocation factor is the model point net premium reserve divided by the aggregate net premium reserve.

4 RBC C3 PHASE III CAPITAL

4.1 *Applicability*

4.1.1 *Life insurance*

This chapter discusses the PBA capital amount for life insurance products, as defined by RBC C3 phase III requirements.

4.1.2 *Fixed, payout, and indexed annuities*

PBA capital requirements are currently undefined for fixed, payout, and indexed deferred annuity products. The US Library projects apply the RBC C3 phase III stochastic calculations for these product types.

4.2 *Starting assets*

Stochastic capital calculations start each valuation with a percent of assets which operates the same as for stochastic reserve calculations.

4.3 *Stochastic capital*

The stochastic capital is calculated by projecting the model over a number of valuation projection task loops, calculating the 'scenario amount' for each loop, then finally calculating the CTE. Specify the CTE percent, such as 0.09 (90%), using the `cte_pct_stoch_cap` variable.

4.3.1 *Scenario rates*

The scenarios are the same as for the stochastic reserve. See the VM-20 reserve chapter, above.

4.3.2 *Working reserve*

The working reserve for the stochastic capital calculation is the cash value.

4.3.3 *Target capital*

Target capital is ignored for the stochastic capital calculation.

4.3.4 *Taxes*

Income taxes are included in the stochastic capital calculations. Specify the stochastic capital tax definition to define the tax reserves for the projection:

- Calculate tax reserve – Taxes are calculated the same as for normal projections; the tax reserve approximation described below does not apply.
- Use working reserve – Taxes are calculated using the working reserve as the tax reserve, then adjusting the stochastic capital amount as described in the Tax reserves adjustment section, below.

4.3.5 *Reinsurance*

The stochastic capital can calculate gross or net (including the impact of reinsurance) results, or both results can be calculated together in the same projection task.

When both results are calculated together in the same projection, the starting assets are assumed to be on a net basis.

Specify, using the `life_pba_reins_gross_up_defn` variable on the Company input page in the PBA: Life: General category, whether the starting assets should be adjusted up by a factor equal to $1.0 / (1.0 - \text{reinsurance percent})$ for the gross stochastic reserve calculation.

4.3.6 Hedging

The stochastic capital can calculate hedged (including the effect of hedging) or unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same projection task.

When calculating hedged results, the calculations can incorporate the effect of hedging using information projected by your separate hedging projection. For life insurance products, a mixture of the results from a run off of existing hedges and projection of best estimate hedges can be incorporated. For annuity products, only a projection of best estimate hedges is included.

The files are comma separated value files of monthly information that containing the following columns:

Column	Description
task_loop	The task loop number for the deterministic projection
period	This represents the months from the valuation date
fund_val	Projected fund values
cash_flow	Projected hedging cash flow
mkt_val	Projected hedging market value

The hedge cash flows and market values included in the file are scaled based on the ratio of the fund values in the deterministic projection to the fund values included in the hedge result file. The hedge cash flows use the ratio as of the previous period.

For life insurance products, specify an 'E' factor used to blend the results.

4.3.7 Profit distribution

There are no distribution of profits in the stochastic capital calculations.

4.3.8 Accumulated deficiency

For stochastic capital, the initial accumulated deficiency is the initial working reserve (cash value), less the starting assets and, for variable products, less the separate account liability.

The accumulated deficiency is rolled forward with profits for each month. The previous deficiency is reduced by 'direct profit', then increased (when applicable) by the reinsurance cost and hedging cost. [`def_accum`]

Direct profit is: (1) the liability cash flow, plus (2) interest and investment income, plus (3) policy loan interest, (4) capital gains less the IMR increase, (5) the increase in the separate account assets (when applicable) less the increase in the working reserve, less (6) income taxes.

4.3.9 Scenario amount

The present values of the accumulated deficiencies are calculated using the after tax discount rate equal to 1.05 (105%) times the 1-year risk free scenario rate, then times (1 minus the tax rate). [`disc_rate_mthly`]

The scenario amount for each valuation projection task loop is calculated the same way as for stochastic PBR.

4.3.10 Stochastic amount

The stochastic capital amount is the CTE statistic for the stochastic valuation iterations. The CTE calculation is the same as described above for the stochastic reserve amount.

For life insurance products, specify, using the `life_pba_cte_pct_stoch_cap` variable on the Company input page in the PBA: Life: General category, the CTE percent for stochastic capital calculations (such as 90%).

For annuity products, specify, using the `ann_pba_cte_pct_stoch_cap` variable on the Company input page in the PBA: Annuity: General category, the CTE percent for stochastic capital calculations (such as 90%).

4.4 Tax reserves adjustment

When tax reserves are approximated using the working reserve (cash value) as the tax reserve, the stochastic capital amount is increased (may be negative) by:

$$(\text{initial working reserve} - \text{initial tax reserve}) \times (\text{corporate tax rate}) \times (1 - \text{average ratio})$$

The 'average ratio' is the average of the ratio for each tail scenario of the policies in force at the time when the greatest PV result occurred, divided by the initial policies in force.

4.5 Final RBC C3 amount

The final RBC C3 amount is calculated as the stochastic capital amount less the net statutory reserve including the separate account reserve, but not less than zero.

5 AG 43 RESERVE

5.1 Applicability

This chapter discusses the PBA reserve calculations for variable annuity, as defined by Actuarial Guideline XLIII.

5.2 Starting assets

5.2.1 Circularity problem

The scenario amounts calculated for the stochastic reserve calculations present a circularity problem, since they depend on the starting amount of assets, which must be within an acceptable range of the PBR result.

Thus stochastic PBA calculations must first be run using an assumed amount of starting assets. If the resulting PBR is inconsistent with that assumption, you must adjust the assumed amount of starting assets and recalculate.

The amount of starting assets affects the stochastic amount results. It has no effect on the standard scenario reserve.

5.2.2 Initial valuation

The valuation at the projection date ($t = 0$), begins with the specified starting asset amount for PBA projections:

- Initial asset percent – The percent of existing assets specified when the valuation includes asset projections. This variable thus replaces – for PBA valuations – the Existing asset percent to use variable that normally applies for asset/liability projections.

- Initial assets – The amount of starting assets for a liability-only valuation. (This variable has no effect on an asset/liability valuation, which directly specifies the existing assets being projected.) This amount is adjusted by the initial asset percent, but that variable should typically be left at its 100% (1.00) default value for liability-only valuations.

These initial asset variables have no effect on PBA valuations after the projection date, as explained next.

5.2.3 Subsequent valuations

For PBA valuations after the projection date, the starting asset amount is calculated by the nested projection as a percent of the assets projected as of each PBA valuation date. The percent is the ratio of: (1) the NPR (formulaic statutory reserve for model points issued before the NPR effective date) net of reinsurance, to (2) the projected assets, at the end of the previous month.

5.2.4 Other starting assets

The following assets are projected separately, and are thus considered in addition to the starting assets amount:

- Initial separate account asset
- Initial market value of hedge assets.

5.3 Standard scenario reserve

The standard scenario reserve is calculated by projecting each model point's cash flows using the standard scenario, then calculating its basic adjusted reserve and greatest present value of accumulated net revenue. The model point margins and benefit costs are projected using the prescribed standard scenario. This is a deterministic projection, using a single projection task loop.

5.3.1 Scenario rates

Specify the standard valuation law discount rate and prescribed standard scenario asset class returns as PBA assumptions. These rates override the rates provided by the scenario file. You must specify some scenario file in the `scen_id` variable, but it does not matter what those scenario rates are since they are overridden by the PBA assumptions. In the Run parameter set, specify to run only one Projection task loop for the standard scenario reserve valuation task.

Historical scenario rates must be specified in the `historic_rates_id` variable, which contains rates before the projection date such as for calculation of moving average competitor rates in the early projection periods. See the Scenario tools chapter of the Economic scenarios book for information about creating scenario files.

Discount rates

Present values discounted at the standard valuation law discount rate, which varies by issue year.

Asset class returns

The standard scenario asset class specified for each fund in the 'va_account' submodel must correspond to the asset classes specified in this table.

5.3.2 Standard scenario amount

The standard scenario reserve is the aggregate 'standard scenario amount' for each model point. This is the 'basic adjusted reserve' at the valuation date, plus the greatest present value of accumulated net revenue.

5.3.3 Basic adjusted reserve

The basic adjusted reserve is the CARVM reserve, excluding GMDB's, assuming no penalty free withdrawals are taken. Alternatively, you can specify the pre-calculated basic adjusted reserve on the model point record.

No adjustment is made for aggregate reinsurance arrangements. Customize to include this effect.

5.3.4 Accumulated net revenue

The accumulated net revenue is calculated for each model point:

$$(\text{previous accumulated net revenue} + \text{margin}) \times \text{monthly discount rate} - \text{benefit costs}$$

where:

The margins on the account value are determined each month.

The benefit costs are the benefits paid for each rider (net of reinsurance, when applicable).

5.3.5 Hedging

The standard scenario reserve can calculate hedged (including the effect of hedging) or unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same projection task.

When calculating hedged results, the calculations can incorporate the effect of hedging using information projected by your separate hedging projection.

This file is a comma separated value file of monthly information that contains the following columns:

Column	Description
task_loop	The task loop number for the deterministic projection
period	This represents the months from the valuation date
gmab_if	End of month Guaranteed Minimum Accumulation Benefit (GMAB) in force. This adjusts the basis of the hedging cash flows and market values.
gmwb_if	End of month Guaranteed Minimum Withdrawal Benefit (GMWB) in force. This adjusts the basis of the hedging cash flows and market values.
cash_flow	Projected hedging cash flow
mkt_val	Projected hedging market value

The hedge cash flows and market values included in the file are scaled based on the ratio of the guaranteed benefit inforce amounts in the standard scenario projection to the guaranteed benefit inforce amounts included in the hedge result file.

For the standard scenario reserve amount the value of existing hedges is deducted from the sum of the model point standard scenario reserve amounts.

5.4 Stochastic reserve

The stochastic reserve is calculated by projecting cash flows over a number of stochastic scenarios, calculating the 'scenario amount' for each path, then finally calculating the 70% CTE as explained below.

5.4.1 Scenario rates

The scenario rates for point-in-time (standard) valuations must be specified in the scen_id variable (on the Economy page). These rates cannot be generated on-the-fly. See the Scenario tools chapter of the Economic scenarios book for information about creating scenario files.

For nested projections, the `scen_id` variable (in the Economy input page) specifies the outer loop scenario paths. The projection set loops (specified for the projection set on the Run controller) selects which of these outer loops to process. These rates cannot be generated on-the-fly. See the Scenario tools chapter of the Economic scenarios book for information about creating scenario files.

Historical scenario rates must be specified in the `historic_rates_id` variable, which contains rates before the projection date such as for calculation of moving average competitor rates in the early projection periods.

The `inner_loop_valn_scen_id` variable specifies the inner loop scenario paths, which apply only for valuation tasks. The projection task loops (in the Run parameter set) selects which of these inner loops to process.

Inner loop scenarios can be generated on-the-fly for valuations within the nested projection after the projection date ($t = 0$). See the Economic scenarios generation chapter of the Economic scenarios book.

5.4.2 Working reserve

The working reserve for the stochastic reserve calculation is the cash value.

5.4.3 Target capital

Target capital is ignored for the stochastic reserve calculation.

5.4.4 Taxes

Taxes are excluded from the stochastic reserve calculation.

5.4.5 Hedging

The stochastic reserve can calculate hedged (including the effect of hedging) or unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same projection task.

When calculating hedged results, the calculations can incorporate the effect of hedging using information projected by your separate hedging projection. A mixture of the results from a run off of existing hedges and projection of best estimate hedges can be incorporated.

The files are comma separated value files of monthly information that containing the following columns:

Column	Description
<code>task_loop</code>	The task loop number for the deterministic projection
<code>period</code>	This represents the months from the valuation date
<code>gmab_if</code>	End of month Guaranteed Minimum Accumulation Benefit (GMAB) in force. This adjusts the basis of the hedging cash flows and market values.
<code>gmwb_if</code>	End of month Guaranteed Minimum Withdrawal Benefit (GMWB) in force. This adjusts the basis of the hedging cash flows and market values.
<code>cash_flow</code>	Projected hedging cash flow
<code>mkt_val</code>	Projected hedging market value

The hedge cash flows and market values included in the file are scaled based on the ratio of the guaranteed benefit inforce amounts in the standard scenario projection to the guaranteed benefit inforce amounts included in the hedge result file. The scale factor cannot exceed a specified maximum scale factor.

Specify an 'E' factor used to blend the results.

5.4.6 Profit distribution

There are no distribution of profits in the stochastic reserve calculation.

5.4.7 Accumulated assets

At the valuation date, the calculations begin with the starting asset amount described in the Starting assets section, above. This starting amount is increased by liability cash flows and income from invested assets, and decreased by taxes and the increase in the IMR.

5.4.8 Accumulated deficiency

The 'scenario amount' for each valuation projection task loop is the starting assets plus the greatest present value of the accumulated deficiency at the end of each projection year.

The initial accumulated deficiency is the initial cash value, less the starting assets.

Profits are calculated using the 'working reserve' defined as the cash value.

The accumulated deficiency is rolled-forward with profits for each month. The previous deficiency is reduced by 'direct profit', then increased (when applicable) by the reinsurance cost and hedging cost.

Direct profit is: (1) the liability cash flow, plus (2) interest and investment income, (3) capital gains less the IMR increase, and (4) the increase in separate account assets less the increase in working reserve (cash value).

5.4.9 PV accumulated deficiencies

The present value at the valuation date is calculated for the accumulated deficiency at each 12 months from the valuation date, and at the end of the valuation projection.

The discount rate for each month is the new money discount rate, which is the rate at which positive cash flows are invested, reduced for expected credit losses. Since income taxes are excluded for PBA reserve calculations, this is the before tax rate.

5.4.10 Scenario amount

The 'scenario amount' for each valuation iteration is: (1) the greatest of these PVs, plus (2) the starting assets, (3) initial separate account assets (when applicable), and (4) initial market value of hedging assets (when applicable).

5.4.11 CTE statistic

The 'stochastic amount' is the Conditional Tail Expectation (CTE) statistic for the stochastic valuation iterations. The CTE percent is 0.70 (70%) for stochastic variable annuity reserves [cte_pct].

The stochastic valuation projection task loops are ranked in order by their scenario amount. The 'tail scenarios' are the projection task loops with the highest scenario amounts. The number of tail scenarios is one minus the CTE percent times the number of stochastic scenarios, rounded down to the nearest whole number.

The stochastic amount is the average of the scenario amounts for the tail scenarios.

5.4.12 Multiple reserve calculations

When results both gross and net of reinsurance are being calculated together, the tail scenarios are selected separately for each result. That is, different scenarios may be used for calculating the gross (excluding reinsurance) PBR than are used for calculating the PBR net of reinsurance.

5.5 Final reserve

5.5.1 Principle-based reserve

The final principle-based reserve is either:

- Max of Stochastic and Std Scenario: The AG 43 reserve is the greater of the stochastic amount and the standard scenario amount.
- Std Scenario: The AG 43 reserve equals the standard scenario amount.

5.5.2 Excess reserve

The excess reserve is the difference between the principle-based reserve and the aggregate standard scenario reserve adjusted for hedging.

5.5.3 Model point reserve

The model point reserve is the sum of the model point standard scenario reserve, less an allocated portion of the standard scenario reserve hedging adjustment, plus an allocated portion of the excess reserve. The allocation factor is the model point standard scenario reserve divided by the aggregate standard scenario reserve.

6 RBC C3 PHASE II CAPITAL

6.1 Applicability

This chapter discusses the PBA capital amount for variable annuity products, as defined by RBC C3 phase II requirements.

6.2 Starting assets

Stochastic capital calculations start each valuation with a percent of assets which operates the same as for stochastic reserve calculations.

6.3 Standard scenario capital

The standard scenario capital is calculated by projecting the block of business using the standard scenario, calculating its basic adjusted reserve and greatest present value of accumulated net revenue. The model point margins and benefit costs are projected using the prescribed standard scenario. There is only a single projection task loop for this calculation.

The standard scenario capital amount can be calculated in a single portfolio valuation or using a two step process of a seriatim valuation followed by a portfolio valuation.

When the two step process is used, the model point's standard scenario margins, costs, and working reserves are calculated in the first step and then used in a second step run to determine the standard scenario capital amount in aggregate.

6.3.1 Scenario rates

Specify the standard valuation law discount rate and prescribed standard scenario asset class returns as PBA assumptions. These rates override the rates provided by the scenario file. You must specify some scenario file in the scen_id variable, but it does not matter what those scenario rates are since they are overridden by the PBA assumptions. In the Run parameter set, specify to run only one Projection task loop for the standard scenario capital valuation task.

Historical scenario rates must be specified in the `historic_rates_id` variable, which contains rates before the projection date such as for calculation of moving average competitor rates in the early projection periods. See the Scenario tools chapter of the Economic scenarios book for information about creating scenario files.

Discount rates

Present values are discounted at the 10-year risk free rate plus 0.5%, floored at 3% and capped at 9%. Since income taxes are included for PBA capital calculations, this is the after tax rate.

Asset class returns

The standard scenario asset class specified for each fund in the 'va_account' submodel must correspond to the asset classes specified in this table.

6.3.2 Standard scenario amount

The standard scenario capital is the aggregated 'standard scenario amount' for each model point. The standard scenario amount is the basic adjusted reserve plus the greatest present value of accumulated net revenue.

6.3.3 Basic adjusted reserve

The basic adjusted reserve is the working reserve, which is the cash value.

6.3.4 Accumulated net revenue

The accumulated net revenue calculated for each model point is:

$$(\text{previous accumulated net revenue} + \text{margin}) \times \text{monthly discount rate} - \text{benefit costs}$$

where:

The margins on the account value are determined each month.

The benefit costs are the benefits paid for each rider (net of reinsurance, when applicable).

6.3.5 Hedging

The standard scenario capital can calculate hedged (including the effect of hedging) or unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same projection task.

When calculating hedged results, the calculations can incorporate the effect of hedging using information projected by your separate hedging projection.

This file is a comma separated value file of monthly information that contains the following columns:

Column	Description
task_loop	The task loop number for the deterministic projection
period	This represents the months from the valuation date
gmab_if	End of month Guaranteed Minimum Accumulation Benefit (GMAB) in force. This adjusts the basis of the hedging cash flows and market values.
gmwb_if	End of month Guaranteed Minimum Withdrawal Benefit (GMWB) in force. This adjusts the basis of the hedging cash flows and market values.
cash_flow	Projected hedging cash flow

Column	Description
mkt_val	Projected hedging market value

The hedge cash flows and market values included in the file are scaled based on the ratio of the guaranteed benefit inforce amounts in the standard scenario projection to the guaranteed benefit inforce amounts included in the hedge result file.

For the standard scenario capital amount the value of existing hedges is deducted from the sum of the model point standard scenario capital amounts.

6.4 Stochastic capital

The stochastic capital is calculated by projecting the model over a number of valuation projection task loops, calculating the 'scenario amount' for each loop, then finally calculating the CTE. Specify the CTE percent, such as 0.09 (90%), using the `cte_pct_stoch_cap` variable.

6.4.1 Scenario rates

The scenarios are the same as for the stochastic reserve. See the AG 43 reserve chapter, above.

6.4.2 Working reserve

The working reserve for the stochastic capital calculation is the cash value.

6.4.3 Target capital

Target capital is ignored for the stochastic capital calculation.

6.4.4 Taxes

Income taxes are included in PBA capital calculations, using the working reserve as the tax reserve, then adjusting the stochastic capital amount as described in the Tax reserves adjustment section, below.

6.4.5 Hedging

The stochastic capital can calculate hedged (including the effect of hedging) or unhedged (excluding the effect of hedging) results, or both results can be calculated together in the same projection task.

When calculating hedged results, the calculations can incorporate the effect of hedging using information projected by your separate hedging projection. A mixture of the results from a run off of existing hedges and projection of best estimate hedges can be incorporated.

The files are comma separated value files of monthly information that containing the following columns:

Column	Description
task_loop	The task loop number for the deterministic projection
period	This represents the months from the valuation date
gmab_if	End of month Guaranteed Minimum Accumulation Benefit (GMAB) in force. This adjusts the basis of the hedging cash flows and market values.

Column	Description
gmwb_if	End of month Guaranteed Minimum Withdrawal Benefit (GMWB) in force. This adjusts the basis of the hedging cash flows and market values.
cash_flow	Projected hedging cash flow
mkt_val	Projected hedging market value

The hedge cash flows and market values included in the file are scaled based on the ratio of the guaranteed benefit inforce amounts in the standard scenario projection to the guaranteed benefit inforce amounts included in the hedge result file. The scale factor cannot exceed a specified maximum scale factor.

Specify an 'E' factor used to blend the results.

6.4.6 Profit distribution

There are no distribution of profits in the stochastic capital calculation.

6.4.7 Accumulated deficiency

For PBA capital, the initial (at the projection date) accumulated deficiency is the initial cash value, less the starting assets.

The accumulated deficiency is rolled-forward with profits for each month. The previous deficiency is reduced by direct profit, then increased (when applicable) by the reinsurance cost and hedging cost. [def_accum]

Direct profit is: (1) the liability cash flow, (2) interest and investment income, (3) capital gains less the IMR increase, and (4) the increase in separate account assets less the increase in working reserve, less (5) income taxes.

6.4.8 PV accumulated deficiencies

The present value at the valuation date is calculated for the accumulated deficiency at each 12 months from the valuation date, and at the end of the valuation projection.

The discount rate for each month is the 1-year risk free rate, which is the rate at which positive cash flows are invested, reduced for expected credit losses. Since income taxes are included for PBA capital calculations, this is the after tax rate. [disc_rate_mthly]

6.4.9 Scenario amount

The scenario amount for each valuation projection task loop is calculated the same way as for stochastic PBR.

6.4.10 CTE statistic

Just as for the stochastic PBR, the stochastic capital amount is the CTE statistic for the stochastic valuation results. Specify the CTE percent for the stochastic capital amount, such as 0.90 (90%).

6.4.11 Multiple PBA calculations

When multiple PBA capital amounts are being calculated together, the tail scenarios are ordered separately for each result. For example, different valuation iterations may be used for calculating the gross (excluding reinsurance) PBA capital than are used for calculating the PBA capital net of reinsurance.

6.5 Tax reserves adjustment

For RBC C-3 phase II capital calculations, tax reserves are approximated using the working reserve (cash value) as the tax reserve. The stochastic capital amount is adjusted by increasing (may be negative) by:

$$(\text{initial tax reserve} - \text{initial working reserve}) \times (\text{corporate tax rate}) \times (1 - \text{average ratio})$$

The 'average ratio' is the average of the ratio for each tail scenario of the policies in force at the time when the greatest PV result occurred, divided by the initial policies in force.

6.6 Final RBC C3 amount

The final total asset requirement is either:

- Max of Stochastic and Std Scenario: The total asset requirement is the greater of the stochastic amount and the standard scenario amount.
- Std Scenario: The total asset requirement equals the standard scenario amount.

The final RBC C3 amount is calculated as the total asset requirement amount less the net statutory reserve including the separate account reserve, but not less than zero.

7 LOC ECONOMIC RESERVE

Entering into a LOC financing agreement offsets part of the reserve, thus increasing free surplus. The reserve offset is the difference between the statutory reserve and a letter of credit economic reserve.

The LOC economic reserve can be determined using a gross premium reserve methodology or the required level of primary security determined using the actuarial method described in Actuarial Guideline 48.

Specify the selection for the LOC economic reserve using the 'loc_econ_res_defn' variable in the Letter of Credit: General section of the Company input page.

7.1 GPR economic reserve

7.1.1 GPR methodology

When the LOC economic reserve definition is 'Gross Premium Reserve, the economic reserve is determined using a gross premium reserve methodology, where the gross premium reserve is the present value of model point cash flows, calculated using the discount rate specified for each financing transaction.

Gross premium reserve factors are calculated for each model point, for the projection date and each future period. The GPR amount along with the units/policies calculated in the GPR valuation can be stored by model point, or in aggregate for the model point groupings within a cohort.

When the GPR factors are stored by model point, GPR results are captured for each model point along with the units/policies determined in the GPR valuation. When used in a projection, the GPR is adjusted for the experience units/policies separately for each model point, prior to aggregating into the cohorts. This method is more accurate, since adjusting each model point accounts for the changing mix of business since the GPR valuation was performed.

When the GPR factors are stored by cohort, GPR results are aggregated to its cohort, within each model point report group using the model point grouping capability. When used in a projection, the GPR is adjusted for the experience units/policies for the cohort. This method is more efficient, since it reduces the size of the GPR valuation output file, and adjusts valuation results for each cohort (within each reporting group).

Note – When the GPR calculations are for use in a seriatim projection, the GPR factors must be stored by model point.

7.1.2 Assumptions

The GPR valuation calculations use whatever assumptions you specify for the valuation task. These are typically different from the assumptions for the experience projection task.

7.1.3 Model point GPR percent

The GPR percent variable on the model point record applies when GPR values are calculated for LOC reserve financing.

For example, reserve financing arrangements can apply only to specific policies within the cohort assignment rules. Include a model point by leaving its percent at the 1.0 (100%) default value. Exclude a model point by setting its GPR percent to 0.0. Partially include a model point by specifying the percent to include.

7.1.4 Valuation period

GPR factors apply over the number of periods specified in the valn_yrs project variable (in the Run parameter set) for the GPR valuation.

This value should be long enough to reach to maturity for each model point included in the GPR valuation. GPR cash flows are projected for this number of years from $t = 0$, then discounted to calculate the GPR amount at each month.

7.1.5 GPR interest rate

The GPR interest rate varies by each LOC cohort.

Specify, for each LOC cohort, whether the rate is a specified fixed rate or the asset yield rate using the loc_gpr_fixed_disc_defn variable in the Letter of Credit: Cohorts section on the Company input page.

The monthly discount rate = $(1 + \text{annual rate})^{1/12} - 1$

7.1.6 Discounted liability cash flows

The GPR is the present value of liability cash flows. Cash flows paid out increase the reserve; cash flows received decrease the reserve.

The gross GPR is the present value of liability cash flows ignoring reinsurance. The ceded GPR is the present value of reinsurance cash flows. The net GPR is the gross GPR minus the ceded GPR.

7.2 AG 48 economic reserve

7.2.1 VM-20 methodology

When the LOC economic reserve definition is 'AG 48', the economic reserve is determined using the actuarial method of Actuarial Guideline 48. The actuarial method is the greater of the deterministic reserve, stochastic reserve (when included), and the applicable percentage of the NPR determined using the methodologies of VM-20.

7.2.2 Assumptions

The AG 48 calculations use whatever assumptions you specify for the valuation task. These are typically different from the assumptions for the experience projection task.

7.2.3 VM-20 calculation assumptions

As for VM-20, assumptions may vary between the deterministic and stochastic valuations. Specify the calculation assumptions using the variables in the PBA: Life: General section of the company input page.

Since actuarial guideline 48 is applicable in advance of the adoption of VM-20, the life_pba_res_eff_date variable for AG 48 calculations should be left at the default value of 9999/01. If this variable is changed, then

the statutory reserves will be calculated using the VM-20 methodology as opposed to using the current valuation manual methodology.

7.2.4 AG 48 NPR percentages

The required percentages applied to the VM-20 NPR are referenced from the global 'AG48 Actuarial Method NPR Percentages.xlsm' external source workbook and are referenced through the `ag48_npr_pct` variable, on the XXXProduct input page in the Reserve: NPR category.

7.2.5 Actuarial method

The AG 48 actuarial method is the greater of the deterministic reserve, stochastic reserve (when included), and the applicable percentage of the NPR.

8 PBA AND LOC VALUATION TASKS SETUP

8.1 PBA and LOC valuations task overview

8.1.1 Point-in-time valuations

PBA and LOC valuations may be for a single point-in-time valuation, such as a valuation at December of a calendar year. The valuation tasks are performed as of the projection date (valuation date).

8.1.2 Inner loop valuations

PBA and LOC valuations can also be included as inner loop valuations within a nested projection. For example, inner loop valuations to produce new PBA reserves and capital amounts along an outer loop horizon. The experience projection is dependent on the PBA or LOC valuation tasks being completed for the current time step.

8.1.3 Concurrent valuations

A PBA valuation task can calculate multiple results together. For example:

- A deterministic reserve valuation task could calculate both the gross and net deterministic reserves.
- A stochastic PBA valuation task could calculate both stochastic reserves and capital, and could also calculate results both excluding and including reinsurance.
- Other combinations could apply, such as calculating gross stochastic reserves and capital in one valuation task, and net stochastic reserves and capital in another task.

PBA results calculated together are different from the results calculated separately. The asset/liability valuation task calculating multiple PBA results develops a single asset yield, which is used for each of the PBA results calculated by that task. This is different from separate valuation tasks for each result, where assets are projected separately for each task. For example, the asset cash flows for a single PBA capital calculation includes the impact of taxes, while a single PBR calculation does not.

The asset yield affects liability cash flows when the crediting rate is based on, or constrained by, the asset yield. The asset yield directly affects the deterministic reserve since it is the discount rate for the gross premium reserve.

The single asset yield is calculated using the reserve calculation (when it is being calculated), net of reinsurance (when included). For example, calculating PBA capital and reserves gross and net of reinsurance uses the asset cash flows projected for PBA reserves (thus before tax) net of reinsurance. Calculating only PBA capital gross and net of reinsurance uses the asset cash flows projected for capital (thus after tax) and net of reinsurance.

This use of the same asset yield is the only difference between separate and combined calculations. Even when multiple PBA results are calculated together in a valuation task, each PBA result is a separate calculation.

Of course, combining multiple PBA result calculations in the same projection task decreases calculation time. Assets are projected fewer times – just once for the deterministic reserve valuation task, and once for the stochastic reserve and capital valuation task.

8.1.4 Final valuations

Since PBA and LOC valuations require more than one projection task, an additional ‘final’ valuation projection task can be used to determine the final PBA or LOC valuation results.

8.2 General PBA and LOC valuation setup

The following setup instructions apply for all PBA and LOC valuations.

8.2.1 Input variables

Specify the following variables on the Company input page in the PBA category for PBA valuations and for the LOC economic reserve calculation that uses AG 48:

Variable	Value
Reserve and capital calculation definition	Not Applicable, for no PBA reserves and capital Capital, for capital calculations only Reserves, for reserve calculations only Reserves and Capital, for both reserve and capital calculations together
Reinsurance definition	Gross, for gross calculations only Net, for net calculations only Both, for both gross and net calculations
Reinsurance gross up definition	No, net results are not grossed up for gross of reinsurance calculations Yes, net results are gross up for gross of reinsurance calculations
Hedging definition	Hedged, for hedged calculations only Unhedged, for unhedged calculations only Both, for both hedged and unhedged calculations
Initial assets	Has no effect for asset/liability valuation For liability-only valuations, specify initial invested assets that approximate the PBR result
Initial assets percent	For asset/liability valuations, specify a percent of the initial invested assets that approximates the PBR result

8.2.2 Run page

Create a valuation projection task on a Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

For cell pricing, and for seriatim based valuation tasks, create a projection task on a Run page which uses the product submodel as the target model object.

8.2.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Projection period from	0
Projection period to	0, since the valuation projection is specified by the Valuation years project variable.
Projection period min	0 for portfolio tasks, since there should be no calculations before the projection date. -600 for seriatim tasks, to allow for external source lookup keys calculating before the projection date (such as pol_yr)
Projection period max	At least as long as the calculations for each submodel. For example, UL individual fund calculations must calculate to maturity when used for solving, even when the projection period is shorter. For ALM projections, the asset submodels must calculate 360 months (30 years) longer than the valuation years, in order to calculate the market value of bonds (etc.) purchased at the end of the projection. Setting this value longer than needed allocates (and thus uses) more memory for the projection; however submodels for model point calculations automatically reduce memory use to the maturity period for each model point.
Assumption set restore	Yes for seriatim tasks, and when independent model point processing applies for portfolio tasks; No otherwise
Extract model points	Yes to enable the Extract model points feature specified for the model point file in the Input manager; has no effect when this feature is not used for the model point file <i>Note, this feature cannot be used for inner loop valuations within nested projections, which add an artificial model point to make sure there is some business in force through the entire projection</i>
Sliding windows	Do Not Slide for seriatim runs and when small models or independent model point processing enables portfolio runs to fit within available memory; Slide to reduce memory requirements during a valuation task which does not otherwise fit into available memory

8.2.4 Project variables

Also specify the project variables in the Run parameter set.

Property	Values
Projection date [proj_date]	Valuation date is the end of the month specified as yyyy/mm
Pricing projection? [pricing_defn]	No
Valuation years [valn_yrs]	Valuation calculations project this many years from the valuation date
Project GAAP results? [gaap_defn]	No
Include target capital? [tgt_cap_defn]	Has no effect on valuation tasks
Use independent model point processing? [independent_mp_defn]	Yes for portfolio valuation tasks to project 'independent' model points using seriatim calculations before starting the portfolio calculations; No otherwise; Has no effect on seriatim tasks
Validate data definitions? [validate_data_defn]	No for production runs; Yes for testing
Include decrements? [decrements_defn]	Yes
Discount rates selected [disc_rates_selected]	Typically not used for valuation tasks
Portfolio yield definition [disc_rate_port_yld_defn]	Typically not used for valuation tasks
Valuation periods selected [valn_periods_selected]	Has no effect on point-in-time valuations For nested projections, must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0.

Property	Values
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Has no effect on point-in-time valuations. For nested projections, specify how the system should manage generated scenario files selecting from: Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs that generate large scenario files.

8.2.5 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Fixed discount rate list	Typically not used for valuation tasks
Use column discount rates	No; typically not used for valuation tasks
Output manager	Select the desired output manager to limit the output captured
Model object output	Select All Columns, No Columns, or Output Definitions for the model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions

8.2.6 Run controller projection set

Point-in-time valuation tasks can be organized into a projection set on the Run controller, with appropriate task dependencies established. Then, they can be run together by submitting the projection task set from the Run controller. This is not required; standard projections can be run from the individual Run page.

Valuation tasks for PBA and LOC projections are typically setup within a Run controller projection set and an experience projection task; with appropriate task dependencies established. For PBA or LOC projections with inner loop valuations, the valuation tasks must be combined with appropriate scenario generation tasks and an experience projection task in a Run controller projection set.

8.3 VM-20 valuation setup

The following setup instructions for VM-20 valuations are in addition to the general instructions discussed above.

8.3.1 Applicability

VM-20 valuations apply for universal life, indexed universal life, variable universal life, term life, and traditional life insurance products. The valuations produce a deterministic reserve and the scenario amounts required for the stochastic reserve.

A valuation can be performed for point-in-time valuation of in-force business, or for inner loop valuations of a cell pricing or portfolio projection.

8.3.2 Input variables

Specify the following variables on the Company input page for VM-20 valuations in the PBA category:

Variable	Value
Reserve and capital calculation definition	Reserves, or Reserves and Capital
Hedging best estimate output file deterministic reserve	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under best estimate projections of a hedging program
Hedging best estimate output file stochastic reserve	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under best estimate projections of a hedging program
Initial assets	For liability-only reserve valuations, specify initial invested assets that approximate the PBR result
Initial assets percent	For asset/liability valuations, specify a percent of the initial invested assets that approximates the PBR result

8.3.3 Run page

Create a deterministic reserve valuation projection task on a Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

Create a scenario amount valuation projection task on a Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

For valuations used in a cell pricing projection, create all valuation projection tasks on a Run page which uses the product submodel as the target model object.

8.3.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	Deterministic reserve	Stochastic reserve
Target column	valn_tgt_pba_pol_res	valn_tgt_pba_scen_amt
Projection task loops	1	1-1000 (for example)

8.3.5 Project variables

Also specify the project variables in the Run parameter set.

Property	Deterministic reserve	Stochastic reserve
Include assets? [assets_defn]	Yes (required to calculate the asset yield, which is the discount rate)	Yes (typically), or No for liability-only using assumed asset yield

8.4 VM-22 valuation setup

The following setup instructions for VM-22 valuations are in addition to the general instructions discussed above.

8.4.1 Applicability

VM-22 valuations apply for fixed annuity, fixed indexed annuity, and payout annuity products. The valuations produce the scenario amounts required for the stochastic reserve.

A valuation can be performed for point-in-time valuation of in-force business, or for inner loop valuations of a cell pricing or portfolio projection.

8.4.2 Input variables

Specify the following variables on the Company input page for VM-22 valuations in the PBA category:

Variable	Value
Reserve and capital calculation definition	Reserves, or Reserves and Capital
Hedging run off output file for stochastic reserve	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs from existing hedges projections of a hedging program
Hedging best estimate output file for stochastic reserve	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under best estimate projections of a hedging program
Hedging statutory scale factor max for stochastic reserve	The scale factor applied to the hedging cash flows cannot be greater than this amount.
Hedging e factor for stochastic reserve	For the reserve hedged stochastic amount, the hedged stochastic amount equals the hedging_e_factor times the stochastic amount using best estimate hedge cash flows plus (1 - hedging_e_factor) times the stochastic amount using existing hedge cash flows
Initial assets	For liability-only reserve valuations, specify initial invested assets that approximate the PBR result
Initial assets percent	For asset/liability valuations, specify a percent of the initial invested assets that approximates the PBR result

8.4.3 Run page

Create a scenario amount valuation projection task on a Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

For valuations used in a cell pricing projection, create all valuation projection tasks on a Run page which uses the product submodel as the target model object.

8.4.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	Stochastic reserve
Target column	valn_tgt_pba_scen_amt
Projection task loops	1-1000 (for example)

8.4.5 Project variables

Also specify the project variables in the Run parameter set.

Property	Stochastic reserve
Include assets? [assets_defn]	Yes (typically), or No for liability-only using assumed asset yield

8.5 RBC C3 phase III valuation setup

The following setup instructions for RBC C3 phase III valuations are in addition to the general instructions discussed above.

8.5.1 Applicability

RBC C3 phase III valuations apply for universal life, indexed universal life, variable universal life, term life, and traditional life insurance products and fixed annuity, fixed indexed annuity, and payout annuity products. The valuations produce the scenario amounts required for the stochastic capital.

A valuation can be performed for point-in-time valuation of in-force business, or for inner loop valuations of a cell pricing or portfolio projection.

8.5.2 Input variables

Specify the following variables on the Company input page in the PBA category:

Variable	Value
Reserve and capital calculation definition	Capital, or Reserves and Capital
Hedging run off output file for stochastic capital (annuity only)	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs from existing hedges projections of a hedging program
Hedging best estimate output file for stochastic capital	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under best estimate projections of a hedging program
Hedging statutory scale factor max for stochastic capital (annuity only)	The scale factor applied to the hedging cash flows cannot be greater than this amount

Variable	Value
Hedging e factor for stochastic capital (annuity only)	For the capital hedged stochastic amount, the hedged stochastic amount equals the stochastic amount using best estimate hedge cash flows plus the hedging_e_factor times the maximum of zero and the stochastic amount unhedged less the stochastic amount using best estimate hedge cash flows
Tax definition for stochastic capital calculation	Specify whether to calculate the tax reserve or use the working reserve for the tax reserve in the stochastic capital calculation
Initial assets	For liability-only reserve valuations, specify initial invested assets that approximate the PBR result
Initial assets percent	For asset/liability valuations, specify a percent of the initial invested assets that approximates the PBR result

8.5.3 Run page

Create a scenario amount valuation projection task on a Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

For valuations used in a cell pricing projection, create all valuation projection tasks on a Run page which uses the product submodel as the target model object.

8.5.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	Stochastic capital
Target column	valn_tgt_pba_scen_amt
Projection task loops	1-1000 (for example)

8.5.5 Project variables

Also specify the project variables in the Run parameter set.

Property	Stochastic capital
Include assets? [assets_defn]	Yes (typically), or No for liability-only using assumed asset yield

8.6 AG 43 valuation setup

The following setup instructions for AG 43 valuations are in addition to the general instructions discussed above.

8.6.1 Applicability

AG 43 valuations apply for variable annuity products. The valuations produce the standard scenario reserve and the scenario amounts required for the stochastic reserve.

A valuation can be performed for point-in-time valuation of in-force business, or for inner loop valuations of a cell pricing or portfolio projection.

8.6.2 Input variables

Specify the following variables on the Company input page in the PBA category:

Variable	Value
Reserve and capital calculation definition	Reserves, or Reserves and Capital
Model point projection frequency	Monthly: Model points are processed monthly for PBA valuations. Annual: Model points are processed on an annual basis starting from the projection date for PBA valuations.
Hedging run off output file stochastic reserve	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under run off of existing hedges
Hedging best estimate output file stochastic reserve	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under best estimate projections of a hedging program
Hedging max scaling factor stochastic reserve	The scale factor applied to the hedging cash flows cannot be greater than the specified amount.
Hedging e factor	For the AG 43 hedged stochastic amount, the hedged stochastic amount equals the hedging_e_factor times the stochastic amount using best estimate hedge cash flows plus (1 - hedging_e_factor) times the stochastic amount using existing hedge cash flows.
AG 43 new money discount rate	Has no effect for standard scenario valuation Specify the terms, multiples, additions, defaults, and weights for the AG 43 new money based discount rate
AG 43 standard scenario SVL discount rate	Standard valuation law rate for AG 43 standard scenario discounting
Standard scenario returns	Specify the standard scenario returns
Initial assets	For liability-only valuations, specify initial invested assets that approximate the PBR result
Initial assets percent	For asset/liability valuations, specify a percent of the initial invested assets that approximates the PBR result

8.6.3 Run page

If using a seriatim valuation approach for the standard scenario reserve valuation, create a standard scenario reserve valuation projection task on a Run page which uses the product submodel as the target model object.

If using a portfolio valuation approach for the standard scenario reserve valuation, create a standard scenario reserve valuation projection task on a Run page which uses the company submodel as the target model object.

Create a scenario amount valuation projection task on a Run page which uses the company submodel as the target model object.

For valuations used in a cell pricing projection, create all valuation projection tasks on a Run page which uses the product submodel as the target model object.

8.6.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	Standard scenario reserve	Stochastic reserve
Target column	*valn_tgt_pba_std_scen from the va product submodel, or valn_tgt_pba_pol_res from the company submodel	valn_tgt_pba_scen_amt
Projection task loops	1	1-1000 (for example)

* The task calculating the standard scenario reserve or capital is typically a seriatim valuation, which has the va product type submodel as the top model and is targeting valn_tgt_pba_std_scen, to project each model point separately. However, a portfolio valuation may also be used, targeting valn_tgt_pba_pol_res in the company submodel.

8.6.5 Project variables

Also specify the project variables in the Run parameter set.

Property	Standard scenario reserve	Stochastic reserve
Include assets? [assets_defn]	No,	No, typically, for liability-only valuation using assumed asset yield

8.7 RBC C3 phase II valuation setup

The following setup instructions for RBC C3 phase II valuations are in addition to the general instructions discussed above.

8.7.1 Applicability

RBC C3 phase II valuations apply for variable annuity products. The valuations produce the standard scenario capital and the scenario amounts required for the stochastic capital.

A valuation can be performed for point-in-time valuation of in-force business, or for inner loop valuations of a cell pricing or portfolio projection.

8.7.2 Input variables

Specify the following variables on the Company input page in the PBA category:

Variable	Value
Reserve and capital calculation definition	Capital, or Reserves and Capital
Model point projection frequency	Monthly: Model points are processed monthly for PBA valuations. Annual: Model points are processed on an annual basis starting from the projection date for PBA valuations.
Hedging run off output file stochastic capital	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under run off of existing hedges
Hedging best estimate output file stochastic capital	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under best estimate projections of a hedging program
Hedging max scaling factor stochastic capital	The scale factor applied to the hedging cash flows cannot be greater than the specified amount.
Hedging e factor	For RBC C3 Phase II, the hedged stochastic amount equals the stochastic amount using best estimate hedge cash flows plus the hedging_e_factor times the maximum of zero and the stochastic amount unhedged less the stochastic amount using best estimate hedge cash flows
Standard scenario returns	Specify the standard scenario returns
Initial assets	For liability-only valuations, specify initial invested assets that approximate the PBR result
Initial assets percent	For asset/liability valuations, specify a percent of the initial invested assets that approximates the PBR result
Valuation results for standard scenario capital	When a two step process is used, specify the name of the step 1 standard scenario capital file*

* Specify results from valuations for dependent tasks using the system <column_output_directory> wildcard as opposed to the user specified <output> wildcard. This is required so that RiskAgility FM will reference the appropriate results files when the projection is distributed to a grid.

Additionally, for nested projections with inner loop valuations, specify the <proj_set_loop_num> and <proj_set_sub_loop_num> system wildcards within the file reference so that RiskAgility FM will reference the appropriate inner loop valuation file.

An example file reference for the standard scenario capital step 1 results is

```
"<column_output_directory>PBA Seriatim_Valn Std Scen
Cap@<proj_set_loop_num>_<proj_set_sub_loop_num>company~liab~va~va_pba.csv"
```

8.7.3 Run page

If using a single step approach for the standard scenario capital valuation, create a standard scenario valuation projection task on a Run page which uses the company submodel as the target model object.

If using a two step approach for the standard scenario capital valuation, create a step 1 standard scenario valuation projection task on a Run page which uses the product submodel as the target model object and then create a step 2 standard scenario valuation projection task on a Run page which uses the company submodel as the target model object.

Create a scenario amount valuation projection task on a Run page which uses the company submodel as the target model object.

For all valuations used in a cell pricing projection, create a projection task on a Run page which uses the product submodel as the target model object.

8.7.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	Standard scenario capital	Stochastic capital
Target column	valn_tgt_pba_pol_res from the company submodel valn_tgt_pba_stad_scen from the va submodel for step 1 and then valn_tgt_pba_rbc_std_scen_step_2 from the company submodel for step 2 when the standard scenario capital is calculated using a two step process*	valn_tgt_pba_scen_amt
Projection task loops	1	1-1000 (for example)

* The task calculating the standard scenario capital is typically a portfolio valuation, which has company as the top model. However, when a two step process is used for the standard scenario capital, separate valuation tasks are required for step 1 from step 2, with step 1 being a seriatim valuation task and step 2 being a portfolio projection task.

8.7.5 Project variables

Also specify the project variables in the Run parameter set.

Property	Standard scenario capital	Stochastic capital
Include assets? [assets_defn]	No,	No, typically, for liability-only valuation using assumed asset yield

8.8 LOC economic reserve valuation setup

The following setup instructions for LOC economic reserve valuations are in addition to the general instructions discussed above.

8.8.1 Applicability

LOC economic reserve valuations apply for universal life, indexed universal life, term life, and traditional life insurance products. When the economic reserve definition is Gross Premium Reserve, the valuation

produces gross premium reserves. When the economic reserve definition is AG 48, the valuation produce a deterministic reserve and the scenario amounts required for the stochastic reserve.

A valuation can be performed for point-in-time valuation of in-force business, or for inner loop valuations of a cell pricing or portfolio projection.

8.8.2 Input variables

Specify the follow variables on the Company input page in the Model Setup: Segment category:

Variable	Value
LOC cohorts selected	Comma separated list of Letter of Credit cohorts. For example, Cohort 1,Cohort 2,Cohort 3

Specify the follow variables on the Company input page in the Letter of Credit: General category:

Variable	Value
Letter of credit economic reserve definition	Gross Premium Reserve or AG 48
Store GPR valuation by cohort?	Applies when the economic reserve definition is Gross Premium Reserve Yes: The GPR valuation results calculated for each model point are aggregated to its LOC cohort, within each model point report group. The projection using the valuation results adjusts the gross and ceded GPR by dividing the cohort result by the units projected for the GPR valuation, then multiplying by the units in the current projection. This method is more efficient, since it reduces the size of the GPR valuation output file, and adjusts valuation results for each cohort (within each reporting group). No: The GPR valuation output file stores the results separately for each model point. The projection using the valuation results adjusts (for the projection versus valuation units) the GPR results separately for each model point, prior to aggregating into the LOC cohorts. This method is more accurate, since adjusting each model point accounts for the changing mix of business since the GPR valuation was performed.

Specify the follow variables on the Company input page in the Letter of Credit: Cohorts category:

Variable	Value
LOC cohort issue years	ALL, or comma separated list of model point issue years to include in the cohort
LOC cohort plan codes	ALL, or comma separated list of model point plan codes to include in the cohort
GPR discount rate is a fixed rate?	Applies when the economic reserve definition is Gross Premium Reserve Yes: The gross premium reserve is the present value of liability model point cash flows, discounted at the specified fixed discount rate. No: The discount rate is the asset yield, which is net of investment expenses.

Variable	Value
GPR fixed discount rate:	Applies when the economic reserve definition is Gross Premium Reserve When the Fixed discount rate definition is Yes, this is the fixed discount rate for this gross premium reserve cohort. This is an effective annual rate.

When the LOC economic reserve definition is AG 48, specify the following variables on the Company input page in the PBA category:

Variable	Value
Reserve and capital calculation definition	Not Applicable, or Capital, since PBA reserves do not apply when AG 48 is used for the LOC economic reserve
PBA reserve effective date	Set to default value of 9999/01 since PBA reserves do not apply when AG 48 is used for the LOC economic reserve
Hedging best estimate output file deterministic reserve	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under best estimate projections of a hedging program
Hedging best estimate output file stochastic reserve	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under best estimate projections of a hedging program
Initial assets	For liability-only reserve valuations, specify initial invested assets that approximate the PBR result
Initial assets percent	For asset/liability valuations, specify a percent of the initial invested assets that approximates the PBR result

8.8.3 Run page

When the LOC economic reserve definition is Gross Premium Reserve, create a gross premium reserve valuation projection task on a Run page which uses the product submodel as the target model object.

When the LOC economic reserve definition is AG 48, create a deterministic reserve valuation projection task, and a scenario amount valuation projection task on a Run page which uses the company submodel as the target model object.

For valuations used in a cell pricing projection, create all valuation projection tasks on a Run page which uses the product submodel as the target model object.

8.8.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	GPR economic reserve	AG 48 economic reserve
Target column	valn_tgt_loc_gpr, for the LOC GPR valuation task	Same as the deterministic reserve and stochastic reserve calculations for VM-20
Projection task loops	1	Same as the deterministic reserve and stochastic reserve calculations for VM-20

8.8.5 Project variables

Also specify the project variables in the Run parameter set.

Property	GPR economic reserve	AG 48 economic reserve
Include assets? [assets_defn]	No, the GPR valuation is a seriatim valuation	Same as the deterministic reserve and stochastic reserve calculations for VM-20

8.9 PBA or LOC final valuation setup

The following setup instructions for a final valuation are in addition to the general instructions discussed above.

8.9.1 Applicability

A final PBA or LOC valuation task is only used for point-in-time valuations. It provides a single projection task which can bring together results of several valuation tasks to calculate the point in time PBA or LOC balance sheet.

8.9.2 Input variables

When the final valuation is for the LOC economic reserve, specify the follow variables on the Company input page in the Model Setup: Segment category:

Variable	Value
LOC cohorts selected	Comma separated list of Letter of Credit cohorts. For example, Cohort 1,Cohort 2,Cohort 3

When the final valuation is for the LOC economic reserve, specify the follow variables on the Company input page in the Letter of Credit: General category:

Variable	Value
Letter of credit economic reserve definition	Must be set the same as was set for the LOC economic reserve valuation task

Variable	Value
Store GPR valuation by cohort?	Must be set the same as was set for the LOC economic reserve valuation task
GPR results:	Applies when the LOC economic reserve definition is Gross Premium Reserve Specify the name of the LOC GPR valuation results file*
LOC AG48 results deterministic reserve	Applies when the LOC economic reserve definition is AG 48 Specify the name of the LOC AG 48 deterministic reserve valuation results file*
LOC AG48 results stochastic reserve	Applies when the LOC economic reserve definition is AG 48 Specify the name of the LOC AG 48 scenario amount valuation results file*

When the final valuation is for the LOC economic reserve, specify the follow variables on the Company input page in the Letter of Credit: Cohorts category:

Variable	Value
LOC cohort issue years	Must be set the same as was set for the LOC economic reserve valuation task
LOC cohort plan codes	Must be set the same as was set for the LOC economic reserve valuation task
GPR discount rate is a fixed rate?	N/A
GPR fixed discount rate:	N/A

When the final reserve is for the LOC economic reserve, specify the following variables on the Company input page in the PBA category:

Variable	Value
Reserve and capital calculation definition	Not Applicable, or Capital, since PBA reserves do not apply when AG 48 is used for the LOC economic reserve
PBA reserve effective date	Set to default value of 9999/01 since PBA reserves do not apply when AG 48 is used for the LOC economic reserve

When the final reserve is for a PBA valuation reserve, specify the following variables on the Company input page in the PBA category:

Variable	Value
Reserve and capital calculation definition	Same as used for the individual PBA valuations
PBA reserve definition	Applies for final VM-20, VM-22, and RBC C3 Phase III valuations Specify the final PBA reserve to be: Max of Stochastic and Deterministic and Policy Amount: The PBR for life insurance products is the greatest of the stochastic amount, the deterministic amount, and the policy amount. Max of Deterministic and Policy Amount: The PBR is the greater of the deterministic and policy amounts. Policy Amount: The PBR is the policy amount. This is the net premium reserve (NPR) calculated for each model point.
AG 43 reserve definition	Applies for final AG 43 valuations. Specify the final AG 43 reserve to be: Max of Stochastic and Std Scenario: The AG 43 reserve is the greater of the stochastic amount and the standard scenario amount. Std Scenario: The AG 43 reserve equals the standard scenario amount.
RBC C3 phase 2 capital definition	Applies for final RBC C3 phase II valuations Specify the final RBC total asset requirement to be: Max of Stochastic and Std Scenario: The RBC C3 phase 2 capital amount is the greater of the stochastic amount and the standard scenario amount. Std Scenario: The RBC C3 phase 2 capital amount equals the standard scenario amount.
Hedging run off output file for standard scenario reserves	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under run off of existing hedges for standard scenario reserves
Hedging statutory scale factor max for standard scenario reserves	The scale factor applied to the hedging cash flows cannot be greater than the specified amount.
Hedging run off output file for standard scenario capital	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under run off of existing hedges for standard scenario capital
Hedging statutory scale factor max for standard scenario capital	The scale factor applied to the hedging cash flows cannot be greater than the specified amount.
CTE percent stochastic capital	Specify the CTE percent used for the calculation of the stochastic capital (i.e. 90%)

Variable	Value
Valuation results for standard scenario reserve	Specify the name of the standard scenario reserve valuation results file*
Valuation results for standard scenario capital	Specify the name of the standard scenario capital valuation results file*. For a two step process, this would be the valuation results file of the second step
Valuation results for deterministic reserve	Specify the name of the deterministic reserve valuation results file*
Valuation results for stochastic reserve	Specify the name of the scenario amount valuation results file for the stochastic reserve*
Valuation results for stochastic capital	Specify the name of the scenario amount valuation results file for the stochastic capital*

* Specify results from valuations for dependent tasks using the system <*column_output_directory*> wildcard as opposed to the user specified <*output*> wildcard. This is required so that RiskAgility FM will reference the appropriate results files when the projection is distributed to a grid.

Additionally, for nested projections with inner loop valuations, specify the <*proj_set_loop_num*> and <*proj_set_sub_loop_num*> system wildcards within the file reference so that RiskAgility FM will reference the appropriate inner loop valuation file.

An example file reference for the valuation results for stochastic reserve is "<*column_output_directory*>PBA Portfolio_Valn Scen Amt Res@<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>company~seg~life_pba.csv"

8.9.3 Run page

Create the final valuation projection tasks on a Run page which uses the company submodel as the target model object.

8.9.4 Run parameter set

Specify the following properties in the Run parameter set.

Property	Value
Target column	liab_and_surp, which will target the calculations for the single point-in-time reserve and capital amounts
Projection task loops	1, since the final valuation is performed using a single projection task loop

9 PROXY BOND CREATION SETUP

9.1 Overview

9.1.1 Advance projection task

A special projection task creates asset record model point files as of the projection date. This task is performed before the PBA valuation projection tasks.

9.1.2 Model point file names

Prior to the PBA valuation, this projection task creates an alternative set of bond and mortgage model point records. These files have the same name as the files for the experience projection, except that '_proxy_bond' is inserted at the end of the base file name specified by the user.

Within the data pages, for the assumption sets used for the proxy bonds generation task, specify the bond and mortgage asset model point files containing the individual assets being projected by the investment strategy. For example, these files might be named

<*data*>BondMP_20141231_<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>.csv and <*data*>MtgMP_20141231_<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>.csv.

The proxy bonds generation task then produces these files:

<*data*>BondMP_20141231_proxy_bond_<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>.csv and <*data*>MtgMP_20141231_proxy_bond_<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>.

- The PBA valuations use the proxy bonds when the PBA valuation tasks specify these special file names.
- The PBA valuations use the individual assets when the PBA valuation tasks specify the normal file names that apply for the experience projection.

The above example is for a nested projection. The same process applies for a point-in-time valuation, except that '_<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>.' is removed from the file names.

9.2 Proxy bonds creation task setup

9.2.1 Input variables

In addition to the input for normal asset modeling, specify the follow variables on the Investment input page in the Asset Run Control category:

Variable	Value
Proxy bond definition	Not Applicable: No proxy bond will be created. Single Non Callable: One non-callable proxy bond represents all bond assets. Callable Separate: One non-callable and one callable proxy bond will be created.

Also, specify the follow variables on the Investment input page in the Asset Sub Portfolios: Definitions category:

Variable	Value
Asset sub portfolio proxy bond category	When PBA valuations replace the projected assets with a proxy bond for each asset sub portfolio, the proxy bond created for this asset sub portfolio has this bond asset category.
Asset sub portfolio proxy bond sale class	When PBA valuations replace the projected assets with a proxy bond for each asset sub portfolio, the proxy bond created for this asset sub portfolio has this sales classification, which determines sales priority and GAAP asset values. The choices are Not Assigned, Available for Sale, Trading, and Held to Maturity.

9.2.2 Run page

Create the proxy bond creation task on a Run page which uses the company submodel as the target model object.

9.2.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	proxy_bond_generation (the top model must be the company submodel)
Projection period from	0
Projection period to	0, since the proxy bond generation is similar to a valuation task and must capture the asset information at the projection date
Projection period min	0
Projection period max	0
Projection task loops	1 (has no effect)
Assumption set restore	Has no effect
Extract model points	No, since the task must process all asset model points
Sliding windows	Has no effect

9.2.4 Project variables

Also specify the project variables in the Run parameter set.

Property	Values
Projection date [proj_date]	Must be the same as for the valuations using the proxy bonds
All other project variables	Has no effect
Valuation periods selected [valn_periods_selected]	Does not affect point-in-time (standard) projections Must be the same as for the nested valuations using the proxy bonds
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Has no effect on standard projections For nested projections, specify how the system should manage generated scenario files selecting from: Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs that generate large scenario files.

9.2.5 Output parameter set

The Output parameter set (specified for the proxy bond creation task on its Run page) should specify no output. The 'output' from this task is the proxy bond model point files.

10 PBA AND LOC PROJECTION TASKS SETUP

10.1 Standard versus nested projections

When a PBA or LOC projection is for a block of inforce only business where there is no desired to perform a full re-evaluation of the PBA reserves and capital or LOC economic reserve after the projection date, a standard projection may be used for PBA or LOC cell pricing and portfolio projections.

However, when a PBA or LOC projection is for a model that includes new business, or when there is a desire to perform a full re-evaluation of the PBA reserves and capital or LOC economic reserve after the projection date, a nested projection must be used for PBA or LOC cell pricing and portfolio projections.

10.2 Valuation and experience task dependency

PBA or LOC cell pricing or portfolio projections require valuation tasks and an experience projection. The valuation tasks must be run in advance of the experience projection and the results of the valuation tasks referenced as input into the experience projection.

When the PBA or LOC cell pricing or portfolio projection is a standard projection, the valuation tasks can also be run as standard projections and processed in advance of the experience projection, either in the same projection set or in separate projection sets.

But, when the PBA or LOC cell pricing or portfolio projection is a nested projection, the PBA or LOC economic reserve valuations become inner loop valuations and must be included together with the experience projection in the nested projection Run controller projection set with appropriate dependencies specified. The PBA or LOC cell pricing or projection is then driven by the projection set loop and sub loops.

10.3 PBA and LOC experience projection task setup

The following setup instructions apply for experience projections which include PBA or LOC calculations.

10.3.1 Input variables

When the experience projection is for an LOC projection, specify the follow variables on the Company input page in the Model Setup: Segment category:

Variable	Value
LOC cohorts selected	Comma separated list of Letter of Credit cohorts. For example, Cohort 1,Cohort 2,Cohort 3

When the experience projection is for an LOC projection, specify the follow variables on the Company input page in the Letter of Credit: General category:

Variable	Value
Letter of credit economic reserve definition	Must be set the same as was set for the LOC economic reserve valuation task
Store GPR valuation by cohort?	Must be set the same as was set for the LOC economic reserve valuation task
GPR results:	Applies when the LOC economic reserve definition is Gross Premium Reserve Specify the name of the LOC GPR valuation results file*
LOC AG48 results deterministic reserve	Applies when the LOC economic reserve definition is AG 48 Specify the name of the LOC AG 48 deterministic reserve valuation results file*
LOC AG48 results stochastic reserve	Applies when the LOC economic reserve definition is AG 48 Specify the name of the LOC AG 48 scenario amount valuation results file*

When the experience projection is for an LOC projection, specify the follow variables on the Company input page in the Letter of Credit: Cohorts category:

Variable	Value
LOC cohort issue years	Must be set the same as was set for the LOC economic reserve valuation task
LOC cohort plan codes	Must be set the same as was set for the LOC economic reserve valuation task

Variable	Value
GPR discount rate is a fixed rate?	N/A
GPR fixed discount rate:	N/A

When the experience projection is for an LOC projection, specify the following variables on the Company input page in the PBA category:

Variable	Value
Reserve and capital calculation definition	Not Applicable, or Capital, since PBA reserves do not apply when AG 48 is used for the LOC economic reserve
PBA reserve effective date	Set to default value of 9999/01 since PBA reserves do not apply when AG 48 is used for the LOC economic reserve

When the experience projection is for a PBA projection, specify the following variables on the Company input page in the PBA category:

Variable	Value
Reserve and capital calculation definition	Same as used for the individual PBA valuations
PBA reserve definition	Applies for experience projections with VM-20, VM-22, and RBC C3 Phase III valuations Specify the final PBA reserve to be: Max of Stochastic and Deterministic and Policy Amount: The PBR for life insurance products is the greatest of the stochastic amount, the deterministic amount, and the policy amount. Max of Deterministic and Policy Amount: The PBR is the greater of the deterministic and policy amounts. Policy Amount: The PBR is the policy amount. This is the net premium reserve (NPR) calculated for each model point.
AG 43 reserve definition	Applies for experience projections with AG 43 valuations. Specify the final AG 43 reserve to be: Max of Stochastic and Std Scenario: The AG 43 reserve is the greater of the stochastic amount and the standard scenario amount. Std Scenario: The AG 43 reserve equals the standard scenario amount.

Variable	Value
RBC C3 phase 2 capital definition	Applies for experience projections with RBC C3 phase II valuations Specify the final RBC total asset requirement to be: Max of Stochastic and Std Scenario: The RBC C3 phase 2 capital amount is the greater of the stochastic amount and the standard scenario amount. Std Scenario: The RBC C3 phase 2 capital amount equals the standard scenario amount.
Hedging run off output file for standard scenario reserves	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under run off of existing hedges for standard scenario reserves
Hedging statutory scale factor max for standard scenario reserves	The scale factor applied to the hedging cash flows cannot be greater than the specified amount.
Hedging run off output file for standard scenario capital	Specify the pre-generated hedge cash flows and market values to use for determining hedge costs under run off of existing hedges for standard scenario capital
Hedging statutory scale factor max for standard scenario capital	The scale factor applied to the hedging cash flows cannot be greater than the specified amount.
CTE percent stochastic capital	Specify the CTE percent used for the calculation of the stochastic capital (i.e. 90%)
Valuation results for standard scenario reserve	Specify the name of the standard scenario reserve valuation results file*
Valuation results for standard scenario capital	Specify the name of the standard scenario capital valuation results file*. For a two step process, this would be the valuation results file of the second step
Valuation results for deterministic reserve	Specify the name of the deterministic reserve valuation results file*
Valuation results for stochastic reserve	Specify the name of the scenario amount valuation results file for the stochastic reserve*
Valuation results for stochastic capital	Specify the name of the scenario amount valuation results file for the stochastic capital*

* Specify results from valuations for dependent tasks using the system <*column_output_directory*> wildcard as opposed to the user specified <*output*> wildcard. This is required so that RiskAgility FM will reference the appropriate results files when the projection is distributed to a grid.

Additionally, for nested projections with inner loop valuations, specify the <*proj_set_loop_num*> and <*proj_set_sub_loop_num*> system wildcards within the file reference so that RiskAgility FM will reference the appropriate inner loop valuation file.

An example file reference for the valuation results for stochastic reserve is "<*column_output_directory*>PBA Portfolio_Valn Scen Amt Res@<*proj_set_loop_num*>_<*proj_set_sub_loop_num*>company~seg~life_pba.csv"

Specify GAAP cohorts the same as for the GAAP valuation tasks.

10.3.2 Run page

For PBA or LOC portfolio projections, specify the experience projection on a Run page with the Company submodel as the target model object.

For PBA or LOC cell pricing, specify the experience projection of a Run page with the product submodel as the target model object.

10.3.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	The desired targeted amount, such as distrib_earnings
Projection period from	0
Projection period to	For standard projections, the desired projection period For nested projections, at least as long as the maximum time between time steps
Projection period min	-600 for seriatim tasks, to allow for external source lookup keys calculating before the projection date (such as pol_yr)
Projection period max	At least as long as the calculations for each submodel
Extract model points	No, for production tasks
Sliding windows	Do Not Slide for seriatim runs and when small models or independent model point processing enables portfolio runs to fit within available memory; Slide to reduce memory requirements during a projection task which does not otherwise fit into available memory

10.3.4 Project variables

Also specify the project variables in the Run parameter set.

Property	Values
Projection date	Projection date is the end of the month specified as yyyy/mm
Pricing projection?	Yes or No
Valuation years	Has no effect on projection tasks which are not performing valuations
Include assets?	Yes or No
Include target capital?	Yes typically
Use independent model point processing?	Yes or No for portfolio projections; has no effect on seriatim projections
Validate data definitions?	No for production runs; Yes for testing
Include decrements?	Yes
Discount rates selected	Specify the stochastic discount rates to use; these must match the disc_rate_id lookup key values used to specify these rates on the Company input page
Portfolio yield definition	Before tax (asset yield rate), or After tax (multiplied by 1 – tax rate)
Valuation periods selected	Has no effect on standard projections For nested projections, must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0. This must match the value for the related PBA and LOC valuation tasks.

Property	Values
Save generated time-step scenario files? [save_time_step_scen_files_defn]	Has no effect on standard projections For nested projections, specify how the system should manage generated scenario files selecting from: Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs that generate large scenario files.

10.3.5 Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Fixed discount rate list	For example, 8, 10, 12 for present values at 8%, 10%, and 12%
Use column discount rates	Yes to use the stochastic discount rates named in the Discount rates selected project variable
Output manager	Select the desired output manager to limit the output captured
Model object output	Select All Columns, No Columns, or Output Definitions for the model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions

10.4 Run controller projection set

PBA and LOC standard projections should typically be set up within a projection set on the Run controller, so the entire set of tasks can be run in one step. The PBA or LOC projection task specifies that the valuation tasks are dependent tasks, which must be completed before the projection task begins.

PBA and LOC nested projections require the PBA or LOC economic reserve valuation tasks to be specified within the same projection set (on the Run controller) as the PBA or LOC experience projection; with the experience projection dependent on their completion.

10.5 AG 48 stand-alone reserve calculation

In some instances, it may be desirable to calculate an AG 48 reserve as a reporting item only, without including letter of credit reserve financing. This stand-alone AG 48 reserve can be projected and used in an external analysis to serve a business need.

When the experience projection should calculate the AG 48 reserve without calculating an explicit LOC, follow the steps described above in sections 8.1 through 10.4, assuming an LOC economic reserve calculation that uses AG 48. Then, specify the following variable on the Company input page in the Model Setup: Segment category:

Variable	Value
LOC cohorts selected	NONE

11 SUBMODELS STRUCTURE

11.1 PBA submodels

The PBA submodels are user defined array instance of the `life_pba_pba`, the `va_pba_pba`, or the `ann_pba_pba` model class. They contain the variables and formulas required to calculate the principle-based capital and reserve amounts for life products (where appropriate).

PBA submodels are required at two levels of the model tree to produce the required PBA results.

Portfolio level

The `life_pba`, `va_pba`, and `ann_pba` submodels calculate portfolio level PBA results. They are children of the segment submodel.

Product level

A product version, such as `ul_life_pba`, calculates individual model point PBA results. They are children of the product submodel.

11.2 Formulas

Formulas for calculating PBA results are contained in the PBA submodels. A combination of columns, scalars, and external functions are used to provide the required functionality.

The formulas in the PBA model classes have been written as reusable formulas across both portfolio projections and seriatim projection, as well as for multiple product types.

This reusability has been accomplished using 'blocks' of code which apply either in the product context or in the segment context. These blocks of code are organized using conditions on compiler directives and unique model pointer names. The compiler directive conditions reference the compiler macros for the product type, such as `'__UL_LIAB_H_'`, and the unique model pointer conditions reference `'life_pba'`, and for the `ul` product instance, `'ul_life_pba'`, so that they are only executed when the 'this' pointer equals the unique model pointer name.

Most formulas in the PBA submodels are very straightforward. The key columns of the PBA submodel are as follows:

pol_amt (life pba only)

The `'pol_amt'` column returns, at $t = 0$ only, the greater of the gross premium reserve and the cash value. The gross premium reserve column drives the calculations of the present values of benefits, expenses, premiums, and reinsurance cash flows which use a recursive approach. All of these columns reference product level values.

std_scen_amt (va pba only)

For capital calculations, the `'std_scen_amt'` column returns, at $t = 0$ only, the working reserve plus the greater of zero and the greatest present value of accumulated net revenue. This amount is calculated in aggregate expect for inner loop valuations for VA cell pricing.

For reserve calculations, the `'std_scen_amt'` column returns, at $t = 0$ only, the basic adjusted reserve plus the greater of zero and the greatest present value of accumulated net revenue. The basic adjusted reserve is the basic reserve using CARVM methodology.

scen_amt

The 'scen_amt' column returns, at t = 0 only, the scenario greatest value plus starting assets. The scenario greatest value column drives the calculation of the accumulated deficiencies using a recursive approach. A deficiency roll-forward approach is used to build the accumulated deficiency amounts, as opposed to specifying the accumulated deficiency as the working reserve less accumulated assets.

pol_res (life pba only)

The 'pol_res' column returns, at t = 0 only, the PBA life deterministic reserve for the model point determined at the last valuation. This column, which is executed only in the experience projection, opens the column output file referenced for the life_pba_valn_results_det_res variable and retrieves the t = 0 'pol_amt' which has been written to the column output file. Results are filtered by a group name which includes the 'mp_id'.

std_scen_res (va pba only)

The 'std_scen_res' column returns, at t = 0 only, the VA PBA standard scenario amount determined at the last valuation. This column, which is executed only in the experience projection, opens the column output file referenced for the va_pba_valn_results_std_scen_res or va_pba_valn_results_std_scen_cap variable (depending on whether a reserve or capital amount is being retrieved), and retrieves the t = 0 'pol_amt' which has been written to the column output file. Results are filtered by a group name which includes the 'mp_id' for the standard scenario reserve calculations.

stoch_amt

The 'stoch_amt' column returns, at t = 0 only, the PBA life capital or reserve stochastic amount, adjusted for any required tax adjustment. Similarly to the 'pol_res' column, this column, which is executed only in the experience projection, opens the column output file referenced for the xxx_pba_valn_results_stoch_res or xxx_pba_valn_results_stoch_cap variables and retrieves the t = 0 'scen_amt' and 'tax_adj_ratio' which has been written to the column output file. Results are filtered by projection task loop and the group name that represents the PBA calculation type. These amounts are moved into a two dimensional array and provided to a function which determines the CTE amount and tax average ratio of policies in force. Finally, the tax reserve adjustment is calculated and added to the CTE amount which is returned as the column result.

11.3 Seriatim versus portfolio runs

Seriatim projection

When the product type submodel is the top model, all PBA calculations occur in a product version PBA submodel, such as ul_life_pba. The product version submodel is a child of the product type submodels. In this context, the PBA submodel references variables and results from the product submodel and determine single model point results.

Portfolio projection

For portfolio projections (which include business segment calculations), PBA calculations occur in both the product version PBA submodel and in an aggregate PBA submodel.

In this configuration, the product version of the PBA submodel calculates only individual model point amounts, while the segment level PBA submodel determines amounts in aggregate. The product version PBA submodel continues to reference variables and results from the product level PBA submodels, while the segment level PBA submodel references aggregated results from the segment submodel, or calls the segment sum_over_liabilities to aggregate model point results. There are also cases where the segment level PBA submodel aggregates results from the product level PBA submodel, using a sum_over_pba external function.

11.4 Valuation versus projection runs

Valuations

During a valuation projection, the PBA submodels provide for calculating the policy amount for the PBA deterministic reserve and the scenario amount for both PBA capital and reserves.

The policy amount for the PBA deterministic reserve utilizes a seriatim projection for the single product type. Only the product version of the PBA submodel is included in this projection and all results are single model point results.

The scenario amount calculation is a portfolio projection driven from the segment submodel. Both the product version and aggregate life_pba submodels are required for this calculation. Segment is the top model, but very few columns from segment are actually required, since most calculations needed to determine the scenario amount are included in the PBA submodel. For both PBA capital and reserve valuations, the submodel structure does not provide for calculating the stochastic amount nor the final capital or reserve.

Projections

During a projection, which could be a seriatim cell pricing or a portfolio liability-only or asset/liability model projection, the PBA submodels calculate the valuation amounts as discussed above. The PBA submodels also calculate the final stochastic amount for both PBA capital and reserves. This is allowable within the nested projection because the structure of the PBA submodel provides for the results of PBA submodels from the valuation runs to be used as input into the experience projection.

11.5 Multiple PBA calculations together

Multiple PBA calculations within a single valuation or projection are possible because the PBA submodels are set up as working submodels.

Since there are no instances of the PBA submodels defined through the model parameters, the initial size for all PBA submodel arrays is zero. A single element of the life_pba is created from within the 'startup' column of segment. When each 'startup' of the models points is executed, a single element of the product version PBA submodel is created. No additional setup of the PBA submodels is completed at this stage. It is through other columns for the model, either processed as a target column or called from other columns, which all setup of data and execution of formulas in the PBA submodels occur.

Valuations

For valuations, the 'valn_tgt_pba' columns of the top model setup and drive the appropriate PBA submodel valuation calculations. For $t = 0$, the target column processes a loop for each PBA calculation. The number of loops processed is based on the run control selections in the PBA submodel. For example, multiple calculation loops apply when PBA capital and reserves, gross and net of reinsurance, are calculated together.

The first step in the loop is to set the run control and index variables as required. Then the appropriate PBA submodel columns are executed for the PBA calculation type. At the end of each process, results are written to the column output file using the 'writeModel' function, specifying an appropriate group name. The PBA submodel results are reset using resetValues(), and the loop moves on to the next PBA calculation type.

As explained above a single asset yield is calculated for asset/liability projections calculating multiple PBA results together. This is because the segment and asset submodel results are retained across PBA calculation loops. Thus the accuracy of the asset yield for each PBA calculation is sacrificed in order to achieve a significant reduction in runtime.

Projections

For projections, specific columns of the top model (or product submodel for deterministic reserves), such as 'stat_res_stoch_res', 'rbc_c3', 'stat_res_mp', setup and drive the appropriate PBA submodel final calculations. For $t = 0$, these columns setup the specific PBA calculation type appropriate for their result and

executes the required PBA submodel column. For example, 'stat_res_stoch_res' sets up the PBA submodel to be a gross stochastic reserve submodel and references the 'stoch_amt' column. That column retrieves the scenario amounts and tax adjustment ratios amounts from output files created during the valuations and performs the CTE calculation to arrive at the stochastic amount.

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C06 HEDGING

This book discusses hedging valuations and hedging results in experience projections. This includes:

- Use of pre-generated hedging results for all products, excluding fixed indexed annuity, variable annuity, and indexed universal life
- Index crediting hedging for fixed indexed annuity and indexed universal life products, and
- Guaranteed benefit hedging for variable annuity products

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1 HEDGING FUNDAMENTALS

1.1 Terminology

Hedge group – A grouping used in calculating hedge results. Results from the liability model points are captured into the hedge group using a number of criteria which vary based on the hedging method. Results from the hedge asset model points are captured into the hedge group based on the matching of the hedge group code of the asset to the hedge group identification.

Hedge valuation – A projection to develop present values of claims and charges or the economic reserve amounts for variable annuity guaranteed benefits.

Point-in-time valuation – A valuation performed at the current valuation date ($t = 0$).

Nested valuation – A valuation performed at each inner loop valuation date during a nested projection.

Hedge projection – A projection task which includes use of pre-generated hedging results, tracking of hedges for fixed indexed annuity and indexed universal life index crediting, or variable annuity Greek hedging. For fixed indexed annuity and indexed universal life index crediting, the hedge projection may or may not include actual assets.

1.2 Hedging calculation types

1.2.1 Hedging valuations

Hedge valuations are projection tasks which are driven by a specific 'valn_tgt_hedge_xxx' target column. There are two hedge valuation projections; a hedge shock valuation projection and a hedge final valuation projection.

The hedge shock valuation projection calculates time 0 present value claims and charges for each hedge group for each scenario of the baseline and each economic shock.

The hedge final valuation projection calculates time 0 economic reserves (average of pv claims less pv charges) amounts for each hedge group for the baseline and each economic shock, using the results of the hedge shock valuation projection.

Hedge valuation projections are liability-only stochastic calculations and are always portfolio projections. Hedge valuations apply to the variable application only.

1.2.2 Hedging projections

Hedging results are included in experience projections when predefined hedge amounts are included, hedge positions are established and tracked in liability-only models or when hedging strategies are included in asset/liability model projections.

For variable annuities, the hedging projection is a special economic statement projection which is driven by a specific 'proj_tgt_hedge_stmt' target column. This special projection calculates a hedged and unhedged economic statement for the guaranteed benefits of the variable annuity product.

1.3 Hedge groups

1.3.1 Applicability

Hedge group affect the calculations and reporting of hedge results. Each hedge group produces a set of hedge results from the projection date for the results collected into the hedge group.

There is no limit to the number of hedge groups. Hedge group definitions must match between the hedging experience projection and each of the hedge valuations supporting it.

1.3.2 Hedge group names

The hedge group IDs must be specified in the `hedge_grps_selected` variable on the Company input page under the Model Setup: Segment category. Each hedge group ID must be separated by a comma only, with no spaces between the hedge group names. (The hedge group names themselves can include spaces.)

The following example shows that the hedge groups are Group 1, Group 2, Group 3, and Group 4.

VarID	assump_set_id seg_id	Baseline Corp	Baseline Seg
Model Setup: Segment			
<code>hedge_grps_selected</code>			Group 1,Group 2,Group 3,Group 4

Note that there are no hedge groups for the corporate segment; hedge groups only apply business segments.

Formulas use these hedge group names to set the hidden variable lookup key `hedge_grp_id`, which is the link to external source tables for variables defining values for each hedge group. Of course, the hedge group names in the external sources must match the specified cohort IDs.

1.3.3 Hedge group assignments

Results of model points are assigned to hedge groups using a number of criteria, which vary based on whether the hedge groups are for fixed indexed annuity, indexed universal life, or variable annuity hedge groups. Specify hedge group assignment criteria in the Hedge Groups category of the Hedging input page:

Variable Name	Value
<code>hedge_grp_plan_codes</code>	A comma separated list of plan codes, with no spaces between the names, such as 'Plan A,Plan B,Plan C'. Applies to all products.
<code>hedge_grp_gmab_defn</code>	'Yes' to include the GMAB claims and charges of in the hedge group results. 'No' otherwise This only applies to variable annuity products.
<code>hedge_grp_gmdb_defn</code>	'Yes' to include the GMDB claims and charges of in the hedge group results. 'No' otherwise This only applies to variable annuity products.
<code>hedge_grp_gmib_defn</code>	'Yes' to include the GMIB claims and charges of in the hedge group results. 'No' otherwise This only applies to variable annuity products.

Variable Name	Value
hedge_grp_gmwb_defn	'Yes' to include the GMWB claims and charges of in the hedge group results. 'No' otherwise This only applies to variable annuity products.
hedge_grp_crediting_defn	The index crediting method linked to the hedge group. For example, select Point to Point to link the point to point index accounts to the hedge group results. This only applies to fixed indexed annuity and indexed universal life products.
hedge_grp_crediting_mths	The crediting period linked to the hedge group. For example, specify 12 to link the one year index crediting accounts to the hedge group. This only applies to fixed indexed annuity and indexed universal life products.
hedge_grp_dyn_lever	The dynamic lever linked to the hedge group. For example, select Dynamic Cap to link the dynamic cap index accounts to the hedge group. This only applies to fixed indexed annuity products.
hedge_grp_index	The index crediting index linked to the hedge group. For example, specify 'US' to link the 'US' index accounts to the hedge group. This only applies to fixed indexed annuity and indexed universal life products.

All the variables discussed above are typically linked to external source ranges of the Hedging.xlsm workbook using the hidden variable `hedge_grp_id` as the column and the variable name as the row lookup key. Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

Model points and index accounts are assigned by testing each hedge group in the order listed in the `hedge_grps_selected` variable. When the model point and index account matches the definition for the first hedge group, it is assigned to that cohort. Otherwise the model point and index account is tested against the definition for the second cohort, and so on until a match is found.

Every model point must be assigned to a hedge group. Specify "ALL" to assign all other plan codes to a hedge group.

1.3.4 Hedge group variables

In addition to the hedge group assignment criteria discussed above, specify the following hedge group specific variables in the Hedge Groups category of the Hedging input page:

Variable Name	Value
hedge_grp_rebal_freq	The period between each rebalance of the hedge position for index credit hedging. A value of 1 would execute hedge rebalancing every months from the projection date. A value of 12 would execute hedge rebalancing every 12 months from the projection date. This only applies to fixed indexed annuity and indexed universal life products.
hedge_grp_asset_id	The asset identifier for the purchase asset which is used for the index crediting hedging strategy. This identifier must match an identifier for an index option asset which has a portfolio definition of Hedging. This only applies to fixed index annuity and indexed universal life products.

The variables discussed above are also typically linked to external source ranges of the Hedging.xlsm workbook using the hidden variable `hedge_grp_id` as the column and the variable name as the row lookup key. Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

2 PRE-GENERATED HEDGING

2.1 Applicability

Pre-generated hedging results may be included in the segment level calculations for all products, excluding fixed indexed annuity, variable annuity, and indexed universal life.

Fixed indexed annuity, variable annuity, and indexed universal life are excluded from the pre-generated hedging process since they contain integrated hedging capabilities as part of the standard asset liability modeling capabilities.

2.2 Hedging file

When using pre-generated hedge results for segment projections, the hedge cash flows and market values created by a separate hedge projection must be provided to the model in a file of monthly information that contains the following columns:

Column	Description
task_loop	The task loop number for the deterministic projection
period	This represents the months from the valuation date
cash_flow	Projected hedging cash flow
mkt_val	Projected hedging market value

Separate files are specified for gross of reinsurance and net of reinsurance results and are specified in the `hedge_pregen_gross_results` and `hedge_pregen_net_results` variables on the Hedging input page.

2.3 Financial statement presentation

2.3.1 Cash flows

Pre-generated hedge results are incorporated into the cash flows through the hedge cash flow captured from the pre-generated hedge result file. When reinsurance is included in the model, the net of reinsurance hedge results are used.

2.3.2 Income statement

Pre-generated hedge results are incorporated into the statutory income statement through the hedge cost, which is calculated as the negative of the hedge cash flows plus the change in the hedge market value. When reinsurance is included in the model, the net of reinsurance hedge results are used.

The hedge cost is calculated on a before sale basis.

2.3.3 Balance sheet

Pre-generated hedge results are incorporated into the statutory balance sheet through the hedge market value. When reinsurance is included in the model, the net of reinsurance hedge results are used.

3 INDEX CREDIT HEDGING

3.1 Applicability

Index credit hedging is applicable to fixed indexed annuity and indexed universal life products.

For liability-only models (both seriatim and portfolio) index credit hedging calculation apply for each index where the `hedge_defn` variable is set to Yes. Index crediting hedging is available for all index crediting methods.

For asset/liability models, index crediting hedging calculations apply when index crediting hedge groups are included in the model. Since real equity index option assets are used, only to the point to point and moving average index methods are included in the hedging calculations.

3.2 Hedge groups

Hedge groups apply only when performing index credit hedging within asset/liability models. For liability-only models, the index crediting is performed separately for each index bucket within each model point.

3.2.1 FIA hedge groups

Any number of hedge groups can be defined for the model. The definitions for each hedge group includes the model point plan codes to include, the indexing method type of the hedge group, the rebalancing frequency for the hedge group, and the id of a purchase equity index option.

Similar to the structure of the fixed indexed annuity indexed fund accounts, the hedge group submodel is a one dimensional array.

The values from the indexed fund accounts for each model point are collected into the hedging groups, through the mapping of plan codes and indexing method.

3.2.2 IUL hedge groups

Any number of hedge groups can be defined for the model. The definitions for each hedge group includes the model point plan codes to include, the indexing method type of the hedge group, the rebalancing frequency for the hedge group, and the id of a purchase equity index option.

Similar to the structure of each indexed universal life indexed fund account, the hedge group submodel is a two dimensional array with the first dimension being the hedge group and the second dimension being the number of buckets required to support the indexing method, such as 12 buckets for a 12 month point to point crediting method. This creates a hedge group bucket structure.

The values from the indexed fund account buckets for each model point are collected into the hedging groups, through the mapping of plan codes and indexing method.

3.3 Liability calculations

3.3.1 Notional amount to hedge

The notional amount to hedge for an index fund account bucket (where there is just 1 bucket for fixed indexed annuity) is the fund value at the beginning of the index crediting period times the index percent to hedge times the participation rate.

For asset/liability models, the notional amount to hedge for the hedge group is the aggregation of the unitized notional amounts to hedge for the appropriate indexed fund account bucket (where there is just the 1 bucket for fixed indexed annuity) for the model points mapped into the hedge group.

3.3.2 Option budget amount

The option budget amount for an index fund account bucket is the notional amount to hedge (excluding the participation rate adjustment) times the option budget rate and is determined at the beginning of the indexing period.

For asset/liability models, the option budget amount for the hedge group is the aggregation of the unitized option budget amounts for the appropriate indexed fund account bucket for the model points mapped into the hedge group.

3.3.3 Long strike

For point-to-point and moving average index crediting methods, a long strike for an index fund account bucket is the index value that represents the guaranteed crediting on the index account over the index crediting period. This amount is determined at the beginning of the index crediting period.

For asset/liability models, the long strike for the hedge group is the weighted long strike for the appropriate indexed fund account bucket.

3.3.4 Cap rate

When dynamic cap features are included, the cap rates for an index fund account bucket is determined using a solving routine that ensures that the option budget is not exceeded. This rate is determined at the beginning of the index crediting period.

For asset/liability models, the cap rate for the hedge group is the weighted cap rate for the assigned indexed fund account bucket and is determined at the beginning of the indexing period.

When the participation rate for the indexed fund account is less than 100%, the cap rate for the hedge group is the weighted cap rate of the indexed fund account bucket after adjustment for the participation rate. This is necessary because the cap rate for the purchased equity index option must be adjusted to offset for the participation rate adjusted notional amount to hedge.

3.3.5 Short strike

When dynamic cap features are included, for point-to-point and moving average index crediting methods, the short strike for an index fund account bucket is the index value times $(1.0 + \text{cap rate})$. This amount is determined at the beginning of the index crediting period.

For asset/liability models, the short strike for the hedge group is also the index value at the period times $(1.0 + \text{cap rate})$. This is the maximum payout index value for the hedge asset to be purchased in the period.

3.4 Hedge strategies

3.4.1 Overview

Index crediting is hedged by maintaining the appropriate notional amount of call options on the index expiring at the end of the index crediting period.

In liability-only models, hedge cash flows and market values are calculated within each index fund account bucket to hedge the index crediting of that single index fund account bucket. The hedge is automatically rebalanced each month for decrements of the notional amounts.

In asset/liability models, an existing portfolio of index options along with the definitions for the purchase index options is established for each hedge group. Rebalancing is used to adjust the hedge position for decrements of the liability model points.

3.4.2 Liability-only model index credit hedging

At the start of the projection, for each in force index fund account buckets, the notional amount, strike price, and market value for a balanced hedge is established.

At the beginning of each index crediting period for each index fund account bucket, a new hedge for the notional amount to hedge is established at the beginning of the month.

The market value of the hedge is calculated each month using closed form 'Black-Scholes' functions.

Hedge payoffs occur at the maturity of the hedge, i.e. at the index crediting period. The payoff is the market value of the hedge before expiry.

Hedge sales occur in each month in order to maintain the appropriate amount of hedge for the required notional amount to hedge. The notional amount required equals the notional amount to hedge less reductions in the fund value due to charges, transfers, and decrements.

3.4.3 Asset/liability model index credit hedging

An existing index crediting hedge position is established by including existing equity index option assets for the hedge groups in the existing index option model point file. Each asset record must use a portfolio definition of Hedging and be mapped to the hedge group using the `hdg_grp_code_mp` field of the existing equity index option asset record. The `hdg_grp_code_mp` must be structured in the form `[hedge_grp_id]` for index options linked to hedge groups for fixed indexed annuities and in the form `[hedge_grp_id][Bucket_xx]` where `xx` is the bucket number to which the existing index option is linked for index options linked to hedge groups for indexed universal life products.

Hedge assets for the notional amount to hedge with a strike equal to the long strike and a maximum payout index value equal to the short strike are purchased in each hedge group at the beginning of the month at the beginning of an indexing period. Purchasing at the beginning of the month is equivalent to purchasing at the end of the previous month and thus aligns the asset maturity with the end of the month when the index crediting will be paid. When purchased, the hedge asset is linked to the hedge group by setting the index option `hdg_grp_code` to an appropriate character string in a format similar to the format used for existing assets.

Hedge payoffs occur at the maturity of the hedge assets. The hedge asset payoff is the difference between the long option and the short option. The payoff of the long option is the index units owned times the positive

difference between the strike price and the index value and the payoff of the short option is the index units owned times the positive difference between the index value and the short strike.

Hedge rebalancing occurs at the end of the month per the hedge rebalancing frequency. A pro rata amount of hedge assets are sold so as to equalize the notional amount of the hedge assets with the notional amount required. The notional amount required equals the notional amount of the hedge assets in the portfolio reduced by the hedged fund value which has decreased due to mortality and lapse.

3.4.4 Asset/liability model hedge strategy auditing

Since index crediting hedging involves a number of interim calculation to determine the appropriate purchases and sales of hedge assets, understanding the results can be challenging.

Index crediting hedging calculations can be examined more closely by using the investment strategy detail report which can be created by using the following variables in the Asset Run Control category of the Investment input page:

Variable Name	Value
asset_detail_rpt_inv_strat_defn	'Yes' to create the investment strategy detail report. 'No' otherwise
asset_detail_rpt_start_period	The starting period for the investment strategy detail.
asset_detail_rpt_final_period	The final period for the investment strategy detail.

When the detail report is activated, a special text file will be created in the column output file folder that contains details on the purchases and sales performed each month from the Detail report start period to the Detail report end period.

3.5 Financial statement presentation

3.5.1 Cash flows

Index crediting hedge results are incorporated into the cash flows as the sum of the hedge option payments and hedge sale amounts, less the hedge investment amount, less any applicable hedge transaction costs. The hedge option payments and hedge sale amounts are end of month cash flows and the hedge investment amount is a beginning of month cash flow.

For liability-only models, segment level hedge cash flows equal the total over the liability model points included in the segment.

For asset/liability models, segment level hedge cash flows equal the total over all hedge groups included in the segment.

3.5.2 Income statement

Index crediting hedge results are incorporated into the statutory income statement through the hedge cost, which is calculated as the negative of the hedge cash flows plus the change in the hedge market value. For asset/liability models, the hedge cost is calculated on a before sale basis.

For liability-only models, segment level hedge costs equal the total over the liability model points included in the segment.

For asset/liability models, segment level hedge costs equal the total over all hedge groups included in the segment.

3.5.3 *Balance sheet*

Index credit hedge results are incorporated into the statutory balance sheet through the market value of the hedge assets.

For liability-only models, segment level hedge market values equal the total over the liability model points in the segment.

For asset/liability models, segment level hedge market values equal the total over all hedge groups included in the segment.

4 VARIABLE ANNUITY HEDGING

4.1 Applicability

Variable annuity hedging is applicable to variable annuity products. Hedging calculations apply to all of the guaranteed minimum benefit types.

Hedging valuations can be performed to produce point-in-time liability economic reserves and shock option values. Inner loop hedging valuation results, produced by a nested projection, can be utilized in standard asset/liability models to evaluate any number of hedging strategies.

Variable annuity hedging during an asset/liability model requires creating separate valuation tasks and an experience asset/liability model projection tasks and linking them together through a two-step process. This is discussed in the VA hedging projection setup section below.

4.2 Hedge groups

4.2.1 *VA hedge groups*

Any number of hedge groups can be defined for the model. The definitions for each hedge group includes the model point plan codes to include and selections for each of the guaranteed benefit types.

For variable annuity hedging, the hedge group submodel is a one dimensional array.

The values from the variable annuity model points are collected into the hedging groups, through the mapping of plan codes and the selections of the guaranteed benefit types included.

4.3 Liability calculations

4.3.1 *Economic and assumption shocks*

Economic and assumption shocks are required for hedging valuations and hedging projections of the guaranteed benefits of variable annuity products. For purposes of all discussions, a non-shocked 'baseline' risk neutral scenario is also considered a shock.

Any number of hedge shocks can be included for each of equity prices, equity volatilities, interest rates, and policyholder lapses. Then, those shocks can be linked together into any number of combination shocks. For example, you could design a combination shock that activates an equity price shock up for all indexes, a parallel shock for interest rates, and a shock of the policyholder lapse rate. A baseline shock is a combination shock where there are no equity price, equity volatility, interest rate, or policyholder shocks active.

Within a point-in-time valuation context, there are no restrictions on the number or setup of the shocks. However, within a projection context, where certain Greek hedging strategies are included, specific shocks must be included in order for the system to be able to determine the appropriate liability and asset Greek values and execute the shock strategies. For example, if a projection is setup to execute an index specific delta hedging strategy, equity price shocks up and down for each index must be included in the shock definitions.

Specify the economic and assumption shocks in the Hedge Valuation: Control category of the Hedging input page:

Variable Name	Value
shock_freq	An integer value representing the frequency for the shock generation within a nested projection. For example, 1 is specified for the generation to occur monthly within the nested projection horizon, 3 for the generation to occur quarterly, etc. This shock frequency typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.
eqt_price_shock_defn	'Yes' if an equity asset price shock is included in the combination shock element. 'No' otherwise. The equity price shock definition typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.
eqt_price_shock_id	Text representing the equity shock to apply for the combination shock element. The value for this variable must match an identifier of the defined equity price shocks. The equity price shock identifier typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.

Variable Name	Value
eqt_price_shock	A floating point value representing the shock amounts for the equity price shock. The shock amounts may be either positive, representing an up shock, or negative, representing a down shock, and are expressed as decimal values. (0.01 = 1%). The equity price shock typically varies by the index name ('US') and the equity price shock identifier. This variable is commonly linked to an external source using the hidden hedge_grp_index_name variable of the hedge group submodel as the column lookup key and the eqt_price_shock_id variable as the row lookup key.
eqt_vol_shock_defn	'Yes' if an equity volatility shock is included in the combination shock element. 'No' otherwise. The equity volatility shock definition typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.
eqt_vol_shock_id	Text representing the equity volatility shock to apply for the combination shock element. The value for this variable must match an identifier of the defined equity volatility shocks. The equity volatility shock identifier typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.

Variable Name	Value
eqt_vol_shock	A floating point value representing the shock amounts for the equity volatility shock. The shock amounts may be either positive, representing an up shock, or negative, representing a down shock, and are expressed as decimal values. (0.01 = 1%). The equity volatility shock typically varies by the index name ('US'), rate term, and the equity volatility shock identifier. This variable is commonly linked to an external source using the hidden hedge_grp_index_name and hedge_grp_rate_term variables and of the hedge group submodel as the column lookup keys and the eqt_vol_shock_id variable as the row lookup key. The previous value is typically used for missing rate terms.
int_rate_shock_defn	'Yes' if an interest rate shock is included in the combination shock element. 'No' otherwise. The interest rate shock definition typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.
int_rate_shock_id	Text representing the interest rate shock to apply for the combination shock element. The value for this variable must match an identifier of the defined interest rate shocks. The interest rate shock identifier typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.

Variable Name	Value
int_rate_shock	A floating point value representing the shock amounts for the interest rate shock. The shock amounts may be either positive, representing an up shock, or negative, representing a down shock, and are expressed as decimal values. (0.01 = 1%). The interest rate shock typically varies by the yield curve id ('Govt'), rate term, and the interest rate shock identifier. This variable is commonly linked to an external source using the hidden hedge_grp_yld_curve_id and hedge_grp_rate_term variables of the hedge group submodel as the column lookup keys and the int_rate_shock_id variable as the row lookup key. Interpolation is typically used for missing rate terms.
phldr_shock_defn	'Yes' if a policyholder shock is included in the combination shock element. 'No' otherwise. The policyholder shock definition typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.
phldr_shock_id	Text representing the policyholder shock to apply for the combination shock element. The only valid value for this variable is "Lapse". The policyholder shock identifier typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.

Variable Name	Value
phldr_vol_shock	A floating point value representing the shock amounts for the policyholder shock. The shock amounts may be either positive, representing an up shock, or negative, representing a down shock, and are expressed as decimal values. (0.01 = 1%). The policyholder shock typically varies by the decrement identifier ('Lapse'), and the policyholder shock identifier. This variable is commonly linked to an external source using the hidden decrem_id variable of the hedge group submodel as the column lookup key and the phldr_shock_id variable as the row lookup key.
scen_file_name_addn	Text representing the addition to the scenario name for the shock in the scen_file_name_addn variable. This text will be appended to the base name of the scenario file, historical rates file, or the inner loop scenario file to produce the name of the shock file. Best practice is to specify a name addition that will create a shocked scenario file which adheres to the scenario file naming convention discussed in the C02 Economic scenarios book. For example, 'Baseline' would be an appropriate value for the scen_file_name_addn variable for the shock which represents the baseline scenarios, and 'ParallelIntRateUp' would be an appropriate value for the scen_file_name_addn for the shock which represent a parallel up 1 for the interest rates. The scenario file name addition typically varies by the combination shock id. This variable is commonly linked to an external source using the variable name as the column lookup key and the hidden shock_id code variable of the hedge group submodel as the row lookup key.

All the variables discussed above are typically linked to external source ranges of the Hedging.xlsm workbook using appropriate column and row lookup keys.

For assumption sets which would be used for point-in-time hedging valuations, the *HedgeValnControl* range is typically used while the *HedgeProjControl* range is used for assumption sets that would be used for valuations included in hedge projections.

Given the nature of the shock definitions, separate ranges are required for the combination shock definitions, and each of the individual shock type definitions. The shock identifier setup in each of the individual shock type definitions are the valid values for the xxx_shock_id entries of the combination shock definitions.

Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

Shock identifiers naming considerations

The system does not automatically restrict the names specified for the individual shock identifiers or the combination shock identifiers. Within a point-in-time valuation context, there are no restrictions. You may name the individual shock types and combination shocks using any naming convention you choose.

However, within a projection context, the shock identifiers must adhere to a specified naming convention in order for the model to be able to use them properly in the hedge projection.

For example, a projection with an index specific delta hedge strategy must include a baseline shock and equity price shocks up and down for each of the indexes to be included in the delta hedging strategy. Thus, the model expects to find each of these shocks in the individual shock type and combination shock definitions, adhering to a specific naming convention.

Using the Pascal convention for all shock identifiers, the expected naming convention is as follows:

Shock type	Convention
Baseline	Baseline
Equity price shock	[IndexName]EqPrice[Direction][Amount] where: • [IndexName] is either Parallel or the identifier of the index (e.g. US) • [Direction] is Up or Down • [Amount] is any text identifying the amount of the shock (e.g. 1Pct) Per the above, sample equity price shock identifiers are as follows: ParallelEqPriceUp5Pct USEqPriceDown1Pct
Equity volatility shock	[IndexName]EqVol[Term][Direction][Amount] where: • [IndexName] is either Parallel or the identifier of the index (e.g. US) • [Term] is either AllTerms or expressed as aYr or aMth (e.g. 1Yr, 6Mth) • [Direction] is Up or Down • [Amount] is any text identifying the amount of the shock (e.g. 1Pct) Per the above, sample equity volatility shock identifiers are as follows: ParallelEqVolAllTermsUp5Pct USEqVol1YrDown1Pct

Shock type	Convention
Interest rate shock	[TermName]IntRate[Direction][Amount] where: • [TermName] is either Parallel or expressed as aYr or aMth (e.g. 1Yr, 6Mth) • [Direction] is Up or Down • [Amount] is any text identifying the amount of the shock (e.g. 1Pct, 20bps) Per the above, sample interest rate shock identifiers are as follows: ParallelIntRateUp1Pct 1YrIntRateDown20bps

The group names in the output of a shock valuation run include the names specified for each of the combination shock identifiers.

Key rate shock considerations

Key rate shocks of specific interest rate terms require the setup of combination and individual interest rate shocks for each of the desired key rate shocks with the correct parameterization of the shock amount for each term of the yield curve.

As an example, suppose you want to include a 10 basis point key rate up shock that shocks the 1 year, 5 year and 10 year terms of the normal 10 term yield curve. The following individual interest rate shocks are required:

- 1YrIntRateUp10bps, where the shocks for the 0.25, 0.5, and 1.0 year rate terms are 0.1%, the shock for the 2 year rate term is 0.075%, the shock for the 3 year rate term is 0.05% and the shock for the 5 year, 7 year, 10 year, 20 year, and 30 year terms are all 0.0%.
- 5YrIntRateUp10bps, where the shocks for the 0.25, 0.5 and 1.0 year rate terms are 0.0%, the shock for the 2 year rate term is 0.025%, the shock for the 3 year rate term is 0.05%, the shock for the 5 year term is 0.1%, the shock for the 7 year rate term is 0.05%, and the shock for the 10, 20, and 30 year terms are all 0.0%.
- 10YrIntRateUp10bps where the shocks for the 0.25, 0.5, 1.0, 2.0, 3.0, and 5.0 year rate terms are 0.0%, the shock for the 7 year rate terms is 0.05%, and the shock for the 10, 20, and 30 year rate terms are all 0.1%.

As required, the combination of these three individual shocks create a parallel 10 basis point shock of yield curve.

Shock scenarios files

Each shock identifier in the combination shock definitions requires a reference to an appropriate scenario file. The scenario file to use for that shock is a scenario file that is named using the base name of the scenario file specified in the Economy input page plus an underscore plus the scen_file_name_addn variable. For example, name for a scenario file name for a parallel interest rate shock up 10 bps could be 201512MonthlyScenarios_ParallelIntRateUp10bps.

Scenario files required for the shocks may be created by another system or created manually in advance and included in the appropriate location for the scenario files. Or the scenario files may be created through a risk neutral stochastic generation tasks of the U.S. Library. Please refer to the C02 Economic scenarios book for information related to risk neutral stochastic generation.

4.3.2 Variable annuity projection

Hedging for variable annuity guaranteed benefits requires a projection of the guaranteed benefits over a range of scenarios for a set of economic and assumption shocks.

As discussed above, specify the set of economic and assumptions shocks to run in the Hedge Valuation: Control category of the Hedging input page.

In addition, specify the number of scenarios to run for each shock and the scenario selection method using the following variables in the Hedge Valuation: General category of the Hedging input page:

Variable Name	Value
valn_scens_to_run	An integer number representing the number of scenarios from the scenario file to run for each model point.
valn_scens_defn	'Model Point Specific' if the starting scenario for the model point is determined using the value on the model point record. 'Block Specific' if all model points start with scenario 1 of the scenario file. When the scenario selection method is 'Model Point Specific', the starting scenario for the model point is adjusted to align with the number of scenarios available in the scenario file when the value specified on the model point record exceed the value specified for the gen_scenario variable on the Economy input page.

The model executes a full projection of variable annuity index account values, guaranteed benefit base amounts, and guaranteed benefits claims and charges for each scenario of each shock starting at the projection date. The projection period is 12 times the valn_yrs variable of the run parameter set.

For any shock including an equity price shocks, the starting fund value for each index is the specified fund value time (1.0 + the asset price shock for the index). For any shock including a policyholder lapse shock, the lapse rate used is the base lapse rate times (1.0 + the lapse shock amount). And, for any shock including an interest rate or equity volatility shock, the projection uses interest rates and index growth rates from the shocked scenario files.

4.3.3 Present values claims and charges

Present values of the guaranteed benefits claims and charges at time 0 are determined for each scenario of each shock for each hedge group. Either a deterministic rate, competitor based rate, or the generated beta rate may be used as the discount rate for the present values. The discount rate is an effective annual rate.

Specify the discount rate definitions in following variables of the Hedge Valuation: General category of the Hedging input page:

Variable Name	Value
hedge_disc_rate_defn	'Deterministic' if the discount rate is specified as a fixed rate 'Competitor Rate' if the discount rate is a specified competitor rate. 'Beta Rate' if the discount rate is the beta rate developed during the risk neutral scenario generation.
hedge_disc_rate_fixed	A floating-point number effective annual rate.
hedge_disc_rate_comp_id	The character identifier of the competitor rate to use for the discount rate.

Each hedge group includes the present values of the guaranteed benefit claims and charges for the model points and guaranteed benefit types assigned to the hedge groups.

4.3.4 *Economic reserve*

The economic reserve for each hedge group is determined as

$$\text{Average PV (claims)} - \text{Average PV (charges)}$$

where the average is over all scenarios run for the baseline run.

4.3.5 *Shock option values*

The shock option values for each shock in each hedge group is determined as

$$\text{Average PV (claims)} - \text{Average PV (charges)}$$

where the average is over all scenarios run for the shock.

4.3.6 *Liability greeks*

Dollar based Greek values for the liabilities are calculated from the liability option values and economic reserve as follows:

Delta

$$\text{liab_opt_val_eqt_price_up} - \text{liab_opt_val_eqt_price_down}$$

Gamma

$$\text{liab_opt_val_eqt_price_up} + \text{liab_opt_val_eqt_price_down} - 2.0 * \text{econ_res}$$

Vega

$$\text{liab_opt_val_eqt_vol_up} - \text{liab_opt_val_eqt_vol_down}$$

Rho

$$\text{liab_opt_val_int_rate_up} - \text{liab_opt_val_int_rate_down}$$
Rho convexity

$$\text{liab_opt_val_int_rate_up} + \text{liab_opt_val_int_rate_down} - 2.0 * \text{econ_res_baseline}$$

A Greek is calculated for each mix of shocks. For example, when there is both parallel equity price shocks and index specific price shocks included, the model calculate delta and gamma for the parallel case and for each index included in the shock calculations. Similarly, when there is both parallel interest rate shocks and key rate interest rate shocks included, the model calculates rho and rho convexity for the parallel case and for each key rate included in the shock calculations.

Important! Greek values for the liabilities are calculated only during hedge projections, where the model has knowledge of the number and structure of the shocks included. No Greeks are calculated in the model for point-in-time valuations.

4.4 Hedge strategies**4.4.1 Overview**

Variable annuity guaranteed benefits are hedged by maintaining the appropriate mix of hedge assets to align the hedge asset and liability Greek measures, such as delta, gamma, vega, rho, and rho convexity. Hedging is performed separately for each hedge group.

For each hedge strategy, an existing portfolio of hedge assets may be specified along with the definitions for the purchase hedge assets. Rebalancing is used to adjust the hedge position for changes in the Greek values of the liabilities.

Strict order of processing

In order to properly account for Greek residuals, the hedge strategies must be processed in the following order; gamma, vega, delta, rho convexity, and then rho.

This strict ordering is required because the hedge assets sold or purchased for matching one Greek can lead to a mismatch in another Greek, if the hedge positions were evaluated in isolation. For example, the index options purchased for gamma hedging contribute an amount of delta to the asset portfolio. Thus, the assets used to hedge gamma must be considered when considering the asset sales and purchases for delta hedging.

Initial hedge positions

An initial hedge position is established by including existing equity index option assets or interest rate derivatives for the hedge groups in the existing index option or interest rate derivative model point file. Each asset record must use a portfolio definition of Hedging and be mapped to the hedge group and specific hedge strategies using the `hdg_grp_code_mp` field of the existing asset record.

The `hdg_grp_code_mp` must be in the form `[hedge_grp_id][GreekType][IndexName/KeyRate][IndexTerm]` where:

- `[hedge_grp_id]` is the name of the hedge group,
- `[GreekType]` is Delta, Gamma, Vega, Rho, or RhoConvexity,
- `[IndexName/KeyRate]` is Parallel, the name of the Index (such as US), or the name of the key rate (such as 1Yr)
- `[IndexTerm]` which applies for Vega hedges only is AllTerms or the name of the index volatility term (such as 1Yr)

When existing hedges are specified for the hedge group, the application assumes the hedge position of those assets is appropriate for the projection date. However, when no existing hedges are specified for the hedge group, hedge assets are purchased at the projection date to develop a balanced hedge position.

4.4.2 Asset greeks

Dollar based Greek values for the assets are calculated by shocking the initial yield curve and volatilities and executing market value calculations to determine option values for the asset for the shock type and then applying the same formulas used for the liabilities.

For example, to calculate the gamma for an equity index option, the baseline option value is calculated first as the market value of the index option which no shock. Then, the market value calculation is executed with an index up shock to calculate the option value for the up shock. Finally, the market value calculation is executed with an index down shock to calculate the option value for the down shock.

With these three values, the gamma of the index option is calculated using the same formula as used for the liability gamma calculation.

4.4.3 Gamma hedging

Gamma hedging provides for matching the dollar gamma of a portfolio of equity index put options contracts to the dollar gamma of the liabilities.

Specify the gamma hedge strategies in the Hedge Strategy: Gamma/Vega Hedging category of the Hedging input page.

Variable Name	Value
hedge_grp_gamma_vega_hedge_defn	'Yes' to include gamma hedging in the projection. 'No' otherwise The gamma hedging definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_type_defn	'Gamma' The gamma hedging type definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_strat_defn	'Equity Options' is the only choice The gamma hedging strategy definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_gamma_vega_hedge_liab_shock_defn	'Parallel' to hedge to the parallel gamma 'Index Specific' to hedge to the index specific gamma The gamma hedging liability shock definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_rebal_freq	Specify the period between each rebalance of the hedge position for gamma hedging. A value of 1 would execute hedge rebalancing every months from the projection date. A value of 12 would execute hedge rebalancing every 12 months from the projection date. The gamma hedging rebalance frequency typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_transaction_costs_bps	A floating-point value representing the basis costs for purchasing and selling gamma hedge assets. This amount is applied against the absolute value of the notional amount purchased or sold. The gamma hedging transaction cost typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_gamma_vega_hedge_tolerance _pct_liab_gamma_vega	A floating-point value representing the percentage of liability gamma to use in determining a hedge mismatch tolerance. The gamma hedging tolerance percent liability gamma typically varies by hedge group and index, where 'Parallel' is treated as a valid index name, and index term, where AllTerms is the only valid selection for the index term for gamma hedging. This variable is commonly linked to an external source using the hidden hedge_grp_id, hedge_grp_index_name, and hedge_grp_index_term code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_tolerance_dollar	A floating-point value representing a fixed dollar amount to use in determining a hedge mismatch tolerance. The gamma hedging tolerance dollar typically varies by hedge group and index, where 'Parallel' is treated as a valid index name, and index term, where AllTerms is the only valid selection for the index term for gamma hedging. This variable is commonly linked to an external source using the hidden hedge_grp_id, hedge_grp_index_name, and hedge_grp_index_term code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_pct	A floating-point value representing the percent of the liability gamma to hedge, 1.0 = 100% The gamma hedging percent typically varies by hedge group and index, where 'Parallel' is treated as a valid index name, and index term, where AllTerms is the only valid selection for the index term for gamma hedging. This variable is commonly linked to an external source using the hidden hedge_grp_id, hedge_grp_index_name, and hedge_grp_index_term code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_gamma_vega_hedge_asset_id	The asset identifier for the purchase asset which is used for the gamma hedging strategy. This identifier must match an identifier for an index option asset which has a portfolio definition of Hedging. The gamma hedging asset identifier typically varies by hedge group and index, where 'Parallel' is treated as a valid index name, and index term, where AllTerms is the only valid selection for the index term for gamma hedging. This variable is commonly linked to an external source using the hidden hedge_grp_id, hedge_grp_index_name, and hedge_grp_index_term code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

All the variables discussed above are typically linked to external source ranges of the Hedging.xlsm workbook using appropriate column and row lookup keys.

Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

When gamma hedging is included, the hedging process is as follows:

First, determine if the gamma hedge position must be adjusted. This is accomplished by calculating the gamma for the existing hedges and comparing those amounts to the gamma of the liabilities. The hedge position requires adjustment when the asset gamma falls outside of the liability gamma boundaries determined using the specified tolerances. The determination is made separately for each hedge method per the gamma hedge shock definition.

Next, sell existing hedges as required to adjust the gamma of the assets to move toward the gamma of the liabilities. This can result in selling short or long positions, and all or just a portion of the hedge asset. The sales process is executed separately for each hedge method per the gamma vega hedge shock definition.

Finally, if required, purchase new hedges as required to adjust the gamma of the assets to match the gamma of the liabilities. The first step in the purchase process is to determine the gamma of a proposed purchase hedge asset, using a virtual purchase of a hedge with a notional amount of one million. The final notional amount to purchase of the hedge asset is then determined as one million times the ratio of the gamma needed and the gamma of the proposed hedge asset. The hedge asset can then be purchased and added to the asset portfolio. Again, the purchase process is executed separately for each hedge method per the gamma vega hedge shock definition.

4.4.4 Vega hedging

Vega hedging provides for matching the dollar vega of a portfolio of equity index put options contracts to the dollar vega of the liabilities.

Specify the vega hedge strategies in the Hedge Strategy: Gamma/Vega Hedging category of the Hedging input page.

Variable Name	Value
hedge_grp_gamma_vega_hedge_defn	'Yes' to include vega hedging in the projection. 'No' otherwise The vega hedging definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_type_defn	'Vega' The vega hedging type definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_strat_defn	'Equity Options' is the only choice The vega hedging strategy definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_liab_shock_defn	'Parallel' to hedge to the parallel vega 'Index Specific' to hedge to the index specific vega 'Index Term Specific' to hedge to the index and term specific vega The vega hedging liability shock definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_gamma_vega_hedge_rebal_freq	Specify the period between each rebalance of the hedge position for vega hedging. A value of 1 would execute hedge rebalancing every months from the projection date. A value of 12 would execute hedge rebalancing every 12 months from the projection date. The vega hedging rebalance frequency typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_transaction_costs_bps	A floating-point value representing the basis costs for purchasing and selling vega hedge assets. This amount is applied against the absolute value of the notional amount purchased or sold. The vega hedging transaction cost typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_tolerance_pct_liab_gamma_vega	A floating-point value representing the percentage of liability vega to use in determining a hedge mismatch tolerance. The vega hedging tolerance percent liability vega typically varies by hedge group and index, where 'Parallel' is treated as a valid index name, and index term, where AllTerms is treated as a valid index term. This variable is commonly linked to an external source using the hidden hedge_grp_id, hedge_grp_index_name, and hedge_grp_index_term code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_gamma_vega_hedge_tolerance_dollar	A floating-point value representing a fixed dollar amount to use in determining a hedge mismatch tolerance. The vega hedging tolerance dollar typically varies by hedge group and index, where 'Parallel' is treated as a valid index name, and index term, where AllTerms is treated as a valid index term. This variable is commonly linked to an external source using the hidden hedge_grp_id, hedge_grp_index_name, and hedge_grp_index_term code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_pct	A floating-point value representing the percent of the liability vega to hedge, 1.0 = 100% The vega hedging percent typically varies by hedge group and index, where 'Parallel' is treated as a valid index name, and index term, where AllTerms is treated as a valid index term. This variable is commonly linked to an external source using the hidden hedge_grp_id, hedge_grp_index_name, and hedge_grp_index_term code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_gamma_vega_hedge_asset_id	The asset identifier for the purchase asset which is used for the vega hedging strategy. This identifier must match an identifier for an index option asset which has a portfolio definition of Hedging. The gamma hedging asset identifier typically varies by hedge group and index, where 'Parallel' is treated as a valid index name, and index term, where AllTerms is treated as a valid index term. This variable is commonly linked to an external source using the hidden hedge_grp_id, hedge_grp_index_name, and hedge_grp_index_term code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

All the variables discussed above are typically linked to external source ranges of the Hedging.xlsm workbook using appropriate column and row lookup keys.

Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

When gamma hedging is included, the hedging process is as follows:

First, determine if the vega hedge position must be adjusted. This is accomplished by calculating the vega for the existing hedges and comparing those amounts to the vega of the liabilities. The hedge position requires adjustment when the asset vega falls outside of the liability vega boundaries determined using the specified tolerances. The determination is made separately for each hedge method per the gamma vega hedge shock definition.

Next, sell existing hedges as required to adjust the vega of the assets to move toward the vega of the liabilities. This can result in selling short or long positions, and all or just a portion of the hedge asset. The sales process is executed separately for each hedge method per the gamma vega hedge shock definition.

Finally, if required, purchase new hedges asset as required to adjust the vega of the assets to match the vega of the liabilities. The first step in the purchase process is to determine the vega of a proposed purchase hedge asset, using a virtual purchase of a hedge with a notional amount of one million. The final notional amount to purchase of the hedge asset is then determined as one million times the ratio of the vega needed and the vega of the proposed hedge asset. The hedge asset can then be purchased and added to the asset portfolio. Again, the purchase process is executed separately for each hedge method per the gamma vega hedge shock definition.

4.4.5 Delta hedging

Delta hedging provides for matching the dollar delta of a portfolio of equity futures contracts to the dollar delta of the liabilities.

Specify the delta hedge strategies in the Hedge Strategy: Delta Hedging category of the Hedging input page.

Variable Name	Value
hedge_grp_delta_hedge_defn	'Yes' to include delta hedging in the projection. 'No' otherwise The delta hedging definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_delta_hedge_strat_defn	'Equity Futures' is the only choice The delta hedging strategy definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_delta_hedge_liab_shock_defn	'Parallel' to hedge to the parallel delta 'Index Specific' to hedge to the index specific deltas The delta hedging liability shock definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_delta_hedge_rebal_freq	Specify the period between each rebalance of the hedge position for delta hedging. A value of 1 would execute hedge rebalancing every months from the projection date. A value of 12 would execute hedge rebalancing every 12 months from the projection date. The delta hedging rebalance frequency typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_delta_hedge_transaction_costs_bps	A floating-point value representing the basis costs for purchasing and selling delta hedge assets. This amount is applied against the absolute value of the notional amount purchased or sold. The delta hedging transaction cost typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_delta_hedge_tolerance_pct_liab_delta	A floating-point value representing the percentage of liability delta to use in determining a hedge mismatch tolerance. The delta hedging tolerance percent liability delta typically varies by hedge group and index, where 'Parallel' is treated as a valid index name. This variable is commonly linked to an external source using the hidden hedge_grp_id and hedge_grp_index_name code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_delta_hedge_tolerance_dollar	A floating-point value representing a fixed dollar amount to use in determining a hedge mismatch tolerance. The delta hedging tolerance dollar typically varies by hedge group and index, where 'Parallel' is treated as a valid index name. This variable is commonly linked to an external source using the hidden hedge_grp_id and hedge_grp_index_name code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_delta_hedge_pct	A floating-point value representing the percent of the liability delta to hedge, 1.0 = 100% The delta hedging percent typically varies by hedge group and index, where 'Parallel' is treated as a valid index name. This variable is commonly linked to an external source using the hidden hedge_grp_id and hedge_grp_index_name code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_delta_hedge_asset_id	The asset identifier for the purchase asset which is used for the delta hedging strategy. This identifier must match an identifier for an index option asset which has a portfolio definition of Hedging. The delta hedging asset identifier typically varies by hedge group and index, where 'Parallel' is treated as a valid index name. This variable is commonly linked to an external source using the hidden hedge_grp_id and hedge_grp_index_name code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

All the variables discussed above are typically linked to external source ranges of the Hedging.xlsm workbook using appropriate column and row lookup keys.

Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

When delta hedging is included, the hedging process is as follows:

First, determine if the delta hedge position must be adjusted. This is accomplished by calculating the delta for the existing hedge assets and comparing those amounts to the delta of the liabilities. The hedge position requires adjustment when the asset delta falls outside of the liability delta boundaries determined using the specified tolerances. The determination is made separately for each hedge method per the delta hedge shock definition.

Next, sell existing hedges as required to adjust the delta of the assets to move toward the delta of the liabilities. This can result in selling short or long positions, and all or just a portion of the hedge asset. The sales process is executed separately for each hedge method per the delta hedge shock definition.

Finally, if required, purchase new hedges asset as required to adjust the delta of the assets to match the delta of the liabilities. The first step in the purchase process is to determine the delta of a proposed purchase hedge asset, using a virtual purchase of a hedge with a notional amount of one million. The final notional amount to purchase of the hedge asset is then determined as one million times the ratio of the delta needed and the delta of the proposed hedge asset. The hedge asset can then be purchased and added to the asset portfolio. Again, the purchase process is executed separately for each hedge method per the delta hedge shock definition.

Since the hedge asset used for delta hedging is an index future, the sales and purchases of the delta hedge assets do not adversely affect the gamma or vega hedge positions.

4.4.6 Rho convexity hedging

Rho convexity hedging provides for matching the dollar rho convexity of a portfolio of interest rate floors to the dollar rho convexity of the liabilities.

Specify the rho convexity hedge strategies in the Hedge Strategy: Rho Convexity Hedging category of the Hedging input page.

Variable Name	Value
hedge_grp_rho_convex_hedge_defn	'Yes' to include rho convexity hedging in the projection. 'No' otherwise The rho convexity hedging definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_rho_convex_hedge_strat_defn	'Interest Rate Floors' is the only choice The rho convexity hedging strategy definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_rho_convex_hedge_liab_shock_defn	'Parallel' to hedge to the parallel rho convexity 'Key Rate Specific' to hedge to the key rate specific rho convexity The rho convexity hedging liability shock definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_rho_convex_hedge_rebal_freq	Specify the period between each rebalance of the hedge position for rho convexity hedging. A value of 1 would execute hedge rebalancing every months from the projection date. A value of 12 would execute hedge rebalancing every 12 months from the projection date. The rho convexity hedging rebalance frequency typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_rho_convex_hedge_transaction_costs_bps	A floating-point value representing the basis costs for purchasing and selling rho convexity hedge assets. This amount is applied against the absolute value of the notional amount purchased or sold. The rho convexity hedging transaction cost typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_rho_convex_hedge_tolerance_pct_liab_rho_convex	A floating-point value representing the percentage of liability rho to use in determining a hedge mismatch tolerance. The rho convexity hedging tolerance percent liability rho convexity typically varies by hedge group and key rate, where 'Parallel' is treated as a valid key rate. This variable is commonly linked to an external source using the hidden hedge_grp_id, and hedge_grp_key_rate code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_rho_convex_hedge_tolerance_dollar	A floating-point value representing a fixed dollar amount to use in determining a hedge mismatch tolerance. The rho convexity hedging tolerance dollar typically varies by hedge group and key rate, where 'Parallel' is treated as a valid key rate. This variable is commonly linked to an external source using the hidden hedge_grp_id, and hedge_grp_key_rate code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_rho_convex_hedge_pct	A floating-point value representing the percent of the liability rho convexity to hedge, 1.0 = 100% The rho convexity hedging percent typically varies by hedge group and key rate, where 'Parallel' is treated as a valid key rate. This variable is commonly linked to an external source using the hidden hedge_grp_id, and hedge_grp_key_rate code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_rho_convex_hedge_asset_id	The asset identifier for the purchase asset which is used for the rho hedging strategy. This identifier must match an identifier for an index option asset which has a portfolio definition of Hedging. The rho convexity hedging asset identifier typically varies by hedge group and key rate, where 'Parallel' is treated as a valid key rate. This variable is commonly linked to an external source using the hidden hedge_grp_id, and hedge_grp_key_rate code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

All the variables discussed above are typically linked to external source ranges of the Hedging.xlsm workbook using appropriate column and row lookup keys.

Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

When rho convexity hedging is included, the hedging process is as follows:

First, determine if the rho convexity hedge position must be adjusted. This is accomplished by calculating the rho convexity for the existing hedge assets and comparing those amounts to the rho convexity of the liabilities. The hedge position requires adjustment when the asset rho convexity falls outside of the liability rho convexity boundaries determined using the specified tolerances. The determination is made separately for each hedge method per the rho convexity hedge shock definition.

Next, sell existing hedges as required to adjust the rho convexity of the assets to move toward the rho convexity of the liabilities. This can result in selling short or long positions, and all or just a portion of the hedge asset. The sales process is executed separately for each hedge method per the rho convexity hedge shock definition.

Finally, if required, purchase new hedges asset as required to adjust the rho convexity of the assets to match the rho convexity of the liabilities. The first step in the purchase process is to determine the rho convexity of a proposed purchase hedge asset, using a virtual purchase of a hedge with a notional amount of one million. The final notional amount to purchase of the hedge asset is then determined as one million times the ratio of the rho convexity needed to the rho convexity of the proposed hedge asset. The hedge asset can then be purchased and added to the asset portfolio. Again, the purchase process is executed separately for each hedge method per the rho convexity hedge shock definition.

4.4.7 Rho hedging

Rho hedging provides for matching the dollar rho of a portfolio of interest rate swaps to the dollar rho of the liabilities.

Specify the rho hedge strategies in the Hedge Strategy: Rho Hedging category of the Hedging input page.

Variable Name	Value
hedge_grp_rho_hedge_defn	'Yes' to include rho hedging in the projection. 'No' otherwise The rho hedging definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_rho_hedge_strat_defn	'Interest Rate Swaps' is the only choice The rho hedging strategy definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_rho_hedge_liab_shock_defn	'Parallel' to hedge to the parallel rho 'Key Rate Specific' to hedge to the key rate specific rho The rho hedging liability shock definition typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_rho_hedge_rebal_freq	Specify the period between each rebalance of the hedge position for rho hedging. A value of 1 would execute hedge rebalancing every months from the projection date. A value of 12 would execute hedge rebalancing every 12 months from the projection date. The rho hedging rebalance frequency typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_rho_hedge_transaction_costs_bps	A floating-point value representing the basis costs for purchasing and selling rho hedge assets. This amount is applied against the absolute value of the notional amount purchased or sold. The rho hedging transaction cost typically varies by hedge group. This variable is commonly linked to an external source using the hidden hedge_grp_id code variable of the hedge group submodel as the column lookup key and the variable name as the row lookup key.
hedge_grp_rho_hedge_tolerance_pct_liab_rho	A floating-point value representing the percentage of liability rho to use in determining a hedge mismatch tolerance. The rho hedging tolerance percent liability rho typically varies by hedge group and key rate, where 'Parallel' is treated as a valid key rate. This variable is commonly linked to an external source using the hidden hedge_grp_id, and hedge_grp_key_rate code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_rho_hedge_tolerance_dollar	A floating-point value representing a fixed dollar amount to use in determining a hedge mismatch tolerance. The rho hedging tolerance dollar typically varies by hedge group and key rate, where 'Parallel' is treated as a valid key rate. This variable is commonly linked to an external source using the hidden hedge_grp_id, and hedge_grp_key_rate code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

Variable Name	Value
hedge_grp_rho_hedge_pct	A floating-point value representing the percent of the liability rho to hedge, 1.0 = 100% The rho hedging percent typically varies by hedge group and key rate, where 'Parallel' is treated as a valid key rate. This variable is commonly linked to an external source using the hidden hedge_grp_id, and hedge_grp_key_rate code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.
hedge_grp_rho_hedge_asset_id	The asset identifier for the purchase asset which is used for the rho hedging strategy. This identifier must match an identifier for an index option asset which has a portfolio definition of Hedging. The rho hedging asset identifier typically varies by hedge group and key rate, where 'Parallel' is treated as a valid key rate. This variable is commonly linked to an external source using the hidden hedge_grp_id, and hedge_grp_key_rate code variables of the hedge submodel as the column lookup keys and the variable name as the row lookup key.

All the variables discussed above are typically linked to external source ranges of the Hedging.xlsm workbook using appropriate column and row lookup keys.

Please refer to the Hedging.xlsm workbook provided in the \ProjectData\Template\Assumptions folder and the Hedging input page of the Template IM to see examples of external source ranges and lookup definitions for each of these variables.

When rho hedging is included, the hedging process is as follows:

First, determine if the rho hedge position must be adjusted. This is accomplished by calculating the rho for the existing hedge assets and comparing those amounts to the rho of the liabilities. The hedge position requires adjustment when the asset rho falls outside of the liability rho boundaries determined using the specified tolerances. The determination is made separately for each hedge method per the rho hedge shock definition.

Next, sell existing hedges as required to adjust the rho of the assets to move toward the rho of the liabilities. This can result in selling short or long positions, and all or just a portion of the hedge asset. The sales process is executed separately for each hedge method per the rho hedge shock definition.

Finally, if required, purchase new hedges asset as required to adjust the rho of the assets to match the rho of the liabilities. The first step in the purchase process is to determine the rho of a proposed purchase hedge asset, using a virtual purchase of a hedge with a notional amount of one million. The final notional amount to purchase of the hedge asset is then determined as one million times the ratio of the rho needed to the rho of the proposed hedge asset. The hedge asset can then be purchased and added to the asset portfolio. Again, the purchase process is executed separately for each hedge method per the rho hedge shock definition.

4.4.8 Strategy auditing

Since variable annuity guaranteed benefit hedging involves a number of interim calculation to determine the appropriate purchases and sales of hedge assets, understanding the results can be challenging.

Variable annuity guaranteed benefit hedging calculations can be examined more closely by using the investment strategy detail report which can be created by using the following variables in the Asset Run Control category of the Investment input page:

Variable Name	Value
asset_detail_rpt_inv_strat_defn	'Yes' to create the investment strategy detail report. 'No' otherwise
asset_detail_rpt_start_period	Specify the starting period for the investment strategy detail.
asset_detail_rpt_final_period	Specify the final period for the investment strategy detail.

When the detail report is activated, a special text file will be created in the column output file folder that contains details on the purchases and sales performed each month from the Detail report start period to the Detail report end period.

4.5 Financial statement presentation

4.5.1 Cash flows

Variable annuity hedging results are incorporated into the cash flows as the sum of the hedge option payments and hedge sale amounts, less the hedge investment amount, and hedge expenses. The hedge option payments and hedge sale amounts are end of month cash flows and the hedge investment amount is a beginning of month cash flow.

Each item is calculated separately for each hedge strategy. For example, there is a separate delta_hedge_inv_amt from gaam_hedge_inv_amt, etc. The amounts all come together into a single hedge_cash_flow column in the hedge group submodel.

Segment level hedge cash flows equal the total over all hedge groups included in the segment.

4.5.2 Income statement

Variable annuity hedging results are incorporated into an economic income statement through the calculation of an unhedged economic profit and a hedged economic profit.

The unhedged economic profit equals the interest on the beginning of month economic reserve plus the guaranteed benefit charges (referred to as the hedge cost charge) less the guaranteed benefit claims, less the change in the economic reserve.

The hedged economic profit equals the profit generated from each of the hedging strategies, plus interest on the beginning of month economic cash plus the guaranteed benefit charges (referred to as the hedge cost charge), less the guaranteed benefit claims, less the change in the economic reserve.

The profit from each of the hedging strategy equals the hedge payment, plus the cash from hedge sales, less the cash required for hedge purchases, less the cash required for hedge transaction costs, plus the increase in the hedge market value.

4.5.3 Balance sheet

Variable annuity hedging results are incorporated into an economic balance sheet with assets equal to the market value of hedge assets and the economic cash account, and the liabilities equal to the economic reserve.

The economic cash account equals the difference between the economic reserve and the hedge market values.

5 VA HEDGING VALUATION TASKS SETUP

This chapter gives instructions for setting up hedge valuation projection tasks.

5.1 Applicability

Hedge valuations apply only to variable annuity hedging, where the economic reserve and option values must be determined in a projection task running the baseline shock risk neutral scenarios.

5.2 Hedge valuation tasks overview

5.2.1 Point-in-time valuations

Hedge valuations may be for a single point-in-time valuation, such as for nightly hedging valuations. The valuation projection tasks are performed as of the projection date (valuation date).

5.2.2 Inner loop valuations

Hedge valuations can also be included as inner loop valuations within a nested projection. For example, inner loop valuations to produce new liability economic reserves and Greeks for hedge projections. The experience projection is dependent on the hedge valuation tasks being completed and the nested projection results being combined together into a single output file.

5.2.3 Shock scenario generation

The model provides for generating risk neutral baseline and shocked scenarios for use in the hedging valuations.

Please refer to chapter 6 of the C02 Economic scenarios book for information regarding risk neutral scenario generation tasks.

5.2.4 All shocks run together

The model provides for running all shocks in a single valuation projection task. This is accomplished by defining the projection task loop of the shock valuation tasks as a combination of the shocks to run and the scenarios to run for each shock.

For example, if a hedging valuation is running a baseline and 10 shocks, and each run is for 1,000 scenarios, then the projection task loop for the valuation projection task should be 1-11,000. Each projection task loop can be executed as an independent run.

Use the distributed processing capabilities of RiskAgility FM to distribute the groups of the projection task loops to a grid, thereby completing the total run in a fraction of the amount of time that would be required if the projection were to be run on a single computer core.

5.2.5 Final valuation

The results of the shock valuation task are the time zero present value of claims and charges for each shock included in the valuation run. Calculate the economic reserve and liability option values for each hedge group using a hedging final valuation task.

The hedge final valuation task is a single projection task loop run which develops the economic reserve and option values using the output file produced by the hedge shock valuation task.

When included in a nested projection, the final valuation task also performs the projection of the model points to the end of the time step and captures the state of the world for the model points. This is a special capability of a valuation final projection task which does not apply to final valuation tasks for PBA or GAAP.

5.2.6 Run controller projection set

Point-in-time valuation tasks are typically organized into a projection set on the Run controller, with appropriate task dependencies established such that the shock valuation tasks is dependent on a scenario generation task, and the final valuation task is dependent on the shock valuation task. Then, the entire projection set can be submitting from the Run controller. This is not required; standard projections can be run from the individual Run page but the dependency of the final task on the valuation task must be considered in any individual task submission process.

Valuation tasks intended for use with hedge projections must be combined with each other and appropriate scenario generation tasks and setup within a Run controller projection set, with appropriate dependencies established.

5.3 Hedge shock valuation task setup

The following setup instructions apply for hedge valuation tasks.

5.3.1 Input variables

Specify hedge groups to be included in the segment assumptions, on the Company input page under the Model Setup: Segment category.

Specify general historic rates and projection scenario files to use, on the Economy input page under the Scenarios : Rates category.

Specify the variable annuity hedge groups, on the Hedging input page under the Hedge Groups category.

Specify the hedge scenarios treatment and hedging discount rate, on the Hedging input page under the Hedge Valuation: General category

Specify the hedge shocks, on the Hedging input page under the Hedge Valuation: Control category.

Specify variable annuity product specifications and assumptions, on the various VA[XXX] input pages.

Specify the in force variable annuity model points in the variable annuity model point files.

5.3.2 Run page

Create a projection task on a 'Hedging Portfolio' Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

5.3.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_hedge_shock of the company submodel
Projection period from	0
Projection period to	0, since the valuation projection is specified by the valn_yrs project variable
Projection period min	-600, to allow for external source lookup keys calculating before the projection date (such as pol_yr)

Property	Values
Projection period max	At least as long as the calculations for each submodel
Projection task loops	1-N where N is the number of projection task loops required based on the number of shocks and scenarios run for each shock.
Wildcard set	Select the appropriate wildcard set for accessing the external assumptions
Distribution Split	Specify a Target Split for Distribution value for the minimum number of Projection Loops that are allocated to a Grid Task in a Distributed Processing environment. Even if the number of cores on the grid exceed the number of projection task loops, the optimal value for this setting may not be 1, since there is overhead in the distribution process as well as in the loading of scenarios in the model.

5.3.4 Project variables

Also specify the project variables in the Run parameter set.

Property	Values
Projection date	Valuation date specified as yyyy/mm/dd. When the 'dd' is omitted the valuation is assumed to be at the end of the month.
Valuation years	The number of years from the projection date to run the shock projection
Use independent model point processing?	Typically Yes, since the hedging valuation is a liability-only projection and memory may be an issue for very large number of model points
Validate data definitions?	No for production runs; Yes for testing
Include decrements?	Yes
Valuation periods selected	Has no effect on point-in-time valuations For valuation within nested projections, must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0.

Property	Values
Save generated time-step scenario files?	Has no effect on point-in-time valuations. For valuations within nested projections, specify how the system should manage generated scenario files selecting from: Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs that generate large scenario files.

5.3.5 Output parameter set

Specify the following output parameters in the Output parameter set.

Property	Values
Column output from	0
Column output to	0; unless additional periods are desired for reviewing results
Output manager	Select the 'Valn' output manager to limit the output captured
Model object output	Select All Columns or the 'Hedge' Output Definitions for the hedge_grp submodel plus any other model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions

5.4 Hedge final valuation setup

5.4.1 Input variables

Specify the input variables similarly to the hedge shock valuation task setup. In addition, you must specify hedge group results from the hedge shock valuations in the valn_hedge_results variable on the Hedging input page under the Hedge Valuation: General category using the system <*column_output_directory*> wildcard.

An example file reference for a point-in-time is <*column_output_directory*>Hedging Portfolio_Shock Valn~company~seg~hedge_grp.csv.

5.4.2 Run page

Create a projection task on a 'Hedging Portfolio' Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

5.4.3 Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	valn_tgt_hedge_final of the company submodel
Projection period from	0
Projection period to	0, when this final valuation task is used for a point-in-time valuation For nested projections, at least as long as the maximum time between time steps
Projection period min	-600, to allow for external source lookup keys calculating before the projection date (such as pol_yr)
Projection period max	At least as long as the calculations for each submodel
Projection task loops	1 since this is a single projection task loop run
Wildcard set	Select the appropriate wildcard set for accessing the external assumptions

5.4.4 Project variables

Also specify the project variables in the Run parameter set.

Property	Values
Projection date	Valuation date specified as yyyy/mm/dd. When the 'dd' is omitted the valuation is assumed to be at the end of the month
Use independent model point processing?	Typically Yes, since the hedging valuation is a liability-only projection and memory may be an issue for very large number of model points
Validate data definitions?	No for production runs; Yes for testing
Include decrements?	Yes
Valuation periods selected	Has no effect on point-in-time valuations For valuations within nested projections, must specify the valuation period for each projection set sub loop as a comma separated list of individual numbers or dash separated ranges with no spaces. For example, 0,12,24,36 specifies that sub loop 1 is the valuation at the projection date, sub loop 2 is the valuation at projection month 12, sub loop 3 is the valuation at projection month == 24, and so on. Alternatively, 0-36 specifies that there would be 37 sub loops run, where sub loop 1 is the valuation at the projection date, sub loops 2 is the valuation at projection month 1, and so on. The first valuation period must always be 0.

Property	Values
Save generated time-step scenario files?	Has no effect on point-in-time valuations. For valuations within nested projections, specify how the system should manage generated scenario files selecting from: Yes: Each generated scenario file created for each time-step are created as new files and are preserved. No: The generated scenario files created for each time-step use the same file name. Only the scenario files generated for the final time-step is available following the run. Specifying No for this project variable can help to save a significant amount of disk space for runs that generate large scenario files.

5.4.5 Output parameter set

Specify the following output parameters in the Output parameter set.

Property	Values
Column output from	0
Column output to	0; unless additional periods are desired for reviewing results
Output manager	Select the 'Valn' output manager to limit the output captured
Model object output	Select All Columns or the 'Hedge' Output Definitions for the hedge_grp submodel plus any other model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions

6 VA HEDGING PROJECTION TASK SETUP

6.1 Hedge projection structure

6.1.1 Overview

A variable annuity hedging projection involves economic statement calculations based on liability economic reserves calculated monthly and liability Greeks at the hedge rebalancing frequency, such as quarterly, along a projection horizon, say 30 years, for a number of outer loop scenarios.

Since the economic reserves and liability Greeks change over time along the outer loop, new risk neutral shock scenarios must be generated, economic reserves and liability option values calculated, using liability model points which are aged through time along the outer loops. A nested projection is required to accomplish these valuation calculations.

Typically, an experience projection using the valuation task results would be included in the nested projection to perform the hedging strategies through the same time steps along the outer loop. However, since the economic statements do not consider the investment asset portfolio and the variable annuity values are assumed to be independent of any asset yields, there is no need to link the experience projection task with the valuation tasks inside a single nested projection.

6.1.2 Two step process

Hedging projections require two steps to complete.

The first step is to calculate the economic reserve and liability option values through the entire projection horizon over the desired outer loop scenarios using a **nested projection**. The nested projection processes the outer loop scenarios within which the inner loop shock final valuations are executed to calculate the economic reserves and liability option values. As with all nested projections, it is critical that all projection tasks included in the nested projection use the same `valn_periods_selected` setting, ensuring that the results of each projection task are aligned.

The second step is to calculate the economic statements, using results from step 1 and including the execution of hedge strategies, through the entire projection horizon over the desired outer loop scenarios using a **standard** projection. The standard projection processes the outer loop scenarios, reading the results for the economic reserves and liability option values from the step 1 projection results, but calculating results for the hedge assets, hedge strategies, and economic statements

6.2 Hedge projection step 1 setup

As discussed, the first step of the hedge projection is to calculate the economic reserve and liability option values through the entire projection horizon over the desired outer loop scenarios using a nested projection. The nested projection is structured using three separate projection tasks; a shock scenario generation task, a hedge shock valuation task, and a hedge final valuation task.

The following describes the setup of the individual task and run controller for the nested projection.

6.2.1 Shock scenario generation task

Follow the instructions of chapter 6 of the C02 Economic scenarios book to setup a risk neutral shock scenario generation task.

6.2.2 Shock valuation task

Follow the instructions above to setup the hedge shock valuation task for the nested projection.

6.2.3 Final valuation task

Follow the instructions above to setup the hedge final valuation task for the nested projection.

6.2.4 Run controller projection set

Setup a projection set in the Run controller, with a name such as 'Hedging Projection – Step 1'. Specify that the projection set should have independent loops and include sub loops.

Add the projection tasks to the 'Sub Loop' section of the projection set such that the scenario generation is first, the shock valuation task is second, and is dependent on the shock scenario generation, and the final valuation task is last and is dependent on the shock valuation task.

Set the projection set sub loops to the number of time steps to process and the projection set loops to the outer loop scenarios to process.

A sample projection set for step 1 of a hedging portfolio projection for outer loop scenarios 1-10 with inner loop valuations at time periods 0 to 36 (1 to 37 sub loops) is setup as follows:

D1. Hedge Portfolio Projection - Step 1 Last Run:

Projection Set Loops: ☒ Independent Loops ☒ Include Sub Loops

Projections before Sub Loop:

Projection	Dependencies
Enter Projection name	

Sub Loop: Projection Set Sub Loops:

Projection	Dependencies
Hedge Valn Shock Gen Proj	Hedge Valn Shock Gen Proj
Proj Shock Valn IMP	Proj Shock Valn IMP
Proj Final Valn IMP	
Enter Projection name	

Projections after Sub Loop:

Projection	Dependencies
Enter Projection name	

6.3 Hedge projection step 2 setup

As discussed, the second step is to calculate the economic statements, using results from step 1 and including the execution of hedge strategies, through the entire projection horizon over the desired outer loop scenarios

The following describes the setup of the individual tasks and run controller for the nested projection.

6.3.1 Combine Time step results

Since hedge projection step 1 is a nested projection, the final valuation results are split into separate output files for each outer loop and time step.

For example, a column output file for projection set loop 1, sub loop 3 of the final valuation task discussed above would be named "Heging Portfolio_Proj Final Valn IMP@1_3~company~hedge_grp.csv".

These output files cannot be used directly in the step 2 projection. They must first be combined into a new output file which has the same structure as an output file from a standard projection. This is accomplished using a 'Combine Timesteps' Other Task which executes the combine time step utility.

Specify the 'Combine Timesteps' Other Task properties as follows:

Property	Values
Other Task Name	Combine Timesteps
Program File	Specify <*global*>Combine TimeStep.exe Due to a limitation in RiskAgility FM, if the task is to be executed on a grid, the path to the program must be specified as the appropriate UNC path.

Property	Values
Optional Parameter	Specify the parameters to the Combine Timestep utility as follows: "[output file reference]" "[combine time step definition file]" "*" For example, the optional parameters for combining the results from the final valuation task discussed above would be "<*output*>Hedging Portfolio_Proj Final Valn IMP@" "<*assump*>Hedge time step definitions.csv" "*" Due to a limitation in RiskAgility FM, if the task is to be executed on a grid, the paths must be specified as the appropriate UNC paths.
Wildcard Set	Specify the wildcard set to use
Grid Compatible	Tick the box to indicate that the Combine Timestep Other Task can be run on the grid.

The combine time step definition file referred to in the option parameters is a comma separated value (CSV) file that contains the definitions of the nested projection outer loops and valuation periods, and whether the combine process is for valuation results or projection results.

For example, the combine time step definition file required for combining the final valuation task discussed above would contain the following information:

	A	B
1	OuterLoops	1-10
2	ValnPeriods	0-36
3	ValnFlag	1

6.3.2 Experience projection task

The following describes the setup of a standard projection task for the experience projection task for step 2 of the hedging projection.

Input variables

Specify hedge groups to be included in the segment assumptions, on the Company input page under the Model Setup: Segment category.

Specify general historic rates and projection scenario files to use, on the Economy input page under the Scenarios : Rates category.

Specify the hedge asset assumptions, including Index option and Interest Rate Derivative purchase asset definitions on the Investment input page.

Specify the variable annuity hedge groups, on the Hedging input page under the Hedge Groups category.

Specify hedge group results from the combined time step hedge final valuations in the valn_hedge_final_results variable on the Hedging input page under the Hedge Valuation: General category using the system <*column_output_directory*> wildcard.

An example file reference would be <*column_output_directory*>combinedTS\Hedging Portfolio_Proj Final Valn IMP~company~seg~hedge_grp.csv.

Specify the hedge shocks, on the Hedging input page under the Hedge Valuation: Control category.

Specify the hedge strategies, on the Hedging input page under the Hedge Strategy categories.

Specify variable annuity product specifications and assumptions, on the various VA[XXX] input pages.
Specify the in force variable annuity model points in the variable annuity model point files.

Specify hedge assets in force at the start of the projection in the index option and interest rate derivative model point files.

Run page

Create a projection task on a 'Hedging Portfolio' Run page which uses the company submodel as the target model object for portfolio based valuation tasks.

Run parameter set

Specify the following properties in the Run parameter set.

Property	Values
Target column	Specify proj_tgt_hedge_stmt from the company submodel
Projection period from	0
Projection period to	Specify the desired projection period
Projection period min	-600, to allow for external source lookup keys calculating before the projection date (such as pol_yr)
Projection period max	At least as long as the calculations for each submodel
Projection task loops	Specify the number of scenarios to run, such as 1-10. Should match the setup of the projection set loops used in step 1.
Sliding windows	Do Not Slide for seriatim runs and when small models or independent model point processing enables portfolio runs to fit within available memory; Slide to reduce memory requirements during a projection task which does not otherwise fit into available memory

Project variables

Also specify the project variables in the Run parameter set.

Property	Values
Projection date	Projection date is the end of the month specified as yyyy/mm
Include assets?	Yes, to include the hedge asset processing
Use independent model point processing?	Yes, typically
Validate data definitions?	No for production runs; Yes for testing
Include decrements?	Yes

Output parameter set

Specify the following properties in the Output parameter set.

Property	Values
Output manager	Select the 'Proj' output manager to limit the output captured
Model object output	Select All Columns or 'Hedge' Output Definitions for the company, segment, and hedge_grp submodels plus any other model objects for which output is desired. When an output manager is specified, selecting Output Definitions restricts the output to include only those columns included in the output definitions

6.3.3 Run controller projection set

Setup a projection set in the Run controller, with a name such as 'Hedging Projection – Step 2'.

Add the Combine Timesteps other task and experience projection task to the projection set. Ensure that the Combine Timesteps other task is first, then the experience projection task, which is set to be dependent on the Combine Timesteps other task.

A sample projection set for step 2 of a hedging portfolio projection is setup as follows:

^ D1. Hedge Portfolio Projection - Step 2 Last Run:

Projection Set Loops: ☒ Independent Loops ☐ Include Sub Loops

Projection	Dependencies
Combine Timesteps	
Proj Statement Enter Projection name	Combine Timesteps

7 SUBMODELS STRUCTURE

The following describes aspects of the various submodels of the U.S. Library which are involved in the hedging calculations.

7.1 company submodel

7.1.1 Variables

There are no variables in the company submodel specific to the hedging valuations and projections.

7.1.2 Formulas

The formulas for setting up and controlling the hedging valuations and projections as well as the results for the company level economic statement are contained in the company submodel.

Several key formulas are:

setup_hedge_valn_shock_lists – Creates a map of the combination of valuation shocks and scenarios to process. This function is call from startup of the company submodel.

setup_hedge_proj_shock_lists – Creates a map of the valuation shocks used in the hedge projection. This function is call from startup of the company submodel.

valn_tgt_hedge_shock – Controls the valuation hedge shock process, calling the valn_tgt_hedge_shock formula of the segment submodels. This is the target column for projection tasks that execute the hedge shock valuations.

valn_tgt_hedge_final – Controls the valuation hedge final process, calling the valn_tgt_hedge_final formula of the segment submodels. This is the target column for projection tasks that execute the hedge valuation final calculation.

proj_tgt_hedge_stmt – Controls the hedge projection process, calling the proj_tgt_hedge_stmt formulas of the segment submodels and the hedge projection results of the company submodel. This is the target column for projection tasks that produce the hedge projection results.

hedge_cash_flow – The hedge cash flows. The company level result is the sum of the hedge cash flows of the segment submodels.

hedge_mkt_val – The hedge asset market value. The company level result is the sum of the hedge market value of the segment submodels.

hedge_cost - The hedge cost, which is the negative of the hedge cash flow plus the hedge market value increase.

econ_prof – The economic profit including the impact of hedging. The company level result is the sum of the economic profit of the segment submodels.

econ_prof_unhedged – The economic profit without hedging. The company level result is the sum of the unhedged economic profit of the segment submodels.

7.2 segment submodel

7.2.1 Variables

The variable for determining which hedge groups are included in the hedging valuations and projections is the hedge_grps_selected variable of the segment submodel.

7.2.2 Formulas

The formulas for setting up the hedge groups and controlling the hedging valuations and projections as well as the results for the segment level economic statement are contained in the segment submodel.

Several key formulas are:

get_hedge_items – Captures the amounts from the pre generated hedging file.

setup_hedge_grps – Sets up the elements of the hedge_grp array based on the hedge groups which are included in the projection. This function is called from initialize of the segment submodel.

valn_tgt_hedge_shock – Controls the valuation hedge shock process, calling the valn_tgt_hedge_shock formula of the hedge group submodels. This formula is executed from the valn_tgt_hedge_shock formula of the company submodel.

valn_tgt_hedge_final – Controls the valuation hedge final process, calling the valn_tgt_hedge_final formula of the hedge group submodels. This formula is executed from the valn_tgt_hedge_final formula of the company submodel.

proj_tgt_hedge_stmt – Controls the hedge projection process, calling the proj_tgt_hedge_stmt formulas of the hedge group submodels.

sum_over_hedge_grps – This function is used to sum the results from the hedge group submodels. A column name at time period is passed to the function.

hedge_cash_flow – The hedge cash flow. For fixed indexed annuity and indexed universal life liability-only models, the hedge cash flow is the sum of the hedge cash flow of the liability model points. For asset/liability models, the hedge cash flow is the sum of the hedge cash flow of the hedge group submodels. For products

other than fixed indexed annuity, indexed universal life, and variable annuity, the hedge cash flow is from the pre generated hedge results.

hedge_mkt_val – The hedge asset market value. For fixed indexed annuity and indexed universal life liability-only models, the hedge market value is the sum of the hedge cash flows of the liability model points. For asset/liability models, the hedge market value is the sum of the hedge cash flows of the hedge group submodels. For products other than fixed indexed annuity, indexed universal life, and variable annuity, the hedge market value is from the pre generated hedge results.

hedge_cost - The hedge cost, which is the negative of the hedge cash flow plus the hedge market value increase.

econ_prof – The economic profit including the impact of hedging. The segment level result is the sum of the economic profit of the hedge group submodels.

econ_prof_unhedged – The economic profit without hedging. The segment level result is the sum of the unhedged economic profit of the hedge group submodels.

7.3 rates submodel

7.3.1 Variables

The variables for defining whether a scenario generation is a shock scenario generation task, the scenario generation shocks, generation assumptions, and calibration criteria for the risk neutral shock scenario generation are contained in the rates submodel.

7.3.2 Formulas

The formulas for setting up the scenario generation shocks and assumptions are contained in the rates submodel.

Several key formulas are:

generate – Executes the generate process. This formula is the Target column for all scenario generation tasks.

setup_generation_parameters – Sets up the generation parameters, passing the necessary data to the Economy library. This is called from the generate formula.

setup_economy_data – Sets up the generation economy data, passing the necessary data to the Economy library. This is called from the generate formula.

setup_calibration_criteria – Sets up the generation calibration criteria, passing the necessary data to the Economy library. This is called from the generate formula.

setup_shock_gen_list - Creates a map of the scenario generation shocks. This function is call from the generate formula of the rates submodel. This is called from the generate formula.

7.4 hedge_grp submodel

7.4.1 Multiple instances

For fixed indexed annuity and indexed universal life asset/liability models, and for all variable annuity hedging, the hedge group submodel is used to control the processing of and report for model points that are collected into the hedge groups.

The hedge grp submodel is a two dimensional user defined array of the hedge_grp_comp model class.

The size of the first dimension is number of hedge groups included in the model, as defined by the segment hedge_grps_selected variable.

The second dimension size varies based on the application. In the fixed indexed annuity application, the second dimension is always one. In the indexed universal life application, the size of the second dimension is the number of buckets required based on the index crediting periods. In the variable annuity application, the second dimension is one for the hedging shock valuation and hedge projection tasks but is sized to equal the number of shocks processed for the hedging valuation final task, thus allowing for more efficient processing and reporting of the shock final results.

7.4.2 Variables

The variables that define the pre generated hedge amounts, hedge group definitions, and variable annuity hedge valuation and hedge projection strategies are contained in the hedge group submodel.

7.4.3 Formulas

The formulas for initializing the hedge groups and controlling the hedging valuations and projections as well as the results for the hedge group level economic statement are contained in the hedge group submodel.

Several key general formulas are:

initialize – Initializes the hedge group submodel element and maps the asset and liabilities to the hedge group. This formula is called from the `setup_hedge_grps` formula of the segment submodel.

hedge_cash_flow – The hedge cash flow, including hedge asset payments, hedge asset sales, hedge asset investment amounts, and hedge asset transaction costs.

hedge_mkt_val – The hedge asset market value, which is the sum of the end of period market value of all hedge assets mapped to the hedge group.

hedge_cost – The hedge cost, which is the negative of the hedge cash flow plus the hedge market value increase.

Several key formulas for fixed index annuity and indexed universal life hedging are:

notional_amt_to_hedge – The notional amount required for hedging index crediting, which is the sum of the notional amount to hedge of the liability model point index fund accounts mapped to the hedge group.

long_strike – The long strike of the index option to be purchased to hedge the index crediting of the liability model point index fund accounts mapped to the hedge group. This is a weighted amount based on notional amount to hedge.

cap_rate – The cap rate for the index option to be purchased to hedge the index crediting of the liability model point index fund accounts mapped to the hedge group. This is a weighted amount based on notional amount to hedge.

short_strike – The maximum payout index for the index option to be purchased to hedge the index crediting of the liability model point index fund accounts mapped to the hedge group. This is equal to the beginning of month index value times one plus the cap rate.

hedge_inv_amt_bom – Executes the hedge investment process and returns the amount invested in an index option to hedge the index crediting of the liability model point index fund accounts mapped to the hedge group.

hedge_notional_amt_available – The notional amount before sale of the hedge assets mapped to the hedge group.

notional_amt_required – The notional amount required to maintain a balance hedge, which is the hedge notional amount available less the fund value decrements of the liability model point index fund accounts mapped to the hedge group.

hedge_sale_amt – Executes the hedge sales process and returns the cash from sales of hedge assets due to rebalancing. The amount sold ensures the remaining notional amount of the hedge assets equals the notional amount required.

Several key formulas for variable annuity hedging are:

valn_tgt_hedge_shock – Controls the valuation hedge shock process, setting up the model point loop and calling the `calc_pv_claims_charges` formula. This formula is executed from the `valn_tgt_hedge_shock` formula of the segment submodel.

valn_tgt_hedge_final – Controls the valuation hedge final process, calling the present value of claims and charges and economic reserve at time 0. This formula is executed from the `valn_tgt_hedge_final` formula of the company submodel.

proj_tgt_hedge_stmt – Controls the hedge projection process, force calling the formulas required for the hedge projection. This formula is executed from the `proj_tgt_hedge_stmt` formula of the segment submodel.

calc_pv_claims_charges – Run a monthly process over a single variable annuity model point to calculate the present value of the claims and charges

get_hedge_valn_results – Retrieves the results from the hedge shock valuation projection task output files for use in the hedge valuation final projection task calculations.

get_hedge_valn_final_results – Retrieves the results from the hedge valuation final projection task output files for use in the hedge projection calculations.

calc_asset_greeks – Calculates the Greeks (delta, vega, gamma, rho, and rho convexity) for the hedge assets included in the portfolio and for a notional amount of a possible purchase hedge assets.

[Greek]_hedge_strat – Determines if the hedge strategy for the Greek (delta, vega, gamma, rho, and rho convexity) needs to be executed for the period.

[Greek]_hedge_sale_amt – Executes the hedge sales process for the Greek (delta, vega, gamma, rho, and rho convexity) and returns the cash from the hedge asset sale.

[Greek]_inv_amt_eom – Executes the hedge investment process for the Greek (delta, vega, gamma, rho, and rho convexity) and returns the amount required for the hedge asset purchase.

econ_prof – The economic profit including the impact of hedging.

econ_prof_unhedged – The economic profit without hedging.

7.5 inv_strategy submodel

7.5.1 Variables

The variables for defining the hedge assets are contained in the `inv_strategy` submodel.

7.5.2 Formulas

The formulas for executing the purchases and sales of the hedge assets are contained in the `inv_strategy` submodel.

Several key formulas are:

sell_assets_hedges – Executes the sales process for index crediting hedges and returns the cash received from the sale.

buy_assets_hedging – Executes the purchase process for both the index crediting and Greek hedging. This function finds the purchase asset to buy and calls the `buy_purch_asset_hedging` function and then returns the cash amount invested in the hedge asset.

buy_purch_asset_hedging – Creates a new element and sets up the purchase hedge asset, calling the appropriate functions from the individual asset type submodels, and then return the cash amount invested in the individual hedge asset.

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L01 TERM LIFE

This book discusses term life insurance products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

- Product – Benefits, premiums, cash values, and factor calculations definitions
- Valuation – Statutory and tax reserves, PBA net premium reserve, and actuarial math library audit report option
- Expense – Commission and expense rates, premium tax rate
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse rates and term conversions

Each of these input page types above begins with 'Term', for use in projections that combine multiple product types.

Additionally, variables defining reinsurance and rider specifications and assumptions are associated with input pages which are common across products. These are:

- Reinsurance – Reinsurance treaty definitions and rates.
- Riders – Riders specifications and assumptions
- RidersValuation – Riders statutory and tax reserve definitions

1.2 Data pages

Variables for each term model point are specified on the TermData page. These variables are discussed in the Model point records chapter, below.

Variables for each reinsurance record are specified on the TermReinsData page. These variables are the same for the term life, traditional life, universal life, variable universal life, indexed universal life, and long term care product types. These variables are discussed in the Reinsurance model point records chapter, below.

Variable for each rider record are specified on the TermRidersData page. These variables are the same for term life, traditional life, universal life, variable universal life, and indexed universal life products. These variables are discussed in the L13 Riders book.

2 PRODUCT BENEFITS AND VALUES

The following explains the basic features for term life products. Permanent life products are discussed in the Traditional life book.

2.1 Product variations

While the standard features are flexible for defining almost any product configuration, the following are common types of term life products.

2.1.1 ART

Annually renewable term (ART) products are also called yearly renewable term (YRT) products. Specify the annual premium per unit as rates changing by policy year. For example, premiums can vary by attained age.

2.1.2 Level term

The premium is level for each year of coverage, which expires at the end of the level term years specified on the model point record. Specify the premium in the level premium per unit variable, which varies premiums by benefit term (level term years) rather than by policy year.

2.1.3 Level term/ART

Following the level term years specified on the model point record, the policy provides ART coverage. The level term premium applies until the end of the level term years, then the annual premium per unit applies for the remaining policy years.

2.1.4 ROP term

Return of premium (ROP) term products provide level term coverage with a cash value that returns cumulative premiums at the end of the level term years. Surrenders before that time receive cumulative premiums times a percent that varies by policy year.

- ROP term products automatically provide this feature; the level term premium per unit includes the premium for the ROP feature.
- ROP term riders add this feature as an option; the ROP rider premium is paid in addition to the level term premium.
- ROP/ART products and riders continue ART coverage after the level term years.

ROP products and their riders are discussed in a separate chapter, below.

2.1.5 Renewable term

Renewable term has a level premium for each renewable term period, such as 10 years. The premium increases at the beginning of each renewal period, and shock lapses occur based on the size of the premium increase.

2.1.6 Decreasing term

Decreasing term products have the sum assured declining by policy year or months since issue. The premium is typically level, and can stop before the coverage ends. For example, mortgage decreasing term products.

2.2 Benefit term

2.2.1 Maturity

The maturity (expiry) date is either a maximum attained age or policy anniversary, but not greater than the statutory reserve mortality table omega age. When this value is zero (default value), coverage expires at the omega age.

2.2.2 Level term maturity

When the maturity definition is Level Term Years, the policy expires after the level term years specified on the model point record. When the level term years (model point variable) is zero, the model point is skipped.

2.2.3 Renewable term maturity

For renewable term products, the last renewable term period is the period which reaches or extends past the maturity date. The last renewable term period can be shortened so that it does not extend past maturity.

2.3 Death benefits

2.3.1 *Sum assured*

The death benefit is the sum assured. While this is normally a level amount, such as \$1,000 per unit, the sum assured can vary annually (using the `pol_yr` column) or monthly (using the `mths_since_issue` column).

2.3.2 *Unearned premium*

The death benefit is increased by the annualized gross premium, times the remaining paid-to months, divided by 12.

2.4 Premiums

2.4.1 *Premium term*

Premiums are paid until either a policy year or attained age. When this value is zero (default value), premium is paid to maturity.

2.4.2 *Premium mode*

Specify the premium mode on the model point record. Premiums can be paid annually, semi-annually, quarterly, monthly, or special monthly (such as electronic transfer or pre-authorized check, with expense rate different from 'monthly' mode).

2.4.3 *Annual premium*

The annual premium per unit variable specifies the premium for each policy year. The exception is that, during the level term years, this premium is replaced by the level term premium (discussed below).

For renewable term products, the premium increases must occur in the first year of each renewal period.

The specified annual premium per unit may be increased by a multiple following the level term period.

2.4.4 *Level term premium*

During the level term years, the annual premium is replaced by the level term premium per unit.

Using annual premium variable in all policy years

When the level term premium variable is zero (default), the annual premium variable applies in all policy years, even during the level term years. Using this input method:

- A different plan code must be defined for each level term period (such as one plan code for 10 years, another for 20 years, and another for 30 years).
- The level term years variable on the model point record must match the plan code definition. For example level term years must be 10 for the '10 years' plan code.
- The premiums in the annual premium variable must be level during the level term years.

2.4.5 *ROP term rider premium*

When the model point includes the ROP term rider, the level term premium per unit is increased by the ROP rider premium. This is either an additional premium per unit, or a multiple of the level term premium per unit.

2.4.6 *Substandard premium*

The annual premium (and level term premium) per unit are adjusted by the substandard multiple specified on the model point record.

2.4.7 *Modal loading*

When premiums are paid more frequently than annually, the annualized premium is the annual premium times the annualized modal loading multiple varying by premium mode. For example, an annualized modal loading multiple of 1.08 means that monthly premiums are 9.0% ($1.08 / 12$) of the annual premium per unit.

Vary the modal loading multiple by premium mode by linking to an external source varying by the prem_mode model point variable. For example:

prem_mode	
Monthly	1.08
Special Monthly	1.08
Quarterly	1.06
Semi-annual	1.02
Annual	1.00

2.4.8 *Policy fee*

The premium is increased by the annualized policy fee varying by premium mode. For example:

prem_mode	
Monthly	43.20
Special Monthly	43.20
Quarterly	42.40
Semi-annual	40.80
Annual	40.00

2.4.9 *Guaranteed premium*

Premium pattern

Cash value and reserve net premiums are based on the pattern of guaranteed premiums by policy year, which can be different from the pattern of current premiums. For this purpose, the guaranteed premiums are used only to specify the gross premium pattern. Thus product pricing projections can calculate cash values and reserves even before the actual premiums have been determined.

Guaranteed level term premiums can vary by policy year, even though the current premium is level.

(See below that cash values and reserves are calculated using only the guaranteed premium per unit, without adding the policy fee or modal loading.)

Deficiency reserves

Deficiency reserves compare the reserve net premium to the greater of the current premium and the guaranteed premium for the annual premium mode. For this purpose, the guaranteed premiums must be known. When the current premiums are guaranteed, simply copy the current premium table to the guaranteed premium external source input. When the guaranteed premium per unit is less than the annual premium per unit, the guaranteed premium is the annual premium. Thus you can leave this variable at its 1.0 default value.

(See in the Deficiency reserves section in the Statutory and tax reserves chapter, below, that the guaranteed premium can include both the policy fee and modal loading, the policy fee only, or neither.)

2.4.10 *DAC tax capitalized premium percent*

Specify (with the Premiums input) the percent of each premium which is capitalized for the capitalized premium tax, which is also called the DAC tax. See the Taxes chapter in the Profit calculations book.

2.5 Cash value

Cash values are calculated using unitary calculations from issue through maturity. ROP term cash values are discussed in the ROP term chapter, below.

2.5.1 Cash value methods

The following cash value methods are available:

- Not applicable – cash values are zero
- SNFL minimum (both 1980 and pre-1980 versions of the U.S. standard nonforfeiture law)
- Net level premium – no expense allowance
- Full preliminary term – zero terminal cash value for the first policy year
- Table – specify net premiums and terminal cash values per unit for each policy year, starting with 0 (issue date) for the terminal cash values.

2.5.2 Death benefit varying within policy year

When the death benefit varies within a policy year, the death benefits used for cash value calculations are the average of the death benefit as of the end of each month within the policy year. For example, when the sum insured varies monthly.

2.5.3 Premium pattern

Cash value net premiums are the level percent of gross premiums calculated using the specified cash value method. For product pricing work, before the gross premiums have been determined, specify the premium pattern in the guaranteed premium variables. For example, the default value of '1' for all policy years specifies that premiums are level.

The cash value method applies to the pattern of guaranteed premiums without policy fees and modal loadings.

2.5.4 Cash value basis

Mortality

Cash value mortality is a base table only.

The substandard multiple (on the model point record) does not apply to cash value mortality.

Interest

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeNonforfeiture range in the Valuation Interest Rates Excel workbook.

Function timing

Function timing choices are:

- **Curtate** – Annual premiums and EOY claims
- **Continuous** – Continuous premiums and time-of-death claims
- **Semicontinuous** – Annual premiums and time-of-death claims.

The continuous premium multiple and the immediate claim multiple are defined in the Function timing section in the Valuation options chapter of the Actuarial math library appendix.

2.5.5 *Nonforfeiture exception rules*

The following nonforfeiture exception rules can apply. When the exception rule is met, the cash value is zero for all policy years.

- **Term 15/65:** Cash values are zero for: (1) Level term policies of 15 years or less expiring before age 66, (2) Decreasing term policies with a net premium less than the net premium for a 15 year level term policy issued at the same age and same initial amount, and (3) when the value of the nonforfeiture benefit at the beginning of each policy year would be less than 2.5% of the sum assured.
- **Term 20/70:** Cash values are zero for: (1) Level term policies of 20 years or less expiring before age 71, (2) Decreasing term policies with a net premium less than for a 20 year level term policy expiring before age 71 issued at the same issue age and same initial amount, and (3) the 2.5% test.
- **Term 30/80:** Similarly, using policies of 30 years or less expiring before age 81, with the 2.5% test.

2.5.6 *Mid-year cash values*

The mid-year cash value is the paid-to-date interpolated terminal cash value, but not less than zero. The paid-to-date is the end of the month before the next premium payment.

After the premium term, the terminal cash values are interpolated monthly.

3 ROP TERM

This chapter discusses the special ROP cash value feature. ROP term reserves are discussed in the Statutory and tax reserves section, below.

3.1 Product versus rider

3.1.1 *Product type designation*

Level term products (when Level term years on the model point record is greater than the zero default value) can include the ROP cash value feature:

- **Product** – When the ROP term definition (Level term includes ROP?) is Yes, this is an ROP term product. The level term premium per unit includes the premium for this feature. The ROP rider premium per unit has no effect.
- **Rider** – When the definition is No, a model point includes the ROP rider when the ROP rider definition (Include ROP rider?) on the model point record is Yes. The ROP rider premium is added to the level term premium per unit.

3.1.2 *Additional premium*

For ROP term riders, the ROP term premium is paid in addition to the level term premium. See the Premiums section in the Product benefits and values chapter, above.

The same commission rates apply for the ROP premium as for the base annual premium.

3.2 Endowment amount

The return of premiums amount is paid as a pure endowment to policyholders persisting to the end of the level term period. The endowment amount is the cumulative annualized premiums paid since issue, excluding any substandard premium. Thus the endowment includes modal loadings and policy fees.

3.3 Cash value

The cash value can either be calculated using AG 45 (discussed below), or can be a percent of the cumulative level term and ROP rider premiums paid since issue. The cumulative premiums include the modal loading and policy fee, but exclude any substandard premium.

You can specify a percent varying by policy year by linking to an external source table using the `pol_yr` column as a lookup key. For example, when the percent is 10% for policy year 5 and 20% for policy year 6, surrenders 60 through 71 months since issue receive 10% of cumulative premiums. Surrenders 72 through 83 months since issue receive 20% of cumulative premiums.

You can vary the percent monthly using the `mths_since_issue` column.

3.4 Actuarial guideline 45

Actuarial Guideline XLV (AG 45, formerly called actuarial guideline CCC) defines the minimum cash value during the level term period. Instead of specifying cash value percentages in a table, the cash value at the end of each policy year is calculated as explained below. The mid-year cash value is the paid-to-date interpolated terminal cash value.

The minimum cash value at the end of each policy year is: (1) PV of the endowment benefit at the end of the level term period, less (2) PV of future cash value net premiums to the end of the level term period using the specified expense allowance.

- The endowment benefit for this calculation assumes annual premium payments, including the policy fee. Thus only the modal loading is excluded from this calculation.
- The cash value renewal net premiums are a level percent of annual premium payments, excluding the policy fee. Thus both the modal loading and policy fee is excluded from this calculation.
- The expense allowance is (1) 1% of the average death benefit at the beginning of the first ten policy years, plus (2) 125% of the minimum of the net level premium for the level term plan of insurance plus the pure endowment amount and 4% of the average death benefit at the beginning of the first ten policy years.
- Any substandard premium is excluded from these calculations.

The minimum cash value is the greater of this result calculated using: (1) current premiums, and (2) guaranteed premiums.

4 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves. PBA reserves and capital are discussed in the PBA book.

4.1 Reserve methods

4.1.1 Terminal reserves

The following reserve methods are available:

- Not applicable – reserves are zero, and don't add to projection time.
- 1980 CRVM, and pre-1980 CRVM
- Net level premium – no expense allowance
- Full preliminary term – zero terminal reserve for the first policy year
- NY 147 amendment 5 – expenses allowances are calculated using the requirement of the 5th amendment of NY regulation 147

- Table – net premiums and terminal reserves per unit are specified for each policy year, starting with 0 (issue date) for the terminal reserves.

4.1.2 **Death benefit varying within policy year**

When the death benefit varies within a policy year, the death benefits used for reserve calculations are the average of the death benefit as of the end of each month within the policy year. For example, when the sum insured varies monthly.

4.1.3 **Premium pattern**

Reserve net premiums are the level percent of gross premiums calculated using the specified reserve method. For product pricing work, before the gross premiums have been determined, specify the premium pattern in the guaranteed premium variables. For example, the default value of '1' for all policy years specifies that premiums are level.

The reserve method applies to the pattern of guaranteed premiums without policy fees and modal loadings.

4.2 **Reserve basis**

4.2.1 **Mortality**

Reserve mortality is a base rate, times the multiple, then plus the addition. Lookups for reserve mortality tables should be setup similar to the experience mortality rates since these rates are read for both the first and second lives.

4.2.2 **Interest**

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeStatValn and LifeTaxValn ranges in the Valuation Interest Rates Excel workbook.

4.2.3 **Function timing**

Function timing can be:

- **Curtate** – Annual premiums and claims paid at the end of the policy year
- **Continuous** – Continuous premiums and immediate payment of claims
- **Semicontinuous** – Annual premiums and immediate payment of claims.

The continuous premium multiple and the immediate claim multiple are defined in the Function timing section in the Valuation options chapter of the Actuarial math library appendix.

4.3 **Regulation XXX**

NAIC model regulation XXX applies to traditional life policies issued in 2000 and after, although you can apply it to earlier issues. NY regulation 147, which is a variation of the early XXX regulation, is also available.

4.3.1 **Segmented reserves comparison**

Regulation XXX reserves are the greater of the unitary and segmented reserves:

- **Unitary** – Reserve net premiums are a level percent of gross premiums over the entire benefit period. The reserve expense allowance is calculated using the entire plan of insurance.
- **Segmented** – Reserve net premiums are a different percent of gross premiums for each segment (discussed below) of the benefit term. The reserve expense allowance is calculated using only the

benefits and premiums during the first segment. The net level premium method applies after the first segment.

The reserve is the greater of the mid-year reserves calculated separately using the unitary and segmented definitions.

The reserve net premium, for calculating deferred net premiums and unearned net premiums, is the net premium for either the unitary or segmented reserve, whichever determines the reserve for the current month.

4.3.2 Segments definition

The levelized premium segmentation method divides the benefit period into segments when the ratios, $G(t)$, of successive guaranteed premiums exceed the ratios, $R(t)$, of successive reserve basis mortality rates. The first segment ends prior to the policy year (designated by 't') when $G(t)$ first exceeds $R(t)$. The next segment extends from the end of the first segment until the next segment starts at the beginning of the policy year when $G(t)$ again exceeds $R(t)$, and so on.

The mortality rate ratios, $R(t)$, can (option choice) be multiplied by 101% or 99% to avoid segments based on insignificant excess values. However, $R(t)$ cannot be less than 1.

Segments are determined using reserve mortality variables `seg_xxx_mort_base`, `seg_xxx_select_input_1`, and `seg_xxx_mort_improve_pct`. These variables typically contain the same tables as for deficiency reserve mortality for the first segment, but excluding the deficiency reserve X factors (discussed below). Separate variables are provided so that XXX deficiency reserves can use mortality varying by the 2001 CSO rate classes, while the segments are determined using the version of the table which does not vary by rate class, as allowed by actuarial guideline XLVI (AG 46).

The same segments apply for statutory, deficiency, and tax reserves.

4.3.3 Select mortality factors

Select mortality factors apply both for the determination of the segments and for the calculation of the segmented reserves. In addition, a table of 'X factor' multiples can apply to the deficiency mortality rates for calculating deficiency reserves during the first segment, but does not apply for determining the segments.

2001 CSO mortality – The Global folder includes the 25-year select multiples that apply for 2001 CSO mortality; see the Mortality Select Multiples Excel workbook. When using these select mortality multiples, the base mortality variable must specify the ultimate rates. When the 2001 CSO select rates apply past the first segment, the select and ultimate mortality rates can be specified as the base variable, leaving the select factor variables at their default values of 1.0 (100%). However, when 'X factor' multiples are used, the base mortality variable must specify the ultimate rates with separate select factors specified so that it is the combined select factors which are constrained to 1.0 (100%) as opposed to just the 'X factor' multiple being constrained.

1980 CSO mortality – The Global folder includes 10-year, 15-year, and 20-year select factors which apply to the 1980 CSO ultimate mortality rates; see the Mortality Select Multiples Excel workbook. For 1980 CSO mortality, the reserve mortality variable must specify the ultimate rates, then the select factor variables (when select mortality is elected for reserves) must specify the following:

The 10-year select factors are implicit in the 1980 CSO select and ultimate mortality rates. This select factors apply for subsequent segments when the first segment is shorter than 10 years.

The following variations for applying special select factors during the first segment:

- Regulation XXX can apply either the 10- or 20-year select factors during the first segment.
- For NY 147, the first segment uses 15-year select factors adjusted by a multiple of 150% (120% for deficiency reserves), with the resulting select factor rounded to the nearest whole percent and capped at 100%. These select factor multiples apply only when reserves segment definition is NY 147; these variables are ignored for regulation XXX.

4.3.4 Mortality improvement

Mortality improvement can apply both for the determination of the segments and for the calculation of the segmented reserves, such as is allowed for under the 5th amendment of NY regulation 147. Mortality improvement applies from a specified improvement base year.

When included, mortality improvement multiples are applied to the select factor adjusted mortality rates in the first segment only.

4.4 ROP term reserves

4.4.1 Reserve methods

Policies including ROP term have reserves calculated either as:

- Greater of base term reserve and cash value
- Base term reserve plus cash value
- Pure endowment method – regulation XXX (or NY regulation 147) including the return of premiums paid at the end of the level term period
- Unusual cash values method – (discussed separately below)
- The greatest of these methods.

Comparisons of the greater of multiple methods are performed using the mid-year reserve values.

4.4.2 Unusual guaranteed cash values

The cash value at the end of a policy year is 'unusual' when the increase over the previous year exceeds the sum of:

- 110% of that year's gross premium, and
- 110% of the nonforfeiture interest rate times [prior year cash value + current year gross premium].

For the policy years through the first unusual cash value, the reserve cannot be less than a term policy over those years and endowing for the unusual cash value amount.

For the policy years until each subsequent unusual cash value, the reserve cannot be less than the net level premium method reserve for a term policy over those years and endowing for the unusual cash value amount, where the net premium percent is the present value of benefits less the previous unusual cash value.

For the final policy years after the last unusual cash value, the reserve cannot be less than the net level premium method reserve for a term policy over those years and endowing for the cash value at maturity, where the net premium percent is calculated using the present value of benefits less the previous unusual cash value.

4.5 Mid-year reserve

Reserves between policy anniversaries can be:

- Annual interpolated mean – Average terminal reserve, plus one-half of the annual reserve net premium. A deferred premium asset is established.
- Annual interpolated terminal – Average terminal reserves at the beginning and end of the policy year. Reserves are the same for each month within a policy year. An unearned premium reserve is established.

- Monthly interpolated mean – Monthly interpolated terminal reserves, plus the portion of the annual reserve net premium that applies to future months. A deferred premium asset is established.
- Monthly interpolated terminal– Monthly interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.
- Daily interpolated mean – Daily interpolated terminal reserves, plus the portion of the annual reserve net premium that applies to the remaining number of days of the year. A deferred premium asset is established.
- Daily interpolated terminal – Daily interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.

When the reserve function timing is Continuous, the mid-year reserve net premium is a discounted continuous net premium.

4.6 Actuarial guideline 32

Actuarial guideline XXXII (AG 32) requires additional reserves for policies which pay death claims at the time of death, but have statutory reserves calculated assuming death claims are paid at the end of the policy year. The additional reserve is:

guideline 32 percent (input) × statutory reserve interest rate × statutory reserve

This addition applies only when the reserve claims timing assumption is curtate (claims paid at the end of the policy year). The addition can be ignored when reserves are more conservative than the minimum statutory formula reserves. Specify the percent as 0.333 (33.3%, one-third) when claims are paid 'upon due notice', and 0.50 (50%) when claims are paid with interest from the date of death.

The additional reserve may not be included in tax reserves in cases such as for issues before 1994, when AG 32 first became effective. Thus this input variable is included separately for tax reserves.

The AG 32 reserve is added to the statutory reserve before calculating the deficiency reserve. The alternative minimum reserve is not affected by this adjustment. That is, the AG 32 reserve increases the basic reserve only, thus decreasing the deficiency reserve.

4.7 Deficiency reserve

4.7.1 1976 Standard valuation law method

The deficiency reserve is the 'alternative minimum' reserve less the statutory basic reserve, calculated using mid-year reserves, when at least one future alternative minimum reserve net premium exceeds the guaranteed gross premium. The alternative minimum reserve is calculated using the lower of the guaranteed gross premium and reserve net premium in each year. Specify the deficiency reserve mortality and interest basis separately from the statutory reserve basis.

As discussed above, the statutory basic reserve includes the AG 32 reserve, while the alternative minimum reserve is not affected by AG 32.

4.7.2 Guaranteed gross premium

The alternative minimum reserve is calculated using the guaranteed premium in policy years when the deficiency reserve premium would be higher. The guaranteed premium can include the policy fee and modal loading (annualized premium), the policy fee only (annual mode premium), or neither (the per unit premium).

When the guaranteed premium per unit is less than the annual premium per unit, the guaranteed premium is the annual premium. Thus you can leave this variable at its default value (1.0) when guaranteed premiums do not apply.

4.7.3 Regulation XXX

Regulation XXX (or NY regulation 147) applies the same for both basic and deficiency reserves. Specify deficiency reserve mortality separately from the statutory mortality. As explained in the Regulation XXX section above, deficiency reserve mortality includes a table of 'X factor' multiples that apply during the first segment.

Also as explained in the Regulation XXX section, segments are determined using reserve mortality variables `seg_xxx_mort_base`, `seg_xxx_select_input_1`, and `seg_xxx_mort_improve_pct`. These variables typically contain the same tables as for deficiency reserve mortality for the first segment, but excluding the deficiency reserve X factors. Separate variables are provided so that XXX deficiency reserves can use mortality varying by the 2001 CSO rate classes, while the segments are determined using the version of the table which does not vary by rate class, as allowed by actuarial guideline XLVI (AG 46).

When the mid-year statutory basic reserve is determined by the segmented reserve, the alternative minimum reserve is calculated using the segmented method. Otherwise, the alternative minimum reserve is calculated using the unitary method.

4.8 Additional reserve

Additional statutory reserves can be specified as an amount per unit. For example, vary these amounts monthly by linking to an external source using the `mths_since_issue` column as a lookup key.

The statutory additional reserve can be added to tax reserves as well.

The additional reserve is added after the determination of the deficiency reserve, and before the consideration of the tabular cost of insurance and cash value floors (discussed below).

When detail reports split out reserve values, 'additional reserves' include the increase over basic reserves due to the reserve floors.

4.9 Tabular cost of insurance floor

The reserve before adding the substandard reserve is subject to a minimum based on the tabular cost of insurance. That is, this floor applies to the sum of: (1) basic reserve (including the AG 32 reserve), (2) deficiency reserve, and (3) additional reserve.

The floor is the tabular cost of insurance (c_x) times the sum assured, times the following fraction for each mid-year reserve method:

- Annual interpolated mean – $\frac{1}{2} c_x$.
- Monthly interpolated mean – c_x times the fraction of the policy year remaining. For example, this floor at the end of policy month 1 is $\frac{11}{12} c_x$.
- Annual interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment. For example, this floor at the end of policy month 1 is $\frac{11}{12} c_x$ for annual premiums, and $\frac{2}{12} c_x$ for quarterly premiums.
- Monthly interpolated terminal– Same as above.
- Daily interpolated mean – c_x times the fraction of the policy year remaining measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $\frac{344}{365} c_x$.
- Daily interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $\frac{344}{365} c_x$ for annual premiums, and $\frac{71}{365} c_x$ for quarterly premiums.

4.10 Substandard reserve

The substandard reserve increases both the statutory and tax reserve. That is, substandard statutory reserve is tax deductible.

The substandard reserve is any excess of: (1) the mid-year statutory reserve calculated using substandard mortality, less (2) the mid-year statutory reserve calculated ignoring substandard mortality. The substandard multiple does not affect the AG 32 and alternative minimum reserve calculations.

The substandard reserve is added after the deficiency reserve and consideration of the tabular cost of insurance floor, but before consideration of the cash value floor.

4.11 Cash value floor

The total mid-year reserve is not less than the cash value.

4.12 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record sets the tax reserve to be the statutory reserve, including the comparison to the tabular cost of insurance floor using the statutory basis, and including the statutory AG 32 reserve, but excluding any deficiency reserve.

The same method choices are available for tax reserves, using the tax reserve interest and mortality basis. The function timing definition, and the mid-year reserve method, is the same as for statutory reserves.

When regulation XXX (or NY regulation 147) applies for tax reserves, the same segments apply as for statutory reserves.

The statutory additional reserve can be tax deductible.

The tabular cost of insurance floor, calculated using the tax reserve mortality and interest, is considered after addition of the additional reserve.

The substandard reserve is tax deductible, and thus added to the tax reserve.

The final mid-year tax reserve is not greater than the mid-year statutory reserve, and not less than the mid-year cash value.

4.13 Deferred premium asset

4.13.1 *Balance sheet*

When the mid-year reserve method is annual, monthly, or daily interpolated mean, a deferred premium asset offsets the formula's assumption that premiums are paid annually. This does not apply for other mid-year reserve methods, which recognize that premiums are paid modally.

The balance sheet includes modal reserve net premiums due during the remainder of the current policy year, but not greater than the deferred gross premiums. Deferred gross premiums exclude any substandard premium.

This is a non-invested asset. It is excluded from calculations of the amount to invest. The asset yield calculation removes the deferred net premium asset from the amount of assets generating investment income.

4.13.2 *Profit effect*

Book profit includes the increase in the deferred net premium asset. However, this is split into two parts, which add to the deferred net premiums increase:

- **Deferred gross premiums increase** – Premium income includes the increase in deferred gross premiums, which is calculated ignoring any substandard premium.
- **Statutory loading increase** – The statutory loading is the deferred gross premiums less deferred net premiums. When regulation XXX (or NY 147) applies, the net premium for this calculation is either the unitary or segmented net premium, using the result that determines the current reserve. For ROP term reserves, the net premium is similarly from the reserve method that determines the current reserve.

The deferred net premiums are not affected by reserve floors. That is, the statutory loading does not change when the reserve is increased due to the tabular cost of insurance or cash value floors.

Substandard premiums do not affect the statutory loading. The amount of deferred gross premiums excludes any substandard premium. Thus substandard premiums increase book profit as they are paid.

4.13.3 Taxable income

Taxable income similarly includes the tax loading increase, which is calculated the same way as statutory loading, but using the tax reserve basis.

4.14 Unearned premium reserve

4.14.1 Balance sheet

When the mid-year reserve method is annual, monthly, or daily interpolated terminal, an unearned premium reserve is established. The unearned premium reserve is the portion of the modal valuation net premium that has been collected but not yet earned.

The balance sheet includes the statutory unearned premium reserve as a separate line item from the statutory reserve. Unearned net premiums exclude any substandard premium. The unearned premium reserve is set to zero if the reserve being held is the tabular cost of insurance or cash value floor.

4.14.2 Profit effect

Book profit includes the increase in the unearned premium reserve.

4.14.3 Taxable income

Taxable income similarly includes the unearned premium reserve increase, which is calculated the same way as statutory unearned premium reserve increase, but using the tax reserve basis.

4.15 Gross premium reserve

Letter of credit (LOC) reserve financing creates a non-invested LOC asset that offsets the 'redundant' part of the statutory reserve, based on the calculation of the gross premium reserve (GPR) for each model point assigned to a reserve financing transaction. Book profit is increased by the LOC asset increase (offsetting part of the statutory reserve increase), and is decreased by LOC costs (interest paid on the LOC assets). See the Gross premium reserves chapter in the PBA book.

5 ACTUARIAL ASSUMPTIONS

5.1 Commissions

5.1.1 Premium commission

The percent of premium commission is the commission rate for each policy year, times the premium payments as they are paid. The policy fee and modal loading can be commissionable (included) or noncommissionable (excluded).

5.1.2 Annualized commission

A specified percent of the total first year premium commission is paid at issue. No further premium commissions are paid until the cumulative earned commission exceeds the annualized amount. From then on, premium commissions are paid as usual. This determination of commission payments is made for continuing policies, not taking into account terminated policies.

5.1.3 Commission chargeback

A portion of the premium commission can be charged back in the event of lapse or surrender during the first three years after issue. The chargeback percent applies either to: (1) total first year premium commissions paid to date, or (2) the annualized commission amount. The chargeback percent typically varies by month, by linking to the external source using the `mths_since_issue` column as a lookup key.

The commission subject to chargeback is calculated per unit in force, and thus is not reduced by decrements. This amount applies to the units lapsing and surrendering each month.

5.1.4 Commission overrides and agent terminations

The specified commission rates are the average commission rates, taking into account average writing agent commissions, commission overrides, and the effect of agent terminations.

5.2 Expenses

5.2.1 Expense adjustment multiple

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

5.2.2 Acquisition expenses

Acquisition expenses incurred at issue can include:

- Amount per policy
- Amount per unit
- Percent of the annualized first year premium

Acquisition expenses could vary by underwriting class, which is discussed in the Mortality section, below.

5.2.3 Maintenance expenses

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

- Amount per policy, separate between premium paying and paid-up
- Amount per unit, separate between premium paying and paid-up
- Percent of the annualized premium

BOM maintenance expense

An additional maintenance expense is incurred at the beginning of each month, when commissions are paid:

- Percent of each commission payment, ignoring commission chargeback.

5.2.4 Transaction expenses

Maintenance expenses include the following expenses related to administrative activities:

- Expense per death claim policy
- Expense per death claim unit

5.2.5 Reinsurance administration expenses

Reinsurance administration expenses are discussed in the Reinsurance chapter, below.

5.2.6 Inflation

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

Note: The average size for future issues can be inflated (to offset expense inflation), by customizing the scalar `units_per_pol` to divide the result by `infl_cumul(commencement_period)` for model points issued after the start of the projection. This would reduce the number of policies issued, with no effect on the units, sum assured, and premium issued for the model point.

5.2.7 Premium tax

Premium taxes are reported with expenses. The tax is a percent of each premium payment.

Inflation, and the expense adjustment multiple, do not apply to premium tax.

5.2.8 Fixed expenses

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

5.3 Mortality

5.3.1 Input variables

'Mortality variables' for term life products include mortality base, multiple, and addition rates, mortality improvement, and YRT reinsurance rates. These variables work the same as for traditional life products, where they are read twice – first for life 1 and then again for life 2. While term life does not include the feature for joint life products, the formulas work the same for consistency.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys `mort_issue_age`, `mort_gender`, `mort_smoking_class`, and `mort_rate_class` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

5.3.2 Mortality adjustment multiple

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

5.3.3 Underwriting class

Life insurance products can use the `underwriting_class` scalar to vary product assumptions such as mortality rates and acquisition expenses. The underwriting class is calculated by comparing the policy size against the nonmedical and paramedical underwriting limits specified with the experience mortality input. The underwriting class has a value of either:

- Nonmedical – when `ins_issued_per_pol` $\leq$ `nonmedical_limit`

- Paramedical – when `nonmedical_limit < ins_issued_per_pol <= paramedical_limit`
- Medical – when `ins_issued_per_pol > paramedical_limit`.

5.3.4 **Rate class**

Mortality rates varying by rate class require special treatment. The model point variable (`rate_class_mp`) can have a large number of options specific to the company's business. But the 2001 CSO mortality tables varying by rate class only define rates for Standard, Preferred, and Super. Thus there are two different rate class hidden variables:

- **Company rate classes** – The company's rate classes are in `mort_rate_class`. For example, there might be a dozen different classes for this scalar. Experience mortality and YRT reinsurance rates vary by this lookup key.
- **Reserve rate classes** – The 2001 CSO mortality rate classes are in `mort_rate_class_res`. The company's different classes are mapped into the three rate classes prescribed for reserve mortality rates. Customize the formula for this scalar, to define how each company rate class is translated into the three reserve rate classes.

5.3.5 **Base mortality**

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

5.3.6 **Mortality improvement**

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$. When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

5.3.7 **Mortality table adjustments**

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

The mortality multiple may be specified to not apply following the level term period. This would be appropriate when the mortality multiples represent adjustments for underwriting which is no longer applicable in the post-level term period.

5.3.8 **Substandard mortality**

The substandard multiple on the model point record applies to the mortality rate after the multiple and addition. This same substandard multiple applies for premiums and YRT reinsurance rates.

5.3.9 **Mortality anti-selection multiple**

Mortality anti-selection can either be calculated using shock lapse mortality, or specified as fixed rates possibly varying by premium increase and years since increase.

A mortality anti-selection multiple applies after the multiple, addition, and substandard multiple have been applied to calculate the mortality rates. This multiple is for specifying the effect of mortality anti-selection using a multiple calculated outside of the projection.

5.3.10 Shock lapse mortality anti-selection

'Shock lapse' mortality anti-selection can be calculated using the Dukes-MacDonald method:

- Mortality at normal lapses (w_1) follows the normal rates which would apply if there were no shock lapse. For subsequent shocks, 'normal' mortality is the anti-selection rates calculated for the previous shock event.
- A specified shock lapse (w_2) occurs at the beginning of a policy year, as a reaction to policyholders having to pay a higher premium. The shock lapse from a premium increase can be spread over several policy years (see the shock lapse rates discussion, below).
- A percent (effectiveness percent, E) of shock lapses (w_2) have mortality the same as for a new issue at the current attained age.
- The remainder ($1 - E$) of the shock lapses have the same anti-selection mortality as those persisting past the shock lapse event.
- Anti-selection mortality is calculated so that the total combined groups have the same deaths as if the shock lapse had not occurred.
- Shock lapses occur at the beginning of each policy year; these are separate from normal lapses since shock lapses are associated with the premium increase which occurs at the beginning of the year.
- Deaths occur at the end of each policy year.
- Each shock lapse event results in anti-selection mortality being calculated to maturity for the continuing policyholders. This is the new 'normal' mortality for the next shock lapse event.

Deaths calculation

The population at the time of the shock lapse (L_t) is thus split into three groups:

- Group 1, the population remaining, is $L_t \times (1 - w_2 - w_1)$, having anti-selection (ongoing) mortality
- Group 2, the 'effective' shock lapses, is $L_t \times (w_2) \times E$, having new issue mortality (select mortality beginning from the shock lapse event)
- Group 3, the other shock lapses, is $L_t \times (w_2) \times (1 - E) + L_t \times w_1$, having anti-selection mortality (the same as for the remaining population).

Anti-selection multiple

The deaths for groups 1 and 3 are calculated by subtracting the group 2 deaths from the total deaths. The mortality rates for groups 1 and 3 are then developed from those deaths applied to the starting number of lives. The mortality anti-selection multiple for each policy year is the calculated mortality rate, divided by the point-in-scale mortality rate.

Note – Calculating group 2 mortality requires reading the mortality variables as if 'issue age' were the shock lapse attained age, starting from policy year 1 instead of the actual year. The formula sets the `mort_issue_age` hidden variable lookup key for reading the external source table.

Shock lapse table

Specify shock lapses in addition to the normal lapse rates, as discussed below. You also specify whether this table increases mortality rates, in addition to increasing lapses.

5.3.11 Monthly mortality

Deaths occur at the end of each month, prior to other decrements and policyholder transactions. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

5.4 Lapses and surrenders

A termination is a 'surrender' when there is a cash value, and a 'lapse' otherwise.

Lapse rates apply to units after mortality decrements, which are processed before all other decrements.

5.4.1 Lapse adjustment multiple

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date.

5.4.2 Policy year lapse rates

The base lapse rate is adjusted by a multiple.

5.4.3 Monthly lapses

The annual lapse rate after all adjustments and excess lapses is constrained to be not more than 1.0 (100%).

Lapses and surrenders occur either: (1) at the end of each month, or (2) modally (just before the next premium is due). Lapses occur monthly after the last premium has been paid, for example in the final years of a decreasing term product.

Each of the above lapse rate adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly or modal lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/\text{event frequency})}$$

5.4.4 Skewed lapses

Skewed lapses allocate a specific portion of annual lapses to each month within each policy year. The skew percentages are normalized to assure that they add to 1.0 (100%) for each policy year. Lapse skewing can be specified differently for each policy year.

When skewed lapses apply, the exponent in the monthly/modal lapse rate formula is replaced by the sum of the skew percentages for the months included in the decrement time step. For example, when the lapse applies for months with monthly skew percentages adding up to 0.4 (40%), the lapse rate is:

$$1 - (1 - \text{annual lapse rate})^{0.4}$$

5.4.5 Shock lapse

Shock lapses are added to the normal lapse rate based on the premium increase after the level term years. As discussed above, the shock lapse assumption increases the mortality rates when the Dukes-MacDonald mortality anti-selection is used.

The shock lapse rate varies by the premium increase percent at the end of the level term years, and at the end of each renewable term period. The ART premium increases following the level term years are not considered in determining the size of the shock lapses. For example, the premium increase following the end of a 20-year level term period is the increase for year 21; the ART premium increases for years 22 and after do not generate a new shock lapse event.

Shock lapses occur just before the beginning of the policy year; this percent of policyholders declines to continue coverage at the increased premium rate.

You can spread the shock lapse over several policy years following the increase. In the example below, a premium increase of 80% results in a 16% shock lapse at the time of the increase, plus another 4% shock lapse one year later.

yrs_since_incr	0	1
prem_incr_pct		
0.00	0.00	0.00
0.40	0.08	0.02
0.80	0.16	0.04
1.20	0.32	0.08

The shock lapse formula sets two hidden variable lookup keys, which must be used to link to the external source table:

- prem_incr_pct – The percent increase in premium
- yrs_since_incr – ‘0’ for the shock lapse at the time of the increase, ‘1’ for the increase a year later, etc.

5.5 Term conversions

Term products can give the right to convert to a permanent product without being subject to underwriting.

5.5.1 Conversion rate

A specified percent of policies in force after lapses converts at the end of each policy year, including maturity of the term policy.

The term conversion rate is constrained to be not more than [1.0 (100%) less the monthly lapse rate].

5.5.2 Cost of conversion

The specified term conversion cost per unit is a cash flow. This generally represents the present value of profits on a newly issued policy, less the present value of profits on a policy issued from a term conversion.

Term conversion costs are not included in coinsurance benefits. Reinsurance effects should be considered in estimating the term conversion cost specified for the projection.

6 RIDERS

6.1 Multiple riders

Multiple riders can apply to a model point.

6.1.1 *Riders input*

Each rider creates an instance of the term_irider submodel. The input for each variable must be linked to an external source containing values varying by the rider names.

6.1.2 *Rider model point input*

The rider model point record identifies which riders apply to each model point.

6.2 Rider types

6.2.1 *Accidental death benefit*

Please refer to the L13 Riders book for a discussion of the accidental death benefit riders which can be linked to term life model points.

6.2.2 *Waiver of premium*

Please refer to the L13 Riders book for a discussion of the waiver of premium riders which can be linked to term life model points.

7 REINSURANCE

7.1 Multiple reinsurance treaties

Multiple reinsurance treaties can apply to a model point.

7.1.1 *Treaty input*

Each reinsurance treaty creates an instance of the term_ireins submodel. The input for each variable must be linked to an external source containing values varying by the treaty names.

The exception is that the YRT retention maximum per policy applies for all treaties combined, and thus does not vary by treaty. For example, it might vary by plan code.

Each treaty can be either a YRT, coinsurance, or modified coinsurance treaty. Each of these types is discussed in a separate section, below.

7.1.2 *Reinsurance model point input*

The reinsurance model point record identifies which treaties apply to each model point, as well as the ceded percent or amount for each treaty. The reinsurance model point layout is discussed in the Reinsurance model point records layout section, below.

7.2 YRT reinsurance

Yearly renewable term (YRT) reinsurance cedes a portion of the net amount at risk (NAR). Reinsurance premiums are calculated using YRT rates applied to the ceded coverage amount.

7.2.1 *Ceded amount*

The ceded amount is a percent of the net amount at risk (NAR) at the beginning of each policy year, which is the sum assured less the cash value at the end of the previous month.

Using reinsurance model point variables

The ceded percent can be specified on the reinsurance model point record, either directly or as the initial ceded death benefit amount, which is converted to the ceded percent (i.e. ceded death benefit amount / death benefit at the projection date.) This is proportionate YRT reinsurance, where the ceded percent is fixed.

Using treaty variables

The ceded percent can be calculated using the reinsurance input variables that vary by treaty code. The ceded amount is a specified percent of any excess of the NAR over the retention threshold per policy. For example:

- First dollar quota share – The threshold is zero.
- Excess retention – The ceded percent applies to the excess of the NAR over the specified threshold.

The ceded percent varies over time, depending on the changing relationship between the NAR and the threshold.

7.2.2 *Maximum YRT retention per policy*

The YRT reinsurance treaty amounts for a model point are increased when the combined ceded amounts would be less than [the NAR less the maximum retention amount per policy]. The YRT retention maximum per policy applies for all treaties combined, and thus does not vary by treaty. For example, it might vary by plan code.

This adjustment applies only for YRT treaties that have the ceded amount calculated using the treaty variables. Treaties that have the ceded percent/amount specified on the model point record are not adjusted.

When the sum of the ceded amounts is too low, the initial (at the projection date, or issue date for new business model points) ceded amount for each treaty is increased in proportion to the individual ceded amounts.

7.2.3 *YRT rate*

The YRT reinsurance rate for each treaty is a base rate, times a multiple. These are mortality variables, which operate the same as for mortality rates. See the Mortality variables section in the Input organization chapter, above.

The YRT premium can be zero for the first policy year by specifying the multiple as zero for the first policy year.

7.2.4 *Reinsurance premium*

YRT reinsurance premiums are paid either annually at the beginning of each policy year, or monthly.

The annual YRT reinsurance premium per policy is:

$$\text{ceded amount} \times (\text{annual YRT rate}) + \text{reinsurance policy fee}$$

When YRT premiums are paid monthly, one twelfth of this amount is paid at the beginning of each month.

The reinsurer refunds the unearned YRT reinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual YRT premium, times the remaining months in the year divided by 12.

7.2.5 Reinsurance benefits

The reinsurance benefit paid to the ceding company each month is the ceded amount times the monthly experience mortality rate. The ceded amount is the amount calculated at the time of each YRT reinsurance premium.

7.2.6 Reinsurance reserve

The formulaic statutory and tax reserve credit for YRT reinsurance is the tabular cost of insurance floor for the ceded amount.

7.3 Coinsurance

Coinsurance treaties cede a portion of the entire policy, including transferring the ceded reserve to the reinsurer. Reinsurance benefits include all benefits provided by the policy.

7.3.1 Coinsurance percent

When the coinsurance percent is calculated using the input for the treaty, the percent is x% of the sum insured at issue up to the threshold per policy specified for each treaty, plus y% of any excess over the threshold. When the coinsurance percent is specified by model point, this treaty input is ignored. The coinsurance percent remains constant to maturity.

The coinsurance percent applies to the entire policy, including the deficiency reserve and the additional reserve.

The coinsurance benefits paid to the ceding company each month are the projected policy benefits, times the coinsurance percent.

7.3.2 Coinsurance premium

Coinsurance premiums are paid at the same premium frequency as the model point.

Coinsurance premiums are based on the gross premium, including modal loading and policy fee.

The reinsurer refunds the unearned coinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual amount times the remaining months in the year divided by 12.

7.3.3 Premium tax

The reinsurer reimburses for the coinsurance percent times premium taxes paid by the ceding company.

7.3.4 Benefits

The reinsurer reimburses for the coinsurance percent times all benefits paid by the ceding company. Reimbursements are split between death benefits and other benefits, for reporting purposes.

7.3.5 Expense allowance

The reinsurer pays an expense allowance to the ceding company at the reinsurance premium frequency. The expense allowance is the annualized reinsurance premium times a specified percent.

The unearned current year expense allowance is refunded to the reinsurer when the policy terminates before the next reinsurance premium payment for any reason. The unearned expense allowance is the last amount paid, times the ratio of the number of months until the next reinsurance premium, divided by [12 divided by the reinsurance premium frequency].

7.3.6 Reinsurance reserve

The reinsurance reserve credit for coinsurance treaties is the total statutory reserve, times the coinsurance percent.

7.4 Modified coinsurance

Modified coinsurance (ModCo) treaties operate as described above for coinsurance treaties, except that the reinsurance reserve is retained by the ceding company. The ceding company reimburses the reinsurer for interest on the reinsurance reserve which is backed by assets invested by the ceding company.

The adjustments occur monthly, so that nested projections match standard projection results. The adjustment amount is the current reinsurance reserve, less the reinsurance reserve as of the previous adjustment date accumulated with interest.

The ModCo reserve adjustment interest rate is the asset yield rate.

7.5 Reinsurance administration expenses

Reinsurance administration expenses are incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include:

- Amount per policy
- Percent of the ceded amount.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Coinurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for coinsurance.

8 FACTORS CALCULATIONS

8.1 Values calculated

8.1.1 Special projection type

A projection type calculates 'factors' when the product submodel is the top model and the target column is factors_tgt. This target column directs the calculation of each item specified (discussed below). The target column calculates results for each policy year; thus the Projection period from and to properties have no effect on factors calculations.

The Assumption Set Restore property (in the run parameters object) must be Yes, since this is a seriatim projection. Project variables such as pricing definition and profit distribution mode do not apply for this special type of projection.

8.1.2 Factor types

Specify at the bottom of the Product input page which factors to calculate:

- **Cash values per unit** – terminal cash values and net premiums. Cash values are calculated automatically for ROP term products when they are needed for reserve factor calculations.
- **Statutory reserves per unit** – segmented and unitary terminal values and net premiums, plus the substandard reserve values

- **Alternative minimum reserves per unit** – segmented and unitary terminal values and net premiums
- **Tax reserves per unit** – segmented and unitary terminal values and net premiums.

8.1.3 Model points

Factors calculations systematically process for each model point. Set up a model point for each issue age to be calculated, within each combination of insured characteristics (sex, smoking class, rate class, etc.). For example, you might set up model points for issue ages 0 through 80 for male nonsmokers, then repeating these model points for male smokers, then female nonsmokers, then finally female smokers.

Best practice is to create the report group for each model point so that it identifies the model point characteristics. For example, the report group might be PlanA-M-NS-ST-35 for the model point calculating factors for plan code PlanA, for a male nonsmoker standard risk class issue age 35. Then the report groups identify the meaning of each model point.

The same as for Pricing cell projections, each model point must be issued on the first day of the projection. For example, when the projection date is 2013/12 (end of December, 2013), the issue date must be 2014/01/01 (January 1, 2014).

8.2 Factors reports

8.2.1 term_py submodel output

The factors calculation output is contained in the term_py submodel, which has column results by policy year.

8.2.2 Reporting tool

The US library reporting tool includes special reports for these term life factors:

- **Nonforfeiture values** – The death benefit, premium pattern, cash value interest rate, and cash value net premium and terminal value for each policy year.
- **Policy reserves** – In addition to the segmented and unitary net premiums and terminal reserves, this report shows (for each policy year) the death benefit, premium pattern, interest rate, mortality rate, and segment breaks. Statutory, alternative minimum, tax, and substandard reserve results are shown on the report.
- **ROP term nonforfeiture values** – Includes the ROP term endowment benefit.
- **ROP term policy reserves** – Includes the endowment and unusual cash value amounts used in calculating reserves for the Pure endowment or Unusual cash value methods.

To produce these reports, select reporting tool parameters for Pricing reports, check the reports to produce in the Policy Year Reports tab, and check which model points to produce reports for in the Liability Groups tab.

8.3 Factors projection setup

Specify the following settings in the run parameters for the projection task calculating factors:

Property	Value
Target Colum	term->factors_tgt
Projection Period From	0
Projection Period To	0 (since calculations are driven by factors_tgt)

Property	Value
Projection Period Min	0
Projection Period Max	Higher than the maximum maturity months, such as 1212 for 101 years
Projection Loops	1
Assumption Set Restore	Yes
proj_date	yyyy/mm, the end of the month before the issue date

The only special variables to specify on the input pages are the options at the bottom of the Product input page, specifying which of the factor types to calculate.

9 MODEL POINT RECORDS

9.1 Model point record layout

The model point record layout for the term life product type is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	up to 125 characters (assigns model point to report group)
B	seg_id	segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date	yyyy/mm/dd or mm/dd/yyyy
F	disab_date_mp	yyyy/mm/dd or mm/dd/yyyy (set to 1/1/9999 if policy is not disabled)
G	plan_code	up to 20 characters (link to assumptions by plan)
H	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
I	tax_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
J	issue_age_mp	positive integer
K	gender_mp	Male, Female, Unisex

Excel Column	Variable	Value
L	smoking_class_mp	Nonsmoker, Smoker, Aggregate
M	rate_class_mp	Standard, Preferred, Super, [extend with user additions]
N	substd_mult_mp	substandard multiple (1.0 for 100%)
O	prem_mode	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
P	level_term_yrs	level term years (level term premium per unit replaces annual premium per unit through this period) – this variable is required for level term and ROP term only
Q	rop_rider_defn	Yes to add ROP feature to a level term plan code that is not an ROP product. The ROP rider premium per unit is added to the level premium per unit. No otherwise
R	gpr_pct_mp	percent of this model point included in Gross Premium Reserve calculations (1.0 for 100%)
S	init_sum_assured_mp	initial sum assured for mp (0 = calculate from initial units)
T	init_policies	initial policies for mp (required)
U	init_units	initial units for mp (0 = calculate from initial sum assured)

9.2 Reinsurance model point record layout

The reinsurance model point record identifies which treaties apply for each model point, using the linked variable, mp_id. The reinsurance model point record can have any number of reinsurance treaties per model point. It also specifies the ceded percent or amount for each treaty as described in the table below. If the reinsurance model points are omitted from the reinsurance model point file, then the liability model point will not include reinsurance.

Warning! The order of reinsurance model points must be in the same order as the liability model points.

The reinsurance model point record layout is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	mp_id	Up to 12 characters. Links to liability data file mp_id. This value can be listed any number of times to indicate the number of treaties applied to the policy.
B	treaty_code	A user defined input that references the treaty code in the Resinurance.xlsm workbook.

Excel Column	Variable	Value
C	ceded_amt_defn	Percent - the ceded amount should be input as a percent from 0 to 1 Dollar Amount - the ceded amount should be input as the ceded death benefit amount Treaty Input - the ceded amount should be input as 0, and the ceded amount will be calculated using the treaty variables
D	ceded_amt	See above. Input the ceded percent, the ceded dollar amount, or 0 if the ceded amount is specified by the treaty.

9.3 Units basis

Product values are calculated per unit in force, then multiplied by the current number of units. For example, the death benefit, cash value, and reserves per unit.

9.3.1 Initial units

The initial (at the projection date) policies must be specified on each model point record, along with either or both of the initial units and initial sum assured. These are the aggregate values for the policies collected into each model point.

- When the initial units are not specified on the model point (this variable is left at its zero default value), the initial units are the initial sum assured divided by the specified sum assured per unit.
- When the initial sum assured is not specified on the model point (this variable is left at its zero default value), the initial units variable applies directly.
- When both the initial units and initial sum assured are specified on the model point record, the initial units variable applies. The Run log contains a warning if the initial sum assured is more than [\$1 times the initial units] different from the initial units times the specified sum assured per unit.
- When both the initial units and initial sum assured are zero, the model point is skipped and a warning is issued in the Run log.

The initial policies must be specified for each model point. The average size policy is the initial units divided by the initial policies, and the initial sum assured divided by the initial policies.

9.3.2 Pricing cells

For model points issued at the start of a seriatim pricing projection, the initial policies, units, and sum assured fields define the policy size. When the model point specifies one policy, the initial units and sum assured are average size amounts.

10 SUBMODELS STRUCTURE

10.1 term submodel

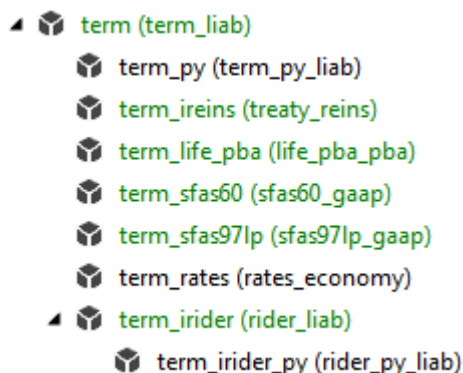
The term submodel calculates the model point's contribution to the liability income statement and balance sheet items for term life products.

The term submodel calculates model point results using values per unit in force, which are then multiplied by the number of units. Results per unit in force have '_per_unit' added to the formula name. Formula names without this addition are results per model point (or are a special result like a ratio).

10.1.1 *Seriatim versus portfolio runs*

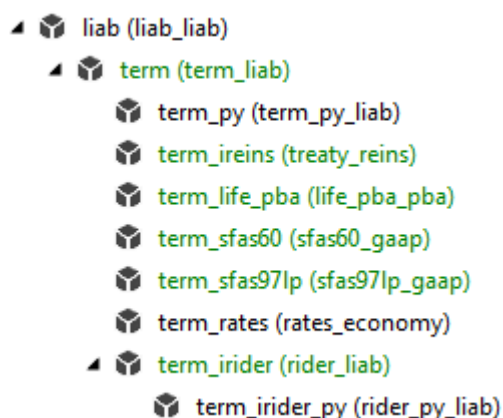
Seriatim projection

When term is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the term submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings.



Portfolio projection

For portfolio projections (which include company and business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the term submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the term submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the term submodel the same columns as for other product types. This makes the term life product type interchangeable with other product types fitting into this business segment structure.

10.1.2 *Variables*

Input variables for specifying term life products are located in the term submodel, except as noted below for the variables in term_py and term_ireins.

10.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when term is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the term submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only term is the top model, (2) when term is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when term is run as the top model and when term is included in portfolio projections. References to company, seg and liab are invalid when term is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) temporary tables are sized for DAC tax calculations and competitor rates, (3) projection flags are created and initialized, (4) reporting groups are set up, (5) term_ireins elements are created, (6) model point data validations are processed, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each projection task loop, immediately after startup when term is the top model. When company is the top model, initialize is executed by calls from seg->initialize which executes term->initialize for each model point belonging to the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when term is running as the top model. This includes executing the initialization of economic scenario rates and the setup of competitor rates. Other initializations are required in both cases.

The main use of initialize is to calculate the cash value and reserve factors for each model point using the actuarial math library. Since these initializations are not required for all calculation types for all product features, conditions are checked to shortcut the initialization process, thus reducing the runtime of the model point.

This cash value and reserves are initialized in the following order:

1. Base cash values
2. ROP base cash values
3. Base statutory reserves
4. Base statutory minimum reserves
5. ROP statutory pure endowment method reserves
6. ROP statutory unusual cash value method reserves
7. Base tax reserves
8. ROP tax pure endowment method reserves
9. ROP tax unusual cash value method reserves.

Other formulas

Other formulas in the term submodel are relatively straightforward. Results which are per unit in force add '_per_unit' to the end of the formula name. Results without '_per_unit' are values for the model point.

Most of the profit, balance sheet, and GAAP columns are only calculated when term is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

10.2 term_py submodel

The term_py submodel is a standalone instance of the term_py_liab model class. It contains the variables and formulas required to calculate per unit in force results which only vary by policy year. Thus 't' in this submodel represents policy year, unlike the term and term_ireins submodels where 't' represents months since the projection date. The policy year values are calculated from issue.

Since all results are per unit, this submodel does not include '_per_unit' in the formula names.

10.2.1 Variables

Variables located in the term_py submodel are only used in formulas within this submodel. These are generally for values calculated from issue and varying by policy year, such as annual premiums, and cash value and reserve terminal values and net premiums.

10.2.2 Formulas

Formulas for calculating policy year per unit in-force results are included in the term_py submodel.

Columns are generally calculated from issue, even for in force model points. As discussed in the term submodel section, the reserve net premium and terminal values of term_py are not calculated on demand through the target column. Instead, these results are calculated in advance through forced calls from the 'initialize' column of term.

Cash value and reserve terminal values and net premiums are calculated by calling the Actuarial math library. See the Actuarial math library section below.

10.3 term_irider submodel

The term_irider submodel is a dependent data driven array containing the variables and formulas required to calculate the rider cash flows and reserves for term life model points.

Please refer to the L13 Riders book.

10.4 term_ireins submodel

The term_ireins submodel is a dependent data driven array containing the variables and formulas required to calculate the ceded amount and reinsurance cash flows for term life model points.

10.4.1 Multiple instances

Each model point carries multiple instances of term_ireins. The number of instances is determined by the number of treaties specified on the reinsurance model point record.

Elements of term_ireins are created during the startup processing of the model point. Conditions are evaluated in startup of term and the term_ireins user defined array is resized as required. The characteristic for each term_ireins element vary based on the treaty_code.

The ceded amount and reinsurance cash flows for multiple treaties are aggregated using the sum_over_reinsurance function in the term submodel. This function can aggregate the values for all treaties, or just the treaties of a specific type of reinsurance, such as YRT.

10.4.2 Variables

Variables for the reinsurance type, the reinsurance expenses, and for calculating the ceded amount (when it is not specified on the reinsurance model point record), are located in the term_ireins submodel. These variables are contained in the term_ireins submodel so that they vary by treaty.

10.4.3 Formulas

Formulas for calculating the ceded amount and the reinsurance cash flows are contained in the term_ireins submodel.

Since term_ireins is defined through the model parameters, the size of the term_ireins array is based on the number of treaties specified for each model point on the reinsurance model point file.

10.5 term_rates submodel

The term_rates submodel is identical to the rates submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

10.6 term_life_pba submodel

The term_life_pba submodel is identical to the life_pba submodel which applies for portfolio projections.

Please refer to the C05 PBA book.

10.7 term_sfas60 submodel

The term_sfas60 submodel is identical to the sfas60 submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

10.8 term_sfas97lp submodel

The term_sfas97lp submodel is identical to the sfas97lp submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

10.9 Actuarial math library

10.9.1 AML DLL

The Actuarial math library (AML) provides standard actuarial calculations for traditional reserve and cash value methods. See the Actuarial math library liability book.

The term submodel calls the AML to calculate reserve and cash value amounts per unit in force. Columns in the term_py submodel store the terminal value and net premium for each policy year for each cash value and reserve basis used in the projection.

10.9.2 term->startup

The term->startup column specifies the name extension for the audit trail comma separated values file (when requested).

10.9.3 *term->initialize*

The term->initialize column initiates the call to the AML calculations for each set of cash value and reserve results. Each set of AML results is calculated by calling the aml_driver in the term_py submodel. A separate call can be made for each of the following results:

- Cash values – Base plan cash values
- Statutory reserves – Base plan statutory reserves
- Statutory minimum reserves – Base plan minimum reserves for calculating deficiency reserves
- Tax reserves – Base plan tax reserves

10.9.4 *AML driver*

The aml_driver in the term_py submodel sets up and calls the AML for each set of cash value and reserve results. This driver is called by the term->initialize column for each set of results.

This driver sets up the call to the AML using the appropriate input for each type of calculations. Each call to the driver calls these functions in the term_py submodel:

- setup_aml_initial_data – Creates the interfaces for using the AML.
- setup_aml_res_defn – Specifies the cash value or reserve method (such as NLP, FPT, CRVM, SNFL, etc.), and the function type (curtate, continuous, or semi-continuous).
- setup_aml_policy_data – Specifies values for each policy year, such as the guaranteed premiums, death benefits, endowment benefits, valuation interest rates, mortality rates, plus the issue age, maturity age, premium term years, and expense allowance grading years.

The operation of these setup functions varies by the type of calculations being made. For example, the policy data passes the cash value interest and mortality rates when calculating cash values, and the statutory rates when calculating statutory reserves.

10.9.5 *Calculation results*

The AML calculation results are stored for each policy year in columns in the term_py submodel. The formula for each column simply gets the AML calculation results for each case. These columns include:

- Terminal cash value and reserves and net premiums for each basis
- Valuation interest and mortality rate for each basis
- Tabular cost of insurance for each basis

The aml_driver calls the appropriate columns after it calls the AML. For example, the cash value basis columns are called when the driver has just calculated cash value results. Then the driver calls the statutory reserve basis columns when those results have just been calculated by the AML. Thus each of these columns stores the AML results using the appropriate valuation basis.

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L02 TRADITIONAL LIFE

This book discusses traditional life insurance products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

- Product – Benefits, premiums, cash values, policy loan definitions, and factor calculations definitions
- Valuation – Statutory and tax reserves and actuarial math library audit report option
- Dividends – Annual and terminal dividend scales, dividend option definitions
- Expense – Commission and expense rates, premium tax rate
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse rates and policy loan use

Each of these input page types above begins with 'Trad', for use in projections that combine multiple product types.

Additionally, variables defining reinsurance and rider specifications and assumptions are associated with input pages which are common across products. These are:

- Reinsurance – Reinsurance treaty definitions and rates.
- Riders – Riders specifications and assumptions
- RidersValuation – Riders statutory and tax reserve definitions

1.2 Data pages

Variables for each indexed universal life model point are specified on the TradData page. These variables are discussed in the Model point records chapter, below.

Variables for each reinsurance record are specified on the TradReinsData page. These variables are the same for the term life, traditional life, universal life, variable universal life, indexed universal life, and long term care product types. These variables are discussed in the Reinsurance model point records chapter, below.

Variable for each rider record are specified on the TradRidersData page. These variables are the same for term life, traditional life, universal life, variable universal life, and indexed universal life products. These variables are discussed in the L13 Riders book.

2 PRODUCT BENEFITS AND VALUES

The following explains the basic features for traditional life products, also called permanent life products. Term life products are discussed in the Term life book.

2.1 Benefit term

The maturity date is either an attained age or policy anniversary, but not greater than the statutory reserve mortality table omega age. When this value is zero (default value), coverage expires at the omega age.

2.2 Death benefit

2.2.1 Sum assured

The death benefit is the sum assured. While this is normally a level amount, such as \$1,000 per unit, the sum assured can vary annually (using the `pol_yr` column) or monthly (using the `mths_since_issue` column).

2.2.2 Unearned premium

The death benefit is increased by the annualized gross premium, times the remaining paid-to months, divided by 12.

2.2.3 Dividend option amounts

Dividend option amounts increase the death benefit. See the Policyholder dividends chapter, below.

2.3 Endowment benefit

The policy can endow at maturity. The maturity amount can be the sum assured, or a specified amount per unit.

Dividend option amounts increase the maturity benefit. See the Policyholder dividends chapter, below.

2.4 Premiums

2.4.1 Premium term

When the premium term variable on the model point record is different from zero, premiums are paid to this policy anniversary. This variable overrides the definition on the Product input page.

When the premium term variable on the model point record is zero, premiums are paid until either an attained age or policy year. When this value is zero (default value), premiums are paid to maturity.

2.4.2 Premium mode

Specify the premium mode on the model point record. Premiums can be paid annually, semi-annually, quarterly, monthly, or special monthly (such as electronic transfer or pre-authorized check, with expense rate different from 'monthly' mode).

2.4.3 Annual premium

Specify the annual premium either as:

- **Annual premium per unit** – The annual premium per unit variable value is increased by the modal loading and the policy fee.
- **Initial annualized premium** – The premium on the model point record for the policies collected into the model point, including the modal loading and the policy fee. This premium is level for all policy years, and is treated as being guaranteed.

2.4.4 Modal loading

When premiums are paid more frequently than annually, the annualized premium is the annual premium times the annualized modal loading multiple varying by premium mode. For example, an annualized modal loading multiple of 1.08 means that monthly premiums are 9.0% ($1.08 / 12$) of the annual premium per unit.

Vary the modal loading multiple by premium mode by linking to an external source varying by the `prem_mode` model point variable. For example:

prem_mode	
Monthly	1.08
Special Monthly	1.08
Quarterly	1.06
Semi-annual	1.02
Annual	1.00

2.4.5 Policy fee

The premium is increased by the annualized policy fee varying by premium mode. For example:

prem_mode	
Monthly	43.20
Special Monthly	43.20
Quarterly	42.40
Semi-annual	40.80
Annual	40.00

2.4.6 Guaranteed premium

When the annualized premium is specified on the model point record, so that premiums are level, that premium is treated as being guaranteed. The following discussion applies when the annual premium is specified using the annual premium per unit variable.

Premium pattern

Cash value and reserve net premiums are based on the pattern of guaranteed premium rates by policy year, which can be different from the pattern of current premiums. For this purpose, the guaranteed premiums are used only to specify the gross premium pattern. Thus product pricing projections can calculate cash values and reserves even before the actual premiums have been determined.

(See below that cash values and reserves are calculated using only the guaranteed premium per unit, without adding the policy fee or modal loading.)

Deficiency reserves

Deficiency reserves compare the reserve net premium to the greater of the current premium and the guaranteed premium for the annual premium mode. For this purpose, the guaranteed premiums must be known. When the current premiums are guaranteed, simply copy the current premium table to the guaranteed premium external source input. When the guaranteed premium per unit is less than the annual premium per unit, the guaranteed premium is the annual premium. Thus you can leave this variable at its 1.0 default value.

(See in the Deficiency reserves section in the Statutory and tax reserves chapter, below, that the guaranteed premium can include both the policy fee and modal loading, the policy fee only, or neither.)

2.4.7 Single premium

Specify on the model point record the total single premium for the policies collected into the model point.

Single premium policies

When the annual premium is zero, this is a single premium policy. Thus single premium plans can be specified either as:

- Zero annual premium, defining the premium as the single premium; the premium term has no effect.
- Premium term of 1 year, with Annual premium mode, and defining the premium as the annual premium.

Additional single premium paid at issue

When there is also an annual premium, the single premium is an additional amount paid at issue. This amount purchases a paid up addition using the same purchase rates which apply for policyholder dividends purchasing PUA. This PUA purchase at issue applies regardless of the dividend option specified for the model point.

This purchase of PUA at issue is made only for new business model points, which are issued during the projection. For in-force model points, the initial (at the projection date) amount is specified on the model point record.

2.4.8 DAC tax capitalized premium percent

Specify (with the Premiums input) the percent of each premium which is capitalized for the capitalized premium tax, which is also called the DAC tax. See the Taxes chapter in the Profit calculations book.

2.5 Cash value**2.5.1 Cash value methods**

The following cash value methods are available:

- Not applicable – base plan cash values are zero. The PUA net single premiums (NSP) are the PUA purchase rates.
- SNFL minimum (both 1980 and pre-1980 versions of the U.S. standard nonforfeiture law)
- Net level premium – no expense allowance
- Full preliminary term – zero terminal cash value for the first policy year
- Table – specify net premiums and terminal cash values per unit for each policy year, starting with 0 (issue date) for the terminal cash values. The PUA NSPs are the PUA purchase rates.

2.5.2 Death benefit varying within policy year

When the death benefit varies within a policy year, the death benefits used for cash value calculations are the average of the death benefit as of the end of each month within the policy year. For example, when the sum insured varies monthly.

2.5.3 Premium pattern

Cash value net premiums are the level percent of gross premiums calculated using the cash value method. For product pricing work, before the gross premiums have been determined, specify the premium pattern in the guaranteed premium variables. For example, the default value of '1' for all policy years specifies that premiums are level.

The cash value method applies to the pattern of guaranteed premiums without policy fees and modal loadings.

2.5.4 Cash value basis**Mortality**

Cash value mortality is a base table only.

Mortality variables are read for both the first and second lives. See the Mortality variables section in the Input organization chapter, above.

Interest

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeNonforfeiture range in the Valuation Interest Rates Excel workbook.

Function timing

Function timing can be:

- **Curtate** – Annual premiums and claims paid at the end of the policy year
- **Continuous** – Continuous premiums and immediate payment of claims
- **Semicontinuous** – Annual premiums and immediate payment of claims.

The continuous premium multiple and the immediate claim multiple are defined in the Function timing section in the Valuation options chapter of the Actuarial math library appendix.

2.5.5 Graded expense allowance

The expense allowance can grade to zero between two policy anniversaries. Net premiums are the cash value method's percent of guaranteed premiums before the grading, the net level premium percent after the grading, and the calculated grading percent in between.

For example, specifying grading between the 15th and the 20th policy anniversaries results in the selected cash value method applying for the first 15 policy years, and the net level premium method applying from the 20th anniversary to maturity. Between these two anniversaries, net premiums are the level percent of the guaranteed premiums which advances the 15th year terminal cash value to the 20th year terminal cash value.

Specify to grade the expense allowance over a period shorter than the premium term by grading from issue (policy anniversary zero) to policy anniversary 'n'. Specify the zero default value for these anniversaries to grade the expense allowance over the premium term.

2.5.6 Mid-year cash values

The mid-year cash value is the paid-to-date interpolated terminal cash value, but not less than zero. The paid-to-date is the end of the month before the next premium payment.

After the premium term, the terminal cash values are interpolated monthly. Monthly interpolation applies from issue for single premium products.

2.6 Nonforfeiture options

2.6.1 Reduced paid up policies

Each in-force model point indicates whether it contains policies that are 'active' or 'reduced paid-up'. Reduced paid up (RPU) policies are projected the same as for active policies, except that:

- No premiums are paid,
- No lapses occur,
- Policyholder dividends do not apply, and
- There can be no policy loan.

RPU policies are 'paid up.' For example, paid up expense rates apply for all RPU policies.

RPU policies can have different assumptions than for active policies, using the model point variable `prem_status_defn` (with choices of Active and RPU) as a lookup key.

2.6.2 Reduced paid up factors

Factors calculations include the RPU amount related to the cash value at the end of each policy year. See the Factors calculations chapter, below.

The reduced paid up (RPU) nonforfeiture option reduces benefits to the percent of normal benefits which has the same present value as the cash value at the time when premiums are suspended, discounting using the

cash value basis. The Factors calculation result is the RPU percent for decrementing at the end of the policy year, then times the sum assured per unit.

2.6.3 Extended term insurance factors

Factors calculations include the extended term insurance (ETI) amount related to the cash value at the end of each policy year. See the Factors calculations chapter, below.

The ETI nonforfeiture option continues the normal death benefit coverage for the time which has the same present value as the cash value, using the cash value basis. The Factors calculation ETI years is the linear interpolation between the full number of years of coverage with present value less than and greater than or equal to the cash value at the end of the policy year.

When the present value of normal death benefits to maturity is less than the cash value, the excess purchases an endowment at maturity using the ETI basis.

2.7 Policy loans

2.7.1 Policy loan amount

Policy loans apply equally to all policies within a model point. For in-force policies, specify on the model point record the initial (at the projection date) policy loan for the policies collected into the model point.

The accumulated policy loan at the end of each month is the balance at the end of the previous month, brought forward with one month's interest at the policy loan rate.

The outstanding loan amount can increase at the end of each policy year, by a percent of the base plan cash value. The increase is reduced when the maximum policy loan would be exceeded.

2.7.2 Maximum policy loan

The policy lapses when the outstanding loan amount exceeds the cash surrender value, including the cash value from dividend options.

2.7.3 Policy loan cash flows

An adjustment increase is a new policy loan and a negative cash flow. An adjustment decrease is a policy loan repayment and a positive cash flow.

The elimination of the policy loan at death and surrender is an 'adjustment' in the loan balance to zero, and thus is a positive loan cash flow which offsets part of the decrement cash flow.

2.7.4 Policy loan rate

The policy loan interest rate is the greater of: (1) a fixed rate and (2) an indexed rate equal to a competitor rate. Interest is added to the loan at the end of each month.

When policy loan interest is payable in advance, policy forms may specify the loan rate as a discount rate. Convert this discount rate to the annual effective annual rate. For example, a 7.4% discount rate payable in advance is an effective annual rate of 8.0% [$= 7.4\% \div (1 - 7.4\%)$].

2.7.5 Policy loan expenses

Policy loan expenses are discussed in the Expenses section, below.

3 POLICYHOLDER DIVIDENDS

3.1 Timing of dividend transactions

3.1.1 *Continuing policies*

In actual practice, policyholder dividends are paid at the end of each policy year. When the dividend is not paid out in cash, it is applied to the selected dividend option on the policy anniversary, which is the start of the next policy year. These two transactions (paying the dividend and applying to dividend options) occur in the same calendar month. For example, policies issued in January have their issue dates scattered throughout the days in January, so the end of one policy year is a day in January and the beginning of the next policy year is the next day in January.

US library projections, however, use the simplifying assumption that all of these policies are issued on January 1. Thus the dividend payment at the end of the policy year would occur in December (December 31), while the dividend would be applied to dividend options in January (January 1). Thus these dividend transactions, which occur together in actual practice, would be separated into different projection months (and even different calendar years in this case of January issues) for the model results.

The traditional life project solves this problem by treating the dividend transactions together at the beginning of the next policy year. The dividend timing transactions are:

- Dividend liability at the end of the policy year is the annual dividend that is to be paid (and plus the terminal dividend liability).
- The dividend is paid immediately at the start of the next policy year, and is thus treated as a beginning of month cash flow.
- The payment is applied to the dividend option at this same time, and is also a BOM cash flow.
- For policies dying, lapsing, surrendering, and maturing at the end of a policy month, the annual dividend is paid at that time, as an end of month cash flow.

In this way, the dividend is both paid and applied to the dividend option at the same time (and in the same month of the projection), just as occurs in actual practice.

Policy year profits are correct, even though the dividend payment cash flow occurs at the beginning of the next policy year. This is because the policy year book profit is reduced by the dividend liability set up for the annual dividend that is to be paid.

3.1.2 *Terminating policies*

For policies dying during the policy year, a pro-rata portion of the annual dividend may be paid with the terminal dividend.

For policies surrendering, lapsing, and maturing at the end of a policy year, the annual dividend is paid with the terminal dividend, since the policy is no longer in force at the beginning of the next month.

3.2 Dividend liability

The statutory balance sheet includes a dividend liability for:

- The annual dividend to be paid on the next policy anniversary, plus
- The terminal dividend (discussed below) that would be paid if all policies lapse at the next policy anniversary.

When the annual dividend is paid on the policy anniversary (assumed in the model projection to occur at the beginning of the next policy year), a new dividend liability is set up for the annual dividend to be paid on the

next policy anniversary. Thus book profit is reduced by this increase in the dividend liability, prior to the cash flow for the actual payment of the dividend.

The annual dividend scale for the next calendar year is declared each December. Thus the dividend liability changes each December, to the 'declared' annual dividend.

When the annual dividend is paid at the policy anniversary during the calendar year, the dividend liability changes to the annual dividend that would be paid at the next policy anniversary using the current dividend scale. This is the 'estimated' annual dividend, which will be replaced by the declared annual dividend when the new scale is determined in December.

3.3 Terminal dividend

The terminal dividend is paid at the time of lapse or surrender. The terminal dividend for a policy year is paid on lapse or surrender any time during that policy year.

Terminal dividends are not paid at death or maturity.

The dividend liability includes the current terminal dividend amount, assuming that all policies will lapse at the next policy anniversary.

3.4 Annual dividend

3.4.1 Dividend scale

The policyholder dividend 'scale' specifies the annual dividend amount per unit to be paid for each policy year. A new dividend scale is declared at the end of each calendar year. The scale can be defined as either:

- Annual dividend amounts per unit (the current dividend scale),
- Dividends calculated using contribution components (calculated dividend scale), or
- Component adjustments to the input dividend scale (adjusted current dividend scale), such as adjusting for changes in the dividend interest rate since that scale was declared.

3.4.2 Estimated and declared dividends

At the beginning of each policy year, the dividend liability changes to use the annual dividend that would be paid at the next policy anniversary using the current dividend scale. This is sometimes called the 'estimated dividend', since the dividend that will actually be paid at the next policy anniversary will use the new dividend scale declared at the end of the current calendar year.

At the end of each calendar year, the dividend liability changes to use the new annual dividend scale declared on that date. This is sometimes called the 'declared dividend', to distinguish it from the estimated dividend used to set the dividend liability for the first part of the policy year.

Thus the liability for the annual dividend changes twice each year: (1) when the model point enters a new policy year, and (2) when the new dividend scale is declared at the end of the calendar year.

3.4.3 Allow decrease in dividend scale?

You can specify that the base plan (that is, ignoring the annual dividend from dividend option amounts) annual dividend amount cannot decrease from one policy year to the next. This practice is sometimes called pegging the current dividend scale. This check is ignored when the annual dividend scale is fixed (by specifying only a table of annual dividends varying by policy year).

When the new dividend scale would produce a lower result, the previous dividend amount continues to apply until the new scale produces an amount at the end of a future policy year that is higher than the last dividend paid. For example, when annual dividend paid at the fifth policy anniversary was \$1.20 per unit, but the new

dividend scale is \$1.15 at the sixth anniversary and \$1.30 at the seventh anniversary, \$1.20 is paid at the sixth policy anniversary.

Specify on the model point record the annual dividend paid at the last policy anniversary before the projection date, for the policies collected into the model point. When you leave this variable at its zero default value, this feature applies only for dividend scales calculated during the projection.

3.5 Contribution components

3.5.1 Standard components

The standard features include the interest and mortality components. You can customize to add components for expenses and other contributions to the dividend scale.

Policyholder dividends calculated using contribution components are based on the sum of the components. Each component is a contribution replacing statutory reserve or cash value rates with results based on current experience.

When the components are adjustments to the input dividend scale, the statutory reserve or cash value rates are replaced with the former dividend rates which were used to calculate that scale. Thus the components adjust the input scale for changes in interest and mortality rates since that scale was declared.

3.5.2 Interest component

The interest component of the base plan dividend is: (1) the dividend asset base, times (2) the dividend interest rate minus the dividend valuation interest rate. The dividend asset base and the dividend valuation interest rate may be based on the statutory reserve or cash value valuation basis. When the contribution components are being used as adjustments to the input dividend scale, the dividend valuation interest rate is replaced by the interest rate used to determine that scale.

The dividend asset base is either the initial (previous terminal plus reserve net premium) formulaic statutory reserve or the previous terminal cash value.

The dividend valuation interest rate is either the statutory interest rate or the cash value interest rate.

The dividend interest rate is determined on the date when the dividend scale is declared. Specify on the model point record the initial dividend interest rate, which is used to calculate the annual dividend to be paid on policy anniversaries in the first calendar year, and to estimate the dividend for the remainder of that first calendar year. When this value is left at zero (default value), the initial dividend interest rate is the dividend interest rate formula applied at the projection date.

The dividend interest rate strategy defines the dividend interest rate at the end of each calendar year, for use in the 'interest component' of the annual dividend calculation.

The dividend interest rate can be defined either as:

- Fixed rate – Competitive constraints do not apply.
- Asset yield basis – The rate is the asset yield less a spread, subject to competitive constraints.
- Competitor rate basis – The rate is a competitor rate less a spread, subject to competitive constraints.

When the dividend interest rate basis is a competitor rate or the asset yield, the rate is first constrained to be not more than a maximum percent above, and not less than a maximum percent below, a specified competitor rate.

Further (thus adjusting the above constraints), the rate is not greater than the asset yield less a minimum spread. When the minimum spread below the asset yield is 1.0 (100%) default value, this constraint does not apply, and model points defining the rate using a competitor rate are eligible for independent model point processing within asset/liability projections.

Finally, the rate is not less than the cash value or statutory reserve interest rate, based on the dividend interest component valuation basis.

Direct recognition of policy loans varies dividends by the amount of the policy loan. The interest component is calculated in two parts, one for the unloaned part and one for the loaned part. The dividend interest rate applies to the unloaned dividend asset base. The dividend interest rate for the loaned part of the asset base is the policy loan interest rate.

3.5.3 Mortality component

The dividend contribution components calculation may include an adjustment for mortality.

The mortality component of the base plan dividend is: (1) the dividend net amount at risk, times (2) the dividend valuation mortality rate minus the experience mortality rate, times (3) a dividend mortality component multiple. The dividend net amount at risk and the dividend valuation mortality rate may be based on the statutory reserve or cash value valuation basis. When the contribution components are being used as adjustments to the input dividend scale, it is assumed that the basis for the net amount at risk and dividend valuation mortality rate are consistent with the basis used to determine that scale.

The dividend net amount at risk is the death benefit less either the initial (previous terminal plus reserve net premium) statutory reserve or the previous terminal cash value.

The dividend valuation mortality rate is either the statutory reserve or cash value mortality rate.

3.6 Dividend options

3.6.1 Future dividends

Each model point elects one of the following dividend options, which are discussed in separate sections below:

- Cash – pays the annual dividend out as a cash flow at the end of each policy year
- Accumulate
- Paid up additions (PUA)
- OYT/PUA rider.

Annual dividends paid during the projection are applied to this dividend option.

3.6.2 Initial dividend option amounts

In-force model points can have existing amounts for dividend options different from the option that applies for future dividends. This assures that existing dividend option amounts are recognized in the projection, thus making it easier to validate initial policy values while combining multiple policies together in a single model point.

For example, a model point paying future dividends in cash can have existing dividend accumulations and PUA face amount. Those existing dividend option amounts are projected the same as for model points using those dividend options, except that future dividends are not added to these amounts.

The exception is for the OYT/PUA rider. The initial OYT face amount affects the projection only when this is the dividend option selected for the model point.

Specify on the model point record the initial (at the projection date) accumulations and PUA face amount for the policies collected into the model point.

3.7 Accumulate option

Dividend accumulations, also called 'dividends on deposit', accrue interest at a specified effective annual rate.

The accumulation amount is paid at death, surrender, and maturity. The statutory and tax reserve is the accumulations amount.

3.8 Paid up additions option

3.8.1 PUA amount

Paid up additions (PUA) are additional level amounts of sum assured. The PUA sum assured is paid as an endowment at maturity when the base plan endows at maturity; otherwise the PUA sum assured is paid only on death.

The amount of PUA purchased on each policy anniversary is: (1) the annual dividend, divided by (2) the purchase rate net single premium (NSP) per \$1 PUA sum assured. PUA purchase rates can be calculated using a specified interest rate and statutory reserve mortality, or can be specified.

PUA purchase rates table

PUA purchase rates are specified for each policy anniversary. The PUA NSP at issue is specified for policy year 0.

3.8.2 PUA cash value

PUA cash values are the NSP calculated using the cash value mortality and interest basis. Mid-year cash values are linearly interpolated between the policy anniversary values.

When the base plan cash value is specified using an input table, the PUA cash values are the PUA purchase rates.

3.8.3 PUA reserves

PUA reserves are the PUA amount times the NSP calculated using the reserve mortality and interest basis. Mid-year PUA reserves use the method that applies for base plan reserves, with a reserve premium of zero.

When the base plan statutory or tax reserve is specified using an input table, the PUA NSPs are the PUA purchase rates.

3.8.4 Dividends on PUA

Dividends on PUA are paid in addition to the annual dividend from the base policy. The calculation is the same as applies for the base plan dividend:

- Input dividend scale per \$1 PUA amount,
- Contribution components calculated using the rates that apply for base plan dividends, or
- Component adjustments to the input dividend scale.

3.9 OYT/PUA rider

3.9.1 OYT death benefit

The one year term (OYT) rider increases the death benefit by the OYT face amount. Specify on the model point record the OYT face amount for the policies collected into the model point.

The model point record specifies the following options for determining the OYT amount:

- **Cash Value** – The OYT death benefit is the base plan terminal cash value. Paying the cash value in addition to the base plan death benefit is sometimes called the ‘fifth dividend option’. The PUA death benefit is also paid. The OYT face amount variable on the model point record has no effect.
- **OYT Less PUA** – The OYT sum assured is decreased by the PUA death benefit, so the total death benefit stays level. This is sometimes referred to as making the PUA to be ‘inside additions’. See figure 1 below.
- **OYT Plus PUA** – The OYT sum assured is paid in addition to the paid up additions death benefit; thus paid up additions increase the death benefit. This is sometimes referred to as making the PUA to be ‘outside additions’. See figure 2 below.

The following diagrams illustrate the OYT sum assured options:

Figure 1 – ‘OYT Less PUA’ option

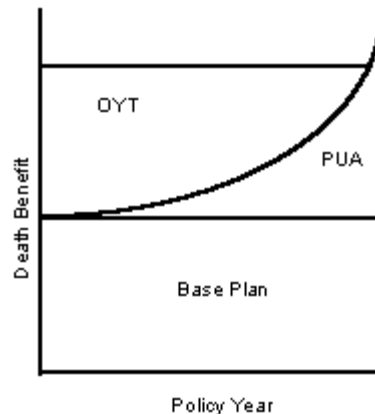
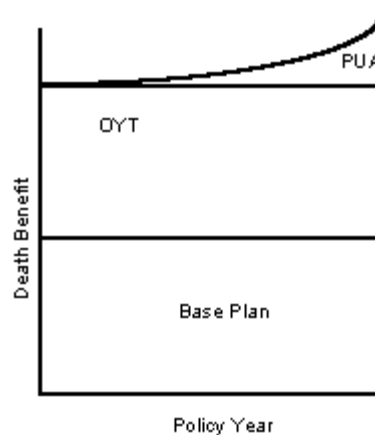


Figure 2 – ‘OYT Plus PUA’ option



3.9.2 Additional premium

The rider can specify an additional premium for the OYT coverage. This applies only when the OYT rider definition is OYT Less PUA or OYT Plus PUA. The additional premium does not apply when the OYT rider definition is Cash Value.

This is a level premium, specified either as: (1) an annual premium per \$1 of OYT (before any reduction by PUA) in product specifications, or (2) the annualized model point OYT premium on the model point record.

Commissions include a percent of this premium, using the base plan commission rate.

3.9.3 OYT cost

The cost of the OYT coverage is determined at the beginning of each policy year, equal to the OYT death benefit times the specified cost per \$1 death benefit. Specify this cost using an external source table.

The available amount to pay this cost is: (1) the annual dividend plus the annual PUA dividend, plus (2) the annualized additional OYT premium less modal loading.

Any excess over the cost is applied to purchase PUA, using the same process as applying the annual dividend to this dividend option. Any shortfall is raised by surrendering PUA.

For the OYT Less PUA death benefit option, purchasing and surrendering PUA changes the OYT cost since the PUA amount affects the OYT death benefit. When the available amount exceeds the OYT cost using the previous PUA, the PUA amount purchased is:

$$\frac{\text{Available amount} - [(\text{OYT} - \text{Previous PUA}) \times \text{OYT cost per \$1}]}{(\text{PUA purchase NSP per \$1} - \text{OYT cost per \$1})}$$

When the available amount is less than the OYT cost using the previous PUA, the PUA amount surrendered is:

$$\frac{[(\text{OYT} - \text{Previous PUA}) \times \text{OYT cost per \$1}] - \text{Available amount}}{(\text{PUA cash value NSP per \$1} - \text{OYT cost per \$1})}$$

3.9.4 Reduced OYT coverage

When all PUA has been surrendered, and the amount available to pay the OYT cost is still too low, the OYT death benefit is reduced to the amount which can be purchased by the amount available. This reduction in the current OYT coverage applies to all OYT rider definitions.

When the model point is in this situation at the projection date, specify the initial OYT face amount on the model point record. This field can be left at its zero default value for the model points with the full OYT coverage in effect.

3.9.5 OYT reserves

There is no cash value from the OYT coverage.

The OYT reserve is the statutory one-year term premium, times the remaining fraction of the policy year*. This treats the OYT cost as being paid at the beginning of each policy year, when the OYT and PUA amounts are determined. *For Mean reserves, the multiple is 0.5.

When the base plan statutory or tax reserve is specified using an input table, you must still specify the reserve basis variables, since they are needed to calculate the OYT reserve.

3.10 Vanishing premium option

3.10.1 Premiums paid by dividend amounts

The model point record can specify a 'vanish year' policy anniversary (default of 0 means this feature does not apply). Beginning on this policy anniversary, premiums are paid using dividend amounts when this is sufficient to cover the entire premium.

This vanishing premium feature applies only for model points having the Accumulate or PUA dividend option. Vanishing premium variables are ignored when dividends are paid in cash, or when the model point includes the OYT/PUA rider.

The amount available to pay the premium is the annual dividend plus the annual PUA dividend. When this amount exceeds the premium amount, the excess is applied to the dividend option.

When the amount available is less than the premium amount, dividend accumulations are first withdrawn to make up the difference, then PUA is surrendered as needed.

When the PUA cash value is too small to pay the full premium, this vanishing option does not apply for that policy year.

3.10.2 *Current vanish status*

Specify on the model point record whether premiums are currently vanished at the projection date.

4 JOINT LIFE PRODUCTS

Joint life products have a second insured, in addition to life 1. The product is a joint life product when the life status variable is First To Die or Frasierized Last Survivor. When the life status is Single Life, the second life variables do not affect the projection.

4.1 Second life characteristics

4.1.1 *Life 2 model point variables*

Specify insured characteristics model point variables for life 2 in addition to life 1. Life 2 characteristics include issue age, gender, smoking class, and rate class. These insured characteristics are particularly important for mortality variables, which are the rates for: (1) experience, cash value, and reserve mortality, and (2) YRT reinsurance rates.

See the Mortality variables section in the Input organization chapter, above.

4.1.2 *Life 1 and life 2 scalars*

Since issue_age is used to calculate values such as the years to maturity, 'life 1' must be:

- the older of the two insureds for first to die model points, and
- the younger of the two insureds for Frasierized last survivor model points.

The order of the two lives on the model point record is handled automatically, by converting the life 1 and life 2 model point variables into scalars, instead of using the model point variables directly:

- For first to die model points with life 1 younger than life 2, the life 1 scalars return the life 2 model point variables, and the life 2 scalars return the life 1 model point variables.
- For Frasierized last survivor model points with life 1 older than life 2, the scalars also switch the values this way.
- Otherwise, the life 1 scalars return the life 1 model point variables, and the life 2 scalars return the life 2 model point variables.

For example, the scalars issue_age and gender normally have the values issue_age_mp and gender_mp, but return the values issue_age_mp_life2 and gender_mp_life2 when the lives must be switched. Since formulas reference the scalars, they are automatically based on the correct life.

4.2 First to die

For first to die products, the death benefit is paid on the first death; the policy terminates at that time. Both insureds are alive at the projection date.

The annual mortality or YRT rate is:

$$q_1 + q_2 - q_1 \times q_2$$

4.3 Frasierized last survivor

For last survivor products, the death benefit is paid when both insureds have died. Since there is no change in status when the first life dies, the projection assumes there is no knowledge about whether life 1 or life 2 is still alive at the projection date, except that at least one of them must be alive since the policy is still in force.

The mortality and YRT rates are Frasierized from issue, as described below.

4.4 Frasierization

Since Frasierized last survivor products continue to operate as normal until the second death, the projection assumes there is no information whether only one life is surviving at the projection date. The mortality rates for life 1 and life 2, plus the joint life contagion effect, are combined into a Frasierized rate equal to the sum of:

$$\begin{aligned} &P(\text{both alive BOY}) \times P(\text{both die during the year} | \text{both alive BOY}) \\ &\quad + P(\text{only life 1 alive BOY}) \times P(\text{life 1 dies during the year} | \text{only life 1 alive BOY}) \\ &\quad + P(\text{only life 2 alive BOY}) \times P(\text{life 2 dies during the year} | \text{only life 2 alive BOY}) \end{aligned}$$

This formula Frasierizes the annual mortality rates, then the Frasierized annual rate is converted to a monthly rate as usual. This method commonly applies for reserve mortality and YRT rates. Annual experience mortality rates are also Frasierized prior to calculating the monthly mortality rate, so that calculations are simplified by producing monthly mortality rates which are constant within each policy year.

4.4.1 Joint life mortality addition

For experience mortality, the annual probability that both lives die together (or within some short interval) is increased by a joint life mortality rate addition, which is sometimes called a contagion effect. The specified addition is reduced when the joint death rate would exceed either the first or second life mortality rate. That is:

$$q_1 \times q_2 + \text{adjusted addition} \leq \min(q_1, q_2)$$

4.4.2 Minimum YRT rate

The Frasierized annual YRT reinsurance rate is subject to a minimum floor, such as .00012 (\$.12 per thousand).

5 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves. PBA reserves and capital are discussed in the PBA book.

5.1 Reserve methods

5.1.1 Terminal reserves

The following reserve methods are available:

- Not applicable – reserves are zero, and don't add to projection time.
- 1980 CRVM, and pre-1980 CRVM
- Net level premium – no expense allowance
- Full preliminary term – zero terminal reserve for the first policy year

- Table – net premiums and terminal reserves per unit are specified for each policy year, starting with 0 (issue date) for the terminal reserves. The PUA net single premiums (NSP) are the PUA purchase rates.

5.1.2 *Death benefit varying within policy year*

When the death benefit varies within a policy year, the death benefits used for reserve calculations are the average of the death benefit as of the end of each month within the policy year. For example, when the sum insured varies monthly.

5.1.3 *Premium pattern*

Reserve net premiums are the level percent of gross premiums calculated using the specified reserve method. For product pricing work, before the gross premiums have been determined, specify the premium pattern in the guaranteed premium variables. For example, the default value of '1' for all policy years specifies that premiums are level.

The reserve method applies to the pattern of guaranteed premiums without policy fees and modal loadings.

5.2 *Reserve basis*

5.2.1 *Mortality*

Reserve mortality is a base rate, times the multiple, then plus the addition. Lookups for reserve mortality tables should be setup similar to the experience mortality rates since these rates are read for both the first and second lives.

5.2.2 *Interest*

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeStatValn and LifeTaxValn ranges in the Valuation Interest Rates Excel workbook.

5.2.3 *Function timing*

Function timing can be:

- **Curtate** – Annual premiums and claims paid at the end of the policy year
- **Continuous** – Continuous premiums and immediate payment of claims
- **Semicontinuous** – Annual premiums and immediate payment of claims.

The continuous premium multiple and the immediate claim multiple are defined in the Function timing section in the Valuation options chapter of the Actuarial math library appendix.

5.2.4 *Graded expense allowance*

The expense allowance can grade to zero between two policy anniversaries. Net premiums are the reserve method's percent of guaranteed premiums before the grading, the net level premium percent after the grading, and the calculated grading percent in between.

For example, specifying grading between the 15th and the 20th policy anniversaries results in the selected reserve method applying for the first 15 policy years, and the net level premium method applying from the 20th anniversary to maturity. Between these two anniversaries, net premiums are the level percent of the guaranteed premiums which advances the 15th year terminal reserve to the 20th year terminal reserve.

Specify to grade the expense allowance over a period shorter than the premium term by grading from issue (policy anniversary zero) to policy anniversary 'n'.

Specifying the zero default value for these anniversaries results in the expense allowance being amortized to zero by the end of the premium term.

5.3 Mid-year reserve

Reserves between policy anniversaries can be:

- Annual interpolated mean – Average terminal reserve, plus one-half of the annual reserve net premium. A deferred premium asset is established.
- Annual interpolated terminal – Average terminal reserves at the beginning and end of the policy year. Reserves are the same for each month within a policy year. An unearned premium reserve is established.
- Monthly interpolated mean – Monthly interpolated terminal reserves, plus the portion of the annual reserve net premium that applies to future months. A deferred premium asset is established.
- Monthly interpolated terminal – Monthly interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.
- Daily interpolated mean – Daily interpolated terminal reserves, plus the portion of the annual reserve net premium that applies to the remaining number of days of the year. A deferred premium asset is established.
- Daily interpolated terminal – Daily interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.

When the reserve function timing is Continuous, the mid-year reserve net premium is a discounted continuous net premium.

5.4 Actuarial guideline 32

Actuarial guideline XXXII (AG 32) requires additional reserves for policies which pay death claims at the time of death, but have statutory reserves calculated assuming death claims are paid at the end of the policy year. The additional reserve is:

guideline 32 percent (input) $\times$ statutory reserve interest rate $\times$ statutory reserve

This addition applies only when the reserve claims timing assumption is curtate (claims paid at the end of the policy year). The addition can be ignored when reserves are more conservative than the minimum statutory formula reserves. Specify the percent as 0.333 (33.3%, one-third) when claims are paid 'upon due notice', and 0.50 (50%) when claims are paid with interest from the date of death.

The additional reserve may not be included in tax reserves in cases such as for issues before 1994, when AG 32 first became effective. Thus this input variable is included separately for tax reserves.

The AG 32 reserve is added to the statutory reserve before calculating the deficiency reserve. The alternative minimum reserve is not affected by this adjustment. That is, the AG 32 reserve increases the basic reserve only, thus decreasing the deficiency reserve.

5.5 Deficiency reserve

5.5.1 1976 Standard valuation law method

The deficiency reserve is the 'alternative minimum' reserve less the statutory basic reserve, calculated using mid-year reserves, when at least one future alternative minimum reserve net premium exceeds the guaranteed gross premium. The alternative minimum reserve is calculated using the lower of the guaranteed gross premium and reserve net premium in each year. Specify the deficiency reserve mortality and interest basis separately from the statutory reserve basis.

The statutory basic reserve includes the AG 32 reserve, while the alternative minimum reserve is not affected by AG 32.

5.5.2 **Guaranteed gross premium**

The alternative minimum reserve is calculated using the guaranteed premium in policy years when the deficiency reserve premium would be higher. The guaranteed premium can include the policy fee and modal loading (annualized premium), the policy fee only (annual mode premium), or neither (the per unit premium).

When the guaranteed premium per unit is less than the annual premium per unit, the guaranteed premium is the annual premium. Thus you can leave this variable at its default value (1.0) when guaranteed premiums do not apply.

5.6 **Additional reserve**

Additional statutory reserves can be specified as an amount per unit. For example, vary these amounts monthly by linking to an external source using the `mths_since_issue` column as a lookup key.

The statutory additional reserve can be added to tax reserves as well.

The additional reserve is added after the determination of the deficiency reserve, and before the consideration of the tabular cost of insurance and cash value floors (discussed below).

When detail reports split out reserve values, 'additional reserves' include the increase over basic reserves due to the reserve floors.

5.7 **Tabular cost of insurance floor**

The reserve is subject to a minimum based on the tabular cost of insurance. That is, this floor applies to the sum of: (1) basic reserve (including the AG 32 reserve), (2) deficiency reserve, and (3) additional reserve.

The floor is the tabular cost of insurance (c_x) times the sum assured, times the following fraction for each mid-year reserve method:

- Annual interpolated mean – $\frac{1}{2} c_x$.
- Monthly interpolated mean – c_x times the fraction of the policy year remaining. For example, this floor at the end of policy month 1 is $\frac{11}{12} c_x$.
- Annual interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment. For example, this floor at the end of policy month 1 is $\frac{11}{12} c_x$ for annual premiums, and $\frac{2}{12} c_x$ for quarterly premiums.
- Monthly interpolated terminal– Same as above.
- Daily interpolated mean – c_x times the fraction of the policy year remaining measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $\frac{344}{365} c_x$.
- Daily interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $\frac{344}{365} c_x$ for annual premiums, and $\frac{71}{365} c_x$ for quarterly premiums.

5.8 **Cash value floor**

The total mid-year reserve is not less than the cash value.

5.9 Dividend options reserve

The base plan reserve is first calculated ignoring dividend option amounts. Then the following dividend option reserves are added to the base plan reserve:

- Dividend accumulations – The accumulation amount.
- Paid up additions – The PUA face amount times the mid-year net single premium calculated using the reserve basis. For Mean reserves, the NSP is the average of the NSP at the beginning and end of the policy year. For other mid-year reserve methods, the NSP is interpolated using the policy month. When the base plan statutory or tax reserve is specified using an input table, the PUA NSPs are the PUA purchase rates.
- One year term – The OYT face amount times a fraction of the tabular cost of insurance (cx). For Mean reserves, this is $\frac{1}{2}$ cx. For other mid-year reserve methods, this is cx times the fraction of the policy year remaining. For example, this is $\frac{11}{12}$ cx at the end of policy month 1. This treats the OYT cost as being paid at the beginning of each policy year, when the OYT and PUA amounts are determined. When the base plan statutory or tax reserve is specified using an input table, you must still specify the reserve basis variables, since they are needed to calculate the OYT reserve.

5.10 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record sets the tax reserve to be the statutory reserve, including the comparison to the tabular cost of insurance floor using the statutory basis, and including the statutory AG 32 reserve, but excluding any deficiency reserve.

The same method choices are available for tax reserves, using the tax reserve interest and mortality basis. The function timing definition, and the mid-year reserve method, is the same as for statutory reserves.

The statutory additional reserve can be tax deductible.

The tabular cost of insurance floor, calculated using the tax reserve mortality and interest, is considered after addition of the additional reserve.

The final mid-year tax reserve is not greater than the mid-year statutory reserve, and not less than the mid-year cash value.

5.11 Deferred premium asset

5.11.1 Balance sheet

When the mid-year reserve method is annual, monthly, or daily interpolated mean, a deferred premium asset offsets the formula's assumption that premiums are paid annually. This does not apply for other mid-year reserve methods, which recognize that premiums are paid modally.

The balance sheet includes modal reserve net premiums due during the remainder of the current policy year, but not greater than the deferred gross premiums.

This is a non-invested asset. It is excluded from calculations of the amount to invest. The asset yield calculation removes the deferred net premium asset from the amount of assets generating investment income.

5.11.2 Profit effect

Book profit includes the increase in the deferred net premium asset. However, this is split into two parts, which add to the deferred net premiums increase:

- **Deferred gross premiums increase** – Premium income includes the increase in deferred gross premiums.

- **Statutory loading increase** – The statutory loading is the deferred gross premiums less deferred net premiums.

The deferred net premiums are not affected by reserve floors. That is, the statutory loading does not change when the reserve is increased due to the tabular cost of insurance or cash value floors.

5.11.3 Taxable income

Taxable income similarly includes the tax loading increase, which is calculated the same way as statutory loading, but using the tax reserve basis.

5.12 Unearned premium reserve

5.12.1 Balance sheet

When the mid-year reserve method is annual, monthly, or daily interpolated terminal, an unearned premium reserve is established. The unearned premium reserve is the portion of the modal valuation net premium that has been collected but not yet earned.

The balance sheet includes the statutory unearned premium reserve as a separate line item from the statutory reserve. The unearned premium reserve is set to zero if the reserve being held is the tabular cost of insurance or cash value floor.

5.12.2 Profit effect

Book profit includes the increase in the unearned premium reserve.

5.12.3 Taxable income

Taxable income similarly includes the unearned premium reserve increase, which is calculated the same way as statutory unearned premium reserve increase, but using the tax reserve basis.

5.13 Gross premium reserve

Letter of credit (LOC) reserve financing creates a non-invested LOC asset that offsets the 'redundant' part of the statutory reserve, based on the calculation of the gross premium reserve (GPR) for each model point assigned to a reserve financing transaction. Book profit is increased by the LOC asset increase (offsetting part of the statutory reserve increase), and is decreased by LOC costs (interest paid on the LOC assets). See the Gross premium reserves chapter in the PBA book.

6 ACTUARIAL ASSUMPTIONS

6.1 Commissions

6.1.1 Premium commission

The percent of premium commission is the commission rate for each policy year, times the premium payments as they are paid. The policy fee and modal loading can be commissionable (included) or noncommissionable (excluded).

6.1.2 Annualized commission

A specified percent of the total first year premium commission is paid at issue. No further premium commissions are paid until the cumulative earned commission exceeds the annualized amount. From then on, premium commissions are paid as usual. This determination of commission payments is made for continuing policies, not taking into account terminated policies.

6.1.3 Commission chargeback

A portion of the premium commission can be charged back in the event of lapse or surrender during the first three years after issue. The chargeback percent applies either to: (1) total first year premium commissions paid to date, or (2) the annualized commission amount. The chargeback percent typically varies by month, by linking to the external source using the `mths_since_issue` column as a lookup key.

The commission subject to chargeback is calculated per unit in force, and thus is not reduced by decrements. This amount applies to the units lapsing and surrendering each month.

6.1.4 Commission overrides and agent terminations

The specified commission rates are the average commission rates, taking into account average writing agent commissions, commission overrides, and the effect of agent terminations.

6.2 Expenses

6.2.1 Expense adjustment multiple

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

6.2.2 Acquisition expenses

Acquisition expenses incurred at issue can include:

- Amount per policy
- Amount per unit
- Percent of the annualized first year premium

Acquisition expenses could vary by underwriting class, which is discussed in the Mortality section, below.

6.2.3 Maintenance expenses

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

- Amount per policy, separate between premium paying and paid-up
- Amount per unit, separate between premium paying and paid-up
- Percent of the annualized premium

BOM maintenance expense

An additional maintenance expense is incurred at the beginning of each month, when commissions are paid:

- Percent of each commission payment, ignoring commission chargeback.

6.2.4 Transaction expenses

Maintenance expenses include the following expenses related to administrative activities:

- Expense per death claim policy
- Expense per death claim unit.

6.2.5 Reinsurance administration expenses

Reinsurance administration expenses are discussed in the Reinsurance chapter, below.

6.2.6 Inflation

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

Note: The average size for future issues can be inflated (to offset expense inflation), by customizing the scalar `units_per_pol` to divide the result by `infl_cumul(commencement_period)` for model points issued after the start of the projection. This would reduce the number of policies issued, with no effect on the units, sum assured, and premium issued for the model point.

6.2.7 Policy loan expenses

Policy loan expenses are a percent of the loan balance at the end of the previous month. One twelfth of the annual rate applies each month.

This amount is a negative cash flow at the end of each month, and is thus subtracted from direct cash flows.

Policy loan expenses reduce book profit, by reducing investment income from policy loans. Thus this expense is treated as an investment expense for statutory and GAAP profit reporting. It is not included with other expenses on profit reports.

However, PBA deterministic reserves include this expense in the present value of expenses. This is a different treatment of policy loan expenses than applies for profit calculations.

6.2.8 Premium tax

Premium taxes are reported with expenses. The tax is a percent of each premium payment.

Inflation, and the expense adjustment multiple, do not apply to premium tax.

6.2.9 Fixed expenses

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

6.3 Mortality

6.3.1 Input variables

'Mortality variables' for traditional life products include mortality base, multiple, and addition rates, mortality improvement, and YRT reinsurance rates. These variables are read twice – first for life 1 and then again for life 2.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys `mort_issue_age`, `mort_gender`, `mort_smoking_class`, `mort_rate_class`, `mort_pol_yr`, `mort_cal_yr`, and `mort_term_conversion_defn` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

6.3.2 Mortality adjustment multiple

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

6.3.3 Term conversion adjustments

Term products often offer the option to convert to a permanent product (i.e. traditional life) prior to the end of a level term period without additional underwriting.

Specify the original term issue date and whether the policy was issued as a result of a term conversion on the model point record. For such policies, the base mortality rates, mortality improvement factors, and mortality table adjustments will be calculated using a point-in-scale method based on the original issue date of the term policy. Additionally, a separate set of base mortality rates and mortality table adjustments may apply to the term conversion cohort of policies if specified.

Term conversion adjustments do not apply to joint life policies.

6.3.4 Underwriting class

Life insurance products can use the `underwriting_class` scalar to vary product assumptions such as mortality rates and acquisition expenses. The underwriting class is calculated by comparing the policy size against the nonmedical and paramedical underwriting limits specified with the experience mortality input. The underwriting class has a value of either:

- Nonmedical – when `ins_issued_per_pol` $\leq$ `nonmedical_limit`
- Paramedical – when `nonmedical_limit` $<$ `ins_issued_per_pol` $\leq$ `paramedical_limit`
- Medical – when `ins_issued_per_pol` $>$ `paramedical_limit`.

6.3.5 Rate class

Mortality rates varying by rate class require special treatment. The model point variable (`rate_class_mp`) can have a large number of options specific to the company's business. But the 2001 CSO mortality tables varying by rate class only define rates for Standard, Preferred, and Super. Thus there are two different rate class hidden variables:

- **Company rate classes** – The company's rate classes are in `mort_rate_class`. For example, there might be a dozen different classes for this scalar. Experience mortality and YRT reinsurance rates vary by this lookup key.
- **Reserve rate classes** – The 2001 CSO mortality rate classes are in `mort_rate_class_res`. The company's different classes are mapped into the three rate classes prescribed for reserve mortality rates. Customize the formula for this scalar, to define how each company rate class is translated into the three reserve rate classes.

6.3.6 Base mortality

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

6.3.7 Mortality improvement

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$. When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

6.3.8 Mortality table adjustments

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

6.3.9 *Joint life mortality*

See the Joint life products chapter, below.

6.3.10 *Monthly mortality*

Deaths occur at the end of each month, prior to other decrements and policyholder transactions. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

6.4 *Lapses and surrenders*

A termination is a 'surrender' when there is a base plan cash value, and a 'lapse' otherwise. This determination ignores any cash value from dividend options.

Lapse rates apply to units after mortality decrements, which are processed before all other decrements.

6.4.1 *Lapse adjustment multiple*

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date.

6.4.2 *Policy year lapse rates*

The base lapse rate is adjusted by a multiple. Separate lapse rates apply for premium paying policies versus paid-up policies.

6.4.3 *Monthly lapses*

The annual lapse rate after all adjustments is constrained to be not more than 1.0 (100%).

Lapses and surrenders occur either: (1) at the end of each month, or (2) modally (just before the next premium is due). Lapses occur monthly after the last premium has been paid.

Each of the above lapse rate adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly or modal lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/\text{event frequency})}$$

6.4.4 *Skewed lapses*

Skewed lapses allocate a specific portion of annual lapses to each month within each policy year. The skew percentages are normalized to assure that they add to 1.0 (100%) for each policy year. Lapse skewing can be specified differently for each policy year.

When skewed lapses apply, the exponent in the monthly/modal lapse rate formula is replaced by the sum of the skew percentages for the months included in the decrement time step. For example, when the lapse applies for months with monthly skew percentages adding up to 0.4 (40%), the lapse rate is:

$$1 - (1 - \text{annual lapse rate})^{0.4}$$

7 *RIDERS*

7.1 *Multiple riders*

Multiple riders can apply to a model point.

7.1.1 Riders input

Each rider creates an instance of the trad_rider submodel. The input for each variable must be linked to an external source containing values varying by the rider names.

7.1.2 Rider model point input

The rider model point record identifies which riders apply to each model point.

7.2 Rider types

7.2.1 Accidental death benefit

Please refer to the L13 Riders book for a discussion of the accidental death benefit riders which can be linked to traditional life model points.

7.2.2 Waiver of premium

Please refer to the L13 Riders book for a discussion of the waiver of premium riders which can be linked to traditional life model points.

8 REINSURANCE

8.1 Multiple reinsurance treaties

Multiple reinsurance treaties can apply to a model point.

8.1.1 Treaty input

Each reinsurance treaty creates an instance of the trad_ireins submodel. The input for each variable must be linked to an external source containing values varying by the treaty names.

The exception is that the YRT retention maximum per policy applies for all treaties combined, and thus does not vary by treaty. For example, it might vary by plan code.

Each treaty can be either a YRT, coinsurance, or modified coinsurance treaty. Each of these types is discussed in a separate section, below.

8.1.2 Reinsurance model point input

The reinsurance model point record identifies which treaties apply to each model point, as well as the ceded percent or amount for each treaty. The reinsurance model point layout is discussed in the Reinsurance model point records layout section, below.

8.2 YRT reinsurance

Yearly renewable term (YRT) reinsurance cedes a portion of the net amount at risk (NAR). Reinsurance premiums are calculated using YRT rates applied to the ceded coverage amount.

8.2.1 Ceded amount

The ceded amount is a percent of the net amount at risk (NAR) at the beginning of each policy year, which is the sum assured less the cash value at the end of the previous month.

Using reinsurance model point variables

The ceded percent can be specified on the reinsurance model point record, either directly or as the initial ceded base death benefit amount, which is converted to the ceded percent (i.e. ceded base death benefit

amount / base death benefit at the projection date, where base death benefit excludes any additional death benefit from dividend options.) This is proportionate YRT reinsurance, where the ceded percent is fixed.

Using treaty variables

The ceded percent can be calculated using the reinsurance input variables that vary by treaty code. The ceded amount is a specified percent of any excess of the NAR over the retention threshold per policy. For example:

- First dollar quota share – The threshold is zero.
- Excess retention – The ceded percent applies to the excess of the NAR over the specified threshold.

The ceded percent varies over time, depending on the changing relationship between the NAR and the threshold.

8.2.2 Maximum YRT retention per policy

The YRT reinsurance treaty amounts for a model point are increased when the combined ceded amounts would be less than [the NAR less the maximum retention amount per policy]. The YRT retention maximum per policy applies for all treaties combined, and thus does not vary by treaty. For example, it might vary by plan code.

This adjustment applies only for YRT treaties that have the ceded amount calculated using the treaty variables. Treaties that have the ceded percent/amount specified on the model point record are not adjusted.

When the sum of the ceded amounts is too low, the initial (at the projection date, or issue date for new business model points) ceded amount for each treaty is increased in proportion to the individual ceded amounts.

8.2.3 YRT rate

The YRT reinsurance rate for each treaty is a base rate, times a multiple. These are mortality variables, which operate the same as for mortality rates. See the Mortality variables section in the Input organization chapter, above.

The YRT premium can be zero for the first policy year by specifying the multiple as zero for the first policy year.

Term conversion adjustments

Term products often offer the option to convert to a permanent product (i.e. traditional life) prior to the end of a level term period without additional underwriting.

Specify the original term issue date and whether the policy was issued as a result of a term conversion on the model point record. For such policies, the YRT base rate and YRT multiple will be calculated using a point-in-scale method based on the original issue date of the term policy. Additionally, a separate set of YRT base rates and YRT multiples may apply to the term conversion cohort of policies if specified.

YRT term conversion adjustments may be turned off using the `yrt_term_conversion_defn` variable (on the LifeReinsurance input page).

YRT term conversion adjustments do not apply to joint life policies.

8.2.4 Reinsurance premium

YRT reinsurance premiums are paid either annually at the beginning of each policy year, or monthly.

The annual YRT reinsurance premium per policy is:

$$\text{ceded amount} \times (\text{annual YRT rate}) + \text{reinsurance policy fee}$$

When YRT premiums are paid monthly, one twelfth of this amount is paid at the beginning of each month.

The reinsurer refunds the unearned YRT reinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual YRT premium, times the remaining months in the year divided by 12.

8.2.5 Reinsurance benefits

The reinsurance benefit paid to the ceding company each month is the ceded amount times the monthly experience mortality rate. The ceded amount is the amount calculated at the time of each YRT reinsurance premium.

8.2.6 Reinsurance reserve

The formulaic statutory and tax reserve credit for YRT reinsurance is the tabular cost of insurance floor for the ceded amount.

8.3 Coinsurance

Coinsurance treaties cede a portion of the entire policy, including transferring the ceded reserve to the reinsurer. Reinsurance benefits include all benefits provided by the policy.

8.3.1 Coinsurance percent

When the coinsurance percent is calculated using the input for the treaty, the percent is x% of the sum insured at issue up to the threshold per policy specified for each treaty, plus y% of any excess over the threshold. When the coinsurance percent is specified by model point, this treaty input is ignored. The coinsurance percent remains constant to maturity.

The coinsurance percent applies to the entire policy, including the deficiency reserve and the additional reserve.

The coinsurance benefits paid to the ceding company each month are the projected policy benefits, times the coinsurance percent.

8.3.2 Coinsurance premium

Coinsurance premiums are paid at the same premium frequency as the model point.

Coinsurance premiums are based on the gross premium, including modal loading and policy fee.

The reinsurer refunds the unearned coinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual amount times the remaining months in the year divided by 12.

8.3.3 Premium tax

The reinsurer reimburses for the coinsurance percent times premium taxes paid by the ceding company.

8.3.4 Benefits

The reinsurer reimburses for the coinsurance percent times all benefits paid by the ceding company. Reimbursements are split between death benefits and other benefits, for reporting purposes.

8.3.5 Expense allowance

The reinsurer pays an expense allowance to the ceding company at the reinsurance premium frequency. The expense allowance is the annualized reinsurance premium times a specified percent.

The unearned current year expense allowance is refunded to the reinsurer when the policy terminates before the next reinsurance premium payment for any reason. The unearned expense allowance is the last amount

paid, times the ratio of the number of months until the next reinsurance premium, divided by [12 divided by the reinsurance premium frequency].

8.3.6 *Reinsurance reserve*

The reinsurance reserve credit for coinsurance treaties is the total statutory reserve, times the coinsurance percent.

8.4 Modified coinsurance

Modified coinsurance (ModCo) treaties operate as described above for coinsurance treaties, except that the reinsurance reserve is retained by the ceding company. The ceding company reimburses the reinsurer for interest on the reinsurance reserve which is backed by assets invested by the ceding company.

The adjustments occur monthly, so that nested projections match standard projection results. The adjustment amount is the current reinsurance reserve, less the reinsurance reserve as of the previous adjustment date accumulated with interest.

The ModCo reserve adjustment interest rate is the asset yield rate.

8.5 Reinsurance administration expenses

Reinsurance administration expenses are incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include:

- Amount per policy
- Percent of the ceded amount.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Coinurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for coinsurance.

9 FACTORS CALCULATIONS

9.1 Values calculated

9.1.1 *Special projection type*

A projection type calculates 'factors' when the product submodel is the top model and the target column is factors_tgt. This target column directs the calculation of each item specified (discussed below). The target column calculates results for each policy year; thus the Projection period from and to properties have no effect on factors calculations.

The Assumption Set Restore property (in the run parameters object) must be Yes, since this is a seriatim projection. Project variables such as pricing definition and profit distribution mode do not apply for this special type of projection.

9.1.2 *Factor types*

Specify at the bottom of the Product input page which factors to calculate:

- Cash values per unit – net premium and terminal values

- Statutory reserves per unit – net premium and terminal values
- Alternative minimum reserves per unit – net premium and terminal values
- Tax reserves per unit – net premium and terminal values
- PUA values – net single premiums per \$1 of PUA face amount, for cash values, statutory reserves, tax reserves, and purchase rates.

9.1.3 *Model points*

Factors calculations systematically process for each model point. Set up a model point for each issue age to be calculated, within each combination of insured characteristics (sex, smoking class, rate class, etc.). For example, you might set up model points for issue ages 0 through 80 for male nonsmokers, then repeating these model points for male smokers, then female nonsmokers, then finally female smokers.

Best practice is to create the report group for each model point so that it identifies the model point characteristics. For example, the report group might be PlanA-M-NS-ST-35 for the model point calculating factors for plan code PlanA, for a male nonsmoker standard risk class issue age 35. Then the report groups identify the meaning of each model point.

The same as for Pricing cell projections, each model point must be issued on the first day of the projection. For example, when the projection date is 2013/12 (end of December, 2013), the issue date must be 2014/01/01 (January 1, 2014).

9.2 Factors reports

9.2.1 *trad_py submodel output*

The factors calculation output is contained in the trad_py submodel, which has column results by policy year.

9.2.2 *Reporting tool*

The US library reporting tool includes special reports for these traditional life factors:

- **Nonforfeiture values** – The death benefit, premium pattern, cash value interest rate, and cash value net premium and terminal value for each policy year.
- **Policy reserves** – In addition to the segmented and unitary net premiums and terminal reserves, this report shows (for each policy year) the death benefit, premium pattern, interest rate, mortality rate, and segment breaks. Statutory, alternative minimum, and tax reserve results are shown on the report.
- **PUA values** – The PUA net single premiums per \$1 of PUA face amount, for cash values, statutory reserves, tax reserves, and purchase rates.

To produce these reports, select reporting tool parameters for Pricing reports, check the reports to produce in the Policy Year Reports tab, and check which model points to produce reports for in the Liability Groups tab.

9.3 Factors projection setup

Specify the following settings in the run parameters for the projection task calculating factors:

Property	Value
Target Colum	trad->factors_tgt
Projection Period From	0
Projection Period To	0 (since calculations are driven by factors_tgt)
Projection Period Min	0
Projection Period Max	Higher than the maximum maturity months, such as 1452 for 121 years
Projection Loops	1
Assumption Set Restore	Yes
proj_date	yyyy/mm, the end of the month before the issue date

The only special variables to specify on the input pages are the options at the bottom of the Product input page, specifying which of the factor types to calculate.

10 MODEL POINT RECORDS

10.1 Model point record layout

The model point record layout for the traditional life product type is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	up to 125 characters (assigns model point to report group)
B	seg_id	business segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date_mp	yyyy/mm/dd or mm/dd/yyyy
F	disab_date_mp	yyyy/mm/dd or mm/dd/yyyy (set to 1/1/9999 if policy is not disabled)
G	term_conversion_defn_mp	Yes (calculate term conversion adjustments), No
H	term_issue_date_mp	None (when original term issue date is unavailable) yyyy/mm/dd or mm/dd/yyyy
I	plan_code	up to 20 characters (link to assumptions by plan)

Excel Column	Variable	Value
J	stat_valn_code	up to 20 characters (link to valuation basis and assumption by valuation code)
K	tax_valn_code	up to 20 characters (link to valuation basis and assumption by valuation code)
L	issue_age_mp	positive integer
M	gender_mp	Male, Female, Unisex
N	smoking_class_mp	Nonsmoker, Smoker, Aggregate
O	rate_class_mp	Standard, Preferred, Super, [extend with user additions]
P	issue_age_mp_life2	positive integer
Q	gender_mp_life2	Male, Female, Unisex
R	smoking_class_mp_life2	Nonsmoker, Smoker, Aggregate
S	rate_class_mp_life2	Standard, Preferred, Super, [extend with user additions]
T	prem_status_defn	Active, RPU
U	prem_mode	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
V	prem_term_mp	premium years (0 = premium term is specified on the Product input page)
W	init_annualzd_prem	initial annualized premium for mp
X	single_prem	additional single premium for mp
Y	gpr_pct_mp	percent of this model point included in Gross Premium Reserve calculations (1.0 for 100%)
Z	init_sum_assured_mp	initial sum assured for mp (0 = calculate from initial units)
AA	init_policies	initial policies for mp (required)
AB	init_units	initial units for mp (0 = calculate from initial sum assured)
AC	init_pol_loan	initial policy loan for mp
AD	init_div_int_rate	initial dividend interest rate, such as 0.0500 for 5%, for determining dividend amounts before a new rate is declared at the end of the first calendar year; when this value is zero (default) the rate is calculated by applying the dividend interest rate formula at the projection date

Excel Column	Variable	Value
AE	prev_annual_div	base policy dividend for mp paid at the last policy anniversary. When this value is zero, the check against decreasing the dividend for the next policy year applies only for future dividends.
AF	div_opt_defn	Cash, Accumulate, PUA, OYT PUA
AG	init_div_accum	initial dividend accumulations for mp
AH	init_pua_face	initial paid up additions face amount for mp
AI	init_oym_face	initial OYT rider coverage amount for mp, when only part of the specified OYT amount has been purchased due to the amount available to pay the OYT cost being too low. This field can be left at its default of 0 when the full OYT coverage is in effect. This field applies for all OYT definitions, including Cash Value.
AJ	oym_defn	Cash Value, OYT Less PUA, OYT Plus PUA
AK	oym_face	one year term rider face amount for mp when the OYT definition is OYT Less PUA or OYT Plus PUA. It has no effect when the definition is Cash Value. This is the OYT face amount; the actual OYT coverage can be less in policy years when the amount available to pay the OYT cost is not sufficient to pay for the full coverage amount.
AL	annualzd_prem_oym	one year term rider annualized additional premium for mp
AM	vanish_yr	premiums vanish beginning this policy year (if possible); 0 = premiums do not vanish
AN	init_vanish_flag	1 when premiums are currently vanished; 0 otherwise
AO	prev_annual_div_pua	0.00 (to carry previous time step annual pua dividend value only).

10.2 Reinsurance model point record layout

The reinsurance model point record identifies which treaties apply for each model point, using the linked variable, mp_id. The reinsurance model point record can have any number of reinsurance treaties per model point. It also specifies the ceded percent or amount for each treaty as described in the table below. If the reinsurance model points are omitted from the reinsurance model point file, then the liability model point will not include reinsurance.

Warning! The order of reinsurance model points must be in the same order as the liability model points.

The reinsurance model point record layout is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	mp_id	Up to 12 characters. Links to liability data file mp_id. This value can be listed any number of times to indicate the number of treaties applied to the policy.
B	treaty_code	A user defined input that references the treaty code in the Resinsurance.xlsm workbook.
C	ceded_amt_defn	Percent - the ceded amount should be input as a percent from 0 to 1 Dollar Amount - the ceded amount should be input as the ceded death benefit amount Treaty Input - the ceded amount should be input as 0, and the ceded amount will be calculated using the treaty variables
D	ceded_amt	See above. Input the ceded percent, the ceded dollar amount, or 0 if the ceded amount is specified by the treaty.

10.3 Units basis

Product values are calculated per unit in force, then multiplied by the current number of units. For example, the death benefit, cash value, and reserves per unit.

10.3.1 Initial units

The initial (at the projection date) policies must be specified on each model point record, along with either or both of the initial units and initial sum assured. These are the aggregate values for the policies collected into each model point.

- When the initial units are not specified on the model point (this variable is left at its zero default value), the initial units are the initial sum assured divided by the specified sum assured per unit.
- When the initial sum assured is not specified on the model point (this variable is left at its zero default value), the initial units variable applies directly.
- When both the initial units and initial sum assured are specified on the model point record, the initial units variable applies. The Run log contains a warning if the initial sum assured is more than [\$1 times the initial units] different from the initial units times the specified sum assured per unit.
- When both the initial units and initial sum assured are zero, the model point is skipped and a warning is issued in the Run log.

The initial policies must be specified for each model point. The average size policy is the initial units divided by the initial policies, and the initial sum assured divided by the initial policies.

10.3.2 Pricing cells

For model points issued at the start of a seriatim pricing projection, the initial policies, units, and sum assured fields define the policy size. When the model point specifies one policy, the initial units and sum assured are average size amounts.

11 SUBMODELS STRUCTURE

11.1 trad submodel

The trad submodel calculates the model point's contribution to the liability income statement and balance sheet items for traditional life products.

The trad submodel calculates model point results using values per unit in force, which are then multiplied by the number of units. Results per unit in force have '_per_unit' added to the formula name. Formula names without this addition are results per model point (or are a special result like a ratio).

11.1.1 *Seriatim versus portfolio runs*

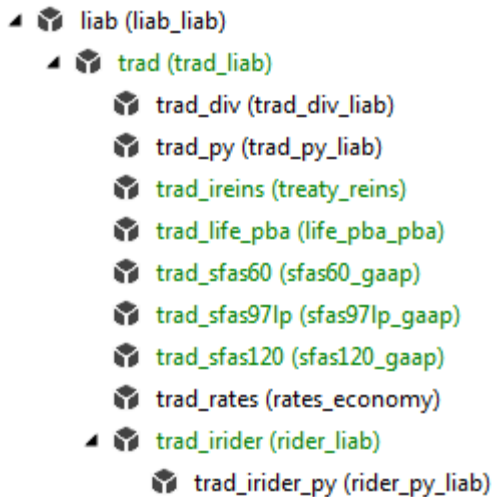
Seriatim projection

When trad is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the trad submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings.



Portfolio projection

For portfolio projections (which include company and business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the trad submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the trad submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the trad submodel the same columns as for other product types. This makes the traditional life product type interchangeable with other product types fitting into this business segment structure.

11.1.2 Variables

Variables for traditional life products are located in the trad submodel, except as noted below for the variables in trad_py, trad_div, and trad_ireins.

11.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when trad is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the trad submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only trad is the top model, (2) when trad is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when trad is run as the top model and when trad is included in portfolio projections. References to company, segment and liab are invalid when trad is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) temporary tables are sized for DAC tax calculations and competitor rates, (3) projection flags are created and initialized, (4) reporting groups are set up, (5) trad_ireins elements are created, (6) model point data validations are processed, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each projection task loop, immediately after startup when trad is the top model. When company is the top model, initialize is executed by calls from seg->initialize which executes trad->initialize for each model point belonging to the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when trad is running as the top model. This includes executing the initialization of economic scenario rates and the setup of competitor rates. Other initializations are required in both cases.

The main use of initialize is to calculate the cash value and reserve factors for each model point using the actuarial math library. Since these initializations are not required for all calculation types for all product features, conditions are checked to shortcut the initialization process, thus reducing the runtime of the model point.

This cash value and reserves are initialized in the following order:

1. Base cash values
2. Base statutory reserves
3. Base statutory minimum reserves
4. Base tax reserves

Other formulas

Other formulas in the trad submodel are relatively straightforward. Results which are per unit in force add '_per_unit' to the end of the formula name. Results without '_per_unit' are values for the model point.

Most of the profit, balance sheet, and GAAP columns are only calculated when trad is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

11.2 trad_div submodel

The trad_div submodel contains the variables and formulas to calculate policyholder dividend values per unit in force. This is a monthly submodel, the same as for the trad submodel.

Since all results are per unit, this submodel does not include '_per_unit' in the formula names.

11.2.1 Variables

Variables for policyholder dividends are located in the trad_div submodel. The exception is that variables on the model point record are located in the trad submodel, such as div_opt_defn and init_pua_face.

11.2.2 Formulas

Formulas for policyholder dividend calculations per unit in force are located in the trad_div submodel.

The formulas in this submodel are calculated only when they are called by formulas in the trad submodel.

11.3 trad_py submodel

The trad_py submodel is a standalone instance of the trad_py/liab model class. It contains the formulas and variables required to calculate per unit in force results which only vary by policy year. Thus 't' in this submodel represents policy year, unlike the trad, trad_div, and trad_ireins submodels where 't' represents months since the projection date. The policy year values are calculated from issue.

Since all results are per unit, this submodel does not include '_per_unit' in the formula names.

11.3.1 Variables

Variables located in the trad_py submodel are only used in formulas within this submodel. These are generally for values calculated from issue and varying by policy year, such as annual premiums, and cash value and reserve terminal values and net premiums.

11.3.2 Formulas

Formulas for calculating policy year per unit in-force results are included in the trad_py submodel.

Columns are generally calculated from issue, even for in force model points. As discussed in the trad submodel section, the reserve net premium and terminal values of trad_py are not calculated on demand through the target column. Instead, these results are calculated in advance through forced calls from the 'initialize' column of trad.

Cash value and reserve terminal values and net premiums are calculated by calling the Actuarial math library. See the Actuarial math library section below.

11.4 trad_irider submodel

The trad_irider submodel is a dependent data driven array instance of the rider_liab model class containing the variables and formulas required to calculate the rider cash flows and reserves for traditional life model points.

Please refer to the L13 Riders book.

11.5 trad_ireins submodel

The trad_ireins submodel is a dependent data driven array instance of the treaty_reins model class. It contains the variables and formulas required to calculate the ceded amount and reinsurance cash flows for traditional life model points.

11.5.1 Multiple instances

Each model point carries multiple instances of trad_ireins. The number of instances is determined by the number of treaties specified on the reinsurance model point record.

Elements of trad_ireins are created during the startup processing of the model point. Conditions are evaluated in startup of trad and the trad_ireins user defined array is resized as required. The characteristic for each trad_ireins element is specified by treaty code.

The ceded amount and reinsurance cash flows for multiple treaties are aggregated using the sum_over_reinsurance function in the trad submodel. This function can aggregate the values for all treaties, or just the treaties of a specific type of reinsurance, such as YRT.

11.5.2 Variables

Variables for the reinsurance type, the reinsurance expenses, and for calculating the ceded amount (when it is not specified on the reinsurance model point record), are located in the trad_ireins submodel. These variables are contained in the trad_ireins submodel so that they vary by treaty.

11.5.3 Formulas

Formulas for calculating the ceded amount and the reinsurance cash flows are contained in the trad_ireins submodel.

Since trad_ireins is defined through the model parameters, the size of the trad_ireins array is based on the number of treaties specified for each model point on the reinsurance model point file.

11.6 trad_rates submodel

The trad_rates submodel is identical to the rates submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

11.7 trad_life_pba submodel

The trad_life_pba submodel is identical to the life_pba submodel which applies for portfolio projections. Please refer to the C05 PBA book.

11.8 trad_sfas60 submodel

The trad_sfas60 submodel is identical to the sfas60 submodel which applies for portfolio projections. Please refer to the C04 GAAP accounting book.

11.9 trad_sfas97lp submodel

The trad_sfas97lp submodel is identical to the sfas97lp submodel which applies for portfolio projections. Please refer to the C04 GAAP accounting book.

11.10 Actuarial math library

11.10.1 AML DLL

The Actuarial math library (AML) provides standard actuarial calculations for traditional reserve and cash value methods. See the Actuarial math library liability book.

The trad submodel calls the AML to calculate reserve and cash value amounts per unit in force. Columns in the trad_py submodel store the terminal value and net premium for each policy year for each cash value and reserve basis used in the projection.

11.10.2 trad->startup

The trad->startup column specifies the name extension for the audit trail comma separated values file (when requested).

11.10.3 trad->initialize

The trad->initialize column initiates the call to the AML calculations for each set of cash value and reserve results. Each set of AMLresults is calculated by calling the aml_driver in the trad_py submodel. A separate call can be made for each of the following results:

- Cash values – Base plan cash values
- PUA cash values – Cash values per \$1 PUA
- PUA NSP purchase – Purchase rates per \$1 PUA
- ETI and RPU values – Base plan extended term insurance years and reduced paid-up sum assured for surrender at the end of each policy year
- Statutory reserves – Base plan statutory reserves
- PUA statutory reserves – Statutory reserves per \$1 PUA
- Statutory minimum reserves – Base plan minimum reserves for calculating deficiency reserves
- Tax reserves – Base plan tax reserves
- PUA tax reserves – Tax reserves per \$1 PUA

11.10.4 AML driver

The `aml_driver` in the `trad_py` submodel sets up and calls the AML for each set of cash value and reserve results. This driver is called by the `trad->initialize` column for each set of results.

This driver sets up the call to the AML using the appropriate input for each type of calculations. Each call to the driver calls these functions in the `trad_py` submodel:

- `setup_aml_initial_data` – Creates the interfaces for using the AML, and sets the definition for whether mortality rates are calculated for a single life or joint life product.
- `setup_aml_res_defn` – Specifies the cash value or reserve method (such as NLP, FPT, CRVM, SNFL, etc.), and the function type (curtate, continuous, or semi-continuous).
- `setup_aml_policy_data` – Specifies values for each policy year, such as the guaranteed premiums, death benefits, endowment benefits, valuation interest rates, mortality rates (for each life), plus the issue age, maturity age, premium term years, and expense allowance grading years.

The operation of these setup functions varies by the type of calculations being made. For example, the policy data passes the cash value interest and mortality rates when calculating cash values, and the statutory rates when calculating statutory reserves.

11.10.5 Calculation results

The AML calculation results are stored for each policy year in columns in the `trad_py` submodel. The formula for each column simply gets the AML calculation results for each case. These columns include:

- Terminal cash value and reserves and net premiums for each basis
- ETI years (fractional) and endowment amount
- RPU amount
- Net single premium per \$1 PUA for each basis
- Valuation interest and mortality rate for each basis
- Tabular cost of insurance for each basis

The `aml_driver` calls the appropriate columns after it calls the AML. For example, the cash value basis columns are called when the driver has just calculated cash value results. Then the driver calls the statutory reserve basis columns when those results have just been calculated by the AML. Thus each of these columns stores the AML results using the appropriate valuation basis.

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L03 UNIVERSAL LIFE

This book discusses flexible premium universal life products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

- Product – Benefits, premiums, secondary guarantee, surrender charge, policy loan definitions, and factor calculations definitions
- Fund – Crediting rate, and current, guaranteed, and shadow fund persistency bonus/refund, cost of insurance, and loads
- Valuation – Statutory and tax reserves, and PBA net premium reserve definitions
- Expense – Commission and expense rates, premium tax rate
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse rates, withdrawal use, dynamic premium payments, and policy loan use

Each of these input page types begins with 'UL', for use in projections that combine multiple product types.

Additionally, variables defining reinsurance and rider specifications and assumptions are associated with input pages which are common across products. These are:

- Reinsurance – Reinsurance treaty definitions and rates.
- Riders – Riders specifications and assumptions
- RidersValuation – Riders statutory and tax reserve definitions

1.2 Data pages

Variables for each universal life model point are specified on the ULData page. These variables are discussed in the Model point records chapter, below.

Variables for each reinsurance record are specified on the ULReinsData page. These variables are the same for the term life, traditional life, universal life, variable universal life, indexed universal life, and long term care product types. These variables are discussed in the Reinsurance model point records chapter, below.

Variable for each rider record are specified on the ULRidersData page. These variables are the same for term life, traditional life, universal life, variable universal life, and indexed universal life products. These variables are discussed in the L13 Riders book.

2 PRODUCT BENEFITS AND VALUES

The following explains the basic features for flexible premium universal life products, which have no underlying traditional life type guaranteed values. These are often referred to as accumulation UL products, or current assumption UL products.

2.1 Benefit term

2.1.1 Maturity age

The benefit term is an attained age specified by product. When the specified benefit term is zero (default value), the benefit term is age 121.

2.1.2 **GMP maturity age**

The guaranteed maturity premium (GMP) for UL CRVM reserve calculations is paid to the statutory reserve base mortality omega age, if earlier than the maturity age. When statutory reserve mortality is not specified, the GMP is assumed to be paid to the maturity age.

2.1.3 **IRS restrictions maturity age**

IRS restriction premiums are calculated using a separate maturity attained age, such as 95 or 100, which is specified in the `factors_tgt_age` variable for the Factors calculation type. IRS restrictions are not calculated during projections.

This maturity age applies for factors calculations for: (1) guideline single premium (GSP), (2) guideline level premium (GLP), (3) modified endowment contract (MEC) premium, and (4) cash value accumulation test (CVAT) net single premiums.

2.2 **Extended maturity option**

2.2.1 **When formulaic reserves apply**

When the policy is still in force at the statutory reserve mortality omega age, the extended maturity option (EMO) continues the policy until the maturity age. For example, products valued using 1980 CSO mortality continue in force under the EMO feature after attained age 100 when the maturity age is higher, such as 120.

Specify the death benefit by product:

- The fund value, but not less than zero
- The sum assured, but not less than the fund value.

There are no further premiums, loads, or charges during the EMP period. The fund value continues to increase with credited interest. The formulaic reserve is the current death benefit.

2.2.2 **When PBA reserves apply**

The EMO feature is ignored when PBA reserves are being calculated. These model points have reserves calculated using a modern mortality table, which extends to attained age 121 or later.

2.3 **Death benefits**

2.3.1 **Basic death benefit**

Specify the death benefit option on the model point record:

- Sum assured
- Sum assured plus the fund value, treating negative fund values as zero.
- Sum assured plus return of premiums (ROP) equal to cumulative premiums less withdrawals since issue. For in-force model points, specify on the model point record the initial (at the projection date) cumulative premiums, and cumulative withdrawals, for the policies collected into the model point.

Specify the sum assured per unit for the product; this input could vary such as by policy year or months since issue. When the initial sum assured is specified on the model point record, this sum assured specified for the product specifies the pattern of changes in the sum assured between policy years. See the Units basis section in the Model point records chapter, below.

2.3.2 **Minimum death benefit**

The death benefit is not less than a minimum percent of the fund value, sometimes called the corridor factor. Specify the minimum death benefit option either by product or by model point:

- For the Guideline premium test, the Global directory contains the Guideline Premium Test Corridor workbook containing the typical percentages of 250% through attained age 40, then grading to 105% for attained ages 75 through 90, then grading to 100% at attained age 95.
- For the Cash value accumulation test (CVAT), also called the Cash value net single premium test, specify percentages varying by attained age and typically also by sex and smoking class. The Factors calculation type can be used to calculate the percentages; see the Solving calculations chapter, below.

2.3.3 Death benefit paid versus in force

Death benefits paid at the end of each month are based on the fund value before policyholder transactions are processed. Thus the death benefit is calculated using the fund value before withdrawals.

Death benefits in force at the end of the month are based on the fund value at the end of the month, after withdrawals have been taken.

2.4 Endowment benefits

The remaining fund value is paid to policyholders reaching the maturity age.

2.5 Withdrawal benefits

2.5.1 Partial withdrawal provisions

Policyholders can withdraw cash from the fund value without surrendering the policy. This is sometimes referred to as a partial surrender, or cash withdrawal provision.

Partial withdrawals are processed at the end of each month, for policies persisting into the next month. Thus death and surrender benefits paid are based on the fund value before withdrawals, while death benefits and cash values in force at the end of the month are based on fund values after withdrawals.

Partial withdrawals apply to all policies within a model point equally.

Surrender charges

The percent of fund surrender charge applies for withdrawals.

Other surrender charges are pro-rated using the ratio of the withdrawal amount paid to the cash value before the withdrawal. The outstanding surrender charge amounts are then reduced by the proportion already collected on withdrawals. Specify the initial multiple which applies to these surrender charges, which vary by product, on the model point record.

The projection ignores partial withdrawal fees, assuming that they approximately cover administration expenses, and are not a significant element of profits.

Sum assured

Partial withdrawals do not increase the net amount at risk:

- When the death benefit option is the sum assured, the sum assured decreases by the withdrawal amount plus the related surrender charge.
- This sum assured reduction does not apply when the death benefit option is the sum assured plus the fund value or plus the return of premiums, since the death benefit automatically decreases for the withdrawal.

2.5.2 Partial withdrawal use

The partial withdrawal amount is an annual percent of the fund value. One twelfth of the percent applies at the end of each month. Thus the withdrawal amount is different each month, as the withdrawal basis changes. The withdrawal amount is limited to the cash value less the policy loan.

Partial withdrawals stop for model points in minimally funded status. See the Dynamic premiums section, below.

For the multiple shadow fund design, withdrawals are taken out from each shadow funds. See the Multiple shadow funds section in the Secondary Guarantee chapter, below.

Dynamic partial withdrawals (related to the competitiveness of the crediting rate) are assumed to be included in excess lapses.

2.5.3 Cumulative withdrawals

For in-force model points which use the Plus ROP death benefit option or have a preferred policy loan, specify on the model point record the initial (at the projection date) cumulative withdrawals for the policies collected into the model point.

2.6 Premiums

2.6.1 Premium term

Specify the premium term on the model point record. This is the policy anniversary when premium payments stop. When this value is zero, premiums are paid to the GMP maturity age.

2.6.2 Contractual premium term

Specify the contractual premium term either as an attained age or policy anniversary. When the contractual premium term is zero (default value), the contractual premium term is to the maturity age, but not greater than age 121.

The contractual premium term applies only for calculating: (1) GMP for UL CRVM reserves, (2) IRS restriction Guideline Level Premium (GLP). It has no effect on projected premium payments.

2.6.3 Premium mode

Specify the premium mode on the model point record. Premiums can be paid annually, semi-annually, quarterly, monthly, or special monthly (such as electronic transfer or pre-authorized check, with expense rate different from the monthly mode).

2.6.4 Annual premium

Specify the annual premium for the product either as:

- Annual premium per unit – The Annual premium per unit specified for the product, plus the policy fee. Premium suspensions apply from issue.
- Initial annualized premium – The premium on the model point record for the policies collected into the model point, including the policy fee. Premium suspensions apply beginning at the next policy anniversary, since this is the actual premium at the projection date.

2.6.5 Policy fee

The policy fee can be deducted as a load from each premium payment. The deduction from each premium payment is the annual policy fee divided by the number of premium payments per year.

Percent of premium commissions can include or exclude the policy fee.

The premium suspension multiple applies to the entire annual premium, including the addition for the policy fee. However, the policy fee itself is not reduced for its use in: (1) fund value load, (2) target premium, and (3) required premium for the secondary guarantee cumulative premiums test.

2.6.6 Modal loading

Modal loadings do not apply for flexible premium products, since the fund value simply accrues less interest when premiums are paid later than at the beginning of each year. Of course, you can vary the annual

premium per unit and policy fee by premium mode by using the model point variable `prem_mode` as a lookup key for linking to the external source.

However a monthly modal loading multiple applies for the cumulative premiums test secondary guarantee. See the Secondary guarantee chapter, below.

2.6.7 Additional single premium

An additional single premium can apply at issue, sometimes called a 'dump in single premium'. Specify on the model point record the total single premium for the policies collected into the model point. This affects new business model points only, since in-force model points start from initial values rather than calculating from issue.

2.6.8 Single premium policies

For a single premium plan, define the premium either as:

- Zero annual premium, defining the premium as the additional single premium. Since the annual premium is zero, the premium term has no effect.
- Premium term of 1 year, with Annual premium mode, and defining the premium as the annual premium.

2.6.9 Premium suspensions

Premiums are reduced by a premium suspension percent at the beginning of each policy year. The specified annual premium is multiplied by the cumulative premium suspension multiple. The suspension rate specified for the first policy year is ignored.

The reduced premium is assumed to be paid the same for all policies.

The premium suspension multiple applies to the entire annual premium, including the addition for the policy fee. However, the policy fee itself is not reduced for its use in: (1) fund value load, (2) target premium, and (3) required premium for the secondary guarantee cumulative premiums test.

Premium suspensions start date

When the model point record specifies the initial annualized premium, premium suspensions apply only for future policy anniversaries, i.e., after the projection date.

When the premium is specified as an annual premium per unit for the product, however, premium suspensions are calculated from issue.

2.6.10 Cumulative premiums

For in-force model points which use the Plus ROP death benefit option or have a preferred policy loan, specify on the model point record the initial (at the projection date) cumulative premiums for the policies collected into the model point.

2.6.11 Target premium

The target premium can be used to define values as percentages of target and excess premiums, such as for: (1) product loads for the current fund value, (2) surrender charges, and (3) commissions.

The policy fee is added to the target premium, except for the purpose of calculating commissions when the policy fee is not commissionable. The policy fee added to the target premium is not reduced by premium suspensions.

(The target premium for shadow fund loads is a separate input variable.

2.6.12 DAC tax capitalized premium percent

Specify (in the Product input page, under the Premiums category) the percent of each premium which is capitalized for the capitalized premium tax, which is also called the DAC tax. See the Taxes chapter in the Profit calculations book.

2.6.13 IRS premium restrictions

Tax regulations include 'definition of life insurance' premium restrictions. Projections ignore these rules; it is up to the user to specify assumed premiums consistent with the rules.

The Factors calculation type can calculate these IRS restriction premium rates: (1) guideline single premium (GSP), (2) guideline level premium (GLP), (3) modified endowment contract (MEC) premium, and (4) cash value accumulation test (CVAT) net single premiums. See the Solving calculations chapter, below.

2.7 Dynamic premiums

Premium payments can change based on policy funding conditions.

Fully funded secondary guarantee

Premium payments can stop when the secondary guarantee is fully funded. See the Premium adjustment section in the Secondary guarantee chapter, below.

Minimally funded policy

Premium payments can increase when needed to prevent the policy from lapsing. Premiums are then paid annually at the beginning of each policy year, regardless of the model point's premium mode.

When there is no secondary guarantee, this is the annual premium that keeps the current fund value* above zero at the end of each policy year. This annual premium is calculated the same as for shadow fund required annual premiums, but using the current fund COI and load rates, and the guaranteed interest rate. Interest crediting above the guaranteed rate during the previous year reduces the next premium payment. *When the policy lapses when the cash value becomes zero or negative, this is the premium that keeps the cash value above zero.

When a secondary guarantee applies, this is the annual premium that either keeps the current fund value (or cash value) positive, or keeps the secondary guarantee in force, whichever is lower. The annual premium based on the secondary guarantee is discussed in the Premium adjustment section in the Secondary guarantee chapter, below.

This test is made at the beginning of each policy year. The policy stays in force when the fund-related lapse feature would otherwise terminate it during the year. A 'catch up' premium is paid at the beginning of the next policy year, in addition to the ongoing minimal funding annual premium, to restore the negative value that accumulated during the prior year.

Partial withdrawals and policy loan increases stop for model points in minimally funded status.

2.8 Product loads

2.8.1 Input variables

Input variables for product loads differ between current, guaranteed, and shadow fund values.

The Fund input page includes a separate set of input variables for current, guaranteed, and shadow fund loads. For example, input variable `load_per_pol_curr` specifies the current fund per policy load and `load_per_pol_gtd` specifies the guaranteed per policy load. The formulas, however, use the same code variable, `load_per_pol`, in each case, but each input variable is associated with this code variable in the appropriate instance of the submodel.

2.8.2 Premium loads

The following loads are deducted from each premium payment:

- Percent of premium (including the additional single premium). The target premium percent applies to the premiums paid during the policy year until the premiums exceed the target premium, then the excess premium percent applies to premiums paid during the remainder of the policy year.

- Policy fee – The policy fee can be deducted as a load; see the Premiums section, above. The amount deducted from each premium is the policy fee divided by the number of premium payments per year. The policy fee is not reduced for premium suspensions for this purpose.

2.8.3 Non-premium loads

The following loads are deducted from the fund value at the beginning of each month; one twelfth of the load rate applies each month:

- Amount per policy
- Amount per unit
- Percent of fund value – one twelfth of the annual percent of fund value at the end of the previous month is deducted at the beginning of each month.

2.9 Cost of insurance charge

2.9.1 Input variables

Input variables for cost of insurance charges differ between current, guaranteed, and shadow fund values.

The Fund input page includes a separate set of input variables for current, guaranteed, and shadow fund cost of insurance charges. For example, input variable `coi_rate_base_curr` specifies the current fund COI rates and `coi_rate_base_gtd` specifies the guaranteed COI rates, and `coi_rate_base_sf` specifies the shadow fund COI rates. The formulas, however, use the same code variable, `coi_rate_base`, in each case, but each input variable is associated with this code variable in the appropriate instance of the submodel.

These variables are special since they are read twice – first for life 1 and then again for life 2. This is accomplished by using special lookup keys.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. 'Insured characteristics' (such as issue age, gender, smoking class, and rate class) are set to either the life 1 or life 2 values, as needed. The external source uses lookup keys `mort_gender`, `mort_smoking_class`, `mort_rate_class`, and `mort_issue_age` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

2.9.2 Charge amount

The COI charge is calculated and deducted at the beginning of each month, after first adding premiums and deducting loads. The COI rate applies to the net amount at risk (see Net amount at risk topic). The death benefit takes the minimum death benefit (corridor factor) into account.

2.9.3 Net amount at risk

The net amount at risk includes a sum assured component and a fund value component.

$$\text{NAR} = \text{SumAssured} * \text{DiscountAdj} - \text{FundValue} * \text{FVAdj}$$

Specify the discount rate used to adjust the sum assured component:

- None – No adjustments to the sum assured.
- Current Rate – The sum assured is discounted by the monthly current interest rate. The 'current interest rate' is the interest rate associated with the type of fund. So for the current fund, the discount factor is based on the current credited interest rate. For the guaranteed fund, the discount factor is based on the guaranteed interest rate. For the shadow fund (if applicable), the discount factor is based on the shadow credited interest rate.
- Guaranteed Rate – The sum assured is discounted by the monthly guaranteed interest rate.

- **Half Period Current Rate** – The sum assured is discounted by a half of month current interest rate. The 'current interest rate' is the interest rate associated with the type of fund. So for the current fund, the discount factor is based on the current credited interest rate. For the guaranteed fund, the discount factor is based on the guaranteed interest rate. For the shadow fund (if applicable), the discount factor is based on the shadow credited interest rate.
- **Half Period Guaranteed Rate** – the sum assured is discounted by a half of month guaranteed interest rate.

Specify the fund value component:

- **Prior Month Fund Value** – The ending fund value of the previous month
- **Beginning Fund Value With Premium Less Load** – The ending fund value of the previous month plus the current month's premium less loads.
- **Beginning Fund Value With Premium Less All Other Charges** – The ending fund value of the previous month plus the current month's premium less loads and all other charges (except the COI charge, since this is calculated based on the net amount at risk). Note, the only other available charge other than loads is the rider charge, which applies to the current fund only. So this option is the same as Beginning Fund Value With Premium Less Load for guaranteed and shadow funds.
- **Prior Month Cash Value** – The ending cash value of the previous month.

Specify the fund value adjustment:

- **None** – No adjustments to the fund value component
- **COI Rate** – Multiply the fund value by $(1 - \text{Monthly COI Rate})$.

When the fund value is negative, the net amount at risk is calculated using zero fund value. See the Secondary guarantee chapter, below.

2.9.4 COI rate

Specify COI rates either as annualized rates or as annual mortality rates. The annual rate is adjusted by a multiple, which can vary by policy year or attained age. The adjusted annual rate is then converted to a monthly COI rate using either an annualized, linear, or exponential formula.

Annualized formula:

$$\text{monthly COI rate} = \text{annualized COI rate}/12$$

Linear formula:

$$\text{monthly COI rate} = \frac{(\text{annual mortality rate}/12)}{(1 - \text{annual mortality rate}/12)}$$

Exponential formula:

$$\text{monthly COI rate} = \frac{[1 - (1 - \text{annual mortality rate})^{(1/12)}]}{(1 - \text{annual mortality rate})^{(1/12)}}$$

The substandard multiple on the model point record applies to the annualized COI rate, or to the annual mortality rate, before conversion to a monthly COI rate. This is the same substandard multiple as applies for experience mortality rates.

This multiple applies the same to the current and guaranteed COI rates. Of course the COI rate cannot exceed 1.0 (100%).

2.10 Interest crediting

2.10.1 Crediting rate

Interest is credited to the fund value at the end of each month. The crediting rate varies between each fund calculation type:

- The base crediting rate for the current fund value is determined according to the company's crediting strategy; see the Crediting strategy chapter, below.
- The guaranteed fund value receives the guaranteed crediting rate, ignoring any temporary guarantee of the crediting rate at issue, and ignoring the remaining time in the current crediting rate declaration period.
- Specify the shadow fund crediting rate on the Product input page, under the Secondary Guarantee category.

Initial crediting rate

For in-force model points, specify the initial (at the projection date) base crediting rate on the model point record. When this variable is the zero default value, the initial crediting rate is calculated using the crediting strategy.

2.10.2 Additional interest

The base crediting rate for the current fund value can be increased by an additional interest credit.

The crediting rate can be increased by an additional interest credit when the policy's unloaned fund value crosses the first and second threshold per policy:

- Ratcheted additional interest credits the addition on the part of the unloaned fund value over the threshold.
- Banded additional interest credits the addition on the entire unloaned fund value when the threshold is reached.

The additional crediting rates are the total addition to the base crediting rate for each threshold. That is, the addition for the second band is the band 2 rate, not the sum of the band 1 and band 2 rates.

The additional interest credit and ratcheting threshold for the secondary guarantee accumulation are separate variables. See the Secondary guarantee chapter, below.

2.10.3 Guaranteed crediting rate

The minimum crediting rate is a specified fixed rate. This is the crediting rate for the guaranteed fund.

2.10.4 Initial guarantee of crediting rate at issue

The crediting rate at issue can be guaranteed for a specified number of months after issue. This guarantee applies only for determining the base crediting rate. It is ignored for discounting the sum assured for determining the COI net amount at risk. It is also ignored for calculating the guaranteed fund.

For in-force projections, the initial crediting rate is used as this temporary guarantee, since the crediting rate at issue is not known.

Nested projection differences

Warning! This feature can cause nested projections to produce different results than the equivalent standard projection. For the standard projection, the initial crediting rate on the model point record is guaranteed for the first months from issue. When a higher current crediting rate is subsequently declared, the model point record for a new time step contains this current rate as the initial crediting rate, which is now treated as being guaranteed during these first months after issue.

For example, a model point has 5.0% initial crediting rate for the standard projection, which is guaranteed for the first two policy years. The interest crediting strategy then increases the current rate to 5.5%. When a new time step begins, the model point record will contain this 5.5% initial crediting rate, which is the crediting rate at the start of this time step. Say the standard projection's crediting strategy reduces the current crediting rate to 5.25% before the second policy anniversary. The nested projection that has 5.5% initial crediting rate does not reduce the current crediting rate to 5.25%, because the 5.5% rate is treated as the temporary guaranteed rate.

If this feature is significant for your block of business, you can make nested projections match standard projections by adding a model point field for the crediting rate at issue. Customize the formula for the adjusted guaranteed crediting rate, so that it is based on the new model point variable, rather than on the initial crediting rate.

2.10.5 Monthly rate

The crediting rate is an effective annual interest rate, which is converted to the equivalent monthly rate:

$$(1 + \text{annual rate})^{1/12} - 1$$

2.10.6 Crediting on loaned fund value

See the Policy loans section, below.

2.11 Persistency bonus and refund

2.11.1 Input variables

Input variables for persistency bonuses and refunds differ between current, guaranteed, and shadow fund values.

The Fund input page includes a separate set of input variables for current, guaranteed, and shadow fund persistency bonuses and refunds. For example, input variable `persist_bon_input_curr` specifies the current fund persistency bonus, `persist_bon_input_gtd` specifies the guaranteed persistency bonus, and `persist_bon_input_sf` specified the shadow fund persistency bonus. The formulas, however, use the same code variable, `persist_bon_input`, in each case, but each input variable is associated with this code variable in the appropriate instance of the submodel.

2.11.2 Timing

Persistency bonuses and refunds increase the fund value at the end of the policy year. The fund value increases before deaths, withdrawals, and lapses occur, so the amount applies to the units in force at the end of the previous month.

2.11.3 Persistency bonus

Persistency bonuses increase the fund value by a specified amount per unit. You can customize to add other bonus definitions.

2.11.4 Persistency refund

Persistency refunds increase the fund value by returning a percent of cumulative product loads through the previous month, less previous refunds. You can customize to add other refund definitions. Specify on the

model point record the initial (at the projection date) persistency refund basis for the policies collected into the model point.

2.12 Fund value

2.12.1 Multiple fund values

Up to three sets of fund value rates can apply for a particular model point. See the `ul_ifund` submodel section in the Submodels structure chapter, below. A separate instance of the `ul_ifund` submodel applies for each type of calculation:

- Current – all cases (`ul_ifund_curr`)
- Guaranteed – when guaranteed rates are needed for reserve calculations (`ul_ifund_gtd`)
- Secondary Guarantee – for shadow fund values (`ul_ifund_sf`).

2.12.2 `ul_ifund` submodel output

The formulas in the `ul_ifund` submodel have separate values for each instance of this submodel. When output is created for this submodel, select either the current, guaranteed, or shadow fund instance.

The `ul_ifund` submodels iterate to solve for values such as the guaranteed maturity premium. See the Solving calculations chapter, below. When the `solving_display_defn` variable (on the Product input page in the Premiums category) is All Solves, the `ul_ifund` output includes the final values for each solving calculation.

Warning! Since the `ul_ifund` submodel is reset for each solving calculation, the final results apply only for the last time this submodel was calculated. For example, the `ul_ifund_gtd` values are not the guaranteed fund values calculated from issue; they could be the calculations starting from the current fund value at the last time when UL CRVM reserve calculations were processed.

To see a particular set of results, customize to add code to write out the submodel results at the time that calculation has just been performed. Search for the 'writeModel' command in the `ul->initialize` formula to see how this is done when the `solving_display_defn` calls for writing out the submodel values.

2.12.3 Current fund

For in-force model points, specify on the model point record the initial (at the projection date) current fund value for the policies collected into the model point.

The previous month's fund value is adjusted by the following amounts which occur as of the beginning of each month: (1) premium payment addition, (2) loads deduction and (3) charges deduction. The COI charge is calculated using the beginning of month fund value before charges are subtracted.

Interest crediting applies to this beginning of month fund value after the premium and deductions have been processed. Credited interest is added to the fund value at the end of the month.

This end of month fund value before withdrawals is used to calculate death benefits paid for the month, since the mortality decrement occurs before policyholder transactions and at the end of the month. Surrender benefits are also calculated using this fund value before withdrawals, including related surrender charges.

Finally, withdrawals are processed, reducing the final fund value at the end of the month. It is this fund value after withdrawals which is used to calculate reserves, and the death benefit and cash value in force at the end of each month. Policy loan transactions are also based on this fund value after withdrawals.

2.12.4 Guaranteed fund

Guaranteed fund values are projected when needed for solving calculations. See the Solving calculations chapter, below.

The guaranteed fund is also used during the projection, when it is updated to the current fund value at a reserve valuation date, then projected forward using premium payment assumptions for the reserve

calculation. This occurs for UL CRVM reserves, and PBA NPR base and secondary guarantee reserves. When these calculations apply, the `ul_ifund_gtd` output does not represent guaranteed fund calculations from issue. Instead, set the `solving_display_defn` variable (on the Product input page in the Premiums category) to All Solves to save each type of solving fund result in the output database.

Flag and indicator values control each type of solving, such as the timing and amount of premium payments. See the Submodels structure chapter, below.

2.12.5 Shadow fund

In the `ul_ifund_sf` instance of the `ul_ifund` submodel, the fund value columns are the shadow fund values. When the product includes multiple shadow funds special input and columns calculate each component of the shadow fund value. See the Secondary guarantee chapter, below.

Similarly to the guaranteed fund, shadow fund values are also projected when needed for solving calculations. See the Solving calculations chapter, below.

For PBA NPR secondary reserve calculations, the shadow fund is used to solve for the new shadow fund level premium as of each valuation date. This calculation begins by updating to the current shadow fund value calculated for the projection, then calculating future values assuming the solved shadow fund level premium is being paid.

Flag and indicator values control each type of solving, such as the timing and amount of premium payments. See the Submodels structure chapter, below.

2.13 Cash value

The cash value is the fund value less the surrender charge.

2.13.1 Fund value for surrender charge

The fund value for calculating the surrender benefit is the current fund value.

2.13.2 Surrender charge

Specify surrender charges as:

- Amount per unit
- Percent target premium
- Percent of fund value.

The surrender charge rates can vary annually (using the `pol_yr` column) or monthly (using the `mths_since_issue` column).

Surrender charges (other than the percent of fund value charge) are reduced by the proportion of the charges which have already been collected during partial withdrawals. Specify on the model point record initial (at the projection date) multiple which applies to these surrender charge rates.

2.14 Policy loan

2.14.1 Policy loan amount

Policy loans apply equally to all policies within a model point. For in-force policies, specify on the model point record the initial (at the projection date) policy loan for the policies collected into the model point.

The accumulated policy loan at the end of each month is the balance at the end of the previous month, brought forward with one month's interest at the policy loan rate.

The outstanding loan amount is increased at the end of each policy year, by a percent of the total fund value.

Policy loan increases stop for model points in minimally funded status. See the Dynamic premiums section, above.

Policy loan input is ignored for multiple shadow fund products.

2.14.2 Maximum policy loan

The outstanding loan amount is reduced at the end of any month when it would exceed the cash value.

2.14.3 Policy loan cash flows

An adjustment increase is a new policy loan and a negative cash flow. An adjustment decrease is a policy loan repayment and a positive cash flow.

The elimination of the policy loan at death and surrender is an 'adjustment' in the loan balance to zero, and thus is a positive loan cash flow which offsets part of the decrement cash flow.

2.14.4 Policy loan rate

The policy loan interest rate is the greater of: (1) a fixed rate and (2) an indexed rate equal to a competitor rate. Interest is added to the loan at the end of each month.

When policy loan interest is payable in advance, policy forms may specify the loan rate as a discount rate. Convert this discount rate to the annual effective annual rate. For example, a 7.4% discount rate payable in advance must be input as an effective annual rate of 8.0% [$= 7.4\% \div (1 - 7.4\%)$].

2.14.5 Loaned fund crediting rate

The crediting rate on the loaned fund value can be:

- Unloaned rate – The same crediting rate applies to the entire fund value. However, the additional interest credit only applies to the unloaned fund value.
- Fixed rate – A specified crediting rate.
- Guaranteed rate – The guaranteed crediting rate, not including any temporary guarantee of the crediting rate at issue.
- Fixed spread – The policy loan interest rate less a specified spread.

2.14.6 Preferred loans

The loan can be split into two parts. The 'preferred' part of the loan is the amount up to the cash value or fund value, less cumulative premiums paid net of cumulative partial withdrawals. The crediting rate on the preferred loan is the policy loan interest rate less a specified preferred loan spread. A special case of a preferred loan is a zero net cost loan, sometimes called a wash loan, where the preferred loan spread is zero.

The remainder of the loan is a non-preferred policy loan, which has a crediting rate as discussed above.

For in-force model points which have an initial (at the projection date) policy loan for products with the preferred policy loan feature, specify on the model point record the initial cumulative premiums and cumulative withdrawals for the policies collected into the model point.

2.14.7 Policy loan expenses

Policy loan expenses are discussed in the Expenses section, below.

3 FUND VALUES

3.1 Fixed fund value crediting

3.1.1 Overview

The fixed fund value is increased for credited interest at the end of each month. The crediting rate for the fixed fund value is the larger of the policy's guaranteed crediting rate and the rate determined through the crediting strategy.

3.1.2 Initial crediting rate

Specify the initial (at projection date) crediting rate for the fixed fund value on the model point record. When the value on the model point is zero, the initial crediting rate is determined based on the crediting strategy.

The initial crediting rate applies until the crediting rate is reset through the application of the crediting strategy.

3.1.3 Crediting strategy declaration

Declaration timing

The crediting rate can change at the beginning of a month, policy year, calendar year, or calendar quarter.

Declaration period

The crediting rate changes according to the declaration period, which is one or more cycles of the specified period. A different number of cycles can apply at issue than for ongoing crediting rate changes. For example, the crediting rate at issue could apply for the first two policy years, then changing every policy year thereafter.

3.1.4 Crediting strategy rate basis

Fixed rate

The crediting rate can be a fixed rate, such as for calculating illustration values. Crediting rate constraints do not apply.

Asset yield basis

You can specify the crediting rate as the asset yield less a spread, subject to the crediting rate constraints. This definition makes the model points dependent on asset results, so independent model point processing cannot apply for asset/liability projections.

The asset yield includes the effects of IMR amortization, defaults, and borrowing costs.

For liability-only projections, you specify the asset yield as an assumption for the projection. Independent model point processing can still apply, since the asset yield is not affected by liability results.

Competitor rate basis

You can specify the crediting rate as a competitor rate less a spread, subject to the crediting rate constraints.

When the constraints do not include the asset yield, independent model point processing can apply even for asset/liability projections. For example, this approach could take the view that the crediting rate is primarily driven by market forces, and that appropriate investment strategies must be determined in order to achieve the desired spread from the asset yield.

3.1.5 Crediting strategy constraints

The following constraints apply when the crediting rate basis is the asset yield or a competitor rate. Crediting rate constraints do not apply when the crediting rate is a fixed rate.

The crediting rate is first constrained to be not more than a maximum percent above, and not less than a maximum percent below, a specified competitor rate.

Further (thus adjusting the above constraints), the crediting rate is not greater than [the asset yield less a minimum spread]. When the minimum spread below the asset yield is 1.0 (100%) default value, this constraint does not apply, and model points defining the crediting rate using a competitor rate are eligible for independent model point processing within asset/liability projections.

Finally, the crediting rate is not less than the guaranteed crediting rate in all cases, even when the crediting rate is defined as a fixed rate. This includes the temporary guarantee of the crediting rate at issue.

4 SECONDARY GUARANTEE

The secondary guarantee keeps the death benefit in force through a specified policy anniversary even when the fund value is zero or negative. These products are sometimes called death benefit guarantee universal life (DBGUL), secondary guarantee universal life (SGUL), universal life secondary guarantee (ULSG), no-lapse guarantee (NLG), and protection universal life products. Products without the guarantee are sometimes called accumulation universal life, or current assumption universal life products.

4.1 General

4.1.1 Secondary guarantee years

Specify the secondary guarantee years (benefit term) as part of the product definition. This is the policy anniversary when the guarantee expires. When this variable is the default value of zero, the secondary guarantee continues to maturity.

The secondary guarantee is in effect through these years, as long as the policyholder meets requirements using either the shadow fund test or the cumulative premiums test (discussed in separate sections below). When the test is violated, the secondary guarantee continues in effect until the end of the secondary grace period (discussed in a separate section below), as long as the current fund value is positive to keep the policy in force.

4.1.2 Secondary guarantee premium term

To keep the guarantee in force, the policyholder must pay premiums high enough to meet the test through the secondary guarantee premium term. Specify the secondary guarantee premium term attained age as part of the product definition. This age cannot exceed the Extended Maturity Option age, since premiums cannot be paid during the EMO period. When this variable is the default value of zero, the premium requirements continue to the end of the secondary guarantee benefit years (but not past the EMO age).

The secondary guarantee funding status at the end of this premium term applies for the remainder of the secondary guarantee benefit years. When premium requirements have been met as of the end of this premium term, the guarantee is fully funded. When premium requirements have not been met as of the end of this premium term, the guarantee is no longer in effect. Thus the secondary guarantee 'catch up' provision applies only during the secondary guarantee premium term.

4.1.3 Extended maturity option

When the secondary guarantee test has been satisfied at the EMO attained age, the guarantee continues through the EMO period.

4.1.4 Input variables

Most input variables related to the secondary guarantee are organized on the Product input page, in the Secondary Guarantee category. These define the guarantee terms, premium and lapse adjustments, rules for calculating negative fund values, secondary guarantee test interest crediting (including ratcheting threshold), and shadow fund target premium (for premium loads varying between target and excess premiums).

As discussed below, shadow fund load, COI, and persistency bonus and refund rates are specified on the Fund input page.

4.2 Cumulative premiums test

4.2.1 General

The cumulative premiums test is that accumulated premiums paid, less the policy loan, cannot be less than the accumulated required premiums. The accumulation of premiums paid is reduced by partial withdrawal payments, but not by the surrender charge reduction to the fund value.

Specify the required premiums per unit on the Product input page, under the Secondary Guarantee category. Specifying required premiums varying by policy year is sometimes called an annually renewable term (ART) design. Specifying a level premium is sometimes called a stipulated level premium design.

The required premiums are assumed to be paid monthly. The required annual premium per unit is increased by the policy fee, then multiplied by the specified annualized modal loading multiple for the monthly premium mode.

4.2.2 Interest accumulation

Both the actual and required premiums accumulate at the secondary guarantee interest rate specified on the Product page, under the Secondary Guarantee category.

For in-force model points, specify on the model point record the initial (at the projection date) accumulation of actual premiums for the policies collected into the model point. The initial accumulation of required premiums from issue is calculated during the projection, using the units per policy at the projection date.

4.2.3 Interest ratcheting

The cumulative premiums accumulation rate is ratcheted, adding an additional credit (can be negative) to the accumulation rate for the excess accumulation over a per unit threshold varying by months since issue. The threshold is the accumulation of the level annual premium which satisfies the cumulative premiums test at the end of the secondary guarantee premium term, before adding interest for the current month.

4.2.4 Policy loan

The policy loan is taken into account in determining whether the secondary guarantee is in the grace period. The policy loan is ignored in other calculations, however, such as fully or minimally funding the secondary guarantee. Reserves are not affected by the policy loan.

4.3 Shadow fund test

4.3.1 Shadow fund accumulation

The shadow fund test is that the shadow fund, less the policy loan, must be positive.

The shadow fund accumulates at the secondary guarantee interest rate specified on the Product input page, under the Secondary Guarantee category. The policy loan does not affect shadow fund interest crediting.

The shadow fund uses the shadow fund COI, load, and persistency bonus/refund rates specified for the on the Fund input page, under the Shadow Fund category. Percent of premium loads are banded using the shadow fund target premium, not the target premium which applies for the current fund. The shadow fund is reduced by partial withdrawals the same as for the current fund.

When the shadow fund is zero or negative, 'catch up' interest, COI, and load rates apply. These special catch up rates are used to calculate the required annual premiums for reserve calculations. The shadow fund rates that apply when the shadow fund is positive are used to calculate the fully funded single premium shadow fund, and the annual premium for minimally funding the shadow fund to keep the secondary guarantee in force.

COI charges are calculated using the death benefit based on the shadow fund, including applying the minimum death benefit percentages to the shadow fund value.

For in-force model points, specify on the model point record the initial (at the projection date) shadow fund value for the policies collected into the model point.

4.3.2 Input variables

Product input page

The Product input page contains input variables for the secondary guarantee interest rates (including interest ratcheting), which are the same as apply for the cumulative premiums test, plus the shadow fund catch up interest accumulation rate. This input page also includes the shadow fund target premium definition, whether the minimum death benefit percentages apply to the shadow fund, and the premium for calculating the ratcheted interest threshold.

Fund input page

COI, load, and persistency bonus/refund variables are specified on the Fund input page, so that they can vary between current, guaranteed, and shadow fund calculations. See the Fund variables section in the Input organization chapter, above.

This input page also includes input variables for the special 'catch up' COI rates. This variable applies only for shadow fund calculations, and only when the shadow fund is negative and for calculating the required annual shadow fund premiums for calculating the secondary reserve. The catch up load rate variables are only for multiple shadow fund products (discussed separately below). They do not affect single shadow fund products.

4.3.3 Ratcheted interest crediting

The single shadow fund crediting rate is ratcheted, adding an additional credit (can be negative) to the secondary guarantee interest rate for the excess shadow fund value over a per unit threshold varying by months since issue. This threshold is the shadow fund accumulation assuming the specified secondary guarantee threshold premium (plus the policy fee) is paid annually, before interest is credited for the current month.

When the specified threshold premium is zero, the threshold premium is the level annual premium which produces a zero shadow fund at the end of the secondary guarantee, using the interest and COI rates that apply when the shadow fund is positive. See the Solving calculations chapter, below.

4.3.4 Shadow fund target premium

Percent of premium loads can be banded using the shadow fund target premium, which is different from the target premium which applies for the current fund. The shadow fund target premium per unit is specified as part of the product definition, plus the policy fee.

The shadow fund target premium for each policy year can be the greater of: (1) the specified target premium, and (2) the minimum required annual premium calculated for regulation XXX reserves. This assures that the percent of target premium load percent applies for all calculations of the required annual premium, in order to avoid creating multiple regulation XXX reserve segments.

4.3.5 Policy loan

Interest is credited on the loaned shadow fund value, the same as for the loaned current fund value.

The policy loan is taken into account in determining whether the secondary guarantee is in the grace period. The policy loan is ignored in other calculations, however, such as fully or minimally funding the secondary guarantee. Reserves are not affected by the policy loan.

4.4 Multiple shadow funds

4.4.1 Shadow fund sub-accounts

The shadow fund value can be the sum of multiple sub-accounts: (Note the following gives names to the accounts, instead of numbering them, since products that don't split out first year premiums use only SF2 and SF3, which would be confusing numbering.)

- **First year premium shadow fund** – Premiums paid in the first policy year can accumulate in a special 'first year premium' sub-account.
- **Main shadow fund** – When the beginning total shadow fund is zero or positive, premiums are added to the 'main' shadow fund. The exception is that premiums paid in the first year can be split out into the first year premium sub-account.
- **Catch up shadow fund** – When the beginning total shadow fund is negative, premiums are added to the 'catch up' shadow fund.

The secondary guarantee operates as usual using the total shadow fund, which is the sum of their three sub-accounts.

4.4.2 Product loads

Percent of premium load

The percent of premium load is calculated using the load rates for the sub-account the premium is allocated to. When the beginning total shadow fund is zero or positive, the main load rates apply; when there is a first year premium sub-account, the main load rates simply vary between the first and subsequent policy years. When the beginning total shadow fund is negative, the catch up load rates apply.

Per policy and per unit load

When the beginning total shadow fund is zero or positive, the 'main' shadow fund per policy and per unit load rates apply. When the beginning total shadow fund is negative, the catch up per policy and per unit load rates apply.

Percent of fund load

Each sub-account is reduced by the percent of fund load at the beginning of each month, before premiums are paid, the same as for the current fund. The exception is that this load does not apply to the catch up shadow fund.

4.4.3 COI charges

The COI charge is calculated and deducted at the beginning of each month. The COI rate applies to the total net amount at risk

When the total shadow fund before the COI charge is positive, the COI charge is calculated using the COI rate for the main shadow fund sub-account. When the total shadow fund before the COI charge is negative, the COI charge is calculated using the catch up shadow fund COI rate.

When the total shadow fund is negative, the COI net amount at risk uses zero shadow fund value.

4.4.4 Allocation of deduction

The percent of fund load is deducted from each shadow fund sub-account, and premiums (net of percent of premium loads) are allocated as described above.

The total of the COI charge and per policy and per unit loads is then deducted at the beginning of each month, allocated between the sub-account values in this order:

- Reduce the catch up sub-account until it is zero, then
- Reduce the main sub-account until it is zero, then

- Reduce the first year premium sub-account until it is zero, then finally
- Deduct any remainder from the catch up sub-account.

4.4.5 Interest crediting

Each of the three shadow fund sub-accounts accumulates at its own interest crediting rate:

- The first year premium sub-account accumulates at the special 'first year premium' shadow fund crediting rate specified on the Product page, under the Secondary Guarantee category.
- The main sub-account accumulates at the secondary guarantee interest rate that applies for single shadow fund products.
- The catch up sub-account accumulates at the catch up shadow fund crediting rate that applies for single shadow fund products.

Ratcheted interest crediting does not apply for multiple shadow fund products.

When the catch up sub-account is negative, interest crediting either continues or stops the same as for the current fund.

4.4.6 Persistency bonus and refund

Persistency bonuses and refunds that apply to the shadow fund are added to the main shadow fund.

4.4.7 Minimum annual premiums

The minimum ART premiums for secondary guarantee reserves are calculated using the catch up COI, load, and interest crediting rates.

The minimum annual premiums for adjusting premium payments to prevent the policy from lapsing are calculated using the first year premium (first year) and main (subsequent years) rates.

4.4.8 Policy loan

Policy loan input is ignored for multiple shadow fund products.

4.4.9 Initial sub-account values

The model point variable `init_shadow_fund_val` specifies the total shadow fund at the projection date. Specify the initial catch up sub-account value in model point variable `init_shadow_fund_catchup`. Specify the initial first year premium sub-account value in model point variable `init_shadow_fund_first_yr_prem`.

4.5 Negative fund value

When the fund value is negative, it is treated as zero for product value calculations:

- Benefit values use zero fund value.
- Interest crediting can either: (1) continue to apply, making the fund value more negative, or (2) stop when the fund value is negative. When there is no interest crediting, the fund value is treated as zero for calculating the average crediting rate for GAAP calculations.
- Loads continue to be deducted.
- COI net amount at risk uses zero fund value. COI charges continue to be deducted.
- The YRT reinsurance net amount at risk uses zero fund value.
- The GAAP basic benefit reserve is zero.

4.6 Secondary guarantee grace period

Specify the secondary guarantee grace period months on the Product input page, under the Secondary Guarantee category.

4.6.1 Positive fund value

When the current fund value is positive, the secondary guarantee remains in effect until the test has been violated for the number of grace period months. At the end of the grace period, the secondary guarantee terminates even though the policy continues to be in force without the guarantee. Thus some model points have no secondary guarantee, even though it is specified for the product.

The grace period is restored when the test is satisfied before the current grace period ends. That is, the next time the secondary guarantee test is violated, the full grace period months apply again.

When the guarantee is already violated at the projection date, specify on the model point record the initial (at the projection date) months that the guarantee has been in the grace period. Any number greater than or equal to the grace period months indicates that the model point no longer includes the secondary guarantee.

4.6.2 Catch up provision

When the secondary guarantee is in this grace period, the secondary guarantee reserve includes a calculation of the 'catch up' amount needed to restore the guarantee.

When premiums are increased to minimally fund the guarantee, the catch up amount is paid as a premium at the projection date. See the Premium adjustment section below.

4.6.3 Negative fund value

This grace period does not extend the secondary guarantee when the current fund value is negative. Fund-related lapse occurs immediately when the secondary guarantee test is violated.

4.7 Premium adjustment

Premium adjustments are evaluated at the beginning of each policy year, when the annual premium is determined for the year.

4.7.1 Fully funded secondary guarantee

Premium payments can stop when the secondary guarantee is fully funded, which occurs when:

- For the shadow fund, the shadow fund value equals or exceeds the required payments single premium calculated for AXXX reserve step 4.
- For the cumulative premiums test, the accumulated premiums, without further premium payments, equals or exceeds the accumulated required premiums at the end of the secondary guarantee period.

4.7.2 Minimally funded policy

Premium payments can increase when needed to prevent the policy from lapsing. See the Dynamic premiums section in the Product benefits and values chapter, above.

The annual premium to keep the secondary guarantee in effect is:

- For the shadow fund, the premium needed to produce a zero shadow fund value at the end of the policy year. This calculation uses the COI, load, and interest crediting rates that apply when the shadow fund is positive.
- For the cumulative premiums test, the premium which meets the required premiums accumulation at the end of the policy year.

4.8 Lapse adjustment

The lapse rate is adjusted when the fund value is negative, i.e., when the secondary guarantee is keeping the policy in force:

- When the secondary guarantee is fully funded (see above), the lapse rate is multiplied by an adjustment percent.
- When premiums are still required to keep the secondary guarantee in effect until its expiration, the lapse rate is multiplied by a different adjustment percent.

4.9 In-force statistics

Units in force at the end of each month are calculated for these special situations:

- SG in the money – When the fund value* is zero or negative, meaning that the secondary guarantee is keeping the policy in force. *This test is based on the cash value, when the fund-related lapse feature tests against the cash value instead of the fund value (see the Lapses and surrenders topic in the Product benefits and values chapter, above.)
- SG minimally funded – When premiums are increased to keep the secondary guarantee in force (see the Premium adjustment section, above.)
- SG fully funded – When premiums are stopped because the secondary guarantee is paid up (see the Premium adjustment section, above.)

5 JOINT LIFE PRODUCTS

Joint life products have a second insured, in addition to life 1. The product is a joint life product when the life status variable is First To Die or Frasierized Last Survivor. When the life status is Single Life, the second life variables do not affect the projection.

5.1 Second life characteristics

5.1.1 Life 2 model point variables

Specify insured characteristics model point variables for life 2 in addition to life 1. Life 2 characteristics include issue age, gender, smoking class, and rate class. These insured characteristics are particularly important for mortality variables, which are the rates for: (1) experience, cash value, and reserve mortality, and (2) YRT reinsurance rates.

See the Mortality variables section in the Input organization chapter, above.

5.1.2 Life 1 and life 2 scalars

Since issue_age is used to calculate values such as the years to maturity, 'life 1' must be:

- the older of the two insureds for first to die model points, and
- the younger of the two insureds for Frasierized last survivor model points.

The order of the two lives on the model point record is handled automatically, by converting the life 1 and life 2 model point variables into scalars, instead of using the model point variables directly:

- For first to die model points with life 1 younger than life 2, the life 1 scalars return the life 2 model point variables, and the life 2 scalars return the life 1 model point variables.
- For Frasierized last survivor model points with life 1 older than life 2, the scalars also switch the values this way.

- Otherwise, the life 1 scalars return the life 1 model point variables, and the life 2 scalars return the life 2 model point variables.

For example, the scalars `issue_age` and `gender` normally have the values `issue_age_mp` and `gender_mp`, but return the values `issue_age_mp_life2` and `gender_mp_life2` when the lives must be switched. Since formulas reference the scalars, they are automatically based on the correct life.

5.2 First to die

For first to die products, the death benefit is paid on the first death; the policy terminates at that time. Both insureds are alive at the projection date.

The annual mortality/COI/YRT rate is:

$$q_1 + q_2 - q_1 \times q_2$$

5.3 Frasierized last survivor

For last survivor products, the death benefit is paid when both insureds have died. Since there is no change in status when the first life dies, the projection assumes there is no knowledge about whether life 1 or life 2 is still alive at the projection date, except that at least one of them must be alive since the policy is still in force.

The mortality/COI/YRT rates are Frasierized from issue, as described below.

5.4 Frasierization

Since Frasierized last survivor products continue to operate as normal until the second death, the projection assumes there is no information whether only one life is surviving at the projection date. The mortality rates for life 1 and life 2, plus the joint life contagion effect, are combined into a Frasierized rate equal to the sum of:

$$\begin{aligned} &P(\text{both alive BOY}) \times P(\text{both die during the year} | \text{both alive BOY}) \\ &\quad + P(\text{only life 1 alive BOY}) \times P(\text{life 1 dies during the year} | \text{only life 1 alive BOY}) \\ &\quad + P(\text{only life 2 alive BOY}) \times P(\text{life 2 dies during the year} | \text{only life 2 alive BOY}) \end{aligned}$$

This formula Frasierizes the annual mortality rates, then the Frasierized annual rate is converted to a monthly rate as usual. This method commonly applies for reserve mortality and YRT rates. Annual experience mortality rates are also Frasierized prior to calculating the monthly mortality rate, so that calculations are simplified by producing monthly mortality rates which are constant within each policy year.

5.4.1 Joint life mortality addition

For experience mortality, the annual probability that both lives die together (or within some short interval) is increased by a joint life mortality rate addition, which is sometimes called a contagion effect. The specified addition is reduced when the joint death rate would exceed either the first or second life mortality rate. That is:

$$q_1 \times q_2 + \text{adjusted addition} \leq \min(q_1, q_2)$$

5.4.2 Minimum YRT rate

The Frasierized annual YRT reinsurance rate is subject to a minimum floor, such as .00012 (\$.12 per thousand).

6 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves calculated by model point. PBA reserves are discussed in the PBA book, including the net premium reserve (NPR).

6.1 Reserve basis

6.1.1 Mortality

Reserve mortality is a base rate, times the multiple, then plus the addition. Lookups for reserve mortality tables should be setup similar to the experience mortality rates since these rates are read for both the first and second lives.

The base statutory mortality rates affect the projection even when reserves are a percent of the fund or cash value, since the omega age specifies the GMP maturity age.

Substandard multiple

The substandard multiple (on model point records) does not affect reserves. It applies for experience mortality, COI charges, and YRT reinsurance premiums.

6.1.2 Interest

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeStatValn and LifeTaxValn ranges in the Valuation Interest Rates Excel workbook.

6.1.3 Function timing

Function timing options affect traditional reserve calculations, such as the CRVM reserve (for the UL CRVM method) and the XXX reserve (for the AXXX reserve):

- **Curtate** – Annual premiums and claims paid at the end of the policy year
- **Continuous** – Continuous premiums and time-of-death claims
- **Semicontinuous** – Annual premiums and time-of-death claims.

Premium timing multiple

When premiums are assumed to be paid continuously, the curtate reserve formulas are adjusted by the following 'premium timing' multiple:

$$\frac{(\delta - d)}{\delta^2} + \frac{(i - \delta)}{\delta^2} \times \frac{(1 - q)}{(1 + i)}$$

Claims timing multiple

When claims are assumed to be paid immediately, the curtate reserve formulas are adjusted by the following 'claims timing' multiple:

$$\frac{i}{\delta}$$

Reserve for immediate payment of claims

Actuarial guideline XXXII (AG 32) requires additional reserves for policies which pay death claims at the time of death, but have statutory reserves calculated assuming death claims are paid at the end of the policy year. This can be approximated by using the Semicontinuous function timing option.

6.2 UL CRVM reserve

6.2.1 Percent of fund approximation

The UL CRVM reserve can be approximated using percentages of the fund value or cash value. This approximation replaces the UL CRVM reserve. The secondary guarantee reserve, if applicable, is still calculated and compared against this amount.

6.2.2 GMP and GMF

The UL CRVM reserve is based on the guaranteed maturity fund (GMF), which is calculated using the guaranteed maturity premium (GMP) which endows the policy for the sum assured at the GMP maturity age:

- The GMP maturity age is the statutory reserve base mortality omega age, when earlier than the maturity age.
- The GMP is paid for the contractual premium term, which may be different from the model point's premium term.
- The GMP is paid using the model point's premium mode.

The GMF is projected by calculating the GMP which endows the fund value at the GMP maturity age using guaranteed loads, COI rates, and interest crediting. The GMF is then projected using the GMP and these guaranteed rates. When specified by the `solving_display_defn` variable (on the Product input page in the Premiums category), the `ul_ifund` output includes this fund value per unit.

The GMF is recalculated (in the situations described below) by starting from the current fund value, then projecting using guaranteed rates and assuming that the GMP is paid in the future. When the reserve method is UL CRVM, the `ul_ifund` output includes this fund value per unit. This does not apply when the reserve method is Approximate UL CRVM, which uses only the GMF values.

6.2.3 UL CRVM

Reserve basis

The statutory mortality variables for UL CRVM reserves (`stat_mort_base`, `stat_mort_mult`, and `stat_mort_addn`) are separate from the mortality variables for XXX reserves. The same applies for tax reserves. The deficiency mortality variables for alternative minimum UL CRVM reserves (`def_res_mort_base`, `def_res_mort_mult`, and `def_res_mort_addn`) are separate from the mortality variables for XXX deficiency reserves.

CRVM reserve [`ul_py->stat_crvm_terminal`]

The traditional Commissioners Reserve Valuation Method (CRVM) reserve is calculated for the plan of insurance endowing at the GMP maturity age, using the same expense allowance as for the corresponding traditional plan.

Benefit amounts [`array dth_ben_tbl_crvm`]

The benefits for calculating the traditional CRVM reserve are the projected average EOM death benefit per unit amounts for each policy year, using guaranteed rates and charges and GMP premium payments.

The sum assured is paid as an endowment amount at the GMP maturity age.

UL CRVM reserve [`ul->stat_res_ulcrvm_per_unit`]

The ratio 'r' is the current fund value divided by the GMF. When $r \leq 1$, the reserve is the traditional CRVM reserve times r.

When $r > 1$, the current fund value is projected to maturity assuming: (1) future premiums equal to the GMP, and (2) guaranteed rates.

The average death benefit in each policy year from this projection is compared to the death benefits from the GMF projection from issue. Any excess death benefit, and any excess maturity value, over the GMF projection from issue is discounted using statutory reserve mortality and interest.

The reserve is: (1) the traditional CRVM reserve, plus (2) the present value of these excess benefits.

6.2.4 **Approximate UL CRVM**

The following approximation can be appropriate for policies which are not heavily funded.

The traditional CRVM reserve and r are calculated from issue the same as described above. When $r \leq 1$, the reserve is the CRVM reserve times r ; there is no approximation in this case.

When $r > 1$, the reserve is: (1) the traditional CRVM reserve, plus (2) the excess of the current fund value over the GMF. This approximation thus reduces calculation time by not recalculating the GMF.

6.3 **Secondary guarantee reserve**

For products with a secondary guarantee, the reserve [ul->stat_res_per_unit_prelim] is the greater of: (1) the UL CRVM reserve, and (2) the secondary guarantee reserve defined by regulation XXX and AG 38.

6.3.1 **Exception**

The secondary guarantee reserve does not apply when the: (1) secondary guarantee benefit term is five years or less, (2) the annual required premium for the first policy year is not less than the statutory net level premium method reserve premium for the guarantee, and (3) the current surrender charge is not less than the annual required premium for the first policy year.

6.3.2 **Regulation XXX**

Reserve basis

The statutory mortality variables for XXX reserves (stat_xxx_mort_base, stat_xxx_mort_mult, and stat_xxx_mort_addn) are separate from the mortality variables for ULCRVM reserves. The same applies for tax reserves. As discussed below, these must be ultimate mortality rates for 1980 CSO mortality, since select multiples (stat_xxx_select_input_1 for the first segment, and stat_xxx_select_input_2 for subsequent segments) apply to them. For 2001 CSO mortality, these can be the select and ultimate rates, then leaving the select multiple variables at their 1.0 default value.

Deficiency reserve (alternative minimum) mortality (def_res_xxx_mort_base) applies for XXX deficiency reserves. As also discussed below, separate variables (seg_xxx_mort_base and seg_xxx_select_input_1) specify the mortality rates for determining the XXX segments.

Segmented reserve

Reserve net premiums are a different percent of the annual required premiums for each segment of the secondary guarantee premium term. The division into segments is discussed below. Reserves are calculated applying the Commissioners Reserve Valuation Method (CRVM) expense allowance to the first segment, with no further expense allowances for the other segments.

Annual required premiums [ul_py->sg_art_prem]

The annual required premiums are the minimum premium in each year which that will satisfy the secondary guarantee test.

- For the shadow fund, this is the premium paid at the beginning of each policy year which results in zero shadow fund value at the end of the year. [array shadow_fund_art_prem]
- For the cumulative premiums test, these are the required premiums specified for the guarantee.

Segmented definition

The levelized premium segmentation method divides the benefit period into segments when the ratios, $G(t)$, of successive required premiums exceed the ratios, $R(t)$, of successive deficiency reserve basis mortality rates but excluding the X factor. The first segment ends prior to the policy year (designated by 't') when $G(t)$ first exceeds $R(t)$. The next segment extends from the end of the first segment until the next segment starts at the beginning of the policy year when $G(t)$ again exceeds $R(t)$, and so on.

The mortality rate ratios, $R(t)$, can be multiplied by 101% or 99% to avoid segments based on insignificant excess values. However, $R(t)$ cannot be less than 1.

Segments are determined using reserve mortality variables `seg_xxx_mort_base` and `seg_xxx_select_input_1`. These variables typically contain the same tables as for deficiency reserve mortality for the first segment, but excluding the deficiency reserve X factors (discussed below). Separate variables are provided so that XXX deficiency reserves can use mortality varying by the 2001 CSO rate classes, while the segments are determined using the version of the table which does not vary by rate class, as allowed by actuarial guideline XLVI (AG 46).

The same segments apply for statutory, deficiency, and tax reserves.

Select mortality factors

Select mortality factors apply both for the determination of the segments and for the calculation of the segmented reserves. In addition, 'X factor' multiples can apply to the deficiency mortality rates for calculating deficiency reserves during the first segment, but does not apply for determining the segments.

2001 CSO mortality – The Global folder includes the 25-year select multiples that apply for 2001 CSO mortality; see the Mortality Select Multiples Excel workbook. When using these select mortality multiples, the base mortality variable must specify the ultimate rates. When the 2001 CSO select rates apply past the first segment, the select and ultimate mortality rates can be specified as the base variable, leaving the select factor variables at their default values of 1.0 (100%). However, when 'X factor' multiples are used, the base mortality variable must specify the ultimate rates with separate select factors specified so that it is the combined select factors which are constrained to 1.0 (100%) as opposed to just the 'X factor' multiple being constrained.

1980 CSO mortality – The Global folder includes 10-year, 15-year, and 20-year select factors which apply to the 1980 CSO ultimate mortality rates; see the Mortality Select Multiples Excel workbook. For 1980 CSO mortality, the reserve mortality variable must specify the ultimate rates, then the select factor variables (when select mortality is elected for reserves) must specify the following:

The 10-year select factors are implicit in the 1980 CSO select and ultimate mortality rates. This select factors apply for subsequent segments when the first segment is shorter than 10 years.

The following variations for applying special select factors during the first segment:

- Regulation XXX can apply either the 10- or 20-year select factors during the first segment.
- For NY 147, the first segment uses 15-year select factors adjusted by a multiple of 150% (120% for deficiency reserves), with the resulting select factor rounded to the nearest whole percent and capped at 100%. These select factor multiples apply only when reserves segment definition is NY 147; these variables are ignored for regulation XXX.

Benefit amounts [array `dth_ben_tbl_xxx`]

The benefits for these regulation XXX calculations are the projected average EOM death benefit per unit amounts for each policy year, using guaranteed rates and charges and assuming that the annual required premiums are paid at the beginning of each policy year.

Only the death benefits apply; there is no endowment amount at the end of the guarantee period.

Mortality improvement

Mortality improvement can apply for the determination of the segments and the calculation of the segmented, basic, and tax reserves, such as is allowed for under the 6th amendment of NY regulation 147. Mortality improvement applies from a specified improvement base year. When included, mortality improvement multiples are applied to the select factor adjusted mortality rates in the first segment only.

Lapse rates

Under the 5th and 6th amendment of NY regulation 147, lapse rates may be included in the present value calculations in steps 2 and 5 of Actuarial guideline XXXVIII (AG 38), as described in section 6.3.3. The Global folder includes the maximum lapse rates allowed, contingent on issue year, issue age, and policy year; see the NY147 Lapse Rates Excel workbook.

Catch up provision reduction

The catch up provision applies when the secondary guarantee test has been violated for less than the grace period months (and the current fund value is positive to keep the policy in force). The regulation XXX reserve is reduced by the 'catch up premium' which would reinstate the guarantee:

- For the shadow fund, this is the negative shadow fund.
- For the cumulative premiums test, this is the excess of the accumulated required premiums over the accumulated actual premiums.

The reduction is prorated between the basic and deficiency reserve.

6.3.3 Actuarial guideline 38

Actuarial guideline XXXVIII (AG 38), which is often referred to as guideline AXXX, defines the calculation of secondary guarantee reserves for flexible premium universal life products issued after 2002. The XXX reserve is increased by additional calculation steps, resulting in 9 calculation steps as follows:

Step 1: Annual required premiums [ul_py->sg_art_prem]

This step is the same as for regulation XXX, described above. For shadow fund guarantees, these are the minimum premiums required by the contract, using the shadow fund premium load. That is, they are not affected by the 7% premium load restriction which applies for later steps for AG 38-8B and 8C.

Step 2: Segmented reserves for the guarantee period [statutory: ul_py->stat_xxx_terminal and ul->stat_res_xxx_per_unit] [including deficiency reserve: ul_py->stat_xxx_min_terminal and ul->stat_res_xxx_min_per_unit]

This step is the same as for regulation XXX, described above. AG 38-8C uses lapse rates in this reserve calculation.

The 'catch up' provision amount is prorated to reduce the basic and deficiency reserves.
[ul->stat_res_xxx_reduced_per_unit and ul->def_res_xxx_reduced_per_unit]

Step 3: Excess premiums paid [ul->axxx_excess_prem_per_unit]

The amount of actual premiums paid in excess of the required premiums is:

- For the shadow fund test, the current shadow fund.
- For the cumulative premiums test, the accumulated actual premiums less the accumulated required premiums.

When the 'excess' is negative, this is the 'catch up provision' amount which reduces the regulation XXX reserve from step 2.

Step 4: Required payments single premium [ul->sg_req_single_prem_per_unit (before gross up)]

The single payment necessary at the valuation date to fully fund the guarantee through the secondary guarantee benefit term, assuming that required premiums have been paid in the past, is calculated as follows:

- For the shadow fund, the array `shadow_fund_single_prem_net` contains the shadow fund which would result in zero shadow fund at the end of the secondary guarantee period. For AG 38-8A, this amount is grossed up by dividing by [100% minus the shadow fund premium load percent]. For AG 38-8B and 8C, the amount is grossed up by [one minus the prescribed 7% load].
- For the cumulative premiums test, the ending accumulation of required premiums is rolled backward to the beginning of each month, assuming that no premiums are paid. The single premium is the single premium value less the accumulated required premiums at the valuation date. For AG 38-8B and 8C, the amount is grossed up by [one minus the prescribed 7% load].

Then the calculated 'pre-funding ratio' is any excess premiums paid (step 3), divided by the single premium. [ul->axxx_pre_funding_ratio]. This ratio is floored at zero, and capped at 1.

Step 5: Present value of benefits (net single premium) [ul->stat_axxx_nsp_per_unit]

The present value of benefits through the secondary guarantee benefit term is calculated using statutory reserve mortality. The benefit is the same as discussed above for regulation XXX calculations. These values are calculated as of the end of each policy year [ul_py->stat_xxx_nsp], then linear interpolation applies for other months. AG 38-8C allows including lapse rates in this calculation.

Step 6: Net amount of additional premiums [ul->stat_axxx_addl_prem_per_unit]

Any excess of the present value of benefits (step 5) over the basic and deficiency reserve (step 2) is multiplied by the pre-funding ratio (step 4).

Step 7: Reduced deficiency reserve [ul->def_res_axxx_reduced_per_unit]

When the required payments single premium (step 4) is greater than the excess premiums paid (step 3), the deficiency reserve (step 2) is multiplied by [one minus the pre-funding ratio floored at zero] (step 4).

Step 8: Minimum reserve [ul->stat_res_axxx_min_per_unit]

The minimum reserve before the surrender charge reduction is the lower of: (1) present value of benefits (step 5), and (2) the XXX basic and deficiency reserve (step 2) plus the net amount of additional premiums (step 6). This amount is then reduced by the surrender charge reduction (step 8b, below). However, the surrender charge reduction cannot reduce the minimum reserve below the XXX basic and deficiency reserve (step 2).

Step 8b: Surrender charge reduction [ul->axxx_surr_chg_per_unit]

The final reserve [ul->stat_res_axxx_per_unit] is the minimum reserve, reduced for the surrender charge:

- For AG 38-8A, the reduction is the fund value less the cash surrender value.
- For AG 38-8B and 8C, the surrender charge reduction is multiplied by the ratio of: (1) the net level premium for the secondary guarantee years, divided by (2) the net level premium for whole life insurance.
- For AG 38-8B, there is no reduction for surrender charges when the pre-funding ratio exceeds [one minus the prescribed 7% load].

Step 9: Increased basic reserve

The minimum reserve calculated in step 8 applies for the projection. For reporting purposes, the basic reserve is the minimum reserve (step 8) minus the reduced deficiency reserve (step 7).

6.3.4 Actuarial guideline 38 variations

Actuarial guideline 38 includes the following variations:

AG 38-8A

AG 38-8A (sometimes called the 'original version') applies for policies issued from January 1, 2003 through June 30, 2005.

AG 38-8B

AG 38-8B (sometimes called the 'first revision') applies for policies issued July 1, 2005 through December 31, 2006:

- The percent of premium load allowance used in grossing up the required payments single premium (step 4 as described below) is 7%. This is one minus the variable `sg_funding_ratio`, which has a default of 93%.
- The surrender charge reduction is eliminated when the secondary guarantee is fully funded.
- The surrender charge reduction is reduced when the secondary guarantee period is less than whole life coverage.

AG 38-8C

AG 38-8C (sometimes called the 'second revision') applies for policies issued in 2007 through December 31, 2012:

- The provisions of AG 38-8B continue to apply, except that the surrender charge reduction is restored when the pre-funding ratio exceeds [one minus the 7% prescribed premium load]. That is, this provision reverts back to the AG 38-8A version.
- Lapse rates may be included in the present value calculations in steps 2 and 5 as described below. The Global folder includes the maximum lapse rates; see the AG38 Lapse Rates Excel workbook.

AG 38-8D

AG38 applies for certain policies issued from July 1, 2005 through December 31, 2012, or any valuation date thereafter. Additional reserves may be required should the deterministic reserve calculated under a VM-20 methodology with modifications exceed the reserve calculated under AG 38-8B or 8C. Specify in variable `stat_res_ag38_8d_defn` whether AG38-8D applies for the product, and in variable `stat_res_ag38_defn` the base valuation method (e.g. AG38-8B or 8C).

The reserve calculated for the model point is for the base valuation method (e.g. AG 38-8B or 8C). The aggregate reserve is the greater of the special VM-20 deterministic reserve (discussed below), and the sum of the reserves calculated by model point ignoring reinsurance.

The additional reserve can be tax deductible.

AG38-8D has no impact on the reinsurance reserve credit applicable to policies with YRT reinsurance. For policies with coinsurance or modified coinsurance, the ceded reinsurance reserve is the treaty percent applied to the gross reserve calculated by model point plus the treaty percent applied to the additional reserve (if applicable) calculated under the special VM-20 deterministic reserve methodology.

The modified VM-20 deterministic reserve calculation assumes that:

- The starting assets amount is the sum of the statutory reserves calculated by model point.
- For a liability only projection, the asset yield assumed in the reserve calculations is defined using the competitor rate basis specified for the crediting strategy. The deterministic scenario is defined in a scenario rates file, adjusted by a specified multiple and addition. This asset yield is used to define the crediting rate and discount rate in the inner loop valuation projection. The asset yield is assumed to already account for the asset yield requirements specified under the Primary Reserve Methodology (Section 8Da.2.a) of AG 38.
- For an asset/liability projection, specify the issue year yields for A-rated bonds by issue year from 2005 to 2012 in the input variable `ag38_8d_issue_yld_rate`. At the start of each valuation period, weighted average rates are calculated using the specified bond yields in combination with the model point reserves by issue year, calculated at each future valuation date. The starting assets earn the actual portfolio net asset yield capped with rates calculated above. All rates are net of defaults.

- For an asset/liability projection, new investments made during the valuation are invested as cash earning a fixed rate specified as a competitor rate at the valuation date and capped at 7%.

The special VM-20 reserve is calculated the same as for PBA reserves, using monthly cash flows and discounting.

For nested projections, the aggregate AG38-8D reserve between valuation dates is the sum of the model point reserves times the ratio of the aggregate Ag38-8D reserve to that sum on the previous valuation date. For an asset/liability projection, the assets are projected using the asset portfolio and investment strategy specified for the projection. The new valuation rates are reflected in the competitor rates and scenario rates files.

AG 38-8E

AG38-8E applies to policies issued in 2013 or later. Specify whether the product uses method I or method II.

For Method I, the following steps are revised from 8C:

- Step 1: For policies with policy design #3, specify (stat_res_ag38_min_prem_method_1_per_unit) the vectors of minimum gross annual premiums. When this variable is zero (default value), these premiums are calculated using the COI, load, and crediting rate assumptions used in the minimum gross annual premium calculations.
- Step 4: The pre-funding ratio in this step is not less than -1, and not more than 1.
- Step 6: When variable stat_res_ag38_min_prem_per_unit is zero, or the step 3 amount is positive, then this step is the same as under 8C. When the step 3 amount is negative, the 'net amount of additional premium' (ul->stat_axxx_addl_prem_per_unit) is the ratio from step 4 ul->axxx_pre_funding_ratio times the basic reserve from step 2 (stat_res_xxx_per_unit), less the deficiency reserve calculated in step2, plus the following amount:
 - For shadow fund guarantees: The deficiency reserve at issue calculated using 100% X, multiplied by one minus the ratio (floored at zero) of the shadow fund divided by the required single premium from step 4. This amount is floored at zero.
 - For cumulative premium test guarantees: The deficiency reserve at issue calculated using 100% X, multiplied by 1 minus the ratio (floored at zero) of the amount of cumulative premiums paid, divided by the required single premium from step 4. This amount is floored at zero.
- Step 8: Same as under 8C, but adjusting the reserve floor. When the step 3 amount is positive, the reserve floor is the sum of the step 2 basic and deficiency reserve without adjustment. When the step 3 amount is negative, then the reserve floor is the sum of the step 2 basic and deficiency reserves and the amount from step 6.

For Method II, the following steps are revised from 8C:

- Step 1: The minimum gross annual premiums is defined the same as for Method I. (Note that there is no dynamic functionality which determines the lowest schedule of minimum gross premiums that keeps the policy in force over the life of the secondary guarantee period and that produces the greatest deficiency reserve at issue.)
- Step 3: For shadow fund guarantees, this is the shadow fund minus the shadow fund that would have resulted from paying minimum gross premiums from issue. For cumulative premium test guarantees, this is the cumulative premiums paid minus the cumulative premium that would have resulted from paying the minimum gross premium from issue. This result can be negative.
- Step 4: Same under 8C but with the following adjustment in the single premium (ul->axxx_req_pmt_single_prem_per_unit before gross up):
 - For shadow fund guarantees: When step 3 is positive, the single premium is reduced by the shadow fund (floored at zero) based on paying the minimum gross premium from issue. When step 3 is

negative, the single premium is replaced by the shadow fund value (floored at zero) based on paying the minimum gross premium from issue.

- For cumulative premium test guarantees: When step 3 is positive, the single premium is reduced by the excess of (1) accumulated premiums based on paying the minimum gross premium from issue over (2) the cumulative premium requirements. When step 3 is negative, the single premium is replaced by the excess of (1) accumulated premiums based on paying the minimum gross premium from issue over (2) the accumulated required premiums.
- After the adjustment, the amount is grossed up by the `sg_funding_ratio`, then the `axxx_pre_funding_ratio` is calculated as described in 8C.
- Step 6: When step 3 is positive, this step is the same as under 8C. When step 3 is negative, the 'net amount of additional premium' (`ul->stat_axxx_addl_prem_per_unit`) is the ratio from step 4 (`ul->axxx_pre_funding_ratio`) times the basic reserve from step 2 (`stat_res_xxx_per_unit`), less the deficiency reserve calculated in step 2, plus the following amount:
 - For shadow fund guarantees: the deficiency reserve at issue calculated using 100% X, times one minus the ratio (floored at zero) of the shadow fund divided by the minimum amount in the shadow fund that would fully fund the guarantee. This amount is floored at zero.
 - For cumulative premium test guarantees: The deficiency reserve at issue calculated using 100% X, times 1 minus the ratio (floored at zero) of accumulated premiums paid, divided by the minimum accumulated premiums required to fully fund the guarantee. This amount is floored at zero.
- Step 7: The result is floored at zero
- Step 8: Same as under 8C, but adjusting the reserve floor. When step 3 is positive, the reserve floor is the sum of the step 2 basic and deficiency reserves without adjustment. When step 3 is negative, then the reserve floor is the sum of the step 2 basic and deficiency reserves and the amount from step 6.

6.4 Substandard reserve

Substandard reserves are not included in the projection. The substandard multiple (on model point records) does not affect reserves. It applies for experience mortality, COI charges, and YRT reinsurance premiums.

However, stochastic and deterministic PBR calculations include the substandard multiple in the PBA mortality rate, since it is part of the expected experience for the policy.

6.5 Mid-year reserve

Reserves between policy anniversaries can be:

- Monthly interpolated terminal – Monthly interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.
- Daily interpolated terminal – Daily interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.

When the reserve function timing is Continuous, the mid-year reserve net premium is a discounted continuous net premium.

6.6 Deficiency reserve

The deficiency reserve is the 'alternative minimum' reserve less the statutory basic reserve, calculated using mid-year reserves. When secondary guarantee reserves apply, the 'alternative minimum' reserve is the greater of the UL CRVM alternative minimum reserve and the secondary guarantee alternative minimum reserve.

6.6.1 **UL CRVM alternative minimum reserve**

When basic deficiency reserves are requested, a UL CRVM alternative minimum reserve is calculated using the lower of the GMP and reserve net premium in each year.

Specify the UL CRVM deficiency reserve mortality and interest basis separately from the statutory reserve basis.

When a percent of fund approximation is used for the UL CRVM reserve, no UL CRVM alternative minimum reserve is calculated.

6.6.2 **Secondary guarantee alternative minimum reserve**

When secondary guarantee deficiency reserves are requested, a secondary guarantee minimum reserve is calculated using regulation XXX and actuarial guideline 38 requirements.

Regulation XXX

Regulation XXX (or NY regulation 147) applies the same for both basic and deficiency reserves. Specify deficiency reserve mortality separately from the statutory mortality. As explained in the secondary guarantee reserve Regulation XXX topic, deficiency reserve mortality includes a table of 'X factor' multiples that apply during the first segment.

Also as explained in the secondary guarantee reserve Regulation XXX topic, segments are determined using reserve mortality variables `seg_xxx_mort_base`, `seg_xxx_select_input_1`, and `seg_xxx_mort_improve_pct`. These variables typically contain the same tables as for deficiency reserve mortality for the first segment, but excluding the deficiency reserve X factors. Separate variables are provided so that XXX deficiency reserves can use mortality varying by the 2001 CSO rate classes, while the segments are determined using the version of the table which does not vary by rate class, as allowed by actuarial guideline XLVI (AG 46).

Actuarial guideline 38

When actuarial guideline 38 applies, the actuarial guideline 38 net single premium, additional premium, and surrender charge amounts are calculated as discussed in the secondary guarantee section. These amounts are used to adjust the XXX alternative minimum reserve to arrive at the actuarial guideline 38 minimum reserve.

6.7 **Additional reserve**

Additional statutory reserves can be specified as an amount per unit. [`ul->stat_res_addl_per_unit`]

The additional reserve is not considered during the calculation of deficiency reserves.

The additional reserve can be added to the final reserve (after the comparison between the ULCRVM and secondary guarantee reserves), or only to the ULCRVM reserve (before the comparison to the secondary guarantee reserve). The additional reserve is added before comparisons to the cash value and tabular cost of insurance floor.

When detail reports split out reserve values, 'additional reserves' include the increase over basic reserves due to minimum reserves (such as cash value and the tabular cost of insurance floor). [`ul->stat_res_addl`]

The coinsurance reserve credit can exclude the additional reserve.

6.8 **Tabular cost of insurance floor**

The reserve is subject to a minimum based on the tabular cost of insurance. That is, this floor applies to the sum of: (1) basic reserve (including the AG 38 reserve), (2) deficiency reserve, and (3) additional reserve.

The floor is the tabular cost of insurance (c_x) times the sum assured, times the following fraction for each mid-year reserve method:

- Monthly interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment. For example, this floor at the end of policy month 1 is $11/12 c_x$ for annual premiums, and $2/12 c_x$ for quarterly premiums.
- Daily interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $344/365 c_x$ for annual premiums, and $71/365 c_x$ for quarterly premiums.

6.9 Cash value floor

Finally, the mid-year reserve is not less than the cash surrender value. [ul->stat_res_per_unit]

6.10 EMO reserve

When the Extended Maturity Option is in effect, the reserve is the current death benefit.

6.11 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record sets the tax reserve to be the statutory reserve, including the comparison to tabular cost of insurance floor using the statutory basis, but excluding any deficiency reserve.

Tax reserves are calculated by the same methods, using the tax reserve interest and mortality basis. The secondary guarantee reserve uses the same segments that apply for statutory reserves.

The statutory additional reserve can be added to tax reserves. That is, the additional reserve can be deductible.

The tax reserve is not greater than the statutory reserve, including the additional reserve even when it is not included in the tax reserve.

6.12 Unearned premium reserve

6.12.1 Balance sheet

When the mid-year reserve method is monthly or daily interpolated terminal, an unearned premium reserve is established. The unearned premium reserve is the portion of the modal valuation net premium that has been collected but not yet earned.

The balance sheet includes the statutory unearned premium reserve as a separate line item from the statutory reserve. The unearned premium reserve is set to zero if the reserve being held is the tabular cost of insurance or cash value floor.

6.12.2 Profit effect

Book profit includes the increase in the unearned premium reserve.

6.12.3 Taxable income

Taxable income similarly includes the unearned premium reserve increase, which is calculated the same way as statutory unearned premium reserve increase, but using the tax reserve basis.

6.13 Gross premium reserve

Letter of credit (LOC) reserve financing creates a non-invested LOC asset that offsets the 'redundant' part of the statutory reserve, based on the calculation of the gross premium reserve (GPR) for each model point assigned to a reserve financing transaction. Book profit is increased by the LOC asset increase (offsetting part of the statutory reserve increase), and is decreased by LOC costs (interest paid on the LOC assets). See the Gross premium reserves chapter in the PBA book.

7 ACTUARIAL ASSUMPTIONS

7.1 Commissions

7.1.1 Premium commission

Percent of premium commissions apply the target premium percent to the premiums paid during the policy year (including the additional single premium) until the premiums exceed the target premium. Then the excess premium percent applies to premiums paid during the remainder of the policy year.

When the policy fee is not commissionable, the policy fee part of the premium is excluded from this calculation, and the target premium excludes the policy fee.

7.1.2 Fund value commission

One twelfth of the annual percent of unloaned fund value at the end of the previous month is paid at the beginning of each month.

7.1.3 Annualized commission

A specified percent of the total first year target premium commission can be paid at issue. No further premium commissions are paid until the cumulative earned commission exceeds the annualized amount. From then on, premium commissions are paid as usual. This determination of commission payments is made for continuing policies, not taking into account terminated policies.

Other types of commissions are not affected by the annualization of commissions.

7.1.4 Commission chargeback

A portion of the premium commission can be charged back in the event of lapse or surrender during the first three years after issue. The chargeback percent applies either to: (1) total first year premium commissions paid to date, or (2) the annualized commission amount. The chargeback percent typically varies by month, by linking to the external source using the `mths_since_issue` column as a lookup key.

Other types of commissions are not subject to chargeback.

The commission subject to chargeback is calculated per unit in force, and thus is not reduced by decrements. This amount applies to the units lapsing and surrendering each month.

For in-force model points, specify on the model point record the initial (at the projection date) commissions subject to chargeback for the policies collected into the model point. When this variable is negative (the default value is -1), the chargeback amount is calculated from issue assuming that premiums have been paid as specified.

7.1.5 Commission overrides and agent terminations

The specified commission rates are the average commission rates, taking into account average writing agent commissions, commission overrides, and the effect of agent terminations.

7.2 Expenses

7.2.1 Expense adjustment multiple

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

7.2.2 Acquisition expenses

Acquisition expenses incurred at issue can include:

- Amount per policy
- Amount per unit
- Percent of target premium
- Percent of the total first year premium (including the additional single premium).

Acquisition expenses could vary by underwriting class, which is discussed in the Mortality section, below.

7.2.3 Maintenance expenses

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Maintenance expenses can include:

- Amount per policy, separate between premium paying and paid up
- Amount per unit, separate between premium paying and paid up
- Percent of the annualized premium (excluding the additional single premium)
- Percent of the total fund value at the end of the previous month.

BOM maintenance expense

An additional maintenance expense is incurred at the beginning of each month, when commissions are paid:

- Percent of each commission payment, ignoring commission chargeback.

7.2.4 Transaction expenses

Maintenance expenses include the following expenses related to administrative activities:

- Expense per death claim policy
- Expense per death claim unit.

7.2.5 Reinsurance administration expenses

Reinsurance administration expenses are discussed in the Reinsurance chapter, below.

7.2.6 Inflation

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

Note: The average size for future issues can be inflated (to offset expense inflation), by customizing the scalar `units_per_pol` to divide the result by `infl_cumul(commencement_period)` for model points issued after the start of the projection. This would reduce the number of policies issued, with no effect on the units, sum assured, and premium issued for the model point.

7.2.7 Policy loan expenses

Policy loan expenses are a percent of the loan balance at the end of the previous month. One twelfth of the annual rate applies each month.

This amount is a negative cash flow at the end of each month, and is thus subtracted from direct cash flows.

Policy loan expenses reduce book profit, by reducing investment income from policy loans. Thus this expense is treated as an investment expense for statutory and GAAP profit reporting. It is not included with other expenses on profit reports.

However, PBA deterministic reserves include this expense in the present value of expenses. This is a different treatment of policy loan expenses than applies for profit calculations.

7.2.8 Premium tax

Premium taxes are reported with expenses. The tax is a percent of each premium payment, including the additional single premium paid at issue.

Inflation and the expense adjustment multiple do not apply to premium tax.

7.2.9 Fixed expenses

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

7.3 Mortality

7.3.1 Input variables

Mortality variables for universal life products include mortality base, multiple, and addition rates, as well as mortality improvement rates. A single set of input variables is included in the model but these variables are special since they are read twice – first for life 1 and then again for life 2 (when applicable). This is accomplished by using special lookup keys.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. 'Insured characteristics' (such as issue age, gender, smoking class, and rate class) are set to either the life 1 or life 2 values, as needed. The external source uses lookup keys `mort_issue_age`, `mort_gender`, `mort_smoking_class`, `mort_rate_class`, `mort_pol_yr`, `mort_cal_yr`, and `mort_term_conversion_defn` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

7.3.2 Mortality adjustment multiple

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

7.3.3 Term conversion adjustments

Term products often offer the option to convert to a permanent product (i.e. universal life) prior to the end of a level term period without additional underwriting.

Specify the original term issue date and whether the policy was issued as a result of a term conversion on the model point record. For such policies, the base mortality rates, mortality improvement factors, and mortality table adjustments will be calculated using a point-in-scale method based on the original issue date of the term policy. Additionally, a separate set of base mortality rates and mortality table adjustments may apply to the term conversion cohort of policies if specified.

Term conversion adjustments do not apply to joint life policies.

7.3.4 Underwriting class

Life insurance products can use the `underwriting_class` scalar to vary product assumptions such as mortality rates and acquisition expenses. The underwriting class is calculated by comparing the policy size against the nonmedical and paramedical underwriting limits specified with the experience mortality input. The underwriting class has a value of either:

- Nonmedical – when `ins_issued_per_pol` $\leq$ `nonmedical_limit`
- Paramedical – when `nonmedical_limit` $<$ `ins_issued_per_pol` $\leq$ `paramedical_limit`
- Medical – when `ins_issued_per_pol` $>$ `paramedical_limit`.

7.3.5 Rate class

Mortality rates varying by rate class require special treatment. The model point variable (`rate_class_mp`) can have a large number of options specific to the company's business. But the 2001 CSO mortality tables varying by rate class only define rates for Standard, Preferred, and Super issues. Thus there are two different rate class scalars – `rate_class` contains the company rate classes, and `rate_class_res` maps the company rate classes into the reserve rate classes. These same considerations apply for the second life variable (`rate_class_mp_life2`) and scalar (`rate_class_life2`).

7.3.6 Base mortality

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

7.3.7 Mortality improvement

The base rate is then adjusted by mortality improvements. Negative improvement rates specify worsening mortality due to anti-selection after issue.

Improvement starts after the specified base year, which is either the issue year or a specified calendar year. The improvement can end after a specified policy or calendar year.

When improvement rates apply by policy year, the mortality rate improves at the end of each policy year by the rate specified for that year. When mortality rates apply by attained age, the mortality rates for each age improve by the rate specified for that age. The following example shows the mortality improvement multiple for a policy issued in 2008 with mortality improvement base year of 2010, thus improvements begin at the end of the third policy year:

Policy Year	Value	Attained Age Method		Policy Year Method	
		Multiple	Calculation	Multiple	Calculation
1	0.05	1		1	
2	0.07	1		1	
3	-0.05	1		1	
4	0.025	.0975	$1-.025$	1.05	$1-(-.05)$
5	0.075	.855625	$(1-.075)^2$	1.02375	$* (1-.025)$
6	0.10	.729	$(1-.10)^3$	0.946969	$* (1-.075)$
7	0.125	.586182	$(1-.125)^4$	0.852272	$* (1-.10)$

7.3.8 Mortality table adjustments

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

7.3.9 Substandard mortality

The substandard multiple on the model point record applies to the mortality rate after the multiple and addition. This same substandard multiple applies for COI and YRT reinsurance rates.

7.3.10 Joint life mortality

See the Joint life products chapter, below.

7.3.11 Monthly mortality

Deaths occur at the end of each month, prior to other decrements and policyholder transactions. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{1/12}$$

7.4 Lapses and surrenders

A termination is a 'surrender' when there is a cash value, and a 'lapse' otherwise.

Lapse rates apply to units after mortality decrements, which are processed before all other decrements.

7.4.1 Lapse adjustment multiple

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

7.4.2 Policy year lapse rates

The base lapse rate is adjusted by a multiple.

7.4.3 Excess lapses

Lapse rates can be dynamically adjusted according to policyholder behavior related to the competitive position of the base crediting rate as interest rates change. Excess lapses can occur when the base crediting rate is below a specified competitor rate. When the crediting rate is higher than the competitor rate, excess lapses can be negative.

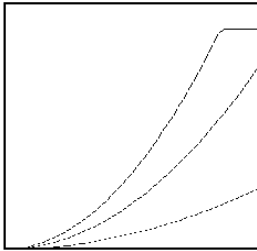
Maximum and minimum lapse rate variables in the excess lapse features affect the excess lapse calculations only. They do not limit the specified lapse rates. For example, the maximum lapse rate variable can reduce the excess lapse rate to zero, but cannot reduce the total lapse rate to be less than the specified lapse rate adjusted by the calendar date multiple.

Excess lapses can be calculated using either an exponential or arctangent formula.

7.4.4 Exponential excess lapses

The exponential formula produces results that look like figure 1, where the x-axis is the difference between the competitor rate and the crediting rate. The highest excess lapses in the graph are for a policy with no surrender charge. The lowest excess lapses represent a policy with a large surrender charge.

Figure 1 Exponential formula



There are two versions of the exponential formula:

Increased lapses:

$$\begin{aligned}
 &\text{When competitor rate} > \text{crediting rate} + \text{threshold} \\
 &\text{difference} = (\text{competitor rate} - \text{crediting rate} - \text{threshold}) \times 100 \\
 &\text{surrender charge ratio} = 1 - \left(\frac{\text{cash value}}{\text{fund value}} \right) \\
 &\text{excess lapse} = \frac{\left[\text{multiple} \times (\text{difference}^{\text{exponent}}) \times (1 - \text{surrender charge multiple} \times \text{SC ratio}) \right]}{100} \\
 &\quad - \text{surrender charge subtraction} \times \text{SC ratio} \\
 &\text{excess lapse} \leq \max(0, \text{maximum lapse rate} - \text{planned lapse rate})
 \end{aligned}$$

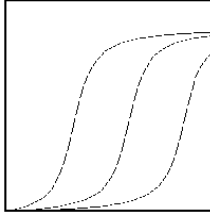
Decreased lapses:

$$\begin{aligned}
 &\text{When crediting rate} > \text{competitor rate} + \text{threshold} \\
 &\text{difference} = (\text{crediting rate} - \text{competitor rate} - \text{threshold}) \times 100 \\
 &\text{lapse reduction} = \frac{\left[\text{multiple} \times (\text{difference}^{\text{exponent}}) \right]}{100} \\
 &\text{lapse reduction} \leq \max(0, \text{planned lapse rate} - \text{minimum lapse rate})
 \end{aligned}$$

7.4.5 Arctangent excess lapses

The arctangent formula produces results that look like figure 2. The highest excess lapses in the graph are for a policy with no surrender charge. The lowest curve has a large surrender charge.

Figure 2 Arctangent formula



difference = competitor rate – crediting rate

$$\text{surrender charge ratio} = 1 - \left(\frac{\text{cash value}}{\text{fund value}} \right)$$

$$\text{excess lapse} = \frac{[(\text{maximum total lapse} - \text{base lapse}) \times (A - B)]}{(1 - B)}$$

total lapse = base lapse + excess lapse,

$\leq$ maximum total lapse, $\geq$ minimum total lapse

$$A = \frac{\arctan[100 \times \text{multiple} \times (\text{difference} - \text{SC ratio} \times \text{SC multiple} - \text{threshold})]}{(\pi/2)}$$

$$B = \frac{\arctan[-100 \times \text{multiple} \times (\text{SC ratio} \times \text{SC multiple} + \text{threshold})]}{(\pi/2)}$$

When competitor rate < crediting rate : SC multiple = 0

The arctangent excess lapse formula shown above is a version of the following formula familiar to some users:

$$\text{total lapse rate} = a + b \times \arctan \left[m \times \left(\frac{\text{competitor rate} - \text{crediting rate} - \text{SC ratio}}{c} \right) - n \right]$$

Base lapse rates = the result of the above formula when the crediting rate equals the competitor rate.

The formulas are identical when the parameters are converted as follows:

$$\begin{aligned}\text{Threshold} &= \frac{n}{m} \\ \text{Multiple} &= \frac{m}{100} \\ \text{SC multiple} &= \frac{1}{c} \\ \text{Maximum lapse rate} &= a + b \times \left(\frac{\pi}{2}\right)\end{aligned}$$

7.4.6 Monthly lapses

The annual lapse rate after all adjustments and excess lapses is constrained to be not more than 1.0 (100%).

Lapses and surrenders occur either: (1) at the end of each month, or (2) modally (just before the next premium is due). Lapses occur monthly after the last premium has been paid.

Each of the above lapse rate adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly or modal lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/\text{event frequency})}$$

Modal lapse timing does not apply to fund-related lapse, which is discussed below.

7.4.7 Skewed lapses

Skewed lapses allocate a specific portion of annual lapses to each month within each policy year. The skew percentages are normalized to assure that they add to 1.0 (100%) for each policy year. Lapse skewing can be specified differently for each policy year.

When skewed lapses apply, the exponent in the monthly/modal lapse rate formula is replaced by the sum of the skew percentages for the months included in the decrement time step. For example, when the lapse applies for months with monthly skew percentages adding up to 0.4 (40%), the lapse rate is:

$$1 - (1 - \text{annual lapse rate})^{0.4}$$

7.4.8 Fund-related lapse

The model point lapses when:

- Either (option choice) the cash value or the fund value becomes zero or negative, and
- The secondary guarantee test condition has been violated. However, lapse does not occur when premiums are adjusted to minimally fund the guarantee.

Fund-related lapse does not occur during the specified number of 'exception years' after issue.

8 RIDERS

8.1 Multiple riders

Multiple riders can apply to a model point.

8.1.1 *Riders input*

Each rider creates an instance of the `ul_rider` submodel. The input for each variable must be linked to an external source containing values varying by the rider names.

8.1.2 *Rider model point input*

The rider model point record identifies which riders apply to each model point.

8.2 Rider types

8.2.1 *Accidental death benefit*

Please refer to the L13 Riders book for a discussion of the accidental death benefit riders which can be linked to universal life model points.

8.2.2 *Waiver of premium*

Please refer to the L13 Riders book for a discussion of the waiver of premium riders which can be linked to universal life model points.

8.2.3 *Waiver of COI and other charges*

Please refer to the L13 Riders book for a discussion of the waiver of COI and other charges riders which can be linked to universal life model points.

9 REINSURANCE

9.1 Multiple reinsurance treaties

Multiple reinsurance treaties can apply to a model point.

9.1.1 *Treaty input*

Each reinsurance treaty creates an instance of the `ul_ireins` submodel. The input for each variable must be linked to an external source containing values varying by the treaty names.

The exception is that the YRT retention maximum per policy applies for all treaties combined, and thus does not vary by treaty. For example, it might vary by plan code.

Each treaty can be either a YRT, coinsurance, or modified coinsurance treaty. Each of these types is discussed in a separate section, below.

Each of these types is discussed in a separate section, below.

9.1.2 *Reinsurance model point input*

The reinsurance model point record identifies which treaties apply to each model point, as well as the ceded percent or amount for each treaty. The reinsurance model point layout is discussed in the Reinsurance model point records layout section, below.

9.2 YRT reinsurance and coinsurance of mortality risk

Yearly renewable term (YRT) reinsurance cedes a portion of the net amount at risk (NAR). Reinsurance premiums are calculated using YRT rates applied to the ceded coverage amount.

Coinurance of mortality risk is similar to YRT reinsurance, but with reinsurance premiums based on the policy's cost of insurance rates.

9.2.1 Ceded amount

The ceded amount is a percent of the net amount at risk (NAR) at the beginning of each month, using the fund value at the end of the previous month. This is the total fund value. When the unloaned fund value is negative, the NAR is calculated using zero unloaned fund value.

Using reinsurance model point variables

The ceded percent can be specified on the reinsurance model point record, either directly or as the initial ceded death benefit amount, which is converted to the ceded percent (i.e. ceded death benefit amount / death benefit at the projection date.) This is proportionate YRT reinsurance, where the ceded percent is fixed.

Using treaty variables

The ceded percent can be calculated using the reinsurance input variables that vary by treaty code. The ceded amount is a specified percent of any excess of the NAR over the retention threshold per policy. For example:

- **First dollar quota share** – The threshold is zero.
- **Excess retention** – The ceded percent applies to the excess of the NAR over the specified threshold.

The ceded percent varies over time, depending on the changing relationship between the NAR and the threshold.

9.2.2 Maximum YRT retention per policy

The YRT reinsurance treaty amounts for a model point are increased when the combined ceded amounts would be less than [the NAR less the maximum retention amount per policy]. Specify the maximum retention amount in the iul submodel; this variable applies for the product instead of the ul_ireins variables that vary by treaty.

This adjustment applies only for YRT treaties that have the ceded amount calculated using the treaty variables. Treaties that have the ceded percent/amount specified on the model point record are not adjusted.

When the sum of the ceded amounts is too low, the initial (at the projection date, or issue date for new business model points) ceded amount for each treaty is increased in proportion to the individual ceded amounts.

9.2.3 YRT rate

The YRT reinsurance rate for each treaty is a base table, times a multiple. These variables typically operate the same as for mortality rates.

The YRT premium can be zero for the first policy year by specify the multiple using a Select and Ultimate table style with zero rates for select duration 0.

Term conversion adjustments

Term products often offer the option to convert to a permanent product (i.e. universal life) prior to the end of a level term period without additional underwriting.

Specify the original term issue date and whether the policy was issued as a result of a term conversion on the model point record. For such policies, the YRT base rate and YRT multiple will be calculated using a point-in-

scale method based on the original issue date of the term policy. Additionally, a separate set of YRT base rates and YRT multiples may apply to the term conversion cohort of policies if specified.

YRT term conversion adjustments may be turned off using the `yrt_term_conversion_defn` variable (on the LifeReinsurance input page).

YRT term conversion adjustments do not apply to joint life policies.

Substandard multiple

The substandard multiple on the model point record applies to the YRT rate. This is the same substandard multiple as applies for experience mortality rates.

9.2.4 Reinsurance premium

YRT reinsurance premiums are paid either annually at the beginning of each policy year, or monthly.

The YRT reinsurance premium per policy is:

$$\text{ceded amount} \times (\text{annual YRT rate}) + \text{reinsurance policy fee}$$

When YRT premiums are paid monthly, one twelfth of this amount is paid at the beginning of each month.

The reinsurer refunds the unearned YRT reinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual YRT premium, times the remaining months in the year divided by 12.

9.2.5 Reinsurance benefits

The reinsurance benefit paid to the ceding company each month is the ceded amount times the monthly experience mortality rate. The ceded amount is the amount calculated at the time of each YRT reinsurance premium.

9.2.6 Reinsurance reserve

The formulaic statutory and tax reserve credit for YRT reinsurance is the tabular cost of insurance floor for the ceded amount.

9.3 Coinsurance

Coinsurance treaties cede a portion of the entire policy, including transferring the ceded reserve to the reinsurer. Reinsurance benefits include all benefits provided by the policy.

9.3.1 Coinsurance percent

When the coinsurance percent is calculated using the input for the treaty, the percent is x% of the sum insured at issue up to the threshold per policy specified for each treaty, plus y% of any excess over the threshold. When the coinsurance percent is specified by model point, this treaty input is ignored. The coinsurance percent remains constant to maturity.

The coinsurance percent applies to the entire policy, including the deficiency reserve and the additional reserve.

The coinsurance benefits paid to the ceding company each month are the projected policy benefits, times the coinsurance percent.

9.3.2 Coinsurance premium

Coinsurance premiums are paid at the same premium frequency as the model point.

Coinsurance premiums are based on the gross premium, including policy fee.

The reinsurer refunds the unearned coinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual amount times the remaining months in the year divided by 12.

9.3.3 Premium tax

The reinsurer reimburses for the coinsurance percent times premium taxes paid by the ceding company.

9.3.4 Benefits

The reinsurer reimburses for the coinsurance percent times all benefits paid by the ceding company. Reimbursements are split between death benefits and other benefits, for reporting purposes.

9.3.5 Expense allowance

The reinsurer pays an expense allowance to the ceding company at the reinsurance premium frequency. The expense allowance is the annualized reinsurance premium times a specified percent.

The unearned current year expense allowance is refunded to the reinsurer when the policy terminates before the next reinsurance premium payment for any reason. The unearned expense allowance is the last amount paid, times the ratio of the number of months until the next reinsurance premium, divided by [12 divided by the reinsurance premium frequency].

9.3.6 Reinsurance reserve

The reinsurance reserve credit for coinsurance treaties is the total statutory reserve, times the coinsurance percent.

9.4 Modified coinsurance

Modified coinsurance (ModCo) treaties operate as described above for coinsurance treaties, except that the reinsurance reserve is retained by the ceding company. The ceding company reimburses the reinsurer for interest on the reinsurance reserve which is backed by assets invested by the ceding company.

The adjustments occur monthly, so that nested projections match standard projection results. The adjustment amount is the current reinsurance reserve, less the reinsurance reserve as of the previous adjustment date accumulated with interest.

The ModCo reserve adjustment interest rate is the asset yield rate.

9.5 Reinsurance administration expenses

Reinsurance administration expenses are assumed to be incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include:

- Amount per policy
- Percent of the ceded amount.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Modified coinsurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for reinsurance.

10 SOLVING CALCULATIONS

Solving calculation results are amounts per unit in force.

10.1 On-the-fly calculations

10.1.1 Values calculated

The following values are calculated as needed during projections. Calculations use the death benefit option and minimum death benefit (corridor) percent which applies for the model point. Partial withdrawals and policy loans are ignored.

Result	Interest	COI	Loads	Premium Mode	Death Benefit
Shadow fund annual required premiums, for AXXX step 1, for each policy year [array shadow_fund_art_prem]	Secondary guarantee interest rate	Secondary guarantee	Secondary guarantee	Annual	Shadow fund
Shadow fund level premium at issue, for PBA NPR, and interest ratcheting threshold (when the input threshold premium is zero default value) [solved_shadow_fund_level_prem]	Secondary guarantee interest rate	Secondary guarantee	Secondary guarantee	Annual	Shadow fund
Shadow fund single premium net of premium load at the projection start, for AXXX step 4 and for PBA NPR [solved_shadow_fund_single_prem_net]	Secondary guarantee interest rate with ratcheting	Secondary guarantee	Secondary guarantee	Annual	Shadow fund
Guaranteed maturity premium (GMP), for UL CRVM, endowing at the sum assured [solved_gmp]	Guaranteed*	Guaranteed	Guaranteed	Modal	Guaranteed fund
Guaranteed level premium, for PBA NPR, with zero endowment [solved_guar_level_prem]	Guaranteed*	Guaranteed	Guaranteed	Annual	Guaranteed fund

*The guaranteed interest rate ignores any initial guarantee of the crediting rate at issue.

10.1.2 Solving iterations

When the closed form solution does not apply (see below), these results are solved by iterating the appropriate instance of the ul_ifund submodel, projecting using successive trial premiums. [solving_driver function]

10.1.3 Displaying solving results

Use the solving_display_defn variable (on the Product input page in the Premiums category) to have the Run Log show the series of solving guesses and related fund value results. The output for the ul_ifund submodel then includes the column values for each solving fund projection using the final solved premium result.

This variable should of course be set to No for production runs, in order to improve calculation time and to reduce the size of the output.

10.2 Closed form solutions

10.2.1 Available formulas

Closed form solution calculations can replace the solving iterations to reduce calculation time significantly. Closed form solutions are available for these results:

- Shadow fund ART premium for each policy year
- Shadow fund level premium at issue – calculated under the assumption that the shadow fund level premium will always be less than the product target premium and the base shadow fund loads and cost of insurance rates apply.
- Cumulative premiums test level premium at issue – always calculated using a closed form solution, since solving iterations are never needed under the assumption that interest ratcheting does not apply when this premium is paid.
- Shadow fund single premium (net of premium load)
- Cumulative premiums test single premium – always calculated using a closed form solution, including the application of interest ratcheting.
- Guaranteed maturity premium (GMP) – calculated under the assumption that the GMP will always exceed the product target premium.
- Guaranteed level premium

10.2.2 Solving methods

The following types of solving methods are used:

- **Accumulation factors** – Calculations for level premium results are based on an approach which creates accumulation factors for the various elements which develop the ending fund value. The closed form solution for the premium is a formula which sums the target fund value with factors for charges and loads and divides by a factor for the premium. These solutions were developed using techniques presented in the paper 'Calculating Funding Premiums For Universal Life Insurance' by Calvin D. Cherry, on pages 20-27 of the North American Actuarial Journal, Volume 4, Number 2.
- **Fund rollback** – The ending target fund value is rolled back each month, first removing credited interest (including ratcheting), then removing charges, then removing loads, then finally adding the premium load to the resulting 'fund value' at the moment before the solving date. This method applies for shadow fund solving that has a single premium paid at the solving date: (1) the shadow fund single premium (net of premium load) at the projection date, and (2) the shadow fund ART premium for each policy year.
- **Cumulative premiums test** – Calculations for cumulative premium test secondary guarantees are straightforward, either using annuity factors or premium accumulation rollback. Thus solving iterations never apply for these calculations.

10.2.3 Using solving versus closed form solution

For results that can be calculated either using solving iterations or a closed form solution, you can choose which calculation method applies. Solving iterations return the exact result, but increase the calculation time. Closed form solutions are significantly faster, but are not appropriate in all cases.

Use the model point variable `closed_form_prem_defn` to control which type of calculation applies:

- **When Appropriate** – Closed form solutions are used as much as possible; see the Validation topic, below. The closed form result is tested against fund value calculations; when this test fails, solving calculations replace the closed form result to assure accuracy for the projection.
- **No** – Solving iterations apply in all cases. For example, this option can be used to compare against closed form solution results.

- **Yes** – Use closed form solutions in all cases. When the closed form result is tested (against fund value calculations), the Run Log turns yellow and contains a warning when the ending fund value does not match the target value within tolerance limits. This choice reduces calculation time when the accuracy provided by solving iterations is not important for the projection. You should evaluate the loss of accuracy by comparing against a test projection using the When Appropriate choice.

10.2.4 Validation for using closed form solution

The When Appropriate choice is validated against the model point's product definition, in order to determine whether each closed form solution can be used. Solving iterations are used when the product has specifications which are likely to produce a closed form result different from the solving result. The following product specifications cause solving iterations to be used for the When Appropriate choice:

- Plus ROP death benefit – applies for all results
- Persistency bonus or refund – applies for all results
- Non-level percent of premium load different between target and excess premiums when premiums are paid more frequently than annually – applies for level premium solving (GMP and guaranteed level premium) only
- Percent of fund value load – applies for level premium solving (GMP, guaranteed level premium, and guaranteed shadow fund level premium) only
- Minimum death benefit percent (corridor factor) is greater than 1.0 at the GMP maturity age – applies for GMP calculation only
- Plus Fund death benefit – applies for GMP calculation only.

For example, when a persistency bonus/refund applies for the shadow fund but is not guaranteed, the GMP can be calculated using the closed form solution, but the shadow fund annual and single premiums must be solved for.

10.2.5 Displaying closed form results

Use the solving_display_defn variable (on the Product input page in the Premiums category) to have the Run Log show the closed form solution results. When the closed form solution does not produce the expected ending fund value, and closed_form_prem_defn is When Appropriate, the resulting solving calculations are also displayed and the final solving iteration values are included in the ul_ifund output results.

This variable should of course be set to No for production runs, in order to improve calculation time and to reduce the size of the output.

10.3 Solving result arrays

Following the solving calculations discussed above, the following array values are calculated for use during the projection:

Result	Frequency	Values
Shadow fund annual required premiums [shadow_fund_art_prem]	Each policy year	Direct solving results
XXX reserve death benefits [dth_ben_tbl_xxx]	Each policy year	Average of monthly death benefits paying SG ART premiums using guaranteed fund

Result	Frequency	Values
Shadow fund crediting addition threshold [shadow_fund_crediting_addn_thresh]	Each policy month	Shadow fund accumulation using input shadow fund threshold premium if specified, or solved shadow fund level premium at issue when this input is the zero default value
Shadow fund single premium net of premium load [shadow_fund_single_prem_net]	Projection months	Shadow fund accumulation using solved shadow fund single premium net of policy loads
Guaranteed maturity fund [guar_maturity_fund]	Projection months	Guaranteed fund accumulation using solved GMP
Guaranteed maturity COI charge [guar_maturity_coi_chg]	Projection months from issue	Guaranteed fund COI charge using solved GMP, used for waiver of COI benefits
CRVM reserve death benefits [dth_ben_tbl_crv]	Each policy year	Average of monthly death benefits paying GMP using guaranteed fund
Guaranteed level premium fund [guar_level_prem_fund]	Projection months	Guaranteed fund accumulation using guaranteed level premium
NPR base reserve death benefits [dth_ben_tbl_npr_base]	Each policy year	Average of monthly death benefits paying guaranteed level premium using guaranteed fund
Guaranteed SG level premium fund [guar_sg_level_prem_fund]	End of each policy year	Guaranteed fund accumulation using guaranteed SG level premium
NPR SG reserve death benefits [dth_ben_tbl_npr_sg]	Each policy year	Average of monthly death benefits paying guaranteed SG level premium using guaranteed fund
Shadow fund using shadow fund level premium at issue [shadow_fund_level_prem]	Projection months	Shadow fund accumulation using shadow fund level premium at issue

10.4 Factors calculations

A projection type calculates 'factors' when the product submodel is the top model and the target column is factors_tgt. This target column directs the calculation of each item specified (discussed below). The target column calculates results for each policy year; thus the Projection period from and to properties have no effect on factors calculations.

The Assumption Set Restore property (in the run parameters object) must be Yes, since this is a seriatim projection. Project variables such as pricing definition do not apply for this special type of projection.

10.4.1 Values calculated

Factors calculations use iterative solving (closed form solutions do not apply) to find the following values for each model point:

Result	Interest	COI	Loads	Premium Mode	Death Benefit
Base fund level premium (to produce specified fund value per unit at target duration)	Specified	Current or guaranteed	Current, guaranteed, or None	Modal	Actual
Shadow fund level premium (zero shadow fund at SG expiration)	Secondary guarantee interest rate	Shadow fund	Shadow fund	Modal	Actual
Cumulative premiums test level premium (accumulates to ending required premiums accumulation)	Secondary guarantee interest rate	N/A	N/A	Modal	N/A
Guaranteed maturity premium (GMP)	Guaranteed*	Guaranteed	Guaranteed	Modal	Actual
Guideline single premium (GSP)	Specified** (6%)	Guaranteed or current	Current or guaranteed	N/A	Sum assured
Guideline level premium (GLP)	Specified** (4%)	Guaranteed or current	Current or guaranteed	Annual	Actual
7-pay MEC premium	Specified** (4%)	Guaranteed or current	None	Annual	Sum assured
Cash value accumulation test (CVAT) net single premiums, and the related minimum death benefit percentages	Guaranteed*	Guaranteed	None	N/A	Sum assured without corridor

*The guaranteed interest rate ignores any initial guarantee of the crediting rate at issue.

**The interest rate for IRS premium restrictions is the greater of the specified interest rate and the guaranteed crediting rate.

The solving for the shadow fund level premium uses the specified target premium, ignoring the floor equal to the required annual premiums. This solving further ignores the shadow fund interest crediting addition threshold when it is specified as a level annual premium for calculating the threshold during the projection.

10.4.2 Policy fee

The policy fee is included in calculating the target premium and secondary guarantee level premium. It is excluded for GMP, GLP, GSP, MEC, and CVAT, except for the charging of the policy fee as a product load.

10.4.3 Model points

Factors calculations systematically process for each model point. Set up a model point for each issue age to be calculated, within each combination of insured characteristics (sex, smoking class, rate class, etc.). For example, you might set up model points for issue ages 0 through 80 for male nonsmokers, then repeating these model points for male smokers, then female nonsmokers, then finally female smokers.

When Annual mode premiums are to be calculated for factors using modal premiums, set the model point premium mode to Annual.

Best practice is to create the report group for each model point so that it identifies the model point characteristics. For example, the report group might be PlanA-M-NS-ST-35 for the model point calculating factors for plan code PlanA, for a male nonsmoker standard risk class issue age 35. Then the report groups identify the meaning of each model point.

The same as for Pricing cell projections, each model point must be issued on the first day of the projection. For example, when the projection date is 2013/12 (end of December, 2013), the issue date must be 2014/01/01 (January 1, 2014).

10.4.4 *ul_py submodel output*

The factors calculation output is contained in the *ul_py* submodel, which has column results by policy year.

10.4.5 *Factors premium reporting*

The solved premium is reported in the column *factor_prem* in the *ul_ifund* submodel. The standard features round this result up to the next whole penny per unit.

The output file includes this value for every model point. Since the premium result shows only for the commencement period ($t = 0$) in this column, the output from and to properties should both be '0'.

10.4.6 *Cash value test reporting*

The solved cash value test net single premium results are reported in the column *factor_cvat_min_dth_ben_factors* in the *ul_ifund* submodel. The standard features convert the result into the corresponding minimum death benefit multiple, equal to the sum assured divided by the fund value at the end of the previous policy year. These fund values are projected assuming the solved single premium is paid at issue.

Since these values vary over time, the output to property must be the number of months to maturity of the youngest issue age. As for the factors premium reporting, use a tool to query the output database in order to retrieve the values for every model point.

10.5 Factors projection setup

10.5.1 *Input variables*

The special variables for factors calculations are at the bottom of the Product input page.

factors_ calc_num	factors_defn	factors_include_defn	factors_fund_val	factors_tgt_dur_defn	factors_tgt_dur	factors_prem_term_defn	factors_prem_term	factors_co_rate_defn	factors_load_rate_defn	factors_int_rate
1	Level Premium	Yes	1,000.00	GMP Maturity Age	0	Model Point Premium Term	0	Current	Current	7.00%
2	Shadow Fund Level Premium	Yes		Secondary Guarantee Years	0	Model Point Premium Term	0	Shadow Fund	Shadow Fund	
3	Cumulative Premium Test Level Premium	Yes								
4	Guaranteed Maturity Premium	Yes	1,000.00	GMP Maturity Age	0	Contract Premium Term	0	Guaranteed	Guaranteed	
5	Guideline Single Premium	Yes	1,000.00	Specified Age	95	Specified Premium Term	1	Current	Current	6.00%
6	Guideline Level Premium	Yes	1,000.00	Specified Age	95	Contract Premium Term	0	Current	Current	4.00%
7	Seven Pay MEC Premium	Yes	1,000.00	Specified Age	95	Specified Premium Term	7	Current Not Applicable		4.00%
8	Cash Value Test NSP	No	1,000.00	Specified Age	95	Specified Premium Term	1	Guaranteed Not Applicable		

For each factors type, specify whether to include these calculations, then specify the variables that control that type:

- *factors_test_num* – The factors calculations progress through the different factors types using this number as the lookup key.
- *factors_defn* – The result to be calculated; this option list variable is used as a condition in formulas.
- *factors_include_defn* – Yes to include this calculation, No to exclude it (these lines can be deleted from the external source input if desired).
- *factors_tgt_fund_val* – Target fund value per unit.

- `factors_tgt_dur_defn` and `factors_tgt_dur` – The target duration is a specified attained age or policy anniversary, maturity age, GMP maturity age, or secondary guarantee premium term.
- `factors_prem_term_defn` and `factors_prem_term` – The premium term is either a specified attained age or policy anniversary, the premium term, or the contractual premium term. When the premium term exceeds the target duration, the solving is of course not affected by premiums after the target fund value duration.
- `factors_coi_rate_defn` – Either the current, guaranteed, or shadow fund COI rates.
- `factors_load_rate_defn` – Either the current, guaranteed, or shadow fund load rates, or using zero loads.
- `factors_int_rate` – The greater of the guaranteed rate and the specified interest rate.

Cumulative premiums test level premium

An exception to the above description is the Cumulative Premium Test Level Premium. This calculation finds the level premium that accumulates to the required premiums accumulation at the end of the secondary guarantee premium term. The only variable that applies is whether this calculation is included. The target duration, premium term, and accumulation interest rates (including the threshold feature) are specified by the cumulative premiums test secondary guarantee definitions. COI rates and loads do not affect this calculation.

10.5.2 Run parameters

Specify the following settings in the run parameters for the projection task calculating factors:

Property	Value
Target Colum	ul->factors_tgt
Projection Period From	0
Projection Period To	0 (since solving calculations are used, this is not a forward projection)
Projection Period Min	0
Projection Period Max	0 for factors premium output; longest months to maturity for cash value test output
Projection Loops	1
Assumption Set Restore	Yes
proj_date	yyyy/mm, the end of the month before the issue date

11 MODEL POINT RECORDS

11.1 Model point record layout

The model point record layout for the universal life product type is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	up to 125 characters (assigns model point to report group)
B	seg_id	segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date_mp	yyyy/mm/dd or mm/dd/yyyy
F	disab_date_mp	yyyy/mm/dd or mm/dd/yyyy (set to 1/1/9999 if policy is not disabled)
G	term_conversion_defn_mp	Yes (calculate term conversion adjustments), No
H	term_issue_date_mp	None (when original term issue date is unavailable) yyyy/mm/dd or mm/dd/yyyy
I	plan_code	up to 20 characters (link to assumptions by plan)
J	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
K	taxt_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
L	issue_age_mp	positive integer
M	gender_mp	Male, Female, Unisex
N	smoking_class_mp	Nonsmoker, Smoker, Aggregate
O	rate_class_mp	Standard, Preferred, Super, [extend with user additions]
P	substd_mult_mp	substandard multiple (1.0 for 100%)
Q	issue_age_mp_life2	positive integer, when applicable, otherwise -1
R	gender_mp_life2	Nonsmoker, Smoker, Aggregate
S	smoking_class_mp_life2	Standard, Preferred, Super, [extend with user additions]
T	rate_class_mp_life2	Active, RPU

Excel Column	Variable	Value
U	substd_mult_mp_life2	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
V	prem_mode	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
W	prem_term_mp	premium years (0 for 'to GMP maturity')
X	init_annualzd_prem	initial annualized premium for mp
Y	single_prem	additional single premium for mp
Z	closed_form_prem_defn	When Appropriate (use solving only when needed), No (force the use of solving calculations), Yes (force the use of closed form solutions)
AA	dth_ben_defn	Sum Assured, Plus Fund, Plus ROP
AB	dth_ben_min_pct_defn_mp	Product Specification (use product defn), Guideline Prem Test, Cash Value Test
AC	gpr_pct_mp	percent of this model point included in Gross Premium Reserve calculations used for reserve financing (1.0 for 100%)
AD	init_sum_assured_mp	initial sum assured for mp (0 = calculate from initial units)
AE	init_policies	initial policies for mp (required)
AF	init_units	initial units for mp (0 = calculate from initial sum assured)
AG	init_curr_fund_val	initial current fund value for mp
AH	init_shadow_fund_val	initial shadow fund value for mp (only for shadow fund test SG)
AI	init_shadow_fund_val_first_yr_prem	Initial upfirst year premium shadow fund value
AJ	init_shadow_fund_val_catch_up	Initial catch up shadow fund value
AK	init_crediting_rate_base	initial base crediting rate (0.05 for 5.0%); when this value is the 0.0 default value the initial rate is calculated using the crediting strategy
AL	init_gross_prem_cumul	initial cumulative premiums for mp

Excel Column	Variable	Value
AM	init_withdrl_cumul	initial cumulative withdrawals for mp
AN	init_sg_prem_accum	initial SG accumulated premiums for mp (only for cumulative premiums test SG)
AO	init_pol_loan	initial policy loan for mp
AP	init_persist_ref_basis	initial persistency refund accumulation for mp
AQ	init_surr_chg_withdrl_reductn_mult	average initial surrender charge reduction multiple from previous withdrawals for mp (1.0 for 100%)
AR	init_sg_mths_in_grace_period	consecutive months that the secondary guarantee conditions have not been met for a model point being kept in force by a positive current fund value
AS	init_comm_subject_to_chargeback	initial commission subject to chargeback for mp (-1 to calculate from issue)
AT	init_sg_funding_indicator	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step

11.2 Reinsurance model point record layout

The reinsurance model point record identifies which treaties apply for each model point, using the linked variable, mp_id. The reinsurance model point record can have any number of reinsurance treaties per model point. It also specifies the ceded percent or amount for each treaty as described in the table below. If the reinsurance model points are omitted from the reinsurance model point file, then the liability model point will not include reinsurance.

Warning! The order of reinsurance model points must be in the same order as the liability model points.

The reinsurance model point record layout is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	mp_id	Up to 12 characters. Links to liability data file mp_id. This value can be listed any number of times to indicate the number of treaties applied to the policy.
B	treaty_code	A user defined input that references the treaty code in the Resinsurance.xlsm workbook.

Excel Column	Variable	Value
C	ceded_amt_defn	Percent - the ceded amount should be input as a percent from 0 to 1 Dollar Amount - the ceded amount should be input as the ceded death benefit amount Treaty Input - the ceded amount should be input as 0, and the ceded amount will be calculated using the treaty variables
D	ceded_amt	See above. Input the ceded percent, the ceded dollar amount, or 0 if the ceded amount is specified by the treaty.

11.3 Units basis

Product values are calculated per unit in force, then multiplied by the current number of units. For example, the death benefit, cash value, and reserves per unit.

11.3.1 Initial units

The initial (at the projection date) policies must be specified on each model point record, along with either or both of the initial units and initial sum assured. These are the aggregate values for the policies collected into each model point.

- When the initial units are not specified on the model point (this variable is left at its zero default value), the initial units are the initial sum assured divided by the specified sum assured per unit.
- When the initial sum assured is not specified on the model point (this variable is left at its zero default value), the initial units variable applies directly.
- When both the initial units and initial sum assured are specified on the model point record, the initial units variable applies. The Run log contains a warning if the initial sum assured is more than [\$1 times the initial units] different from the initial units times the specified sum assured per unit.
- When both the initial units and initial sum assured are zero, the model point is skipped and a warning is issued in the Run log.

The initial policies must be specified for each model point. The average size policy is the initial units divided by the initial policies, and the initial sum assured divided by the initial policies.

11.3.2 Pricing cells

For model points issued at the start of a seriatim pricing projection, the initial policies, units, and sum assured fields define the policy size. When the model point specifies one policy, the initial units and sum assured are average size amounts.

11.4 Required model point variables

11.4.1 Always required

The following model point variables are required in all cases:

- **Model point identification:** group and mp_id
- **Insured characteristics:** issue_age, gender, smoking_class, rate_class, and substd_mult, for life 1 in all cases, and also for life 2 for joint life products

- **Policy specifications:** issue_date_mp, term_conversion_defn_mp, term_issue_date_mp, plan_code, sat_valn_code, tax_valn_code, prem_mode, prem_term_mp (may be default of zero which indicates that premiums are paid in all years), dth_ben_defn, and dth_ben_min_pct_defn_mp (may be default of Product Specifications)
- **Initial policies:** init_policies
- **Units basis fields:** one or both of init_sum_assured and init_units
- **Closed form solution usage:** closed_form_prem_defn
- **Annual premium:** init_annualized_prem is required when the annual premium is specified by model point.
- **Single premium:** single_prem is required for new business model points. It is required for in-force model points in these conditions: (1) the model point is still in the first year and the premium load varies between target and excess premium, or (2) the commission subject to chargeback is being calculated from issue.

11.4.2 In-force model point values

In-force model points are driven by all the 'initial values' variables, which specify the aggregate model point values at the projection date.

New business model points (model points with issue date after the projection date), however, ignore these variables. The exception is that the initial policies must be specified, along with either or both of the initial units and initial sum assured.

The following table shows when each in-force values variable is required for in-force model points:

Variable	Required When
init_curr_fund_val	always required
init_sg_fund_val	products with shadow fund secondary guarantee
init_crediting_rate_base	always required
init_gross_prem_cumul	model points with current or future policy loan, for products with the preferred policy loan feature
init_withdrl_cumul	
init_sg_prem_accum	products with cumulative premiums test secondary guarantee
init_pol_loan	model points with policy loan
init_persist_ref_basis	products with persistency refund
init_surr_chg_withdrl_reductn_mult	model points with previous withdrawals, for products with surrender charges other than percent of fund value (may be left at 1 default to ignore this initial effect)
init_sg_mths_in_grace_period	model points with positive current fund value but unsatisfied secondary guarantee conditions
init_comm_subject_to_chargeback	model points still subject to commission chargeback (may be left at -1 default to calculate from issue)

11.4.3 Variables to ignore

The following model point variables are needed only for nested projections, and should be ignored. These values must be carried forward to the next time step in order to reproduce the results from standard projections:

Variable	Value
init_sg_funding_indicator	use zero default value
init_surr_chg_withdrl_reductn_mult	Note above that this field may be left at 1 default to ignore this initial effect
init_sg_mths_in_grace_period	This field can be ignored when this information is not readily available; the primary purpose is to provide continuity for nested projections
init_comm_subject_to_chargeback	This field can be ignored when this information is not readily available; the primary purpose is to provide continuity for nested projections

12 SUBMODELS STRUCTURE

12.1 ul submodel

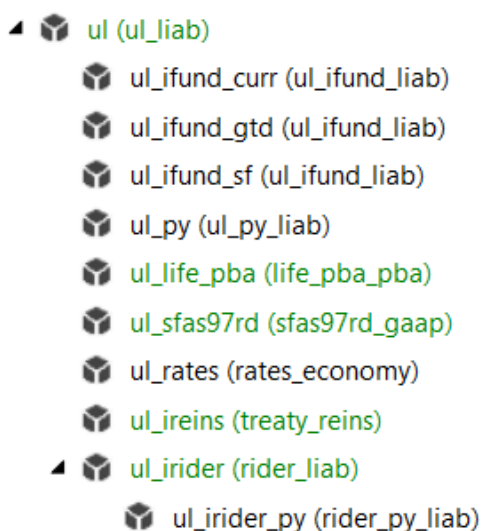
The ul submodel calculates the model point's contribution to the liability income statement and balance sheet items for universal life products.

The ul submodel calculates model point results using values per unit in force, which are then multiplied by the number of units. Results per unit in force have '_per_unit' added to the formula name. Formula names without this addition are results per model point (or are a special result like a ratio).

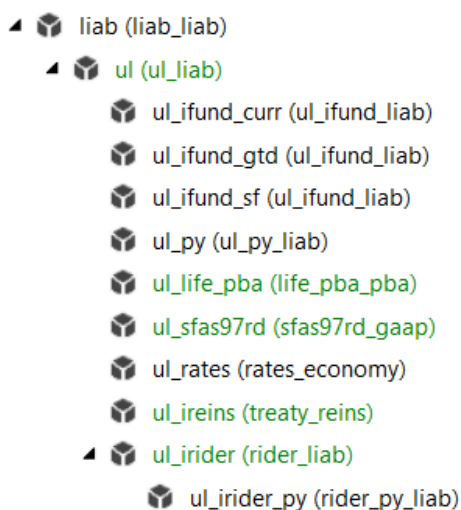
12.1.1 Seriatim versus portfolio runs

Seriatim projection

When ul is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the ul submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings.



For portfolio projections (which include business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the ul submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the ul submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the ul submodel the same columns as for other product types. This makes the universal life product type interchangeable with other product types fitting into this business segment structure.

12.1.2 Variables

Variables for universal life products are located in the ul submodel, except as noted below for the variables in ul_ifund, and ul_py.

12.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when ul is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the ul submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only ul is the top model, (2) when ul is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when ul is run as the top model and when ul is included in portfolio projections. References to company, segment and liab are invalid when ul is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) solving values are initialized, (3) temporary tables are sized for DAC tax calculations and competitor rates, (4) projection flags are created and initialized, (5) reporting groups are set up, (6) ul_ifund_curr, ul_ifund_gtd, and ul_ifund_sf elements are setup, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each projection task loop, immediately after startup when ul is the top model. When company is the top model, initialize is executed by calls from seg->initialize which executes ul->initialize for each model point in the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when ul is running as the top model. This includes executing the initialization of economic scenario rates and the setup of competitor rates. Other initializations are required in both cases.

The main use of initialize is to orchestrate the various solving calculations for each model point, and to precalculate the per unit reserve amounts using the actuarial math library. Since these initializations are not required for all calculation types for all product features, conditions are checked to shortcut the initialization process, thus reducing the runtime of the model point.

The solving calculations, and the filling of the associated array values, are discussed in the Solving calculations chapter, above.

The reserves are initialized in the following order:

1. Base statutory reserves
2. Base statutory minimum reserves
3. XXX statutory reserves
4. XXX statutory minimum reserves
5. Base tax reserves
6. XXX tax reserves.

solving_driver

The 'solving_driver' function is used to iterate the ul_ifund submodel to solve for results when the closed form solution is not used, or produces an ending fund value that does not match the target value within the tolerance limit. See the Solving calculations chapter, above.

Each call to this solving_driver function includes the appropriate fund_accum_defn for ul_ifund. See the ul_ifund submodel section, below.

Following some initial setup, a 'while' loop is processed. Within each pass of the loop, a trial premium is used to project `ul_ifund`, then the projected fund value is compared to the solving target. A new trial premium is used to re-project `ul_ifund` until the solution has been determined within a tolerance limit.

Details of the solving process and results from the final solve can be captured when `solving_display_defn` (on the Product input page, under the Premiums category) is All Solves or GMF Only. The solve guesses are written to the Run Log, and the final solving fund accumulation is written to the column output using a `writeModel` statement. This option should be No for production runs.

calc_prem

The function 'calc_prem' calculates the closed form solutions for premium results using the Accumulation factors or Fund rollback methods. See the Solving calculations chapter, above. Each call to this `calc_prem` function includes the `fund_accum_defn` which would have applied for solving for the result using iterations of `ul_ifund`.

The closed form results are written to the Run Log when `solving_display_defn` (on the Product input page, under the Premiums category) is All Solves or GMF Only. This option should be No for production runs.

stat_res_ulcrvm_per_unit

Columns 'stat_res_ulcrvm_per_unit' (and the corresponding column for tax reserves) are special, in the sense that they include a future projection of the `guaranteed_fund` instance of `ul_ifund`. This future projection requires that these columns reset and recalculate the `guaranteed_fund` instance of `ul_ifund`. When UL CRVM reserves require the GMF to be recalculated, this instance of `ul_ifund` is recalculated beginning from the current fund value, using guaranteed rates and assuming the GMP is paid in the future.

PBA NPR calculation columns

The column `stat_res_per_unit` orchestrates the calculation timing of columns for PBA NPR calculations. At each exact calculation period, the `guaranteed_fund` instance of the `ul_ifund` submodel is rebased to start from the current fund value. These `guaranteed_fund` values (paying the `guaranteed_level_premium`) are used to calculate the present value of benefits less premiums as of the valuation date. Then that column [`stat_res_npr_base_pv_per_unit`] is called to roll this present value forward for each month until the next exact calculation period.

Similarly at each exact calculation period, the reserve lapse rate is calculated in column `stat_res_npr_sg_lapse_rate`, and is stored in variable `stat_res_npr_sg_lapse_rate_saved`. Using the stored values, the net single premium is calculated as of the valuation date. This calculation also includes rebasing the `ul_ifund` submodel to start from the current fund value, then projecting forward using the revised `guaranteed SG level premium per unit`. Then that column [`stat_res_npr_nsp_per_unit`] is called to roll this present value forward for each month until the next exact calculation period.

The final NPR reserve calculations are performed each month in columns `stat_res_npr_base_per_unit` and `stat_res_npr_sg_per_unit`. These columns use the present value results which have been either calculated using the exact method, or rolled forward between valuation dates.

Other formulas

Other formulas in the `ul` submodel are relatively straightforward. Results which are per unit in force add '_per_unit' to the end of the formula name. Results without '_per_unit' are values for the model point.

Most of the profit, balance sheet, and GAAP columns are only calculated when `ul` is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

12.2 ul_ifund_xxx submodels

The ul_ifund_xxx submodels are three instances of the ul_ifund_liab model class. They contain the variables and formulas required to calculate per unit in force individual fund values for universal life model points. Since all results are per unit in force, this submodel does not include ‘_per_unit’ in the formula names.

12.2.1 Multiple instances

Each model point carries three unique instances of the ul_ifund_liab model class:

- ul_ifund_curr, for the current fund value, which is always present.
- ul_ifund_gtd, for the guaranteed fund, which is projected for solving calculations and for creating related array values (see the Solving calculations chapter, above), and when calculating present values for UL CRVM reserves and for PBA NPR base and secondary guarantee reserves.
- ul_ifund_sf, for the shadow fund value, which is also projected for solving calculations. This instance is not created when the product uses the cumulative premiums test, or has no secondary guarantee.

These unique elements of ul_ifund are set up during the startup processing of the model point.

Formula references to the unique instances utilize the unique model reference pointer. References for each fund value basis follow this example:

- Current: sm_ul_ifund_curr->fund_val_aft_withdrl
- Guaranteed: sm_ul_ifund_gtd->fund_val_aft_withdrl
- Shadow fund: sm_ul_ifund_sf->fund_val_aft_withdrl

12.2.2 Working submodel

A specific ul_ifund instance may be a ‘working’ submodel which is reset at various times as required for solving calculations, and for reserve projections starting from the current fund value. Accordingly the values in these columns at the end of a projection apply only for the last time this submodel was calculated. For example, the ul_ifund_gtd values are not the guaranteed fund values calculated from issue; they could be the calculations starting from the current fund value at the last time when UL CRVM reserve calculations were processed.

To see a particular set of results, customize to add code to write out the submodel results at the time that calculation has just been performed. Search for the ‘writeModel’ command in the ul->initialize formula to see how this is done when the solving_display_defn calls for writing out the submodel values.

12.2.3 Variables

Variables for COI rates, product loads, and persistency bonuses and refunds are located in the ul_ifund submodel. These variables are contained in the ul_ifund submodel so that they can vary between current, guaranteed, and shadow fund calculations. See the Fund variables section in the Input organization chapter, above.

In addition to the variables discussed above, two variables in ul_ifund are hidden from the data entry but are critically important to the successful operation of the ul_ifund elements:

fund_val_basis

The ‘fund_val_basis’ variable is an option choice variable with choices of Current, Guaranteed, and Shadow Fund. This variable is used as a condition in formulas, most commonly for determining which input rates to use. After each ul_ifund instance is setup inside ‘startup’ of ul, this variable of the ul_ifund instance is set.

fund_accum_defn

A second ul_ifund variable, ‘fund_accum_defn’ is used as a condition in formulas, most commonly for determining which type of calculation is currently being iterated. This is also a choice list variable with choices which provide for execution of specific formulas for each fund accumulation definition type:

- **Current Fund** – Formulas for the current fund value.
- **Guaranteed Fund** – Formulas for calculating: (1) average death benefit for XXX reserves assuming the secondary guarantee annual required premiums are paid (when `dth_ben_tbl_xxx_flag`), (2) guaranteed maturity premium and fund value (when `pba_npr_indicator` is 0), (3) average death benefit for CRVM reserves assuming the GMP is paid (when `pba_npr_indicator` is 0), (4) guaranteed level premium and fund value (when `pba_npr_indicator` is 1), (5) average death benefit for NPR base reserves assuming the guaranteed level premium is paid (when `pba_npr_indicator` is 1), (6) guaranteed SG level premium fund value (when `pba_npr_indicator` is 2), and (7) average death benefit for NPR SG reserves assuming the guaranteed SG level premium is paid (when `pba_npr_indicator` is 2).
- **Shadow Fund** – Formulas for the shadow fund value.
- **Shadow Fund ART Premium** – Formulas projecting the shadow fund value during the solving process for the shadow fund required premium for each policy year.
- **Shadow Fund Level Premium** – Formulas projecting the shadow fund value paying the level premium that results in zero ending shadow fund value. These calculations apply at issue, to calculate the shadow fund level premium at issue. They apply again at each NPR valuation date, using shadow fund values starting from the current shadow fund at the valuation date.
- **Shadow Fund Threshold** – Formulas projecting the shadow fund value during the development of the shadow fund interest ratcheting threshold per unit for each month, assuming that the specified secondary guarantee threshold premium per unit (plus the policy fee) is paid at the beginning of each policy year. When this input variable is left at its zero default value, the threshold amounts are calculated using the solved shadow fund level premium.
- **Shadow Fund Single Premium** – Formulas projecting the shadow fund value during the solving process for the shadow fund single premium, and the subsequent rolling forward to calculate this result for each projection month.

12.2.4 Formulas

Formulas for projecting universal life model point fund values are contained in the `ul_ifund` submodel. A combination of columns, scalars, and external functions are used to provide the required functionality.

Startup

The startup column of each `ul_ifund` instance is called during the setup of the model.

The `guaranteed_fund` and `shadow_fund` instances of the `ul_ifund` submodel can be reset and recalculated multiple times during the projection. This is controlled by `ul` columns like `initialize`, `stat_res_ulcrvm_per_unit`, `stat_res_per_unit` (which calls `sg_level_prem_per_unit`, `stat_res_npr_base_pv_per_unit`, and `stat_res_npr_sg_per_unit`), and by the solving driver function, where `ul_ifund` is reset or rebased, then re-projected. As discussed above, the `ul_ifund` formulas include variations for different values of `fund_val_basis` and `fund_accum_defn`.

12.3 ul_irider submodel

The `ul_irider` submodel is a dependent data driven array instance of the `rider_liab` model class containing the variables and formulas required to calculate the rider cash flows and reserves for universal life model points.

Please refer to the L13 Riders book.

12.4 ul_ireins submodel

The `ul_ireins` submodel is a dependent data driven array of the `treaty_reins` model class containing the variables and formulas required to calculate the ceded amount and reinsurance cash flows for universal life model points.

12.4.1 *Multiple instances*

Each model point carries multiple instances of `ul_ireins`. The number of instances is determined by the number of treaties specified on the reinsurance model point record.

Elements of `ul_ireins` are created during the startup processing of the model point. Conditions are evaluated in startup of `ul` and the `ul_ireins` user defined array is resized as required. The characteristic for each `ul_ireins` element is specified by the treaty code.

The ceded amount and reinsurance cash flows for multiple treaties are aggregated using the `sum_over_reinsurance` function in the `ul` submodel. This function can aggregate the values for all treaties, or just the treaties of a specific type of reinsurance, such as YRT.

12.4.2 *Variables*

Variables for the reinsurance type, the reinsurance expenses, and for calculating the ceded amount (when it is not specified on the reinsurance model point record), are located in the `ul_ireins` submodel. These variables are contained in the `ul_ireins` submodel so that they vary by treaty.

12.4.3 *Formulas*

Formulas for calculating the ceded amount and the reinsurance cash flows are contained in the `iul_ireins` submodel.

Since `ul_ireins` is defined through the model parameters, the size of the `ul_ireins` array is based on the number of treaties specified for each model point on the reinsurance model point file.

12.5 `ul_py` submodel

The `ul_py` submodel is a standalone instance of the `ul_py_liab` model class. It contains the formulas and variables required to calculate per unit in force results which only vary by policy year. Thus 't' in this submodel represents policy year, unlike the `ul` submodel where 't' represents months since the projection date. The policy year values are calculated from issue.

Since all results are per unit, this submodel does not include '`_per_unit`' in the formula names.

12.5.1 *Variables*

Variables located in the `ul_py` submodel are only used in formulas within this submodel. These are generally for values calculated from issue and varying by policy year, such as annual premiums, and cash value and reserve terminal values and net premiums.

12.5.2 *Formulas*

Formulas for calculating policy year per unit in-force results are included in the `ul_py` submodel. Formulas for `ul_py` are used to calculate annual and target premiums, plus reserve values by policy year.

The `ul_py` submodel also includes formulas which Frasierize the experience mortality and YRT reinsurance rates, which are Frasierized using annual rates. COI rates, however, are Frasierized in the `ul_ifund` submodel since these rates vary between the current, guaranteed, and shadow fund rates bases.

Columns are generally calculated beginning at policy year 1, even for in force model points. As discussed in the `ul` submodel section, the reserve terminal values and net premiums are calculated by calling the Actuarial math library. See the Actuarial math library section below.

12.6 `ul_rates` submodel

The `ul_rates` submodel is identical to the rates submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

12.7 ul_life_pba submodel

The ul_life_pba submodel is identical to the life_pba submodel which applies for portfolio projections. This submodel is used for principle-based approach (PBA) capital and reserve calculations, and for gross premium reserve (GPR) calculations.

Please refer to the C05 PBA book.

12.8 ul_sfas97rd submodel

The ul_sfas97rd submodel is identical to the sfas97rd submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

12.9 Actuarial math library

12.9.1 AML DLL

The Actuarial math library (AML) provides standard actuarial calculations for traditional reserve methods. See the Actuarial math library liability book.

The ul submodel calls the AML to calculate reserve amounts per unit in force. Columns in the ul_py submodel store the terminal value and net premium for each policy year for each reserve basis used in the projection.

12.9.2 ul->startup

The ul->startup column specifies the name extension for the audit trail comma separated values file (when requested).

12.9.3 ul->initialize

The ul->initialize column initiates the call to the AML calculations for each set of reserve results. Each set of AML results is calculated by calling the aml_driver in the ul_py submodel. A separate call can be made for each of the following results:

- Statutory base reserves – Base plan CRVM statutory reserves
- Statutory minimum reserves – Base plan alternative minimum CRVM reserves for calculating deficiency reserves
- Statutory XXX reserves – Base plan Regulation XXX reserves
- Statutory minimum XXX reserves – Base plan Regulation XXX minimum reserves for calculating XXX deficiency reserves
- Tax base reserves – Base plan CRVM tax reserves
- Tax XXX reserves – Base plan Regulation XXX tax reserves
- Base NPR reserves – Base NPR reserves
- SG NPR reserves – Secondary guarantee NPR reserves

12.9.4 AML driver

The aml_driver in the ul_py submodel sets up and calls the AML for each set of reserve results. This driver is called by the ul->initialize column for each set of results.

This driver sets up the call to the AML using the appropriate input for each type of calculations. Each call to the driver calls these functions in the ul_py submodel:

- setup_aml_initial_data – Creates the interfaces for using the AML.

- `setup_aml_res_defn` – Specifies the expense allowance and segmentation method (1980 CRVM, Regulation XXX), and the function type (curtate, continuous, or semi-continuous).
- `setup_aml_policy_data` – Specifies values for each policy year, such as the reserve premiums, death benefits, valuation interest rates, mortality rates, plus the issue age, maturity age, premium term years, and expense allowance grading years.

The operation of these setup functions varies by the type of calculations being made. For example, the policy data passes the statutory rates when calculating statutory reserves, and the tax rates when calculating tax reserves.

12.9.5 Calculation results

The AML calculation results are stored for each policy year in columns in the `ul_py` submodel. The formula for each column simply gets the AML calculation results for each case. These columns include:

- Terminal reserves and net premiums for each basis
- Valuation interest and mortality rate for each basis
- Tabular cost of insurance for each basis

The `aml_driver` calls the appropriate columns after it calls the AML. For example, the statutory basis columns are called when the driver has just calculated statutory results. Then the driver calls the tax reserve basis columns when those results have just been calculated by the AML. Thus each of these columns stores the AML results using the appropriate valuation basis.

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L04 VARIABLE UNIVERSAL LIFE

This book discusses variable universal life products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

- Product – Benefits, premiums, secondary guarantee, surrender charge, policy loan definitions, and factor calculations definitions
- Fund – Crediting rate, and current, guaranteed, and AG 37 persistency bonus/refund, cost of insurance, and loads
- Valuation – Statutory and tax reserve definitions
- Expense – Commission and expense rates, premium tax rate
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse rates, withdrawal use, dynamic premium payments, and policy loan use

Each of these input page types begins with 'Vul', for use in projections that combine multiple product types.

Additionally, variables defining reinsurance and rider specifications and assumptions are associated with input pages which are common across products. These are:

- Reinsurance – Reinsurance treaty definitions and rates.
- Riders – Riders specifications and assumptions
- RidersValuation – Riders statutory and tax reserve definitions

1.2 Data pages

Variables for each variable universal life model point are specified on the VulData page. These variables are discussed in the Model point records chapter, below.

Variables for each reinsurance record are specified on the VulReinsData page. These variables are the same for the term life, traditional life, universal life, variable universal life, indexed universal life, and long term care product types. These variables are discussed in the Reinsurance model point records chapter, below.

Variable for each rider record are specified on the VulRidersData page. These variables are the same for term life, traditional life, universal life, variable universal life, and indexed universal life products. These variables are discussed in the L13 Riders book.

2 PRODUCT BENEFITS AND VALUES

2.1 Benefit term

2.1.1 Maturity age

The benefit term is a maximum attained age. When the specified benefit term is zero (default value), the benefit term is age 121.

2.1.2 **GMP maturity age**

The guaranteed maturity premium (GMP) for UL CRVM reserve calculations is paid to the statutory reserve base mortality table omega age, if earlier than the maturity age. When this table is not specified, the GMP is assumed to be paid to the maturity age.

2.1.3 **IRS restrictions maturity age**

IRS restriction premiums are calculated using a separate maturity attained age, such as 95 or 100, which is specified in the `factors_tgt_age` variable for the Factors calculation type. IRS restrictions are not calculated during projections.

This maturity age applies for factors calculations for: (1) guideline single premium (GSP), (2) guideline level premium (GLP), (3) modified endowment contract (MEC) premium, and (4) cash value accumulation test (CVAT) net single premiums.

2.2 **Extended maturity option**

2.2.1 **When formulaic reserves apply**

When the policy is still in force at the statutory reserve mortality table omega age, the extended maturity option (EMO) continues the policy until the maturity age. For example, products valued using the 1980 CSO mortality table continue in force under the EMO feature after attained age 100 when the maturity age is higher, such as 120.

Specify the death benefit by product:

- The fund value, but not less than zero
- The sum assured, but not less than the fund value.

There are no further premiums, loads, or charges during the EMP period. The fund value continues to increase with credited interest. The formulaic reserve is the current death benefit.

2.2.2 **When PBA reserves apply**

The EMO feature is ignored when PBA reserves are being calculated. These model points have reserves calculated using a modern mortality table, which extends to attained age 121 or later.

2.3 **Death benefits**

2.3.1 **Basic death benefit**

Specify the death benefit option on the model point record:

- Sum assured
- Sum assured plus the fund value, treating negative fund values as zero.
- Sum assured plus return of premiums (ROP) equal to cumulative premiums less withdrawals since issue. For in-force model points, specify on the model point record the initial (at the projection date) cumulative premiums, and cumulative withdrawals, for the policies collected into the model point.

Specify the sum assured per unit for the product. When the initial sum assured is specified on the model point record, this sum assured specified for the product specifies the pattern of changes in the sum assured between policy years. See the Units basis section in the Model point records chapter, below.

2.3.2 **Minimum death benefit**

The death benefit is not less than a minimum percent of the fund value, sometimes called the corridor factor. Specify the minimum death benefit option either by product or by model point:

- For the Guideline premium test, the percentages are prescribed as 250% through attained age 40, then grading to 105% for attained ages 75 through 90, then grading to 100% at attained age 95. The guideline test percentages are the default value for this input variable. You can customize to use your own table instead of these default values. (For example, some companies continue the 105% through age 95, then grading to 100% at age 100. Some companies continue the 101% from ages 94 through 100.)
- For the Cash value accumulation test (CVAT), also called the Cash value net single premium test, specify percentages varying by attained age and typically also by sex and smoking class. The Factors calculation type can be used to calculate the percentages; see the Solving calculations chapter, below.

2.3.3 *Death benefit paid versus in force*

Death benefits paid at the end of each month are based on the fund value before policyholder transactions are processed. Thus the death benefit is calculated using the fund value before withdrawals.

Death benefits in force at the end of the month are based on the fund value at the end of the month, after withdrawals have been taken.

2.4 Endowment benefits

The remaining fund value is paid to policyholders reaching the maturity age.

2.5 Withdrawal benefits

2.5.1 *Partial withdrawal provisions*

Policyholders can withdraw cash from the fund value without surrendering the policy. This is sometimes referred to as a partial surrender, or cash withdrawal provision.

Partial withdrawals are processed at the end of each month, for policies persisting into the next month. Thus death and surrender benefits paid are based on the fund value before withdrawals, while death benefits and cash values in force at the end of the month are based on fund values after withdrawals.

Partial withdrawals apply to all policies within a model point equally.

Surrender charges

The percent of fund surrender charge applies for withdrawals. The surrender charge rates can vary annually (using the `pol_yr` column) or monthly (using the `mths_since_issue` column).

Surrender charges (other than the percent of fund value charge) are reduced by the proportion of the charges which have already been collected during partial withdrawals. Specify on the model point record initial (at the projection date) multiple which applies to these surrender charge rates.

The projection ignores partial withdrawal fees, assuming that they approximately cover administration expenses, and are not a significant element of profits.

Sum assured

Partial withdrawals do not increase the net amount at risk:

- When the death benefit option is the sum assured, the sum assured decreases by the withdrawal amount plus the related surrender charge.
- This sum assured reduction does not apply when the death benefit option is the sum assured plus the fund value or plus the return of premiums, since the death benefit automatically decreases for the withdrawal.

2.5.2 *Partial withdrawal use*

The partial withdrawal amount is an annual percent of the total fund value, including the DCA account. One twelfth of the percent applies at the end of each month. Thus the withdrawal amount is different each month, as the withdrawal basis changes. The total partial withdrawal is capped to not exceed the unloaned cash value.

Dynamic partial withdrawals (related to the competitiveness of the variable account growth rates) are assumed to be included in excess lapses.

2.5.3 *Cumulative withdrawals*

For in-force model points which use the Plus ROP death benefit option or have a preferred policy loan, specify on the model point record the initial (at the projection date) cumulative withdrawals for the policies collected into the model point.

2.6 **Premiums**

2.6.1 *Premium term*

Specify the premium term on the model point record. This is the policy anniversary when premium payments stop. When this value is zero, premiums are paid to the GMP maturity age.

2.6.2 *Contractual premium term*

Specify the contractual premium term for the product, either as an attained age or policy anniversary (duration). When the contractual premium term is zero (default value), the contractual premium term is age 121 or the maturity age, whichever comes first.

The contractual premium term applies only for calculating: (1) GMP for UL CRVM reserves, (2) IRS restriction Guideline Level Premium (GLP). It has no effect on projected premium payments.

2.6.3 *Premium mode*

Specify the premium mode on the model point record. Premiums can be paid annually, semi-annually, quarterly, monthly, or special monthly (such as electronic transfer or pre-authorized check, with expense rate different from the monthly mode).

2.6.4 *Annual premium*

Specify the annual premium either as:

- **Annual premium per unit** – The Annual premium per unit specified for the product, plus the policy fee. Premium suspensions apply from issue.
- **Initial annualized premium** – The premium on the model point record for the policies collected into the model point, including the policy fee. Premium suspensions apply beginning at the next policy anniversary, since this is the actual premium at the projection date.

2.6.5 *Policy fee*

The policy fee can be deducted as a load from each premium payment. The deduction from each premium payment is the annual policy fee divided by the number of premium payments per year.

Percent of premium commissions can include or exclude the policy fee.

The premium suspension multiple applies to the entire annual premium, including the addition for the policy fee. However, the policy fee itself is not reduced for its use in: (1) fund value load, (2) target premium, and (3) required premium for the GMDB test.

2.6.6 *Modal loading*

Modal loadings do not apply for flexible premium products, since the fund value simply accrues less interest when premiums are paid later than at the beginning of each year. Of course, you can vary the annual premium per unit and policy fee by premium mode by using the model point variable `prem_mode` as a lookup key for linking to the external source.

However a monthly modal loading multiple applies for the guaranteed minimum death benefit. See the Guaranteed minimum death benefit chapter, below.

2.6.7 *Additional single premium*

An additional single premium can apply at issue, sometimes called a 'dump in single premium'. Specify on the model point record the total single premium for the policies collected into the model point. This affects new business model points only, since in-force model points start from initial values rather than calculating from issue.

2.6.8 *Single premium policies*

For a single premium plan, define the premium either as:

- Zero annual premium, defining the premium as the additional single premium. Since the annual premium is zero, the premium term has no effect.
- Premium term of 1 year, with Annual premium mode, and defining the premium as the annual premium.

2.6.9 *Premium suspensions*

Premiums are reduced by a premium suspension percent at the beginning of each policy year. The specified annual premium is multiplied by the cumulative premium suspension multiple. The suspension rate specified for the first policy year is ignored.

The reduced premium is assumed to be paid the same for all policies.

The premium suspension multiple applies to the entire annual premium, including the addition for the policy fee. However, the policy fee itself is not reduced for its use in: (1) fund value load, (2) target premium, and (3) required premium for the secondary guarantee cumulative premiums test.

Premium suspensions start date

When the model point record specifies the initial annualized premium, premium suspensions apply only for future policy anniversaries, i.e., after the projection date.

When the premium is specified as an annual premium per unit, however, premium suspensions are calculated from issue.

2.6.10 *Dynamic premiums*

Premium payments can change when the GMDB is fully or minimally funded. See the Guaranteed minimum death benefit chapter, below.

2.6.11 *Cumulative premiums*

For in-force model points which use the Plus ROP death benefit option or have a preferred policy loan, specify on the model point record the initial (at the projection date) cumulative premiums for the policies collected into the model point.

2.6.12 *Target premium*

The target premium can be used to define values as percentages of target and excess premiums, such as for: (1) product loads for the current fund value, (2) surrender charges, and (3) commissions.

The policy fee is added to the target premium, except for the purpose of calculating commissions when the policy fee is not commissionable. The policy fee added to the target premium is not reduced by premium suspensions.

2.6.13 DAC tax capitalized premium percent

Specify (in the Product input page, under the Premiums category) the percent of each premium which is capitalized for the capitalized premium tax, which is also called the DAC tax. See the Taxes chapter in the Profit calculations book.

2.6.14 IRS premium restrictions

Tax regulations include 'definition of life insurance' premium restrictions. Projections ignore these rules; it is up to the user to specify assumed premiums consistent with the rules.

The Factors calculation type can calculate these IRS restriction premium rates: (1) guideline single premium (GSP), (2) guideline level premium (GLP), (3) modified endowment contract (MEC) premium, and (4) cash value accumulation test (CVAT) net single premiums. See the Solving calculations chapter, below.

2.7 Product loads

Loads are calculated in aggregate (such as per unit in force), then deducted on a pro-rata basis from the fixed and variable fund values. See the Fund values chapter below.

2.7.1 Input variables

Input variables for product loads differ between current, guaranteed, and actuarial guideline 37 fund values.

The Fund input page includes a separate set of input variables for current, guaranteed, and actuarial guideline 37 loads. For example, input variable `load_per_pol_curr` specifies the current fund per policy load, `load_per_pol_gtd` specifies the guaranteed per policy load, and `load_per_pol_ag37` specifies the actuarial guideline 37 per policy load. The formulas, however, use the same code variable, `load_per_pol`, in each case, but each input variable is associated with this code variable in the appropriate instance of the submodel.

2.7.2 Premium loads

The following loads are deducted from each premium payment:

- Percent of premium (including the additional single premium) – The target premium percent applies to the premiums paid during the policy year until the premiums exceed the target premium, then the excess premium percent applies to premiums paid during the remainder of the policy year.
- Policy fee – The policy fee can be deducted as a load; see the Premiums section, above. The amount deducted from each premium is the policy fee divided by the number of premium payments per year. The policy fee is not reduced for premium suspensions for this purpose.

2.7.3 Non-premium loads

The following loads are deducted from the fund value at the beginning of each month; one twelfth of the load rate applies each month:

- Amount per policy
- Amount per unit
- Percent of fund value – one twelfth of the annual percent of fund value at the end of the previous month is deducted at the beginning of each month. This load can apply either to the variable fund value only, or to the total fund value (variable, unloaned fixed, loaned fixed, and DCA account).

2.7.4 *Mortality and expense load*

The mortality and expense load is deducted from the fund value at the beginning of each month; one twelfth of the load rate applies each month. The load applies to the variable fund value at the beginning of the month after reduction for the premium and non-premium loads.

The charge rate is ratcheted when the variable fund value exceeds the specified threshold per policy. The first charge rate applies up to the threshold; the second charge rate applies to any amount of the fund value in excess of the threshold.

Specify the the base charge rate plus an addition (typically negative) that applies over the threshold.

2.7.5 *Investment management fee*

The investment management fee reduces the growth rate for each variable fund value; it is not a product load. See the Variable fund values section in the Fund values chapter, below.

2.8 *Cost of insurance charge*

The COI charge is calculated and deducted at the beginning of each month, after first adding premiums and deducting loads. COI charges are calculated in aggregate (for the policy), then deducted on a pro-rata basis from the fixed and variable fund values. See the Fund values chapter below.

2.8.1 *Input variables*

Input variables for cost of insurance charges differ between current, guaranteed, and actuarial guideline 37 fund values.

The Fund input page includes a separate set of input variables for current, guaranteed, and actuarial guideline 37 cost of insurance charges. For example, input variable `coi_rate_base_curr` specifies the current fund COI rates, `coi_rate_base_gtd` specifies the guaranteed COI rates, and `coi_rate_base_ag37` specifies the actuarial guideline 37 COI rates. The formulas, however, use the same code variable, `coi_rate_base`, in each case, but each input variable is associated with this code variable in the appropriate instance of the submodel.

These variables are special since they are read twice – first for life 1 and then again for life 2. This is accomplished by using special lookup keys.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. ‘Insured characteristics’ (such as issue age, gender, smoking class, and rate class) are set to either the life 1 or life 2 values, as needed. The external source uses lookup keys `mort_gender`, `mort_smoking_class`, `mort_rate_class`, and `mort_issue_age` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

2.8.2 *Net amount at risk COI charge*

Typically, the COI charge is calculated as a percent of net amount at risk. The COI rate applies to the net amount at risk (see Net amount at risk topic). The death benefit takes the minimum death benefit (corridor factor) into account.

2.8.3 *Net amount at risk*

The net amount at risk includes a sum assured component and a fund value component.

$$\text{NAR} = \text{SumAssured} * \text{DiscountAdj} - \text{FundValue} * \text{FVAdj}$$

Specify the discount rate used to adjust the sum assured component:

- None – No adjustments to the sum assured.
- Current Rate – The sum assured is discounted by the monthly current interest rate. The 'current interest rate' is the interest rate associated with the type of fund. So for the current fund, the discount factor is based on the current credited interest rate. For the guaranteed fund, the discount factor is based on the guaranteed interest rate. For the AG 37 fund, the discount factor is based on the specified AG 37 statutory or tax interest rate.
- Guaranteed Rate – The sum assured is discounted by the monthly guaranteed interest rate.
- Half Period Current Rate – The sum assured is discounted by a half of month current interest rate. The 'current interest rate' is the interest rate associated with the type of fund. So for the current fund, the discount factor is based on the current credited interest rate. For the guaranteed fund, the discount factor is based on the guaranteed interest rate. For the AG 37 fund, the discount factor is based on the specified AG 37 statutory or tax interest rate.
- Half Period Guaranteed Rate – the sum assured is discounted by a half of month guaranteed interest rate.

Specify fund value component (variable, unloaned fixed, loaned fixed, and DCA):

- Prior Month Fund Value – The ending fund value of the previous month
- Beginning Fund Value With Premium Less Load – The ending fund value of the previous month plus the current month's premium less loads.
- Beginning Fund Value With Premium Less All Other Charges – The ending fund value of the previous month plus the current month's premium less loads and all other charges (except the COI charge, since this is calculated based on the net amount at risk). Note, the only other available charges other than loads are rider charge and mortality and expense charges, which apply to the current fund only. So this option is the same as Beginning Fund Value With Premium Less Load for guaranteed fund and AG 37 fund.
- Prior Month Cash Value – The ending cash value of the previous month.

Specify the fund value adjustment:

- None – No adjustments to the fund value component
- COI Rate – Multiply the fund value by $(1 - \text{Monthly COI Rate})$.

When the fund value is negative, the net amount at risk is calculated using zero fund value. See the Guaranteed minimum death benefit chapter, below.

2.8.4 COI rate

Specify COI rates either as annualized rates or as annual mortality rates. The annual rate is adjusted by a multiple, which can vary by policy year or attained age. The adjusted annual rate is then converted to a monthly COI rate using either an annualized, linear, or exponential formula.

Annualized formula:

$$\text{monthly COI rate} = \text{annualized COI rate}/12$$

Linear formula:

$$\text{monthly COI rate} = \frac{(\text{annual mortality rate}/12)}{(1 - \text{annual mortality rate}/12)}$$

Exponential formula:

$$\text{monthly COI rate} = \frac{1 - (1 - \text{annual mortality rate})^{(1/12)}}{(1 - \text{annual mortality rate})^{(1/12)}}$$

Substandard multiple

The substandard multiple on the model point record applies to the annualized COI rate, or to the annual mortality rate, before conversion to a monthly COI rate. This is the same substandard multiple as applies for experience mortality rates.

This multiple applies the same to the current and guaranteed COI rates. Of course the COI rate cannot exceed 1.0 (100%).

2.8.5 Percent of fund COI charge

Some older single premium VUL products define the current COI charge as a percent of total (variable, unloaned fixed, loaned fixed, and DCA) fund value, but not greater than the COI charge calculated using the net amount at risk approach and guaranteed rates.

The percent is treated as an effective annual rate, so the monthly percent is:

$$(1 + \text{annual rate})^{1/12} - 1$$

2.9 Interest crediting

Interest is credited to the fund value at the end of each month.

2.9.1 Guaranteed rate

The minimum crediting rate for the fixed fund is a fixed rate. This is the crediting rate for the guaranteed fund.

2.9.2 Crediting rate

The crediting rate varies between each fund calculation type:

- The crediting rate for the current fund value is determined according to the company's crediting strategy for the unloaned fixed fund value
- The policy loan crediting approach is used for the loaned fund value
- A DCA account accumulates at a fixed rate
- Total returns on scenario indexes apply for variable fund values; see the Fund Values chapter, below
- The guaranteed fund value receives the guaranteed crediting rate.

2.10 Persistency bonus**2.10.1 Input variables**

Input variables for persistency bonuses and refunds differ between current, guaranteed, and actuarial guideline 37 fund values.

The Fund input page includes a separate set of input variables for current, guaranteed, and actuarial guideline 37 persistency bonuses and refunds. For example, input variable `persist_bon_cur` specifies the current fund persistency bonus, `persist_bon_gtd` specifies the guaranteed persistency bonus, and `persist_bon_ag37` specifies the actuarial guideline 37 persistency bonus. The formulas, however, use the

same code variable, `persist_bon_input`, in each case, but each input variable is associated with this code variable in the appropriate instance of the submodel.

2.10.2 Timing

Persistency bonuses increase the unloaned fund value by a specified percent at the end of the policy year. The fund value increases before deaths, withdrawals, and lapses occur, so the amount applies to the units in force at the end of the previous month.

2.11 Fund value

2.11.1 Premiums, loads, and charges

Premiums are credited to the fund value at the beginning of the month of each premium payment.

The allocation of premiums between the fixed account and the separate account is discussed in the Fund values chapter below.

2.11.2 Loads and charges

Loads and charges are deducted from the fund value at the beginning of the month after the addition of premium payments.

The allocation of loads and charges between the fixed account and the separate account is discussed in the Fund values chapter below.

2.11.3 Interest crediting

Interest is credited to the fund value at the end of each month. The total amount credited to the fund value is the sum of the amount credited to the fixed account and the separate account.

The amount credited to the fixed fund value and the indexed fund value is discussed in the Fund values chapter below.

2.11.4 Partial withdrawals

Partial withdrawals are deducted from the fund value at the end of the month, after interest has been credited.

The deduction from the fund value is the amount withdrawn plus the surrender charge on the partial withdrawal.

2.12 Cash value

The cash value is the total fund value less the surrender charge.

2.12.1 Fund value for surrender charge

The fund value for calculating the surrender benefit is the fixed account fund value plus the separate account fund value.

2.12.2 Surrender charge

Specify surrender charges as:

- Amount per unit
- Percent target premium
- Percent of fund value – This is a percent of the total (variable, unloaned fixed, loaned fixed, and DCA). fund value. That is, the surrender charge is the same as if the policy loan were repaid, allocating the repayment to the fixed fund value.

The surrender charge rates can vary annually (using the `pol_yr` column) or monthly (using the `mths_since_issue` column).

Surrender charges (other than the percent of fund value charge) are reduced by the proportion of the charges which have already been collected during partial withdrawals. Specify on the model point record initial (at the projection date) multiple which applies to these surrender charge rates.

2.13 Policy loan

Policy loan expenses are discussed in the Expenses section, below.

2.13.1 Policy loan amount

Policy loans apply equally to all policies within a model point. For in-force policies, specify on the model point record the initial (at the projection date) policy loan for the policies collected into the model point.

The accumulated policy loan at the end of each month is the balance at the end of the previous month, brought forward with one month's interest at the policy loan rate.

The outstanding loan amount is increased at the end of each policy year, by a percent of the total fund value.

The outstanding loan amount is reduced at the end of any month when it would exceed the cash value.

2.13.2 Policy loan cash flows

An adjustment increase is a new policy loan and a negative cash flow. An adjustment decrease is a policy loan repayment and a positive cash flow.

The elimination of the policy loan at death and surrender is an 'adjustment' in the loan balance to zero, and thus is a positive loan cash flow which offsets part of the decrement cash flow.

2.13.3 Policy loan rate

The policy loan interest rate is the greater of: (1) a fixed rate and (2) an indexed rate equal to a competitor rate. Interest is added to the loan at the end of each month.

When policy loan interest is payable in advance, policy forms may specify the loan rate as a discount rate. Convert this discount rate to the annual effective annual rate. For example, a 7.4% discount rate payable in advance must be input as an effective annual rate of 8.0% [$= 7.4\% \div (1 - 7.4\%)$].

2.13.4 Policy loan crediting rate

The crediting rate on the loaned fund value can be:

- **Unloaned rate** – The same crediting rate applies to the unloaned fixed fund value.
- **Fixed rate** – A specified crediting rate.
- **Guaranteed rate** – The guaranteed crediting rate.
- **Fixed spread** – The policy loan interest rate less a specified spread.

2.13.5 Preferred loans

The loan can be split into two parts. The 'preferred' part of the loan is the amount up to the cash value or total (variable, unloaned fixed, loaned fixed, and DCA) fund value, less cumulative premiums paid net of cumulative partial withdrawals. The crediting rate on the preferred loan is the policy loan interest rate less a specified preferred loan spread. A special case of a preferred loan is a zero net cost loan, sometimes called a wash loan, where the preferred loan spread is zero.

The remainder of the loan is a non-preferred policy loan, which has a crediting rate as discussed above.

For in-force model points which have an initial (at the projection date) policy loan for products with the preferred policy loan feature, specify on the model point record the initial cumulative premiums and cumulative withdrawals for the policies collected into the model point.

3 FUND VALUES

3.1 Policy fund values

3.1.1 Current fund

A current fund is projected for experience calculations.

For in-force model points, specify on the model point record the initial (at the projection date) current fund value, which is the sum of the specified fixed fund value, DCA fund value, and separate accounts values, for the policies collected into the model point.

3.1.2 Guaranteed fund

Guaranteed fund values are projected when needed for solving calculations. See the Solving calculations chapter, below.

The guaranteed fund is also used during the projection, when it is updated to the current fund value at a reserve valuation date, then projected forward using premium payment assumptions for the reserve calculation. This occurs for UL CRVM reserves. When these calculations apply, the `vul_ifund_gtd` output does not represent guaranteed fund calculations from issue. Instead, set the `solving_display_defn` variable (on the Product input page in the Premiums category) to All Solves to save each type of solving fund result in the output database.

Flag and indicator values control each type of solving, such as the timing and amount of premium payments. See the Submodels structure chapter, below.

3.1.3 AG 37 fund

Actuarial guideline 37 fund values are projected when needed for reserving calculations.

3.1.4 `vul_ifund` submodel output

The formulas in the `vul_ifund` submodel have separate values for each instance of this submodel. When output is created for this submodel, select either the current, guaranteed, or ag37 fund instance.

The `vul_ifund` submodels iterate to solve for values such as the guaranteed maturity premium. See the Solving calculations chapter, below. When the `solving_display_defn` variable (on the Product input page in the Premiums category) is All Solves, the `iul_ifund` output includes the final values for each solving calculation.

Warning! Since the `vul_ifund` submodel is reset for each solving calculation, the final results apply only for the last time this submodel was calculated. For example, the `vul_ifund_gtd` values are not the guaranteed fund values calculated from issue; they could be the calculations starting from the current fund value at the last time when UL CRVM reserve calculations were processed.

To see a particular set of results, customize to add code to write out the submodel results at the time that calculation has just been performed. Search for the 'writeModel' command in the `iul->initialize` formula to see how this is done when the `solving_display_defn` calls for writing out the submodel values.

3.2 Fund value structure

3.2.1 Fund types

The current fund value can be split between the following fund types:

- **DCA fund value** – The amount of current fund value which is to yet to be allocated to the separate account.
- **Fixed fund value** – The amount of current fund value held in the general account, earning interest based on the crediting strategy.
- **Separate account** – The amount of the current fund value invested in funds earning interest based on the change in the value of an index. The standard application provides for the separate account to be split into up to 6 variable funds.

3.2.2 *Input variables*

Input variables for the DCA fund, fixed fund, and separate account fund values are specified on the VulFund input page.

The DCA fund input variables include the DCA crediting rate. This variable commonly varies by plan code.

The fixed fund input variables include the guaranteed crediting rate and crediting strategy. The variables commonly vary by plan code.

The separate account input variables include the separate account index, investment fee refund percent, and the management fee percent. The variables commonly vary by a fund_id, identified as index0 through index5.

3.3 Fund allocations

3.3.1 *Initial fund values*

Fund values are allocated to the fixed account and up to six separate account variable funds.

For in force model points, specify the initial (at the projection date) amount in each account on the model point record.

3.3.2 *Premium allocation*

Each premium payment is allocated between the fund values in proportion to the 'target' percent specified on the model point record for the fixed fund value and each separate account variable fund value. The total of the percentages should add to 1.00 (100%), otherwise the specified values are normalized so they do add to that result. When the target percent fields are all left at their zero default values, the target percentages are the distribution of the initial unloaned fixed and separate account variable fund values, ignoring the DCA fund value.

New business model points should use the target percent variables, in which case the initial fund value variables are ignored. But when the target percent variables are zero, the initial fund value variables can specify the target allocations, the same as for in-force model points.

3.3.3 *Loads and charges allocation*

Percent of premium loads are deducted from each premium payment. Thus this load is allocated to the fund values in the same proportion as the allocation of the premium.

Percent of fund loads are deducted from the fund value, including the DCA fund. Note that, when the percent of fund load is calculated using only the separate account, the allocation of the load is only to the separate account variable funds.

Other loads, and COI charges, are calculated in aggregate (for the policy), then allocated pro-rata in proportion to the fixed and variable fund values, excluding the DCA fund.

3.3.4 *Withdrawals allocation*

Partial withdrawals are allocated as a proportion of the account unloaned fund value to the total unloaned fund value, including the DCA fund.

3.3.5 *Policy loan transfers*

The policy loan is considered to be part of the fixed fund value. Interest is credited separately on the loaned and unloaned fixed fund value. This produces the same result as if the policy loan were treated as being a separate fund value, with the fixed fund value reduced so that it does not include the policy loan.

As the loan grows, funds are transferred from the unloaned fixed fund value and separate account variable fund values using a pro-rata approach so as to maintain the collateral for the loan.

The interest credited on the loaned fund value is added to the unloaned fixed and separate account variable fund values on a pro-rata basis of the account unloaned fund value to the total unloaned fund value, including the DCA fund.

3.3.6 *Rebalancing*

Rebalancing between the unloaned fixed and separate account variable fund values can be included in the projection. At the beginning of each policy year, transfers restore the unloaned fixed and variable fund values, excluding the DCA fund to have the distribution specified for premium allocations.

3.4 *Dollar cost averaging fund value*

Variable universal life can include the dollar cost averaging (DCA) account. This is a fixed account that transfers gradually to be allocated to the separate account funds.

3.4.1 *Initial DCA amount*

VUL model points include a dollar cost averaging (DCA) fund value. The DCA account applies across all VUL products.

For new issues, a specified percent of the single premium paid at issue is allocated to the DCA fund value. The starting DCA fund value is this allocated single premium, less the related percent of premium load.

For in-force model points, specify on the model point record the initial (at the projection date) DCA fund value for the policies collected into the model point.

3.4.2 *DCA crediting*

The DCA fund value earns a specified fixed crediting rate. The crediting rate is an effective annual interest rate, which is converted to the equivalent monthly rate:

$$(1 + \text{annual rate})^{\frac{1}{12}} - 1$$

The DCA interest crediting is considered to be part of the interest crediting on the fixed fund value.

3.4.3 *DCA transfers*

The DCA fund value transfers to the separate account variable fund values at the end of each of the first 'n' months after issue. Following this specified number of months from issue, the DCA fund value is zero. Specify the months from issue on the model point record.

The amount of each transfer is the current DCA fund value, divided by the remaining transfer months. For example, the entire DCA fund value transfers out in the last month, one-half of the current DCA fund value transfers in the month before that, one-third of the amount transfers in the month before that, and so on.

The amount transferring out of the DCA fund value is allocated to the variable fund values in the same proportion as applies to the premium paid in that month.

3.5 Fixed fund value

3.5.1 Overview

The fixed fund value is increased for credited interest at the end of each month. The crediting rate for the fixed fund value is the larger of the policy's guaranteed crediting rate and the rate determined through the crediting strategy.

When the policy has a loan, interest is credited to the unloaned fixed fund each month using the base crediting rate determined according to the company's crediting strategy. The loaned fund value may have interest crediting different from the remainder of the fixed fund value. See the Policy loan section in the Product benefits and values chapter, above.

3.5.2 Initial crediting rate

Specify the initial (at projection date) crediting rate for the fixed fund value on the model point record. When the value on the model point is zero, the initial crediting rate is determined based on the crediting strategy.

The initial crediting rate applies until the crediting rate is reset through the application of the crediting strategy.

3.5.3 Crediting strategy declaration

Declaration timing

The crediting rate can change at the beginning of a month, policy year, calendar year, or calendar quarter.

Declaration period

The crediting rate changes according to the declaration period, which is one or more cycles of the specified period. A different number of cycles can apply at issue than for ongoing crediting rate changes. For example, the crediting rate at issue could apply for the first two policy years, then changing every policy year thereafter.

3.5.4 Crediting strategy rate basis

Fixed rate

The crediting rate can be a fixed rate, such as for calculating illustration values. Crediting rate constraints do not apply.

Asset yield basis

You can specify the crediting rate as the asset yield less a spread, subject to the crediting rate constraints. This definition makes the model points dependent on asset results, so independent model point processing cannot apply for asset/liability projections.

The asset yield includes the effects of IMR amortization, defaults, and borrowing costs.

For liability-only projections, you specify the asset yield as an assumption for the projection. Independent model point processing can still apply, since the asset yield is not affected by liability results.

Competitor rate basis

You can specify the crediting rate as a competitor rate less a spread, subject to the crediting rate constraints.

When the constraints do not include the asset yield, independent model point processing can apply even for asset/liability projections. For example, this approach could take the view that the crediting rate is primarily driven by market forces, and that appropriate investment strategies must be determined in order to achieve the desired spread from the asset yield.

3.5.5 *Crediting strategy constraints*

The following constraints apply when the crediting rate basis is the asset yield or a competitor rate. Crediting rate constraints do not apply when the crediting rate is a fixed rate.

The crediting rate is first constrained to be not more than a maximum percent above, and not less than a maximum percent below, a specified competitor rate.

Further (thus adjusting the above constraints), the crediting rate is not greater than [the asset yield less a minimum spread]. When the minimum spread below the asset yield is 1.0 (100%) default value, this constraint does not apply, and model points defining the crediting rate using a competitor rate are eligible for independent model point processing within asset/liability projections.

Finally, the crediting rate is not less than the guaranteed crediting rate in all cases, even when the crediting rate is defined as a fixed rate.

3.6 *Separate account*

Each model point can use up to six variable fund values in the separate account. The same variable funds apply across all VUL products.

3.6.1 *Variable funds*

For in-force model points, specify on the model point record the initial (at the projection date) variable fund values for the policies collected into the model point.

3.6.2 *Variable fund growth*

Variable fund values fluctuate with the change in the funds' asset market values.

The scenario index for each of the variable fund values must be contained in the scenario file. See the Economic scenarios book. Scenario growth rates are returned from the Economy DLL as a monthly rate, which can be negative.

3.6.3 *Investment management fee*

Specify the investment management fee in the product definitions as an effective annual rate. This fee reduces the separate account fund growth rate; it is not counted as a product 'load'.

3.6.4 *Investment fee refund*

The insurer may receive a portion of the investment fee as revenue.

The refund rate is specified as an annual rate. One twelfth of the annual rate is applied against the fund value at the beginning of month after premiums and reductions for loads and cost of insurance changes to determine the refund amount.

4 *GUARANTEED MINIMUM DEATH BENEFIT*

4.1 *General*

4.1.1 *GMDB years*

The Guaranteed minimum death benefit (GMDB) keeps the death benefit in force through a specified policy anniversary even when the unloaned fund value is zero or negative, as long as the policyholder meets requirements using the GMDB premiums test.

This is the policy anniversary when the guarantee expires. After the end of this policy year, the policy lapses when the unloaned fund value becomes zero or negative. When this variable is zero (default value), the GMDB continues to maturity.

When the GMDB test has been satisfied at the EMO attained age, the guarantee continues through the EMO period.

4.1.2 GMDB premiums test

The GMDB keeps the policy in force through a specified policy anniversary, as long as accumulated premiums paid, less accumulated withdrawals taken, less the policy loan, are not less than the accumulated required premiums. The accumulation of premiums paid is reduced by partial withdrawal payments, but not by the surrender charge reduction to the fund value.

Required GMDB premium

The required GMDB premiums are assumed to be paid monthly. The required annual premium per unit is increased by the policy fee, then multiplied by the specified annualized modal loading multiple for the monthly premium mode.

Interest accumulation

Both the actual and required premiums accumulate at the GMDB interest rate.

For in-force model points, specify on the model point record the initial (at the projection date) accumulation of actual premiums less accumulation of actual withdrawals for the policies collected into the model point. The initial accumulation of required premiums from issue is calculated during the projection, using the units per policy at the projection date.

Policy loan

As stated above, the GMDB test requires the accumulated premiums, less the policy loan, to exceed the accumulated required GMDB premiums. The policy loan is taken into account in determining whether the GMDB is in the grace period.

The policy loan is ignored in other calculations, however, such as fully or minimally funding the GMDB. Reserves are not affected by the policy loan.

4.1.3 Negative fund value

When the unloaned fund value is negative, it is treated as zero for product value calculations:

- Benefit values use zero unloaned fund value.
- Interest crediting on the unloaned fixed fund can either: (1) continue to apply, making the unloaned fixed fund value more negative, or (2) stop when the unloaned fixed fund value is negative. When there is no interest crediting, the unloaned fixed fund value is treated as zero for calculating the average crediting rate for GAAP calculations.
- Loads continue to be deducted.
- COI net amount at risk uses zero unloaned fund value. COI charges continue to be deducted.
- The YRT reinsurance net amount at risk uses zero unloaned fund value.
- The GAAP basic benefit reserve used a zero unloaned fund value.

4.2 GMDB grace period

4.2.1 Positive fund value

When the current fund value is positive, the GMDB remains in effect until the test has been violated for the specified number of grace period months. At the end of the grace period, the GMDB terminates even though

the policy continues to be in force without the guarantee. Thus some model points have no GMDB, even though it is specified for the product.

The grace period is restored when the test is satisfied before the current grace period ends. That is, the next time the GMDB test is violated, the full grace period months apply again.

When the guarantee is already violated at the projection date, specify on the model point record the initial (at the projection date) months that the guarantee has been in the grace period. Any number greater than or equal to the grace period months indicates that the model point no longer includes the GMDB.

4.2.2 Catch up provision

When the GMDB is in this grace period, the GMDB reserve includes a calculation of the 'catch up' amount needed to restore the guarantee.

When premiums are increased to minimally fund the guarantee, the catch up amount is paid as a premium at the projection date. See the Premium adjustment section below.

4.2.3 Negative fund value

This grace period does not extend the GMDB when the current fund value is negative. Fund-related lapse occurs immediately when the GMDB test has been violated.

4.3 Premium adjustment

Premium adjustments are evaluated at the beginning of each policy year, when the annual premium is determined for the year.

4.3.1 Fully funded GMDB

Premium payments can stop when the GMDB is fully funded, which occurs when the accumulated premiums, without further premium payments, equals or exceeds the accumulated required premiums at the end of the GMDB period.

4.3.2 Minimally funded GMDB

When the GMDB conditions would be violated, premium payments can increase to be the minimum amounts to keep the guarantee in effect. The annual premium each policy year is then the premium which meets the required premiums accumulation at the end of the policy year.

When the guarantee is in the grace period at the projection date, the catch-up amount is assumed to be paid at the beginning of the first policy anniversary after the projection date. The annual premium payments, which are paid modally, are the same annually required premiums as in years when there is no catch-up payment. For example, when the initial (at the projection date) accumulated premiums is less than the accumulation of required premiums, a special single premium is paid at the first policy anniversary, then the annually required premium is paid modally.

Lapses due to negative fund value are ignored when premiums are adjusted for minimally funded status.

Partial withdrawals and policy loan increases stop for model points in minimally funded status.

4.4 Lapse adjustment

The lapse rate is adjusted when the unloaned fund value is negative, i.e., when the GMDB is keeping the policy in force:

- When the GMDB is fully funded (see above), the lapse rate is multiplied by an adjustment percent.
- When premiums are still required to keep the GMDB in effect until its expiration, the lapse rate is multiplied by a different adjustment percent.

4.5 In-force statistics

Units in force at the end of each month are calculated for these special situations:

- **GMDB in the money** – When the fund value* is zero or negative, meaning that the GMDB is keeping the policy in force. *This test is based on the cash value, when the fund-related lapse feature tests against the cash value instead of the fund value (see the Lapses and surrenders topic in the Product benefits and values chapter, above.)
- **GMDB minimally funded** – When premiums are increased to keep the GMDB in force (see the Premium adjustment section, above.)
- **GMDB fully funded** – When premiums are stopped because the GMDB is paid up (see the Premium adjustment section, above.)

5 JOINT LIFE PRODUCTS

Joint life products have a second insured, in addition to life 1. The product is a joint life product when the life status variable is First To Die or Frasierized Last Survivor. When the life status is Single Life, the second life variables do not affect the projection.

5.1 Second life characteristics

5.1.1 Life 2 model point variables

Specify insured characteristics model point variables for life 2 in addition to life 1. Life 2 characteristics include issue age, gender, smoking class, and rate class. These insured characteristics are particularly important for mortality variables, which are the rates for: (1) experience, cash value, and reserve mortality, and (2) YRT reinsurance rates.

See the Mortality variables section in the Input organization chapter, above.

5.1.2 Life 1 and life 2 scalars

Since `issue_age` is used to calculate values such as the years to maturity, 'life 1' must be:

- the older of the two insureds for first to die model points, and
- the younger of the two insureds for Frasierized last survivor model points.

The order of the two lives on the model point record is handled automatically, by converting the life 1 and life 2 model point variables into scalars, instead of using the model point variables directly:

- For first to die model points with life 1 younger than life 2, the life 1 scalars return the life 2 model point variables, and the life 2 scalars return the life 1 model point variables.
- For Frasierized last survivor model points with life 1 older than life 2, the scalars also switch the values this way.
- Otherwise, the life 1 scalars return the life 1 model point variables, and the life 2 scalars return the life 2 model point variables.

For example, the scalars `issue_age` and `gender` normally have the values `issue_age_mp` and `gender_mp`, but return the values `issue_age_mp_life2` and `gender_mp_life2` when the lives must be switched. Since formulas reference the scalars, they are automatically based on the correct life.

5.2 First to die

For first to die products, the death benefit is paid on the first death; the policy terminates at that time. Both insureds are alive at the projection date.

The annual mortality/COI/YRT rate is:

$$q_1 + q_2 - q_1 \times q_2$$

5.3 Frasierized last survivor

For last survivor products, the death benefit is paid when both insureds have died. Since there is no change in status when the first life dies, the projection assumes there is no knowledge about whether life 1 or life 2 is still alive at the projection date, except that at least one of them must be alive since the policy is still in force.

The mortality/COI/YRT rates are Frasierized from issue, as described below.

5.4 Frasierization

Since Frasierized last survivor products continue to operate as normal until the second death, the projection assumes there is no information whether only one life is surviving at the projection date. The mortality rates for life 1 and life 2, plus the joint life contagion effect, are combined into a Frasierized rate equal to the sum of:

$$\begin{aligned} &P(\text{both alive BOY}) \times P(\text{both die during the year} | \text{both alive BOY}) \\ &\quad + P(\text{only life 1 alive BOY}) \times P(\text{life 1 dies during the year} | \text{only life 1 alive BOY}) \\ &\quad + P(\text{only life 2 alive BOY}) \times P(\text{life 2 dies during the year} | \text{only life 2 alive BOY}) \end{aligned}$$

This formula Frasierizes the annual mortality rates, then the Frasierized annual rate is converted to a monthly rate as usual. This method commonly applies for reserve mortality and YRT rates. Annual experience mortality rates are also Frasierized prior to calculating the monthly mortality rate, so that calculations are simplified by producing monthly mortality rates which are constant within each policy year.

5.4.1 Joint life mortality addition

For experience mortality, the annual probability that both lives die together (or within some short interval) is increased by a joint life mortality rate addition, which is sometimes called a contagion effect. The specified addition is reduced when the joint death rate would exceed either the first or second life mortality rate. That is:

$$q_1 \times q_2 + \text{adjusted addition} \leq \min(q_1, q_2)$$

5.4.2 Minimum YRT rate

The Frasierized annual YRT reinsurance rate is subject to a minimum floor, such as .00012 (\$.12 per thousand).

6 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves calculated by model point. PBA reserves are discussed in the PBA book, including the net premium reserve (NPR).

6.1 Separate and general accounts

6.1.1 Separate account reserve

The separate account reserve is the sum of the variable fund values.

6.1.2 General account reserve

The reserve held in the general account is the total reserve, less the separate account reserve. The general account reserve can be negative when the total reserve is less than the variable fund values.

The GMDB reserve is held in the general account.

6.2 Reserve basis

6.2.1 Mortality

Reserve mortality is a base rate, times the multiple, then plus the addition. Lookups for reserve mortality tables should be setup similar to the experience mortality rates since these rates are read for both the first and second lives.

The base statutory mortality rates affect the projection even when reserves are a percent of the fund or cash value, since the omega age specifies the GMP maturity age.

Substandard multiple

The substandard multiple (on model point records) does not affect reserves. It applies for experience mortality, COI charges, and YRT reinsurance premiums.

6.2.2 Interest

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeStatValn and LifeTaxValn ranges in the Valuation Interest Rates Excel workbook.

6.2.3 Function timing

Function timing options affect traditional reserve calculations, such as the CRVM reserve (for the UL CRVM method) and the XXX reserve (for the AXXX reserve):

- **Curtate** – Annual premiums and claims paid at the end of the policy year
- **Continuous** – Continuous premiums and time-of-death claims
- **Semicontinuous** – Annual premiums and time-of-death claims.

Premium timing multiple

When premiums are assumed to be paid continuously, the curtate reserve formulas are adjusted by the following 'premium timing' multiple:

$$\frac{(\delta - d)}{\delta^2} + \frac{(i - \delta)}{\delta^2} \times \frac{(1 - q)}{(1 + i)}$$

Claims timing multiple

When claims are assumed to be paid immediately, the curtate reserve formulas are adjusted by the following 'claims timing' multiple:

$$\frac{i}{\delta}$$

Reserve for immediate payment of claims

Actuarial guideline XXXII (AG 32) requires additional reserves for policies which pay death claims at the time of death, but have statutory reserves calculated assuming death claims are paid at the end of the policy year. This can be approximated by using the Semicontinuous function timing option.

6.3 UL CRVM reserve

6.3.1 Percent of fund approximation

The UL CRVM reserve can be approximated using percentages of the fund value or cash value. This is the total (variable, unloaned fixed, loaned fixed, and DCA) fund value. The GMDB reserve, if applicable, is still calculated and compared against this amount.

6.3.2 GMP and GMF

The UL CRVM reserve is based on the guaranteed maturity fund (GMF), which is calculated using the guaranteed maturity premium (GMP) which endows the policy for the sum assured at the GMP maturity age:

- The GMP maturity age is the statutory reserve base mortality table omega age, when earlier than the maturity age.
- The GMP is paid for the contractual premium term, which may be different from the model point's premium term.
- The GMP is paid using the model point's premium mode.

The GMF is projected by calculating the GMP which endows the fund value at the GMP maturity age using guaranteed loads, COI rates, and interest crediting. The GMF is then projected using the GMP and these guaranteed rates. When specified by the `solving_display_defn` variable (in the Product input page in the Premiums category), the `vul_ifund` output includes this fund value per unit.

The GMF is recalculated (in the situations described below) by starting from the current fund value, then projecting using guaranteed rates and assuming that the GMP is paid in the future. When the reserve method is UL CRVM, the `vul_ifund` output includes this fund value per unit. This does not apply when the reserve method is Approximate UL CRVM, which uses only the GMF values.

6.3.3 UL CRVM

Reserve basis

The statutory mortality variables for UL CRVM reserves (`stat_mort_base`, `stat_mort_mult`, and `stat_mort_addn`) are separate from the mortality variables for XXX reserves. The same applies for tax reserves. The deficiency mortality variables for alternative minimum UL CRVM reserves (`def_res_mort_base`, `def_res_mort_mult`, and `def_res_mort_addn`) are separate from the mortality variables for XXX deficiency reserves.

CRVM reserve [`vul_py->stat_crvm_terminal`]

The traditional Commissioners Reserve Valuation Method (CRVM) reserve is calculated for the plan of insurance endowing at the GMP maturity age, using the same expense allowance as for the corresponding traditional plan.

Benefit amounts [`array dth_ben_tbl_crvm`]

The benefits for calculating the traditional CRVM reserve are the projected average EOM death benefit per unit amounts for each policy year, using guaranteed rates and charges and GMP premium payments.

The sum assured is paid as an endowment amount at the GMP maturity age.

UL CRVM reserve [`vul->stat_res_ulcrvm_per_unit`]

The ratio 'r' is the current total (variable, unloaned fixed, loaned fixed, and DCA) fund value divided by the GMF. When $r \leq 1$, the reserve is the traditional CRVM reserve times r.

When $r > 1$, the current total fund value is projected to maturity assuming: (1) future premiums equal to the GMP, and (2) guaranteed rates.

The average death benefit in each policy year from this projection is compared to the death benefits from the GMF projection from issue. Any excess death benefit, and any excess maturity value, over the GMF projection from issue is discounted using statutory reserve mortality and interest.

The reserve is: (1) the traditional CRVM reserve, plus (2) the present value of these excess benefits.

6.3.4 Approximate UL CRVM

The following approximation can be appropriate for policies which are not heavily funded.

The traditional CRVM reserve and r are calculated from issue the same as described above. When $r \leq 1$, the reserve is the CRVM reserve times r ; there is no approximation in this case.

When $r > 1$, the reserve is: (1) the traditional CRVM reserve, plus (2) the excess of the current fund value over the GMF. This approximation thus reduces calculation time by not recalculating the GMF.

6.4 GMDB reserve

6.4.1 Approximation

The GMDB reserve can be approximated as either:

- Percentages of the separate account or total fund (variable, unloaned fixed, loaned fixed, and DCA) value or cash value.
- Amounts per unit.

6.4.2 Actuarial guideline 37

Actuarial guideline XXXVII (AG 37) specifies that an additional reserve is required for variable life insurance policies that provide a minimum death benefit guarantee. The main characteristics of the reserve for the GMDB are as follows:

- The reserve is held in the general account so that it will be backed by the general assets of the company.
- The reserve should be adequate to cover, under all but the most extreme circumstances, the GMDB death claims for the next year.
- The reserve should react slowly but steadily to an extended period of poor investment experience of the separate account.
- The reserve should not overreact and cause unnecessary fluctuations in surplus by increasing too rapidly in a sharp market downswing and decreasing too rapidly in a sharp market upswing after a period of poor market experience.

The AG 37 GMDB reserve for each model point is the larger of:

1. One year term reserve to assure coverage of next year's claim, called the One Year Term (OYT) method, and
2. Reserve designed to protect against an extended period of poor investment experience of the separate account, called the Attained Age Level Reserve (AALR) method.

Valuation interest and mortality rates used in these items are specified separately from those used for the basic reserve. A curtate function approach is always used.

One Year Term (OYT) Method

The OYT reserve is the present value of any GMDB claims occurring within the next year, assuming that the separate account fund value drops by one-third and then grows at an assumed investment rate, and assuming that the policyholder will meet requirements to keep the guarantees in force. The claims are valued using the AG 37 reserve mortality rates, discounted at the AG 37 reserve interest rate.

Attained Age Level Reserve (AALR) Method

The AALR produces a gradual increase in the reserve when this is necessary to cover guaranteed minimum death benefit claims arising from an extended period of poor investment performance. The approach is to fund the cost of future guaranteed minimum death benefit claims by level payments over the future premium paying period of the contract.

The AALR is equal to the 'residue' of the previous month's AALR, plus a 'payment', and is calculated as follows:

- The 'residue' is the previous month's AALR accumulated at the AG 37 reserve interest rate (which should be specified as the maximum valuation interest rate), and decreased for the tabular AG 37 reserve mortality based on the excess, if any, of the guaranteed minimum death benefit over the death benefit which otherwise would have been in effect during the preceding month. The result is then divided by the tabular probability of survival. The initial residue at issue is zero.
- The 'payment' amount, which may be positive or negative, and is (1) the present value of any future GMDB benefits, less and residue of the previous AALR, divided by (2) the present value of \$1 paid annually during the guarantee period, beginning on the valuation date. These amounts are projected over the term of the GMDB, assuming that: (1) the fund value grows at the AG 37 reserve interest rate, (2) COI charges are determined using the AG37 COI rates, (3) premiums are paid only as required to prevent the GMDB from terminating, (4) there are no other policy credits and charges, (5) negative unloaned fund values are treated as zero for the purpose of determining the GMDB, and (6) present values are based on the AG 37 reserve mortality and interest rates.

The AALR as of the projection date is specified on the model point record. Neither the AALR nor the 'residue' can be less than zero.

6.5 Substandard reserve

Substandard reserves are not included in the projection. The substandard multiple (on model point records) does not affect reserves. It applies for experience mortality, COI charges, and YRT reinsurance premiums.

However, stochastic and deterministic PBR calculations include the substandard multiple in the PBA mortality rate, since it is part of the expected experience for the policy.

6.6 Mid-year reserve

Reserves between policy anniversaries can be:

- Monthly interpolated terminal – Monthly interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.
- Daily interpolated terminal – Daily interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.

When the reserve function timing is Continuous, the mid-year reserve net premium is a discounted continuous net premium.

6.7 Deficiency reserves

The deficiency reserve is the 'alternative minimum' reserve less the statutory basic reserve (i.e. the statutory UL CRVM reserve), calculated using mid-year reserves, when at least one future alternative minimum reserve net premium exceeds the GMP. The alternative minimum reserve is calculated using the lower of the GMP and reserve net premium in each year.

Specify the deficiency reserve mortality and interest basis separately from the statutory reserve basis.

When a percent of fund approximation is used for the UL CRVM reserve, no UL CRVM alternative minimum reserve is calculated.

6.8 Additional reserve

Additional statutory reserves can be specified as an amount per unit. [vul->stat_res_addl_per_unit]

The additional reserve is not considered during the calculation of deficiency reserves.

The additional reserve can be added to the final reserve (after the comparison between the ULCRVM and secondary guarantee reserves), or only to the ULCRVM reserve (before the comparison to the secondary guarantee reserve). The additional reserve is added before comparisons to the cash value and tabular cost of insurance floor.

When detail reports split out reserve values, 'additional reserves' include the increase over basic reserves due to minimum reserves (such as cash value and the tabular cost of insurance floor). [vul->stat_res_addl]

The coinsurance reserve credit can exclude the additional reserve.

6.9 Tabular cost of insurance floor

The reserve is subject to a minimum based on the tabular cost of insurance. That is, this floor applies to the sum of: (1) basic reserve (including the AG 38 reserve), (2) deficiency reserve, and (3) additional reserve.

The floor is the tabular cost of insurance (c_x) times the sum assured, times the following fraction for each mid-year reserve method:

- Monthly interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment. For example, this floor at the end of policy month 1 is $11/12 c_x$ for annual premiums, and $2/12 c_x$ for quarterly premiums.
- Daily interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $344/365 c_x$ for annual premiums, and $71/365 c_x$ for quarterly premiums.

6.10 Cash value floor

Finally, the mid-year reserve is not less than the cash surrender value. [vul->stat_res_per_unit]

6.11 EMO reserve

When the Extended Maturity Option is in effect, the reserve is the current death benefit.

6.12 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record sets the tax reserve to be the statutory reserve, including the comparison to tabular cost of insurance floor using the statutory basis, but excluding any deficiency reserve.

Tax reserves are calculated by the same methods, using the tax reserve interest and mortality basis. The secondary guarantee reserve uses the same segments as apply for statutory reserves.

The statutory additional reserve can be added to tax reserves. That is, the additional reserve can be deductible.

The tax reserve is not greater than the statutory reserve, including the additional reserve even when it is not included in the tax reserve.

6.13 Unearned premium reserve

6.13.1 Balance sheet

When the mid-year reserve method is monthly or daily interpolated terminal, an unearned premium reserve is established. The unearned premium reserve is the portion of the modal valuation net premium that has been collected but not yet earned.

The balance sheet includes the statutory unearned premium reserve as a separate line item from the statutory reserve. The unearned premium reserve is set to zero if the reserve being held is the tabular cost of insurance or cash value floor.

6.13.2 Profit effect

Book profit includes the increase in the unearned premium reserve.

6.13.3 Taxable income

Taxable income similarly includes the unearned premium reserve increase, which is calculated the same way as statutory unearned premium reserve increase, but using the tax reserve basis.

7 ACTUARIAL ASSUMPTIONS

7.1 Commissions

7.1.1 Premium commission

Percent of premium commissions apply the target premium percent to the premiums paid during the policy year (including the additional single premium) until the premiums exceed the target premium. Then the excess premium percent applies to premiums paid during the remainder of the policy year.

When the policy fee is not commissionable, the policy fee part of the premium is excluded from this calculation, and the target premium excludes the policy fee.

7.1.2 Fund value commission

One twelfth of the annual percent of unloaned fund value at the end of the previous month is paid at the beginning of each month.

7.1.3 Annualized commission

A specified percent of the total first year target premium commission can be paid at issue. No further premium commissions are paid until the cumulative earned commission exceeds the annualized amount. From then on, premium commissions are paid as usual. This determination of commission payments is made for continuing policies, not taking into account terminated policies.

Other types of commissions are not affected by the annualization of commissions.

7.1.4 Commission chargeback

A portion of the premium commission can be charged back in the event of lapse or surrender during the first three years after issue. The chargeback percent applies either to: (1) total first year premium commissions paid to date, or (2) the annualized commission amount. The chargeback percent varies by months since issue.

Other types of commissions are not subject to chargeback.

The commission subject to chargeback is calculated per unit in force, and thus is not reduced by decrements. This amount applies to the units lapsing and surrendering each month.

For in-force model points, specify on the model point record the initial (at the projection date) commissions subject to chargeback for the policies collected into the model point. When this variable is negative (the default value is --1), the chargeback amount is calculated from issue assuming that premiums have been paid as specified.

7.1.5 Commission overrides and agent terminations

The specified commission rates are the average commission rates, taking into account average writing agent commissions, commission overrides, and the effect of agent terminations.

7.2 Expenses

7.2.1 Expense adjustment multiple

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

7.2.2 Acquisition expenses

Acquisition expenses incurred at issue can include:

- Amount per policy
- Amount per unit
- Percent of target premium
- Percent of the total first year premium (including the additional single premium).

Acquisition expenses could vary by underwriting class, which is discussed in the Mortality section, below.

7.2.3 Maintenance expenses

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units inforce during the month, not at the end of the month after decrements.

Maintenance expenses can include:

- Amount per policy, separate between premium paying and paid up
- Amount per unit, separate between premium paying and paid up
- Percent of the annualized premium (excluding the additional single premium)
- Percent of the total (variable, unloaned fixed, loaned fixed, and DCA) fund value at the end of the previous month.

BOM maintenance expense

An additional maintenance expense is incurred at the beginning of each month, when commissions are paid:

- Percent of each commission payment, ignoring commission chargeback.

7.2.4 Transaction expenses

Maintenance expenses include the following expenses related to administrative activities:

- Expense per death claim policy
- Expense per death claim unit.

7.2.5 Reinsurance administration expenses

Reinsurance administration expenses are discussed in the Reinsurance chapter, below.

7.2.6 Inflation

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

Note: The average size for future issues can be inflated (to offset expense inflation), by customizing the scalar `units_per_pol` to divide the result by `infl_cumul(commencement_period)` for model points issued after the start of the projection. This would reduce the number of policies issued, with no effect on the units, sum assured, and premium issued for the model point.

7.2.7 Policy loan expenses

Policy loan expenses are a percent of the loan balance at the end of the previous month. One twelfth of the annual rate applies each month.

This amount is a negative cash flow at the end of each month, and is thus subtracted from direct cash flows.

Policy loan expenses reduce book profit, by reducing investment income from policy loans. Thus this expense is treated as an investment expense for statutory and GAAP profit reporting. It is not included with other expenses on profit reports.

However, PBA deterministic reserves include this expense in the present value of expenses. This is a different treatment of policy loan expenses than applies for profit calculations.

7.2.8 Premium tax

Premium taxes are reported with expenses. The tax is a percent of each premium payment, including the additional single premium paid at issue.

Inflation and the expense adjustment calendar month multiple do not apply to premium tax.

7.2.9 Fixed expenses

The business segment can include fixed expenses in addition to the product expenses, which are variable in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

7.3 Mortality

7.3.1 Input variables

'Mortality variables' for VUL products include mortality base, multiple, and addition rates, mortality improvement, COI rates, and YRT reinsurance rates. These variables are special since they are read twice – first for life 1 and then again for life 2. This is accomplished by using special lookup keys.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. 'Insured characteristics' (such as issue age, gender, smoking class, and rate class) are set to either the life 1 or life 2 values, as needed. The external source uses lookup keys `mort_issue_age`, `mort_gender`, `mort_smoking_class`, `mort_rate_class`, `mort_pol_yr`, `mort_cal_yr`, and `mort_term_conversion_defn` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

7.3.2 Mortality adjustment multiple

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

7.3.3 Term conversion adjustments

Term products often offer the option to convert to a permanent product (i.e. variable universal life) prior to the end of a level term period without additional underwriting.

Specify the original term issue date and whether the policy was issued as a result of a term conversion on the model point record. For such policies, the base mortality rates, mortality improvement factors, and mortality table adjustments will be calculated using a point-in-scale method based on the original issue date of the term policy. Additionally, a separate set of base mortality rates and mortality table adjustments may apply to the term conversion cohort of policies if specified.

Term conversion adjustments do not apply to joint life policies.

7.3.4 Underwriting class

Life insurance products can use the `underwriting_class` scalar to vary product assumptions such as mortality rates and acquisition expenses. The underwriting class is calculated by comparing the policy size against the nonmedical and paramedical underwriting limits specified with the experience mortality input. The underwriting class has a value of either:

- Nonmedical – when `ins_issued_per_pol` $\leq$ `nonmedical_limit`
- Paramedical – when `nonmedical_limit` $<$ `ins_issued_per_pol` $\leq$ `paramedical_limit`
- Medical – when `ins_issued_per_pol` $>$ `paramedical_limit`.

7.3.5 Rate class

Mortality rates varying by rate class require special treatment. The model point variable (`rate_class_mp`) can have a large number of options specific to the company's business. But the 2001 CSO mortality tables varying by rate class only define rates for Standard, Preferred, and Super. Thus there are two different rate class hidden variables:

- **Company rate classes** – The company's rate classes are in `mort_rate_class`. For example, there might be a dozen different classes for this scalar. Experience mortality and YRT reinsurance rates vary by this lookup key.
- **Reserve rate classes** – The 2001 CSO mortality rate classes are in `mort_rate_class_res`. The company's different classes are mapped into the three rate classes prescribed for reserve mortality rates. Customize the formula for this scalar, to define how each company rate class is translated into the three reserve rate classes.

7.3.6 Base mortality

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

7.3.7 Mortality improvement

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$.

When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

7.3.8 Mortality table adjustments

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

7.3.9 Substandard mortality

The substandard multiple on the model point record applies to the mortality rate after the multiple and addition. This same substandard multiple applies for COI and YRT reinsurance rates.

7.3.10 Joint life mortality

See the Joint life products chapter, below.

7.3.11 Monthly mortality

Deaths occur at the end of each month, prior to other decrements and policyholder transactions. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{1/12}$$

7.4 Lapses and surrenders

A termination is a 'lapse' when there is no policy value, and is a 'surrender' when there is a cash value.

Lapse rates apply to units after mortality decrements, which are processed before all other decrements.

7.4.1 Lapse adjustment multiple

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

7.4.2 Policy year lapse rates

The base lapse rate is adjusted by a multiple.

7.4.3 Excess lapses

Lapse rates can be dynamically adjusted according to policyholder behavior related to the competitive position of the weighted average variable fund values growth rate in the current month, compared to a competitor rate.

Excess lapses can occur when the average growth rate is below a specified competitor rate. When the average growth rate is higher than the competitor rate, excess lapses can be negative.

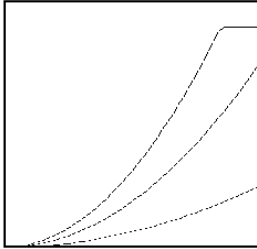
Maximum and minimum lapse rate variables in the excess lapse features affect the excess lapse calculations only. They do not limit the specified lapse rates. For example, the maximum lapse rate variable can reduce the excess lapse rate to zero, but cannot reduce the total lapse rate to be less than the specified lapse rate adjusted by the calendar date multiple.

Excess lapses can be calculated using either an exponential or arctangent formula.

7.4.4 Exponential excess lapses

The exponential formula produces results that look like figure 1, where the x-axis is the difference between the competitor rate and the average growth rate. The highest excess lapses in the above graph are for a policy with no surrender charge. The lowest excess lapses represent a policy with a large surrender charge.

Figure 1 Exponential formula



There are two versions of the exponential formula:

Increased lapses:

$$\begin{aligned}
 &\text{When competitor rate} > \text{avg growth rate} + \text{threshold} \\
 &\text{difference} = (\text{competitor rate} - \text{avg growth rate} - \text{threshold}) \times 100 \\
 &\text{surrender charge ratio} = 1 - \left(\frac{\text{cash value}}{\text{fund value}} \right) \\
 &\text{excess lapse} = \frac{\left[\text{multiple} \times (\text{difference}^{\text{exponent}}) \times (1 - \text{surrender charge multiple} \times \text{SC ratio}) \right]}{100} \\
 &\quad - \text{surrender charge subtraction} \times \text{SC ratio} \\
 &\text{excess lapse} \leq \max(0, \text{maximum lapse rate} - \text{planned lapse rate})
 \end{aligned}$$

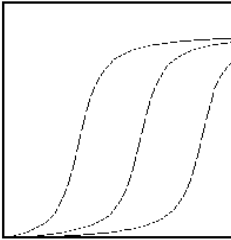
Decreased lapses:

$$\begin{aligned}
 &\text{When avg growth rate} > \text{competitor rate} + \text{threshold} \\
 &\text{difference} = (\text{avg growth rate} - \text{competitor rate} - \text{threshold}) \times 100 \\
 &\text{lapse reduction} = \frac{\left[\text{multiple} \times (\text{difference}^{\text{exponent}}) \right]}{100} \\
 &\text{lapse reduction} \leq \max(0, \text{planned lapse rate} - \text{minimum lapse rate})
 \end{aligned}$$

7.4.5 Arctangent excess lapses

The arctangent formula produces results that look like figure 2. The highest excess lapses in the above graph are for a policy with no surrender charge. The lowest curve has a large surrender charge.

Figure 2 Arctangent formula



difference = competitor rate – avg growth rate

$$\text{surrender charge ratio} = 1 - \left(\frac{\text{cash value}}{\text{fund value}} \right)$$

$$\text{excess lapse} = \frac{[(\text{maximum total lapse} - \text{base lapse}) \times (A - B)]}{(1 - B)}$$

total lapse = base lapse + excess lapse,

$\leq$ maximum total lapse, $\geq$ minimum total lapse

$$A = \frac{\arctan[100 \times \text{multiple} \times (\text{difference} - \text{SC ratio} \times \text{SC multiple} - \text{threshold})]}{(\pi/2)}$$

$$B = \frac{\arctan[-100 \times \text{multiple} \times (\text{SC ratio} \times \text{SC multiple} + \text{threshold})]}{(\pi/2)}$$

When competitor rate < avg growth rate : SC multiple = 0

The arctangent excess lapse formula shown above is a version of the following formula familiar to some users:

$$\text{total lapse rate} = a + b \times \arctan \left[m \times \left(\frac{\text{competitor rate} - \text{avg growth rate} - \text{SC ratio}}{c} \right) - n \right]$$

Base lapse rates = the result of the above formula when the average growth rate equals the competitor rate.

The formulas are identical when the parameters are converted as follows:

$$\begin{aligned}\text{Threshold} &= \frac{n}{m} \\ \text{Multiple} &= \frac{m}{100} \\ \text{SC multiple} &= \frac{1}{c} \\ \text{Maximum lapse rate} &= a + b \times \left(\frac{\pi}{2}\right)\end{aligned}$$

7.4.6 Monthly lapses

The annual lapse rate after all adjustments and excess lapses is constrained to be not more than 1.0 (100%).

Lapses and surrenders occur either: (1) at the end of each month, or (2) modally (just before the next premium is due). Lapses occur monthly after the last premium has been paid.

Each of the above lapse rate adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly or modal lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/\text{event frequency})}$$

Modal lapse timing does not apply to fund-related lapse, which is discussed below.

7.4.7 Fund-related lapse

The model point lapses when:

- Either the cash value or the fund value becomes zero or negative, and
- The GMDB test condition has been violated. However, lapse does not occur when premiums are adjusted to minimally fund the guarantee.

Fund-related lapse does not occur during the specified number of 'exception years' after issue.

8 RIDERS

8.1 Multiple riders

Multiple riders can apply to a model point.

8.1.1 Riders input

Each rider creates an instance of the vul_rider submodel. The input for each variable must be linked to an external source containing values varying by the rider names.

8.1.2 Rider model point input

The rider model point record identifies which riders apply to each model point.

8.2 Rider types

8.2.1 *Accidental death benefit*

Please refer to the L13 Riders book for a discussion of the accidental death benefit riders which can be linked to variable universal life model points.

8.2.2 *Waiver of premium*

Please refer to the L13 Riders book for a discussion of the waiver of premium riders which can be linked to variable universal life model points.

8.2.3 *Waiver of COI and other charges*

Please refer to the L13 Riders book for a discussion of the waiver of COI and other charges riders which can be linked to variable universal life model points.

9 REINSURANCE

9.1 Multiple reinsurance treaties

Multiple reinsurance treaties can apply to a model point.

9.1.1 *Treaty input*

Each reinsurance treaty creates an instance of the vul_ireins submodel. The input for each variable must be linked to an external source containing values varying by the treaty names.

The exception is that the YRT retention maximum per policy applies for all treaties combined, and thus does not vary by treaty. For example, it might vary by plan code.

For variable universal life, each treaty can be either a YRT or modified coinsurance treaty. ***Coinsurance treaties are not applicable to variable products since the separate account assets are not under management of the ceding company. An error message is produced if a coinsurance treaty is specified for a variable universal life model point.***

Each of these types is discussed in a separate section, below.

9.1.2 *Reinsurance model point input*

The reinsurance model point record identifies which treaties apply to each model point, as well as the ceded percent or amount for each treaty. The reinsurance model point layout is discussed in the Reinsurance model point records layout section, below.

9.2 YRT reinsurance and coinsurance of mortality risk

Yearly renewable term (YRT) reinsurance cedes a portion of the net amount at risk (NAR). Reinsurance premiums are calculated using YRT rates applied to the ceded coverage amount.

Coinsurance of mortality risk is similar to YRT reinsurance, but with reinsurance premiums based on the policy's cost of insurance rates.

9.2.1 *Ceded amount*

The ceded amount is a percent of the net amount at risk (NAR) at the beginning of each month, using the fund value at the end of the previous month. This is the total (variable, unloaned fixed, loan fixed, and DCA) fund value. When the unloaned fund value is negative, the NAR is calculated using zero unloaned fund value.

Using reinsurance model point variables

The ceded percent can be specified on the reinsurance model point record, either directly or as the initial ceded death benefit amount, which is converted to the ceded percent (i.e. ceded death benefit amount / death benefit at the projection date.) This is proportionate YRT reinsurance, where the ceded percent is fixed.

Using treaty variables

The ceded percent can be calculated using the reinsurance input variables that vary by treaty code. The ceded amount is a specified percent of any excess of the NAR over the retention threshold per policy. For example:

- **First dollar quota share** – The threshold is zero.
- **Excess retention** – The ceded percent applies to the excess of the NAR over the specified threshold.

The ceded percent varies over time, depending on the changing relationship between the NAR and the threshold.

9.2.2 *Maximum YRT retention per policy*

The YRT reinsurance treaty amounts for a model point are increased when the combined ceded amounts would be less than [the NAR less the maximum retention amount per policy]. Specify the maximum retention amount in the trad submodel; this variable applies for the product instead of the vul_ireins variables that vary by treaty.

This adjustment applies only for YRT treaties that have the ceded amount calculated using the treaty variables. Treaties that have the ceded percent/amount specified on the model point record are not adjusted.

When the sum of the ceded amounts is too low, the initial (at the projection date, or issue date for new business model points) ceded amount for each treaty is increased in proportion to the individual ceded amounts.

9.2.3 *YRT rate*

The YRT reinsurance rate for each treaty is a base table, times a multiple. These are 'mortality table' variables, which operate the same as for mortality rates.

The YRT premium can be zero for the first policy year by specify the multiple using a Mortality Selector (or Select and Ultimate) table style with zero rates for select duration 0.

Term conversion adjustments

Term products often offer the option to convert to a permanent product (i.e. variable universal life) prior to the end of a level term period without additional underwriting.

Specify the original term issue date and whether the policy was issued as a result of a term conversion on the model point record. For such policies, the YRT base rate and YRT multiple will be calculated using a point-in-scale method based on the original issue date of the term policy. Additionally, a separate set of YRT base rates and YRT multiples may apply to the term conversion cohort of policies if specified.

YRT term conversion adjustments may be turned off using the yrt_term_conversion_defn variable (on the LifeReinsurance input page).

YRT term conversion adjustments do not apply to joint life policies.

Substandard multiple

The substandard multiple on the model point record applies to the YRT rate. This is the same substandard multiple as applies for experience mortality rates.

9.2.4 **Reinsurance premium**

YRT reinsurance premiums are paid either annually at the beginning of each policy year, or monthly.

The YRT reinsurance premium per policy is:

$$\text{ceded amount} \times (\text{annual YRT rate}) + \text{reinsurance policy fee}$$

When YRT premiums are paid monthly, one twelfth of this amount is paid at the beginning of each month.

The reinsurer refunds the unearned YRT reinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual YRT premium, times the remaining months in the year divided by 12.

9.2.5 **Reinsurance benefits**

The reinsurance benefit paid to the ceding company each month is the ceded amount times the monthly experience mortality rate. The ceded amount is the amount calculated at the time of each YRT reinsurance premium.

9.2.6 **Reinsurance reserve**

The formulaic statutory and tax reserve credit for YRT reinsurance is the tabular cost of insurance floor for the ceded amount.

9.3 **Modified coinsurance**

Modified coinsurance (ModCo) treaties operate similarly to coinsurance treaties, except that the reinsurance reserve is retained by the ceding company. The ceding company reimburses the reinsurer for interest on the reinsurance reserve which is backed by assets invested by the ceding company.

The adjustments occur monthly, so that nested projections match standard projection results. The adjustment amount is the current reinsurance reserve, less the reinsurance reserve as of the previous adjustment date accumulated with interest.

The ModCo reserve adjustment interest rate is the asset yield rate.

Modified coinsurance treaties cede a portion of the entire policy, including transferring the ceded reserve to the reinsurer. Reinsurance benefits include all benefits provided by the policy.

9.3.1 **Coinsurance percent**

When the coinsurance percent is calculated using the input for the treaty, the percent is x% of the sum insured at issue up to the threshold per policy specified for each treaty, plus y% of any excess over the threshold. When the coinsurance percent is specified by model point, this treaty input is ignored. The coinsurance percent remains constant to maturity.

The coinsurance percent applies to the entire policy, including the deficiency reserve and the additional reserve.

The coinsurance benefits paid to the ceding company each month are the projected policy benefits, times the coinsurance percent.

9.3.2 **Coinsurance premium**

Coinsurance premiums are paid at the same premium frequency as the model point. Coinsurance premiums are based on the gross premium, including policy fee.

The reinsurer refunds the unearned coinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual amount times the remaining months in the year divided by 12.

9.3.3 Premium tax

The reinsurer reimburses for the coinsurance percent times premium taxes paid by the ceding company.

9.3.4 Benefits

The reinsurer reimburses for the coinsurance percent times all benefits paid by the ceding company. Reimbursements are split between death benefits and other benefits for reporting purposes.

9.3.5 Expense allowance

The reinsurer pays an expense allowance to the ceding company at the reinsurance premium frequency. The expense allowance is the annualized reinsurance premium times a specified percent.

The unearned current year expense allowance is refunded to the reinsurer when the policy terminates before the next reinsurance premium payment for any reason. The unearned expense allowance is the last amount paid, times the ratio of the number of months until the next reinsurance premium, divided by [12 divided by the reinsurance premium frequency].

9.3.6 Reinsurance reserve

There is no reserve credit for modified coinsurance.

9.4 Reinsurance administration expenses

Reinsurance administration expenses are assumed to be incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include:

- Amount per policy
- Percent of the ceded amount.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Modified coinsurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for reinsurance.

10 SOLVING CALCULATIONS

Solving calculation results are amounts per unit in force.

10.1 On-the-fly calculations

10.1.1 Values calculated

The following value is calculated as needed during projections. Calculations use the death benefit option and minimum death benefit (corridor) percent table which applies for the model point. Partial withdrawals and policy loans are ignored.

Result	Interest	COI	Loads	Premium mode	Death benefit
Guaranteed maturity premium (GMP), for UL CRVM, endowing at the sum assured [solved_gmp]	Guaranteed*	Guaranteed	Guaranteed	Modal	Guaranteed fund

10.1.2 Solving iterations

When the closed form solution does not apply (see below), these results are solved by iterating the appropriate appropriate instance of the vul_ifund submodel, projecting using successive trial premiums. These calculations are performed by the solving_driver function.

10.1.3 Displaying solving results

Use the variable solving_display_defn (in the Product input page in the Premiums category) to have the Run Log show the series of solving guesses and related fund value results. The output for the vul_ifund submodel then includes the column values for each solving fund projection using the final solved premium result.

This variable should of course be set to No for production runs, in order to improve calculation time and to reduce the size of the output.

10.2 Closed form solutions

10.2.1 Calculation method

Closed form solution calculations can replace the solving iterations to reduce calculation time significantly.

Calculations for level premium results are based on an approach which creates accumulation factors for the various elements which develop the ending fund value. The closed form solution for the premium is a formula which sums the target fund value with factors for charges and loads and divides by a factor for the premium. These solutions were developed using techniques presented in the paper 'Calculating Funding Premiums For Universal Life Insurance' by Calvin D. Cherry, on pages 20-27 of the North American Actuarial Journal, Volume 4, Number 2.

10.2.2 Using solving versus closed form solution

You can choose which calculation method applies. Solving iterations return the exact result, but increase the calculation time. Closed form solutions are significantly faster, but are not appropriate in all cases.

Use the model point variable closed_form_prem_defn to control which type of calculation applies:

- **When Appropriate** – Closed form solutions are used as much as possible; see the Validation topic, below. The closed form result is tested against fund value calculations; when this test fails, solving calculations replace the closed form result to assure accuracy for the projection.
- **No** – Solving iterations apply in all cases. For example, this option can be used to compare against closed form solution results.
- **Yes** – Use closed form solutions in all cases. When the closed form result is tested (against fund value calculations), the Run Log turns yellow and contains a warning when the ending fund value does not match the target value within tolerance limits. This choice reduces calculation time when the accuracy provided by solving iterations is not important for the projection. You should evaluate the loss of accuracy by comparing against a test projection using the When Appropriate choice.

10.2.3 Validation for using closed form solution

The When Appropriate choice is validated against the model point's product definition, in order to determine whether the closed form solution can be used. Solving iterations are used when the product has

specifications which are likely to produce a closed form result different from the solving result. The following product specifications cause solving iterations to be used for the When Appropriate choice:

- Plus ROP death benefit
- Persistency bonus
- Non-level percent of premium load different between target and excess premiums when premiums are paid more frequently than annually
- Percent of fund value load
- Minimum death benefit percent (corridor factor) is greater than 1.0 at the GMP maturity age
- Plus Fund death benefit.

10.2.4 Displaying closed form results

Use the variable `solving_display_defn` (in the Product input page in the Premiums category) to have the Run Log show the closed form solution results. When the closed form solution does not produce the expected ending fund value, and `closed_form_prem_defn` is When Appropriate, the resulting solving calculations are also displayed and the final solving iteration values are included in the `vul_ifund` output results.

This variable should of course be set to No for production runs, in order to improve calculation time and to reduce the size of the output.

10.3 Solving result arrays

Following the solving calculations discussed above, the following array values are calculated for use during the projection:

Result	Frequency	Values
Guaranteed maturity fund [<code>guar_maturity_fund</code>]	Projection months	Guaranteed fund accumulation using solved GMP
Guaranteed maturity COI charge [<code>guar_maturity_coi_chg</code>]	Projection months from issue	Guaranteed fund COI charge using solved GMP, used for waiver of COI benefits
CRVM reserve death benefits [<code>dth_ben_tbl_crvm</code>]	Each policy year	Average of monthly death benefits paying GMP using guaranteed fund

10.4 Factors calculations

A projection type calculates 'factors' when the product submodel is the top model and the target column is `factors_tgt`. This target column directs the calculation of each item specified (discussed below). The target column calculates results for each policy year; thus the Projection period from and to properties have no effect on factors calculations.

The Assumption Set Restore property (in the run parameters object) must be Yes, since this is a seriatim projection. Project variables such as pricing definition do not apply for this special type of projection.

10.4.1 Values calculated

Factors calculations use iterative solving (closed form solutions do not apply) to find the following values for each model point:

Result	Interest	COI	Loads	Premium mode	Death benefit
Base fund level premium (to produce specified fund value per unit at target duration)	Specified	Current or guaranteed	Current or guaranteed	Modal	Actual
GMDB level premium (accumulates to ending required premiums accumulation)	Guarantee interest rate	N/A	N/A	Modal	N/A
Guaranteed maturity premium (GMP)	Guaranteed	Guaranteed	Guaranteed	Modal	Actual
Guideline single premium (GSP)	Specified* (6%)	Guaranteed or current	Current or guaranteed	N/A	Sum assured
Guideline level premium (GLP)	Specified* (4%)	Guaranteed or current	Current or guaranteed	Annual	Actual
7-pay MEC premium	Specified* (4%)	Guaranteed or current	N/A	Annual	Sum assured
Cash value accumulation test (CVAT) net single premiums, and the related minimum death benefit percentages	Guaranteed	Guaranteed	Ignore	N/A	Sum assured without corridor

*The interest rate for IRS premium restrictions is the greater of the specified interest rate and the guaranteed crediting rate.

10.4.2 Policy fee

The policy fee is included in calculating the target premium and GMDB level premium. It is excluded for GMP, GLP, GSP, MEC, and CVAT, except for the charging of the policy fee as a product load.

10.4.3 Model points

Factors calculations systematically process for each model point. Set up a model point for each issue age to be calculated, within each combination of insured characteristics (sex, smoking class, rate class, etc.). For example, you might set up model points for issue ages 0 through 80 for male nonsmokers, then repeating these model points for male smokers, then female nonsmokers, then finally female smokers.

When Annual mode premiums are to be calculated for factors using modal premiums, set the model point premium mode to Annual.

Best practice is to create the report group for each model point so that it identifies the model point characteristics. For example, the report group might be PlanA-M-NS-ST-35 for the model point calculating factors for plan code PlanA, for a male nonsmoker standard risk class issue age 35. Then the report groups identify the meaning of each model point.

The same as for Pricing cell projections, each model point must be issued on the first day of the projection. For example, when the projection date is 2013/12 (end of December, 2013), the issue date must be 2014/01/01 (January 1, 2014).

10.4.4 Factors premium reporting

The solved premium is reported in the column `factor_prem` in the `vul_ifund` submodel. The standard features round this result up to the next whole penny per unit.

The output file includes this value for every model point. Since the premium result shows only for the commencement period ($t = 0$) in this column, the output from and to properties should both be '0'.

10.4.5 Cash value test reporting

The solved cash value test net single premium results are reported in the column `factor_cvat_min_dth_ben_factors` in the `vul_ifund` submodel. The standard features convert the result into the corresponding minimum death benefit multiple, equal to the sum assured divided by the fund value at the end of the previous policy year. These fund values are projected assuming the solved single premium is paid at issue.

Since these values over time, the output to property must be the number of months to maturity of the youngest issue age. As for the factors premium reporting, use a tool to query the output database in order to retrieve the values for every model point.

10.5 Factors projection setup

10.5.1 Input variables

The special variables for factors calculations are at the bottom of the Product input page.

- `factors_test_num` – The factors calculations progress through the different factors types using this number as the lookup key.
- `factors_defn` – The result to be calculated; this option list variable is used as a condition in formulas.
- `factors_include_defn` – Yes to include this calculation, No to exclude it (these lines can be deleted from the external source input if desired).
- `factors_tgt_fund_val` – Target fund value per unit.
- `factors_tgt_dur_defn` and `factors_tgt_dur` – The target duration is a specified attained age or policy anniversary, maturity age, or GMP maturity age.
- `factors_prem_term_defn` and `factors_prem_term` – The premium term is either a specified attained age or policy anniversary, the premium term, or the contractual premium term. When the premium term exceeds the target duration, the solving is of course not affected by premiums after the target fund value duration.
- `factors_coi_rate_defn` – Either the current or guaranteed COI rates.
- `factors_load_rate_defn` – Either the current or guaranteed or using zero loads.
- `factors_int_rate` – The greater of the guaranteed rate and the specified interest rate.

10.5.2 Run parameters

Specify the following settings in the run parameters for the projection task calculating factors:

Property	Value
Target Colum	<code>vul->factors_tgt</code>
Projection Period From	0

Property	Value
Projection Period To	0 (since solving calculations are used, this is not a forward projection)
Projection Period Min	0
Projection Period Max	0 for factors premium output; longest months to maturity for cash value test output
Projection Loops	1
Assumption Set Restore	Yes
proj_date	yyyy/mm, the end of the month before the issue date

11 MODEL POINT RECORDS

11.1 Model point record layout

The model point record layout for the variable universal life product type is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Values
A	group	up to 125 characters (assigns model point to report group)
B	seg_id	segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date_mp	yyyy/mm/dd or mm/dd/yyyy
F	disab_date_mp	yyyy/mm/dd or mm/dd/yyyy (set to 1/1/9999 if policy is not disabled)
G	term_conversion_defn_mp	Yes (calculate term conversion adjustments), No
H	term_issue_date_mp	None (when original term issue date is unavailable) yyyy/mm/dd or mm/dd/yyyy
I	plan_code	up to 20 characters (link to assumptions by plan)
J	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)

Excel Column	Variable	Values
K	tax_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
L	issue_age_mp	positive integer
M	gender_mp	Male, Female, Unisex
N	smoking_class_mp	Nonsmoker, Smoker, Aggregate
O	rate_class_mp	Standard, Preferred, Super, [extend with user additions]
P	substd_mult_mp	substandard multiple (1.0 for 100%)
Q	issue_age_mp_life2	life 2 values, when applicable
R	gender_mp_life2	
S	smoking_class_mp_life2	
T	rate_class_mp_life2	
U	substd_mult_mp_life2	
V	prem_mode	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
W	prem_term_mp	premium years (0 for 'to GMP maturity')
X	init_annualzd_prem	initial annualized premium for mp
Y	single_prem	additional single premium for mp
Z	dca_pct_single_prem	percentage of single premium paid at issue allocated to the DCA account
AA	dca_mths	number of months from issue to perform DCA transfers
AB	closed_form_prem_defn	When Appropriate (use solving only when needed), No (force the use of solving calculations), Yes (force the use of closed form solutions)
AC	dth_ben_defn	Sum Assured, Plus Fund, Plus ROP
AD	dth_ben_min_pct_defn_mp	Product Specification (use product definition), Guideline Prem Test, Cash Value Test
AE	init_sum_assured_mp	initial sum assured for mp (0 = calculate from initial units)

Excel Column	Variable	Values
AF	init_policies	initial policies for mp (required)
AG	init_units	initial units for mp (0 = calculate from initial sum assured)
AH	init_av_dca	initial DCA account value for mp
AI	init_av_fixed	initial unloaned fixed account value for mp
AJ	init_av_index0	initial account value represented by index 0 for mp
AK	init_av_index1	initial account value represented by index 1 for mp
AL	init_av_index2	initial account value represented by index 2 for mp
AM	init_av_index3	initial account value represented by index 3 for mp
AN	init_av_index4	initial account value represented by index 4 for mp
AO	init_av_index5	initial account value represented by index 5 for mp
AP	prem_pct_fixed	percentage of premium to allocate to the fixed account
AQ	prem_pct_index0	percentage of premium to allocate to the account for index 0
AR	prem_pct_index1	percentage of premium to allocate to the account for index 1
AS	prem_pct_index2	percentage of premium to allocate to the account for index 2
AT	prem_pct_index3	percentage of premium to allocate to the account for index 3
AU	prem_pct_index4	percentage of premium to allocate to the account for index 4
AV	prem_pct_index5	percentage of premium to allocate to the account for index 5
AW	init_crediting_rate_base	initial base crediting rate (0.05 for 5.0%); when this value is the 0.0 default value the initial rate is calculated using the crediting strategy
AX	init_gross_prem_cumul	initial cumulative premiums for mp
AY	init_withdrl_cumul	initial cumulative withdrawals for mp

Excel Column	Variable	Values
AZ	init_sg_prem_accum	initial GMDb accumulated premiums for mp
BA	init_gmdb_mths_in_grace_period	consecutive months that the GMDb conditions have not been met for a model point being kept in force by a positive current fund value
BB	init_gmdb_funding_indicator	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BC	init_stat_aalr	initial statutory GMDb AG 37 attained age level reserve for mp
BD	init_tax_aalr	initial tax GMDb AG 37 attained age level reserve for mp
BE	init_pol_loan	initial policy loan for mp
BF	init_surr_chg_withdrl_reductn_mult	average initial surrender charge reduction multiple from previous withdrawals for mp (1.0 for 100%)
BG	init_comm_subject_to_chargeback	initial commission subject to chargeback for mp (-1 to calculate from issue)

11.2 Reinsurance model point record layout

The reinsurance model point record identifies which treaties apply for each model point, using the linked variable, mp_id. The reinsurance model point record can have any number of reinsurance treaties per model point. It also specifies the ceded percent or amount for each treaty as described in the table below. If the reinsurance model points are omitted from the reinsurance model point file, then the liability model point will not include reinsurance.

Warning! The order of reinsurance model points must be in the same order as the liability model points.

The reinsurance model point record layout is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	mp_id	Up to 12 characters. Links to liability data file mp_id. This value can be listed any number of times to indicate the number of treaties applied to the policy.
B	treaty_code	A user defined input that references the treaty code in the Resinsurance.xlsm workbook.

Excel Column	Variable	Value
C	ceded_amt_defn	Percent - the ceded amount should be input as a percent from 0 to 1 Dollar Amount - the ceded amount should be input as the ceded death benefit amount Treaty Input - the ceded amount should be input as 0, and the ceded amount will be calculated using the treaty variables
D	ceded_amt	See above. Input the ceded percent, the ceded dollar amount, or 0 if the ceded amount is specified by the treaty.

11.3 Units basis

Product values are calculated per unit in force, then multiplied by the current number of units.

11.3.1 Initial units

The initial policies must be specified on each model point record, along with either or both of the initial units and initial sum assured. These are the aggregate values for the policies collected into each model point.

When the initial units are not specified on the model point (this variable is left at its zero default value), the initial units are the initial sum assured divided by the sum assured per unit specified for the product.

When the initial sum assured is not specified on the model point (this variable is left at its zero default value), the initial units variable must be specified. Otherwise the model point is skipped, and a warning is issued in the Run log. When only one of these variables is specified, the sum assured per unit is the variable specified for the product.

When both the initial units and initial sum assured are specified on the model point record, however, the sum assured per unit is the initial sum assured divided by the initial units, then multiplied by the change in the sum assured per unit variable as the projection progresses. For example, the initial sum assured per unit for an in-force model point can be less than the product variable value when partial withdrawals have reduced the sum assured before the projection date.

The initial policies must be specified for each model point. The average size policy is the initial units divided by the initial policies, and the initial sum assured divided by the initial policies.

11.3.2 Pricing cells

For model points issued at the start of a serial pricing projection, the initial policies, units, and sum assured fields define the policy size. When the model point specifies one policy, the initial units and sum assured are average size amounts.

11.4 Required model point variables

11.4.1 Always required

The following model point variables are required in all cases:

- **Model point identification:** group and mp_id
- **Insured characteristics:** issue_age, gender, smoking_class, rate_class_mp, and substd_mult, for life 1 in all cases, and also for life 2 for joint life products

- **Policy specifications:** issue_date_mp, term_conversion_defn_mp, term_issue_date_mp, plan_code, stat_valn_code, tax_valn_code, prem_mode, prem_term_mp (may be default of zero which indicates that premiums are paid in all years), dth_ben_defn, and dth_ben_min_pct_defn_mp
- **Initial policies:** init_policies
- **Units basis fields:** one or both of init_sum_assured and init_units
- **Closed form solution usage:** closed_form_prem_defn
- **Annual premium:** init_annualized_prem is required when the annual premium is specified by model point.
- **Single premium:** single_prem is required for new business model points. It is required for in-force model points in these conditions: (1) the model point is still in the first year and the premium load varies between target and excess premium, or (2) the commission subject to chargeback is being calculated from issue.
- **AV/premium allocations:** One or both of the account value or premium allocation fields for each account. For inforce model points, account value allocations must be specified. When both are specified, the premium allocations are based on the premium allocation fields. Otherwise the premium allocations are determined using the initial account value allocations. For new issue model points, either set of fields can be specified.

11.4.2 In-force model point values

In-force model points are driven by all the 'initial values' variables, which specify the aggregate model point values at the projection date.

New business model points (model points with issue date after the projection date), however, ignore these variables. The exception is that the initial policies must be specified, along with either or both of the initial units and initial sum assured.

The following table shows when each in-force values variable is required for in-force model points:

Variable	Required When
init_av_xxx	always required for the accounts which are used
init_crediting_rate_base	always required
init_gross_prem_cumul	model points with current or future policy loan, for products with the preferred policy loan feature
init_withdrl_cumul	
init_gmdb_accum	products with GMDB
init_stat_aalr	products with GMDB AG 37 statutory reserves
init_tax_aalr	products with GMDB AG 37 tax reserves
init_pol_loan	model points with policy loan
init_surr_chg_withdrl_reductn_mult	model points with previous withdrawals, for products with surrender charges other than percent of fund value (may be left at 1 default to ignore this initial effect)

Variable	Required When
init_gmdb_mths_in_grace_period	model points with positive current fund value but unsatisfied GMDB conditions
init_comm_subject_to_chargeback	model points still subject to commission chargeback (may be left at -1 default to calculate from issue)

11.4.3 Variables to ignore

The following variables are needed only for nested projections, and should be ignored. These values must be carried forward to the next time step in order to reproduce the results from standard projections:

Variable
init_gmdb_funding_indicator
init_surr_chg_withdrl_reductn_mult
init_gmdb_mths_in_grace_period
init_comm_subject_to_chargeback

12 SUBMODELS STRUCTURE

12.1 vul submodel

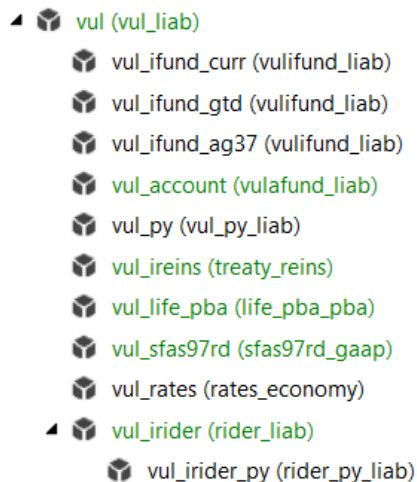
The vul submodel calculates the model point's contribution to the liability income statement and balance sheet items for variable universal life products.

The vul submodel calculates model point results using values per unit in force, which are then multiplied by the number of units. Results per unit in force have '_per_unit' added to the formula name. Formula names without this addition are results per model point (or are a special result like a ratio).

12.1.1 Seriatim versus portfolio runs

Seriatim projection

When vul is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the vul submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings



Portfolio projection

For portfolio projections (which include business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the vul submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the vul submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the vul submodel the same columns as for other product types. This makes the variable universal life product type interchangeable with other product types fitting into this business segment structure.

12.1.2 Variables

Variables for variable universal life products are located in the vul submodel, except as noted below for the variables in vul_ifund, vul_account, and vul_py.

12.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when vul is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the vul submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only vul is the top model, (2) when vul is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when vul is run as the top model and when vul is included in projections with segment as the top model. References to company, segment and liab are invalid when vul is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) solving values are initialized, (3) temporary tables are sized for DAC tax calculations and competitor rates, (4) projection flags are created and initialized, (5) reporting groups are set up, (6) vul_ifund elements are setup, (7) vul_account elements are created, (8) vul_ireins elements are created, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each projection task loop, immediately after startup when vul is the top model. When segment is the top model, initialize is executed by calls from seg->initialize which executes vul->initialize for each model point in the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when ul is running as the top model. This includes executing the initialization of economic scenario rates and the setup of competitor rates. Other initializations are required in both cases.

The main use of initialize is to orchestrate the various solving calculations for each model point, and to precalculate the per unit reserve amounts using the actuarial math library. Since these initializations are not required for all calculation types for all product features, conditions are checked to shortcut the initialization process, thus reducing the runtime of the model point.

The solving calculations, and the filling of the associated array values, is discussed in the Solving calculations chapter, above.

The reserves are initialized in the following order:

1. Base statutory reserves
2. Base statutory minimum reserves
3. Base tax reserves.

solving_driver

The 'solving_driver' function is used to iterate the vul_ifund submodel to solve for results when the closed form solution is not used, or produces an ending fund value that does not match the target value within the tolerance limit. See the Solving calculations chapter, above.

Each call to this solving_driver function includes the appropriate fund_accum_defn for vul_ifund. See the vul_ifund submodel section, below.

Following some initial setup, a 'while' loop is processed. Within each pass of the loop, a trial premium is used to project vul_ifund, then the projected fund value is compared to the solving target. A new trial premium is used to re-project vul_ifund until the solution has been determined within a tolerance limit.

Details of the solving process and results from the final solve can be captured when solving_display_defn (in the Product input page in the Premiums category) is Yes. The solve guesses are written to the Run Log, and

the final solving fund accumulation is written to the column output using a writeModel statement. This option should be No for production runs.

calc_prem

The function 'calc_prem' calculates the closed form solutions for premium results using the Accumulation factors or Fund rollback methods. See the Solving calculations chapter, above. Each call to this calc_prem function includes the fund_accum_defn which would have applied for solving for the result using iterations of vul_ifund.

The closed form results are written to the Run Log when solving_display_defn (in the Product input page in the Premiums category) is Yes. This option should be No for production runs.

stat_res_ulcrvm_per_unit

Columns 'stat_res_ulcrvm_per_unit' (and the corresponding column for tax reserves) are special, in the sense that they include a future projection of the guaranteed_fund instance of vul_ifund. This future projection requires that these columns reset and recalculate the guaranteed_fund instance of vul_ifund. When UL CRVM reserves require the GMF to be recalculated, this instance of vul_ifund is recalculated beginning from the current fund value, using guaranteed rates and assuming the GMP is paid in the future.

stat_ag37_aalr_pv_gmdb_per_unit, stat_ag37_oyt_res_per_unit

Columns 'stat_ag37_aalr_pv_gmdb_per_unit', 'stat_ag37_oyt_res_per_unit' (and the corresponding columns for tax reserves) are also special. These require a future projection of the AG 37 instance of vul_ifund. This future projection requires that these columns reset and recalculate the AG 37 instance of vul_ifund. When VUL AG 37 reserves require the AALR and OYT amounts to be recalculated, this instance of vul_ifund is recalculated beginning from the current fund value, using AG 37 rates and projection assumptions, including payment of the catch-up premium.

Other formulas

Other formulas in the vul submodel are relatively straightforward. Results which are per unit in force add '_per_unit' to the end of the formula name. Results without '_per_unit' are values for the model point.

Most of the profit, balance sheet, and GAAP columns are only calculated when vul is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

12.2 vul_ifund_xxx submodels

The vul_ifund_xxx submodels are three instances of the vul_ifund_liab model class. They contain the variables and formulas required to calculate per unit in force individual fund values for variable universal life model points. Since all results are per unit in force, this submodel does not include '_per_unit' in the formula names.

12.2.1 Multiple instances

Each model point carries three unique instances of the vul_ifund_liab model class:

- vul_ifund_curr, for the current fund value, which is always present.
- vul_ifund_gtd, for the guaranteed fund, which is projected for solving calculations and for creating related array values (see the Solving calculations chapter, above), and when calculating present values for UL CRVM reserves.
- vul_ifund_ag37, for the AG 37 fund value, which is projected for determining the attained age level reserve and one year term reserve.

These unique elements of vul_ifund are setup during the startup processing of the model point.

Formula references to the unique instances utilize the unique model reference pointer. References for each fund value basis follow this example:

- Current: `sm_vul_ifund_curr->fund_val_aft_withdrl`
- Guaranteed: `sm_vul_ifund_gtd->fund_val_aft_withdrl`
- AG 37: `sm_vul_ifund_ag37->fund_val_aft_withdrl`

12.2.2 Working submodel

A specific `vul_ifund` instance may be a 'working' submodel which is reset at various times as required for solving calculations, and for reserve projections starting from the current fund value. Accordingly the values in these columns are the end of a projection apply only for the last time this submodel was calculated. For example, the `vul_ifund_gtd` values are not the guaranteed fund values calculated from issue; they could be the calculations starting from the current fund value at the last time when UL CRVM reserve calculations were processed.

To see a particular set of results, customize to add code to write out the submodel results at the time that calculation has just been performed. Search for the 'writeModel' command in the `vul->initialize` formula to see how this is done when the `solving_display_defn` calls for writing out the submodel values

12.2.3 Variables

Variables for COI rates, product loads, and persistency bonuses are located in the `vul_ifund` submodel. These variables are contained in the `vul_ifund` submodel so that they can vary between current, guaranteed, and AG 37 fund calculations.

In addition to the variables discussed above, two variables in `vul_ifund` are hidden from the data entry but are critically important to the successful operation of the `vul_ifund` elements:

fund_val_basis

The 'fund_val_basis' variable is an option choice variable with choices of Current, Guaranteed, and AG 37e. These choices provide for different bases for the COI, product loads, and persistency bonus product features.

After each `vul_ifund` element is setup inside 'startup' of `vul`, this variable of the `vul_ifund` element is set.

fund_accum_defn

While the `fund_val_basis` variable is used to map to the data, a second `vul_ifund` variable, 'fund_accum_defn' is used to condition the formulas.

The `fund_accum_defn` variable is also a choice list variable with choices which provide for execution of specific formulas for each fund accumulation definition type:

- Current Fund – Formulas for the current fund value.
- Guaranteed Fund – Formulas for calculating: (1) guaranteed maturity premium and fund value, (2) average death benefit for CRVM reserves assuming the GMP is paid, (3) guaranteed level premium and fund value,
- AG 37 Fund – Formulas for the fund value for AG 37 AALR and OYT calculations.

12.2.4 Formulas

Formulas for projecting variable universal life model point fund values are contained in the `vul_ifund` submodel. A combination of columns, scalars, and external functions are used to provide the required functionality.

Startup

The startup column of each vul_ifund instance is called during the setup of the model.

The guaranteed_fund and ag_37_fund instances of the vul_ifund submodel can be reset and recalculated multiple times during the projection. This is controlled by vul columns like initialize, stat_res_ulcrvm_per_unit, stat_ag37_aalr_pv_gmdb_per_unit, stat_ag37_oyt_res_per_unit, and by the solving driver function, where vul_ifund is reset or rebased, then re-projected. As discussed above, the vul_ifund formulas include variations for different values of fund_accum_defn.

Although there are instances where fund_val_basis could be used in the formulas, best practice is to use fund_accum_defn exclusively in the formulas and let fund_val_basis be used only to map the appropriate data to the vul_ifund elements.

12.3 vul_account submodel

The vul_account submodel is a user defined array instance of the vulafund_liab model class and contains all variables and formulas required to calculate per unit fixed, DCA, and separate account variable fund values for variable universal life model points.

12.3.1 Multiple instances

Each model point carries between one and eight instances of vul_account. There is always a fixed fund, and a DCA fund when DCA is included. The separate account variable funds are distinguished by fund_id, which has values of:

- Index0, for the first separate account variable fund value.
- Index1, for the second separate account variable fund value.
- Index2, for the third separate account variable fund value.
- Index3, for the fourth separate account variable fund value.
- Index4, for the fifth separate account variable fund value.
- Index5, for the sixth separate account variable fund value.

Elements of vul_account are created during the startup processing of the model point. Conditions are evaluated in startup of vul and the vul_account user defined array is resized as required. The characteristic for each vul_account element is specified and the assumptions are read using the ms_rereadGrid() system function.

Specific instances of the vul_account submodel are aggregated using the 'sum_over_funds' function in the vul submodel.

The vul_account array element numbers range from 0 to 7. The actual integer value for each array depends on which accounts exist for a particular model point. Element numbers, beginning with element 0, are assigned to each account in the following order:

- **DCA fund** – the first element number is assigned to the DCA fund, if applicable.
- **Fixed fund** – the next lowest element number is assigned to the fixed fund, if applicable.
- **Separate account variable funds** – the separate account variable funds are then assigned to element numbers in order. Index0 would be assigned to the next lowest element number and Index5 would be assigned to the highest element number.

12.3.2 Variables

Variables for the index name, investment management fee, and investment fee refunds for the separate account variable funds are contained in the vul_account submodel. These variables are contained in the vul_account submodel so that they can vary by variable fund..

12.3.3 *Formulas*

Formulas for projecting variable universal life model point accounts are contained in the `vul_account` submodel. A combination of columns, scalars, and external functions are used to provide the required functionality.

Startup

Since there are no instances of `vul_account` defined through the model parameters, the initial size of the `vul_account` array is zero, and the 'startup' column is not called automatically. The startup column is called when the `vul_account` array is resized using the `sm_vul_account.resize(count)` statement from within the startup column of the `va` submodel.

12.4 `vul_irider`

The `vul_irider` submodel is a dependent data driven array instance of the `rider_liab` model class containing the variables and formulas required to calculate the rider cash flows and reserves for variable universal life model points.

Please refer to the L13 Riders book.

12.5 `vul_ireins` submodel

The `vul_ireins` submodel is a dependent data driven array instance of the `treaty_reins` model class containing the variables and formulas required to calculate the ceded amount and reinsurance cash flows for variable universal life model points.

12.5.1 *Multiple instances*

Each model point carries multiple instances of `vul_ireins`. The number of instances is determined by the number of treaties specified on the reinsurance model point record.

Elements of `vul_ireins` are created during the startup processing of the model point. Conditions are evaluated in startup of `vul` and the `vul_ireins` user defined array is resized as required. The characteristic for each `vul_ireins` element is specified by the treaty code.

The ceded amount and reinsurance cash flows for multiple treaties are aggregated using the `sum_over_reinsurance` function in the `vul` submodel. This function can aggregate the values for all treaties, or just the treaties of a specific type of reinsurance, such as YRT.

12.5.2 *Variables*

Variables for the reinsurance type, the reinsurance expenses, and for calculating the ceded amount (when it is not specified on the reinsurance model point record), are located in the `vul_ireins` submodel. These variables are contained in the `vul_ireins` submodel so that they vary by treaty.

12.5.3 *Formulas*

Formulas for calculating the ceded amount and the reinsurance cash flows are contained in the `vul_ireins` submodel.

Since `vul_ireins` is defined through the model parameters, the size of the `vul_ireins` array is based on the number of treaties specified for each model point on the reinsurance model point file.

12.6 `vul_py` submodel

The `vul_py` submodel is a standalone instance of the `vul_py_liab` model class. It contains the variables and formulas required to calculate per unit in force results which only vary by policy year. Thus, 't' in this

submodel represents policy year, unlike the vul, vul_ifund, vul_account, and vul_ireins submodels where 't' represents months since the projection date. The policy year values are calculated from issue. Since all results are per unit, this submodel does not include '_per_unit' in the formula names.

12.6.1 Variables

Variables located in the vul_py submodel are only used in formulas within this submodel. These are generally for values calculated from issue and varying by policy year, such as annual premiums, and cash value and reserve terminal values and net premiums.

12.6.2 Formulas

Formulas for calculating policy year per unit in force results are included in the vul_py submodel. Formulas for vul_py are used to calculate annual and target premiums, plus reserve values by policy year.

The vul_py submodel also includes formulas which Frasierizes the experience mortality, which are Frasierized using annual rates. COI rates, however, are Frasierized in the vul_ifund submodel since these rates vary between the current, guaranteed, and AG 37 fund rates bases.

Columns are generally calculated beginning at policy year 1, even for in force model points. As discussed in the vul submodel section, the reserve terminal values and net premiums are calculated by calling the Actuarial math library. See the Actuarial math library section below.

12.7 vul_rates submodel

The vul_rates submodel is identical to the rates submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

12.8 vul_life_pba submodel

The vul_life_pba submodel is identical to the life_pba submodel which applies for portfolio projections. This submodel is used for principle-based approach (PBA) capital and reserve calculations.

Please refer to the C05 PBA book.

12.9 vul_sfas97rd submodel

The vul_sfas97rd submodel is identical to the sfas97rd submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

12.10 Actuarial math library

12.10.1 AML DLL

The Actuarial math library (AML) provides standard actuarial calculations for traditional reserve methods. See the Actuarial math library liability book.

The vul submodel calls the AML to calculate reserve amounts per unit in force. Columns in the vul_py submodel store the terminal value and net premium for each policy year for each reserve basis used in the projection.

12.10.2 vul->startup

The vul->startup column specifies the name extension for the audit trail comma separated values file (when requested).

12.10.3 *vul->initialize*

The *vul->initialize* column initiates the call to the AML calculations for each set of reserve results. Each set of AML results is calculated by calling the *aml_driver* in the *vul_py* submodel. A separate call can be made for each of the following results:

- Statutory base reserves – Base plan CRVM statutory reserves
- Statutory minimum reserves – Base plan alternative minimum CRVM reserves for calculating deficiency reserves
- Tax base reserves – Base plan CRVM tax reserves

12.10.4 *AML driver*

The *aml_driver* in the *vul_py* submodel sets up and calls the AML for each set of reserve results. This driver is called by the *vul->initialize* column for each set of results.

This driver sets up the call to the AML using the appropriate input for each type of calculations. Each call to the driver calls these functions in the *vul_py* submodel:

- *setup_aml_initial_data* – Creates the interfaces for using the AML.
- *setup_aml_res_defn* – Specifies the expense allowance method (1980 CRVM), and the function type (curtate, continuous, or semi-continuous).
- *setup_aml_policy_data* – Specifies values for each policy year, such as the reserve premiums, death benefits, valuation interest rates, mortality rates, plus the issue age, maturity age, premium term years, and expense allowance grading years.

The operation of these setup functions varies by the type of calculations being made. For example, the policy data passes the statutory rates when calculating statutory reserves, and the tax rates when calculating tax reserves.

12.10.5 *Calculation results*

The AML calculation results are stored for each policy year in columns in the *vul_py* submodel. The formula for each column simply gets the AML calculation results for each case. These columns include:

- Terminal reserves and net premiums for each basis
- Valuation interest and mortality rate for each basis
- Tabular cost of insurance for each basis

The *aml_driver* calls the appropriate columns after it calls the AML. For example, the statutory basis columns are called when the driver has just calculated statutory results. Then the driver calls the tax reserve basis columns when those results have just been calculated by the AML. Thus each of these columns stores the AML results using the appropriate valuation basis.

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L05 INDEXED UNIVERSAL LIFE

This book discusses flexible premium indexed universal life products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are specified on these input pages:

- Product – Benefits, premiums, secondary guarantee, surrender charge, policy loan definitions, and factor calculation definitions
- Fund – Index fund definitions, fixed crediting rate, cost of insurance, loads, and index accounts
- Valuation – Statutory and tax reserves, and PBA net premium reserve definitions
- Expense – Commission and expense rates, premium tax rate
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse rates, withdrawal use, dynamic premium payments, and policy loan use

Each of these input page types begins with 'lul', for use in projections that combine multiple product types.

Additionally, variables defining reinsurance and rider specifications and assumptions are associated with input pages which are common across products. These are:

- Reinsurance – Reinsurance treaty definitions and rates.
- Riders – Riders specifications and assumptions
- RidersValuation – Riders statutory and tax reserve definitions

1.2 Data pages

Variables for each indexed universal life model point are specified on the lulData page. These variables are discussed in the Model point records chapter, below.

Variables for each reinsurance record are specified on the lulReinsData page. These variables are the same for the term life, traditional life, universal life, variable universal life, indexed universal life, and long term care product types. These variables are discussed in the Reinsurance model point records chapter, below.

Variable for each rider record are specified on the lulRidersData page. These variables are the same for term life, traditional life, universal life, variable universal life, and indexed universal life products. These variables are discussed in the L13 Riders book.

2 PRODUCT BENEFITS AND VALUES

The following explains the basic features for flexible premium indexed universal life products, which have no underlying traditional life type guaranteed values. These are often referred to as equity linked UL products.

2.1 Benefit term

2.1.1 Maturity age

The benefit term is an attained age specified by product. When the specified benefit term is zero (default value), the benefit term is age 121.

2.1.2 GMP maturity age

The guaranteed maturity premium (GMP) for UL CRVM reserve calculations is paid to the statutory reserve base mortality omega age, if earlier than the maturity age. When statutory reserve mortality is not specified, the GMP is assumed to be paid to the maturity age.

2.1.3 **IRS restrictions maturity age**

IRS restriction premiums are calculated using a separate maturity attained age, such as 95 or 100, which is specified in the `factors_tgt_age` variable for the Factors calculation type. IRS restrictions are not calculated during projections.

This maturity age applies for factors calculations for: (1) guideline single premium (GSP), (2) guideline level premium (GLP), (3) modified endowment contract (MEC) premium, and (4) cash value accumulation test (CVAT) net single premiums.

2.2 **Extended maturity option**

2.2.1 **When formulaic reserves apply**

When the policy is still in force at the statutory reserve mortality omega age, the extended maturity option (EMO) continues the policy until the maturity age. For example, products valued using 1980 CSO mortality continue in force under the EMO feature after attained age 100 when the maturity age is higher, such as 120.

Specify the death benefit by product:

- The fund value, but not less than zero
- The sum assured, but not less than the fund value.

There are no further premiums, loads, or charges during the EMP period. The fund value continues to increase with credited interest. The formulaic reserve is the current death benefit.

2.2.2 **When PBA reserves apply**

The EMO feature is ignored when PBA reserves are being calculated. These model points have reserves calculated using a modern mortality table, which extends to attained age 121 or later.

2.3 **Death benefits**

2.3.1 **Basic death benefit**

Specify the death benefit option on the model point record:

- Sum assured
- Sum assured plus the fund value, treating negative fund values as zero.
- Sum assured plus return of premiums (ROP) equal to cumulative premiums less withdrawals since issue. For in-force model points, specify on the model point record the initial (at the projection date) cumulative premiums, and cumulative withdrawals, for the policies collected into the model point.

Specify the sum assured per unit for the product; this input could vary such as by policy year or months since issue. When the initial sum assured is specified on the model point record, this sum assured specified for the product specifies the pattern of changes in the sum assured between policy years. See the Units basis section in the Model point records chapter, below.

2.3.2 **Minimum death benefit**

The death benefit is not less than a minimum percent of the fund value, sometimes called the corridor factor. Specify the minimum death benefit option either by product or by model point:

- For the Guideline premium test, the Global directory contains the Guideline Premium Test Corridor workbook containing the typical percentages of 250% through attained age 40, then grading to 105% for attained ages 75 through 90, then grading to 100% at attained age 95.
- For the Cash value accumulation test (CVAT), also called the Cash value net single premium test, specify percentages varying by attained age and typically also by sex and smoking class. The Factors calculation type can be used to calculate the percentages; see the Solving calculations chapter, below.

2.3.3 Death benefit paid versus in force

Death benefits paid at the end of each month are based on the fund value before policyholder transactions are processed. Thus the death benefit is calculated using the fund value before withdrawals.

Death benefits in force at the end of the month are based on the fund value at the end of the month, after withdrawals have been taken.

2.4 Endowment benefits

The remaining fund value is paid to policyholders reaching the maturity age.

2.5 Withdrawal benefits

2.5.1 Partial withdrawal provisions

Policyholders can withdraw cash from the fund value without surrendering the policy. This is sometimes referred to as a partial surrender, or cash withdrawal provision.

Partial withdrawals are processed at the end of each month, for policies persisting into the next month. Thus death and surrender benefits paid are based on the fund value before withdrawals, while death benefits and cash values in force at the end of the month are based on fund values after withdrawals.

Partial withdrawals apply to all policies within a model point equally.

Surrender charges

The percent of fund surrender charge applies for withdrawals.

Other surrender charges are pro-rated using the ratio of the withdrawal amount paid to the cash value before the withdrawal. The outstanding surrender charge amounts are then reduced by the proportion already collected on withdrawals. Specify the initial multiple which applies to these surrender charges, which vary by product, on the model point record.

The projection ignores partial withdrawal fees, assuming that they approximately cover administration expenses, and are not a significant element of profits.

Sum assured

Partial withdrawals do not increase the net amount at risk:

- When the death benefit option is the sum assured, the sum assured decreases by the withdrawal amount plus the related surrender charge.
- This sum assured reduction does not apply when the death benefit option is the sum assured plus the fund value or plus the return of premiums, since the death benefit automatically decreases for the withdrawal.

2.5.2 Partial withdrawal use

The partial withdrawal amount is an annual percent of the fund value. One twelfth of the percent applies at the end of each month. Thus the withdrawal amount is different each month, as the withdrawal basis changes. The withdrawal amount is limited to the cash value less the policy loan.

Partial withdrawals stop for model points in minimally funded status. See the Dynamic premiums section, below.

Dynamic partial withdrawals (related to the competitiveness of the crediting rate) are assumed to be included in excess lapses.

2.5.3 Cumulative withdrawals

For in-force model points which use the Plus ROP death benefit option or have a preferred policy loan, specify on the model point record the initial (at the projection date) cumulative withdrawals for the policies collected into the model point.

2.6 Premiums

2.6.1 Premium term

Specify the premium term on the model point record. This is the policy anniversary when premium payments stop. When this value is zero, premiums are paid to the GMP maturity age.

2.6.2 Contractual premium term

Specify the contractual premium term either as an attained age or policy anniversary. When the contractual premium term is zero (default value), the contractual premium term is to the maturity age, but not greater than age 121.

The contractual premium term applies only for calculating: (1) GMP for UL CRVM reserves, (2) IRS restriction Guideline Level Premium (GLP). It has no effect on projected premium payments.

2.6.3 Premium mode

Specify the premium mode on the model point record. Premiums can be paid annually, semi-annually, quarterly, monthly, or special monthly (such as electronic transfer or pre-authorized check, with expense rate different from the monthly mode).

2.6.4 Annual premium

Specify the annual premium for the product either as:

- Annual premium per unit – The Annual premium per unit specified for the product, plus the policy fee. Premium suspensions apply from issue.
- Initial annualized premium – The premium on the model point record for the policies collected into the model point, including the policy fee. Premium suspensions apply beginning at the next policy anniversary, since this is the actual premium at the projection date.

2.6.5 Policy fee

The policy fee can be deducted as a load from each premium payment. The deduction from each premium payment is the annual policy fee divided by the number of premium payments per year.

Percent of premium commissions can include or exclude the policy fee.

The premium suspension multiple applies to the entire annual premium, including the addition for the policy fee. However, the policy fee itself is not reduced for its use in: (1) fund value load, (2) target premium, and (3) required premium for the secondary guarantee cumulative premiums test.

2.6.6 Modal loading

Modal loadings do not apply for flexible premium products, since the fund value simply accrues less interest when premiums are paid later than at the beginning of each year. Of course, you can vary the annual premium per unit and policy fee by premium mode by using the model point variable `prem_mode` as a lookup key for linking to the external source.

However a monthly modal loading multiple applies for the cumulative premiums test secondary guarantee. See the Secondary guarantee chapter, below.

2.6.7 Additional single premium

An additional single premium can apply at issue, sometimes called a 'dump in single premium'. Specify on the model point record the total single premium for the policies collected into the model point. This affects new business model points only, since in-force model points start from initial values rather than calculating from issue.

2.6.8 Single premium policies

For a single premium plan, define the premium either as:

- Zero annual premium, defining the premium as the additional single premium. Since the annual premium is zero, the premium term has no effect.

- Premium term of 1 year, with Annual premium mode, and defining the premium as the annual premium.

2.6.9 Premium suspensions

Premiums are reduced by a premium suspension percent at the beginning of each policy year. The specified annual premium is multiplied by the cumulative premium suspension multiple. The suspension rate specified for the first policy year is ignored.

The reduced premium is assumed to be paid the same for all policies.

The premium suspension multiple applies to the entire annual premium, including the addition for the policy fee. However, the policy fee itself is not reduced for its use in: (1) fund value load, (2) target premium, and (3) required premium for the secondary guarantee cumulative premiums test.

Premium suspensions start date

When the model point record specifies the initial annualized premium, premium suspensions apply only for future policy anniversaries, i.e., after the projection date.

When the premium is specified as an annual premium per unit for the product, however, premium suspensions are calculated from issue.

2.6.10 Cumulative premiums

For in-force model points which use the Plus ROP death benefit option or have a preferred policy loan, specify on the model point record the initial (at the projection date) cumulative premiums for the policies collected into the model point.

2.6.11 Target premium

The target premium can be used to define values as percentages of target and excess premiums, such as for: (1) product loads for the current fund value, (2) surrender charges, and (3) commissions.

The policy fee is added to the target premium, except for the purpose of calculating commissions when the policy fee is not commissionable. The policy fee added to the target premium is not reduced by premium suspensions.

2.6.12 DAC tax capitalized premium percent

Specify (in the Product input page, under the Premiums category) the percent of each premium which is capitalized for the capitalized premium tax, which is also called the DAC tax. See the Taxes chapter in the Profit calculations book.

2.6.13 IRS premium restrictions

Tax regulations include 'definition of life insurance' premium restrictions. Projections ignore these rules; it is up to the user to specify assumed premiums consistent with the rules.

The Factors calculation type can calculate these IRS restriction premium rates: (1) guideline single premium (GSP), (2) guideline level premium (GLP), (3) modified endowment contract (MEC) premium, and (4) cash value accumulation test (CVAT) net single premiums. See the Solving calculations chapter, below.

2.7 Dynamic premiums

Premium payments can change based on policy funding conditions.

Fully funded secondary guarantee

Premium payments can stop when the secondary guarantee is fully funded. See the Premium adjustment section in the Secondary guarantee chapter, below.

Minimally funded policy

Premium payments can increase when needed to prevent the policy from lapsing. Premiums are then paid annually at the beginning of each policy year, regardless of the model point's premium mode.

When there is no secondary guarantee, this is the annual premium that keeps the current fund value* above zero at the end of each policy year. This annual premium is calculated using the current fund COI and load rates, and the guaranteed interest rate. Interest crediting above the guaranteed rate during the previous year reduces the next premium payment. *When the policy lapses when the cash value becomes zero or negative, this is the premium that keeps the cash value above zero.

When a secondary guarantee applies, this is the annual premium that either keeps the current fund value (or cash value) positive, or keeps the secondary guarantee in force, whichever is lower. The annual premium based on the secondary guarantee is discussed in the Premium adjustment section in the Secondary guarantee chapter, below.

This test is made at the beginning of each policy year. The policy stays in force when the fund-related lapse feature would otherwise terminate it during the year. A 'catch up' premium is paid at the beginning of the next policy year, in addition to the ongoing minimal funding annual premium, to restore the negative value that accumulated during the prior year.

Partial withdrawals and policy loan increases stop for model points in minimally funded status.

2.8 Product loads

2.8.1 Input variables

Input variables for product loads differ between current and guaranteed fund values.

The Fund input page includes a separate set of input variables for current and guaranteed loads. For example, input variable `load_per_pol_curr` specifies the current fund per policy load and `load_per_pol_gtd` specifies the guaranteed per policy load. The formulas, however, use the same code variable, `load_per_pol`, in each case, but each input variable is associated with this code variable in the appropriate instance of the submodel.

2.8.2 Premium loads

The following loads are deducted from each premium payment:

- Percent of premium (including the additional single premium). The target premium percent applies to the premiums paid during the policy year until the premiums exceed the target premium, then the excess premium percent applies to premiums paid during the remainder of the policy year.
- Policy fee – The policy fee can be deducted as a load; see the Premiums section, above. The amount deducted from each premium is the policy fee divided by the number of premium payments per year. The policy fee is not reduced for premium suspensions for this purpose.

2.8.3 Non-premium loads

The following loads are deducted from the fund value at the beginning of each month; one twelfth of the load rate applies each month:

- Amount per policy
- Amount per unit
- Percent of fund value – one twelfth of the annual percent of fund value at the end of the previous month is deducted at the beginning of each month.

2.9 Cost of insurance charge

2.9.1 Input variables

Input variables for cost of insurance charges differ between current and guaranteed fund values.

The Fund input page includes a separate set of input variables for current and guaranteed cost of insurance charges. For example, input variable `coi_rate_base_curr` specifies the current fund COI rates and `coi_rate_base_gtd` specifies the guaranteed COI rates. The formulas, however, use the same code variable,

coi_rate_base, in each case, but each input variable is associated with this code variable in the appropriate instance of the submodel.

These variables are special since they are read twice – first for life 1 and then again for life 2. This is accomplished by using special lookup keys.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. 'Insured characteristics' (such as issue age, gender, smoking class, and rate class) are set to either the life 1 or life 2 values, as needed. The external source uses lookup keys mort_gender, mort_smoking_class, mort_rate_class, and mort_issue_age instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

2.9.2 Charge amount

The COI charge is calculated and deducted at the beginning of each month, after first adding premiums and deducting loads. The COI rate applies to the net amount at risk (see Net amount at risk topic).. The death benefit takes the minimum death benefit (corridor factor) into account.

2.9.3 Net amount at risk

The net amount at risk includes a sum assured component and a fund value component.

$$\text{NAR} = \text{SumAssured} * \text{DiscountAdj} - \text{FundValue} * \text{FVAdj}$$

Specify the discount rate used to adjust the sum assured component:

- None – No adjustments to the sum assured.
- Current Rate – The sum assured is discounted by the monthly current interest rate. The 'current interest rate' is the interest rate associated with the type of fund. So for the current fund, the discount factor is based on the current credited interest rate. For the guaranteed fund, the discount factor is based on the guaranteed interest rate.
- Guaranteed Rate – The sum assured is discounted by the monthly guaranteed interest rate.
- Half Period Current Rate – The sum assured is discounted by a half of month current interest rate. The 'current interest rate' is the interest rate associated with the type of fund. So for the current fund, the discount factor is based on the current credited interest rate. For the guaranteed fund, the discount factor is based on the guaranteed interest rate.
- Half Period Guaranteed Rate – the sum assured is discounted by a half of month guaranteed interest rate.

Specify the fund value component:

- Prior Month Fund Value – The ending fund value of the previous month
- Beginning Fund Value With Premium Less Load – The ending fund value of the previous month plus the current month's premium less loads.
- Beginning Fund Value With Premium Less All Other Charges – The ending fund value of the previous month plus the current month's premium less loads and all other charges (except the COI charge, since this is calculated based on the net amount at risk). Note, the only other available charge other than loads is the rider charge, which applies to the current fund only. So this option is the same as Beginning Fund Value With Premium Less Load for guaranteed fund.
- Prior Month Cash Value – The ending cash value of the previous month.

Specify the fund value adjustment:

- None – No adjustments to the fund value component
- COI Rate – Multiply the fund value by (1 – Monthly COI Rate).

When the fund value is negative, the net amount at risk is calculated using zero fund value. See the Secondary guarantee chapter, below.

2.9.4 COI rate

Specify COI rates either as annualized rates or as annual mortality rates. The annual rate is adjusted by a multiple, which can vary by policy year or attained age. The adjusted annual rate is then converted to a monthly COI rate using either an annualized, linear, or exponential formula.

Annualized formula:

$$\text{monthly COI rate} = \text{annualized COI rate}/12$$

Linear formula:

$$\text{monthly COI rate} = \frac{(\text{annual mortality rate}/12)}{(1 - \text{annual mortality rate}/12)}$$

Exponential formula:

$$\text{monthly COI rate} = \frac{[1 - (1 - \text{annual mortality rate})^{(1/12)}]}{(1 - \text{annual mortality rate})^{(1/12)}}$$

Substandard multiple

The substandard multiple on the model point record applies to the annualized COI rate, or to the annual mortality rate, before conversion to a monthly COI rate. This is the same substandard multiple as applies for experience mortality rates.

This multiple applies the same to the current and guaranteed COI rates. Of course the COI rate cannot exceed 1.0 (100%).

2.10 Fund value

2.10.1 Premiums

Premiums are credited to the fund value at the beginning of the month of each premium payment.

The allocation of premiums between the fixed fund and the indexed fund value is discussed in the Fund values chapter below.

2.10.2 Loads and charges

Loads and charges are deducted from the fund value at the beginning of the month after the addition of premium payments.

The allocation of loads and charges between the fixed fund and the indexed fund value is discussed in the Fund values chapter below.

2.10.3 Interest crediting

Interest is credited to the fund value at the end of each month. The total amount credited to the fund value is the sum of the amount credited to the fixed fund and the indexed fund.

The amount credited to the fixed fund value and the indexed fund value is discussed in the Fund values chapter below.

2.10.4 Partial withdrawals

Partial withdrawals are deducted from the fund value at the end of the month, after interest has been credited.

The deduction from the fund value is the amount withdrawn plus the surrender charge on the partial withdrawal.

2.11 Cash value

The cash value is the fund value less a surrender charge.

2.11.1 Fund value for surrender charge

The fund value for calculating the surrender benefit is the fixed fund value plus the indexed fund values. There are no partial excess indexing credits for indexed buckets that have not reached the end of their indexing period.

2.11.2 Surrender charge

The cash value is the fund value less the surrender charge. Specify surrender charges as:

- Amount per unit
- Percent target premium
- Percent of fund value.

The surrender charge rates can vary annually (using the `pol_yr` column) or monthly (using the `mths_since_issue` column).

Surrender charges (other than the percent of fund value charge) are reduced by the proportion of the charges which have already been collected during partial withdrawals. Specify on the model point record initial (at the projection date) multiple which applies to these surrender charge rates.

2.12 Policy loan

2.12.1 Policy loan definition

Policy loans may be treated as “Fixed Fund” or “Indexed Fund” (model point choice list) policy loans.

When the policy loan is an “Indexed Fund” policy loan:

- The policy loan rate is a fixed rate,
- There is no separation of policy loans into preferred and non preferred loans,
- The policy loan is restricted to not exceed the cash value but can exceed the fixed fund value, and
- Crediting is not affected by the policy loan such that the fixed fund value is treated as all being unloaned and the indexed fund values receive guaranteed and excess interest the same as if there were no loan.

When the policy loan is a “Fixed Fund” policy loan:

- The policy loan rate is a fixed rate,
- Policy loans can be separated into preferred and non preferred loans,
- The policy loan cannot exceed the fixed fund value and funds are transferred from the indexed funds to the fixed fund when a policy loan would exceed the fixed fund value, and
- The crediting rate on the loaned fund is either (1) the unloaned fund crediting rate, (2) a fixed rate, (3) the guaranteed crediting rate on the fixed fund, or (4) the policy loan rate less a fixed spread.

2.12.2 Policy loan amount

Policy loans apply equally to all policies within a model point. For in-force policies, specify on the model point record the initial (at the projection date) policy loan for the policies collected into the model point.

The accumulated policy loan at the end of each month is the balance at the end of the previous month, brought forward with one month's interest at the policy loan rate.

The outstanding loan amount is increased at the end of each policy year, by a percent of the total fund value, to reflect assumed new loan activity.

Policy loan increases stop for model points in minimally funded status. See the Dynamic premiums section, above.

2.12.3 Maximum policy loan

The outstanding loan amount is reduced at the end of any month when it would exceed the cash value.

2.12.4 Policy loan cash flows

An adjustment increase is a new policy loan and a negative cash flow. An adjustment decrease is a policy loan repayment and a positive cash flow.

The elimination of the policy loan at death and surrender is an 'adjustment' in the loan balance to zero, and thus is a positive loan cash flow which offsets part of the decrement cash flow.

2.12.5 Policy loan rate

The policy loan interest rate is a fixed rate. Interest is added to the loan at the end of each month.

When policy loan interest is payable in advance, policy forms may specify the loan rate as a discount rate. Convert this discount rate to the annual effective annual rate. For example, a 7.4% discount rate payable in advance must be input as an effective annual rate of 8.0% [$= 7.4\% \div (1 - 7.4\%)$].

2.12.6 Loaned fund crediting rate

When the policy loan is a "Fixed Fund" policy loan, the crediting rate on the loaned fund value can be:

- Unloaned rate – The same crediting rate applies to the entire fund value. However, the additional interest credit only applies to the unloaned fund value.
- Fixed rate – A specified crediting rate.
- Guaranteed rate – The guaranteed crediting rate, not including any temporary guarantee of the crediting rate at issue.
- Fixed spread – The policy loan interest rate less a specified spread.

2.12.7 Preferred loans

When the policy loan is a "Fixed Fund" policy loan, the loan can be split into two parts. The 'preferred' part of the loan is the amount up to the cash value or fund value, less cumulative premiums paid. The crediting rate on the preferred loan is the policy loan interest rate less a specified preferred loan spread. A special case of a preferred loan is a zero net cost loan, sometimes called a wash loan, where the preferred loan spread is zero.

The remainder of the loan is a non-preferred policy loan, which has a crediting rate as discussed above.

For in-force model points which have an initial (at the projection date) policy loan for products with the preferred policy loan feature, specify on the model point record the initial cumulative premiums and cumulative withdrawals for the policies collected into the model point.

2.12.8 Policy loan expenses

Policy loan expenses are discussed in the Expenses section, below.

3 FUND VALUES

3.1 Policy fund values

3.1.1 Current fund

A current fund is projected for experience calculations.

For in-force model points, specify on the model point record the initial (at the projection date) current fund value, which is the sum of the specified fixed fund value and indexed fund accounts values, for the policies collected into the model point.

3.1.2 **Guaranteed fund**

Guaranteed fund values are projected when needed for solving calculations. See the Solving calculations chapter, below.

The guaranteed fund is also used during the projection, when it is updated to the current fund value at a reserve valuation date, then projected forward using premium payment assumptions for the reserve calculation. This occurs for UL CRVM reserves, and PBA NPR base and secondary guarantee reserves. When these calculations apply, the `iul_ifund_gtd` output does not represent guaranteed fund calculations from issue. Instead, set the `solving_display_defn` variable (on the Product input page in the Premiums category) to All Solves to save each type of solving fund result in the output database.

Flag and indicator values control each type of solving, such as the timing and amount of premium payments. See the Submodels structure chapter, below.

3.1.3 ***iul_ifund* submodel output**

The formulas in the `iul_ifund` submodel have separate values for each instance of this submodel. When output is created for this submodel, select either the current or guaranteed fund instance.

The `iul_ifund` submodels iterate to solve for values such as the guaranteed maturity premium. See the Solving calculations chapter, below. When the `solving_display_defn` variable (on the Product input page in the Premiums category) is All Solves, the `iul_ifund` output includes the final values for each solving calculation.

Warning! Since the `iul_ifund` submodel is reset for each solving calculation, the final results apply only for the last time this submodel was calculated. For example, the `iul_ifund_gtd` values are not the guaranteed fund values calculated from issue; they could be the calculations starting from the current fund value at the last time when UL CRVM reserve calculations were processed.

To see a particular set of results, customize to add code to write out the submodel results at the time that calculation has just been performed. Search for the 'writeModel' command in the `iul->initialize` formula to see how this is done when the `solving_display_defn` calls for writing out the submodel values.

3.2 **Fund value structure**

3.2.1 **Fund types**

The current fund value can be split between the following fund types:

- **Fixed fund value** – The amount of current fund value earning interest based on the crediting strategy.
- **Indexed fund value** – The amount of the current fund value earning interest based on the growth of an index. The standard application provides for the indexed fund value to be split into up to 5 indexed fund accounts.

3.2.2 **Input variables**

Input variables for the fixed fund and indexed fund values are specified on the `IulFund` input page.

The fixed fund input variables include the guaranteed crediting rate and crediting strategy. The variables commonly vary by plan code.

The indexed account input variables include the index crediting definition, crediting (or indexing) months, the crediting index and lookback intervals, the excess crediting fund base definition, the crediting floor, participation rate, and initial and dynamic cap rate parameters. The variables commonly vary by a `fund_id`, identified as `index0` through `index4`.

3.2.3 **Indexed fund account buckets**

Each indexed fund account has a bucket for each month of its indexing months when premiums are paid. For example, a one year indexing period with a monthly premium model point has 12 buckets. Each bucket is identified by a bucket number and bucket age. The bucket number is fixed when the bucket is created but the bucket age increases as the bucket moves to the end of the indexing period.

When created, the bucket numbers are relative to the projection date. For example, for a 12 month indexing period, bucket 0 is the bucket which will receive premiums in projection months 1, 13, 25, etc., and has a bucket age of 11 at the projection date and 0 in the first month of the projection. Bucket 11 is the bucket which will receive premiums in projection months 12, 24, 36, etc., so it has a bucket age of 0 at the projection date and 1 in the first month of the projection.

3.3 Allocations and transfers

3.3.1 Initial fund values

Specify the initial (at projection date) current fund value of the model point (separate by fixed and each indexed fund value account) on the model point record.

Since the history of each indexed fund account bucket is not provided as input to the model, the fund values specified for each indexed account are treated as belonging to a single bucket for the index where the choice of the bucket is either (model point option choice):

- New Bucket – The in force fund value for the index is allocated to the bucket which will also receive the premiums paid in the first month of the projection. The model point is assumed to contain each bucket's fund values accumulated to the projection date at the option budget rate.
- Policy Anniversary – The in force fund value for the index is allocated to the bucket which will mature at the end of the current policy year. The model point contains the sum of the indexed bucket's fund values at the projection date.

3.3.2 Premiums

Specify the percentage (0.5 for 50%) allocations of future premiums to the fixed fund value and indexed fund accounts on the model point record.

The total of these percentages should add to 1.00 (100%), otherwise they are normalized.

3.3.3 Loads and charges

Loads and COI charges are calculated in aggregate (for the policy), then allocated to the model point's fixed and indexed fund value in the following priority:

- The entire deduction is allocated to the fixed fund value, until it is zero.
- Any remaining deduction is allocated between the indexed fund value accounts on a pro-rata basis, in proportion to the fund in each indexed fund value account.
- The deduction from each indexed fund value account is allocated to the most recent indexed fund account bucket (LIFO basis), until it is zero. Any remaining deduction is allocated to the next most recent bucket, and so on.

3.3.4 Partial withdrawals

Partial withdrawals are allocated the same as for loads and COI charges. That is, the withdrawal is taken from the fixed fund value until it is exhausted. Any remaining amount is pro-rated to the indexed fund value accounts, taken from the most recent indexed fund account bucket (LIFO basis).

3.3.5 Policy loans

When the policy loan is a Fixed Fund loan, the fixed fund value must be as large as the policy loan.

When this is not the case, the resulting transfer to the fixed fund value is allocated between the indexed fund value accounts on a pro-rata basis (in proportion to the fund in each indexed fund value) and taken from the most recent indexed account bucket (LIFO basis).

3.3.6 Fixed fund rebalancing

Since the fixed fund value is the first source of funds for covering the non premium loads, charges, and policy loan growth (for Fixed Fund policy loans), you may select to have the funds in the fixed fund value increased

at the beginning of each policy year to hold an amount to cover the expected non premium loads, charges, and policy loan growth.

Specify to perform this rebalancing by setting the fund_val_rebal_defn to "Rebalance".

The estimate of the amount required to cover the non premium loads, charges, and policy loan growth for the policy year is calculated as the present value of the expected amounts for the policy year, assuming no additional premium payments, discounted at the monthly crediting rate.

When all funds are included in the indexed fund value, this rebalance typically results in a transfer of funds into the fixed fund value in policy month 1 and a zero balance in the fixed fund value at the end of policy month 12.

3.3.7 Indexed fund account rollover

At the end of the indexing period for each indexed fund account bucket, the fund value of that bucket automatically rolls over as a new indexed fund account bucket and is added to premiums received for that month.

3.4 Fixed fund value crediting

3.4.1 Overview

The fixed fund value is increased for credited interest at the end of each month. The crediting rate for the fixed fund value is the larger of the policy's guaranteed crediting rate and the rate determined through the crediting strategy.

When the policy loan is a Fixed Fund policy loan, interest credited to the fixed fund value is calculated separately for the loaned and unloaned parts of the fund value. Interest crediting on the loaned fund value is discussed in the Policy loan section in the Product benefits and values chapter, above.

When the policy loan is an Indexed Fund policy loan, the loaned fund value is ignored. That is, interest is credited to the fixed fund value as if the policy loan did not exist.

3.4.2 Initial crediting rate

Specify the initial (at projection date) crediting rate for the fixed fund value on the model point record. When the value on the model point is zero, the initial crediting rate is determined based on the crediting strategy.

The initial crediting rate applies until the crediting rate is reset through the application of the crediting strategy.

3.4.3 Crediting strategy declaration

Declaration timing

The crediting rate can change at the beginning of a month, policy year, calendar year, or calendar quarter.

Declaration period

The crediting rate changes according to the declaration period, which is one or more cycles of the specified period. A different number of cycles can apply at issue than for ongoing crediting rate changes. For example, the crediting rate at issue could apply for the first two policy years, then changing every policy year thereafter.

3.4.4 Crediting strategy rate basis

Fixed rate

The crediting rate can be a fixed rate, such as for calculating illustration values. Crediting rate constraints do not apply.

Asset yield basis

You can specify the crediting rate as the asset yield less a spread, subject to the crediting rate constraints. This definition makes the model points dependent on asset results, so independent model point processing cannot apply for asset/liability projections.

The asset yield includes the effects of IMR amortization, defaults, and borrowing costs.

For liability-only projections, you specify the asset yield as an assumption for the projection. Independent model point processing can still apply, since the asset yield is not affected by liability results.

Competitor rate basis

You can specify the crediting rate as a competitor rate less a spread, subject to the crediting rate constraints.

When the constraints do not include the asset yield, independent model point processing can apply even for asset/liability projections. For example, this approach could take the view that the crediting rate is primarily driven by market forces, and that appropriate investment strategies must be determined in order to achieve the desired spread from the asset yield.

3.4.5 Crediting strategy constraints

The following constraints apply when the crediting rate basis is the asset yield or a competitor rate. Crediting rate constraints do not apply when the crediting rate is a fixed rate.

The crediting rate is first constrained to be not more than a maximum percent above, and not less than a maximum percent below, a specified competitor rate.

Further (thus adjusting the above constraints), the crediting rate is not greater than [the asset yield less a minimum spread]. When the minimum spread below the asset yield is 1.0 (100%) default value, this constraint does not apply, and model points defining the crediting rate using a competitor rate are eligible for independent model point processing within asset/liability projections.

Finally, the crediting rate is not less than the guaranteed crediting rate in all cases, even when the crediting rate is defined as a fixed rate.

3.5 Indexed fund value crediting

3.5.1 Overview

The indexed fund value is increased for credited interest at the end of each month. The amount credited is the sum of the crediting to each indexed fund account bucket.

When the crediting floor for the index is greater than 0%, the indexed fund account bucket is increased for credited interest each month similar to the fixed fund value, except that (1) the crediting rate is the index crediting floor rate, and (2) the indexed fund account bucket crediting is never affected by the policy loan.

At the end of the indexing period for the bucket, the indexed fund account bucket is increased for credited interest based on the indexing crediting method. Index crediting applies when the credited interest based on the indexing method (discussed below) is greater than the amount of interest credited to the indexed fund account bucket based on the index crediting floor.

The index crediting applies to either (option choice):

- The indexed account bucket fund value at the end of the indexing period, reduced by the interest credited based on the index crediting floor during the indexing period, or
- The average of the indexed account bucket fund value at the beginning and end of the indexing period, where the indexed account bucket fund value at the end of the period is reduced by the interest credited based on the index crediting floor during the indexing period.

The amount of the index crediting is subject to index participation rates and cap rates (discussed below).

No interest is credited to the indexed fund account bucket if the indexed fund account bucket fund value before interest for the month is less than zero.

3.5.2 Participation rate

The amount of the index crediting is subject to index participation rates.

Each indexed fund account bucket has its own participation rate. The participation rate is declared at the start of each indexing period for each indexed fund account bucket. The standard application provides for specifying participation rate by calendar date for each index.

Specify the participation rate for the inforce indexed account bucket as participation rates at the projection start date for each index in the assumptions.

3.5.3 *Cap rates*

The amount of the index crediting is subject to index cap rates.

Each indexed fund account bucket has its own indexing cap rate. The indexing cap may be subject to a minimum and maximum cap rate for the index. The cap rate is declared at the start of each indexing period for each indexed fund account bucket. The standard application provides for solving for new cap rates at the beginning of each indexing period (discussed below)

Specify the cap rates for the inforce indexed account bucket as initial cap rates by index and bucket age in the assumptions.

3.5.4 *Indexing method*

The indexing method is one of (option choice):

- **Point to point** – The index crediting rate is the participation rate times (1) the index value at the end of the ending period divided by (2) the index value at the beginning of the indexing period (3) minus 1.0. The resulting amount is floored at zero and capped by the indexed fund account bucket cap rate.
- **Moving average** – The index crediting rate is the participation rate times (1) the average of the index value over a specified look-back sampling interval during the indexing period, divided by (2) the index value at the beginning of the indexing period, (3) minus 1.0. The resulting amount is floored at zero and capped by the indexed fund account bucket cap rate. This indexing method operates similarly to an Asian index option asset. When the look-back sampling interval is annual, and the indexing period is also annual, this is the same as the point to point indexing method.
- **Monthly sum cap** – The index crediting is the sum over each month in the indexing period of the lower of (1) the index value at the end of the month divided by the index value at the start of the month, minus 1.0, and (2) a monthly indexed fund account bucket cap rate. The resulting amount is floored at zero and the participation rate has no effect for this method.

3.5.5 *Scenario index*

The index calculations for the indexing method are based on changes in a specified economic index. The index must be contained in the scenario rates file used for the projection. For example, the index could be the 'US' equity index contained in the scenario rates. Please refer to the C02 Economic scenarios book.

Note that past index values are produced from growth rates provided in the historic rates file specified for the projection. During the first year of the projection, index changes compare the index scenario rate against the historic rate at the beginning of the indexing period.

3.6 Indexed account bucket calculations

3.6.1 *Option budget*

At the beginning of each indexing period, the option budget is determined as a rate using the formula $\max(0.0, (A) - (B) - (C))$, where:

- (A) an assumed earned rate over the indexing period based on the asset yield on invested assets (assets excluding any hedge assets) starting the month of the new indexing period, less a fixed spread,
- (B) the amount of guaranteed interest crediting to the indexed fund account bucket over the indexing period based on the crediting floor rate, discounted back to the start of the indexing period at a risk free rate for a specified quality grade,

(C) a specified hedging fee.

For new business projection where you have selected to use a “Proxy Yield” approach for the initial option budget, the assumed earned rate is approximated using the yield rate on a specified quality grade and maturity term. Since the option budget is assumed to be at the beginning of the month, the scenario rates used are the rates at the end of the previous month.

3.6.2 *Dynamic cap calculations*

The indexing cap rate dynamically adjusts to a new value at the creation of a new indexed fund account bucket or rollover of an existing indexed fund account bucket to a new indexing period. This calculation is assumed to occur at the beginning of the month, using beginning of month values for asset yields and scenario rates.

A cap rate is derived based on an available option budget and the cost of a long call option to cover the upside equity participation. The solved for cap rate can be constrained to be above a minimum cap rate and below a maximum cap rate. The final cap rate can also be rounded.

Important! The application has been designed to produce warning messages in the run log when the available option budget is zero or the solved for cap rate has been constrained by the minimum or maximum rates. Since these are typically not errors in the calculations, the run log is not changed to Yellow when these warning messages are produced.

For new business projections, you may select to provide the initial cap rate as “User Input” as opposed to having the application solve for an initial cap rate.

3.6.3 *Point to point indexing cap solving*

For the point to point indexing method, a call spread is purchased to cover the upside equity participation.

The long option cost is the market value at the start of the indexing period of the long European call option to cover the upside equity participation. The option period is the indexing period and the option strike price is the current index value, accumulated to the end of the indexing period at the index crediting floor rate.

When the long option cost exceeds the available option budget, an out-of-the money European call option is sold to lower the option cost by forgoing proceeds in excess of the cap. This short option has the same properties as the long option except for the strike price.

The strike price of the out-of-the money option is solved for so that the proceeds from the short position plus the option budget are enough to offset the long option cost:

$$\text{long optioncost} - \text{short optioncost} = \text{optionbudget}$$

The dynamic cap is then:

$$\text{cap} = (\text{solved strike value} / \text{at - the - money strike value}) - 1$$

As discussed above, the solved cap rate is then constrained to be between the minimum and maximum cap rates specified for the index and may be rounded.

3.6.4 *Moving average indexing cap solving*

For the moving average indexing method, a call spread is purchased to cover the upside equity participation.

The process is the same as for the point to point method except the call option to purchase is a fixed strike (average price) Asian option.

3.6.5 *Monthly sum cap indexing cap solving*

For the monthly sum cap indexing method, a monthly sum cap option is purchased to cover the upside monthly sum cap crediting. The cap rate for the option is determined by finding the cap rate for a monthly sum cap option which can be funded by the option budget.

The option cost is the market value at the start of the indexing period of the monthly sum cap option, adjusting for the index crediting floor rate. A closed form Black Scholes based formula is used to determine the market value of the monthly sum cap option.

The solved cap rate is then constrained to be between the minimum and maximum cap rates specified for the index and may be rounded.

4 SECONDARY GUARANTEE

The secondary guarantee keeps the death benefit in force through a specified policy anniversary even when the fund value is zero or negative. These products are sometimes called death benefit guarantee indexed universal life (DBGIUL), secondary guarantee indexed universal life (SGIUL), indexed universal life secondary guarantee (IULSG), no-lapse guarantee (NLG), and protection indexed universal life products. Products without the guarantee are sometimes called accumulation indexed universal life, or current assumption indexed universal life products.

4.1 General

4.1.1 Secondary guarantee years

Specify the secondary guarantee years (benefit term) as part of the product definition. This is the policy anniversary when the guarantee expires. When this variable is the default value of zero, the secondary guarantee continues to maturity.

The secondary guarantee is in effect through these years, as long as the policyholder meets requirements using the cumulative premiums test (discussed in separate sections below). When the test is violated, the secondary guarantee continues in effect until the end of the secondary grace period (discussed in a separate section below), as long as the current fund value is positive to keep the policy in force.

4.1.2 Secondary guarantee premium term

To keep the guarantee in force, the policyholder must pay premiums high enough to meet the test through the secondary guarantee premium term. Specify the secondary guarantee premium term attained age as part of the product definition. This age cannot exceed the Extended Maturity Option age, since premiums cannot be paid during the EMO period. When this variable is the default value of zero, the premium requirements continue to the end of the secondary guarantee benefit years (but not past the EMO age).

The secondary guarantee funding status at the end of this premium term applies for the remainder of the secondary guarantee benefit years. When premium requirements have been met as of the end of this premium term, the guarantee is fully funded. When premium requirements have not been met as of the end of this premium term, the guarantee is no longer in effect. Thus the secondary guarantee 'catch up' provision applies only during the secondary guarantee premium term.

4.1.3 Extended maturity option

When the secondary guarantee test has been satisfied at the EMO attained age, the guarantee continues through the EMO period.

4.1.4 Input variables

Most input variables related to the secondary guarantee are organized on the Product input page, in the Secondary Guarantee category. These define the guarantee terms, premium and lapse adjustments, rules for calculating negative fund values, secondary guarantee test interest crediting (including ratcheting threshold), and secondary guarantee required premiums.

4.2 Cumulative premiums test

4.2.1 General

The cumulative premiums test is that accumulated premiums paid, less the policy loan, cannot be less than the accumulated required premiums. The accumulation of premiums paid is reduced by partial withdrawal payments, but not by the surrender charge reduction to the fund value.

Specify the required premiums per unit on the Product input page, under the Secondary Guarantee category. Specifying required premiums varying by policy year is sometimes called an annually renewable term (ART) design. Specifying a level premium is sometimes called a stipulated level premium design.

The required premiums are assumed to be paid monthly. The required annual premium per unit is increased by the policy fee, then multiplied by the specified annualized modal loading multiple for the monthly premium mode.

4.2.2 Interest accumulation

Both the actual and required premiums accumulate at the secondary guarantee interest rate specified on the Product page, under the Secondary Guarantee category.

For in-force model points, specify on the model point record the initial (at the projection date) accumulation of actual premiums for the policies collected into the model point. The initial accumulation of required premiums from issue is calculated during the projection, using the units per policy at the projection date.

4.2.3 Interest ratcheting

The cumulative premiums accumulation rate is ratcheted, adding an additional credit (can be negative) to the accumulation rate for the excess accumulation over a per unit threshold varying by months since issue. The threshold is the accumulation of the level annual premium which satisfies the cumulative premiums test at the end of the secondary guarantee premium term, before adding interest for the current month.

4.2.4 Policy loan

The policy loan is taken into account in determining whether the secondary guarantee is in the grace period. The policy loan is ignored in other calculations, however, such as fully or minimally funding the secondary guarantee. Reserves are not affected by the policy loan.

4.3 Negative fund value

When the fund value is negative, it is treated as zero for product value calculations:

- Benefit values use zero fund value.
- Interest crediting can either: (1) continue to apply, making the fund value more negative, or (2) stop when the fund value is negative. When there is no interest crediting, the fund value is treated as zero for calculating the average crediting rate for GAAP calculations.
- Loads continue to be deducted.
- COI net amount at risk uses zero fund value. COI charges continue to be deducted.
- The YRT reinsurance net amount at risk uses zero fund value.
- The GAAP basic benefit reserve is zero.

4.4 Secondary guarantee grace period

Specify the secondary guarantee grace period months on the Product input page, under the Secondary Guarantee category.

4.4.1 Positive fund value

When the current fund value is positive, the secondary guarantee remains in effect until the test has been violated for the number of grace period months. At the end of the grace period, the secondary guarantee terminates even though the policy continues to be in force without the guarantee. Thus some model points have no secondary guarantee, even though it is specified for the product.

The grace period is restored when the test is satisfied before the current grace period ends. That is, the next time the secondary guarantee test is violated, the full grace period months apply again.

When the guarantee is already violated at the projection date, specify on the model point record the initial (at the projection date) months that the guarantee has been in the grace period. Any number greater than or equal to the grace period months indicates that the model point no longer includes the secondary guarantee.

4.4.2 Catch up provision

When the secondary guarantee is in this grace period, the secondary guarantee reserve includes a calculation of the 'catch up' amount needed to restore the guarantee.

When premiums are increased to minimally fund the guarantee, the catch up amount is paid as a premium at the projection date. See the Premium adjustment section below.

4.4.3 Negative fund value

This grace period does not extend the secondary guarantee when the current fund value is negative. Fund-related lapse occurs immediately when the secondary guarantee test is violated.

4.5 Premium adjustment

Premium adjustments are evaluated at the beginning of each policy year, when the annual premium is determined for the year.

4.5.1 Fully funded secondary guarantee

Premium payments can stop when the secondary guarantee is fully funded, which occurs when the accumulated premiums, without further premium payments, equals or exceeds the accumulated required premiums at the end of the secondary guarantee period.

4.5.2 Minimally funded policy

Premium payments can increase when needed to prevent the policy from lapsing. See the Dynamic premiums section in the Product benefits and values chapter, above.

The annual premium to keep the secondary guarantee in effect is the premium which meets the required premiums accumulation at the end of the policy year.

4.6 Lapse adjustment

The lapse rate is adjusted when the fund value is negative, i.e., when the secondary guarantee is keeping the policy in force:

- When the secondary guarantee is fully funded (see above), the lapse rate is multiplied by an adjustment percent.
- When premiums are still required to keep the secondary guarantee in effect until its expiration, the lapse rate is multiplied by a different adjustment percent.

4.7 In-force statistics

Units in force at the end of each month are calculated for these special situations:

- SG in the money – When the fund value is zero or negative, meaning that the secondary guarantee is keeping the policy in force. This test is based on the cash value, when the fund-related lapse feature tests against the cash value instead of the fund value.
- SG minimally funded – When premiums are increased to keep the secondary guarantee in force (see the Premium adjustment section, above.)
- SG fully funded – When premiums are stopped because the secondary guarantee is paid up (see the Premium adjustment section, above.)

5 JOINT LIFE PRODUCTS

Joint life products have a second insured, in addition to life 1. The product is a joint life product when the life status variable is First To Die or Frasierized Last Survivor. When the life status is Single Life, the second life variables do not affect the projection.

5.1 Second life characteristics

5.1.1 Life 2 model point variables

Specify insured characteristics model point variables for life 2 in addition to life 1. Life 2 characteristics include issue age, gender, smoking class, and rate class. These insured characteristics are particularly important for mortality variables, which are the rates for: (1) experience, cash value, and reserve mortality, and (2) YRT reinsurance rates.

See the Mortality variables section in the Input organization chapter, above.

5.1.2 Life 1 and life 2 scalars

Since issue_age is used to calculate values such as the years to maturity, 'life 1' must be:

- the older of the two insureds for first to die model points, and
- the younger of the two insureds for Frasierized last survivor model points.

The order of the two lives on the model point record is handled automatically, by converting the life 1 and life 2 model point variables into scalars, instead of using the model point variables directly:

- For first to die model points with life 1 younger than life 2, the life 1 scalars return the life 2 model point variables, and the life 2 scalars return the life 1 model point variables.
- For Frasierized last survivor model points with life 1 older than life 2, the scalars also switch the values this way.
- Otherwise, the life 1 scalars return the life 1 model point variables, and the life 2 scalars return the life 2 model point variables.

For example, the scalars issue_age and gender normally have the values issue_age_mp and gender_mp, but return the values issue_age_mp_life2 and gender_mp_life2 when the lives must be switched. Since formulas reference the scalars, they are automatically based on the correct life.

5.2 First to die

For first to die products, the death benefit is paid on the first death; the policy terminates at that time. Both insureds are alive at the projection date.

The annual mortality/COI/YRT rate is:

$$q_1 + q_2 - q_1 \times q_2$$

5.3 Frasierized last survivor

For last survivor products, the death benefit is paid when both insureds have died. Since there is no change in status when the first life dies, the projection assumes there is no knowledge about whether life 1 or life 2 is still alive at the projection date, except that at least one of them must be alive since the policy is still in force.

The mortality/COI/YRT rates are Frasierized from issue, as described below.

5.4 Frasierization

Since Frasierized last survivor products continue to operate as normal until the second death, the projection assumes there is no information whether only one life is surviving at the projection date. The mortality rates for life 1 and life 2, plus the joint life contagion effect, are combined into a Frasierized rate equal to the sum of:

$$\begin{aligned} &P(\text{both alive BOY}) \times P(\text{both die during the year} | \text{both alive BOY}) \\ &\quad + P(\text{only life 1 alive BOY}) \times P(\text{life 1 dies during the year} | \text{only life 1 alive BOY}) \\ &\quad + P(\text{only life 2 alive BOY}) \times P(\text{life 2 dies during the year} | \text{only life 2 alive BOY}) \end{aligned}$$

This formula Frasierizes the annual mortality rates, then the Frasierized annual rate is converted to a monthly rate as usual. This method commonly applies for reserve mortality and YRT rates. Annual experience mortality rates are also Frasierized prior to calculating the monthly mortality rate, so that calculations are simplified by producing monthly mortality rates which are constant within each policy year.

5.4.1 Joint life mortality addition

For experience mortality, the annual probability that both lives die together (or within some short interval) is increased by a joint life mortality rate addition, which is sometimes called a contagion effect. The specified addition is reduced when the joint death rate would exceed either the first or second life mortality rate. That is:

$$q_1 \times q_2 + \text{adjusted addition} \leq \min(q_1, q_2)$$

5.4.2 Minimum YRT rate

The Frasierized annual YRT reinsurance rate is subject to a minimum floor, such as .00012 (\$.12 per thousand).

6 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves calculated by model point. PBA reserves are discussed in the PBA book, including the net premium reserve (NPR).

6.1 Reserve basis

6.1.1 Mortality

Reserve mortality is a base rate, times the multiple, then plus the addition.

Lookups for reserve mortality tables should be set up similar to the experience mortality rates since these rates are read for both the first and second lives.

The base statutory mortality rates affect the projection even when reserves are a percent of the fund or cash value, since the omega age specifies the GMP maturity age.

Substandard multiple

The substandard multiple (on model point records) does not affect reserves. It applies for experience mortality, COI charges, and YRT reinsurance premiums.

6.1.2 Interest

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeStatValn and LifeTaxValn ranges in the Valuation Interest Rates Excel workbook.

6.1.3 Function timing

Function timing options affect traditional reserve calculations, such as the CRVM reserve (for the UL CRVM method) and the XXX reserve (for the AXXX reserve):

- **Curtate** – Annual premiums and claims paid at the end of the policy year
- **Continuous** – Continuous premiums and time-of-death claims
- **Semicontinuous** – Annual premiums and time-of-death claims.

Premium timing multiple

When premiums are assumed to be paid continuously, the curtate reserve formulas are adjusted by the following 'premium timing' multiple:

$$\frac{(\delta - d)}{\delta^2} + \frac{(i - \delta)}{\delta^2} \times \frac{(1 - q)}{(1 + i)}$$

Claims timing multiple

When claims are assumed to be paid immediately, the curtate reserve formulas are adjusted by the following 'claims timing' multiple:

$$\frac{i}{\delta}$$

Reserve for immediate payment of claims

Actuarial guideline XXXII (AG 32) requires additional reserves for policies which pay death claims at the time of death, but have statutory reserves calculated assuming death claims are paid at the end of the policy year. This can be approximated by using the Semicontinuous function timing option.

6.2 UL CRVM reserve

6.2.1 Percent of fund approximation

The UL CRVM reserve can be approximated using percentages of the fund value or cash value. This approximation replaces the UL CRVM reserve. The secondary guarantee reserve, if applicable, is still calculated and compared against this amount.

6.2.2 GMP and GMF

The UL CRVM reserve is based on the guaranteed maturity fund (GMF), which is calculated using the guaranteed maturity premium (GMP) which endows the policy for the sum assured at the GMP maturity age:

- The GMP maturity age is the statutory reserve base mortality omega age, when earlier than the maturity age.
- The GMP is paid for the contractual premium term, which may be different from the model point's premium term.
- The GMP is paid using the model point's premium mode.

The GMF is projected by calculating the GMP which endows the fund value at the GMP maturity age using guaranteed loads, COI rates, and interest crediting. The GMF is then projected using the GMP and these guaranteed rates. When specified by the `solving_display_defn` variable (on the Product input page in the Premiums category), the `iul_ifund` output includes this fund value per unit.

The GMF is recalculated (in the situations described below) by starting from the current fund value, then projecting using guaranteed rates and assuming that the GMP is paid in the future. When the reserve method is UL CRVM, the `iul_ifund` output includes this fund value per unit. This does not apply when the reserve method is Approximate UL CRVM, which uses only the GMF values.

6.2.3 **UL CRVM**

Reserve basis

The statutory mortality variables for UL CRVM reserves (stat_mort_base, stat_mort_mult, and stat_mort_addn) are separate from the mortality variables for XXX reserves. The same applies for tax reserves. The deficiency mortality variables for alternative minimum UL CRVM reserves (def_res_mort_base, def_res_mort_mult, and def_res_mort_addn) are separate from the mortality variables for XXX deficiency reserves.

CRVM reserve [iul_py->stat_crvm_terminal]

The traditional Commissioners Reserve Valuation Method (CRVM) reserve is calculated for the plan of insurance endowing at the GMP maturity age, using the same expense allowance as for the corresponding traditional plan.

Benefit amounts [array dth_ben_tbl_crvm]

The benefits for calculating the traditional CRVM reserve are the projected average EOM death benefit per unit amounts for each policy year, using guaranteed rates and charges and GMP premium payments.

The sum assured is paid as an endowment amount at the GMP maturity age.

UL CRVM reserve [iul->stat_res_ulcrvm_per_unit]

The ratio 'r' is the current fund value divided by the GMF. When $r \leq 1$, the reserve is the traditional CRVM reserve times r.

When $r > 1$, the current fund value is projected to maturity assuming: (1) future premiums equal to the GMP, and (2) guaranteed rates.

The average death benefit in each policy year from this projection is compared to the death benefits from the GMF projection from issue. Any excess death benefit, and any excess maturity value, over the GMF projection from issue is discounted using statutory reserve mortality and interest.

The reserve is: (1) the traditional CRVM reserve, plus (2) the present value of these excess benefits.

6.2.4 **Approximate UL CRVM**

The following approximation can be appropriate for policies which are not heavily funded.

The traditional CRVM reserve and r are calculated from issue the same as described above. When $r \leq 1$, the reserve is the CRVM reserve times r; there is no approximation in this case.

When $r > 1$, the reserve is: (1) the traditional CRVM reserve, plus (2) the excess of the current fund value over the GMF. This approximation thus reduces calculation time by not recalculating the GMF.

6.2.5 **Actuarial guideline 36**

RiskAgility FM U.S. Library does not include the methodologies described in actuarial guideline 36.

6.3 **Secondary guarantee reserve**

For products with a secondary guarantee, the reserve [iul->stat_res_per_unit_prelim] is the greater of: (1) the UL CRVM reserve, and (2) the secondary guarantee reserve defined by regulation XXX and AG 38.

6.3.1 **Exception**

The secondary guarantee reserve does not apply when the: (1) secondary guarantee benefit term is five years or less, (2) the annual required premium for the first policy year is not less than the statutory net level premium method reserve premium for the guarantee, and (3) the current surrender charge is not less than the annual required premium for the first policy year.

6.3.2 Regulation XXX

Reserve basis

The statutory mortality variables for XXX reserves (stat_xxx_mort_base, stat_xxx_mort_mult, and stat_xxx_mort_addn) are separate from the mortality variables for UL CRVM reserves. The same applies for tax reserves. As discussed below, these must be ultimate mortality rates for 1980 CSO mortality, since select multiples (stat_xxx_select_input_1 for the first segment, and stat_xxx_select_input_2 for subsequent segments) apply to them. For 2001 CSO mortality, these can be the select and ultimate rates, then leaving the select multiple variables at their 1.0 default value.

Deficiency reserve (alternative minimum) mortality (def_res_xxx_mort_base) applies for XXX deficiency reserves. As also discussed below, separate variables (seg_xxx_mort_base and seg_xxx_select_input_1) specify the mortality rates for determining the XXX segments.

Segmented reserve

Reserve net premiums are a different percent of the annual required premiums for each segment of the secondary guarantee premium term. The division into segments is discussed below. Reserves are calculated applying the Commissioners Reserve Valuation Method (CRVM) expense allowance to the first segment, with no further expense allowances for the other segments.

Annual required premiums [iul_py->sg_art_prem]

The annual required premiums are the minimum premium in each year which that will satisfy the secondary guarantee test. For the cumulative premiums test, these are the required premiums specified for the guarantee.

Segmented definition

The levelized premium segmentation method divides the benefit period into segments when the ratios, $G(t)$, of successive required premiums exceed the ratios, $R(t)$, of successive deficiency reserve basis mortality rates but excluding the X factor. The first segment ends prior to the policy year (designated by 't') when $G(t)$ first exceeds $R(t)$. The next segment extends from the end of the first segment until the next segment starts at the beginning of the policy year when $G(t)$ again exceeds $R(t)$, and so on.

The mortality rate ratios, $R(t)$, can be multiplied by 101% or 99% to avoid segments based on insignificant excess values. However, $R(t)$ cannot be less than 1.

Segments are determined using reserve mortality variables seg_xxx_mort_base and seg_xxx_select_input_1. These variables typically contain the same tables as for deficiency reserve mortality for the first segment, but excluding the deficiency reserve X factors (discussed below). Separate variables are provided so that XXX deficiency reserves can use mortality varying by the 2001 CSO rate classes, while the segments are determined using the version of the table which does not vary by rate class, as allowed by actuarial guideline XLVI (AG 46).

The same segments apply for statutory, deficiency, and tax reserves.

Select mortality factors

Select mortality factors apply both for the determination of the segments and for the calculation of the segmented reserves. In addition, 'X factor' multiples can apply to the deficiency mortality rates for calculating deficiency reserves during the first segment, but does not apply for determining the segments.

2001 CSO mortality – The Global folder includes the 25-year select multiples that apply for 2001 CSO mortality; see the Mortality Select Multiples Excel workbook. When using these select mortality multiples, the base mortality variable must specify the ultimate rates. When the 2001 CSO select rates apply past the first segment, the select and ultimate mortality rates can be specified as the base variable, leaving the select factor variables at their default values of 1.0 (100%). However, when 'X factor' multiples are used, the base mortality variable must specify the ultimate rates with separate select factors specified so that it is the combined select factors which are constrained to 1.0 (100%) as opposed to just the 'X factor' multiple being constrained.

1980 CSO mortality – The Global folder includes 10-year, 15-year, and 20-year select factors which apply to the 1980 CSO ultimate mortality rates; see the Mortality Select Multiples Excel workbook. For 1980 CSO mortality, the reserve mortality variable must specify the ultimate rates, then the select factor variables (when select mortality is elected for reserves) must specify the following:

The 10-year select factors are implicit in the 1980 CSO select and ultimate mortality rates. This select factors apply for subsequent segments when the first segment is shorter than 10 years.

The following variations for applying special select factors during the first segment:

- Regulation XXX can apply either the 10- or 20-year select factors during the first segment.
- For NY 147, the first segment uses 15-year select factors adjusted by a multiple of 150% (120% for deficiency reserves), with the resulting select factor rounded to the nearest whole percent and capped at 100%. These select factor multiples apply only when reserves segment definition is NY 147; these variables are ignored for regulation XXX.

Benefit amounts [array dth_ben_tbl_xxx]

The benefits for these regulation XXX calculations are the projected average EOM death benefit per unit amounts for each policy year, using guaranteed rates and charges and assuming that the annual required premiums are paid at the beginning of each policy year.

Only the death benefits apply; there is no endowment amount at the end of the guarantee period.

Mortality improvement

Mortality improvement can apply for the determination of the segments and the calculation of the segmented, basic, and tax reserves, such as is allowed for under the 6th amendment of NY regulation 147. Mortality improvement applies from a specified improvement base year. When included, mortality improvement multiples are applied to the select factor adjusted mortality rates in the first segment only.

Lapse rates

Under the 5th and 6th amendment of NY regulation 147, lapse rates may be included in the present value calculations in steps 2 and 5 of Actuarial guideline XXXVIII (AG 38), as described in section 6.3.3. The Global folder includes the maximum lapse rates allowed, contingent on issue year, issue age, and policy year; see the NY147 Lapse Rates Excel workbook.

Catch up provision reduction

The catch up provision applies when the secondary guarantee test has been violated for less than the grace period months (and the current fund value is positive to keep the policy in force). The regulation XXX reserve is reduced by the 'catch up premium' which would reinstate the guarantee. For the cumulative premiums test, this is the excess of the accumulated required premiums over the accumulated actual premiums.

The reduction is prorated between the basic and deficiency reserve.

6.3.3 Actuarial guideline 38

Actuarial guideline XXXVIII (AG 38), which is often referred to as guideline AXXX, defines the calculation of secondary guarantee reserves for flexible premium universal life products issued after 2002. The XXX reserve is increased by additional calculation steps, resulting in 9 calculation steps as follows:

Step 1: Annual required premiums [iul_py->sg_art_prem]

This step is the same as for regulation XXX, described above.

Step 2: Segmented reserves for the guarantee period [statutory: iul_py->stat_xxx_terminal and iul->stat_res_xxx_per_unit] [including deficiency reserve: iul_py->stat_xxx_min_terminal and iul->stat_res_xxx_min_per_unit]

This step is the same as for regulation XXX, described above. AG 38-8C uses lapse rates in this reserve calculation.

The 'catch up' provision amount is prorated to reduce the basic and deficiency reserves.
[iul->stat_res_xxx_reduced_per_unit and ul->def_res_xxx_reduced_per_unit]

Step 3: Excess premiums paid [iul->axxx_excess_prem_per_unit]

The amount of actual premiums paid in excess of the required premiums is the accumulated actual premiums less the accumulated required premiums.

When the 'excess' is negative, this is the 'catch up provision' amount which reduces the regulation XXX reserve from step 2.

Step 4: Required payments single premium [iul->sg_req_single_prem_per_unit (before gross up)]

The single payment necessary at the valuation date to fully fund the guarantee through the secondary guarantee benefit term, assuming that required premiums have been paid in the past, is calculated as the ending accumulation of required premiums is rolled backward to the beginning of each month, assuming that no premiums are paid. The single premium is the single premium value less the accumulated required premiums at the valuation date. For AG 38-8B and 8C, the amount is grossed up by [one minus the prescribed 7% load].

Then the calculated 'pre-funding ratio' is any excess premiums paid (step 3), divided by the single premium. [iul->axxx_pre_funding_ratio]. This ratio is floored at zero, and capped at 1.

Step 5: Present value of benefits (net single premium) [iul->stat_axxx_nsp_per_unit]

The present value of benefits through the secondary guarantee benefit term is calculated using statutory reserve mortality. The benefit is the same as discussed above for regulation XXX calculations. These values are calculated as of the end of each policy year [iul_py->stat_xxx_nsp], then linear interpolation applies for other months. AG 38-8C allows including lapse rates in this calculation.

Step 6: Net amount of additional premiums [iul->stat_axxx_addl_prem_per_unit]

Any excess of the present value of benefits (step 5) over the basic and deficiency reserve (step 2) is multiplied by the pre-funding ratio (step 4).

Step 7: Reduced deficiency reserve [iul->def_res_axxx_reduced_per_unit]

When the required payments single premium (step 4) is greater than the excess premiums paid (step 3), the deficiency reserve (step 2) is multiplied by [one minus the pre-funding ratio floored at zero] (step 4).

Step 8: Minimum reserve [iul->stat_res_axxx_min_per_unit]

The minimum reserve before the surrender charge reduction is the lower of: (1) present value of benefits (step 5), and (2) the XXX basic and deficiency reserve (step 2) plus the net amount of additional premiums (step 6). This amount is then reduced by the surrender charge reduction (step 8b, below). However, the surrender charge reduction cannot reduce the minimum reserve below the XXX basic and deficiency reserve (step 2).

Step 8b: Surrender charge reduction [iul->axxx_surr_chg_per_unit]

The final reserve [iul->stat_res_axxx_per_unit] is the minimum reserve, reduced for the surrender charge:

- For AG 38-8A, the reduction is the fund value less the cash surrender value.
- For AG 38-8B and 8C, the surrender charge reduction is multiplied by the ratio of: (1) the net level premium for the secondary guarantee years, divided by (2) the net level premium for whole life insurance.
- For AG 38-8B, there is no reduction for surrender charges when the pre-funding ratio exceeds [one minus the prescribed 7% load].

Step 9: Increased basic reserve

The minimum reserve calculated in step 8 applies for the projection. For reporting purposes, the basic reserve is the minimum reserve (step 8) minus the reduced deficiency reserve (step 7).

6.3.4 Actuarial guideline 38 variations

Actuarial guideline 38 includes the following variations:

AG 38-8A

AG 38-8A (sometimes called the 'original version') applies for policies issued from January 1, 2003 through June 30, 2005.

AG 38-8B

AG 38-8B (sometimes called the 'first revision') applies for policies issued July 1, 2005 through December 31, 2006:

- The percent of premium load allowance used in grossing up the required payments single premium (step 4 as described below) is 7%. This is one minus the variable `sg_funding_ratio`, which has a default of 93%.
- The surrender charge reduction is eliminated when the secondary guarantee is fully funded.
- The surrender charge reduction is reduced when the secondary guarantee period is less than whole life coverage.

AG 38-8C

AG 38-8C (sometimes called the 'second revision') applies for policies issued in 2007 through December 31, 2012:

- The provisions of AG 38-8B continue to apply, except that the surrender charge reduction is restored when the pre-funding ratio exceeds [one minus the 7% prescribed premium load]. That is, this provision reverts back to the AG 38-8A version.
- Lapse rates may be included in the present value calculations in steps 2 and 5 as described below. The Global folder includes the maximum lapse rates; see the AG38 Lapse Rates Excel workbook.

AG 38-8D

AG38 applies for certain policies issued from July 1, 2005 through December 31, 2012, or any valuation date thereafter. Additional reserves may be required should the deterministic reserve calculated under a VM-20 methodology with modifications exceed the reserve calculated under AG 38-8B or 8C. Specify in variable `stat_res_ag38_8d_defn` whether AG38-8D applies for the product, and in variable `stat_res_ag38_defn` the base valuation method (e.g. AG38-8B, 8C).

The reserve calculated for the model point is for the base valuation method (e.g. AG 38-8B, 8C). The aggregate reserve is the greater of the special VM-20 deterministic reserve (discussed below), and the sum of the reserves calculated by model point ignoring reinsurance.

The additional reserve can be tax deductible.

AG38-8D has no impact on the reinsurance reserve credit applicable to policies with YRT reinsurance. For policies with coinsurance or modified coinsurance, the ceded reinsurance reserve is the treaty percent applied to the gross reserve calculated by model point plus the treaty percent applied to the additional reserve (if applicable) calculated under the special VM-20 deterministic reserve methodology.

The modified VM-20 deterministic reserve calculation assumes that:

- The starting assets amount is the sum of the statutory reserves calculated by model point.
- For a liability only projection, the asset yield assumed in the reserve calculations is defined using the competitor rate basis specified for the crediting strategy. The deterministic scenario is defined in a scenario rates file, adjusted by a specified multiple and addition. This asset yield is used to define the crediting rate and discount rate in the inner loop valuation projection. The asset yield is assumed to already account for the asset yield requirements specified under the Primary Reserve Methodology (Section 8Da.2.a) of AG 38.
- For an asset/liability projection, specify the issue year yields for A-rated bonds by issue year from 2005 to 2012 in the input variable `ag38_8d_issue_yld_rate`. At the start of each valuation period, weighted average rates are calculated using the specified bond yields in combination with the model point reserves by issue year, calculated at each future valuation date. The starting assets earn the actual portfolio net asset yield capped with rates calculated above. All rates are net of defaults.

- For an asset/liability projection, new investments made during the valuation are invested as cash earning a fixed rate specified as a competitor rate at the valuation date and capped at 7%.

The special VM-20 reserve is calculated the same as for PBA reserves, using monthly cash flows and discounting.

For nested projections, the aggregate AG38-8D reserve between valuation dates is the sum of the model point reserves times the ratio of the aggregate Ag38-8D reserve to that sum on the previous valuation date. For an asset/liability projection, the assets are projected using the asset portfolio and investment strategy specified for the projection. The new valuation rates are reflected in the competitor rates and scenario rates files.

AG 38-8E

AG38-8E applies to policies issued in 2013 or later. Specify whether the product uses method I or method II.

For Method I, the following steps are revised from 8C:

- Step 1: For policies with policy design #3, specify (stat_res_ag38_min_prem_method_1_per_unit) the vectors of minimum gross annual premiums. When this variable is zero (default value), these premiums are calculated using the COI, load, and crediting rate assumptions used in the minimum gross annual premium calculations.
- Step 4: The pre-funding ratio in this step is not less than -1, and not more than 1.
- Step 6: When variable stat_res_ag38_min_prem_per_unit is zero, or the step 3 amount is positive, then this step is the same as under 8C. When the step 3 amount is negative, the 'net amount of additional premium' (iul->stat_axxx_addl_prem_per_unit) is the ratio from step 4 iul->axxx_pre_funding_ratio times the basic reserve from step 2 (stat_res_xxx_per_unit), less the deficiency reserve calculated in step 2, plus the deficiency reserve at issue calculated using 100% X, multiplied by 1 minus the ratio (floored at zero) of the amount of cumulative premiums paid, divided by the required single premium from step 4. This amount is floored at zero.
- Step 8: Same as under 8C, but adjusting the reserve floor. When the step 3 amount is positive, the reserve floor is the sum of the step 2 basic and deficiency reserve without adjustment. When the step 3 amount is negative, then the reserve floor is the sum of the step 2 basic and deficiency reserves and the amount from step 6.

For Method II, the following steps are revised from 8C:

- Step 1: The minimum gross annual premiums is defined the same as for Method I. (Note that there is no dynamic functionality which determines the lowest schedule of minimum gross premiums that keeps the policy in force over the life of the secondary guarantee period and that produces the greatest deficiency reserve at issue.)
- Step 3: This is the cumulative premiums paid minus the cumulative premium that would have resulted from paying the minimum gross premium from issue. This result can be negative.
- Step 4: Same under 8C but with the following adjustment in the single premium (iul->axxx_req_pmt_single_prem_per_unit before gross up):
 - When step 3 is positive, the single premium is reduced by the excess of (1) accumulated premiums based on paying the minimum gross premium from issue over (2) the cumulative premium requirements. When step 3 is negative, the single premium is replaced by the excess of (1) accumulated premiums based on paying the minimum gross premium from issue over (2) the accumulated required premiums.
 - After the adjustment, the amount is grossed up by the sg_funding_ratio, then the axxx_pre_funding_ratio is calculated as described in 8C.
- Step 6: When step 3 is positive, this step is the same as under 8C. When step 3 is negative, the 'net amount of additional premium' (iul->stat_axxx_addl_prem_per_unit) is the ratio from step 4 (iul->axxx_pre_funding_ratio) times the basic reserve from step 2 (stat_res_xxx_per_unit), less the

deficiency reserve calculated in step 2, plus the deficiency reserve at issue calculated using 100% X, times 1 minus the ratio (floored at zero) of accumulated premiums paid, divided by the minimum accumulated premiums required to fully fund the guarantee. This amount is floored at zero.

- Step 7: The result is floored at zero
- Step 8: Same as under 8C, but adjusting the reserve floor. When step 3 is positive, the reserve floor is the sum of the step 2 basic and deficiency reserves without adjustment. When step 3 is negative, then the reserve floor is the sum of the step 2 basic and deficiency reserves and the amount from step 6.

6.4 Substandard reserve

Substandard reserves are not included in the projection. The substandard multiple (on model point records) does not affect reserves. It applies for experience mortality, COI charges, and YRT reinsurance premiums.

However, stochastic and deterministic PBR calculations include the substandard multiple in the PBA mortality rate, since it is part of the expected experience for the policy.

6.5 Mid-year reserve

Reserves between policy anniversaries can be:

- Monthly interpolated terminal – Monthly interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.
- Daily interpolated terminal – Daily interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.

When the reserve function timing is Continuous, the mid-year reserve net premium is a discounted continuous net premium.

6.6 Deficiency reserve

The deficiency reserve is the 'alternative minimum' reserve less the statutory basic reserve, calculated using mid-year reserves. When secondary guarantee reserves apply, the 'alternative minimum' reserve is the greater of the UL CRVM alternative minimum reserve and the secondary guarantee alternative minimum reserve.

6.6.1 UL CRVM alternative minimum reserve

When basic deficiency reserves are requested, a UL CRVM alternative minimum reserve is calculated using the lower of the GMP and reserve net premium in each year.

Specify the UL CRVM deficiency reserve mortality and interest basis separately from the statutory reserve basis.

When a percent of fund approximation is used for the UL CRVM reserve, no UL CRVM alternative minimum reserve is calculated.

6.6.2 Secondary guarantee alternative minimum reserve

Regulation XXX

Regulation XXX (or NY regulation 147) applies the same for both basic and deficiency reserves. Specify deficiency reserve mortality separately from the statutory mortality. As explained in the secondary guarantee reserve Regulation XXX topic, deficiency reserve mortality includes a table of 'X factor' multiples that apply during the first segment.

Also as explained in the secondary guarantee reserve Regulation XXX topic, segments are determined using reserve mortality variables `seg_xxx_mort_base`, `seg_xxx_select_input_1`, and `seg_xxx_mort_improve_pct`. These variables typically contain the same tables as for deficiency reserve mortality for the first segment, but

excluding the deficiency reserve X factors. Separate variables are provided so that XXX deficiency reserves can use mortality varying by the 2001 CSO rate classes, while the segments are determined using the version of the table which does not vary by rate class, as allowed by actuarial guideline XLVI (AG 46).

Actuarial guideline 38

When actuarial guideline 38 applies, the actuarial guideline 38 net single premium, additional premium, and surrender charge amounts are calculated as discussed in the secondary guarantee section. These amounts are used to adjust the XXX alternative minimum reserve to arrive at the actuarial guideline 38 minimum reserve.

6.7 Additional reserve

Additional statutory reserves can be specified as an amount per unit. [iul->stat_res_addl_per_unit]

The additional reserve is not considered during the calculation of deficiency reserves.

The additional reserve can be added to the final reserve (after the comparison between the UL CRVM and secondary guarantee reserves), or only to the UL CRVM reserve (before the comparison to the secondary guarantee reserve). The additional reserve is added before comparisons to the cash value and tabular cost of insurance floor.

When detail reports split out reserve values, 'additional reserves' include the increase over basic reserves due to minimum reserves (such as cash value and the tabular cost of insurance floor). [iul->stat_res_addl]

The coinsurance reserve credit can exclude the additional reserve.

6.8 Tabular cost of insurance floor

The reserve is subject to a minimum based on the tabular cost of insurance. That is, this floor applies to the sum of: (1) basic reserve (including the AG 38 reserve), (2) deficiency reserve, and (3) additional reserve.

The floor is the tabular cost of insurance (c_x) times the sum assured, times the following fraction for each mid-year reserve method:

- Monthly interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment. For example, this floor at the end of policy month 1 is $11/12 c_x$ for annual premiums, and $2/12 c_x$ for quarterly premiums.
- Daily interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $344/365 c_x$ for annual premiums, and $71/365 c_x$ for quarterly premiums.

6.9 Cash value floor

Finally, the mid-year reserve is not less than the cash surrender value. [iul->stat_res_per_unit]

6.10 EMO reserve

When the Extended Maturity Option is in effect, the reserve is the current death benefit.

6.11 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record sets the tax reserve to be the statutory reserve, including the comparison to tabular cost of insurance floor using the statutory basis, but excluding any deficiency reserve.

Tax reserves are calculated by the same methods, using the tax reserve interest and mortality basis. The secondary guarantee reserve uses the same segments that apply for statutory reserves.

The statutory additional reserve can be added to tax reserves. That is, the additional reserve can be deductible.

The tax reserve is not greater than the statutory reserve, including the additional reserve even when it is not included in the tax reserve.

6.12 Unearned premium reserve

6.12.1 Balance sheet

When the mid-year reserve method is monthly or daily interpolated terminal, an unearned premium reserve is established. The unearned premium reserve is the portion of the modal valuation net premium that has been collected but not yet earned.

The balance sheet includes the statutory unearned premium reserve as a separate line item from the statutory reserve. The unearned premium reserve is set to zero if the reserve being held is the tabular cost of insurance or cash value floor.

6.12.2 Profit effect

Book profit includes the increase in the unearned premium reserve.

6.12.3 Taxable income

Taxable income similarly includes the unearned premium reserve increase, which is calculated the same way as statutory unearned premium reserve increase, but using the tax reserve basis.

6.13 Gross premium reserve

Letter of credit (LOC) reserve financing creates a non-invested LOC asset that offsets the 'redundant' part of the statutory reserve, based on the calculation of the gross premium reserve (GPR) for each model point assigned to a reserve financing transaction. Book profit is increased by the LOC asset increase (offsetting part of the statutory reserve increase), and is decreased by LOC costs (interest paid on the LOC assets). See the Gross premium reserves chapter in the PBA book.

7 ACTUARIAL ASSUMPTIONS

7.1 Commissions

7.1.1 Premium commission

Percent of premium commissions apply the target premium percent to the premiums paid during the policy year (including the additional single premium) until the premiums exceed the target premium. Then the excess premium percent applies to premiums paid during the remainder of the policy year.

When the policy fee is not commissionable, the policy fee part of the premium is excluded from this calculation, and the target premium excludes the policy fee.

7.1.2 Fund value commission

One twelfth of the annual percent of unloaned fund value at the end of the previous month is paid at the beginning of each month. The deduction of the loan amount from the total fund value for the fund value commission applies for both Indexed Fund and Fixed Fund policy loan types.

7.1.3 Annualized commission

A specified percent of the total first year target premium commission can be paid at issue. No further premium commissions are paid until the cumulative earned commission exceeds the annualized amount. From then on, premium commissions are paid as usual. This determination of commission payments is made for continuing policies, not taking into account terminated policies.

Other types of commissions are not affected by the annualization of commissions.

7.1.4 Commission chargeback

A portion of the premium commission can be charged back in the event of lapse or surrender during the first three years after issue. The chargeback percent applies either to: (1) total first year premium commissions paid to date, or (2) the annualized commission amount. The chargeback percent typically varies by month, by linking to the external source using the `mths_since_issue` column as a lookup key.

Other types of commissions are not subject to chargeback.

The commission subject to chargeback is calculated per unit in force, and thus is not reduced by decrements. This amount applies to the units lapsing and surrendering each month.

For in-force model points, specify on the model point record the initial (at the projection date) commissions subject to chargeback for the policies collected into the model point. When this variable is negative (the default value is -1), the chargeback amount is calculated from issue assuming that premiums have been paid as specified.

7.1.5 Commission overrides and agent terminations

The specified commission rates are the average commission rates, taking into account average writing agent commissions, commission overrides, and the effect of agent terminations.

7.2 Expenses

7.2.1 Expense adjustment multiple

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

7.2.2 Acquisition expenses

Acquisition expenses incurred at issue can include:

- Amount per policy
- Amount per unit
- Percent of target premium
- Percent of the total first year premium (including the additional single premium).

Acquisition expenses could vary by underwriting class, which is discussed in the Mortality section, below.

7.2.3 Maintenance expenses

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Maintenance expenses can include:

- Amount per policy, separate between premium paying and paid up
- Amount per unit, separate between premium paying and paid up
- Percent of the annualized premium (excluding the additional single premium)
- Percent of the total or indexed (option choice) fund value at the end of the previous month.

BOM maintenance expense

An additional maintenance expense is incurred at the beginning of each month, when commissions are paid:

- Percent of each commission payment, ignoring commission chargeback.

7.2.4 Transaction expenses

Maintenance expenses include the following expenses related to administrative activities:

- Expense per death claim policy
- Expense per death claim unit.

7.2.5 Reinsurance administration expenses

Reinsurance administration expenses are discussed in the Reinsurance chapter, below.

7.2.6 Inflation

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

Note: The average size for future issues can be inflated (to offset expense inflation), by customizing the scalar `units_per_pol` to divide the result by `infl_cumul(commencement_period)` for model points issued after the start of the projection. This would reduce the number of policies issued, with no effect on the units, sum assured, and premium issued for the model point.

7.2.7 Policy loan expenses

Policy loan expenses are a percent of the loan balance at the end of the previous month. One twelfth of the annual rate applies each month.

This amount is a negative cash flow at the end of each month, and is thus subtracted from direct cash flows.

Policy loan expenses reduce book profit, by reducing investment income from policy loans. Thus this expense is treated as an investment expense for statutory and GAAP profit reporting. It is not included with other expenses on profit reports.

However, PBA deterministic reserves include this expense in the present value of expenses. This is a different treatment of policy loan expenses than applies for profit calculations.

7.2.8 Premium tax

Premium taxes are reported with expenses. The tax is a percent of each premium payment, including the additional single premium paid at issue.

Inflation and the expense adjustment multiple do not apply to premium tax.

7.2.9 Fixed expenses

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

7.3 Mortality

7.3.1 Input variables

Mortality variables for indexed universal life products include mortality base, multiple, and addition rates, as well as mortality improvement rates. A single set of input variables is included in the model but these variables are special since they are read twice – first for life 1 and then again for life 2 (when applicable). This is accomplished by using special lookup keys.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. 'Insured characteristics' (such as issue age, gender, smoking class, and rate class) are set to either the life 1 or life 2 values, as needed. The external source uses lookup keys `mort_issue_age`, `mort_gender`, `mort_smoking_class`, `mort_rate_class`, `mort_pol_yr`, `mort_cal_yr`, and `mort_term_conversion_defn` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

7.3.2 **Mortality adjustment multiple**

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

7.3.3 **Term conversion adjustments**

Term products often offer the option to convert to a permanent product (i.e. indexed universal life) prior to the end of a level term period without additional underwriting.

Specify the original term issue date and whether the policy was issued as a result of a term conversion on the model point record. For such policies, the base mortality rates, mortality improvement factors, and mortality table adjustments will be calculated using a point-in-scale method based on the original issue date of the term policy. Additionally, a separate set of base mortality rates and mortality table adjustments may apply to the term conversion cohort of policies if specified.

Term conversion adjustments do not apply to joint life policies.

7.3.4 **Underwriting class**

Life insurance products can use the `underwriting_class` scalar to vary product assumptions such as mortality rates and acquisition expenses. The underwriting class is calculated by comparing the policy size against the nonmedical and paramedical underwriting limits specified with the experience mortality input. The underwriting class has a value of either:

- Nonmedical – when `ins_issued_per_pol` $\leq$ `nonmedical_limit`
- Paramedical – when `nonmedical_limit` $<$ `ins_issued_per_pol` $\leq$ `paramedical_limit`
- Medical – when `ins_issued_per_pol` $>$ `paramedical_limit`.

7.3.5 **Rate class**

Mortality rates varying by rate class require special treatment. The model point variable (`rate_class_mp`) can have a large number of options specific to the company's business. But the 2001 CSO mortality tables varying by rate class only define rates for Standard, Preferred, and Super issues. Thus there are two different rate class scalars – `rate_class` contains the company rate classes, and `rate_class_res` maps the company rate classes into the reserve rate classes. These same considerations apply for the second life variable (`rate_class_mp_life2`) and scalar (`rate_class_life2`).

7.3.6 **Base mortality**

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

7.3.7 **Mortality improvement**

The base rate is then adjusted by mortality improvements. Negative improvement rates specify worsening mortality due to anti-selection after issue.

Improvement starts after the specified base year, which is either the issue year or a specified calendar year. The improvement can end after a specified policy or calendar year.

When improvement rates apply by policy year, the mortality rate improves at the end of each policy year by the rate specified for that year. When mortality rates apply by attained age, the mortality rates for each age improve by the rate specified for that age. The following example shows the mortality improvement multiple for a policy issued in 2008 with mortality improvement base year of 2010, thus improvements begin at the end of the third policy year:

Policy Year	Value	Attained Age Method		Policy Year Method	
		Multiple	Calculation	Multiple	Calculation
1	0.05	1		1	
2	0.07	1		1	
3	-0.05	1		1	
4	0.025	.0975	1-.025	1.05	1-(-.05)
5	0.075	.855625	(1-.075)^2	1.02375	* (1-.025)
6	0.10	.729	(1-.10)^3	0.946969	* (1-.075)
7	0.125	.586182	(1-.125)^4	0.852272	* (1-.10)

7.3.8 Mortality table adjustments

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

7.3.9 Substandard mortality

The substandard multiple on the model point record applies to the mortality rate after the multiple and addition. This same substandard multiple applies for COI and YRT reinsurance rates.

7.3.10 Joint life mortality

See the Joint life products chapter, below.

7.3.11 Monthly mortality

Deaths occur at the end of each month, prior to other decrements and policyholder transactions. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{1/12}$$

7.4 Lapses and surrenders

A termination is a 'surrender' when there is a cash value, and a 'lapse' otherwise.

Lapse rates apply to units after mortality decrements, which are processed before all other decrements.

7.4.1 Lapse adjustment multiple

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

7.4.2 Policy year lapse rates

The base lapse rate is adjusted by a multiple.

7.4.3 Excess lapses

Lapse rates can be dynamically adjusted according to policyholder behavior related to the competitive position of the crediting as economic condition change.

Excess lapses can occur when a crediting cap rate for the index with the largest fund value is below a new money cap rate calculated using a specified competitor rate option budget. The crediting cap rate used in the excess lapse comparison can be (option choice):

- Average – The weighted average of the cap rates for all indexed fund account fund buckets of the index with the largest fund value
- New Money – The new money cap rate for the index with the largest fund value
- Maximum Rate – The larger of the average and new money cap rates
- Minimum Rate – The smaller of the average and new money cap rates

When the cap rate for the index with the largest fund value is above the competitor cap rate, excess lapses can be negative.

When a model point has all funds in the fixed account, the excess lapse calculations are the same as for a flexible premium universal life product, where the comparison is between the fixed account crediting rate and a specified competitor rate.

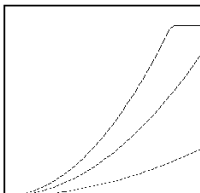
Maximum and minimum lapse rate variables in the excess lapse features affect the excess lapse calculations only. They do not limit the specified lapse rates. For example, the maximum lapse rate variable can reduce the excess lapse rate to zero, but cannot reduce the total lapse rate to be less than the specified lapse rate adjusted by the calendar date multiple.

Excess lapses can be calculated using either an exponential or arctangent formula.

7.4.4 Exponential excess lapses

The exponential formula produces results that look like figure 1, where the x-axis is the difference between the competitor rate and the crediting rate. The highest excess lapses in the graph are for a policy with no surrender charge. The lowest excess lapses represent a policy with a large surrender charge.

Figure 1 Exponential formula



There are two versions of the exponential formula:

Increased lapses:

$$\begin{aligned}
 &\text{When competitor cap (rate)} > \text{crediting cap (rate)} + \text{threshold} \\
 &\text{difference} = (\text{competitor cap (rate)} - \text{crediting cap (rate)} - \text{threshold}) \times 100 \\
 &\text{surrender charge ratio} = 1 - \left(\frac{\text{cash value}}{\text{fund value}} \right) \\
 &\text{excess lapse} = \frac{\left[\text{multiple} \times (\text{difference}^{\text{exponent}}) \times (1 - \text{surrender charge multiple} \times \text{SC ratio}) \right]}{100} \\
 &\quad - \text{surrender charge subtraction} \times \text{SC ratio} \\
 &\text{excess lapse} \leq \max(0, \text{maximum lapse rate} - \text{planned lapse rate})
 \end{aligned}$$

Decreased lapses:

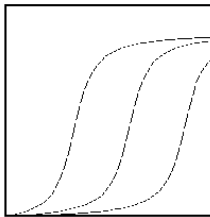
When crediting cap (rate) > competitor cap (rate) + threshold
 difference = (crediting cap (rate) – competitor cap (rate) – threshold) × 100

$$\text{lapse reduction} = \frac{\left[\text{multiple} \times (\text{difference})^{\text{exponent}} \right]}{100}$$

$$\text{lapse reduction} \leq \max(0, \text{planned lapse rate} - \text{minimum lapse rate})$$

7.4.5 Arctangent excess lapses

The arctangent formula produces results that look like figure 2. The highest excess lapses in the graph are for a policy with no surrender charge. The lowest curve has a large surrender charge.

Figure 2 Arctangent formula

difference = competitor cap (rate) – crediting cap (rate)
 surrender charge ratio = $1 - \left(\frac{\text{cash value}}{\text{fund value}} \right)$

$$\text{excess lapse} = \frac{\left[(\text{maximum total lapse} - \text{base lapse}) \times (A - B) \right]}{(1 - B)}$$

 total lapse = base lapse + excess lapse,
 $\leq \text{maximum total lapse}, \geq \text{minimum total lapse}$

$$A = \frac{\arctan \left[100 \times \text{multiple} \times (\text{difference} - \text{SC ratio} \times \text{SC multiple} - \text{threshold}) \right]}{\left(\frac{\pi}{2} \right)}$$

$$B = \frac{\arctan \left[-100 \times \text{multiple} \times (\text{SC ratio} \times \text{SC multiple} + \text{threshold}) \right]}{\left(\frac{\pi}{2} \right)}$$

 When competitor cap (rate) < crediting cap (rate) : SC multiple = 0

The arctangent excess lapse formula shown above is a version of the following formula familiar to some users:

$$\text{total lapse rate} = a + b \times \arctan \left[m \times \left(\frac{\text{competitor cap (rate)} - \text{crediting cap (rate)} - \text{SC ratio}}{c} \right) - n \right]$$

Base lapse rates = the result of the above formula when the crediting rate equals the competitor rate.

The formulas are identical when the parameters are converted as follows:

$$\begin{aligned}\text{Threshold} &= \frac{n}{m} \\ \text{Multiple} &= \frac{m}{100} \\ \text{SC multiple} &= \frac{1}{c} \\ \text{Maximum lapse rate} &= a + b \times \left(\frac{\pi}{2}\right)\end{aligned}$$

7.4.6 Monthly lapses

The annual lapse rate after all adjustments and excess lapses is constrained to be not more than 1.0 (100%).

Lapses and surrenders occur either: (1) at the end of each month, or (2) modally (just before the next premium is due). Lapses occur monthly after the last premium has been paid.

Each of the above lapse rate adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly or modal lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/\text{event frequency})}$$

Modal lapse timing does not apply to fund-related lapse, which is discussed below.

7.4.7 Skewed lapses

Skewed lapses allocate a specific portion of annual lapses to each month within each policy year. The skew percentages are normalized to assure that they add to 1.0 (100%) for each policy year. Lapse skewing can be specified differently for each policy year.

When skewed lapses apply, the exponent in the monthly/modal lapse rate formula is replaced by the sum of the skew percentages for the months included in the decrement time step. For example, when the lapse applies for months with monthly skew percentages adding up to 0.4 (40%), the lapse rate is:

$$1 - (1 - \text{annual lapse rate})^{0.4}$$

7.4.8 Fund-related lapse

The model point lapses when:

- Either (option choice) the cash value or the fund value becomes zero or negative, and
- The secondary guarantee test condition has been violated. However, lapse does not occur when premiums are adjusted to minimally fund the guarantee.

Fund-related lapse does not occur during the specified number of 'exception years' after issue.

8 RIDERS

8.1 Multiple riders

Multiple riders can apply to a model point.

8.1.1 *Riders input*

Each rider creates an instance of the iul_rider submodel. The input for each variable must be linked to an external source containing values varying by the rider names.

8.1.2 *Rider model point input*

The rider model point record identifies which riders apply to each model point.

8.2 Rider types

8.2.1 *Accidental death benefit*

Please refer to the L13 Riders book for a discussion of the accidental death benefit riders which can be linked to indexed universal life model points.

8.2.2 *Waiver of premium*

Please refer to the L13 Riders book for a discussion of the waiver of premium riders which can be linked to indexed universal life model points.

8.2.3 *Waiver of COI and other charges*

Please refer to the L13 Riders book for a discussion of the waiver of COI and other charges riders which can be linked to indexed universal life model points.

9 REINSURANCE

9.1 Multiple reinsurance treaties

Multiple reinsurance treaties can apply to a model point.

9.1.1 *Treaty input*

Each reinsurance treaty creates an instance of the iul_ireins submodel. The input for each variable must be linked to an external source containing values varying by the treaty names.

The exception is that the YRT retention maximum per policy applies for all treaties combined, and thus does not vary by treaty. For example, it might vary by plan code.

Each treaty can be either a YRT, coinsurance, or modified coinsurance treaty. Each of these types is discussed in a separate section, below.

Each of these types is discussed in a separate section, below.

9.1.2 *Reinsurance model point input*

The reinsurance model point record identifies which treaties apply to each model point, as well as the ceded percent or amount for each treaty. The reinsurance model point layout is discussed in the Reinsurance model point records layout section, below.

9.2 YRT reinsurance and coinsurance of mortality risk

Yearly renewable term (YRT) reinsurance cedes a portion of the net amount at risk (NAR). Reinsurance premiums are calculated using YRT rates applied to the ceded coverage amount.

Coinurance of mortality risk is similar to YRT reinsurance, but with reinsurance premiums based on the policy's cost of insurance rates.

9.2.1 Ceded amount

The ceded amount is a percent of the net amount at risk (NAR) at the beginning of each month, using the fund value at the end of the previous month. This is the total fund value. When the unloaned fund value is negative, the NAR is calculated using zero unloaned fund value.

Using reinsurance model point variables

The ceded percent can be specified on the reinsurance model point record, either directly or as the initial ceded death benefit amount, which is converted to the ceded percent (i.e. ceded death benefit amount / death benefit at the projection date.) This is proportionate YRT reinsurance, where the ceded percent is fixed.

Using treaty variables

The ceded percent can be calculated using the reinsurance input variables that vary by treaty code. The ceded amount is a specified percent of any excess of the NAR over the retention threshold per policy. For example:

- **First dollar quota share** – The threshold is zero.
- **Excess retention** – The ceded percent applies to the excess of the NAR over the specified threshold.

The ceded percent varies over time, depending on the changing relationship between the NAR and the threshold.

9.2.2 Maximum YRT retention per policy

The YRT reinsurance treaty amounts for a model point are increased when the combined ceded amounts would be less than [the NAR less the maximum retention amount per policy]. Specify the maximum retention amount in the iul submodel; this variable applies for the product instead of the iul_ireins variables that vary by treaty.

This adjustment applies only for YRT treaties that have the ceded amount calculated using the treaty variables. Treaties that have the ceded percent/amount specified on the model point record are not adjusted.

When the sum of the ceded amounts is too low, the initial (at the projection date, or issue date for new business model points) ceded amount for each treaty is increased in proportion to the individual ceded amounts.

9.2.3 YRT rate

The YRT reinsurance rate for each treaty is a base table, times a multiple. These variables typically operate the same as for mortality rates.

The YRT premium can be zero for the first policy year by specify the multiple using a Select and Ultimate table style with zero rates for select duration 0.

Term conversion adjustments

Term products often offer the option to convert to a permanent product (i.e. indexed universal life) prior to the end of a level term period without additional underwriting.

Specify the original term issue date and whether the policy was issued as a result of a term conversion on the model point record. For such policies, the YRT base rate and YRT multiple will be calculated using a point-in-scale method based on the original issue date of the term policy. Additionally, a separate set of YRT base rates and YRT multiples may apply to the term conversion cohort of policies if specified.

YRT term conversion adjustments may be turned off using the `yrt_term_conversion_defn` variable (on the LifeReinsurance input page).

YRT term conversion adjustments do not apply to joint life policies.

Substandard multiple

The substandard multiple on the model point record applies to the YRT rate. This is the same substandard multiple as applies for experience mortality rates.

9.2.4 Reinsurance premium

YRT reinsurance premiums are paid either annually at the beginning of each policy year, or monthly.

The YRT reinsurance premium per policy is:

$$\text{ceded amount} \times (\text{annual YRT rate}) + \text{reinsurance policy fee}$$

When YRT premiums are paid monthly, one twelfth of this amount is paid at the beginning of each month.

The reinsurer refunds the unearned YRT reinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual YRT premium, times the remaining months in the year divided by 12.

9.2.5 Reinsurance benefits

The reinsurance benefit paid to the ceding company each month is the ceded amount times the monthly experience mortality rate. The ceded amount is the amount calculated at the time of each YRT reinsurance premium.

9.2.6 Reinsurance reserve

The formulaic statutory and tax reserve credit for YRT reinsurance is the tabular cost of insurance floor for the ceded amount.

9.3 Coinsurance

Coinsurance treaties cede a portion of the entire policy, including transferring the ceded reserve to the reinsurer. Reinsurance benefits include all benefits provided by the policy.

9.3.1 Coinsurance percent

When the coinsurance percent is calculated using the input for the treaty, the percent is x% of the sum insured at issue up to the threshold per policy specified for each treaty, plus y% of any excess over the threshold. When the coinsurance percent is specified by model point, this treaty input is ignored. The coinsurance percent remains constant to maturity.

The coinsurance percent applies to the entire policy, including the deficiency reserve and the additional reserve.

The coinsurance benefits paid to the ceding company each month are the projected policy benefits, times the coinsurance percent.

9.3.2 Coinsurance premium

Coinsurance premiums are paid at the same premium frequency as the model point.

Coinsurance premiums are based on the gross premium, including policy fee.

The reinsurer refunds the unearned coinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual amount times the remaining months in the year divided by 12.

9.3.3 **Premium tax**

The reinsurer reimburses for the coinsurance percent times premium taxes paid by the ceding company.

9.3.4 **Benefits**

The reinsurer reimburses for the coinsurance percent times all benefits paid by the ceding company. Reimbursements are split between death benefits and other benefits, for reporting purposes.

9.3.5 **Expense allowance**

The reinsurer pays an expense allowance to the ceding company at the reinsurance premium frequency. The expense allowance is the annualized reinsurance premium times a specified percent.

The unearned current year expense allowance is refunded to the reinsurer when the policy terminates before the next reinsurance premium payment for any reason. The unearned expense allowance is the last amount paid, times the ratio of the number of months until the next reinsurance premium, divided by [12 divided by the reinsurance premium frequency].

9.3.6 **Reinsurance reserve**

The reinsurance reserve credit for coinsurance treaties is the total statutory reserve, times the coinsurance percent.

9.4 **Modified coinsurance**

Modified coinsurance (ModCo) treaties operate as described above for coinsurance treaties, except that the reinsurance reserve is retained by the ceding company. The ceding company reimburses the reinsurer for interest on the reinsurance reserve which is backed by assets invested by the ceding company.

The adjustments occur monthly, so that nested projections match standard projection results. The adjustment amount is the current reinsurance reserve, less the reinsurance reserve as of the previous adjustment date accumulated with interest.

The ModCo reserve adjustment interest rate is the asset yield rate.

9.5 **Reinsurance administration expenses**

Reinsurance administration expenses are assumed to be incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include:

- Amount per policy
- Percent of the ceded amount.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Modified coinsurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for reinsurance.

10 SOLVING CALCULATIONS

Solving calculation results are amounts per unit in force.

10.1 On-the-fly calculations

10.1.1 Values calculated

The following values are calculated as needed during projections. Calculations use the death benefit option and minimum death benefit (corridor) percent which applies for the model point. Partial withdrawals and policy loans are ignored.

Result	Interest	COI	Loads	Premium Mode	Death Benefit
Guaranteed maturity premium (GMP), for UL CRVM, endowing at the sum assured [solved_gmp]	Guaranteed*	Guaranteed	Guaranteed	Modal	Guaranteed fund
Guaranteed level premium, for PBA NPR, with zero endowment [solved_guar_level_prem]	Guaranteed*	Guaranteed	Guaranteed	Annual	Guaranteed fund

*The guaranteed interest rate ignores any initial guarantee of the crediting rate at issue.

10.1.2 Solving iterations

When the closed form solution does not apply (see below), these results are solved by iterating the appropriate instance of the iul_ifund submodel, projecting using successive trial premiums. [solving_driver function]

10.1.3 Displaying solving results

Use the solving_display_defn variable (on the Product input page in the Premiums category) to have the Run Log show the series of solving guesses and related fund value results. The output for the ul_ifund submodel then includes the column values for each solving fund projection using the final solved premium result.

This variable should of course be set to No for production runs, in order to improve calculation time and to reduce the size of the output.

10.2 Closed form solutions

10.2.1 Available formulas

Closed form solution calculations can replace the solving iterations to reduce calculation time significantly. Closed form solutions are available for these results:

- Cumulative premiums test level premium at issue – always calculates using a closed form solution, since solving iterations are never needed under the assumption that interest ratcheting does not apply when this premium is paid.
- Cumulative premiums test single premium – always calculates using a closed form solution, including the application of interest ratcheting.
- Guaranteed maturity premium (GMP)
- Guaranteed level premium

10.2.2 Solving methods

The following types of solving methods are used:

- **Accumulation factors** – Calculations for level premium results are based on an approach which creates accumulation factors for the various elements which develop the ending fund value. The closed form solution for the premium is a formula which sums the target fund value with factors for charges and loads and divides by a factor for the premium. These solutions were developed using techniques presented in the paper 'Calculating Funding Premiums For Universal Life Insurance' by Calvin D. Cherry, on pages 20-27 of the North American Actuarial Journal, Volume 4, Number 2.
- **Fund rollback** – The ending target fund value is rolled back each month, first removing credited interest (including ratcheting), then removing charges, then removing loads, then finally adding the premium load to the resulting 'fund value' at the moment before the solving date.
- **Cumulative premiums test** – Calculations for cumulative premium test secondary guarantees are straightforward, either using annuity factors or premium accumulation rollback. Thus solving iterations never apply for these calculations.

10.2.3 Using solving versus closed form solution

For results that can be calculated either using solving iterations or a closed form solution, you can choose which calculation method applies. Solving iterations return the exact result, but increase the calculation time. Closed form solutions are significantly faster, but are not appropriate in all cases.

Use the model point variable `closed_form_prem_defn` to control which type of calculation applies:

- **When Appropriate** – Closed form solutions are used as much as possible; see the Validation topic, below. The closed form result is tested against fund value calculations; when this test fails, solving calculations replace the closed form result to assure accuracy for the projection.
- **No** – Solving iterations apply in all cases. For example, this option can be used to compare against closed form solution results.
- **Yes** – Use closed form solutions in all cases. When the closed form result is tested (against fund value calculations), the Run Log turns yellow and contains a warning when the ending fund value does not match the target value within tolerance limits. This choice reduces calculation time when the accuracy provided by solving iterations is not important for the projection. You should evaluate the loss of accuracy by comparing against a test projection using the When Appropriate choice.

10.2.4 Validation for using closed form solution

The When Appropriate choice is validated against the model point's product definition, in order to determine whether each closed form solution can be used. Solving iterations are used when the product has specifications which are likely to produce a closed form result different from the solving result. The following product specifications cause solving iterations to be used for the When Appropriate choice:

- Plus ROP death benefit – applies for all results
- Non-level percent of premium load different between target and excess premiums when premiums are paid more frequently than annually – applies for level premium solving (GMP and guaranteed level premium) only
- Percent of fund value load – applies for level premium solving (GMP and guaranteed level premium) only
- Minimum death benefit percent (corridor factor) is greater than 1.0 at the GMP maturity age – applies for GMP calculation only
- Plus Fund death benefit – applies for GMP calculation only.

10.2.5 Displaying closed form results

Use the `solving_display_defn` variable (on the Product input page in the Premiums category) to have the Run Log show the closed form solution results. When the closed form solution does not produce the expected ending fund value, and `closed_form_prem_defn` is When Appropriate, the resulting solving calculations are also displayed and the final solving iteration values are included in the `iul_ifund` output results.

This variable should of course be set to No for production runs, in order to improve calculation time and to reduce the size of the output.

10.3 Solving result arrays

Following the solving calculations discussed above, the following array values are calculated for use during the projection:

Result	Frequency	Values
XXX reserve death benefits [dth_ben_tbl_xxx]	Each policy year	Average of monthly death benefits paying SG ART premiums using guaranteed fund
Guaranteed maturity fund [guar_maturity_fund]	Projection months	Guaranteed fund accumulation using solved GMP
Guaranteed maturity COI charge [guar_maturity_coi_chg]	Projection months from issue	Guaranteed fund COI charge using solved GMP, used for waiver of COI benefits
CRVM reserve death benefits [dth_ben_tbl_crvm]	Each policy year	Average of monthly death benefits paying GMP using guaranteed fund
Guaranteed level premium fund [guar_level_prem_fund]	Projection months	Guaranteed fund accumulation using guaranteed level premium
NPR base reserve death benefits [dth_ben_tbl_npr_base]	Each policy year	Average of monthly death benefits paying guaranteed level premium using guaranteed fund
Guaranteed SG level premium fund [guar_sg_level_prem_fund]	End of each policy year	Guaranteed fund accumulation using guaranteed SG level premium
NPR SG reserve death benefits [dth_ben_tbl_npr_sg]	Each policy year	Average of monthly death benefits paying guaranteed SG level premium using guaranteed fund

10.4 Factors calculations

A projection type calculates 'factors' when the product submodel is the top model and the target column is factors_tgt. This target column directs the calculation of each item specified (discussed below). The target column calculates results for each policy year; thus the Projection period from and to properties have no effect on factors calculations.

The Assumption Set Restore property (in the run parameters object) must be Yes, since this is a seriatim projection. Project variables such as pricing definition do not apply for this special type of projection.

10.4.1 Values calculated

Factors calculations use iterative solving (closed form solutions do not apply) to find the following values for each model point:

Result	Interest	COI	Loads	Premium Mode	Death Benefit
Base fund level premium (to produce specified fund value per unit at target duration)	Specified	Current or guaranteed	Current, guaranteed, or None	Modal	Actual

Result	Interest	COI	Loads	Premium Mode	Death Benefit
Cumulative premiums test level premium (accumulates to ending required premiums accumulation)	Secondary guarantee interest rate	N/A	N/A	Modal	N/A
Guaranteed maturity premium (GMP)	Guaranteed*	Guaranteed	Guaranteed	Modal	Actual
Guideline single premium (GSP)	Specified** (6%)	Guaranteed or current	Current or guaranteed	N/A	Sum assured
Guideline level premium (GLP)	Specified** (4%)	Guaranteed or current	Current or guaranteed	Annual	Actual
7-pay MEC premium	Specified** (4%)	Guaranteed or current	None	Annual	Sum assured
Cash value accumulation test (CVAT) net single premiums, and the related minimum death benefit percentages	Guaranteed*	Guaranteed	None	N/A	Sum assured without corridor

*The guaranteed interest rate ignores any initial guarantee of the crediting rate at issue.

**The interest rate for IRS premium restrictions is the greater of the specified interest rate and the guaranteed crediting rate.

10.4.2 Policy fee

The policy fee is included in calculating the target premium. It is excluded for GMP, GLP, GSP, MEC, and CVAT, except for the charging of the policy fee as a product load.

10.4.3 Model points

Factors calculations systematically process for each model point. Set up a model point for each issue age to be calculated, within each combination of insured characteristics (sex, smoking class, rate class, etc.). For example, you might set up model points for issue ages 0 through 80 for male nonsmokers, then repeating these model points for male smokers, then female nonsmokers, then finally female smokers.

When Annual mode premiums are to be calculated for factors using modal premiums, set the model point premium mode to Annual.

Best practice is to create the report group for each model point so that it identifies the model point characteristics. For example, the report group might be PlanA-M-NS-ST-35 for the model point calculating factors for plan code PlanA, for a male nonsmoker standard risk class issue age 35. Then the report groups identify the meaning of each model point.

The same as for Pricing cell projections, each model point must be issued on the first day of the projection. For example, when the projection date is 2013/12 (end of December, 2013), the issue date must be 2014/01/01 (January 1, 2014).

10.4.4 iul_py submodel output

The factors calculation output is contained in the iul_py submodel, which has column results by policy year.

10.4.5 Factors premium reporting

The solved premium is reported in the column factor_prem in the iul_ifund submodel. The standard features round this result up to the next whole penny per unit.

The output file includes this value for every model point. Since the premium result shows only for the commencement period ($t = 0$) in this column, the output from and to properties should both be '0'.

10.4.6 Cash value test reporting

The solved cash value test net single premium results are reported in the column factor_cvat_min_dth_ben_factors in the iul_ifund submodel. The standard features convert the result into the corresponding minimum death benefit multiple, equal to the sum assured divided by the fund value at the end of the previous policy year. These fund values are projected assuming the solved single premium is paid at issue.

Since these values vary over time, the output to property must be the number of months to maturity of the youngest issue age. As for the factors premium reporting, use a tool to query the output database in order to retrieve the values for every model point.

10.5 Factors projection setup

10.5.1 Input variables

The special variables for factors calculations are at the bottom of the Product input page.

For each factors_calc_num, specify the following:

- factors_defn – The result to be calculated; this option list variable is used as a condition in formulas.
- factors_include_defn – Yes to include this calculation, No to exclude it (these lines can be deleted from the external source input if desired).
- factors_tgt_fund_val – Target fund value per unit.
- factors_tgt_dur_defn and factors_tgt_dur – The target duration is a specified attained age or policy anniversary, maturity age, GMP maturity age, or secondary guarantee premium term.
- factors_prem_term_defn and factors_prem_term – The premium term is either a specified attained age or policy anniversary, the premium term, or the contractual premium term. When the premium term exceeds the target duration, the solving is of course not affected by premiums after the target fund value duration.
- factors_coi_rate_defn – Either the current or guaranteed COI rates.
- factors_load_rate_defn – Either the current or guaranteed fund load rates, or using zero loads.
- factors_int_rate – The greater of the guaranteed rate and the specified interest rate.

Cumulative premiums test level premium

An exception to the above description is the Cumulative Premium Test Level Premium. This calculation finds the level premium that accumulates to the required premiums accumulation at the end of the secondary guarantee premium term. The only variable that applies is whether this calculation is included. The target duration, premium term, and accumulation interest rates (including the threshold feature) are specified by the cumulative premiums test secondary guarantee definitions. COI rates and loads do not affect this calculation.

10.5.2 Run parameters

Specify the following settings in the run parameters for the projection task calculating factors:

Property	Value
Target Colum	iul->factors_tgt
Projection Period From	0

Property	Value
Projection Period To	0 (since solving calculations are used, this is not a forward projection)
Projection Period Min	0
Projection Period Max	0 for factors premium output; longest months to maturity for cash value test output
Projection Loops	1
Assumption Set Restore	Yes
proj_date	yyyy/mm, the end of the month before the issue date

11 MODEL POINT RECORDS

11.1 Model point record layout

The model point record layout for the universal life product type is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	up to 125 characters (assigns model point to report group)
B	seg_id	segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date	yyyy/mm/dd or mm/dd/yyyy
F	disab_date_mp	yyyy/mm/dd or mm/dd/yyyy (set to 1/1/9999 if policy is not disabled)
G	term_conversion_defn_mp	Yes (calculate term conversion adjustments), No
H	term_issue_date_mp	None (when original term issue date is unavailable) yyyy/mm/dd or mm/dd/yyyy
I	plan_code	up to 20 characters (link to assumptions by plan)
J	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
K	tax_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)

Excel Column	Variable	Value
L	issue_age_mp	positive integer
M	gender_mp	Male, Female, Unisex
N	smoking_class_mp	Nonsmoker, Smoker, Aggregate
O	rate_class_mp	Standard, Preferred, Super, [extend with user additions]
P	substd_mult_mp	substandard multiple (1.0 for 100%)
Q	issue_age_mp_life2	positive integer, when applicable, otherwise -1
R	gender_mp_life2	Male, Female, Unisex
S	smoking_class_mp_life2	Nonsmoker, Smoker, Aggregate
T	rate_class_mp_life2	Standard, Preferred, Super, [extend with user additions]
U	substd_mult_mp_life2	substandard multiple (1.0 for 100%)
V	prem_mode	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
W	prem_term_mp	premium years (0 for 'to GMP maturity')
X	init_annualzd_prem	initial annualized premium for mp
Y	single_prem	additional single premium for mp
Z	closed_form_prem_defn	When Appropriate (use solving only when needed), No (force the use of solving calculations), Yes (force the use of closed form solutions)
AA	dth_ben_defn	Sum Assured, Plus Fund, Plus ROP
AB	dth_ben_min_pct_defn_mp	Product Specification (use product defn), Guideline Prem Test, Cash Value Test
AC	gpr_pct_mp	percent of this model point included in Gross Premium Reserve calculations used for reserve financing (1.0 for 100%)
AD	init_sum_assured_mp	initial sum assured for mp (0 = calculate from initial units)
AE	init_policies	initial policies for mp (required)
AF	init_units	initial units for mp (0 = calculate from initial sum assured)

Excel Column	Variable	Value
AG	init_fixed_fund_val	initial fixed fund value for mp
AH	init_fund_val_bucket_defn	New Bucket, Policy Anniversary
AI	init_fund_val_index0	initial fund value in index0 for mp
AJ	init_fund_val_index1	initial fund value in index1 for mp
AK	init_fund_val_index2	initial fund value in index2 for mp
AL	init_fund_val_index3	initial fund value in index3 for mp
AM	init_fund_val_index4	initial fund value in index4 for mp
AN	prem_pct_fixed	percent of premium allocated to the fixed fund (1.0 for 100%)
AO	prem_pct_index0	percent of premium allocated to index0 (1.0 for 100%)
AP	prem_pct_index1	percent of premium allocated to index1 (1.0 for 100%)
AQ	prem_pct_index2	percent of premium allocated to index2 (1.0 for 100%)
AR	prem_pct_index3	percent of premium allocated to index3 (1.0 for 100%)
AS	prem_pct_index4	percent of premium allocated to index4 (1.0 for 100%)
AT	init_crediting_rate_base	initial base crediting rate for the fixed fund (0.05 for 5.0%); when this value is the 0.0 default value the initial rate is calculated using the crediting strategy
AU	init_credited_int_guar_index0	Initial cumulative guaranteed interest credited for mp to index0 during the current indexing period
AV	init_credited_int_guar_index1	Initial cumulative guaranteed interest credited for mp to index1 during the current indexing period
AW	init_credited_int_guar_index2	Initial cumulative guaranteed interest credited for mp to index2 during the current indexing period
AX	init_credited_int_guar_index3	Initial cumulative guaranteed interest credited for mp to index3 during the current indexing period
AY	init_credited_int_guar_index4	Initial cumulative guaranteed interest credited for mp to index4 during the current indexing period

Excel Column	Variable	Value
AZ	init_gross_prem_cumul	initial cumulative premiums for mp
BA	init_withdrl_cumul	initial cumulative withdrawals for mp
BB	init_sg_prem_accum	initial SG accumulated premiums for mp (only for cumulative premiums test SG)
BC	pol_loan_defn	Fixed Fund, Indexed Fund
BD	init_pol_loan	initial policy loan for mp
BE	init_host_contract_val	initial SFAS 133 host contract value
BF	init_host_accrual_rate	initial SFAS 133 host accrual rate (0.05 for 5%)
BG	init_surr_chg_withdrl_reductn_mult	average initial surrender charge reduction multiple from previous withdrawals for mp (1.0 for 100%)
BH	init_sg_mths_in_grace_period	consecutive months that the secondary guarantee conditions have not been met for a model point being kept in force by a positive current fund value
BI	init_comm_subject_to_chargeback	initial commission subject to chargeback for mp (-1 to calculate from issue)
BJ	init_sg_funding_indicator	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step

11.2 Reinsurance model point record layout

The reinsurance model point record identifies which treaties apply for each model point, using the linked variable, mp_id. The reinsurance model point record can have any number of reinsurance treaties per model point. It also specifies the ceded percent or amount for each treaty as described in the table below. If the reinsurance model points are omitted from the reinsurance model point file, then the liability model point will not include reinsurance.

Warning! The order of reinsurance model points must be in the same order as the liability model points.

The reinsurance model point record layout is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	mp_id	Up to 12 characters. Links to liability data file mp_id. This value can be listed any number of times to indicate the number of treaties applied to the policy.
B	treaty_code	A user defined input that references the treaty code in the Resinurance.xlsm workbook.
C	ceded_amt_defn	Percent - the ceded amount should be input as a percent from 0 to 1 Dollar Amount - the ceded amount should be input as the ceded death benefit amount Treaty Input - the ceded amount should be input as 0, and the ceded amount will be calculated using the treaty variables
D	ceded_amt	See above. Input the ceded percent, the ceded dollar amount, or 0 if the ceded amount is specified by the treaty.

11.3 Units basis

Product values are calculated per unit in force, then multiplied by the current number of units. For example, the death benefit, cash value, and reserves per unit.

11.3.1 Initial units

The initial (at the projection date) policies must be specified on each model point record, along with either or both of the initial units and initial sum assured. These are the aggregate values for the policies collected into each model point.

- When the initial units are not specified on the model point (this variable is left at its zero default value), the initial units are the initial sum assured divided by the specified sum assured per unit.
- When the initial sum assured is not specified on the model point (this variable is left at its zero default value), the initial units variable applies directly.
- When both the initial units and initial sum assured are specified on the model point record, the initial units variable applies. The Run log contains a warning if the initial sum assured is more than [\$1 times the initial units] different from the initial units times the specified sum assured per unit.
- When both the initial units and initial sum assured are zero, the model point is skipped and a warning is issued in the Run log.

The initial policies must be specified for each model point. The average size policy is the initial units divided by the initial policies, and the initial sum assured divided by the initial policies.

11.3.2 Pricing cells

For model points issued at the start of a seriatim pricing projection, the initial policies, units, and sum assured fields define the policy size. When the model point specifies one policy, the initial units and sum assured are average size amounts.

11.4 Required model point variables

11.4.1 Always required

The following model point variables are required in all cases:

- **Model point identification:** group and mp_id
- **Insured characteristics:** issue_age, gender, smoking_class, rate_class, and substd_mult, for life 1 in all cases, and also for life 2 for joint life products
- **Policy specifications:** issue_date_mp, term_conversion_defn_mp, term_issue_date_mp, plan_code, stat_valn_code, tax_valn_code, prem_mode, prem_term_mp (may be default of zero which indicates that premiums are paid in all years), dth_ben_defn, and dth_ben_min_pct_defn_mp (may be default of Product Specifications)
- **Initial policies:** init_policies
- **Units basis fields:** one or both of init_sum_assured and init_units
- **Closed form solution usage:** closed_form_prem_defn
- **Annual premium:** init_annualized_prem is required when the annual premium is specified by model point.
- **Single premium:** single_prem is required for new business model points. It is required for in-force model points in these conditions: (1) the model point is still in the first year and the premium load varies between target and excess premium, or (2) the commission subject to chargeback is being calculated from issue.

11.4.2 In-force model point values

In-force model points are driven by all the 'initial values' variables, which specify the aggregate model point values at the projection date.

New business model points (model points with issue date after the projection date), however, ignore these variables. The exception is that the initial policies must be specified, along with either or both of the initial units and initial sum assured.

The following table shows when each in-force values variable is required for in-force model points:

Variable	Required When
init_fund_val_xxx	always required for the accounts which are used
prem_pct_xxx	always required for the accounts which are used
init_crediting_rate_base	always required
init_credited_int_guar_xxx	recommended but NOT required, when left at 0.0 default, amount is estimated from guaranteed credited interest in first projection month
init_gross_prem_cumul	model points with Plus ROP death benefits
init_withdrl_cumul	model points with current or future policy loan, for products with the preferred policy loan feature
init_sg_prem_accum	products with secondary guarantee
init_pol_loan	model points with policy loan
init_surr_chg_withdrl_reductn_mult	model points with previous withdrawals, for products with surrender charges other than percent of fund value (may be left at 1 default to ignore this initial effect)
init_sg_mths_in_grace_period	model points with positive current fund value but unsatisfied secondary guarantee conditions

Variable	Required When
init_comm_subject_to_chargeback	model points still subject to commission chargeback (may be left at -1 default to calculate from issue)

11.4.3 Variables to ignore

The following model point variables are needed only for nested projections, and should be ignored. These values must be carried forward to the next time step in order to reproduce the results from standard projections:

Variable	Value
init_sg_funding_indicator	use zero default value
init_surr_chg_withdrl_reductn_mult	Note above that this field may be left at 1 default to ignore this initial effect
init_sg_mths_in_grace_period	This field can be ignored when this information is not readily available; the primary purpose is to provide continuity for nested projections
init_comm_subject_to_chargeback	This field can be ignored when this information is not readily available; the primary purpose is to provide continuity for nested projections

12 SUBMODELS STRUCTURE

12.1 iul submodel

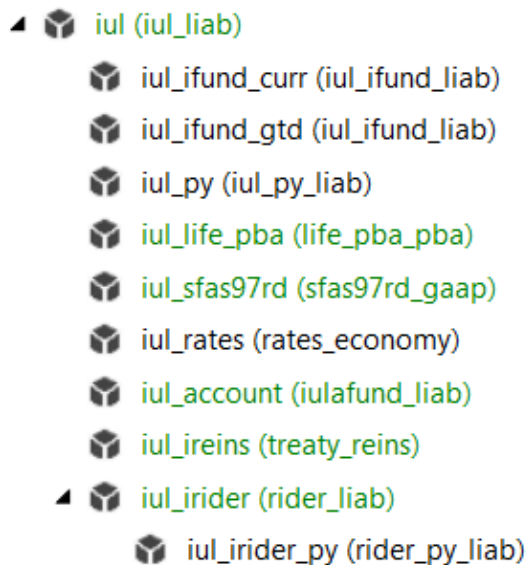
The iul submodel calculates the model point's contribution to the liability income statement and balance sheet items for indexed universal life products.

The iul submodel calculates model point results using values per unit in force, which are then multiplied by the number of units. Results per unit in force have '_per_unit' added to the formula name. Formula names without this addition are results per model point (or are a special result like a ratio).

12.1.1 Seriatim versus portfolio runs

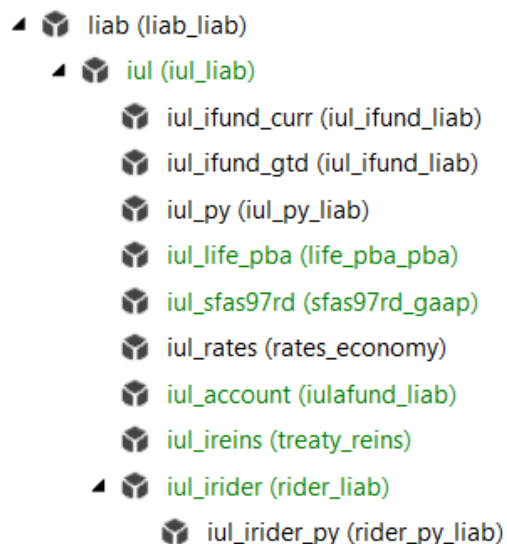
Seriatim projection

When iul is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the iul submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings.



Portfolio projection

For portfolio projections (which include business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the iul submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the iul submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the iul submodel the same columns as for other product types. This makes the indexed universal life product type interchangeable with other product types fitting into this business segment structure.

12.1.2 Variables

Variables for indexed universal life products are located in the iul submodel, except as noted below for the variables in iul_ifund, iul_account, and iul_py.

12.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when iul is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the iul submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only iul is the top model, (2) when iul is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when iul is run as the top model and when iul is included in portfolio projections. References to company, segment and liab are invalid when iul is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) solving values are initialized, (3) temporary tables are sized for DAC tax calculations and competitor rates, (4) projection flags are created and initialized, (5) reporting groups are set up, (6) iul_ifund_curr, iul_ifund_gtd, and iul_account elements are setup, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each projection task loop, immediately after startup when iul is the top model. When company is the top model, initialize is executed by calls from seg->initialize which executes iul->initialize for each model point in the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when iul is running as the top model. This includes executing the initialization of economic scenario rates and the setup of competitor rates. Other initializations are required in both cases.

The main use of initialize is to orchestrate the various solving calculations for each model point, and to precalculate the per unit reserve amounts using the actuarial math library. Since these initializations are not required for all calculation types for all product features, conditions are checked to shortcut the initialization process, thus reducing the runtime of the model point.

The solving calculations, and the filling of the associated array values, are discussed in the Solving calculations chapter, above.

The reserves are initialized in the following order:

1. Base statutory reserves
2. Base statutory minimum reserves
3. XXX statutory reserves
4. XXX statutory minimum reserves
5. Base tax reserves
6. XXX tax reserves.

solving_driver

The 'solving_driver' function is used to iterate the iul_ifund submodel to solve for results when the closed form solution is not used, or produces an ending fund value that does not match the target value within the tolerance limit. See the Solving calculations chapter, above.

Each call to this solving_driver function includes the appropriate fund_accum_defn for iul_ifund. See the iul_ifund submodel section, below.

Following some initial setup, a 'while' loop is processed. Within each pass of the loop, a trial premium is used to project iul_ifund, then the projected fund value is compared to the solving target. A new trial premium is used to re-project iul_ifund until the solution has been determined within a tolerance limit.

Details of the solving process and results from the final solve can be captured when solving_display_defn (on the Product input page, under the Premiums category) is All Solves or GMF Only. The solve guesses are

written to the Run Log, and the final solving fund accumulation is written to the column output using a writeModel statement. This option should be No for production runs.

calc_prem

The function 'calc_prem' calculates the closed form solutions for premium results using the Accumulation factors or Fund rollback methods. See the Solving calculations chapter, above. Each call to this calc_prem function includes the fund_accum_defn which would have applied for solving for the result using iterations of iul_ifund.

The closed form results are written to the Run Log when solving_display_defn (on the Product input page, under the Premiums category) is All Solves or GMF Only. This option should be No for production runs.

stat_res_ulcrvm_per_unit

Columns 'stat_res_ulcrvm_per_unit' (and the corresponding column for tax reserves) are special, in the sense that they include a future projection of the guaranteed_fund instance of iul_ifund. This future projection requires that these columns reset and recalculate the guaranteed_fund instance of iul_ifund. When IUL CRVM reserves require the GMF to be recalculated, this instance of iul_ifund is recalculated beginning from the current fund value, using guaranteed rates and assuming the GMP is paid in the future.

PBA NPR calculation columns

The column stat_res_per_unit orchestrates the calculation timing of columns for PBA NPR calculations. At each exact calculation period, the guaranteed fund instance of the iul_ifund submodel is rebased to start from the current fund value. These guaranteed fund values (paying the guaranteed level premium) are used to calculate the present value of benefits less premiums as of the valuation date. Then that column [stat_res_npr_base_pv_per_unit] is called to roll this present value forward for each month until the next exact calculation period.

Similarly at each exact calculation period, the reserve lapse rate is calculated in column stat_res_npr_sg_lapse_rate, and is stored in variable stat_res_npr_sg_lapse_rate_saved. Using the stored values, the net single premium is calculated as of the valuation date. This calculation also includes rebasing the ul_ifund submodel to start from the current fund value, then projecting forward using the revised guaranteed SG level premium per unit. Then that column [stat_res_npr_nsp_per_unit] is called to roll this present value forward for each month until the next exact calculation period.

The final NPR reserve calculations are performed each month in columns stat_res_npr_base_per_unit and stat_res_npr_sg_per_unit. These columns use the present value results which have been either calculated using the exact method, or rolled forward between valuation dates.

Other formulas

Other formulas in the iul submodel are relatively straightforward. Results which are per unit in force add '_per_unit' to the end of the formula name. Results without '_per_unit' are values for the model point.

Most of the profit, balance sheet, and GAAP columns are only calculated when ul is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

12.2 iul_ifund_xxx submodels

The iul_ifund_xxx submodels are two instances of the iul_ifund_liab model class. They contain the variables and formulas required to calculate per unit in force individual fund values for indexed universal life model points. Since all results are per unit in force, this submodel does not include '_per_unit' in the formula names.

12.2.1 Multiple instances

Each model point carries two unique instances of the iul_ifund_liab model class:

- iul_ifund_curr, for the current fund value, which is always present.

- iul_ifund_gtd, for the guaranteed fund, which is projected for solving calculations and for creating related array values (see the Solving calculations chapter, above), and when calculating present values for IUL CRVM reserves and for PBA NPR base and secondary guarantee reserves.

These unique elements of iul_ifund are set up during the startup processing of the model point.

Formula references to the unique instances utilize the unique model reference pointer. References for each fund value basis follow this example:

- Current: sm_iul_ifund_curr->fund_val_aft_withdrl
- Guaranteed: sm_iul_ifund_gtd->fund_val_aft_withdrl

12.2.2 Working submodel

A specific iul_ifund instance may be a 'working' submodel which is reset at various times as required for solving calculations, and for reserve projections starting from the current fund value. Accordingly the values in these columns at the end of a projection apply only for the last time this submodel was calculated. For example, the iul_ifund_gtd values are not the guaranteed fund values calculated from issue; they could be the calculations starting from the current fund value at the last time when IUL CRVM reserve calculations were processed.

To see a particular set of results, customize to add code to write out the submodel results at the time that calculation has just been performed. Search for the 'writeModel' command in the iul->initialize formula to see how this is done when the solving_display_defn calls for writing out the submodel values.

12.2.3 Variables

Variables for COI rates and product loads are located in the iul_ifund submodel. These variables are contained in the iul_ifund submodel so that they can vary between current and guaranteed calculations. See the Fund variables section in the Input organization chapter, above.

In addition to the variables discussed above, two variables in iul_ifund are hidden from the data entry but are critically important to the successful operation of the iul_ifund elements:

fund_val_basis

The 'fund_val_basis' variable is an option choice variable with choices of Current or Guaranteed. This variable is used as a condition in formulas, most commonly for determining which input rates to use. After each iul_ifund instance is setup inside 'startup' of iul, this variable of the iul_ifund instance is set.

fund_accum_defn

A second iul_ifund variable, 'fund_accum_defn' is used as a condition in formulas, most commonly for determining which type of calculation is currently being iterated. This is also a choice list variable with choices which provide for execution of specific formulas for each fund accumulation definition type:

- **Current Fund** – Formulas for the current fund value.
- **Guaranteed Fund** – Formulas for calculating: (1) average death benefit for XXX reserves assuming the secondary guarantee annual required premiums are paid (when dth_ben_tbl_xxx_flag), (2) guaranteed maturity premium and fund value (when pba_npr_indicator is 0), (3) average death benefit for CRVM reserves assuming the GMP is paid (when pba_npr_indicator is 0), (4) guaranteed level premium and fund value (when pba_npr_indicator is 1), (5) average death benefit for NPR base reserves assuming the guaranteed level premium is paid (when pba_npr_indicator is 1), (6) guaranteed SG level premium fund value (when pba_npr_indicator is 2), and (7) average death benefit for NPR SG reserves assuming the guaranteed SG level premium is paid (when pba_npr_indicator is 2).

12.2.4 Formulas

Formulas for projecting indexed universal life model point fund values are contained in the iul_ifund submodel. A combination of columns, scalars, and external functions are used to provide the required functionality.

Startup

The startup column of each iul_ifund instance is called during the setup of the model.

The guaranteed_fund instance of the iul_ifund submodel can be reset and recalculated multiple times during the projection. This is controlled by iul columns like initialize, stat_res_ulcrvm_per_unit, stat_res_per_unit (which calls sg_level_prem_per_unit, stat_res_npr_base_pv_per_unit, and stat_res_npr_sg_per_unit), and by the solving driver function, where iul_ifund is reset or rebased, then re-projected. As discussed above, the iul_ifund formulas include variations for different values of fund_val_basis and fund_accum_defn.

12.3 iul_account submodel

The iul_account submodel is a two dimensional user defined array instance of the iulafund_liab model class and contains all variables and formulas required to calculate per unit fixed and indexed account fund values for indexed universal life model points.

The first dimension of the iul_account submodel is the index and the second dimension is the indexing bucket for the premium payments and rollovers.

12.3.1 Multiple instances and buckets

Each model point carries between one and five elements in the first dimension of iul_account. There is always a fixed account. The indexed accounts are distinguished by fund_id, which has values of:

- Index0, for the first indexed account fund value.
- Index1, for the second indexed account fund value.
- Index2, for the third variable account fund value.
- Index3, for the fourth variable account fund value.
- Index4, for the fifth variable account fund value.

Each index account element of iul_account carries the number of elements in the second dimension, referred to as buckets, required to support the indexing crediting months. The fixed account always has one element in the second dimension. The number of elements in the other indexed account is based on the value of the crediting_mths variable. For example, an index with a value of 12 specified for the crediting_mths variable will create 12 buckets.

Elements of iul_account are created during the startup processing of the model point. Conditions are evaluated in startup of iul and the iul_account user defined array is resized as required.

Specific instances of the iul_account submodel are aggregated using the 'sum_over_funds' and 'sum_over_fund_buckets' functions of the iul submodel.

The first dimension of the iul_account array element numbers range from 0 to 5. The actual integer value for each array depends on which accounts exist for a particular model point. Element numbers, beginning with element 0, are assigned to each account in the following order:

- **Fixed account** – the first element number is assigned to the fixed account.
- **Indexed accounts** – the indexed accounts are then assigned to element numbers in order, as required. For example, if a model point uses Index0 and Index3, Index0 would be assigned element number 1 and Index3 would be assigned element number 2.

The second dimension of the iul_account array element numbers range from 0 to crediting_mths – 1 of the index. Each element is characterized by a bucket number (bucket_num scalar) and bucket age (bucket_age column). The bucket number is defined when the bucket is created. The bucket age is the number of months since the start of the bucket, with the bucket beginning at age 0 and increasing to crediting_mths – 1. When a bucket renews, the bucket age resets to 0.

When created, the bucket numbers are relative to the projection date. For example, for a 12 month indexing period, bucket 0 is the bucket which will receive premiums in projection months 1, 13, 25, etc., and has a

bucket age of 11 at the projection date and 0 in the first month of the projection. Bucket 11 is the bucket which will receive premiums in projection months 12, 24, 36, etc., so it has a bucket age of 0 at the projection date and 1 in the first month of the projection.

12.3.2 Variables

Variables for the crediting rate definition, crediting months, crediting index, lookback interval, excess crediting fund basis, crediting floor, the participation rate, and cap rate parameters for the index accounts are contained in the `iul_account` submodel. These variables are contained in the `iul_account` submodel so that they can vary by index account.

Some `iul_account` variables are set from values of the model point or from values of the `iul` submodels. These variable values are not entered as assumptions for the `iul_account`. Instead, these variable values are set using the `set_iul_account_vars` external function of the `iul` submodel, which is called from startup after the `iul_account` elements have been sized.

12.3.3 Formulas

Formulas for projecting indexed universal life model point accounts are contained in the `iul_account` submodel. A combination of columns, scalars, and external functions are used to provide the required functionality.

Startup

Since there are no instances of `iul_account` defined through the model parameters, the initial size of the `iul_account` two dimensional array is zero, and the 'startup' column is not called automatically. The startup column is called when the `iul_account` array is resized using the `sm_iul_account[account].resize(count)` statement from within the startup column of the `iul` submodel.

12.4 iul_rider submodel

The `iul_rider` submodel is a dependent data driven array instance of the `rider_liab` model class containing the variables and formulas required to calculate the rider cash flows and reserves for indexed universal life model points.

Please refer to the L13 Riders book.

12.5 iul_ireins submodel

The `iul_ireins` submodel is a dependent data driven array of the `treaty_reins` model class containing the variables and formulas required to calculate the ceded amount and reinsurance cash flows for indexed universal life model points.

12.5.1 Multiple instances

Each model point carries multiple instances of `iul_ireins`. The number of instances is determined by the number of treaties specified on the reinsurance model point record.

Elements of `iul_ireins` are created during the startup processing of the model point. Conditions are evaluated in startup of `iul` and the `iul_ireins` user defined array is resized as required. The characteristic for each `iul_ireins` element is specified by the treaty code.

The ceded amount and reinsurance cash flows for multiple treaties are aggregated using the `sum_over_reinsurance` function in the `iul` submodel. This function can aggregate the values for all treaties, or just the treaties of a specific type of reinsurance, such as YRT.

12.5.2 Variables

Variables for the reinsurance type, the reinsurance expenses, and for calculating the ceded amount (when it is not specified on the reinsurance model point record), are located in the iul_ireins submodel. These variables are contained in the iul_ireins submodel so that they vary by treaty.

12.5.3 Formulas

Formulas for calculating the ceded amount and the reinsurance cash flows are contained in the iul_ireins submodel.

Since iul_ireins is defined through the model parameters, the size of the iul_ireins array is based on the number of treaties specified for each model point on the reinsurance model point file.

12.6 iul_py submodel

The iul_py submodel is a standalone instance of the iul_py_liab model class. It contains the formulas and variables required to calculate per unit in force results which only vary by policy year. Thus 't' in this submodel represents policy year, unlike the iul submodel where 't' represents months since the projection date. The policy year values are calculated from issue.

Since all results are per unit, this submodel does not include '_per_unit' in the formula names.

12.6.1 Variables

Variables located in the iul_py submodel are only used in formulas within this submodel. These are generally for values calculated from issue and varying by policy year, such as annual premiums, and cash value and reserve terminal values and net premiums.

12.6.2 Formulas

Formulas for calculating policy year per unit in-force results are included in the iul_py submodel. Formulas for iul_py are used to calculate annual and target premiums, plus reserve values by policy year.

The iul_py submodel also includes formulas which Frasierize the experience mortality rates, which are Frasierized using annual rates. COI rates, however, are Frasierized in the iul_ifund submodel since these rates vary between the current and guaranteed fund rates bases.

Columns are generally calculated beginning at policy year 1, even for in force model points. As discussed in the iul submodel section, the reserve terminal values and net premiums are calculated by calling the Actuarial math library. See the Actuarial math library section below.

12.7 iul_rates submodel

The iul_rates submodel is identical to the rates submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

12.8 iul_life_pba submodel

The iul_life_pba submodel is identical to the life_pba submodel which applies for portfolio projections. This submodel is used for principle-based approach (PBA) capital and reserve calculations, and for gross premium reserve (GPR) calculations.

Please refer to the C05 PBA book.

12.9 iul_sfas97rd submodel

The iul_sfas97rd submodel is identical to the sfas97rd submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

12.10 Actuarial math library

12.10.1 AML DLL

The Actuarial math library (AML) provides standard actuarial calculations for traditional reserve methods. See the Actuarial math library liability book.

The iul submodel calls the AML to calculate reserve amounts per unit in force. Columns in the iul_py submodel store the terminal value and net premium for each policy year for each reserve basis used in the projection.

12.10.2 iul->startup

The iul->startup column specifies the name extension for the audit trail comma separated values file (when requested).

12.10.3 iul->initialize

The iul->initialize column initiates the call to the AML calculations for each set of reserve results. Each set of AML results is calculated by calling the aml_driver in the iul_py submodel. A separate call can be made for each of the following results:

- Statutory base reserves – Base plan CRVM statutory reserves
- Statutory minimum reserves – Base plan alternative minimum CRVM reserves for calculating deficiency reserves
- Statutory XXX reserves – Base plan Regulation XXX reserves
- Statutory minimum XXX reserves – Base plan Regulation XXX minimum reserves for calculating XXX deficiency reserves
- Tax base reserves – Base plan CRVM tax reserves
- Tax XXX reserves – Base plan Regulation XXX tax reserves
- Base NPR reserves – Base NPR reserves
- SG NPR reserves – Secondary guarantee NPR reserves

12.10.4 AML driver

The aml_driver in the iul_py submodel sets up and calls the AML for each set of reserve results. This driver is called by the iul->initialize column for each set of results.

This driver sets up the call to the AML using the appropriate input for each type of calculations. Each call to the driver calls these functions in the iul_py submodel:

- setup_aml_initial_data – Creates the interfaces for using the AML.
- setup_aml_res_defn – Specifies the expense allowance and segmentation method (1980 CRVM, Regulation XXX), and the function type (curtate, continuous, or semi-continuous).
- setup_aml_policy_data – Specifies values for each policy year, such as the reserve premiums, death benefits, valuation interest rates, mortality rates, plus the issue age, maturity age, premium term years, and expense allowance grading years.

The operation of these setup functions varies by the type of calculations being made. For example, the policy data passes the statutory rates when calculating statutory reserves, and the tax rates when calculating tax reserves.

12.10.5 Calculation results

The AML calculation results are stored for each policy year in columns in the iul_py submodel. The formula for each column simply gets the AML calculation results for each case. These columns include:

- Terminal reserves and net premiums for each basis
- Valuation interest and mortality rate for each basis
- Tabular cost of insurance for each basis

The `aml_driver` calls the appropriate columns after it calls the AML. For example, the statutory basis columns are called when the driver has just calculated statutory results. Then the driver calls the tax reserve basis columns when those results have just been calculated by the AML. Thus each of these columns stores the AML results using the appropriate valuation basis.

L06 FIXED ANNUITY

This book discusses fixed deferred annuity products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

- Product – Benefits, premiums, MVA, and surrender charge definitions
- Fund – Interest crediting, loads, fees
- Valuation – Statutory and tax reserve definitions
- Expense – Commission and expense rates, premium tax rate
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse and annuitization rates, withdrawals, dynamic premiums

Each of these input page types begins with 'FA', for use in projections that combine multiple product types.

1.2 Data pages

Variables for each fixed annuity model point are specified on the FADData page. These variables are discussed in the Model point records chapter, below.

2 PRODUCT BENEFITS AND VALUES

The following explains the basic features for fixed deferred annuity products.

2.1 Maturity

The policy terminates at the maximum attained age specified for the product. When the specified maturity age is zero (default value), the maturity age is the omega age of the experience mortality table.

The fund value is paid at maturity.

2.2 Death benefits

2.2.1 Basic death benefit

The death benefit is either the: (1) fund value, or (2) cash value, without any market value adjustment.

2.2.2 Death benefit paid versus in force

Death benefits paid at the end of each month are based on the fund value before policyholder transactions are processed. Thus the death benefit is calculated using the fund value before penalty free withdrawals.

Death benefits in force at the end of the month are based on the fund value at the end of the month, after penalty free withdrawals have been taken.

2.3 Annuity benefits

2.3.1 Current basis purchase rates

Current basis annualized annuity benefits are determined as the fund value less a load and policy fee divided by a current basis immediate annuity purchase rate factor.

The current basis annuity purchase rate factor uses current basis base mortality times a multiple and current basis interest. The current basis interest rate may be a specified fixed rate or a spread from a specified competitor rate, where the spread may vary each benefit year.

2.3.2 Guaranteed basis purchase rates

Guaranteed basis annualized annuity benefits are determined as the fund value divided by a guaranteed basis immediate annuity purchase rate factor.

The guaranteed basis annuity purchase rate factor uses guaranteed basis base mortality times a multiple and a guaranteed basis interest. The guaranteed basis interest rate is always a specified fixed rate.

2.3.3 Projection treatment

Policyholders who annuitize during the projection receive the fund value as a lump sum. There is no calculation of the annuity purchase rates or projection of future annuity benefits.

2.4 Withdrawal benefits

2.4.1 Penalty free withdrawals

The policyholder can withdraw a limited portion of the fund value without surrendering the policy or incurring surrender charges. This is sometimes referred to as a partial surrender or cash withdrawal provision.

Penalty free withdrawals are processed at the end of each month, for policies persisting into the next month; after all other decrements are processed. Thus death and surrender benefits paid are based on the fund value before withdrawals, while death benefits and cash values in force at the end of the month are based on fund values after withdrawals.

2.4.2 Maximum penalty free withdrawal

The maximum annual penalty free withdrawal amount is determined at the beginning of each policy year. The maximum annual withdrawal can be a percent of either:

- Fund value – Fund value at the beginning of the first policy month, after all loads and charges.
- Cumulative premiums less withdrawals – Cumulative premiums paid through the first policy month, less cumulative withdrawals taken. Specify on the model point record the initial (at the projection date) cumulative premiums and cumulative withdrawals for the policies collected into the model point.
- Cumulative interest less withdrawals – Cumulative credited interest less withdrawals over a specified number of months. Specify on the model point record the initial cumulative amount for the policies collected into the model point. The projection grades this amount linearly over the first months of the projection. For example, a maximum using 12 months has the basis for the second month of the projection equal to 11/12 of the initial amount, plus the first month's credited interest plus withdrawal.

Nested projection differences

Warning! The Cumulative interest less withdrawals feature can cause nested projections to produce different results than the equivalent standard projection. For the standard projection, the penalty free withdrawal base is rolled forward from the initial penalty free withdrawal base using credited interest for the current period less withdrawals from the previous period less a reduction of the initial penalty free withdrawal base. This roll forward process within nested projections for time steps after the first will not match the roll forward process in the standard projection.

If this feature is significant for your block of business, you can make nested projections match standard projections by adding model point fields for the reduction factor and the withdrawal amount in the last month of the time step. Customize the formula for the penalty free withdrawal base beginning of month, so that it uses the new model point variable, rather than determining the reduction based on the initial penalty free withdrawal base at the beginning of the time step.

2.4.3 *Penalty free withdrawal use*

The penalty free withdrawal amount taken each policy year is an assumed percent of the maximum amount, times an assumed percent of policyholders taking withdrawals. The maximum withdrawal is calculated as a percent of the beginning fund value for the first month of the policy year. For persisting policyholders, one twelfth of this percent of the fund value is withdrawn at the end of each month during the policy year.

2.5 Premiums

2.5.1 *Premium term*

Specify the premium term on the model point record. This is the policy anniversary when premium payments stop. When this value is zero, premiums are paid to maturity.

2.5.2 *Premium mode*

Specify the premium mode on the model point record. Premiums can be paid annually, semi-annually, quarterly, monthly, or special monthly (such as electronic transfer or pre-authorized check, with expense rate different from the monthly mode).

2.5.3 *Annual premium*

Specify on the model point record the initial (at the projection date) total annualized premium for the policies collected into the model point. Each premium payment is the annualized premium, divided by the number of premium payments per year.

2.5.4 *Single premium*

Specify on the model point record the total single premium for the policies collected into the model point.

2.5.5 *Single premium policies*

For a single premium plan, define the premium either as:

- Zero annual premium, paying only the single premium. Since the annual premium is zero, the premium term has no effect.
- Premium term of 1 year, with Annual premium mode and zero single premium.

2.5.6 *Premium suspensions*

Premiums are reduced by a premium suspension percent at each policy anniversary after the projection date. The specified annual premium is multiplied by the cumulative premium suspension multiple. The suspension rate specified for the first policy year is ignored.

The reduced premium is assumed to be paid the same for all policies.

A negative premium suspension percent is an increase in the premium payments.

2.6 Premium bonus

The fund value is increased by a specified percent of the premium paid at issue. A percent of this amount can be capitalized into SFAS 97 GAAP DAC.

Please refer to the C04 GAAP accounting book.

2.7 Product loads

2.7.1 Administration fee

An amount per policy is deducted from the fund value at the beginning of each policy year.

2.7.2 Percent of premium

A percent of each premium payment is deducted when the premium is added to the fund value.

2.8 Interest crediting

2.8.1 Crediting rate

Interest is credited to the fund value at the end of each month. The base crediting rate is defined in the Crediting strategy chapter, below.

Initial crediting rate

Specify on the model point record the initial (at the projection date) crediting rate.

2.8.2 Additional interest

The base current crediting rate can be increased by an additional interest credit when the policy's fund value crosses the first and second threshold per policy:

- Ratcheted additional interest credits the addition on the part of the fund value over the threshold.
- Banded additional interest credits the addition on the entire fund value when the threshold is reached.

The additional crediting rates are the total addition to the base crediting rate for each threshold. That is, the addition for the second band is the band 2 rate, not the sum of the band 1 and band 2 rates.

2.8.3 Guaranteed crediting rate

The minimum guaranteed rate is a specified fixed rate.

2.8.4 Monthly rate

The crediting rate is an effective annual interest rate, which is converted to the equivalent monthly rate:

$$(1 + \text{annual rate})^{1/12} - 1$$

2.9 Fund value

Specify on the model point record the initial (at the projection date) fund value for the policies collected into the model point.

The previous month's fund value is adjusted by the following amounts which occur as of the beginning of each month: (1) premium payment addition, (2) premium bonus addition, (3) loads subtraction, and (4) charges subtraction.

Interest is credited on this beginning of month fund value. Credited interest is added to the fund value at the end of the month.

The end of month fund value before withdrawals is used to calculate death and surrender benefits paid for the month, since the mortality decrement occurs before policyholder transactions at the end of the month.

Finally, withdrawals are processed, reducing the final fund value at the end of the month. It is this fund value after withdrawals which is used to calculate reserves, and the death benefit and cash value in force at the end of each month.

2.10 Cash value

The cash value is the fund value less the surrender charge. The fund value may be adjusted by the market value adjustment (MVA) provision, which is discussed as a separate topic, below.

2.10.1 Fund value for surrender charge

The fund value for calculating the surrender benefit is the current fund value.

2.10.2 Surrender charge definition

The surrender charge is a percent of either:

- Fund value less remaining penalty free withdrawal entitlement, either before or after the market value adjustment is applied.
- Cumulative premiums paid.

2.10.3 Surrender charge waiver

The surrender charge can be waived during the final month of each crediting rate declaration period, such as for 'CD annuity' products. Lapses can be skewed to be higher in this month. See the Surrenders section, below.

The Bail out provision is discussed separately, below.

2.10.4 Minimum cash value

During a number of months after issue, the cash surrender value before the market value adjustment can be floored at the cumulative premiums less cumulative withdrawals. This is sometimes called the 'return of principal floor'.

2.11 Bail out provision

The surrender charge can be waived when the base crediting rate violates (falls below) the 'bail out' rate.

For in force policies, specify the bail out rate on the model point record. When this variable is zero (default value), the bail out rate is the initial crediting rate.

For new business model points (issued after the projection date), the bail out rate is the initial crediting rate.

2.12 Market value adjustment

2.12.1 Adjustment provisions

The market value adjustment (MVA) multiple adjusts the fund value at the time of surrender, to approximate the change in the market value of the invested assets backing this product. The initial MVA period is a number of years from issue. Each renewal MVA period is similarly a number of years from the end of the previous period; specify zero years when the initial MVA period does not renew.

The MVA multiple applies to the fund value less the remaining penalty free withdrawal amount available.

2.12.2 Interest rate comparison

The MVA adjustment is based on comparing the base crediting rate to a specified competitor rate. The adjustment multiple can be either:

Linear

$$1 + \text{multiple} \times (\text{crediting rate} - \text{competitor rate}) \times \text{MVA years remaining}$$

Exponential

$$\left(\frac{1 + \text{crediting rate}}{1 + \text{competitor rate}} \right)^{\text{MVA years remaining}}$$

The MVA years remaining is the number of months remaining in the MVA period, divided by 12.

2.12.3 MVA surrender charge

The percent of fund surrender charge can be calculated either:

- Before the MVA adjustment – The surrender charge is a percent of the fund value less the remaining penalty free withdrawal entitlement.
- After the MVA adjustment – The surrender charge is a percent of (1) the fund value times the MVA adjustment multiple, less (2) the remaining penalty free withdrawal entitlement.

2.12.4 Constraints

The MVA adjustment is reduced when the cash surrender value would be less than the return of principal minimum cash value.

3 FUND VALUE

3.1 Fund value crediting

3.1.1 Overview

The fund value is increased for credited interest at the end of each month. The crediting rate for the fund value is the larger of the policy's guaranteed crediting rate and the rate determined through the crediting strategy.

3.1.2 Initial crediting rate

The initial base crediting rate, which is specified on the model point record for in-force model points, applies until the first declaration date on or after the projection date.

3.1.3 Crediting strategy declaration

Declaration timing

The crediting rate can change at the beginning of a month, policy year, calendar year, or calendar quarter.

Declaration period

The crediting rate changes according to the declaration period, which is one or more cycles of the specified period. A different number of cycles can apply at issue than for ongoing crediting rate changes. For example, the crediting rate at issue could apply for the first two policy years, then changing every policy year thereafter.

3.1.4 Crediting strategy rate basis

Fixed rate

The crediting rate can be a fixed rate, such as for calculating illustration values. Crediting rate constraints do not apply.

Asset yield basis

You can specify the crediting rate as the asset yield less a spread, subject to the crediting rate constraints. This definition makes the model points dependent on asset results, so independent model point processing cannot apply for asset/liability projections.

The asset yield includes the effects of IMR amortization, defaults, and borrowing costs.

For liability-only projections, you specify the asset yield as an assumption for the projection. Independent model point processing can still apply, since the asset yield is not affected by liability results.

Competitor rate basis

You can specify the crediting rate as a competitor rate less a spread, subject to the crediting rate constraints.

When the constraints do not include the asset yield, independent model point processing can apply even for asset/liability projections. For example, this approach could take the view that the crediting rate is primarily driven by market forces, and that appropriate investment strategies must be determined in order to achieve the desired spread from the asset yield.

3.1.5 Crediting strategy constraints

The following constraints apply when the crediting rate basis is the asset yield or a competitor rate. Crediting rate constraints do not apply when the crediting rate is a fixed rate.

The crediting rate is first constrained to be not more than a maximum percent above, and not less than a maximum percent below, a specified competitor rate.

Further (thus adjusting the above constraints), the crediting rate is not greater than [the asset yield less a minimum spread]. When the minimum spread below the asset yield is 1.0 (100%) default value, this constraint does not apply, and model points defining the crediting rate using a competitor rate are eligible for independent model point processing within asset/liability projections.

Finally, the crediting rate is not less than the guaranteed crediting rate in all cases, even when the crediting rate is defined as a fixed rate. This includes the temporary guarantee of the crediting rate at issue.

3.1.6 Weighted change restriction

The new crediting rate is a percent of the previous crediting rate, plus (1.0 – percent) of the new crediting rate determined as described above. A different percent applies when the new rate is higher than when it is lower.

4 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves. PBA reserves and capital are discussed in the PBA book.

4.1 CARVM reserve

Specify CARVM valuation variables on the FAValuation input page.

4.1.1 Percent of fund approximation

The CARVM reserve can be approximated using percentages of the fund value or cash value. This may be appropriate to improve run time by eliminating these compute-intensive calculations.

The effect of the market value adjustment (MVA) and bailout provisions are discussed separately, below.

4.1.2 Actuarial guideline 33

Fixed annuity reserves can be calculated using the Commissioners' Annuity Reserve Valuation method (CARVM) and Actuarial Guideline XXXIII. The CARVM integrated reserve for the policy is the maximum of the integrated reserve of the cash value stream and, when included, the annuitization stream.

Cash value streams

Cash value streams are always included in the CARVM calculations. The present value of guaranteed benefits is calculated using a different valuation interest rate for each of the following benefits:

- Death benefits,
- Nursing home benefits, when included,
- Penalty free withdrawals, and
- Full cash value surrenders.

There are two projections of the cash value stream guaranteed benefits. One assuming maximum penalty free withdrawals are taken and one assuming no penalty free withdrawals are taken. The maximum integrated reserve for the cash value stream is the greater of the integrated reserve for the two cash value streams.

Annuitization stream

Annuitization streams may be included in the CARVM calculations. The present value of guaranteed benefits is calculated using a different valuation interest rate for each of the following benefits:

- Death benefits,
- Nursing home benefits, when included,
- Penalty free withdrawals, and
- Full annuitization benefit.

For the annuitization stream guaranteed benefits, there are potentially many different streams projected with different types of annuity benefits. Any number of CARVM annuitization tests can be defined, where each test represents a type of annuity to purchase and then any number of the defined tests can be applied. For example, you could project the annuitization stream for three annuitization tests, one purchasing life only annuities, one purchasing 5 year certain and life annuities, and one purchasing 10 year certain and life annuities.

And, within each annuity benefit type stream projection, there are two streams projected. One assuming maximum penalty free withdrawals are taken and one assuming no penalty free withdrawals are taken. The maximum integrated reserve for the annuitization test is the greater of the integrated reserve from these two projections.

Finally, the integrated reserve for the annuitization stream is the maximum of the integrated reserve for each annuitization test.

During the CARVM projection, the full annuitization benefit is determined as the benefit for an annual payment immediate annuity for a specified certain years and a single life (when a life annuity type is used) purchased by the fund value after penalty free withdrawals. The purchase rate for the annuity may be determined using only a guaranteed basis annuity purchase rate or both a current basis annuity purchase rate and a guaranteed basis annuity purchase rate. When the current basis annuity purchase rate is included, the annuity benefit at the valuation date is the greater of the current basis annuity benefit and the guaranteed basis annuity benefit. Annuity benefits for CARVM projection periods after the valuation date are always based on the guaranteed basis purchase rate. The current and guaranteed annuity purchase rate bases are discussed in section 2.3 above.

Since contracts may provide, as contractual guarantees, the use of preferential purchase rates at annuitization to those listed in the contract, you may specify a percentage of the valuation date fund value as

a floor to the annuitization stream integrated reserve to account for this additional guarantee. For example, when such provision applies, you should set the `stat_carvm_annuitzn_min_fund_val_pct` and `tax_carvm_annuitzn_min_fund_val_pct` to the appropriate floor percentage, such as 93%.

4.1.3 CARVM valuation basis

Interest

CARVM valuation interest may vary by the valuation method, cash settlement definition, guarantee duration definition, plan type, and guarantee duration. Separate plan type definitions are provided for cash value elective benefits, annuitization elective benefits, and non-elective benefits.

The Global folder includes an external source workbook containing the maximum standard valuation law valuation rates. The `DefAnnStatValn` and `DefAnnTaxValn` ranges in the Valuation Interest Rates Excel workbook include rates varying by valuation method (Issue Year, Change in Fund), cash settlement definition (With Cash Settlements, Without Cash Settlements), guarantee duration definition (With Guarantees, Without Guarantees, With or Without Guarantees), plan types (Plan A, Plan B, Plan C) and guarantee durations ranges of 0-5, 6-10, 11-20 and 21+.

For the death benefits, nursing home benefits, penalty free withdrawals, and full cash surrender benefits, the guarantee duration is determined as the number of years the guaranteed credited rate exceeds the valuation interest rate for life insurance policies with a guarantee duration in excess of 20 years, as specified in the `stat_carvm_life_int_rate` and `tax_carvm_life_int_rate` variables.

For the annuity benefits, the guarantee duration is determined as the number of years since issue when the annuitization occurs.

Mortality

Reserve mortality is a base rate, times a multiple. Separate reserve mortality is specified for death benefits and annuity benefits.

These variables are read twice when then gender is blended – first for male rates and then again for female rates.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys `mort_issue_age` and `mort_gender` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Using the hidden variable ensures the attained age does not advance with each yearly loop.

Nursing home

Nursing home incidence is a base rate.

The value is read twice when the gender is blended – first for male rates and then again for female rates.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys `mort_issue_age` and `mort_gender` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Using the hidden variable ensures the attained age does not advance with each yearly loop.

4.1.4 CARVM valuation method

The CARVM valuation method may be 'Issue Year' or 'Change in Fund'.

CARVM issue year method

Under the CARVM issue year valuation method, the CARVM guaranteed benefit streams are developed from a projection of a single fund value bucket. Benefits are discounted using the appropriate valuation interest

rate for the policy issue year, cash settlement definition, guarantee duration definition, plan type, and guarantee duration from the issue year method valuation interest rate table.

The reserve for the benefit stream is the maximum discounted value of the issue year fund bucket.

CARVM change in fund method

Under the CARVM change in fund valuation method, the CARVM guaranteed benefit streams are developed from a projection of a series of change in fund value buckets. Benefits are discounted using the appropriate valuation interest rate for the change in fund year, cash settlement definition, guarantee duration definition, plan type, and guarantee duration from the change in fund valuation interest rate table.

The reserve for the benefit stream is the sum of the maximum discounted value for each change in fund value bucket.

The first step in the change in fund valuation process is to develop the calendar year fund value buckets.

For inforce model points, specify the fund value at the end of each calendar year and the projection date on the model point record. Data is required to be formatted as a string of YYYY-Value items separated by a vertical bar. For example, assuming a 2013 issue and a projection date of 3/2017, data would be specified as "2013-100000|2014-120000|2015-115000|2016-117000|2017-119000" and would be interpreted as model point fund values of 100000 at the end of 2013, 120000 at the end of 2014 and 115000 at the end of 2015, 117000 at the end of 2016, and 119000 as of March 2017.

The change in fund buckets are developed by taking the difference between the provided fund values. Negative change in fund value buckets are not allowed. If a negative amount is calculated for a year, that amount is reduced to zero and the negative amount is applied to the previous year's change in fund value reducing that amount. The process continues until there are no negative change in fund value buckets.

For example, given the above provided fund values, the change in fund buckets would be 100000 for year 2013, 15000 for year 2014, 0 for year 2015, 2000 for year 2016, and 2000 for year 2017.

Using these buckets, the guaranteed benefits streams are developed separately for each fund value bucket. The fund value in each fund bucket grows with interest and is reduced by penalty free withdrawals, when included. A last-in-first-out (LIFO) approach is used to reduce the fund value buckets for penalty free withdrawals. That is, penalty free withdrawals assumed in the CARVM projection reduce the fund value from the most recent fund value bucket first, until that bucket is exhausted, then reduces the fund value from the next most recent bucket, etc., until the amount of fund value reduction equals the penalty free withdrawal amount.

Finally, the present values of the guaranteed benefits for the benefit stream are determined separately for each fund value bucket, discounting the guaranteed benefits by the appropriate change in fund valuation interest rate for the cash settlement definition, guarantee duration definition, plan type, guarantee duration, and fund bucket year. Each fund value bucket produces its own maximum discounted value.

For valuations after the projection date, the past end of calendar year fund values required for the change in fund calculations are updated using survivorship. And, the fund value as of the current projection period is the current fund value.

4.1.5 Continuous CARVM

The continuous CARVM method is the same as for CARVM, except that the surrender charge is the lower of the surrender charge for the current month and the next month.

For example, when the surrender charge decreases from 7% in the first policy year to 6% in the second policy year, the continuous CARVM reserve at the end of the first policy year is based on the 6% surrender charge.

4.1.6 CARVM crediting rate

Future policy benefits are projected using the current crediting rate until the end of the current declaration period, then discounting using the unadjusted reserve interest rate. This produces an 'excess interest reserve'. After the current declaration period, benefits are projected using the guaranteed crediting rate.

4.1.7 MVA adjustment

Reserves can be calculated either including or excluding the MVA adjustment. This applies for approximating the reserve as a percent of fund value or cash value, in addition to calculating CARVM reserves. This option applies the same for both statutory and tax reserves.

For purposes of CARVM projections, the competitor rate used in MVA calculations is fixed at its value on the valuation date.

4.1.8 Bailout provision

Current violation

When the bailout rate is violated (waiving the surrender charge) at the valuation date, the surrender charge is ignored for reserve calculations.

Potential future violation

When the bailout provision is not waiving the surrender charge at the valuation date, the surrender charge is referred to as a 'contingent surrender charge'. Actuarial Guideline XIII (AG 13) specifies that the value of guaranteed benefits under AG 33/CARVM is not reduced by contingent surrender charges when both:

- The bailout rate exceeds the whole life statutory reserve interest rate for the model point's year of issue, and
- The guaranteed crediting rate is less than the bailout rate. The current crediting rate is considered to be guaranteed through the end of the current crediting rate declaration period.

Otherwise, the surrender charge is considered to be non-contingent, and thus is included in the reserve calculation.

4.1.9 CARVM calculation frequency

CARVM reserves are calculated as of the projection date, the first month after issue, at the end of each policy year, and at the end of each calendar year. Approximations are used between these calculation dates in order to minimize these compute-intensive calculations.

For the projection months when CARVM reserves are not calculated, the reserve for each model point is approximated by multiplying the previous month's reserve by the ratio of the current month's fund value to the previous month's fund value.

The model point record includes the initial (at the projection date) statutory and tax CARVM reserve amounts. These values are only for use by nested projections. Ignore these variables when preparing model point records for your projection, since the CARVM valuation is calculated at the projection date.

4.1.10 Terminating CARVM projections

CARVM reserve calculations consider benefits projected only through a specified number of years from each valuation date.

4.2 Additional reserve

Additional statutory reserves can be specified as a percent of the end of month fund value.

4.3 Reinsurance reserve

Reserves are calculated both gross of (ignoring) reinsurance and net of reinsurance.

The reinsurance reserve is the ceded reserve, calculated as the difference between the gross statutory reserve and the net statutory reserve.

4.4 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record sets the tax reserve to be the statutory reserve.

Tax reserves are calculated by the same methods, using the tax reserve valuation basis and method.

For policies issued after 1987, the crediting interest calculated in the CARVM projection is capped as required to disallow a deduction for an 'excess interest' reserve. The capping process adjusts the crediting so that any crediting rate guaranteed above the tax valuation rate is only guaranteed to the end of the current taxable year.

The statutory additional reserve can be added to tax reserves. That is, the additional reserve can be deductible.

The tax reserve is not greater than the statutory reserve, including the additional reserve even when it is not included in the tax reserve.

5 ACTUARIAL ASSUMPTIONS

5.1 Commissions

5.1.1 *Premium commission*

A percent of each premium is paid as a commission when the premium is paid.

5.1.2 *Fund value commission*

One twelfth of the annual percent of fund value at the end of the previous month is paid at the beginning of each month.

5.1.3 *Commission overrides and agent terminations*

The specified commission rates are the average commission rates that are assumed to apply, taking into account average writing agent commissions, commission overrides, and the effect of agent terminations.

5.1.4 *Commission chargebacks*

A portion of the premium commission can be charged back in the event of surrender during the first three years after issue. The chargeback percent applies to the total first year premium commissions paid to date. The chargeback percent typically varies by month, by linking to the external source using the `mths_since_issue` column as a lookup key.

Other types of commissions are not subject to chargeback.

The commission subject to chargeback is calculated per unit in force. For in-force model points, specify on the model point record the initial (at the projection date) commissions subject to chargeback for the policies collected into the model point.

5.2 Expenses

5.2.1 Expense adjustment multiple

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

5.2.2 Acquisition expenses

Acquisition expenses incurred at issue can include:

- Amount per policy
- Percent of total first year premium.

5.2.3 Maintenance expenses

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Maintenance expenses include:

- Amount per policy
- Percent of annualized premium
- Percent of the total fund value at the end of the previous month.

5.2.4 Inflation

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

5.2.5 Premium tax

Premium taxes are reported with expenses. The tax is a percent of each premium payment.

Inflation and the expense adjustment multiple do not apply to premium tax.

5.2.6 Fixed expenses

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

5.3 Decrement processing

Death claims are based on the fund value at the end of month before decrements and transactions.

Surrenders and annuitizations are based on the fund value after death claims. Thus lapse and annuitization rates are applied to the policies in force after the mortality decrement. The annuitization rate is reduced if the total lapse and annuitization rate would exceed 100%. The sum of these two rates is the voluntary policy termination rate.

Penalty free withdrawals are then processed for the policies continuing to the next month.

5.4 Mortality

5.4.1 Input variables

'Mortality variables' for annuity products include mortality base, multiple, and addition rates, and mortality improvement. These variables are read twice when blended mortality applies – first for male rates and then again for female rates. See the Blended mortality topic in the Mortality section, below.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys mort_issue_age and mort_gender instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age. Using the hidden variable ensures the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

5.4.2 Mortality adjustment multiple

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

5.4.3 Base mortality

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable). See the Mortality variables section in the Input organization chapter, above.

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

5.4.4 Mortality improvement

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$. When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

5.4.5 Mortality table adjustments

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

5.4.6 Blended mortality

The model point can be a 'blend' between males and females issued at the specified issue age. This occurs when the gender variable is Blended. Specify on the model point record the proportion of issues that are males.

The male and female mortality rates are each determined from the mortality table. The blended mortality rate is:

$$\text{proportion of males} \times \text{male mortality rate} + \text{proportion of females} \times \text{female mortality rate}$$

5.4.7 Monthly mortality

Deaths occur at the end of each month. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

5.5 Surrenders

5.5.1 Lapse adjustment multiple

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date.

5.5.2 Policy year lapse rates

The base lapse rate is adjusted by a multiple.

5.5.3 Excess lapses

Lapse rates can be dynamically adjusted according to policyholder behavior related to the competitive position of the base crediting rate as interest rates change. Excess lapses can occur when the base crediting rate is below a specified competitor rate. When the crediting rate is higher than the competitor rate, excess lapses can be negative.

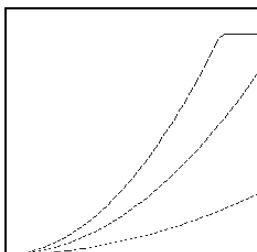
Maximum and minimum lapse rate variables in the excess lapse features affect the excess lapse calculations only. They do not limit the specified lapse rates. For example, the maximum lapse rate variable can reduce the excess lapse rate to zero, but cannot reduce the total lapse rate to be less than the specified lapse rate adjusted by the calendar date multiple.

Excess lapses can be calculated using either an exponential or arctangent formula.

5.5.4 Exponential excess lapses

The exponential formula produces results that look like figure 1, where the x-axis is the difference between the competitor rate and the crediting rate. The highest excess lapses in the above graph are for a policy with no surrender charge. The lowest excess lapses represent a policy with a large surrender charge.

Figure 1 Exponential formula



There are two versions of the exponential formula:

Increased lapses:

$$\begin{aligned}
 &\text{When competitor rate} > \text{crediting rate} + \text{threshold} \\
 &\text{difference} = (\text{competitor rate} - \text{crediting rate} - \text{threshold}) \times 100 \\
 &\text{surrender charge ratio} = 1 - \left(\frac{\text{cash value}}{\text{fund value}} \right) \\
 &\text{excess lapse} = \frac{\left[\text{multiple} \times (\text{difference}^{\text{exponent}}) \times (1 - \text{surrender charge multiple} \times \text{SC ratio}) \right]}{100} \\
 &\quad - \text{surrender charge subtraction} \times \text{SC ratio} \\
 &\text{excess lapse} \leq \max(0, \text{maximum lapse rate} - \text{planned lapse rate})
 \end{aligned}$$

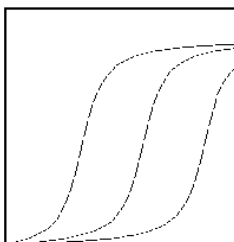
Decreased lapses:

$$\begin{aligned}
 &\text{When crediting rate} > \text{competitor rate} + \text{threshold} \\
 &\text{difference} = (\text{crediting rate} - \text{competitor rate} - \text{threshold}) \times 100 \\
 &\text{lapse reduction} = \frac{\left[\text{multiple} \times (\text{difference}^{\text{exponent}}) \right]}{100} \\
 &\text{lapse reduction} \leq \max(0, \text{planned lapse rate} - \text{minimum lapse rate})
 \end{aligned}$$

5.5.5 Arctangent excess lapses

The arctangent formula produces results that look like figure 2. The highest excess lapses in the above graph are for a policy with no surrender charge. The lowest curve has a large surrender charge.

Figure 2 Arctangent formula



difference = competitor rate – crediting rate

$$\text{surrender charge ratio} = 1 - \left(\frac{\text{cash value}}{\text{fund value}} \right)$$

$$\text{excess lapse} = \frac{[(\text{maximum total lapse} - \text{base lapse}) \times (A - B)]}{(1 - B)}$$

total lapse = base lapse + excess lapse,

$\leq$ maximum total lapse, $\geq$ minimum total lapse

$$A = \frac{\arctan[100 \times \text{multiple} \times (\text{difference} - \text{SC ratio} \times \text{SC multiple} - \text{threshold})]}{\left(\frac{\pi}{2}\right)}$$

$$B = \frac{\arctan[-100 \times \text{multiple} \times (\text{SC ratio} \times \text{SC multiple} + \text{threshold})]}{\left(\frac{\pi}{2}\right)}$$

When competitor rate < crediting rate : SC multiple = 0

The arctangent excess lapse formula shown above is a version of the following formula familiar to some users:

$$\text{total lapse rate} = a + b \times \arctan \left[m \times \left(\frac{\text{competitor rate} - \text{crediting rate} - \text{SC ratio}}{c} \right) - n \right]$$

Base lapse rates = the result of the above formula when the crediting rate equals the competitor rate.

The formulas are identical when the parameters are converted as follows:

$$\text{Threshold} = \frac{n}{m}$$

$$\text{Multiple} = \frac{m}{100}$$

$$\text{SC multiple} = \frac{1}{c}$$

$$\text{Maximum lapse rate} = a + b \times \left(\frac{\pi}{2}\right)$$

5.5.6 Monthly lapses

Surrenders occur at the end of each month. Each of the above lapse rate adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/12)}$$

The monthly lapse rate is applied to the fund value at the end of the month, before penalty free withdrawals are processed.

5.5.7 Skewed lapses

Skewed lapses allocate a specific portion of annual lapses to each month within a policy year. The skew percentages are normalized to assure that they add to 100% for each policy year. For example, lapses could be skewed to occur in the last month of the policy year when surrender charge are waived for CD annuity products.

When skewed lapses apply, the exponent in the monthly lapse rate formula is replaced by the skew percent for the current month. For example, when the lapse applies for a month with skew percent of 40%, the lapse rate is:

$$1 - (1 - \text{annual lapse rate})^{0.4}$$

5.6 Annuitizations

The fund value is paid out as a lump sum at annuitization. The MVA adjustment does not apply.

Annuitizations occur at the end of each month. The annual annuitization rate is reduced when the total of the annual lapse and annuitization rates would exceed 100%.

Finally, the annual rate is converted into the monthly annuitization rate:

$$1 - (1 - \text{annual annuitization rate})^{(1/12)}$$

5.7 Risk based capital risk definition

The risk based capital (RBC) C3 component percent varies between 'low', 'medium', and 'high' risk annuity products. Specify on the Company input page this risk definition for each product, then vary the RBC C3 percent between Low, Medium, and High risks.

6 REINSURANCE

6.1 Coinsurance

Coinsurance treaties cede a portion of the entire policy, including transferring the ceded reserve to be invested by the reinsurer. Reinsurance benefits include all benefits provided by the policy. Reinsurance premiums are based on the coinsurance percent of the gross premium.

Specify the coinsurance percent on the model point record.

Reinsurance premiums are paid at the same premium frequency as the model point. The coinsurance premium is the premium paid times the coinsurance percent.

The reinsurance benefit paid to the ceding company each month is the coinsurance percent times each type of benefit incurred by the ceding company.

The reinsurer reimburses the ceding company for a portion of the ceding company's expenses. The expenses reimbursed are the coinsurance percent times:

- Acquisition expenses
- Maintenance expenses
- Commissions net of commission chargebacks.

The reinsurer reimburses the coinsurance percent times the premium taxes paid by the ceding company.

The ceding company receives a reinsurance reserve credit for coinsurance treaties equal to the coinsurance percent times the formulaic statutory reserve.

6.2 Reinsurance administration expenses

Reinsurance administration expenses are incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include an amount per policy.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Coinsurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for coinsurance.

7 MODEL POINT RECORDS

7.1 Model point record layout

The model point record layout for fixed annuities is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	up to 25 characters (assigns model point to report group)
B	seg_id	segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date	yyyy/mm/dd or mm/dd/yyyy
F	plan_code	up to 20 characters (link to assumptions by plan)
G	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
H	tax_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
I	issue_age	positive integer
J	gender	Male, Female, Blended (blended mortality)
K	male_sex_pct	male percentage (1.0 for 100%). This only applies when gender is Blended.

Excel Column	Variable	Value
L	prem_mode	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
M	prem_term_mp	premium years (0 = omega age from experience mortality table)
N	init_annualzd_prem	initial annualized premium for mp
O	single_prem	single premium for mp
P	reins_pct	ceded coinsurance percent
Q	init_policies	initial policies for mp (required)
R	init_fund_val	initial fund value for model point
S	init_crediting_rate_base	initial base crediting rate
T	init_bail_out_rate	bail out rate (0 = use initial crediting rate)
U	init_p fwd_base	initial penalty free withdrawal base for model point
V	init_p fwd_entitlement	initial penalty free withdrawal entitlement for model point
W	init_gross_prem_cumul	total cumulative premiums paid for model point
X	init_p fwd_cumul	initial cumulative withdrawals for model point
Y	init_comm_subject_to_chargeback	initial commission subject to chargeback for model point
Z	init_carvm_change_in_fund_data	fund values for the end of each prior calendar year and projection date for CARVM change in fund calculations. Data should be provided as a string of YYYY-Value data separated by a vertical bar. For example, assuming a projection date of 3/2017, data specified as “2015-100000 2016-120000 2017-130000” would be interpreted as model point fund values of 100000 at the end of 2015, 120000 at the end of 2016 and 130000 at the end of March 2017.
AA	init_stat_res_carvm	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
AB	init_tax_res_carvm	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step

7.2 Required model point variables

7.2.1 Always required

The following model point variables are required in all cases:

- **Model point identification:** group, seg_id, and mp_id
- **Insured characteristics:** issue_age and gender
- **Policy specifications:** issue_date, plan_code, stat_valn_code, and tax_valn_code
- **Initial policies:** init_policies
- **Annual premium:** init_annualized_prem, prem_mode, and prem_term are required for premium paying model points.
- **Single premium:** single_prem is required for new business model points.
- **Reinsurance:** reins_pct is required for model points with coinsurance.

7.2.2 In-force model point values

In-force model points are driven by all the 'initial values' variables, which specify the aggregate model point values as of the projection date.

New business model points (model points with issue date after the projection date), however, ignore these variables. The exception is that the initial policies must be specified.

The following table shows when each in-force values variable is required for in-force model points:

Variable	Required When
init_fund_val	always required
init_crediting_rate_base	always required
init_bail_out_rate	model points with a bail out provision
init_p fwd_base	model points with penalty free withdrawals based on cumulative interest less withdrawals
init_gross_prem_cumul	model points with surrender charge or penalty free withdrawals based on cumulative premiums, or still subject to the return of principal cash value floor
init_p fwd_cumul	model points with penalty free withdrawals based on cumulative premiums less withdrawals, or still subject to the return of principal cash value floor
init_comm_subject_to_chargeback	model points still subject to commission chargeback
init_carvm_change_in_fund_data	model points using the CARVM change in fund method

7.2.3 Variables to ignore

The following variables are needed only for nested projections, and should be ignored. These values must be carried forward to the next time step in order to reproduce the results from standard projections:

Variable
init_p fwd_entitlement
init_stat_res_carvm
init_tax_res_carvm

8 SUBMODELS STRUCTURE

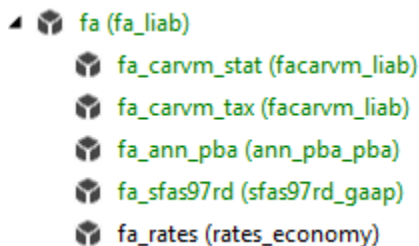
8.1 fa submodel

The fa submodel calculates the model point's contribution to the liability income statement and balance sheet items for fixed annuity products.

8.1.1 *Seriatim versus portfolio projections*

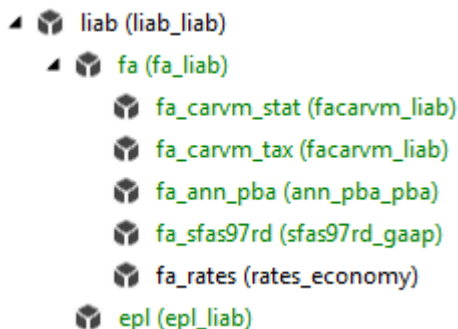
Seriatim projection

When fa is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the fa submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings.



Portfolio projection

For portfolio projections (which include business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the fa submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the fa submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the fa submodel the same columns as for other product types. This makes the fixed annuity product type interchangeable with other product types fitting into this business segment structure.

8.1.2 Variables

Variables for fixed annuity products are contained in the fa submodel.

8.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when fa is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the fa submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only fa is the top model, (2) when fa is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when fa is run as the top model and when fa is included in projections with segment as the top model. References to company, segment and liab are invalid when fa is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) temporary tables are sized for DAC tax calculations and competitor rates, (3) projection flags are created and initialized, reporting groups are set up, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each projection task loop, immediately after startup when fa is the top model. When segment is the top model, initialize is executed by calls from seg->initialize which executes fa->initialize for each model point in the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when fa is running as the top model. This includes executing the initialization of economic scenario rates, setting up the competitor rates. Other initializations are required in both cases.

Other formulas

Other formulas in the fa submodel are relatively straightforward. Most of the profit, balance sheet, and GAAP columns are only calculated when fa is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

8.2 fa_carvm_stat submodel

The fa_carvm_stat submodel is a user defined array instance of the facarvm_liab model class and contains all variables and formulas required to calculate fixed annuity CARVM reserves.

Time 0 in the fa_carvm_stat submodel corresponds to the policy year of the reserve period. Thus, 't' is policy year, unlike the fa submodel where 't' represents months since the projection date. The values in fa_carvm_stat are cleared before beginning the calculation for each reserve period.

8.2.1 Variables

Variables for statutory CARVM reserve parameters are contained in the fa_carvm_stat submodel.

In addition to the variables discussed above, two variables in fa_carvm_stat are hidden from the data entry but are critically important to the successful operation of the fa_carvm_stat elements:

Model_name

The 'model_name' variable is an option choice variable with choices of STAT and TAX. This variable is used to retrieve the appropriate discount rate for statutory and tax reserves, since the facarvm_liab model class is used for both statutory and tax CARVM calculations.

Reserve_period

The 'reserve_period' variable is the period for which the reserve is being calculated. It is set in fa submodel at the beginning of each reserve calculation to the current projection period.

8.2.2 Formulas

Formulas for calculating the statutory CARVM reserves are included in the fa_carvm_stat submodel. Generally, fa_carvm_stat formulas prospectively calculate the present value of the CARVM streams from the reserve period to either the specified CARVM duration or the omega age of the model point.

8.2.3 Multiple instances

Each model point carries up to two instances of fa_carvm_stat, which calculate the CARVM reserves assuming:

- Policyholders do not take penalty free withdrawals
- Policyholders take full penalty free withdrawals.

8.3 fa_carvm_tax submodel

The fa_carvm_tax submodel is a user defined array instance of the facarvm/liab product/purpose and contains all variables and formulas required to calculate fixed annuity tax CARVM reserves.

Time 0 in the fa_carvm_tax submodel corresponds to the policy year of the reserve period. Thus, 't' is policy year, unlike the fa submodel where 't' represents months since the projection date. The values in fa_carvm_tax are cleared before beginning the calculation for each reserve period.

8.3.1 Variables

Variables for tax CARVM reserve parameters are contained in the fa_carvm_tax submodel.

In addition to the variables discussed above, two variables in fa_carvm_tax are hidden from the data entry but are critically important to the successful operation of the fa_carvm_tax elements:

Model_name

The 'model_name' variable is an option choice variable with choices of STAT and TAX. This variable is used to retrieve the appropriate discount rate for statutory and tax reserves, since the facarvm/liab model class is used for both statutory and tax CARVM calculations.

Reserve_period

The 'reserve_period' variable is the period for which the reserve is being calculated. It is set in fa submodel at the beginning of each reserve calculation to the current projection period.

8.3.2 Formulas

Formulas for calculating the tax CARVM reserves are included in the fa_carvm_tax submodel. Formulas for fa_carvm_tax are challenging, and not typically customized.

Generally, fa_carvm_tax formulas prospectively calculate the present value of the CARVM streams from the reserve period to either the specified CARVM duration or the omega age of the model point.

8.3.3 *Multiple instances*

Each model point carries up to two instances of `fa_carvm_tax`, which calculate the CARVM reserves assuming:

- Policyholders do not take penalty free withdrawals
- Policyholders take full penalty free withdrawals.

8.4 `fa_rates` submodel

The `fa_rates` submodel is identical to the `rates` submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

8.5 `fa_ann_pba` submodel

The `fa_ann_pba` submodel is identical to the `ann_pba` submodel which applies for portfolio projections.

Please refer to the C05 PBA book.

8.6 `fa_sfas97rd` submodel

The `fa_sfas97rd` submodel is identical to the `sfas97rd` submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

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L07 VARIABLE ANNUITY

This book discusses variable deferred annuity products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

- Product – Benefits, premiums, benefit charges, surrender charge, and reinsurance
- Fund – Investment fees, growth rates, DCA definitions, CARVM AG34 and PBA asset class definitions
- Valuation – Statutory and tax reserves, and CARVM detail reports option
- Expense – Commission and expense rates, premium tax rate
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse and annuitization rates, withdrawals, dynamic premiums, rebalancing

Each of these input page types begins with 'VA', for use in projections that combine multiple product types.

1.2 Data pages

Variables for each variable annuity model point are specified on the VADData page. These variables are discussed in the Model point records chapter, below.

2 PRODUCT BENEFITS AND VALUES

The following explains the benefits available for variable annuity products.

2.1 Maturity

The policy terminates at the maximum attained age specified for the product. When the specified maturity age is zero (default value), the maturity age is the omega age of the experience mortality table.

The fund value is paid at maturity.

2.2 Death benefits

2.2.1 *Basic death benefit*

Death benefits occur at the end of the month. The death benefit is either the fund value or cash value.

When a guaranteed minimum death benefit or enhanced earnings death benefit rider is included, the death benefit is the greater of this basic death benefit and the rider basis.

2.2.2 *Death benefit paid versus in force*

Death benefits paid at the end of each month are based on the fund value before policyholder transactions are processed. Thus the death benefit is calculated using the fund value before penalty free withdrawals.

Death benefits in force at the end of the month are based on the fund value at the end of the month, after penalty free withdrawals have been taken.

2.3 Annuity benefits

GMIB benefits are discussed in the Guarantee benefits chapter, below.

2.3.1 Current basis purchase rates

Current basis annualized annuity benefits are determined as the fund value less a load and policy fee divided by a current basis immediate annuity purchase rate factor.

The current basis annuity purchase rate factor uses current basis base mortality times a multiple and current basis interest. The current basis interest rate may be a specified fixed rate or a spread from a specified competitor rate, where the spread may vary each benefit year.

2.3.2 Guaranteed basis purchase rates

Guaranteed basis annualized annuity benefits are determined as the fund value divided by a guaranteed basis immediate annuity purchase rate factor.

The guaranteed basis annuity purchase rate factor uses guaranteed basis base mortality times a multiple and a guaranteed basis interest. The guaranteed basis interest rate is always a specified fixed rate.

2.3.3 Projection treatment

Policyholders who annuitize during the projection receive the fund value as a lump sum. There is no calculation of the annuity purchase rates or projection of future annuity benefits.

2.4 Withdrawal benefits

GMWB benefits are discussed in the Guarantee benefits chapter, below.

2.4.1 Penalty free withdrawals

The policyholder can withdraw a limited portion of the fund value without surrendering the policy or incurring surrender charges. This is sometimes referred to as a partial surrender or cash withdrawal provision.

Penalty free withdrawals are processed at the end of each month, for policies persisting into the next month; after all other decrements are processed. Thus death and surrender benefits paid are based on the fund value before withdrawals, while death benefits and cash values in force at the end of the month are based on fund values after withdrawals.

2.4.2 Maximum penalty free withdrawal

The maximum annual penalty free withdrawal amount is determined at the beginning of each policy year. The maximum annual withdrawal can be a percent of either:

- Fund value – Fund value (including the DCA account) at the beginning of the first policy month, after all loads and charges.
- Cumulative premiums less withdrawals – Cumulative premiums paid through the first policy month, less cumulative withdrawals taken. Specify on the model point record the initial (at the projection date) cumulative premiums and cumulative withdrawals for the policies collected into the model point.

2.4.3 Penalty free withdrawal use

The penalty free withdrawals taken each policy year is an assumed percent of the maximum amount, times an assumed percent of policyholders taking withdrawals.

The withdrawal benefit is one twelfth of the annual decrement times the fund value at the end of each month, before all end of month cash flows. Thus the withdrawal amount is different each month, as the withdrawal basis changes.

2.5 Premiums

2.5.1 Premium term

Specify the premium term on the model point record. This is the policy anniversary when premium payments stop. When this value is zero, premiums are paid to maturity.

2.5.2 Premium mode

Specify the premium mode on the model point record. Premiums can be paid annually, semi-annually, quarterly, monthly, or special monthly (such as electronic transfer or pre-authorized check, with expense rate different from the monthly mode).

2.5.3 Annual premium

Specify on the model point record the initial (at the projection date) total annualized premium for the policies collected into the model point. Each premium payment is the annualized premium, divided by the number of premium payments per year.

2.5.4 Single premium

Specify on the model point record the total single premium for the policies collected into the model point.

2.5.5 Single premium policies

For a single premium plan, define the premium either as:

- Zero annual premium, paying only the single premium. Since the annual premium is zero, the premium term has no effect.
- Premium term of 1 year, with Annual premium mode, with zero single premium.

2.5.6 Premium suspensions

Premiums are reduced by a premium suspension percent at each policy anniversary after the projection date. The specified annual premium is multiplied by the cumulative premium suspension multiple. The suspension rate specified for the first policy year is ignored.

The reduced premium is assumed to be paid the same for all policies.

A negative premium suspension percent is an increase in the premium payments.

2.6 Premium bonus

The starting fund value can be increased by a specified percent of the single premium. The premium bonus does not apply to the portion of the premium allocated to the DCA account. This has no effect on commissions and expenses, which continue to be calculated using the premium paid.

2.7 Product loads

2.7.1 Mortality and expense load

Specify the annual mortality and expense load in the Product input page; one twelfth of this rate applies each month. The mortality and expense load only applies to the separate account fund values. The monthly load is:

$$\text{beginning fund value} \times \text{M \& E charge rate} \times (1/12)$$

where:

Beginning fund value is the fund value after the administration fee has been deducted

M & E charge rate is the annual mortality and expense charge percent.

2.7.2 Administration fee

An amount per policy (in the Product input page) is deducted from the fund value at the beginning of each policy year.

2.8 Fund value

2.8.1 Premiums

Premiums are credited to the fund value at the beginning of the month of each premium payment.

The allocation of premiums between the fixed account and the separate account is discussed in the Fund values chapter below.

2.8.2 Loads and charges

Loads and charges are deducted from the fund value at the beginning of the month after the addition of premium payments.

The allocation of loads and charges between the fixed account and the separate account is discussed in the Fund values chapter below.

2.8.3 Interest crediting

Interest is credited to the fund value at the end of each month. The total amount credited to the fund value is the sum of the amount credited to the fixed account and the separate account.

The amount credited to the fixed fund value and the indexed fund value is discussed in the Fund values chapter below.

2.8.4 Penalty free withdrawals

Penalty free withdrawals are deducted from the fund value at the end of the month, after interest has been credited.

The deduction from the fund value is the amount withdrawn.

2.9 Cash value

The cash value is the fund value less the surrender charge.

2.9.1 Fund value for surrender charge

The fund value for calculating the surrender benefit is the fixed account fund value plus the separate account fund value.

2.9.2 Surrender charge

The surrender charge is a percent, varying by policy year, of either:

- Fund value less remaining penalty free withdrawal entitlement, either before or after the market value adjustment is applied, floored by the cumulative premium paid as required by the rules of the Security and Exchange Commission (SEC).
- Cumulative premiums paid.

The penalty free withdrawal entitlement reduces the base before the surrender charge is applied.

3 FUND VALUES

3.1 Fund value structure

3.1.1 Fund types

The current fund value can be split between the following fund types:

- **DCA fund value** – The amount of current fund value which is to yet to be allocated to the separate account.
- **Fixed fund value** – The amount of current fund value held in the general account, earning interest based on the crediting strategy.
- **Separate account** – The amount of the current fund value invested in funds earning interest based on the change in the value of an index. The standard application provides for the separate account to be split into up to 6 variable funds.

3.1.2 Input variables

Input variables for the DCA fund, fixed fund, and separate account fund values are specified on the VAFund input page.

The DCA fund input variables include the DCA crediting rate. This variable commonly varies by plan code.

The fixed fund input variables include the guaranteed crediting rate and crediting strategy. The variables commonly vary by plan code.

The separate account input variables include the separate account index, investment fee refund percent, and the management fee percent. The variables commonly vary by a fund_id, identified as index0 through index5.

3.2 Fund allocations

3.2.1 Initial fund values

Fund values are allocated to the fixed account and up to six separate account variable funds. Specify the initial fund values on the model point record:

- For in force model points, this is the initial (at the projection date) amount in each account.
- For new business model points, this is the percent allocation between accounts. The total of the percentages for all accounts should add to 1.0 (100%), otherwise the specified values are normalized so they do add to that result. The percent for the DCA account should be zero, since this is a product assumption instead of an account allocation.

3.2.2 Premium allocation

Each annual premium payment is allocated using the proportion of fixed and separate account variable fund values specified on the model point record.

3.2.3 Loads and charges allocation

Loads and charges which are a percent of fund value are calculated and deducted separately from each separate account variable fund value. Percent of fund loads and charges do not apply to the fixed fund value.

Loads and charges that are not a percent of fund value, such as the administration fee, are calculated for the model point, then allocated in proportion to the current fund values, including the DCA account.

3.2.4 Withdrawals allocation

Penalty free withdrawals are allocated as a proportion of the account fund value to the total fund value, including the DCA fund.

3.2.5 Rebalancing

Rebalancing between the fixed and separate account variable fund values can be included in the projection.

The fund values can be rebalanced at the end of each calendar month, quarter, half year, or year based on either: (1) target allocation percentages or (2) initial fund value allocation.

Specify the allocation percentages on the model point record. The total of the percentages for all accounts should add to 1.0 (100%), otherwise the specified values are normalized so they do add to that result.

Transfers restore the fixed and variable fund values, excluding the DCA fund to have the desired allocations.

3.3 Dollar cost averaging fund value

Variable annuities can include the dollar cost averaging (DCA) account. This is a fixed account that transfers gradually to be allocated to the separate account funds.

3.3.1 Initial DCA allocation

For new business model points, specify (on the Fund input page) the percent of the single premium to allocate to the DCA account.

For in force policies, specify on the model point record the initial (at the projection date) DCA account value.

3.3.2 DCA crediting

The DCA account is credited monthly using the specified (on the Fund input page) effective annual DCA crediting rate. Interest is added to the DCA account at the beginning of each month. The fund definition for this account must be DCA.

Loads and charges (except for the administration fee) do not apply to the DCA account.

3.3.3 DCA transfers

The DCA account is transferred to the separate account funds in equal installments at the end each 'n' (specified on the model point record) months after issue until the account is exhausted.

Each transfer amount is allocated between the separate account funds in proportion to the allocation percent specified (on the Fund input page) for each separate account fund.

The DCA account is excluded from rebalancing.

3.4 Fixed fund value

3.4.1 Overview

The fixed fund value is increased for credited interest at the end of each month. The crediting rate for the fixed fund value is the larger of the policy's guaranteed crediting rate and the rate determined through the crediting strategy.

3.4.2 Crediting strategy declaration

The fixed account crediting rate changes at the beginning of each month.

3.4.3 *Crediting strategy rate basis*

Asset yield basis

You can specify the crediting rate as the asset yield less a spread. This definition makes the model points dependent on asset results, so independent model point processing cannot apply for asset/liability projections.

The asset yield includes the effects of IMR amortization, defaults, and borrowing costs.

For liability-only projections, you specify the asset yield as an assumption for the projection. Independent model point processing can still apply, since the asset yield is not affected by liability results.

For asset/liability models containing only new business model points, you can establish the initial asset yield (to be the basis for declaring crediting rates for the model start month) by investing from interim cash at the projection date. See the Investing at the projection date section in the Investment asset purchases chapter in the Asset processing book.

Competitor rate basis

You can specify the crediting rate as a competitor rate less a spread.

When the constraints do not include the asset yield, independent model point processing can apply even for asset/liability projections. For example, this approach could take the view that the crediting rate is primarily driven by market forces, and that appropriate investment strategies must be determined in order to achieve the desired spread from the asset yield.

3.5 *Separate account*

Each model point can use up to six variable fund values in the separate account. The same variable funds apply across all VA products.

3.5.1 *Variable funds*

For in-force model points, specify on the model point record the initial (at the projection date) variable fund values for the policies collected into the model point.

3.5.2 *Variable fund growth*

Variable fund values fluctuate with the change in the funds' asset market values.

The scenario index for each of the variable fund values must be contained in the scenario file. See the Economic scenarios book. Scenario growth rates are returned from the Economy DLL as a monthly rate, which can be negative.

3.5.3 *Investment management fee*

Specify the investment management fee in the product definitions as an effective annual rate. This fee reduces the separate account fund growth rate; it is not counted as a product 'load'.

3.5.4 *Investment fee refund*

The insurer may receive a portion of the investment fee as revenue.

The refund rate is specified as an annual rate. One twelfth of the annual rate is applied against the fund value at the beginning of month after premiums and rebalances but before reductions for loads and charges to determine the refund amount.

3.5.5 *Drop and recovery assumptions*

Reserves and capital calculations include assumed drops and subsequent recoveries for each separate account fund.

Reference the Valuation Drop and Recovery Rates.xlsm global external source for the drop and recovery rates which apply for AG 34 and for AG 43 standard scenario calculations.

4 GUARANTEE BENEFITS

4.1 Rider choices

Specify on the model point whether each rider type applies:

- GMAB -- Guaranteed minimum accumulation benefit
- GMDB -- Guaranteed minimum death benefit
- GMWB -- Guaranteed minimum withdrawal benefit
- GMIB -- Guaranteed minimum income benefit
- EEDB -- Enhanced earnings death benefit.

4.2 Guaranteed minimum accumulation benefit

4.2.1 *Benefit amount*

The GMAB increases the fund value at the end of every 'n' policy years during the GMAB benefit years. The GMAB value is the greater of: (1) fund value ratchet and (2) premium accumulation. The benefit is reduced by withdrawals.

Fund value ratchet

Every 'n' policy years, the fund value ratchet increases to the current fund value when it is higher than the former benefit amount. The starting fund value ratchet amount is the premium paid at issue.

For in force policies, specify on the model point record the initial (at the projection date) fund value ratchet amount. When this is not specified, the projection uses the initial fund value.

Premium accumulation

Premiums accumulate at the specified GMAB effective annual interest rate.

For in-force policies, specify on the model point record the initial (at the projection date) GMAB premium accumulation amount.

Reduction for withdrawals

Withdrawals reduce the GMAB amount, either by:

- Dollar for dollar – Reduces the GMAB amount by the withdrawal amount
- Pro rata – Reduces the GMAB amount by the same percent as the withdrawal amount divided by the fund value before the withdrawal.

4.2.2 *GMAB charge*

Specify the GMAB charge annual percent on the Product input page. One twelfth of this percent is deducted from the separate account fund values at the beginning of the month, after the administration fee has been deducted.

The GMAB charge is no longer collected after the GMAB expires.

For in force policies, specify on the model point record the accumulated GMAB charges for AG 39.

4.2.3 **GMAB cost**

The GMAB cost is the GMAB increase in the fund value.

4.3 **Guaranteed minimum death benefit**

4.3.1 **Benefit amount**

The GMDB amount is the greater of: (1) fund value ratchet and (2) premium accumulation. The benefit is reduced by withdrawals.

After a specified maximum attained age, the GMDB either: (1) expires or (2) is frozen at the GMDB amount at the maximum age (but reduced by further withdrawals).

Fund value ratchet

Every 'n' policy years until the maximum attained age, the benefit amount changes either by:

- Ratchet – Increases to the current fund value when it is higher than the previous GMDB amount
- Reset – Changes to the current fund value, even when it is lower.

The starting fund value ratchet is the premium paid at issue. For in force policies, specify on the model point record the initial (at the projection date) GMDB fund value ratchet amount. When this is not specified, the projection uses the initial fund value.

Premium accumulation

Premiums accumulate through a maximum attained age at the specified effective annual interest rate. Premium roll up occurs either: (1) every policy anniversary, or (2) every period.

The amount can be capped at a maximum multiple of the cost basis (cumulative premiums less cumulative withdrawals).

For in force policies, specify on the model point record the initial (at the projection date) GMDB premium accumulation amount.

Reduction for withdrawals

Withdrawals reduce each of the GMDB benefit amounts, either by:

- Dollar for dollar – Reduces the GMDB amounts by the withdrawal amount
- Pro rata – Reduces the GMDB amounts by the same percent as the withdrawal amount divided by the fund value before the withdrawal.

4.3.2 **Fund value charge**

Specify the GMDB charge annual percent on the Product input page. One twelfth of this percent is applied to the separate account fund values at the beginning of the month, after the administration fee has been deducted.

The GMDB charge continues to be collected after the GMDB maximum attained age, even when the benefit vanishes after this time.

4.3.3 **GMDB cost**

The GMDB cost is the GMDB death benefits incurred in excess of the fund value released on death.

4.4 Guaranteed minimum withdrawal benefit

4.4.1 *Guaranteed withdrawals*

The GMWB provides for continuing withdrawals even when the fund value is exhausted. No withdrawals are taken until the end of the wait period.

The guaranteed withdrawals are penalty free. That is, the maximum penalty free withdrawal amount is automatically the greater of the normal maximum and the guaranteed withdrawal amount.

The withdrawals can be guaranteed either for a specified number of years, or for a lifetime annuity.

4.4.2 *Base amount*

The base amount determined is the greater of: (1) fund value ratchet, and (2) accumulation of premiums. The benefit is reduced by withdrawals. After a specified maximum age, the base amount stops accumulating and is reduced by withdrawals taken.

Specify on the model point record the maximum annual withdrawal amount as a percent of the base amount. The maximum withdrawal percent applies either to:

- Premium – Cumulative premiums.
- GMWB Base – The GMWB base at the end of the waiting period.
- GMWB Base Adjustable – The GMWB base, which is allowed to increase but never decrease.

Fund value ratchet

Every 'n' policy years until a maximum attained age, the fund ratchet amount increases to the current fund value when it is higher than the previous amount. The starting fund value ratchet is the premium paid at issue. The fund value ratchet increases by premiums paid in the month they are paid.

For in force policies, specify on the model point record the initial (at the projection date) GMWB fund value ratchet amount. This value must be specified; otherwise the projection uses zero initial value.

Premium accumulation

For the accumulation or roll up benefit, premiums less withdrawals are accumulated either:

- Return – The accumulated premium is the premium paid, without any interest.
- Roll up – Accumulated premiums increase by the specified effective annual interest rate either: (1) every policy anniversary, or (2) every period. The accumulation amount is capped at a maximum multiple of the cost basis (cumulative premiums less cumulative withdrawals).
- Step up – Cumulative premiums increase by the specified percent at the end of the wait period.

For in force policies, specify on the model point record the initial GMWB premium accumulation amount.

4.4.3 *GMWB annuity present value*

The GMWB in-the-moneyness percent, and the GMWB cost, is calculated using the present value of the guaranteed annuity payments. This present value is calculated using the specified GMWB mortality table and effective annual discount rate.

4.4.4 *Dynamic penalty free withdrawals*

When dynamic penalty free withdrawals are included, the withdrawal percent at the beginning of each policy year is:

$$\min(1.0, \text{mult} \times (\max(0.0, \text{ITM}\%)))$$

where:

ITM% is the GMWB in-the-moneyness percent.

The monthly penalty free withdrawal rate is:

$$\min(1.0, \text{GMWBmax withdrawal amount} \times \text{dynamic rate} / \text{fund value})$$

where:

Dynamic rate is the dynamic penalty free withdrawal rate

Fund value is the end of month fund value before claims.

4.4.5 Fund value charge

The GMWB charge is an annual percent of either:

- Fund value basis – Either the separate account only or the entire fund value
- GMWB base amount
- Greater of the fund value basis and the GMWB base amount.

One twelfth of this amount is deducted from the fund values at the beginning of the month. When the GMWB base is less than the fund value, and the percent GMWB charge applies to the GMWB base only, the charge applies to the beginning of month fund value after the administration fee has been deducted. Otherwise, the charge applies to the beginning of month fund value after all loads have been deducted.

The GMWB charge continues to be collected after the GMWB maximum attained age.

For in-force policies, specify on the model point record the initial (at the projection date) accumulated GMWB charges for AG 39.

4.4.6 GMWB cost

The GMWB cost is the GMWB guaranteed annuity present value at the time the fund value is exhausted.

4.5 Guaranteed minimum income benefit

4.5.1 Base amount

The GMIB applies for annuitization after a specified number of policy years. After a specified maximum attained age, the GMIB can either: (1) expire or (2) remain frozen at the GMIB amount at the maximum age.

The GMIB present value is the base amount applied to the GMIB guaranteed life annuity rates (discussed below). The base amount is the greater of: (1) fund value ratchet and (2) premium accumulation. The benefit is reduced by withdrawals.

Fund value ratchet

Every 'n' policy years until a maximum attained age, the fund ratchet amount increases to the current fund value when it is higher than the previous amount. The starting fund value ratchet is the premium paid at issue.

For in force policies, specify on the model point record the initial (at the projection date) GMIB fund value ratchet amount.

Premium accumulation

Premiums accumulate to the GMAB option date at the specified effective annual interest rate. Premium roll ups occur either (1) every policy anniversary, or (2) every period.

The amount can be capped at a maximum multiple of the cost basis (cumulative premiums less cumulative withdrawals).

For in force policies, specify on the model point record the initial (at the projection date) GMIB premium accumulation amount.

Reduction for withdrawals

Withdrawals reduce the GMIB amount either by:

- Dollar for dollar – Reduces the base amount values by the withdrawal amount
- Pro rata – Reduces the GMIB amount by the same percent as the withdrawal amount divided by the fund value before the withdrawal.

4.5.2 Annuity benefit present value

The GMIB present value is:

$$\text{base amount} \times (\text{current annuity factor} / \text{guaranteed annuity factor})$$

where:

Current annuity factor is calculated using current rates

Guaranteed annuity factor is calculated using guaranteed rates.

Specify the current and guaranteed mortality and interest rates for the product.

4.5.3 Fund value charge

Specify the GMIB charge annual rate for the product. One twelfth of this percent is applied to the separate account fund values at the beginning of the month, after the administration fee has been deducted.

The GMIB charge continues to be collected after the GMIB maximum attained age, even when the benefit vanishes after this period.

For in force policies, specify on the model point record the initial (at the projection date) accumulated GMIB charges for AG 39.

4.5.4 GMIB cost

The GMIB cost is any GMIB amount in excess of the fund value released on surrender.

4.6 Enhanced earnings death benefit

4.6.1 Benefit amount

The death benefit increases by a specified percent of the excess of: (1) the greater of the fund value and the GMDB amount, over (2) the cumulative premiums. The specified percent typically reduces when the policyholder reaches a specified attained age. The EEDB can be capped at a specified multiple of cumulative premiums.

4.6.2 Fund value charge

Specify the EEDB charge annual percent for the product. One twelfth of this amount is applied to the separate account fund values at the beginning of the month, after the administration fee has been deducted.

4.6.3 EEDB cost

The EEDB cost is: (1) the death benefits incurred, less (2) the greater of the GMDB and the fund value released on death.

5 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves. PBA reserves and capital are discussed in the PBA book.

5.1 CARVM reserve

Variable annuity standard scenario reserves under Actuarial Guideline XLIII (AG 43) require a calculation of a basic and basic adjusted reserve using the statutory statement valuation requirements of Actuarial Guideline XXXIII (AG 33).

Reserves under Actuarial Guideline XXXIV (AG 34) and XXXIX (AG 39) can still be calculated when PBA reserves are not calculated, such as for tax reserves for policies issued before 2010. These calculations are also discussed here.

Specify CARVM valuation variables on the VAValuation input page.

5.1.1 *Percent of fund approximation*

The basic reserve required under actuarial guideline XLIII (AG 43) or the total CARVM reserve when applying actuarial guidelines XXXIV (AG 34) and XXXIX (AG 39) can be approximated using percentages of the fund value. This may be appropriate to improve run time by eliminating these compute-intensive calculations.

5.1.2 *Actuarial guideline 33*

For policies which require actuarial guideline XLIII (AG 43) standard scenario basic reserves, the basic reserve is the maximum of the integrated reserve for the cash value stream reserve, and when included, the annuitization stream.

The actuarial guideline XXXIII (AG 33) approach also applies to policies which do not require actuarial guideline XLIII (AG 43) reserves and do not include GMDB benefits.

Cash value streams

Cash value streams are always included in the CARVM calculations. The present value of guaranteed benefits is calculated using a different valuation interest rate for each of the following benefits:

- Death benefits,
- Penalty free withdrawals, and
- Full cash value surrenders.

There are two projections of the cash value stream guaranteed benefits. One assuming maximum penalty free withdrawals are taken and one assuming no penalty free withdrawals are taken. The maximum integrated reserve for the cash value stream is the greater of the integrated reserve for the two cash value streams.

Annuitization stream

Annuitization streams may be included in the CARVM calculations. The present value of guaranteed benefits is calculated using a different valuation interest rate for each of the following benefits:

- Death benefits,
- Penalty free withdrawals, and
- Full annuitization benefit.

For the annuitization stream guaranteed benefits, there are potentially many different streams projected with different types of annuity benefits. Any number of CARVM annuitization tests can be defined, where each test represents a type of annuity to purchase and then any number of the defined tests can be applied. For example, you could project the annuitization stream for three annuitization tests, one purchasing life only

annuities, one purchasing 5 year certain and life annuities, and one purchasing 10 year certain and life annuities.

And, within each annuity benefit type stream projection, there are two streams projected. One assuming maximum penalty free withdrawals are taken and one assuming no penalty free withdrawals are taken. The maximum integrated reserve for the annuitization test is the greater of the integrated reserve from these two projections.

Finally, the integrated reserve for the annuitization stream is the maximum of the integrated reserve for each annuitization test.

During the CARVM projection, the full annuitization benefit is determined as the benefit for an annual payment immediate annuity for a specified certain years and a single life (when a life annuity type is used) purchased by the fund value after penalty free withdrawals. The purchase rate for the annuity may be determined using only a guaranteed basis annuity purchase rate or both a current basis annuity purchase rate and a guaranteed basis annuity purchase rate. When the current basis annuity purchase rate is included, the annuity benefit at the valuation date is the greater of the current basis annuity benefit and the guaranteed basis annuity benefit. Annuity benefits for CARVM projection periods after the valuation date are always based on the guaranteed basis purchase rate. The current and guaranteed annuity purchase rate bases are discussed in section 2.3 above.

Since contracts may provide, as contractual guarantees, the use of preferential purchase rates at annuitization to those listed in the contract, you may specify a percentage of the valuation date fund value as a floor to the annuitization stream integrated reserve to account for this additional guarantee. For example, when such provision applies, you should set the `stat_carvm_annuitzn_min_fund_val_pct` and `tax_carvm_annuitzn_min_fund_val_pct` to the appropriate floor percentage, such as 93%.

5.1.3 Actuarial guideline 34

For policies which do not require actuarial guideline XLIII (AG 43) reserves but do include GMDB benefits, actuarial guideline XXXIV (AG 34) reserves apply and the CARVM reserve for the contract is the maximum of the integrated reserve including all benefits.

Integrated reserve

The integrated reserve for actuarial guideline XXXIV (AG 34) is calculated using the actuarial guideline XXXIII (AG 33) approach, where the benefits for each of the cash value and annuitization stream, when included, include all benefits.

Two policy benefit projections are required for the integrated reserve, one with an unreduced account value and one where the initial account value is reduced through an immediate drop.

Projected non-guaranteed death benefits, penalty free withdrawals, full surrender benefits, and annuitization benefits are determined from the projection of an unreduced account value with an assumed return equal to valuation interest rate less charges and fees.

The projected net amount at risk for the guaranteed minimum death benefit is determined by assuming an immediate drop in the supporting asset values, followed by a subsequent recovery based upon a net assumed return. For example, the reduced account value after the immediate drop equals the account value on the valuation date, multiplied by $(1.0 - \text{immediate drop percentage})$. The projected reduced account value "n" years later equals the reduced account value multiplied by $(1.0 + \text{net assumed return})^n$, where the net assumed return is the gross assumed return less fees and charges. The immediate drop percentage and the gross assumed return are those from the appropriate asset class from actuarial guideline XXXIV (AG 34).

Please see the Valuation Drop And Recovery Rates.xlsm global external source for the drop and recovery rates for actuarial guideline XXXIV (AG 34).

GMDB reserve

The reserve for the minimum guaranteed death benefit is the difference between the maximum integrated reserve and the basic reserve.

The basic reserve for actuarial guideline XXXIV (AG 34) is calculated using the actuarial guideline XXXIII (AG 33) approach, where the benefits for each of the cash value and annuitization stream, when included, do not include the guaranteed minimum death benefits.

5.1.4 Actuarial guideline 39

For policies which do not require actuarial guideline XLIII (AG 43) reserves but do include guaranteed living benefits (VAGLBs), such as guaranteed minimum accumulation benefits (GMAB), guaranteed minimum income benefits (GMIB), and guaranteed minimum withdrawal benefits (GMWB), actuarial guideline XXXIX (AG 39) applies.

The reserve for each of the living benefits equals the cumulative charges for the benefit from issue, where the cumulative amount is reduced by 2.5% each quarter beginning in 2008. The initial cumulative charge amount, assumed to already be reduced per the actuarial guideline XXXIX (AG 39) provisions, is specified on the model point record.

This reserve is an addition to the reserve calculated using actuarial guideline XXXIII (AG 33) or actuarial guideline XXXIV (AG 34).

5.1.5 CARVM valuation basis

Interest

CARVM valuation interest may vary by the cash settlement definition, guarantee duration definition, plan type, and guarantee duration. Separate plan type definitions are provided for cash value elective benefits, annuitization elective benefits, and non-elective benefits.

The Global folder includes an external source workbook containing the maximum standard valuation law valuation rates. The DefAnnStatValn and DefAnnTaxValn ranges in the Valuation Interest Rates Excel workbook include rates varying by valuation method (Issue Year, Change in Fund), cash settlement definition (With Cash Settlements, Without Cash Settlements), guarantee duration definition (With Guarantees, Without Guarantees, With or Without Guarantees), plan types (Plan A, Plan B, Plan C) and guarantee durations ranges of 0-5, 6-10, 11-20 and 21+.

Under actuarial guideline XXXIV (AG 34), the valuations rate for streams A, B, and C, where

- Stream A – Death benefits paid in excess of the fund value
- Stream B – Fund values paid at death
- Stream C – Withdrawals and final cash value

are based on the selections of the plan types for each of the streams.

For the death benefits, penalty free withdrawals, and full cash surrender benefits, the guarantee duration is determined as the number of years the fixed account guaranteed credited rate exceeds the valuation interest rate for life insurance policies with a guarantee duration in excess of 20 years, as specified in the `stat_carvm_life_int_rate` and `tax_carvm_life_int_rate` variables.

For the annuity benefits, the guarantee duration is determined as the number of years since issue when the annuitization occurs.

Mortality

Reserve mortality is a base rate, times a multiple. Separate reserve mortality is specified for death benefits, GMDB benefits, and annuity benefits.

Mortality variables are read for both males and females when blended mortality applies.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys `mort_issue_age` and `mort_gender` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Using the hidden variable ensures the attained age does not advance with each yearly loop.

5.1.6 CARVM valuation method

The CARVM valuation method is always the 'Issue Year' method.

Under the CARVM issue year valuation method, the CARVM guaranteed benefit streams are developed from a projection of a single fund value bucket. Benefits are discounted using the appropriate valuation interest rate for the policy issue year, cash settlement definition, guarantee duration definition, plan type, and guarantee duration from the issue year method valuation interest rate table.

The reserve for the benefit stream is the maximum discounted value of the issue year fund bucket.

5.1.7 Continuous CARVM

The continuous CARVM method is the same as for CARVM, except that the surrender charge is the lower of the surrender charge for the current month and the next month.

For example, when the surrender charge decreases from 7% in the first policy year to 6% in the second policy year, the continuous CARVM reserve at the end of the first policy year is based on the 6% surrender charge.

5.1.8 CARVM crediting rate

Future policy benefits are projected using the net assumed return.

The net assumed returns for the fixed and DCA funds are the rates specified in the CARVM valuation basis assumptions.

When projecting the benefits for calculations under actuarial guideline XLIII (AG 43), the net assumed return for the variable funds equals the rate specified in the `va_pba_ag43_std_scen_svl_disc_rate` variable on the Company input page less fees and charges.

When projecting the benefits for calculations under actuarial guideline XXXIII (AG 33) and for the unreduced account value under actuarial guideline XXXIV (AG 34), the net assumed return for the variable funds equals the CARVM valuation interest rate less fees and charges.

When projecting the benefits for the net amount at risk calculations for the reduced account value under actuarial guideline XXXIV (AG 34), the net assumed return for the variable funds equals the gross assumed return specified in the `stat_carvm_ag34_tbl` and `tax_carvm_ag34_tbl` variables less fees and charges. The gross return table should contain the gross assumed returns for the assets classes included in appendix 1 of actuarial guideline XXXIV (AG 34).

5.1.9 CARVM calculation frequency

CARVM reserves are calculated as of the projection date, the first month after issue, at the end of each policy year, and at the end of each calendar year. Approximations are used between these calculation dates in order to minimize these compute-intensive calculations.

For the projection months when CARVM reserves are not calculated, the reserve for each model point is approximated by multiplying the previous month's reserve by the ratio of the current month's fund value to the previous month's fund value.

The model point record includes the initial (at the projection date) statutory and tax CARVM reserve amounts. These values are only for use by nested projections. Ignore these variables when preparing model point records for your projection, since the CARVM valuation is calculated at the projection date.

5.1.10 Terminating CARVM projections

CARVM reserve calculations consider benefits projected only through a specified number of years from each valuation date.

5.2 Reinsurance reserve

Reserves are calculated both gross of (ignoring) reinsurance and net of reinsurance.

When actuarial guideline XXXIV (AG 34) applies, and for contracts which reinsure some or all of the GMDB benefits, the net reserve is calculated using the actuarial guideline XXXIV (AG34) method. During this calculation, the net amount at risk and death benefit streams from un-reduced fund values include the impact of reinsurance, the base benefit streams for penalty free withdrawals, cash surrenders and annuitization are the same as for the gross calculation, and a new benefit, equal to the reinsurance gross premiums, is included.

5.3 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record sets the tax CARVM reserve equal to the statutory CARVM reserve. When actuarial guideline XLIII (AG 43) applies, this setting sets the tax standard scenario reserve equals the statutory standard scenario reserve.

Tax reserves are calculated by the same methods, using the tax reserve valuation basis and method.

The tax reserve is not greater than the statutory reserve.

6 ACTUARIAL ASSUMPTIONS

6.1 Commissions

6.1.1 Premium commission

A percent of each premium is paid as a commission when the premium is paid.

6.1.2 Fund value commission

One twelfth of the annual percent of fund value at the end of the previous month is paid at the beginning of each month.

6.1.3 Commission overrides and agent terminations

The specified commission rates are the average commission rates that are assumed to apply, taking into account average writing agent commissions, commission overrides, and the effect of agent terminations.

6.1.4 Commission chargebacks

A portion of the premium commission can be charged back in the event of lapse or surrender during the first three years after issue. The chargeback percent applies to the total first year premium commissions paid to date and occurs at the end of the month. The chargeback percent varies by months since issue.

Other types of commissions are not subject to chargeback.

The commission subject to chargeback is reduced by decrements. However, the commission subject to chargeback is only reduced by deaths when it is applied to the chargeback percent.

For in-force model points, specify on the model point record the initial (at the projection date) commissions subject to chargeback for the policies collected into the model point.

6.2 Expenses

6.2.1 Acquisition expenses

Acquisition expenses occur at issue and include:

- Amount per policy
- Percent of premium.

6.2.2 Maintenance expenses

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Maintenance expenses include:

- Amount per policy
- Percent of the annualized premium (excluding the single premium)
- Percent of the total fund value at the end of each month. This percent applies to the fund value at the beginning of the current month.

6.2.3 Inflation

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

6.2.4 Expense adjustment multiple

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

6.2.5 Premium tax

Premium taxes are reported with expenses. The tax is a percent of each premium payment.

Inflation and the expense adjustment multiple do not apply to premium tax.

6.2.6 Fixed expenses

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

6.3 Mortality

6.3.1 Input variables

'Mortality variables' for annuity products include mortality base, multiple, and addition rates, and mortality improvement. These variables are read twice when blended mortality applies – first for male rates and then again for female rates. See the Blended mortality topic in the Mortality section, below.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys mort_issue_age and mort_gender instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age. Using the hidden variable ensures the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

6.3.2 Mortality adjustment multiple

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

6.3.3 *Base mortality*

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

6.3.4 *Mortality improvement*

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$. When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

6.3.5 *Mortality table adjustments*

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

6.3.6 *Blended mortality*

The model point can be a 'blend' between males and females issued at the specified issue age. This occurs when the gender variable is Blended. Specify on the model point record the proportion of issues that are males.

The male and female mortality rates are each determined from the mortality table. The blended mortality rate is:

$$\text{proportion of males} \times \text{male mortality rate} + \text{proportion of females} \times \text{female mortality rate}$$

6.3.7 *Monthly mortality*

Deaths occur at the end of each month. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

6.4 *Lapses and surrenders*

6.4.1 *Lapse adjustment multiple*

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date.

6.4.2 *Policy year lapse rates*

The base lapse rate is adjusted by a multiple.

6.4.3 *Dynamic lapses*

Lapse rates can be dynamically adjusted depending on the in-the-moneyness of the following guaranteed benefits:

- Guaranteed minimum accumulation benefit
- Guaranteed minimum income benefit
- Guaranteed minimum withdrawal benefit.

For each benefit, in-the-moneyness is determined by comparing the guaranteed benefit amount to either:

Fund value

The in-the-money percent is:

$$1 - (\text{fund value} / \text{guaranteed amount})$$

where:

Fund value is the end of month fund value, before any claims or dollar cost averaging account transfers occur

Guaranteed amount is the end of month guaranteed account value before claims

Annuity benefit

The in-the-money percent is:

$$(\text{guaranteed benefit} \times \text{annuity factor}) / \text{fund value} - 1$$

where:

Guaranteed benefit is the end of month guaranteed account value after claims

Annuity factor is the present value of guaranteed benefit amount

Fund value is the end of month fund value, before any claims or DCA transfers occur.

Dynamic lapses

For each benefit type, the dynamic lapse rate is:

$$\text{base lapse rate} \times (1.0 - \text{multiple} \times (\text{ITM}\% + \text{addition}))$$

where:

ITM% is the in-the-moneyness percent calculated above.

For policies with multiple riders, the dynamic lapse rate is the minimum of the dynamic rates.

The excess lapse rate is calculated as:

$$\text{dynamic lapse rate} - \text{base lapse rate}$$

Then, a dynamic multiple is calculated as:

$$\min(\max(\text{excess lapse rate} / \text{base lapse rate}, \text{dynamic minimum change}), \text{dynamic maximum change})$$

where:

Dynamic minimum change is the specified minimum ratio of the excess lapse rate to the base lapse rate

Dynamic maximum change is the maximum ratio of the excess lapse rate to the base lapse rate.

Finally, the dynamic lapse rate is:

$$\min(\max((1.0 + \text{dynamic multiple}) \times \text{base lapse rate}, \text{dynamic minimum}), \text{dynamic maximum})$$

6.4.4 Monthly lapses

Lapses and surrenders occur at the end of each month. Finally, the annual rate is converted into the monthly lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/12)}$$

The monthly lapse rate is applied to the fund value at the end of the month, before penalty free withdrawals are processed, and before claims are paid.

6.5 Annuitizations

6.5.1 Annuitization rate

The annual annuitization election rate typically varies either by attained age or policy year; one twelfth of this percent applies each month.

6.5.2 Dynamic annuitization

The annuitization rate can be dynamic. The dynamic annuitization rate is based on the in-the-moneyness of the guaranteed minimum income benefit.

In-the-moneyness is determined by comparing the guaranteed benefit amount to either:

Fund value

The in the money percent is:

$$1 - (\text{fund value} / \text{guaranteed amount})$$

where:

Fund value is the end of month fund value, before any claims or dollar cost averaging account transfers occur

Guaranteed amount is the end of month guaranteed account value before claims.

Annuity benefit

The in-the-money percent is:

$$(\text{guaranteed benefit} \times \text{annuity factor}) / \text{fund value} - 1$$

where:

Guaranteed benefit is the end of month guaranteed account value after claims

Annuity factor is the present value of guaranteed benefit amount

Fund value is the end of month fund value, before any claims or dollar cost averaging account transfers occur.

The dynamic annuitization rate is:

$$\text{base annuitization rate} + \min(\text{ITM}\%, \text{dynamic maximum}) \times \text{base annuitization rate}$$

where:

Dynamic maximum is the maximum multiple.

6.6 Risk based capital risk definition

The risk based capital (RBC) C3 component percent varies between 'low', 'medium', and 'high' risk annuity products. Specify on the Company input page this risk definition for each product, then vary the RBC C3 percent between Low, Medium, and High risks.

7 REINSURANCE

Modified coinsurance can apply to the base policy benefits, the rider benefits, or both base and rider benefits.

7.1 Base only

Reinsurance arrangements covering the base contract are modified coinsurance (ModCo) treaties. Specify on the model point record the portion of the policy ceded to the reinsurer.

Reinsurance benefits

The reinsurance benefits are the gross benefits times the reinsurance percent.

Reinsurance premiums

Reinsurance premiums are paid at the same premium frequency as the model point and are equal to the premium paid times the reinsurance percent.

Reinsurance expenses and commissions

Reinsurance expenses and commissions reimbursed are the expenses and commissions paid by the ceding company that are refunded by the reinsurer. These are positive cash flows for the ceding company. Inflation applies to reinsurance expenses reimbursed.

7.2 Rider only

Reinsurance arrangements covering the riders are YRT treaties.

Reinsurance benefits

The reinsurance benefits are the guaranteed benefits in excess of the base benefits times the reinsurance percent.

Reinsurance premiums

Reinsurance premiums are the reinsurance rider charges. The reinsurance rider charge is one twelfth of the annual charge percent applied to the fund value at the beginning of the month, after the administration fee has been deducted.

For all riders except the GMWB, the reinsurance rider charge only applies to the separate account fund value. The reinsurance GMWB charge can also apply to the fixed fund value.

7.3 Both base and rider

Reinsurance arrangements covering the base contract are modified coinsurance ("ModCo") treaties. The reinsurance arrangements covering the riders are yearly renewable term (YRT) treaties.

7.4 Reinsurance administration expenses

Reinsurance administration expenses are incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include an amount per policy.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Coinsurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for coinsurance.

8 MODEL POINT RECORDS

8.1 Model point record layout

The model point record layout for variable annuities is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Values
A	group	up to 25 characters (assigns model point to report group)
B	seg_id	segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date	yyyy/mm/dd or mm/dd/yyyy
F	plan_code	up to 20 characters (link to assumptions by plan)
G	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
H	tax_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
I	issue_age	positive integer
J	gender	Male, Female, Unisex
K	male_sex_pct	male percent (1.0 for 100%). This variable only applies when gender is Unisex.
L	prem_mode	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
M	prem_term_mp	premium years (0 = omega age from mortality table)
N	init_annualzd_prem	initial annualized premium for mp
O	single_prem	single premium for mp
P	dca_mths	number of months from policy issue to perform the DCA transfers

Excel Column	Variable	Values
Q	reins_pct	ceded reinsurance percent for mp
R	init_policies	initial policies for mp (required)
S	init_av_dca	initial DCA account value
T	init_av_fixed	initial fixed account value
U	init_av_index0	initial separate account value represented by index 0
V	init_av_index1	initial separate account value represented by index 1
W	init_av_index2	initial separate account value represented by index 2
X	init_av_index3	initial separate account value represented by index 3
Y	init_av_index4	initial separate account value represented by index 4
Z	init_av_index5	initial separate account value represented by index 5
AA	init_rebal_pct_fixed	percent of total account value to allocate to the fixed account when accounts are rebalanced
AB	init_rebal_pct_index0	percent of total account value to allocate to the index 0 account when accounts are rebalanced
AC	init_rebal_pct_index1	percent of total account value to allocate to the index 1 account when accounts are rebalanced
AD	init_rebal_pct_index2	percent of total account value to allocate to the index 2 account when accounts are rebalanced
AE	init_rebal_pct_index3	percent of total account value to allocate to the index 3 account when accounts are rebalanced
AF	init_rebal_pct_index4	percent of total account value to allocate to the index 4 account when accounts are rebalanced
AG	init_rebal_pct_index5	percent of total account value to allocate to the index 5 account when accounts are rebalanced
AH	gmab_ind	1 (GMAB is included), 0 (GMAB is not included)
AI	init_gmab_av	initial GMAB fund value ratchet amount
AJ	init_gmab_prem	initial GMAB premium accumulation amount
AK	init_gmab_chg_cumul	initial accumulated amount of GMAB charges for AG39
AL	gmab_sfas133_req_prof_chg	GMAB SFAS133 required profit charge

Excel Column	Variable	Values
AM	gmdb_ind	1 (GMDB is included), 0 (GMDB is not included)
AN	init_gmdb_av	initial GMDB fund value ratchet amount
AO	init_gmdb_prem	initial GMDB premium accumulation amount
AP	gmwb_ind	1 (GMWB is included), 0 (GMWB is not included)
AQ	init_gmwb_av	initial GMWB fund value ratchet amount
AR	init_gmwb_prem	initial GMWB premium accumulation amount
AS	init_gmwb_max_wdl_amt	initial maximum annual withdrawal amount
AT	init_gmwb_chg_cumul	initial accumulated amount of GMWB charges for AG39
AU	gmwb_sfas133_req_prof_chg	GMWB SFAS133 required profit charge
AV	gmib_ind	1 (GMIB is included), 0 (GMIB is not included)
AW	init_gmib_ratchet	initial GMIB ratchet amount
AX	init_gmib_rollup	initial GMIB roll up amount
AY	init_gmib_chg_cumul	initial accumulated amount of GMIB charges for AG39
AZ	gmib_sfas133_req_prof_chg	GMIB SFAS133 required profit charge
BA	eeb_ind	1 (EEDB is included), 0 (EEDB is not included)
BB	init_gross_prem_cumul	total premium paid for in force policies before the valuation date
BC	init_p fwd_cumul	amount of penalty free withdrawals taken by the valuation date. Should be set to 0 for new business
BD	init_p fwd_curr_pol_yr	amount of penalty free withdrawals taken in the year of valuation by the policyholder. Should be set to 0 for new business
BE	init_p fwd_entitlement	amount of penalty free withdrawal entitlement at the valuation date. Should be set to 0 for new business
BF	stat_res_precalc_bar	precalculated statutory basic adjusted reserve used by the AG43 standard scenario calculations
BG	surr_chg_precalc_amortzn_yrs	precalculated statutory surrender charge amortization period used by the AG43 standard scenario calculations
BH	tax_res_precalc_bar	precalculated tax basic adjusted reserve used by the AG43 standard scenario calculations

Excel Column	Variable	Values
BI	tax_surr_chg_precalc_amortzn_yrs	precalculated tax surrender charge amortization period used by the AG43 standard scenario calculations
BJ	init_comm_subject_to_chargeback	amount of commissions subject to chargeback at the valuation date. Should be 0 for new business
BK	init_stat_res_carvm	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BL	init_stat_res_carvm_net	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BM	init_stat_res_carvm_basic	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BN	init_tax_res_carvm	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BO	init_tax_res_carvm_net	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BP	init_tax_res_carvm_basic	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BQ	init_stat_res_std_scen_res	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BR	init_stat_res_std_scen_res_unhedged	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BS	init_stat_res_net_std_scen_res	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BT	init_stat_res_net_std_scen_res_unhedged	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BU	init_tax_res_std_scen_res	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step

Excel Column	Variable	Values
BV	init_tax_res_std_scen_res_unhedge d	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BW	init_tax_res_net_std_scen_res	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BX	init_tax_res_net_std_scen_res_unh edged	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the status for the next time step
BY	hedge_valn_scen_start_num	1; the starting scenario to use for this model point for hedging shock valuations. This field only applies when the valn_scens_defn of the hedging assumptions is set to 'Model Point Specific'

8.2 Required model point variables

8.2.1 Always required

The following model point variables are required in all cases:

- **Model point identification:** group, seg_id, and mp_id
- **Insured characteristics:** issue_age and gender
- **Policy specifications:** issue_date, plan_code, stat_valn_code, and tax_valn_code
- **Initial policies:** init_policies
- **Annual premium:** init_annualized_prem, prem_mode, and prem_term are required for premium paying model points.
- **Single premium:** single_prem is required for new business model points.
- **Reinsurance:** reins_pct is required for model points with base plan reinsurance.
- **Fund value rebalancing proportions:** 'init_av' variables are required when the fund value is rebalanced based on the initial account value. 'init_rebal_pct' variables are required when the fund value is rebalanced based on a specified percent.
- **DCA:** dca_mths is required for model points that have a nonzero DCA account value.
- **GMWB maximum annual withdrawal percent:** gmwb_max_annual_wdl_pct is required for model points with GMWB.
- **SFAS 133 charges:** gmab_sfas133_req_prof_chg, gmwb_sfas133_req_prof_chg, and gmib_sfas133_req_prof_chg are required when calculating SFAS 133 liabilities for rider benefits.

8.2.2 In-force model point values

In-force model points are driven by all the 'initial values' variables, which specify the aggregate model point values as of the projection date.

New business model points (model points with issue date after the projection date), however, ignore these variables. The exception is that the initial policies must be specified.

The following table shows when each in-force values variable is required for in-force model points:

Variable	Required When
init_av_dca	always required (may be left at 0 when no fund value is allocated to the DCA account)
init_av_fixed	always required (may be left at 0 when no fund value is allocated to the fixed account)
init_av_index0	always required (may be left at 0 when no fund value is allocated to this separate account)
init_av_index1	always required (may be left at 0 when no fund value is allocated to this separate account)
init_av_index2	always required (may be left at 0 when no fund value is allocated to this separate account)
init_av_index3	always required (may be left at 0 when no fund value is allocated to this separate account)
init_av_index4	always required (may be left at 0 when no fund value is allocated to this separate account)
init_av_index5	always required (may be left at 0 when no fund value is allocated to this separate account)
init_gmab_av	model points with GMAB
init_gmab_prem	
init_gmab_chg_cumul	
init_gmdb_av	model points with GMDb
init_gmdb_prem	
init_gmwb_av	model points with GMWB
init_gmwb_prem	
init_gmwb_max_wdl_amt	
init_gmwb_chg_cumul	
init_gmib_ratchet	model points with GMIB
init_gmib_rollup	
init_gmib_chg_cumul	
init_gross_prem_cumul	always required

Variable	Required When
init_p fwd_cumul	model points with penalty free withdrawals
init_p fwd_curr_pol_yr	model points with penalty free withdrawals
init_p fwd_entitlement	model points with penalty free withdrawals
init_comm_subject_to_chargeback	model points with commission chargebacks

8.2.3 Variables to ignore

The following variables are needed only for nested projections, and should be ignored. These values must be carried forward to the next time step in order to reproduce the results from standard projections:

Variable
init_stat_res_carvm
init_stat_res_carvm_net
init_stat_res_carvm_basic
init_tax_res_carvm
init_tax_res_carvm_net
init_tax_res_carvm_basic
init_stat_res_std_scen_res
init_stat_res_std_scen_res_unhedged
init_stat_res_net_std_scen_res
init_stat_res_net_std_scen_res_unhedged
init_tax_res_std_scen_res
init_tax_res_std_scen_res_unhedged
init_tax_res_net_std_scen_res
init_tax_res_net_std_scen_res_unhedged

9 SUBMODELS STRUCTURE

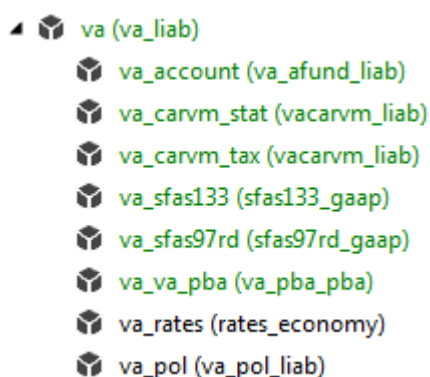
9.1 va submodel

The va submodel calculates the model point's contribution to the liability income statement and balance sheet items for variable annuity products.

9.1.1 *Seriatim versus portfolio runs*

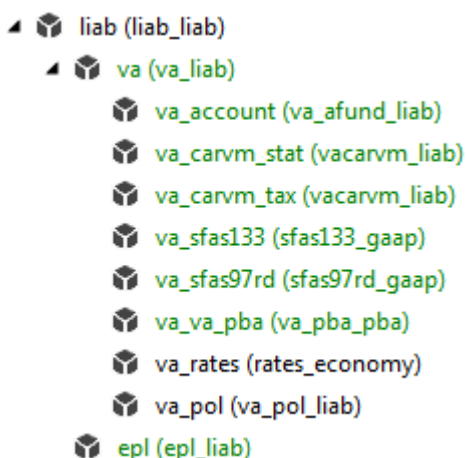
Seriatim projection

When va is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the va submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings.



Portfolio projection

For portfolio projections (which include business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the va submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the va submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the va submodel the same columns as for other product types. This makes the variable annuity product type interchangeable with other product types fitting into this business segment structure.

9.1.2 Variables

Input variables for specifying variable annuity products are contained in the va submodel.

9.1.3 Calculated array

A two dimensional calculated array, named `va_mp_vals`, is declared in the va submodel class. This array is used to store model point cash flows and balances for reference by the columns of the va submodel. The values of this array are calculated through a procedural process (discussed below).

The first dimension of the array represents a calculated value and the second dimension is the projection period. Appropriately named integer values are declared in the externs for each of the calculated values, making it much easier to write and understand the formulas which reference the array.

9.1.4 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when va is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the va submodel include:

startup

The 'startup' column is automatically executed by RiskAgility FM during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only va is the top model, (2) when va is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when va is run as the top model and when va is included in projections with segment as the top model. References to segment and liab are invalid when va is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) temporary tables are sized for DAC tax calculations and competitor rates, (3) projection flags are created and initialized, reporting groups are set up, (4) `va_account` elements are created, (5) resizes and initializes the `va_mp_vals` array, etc.

initialize

The 'initialize' column is automatically executed by RiskAgility FM during model initialization for each model point for each projection task loop, immediately after startup when va is the top model. When company is the top model, initialize is executed by calls from `seg->initialize` which calls `setup_liab_model_points` which executes `va->initialize` for each model point in the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when va is running as the top model. This includes executing the initialization of economic scenario rates, setting up the competitor rates. Other initializations are required in both cases.

mp_processing

The `mp_processing` column controls a **procedural projection** of the model point values for the period. This column **must** be included in each column that is referencing values from the `va_mp_vals` calculated array or the arrays of the policy and account level submodels. This ensures the order of calculation so that the calculated arrays are populated with the correct values when referenced.

proj_mp_one_period

The `proj_mp_one_period` external function executes the procedural process to project the model point one period.

This function is called from the `mp_processed` column for each period.

The order of processing of the model point calculations is as follows:

- Project the per policy values forward to the end of the period
- Project the decrements and survivorship for the period, calling proj_surv_one_period (discussed below)
- Calculate direct cash flows and separate account transfers occurring at the beginning of the period
- Calculate direct cash flows and separate account transfers occurring at the end of the period
- Calculate claims from GMxB riders at the end of the period
- Finalize end of period policy values

In general, each item is calculated by taking a per policy value from the va_pol submodel (discussed below) and multiplying it by an appropriate policies value and, if necessary, decrement rate.

proj_surv_one_period

The proj_surv_one_period external function executes a procedural process to calculate the experience decrements and end of period survivorship for the model point for one period.

This function is called from the proj_mp_one_period function for each period.

Other formulas

Other formulas in the va submodel are relatively straightforward. Most of the profit, balance sheet, and GAAP columns are only calculated when va is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

9.2 va_pol submodel

The va_pol submodel is a standalone instance of the va_pol_liab model class and contains all variables and formulas required to calculate per policy values for variable annuity model points.

The va_pol_liab model class is structured to execute a procedural process to determine the per policy values, which are temporarily stored in class variables for the period and into arrays for reference by columns for reporting.

9.2.1 Variables

There are no variables in the va_pol submodel.

9.2.2 Calculated variables

A number of variables are declared in the va_pol_liab model class to store the current period values of the calculated results. The calculated variables are always set to zero at the beginning of each projection period, and then calculated through the procedural process of the va_pol and va_account submodels (discussed below), such that the calculated variables representing sums over the accounts are aggregated into total amounts as the accounts are processed.

The results of the calculated variables are referenced in the functions of the va_account, va_pol, and va model classes. They are also referenced in the columns of the va_pol model class to provide the values for reporting.

The calculated variables used the same names of the va_pol_liab model class columns with an additional _cv appended to the name. For example, the class variable representing the result for the cash_val column is cash_val_cv.

9.2.3 Calculated array

A two dimensional calculated array, named va_pol_vals, is declared in the va_pol model class. This array is used to store the annualized gross premium amount required for policy level calculations. The values of this array are captured from the calculated variables within the procedural process of the va_pol submodel (discussed below).

The first dimension of the array represents a calculated value and the second dimension is the projection period. Appropriately named integer values are declared in the externs for each of the calculated values, making it much easier to write and understand the formulas which reference the array.

9.2.4 Formulas

Formulas for projecting per policy values for variable annuity model point are contained in the va_pol submodel.

startup

The 'startup' column is automatically executed by RiskAgility FM during model initialization for each model point for each projection task loop. The startup formulas for the policy level calculation allocate and initialize the arrays which collect the per policy values.

proj_pol_one_period

The proj_pol_one_period external function executes the procedural process to calculate per policy cash flow amount and balances for the period.

This function is called from the proj_mp_one_period external function of the va submodel.

initialize_pol

The initialize_pol external function initializes the per policy fund values, annualized premiums, and penalty free withdrawal entitlement amounts at the commencement period based on the amounts specified on the model point record.

This function is called from the proj_pol_one_period function at the commencement period.

initialize_gmxb_base

The initialize_gmxb_base external functions initialize the per policy GMxB base amounts at the commencement period based on the amounts specified on the model point record.

These functions are called from the proj_mp_one_period function at the commencement period. The specific function for the GMxB benefit is only called if that benefit is active.

proj_fund_b_bef_one_period

The proj_fund_b_bef_one_period external function executes the procedural process to calculate the per policy fund value at the beginning of the period, after the addition of premium and premium credits.

This function is called from the proj_pol_one_period function for each period after the commencement period.

proj_gmwb_base_b_bef_one_period

The proj_gmwb_base_b_bef_one_period external functions execute the procedural process to calculate the per policy GMWB base amounts at the beginning of the month after the application of stepup or rollup and the addition of premium. Both the account value and premium based GMWB base amounts are calculated with the final beginning of period base amount being calculated as the maximum.

This function is called from the proj_pol_one_period function for each period after the commencement period when the GMWB benefit is active.

proj_fund_bef_withdrl_one_period

The proj_fund_bef_withdrl_one_period external function executes the procedural process to calculate the per policy fund value at the end of the period, but before adjustments for penalty free withdrawals.

This function is called from the proj_pol_one_period function for each period after the commencement period.

proj_gmxb_base_bef_withdrl_one_period

The proj_gmxb_base_bef_withdrl_one_period external functions execute the procedural process to calculate the per policy GMxB base amounts at the end of the period, but before adjustments for penalty free withdrawals. For the GMWB benefit, this function determines the per policy maximum withdrawal amount.

These functions are called from the proj_pol_one_period function for each period after the commencement period. The specific function for the GMxB benefit is only called if that benefit is active.

calc_gmxb_itm_for_period

The calc_gmxb_itm_for_period external functions execute the procedural process to calculate the per policy GMxB in-the-moneyness factor for the period.

These functions are called from the proj_pol_one_period function for each period after the commencement period. The specific function for the GMxB benefit is only called if that benefit is active.

proj_fund_bef_dca_one_period

The proj_fund_bef_dca_one_period external function executes the procedural process to calculate the per policy fund value at the end of the period after adjustments for penalty free withdrawals but before adjustment for DCA transfers.

This function is called from the proj_pol_one_period function for each period after the commencement period.

proj_gmxb_base_bef_dca_one_period

The proj_gmxb_base_bef_dca_one_period external functions execute the procedural process to calculate the GMxB base amounts at the end of the period after adjustments for penalty free withdrawals. Both the account value and premium based amounts are calculated with the final end of period base amount being calculated as the maximum.

These functions are called from the proj_pol_one_period function for each period after the commencement period. The specific function for the GMxB benefit is only called if that benefit is active.

proj_fund_aft_dca_one_period

The proj_fund_aft_dca_one_period external function executes the procedural process to calculate the per policy fund value at the end of the period after adjustment for DCA transfers.

This function is called from the proj_pol_one_period function for each period after the commencement period.

calc_cash_val_for_period

The calc_cash_val_for_period external function executes the procedural process to calculate the per policy cash value at the end of the period.

This function is called from the proj_pol_one_period function for each period after the commencement period.

9.3 va_account submodel

The va_account submodel is a user defined array instance of the va_afund_liab model class and contains all variables and formulas required to calculate per policy separate and general account fund values for variable annuity model points.

9.3.1 Multiple instances

Each model point carries between one and eight instances of va_account distinguished by fund_id, which has values of:

- Fixed, for the fixed account fund value.
- DCA, for the dollar cost averaging fund value.

- Index0, for the first separate account variable fund value.
- Index1, for the second separate account variable fund value.
- Index2, for the third separate account variable fund value.
- Index3, for the fourth separate account variable fund value.
- Index4, for the fifth separate account variable fund value.
- Index5, for the sixth separate account variable fund value.

Elements of `va_account` are created during the startup processing of the model point. Conditions are evaluated in startup of `va` and the `va_account` user defined array is resized as required.

Specific instances of the `va_account` submodel are aggregated during the procedural processing by summing the results into the appropriate values of the `va_pol` submodel.

The `va_account` array element numbers range from 0 to 7. The actual integer value for each array depends on which accounts exist for a particular model point. Element numbers, beginning with element 0, are assigned to each account in the following order:

- DCA account – the first element number is assigned to the DCA account, if applicable.
- Fixed account – the next lowest element number is assigned to the general account, if applicable.
- Separate account variable funds – the separate account variable funds are then assigned to element numbers in order. Index0 would be assigned to the next lowest element number and Index5 would be assigned to the highest element number.

9.3.2 Variables

Variables for crediting rates, investment fees, and DCA allocations are contained in the `va_account` submodel.

9.3.3 Calculated variables

A number of variables are declared in the `va_afund_liab` model class to store the current period values of the calculated results. The calculated variables are always set to zero at the beginning of each projection period, and then calculated through the procedural process of the `va_account` submodel (discussed below). The results of the calculated variables are referenced in the functions of the `va_account` model classes. They are also referenced in the columns of the `va_afund_liab` model class to provide the values for reporting.

The calculated variables used the same names of the columns with an additional `_cv` appended to the name. For example, the class variable representing the result for the `av_e` column is `av_e_cv`.

9.3.4 Formulas

Formulas for projecting variable annuity model point fund values are contained in the `va_account` submodel.

startup

Since there are no instances of `va_account` defined through the model parameters, the initial size of the `va_account` array is zero. The startup column is called when the `va_account` array is resized using the `sm_va_account.resize(count)` statement from within the startup column of the `va` submodel.

initialize_account_val

The `initialize_account_val` external function initializes the per policy account values at the commencement period based on the amounts specified on the model point record, including adjusting for any initial account value drop for AG43 standard scenario calculations.

This function is called from the `initialize_pol` function of the `va_pol` submodel at the commencement period.

proj_account_b_bef_one_period

The proj_account_b_bef_one_period external function executes the procedural process to calculate the per policy account value at the beginning of the period, after the addition of premium and premium credits.

This function is called from the proj_fund_b_bef_one_period function of the va_pol submodel for each period after the commencement period.

proj_account_bef_withdrl_one_period

The proj_account_bef_withdrl_one_period external function executes the procedural process to calculate the per policy account value at the end of the period, but before adjustments for penalty free withdrawals.

This function is called from the proj_fund_bef_withdrl_one_period function of the va_pol submodel for each period after the commencement period.

proj_account_bef_dca_one_period

The proj_account_bef_dca_one_period external function executes the procedural process to calculate the per policy account value at the end of the period after adjustments for penalty free withdrawals but before adjustment for DCA transfers.

This function is called from the proj_fund_bef_dca_one_period function of the va_pol submodel for each period after the commencement period.

proj_account_dca_transfer_one_period

The proj_account_dca_transfer_one_period external function executes the procedural process to calculate the per policy DCA transfer for the period.

This function is called from the proj_fund_aft_dca_one_period function of the va_pol submodel for each period after the commencement period.

proj_account_aft_dca_one_period

The proj_account_aft_dca_one_period external function executes the procedural process to calculate the per policy account value at the end of the period after adjustment for DCA transfers.

This function is called from the proj_fund_aft_dca_one_period function of the va_pol submodel for each period after the commencement period.

9.4 va_carvm_stat submodel

The va_carvm_stat submodel is a user defined array instance of the vacarvm_liab model class and contains all variables and formulas required to calculate variable annuity statutory CARVM reserves.

Time 0 in the va_carvm_stat submodel corresponds to the policy year of the reserve period. Thus, 't' is policy year, unlike the va, va_pol, and va_account submodels where 't' represents months since the projection date. The values in va_carvm_stat are cleared before beginning the calculation for each reserve period.

9.4.1 Variables

Variables for statutory CARVM reserve parameters are contained in the va_carvm_stat submodel.

In addition to the variables discussed above, two variables in va_carvm_stat are hidden from the data entry but are critically important to the successful operation of the va_carvm_stat elements:

Model_name

The 'model_name' variable is an option choice variable with choices of STAT and TAX. This variable is used to retrieve the appropriate discount rate for statutory and tax reserves, since the vacarvm/liab model class is used for both statutory and tax CARVM calculations.

Reserve_period

The 'reserve_period' variable is the period for which the reserve is being calculated. It is set in va submodel at the beginning of each reserve calculation to the current projection period.

9.4.2 Formulas

Formulas for calculating the statutory CARVM reserves are included in the va_carvm_stat submodel. Formulas for va_carvm_stat are complex, and not typically customized.

Generally, va_carvm_stat formulas prospectively calculate the present value of the CARVM streams from the reserve period to either the specified CARVM duration or the omega age of the model point.

9.4.3 Multiple instances

Each model point carries up to two instances of va_carvm_stat, which calculate the CARVM reserves assuming:

- Policyholders do not take penalty free withdrawals
- Policyholders take full penalty free withdrawals.

9.5 va_carvm_tax submodel

The va_carvm_tax submodel is a user defined array instance of the vacarvm_liab model class and contains all variables and formulas required to calculate variable annuity tax CARVM reserves.

Time 0 in the va_carvm_tax submodel corresponds to the policy year of the reserve period. Thus, 't' is policy year, unlike the va, va_pol, and va_account submodels where 't' represents months since the projection date. The values in va_carvm_tax are cleared before beginning the calculation for each reserve period.

9.5.1 Variables

Variables for tax CARVM reserve parameters are contained in the va_carvm_tax submodel.

In addition to the variables discussed above, two variables in va_carvm_tax are hidden from the data entry but are critically important to the successful operation of the va_carvm_tax elements:

Model_name

The 'model_name' variable is an option choice variable with choices of STAT and TAX. This variable is used to retrieve the appropriate discount rate for statutory and tax reserves, since the vacarvm_liab model class is used for both statutory and tax CARVM calculations.

Reserve_period

The 'reserve_period' variable is the period for which the reserve is being calculated. It is set in va submodel at the beginning of each reserve calculation to the current projection period.

9.5.2 Formulas

Formulas for calculating the tax CARVM reserves are included in the va_carvm_tax submodel. Formulas for va_carvm_tax are complex, and not typically customized.

Generally, va_carvm_tax formulas prospectively calculate the present value of the CARVM streams from the reserve period to either the specified CARVM duration or the omega age of the model point.

9.5.3 Multiple instances

Each model point carries up to two instances of va_carvm_tax, which calculate the CARVM reserves assuming:

- Policyholders do not take penalty free withdrawals

- Policyholders take full penalty free withdrawals.

9.6 va_rates submodel

The va_rates submodel is identical to the rates submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

9.7 va_va_pba submodel

The va_va_pba submodel is identical to the va_pba submodel which applies for portfolio projections.

Please refer to the C05 PBA book.

9.8 va_sfas97rd submodel

The va_sfas97rd submodel is identical to the sfas97rd submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

9.9 va_sfas133 submodel

The va_sfas133 submodel is the sfas133 submodel which applies for SFAS 133 calculations.

Please refer to the C04 GAAP accounting book.

L08 FIXED INDEXED ANNUITY

This book discusses fixed indexed annuity products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

- Product – Benefits, premiums, benefit charges, surrender charge, and reinsurance
- Fund – Indexing definitions and rates, loads
- Valuation – Statutory and tax reserves
- Expense – Commission and expense rates, premium tax rate
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse and annuitization rates, withdrawals

Each of these input page types begins with 'FIA', for use in projections that combine multiple product types.

1.2 Data pages

Variables for each fixed indexed annuity model point are specified on the FIADData page. These variables are discussed in the Model point records chapter, below.

2 PRODUCT BENEFITS AND VALUES

The following explains the **basic features** for fixed indexed annuity products.

2.1 Maturity

The policy terminates at the maximum attained age specified for the product. When the specified maturity age is zero (default value), the maturity age is **the omega age of the experience mortality table**.

The fund value is paid at maturity.

2.2 Death benefits

2.2.1 Basic death benefit

Death benefits occur at the end of the month. The death benefit is either:

- Fund value – the greater of the contract value and the fund value.
- Cash value – the cash value (discussed below).

2.2.2 Death benefit paid versus in force

Death benefits **paid at the end of each month** are based on the fund value **before policyholder transactions are processed**. Thus the death benefit is calculated **using the fund value before penalty free withdrawals**.

Death benefits in force at the end of the month are based on the fund value at the end of the month, after penalty free withdrawals have been taken.

2.3 Annuity benefits

2.3.1 Current basis purchase rates

Current basis annualized annuity benefits are determined as the fund value less a load and policy fee divided by a current basis immediate annuity purchase rate factor.

The current basis annuity purchase rate factor uses current basis base mortality times a multiple and current basis interest. The current basis interest rate may be a specified fixed rate or a spread from a specified competitor rate, where the spread may vary each benefit year.

2.3.2 Guaranteed basis purchase rates

Guaranteed basis annualized annuity benefits are determined as the fund value divided by a guaranteed basis immediate annuity purchase rate factor.

The guaranteed basis annuity purchase rate factor uses guaranteed basis base mortality times a multiple and a guaranteed basis interest. The guaranteed basis interest rate is always a specified fixed rate.

2.3.3 Projection treatment

Policyholders who annuitize during the projection receive the fund value as a lump sum. There is no calculation of the annuity purchase rates or projection of future annuity benefits.

2.4 Withdrawal benefits

GMWB benefits are paid monthly. See the GMWB rider chapter, below.

2.4.1 Penalty free withdrawals

The policyholder can withdraw a limited portion of the fund value without surrendering the policy or incurring surrender charges. This is sometimes referred to as a partial surrender or cash withdrawal provision.

Penalty free withdrawals are processed at the end of each month, for policies persisting into the next month, after all other decrements are processed. Thus death and surrender benefits paid are based on the fund value before withdrawals, while death benefits and cash values in force at the end of the month are based on fund values after withdrawals.

2.4.2 Maximum penalty free withdrawal

The maximum annual penalty free withdrawal amount is determined at the beginning of each policy year. The maximum annual withdrawal can be a percent of either:

- **Fund value** – greater of the contract value and the fund value at the beginning of the first policy month, after all loads and charges are deducted.
- **Cumulative premiums less withdrawals** – cumulative premiums, less cumulative withdrawals taken, within the current indexing period. Specify on the model point record the initial (at the projection date) cumulative premiums* and cumulative withdrawals for the policies collected into the model point.
*Cumulative premiums are provided for customizing to include annual premiums; the standard features provide only for single premium products.

2.4.3 Penalty free withdrawal use

The penalty free withdrawals taken each policy year are an assumed percent of the maximum amount, times an assumed percent of policyholders taking withdrawals. The maximum withdrawal is calculated as a percent of the beginning fund value for the first month of the policy year. One twelfth of this amount is withdrawn at the end of each month during the policy year.

Withdrawals are policyholder transactions, which occur at the end of the month, after decrements have been processed.

2.5 Premiums

The standard features provide for a single premium at issue, with automatic contract renewal at the end of each indexing period. Specify on the model point record the total single premium for the policies collected into the model point.

2.6 Premium bonus

The fund value is increased at issue by a specified percent of the single premium paid. This applies only for the premium paid at issue; it does not apply for the single premium paid for each new indexing period when the policy automatically rolls over at the end of the previous indexing period.

A percent of this amount can be capitalized into SFAS 97 GAAP DAC. See the GAAP accounting book.

2.7 Product loads and charges

2.7.1 Administration fee

A specified amount per policy is deducted from the fund value at the beginning of each policy year.

2.8 Fund value

2.8.1 Premiums

The premium is credited to the fund value at the beginning of the month of issue.

The allocation of premiums between the fixed fund and the indexed fund value is discussed in the Fund values chapter below.

2.8.2 Loads and charges

Loads and charges are deducted from the fund value at the beginning of the month.

The allocation of loads and charges between the fixed fund and the indexed fund value is discussed in the Fund values chapter below.

2.8.3 Interest crediting

Interest is credited to the fund value at the end of each month. The total amount credited to the fund value is the sum of the amount credited to the fixed fund and the indexed fund.

The amount credited to the fixed fund value and the indexed fund value is discussed in the Fund values chapter below.

2.8.4 Penalty free withdrawals

Penalty free withdrawals are deducted from the fund value at the end of the month, after interest has been credited.

The deduction from the fund value is the amount withdrawn.

2.9 Cash value

2.9.1 Surrender benefits

The amount paid at surrender is the **greater of the contract value and the fund value**, each reduced by their respective surrender charge, but not less than the minimum cash value.

2.9.2 Surrender charge

The surrender charge is a percent, varying by policy year, of either:

- Greater of the contract value and the fund value, then less the remaining penalty free withdrawal entitlement
- Single premium paid.

2.9.3 Minimum cash value

During a number of months after issue, the **cash value can be floored at the cumulative premiums* less cumulative withdrawals**. This is sometimes called the 'return of principal floor'. When the specified number of months is zero (the default value), the minimum cash value does not apply. *Cumulative premiums are simply the single premium at the beginning of the current indexing period.

2.9.4 Minimum contract value

The 'minimum contract value' is the greater of the **contract value and the minimum cash value**. That is, the contract value, subject to the return of principal floor.

This value is used to determine the **option budget** amount and the **option strike price**, which are used to determine the **dynamic participation rate** and **the dynamic cap rate**.

3 FUND VALUES

3.1 Policy fund values

3.1.1 Current fund

A current fund **is projected for experience calculations**.

For in-force model points, specify on the model point record the initial (at the projection date) current fund value, which is the sum of the specified fixed fund value and indexed fund accounts values, for the policies collected into the model point.

3.1.2 Contract value

The contract value is **the guaranteed basis accumulation value**. Specify on the model point record the initial (at the projection date) contract value for the policies collected into the model point.

The previous month's contract value is adjusted by the following amounts which occur as of the beginning of each month: (1) **single premium at issue** times **the net consideration percent** (such as 90%), and (2) charges (when applicable) subtraction.

Interest crediting is calculated using this beginning of month accumulation value. **The crediting rate is the guaranteed crediting rate**. The effective annual guaranteed rate is converted to the equivalent monthly rate:

$$(1 + \text{annual rate})^{\frac{1}{12}} - 1$$

Finally, the contract value is decreased by **penalty free withdrawals** at the end of each month.

3.2 Fund value structure

3.2.1 Fund types

The current fund value can be split between the following fund types:

- **Fixed fund value** – The amount of current fund value earning interest based on the crediting strategy.
- **Indexed fund value** – The amount of the current fund value earning interest based on the growth of an index. The standard application provides for the indexed fund value to be split into up to 5 indexed fund accounts.

3.2.2 Input variables

Input variables for the fixed fund and indexed fund values are specified on the FIATFund input page.

The fixed fund input variables include the guaranteed crediting rate and crediting strategy.

The indexed fund account input variables include the index crediting definition, crediting (or indexing) months, the crediting index and look-back intervals, the crediting floor, participation rate, and initial dynamic cap rate and participation rate parameters. The variables commonly vary by a fund_id, identified as index0 through index4.

3.3 Allocations and transfers

3.3.1 Initial fund values

Specify the initial (at projection date) current fund value of the model point (separate by fixed fund value and each indexed fund value account) on the model point record.

- For in force model points, this is the initial (at the projection date) amount in each account.
- For new business model points, this is the percent allocation between accounts. The total of the percentages for all accounts should add to 1.0 (100%), otherwise the specified values are normalized so they do add to that result.

3.3.2 Premium allocation

The single premium is allocated using the proportion of fixed and indexed fund account fund values specified on the model point record.

3.3.3 Loads and charges allocation

Loads and charges which are a percent of fund value are calculated and deducted separately from each fund value.

Loads and charges that are not a percent of fund value, such as the administration fee, are calculated for the model point, then allocated in proportion to the fixed and indexed fund value accounts.

3.3.4 Withdrawals allocation

Penalty free withdrawals are allocated as a proportion of the account fund value to the total fund value.

3.3.5 Rebalancing allocation

Rebalancing between the fixed and indexed account fund values can be included in the projection.

The fund values can be rebalanced at the end of each calendar month, quarter, half year, or year based on either: (1) target allocation percentages or (2) initial fund value allocation.

Specify the allocation percentages on the model point record. The total of the percentages for all accounts should add to 1.0 (100%), otherwise the specified values are normalized so they do add to that result.

Transfers restore the fixed and indexed account fund values to have the desired allocations.

3.4 Fixed fund value crediting

3.4.1 Overview

The fixed fund value is increased for credited interest at **the end of each month**. The crediting rate for the fixed fund value is the larger of the **fixed account minimum crediting rate** and **the rate determined through the crediting strategy**.

Specify the **fixed account minimum crediting rate** on the model point record.

3.4.2 Initial crediting rate

Specify the initial (at projection date) crediting rate for the fixed fund value on the model point record.

The **initial fixed fund crediting rate** can apply for a specified number of months. Specify **the number of months** from issue the initial rate is guaranteed on the model point record.

3.4.3 Crediting strategy

The fixed fund crediting rate is determined using **an asset yield strategy**. The declared crediting rate is the asset yield of investment assets for the month less a specified asset yield spread.

The asset yield includes the effects of **IMR amortization, defaults, and borrowing costs**.

For new business projections, you may select to either enter the initial crediting rate as “User Input” or specify a “Proxy Yield” for the initial asset yield.

For liability-only projections, you specify the asset yield as an assumption for the projection. Independent model point processing can still apply, since the asset yield is not affected by liability results.

3.5 Indexed fund value crediting

3.5.1 Overview

The indexed fund value is **increased for credited interest at the end of each indexing period**. The amount credited is the sum of the crediting to each indexed fund account.

The indexed fund account is increased for credited interest based on the indexing crediting method. Index crediting applies **when the credited interest based on the indexing method** (discussed below) is greater than the **amount of interest credited to the indexed fund account** based on the index crediting floor.

The amount of the index crediting is subject to index participation rates and cap rates (discussed below).

3.5.2 Participation rates

The amount of the index crediting is **subject to index participation rates**.

Each indexed fund account has its own participation rate. The participation rate is declared at the start of each indexing period for each indexed fund account.

The participation rate can dynamically adjust to a new value at the rollover to each new indexing period, constrained to be between a specified minimum and maximum participation rate. When the participation rate adjusts dynamically, the indexing cap (discussed below) does not apply.

Specify on the model point record the initial (at the projection date) participation rate for in-force business, and for new business when the participation rate is not dynamic.

3.5.3 Cap rates

The amount of the index crediting may be subject to index cap rates.

Each indexed fund account has its own indexing cap rate. The cap rate is declared at the start of each indexing period for each indexed fund account.

The cap rate can dynamically adjust to a new value at the rollover to each new indexing period, constrained to be between a specified minimum and maximum cap rate. When the cap rate adjusts dynamically, the participation rate (discussed above) remains fixed at the initial participation rate specified on the model point record.

For each indexed fund account, specify on the model point record the initial (at the projection date) cap rate for in-force business. Also, specify on the model point, the number of months from issue the specified initial cap rate is guaranteed.

3.5.4 Indexing method

The indexing method is one of (option choice):

- **Point to point** – The index crediting rate is the participation rate times (1) the index value at the end of the ending period divided by (2) the index value at the beginning of the indexing period, (3) minus 1.0. The resulting amount is floored at zero and capped by the indexed fund account cap rate.
- **Moving average** – The index crediting rate is the participation rate times (1) the average of the index value over a specified look-back sampling interval during the indexing period, divided by (2) the index value at the beginning of the indexing period, (3) minus 1.0. The resulting amount is floored at zero and capped by the indexed fund account cap rate. This indexing method operates similarly to an Asian index option asset. When the look-back sampling interval is annual, and the indexing period is also annual, this is the same as the point to point indexing method.
- **Monthly sum cap** – The index crediting is the sum over each month in the indexing period of the lower of (1) the index value at the end of the month divided by the index value at the start of the month, minus 1.0, and (2) a monthly indexed fund account cap rate. The resulting amount is floored at zero and the participation rate has no effect for this method.

3.5.5 Scenario index

The index calculations for the indexing method are based on changes in a specified economic index. The index must be contained in the scenario rates file used for the projection. For example, the index could be the 'US' equity index contained in the scenario rates. Please refer to the C02 Economic scenarios book.

Note that past index values are produced from growth rates provided in the historic rates file specified for the projection. During the first year of the projection, index changes compare the index scenario rate against the historic rate at the beginning of the indexing period.

3.6 Indexed fund account calculations

3.6.1 Option budget

At the beginning of each indexing period, the option budget is determined as a rate using the formula $\max(0.0, (A) - (B))$, where:

- (A) an assumed earned rate over the indexing period based on the asset yield on invested assets (assets excluding any hedge assets) starting the month of the new indexing period, less a fixed spread,
- (B) the amount of guaranteed interest crediting to the indexed fund account over the indexing period based on the crediting floor rate, discounted back to the start of the indexing period at a risk free government rate.

For new business projection where you have selected to use a “Proxy Yield” approach for the initial option budget, the assumed earned rate is approximated using the yield rate on a specified quality grade and maturity term. Since the option budget is assumed to be at the beginning of the month, the scenario rates used are the rates at the end of the previous month.

3.6.2 *Dynamic participation rate calculations*

When the `crediting_dyn_lever` is “Dynamic Participation”, the indexing participation rate dynamically adjusts to a new value **at the creation of a new indexed fund account** or rollover of an existing indexed fund account to a new indexing period. This calculation is assumed to occur **at the beginning of the month**, using beginning of month values for asset yields and scenario rates.

At the start of each indexing period, the dynamic participation rate is: (1) **the option budget**, divided by (2) **the option cost**, constrained to be **between the specified minimum and maximum participation rates**.

Important! The application has been designed to produce warning messages in the run log when the available option budget is zero or the solved for participation rate has been constrained by the minimum or maximum rates. Since these are typically not errors in the calculations, the run log is not changed to Yellow when these warning messages are produced.

3.6.3 *Dynamic cap calculations*

When the `crediting_dyn_lever` is “Dynamic Cap”, the indexing cap rate dynamically adjusts to a new value at the creation of a new indexed fund account or rollover of an existing indexed fund account to a new indexing period. This calculation is assumed to occur at **the beginning of the month**, using beginning of month values for asset yields and scenario rates.

A cap rate is derived based on **an available option budget** and **the cost of a long call option to cover the upside equity participation**. The solved for cap rate can be constrained to be above a minimum cap rate and below a maximum cap rate. The final cap rate is rounded to the nearest 0.25%.

Important! The application has been designed to produce warning messages in the run log when the available option budget is zero or the solved for cap rate has been constrained by the minimum or maximum rates. Since these are typically not errors in the calculations, the run log is not changed to Yellow when these warning messages are produced.

For new business projections, you may select to provide the initial cap rate as “User Input” as opposed to having the application solve for an initial cap rate.

3.6.4 *Point to point indexing cap solving*

For the point to point indexing method, a call spread is **purchased to cover the upside equity participation**.

The long option cost is the **market value at the start of the indexing period of the long European call option to cover the upside equity participation**. The option period is the **indexing period** and the **option strike price** is the **current index value**, accumulated to the end of the indexing period at the index crediting floor rate.

When the **long option cost exceeds the available option budget**, an out-of-the money European call option is **sold to lower the option cost by forgoing proceeds in excess of the cap**. This short option has the same properties as the long option except for **the strike price**.

The strike price of the out-of-the money option is solved for so that **the proceeds from the short position plus the option budget are enough to offset the long option cost**:

$$\text{long optioncost} - \text{short optioncost} = \text{optionbudget}$$

The dynamic cap is then:

$$\text{cap} = (\text{solved strike value} / \text{at - the - money strike value}) - 1$$

As discussed above, the solved cap rate is then constrained to be between the minimum and maximum cap rates specified for the index and is rounded to the nearest 0.25%.

3.6.5 Moving average indexing cap solving

For the moving average indexing method, a call spread is purchased to cover the upside equity participation.

The process is the same as for the point to point method except the call option to purchase is a fixed strike (average price) Asian option.

3.6.6 Monthly sum cap indexing cap solving

For the monthly sum cap indexing method, a monthly sum cap option is purchased to cover the upside monthly sum cap crediting. The cap rate for the option is determined by finding the cap rate for a monthly sum cap option which can be funded by the option budget.

The option cost is the market value at the start of the indexing period of the monthly sum cap option, adjusting for the index crediting floor rate. A closed form Black Scholes based formula is used to determine the market value of the monthly sum cap option.

The solved cap rate is then constrained to be between the minimum and maximum cap rates specified for the index and is rounded to the nearest 0.25%.

4 GUARANTEED MINIMUM WITHDRAWAL BENEFIT

4.1 Guaranteed withdrawals

Fixed indexed annuities can include a guaranteed minimum withdrawal benefit (GMWB) rider, which provides for continuing withdrawals even when the fund value is exhausted. No withdrawals are taken until the end of the wait period.

The guaranteed withdrawals are penalty free. That is, the maximum penalty free withdrawal amount is automatically the greater of the normal maximum and the guaranteed withdrawal amount.

The withdrawals can be guaranteed either for a specified number of years, or for a lifetime annuity.

4.2 Base amount

The 'base amount' is the greater of: (1) fund value ratchet, and (2) accumulation of premiums. The benefit is reduced by withdrawals. After a specified maximum age, the base amount stops accumulating and only changes by withdrawals taken.

The base amount accumulates through the indexing periods. That is, the beginning base amount for a new indexing period is the ending base amount for the previous indexing period.

Fund value ratchet

Every 'n' policy years until a maximum attained age, the fund ratchet amount increases to the fund value* when it is higher than the current amount. The starting fund value ratchet is the premium paid at issue. *The contract value for FIA products has no effect on the GMWB rider.

For in-force policies, specify on the model point record the initial (at the projection date) GMWB fund value ratchet amount for the policies collected into the model point.

Premium accumulation

For the accumulation benefit (which is also called a roll up benefit), premiums less withdrawals are accumulated until the policyholder reaches a maximum age.

Premiums can accumulate by either:

- Return – The accumulated premium is the premium paid, without any interest.
- Roll up – Accumulated premiums increase by the **specified interest rate** either: (1) every policy anniversary, or (2) every month.
- Step up – Accumulated premiums increase by the **specified interest rate at the end of the waiting period**.

The roll up premium accumulation is **capped at a maximum multiple of the cost basis** (cumulative premiums less cumulative withdrawals).

For in-force policies, specify on the model point record the initial (at the projection date) GMWB premium accumulation amount for the policies collected into the model point.

4.3 Withdrawal amounts

4.3.1 Maximum annual withdrawal

The maximum annual withdrawal is a percent of either:

- **The single premium**
- **GMWB base at the end of the waiting period**
- **GMWB base, which is allowed to increase but never decrease.**

4.3.2 Dynamic penalty free withdrawal rate

GMWB withdrawal amounts can be dynamic. The dynamic penalty free withdrawal rate calculated **at the beginning of each policy year** is:

$$\min(1.0, \text{mult} \times \text{ITM}\%)$$

where:

ITM% is **the GMWB in-the-moneyness percent**, but not less than zero.

The monthly penalty free withdrawal rate is:

$$\min(1.0, \text{GMWBmax withdrawal amount} \times \text{dynamic rate} / \text{fund value})$$

where:

GMWB max withdrawal amount is the monthly maximum withdrawal amount

Dynamic rate is the dynamic penalty free withdrawal rate

Fund value is the **end of month fund value before claims**.

4.3.3 Penalty free withdrawals

The guaranteed withdrawals are penalty free. That is, the maximum penalty free withdrawal amount is automatically **the greater of the normal maximum and the guaranteed withdrawal amount**.

4.4 Charges

The GMWB charge is deducted from the fund value* at the **beginning of each month**. *The GMWB charge does not affect the contract value.

The charge is a percent of either:

- Fund value – the fund value (not compared to the contract value)
- GMWB base amount
- Greater of the fund value and the GMWB base amount.

One twelfth of this amount is deducted from the fund values at the beginning of the month. If the GMWB base is less than the fund value and the percent GMWB charge applies to the GMWB base only, the charge applies to the beginning of month fund value after the administration fee has been deducted. Otherwise, the charge applies to the beginning of month fund value after all loads have been deducted. The GMWB charge is only collected until the GMWB maximum attained age.

For in-force policies, specify on the model point record the initial (at the projection date) accumulated GMWB charges for AG 39 for the policies collected into the model point.

4.5 Cost

The GMWB cost is the guaranteed withdrawal present value at the time the fund value is exhausted. The present value is taken over either: (1) specified number of years, or (2) the lifetime of the policyholder. Specify on the Product input page the discount rate and mortality table used to calculate the present value of withdrawal benefits.

5 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves. PBA reserves and capital are discussed in the PBA book.

5.1 CARVM reserve

Specify CARVM valuation variables on the FIValuation input page.

5.1.1 Percent of fund approximation

The CARVM reserve can be approximated using percentages of the fund value or cash value. This may be appropriate to improve run time by eliminating these compute-intensive calculations.

The effect of the market value adjustment (MVA) and bailout provisions are discussed separately, below.

5.1.2 Actuarial guideline 33

Fixed indexed annuity reserves can be calculated using the Commissioners' Annuity Reserve Valuation method (CARVM) and Actuarial Guideline XXXIII. The CARVM integrated reserve for the policy is the maximum of the integrated reserve for the cash value stream reserve, and when included, the annuitization stream.

Cash value streams

Cash value streams are always included in the CARVM calculations. The present value of guaranteed benefits is calculated using a different valuation interest rate for each of the following benefits:

- Death benefits,
- Nursing home benefits, when included
- Penalty free withdrawals, including GMWB withdrawals
- Full cash value surrenders, and
- GMWB annuitization benefits.

When **GMWB benefits are not included**, there are two projections of the cash value stream guaranteed benefits. One assuming **maximum penalty free withdrawals** are taken and one assuming **no penalty free withdrawals are taken**. The maximum integrated reserve for the cash value stream is the greater of the integrated reserve for the two cash value streams.

When GMWB benefits are included, there may be any number of GMWB exercise projections of the cash value stream guaranteed benefits, based on whether the GMWB has been exercised and the specified exercise periods for the CARVM projection. For example, if the GMWB benefit has not been exercised, you could project the cash value stream for three GMWB exercise point tests, one exercising 1 year after the waiting period, one exercising 5 years after the waiting period, and one exercising 10 years after the waiting period. There will also always be a projection assuming the GMWB is not exercised. If the GMWB has already been exercised at the valuation date, there is just a single projection of the cash value stream, including the GMWB benefits. The maximum integrated reserve is now the greatest of the integrated reserve from all of the GMWB exercise projections.

Annuitization stream

Annuitization streams may be included in the CARVM calculations. The present value of guaranteed benefits is calculated using a different valuation interest rate for each of the following benefits:

- Death benefits,
- Nursing home benefits, when included,
- Penalty free withdrawals, including GMWB withdrawals
- Full annuitization benefit, and
- GMWB annuitization benefits.

For the annuitization stream guaranteed benefits, there are potentially many different streams projected with different types of annuity benefits. Any number of CARVM annuitization tests can be defined, where each test represents a type of annuity to purchase and then any number of the defined tests can be applied. For example, you could project the annuitization stream for three annuitization tests, one purchasing life only annuities, one purchasing 5 year certain and life annuities, and one purchasing 10 year certain and life annuities.

And, within each annuity benefit type stream projection, there are two streams projected when GMWB benefits are not included. One assuming maximum penalty free withdrawals are taken and one assuming no penalty free withdrawals are taken. The maximum integrated reserve for the annuitization test is the greater of the integrated reserve from these two projections.

However, similar to the cash value stream processing, when GMWB benefits are included, there may be any number of GMWB exercise projections of the annuitization value stream guaranteed benefits and the maximum integrated reserve for the annuitization test is the greatest of the integrated reserve from all the GMWB exercise projections.

Finally, the integrated reserve for the annuitization stream is the maximum of the integrated reserve for each annuitization test.

During the CARVM projection, the full annuitization benefit is determined as the benefit for an annual payment immediate annuity for a specified certain years and a single life (when a life annuity type is used) purchased by the fund value after penalty free withdrawals. The purchase rate for the annuity may be determined using only a guaranteed basis annuity purchase rate or both a current basis annuity purchase rate and a guaranteed basis annuity purchase rate. When the current basis annuity purchase rate is included, the annuity benefit at the valuation date is the greater of the current basis annuity benefit and the guaranteed basis annuity benefit. Annuity benefits for CARVM projection periods after the valuation date are always based on the guaranteed basis purchase rate. The current and guaranteed annuity purchase rate bases are discussed in section 2.3 above.

Since contracts may provide, as contractual guarantees, the use of preferential purchase rates at annuitization to those listed in the contract, you may specify a percentage of the valuation date fund value as a floor to the annuitization stream integrated reserve to account for this additional guarantee. For example, when such provision applies, you should set the `stat_carvm_annuitzn_min_fund_val_pct` and `tax_carvm_annuitzn_min_fund_val_pct` to the appropriate floor percentage, such as 93%.

GMWB benefits treatment

When GMWB benefits are included, the GMWB base amounts are determined based on the GMWB provisions. The GMWB benefits begin at the earlier of the year following the exercise point or when the fund value falls below zero. Of course, if the GMWB benefit has already been exercised at the valuation date, the GMWB benefit is continued within the CARVM projection using the GMWB withdrawal benefit as of the valuation date.

When the fund value is positive in the CARVM projection, the GMWB withdrawals are treated as penalty free withdrawals.

However, should the fund value falls below zero in the CARVM projection, a GMWB annuitization benefit is calculated at that time equal to the benefit for an annual payment finite or life annuity, based on the GMWB provisions. The valuation basis for the GMWB annuity benefit is the valuation rate for a non-elective benefit and the GMWB mortality basis.

5.1.3 Actuarial guideline 35

Statutory CARVM calculations include AG 35, utilizing the MVRM Using Black-Scholes Projection Method approach.

The accumulation rate for the fund value in each CARVM projection period is the option cost for a full hedging call option for the remaining term to expiry divided by the fund value at the valuation date, accumulated at the risk free rate to the end of the index crediting period.

The option cost of the full hedging call option is recalculated at each CARVM projection period, taking into account changes in the index dividend yield, index volatility, and risk free forward rates.

5.1.4 CARVM valuation basis

Interest

CARVM valuation interest may vary by the cash settlement definition, guarantee duration definition, plan type, and guarantee duration. Separate plan type definitions are provided for cash value elective benefits, annuitization elective benefits, and non-elective benefits.

The Global folder includes an external source workbook containing the maximum standard valuation law valuation rates. The `DefAnnStatValn` and `DefAnnTaxValn` ranges in the Valuation Interest Rates Excel workbook include rates varying by valuation method (Issue Year, Change in Fund), cash settlement definition (With Cash Settlements, Without Cash Settlements), guarantee duration definition (With Guarantees, Without Guarantees, With or Without Guarantees), plan types (Plan A, Plan B, Plan C) and guarantee durations ranges of 0-5, 6-10, 11-20 and 21+.

For the death benefits, nursing home benefits, penalty free withdrawals (including GMWB withdrawals), and full cash surrender benefits, the guarantee duration is determined as the number of years the guaranteed credited rate exceeds the valuation interest rate for life insurance policies with a guarantee duration in excess of 20 years, as specified in the `stat_carvm_life_int_rate` and `tax_carvm_life_int_rate` variables.

For the annuity benefits, the guarantee duration is determined as the number of years since issue when the annuitization occurs.

For the GWMB annuitization benefit, the guarantee duration is determined as the number of years since issue when the GMWB annuitization benefit occurs.

Mortality

Reserve mortality is a base rate, times a multiple. Separate reserve mortality is specified for death benefits, GMWB annuitization benefits, and annuity benefits.

Mortality variables are read for both males and females when blended mortality applies.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys mort_issue_age and mort_gender instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age. Using the hidden variable ensures the attained age does not advance with each yearly loop.

Nursing home

Nursing home incidence is a base rate.

The value is read twice when the gender is blended – first for male rates and then again for female rates.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys mort_issue_age and mort_gender instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age. Using the hidden variable ensures the attained age does not advance with each yearly loop.

5.1.5 CARVM valuation method

The CARVM valuation method is always the 'Issue Year' method.

Under the CARVM issue year valuation method, the CARVM guaranteed benefit streams are developed from a projection of a single fund value bucket. Benefits are discounted using the appropriate valuation interest rate for the policy issue year, cash settlement definition, guarantee duration definition, plan type, and guarantee duration from the issue year method valuation interest rate table.

The reserve for the benefit stream is the maximum discounted value of the issue year fund bucket.

5.1.6 Continuous CARVM

The continuous CARVM method is the same as for CARVM, except that the surrender charge is the lower of the surrender charge for the current month and the next month.

For example, when the surrender charge decreases from 7% in the first policy year to 6% in the second policy year, the continuous CARVM reserve at the end of the first policy year is based on the 6% surrender charge.

5.1.7 CARVM crediting rate

Future policy benefits are projected using the current crediting rate until the end of the current declaration period, then discounting using the unadjusted reserve interest rate. This produces an 'excess interest reserve'. After the current declaration period, benefits are projected using the guaranteed crediting rate.

5.1.8 CARVM calculation frequency

For most model points, CARVM reserves are calculated as of the projection date, the first month after issue, at the end of each policy year, and at the end of each calendar year. However, for model points which use the monthly sum cap crediting method, CARVM reserves are calculated monthly since the calculation of the option costs required for the AG 35 calculations are very sensitive to the recent past index values. Approximations are used between these calculation dates in order to minimize these compute-intensive calculations.

For the projection months when CARVM reserves are not calculated, the reserve for each model point is approximated by multiplying the previous month's reserve by the ratio of the current month's fund value to the previous month's fund value.

The model point record includes the initial (at the projection date) statutory and tax CARVM reserve amounts. These values are only for use by nested projections. Ignore these variables when preparing model point records for your projection, since the CARVM valuation is calculated at the projection date.

5.1.9 Terminating CARVM projections

CARVM reserve calculations consider benefits projected only through a specified number of years from each valuation date.

5.2 Reinsurance reserve

Reserves are calculated both gross of (ignoring) reinsurance and net of reinsurance.

The reinsurance reserve is the ceded reserve, calculated as the difference between the gross reserve and the net reserve.

5.3 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record set the tax reserve equal to the statutory reserve.

Tax reserves are calculated by the same methods, using the tax reserve valuation basis and method.

For policies issued after 1987, the crediting interest calculated in the CARVM projection is capped as required to disallow a deduction for an 'excess interest' reserve. The capping process adjusts the crediting so that any crediting rate guaranteed above the tax valuation rate is only guaranteed to the end of the current taxable year.

The tax reserve is not greater than the statutory reserve.

6 ACTUARIAL ASSUMPTIONS

6.1 Commissions

6.1.1 Premium commission

A percent of the single premium is paid as a commission at issue.

6.1.2 Fund value commission

One twelfth of the annual percent of fund value at the end of the previous month is paid at the beginning of each month.

6.1.3 Commission overrides and agent terminations

The specified commission rates are the average commission rates that are assumed to apply, taking into account average writing agent commissions, commission overrides, and the effect of agent terminations.

6.1.4 Commission chargebacks

A portion of the premium commission can be charged back in the event of lapse or surrender during the first three years after issue. The chargeback percent varies by months since issue.

Other types of commissions are not subject to chargeback.

The commission subject to chargeback is an amount per policy in force. For in-force model points, specify on the model point record the initial (at the projection date) commissions subject to chargeback for the policies collected into the model point.

6.2 Expenses

6.2.1 *Expense adjustment multiple*

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

6.2.2 *Acquisition expenses*

Acquisition expenses incurred at issue can include:

- Amount per policy
- Percent of total first year premium.

6.2.3 *Maintenance expenses*

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Maintenance expenses include:

- Amount per policy
- Percent of annualized premium
- Percent of the total fund value at the end of the previous month.

6.2.4 *Inflation*

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

6.2.5 *Premium tax*

Premium taxes are reported with expenses. The tax is a percent of each premium payment.

Inflation and the expense adjustment multiple do not apply to premium tax.

6.2.6 *Fixed expenses*

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

6.3 Decrement processing

Death claims are based on the contract and fund value at the end of month before decrements and transactions.

Surrenders and annuitizations are based on the contract and fund value after death claims. Thus lapse and annuitization rates are applied to the policies in force after the mortality decrement. The annuitization rate is reduced if the total lapse and annuitization rate would exceed 100%. The sum of these two rates is the voluntary policy termination rate.

Penalty free withdrawals are then processed for the policies continuing to the next month.

6.4 Mortality

6.4.1 *Input variables*

'Mortality variables' for annuity products include mortality base, multiple, and addition rates, and mortality improvement. These variables are read twice when blended mortality applies – first for male rates and then again for female rates. See the Blended mortality topic in the Mortality section, below.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys `mort_issue_age` and `mort_gender` instead of using these insured characteristics values directly.

Hidden variable `mort_attained_age` is also required, since it is based on `mort_issue_age`. Using the hidden variable ensures, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

6.4.2 *Mortality adjustment multiple*

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

6.4.3 *Base mortality*

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

6.4.4 *Mortality improvement*

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$. When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

6.4.5 *Mortality table adjustments*

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

6.4.6 *Blended mortality*

The model point can be a 'blend' between males and females issued at the specified issue age. This occurs when the gender variable is Blended. Specify on the model point record the proportion of issues that are males.

The male and female mortality rates are each determined from the mortality table. The blended mortality rate is:

$$\text{proportion of males} \times \text{male mortality rate} + \text{proportion of females} \times \text{female mortality rate}$$

6.4.7 *Monthly mortality*

Deaths occur at the end of each month. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

6.5 Surrenders

6.5.1 *Lapse adjustment multiple*

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date.

6.5.2 *Policy year lapse rates*

The base lapse rate is adjusted by a multiple.

6.5.3 *Dynamic lapses*

Lapse rates can be dynamically adjusted according to policyholder behavior related to the competitive position as economic rates change.

Maximum and minimum lapse rate variables in the dynamic lapse features affect the excess lapse calculations only. They do not limit the specified lapse rates. For example, the maximum lapse rate variable can reduce the excess lapse rate to zero, but cannot reduce the total lapse rate to be less than the specified lapse rate adjusted by the calendar date multiple.

Dynamic lapses are calculated as the sum of equity and non equity components:

Non equity component

The non equity component is an exponential excess lapse formula, using threshold, multiple, exponent, and surrender charge multiple variables:

$$\begin{aligned} &\text{When competitor rate} > \text{option budget} + \text{threshold} \\ &\text{difference} = (\text{competitor rate} - \text{option budget} - \text{threshold}) \\ &\text{surrender charge ratio} = 1 - \left(\frac{\text{cash value}}{\text{fund value}} \right) \\ &\text{non equity component} = \text{multiple} \times (\text{difference}^{\text{exponent}}) \\ &\quad \times (1 - \text{surrender charge multiple} \times \text{SC ratio}) \end{aligned}$$

where

Option budget is the asset yield less a spread. See the Option budget topic in the Fund values chapter, below.

Equity component

The equity component is:

$$\text{equity component} = \text{multiple} \times (\text{competitor rate} - \text{indexing rate})$$

where

Indexing rate is the indexing increase at the end of the prior policy year (may be zero), divided by the fund value at the beginning of the last month of the prior policy year before monthly loads and charges are deducted. In the first year of each indexing period, this calculation applies for the amounts in the last policy year of the previous indexing period. For the first year after issue, the indexing rate is zero.

6.5.4 Monthly lapses

Lapses and surrenders occur at the end of each month, after mortality decrements, and before policyholder transactions have been processed. Each of the above lapse rate adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/12)}$$

The monthly lapse rate is applied to the fund value at the end of the month, before penalty free withdrawals are processed.

6.6 Annuitizations

The fund value is paid out as a lump sum at annuitization. The MVA adjustment does not apply.

Annuitizations occur at the end of each month. The annual annuitization rate is reduced when the total of the annual lapse and annuitization rates would exceed 100%.

Finally, the annual rate is converted into the monthly annuitization rate:

$$1 - (1 - \text{annual annuitization rate})^{(1/12)}$$

6.7 Risk based capital risk definition

The risk based capital (RBC) C3 component percent varies between 'low', 'medium', and 'high' risk annuity products. Specify on the Company input page this risk definition for each product, then vary the RBC C3 percent between Low, Medium, and High risks.

7 REINSURANCE

7.1 Base plan coinsurance

Coinurance treaties cede a portion of the policy, including transferring the ceded reserve to be invested by the reinsurer. Coinsurance coverage can apply to the base benefits, that is, excluding the GMWB rider.

Specify the coinsurance percent on the model point record. Base plan coinsurance does not apply when this percent is zero.

The reinsurance premium is the single premium at the start of each indexing period, times the coinsurance percent.

The reinsurance benefit paid to the ceding company each month is the coinsurance percent times each type of benefit incurred by the ceding company.

The reinsurer reimburses the ceding company for a portion of the ceding company's expenses. The expenses reimbursed are the coinsurance percent times:

- Acquisition expenses
- Maintenance expenses
- Commissions net of commission chargebacks.

The reinsurer reimburses the coinsurance percent times the premium taxes paid by the ceding company.

The ceding company receives a reinsurance reserve credit for coinsurance treaties equal to the coinsurance percent times the formulaic statutory reserve.

7.2 Rider coinsurance

Reinsurance covering the GMWB rider is a coinsurance arrangement. The reinsurance benefits are the guaranteed benefits in excess of the base benefits, times the reinsurance percent specified for the product.

Reinsurance premiums are the monthly reinsurance GMWB charge times the ceded percent. The monthly reinsurance GMWB charge is one twelfth of the annual reinsurance GMWB charge rate, times the GMWB charge base. The GMWB charge base is either the fund value or the GMWB base, after the administration fee has been deducted.

7.3 Reinsurance administration expenses

Reinsurance administration expenses are incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include an amount per policy.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Coinsurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for coinsurance.

8 MODEL POINT RECORDS

8.1 Model point record layout

The model point record layout for fixed indexed annuities is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	up to 25 characters (assigns model point to report group)
B	seg_id	segment ID

Excel Column	Variable	Value
C	mp_id	up to 20 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date	yyyy/mm/dd or mm/dd/yyyy
F	plan_code	up to 20 characters (link to assumptions by plan)
G	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
H	tax_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
I	issue_age	positive integer
J	gender	Male, Female, Blended
K	male_sex_pct	male percent (1.0 for 100%). This variable only applies when gender is Blended.
L	single_prem	single premium (at the start of the current indexing period) for mp
M	reins_pct	base plan ceded coinsurance percent
N	init_policies	initial policies for mp (required)
O	init_contract_val	initial contract value for mp
P	crediting_rate_fixed	fixed crediting rate (when the fund value crediting rate is specified by model point)
Q	crediting_rate_min	minimum crediting rate (when the fund value crediting rate is a fixed rate based on the proxy yield at issue)
R	host_accrual_rate	host accrual rate (see the SFAS 133 liability chapter of the GAAP accounting book)
S	init_host_val	initial host contract value for mp (see the SFAS 133 liability chapter of the GAAP accounting book)
T	init_hedge_budget	initial option budget for mp
U	gmwb_ind	1 (GMWB is included), 0 (excluded)
V	gmwb_exercise_yrs	GMWB waiting period years for the experience projection

Excel Column	Variable	Value
W	init_gmwb_fund_val	initial GMWB fund value ratchet amount for mp
X	init_gmwb_prem	initial GMWB premium accumulation amount for mp
Y	init_gmwb_max_wdl_amt	initial maximum annual withdrawal amount for mp, when the policy is already taking withdrawals
Z	gmwb_sf133_req_prof_chg	GMWB SFAS133 required profit charge
AA	init_withdrl_cumul	initial cumulative withdrawals (during the current indexing period) for mp
AB	init_pfwd_cumulr	initial cumulative penalty free withdrawals
AC	init_pfwd_entitlement	initial penalty free withdrawal entitlement for mp; this variable is only used for nested projections, where the write-state-of-world function updates the status for the next time step
AD	init_fund_val_fixed	initial fund value for fixed account
AE	init_fund_val_index0	initial fund value for index account 0
AF	init_fund_val_index1	initial fund value for index account 1
AG	init_fund_val_index2	initial fund value for index account 2
AH	init_fund_val_index3	initial fund value for index account 3
AI	init_fund_val_index4	initial fund value for index account 4
AJ	init_rebal_pct_fixed	Rebalance percentage for fixed account
AK	init_rebal_pct_index0	Rebalance percentage for index account0
AL	init_rebal_pct_index1	Rebalance percentage for index account1
AM	init_rebal_pct_index2	Rebalance percentage for index account2
AN	init_rebal_pct_index3	Rebalance percentage for index account3
AO	init_rebal_pct_index4	Rebalance percentage for index account4
AP	crediting_rate_guar_mths_fixed	Crediting rate guaranteed months - fixed:
AQ	crediting_rate_guar_mths_index0	Crediting rate guaranteed months - index 0
AR	crediting_rate_guar_mths_index1	Crediting rate guaranteed months – index1

Excel Column	Variable	Value
AS	crediting_rate_guar_mths_index2	Crediting rate guaranteed months – index2
AT	crediting_rate_guar_mths_index3	Crediting rate guaranteed months – index3
AU	crediting_rate_guar_mths_index4	Crediting rate guaranteed months – index4
AV	init_part_rate_index0	Initial participation rate - index0
AW	init_part_rate_index1	Initial participation rate – index1
AX	init_part_rate_index2	Initial participation rate - index2
AY	init_part_rate_index3	Initial participation rate – index3
AZ	init_part_rate_index4	Initial participation rate – index4
BA	init_cap_rate_index0	Initial cap rate - index0
BB	init_cap_rate_index1	Initial cap rate – index1
BC	init_cap_rate_index2	Initial cap rate – index2
BD	init_cap_rate_index3	Initial cap rate – index3
BE	init_cap_rate_index4	Initial cap rate – index4
BF	crediting_cap_min_index0	Minimum cap - index0
BG	crediting_cap_min_index1	Minimum cap – index1
BH	crediting_cap_min_index2	Minimum cap – index2
BI	crediting_cap_min_index3	Minimum cap – index3
BJ	crediting_cap_min_index4	Minimum cap – index4
BK	crediting_cap_max_index0	Maximum cap - index0
BL	crediting_cap_max_index1	Maximum cap – index1
BM	crediting_cap_max_index2	Maximum cap – index2
BN	crediting_cap_max_index3	Maximum cap – index3
BO	crediting_cap_max_index4	Maximum cap – index4
BP	init_comm_subject_to_chargeback	initial commission subject to chargeback for mp

Excel Column	Variable	Value
BQ	init_stat_res_carvm_gmwb	initial statutory GMWB CARVM reserve for mp; this variable is only used for nested projections, where the write-state-of-world function updates the status for the next time step
BR	init_stat_res_carvm_base	initial statutory base CARVM reserve for mp; this variable is only used for nested projections, where the write-state-of-world function updates the status for the next time step
BS	init_tax_res_carvm_gmwb	initial tax GMWB CARVM reserve mp; this variable is only used for nested projections, where the write-state-of-world function updates the status for the next time step
BT	init_tax_res_carvm_base	initial tax base CARVM reserve for mp; this variable is only used for nested projections, where the write-state-of-world function updates the status for the next time step
BU	init_option_value_net	Initial net option value required for CARVM calculations.

8.2 Required model point variables

8.2.1 Always required

The following model point variables are required in all cases:

Model point identification: group, seg_id, and mp_id

Insured characteristics: issue_age and gender

Policy specifications: issue_date, plan_code, stat_valn_code, and tax_valn_code

Initial policies: init_policies

Single premium: single_prem is required for new business model points, and for in-force model points when the surrender charge is a percent of the single premium.

8.2.2 In-force model point values

In-force model points are driven by all the 'initial values' variables, which specify the aggregate model point values as of the projection date.

New business model points (model points with issue date after the projection date), however, ignore these variables. The exception is that the initial policies must be specified, since this is the number of policies to be issued.

The following table shows when each in-force values variable is required for in-force model points:

Variable	Required When
single_prem	for new business model points, and for in-force model points when the surrender charge is a percent of the single premium
reins_pct	for model points which include coinsurance of the base plan
init_policies	always required
init_contract_val	always required
init_part_rate	when the crediting rate definition is point to point or moving average
init_fia_cap	when the crediting rate is adjusted using a dynamic cap rate
fia_cap_min	always required
init_host_val	always required for GAAP calculations
init_hedge_budget	always required (except when crediting rate is adjusted via a dynamic cap rate during a non GAAP projection)
gmwb_ind	model points with GMWB
gmwb_exercise_years	
init_gmwb_fund_val	
init_gmwb_prem	
init_gmwb_max_wdl_amt	
init_withdrl_cumul	model points with penalty free withdrawals based on cumulative premiums less withdrawals, or still subject to the return of principal cash value floor
init_pfwd_cumul	model points with penalty free withdrawals
init_pfwd_entitlement	model points with penalty free withdrawals
init_fund_val_fixed(index*)	Always required for available subaccount
init_rebal_pct_fixed(index*)	When rebalancing is included
crediting_rate_guar_mths_fixed(index*)	Always required for available subaccount
init_part_rate_index*	Always required for available subaccount
init_cap_rate_index*	Always required for available subaccount

Variable	Required When
crediting_cap_min_index*	Always required for available subaccount
crediting_cap_max_index*	Always required for available subaccount
init_comm_subject_to_char geback	model points still subject to commission chargeback

8.2.3 Variables to ignore

The following variables are needed only for nested projections, and should be ignored. These values must be carried forward to the next time step in order to reproduce the results from standard projections:

Variable
init_pfwd_entitlement
init_stat_res_carvm
init_tax_res_carvm

9 SUBMODELS STRUCTURE

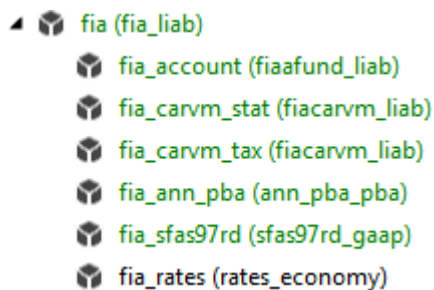
9.1 fia submodel

The fia submodel calculates the model point's contribution to the liability income statement and balance sheet items for fixed indexed annuity products.

9.1.1 Seriatim versus portfolio runs

Seriatim projection

When fia is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the fia submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings.



Portfolio projection

For portfolio projections (which include business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the fia submodel is a data driven array sized to the number of model points included in the model point data file. In

this configuration, the fia submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the fia submodel the same columns as for other product types. This makes the fixed indexed annuity product type interchangeable with other product types fitting into this business segment structure.

9.1.2 Variables

Variables for fixed indexed annuity product parameters are contained in the fia submodel.

9.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when fia is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the fia submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only fia is the top model, (2) when fia is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when fia is run as the top model and when fia is included in projections with segment as the top model. References to company, segment and liab are invalid when fia is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) temporary tables are sized for DAC tax calculations and competitor rates, (3) projection flags are created and initialized, reporting groups are set up, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each projection task loop, immediately after startup when fia is the top model. When segment is the top model, initialize is executed by calls from seg->initialize which executes fia->initialize for each model point in the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when fia is running as the top model. This includes executing the initialization of economic scenario rates, setting up the competitor rates.

Other formulas

Other formulas in the fia submodel are relatively straightforward. Most of the profit, balance sheet, and GAAP columns are only calculated when fia is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

9.2 fia_account submodel

The fia_account submodel is a user defined array instance of the fia_afund_liab model class and contains all variables and formulas required to calculate per policy fixed and indexed account fund values for fixed indexed annuity model points.

9.2.1 Multiple instances

Each model point carries between one and six instances of fia_account distinguished by fund_id, which has values of:

- Fixed, for the fixed account fund value.
- Index0, for the first indexed account fund value.
- Index1, for the second indexed account fund value.
- Index2, for the third indexed account fund value.
- Index3, for the fourth indexed account fund value.
- Index4, for the fifth indexed account fund value.

Elements of fia_account are created during the startup processing of the model point. Conditions are evaluated in startup of fia and the fia_account user defined array is resized as required.

Specific instances of the fia_account submodel are aggregated using the 'sum_over_funds' function in the fia submodel.

The fia_account array element numbers range from 0 to 5. The actual integer value for each array depends on which accounts exist for a particular model point. Element numbers, beginning with element 0, are assigned to each account in the following order:

- Fixed account – the first element number is assigned to the fixed account, if applicable.
- Indexed account – the separate accounts are then assigned to element numbers in order. Index0 would be assigned to the next lowest element number and Index4 would be assigned to the highest element number.

9.2.2 Variables

Variables for crediting rates and lookback sampling intervals are contained in the fia_account submodel.

9.2.3 Formulas

Formulas for projecting variable annuity model point fund values are contained in the fia_account submodel.

Startup

Since there are no instances of fia_account defined through the model parameters, the initial size of the fia_account array is zero. The startup column is called when the fia_account array is resized using the sm_fia_account.resize (count) statement from within the startup column of the fia submodel.

9.3 fia_carvm_stat submodel

The fia_carvm_stat submodel is a user defined array instance of the fiacarvm_liab model class and contains all variables and formulas for required to calculate fixed indexed annuity statutory CARVM reserves.

Time 0 in the fia_carvm_stat submodel corresponds to the policy year of the reserve period. Thus, 't' is policy year, unlike the fia submodel where 't' represents months since the projection date. The values in fia_carvm_stat are cleared before beginning the calculation for each reserve period.

9.3.1 Variables

Variables for statutory CARVM reserve parameters are contained in the fia_carvm_stat submodel.

In addition to the variables discussed above, two variables in fia_carvm_stat are hidden from the data entry but are critically important to the successful operation of the fia_carvm_stat elements:

model_name

The 'model_name' variable is an option choice variable with choices of STAT and TAX. This variable is used to retrieve the appropriate discount rate for statutory and tax reserves, since the fiacarvm_liab model class is used for both statutory and tax CARVM calculations.

reserve_period

The 'reserve_period' variable is the period for which the reserve is being calculated. It is set in fia submodel at the beginning of each reserve calculation to the current projection period.

9.3.2 Formulas

Formulas for calculating the statutory CARVM reserves are included in the fia_carvm_stat submodel.

Generally, fia_carvm_stat formulas prospectively calculate the present value of the CARVM streams from the reserve period to either the specified CARVM duration or the omega age of the model point.

9.3.3 Multiple instances

Each model point carries up to two instances of fia_carvm_stat, which calculate the CARVM reserves assuming:

- Policyholders do not take penalty free withdrawals
- Policyholders take full penalty free withdrawals.

9.4 fia_carvm_tax submodel

The fia_carvm_tax submodel is a user defined array instance of the fiacarvm_liab model class and contains all variables and formulas required to calculate fixed indexed annuity tax CARVM reserves.

Time 0 in the fia_carvm_tax submodel corresponds to the policy year of the reserve period. Thus, 't' is policy year, unlike the fia submodel where 't' represents months since the projection date. The values in fia_carvm_tax are cleared before beginning the calculation for each reserve period.

9.4.1 Variables

Variables for tax CARVM reserve parameters are contained in the fia_carvm_tax submodel.

In addition to the variables discussed above, two variables in fia_carvm_tax are hidden from the data entry but are critically important to the successful operation of the fia_carvm_tax elements:

model_name

The 'model_name' variable is an option choice variable with choices of STAT and TAX. This variable is used to retrieve the appropriate discount rate for statutory and tax reserves, since the `fiacarvm_liab` model class is used for both statutory and tax CARVM calculations.

reserve_period

The 'reserve_period' variable is the period for which the reserve is being calculated. It is set in `fia` submodel at the beginning of each reserve calculation to the current projection period.

9.4.2 Formulas

Formulas for calculating the tax CARVM reserves are included in the `fia_carvm_tax` submodel. Formulas for `fia_carvm_tax` are challenging, and not typically customized.

Generally, `fia_carvm_tax` formulas prospectively calculate the present value of the CARVM streams from the reserve period to either the specified CARVM duration or the omega age of the model point.

9.4.3 Multiple instances

Each model point carries up to two instances of `fia_carvm_tax`, which calculate the CARVM reserves assuming:

- Policyholders do not take penalty free withdrawals
- Policyholders take full penalty free withdrawals.

9.5 fia_rates submodel

The `fia_rates` submodel is identical to the `rates` submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

9.6 fia_ann_pba submodel

The `fia_ann_pba` submodel is identical to the `ann_pba` submodel which applies for portfolio projections.

Please refer to the C05 PBA book.

9.7 fia_sfas97rd submodel

The `fia_sfas97rd` submodel is identical to the `sfas97rd` submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

L09 PAYOUT ANNUITY

This book discusses payout annuity products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

Product – Benefits, premiums, and annuity purchase rate basis

Valuation – Statutory and tax reserves

Expense – Commission and expense rates, premium tax rate

Mortality – Experience mortality rates.

Each of these input page types begins with 'PA', for use in projections that combine multiple product types.

1.2 Data pages

Variables for each payout annuity model point are specified on the PAData and PABenData pages. These variables are discussed in the Model point records chapter, below.

2 PRODUCT BENEFITS AND VALUES

2.1 Policy type

Specify the policy type on the model point record.

2.1.1 Immediate annuity

Immediate annuities provide for annuity benefits for a certain period only, while a primary (life 1) annuitant survives with or without a certain period, or while either of the primary (life 1) annuitant or the secondary (life 2) annuitant survive, with or without a certain period.

Immediate annuity benefits begin within 13 months of the issue date and are typically limited to a single benefit stream making periodic payments

2.1.2 Installment refund annuity

Installment refund annuities provide for the continuation of the annuity benefits following the death of the single (life 1) annuitant until the sum of the annuity benefits equals the single premium. Installment refund annuities must be a single life annuity. There is no defined certain period.

2.1.3 Cash refund annuity

A cash refund annuity pays the excess of the single premium over the cumulative benefits paid as a lump sum at the death of the single (life 1) annuitant. Cash refund annuities must be a single life annuity. There is no defined certain period.

2.1.4 Structured settlement

Structured settlements provide for annuity benefits to be paid through very flexible benefit structures. The benefit structures are often a combination of certain only benefits, life contingent benefits, and lump sums.

2.2 Lives definition

Specify the lives definition on the model point record.

2.2.1 *Not applicable*

All payments are certain.

2.2.2 *Single life*

Life contingent benefits continue while the primary (life 1) annuitant survives.

2.2.3 *Joint and last survivor*

Life contingent benefits while either of the primary (life 1) annuitant or the secondary (life 2) annuitant survive.

The form of the benefit is a joint and x% survivor annuity product, where, after the death of either annuitant, the surviving annuitant receives x% of the benefit. The percent can vary between the primary and secondary annuitant.

The model point record specifies whether each life is living or deceased at the projection date. Thus a joint life annuity can continue to be specified as a joint annuity even after the death of one of the annuitants.

2.3 *Annuity payments*

2.3.1 *Benefit definitions*

Specify the annuity benefit definition on the model point record:

- **Benefit Streams** – The annuity benefits are specified using benefit stream information.
- **Monthly Tables** - The annuity benefits are specified monthly from issue using a table.

Also specify the annuity benefit input definition on the model point record:

- **Actual Benefits** – The benefit amounts specified in the benefit streams or monthly tables are actual benefit payment amount.
- **Notional Benefits** - The benefit amounts specified in the benefit streams or monthly tables are notional amounts. For example, where the benefit is described in terms of a \$1 dollar benefit. This option applies only for new issue model points where the actual benefits should be calculated as a multiple of the notional amount based on the purchase rate specifications and the specified single premium.

Warning! An annuity benefit input definition of 'Notional Benefits' cannot be used with an annuity benefits definition of 'Monthly Tables' for model points which are included in a nested projection. A data validation error will be issued for these cases.

2.3.2 *Benefit streams*

Specify the annuity benefit stream information on the benefit stream model point record:

- **Benefit start date** – The start date of the annuity benefits. Benefits from a stream with start day <= 15 are assumed to occur at the beginning of the month. Benefits from a stream with start day > 15 are assumed to occur at the end of the month.
- **Benefit end date** – The last date of the annuity benefits. The benefit end date is inclusive. That is, a payment is made in the month of the benefit end date.
- **Benefit certain end date** – The last date of certain payments. The benefit certain end date is inclusive. That is, a certain benefit payment is made in the month of the benefit certain end date.
- **Benefit mode** – Annuity payments can be paid monthly, quarterly, semi-annually, or annually.
- **Annualized annuity benefit** – Initial (at the projection date) annualized benefit amount for the policies collected into the model point. This amount must include all benefit increases up through the projection date and, for joint and last survivor annuities, must be the annualized benefit payable when both

annuitants are alive. Also, when the benefit stream start date is after the projection date, this amount is the annualized annuity benefit payable at the start of the benefit stream.

- **Benefit increase definition** – The annualized annuity benefit can stay level, be increased using a simple increase method, or increase at a compound rate.
- **Benefit increase next date** – The next annualized annuity benefit increase date following the projection date. This date must also be after the benefit start date.
- **Benefit increase last date** – The final annualized annuity benefit increase occurs on this date. The benefit increase last date is inclusive. That is, the annualized annuity benefit increases in the month of the benefit increase last date.
- **Benefit increase** – The annualized annuity benefit increase. When a simple increase is used with actual benefits specified, specify the dollar amount increase as an amount greater than \$1. When a simple increase is used with notional benefits, specify a percentage (0.05 = 5%) of the initial (at issue) annualized annuity benefit. When a compound increase is used, always specify a percentage (0.05 = 5%).

SFAS97 limited pay GAAP treatment

The benefit increase next date provided in the benefit stream is a future date and there is no information provided on the benefit stream regarding when increases actually started. For SFAS97 limited pay GAAP calculations, where a projection from issue is required, benefit increases are always assumed to have started on the second anniversary of the benefit stream.

2.3.3 Monthly tables

As an alternative to specifying the benefits as benefit stream information, benefits for the annuity model points may be entered through benefit variables. This approach would be useful for annuity model points which have very unusual benefit amounts that cannot be specified using the benefit stream information.

Specify the monthly tables of benefits using the `ann_ben_certain_bom`, `ann_ben_certain_eom`, `ann_ben_contingent_bom`, and `ann_ben_contingent_eom` variables in the Benefits category on the PAProduct input page.

Each of these variables are typically linked to a comma separated value (CSV) file which contains the benefits for each month since issue. When the CSV file is structured with the model point identification and table name as the rows and months since issue as the columns, the row lookup definitions for each variable are the `mp_id` code variable of the pa submodel and the constant text of the variable name, and the column lookup is the `mths_since_issue` column of the pa sub model.

2.4 Death benefits

Annuity payments are made as scheduled during the certain period and while at least an annuitant is surviving. There is no benefit paid at death, except in the special case of cash refund annuities.

Cash refund annuity

The excess of the single premium over the cumulative benefits paid is paid as a lump sum at the death of the annuitant. Specify on the model point record the initial (at the projection date) cumulative benefits paid for the policies collected into the model point.

2.5 Premium

2.5.1 Single premium

Payout annuity products have only a single premium. Specify on the model point record the single premium for the policies collected into the model point.

For new business model points, this variable is optional since the annuity purchase rate (discussed below) can be used to calculate the single premium.

2.5.2 Policy fee

There is no separate policy fee. Any policy fee is included in the specified single premium.

2.6 Annuity purchase rate

2.6.1 Applicability

For new business model points, the annuity purchase rate information specified for the product may be used to determine either the initial annualized benefit or the single premium. Whether the annuity purchase rate is used is based on the model point record:

When the benefit definition is Notional Benefits and the single premium is specified, the initial annualized annuity benefit is the notional initial annualized annuity benefit amount times the single premium divided by the annuity purchase rate.

When the benefit definition is Actual Benefits and the single premium is left at the zero default value, the single premium is the present value of benefits using the purchase rate assumptions adjusted for the purchase rate loads.

When the benefit definition is Actual Benefits and both the single premium and initial annualized benefit amounts are also specified, no purchase rate calculations occur and both the specified single premium and annualized benefit amounts are used in the experience calculations.

If the benefit definition is Notional Benefits and the single premium is not specified, the model point is skipped and an error message is produced.

The annuity purchase rate net single premium is the present value at issue of the payments based on the actual or notional initial annualized annuity benefit. For installment refund and cash refund annuities, the notional initial annualized annuity benefit is always set to \$1.

2.6.2 Interest basis

The annuity purchase interest rate basis is a fixed rate calculated at the valuation date. The fixed rate is either: (1) a competitor rate less a spread or (2) specified fixed rate, but not less than a specified minimum. This is an effective annual rate.

2.6.3 Mortality basis

The annuity purchase rate mortality basis is a base mortality rate with annual mortality improvements, times a multiple, plus an addition, then plus an amount for substandard lives.

Base mortality rate

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

Mortality Improvements (projection scale)

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a

model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$. When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

Mortality multiple and addition

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

Substandard extra deaths

The base mortality rate after improvements and multiple and addition adjustments is adjusted by a number of substandard extra death specified on the model point record. The substandard extra deaths used for the annuity purchase rate calculation is the same amount used for experience calculations

Blended mortality

The model point can be a 'blend' between males and females issued at the specified issue age. This occurs when the gender variable is Blended. Specify on the model point record the proportion of issues that are males.

The male and female mortality rates are each determined from the mortality table. The blended mortality rate is:

$$\text{proportion of males} \times \text{male mortality rate} + \text{proportion of females} \times \text{female mortality rate}$$

Monthly mortality rate

Deaths occur at the end of each month. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

2.6.4 Loads and fees

Loads include a percent of benefit load, a premium tax percent load, and a policy fee.

When the benefit is calculated, the initial annualized benefit is:

$$\frac{[(\text{premium}) \times (1 - \text{premium tax load \%}) - \text{policy fee}] \times (1 - \text{percent load \%})}{\text{annuity purchase rate net single premium}}$$

When the premium is calculated, the single premium is:

$$\frac{[(\text{annuity purchase rate NSP} \times \text{initial annuity benefit}) / (1 - \text{percent load \%})] + \text{policy fee}}{(1 - \text{premium tax load \%})}$$

2.6.5 Purchase rate calculation

The annuity purchase rate net single premium is the present value at issue of the notional annuity benefit using the specified interest rates, mortality rates, and loads and fees.

For cash and installment refund annuities, the benefit or premium is determined using an iterative process:

The calculation for the cash refund annuity values a single life annuity for the annuity benefit plus decreasing term insurance for a death benefit equal to the greater of zero and the difference between the unknown gross single premium and the cumulative annuity benefits. The iterative process solves for the net single premium based on guesses on the gross single premium.

The calculation for the installment refund annuity values a single life annuity with an unknown period certain for the annuity benefit. The iterative process solves for the net single premium using guesses on the certain period.

3 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves. PBA reserves and capital are discussed in the PBA book.

3.1 CARVM reserve

Payout annuity reserves are calculated using the Commissioners' Annuity Reserve Valuation method (CARVM) and Actuarial Guideline XXXIII (AG 33).

Since the standard payout annuity features do not provide for cash settlement options, the contract reserve is the present value of death and annuity benefits using the specified reserve basis. All payout annuity benefits (death and annuity) are considered to be non-elective benefits under CARVM AG 33.

Minimum reserves for substandard cases required under actuarial guidelines IX-A and IX-C are calculated using constant 'extra deaths' additions to the valuation mortality.

3.2 Reserve basis

3.2.1 Mortality

Reserve mortality is an improved base table, times the multiple, plus the addition, plus a number of substandard extra deaths.

With the exception of the substandard extra deaths, which are specified on the model point record, these variables are special since they are read twice – first for life 1 and then again for life 2. This is accomplished by using special lookup keys.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. 'Insured characteristics' (issue age and gender) are set to either the life 1 or life 2 values, as needed. The external source uses lookup keys mort_issue_age and mort_gender instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age. Using the hidden variable ensures that the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

This reading of mortality variables twice applies for joint annuities, and for blended mortality single life annuities.

3.2.2 Interest

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the PayoutAnnStatValn and PayoutAnnTaxValn ranges in the Valuation Interest Rates Excel workbook.

3.3 Tax reserve

Specifying 'Statutory' as the tax valuation code on the model point record sets the tax reserve equal to the statutory reserve.

Tax reserves are calculated by the same methods, using the tax reserve basis.

The tax reserve is not less than the statutory reserve.

4 ACTUARIAL ASSUMPTIONS

4.1 Commissions

A percent of the single premium is paid as a commission at issue. The specified commission rate is the average rate that is assumed to apply, taking into account average writing agent commissions and commission overrides.

4.2 Expenses

4.2.1 *Expense adjustment multiple*

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

4.2.2 *Acquisition expenses*

Acquisition expenses incurred at issue can include:

- **Amount per policy** – A fixed amount per policy.
- **Percent of single premium** – A percentage of the single premium.

4.2.3 *Maintenance expenses*

The following maintenance expenses are incurred at the end of the month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the policies in force during the month, not at the end of the month after decrements.

- **Amount per policy** – A fixed amount per policy.
- **Percent of the annualized benefit amount** – A percentage of the annualized annuity benefit in force at the beginning of the month. The annualized annuity benefit expense begins when the benefit stream is assumed to start, which may be before benefits begin for streams which pay at the end of the benefit period. The annualized annuity benefit for the policy changes monthly.

4.2.4 *Transaction expenses*

Maintenance expenses include the following expenses related to administrative activities:

- **Amount per benefit payment** – A fixed amount for each payment. This expense is incurred at benefit timing (beginning or end of month), since this is an expense related to each payment.

4.2.5 *Inflation*

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

4.2.6 *Premium tax*

Premium taxes are reported with expenses. The tax is a percent of the single premium payment.

Inflation and the expense adjustment calendar month multiple do not apply to premium tax.

4.2.7 *Fixed expenses*

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

4.3 Mortality

4.3.1 Input variables

'Mortality variables' for payout annuity products include mortality base, mortality improvement rates, for experience and purchase rates, multiples, additions, and substandard extra deaths.

With the exception of the substandard extra deaths, which are specified on the model point record, these variables are special since they are read twice – first for life 1 and then again for life 2. This is accomplished by using special lookup keys.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. 'Insured characteristics' (issue age and gender) are set to either the life 1 or life 2 values, as needed. The external source uses lookup keys mort_issue_age and mort_gender instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age. Using the hidden variable ensures the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

This reading of mortality variables twice applies for joint annuities, and for blended mortality single life annuities.

4.3.2 Mortality adjustment multiple

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date. This variable can be used as a sensitivity test.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

4.3.3 Base mortality

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

4.3.4 Mortality improvement

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$. When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

4.3.5 Mortality table adjustments

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

4.3.6 Substandard adjustment

The base mortality rate after improvements and multiple and addition adjustments is adjusted by a number of substandard extra death specified on the model point record.

4.3.7 *Blended mortality*

The model point can be a 'blend' between males and females issued at the specified issue age. This occurs when the gender variable is Blended. Specify on the model point record the proportion of issues that are males.

The male and female mortality rates are each determined from the mortality table. The blended mortality rate is:

$$\text{proportion of males} \times \text{male mortality rate} + \text{proportion of females} \times \text{female mortality rate}$$

4.3.8 *Monthly mortality*

Deaths occur at the end of each month. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

4.4 Risk based capital risk definition

The risk based capital (RBC) C3 component percent varies between 'low', 'medium', and 'high' risk annuity products. Specify on the Company input page this risk definition for each product, then vary the RBC C3 percent between Low, Medium, and High risks.

5 REINSURANCE

5.1 Coinsurance percent

Coinsurance treaties cede a portion of the entire policy, including transferring the ceded reserve to the reinsurer. Reinsurance benefits include all benefits provided by the policy.

The coinsurance benefits paid to the ceding company each month are the projected policy benefits, times the reinsurance percent specified on the model point record. A model point is not reinsured when the coinsurance percent is zero

5.2 Coinsurance premium

The coinsurance premium is the coinsurance percent times the single premium paid at issue.

5.3 Coinsurance benefits

The reinsurer reimburses for the coinsurance percent times all benefits paid by the ceding company. Reimbursements are split by different benefit types for reporting purposes.

5.4 Coinsurance expense allowance

5.4.1 *Expense reimbursement*

The reinsurer reimburses for the coinsurance percent times:

- Acquisition expenses
- Maintenance expenses
- Commissions

5.4.2 Premium tax

The reinsurer reimburses for the coinsurance percent times premium taxes paid by the ceding company.

5.5 Reinsurance reserve

The reinsurance reserve credit for coinsurance treaties is the total statutory reserve, times the coinsurance percent.

5.6 Reinsurance administration expenses

Reinsurance administration expenses are incurred at the end of each month; one twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

Reinsurance expenses can include an amount per policy.

Inflation, and the expense adjustment multiple, applies to reinsurance administration expenses the same as to maintenance expenses.

Coinsurance does not reduce the expenses calculated by model point. The product expense rates apply to the gross model point amounts, not reduced for coinsurance.

6 MODEL POINT RECORDS

6.1 PA model point record layout

The model point record layout for payout annuities is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	up to 25 characters (assigns model point to report group)
B	seg_id	segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date	yyyy/mm/dd or mm/dd/yyyy
F	plan_code	up to 20 characters (link to assumptions by plan)
G	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
H	tax_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
I	policy_type	Immediate Annuity, Installment Refund Annuity, Cash Refund Annuity, or Structured Settlement

Excel Column	Variable	Value
J	lives_defn	Not Applicable (all payments are certain), Single Life, or Joint And Last Survivor
K	issue_age_mp	positive integer
L	gender_mp	Male, Female, Blended
M	substd_extra_dths	Extra deaths applied to the purchase rate and experience mortality for substandard rating
N	substd_extra_dths_res	Extra deaths applied to the reserve mortality for substandard rating
O	status	Living, Deceased
P	male_sex_pct	male percentage (1.0 for 100%). This only applies when gender is Blended.
Q	issue_age_life2_mp	life 2 characteristics, for joint annuities issue_age_life2_mp must be > -1 when lives_defn is Joint And Last Survivor
R	gender_life2_mp	
S	substd_extra_dths_life2	
T	substd_extra_dths_res_life2	
U	status_life2	
V	single_prem	single premium for mp
W	ann_ben_defn	Benefit Streams, or Monthly Tables When the selection is Benefit Streams, the annuity benefits are specified using dependent data benefit stream model point records (described in the next section). When the selection is Monthly Tables, the annuity benefits are specified monthly from issue using the PABenMthlyTbl.csv file
X	ann_ben_input_defn	Actual Benefits, or Notional Benefits A selection of Notional Benefits specifies that the benefit amounts are a notional amount and the actual benefits should be calculated as a multiple of the notional amount based on the purchase rate specifications and the specified single premium. This selection is only valid for new issue model points.
Y	ann_ben_pct_life1_mp	benefit percent (1.0 for 100%) payable after life 2 is deceased

Excel Column	Variable	Value
Z	ann_ben_pct_life2_mp	benefits percent (1.0 for 100%) payable after life 1 is deceased
AA	reins_pct	ceded coinsurance percent
AB	init_policies	initial policies for mp
AC	init_ann_benefits_cumul	total cumulative benefits paid for mp
AD	gaap_defn	SFAS 91, SFAS 97 Limited Pay
AE	init_surv_life1	initial life 1 survivorship percent
AF	init_surv_life2	initial life 2 survivorship percent

6.2 PA benefit stream model point record layout

The benefit stream model point records utilize the dependent data capability of RiskAgility FM so that the benefit stream model points are linked to the payout annuity model points through a specified keyfield.

At least one benefit stream record must be specified for each payout annuity model point which uses 'Benefit Streams' as the ann_ben_defn selection. When the policy_type selection is Immediate Annuity or Structured Settlement, any number of benefit stream records may be specified for each payout annuity model point. However, when the policy_type is Installment Refund Annuity or Cash Refund Annuity, the benefits streams are limited to a single record.

Warning! The order of the benefit stream model points must be in the same order as the liability model points. And, the benefit stream model points for a liability model point must be grouped together.

The model point record layout for payout annuity benefit streams is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	keyfield	Up to 12 characters. This value MUST match a value of the mp_id field of the payout annuity model point record.
B	ann_ben_id	up to 12 characters unique name
C	ann_ben_start_date_mp	A date (yyyy/mm/dd or mm/dd/yyyy, default = 1/1/1900) representing the first payment date of the benefit stream. Payments from streams with a day <= 15 are treated as beginning of month payments and payments from streams with a day > 15 are treated as end of month payments.

Excel Column	Variable	Value
D	ann_ben_end_date_mp	A date (yyyy/mm/dd or mm/dd/yyyy, default = 1/1/1900) representing the last payment date of the benefit stream. Specify a date with a year of 9999 for a benefits which continue for life
E	ann_ben_certain_date_mp	A date (yyyy/mm/dd or mm/dd/yyyy, default = 1/1/1900) representing the last certain payment date of the benefit stream. Specify a year of 1900 if there are no certain payments in the benefit stream.
F	ann_ben_mode	Monthly, Quarterly, Semi-annual, Annual
G	init_annualzd_ann_ben_mp	A positive amount representing the annualized annuity benefit for the benefit stream as of the projection date payable. For joint and survivor annuities, this is the annualized annuity benefit payable when both annuitants are alive. This amount includes any benefit increases which have occurred prior to the projection date. When the benefit stream start date is after the projection date, this amount is the annualized annuity benefit payable at the start of the benefit stream.
H	ann_ben_incr_defn	Not Applicable, Simple, Compound
I	ann_ben_incr_next_date_mp	A date (yyyy/mm/dd or mm/dd/yyyy, default = 1/1/9999) representing the next increase date (following the projection date) for the annualized annuity benefit of the benefit stream.
J	ann_ben_incr_end_date_mp	A date (yyyy/mm/dd or mm/dd/yyyy, default = 1/1/9999) representing the last increase date for the annualized annuity benefit of the benefit stream. Specify a year of 9999 for increases which continue forever.

Excel Column	Variable	Value
K	ann_ben_incr	A decimal value (range -999999.9999 to 999999.9999) representing the benefit increase. Specify a negative value for benefit decreases. An entry with an absolute value greater than 1.0 is treated as a dollar increase. Otherwise, the increase is treated as a percentage. When the ann_ben_incr_defn is 'Simple' and the ann_ben_input_defn is 'Actual Benefits', specify the dollar amount increase as an amount greater than \$1. When the ann_ben_incr_defn is 'Simple' and the ann_ben_input_defn is 'Notional Benefits', specify a percentage (0.05 = 5%) of the initial (at issue) annualized annuity benefit. When the ann_ben_incr_defn is 'Compound', always specify a percentage (0.05 = 5%).

6.3 Required model point variables

6.3.1 Always required

The following model point variables are required in all cases:

Model point identification: group, seg_id, and mp_id

Insured characteristics: issue_age_mp and gender_mp, status; also male_sex_pct when gender is Blended

Policy specifications: issue_date, policy_type, lives_defn, plan_code, stat_valn_code, tax_valn_code, ann_ben_defn, ann_ben_input_defn, ann_ben_pct_life1_mp

Initial policies: init_policies

Single premium: single_prem is required for in-force model points. It is also required for new issue model points when the single premium is known or is used to determine the initial annualized annuity benefit.

Reinsurance: reins_pct is required for model point with coinsurance

GAAP: gaap_defn is required for all GAAP calculations

6.3.2 Joint annuities

For joint annuities, the second life insured characteristics and policy specifications are also required:

Second life characteristics: issue_age_life2_mp, gender_life2_mp, and status_life2

Policy specifications: ann_ben_pct_life2_mp

6.3.3 In-force model point values

In-force model points are driven by 'initial values' variables, which specify the aggregate model point values at the projection date.

New business model points (model points with issue date after the projection date), however, ignore these variables. The exception is that the initial policies must be specified, along with either or both of the single premium and initial annualized annuity benefit.

The following table shows when each in-force values variable is required for in-force model points:

Variable	Required when
init_ann_benefits_cumul	policy type is installment refund annuity or cash refund annuity

6.3.4 Variables to ignore

The following variables are needed only for nested projections, and should be ignored. These values must be carried forward to the next time step in order to reproduce the results from standard projections:

Variable
init_surv_life1
init_surv_life2

7 SUBMODELS STRUCTURE

7.1 pa submodel

The pa submodel calculates the model point's contribution to the liability income statement and balance sheet items for payout annuity products.

7.1.1 Seriatim versus portfolio runs

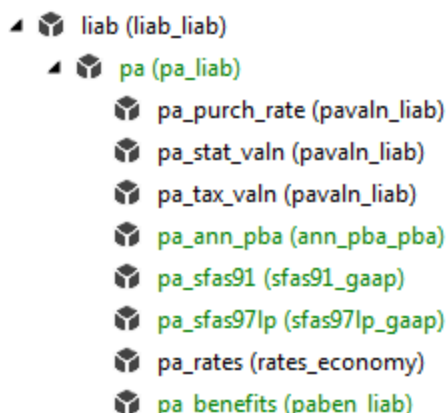
Seriatim projection

When pa is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the pa submodel. Full profit results are calculated for each model point, including PBA, GAAP, investment income, target capital, taxes, and distributable earnings.



Portfolio projection

For portfolio projections (which include business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the pa submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the pa submodel calculates only the model point results which are aggregated for the segment calculations.



PBA, GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the pa submodel the same columns as for other product types. This makes the payout annuity product type interchangeable with other product types fitting into this business segment structure.

7.1.2 Variables

Variables for payout annuity products are located in the pa submodel.

7.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when pa is the top model. These include PBA, GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the pa submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each projection task loop. The startup formulas are split into three sections: (1) when only pa is the top model, (2) when pa is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when pa is run as the top model and when pa is included in projections with segment as the top model. References to company, segment and liab are invalid when pa is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) temporary tables are sized for DAC tax calculations and competitor rates, (4) projection flags are created and initialized, (5) reporting groups are set up, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each projection task loop, immediately after startup when pa is the top model. When segment is the top model, initialize is executed by calls from seg->initialize which executes pa->initialize for each model point in the segment.

Again, conditions using the isMainModel system variable manage the initializations which are only required when pa is running as the top model. This includes executing the initialization of economic scenario rates and the setup of competitor rates. Other initializations are required in both cases.

Other formulas

Other formulas in the pa submodel are relatively straightforward. Most of the profit, balance sheet, and GAAP columns are only calculated when pa is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

7.2 pa_benefits submodel

The pa_benefits submodel is a standalone instance of the pa_ben_liab model class. It contains the variables and formulas required to calculate the undecrement and un 'joint life' adjusted benefits for each benefit stream.

7.2.1 Variables

Variables for benefit streams are contained in the pa_benefits submodel.

7.2.2 Formulas

Formulas to determine the undecrement and un 'joint life' adjusted annualized annuity benefit and beginning and end of month annuity payments are contained in the pa_benefits submodel. Separate results are calculated for certain and contingent benefits.

The annualzd_ann_ben_certain, annualzd_ann_ben_contingent, ann_ben_certain_bom, ann_ben_certain_eom, ann_ben_contingent_bom, and ann_ben_contingent_eom are the key results of this submodel.

7.3 pa_purch_rate submodel

The pa_purch_rate submodel is a standalone instance of the pa_valn_liab model class. It contains the variables and formulas required to calculate the purchase rate for new issue payout annuity model points.

When a new issue payout annuity model point specifies 'Actual Benefits' for the annuity benefit definition without the single premium, the single premium is determined using the annuity purchase rate assumptions and formulas of this submodel.

Likewise, a new issue payout annuity model point specifies the single premium and 'Notional Benefits' for the annuity benefit definition, the actual initial annualized annuity benefit is determined using the annuity purchase rate assumptions and formulas of this submodel.

The annuity purchase rate net single premium is the present value at issue of the payments based on the actual or notional initial annualized annuity benefit. For installment refund and cash refund annuities, the notional initial annualized annuity benefit is always set to \$1.

In order to conserve memory, this submodel is 'shrunk' for in force model points and for new issue model points which do not utilize purchase rate calculations.

7.3.1 Variables

Variables for purchase rate assumptions are contained in the pa_purch_rate submodel.

7.3.2 Formulas

Formulas to determine the purchase rate value of the annuity benefits are contained in the pa_purch_rate submodel.

The valn_amt_certain, valn_amt_life1, valn_amt_life2, valn_amt_joint_life, and valn_amt_term_ins_life1 formulas are the key results of this submodel. Each column result is the actuarial present value of the future annuity benefits for each time period 't' from the projection date using purchase rate interest and mortality assumptions.

Arrays are used to store the total undecrement and un 'joint life' adjusted benefits for the model point. The setup_ann_benefits_array scalar is executed to setup the benefit arrays for the purchase rate calculations. To ensure the arrays are setup before referenced, this scalar is included as a condition in each of the valn_amt_xxx columns.

The value at issue is the combined annuity valuation amounts at the 't' period equivalent to the commencement_period. For example, if the commencement_period of a new 10 year certain and life annuity issue is at $t = 12$, then the at issue value for the annualized annuity benefit of the 10 year certain and life annuity, calculated in the 'pa' submodel, equals $sm_pa_purch_rate \rightarrow valn_amt_certain(12) + sm_pa_purch_rate \rightarrow valn_amt_life1(12)$.

7.4 pa_stat_valn submodel

The pa_stat_valn submodel is a standalone instance of the pa_valn_liab model class. It contains the variables and formulas required to calculate the formulaic statutory reserves for payout annuity model points.

7.4.1 Variables

Variables for statutory valuation assumptions are contained in the pa_stat_valn submodel.

7.4.2 Formulas

Formulas to determine the statutory value of the annuity benefits are contained in the pa_stat_valn submodel.

The valn_amt_certain, valn_amt_life1, valn_amt_life2, valn_amt_joint_life, and valn_amt_term_ins_life1 formulas are the key results of this submodel. Each column result is the actuarial present value of the future annuity benefits for each time period 't' from the projection date using statutory valuation interest and mortality assumptions.

Arrays are used to store the total undecrementated benefits for the model point. The setup_ann_benefits_array scalar is executed to setup the benefit arrays for the reserve calculations. To ensure the arrays are setup before referenced, this scalar is included as a condition in each of the valn_amt_xxx columns.

The final statutory value at any projection period 't' is the combined annuity valuation amounts at the 't' period multiplied by the appropriate experience survivorship factors. For example, the statutory value of an inforce 10 year certain and life annuity for experience projection period 12, calculated in the 'pa' submodel, equals $sm_pa_stat_valn \rightarrow valn_amt_certain(12) + sm_pa_stat_valn \rightarrow valn_amt_life1(12) * surv_life1(12)$.

7.5 pa_tax_valn submodel

The pa_tax_valn submodel is a standalone instance of the pa_valn_liab model class. It contains the variables and formulas required to calculate the formulaic statutory reserves for payout annuity model points.

7.5.1 Variables

Variables for tax valuation assumptions are contained in the pa_tax_valn submodel.

7.5.2 Formulas

Formulas to determine the tax value of the annuity benefits are contained in the pa_tax_valn submodel.

The valn_amt_certain, valn_amt_life1, valn_amt_life2, valn_amt_joint_life, and valn_amt_term_ins_life1 formulas are the key results of this submodel. Each column result is the actuarial present value of the future annuity benefits for each time period 't' from the projection date using tax valuation interest and mortality assumptions.

Arrays are used to store the total undecrementated benefits for the model point. The setup_ann_benefits_array scalar is executed to setup the benefit arrays for the reserve calculations. To ensure the arrays are setup before referenced, this scalar is included as a condition in each of the valn_amt_xxx columns.

The final tax value at any projection period 't' is the combined annuity valuation amounts at the 't' period multiplied by the appropriate experience survivorship factors. For example, the tax value of an in force 10 year certain and life annuity for experience projection period 12, calculated in the 'pa' submodel, equals $sm_pa_tax_valn \rightarrow valn_amt_certain(12) + sm_pa_tax_valn \rightarrow valn_amt_life1(12) * surv_life1(12)$.

7.6 pa_rates submodel

The pa_rates submodel is identical to the rates submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

7.7 pa_ann_pba submodel

The pa_life_pba submodel is identical to the life_pba submodel which applies for portfolio projections. This submodel is used for principle-based approach (PBA) capital and reserve calculations.

Please refer to the C05 PBA book.

7.8 pa_sfas91 submodel

The pa_sfas91 submodel is identical to the sfas91 submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

7.9 pa_sfas97lp submodel

The pa_sfas97lp submodel is identical to the sfas97lp submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

L10 LONG TERM CARE

This book discusses long term care products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the product operation are associated with one of these input pages:

- Product – Benefits, and premiums
- Valuation – Statutory and tax reserves, and the actuarial math library audit report option
- Expense – Commission and expense rates, premium tax rate
- Morbidity – Incidence, transfer, and termination rates, and claim costs for both experience and reserve calculations
- Mortality – Experience mortality rates
- Policyholder – Policyholder behavior assumptions: lapse rates

Each of these input page types begins with 'Ltc', for use in projections that combine multiple product types.

Additionally, variables defining reinsurance assumptions are associated with input pages which are common across products. This input page is:

- Reinsurance – Reinsurance treaty definitions and rates.

1.2 Data pages

Variables for each long term care model point are specified on the LtcData page. These variables are discussed in the Model point records chapter, below.

Variables for each reinsurance record are specified on the LtcReinsData page. These variables are the same for the term life, traditional life, universal life, variable universal life, indexed universal life, and long term care product types. These variables are discussed in the Reinsurance model point records chapter, below.

2 PRODUCT BENEFITS AND VALUES

The following defines the policy benefits and values required for the projection of long term care insurance.

2.1 Coverage types

The standard features provide for three care types: (1) nursing home (NH) facility care, (2) assisted living facility care (ALF), and (3) home health care (HHC). The three care types are projected simultaneously, with transfers between the levels of care.

2.2 Projection definition

Long term care claim projections use either:

- First principles – Claims paid and claim reserves are developed from first principles on monthly basis using specified incidence and termination assumptions. Assumptions include recoveries from, and transfers between, care types.
- First principles approximation – Policy year claim costs are developed on a total policies basis using the specified incidence and termination assumptions from which policy year claims paid and claim reserves are calculated. Monthly values are linearly interpolated, assuming the claim cost for each year applies for the first month of the year.

- Claim costs – Monthly claims paid and claim reserves are developed from claim costs on a total policies basis using specified policy year claim costs and completion factors for months since claim.

2.2.1 Claim exposure

When using the first principles approach, exposure to incidence can apply either to all policies, or only to policies not on claim.

2.2.2 Processing order

Claim incidence occurs at the end of the month and claim payments are assumed to be paid at the end of the month, before terminations.

2.3 Coverage period

Coverage is provided through a specified maturity age. Claims may occur at any time until the specified age. The term for benefits on a claim occurs is defined separately.

2.4 Long term care benefit

2.4.1 Benefit term

The benefit term for long term care insurance may be based either on a number of years or a dollar amount. The benefit term specifications define a policy limit, which may be a combined amount for all care types or additive across the care types.

When the benefit term definition is Dollars, the benefit base represents the pool of money at issue (or at the contingent nonforfeiture date) for the policies included in the model point. Otherwise, the benefit base is the number of years from claim incurral.

The model point record contains a single variable for the Combined policy limit definition. Separate variables by care type apply for the Additive policy limit definition.

The benefit term is the number of months from claim incurral. When the policy limit is a dollar amount, the amount is converted into the equivalent number of months for claim. Variations resulting from benefit use and inflation affect the total benefit amount paid, but not when the benefit ends.

2.4.2 Elimination term

Claim payments commence following the elimination term, which is a number of days since claims incurral. Each care type has a separate elimination term.

The elimination term is converted from days to months by dividing by the number of days in a month, which is $365.25 / 12$.

2.4.3 Maximum daily benefit

Claim payments are based on the maximum daily benefit and the benefit utilization (see below). Each care type has a separate maximum daily benefit.

The maximum daily benefit is either the amount at issue or as of the projection date. The amounts would differ when benefit inflation is included. The definition for the maximum daily benefit is the same for all care types.

The maximum daily benefit is converted to a monthly daily benefit by dividing by the number of days in a month, which is $365.25 / 12$.

2.4.4 *Benefit inflation option (bio)*

The maximum daily benefit for each care type can increase by an inflation rate at the beginning of each policy year. The inflation can be applied on a simple or compound basis.

The benefit inflation can be limited to a specified attained age or a number of policy years. Benefit inflation applies regardless of whether the policy is on claim or not.

The same benefit inflation option provisions apply to all care types.

2.4.5 *Restoration of benefit rider*

Benefits are restored to their full amount after each recovery. When transfers are enabled, and the policy limit definition is Additive, benefits are restored to their full amount upon each transfer from one care type to another. When the policy limit definition is Combined, benefits are not restored upon transfer. There is no limit to the number of restorations.

2.4.6 *Benefit utilization*

When claims are calculated using first principles or first principles approximation, the maximum daily benefit is the maximum amount payable on a daily basis for the care type. Since the actual benefits paid can be less than that amount, the actual daily benefit paid is adjusted by a benefit utilization multiple. Each care type has a separate benefit utilization multiple.

The benefit utilization multiple is a base benefit utilization multiple which varies by months since claim and age at claim, times multiples varying by attained age and maximum daily benefit.

2.5 *Return of premium benefit*

A percent of the cumulative premiums less cumulative benefits paid may be paid upon death, lapse, and to surviving policies on a specified policy anniversary. The percent typically varies by policy year. When there are benefits to those surviving to the end of specified policy years, the return of premium benefit available resets to zero to begin the next policy year, even when the percent payable is less than 100%.

The return of premium option may end at a specified attained age. Reaching this attained age is a maturity event. No future return of premium benefits are paid, even in the event of future death or lapse.

For each model point, the return of premium benefit can be selectively enabled or disabled for each of three contingencies: death, lapse, and maturity. Return of premium benefits are modeled using a premium refund factor and a claims offset factor. These two factors typically vary by policy year, by attained age, or by issue age and policy year.

The premium refund percent is the percent of cumulative premium less cumulative benefits to be refunded upon a covered contingency. The percent is the same regardless of whether ROP is triggered by death, lapse, or maturity.

When the product includes a ROP maturity period, the premium refund percent is usually cyclical and resets to the first year rate after each maturity event. The external source must use the hidden variable lookup key `rop_pol_year`, which is set by formulas each time the variable is called.

The claims offset percent accounts for the effects of claim payments. The ROP benefit depends on the balance of cumulative premiums less cumulative claims, which is called the ROP balance. The ROP balance is not computed from actual paid claims; instead, the ROP balance is assumed to be the product of the claims offset factor and the balance of cumulative premiums. For the purpose of calculating and tracking the cumulative premium balance, claim payments (actual, projected, and waiver of premium) are ignored. When a policy has a return of premium benefit upon maturity, then the cumulative premium balance resets to zero whenever a maturity contingency occurs.

Specify the initial (at the projection date) ROP balance on the model point record. When the user specified an initial balance of zero, then the cumulative premium is calculated from issue. For the purpose of

computing statutory policy reserves for ROP, the cumulative premium is always calculated from policy issue, regardless of any user-specified initial ROP balance.

The ROP benefit available each policy month is the cumulative premium times the claims offset factor times the premium refund factor. The cumulative premium resets to zero after each maturity event.

2.6 Premium waiver benefit

Specify premium waiver definitions on the model point record. Separate definitions are available for each care type.

2.6.1 Additional claim benefit

When claims are calculated using first principles or first principles approximation, premiums can be waived after claim payments begin. The premium waiver benefit is an additional claim benefit; it is not treated as an adjustment to collected premiums.

2.6.2 Elimination term

Premium waiver benefits commence following the elimination term, which is specified as a number of calendar days since claims incurral.

The elimination term is converted from days to months by dividing by the number of days in a month, which is $365.25 / 12$.

2.6.3 Premium waiver benefit

The premium waiver benefit is the gross premium paid for the policies on claim, beginning after the elimination term. The premium waiver benefit ends when the insured dies, recovers, or when the policy benefit limit has been reached.

The premium waiver benefit increases when a premium rate increase occurs.

2.7 Premiums

2.7.1 Premium status

Each model point contains policies either paying premiums, fully paid up, or that have been converted to reduced paid up.

Specify a premium status of Active for premium paying or fully paid up policies. Specify a premium status of RPU for policies converted to reduced paid-up status.

2.7.2 Premium term

Premiums are paid until either a specified policy year or attained age. When the premium term value is zero (default), premiums are paid to maturity.

2.7.3 Premium mode

Specify the premium mode on the model point record. Premiums can be paid annually, semi-annually, quarterly, monthly, or special monthly (such as electronic transfer or pre-authorized check, with expense rate different from 'monthly' mode).

2.7.4 Annual premium

Specify the annual premium for the product either as:

- Annual premium per unit – The Annual premium per unit specified for the product, plus the policy fee. Premium suspensions apply from issue.
- Initial annualized premium – The premium on the model point record for the policies collected into the model point, including the policy fee. Premium suspensions apply beginning at the next policy anniversary, since this is the actual premium at the projection date.

2.7.5 *Modal loading*

When premiums are paid more frequently than annually, the annualized premium is the annual premium times the annualized modal loading multiple. For example, an annualized modal loading multiple of 1.08 means that monthly premiums are 9.0% ($1.08 / 12$) of the annual premium per policy.

2.7.6 *Rider premiums per policy*

Premiums for riders are either an amount per policy or a percent of the base annual premium per policy for the base benefits.

2.7.7 *Refund of unearned premium*

When premiums are paid less frequently than monthly, the unearned premium may be refunded at the beginning of the month to those who died or surrendered in the current month.

The unearned premium for the refund is always a pro-rata amount of the gross premium, assuming the model point was issued on the first day of the issue month and 30 days per month.

The benefit is captured as a reduction to premium income for income statement treatment.

2.8 Premium rate increases

2.8.1 *Premium rate increase percentage*

The premium specified on the model point record or as an annual premium per policy may be increased using a schedule of premium rate increase percentages.

Rate increases occur at the beginning of a policy year and apply to the base premium and rider premium.

2.8.2 *Commissionable status*

Commissions may not be payable on increased premiums.

3 STATUTORY AND TAX RESERVES

3.1 Premium reserves

3.1.1 *Premium reserve basis*

Unearned premium reserves premium basis choices are:

- Gross – The unearned premium reserve is the gross unearned premium
- Net – The unearned premium reserve is the gross unearned premium reserve, times the ratio of the contract reserve net premium to the gross premium.

3.1.2 *Premium waiver exclusion*

The unearned premium reserve may exclude policies on premium waiver.

3.1.3 *Exact day treatment*

The unearned premium reserve is calculated on a pro-rata basis using the number of days from the valuation date to the issue day, assuming premiums have been paid according to the premium frequency, assuming 30 days in each month.

For example, consider a model point paying quarter premiums issued 12/22/2011 with a projection date of 9/30/2012. Further, assume the user has selected to base unearned premium reserves on the gross unearned premium. The unearned premium at the projection date is $82 / 90 * \text{the gross premium}$.

The days provided in the issue date variable are assumed to be the premium due date for all future premiums and is used for all future unearned premium calculations.

3.2 *Contract reserves*

Contract reserves are established as a reserve for the claims expected to be incurred and return of premium benefits. Separate terminal reserves and net premiums are established for claim costs and return of premium benefits.

For claim cost contract reserves, per policy terminal reserves and net premiums are applied to all policies or to policies not on claim based on the claim exposure definition. Return of premium per policy contract reserve terminals and net premiums are also applied to all policies.

3.2.1 *Terminal reserves*

The following reserve methods are available:

- Not applicable – reserves are zero, and don't add to projection time.
- Table – terminal reserves and net premiums per policy are specified by the user
- Net level premium – no expense allowance
- Full preliminary term – zero terminal value for the first policy year
- Two-year full preliminary term – zero terminal reserves for the first two policy years.

Different reserve methods may be used for claim cost contract reserves and return of premium contract reserves.

Specifying 'Statutory' as the tax valuation code on the model point records sets the tax reserve equal to the statutory reserve.

3.2.2 *Claim cost benefits treatment*

Claim cost contract reserves include recognition of long term care benefits, and potentially premium waiver benefits.

Whether to include the premium waiver in the contract reserve is controlled by the same choice for premium waiver benefit cash flow. There is no separate choice list for contract reserve only. Of course, separate premium waiver benefits do not apply when projecting morbidity using Claim Costs. When used, the premium waiver benefit included in the contract reserve is always the gross premium to be waived.

Also, rate increases do not impact the premium waiver benefit in the contract reserving. Thus, the premium waiver benefit projected from issue must remove any previous rate increases from the annualized premium when the premium is specified on the model point.

3.2.3 *Return of premium benefits treatment*

Return of premium reserves includes recognition of return of premium benefits for death, lapse, and maturity.

The ROP benefits per policy for the reserve calculation are the contract ROP percentages of cumulative premiums available for the ROP benefit base for each benefit type for each policy year times a specified claim offset multiple.

For example, assuming an annual premium of \$100, a contract ROP percentage of 80% of cumulative premiums less claims payable to deaths in policy year 2, and a claim offset factor of 0.7, the ROP death benefit per policy for reserving in policy year 2 would be \$112 ($\$200 * 0.8 * 0.7$).

Reserve benefits payable for lapses and maturity are determined in a similar fashion.

3.2.4 Claims costs

Claim costs for contract reserving are the expected incurred claims for claims incurring in a policy year under valuation assumptions.

Claim costs for contract reserving may be specified or, when the claims projection definition is First Principles or First Principles Approximation, developed from specified contract reserving incidence and termination rates. Claim costs must always be specified for contract reserving when the claims projection definition is Claim Costs.

When specified, the claim costs may be specified as an amount per dollar maximum daily benefit or in total. In order to use a total claim cost approach, specify the dollar basis as 0.0.

Contract reserving claims costs are specified or developed for all care types combined.

3.2.5 Mortality

Reserve mortality is a base table, times the multiple, then plus the addition.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys mort_issue_age, mort_gender, mort_smoking_class, and mort_rate_class instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age.

Global tables include industry mortality tables.

Incidence adjustment

When the contract reserving claim costs are developed from first principles and the claim exposure is policies not on claim, the incidence rates are included as a decrement in the contract reserving calculations.

3.2.6 Lapses

Lapses may be included in the contract reserve calculation, subject to specific requirement of the NAIC Health Insurance Reserves Model Regulation.

Specify the reserve lapse definition as Not Applicable, NAIC 1997 Limits, NAIC 2005 Limits, NAIC Limits, or No Limits. When the reserve lapse definition is Not Applicable, no lapses are included in the contract reserve calculation. When the reserve lapse definition is No Limit, the base lapse rates are used in the contract reserve calculation. For all other methods, reserve lapses are included in the contract reserve calculation but are restricted per the following approaches.

When the reserve lapse definition is NAIC 1997 Limits, the reserve lapses are restricted as follows

- For policy years ≤ 4 , the lesser of 80% of the base lapse table and 8%,
- For all future policy years, the lesser of the base lapse table and 4%

When the reserve lapse definition is NAIC 2005 Limits, the reserve lapses are restricted as follows

- For policy year 1, the lesser of 80% of the base lapse table and 6%
- For policy years 2 through 4, the lesser of the 80% of the base lapse table and 4%,

- For all future policy years, the lesser of the base lapse table and 2%

When the reserve lapse definition is NAIC Limits, a total termination rate is first calculated including the mortality rates and the base lapse rates. Then, the total termination rate is restricted as follows:

- For all policy years, the lesser of 80% of the total termination rate and 8%

From this restricted result, the restricted lapse rate is determined as the restricted total termination rates divided by one minus the reserve mortality rate.

3.2.7 Interest

The Global tables include maximum statutory and tax valuation rates by issue year: 'int basis stat life.msv' and 'int basis tax life.msv'.

3.2.8 Function timing

Function timing choices are:

- Curtate – Annual premiums and EOY claim costs
- Continuous – Continuous premiums and immediate payment of claims
- Semicontinuous – Annual premiums and immediate payment of claims

3.2.9 Mid-year reserve

Reserves between policy anniversaries can be:

- Interpolated Terminal – Interpolated terminal reserves.
- Mid Terminal – Average terminal reserves.

3.2.10 Minimum reserve

The contract reserve may be floored at zero or set such that the sum of the calculated contract reserve plus the net unearned premium reserve is not less than the gross unearned premium reserve. If no minimum is applied, the contract reserve may be negative.

3.3 Claim reserves

Claim reserves include a reserve for all incurred but unpaid claims and a claim expense reserve with respect to the estimated expense of settlement of all incurred but unpaid claims.

The base claim reserve is the present value of future claim payments at a specified discount rate.

3.3.1 Benefits treatment

Claim reserves include recognition of long term care benefits, and potentially premium waiver benefits.

Whether to include the premium waiver in the contract reserve is controlled by the same choice for premium waiver benefit cash flow. There is no separate choice list for contract reserve only. Of course, separate premium waiver benefits do not apply when projecting morbidity using Claim Costs. When used, the premium waiver benefit is always the gross premium to be waived. The premium waiver reserve is calculated as a separate line item from the base claim reserve.

Also, rate increases do not impact the premium waiver benefit in the contract reserving. Thus, the premium waiver benefit projected from issue must remove any previous rate increases from the annualized premium when the premium is specified on the model point.

3.3.2 Projected claims

When the claims projection definition is First Principles or First Principles Approximation, termination rates for projecting claims for the claim reserve are the same as those specified for the experience projection.

When the claims projection definition is Claim Costs, the projected claims for the claim reserve are based on the specified experience claim costs and completion factors.

3.3.3 Interest

Claim reserve valuation rates vary by calendar year of claim incurral. Specify the claim reserve valuation interest rate by calendar year of claim using the Calendar Year table style. The claim reserve valuation interest rate may vary by selectors using the Selector Table table style.

For statutory and tax claim reserves, the claim reserve valuation uses the same variables as specified for the contract reserves.

3.4 Claim expense reserve

The claim expense reserve is a percentage of the base claim reserve. This reserve is set up as a reserve for expenses related to the future payment of claims incurred. The percentage should be consistent with the expenses associated with processing disabled life claims. This percentage also applies to the premium waiver reserve.

4 ACTUARIAL ASSUMPTIONS

4.1 Commissions

4.1.1 Premium commission

The percent of premium commission is the commission rate for each policy year, times the premium payments as they are paid. The modal loading can be commissionable (included) or noncommissionable (excluded).

4.1.2 Annualized commission

A specified percent of the total first year periodic premium commission is (specify 0.000000 to exclude this feature) paid at issue. No further periodic premium commissions are paid until the cumulative earned commission exceeds the annualized amount. From then on, commissions are paid as usual. This determination of commission payments is made for continuing policies, not taking into account terminated policies.

4.1.3 Commission chargeback

A portion of the commission can be charged back in the event of lapse during the first three years after issue. The chargeback percent applies either to: (1) total first year commissions paid-to-date, or (2) the annualized commission amount.

The commission subject to chargeback is calculated per policy in force, and thus is not reduced by decrements. This amount applies to the policies lapsing each month.

4.1.4 Commission overrides and agent terminations

There are no separate assumptions for commission overrides and agent terminations, i.e., the specified commission rates are the average commission rates and should already take into account average writing agent commissions, commission overrides, and the effect of agent terminations.

4.2 Expenses

4.2.1 Expense adjustment multiple

Product expense rates (including those expressed as a percent) are adjusted by an adjustment multiple, such as varying by calendar date.

This adjustment does not apply to reinsurance administration expenses and premium taxes.

4.2.2 Acquisition expenses

Acquisition expenses incurred at issue can include:

- Amount per policy
- Percent of the annualized first year premium

4.2.3 Maintenance expenses

The following maintenance expenses are incurred at the end of each month. One twelfth of the annual rate applies each month. The amount of expenses is calculated using the units in force during the month, not at the end of the month after decrements.

- Amount per policy, separate between premium paying and paid-up
- Percent of the annualized premium
- Percent claims paid

4.2.4 Reinsurance administration expenses

Reinsurance administration expenses are discussed in the Reinsurance chapter, below.

4.2.5 Inflation

Product expense rates not expressed as a percent are adjusted by inflation. See the Expenses chapter of the Profit calculations book.

4.2.6 Premium tax

Premium taxes are reported with expenses. The tax is a percent of each premium payment.

Inflation, and the expense adjustment multiple, do not apply to premium tax.

4.2.7 Fixed expenses

The business segment can include fixed expenses in addition to the product expenses, which vary in direct proportion to the amount of business being processed for each model point. Fixed expenses are adjusted by inflation. See the Expenses chapter in the Profit calculations book.

4.3 Mortality

Mortality rates can apply only to either active policies or all policies depending on claim exposure definition. The mortality assumption for policies on claim is assumed to be included in the claim termination assumption.

Deaths occur at the beginning of each month before lapse, premium payment and morbidity.

4.3.1 Input variables

'Mortality variables' for long term care products include mortality base, multiple, and addition rates, mortality improvement, and YRT reinsurance rates.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys mort_issue_age, mort_gender, mort_smoking_class, and mort_rate_class instead of using these insured characteristics values directly.

Hidden variable mort_attained_age is also required, since it is based on mort_issue_age. Further, the attained age does not advance with each yearly loop when calculating the mortality improvement multiple.

4.3.2 Mortality adjustment multiple

Experience mortality rates are adjusted by an adjustment multiple, such as varying by calendar date.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

4.3.3 Rate class

Mortality rates varying by rate class require special treatment. The model point variable (rate_class_mp) can have a large number of options specific to the company's business. But the 2001 CSO mortality tables varying by rate class only define rates for Standard, Preferred, and Super. Thus there are two different rate class hidden variables:

- **Company rate classes** – The company's rate classes are in mort_rate_class. For example, there might be a dozen different classes for this scalar. Experience mortality and YRT reinsurance rates vary by this lookup key.
- **Reserve rate classes** – The 2001 CSO mortality rate classes are in mort_rate_class_res. The company's different classes are mapped into the three rate classes prescribed for reserve mortality rates. Customize the formula for this scalar, to define how each company rate class translates into the three reserve rate classes.

4.3.4 Base mortality

The base mortality rate is read twice, first for life 1 and then again for life 2 (when applicable).

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

4.3.5 Mortality improvement

The base mortality rate is adjusted by mortality improvement percentages, which can be negative. Mortality improvement applies on each policy anniversary.

Mortality improvement percentages vary by attained age and calendar year, beginning on each policy anniversary after the base calendar year. Stop mortality improvement after a certain calendar year by specifying zero improvement rates for that year and after.

Mortality improvement applies from the base year, even when the model point is issued at a later date. For example, when the base year is 2008, and the improvement rate for attained age 45 is .0100 (1.00%), a model point issued at issue age 45 in 2011 has the first year mortality improvement multiple of $(1 - .010)^3$. When the improvement rate for attained age 46 is .0110 through 2011 and .015 beginning in 2012, the second year mortality improvement multiple is $(1 - .011)^3 * (1 - .015)$.

4.3.6 Mortality table adjustments

The base mortality rate after improvements is adjusted by the multiple, then plus the addition (which can be negative).

4.3.7 Mortality anti-selection

Mortality rates can be adjusted for anti-selection following a premium rate increase. The anti-selection multiple varies by the premium increase percent. The multiple increases at the beginning of each policy year following the increase. For example, suppose there is a premium increase in year 21. A 40% premium

increase can result in an increase of 120% to the mortality rate in the first policy year of the increase, 110% increase in the second year, and 105% increases in the third, fourth, and fifth years after the increase.

assump_set_id yrs_since_incr	Serialtim 0	Serialtim 1	Serialtim 2	Serialtim 3	Serialtim 4	Serialtim 5
rate_incr_pct						
0.00	1.00	1.00	1.00	1.00	1.00	1.00
0.40	1.20	1.10	1.05	1.05	1.05	1.00

The mortality anti-selection formula sets two hidden variable lookup keys, which must be used to link to the external source table:

- prem_incr_pct – The percent increase in premium
- yrs_since_incr – '0' for the multiple at the time of the increase, '1' for the multiple a year later, etc.

4.3.8 Monthly mortality

Deaths occur at the end of each month, prior to other decrements and policyholder transactions. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

4.4 Lapses and surrenders

Lapse rates apply to either active policies or all policies depending on claim exposure definition. Lapses occur at the beginning of each month after death, but before premium payment and morbidity.

4.4.1 Lapse adjustment multiple

Lapse rates are adjusted by an adjustment multiple, such as varying by calendar date.

4.4.2 Policy year lapse rates

The base lapse rate is adjusted by a multiple.

4.4.3 Monthly lapses

The annual lapse rate after all adjustments and excess lapses is constrained to be not more than 1.0 (100%).

Lapses and surrenders occur either: (1) at the end of each month, or (2) modally (just before the next premium is due). Lapses occur monthly after the last premium has been paid, for example in the final years of a decreasing term product.

Each of the above lapse rate adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly or modal lapse rate:

$$1 - (1 - \text{annual lapse rate})^{(1/\text{event frequency})}$$

4.4.4 Shock lapse

Shock lapses are added to the normal lapse rate based on the premium increase after the level term years.

The shock lapse rate varies by the premium increase percent at the end of the level term years, and at the end of each renewable term period. The ART premium increases following the level term years are not considered in determining the size of the shock lapses. For example, the premium increase following the end

of a 20-year level term period is the increase for year 21; the ART premium increases for years 22 and after do not generate a new shock lapse event.

Shock lapses occur just before the beginning of the policy year; this percent of policyholders declines to continue coverage at the increased premium rate.

You can spread the shock lapse over several policy years following the increase. In the example below, a premium increase of 80% results in a 16% shock lapse at the time of the increase, plus another 4% shock lapse one year later.

yrs_since_incr	0	1
prem_incr_pct		
0.00	0.00	0.00
0.40	0.08	0.02
0.80	0.16	0.04
1.20	0.32	0.08

The shock lapse formula sets two hidden variable lookup keys, which must be used to link to the external source table:

- prem_incr_pct – The percent increase in premium
- yrs_since_incr – ‘0’ for the shock lapse at the time of the increase, ‘1’ for the increase a year later, etc.

4.5 First principles morbidity

4.5.1 First principles approximation

A full first principles approach to long term care morbidity projections is a compute intensive process, involving development and tracking of a multi-dimensional array of claim information. Depending on the projection bases and features included, the claim array may need to track 12 individual items by incurral period and claim payment duration.

A first principles approximation approach may be utilized to overcome the challenges of a full first principles model. This approach uses the same first principles assumptions as the first principles approach but eliminates the need for many of the compute intensive calculations and the large storage structure for claims. This is accomplished by developing claim costs, claims paid, and claim reserves from the first principles assumptions on a policy year basis and then calculating monthly values from the policy year results using linear interpolation.

Since the first principles approximation is not tracking the claim information on a full monthly basis, the exposure definition is always total policies. When the claims projection definition is set to First Principles Approximation, the same assumptions are used as First Principles. Claim costs, claims paid and claim reserves are developed using first principles on a policy year, total-lives basis then linearly interpolated into monthly values to save runtime and memory. It overwrites the exposure definition input (always assume to be all policies) and the transfer definition input (always assume to be No).

4.5.2 Incidence adjustment calendar date multiple

The annual incidence rates are adjusted by a final multiple varying by calendar date.

4.5.3 Base incidence rates

The base incidence rates are annual rates and vary may vary by policy year, issue age, and/or attained age.

4.5.4 Incidence adjustments

Annual adjustment factors may be applied to the base incidence rates. Up to 6 incident adjustment multiple can be applied.

4.5.5 Incidence adjustment select & ultimate multiple

The annual incidence rates may be adjusted for select and ultimate treatment such that incidence rates may be lower in the early policy years.

4.5.6 Incidence anti-selection

Incidence rates can be adjusted for anti-selection following a premium rate increase.

Specify the incidence anti-selection multiples using the keys `rate_inc_pct` (percent increase in the premium) and `yrs_since_incr` (years since increase). The multiples can further vary by policy and policyholder characteristics.

Adjustments to incidence rates can be spread over a number of policy years following the premium increase. For example, suppose there is a premium increase in year 21. A 40% premium increase can result in an increase of 120% to the incidence rate in the first policy year of the increase, 110% increase in the second year, and 105% increases in the third, fourth, and fifth years after the increase.

4.5.7 Monthly incidence

Each of the above incidence adjustments applies to the annual rate. Finally, the annual rate is constrained to not be below 0.0 or above 1.0 and then converted into the monthly incidence rate. The conversion formula depends on the exposure definition.

When the exposure definition is set to Policies Not On Claim:

$$1 - (1 - \text{annual incidence rate})^{(1/12)}$$

When the exposure definition is set to All Policies:

$$\frac{\text{annual incidence rate}}{12.0}$$

The final monthly incidence rate is then applied to the policies basis to determine the policies going on claim in the projection month.

4.5.8 Termination adjustment calendar date multiple

The monthly termination rates are adjusted by a final multiple varying by calendar date.

4.5.9 Base termination rates

The base termination rates are monthly rates which may vary by months since claim and age at claim using the keys `mths_since_claim` and `claim_age`. The rates may also vary by policy and policyholder characteristics.

4.5.10 Termination adjustments

Monthly adjustment factors may be applied to the base termination rates.

Adjustments may be applied based on policy and policyholder characteristics as follows:

- Marital status

- Benefit term
- Elimination term
- Maximum daily benefit level
- Benefit inflation option
- Issue year

Each of these adjustment multiples may vary by months since claim and age at claim using the keys `mths_since_claim` and `claim_age`.

4.5.11 Termination processing

Terminations are processed monthly for a specified number of years since claim incurral. Following that period, terminations are processed as an annual amount semi-annually.

The table adjusted termination rates are always monthly rates. The annual rate to apply following the monthly period is an annualized rate, calculated using a geometric average of the monthly rates, for the previous 12 months (6 months for the first annual termination).

4.5.12 Care type transfers

Policies on claim in one care type may transfer to another care type as the claim ages. For example, policies on claim under the home health care type may transfer to the assisted living care type. All combinations of transfers are supported. The model does not track the number of transfers, nor the history of transfers, for a claim. There is no limit to the number of times a policy may transfer and no waiting period from one transfer to another. A policy may transfer as often as once per month.

The care type transfer percent is the percent of table adjusted termination rates from a care type that are a transfer to another care type.

The transfer percent may vary by months since claim and age at claim using the keys `mths_since_claim` and `claim_age`.

Care type transfers are treated as an aged claim moving into the new care type for the first time. If the policy limit definition is 'Additive', then the benefit limit for the new care type is assumed restored to its full amount upon each incoming transfer. If the policy limit definition is 'Combined', then benefits are not restored upon transfer because a single benefit limit is shared across all care types.

Warning! If the care type transfer percentages are high (close to 1.0) and form a cycle, and if the policy limit definition is 'Additive', then a significant number of lives on claim may evade benefit limits through repeated transfers. When the transfer percentages are set in this manner, it is recommended to also set the policy limit definition to 'Combined' to avoid this behavior.

4.5.13 Recovery

Policies on claim may recover to an active status. For example, policies in an assisted living facility may recover to active status. Recovery from any care type is supported but recovery is applicable to all or none of the care types together.

The recovery percent is the percent of specified terminations from a care type that become active policies and may vary by months since claim and age at claim using the keys `mths_since_claim` and `claim_age`.

Recovered policies are assumed to restore their full benefit amount.

4.5.14 Death terminations

The final claim termination rate is the table adjusted base termination rates adjusted for the care type transfers and recoveries. The total monthly termination rate is restricted to not be below 0.0 or above 1.0 and then applied to the policies on claim to determine the policies terminating from claim in the projection month.

Since termination rates may represent a percentage recovering, a percentage transferring, and the remaining terminating as death, the percentages are normalized if they exceed 1.0. Each termination percentage is applied against the same exposure of the policies on claim entering the month.

4.6 Claim costs

Claims costs are the total claims incurred for the benefit period for claims occurring in a specific policy year from issue or attained age. Completion factors are used to specify the percentage of total claims paid that has been paid through each month since claim. Thus, monthly claims paid are developed from the monthly claim costs and the completion factors.

Claim costs are applied to the policies after mortality and lapse.

4.6.1 Claim cost adjustment calendar date multiple

The annual claim costs are adjusted by a final multiple varying by calendar date.

4.6.2 Base claim costs

The base claim costs are per policy annual rates and may vary by policy year, issue age, and/or attained age, as well as policy and policyholder characteristics. The base claim costs are specified as a specified amount of initial daily benefit. Define the amount as a specified dollar basis.

The specified claim costs must incorporate the benefit inflation adjustments for increases after claim incurral.

4.6.3 Claim cost table adjustments

Annual adjustment factors may be applied to the base claim costs rates. Up to 6 claim cost adjustment multiples can be applied.

4.6.4 Claim cost adjustment select & ultimate multiple

The annual claim cost rates may be adjusted for select and ultimate treatment such that claim costs may be lower in the early policy years.

4.6.5 Monthly claim costs

Each of the above claim cost adjustments applies to the annual rate. Finally, the annual rate is constrained to not be below 0.0 or above 1.0 and then converted into the monthly claim cost rate:

$$\frac{\text{annual claim cost rate}}{12.0}$$

The final monthly incidence rate is then applied to the claim cost exposure policies to determine the incurred claims in the policy month.

4.6.6 Completion factors

Completion factors are cumulative percentages specified which vary by months since claim by age at claim using the keys mths_since_claim and claim_age. An entry in month 1 represents the cumulative completion for a claim occurring in that same month.

For each month since claim, paid claims are determined as the monthly incurred claim cost at the claim age, multiplied by the increase in the completion factor.

Important: Specify an entry for month 0 since the claim payment in month 1 from incurral is developed based on the change in the completion factor between month 0 and 1.

5 REINSURANCE

5.1 Multiple reinsurance treaties

Multiple reinsurance treaties can apply to a model point.

5.1.1 Treaty input

Each reinsurance treaty creates an instance of the `lrc_ireins` submodel. The input for each variable must be linked to an external source containing values varying by the treaty names.

The exception is that the YRT retention maximum per policy applies for all treaties combined, and thus does not vary by treaty. For example, it might vary by plan code.

Each treaty can be either a YRT, coinsurance, or modified coinsurance treaty. Each of these types is discussed in a separate section, below.

5.1.2 Reinsurance model point input

The reinsurance model point record identifies which treaties apply to each model point, as well as the ceded percent for each treaty. The reinsurance model point layout is discussed in the Reinsurance model point records layout section, below.

5.2 Risk premium (YRT) reinsurance

Although it is less common than other forms of reinsurance, risk premium reinsurance, or YRT reinsurance, can be used on long term care policies.

Yearly renewable term (YRT) reinsurance cedes a portion of the net amount at risk (NAR). Reinsurance premiums are calculated using YRT rates applied to the ceded coverage amount.

The NAR for LTC is the annual long term care benefit, based on the greatest of the specified maximum daily benefit amounts, available less the retained amount.

5.2.1 Ceded amount

The ceded amount is based on the NAR at the beginning of each policy year.

Using reinsurance model point variables

The ceded percent can be specified on the reinsurance model point record. This is proportionate YRT reinsurance, where the ceded percent is fixed.

Using treaty variables

The ceded percent can be calculated using the reinsurance input variables that vary by treaty code. The ceded amount is a specified percent of any excess of the NAR over the retention threshold per policy. For example:

- First dollar quota share – The threshold is zero.
- Excess retention – The ceded percent applies to the excess of the NAR over the specified threshold.

The ceded percent varies over time, depending on the changing relationship between the NAR and the threshold.

5.2.2 YRT rate

The YRT reinsurance rate for each treaty is a base table, times a multiple. These are 'mortality table' variables, which operate the same as for mortality rates. This allows for YRT rates to be entered as attained age, policy year by issues age, and select and ultimate.

For long term care, the YRT rate represents the anticipated claim cost for \$1 annual benefit for a given issue age and policy duration.

The YRT premium can be zero for the first policy year. When rates are YRT rates are entered as attained age rates, this can be accomplished by specifying a zero multiple for the first policy year using a Mortality Selector, Select and Ultimate, or Policy Year by Issue Age table style that allows for specifying with zero rates.

5.2.3 Reinsurance premium

YRT reinsurance premiums are paid annually at the beginning of each policy year. [Customize for monthly premiums.]

The YRT reinsurance premium per policy is:

$$\text{ceded amount} \times (\text{annual YRT rate}) + \text{reinsurance policy fee}$$

The reinsurance policy fee is specified for each treaty. (Note that this policy fee should not vary by premium mode, since all YRT treaties are projected as having annual premiums.)

The reinsurer refunds the unearned YRT reinsurance premium when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual YRT premium, times the remaining months in the year divided by 12.

5.2.4 Reinsurance benefits

The reinsurance benefit paid to the ceding company each month is the long term care benefit times the ceded amount. The ceded amount is the amount calculated at the time of each YRT reinsurance premium.

Reinsurance benefits do not include premium waiver.

5.2.5 Reinsurance reserve

The statutory and tax reserve credit for YRT reinsurance for long term care is equal to the unearned reinsurance premium.

5.3 Coinsurance

Coinsurance treaties cede a portion of the entire policy, including transferring the ceded reserve to the reinsurer. Reinsurance benefits include all long policy benefits.

5.3.1 Coinsurance percent

When the coinsurance percent is calculated using the input for the treaty, the percent is x% of the total policy benefits.

The coinsurance percent applies to the entire policy.

The coinsurance benefits paid to the ceding company each month are the projected policy benefits, times the coinsurance percent.

5.3.2 Coinsurance premium

Coinsurance premiums are paid at the same premium frequency as the model point.

Coinsurance premiums are based on the gross premium, including modal loading.

The unearned coinsurance premium is refunded to the reinsurer when the policy terminates before the end of the policy year for any reason. The unearned premium is the annual amount times the remaining months in the year divided by 12.

5.3.3 Premium tax

The reinsurer reimburses for the coinsurance percent times premium taxes paid by the ceding company.

5.3.4 Benefits

The reinsurer reimburses for the coinsurance percent times the long term care claims paid, premiums waived, and return of premium benefits.

5.3.5 Expense allowance

The reinsurer pays an expense allowance to the ceding company at the reinsurance premium frequency. The expense allowance is the annualized reinsurance premium times a specified percent.

The unearned current year expense allowance is refunded to the reinsurer when the policy terminates before the next reinsurance premium payment for any reason. The unearned expense allowance is the last amount paid, times the ratio of the number of months until the next reinsurance premium, divided by [12 divided by the reinsurance premium frequency].

5.3.6 Reinsurance reserve

The reinsurance reserve credit for coinsurance treaties is the statutory contract reserve, plus the statutory return of premium reserve, plus the statutory unearned premium reserve, plus the statutory base claim reserve, times the coinsurance percent. The claim expense reserves are not included since the ceding company bears the full amount of the claim expenses.

5.4 Modified coinsurance

Modified coinsurance (ModCo) treaties operate as described above for coinsurance treaties, except that the reinsurance reserve is retained by the ceding company. The ceding company reimburses the reinsurer for interest on the reinsurance reserve which is backed by assets invested by the ceding company.

The adjustments occur monthly, so that nested projections match standard projection results. The adjustment amount is the current reinsurance reserve, less the reinsurance reserve as of the previous adjustment date accumulated with interest.

The ModCo reserve adjustment interest rate is the asset yield rate.

5.5 Reinsurance administration expenses

Reinsurance administration expenses are assumed to be incurred at the end of each month; one twelfth of the annual rate applies each month. Reinsurance expenses can include:

- Amount per policy
- Percent of the ceded amount, which is specified as a percentage of the greatest of the maximum daily benefits for each care type.

Inflation, and the expense adjustment multiple (varying by calendar date), applies to reinsurance administration expenses the same as to maintenance expenses.

6 MODEL POINT RECORDS

6.1 Model point record layout

The model point record layout for the long term care product type is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	up to 125 characters (assigns model point to report group)
B	seg_id	segment ID
C	mp_id	up to 12 characters unique name
D	use_mp	Yes (include in projection), No
E	issue_date	yyyy/mm/dd or mm/dd/yyyy
F	plan_code	up to 20 characters (link to assumptions by plan)
G	stat_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
H	tax_valn_code	up to 20 characters (link to valuation basis and assumptions by valuation code)
I	issue_age	positive integer
J	gender	Male, Female, Unisex
K	rate_class	Standard, Preferred, Super, [extend with user additions]
L	tax_defn	Qualified, Non Qualified
M	marital_defn	Married, Single, Initially Married Now Single
N	pmt_style_alf	Reimbursement, Indemnity
O	pmt_style_hhc	Reimbursement, Indemnity
P	pmt_style_nhf	Reimbursement, Indemnity
Q	issue_state	Full name or abbreviation of state (i.e. Florida, FL)
R	sub_plan_1	up to 20 characters
S	sub_plan_2	up to 20 characters
T	prem_status_defn	Active, RPU

Excel Column	Variable	Value
U	prem_term_defn	Duration, Age
V	prem_term_mp	positive integer
W	prem_mode	Annual, Semi-annual, Quarterly, Monthly, Special Monthly
X	prem_waiver_defn_alf	Yes, No
Y	prem_waiver_defn_hhc	Yes, No
Z	prem_waiver_defn_nh	Yes, No
AA	prem_waiver_elim_days_alf	positive integer
AB	prem_waiver_elim_days_hhc	positive integer
AC	prem_waiver_elim_days_nh	positive integer
AD	init_annualzd_prem	annualized premium at the projection date
AE	bio_defn	Not Applicable, Compound, Simple
AF	bio_term_defn	Policy Year, Attained Age
AG	bio_term	positive integer
AH	bio_rate	floating point number
AI	ben_term_defn	Years, Dollars
AJ	pol_limit_defn	Additive, Combined
AK	max_daily_ben_def	Projection, Issue
AL	ben_base_comb	positive number
AM	ben_base_hhc	positive number
AN	elim_days_hhc	positive integer
AO	init_max_daily_ben_hhc	positive number
AP	ben_base_alf	positive number
AQ	elim_days_alf	positive integer

Excel Column	Variable	Value
AR	init_max_daily_ben_alf	positive number
AS	ben_base_nh	positive number
AT	elim_days_nh	positive integer
AU	init_max_daily_ben_nh	positive number
AV	init_policies	initial policies for mp (required)
AW	rop_death_defn	Yes, No
AX	rop_lapse_defn	Yes, No
AY	rop_maturity_defn	Yes, No
AZ	init_rop_ben	positive number
BA	claim_date	yyyy/mm/dd or mm/dd/yyyy
BB	init_policies_on_claim_hhc	positive number
BC	init_stat_claim_res_hhc	positive number
BD	init_stat_prem_waiver_res_hhc	positive number
BE	init_policies_on_claim_alf	positive number
BF	init_stat_claim_res_alf	positive number
BG	init_stat_prem_waiver_res_hhc	positive number
BH	init_policies_on_claim_nh	positive number
BI	init_stat_claim_res_nh	positive number
BJ	init_stat_prem_waiver_res_hhc	positive number
BK	rbc_c2_loss_ratio_curr_yr	positive number
BL	rbc_c2_loss_ratio_prev_yr	positive number

6.2 Reinsurance model point record layout

The reinsurance model point record identifies which treaties apply for each model point, using the linked variable, mp_id. The reinsurance model point record can have any number of reinsurance treaties per model point. It also specifies the ceded percent for each treaty as described in the table below. If the reinsurance model points are omitted from the reinsurance model point file, then the liability model point will not include reinsurance.

Warning! The order of reinsurance model points must be in the same order as the liability model points.

The reinsurance model point record layout is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	mp_id	Up to 12 characters. Links to liability data file mp_id. This value can be listed any number of times to indicate the number of treaties applied to the policy.
B	treaty_code	A user defined input that references the treaty code in the Resinsurance.xlsm workbook.
C	ceded_amt_defn	Percent - the ceded amount should be input as a percent from 0 to 1 Treaty Input - the ceded amount should be input as 0, and the ceded amount will be calculated using the treaty variables
D	ceded_amt	See above. Input the ceded percent, the ceded dollar amount, or 0 if the ceded amount is specified by the treaty.

6.3 Required model point variables

6.3.1 Always required

The following model point variables are required in all cases:

- **Model point identification:** group, seg_id, and mp_id
- **Insured characteristics:** issue_age, gender, rate_class
- **Policy specifications:** issue_date, plan_code, prem_status, prem_term_defn, prem_term_mp (zero indicates that premiums are paid in all years), prem_mode, ben_term_defn, pol_limit_defn, max_daily_ben_def
- **Annual premium:** init_annualized_prem is required when the annual premium is specified by model point.
- **Benefit inflation:** bio_defn must be specified as Yes or No. Other bio_xxx variables should be specified when benefit inflation is included.
- **Care type definitions:** Definitions for at least one care type must be included for each model point. This would include, for example, ben_base_alf, elim_days_alf, and init_max_daily_ben_alf
- **Initial policies:** init_policies

- **Return of premium:** rop_death_defn, rop_lapse_defn, and rop_maturity_defn each must be specified as Yes or No. Other 'rop' variables should be specified when the return of premium feature is included.
- **Risk based capital:** Loss ratios specified in rbc_c2_loss_ratio_curr_yr, and rbc_c2_loss_ratio_prev_yr

6.3.2 On claim data

The following model point variables are required for an on claim case:

- **Claim date:** claim_date
- **Policies on claim:** For example, init_policies_on_claim_alf
- **Statutory claim reserve:** For example, init_stat_claim_res_alf. The claim payment run-off calculated based on the first principles or claim cost claim assumptions for the policies on claim per the average claim date are adjusted by a multiple so that they equate to the specified claim reserve. However, when the calculated starting claim reserve is zero and the entered amount is not, the reserve is released in the first projection month as a paid claim and a warning message is issued. If this value is not included, a starting claim reserve is calculated using the model assumptions.
- **Statutory premium waiver reserve:** For example, init_stat_prem_waiver_res_alf. The premium waiver benefit run-off calculated based on the first principles assumptions for the policies on claim per the average claim date are adjusted by a multiple so that they equate to the specified premium waiver reserve. However, when the calculated starting premium waiver reserve is zero and the entered amount is not, the reserve is released in the first projection month as a premium waiver benefit and a warning message is issued. If this value is not included, a starting premium waiver reserve is calculated using the model assumptions.

7 SUBMODELS STRUCTURE

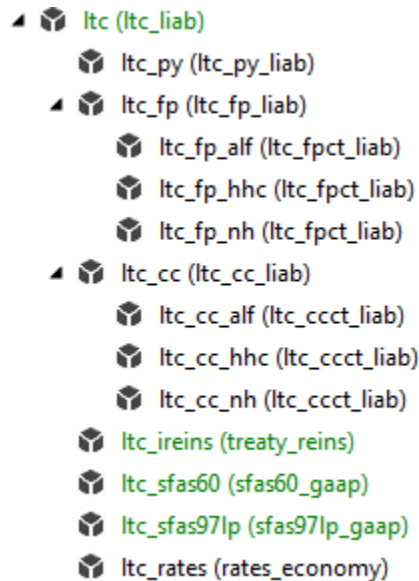
7.1 ltc submodel

The ltc submodel calculates the model point's contribution to the liability income statement and balance sheet items for long term care products.

7.1.1 Seriatim versus portfolio runs

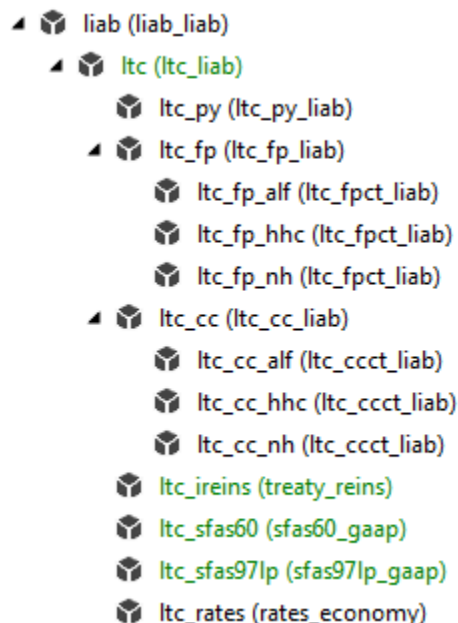
Seriatim projection

When ltc is the top model (seriatim projection or valuation), the following shows the entire submodel tree. In this configuration, each model point is a standalone instance of the ltc submodel. Full profit results are calculated for each model point, including GAAP, investment income, target capital, taxes, and distributable earnings.



Portfolio projection

For portfolio projections (which include business segment calculations), this product submodel tree is connected to the projection through the 'liab' submodel, as shown below. In this configuration, the ltc submodel is a data driven array sized to the number of model points included in the model point data file. In this configuration, the ltc submodel calculates only the model point results which are aggregated for the segment calculations.



GAAP, investment income, target capital, taxes, and distributable earnings are not included in the model point calculations. These results are calculated by aggregating from the ltc submodel the same columns as for other product types. This makes the long term care product type interchangeable with other product types fitting into this business segment structure.

7.1.2 Variables

Input variables for specifying long term care products are located in the ltc submodel, except as noted below for the variables in the ltc_fp, ltc_fp_xxx, ltc_cc, ltc_cc_xxx, and ltc_ireins submodels.

7.1.3 Formulas

Formulas calculating profit results are calculated only for seriatim projections, i.e., when ltc is the top model. These include GAAP, investment income, target capital, taxes, and distributable earnings results.

Key formulas in the ltc submodel include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each iteration. The startup formulas are split into three sections: (1) when only ltc is the top model, (2) when ltc is included within the portfolio structure, and (3) code executed in both cases.

Conditions using the 'isMainModel' system variable differentiate the startup processing when long term care is run as the top model and when long term care is included in projections with segment as the top model. When segment as the top model, references to items in the segment and liab submodel are used to 'pull' down values which each long term care model point requires. References to segment and liab are invalid when long term care is the top model, and must be avoided through the use of conditions checking on the isMainModel system variable.

The startup column does the work to set up each model point: (1) arrays are initialized and sized for use, (2) temporary tables are sized for DAC tax calculations and competitor rates, (3) projection flags are created and initialized, (4) reporting groups are set up, (5) ltc_ireins elements are created, (6) claim projection submodels are created, (7) model point data validations are processed, etc.

Initialize

The 'initialize' column is automatically executed during model initialization for each model point for each iteration, immediately after startup when ltc is the top model. When company is the top model, initialize is executed by calls from seg->initialize which call liab->initialize, which executes ltc->initialize for each model point.

Again, conditions using the isMainModel system variable manage the initializations which are only required when ltc is running as the top model. This includes executing the initialization of economic scenario rates and the setup of competitor rates. Other initializations are required in both cases.

The main use of initialize is to setup input and calculation arrays, drive the initialization of the claim projection submodels, and to calculate the contract reserve factors for each model point using the actuarial math library. Since these initializations are not required for all calculation types for all product features, conditions are checked to shortcut the initialization process, thus reducing the runtime of the model point.

Other formulas

Other formulas in the ltc submodel are relatively straightforward. Results which are per unit in force add '_per_unit' to the end of the formula name. Results without '_per_unit' are values for the model point.

Most of the profit, balance sheet, and GAAP columns are only calculated when ltc is the top model; these columns are the income statement and balance sheet results which are calculated in aggregate during portfolio projections.

7.2 ltc_fp and ltc_fp_xxx submodels

The ltc_fp submodel and ltc_fp_xxx submodels contains the variables and formulas required to calculate claims paid, return of premium benefits, claim reserves, and policy exhibits for individual and then combined care types for long term care model points when the claims projection definition is First Principles or First Principles Approximation.

The ltc_fp submodel is a standalone instance of the ltc_fp/liab model class. Each of ltc_fp_alf, ltc_fp_hhc, and ltc_fp_nh are standalone instances of the ltc_fpct/liab model class.

7.2.1 **Care type instances of ltc_fp_xxx**

Each model point carries only the instances of ltc_fp_xxx it requires for the care types of the model point. This is managed through the ability to 'shrink' the submodel during the startup processing.

A scalar is calculated for each care type within the ltc_fp submodel based on the specification of the maximum daily benefit on the model point record. If the scalar evaluates to 0, the ltc_fp_xxx submodel is 'shrunk' during the startup of ltc_fp and saves model point memory. These flags are used in the formulas to ensure that the ltc_fp_xxx submodels are not referenced when the submodel has been 'shrunk'.

7.2.2 **Variables**

Input variables for specifying long term care first principles assumptions are located in the ltc_fp and ltc_fp_xxx submodels.

The ltc_fp submodel contains the variables which define treatment across all care types. This includes the definitions for monthly terminations period, whether transfers and recovery are included in a full first principles calculation, and the transfer percentages.

The ltc_fp_xxx submodels contain variables for each care type. This includes the benefit utilization multiples, experience incidence and termination assumptions, and contract reserve incidence and termination assumptions.

7.2.3 **Formulas**

Formulas for determining long term care results under the claim costs calculation method are contained in the ltc_fp and ltc_fp_xxx submodels.

Key formulas in these submodels include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each iteration. The startup formulas for the first principles submodels are only executed if a first principles approach is used, and the submodels have not been 'shrunk' by startup of ltc.

Initialize

The 'initialize' column of ltc_fp is executed through a call from the ltc submodel when a first principles approach is used.

The main use of initialize is to setup input and calculation arrays, calling the setup array function, and for the initialize of ltc_fp, to initialize the submodels.

The setup array function of the ltc_fp_xxx submodels is critical to the first principles processing. This function develops input arrays for benefit utilization, incidence rates, and termination rates. When the first principles approximation approach is used, this function develops arrays for the policy year based claim costs, claims paid, and claim reserves. When the full first principles approach is used, this function just initialized the claim and transfer arrays.

'calc' functions and 'populate' columns

The first principles calculations require a number of external functions to drive the development and tracking of a multi-dimensional array of claim information. For example, the functions calc_claim_array, update_claim_array, sum_over_claim_array, and calc_claim_res are key functions for the development, tracking, summations, and calculations over the full first principles results.

Since the arrays must be developed and used in a procedural fashion, special columns are used to drive the process and ensure the arrays are in the proper state before the results are used. These columns include `populate_exper_claim_array`, and `populate_exper_non_transfer_array`.

There are similar functions and driver columns for the first principles approximation approach.

Other formulas

The cash flow and balance sheet formulas in the `ltc_fp` and `ltc_fp_xxx` submodels are relatively straightforward, using 'sum_over' and the 'calc' functions as necessary to retrieve the results from the calculation array and then combine individual care type results together.

7.3 ltc_cc and ltc_cc_xxx submodels

The `ltc_cc` submodel and `ltc_cc_xxx` submodels contains the variables and formulas required to calculate claims paid, return of premium benefits, claim reserves, and policy exhibits for individual and then combined care types for long term care model points when the claims projection definition is Claim Costs.

The `ltc_cc` submodel is a standalone instance of the `ltc_cc/liab` model class. Each of `ltc_cc_alf`, `ltc_cc_hhc`, and `ltc_cc_nh` are standalone instances of the `ltc_ccct/liab` model class.

7.3.1 Care type instances of ltc_cc_xxx

Each model point carries only the instances of `ltc_cc_xxx` it requires for the care types of the model point. This is managed through the ability to 'shrink' the submodel during the startup processing.

A scalar is calculated for each care type is within the `ltc_cc` submodel based on the specification of the maximum daily benefit on the model point record. If the scalar evaluates to 0, the `ltc_cc_xxx` submodel is 'shrunk' during the startup of `ltc_cc` and saves model point memory. These flags are used in the formulas to ensure that the `ltc_cc_xxx` submodels are not referenced when the submodel has been 'shrunk'.

7.3.2 Variables

Input variables for specifying long term care claim costs assumptions are located in the `ltc_cc_xxx` submodels. There are not variables in the `ltc_cc` submodel.

The `ltc_cc_xxx` submodels contain variables for each care type. This includes the assumptions for the per policy claim costs and completion factors.

7.3.3 Formulas

Formulas for determining long term care results under the claim costs method are contained in the `ltc_cc` and `ltc_cc_xxx` submodels.

Key formulas in these submodels include:

Startup

The 'startup' column is automatically executed during model initialization for each model point for each iteration. The startup formulas for the first principles submodels are only executed if a claim costs approach is used, and the submodels have not been 'shrunk' by startup of `ltc`.

Initialize

The 'initialize' column of `ltc_cc` is executed through a call from the `ltc` submodel when the claim costs approach is used.

The main use of initialize is to setup input and calculation arrays, calling the `setup_array` function, and for the initialize of `ltc_cc`, to initialize the submodels.

The `setup_array` function of the `ltc_cc_xxx` submodels is critical to the claim costs processing. This function develops input arrays for the claim costs.

'calc' functions and 'populate' columns

The claim costs calculations are much simpler than the first principles approach and only requires the development, tracking, and summation over a single completion_claim_pmt array. This array contains the future claim payments for each incurral based on the claim costs assumptions and completion factors.

Since the array must be developed and used in a procedural fashion, the special populate_claim_array is used to drive the process and ensure the array is in the proper state before the results are used. The single 'calc_claim_res' function is used to calculate the claim reserve using the values of the completion_claim_array.

Other formulas

The cash flow and balance sheet formulas in the ltc_cc and ltc_cc_xxx submodels are relatively straightforward, using 'sum_over' and the 'calc' functions as necessary to retrieve the results from the calculation array and then combine individual care type results together.

7.4 ltc_ireins submodel

The ltc_ireins submodel is a dependent data driven array instance of the treaty_reins model class. It contains the variables and formulas required to calculate the ceded amount and reinsurance cash flows for ltc model points.

7.4.1 Multiple instances

Each model point carries multiple instances of ltc_ireins. The number of instances is determined by the number of treaties specified on the reinsurance model point record.

Elements of ltc_ireins are created during the startup processing of the model point. Conditions are evaluated in startup of ltc and the ltc_ireins user defined array is resized as required. The characteristic for each ltc_ireins element is specified, and the reinsurance variables are read using the ms_rereadGrid() system function.

The ceded amount and reinsurance cash flows for multiple treaties are aggregated using the sum_over_reinsurance function in the ltc submodel. This function can aggregate the values for all treaties, or just the treaties of a specific type of reinsurance, such as YRT.

7.4.2 Variables

Input variables for specifying the reinsurance type, the reinsurance expenses, and for calculating the ceded amount (when it is not specified on the reinsurance model point record), are located in the vul_ireins submodel. These variables are contained in the ltc_ireins submodel so that they vary by treaty.

7.4.3 Formulas

Formulas for calculating the ceded amount and the reinsurance cash flows are contained in the ltc_ireins submodel.

Since ltc_ireins is defined through the model parameters, the size of the ltc_ireins array is based on the number of treaties specified for each model point on the reinsurance model point file.

7.5 ltc_py submodel

The ltc_py submodel is a standalone instance of the ltc_py/liab model class. It contains the variables and formulas required to calculate per policy in force results which only vary by policy year. Thus 't' in this submodel represents policy year, unlike the ltc and ltc_ireins submodels where 't' represents months since the projection date. The policy year values are calculated from issue.

Since all results are per policy, this submodel does not include '_per_policy' in the formula names.

7.5.1 Variables

There are no variables in the ltc_py submodel. Variables required for the formulas in the ltc_py submodel are referenced from ltc as required using the ltc_py submodel pointer.

7.5.2 Formulas

Formulas for calculating policy year per policy in force results are included in the ltc_py submodel. Formulas for ltc_py are used to calculate annual values, plus reserves by policy year.

Columns are generally calculated beginning at policy year 1, even for in force model points. As discussed in the ltc submodel section, the reserve net premium and terminal values of ltc_py are not calculated on demand through the target column. Instead, these results are calculated in advance through forced calls from the 'initialize' column of ltc.

7.5.3 Actuarial math library

Reserve net premiums and terminal values are calculated by calling the Actuarial math library. See the Actuarial math library appendix book.

7.6 ltc_rates submodel

The ltc_rates submodel is identical to the rates submodel which applies for portfolio projections.

Please refer to the C02 Economic scenarios book.

7.7 ltc_sfas60 submodel

The ltc_sfas60 submodel is identical to the sfas60 submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

7.8 ltc_sfas97lp submodel

The ltc_sfas97lp submodel is identical to the sfas97lp submodel which applies for portfolio projections.

Please refer to the C04 GAAP accounting book.

7.9 Actuarial math library

7.9.1 AML DLL

The Actuarial math library (AML) provides standard actuarial calculations for traditional reserve and cash value methods. See the Actuarial math library appendix book.

The ltc submodel calls the AML to calculate reserve and cash value amounts per unit in force. Columns in the ltc_py submodel store the terminal value and net premium for each policy year for each cash value and reserve basis used in the projection.

7.9.2 term->initialize

The ltc->initialize column initiates the call to the AML calculations for each set of cash value and reserve results. Each set of AML results is calculated by calling the aml_driver in the ltc_py submodel. A separate call can be made for each of the following results:

- Cash values – Base plan cash values
- Statutory reserves – Base plan statutory reserves

- Statutory minimum reserves – Base plan minimum reserves for calculating deficiency reserves
- Tax reserves – Base plan tax reserves

7.9.3 AML driver

The `aml_driver` in the `ltc_py` submodel sets up and calls the AML for each set of cash value and reserve results. This driver is called by the `ltc->initialize` column for each set of results.

This driver sets up the call to the AML using the appropriate input for each type of calculations. Each call to the driver calls these functions in the `ltc_py` submodel:

- `setup_aml_initial_data` – Creates the interfaces for using the AML.
- `setup_aml_res_defn` – Specifies the cash value or reserve method (such as NLP, FPT, CRVM, SNFL, etc.), and the function type (curtate, continuous, or semi-continuous).
- `setup_aml_policy_data` – Specifies values for each policy year, such as the guaranteed premiums, death benefits, endowment benefits, valuation interest rates, mortality rates, plus the issue age, maturity age, premium term years, expense allowance grading years, and the name extension for the audit trail Excel file (when requested).

The operation of these setup functions varies by the type of calculations being made. For example, the policy data passes the cash value interest and mortality rates when calculating cash values, and the statutory rates when calculating statutory reserves.

7.9.4 Calculation results

The AML calculation results are stored for each policy year in columns in the `ltc_py` submodel. The formula for each column simply gets the AML calculation results for each case. These columns include:

- Terminal cash value and reserves and net premiums for each basis
- Valuation interest and mortality rate for each basis
- Tabular cost of insurance for each basis

The `aml_driver` calls the appropriate columns after it calls the AML. For example, the cash value basis columns are called when the driver has just calculated cash value results. Then the driver calls the statutory reserve basis columns when those results have just been calculated by the AML. Thus each of these columns stores the AML results using the appropriate valuation basis.

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L12 EXTERNALLY PROJECTED LIABILITIES

This book discusses externally projected liabilities (EPL) and the epl submodel.

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1 EPL OVERVIEW

1.1 Portfolio projections

Portfolio projections produce results for each business segment. Results for each model point are summed into the segment (seg) submodel, using the 'sum_over_' functions, like sum_over_liabilities.

The Externally projected liabilities (EPL) feature enables some or all of the business segment results to come from a pre-calculated file, rather than from model points calculated during the projection. For example, these could be results projected using some other software system. They could also be results created during a previous USLMS projection.

Of course, the projection results coming from the EPL files must have been projected consistently with the current projection. The EPL files must contain results for each projection period, and for each projection loop. The EPL results must have been projected using the same scenario rates in the case where values vary by economic conditions.

Since EPL values are only summed in the projection, the only input variables are on the EPL model point records, and the EPL files. Variables on the other input pages do not affect EPL results.

EPL files do not apply for seriatim projections.

1.2 EPL model points

1.2.1 Multiple EPL files

The EPL results could be specified in a single EPL file containing results for all the business projected separately from the U.S. Library projection. Alternatively, there could be multiple EPL results. For example, there could be:

- Separate EPL results for each GAAP and GPR cohort
- An EPL file for results coming from multiple different software models
- For GAAP valuations and projections, each model point might need a separate EPL file, such as for SFAS60 where model point factors apply.

Each EPL file is a comma separated values file. The rows are the values for each projection period. The columns are each value to be summed into the business segment.

1.2.2 EPL model point definition

Each EPL values file is specified by an EPL 'model point' record. This specifies:

- The name of the EPL file
- Whether to include this EPL file in the projection
- The assigned business segment, report group, GAAP cohort, and GPR cohort
- Whether to include reinsurance values – ceded results must be in the EPL file when reinsurance applies.
- The percent of the EPL results to use – a multiple applied to each value in the EPL file. The percent can be greater than 1.0 (100%), such as for grossing up the separate projection results to represent the full block of business.

The model point record layout is discussed below.

1.3 Policy values and cash flows

EPL policy values, such as fund values, cash values, and death benefits in force, are results as of the end of each month.

Each EPL cash flow is assumed to occur based on the timing conventions for projected liabilities. For example, premiums are assumed to be paid at the beginning of each month, while death benefits are assumed to be paid at the end of each month.

1.4 Statutory and tax reserves

For projections using formulaic reserves, the EPL file specifies the reserve values at the end of each month.

For projections calculating principle-based reserves and capital, the EPL file provides the policy values and cash flows used in the PBA calculations.

For projections with PBA revaluations, separate EPL values are required for each revaluation period along each outer loop, in addition to the externally projected liability values for the outer loop. This is necessary since the EPL values change at each inner loop branch along each outer loop.

1.5 GAAP accounting

GAAP projections require formulaic based GAAP liabilities and DAC, when applicable, as well as the policy values and cash flows required for cohort valuation and unlocking projections.

For projections with formulaic GAAP reserves, such as for SFAS60, SFAS97 Limited Pay, and SFAS120, the EPL GAAP values are the model point level GAAP liabilities and DAC.

For valuations of GAAP cohorts, such as for SFAS91, SFAS133, SFAS97 Retrospective Deposit, and SFAS120, the EPL file provides the policy values and cash flows representing the model point level contributions to the GAAP cohort.

For projections with unlocking or loss recognition of GAAP liabilities and DAC, separate EPL values are required for each unlocking or loss recognition period along each outer loop, in addition to the externally projected liability values for the outer loops.

GAAP cohort level projections require an approximation regarding the length of the cohort amortization period. This is required since the elapsed months of model point(s) is not included in the EPL file. The projection assumes that the last issue period of the externally projected liability is the projection date.

2 MODEL POINT RECORDS

The EPL model point record layout is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	seg_id	Business segment ID
B	use_mp	Yes (include the EPL file in projection), No
C	pct_to_use_mp	Multiple applied to the EPL file values (1.0 for 100%)
D	reins_defn	Yes (include reinsurance in projection), No

Excel Column	Variable	Value
E	gaap_cohort	GAAP cohort name; Unassigned when GAAP does not apply
F	gpr_cohort	LOC GPR cohort name; Unassigned when GPR does not apply
G	epl_file	EPL file path and name, up to 300 characters, typically using <*data*> wildcard and <*scenario*> wildcard for inner loop valuations within time-step projections

The GAAP and GPR cohort names must be valid names specified in the cohort definitions for the projection.

3 EPL FILES

3.1 File type and name

Each EPL file must be a comma separated value file with an extension of '.csv'. The file must contain results for each projection task loop for the projection. A common practice is to start the file name with 'epl_'.

When inner loop valuations apply for nested projections, the file name must also include the projection set loop and sub loop numbers.

3.2 Record format

Each EPL file contains a header row of item names followed by monthly data with rows for each projection month starting at month 0. The header for each column in the EPL file must match the name of the MoSes column included in the epl submodel for the value represented in the column.

The first two columns must contain the projection task loop number and the projection period. These first two columns must be in ascending order. The projection period must begin at 0 (projection date).

3.3 Flexible contents

Following the first two columns, any number of values may be included in the EPL file, as required to achieve the desired projection results for a representative product type. The order of the columns does not matter.

The items available for use in the EPL file are the columns included in the epl submodel. These are the 'key results' which are summed for business segment calculations using one of the 'sum_over_' functions in seg submodel formulas. For example, sum_over_liabilities aggregates all the model points assigned to the business segment. The function sum_over_sfas60 aggregates the model points assigned to each SFAS60 GAAP cohort within the business segment.

A general rule to apply for determining whether a column should be included is to evaluate whether the column will be used within the 'sum_over_' functions. When considering the columns for GAAP or PBA projections, remember that the full name of the column for the EPL file includes the prefix as discussed in the Column name convention section in the Submodels structure chapter, below.

The file format is flexible; it can contain only the key result columns that apply for the business contained in the EPL file. The 'header row' in the .csv file specifies the variable name being defined. Any columns not included in the file will simply contribute zero values to the projection.

For example, include a column with the header 'deferred_gross_prem' when this value applies, and exclude this column otherwise.

There is no error raised for a missing or misnamed column. The requested value is simply treated as zero.

3.4 Sample EPL file

A sample EPL file structure for a projection of two iterations and two periods is as follows:

task_loop	period	prem_paid	dth_benefits	stat_res_mp	tax_res_mp	cash_val
1	0	0	0	5000.00	4500.00	2500.00
1	1	1000.00	100.00	5800.00	5300.00	3300.00
1	2	1000.00	110.00	6600.00	6100.00	4100.00
2	0	0	0	5000.00	4500.00	2500.00
2	1	1250.00	125.00	5900.00	5400.00	3400.00
2	2	1250.00	150.00	6750.00	6250.00	4250.00

4 SUBMODEL STRUCTURE

4.1 Submodel structure

The epl/liab product/purpose contains the formulas and variables for EPL processing.

The following describes the model tree:

- epl – a data driven array of the epl_liab model class used to determine EPL results. The number of elements in the array is fixed at the number of elements in the EPL model point data file.

4.2 Column name convention

4.2.1 General column names

The columns in the epl submodel generally have the same name as the columns for the various product type submodels. These are the column names which are passed as parameters to the 'sum_over_' functions in the seg submodel. For example, the column name for the premiums paid is prem_paid, the same name of the column for each product type.

However, a special naming convention is required for GAAP and PBA columns.

4.2.2 GAAP column names

GAAP related columns must be separated by gross versus ceded values. These columns are named sfasXXX_gross_{item name} and sfasXXX_ceded_{item name}. For example:

```
sfas60_gross_ben_res_prelim
sfas60_ceded_ben_res_prelim
sfas91_gross_net_res_cash_flow_bom
sfas91_ceded_net_res_cash_flow_bom
sfas97lp_gross_deferred_prof_liab_prelim
sfas97lp_ceded_deferred_prof_liab_prelim
sfas97rd_gross_capzd_acq_exp
sfas97rd_ceded_capzd_acq_exp
```

sfas120_gross_terminal_div
sfas120_ceded_terminal_div
sfas133_gross_gmib_claim
sfas133_ceded_gmib_claim

The 'sum_over_' GAAP functions include logic to prepend sfasXXX_gross or sfasXXX_ceded to the name of the column passed into the function based on the context of the model summation.

4.2.3 PBA column names

Similarly, all PBA related columns must be separated by PBA type. These columns are named XXX_pba_{item name}. For example:

life_pba_working_res
life_pba_reins_res
ann_pba_working_res
ann_pba_reins_res
va_pba_working_res
va_pba_std_scen_ben_costs

The sum_over_pba function includes logic to prepend XXX_pba to the name of the column passed into the function based on the context of the model summation.

L13 RIDERS

This book discusses riders that apply to term life, traditional life, universal life, indexed universal life, and variable universal life products.

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1 INPUT ORGANIZATION

1.1 Input pages

Variables defining the rider operations are associated with one of these input pages:

- Riders – Benefits and premiums definitions
- RidersValuation – Statutory and tax reserves and actuarial math library audit report option

1.2 Data pages

Variables for each rider model point are specified on the [Prod]RiderData page, where [Prod] is the product name. The rider data variables are discussed in the Model point records chapter, below.

2 RIDER BENEFITS AND VALUES

The following explains the basic features for riders.

2.1 Rider variations

While the standard features are flexible for defining many rider configurations, the following are common types of riders which are currently supported by the U.S. Library.

2.1.1 *Accidental death benefit*

A level accidental death benefit (ADB) amount may be specified on the rider data file, and the benefit will be paid based on the set of accidental death incidence rates that are input by the user.

2.1.2 *Waiver of premium*

A waiver of premium rider (WOP) may be specified on the rider data file, and the premium obligations of the policyholder will be waived upon disablement, given a set of disablement rates input by the user. The premium obligations include the base policy premiums and all rider premiums.

The model does not project new disablements or the benefits from already disabled cases in projections. This rider only impacts reserves for point-in-time statutory and tax valuations.

2.1.3 *Waiver of COI's and other charges*

For universal life, variable universal life, and indexed universal life products, a waiver of COI's and other charges rider (WOC) may be specified on the rider data file, and the guaranteed cost of insurance charges and other rider charges of the policyholder will be waived upon disablement, given a set of disablement rates input by the user.

The model does not project new disablements or the benefits from already disabled cases in projections. This rider only impacts reserves for point-in-time statutory and tax valuations.

2.2 Benefit term

2.2.1 *Maturity*

The maturity (expiry) date of the rider is either a maximum attained age or policy anniversary, but not greater than the statutory reserve mortality table omega age of the base policy. When this value is zero (default value), rider coverage expires at the omega age of the base policy.

2.3 Death benefits

2.3.1 *Sum assured*

For accidental death benefit riders (ADB), the death benefit is the sum assured specified on the rider data file. For Accidental Death Benefits, this is a level amount, such as \$1,000 per unit.

2.4 Premiums

For term and traditional life products, the rider premiums are treated as a cash flow, similarly to the base plan premiums.

For universal life, variable universal life, and indexed universal life products, the rider premiums are treated as a charge to the current fund at the beginning of the month.

2.4.1 *Premium term*

Rider premiums are paid until either a policy year or attained age. When this value is zero (default value), premiums are paid to maturity of the rider.

2.4.2 *Premium mode*

The premium mode of the rider is set equal to the premium mode of the base policy.

2.4.3 *Annual premium*

Specify the annual premium per unit for each policy year for the rider in the annual premium per unit variable.

2.4.4 *Substandard premium*

The annual premium per unit is adjusted by the substandard multiple specified on the base model point record.

2.4.5 *Modal loading*

For riders that apply to term and traditional life products, when premiums are paid more frequently than annually, the annualized premium is the annual premium times the annualized modal loading multiple varying by premium mode. The modal loading multiple for the rider premiums is set equal to the modal loading multiple for the base policy.

Modal loading does not apply to universal life, indexed universal life, or variable universal life products.

2.4.6 *Policy fee*

There are no policy fees applied to rider premiums.

2.4.7 *Guaranteed premium*

For applicable rider reserves (ADB, WOP, and WOC), the reserve net premiums are based on the pattern of guaranteed premiums by policy year, which is the same as the pattern of current rider premiums.

Rider reserves are calculated using only the annual premium per unit, without adding the modal loading or substandard multiples.

3 STATUTORY AND TAX RESERVES

The following discusses formulaic reserves for riders.

3.1 Contract reserves

Contract reserves are established as a reserve for the claims expected to be incurred. Separate terminal reserves and net premiums are established for claim costs.

For claim cost contract reserves, per policy terminal reserves and net premiums are applied to all riders regardless of whether the base policy is on claim.

3.1.1 Terminal reserves

The following rider reserve methods are available:

- Not applicable – reserves are zero, and don't add to projection time.
- Net level premium – no expense allowance

3.1.2 Premium pattern

Reserve net premiums are the level percent of gross premiums calculated using the specified reserve method. The reserve method applies to the pattern of guaranteed premiums without modal loading or substandard multiples.

3.1.3 Benefits treatment

For ADB riders, the claims cost benefit is simply the accidental death benefit specified in the rider data file.

For WOP riders, the claims cost benefit is the waived premium that otherwise would have been paid by the policyholder had they not been disabled. The waived premium includes the base policy premium and all rider premiums.

For WOC riders, the claims cost benefit is the waived guaranteed COI charge that otherwise would have been deducted from the fund value had the policyholder not been disabled. The waived COI and other charge benefit includes all other rider charges.

3.1.4 Claims costs definition

Claim costs for contract reserving are the expected incurred claims for claims incurring in a policy year under valuation assumptions. They may be specified using in a table or be calculated within the application. The Table approach is an approximate approach which reduces runtime and complexity.

When the claim cost definition is 'Table', claims costs are specified using the `claim_costs` variable on the RidersValuation input page. Typically, the claim costs vary by benefit code, premium term since disability, years since disability and disablement age. For example, the claim costs could be specified using the mean reserves included in tables h or i of the *Transactions of Society of Actuaries, 1954 Vol 6 No. 14*.

When the claim cost definition is set to 'Calculated', claim costs are determined using a more exact 'first principles' approach where the claim costs are developed from specified contract reserving termination and interest rates.

3.1.5 Mortality

Rider reserve mortality is a base rate, times a multiple, then plus an addition. Rider reserve mortality typically varies similarly to the base plan reserve mortality rates.

Important! Supported riders cover only one life at a time. There is no second life for joint policies.

3.1.6 Incidence

Rider reserve incidence is the probability of an event occurring which triggers the payment of a rider benefit. Rider reserve incidence rates are a base rate, times a multiple, then plus an addition. Incidence rates typically vary similarly to rider reserve mortality rates.

The ADB incidence rate is the accidental death rate while the WOP and WOC incidence rates are the incidence of disability.

3.1.7 Termination

Rider contract reserve termination is the probability that the policyholder leaves the 'disabled life' status, either by becoming 'active' or dying. Termination rates are a base rate, times a multiple, then plus an addition. Termination rates typically vary by gender, elimination period, age of disablement, and months since disablement using external source lookup definitions.

Termination rates only apply to WOP and WOC riders when the claim cost definition is 'Calculated'. Termination rates do not apply if the 'Table' approach is used since those rates are implicit in the specified claim cost factors.

3.1.8 Interest

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeStatValn and LifeTaxValn ranges in the Valuation Interest Rates Excel workbook.

Important! The maximum standard nonforfeiture law does not stipulate a specific set of valuation rates for rider reserves. By default, the US Library uses the maximum standard nonforfeiture law valuation rates by issue year, but any valuation rate can be input by the user.

3.1.9 Function timing

Function timing can be:

- **Curtate** – Annual premiums and claims paid at the end of the policy year
- **Continuous** – Continuous premiums and immediate payment of claims
- **Semicontinuous** – Annual premiums and immediate payment of claims.

The continuous premium multiple and the immediate claim multiple are defined in the Function timing section in the Valuation options chapter of the Actuarial math library appendix.

3.1.10 Mid-year reserve

For riders linked to term and traditional life policies, rider reserves between policy anniversaries can be:

- Annual interpolated mean – Average terminal reserve, plus one-half of the annual reserve net premium. A deferred premium asset is established.
- Annual interpolated terminal – Average terminal reserves at the beginning and end of the policy year. Reserves are the same for each month within a policy year. An unearned premium reserve is established.
- Monthly interpolated mean – Monthly interpolated terminal reserves, plus the portion of the annual reserve net premium that applies to future months. A deferred premium asset is established.
- Monthly interpolated terminal – Monthly interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.
- Daily interpolated mean – Daily interpolated terminal reserves, plus the portion of the annual reserve net premium that applies to the remaining number of days of the year. A deferred premium asset is established.
- Daily interpolated terminal – Daily interpolated terminal reserves at the beginning and end of the policy year. An unearned premium reserve is established.

When the reserve function timing is Continuous, the mid-year reserve net premium is a discounted continuous net premium.

Warning. Interpolate mean reserves should not apply to UL, IUL, or VUL riders because deferred premium assets are not supported by those products.

3.1.11 *Tabular cost of insurance floor*

The rider reserve is subject to a minimum based on the tabular cost of insurance.

The floor is the tabular cost of insurance calculated using the rider incidence rate (c_x) times the sum assured, times the following fraction for each mid-year reserve method:

- Annual interpolated mean – $\frac{1}{2} c_x$.
- Monthly interpolated mean – c_x times the fraction of the policy year remaining. For example, this floor at the end of policy month 1 is $\frac{11}{12} c_x$.
- Annual interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment. For example, this floor at the end of policy month 1 is $\frac{11}{12} c_x$ for annual premiums, and $\frac{2}{12} c_x$ for quarterly premiums.
- Monthly interpolated terminal– Same as above.
- Daily interpolated mean – c_x times the fraction of the policy year remaining measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $\frac{344}{365} c_x$.
- Daily interpolated terminal – c_x times the fraction of the policy year remaining until the next premium payment measured in days. For example, if a policy is issued on 3/10 and the valuation date is 3/31, the floor at the valuation date is $\frac{344}{365} c_x$ for annual premiums, and $\frac{71}{365} c_x$ for quarterly premiums.

3.2 Claim reserves

Disabled life claim reserves apply to WOP and WOC riders. A disabled life claim reserve is calculated if the base policy is disabled at valuation. Disabled life claim reserves are calculated using date of disability, which is defined on the base policy data file.

The base claim reserve is the present value of future claim payments at a specified discount rate.

3.2.1 *Benefits treatment*

For WOP riders, the claim reserve benefit is the waived premium that otherwise would have been paid by the policyholder had they not been disabled. The waived premium includes the base policy premium and all rider premiums.

For WOC riders, the claims cost benefit is the waived guaranteed COI charge that otherwise would have been deducted from the fund value had the policyholder not been disabled. The waived COI and other charge benefit includes all other rider charges.

3.2.2 *Claim reserves definition*

Claim reserves are the present value of remaining claims for disabled lives. They may be specified in a table to be calculated within the application. The Table approach is an approximate approach which reduces runtime and complexity.

When the disabled lives reserves definition is 'Table', claim reserves are specified using the `dlr_res` variable of the RidersValuation input page. Typically, the claim reserves vary by benefit code, premium term since disability, years since disability and disablement age. For example, the claim reserves could be specified using the mean reserves included in tables h or i of the *Transactions of Society of Actuaries, 1954 Vol 6 No. 14*.

When the claim cost definition is set to 'Calculated', claim costs are determined using a more exact 'first principles' approach where the claim costs are developed from specified contract reserving termination and interest rates.

3.2.3 Termination

Rider claim reserve termination is the probability that the policyholder leaves the 'disabled life' status, either by becoming 'active' or dying. Termination rates are a base rate, times a multiple, then plus an addition. Termination rates typically vary by gender, elimination period, age of disablement, and months since disablement using external source lookup definitions.

Termination rates only apply to WOP and WOC riders when 'Calculated' is specified as the disabled lives reserve definition. Termination rates do not apply if the 'Table' approach is used since those rates are implicit in the user-defined claim reserve factors.

3.2.4 Interest

The Global folder includes an external source table containing the maximum standard nonforfeiture law valuation rates by issue year: See the LifeStatValn and LifeTaxValn ranges in the Valuation Interest Rates Excel workbook. The disabled life interest rates may be different than the interest rates used in the contract reserve calculation.

Important! The maximum standard nonforfeiture law does not stipulate a specific set of valuation rates for rider reserves. By default, the US Library uses the maximum standard nonforfeiture law valuation rates by issue year, but any valuation rate can be input by the user.

3.3 Tax reserve

Specifying 'Statutory' as the tax valuation code on the rider model point record sets the tax reserve to be the statutory reserve, including the comparison to the tabular cost of insurance floor using the statutory basis.

The same method choices are available for rider tax reserves as for statutory reserves, using the tax reserve interest, mortality, incidence, and termination basis. The function timing definition, and the mid-year reserve method, is the same as for statutory reserves.

3.4 Deferred premium asset

Deferred premium assets are calculated for riders that are linked to term or traditional life policies. The rider deferred premium assets are included with the base policy deferred premium asset balance sheet item. Please refer to the Deferred premium asset section of the Term or Traditional Life books for information regarding accounting for deferred premium assets.

Riders that are linked to a universal life, variable universal life, or indexed universal life policy do not calculate a deferred premium asset.

3.5 Unearned premium reserve

Unearned premium reserves are calculated for riders that are linked to a life policy. The rider unearned premium reserves are included with the base policy unearned premium reserve balance sheet item. Please refer to the Unearned premium reserve section of the Term Life, Traditional Life, Universal Life, Indexed Universal Life, or Variable Universal Life books for information regarding accounting for unearned premium reserves.

4 ACTUARIAL ASSUMPTIONS

This section only applies to ADB riders. WOP and WOC riders are only supported for point-in-time valuations and are ignored for projections.

4.1 Commissions

Rider commissions are calculated for riders that are linked to a term or traditional life policy. Riders that are linked to a universal life, variable universal life, or indexed universal life policy do not calculate commissions since the rider premiums are accounted as a charge to the current fund.

4.1.1 Premium commission

The rider premium can be commissionable (included) or noncommissionable (excluded). The percent of premium commission is the base policy commission rate for each policy year, times the rider premium payments as they are paid. The modal loading can be commissionable or noncommissionable based on how the base policy commissions are treated. The rider premiums are not subject to commission chargebacks.

4.2 Expenses

4.2.1 Premium tax

For riders linked to term and traditional life policies, the rider premiums are included in the premium taxes, which are reported with expenses. Riders that are linked to universal life, variable universal life, or indexed universal life policies do not calculate a premium tax since the rider premiums are accounted as a charge to the current fund.

4.3 Mortality

4.3.1 Input variables

'Mortality variables' for the applicable riders (ADB) include mortality base, multiple, and addition rates. These variables work the same as for the base plan mortality variables.

Important! Supported riders cover only one life at a time. There is no second life for joint policies. Specify two individual riders and link both of them to the base policy to establish a rider for each life of a joint policy.

The lookup keys are hidden variables which are set by the formula reading the hidden variable. The external source uses lookup keys mort_issue_age, mort_attained_age, mort_gender, mort_smoking_class, mort_rate_class, and mort_pol_yr instead of using these insured characteristics values directly.

Important! The rider issue age and policy year may not be the same as the base plan, since certain riders may be issued later than the base policy.

4.3.2 Mortality adjustment multiple

Experience mortality rates for applicable riders are adjusted by an adjustment multiple, such as varying by calendar date.

The annual mortality rate after all adjustments is constrained to be not less than 0.0, nor more than 1.0 (1000 per 1000).

4.3.3 Rate class

Mortality rates varying by rate class require special treatment, similarly to the base plan. You can read more about how the rate class is set in the Mortality sections of the product books.

4.3.4 **Base mortality**

The base mortality rate is read once for the life in which the rider is specified.

Mortality rates are specified per dollar, so a mortality rate of 0.001 is one death per thousand.

4.3.5 **Mortality table adjustments**

The rider mortality rate is adjusted by the multiple, then plus the addition (which can be negative).

4.3.6 **Substandard mortality**

The substandard multiple on the base plan model point record applies to the mortality rate after the multiple and addition. This same substandard multiple applies for rider premiums.

4.3.7 **Monthly mortality**

Rider deaths when applicable (ADB) occur at the end of each month, prior to other decrements and policyholder transactions. Each of the above mortality adjustments applies to the annual rate. Finally, the annual rate is converted into the monthly mortality rate:

$$1 - (1 - \text{annual mortality rate})^{(1/12)}$$

4.4 **Lapses and surrenders**

Riders are linked to the base policy and only terminate from lapse or surrender when the base policy terminates from lapse or surrender.

However, a rider may expire before the base policy expires if the rider maturity date comes before the base policy maturity date.

5 **REINSURANCE**

Riders cannot be reinsured.

6 **RIDER MODEL POINT RECORDS**

6.1 **Model point record layout**

The model point record layout for riders is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	keyfield	Up to 12 characters. Links to liability data file mp_id. This value can be listed any number of times to indicate the number of riders applied to the policy.
B	rider_code	A user defined input that references the rider code in the Riders.xlsm workbook.
C	issue_date	yyyy/mm/dd or mm/dd/yyyy. This date must be equal to or greater than the base policy issue date.

Excel Column	Variable	Value
D	elim_days	Elimination period for applicable riders (WOP, WOC)
E	ben_code	Benefit code used primarily for Table Method (WOP, WOC)
F	stat_valn_code	Up to 20 characters (link to rider valuation basis and assumptions in Riders – Valuation.xlsm workbook).
G	tax_valn_code	Up to 20 characters (link to rider valuation basis and assumptions in Riders – Valuation.xlsm workbook).
H	init_sum_assured_mp	Initial sum assured for rider record. This amount is in addition to the base policy benefit. This input must contain a value greater than 0 for applicable riders (ADB).

6.2 Units basis

Rider values are calculated per unit in force - such as the rider death benefit, rider premium, and rider reserves per unit, then multiplied by the current number of rider units.

7 SUBMODELS STRUCTURE

7.1 xxx_irider submodel

The xxx_irider submodel (where xxx is a product abbreviation) is a dependent data defined array of the rider_liab model class containing the variables and formulas required to calculate the rider reserves, benefits, and premiums.

The rider_liab model class calculates results using values per unit in force, which are then multiplied by the number of units. Results per unit in force have '_per_unit' added to the formula name. Formula names without this addition are results per rider record (or are a special result like a ratio).

7.1.1 Multiple instances

Each base plan model point carries multiple instances of the xxx_irider submodel. For example, in the Term Life application, the term submodel is the parent of term_irider submodel. The number of instances of the xxx_irider submodel is determined by the number of rider records specified on the rider model point record.

Elements of the rider submodel are created during the startup processing of the model point. The characteristic for each rider submodel element vary based on the rider_code.

The rider reserves and cash flows for multiple rider records are aggregated using the sum_over_rider function in the product submodel. This function aggregates the values for all riders.

The following shows an example Term Life submodel tree. Note – this is just showing a 'term' example, but xxx_irider submodels also exists in the Traditional Life, Universal Life, Variable Universal Life, and Indexed Universal Life applications:

- term (term_liab)
 - term_py (term_py_liab)
 - term_ireins (treaty_reins)
 - term_life_pba (life_pba_pba)
 - term_sfas60 (sfas60_gaap)
 - term_sfas97lp (sfas97lp_gaap)
 - term_rates (rates_economy)
 - term_irider (rider_liab)
 - term_irider_py (rider_py_liab)

7.1.2 Variables

Variables for the rider type, the rider reserves, benefits, and premiums are located in the rider_liab model class, except as noted below for the variables in rider_py_liab. These variables are contained in the rider_liab model class so that they vary by rider.

7.1.3 Formulas

Formulas for calculating the rider reserves, benefits, and premiums are contained in the rider_liab model class, except as noted below for the variables and formulas in the rider_py_liab model class.

Since xxx_irider is defined through the model parameters, the size of the xxx_irider array is based on the number of records specified for each model point on the rider model point file.

7.2 xxx_irider_py submodels

The xxx_irider_py submodel is a standalone instance of the rider_py_liab model class. It contains the variables and formulas required to calculate per unit rider results which only vary by policy year. Thus 't' in this submodel represents policy year, unlike in rider_liab where 't' represents months since the projection date. The policy year values are calculated from rider issue.

Since all results are per unit, this submodel does not include '_per_unit' in the formula names.

7.2.1 Variables

Variables located in the rider_py_liab model class are only used in formulas within the xxx_irider_py submodel. These are generally for values calculated from issue and varying by rider policy year, such as annual premiums, reserve terminal values and net premiums.

7.2.2 Formulas

Formulas for calculating policy year per unit rider results are included in the rider_py_liab model class.

Columns are generally calculated from rider issue, even for in force rider records. The reserve net premium and terminal values of rider_py_liab are not calculated on demand through the target column. Instead, these results are calculated in advance through forced calls from the 'initialize' column of rider_liab.

Reserve terminal values and net premiums are calculated by calling the Actuarial math library. See the Actuarial math library section below.

7.3 Actuarial math library

7.3.1 AML DLL

The Actuarial math library (AML) provides standard actuarial calculations for traditional reserve and cash value methods. See the Actuarial math library liability book.

The rider submodel calls the AML to calculate reserve amounts per unit in force. Columns in the rider_py_liab and rider_liab submodels store the values for each reserve basis used in the projection.

7.3.2 rider_liab->initialize

The rider_liab->initialize column initiates the call to the AML calculations for each set of reserve results. Each set of AML results is calculated by calling the aml_driver in the rider_py_liab model class. A separate call can be made for each of the following results:

- ADB statutory reserves – Accidental death benefit statutory reserves
- ADB tax reserves – Accidental death benefit tax reserves
- WOP active statutory reserves – Waiver of premium/COIs statutory claim costs and contract reserves
- WOP active tax reserves – Waiver of premium/COIs tax claim costs and contract reserves
- WOP disabled statutory reserves – Waiver of premium/COIs statutory claim reserves
- WOP disabled tax reserves – Waiver of premium/COIs tax claim reserves

7.3.3 AML driver

The aml_driver in the rider_py_liab submodel sets up and calls the AML for each set of reserve results. This driver is called by the rider_liab->initialize column for each set of results.

This driver sets up the call to the AML using the appropriate input for each type of calculations. Each call to the driver calls these functions in the rider_py_liab model class:

- setup_aml_initial_data – Creates the interfaces for using the AML (either LifeData for NLP calculations or AnnuityData for claim cost calculations).
- setup_aml_res_defn – Specifies the reserve method (either NLP reserve or claim reserve) and the function type (curtate, continuous, or semi-continuous).
- setup_aml_policy_data – Specifies values for each policy year, such as the guaranteed premiums, death benefits, claim costs, valuation interest rates, mortality rates, plus the issue age, maturity age, premium term years, and the name extension for the audit trail Excel file (when requested).

The operation of these setup functions varies by the type of calculations being made. For example, the policy data passes the statutory interest and mortality rates when calculating statutory reserves, and tax interest and mortality rates for the tax reserves.

7.3.4 Calculation results

The AML calculation results are stored for each policy year in columns in the rider_py_liab model class. The formula for each column simply gets the AML calculation results for each case. These columns include:

- Terminal reserves and net premiums for each basis
- Active life claim costs by policy year for each basis
- Valuation interest and mortality rate for each basis
- Tabular cost of insurance for each basis

The AML calculation results for claim reserves are stored for each period in columns in the rider_liab model class. The formula for each column simply gets the AML calculation results for each case. These columns include:

- Claim reserves for each basis

The aml_driver calls the appropriate columns after it calls the AML. For example, the statutory reserve basis columns are called when the driver has just calculated statutory reserve results. Then the driver calls the statutory reserve basis columns when those results have just been calculated by the AML. Thus each of these columns stores the AML results using the appropriate valuation basis.

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L14 ACTUARIAL MATH LIBRARY

This book discusses the Actuarial math library (AML), which provides standard actuarial calculations for formulaic reserve and cash value methods for the base plan reserves for Term life, Traditional life, Universal life, Indexed universal life, Variable universal life, and Long term care products as well as the claim costs, contract reserves, and claim reserves for riders.

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1 AML OVERVIEW

1.1 Operation

The AML is used when product calculations call for formulaic cash value and reserve calculations. The AML is a separate program, linked to the project for this purpose.

The project organizes the information needed by the AML, such as the pattern of premiums, death benefits, endowment benefits, mortality rates, and so on. The formulas returning the formulaic cash value and reserve values simply contain a function to 'get' the appropriate values.

These calculations occur within the AML DLL, and are thus not part of the project. The AML provides audit reports so that you can see every detail of these calculations, for use in investigating the results returned by the AML.

1.2 Maintenance

The AML features are provided through two files located in the Global directory: ActuarialMath.dll and ActuarialMath.lib.

These files must be for the version of the AML that was released with the version of the project you are using. When the Global directory is being used by multiple projects, you must create a different copy of that directory for each version.

For example, the main Global directory should likely contain the files for the most recent version. Then additional Global_[version number] directories could be set up for older versions. Older versions of projects would then have their <global*> wildcard set to that alternate Global directory.

2 LIFE INSURANCE VALUATIONS

2.1 Valuation definition

Net premiums and terminal values use one of the following valuation definitions:

- Unitary – Present values are calculated from maturity.
- RegulationXXX – Present values are calculated both from maturity (unitary), and within each regulation XXX reserve segment (segmented) using reserve segment mortality. See the Regulation XXX section in the Statutory and tax reserves chapter of the Term life book.
- NY147 – Unitary and segmented present values are calculated using each New York regulation 147 segment definitions and segment mortality. See the Regulation XXX section in the Statutory and tax reserves chapter of the Term life book.
- ActuarialGuideline45 – See the Actuarial guideline 45 section in the ROP term chapter of the Term life book.
- Term – Present values are calculated separately for each renewable term period.
- VM20NPR – Present values are calculated using the PBA net premium reserves methodology defined by VM-20 (chapter 20 of the valuation manual).

2.2 Expense allowance definition

Net premiums use one of the following modified premium expense allowances:

- NetLevel – net level premium (zero expense allowance)
- FullPreliminaryTerm – full preliminary term (zero first year terminal value)
- FullPreliminaryTerm20 – 2-year FPT (zero second year terminal value)

- FullPreliminaryTerm15 – 1.5-year FPT (expense allowance is the average of FPT and 2-year FPT)
- CRVM1980 – 1980 Commissioners Reserve Method
- CRVMPre1980 – pre-1980 CRVM
- NY147Amendment5 – NY 147 Amendment 5
- SNFL1980 – 1980 Standard Nonforfeiture Law
- SNFLPre1980 – pre-1980 SNFL
- UserInput – specified expense allowance per unit

2.3 Period definition

Terminal values are calculated on a policy year basis.

2.4 Function timing

Function timing is defined separately for premiums and for claims.

2.4.1 Premium timing multiple

- Curtate – premium timing multiple is 1.
- Continuous – premiums are assumed to be paid continuously; the curtate reserve formulas are adjusted by the following 'premium timing' multiple:

$$\frac{(\delta - d)}{\delta^2} + \frac{(i - \delta)}{\delta^2} \times \frac{(1 - q)}{(1 + i)}$$

2.4.2 Death claim timing multiple

- Curtate – death claim timing multiple is 1.
- Continuous – death claims are assumed to be paid immediately; the curtate reserve formulas are adjusted by the following 'death claim timing' multiple:

$$\frac{i}{\delta}$$

2.4.3 Claim cost timing multiple

- Curtate – claim cost timing multiple is 1.
- Continuous – claim costs are assumed to be paid continuously; the curtate reserve formulas are adjusted by the following 'claim cost timing' multiple:

$$\frac{(\delta - d)}{\delta^2} + \frac{(i - \delta)}{\delta^2} \times \frac{(1 - q)}{(1 + i)}$$

2.5 Joint life mortality

2.5.1 Single or joint life

The AML treats mortality rates either as:

- Single – the mortality rate is specified for the single insured
- FirstToDie – the annual mortality rate is $q_1 + q_2 - q_1 * q_2$
- FrasierizedLastSurvivor – the annual mortality rate is Frasierized from issue, as described below.

2.5.2 Frasierization

The mortality rates for life 1 and life 2 are combined into a Frasierized rate calculated from issue:

$$\begin{aligned} &P(\text{both alive BOY}) \times P(\text{both die during the year} | \text{both alive BOY}) \\ &\quad + P(\text{only life 1 alive BOY}) \times P(\text{life 1 dies during the year} | \text{only life 1 alive BOY}) \\ &\quad + P(\text{only life 2 alive BOY}) \times P(\text{life 2 dies during the year} | \text{only life 2 alive BOY}) \end{aligned}$$

3 ANNUITY VALUATIONS

3.1 Valuation definition

The valuation definition is always unitary, calculating the present value of annuity benefits from the annuity start period to the annuity maturity period.

3.2 Period definition

Terminal values are calculated on a monthly basis from the annuity start period.

3.3 Function timing

There is no function timing for annuity calculations. All calculations use a curtate approach, and benefits are specified as being paid at the beginning or end of each period.

3.4 Joint life decrements

3.4.1 Single or joint life

The AML treats decrement rates either as:

- Single – the decrement rate is specified for the single insured
- FirstToDie – the monthly decrement rate is $q_1 + q_2 - q_1 * q_2$
- FrasierizedLastSurvivor – the monthly decrement rate is Frasierized from issue, as described below.

3.4.2 Frasierization

The decrement rates for life 1 and life 2 are combined into a Frasierized rate calculated from issue:

$$\begin{aligned} &P(\text{both alive BOY}) \times P(\text{both die during the year} | \text{both alive BOY}) \\ &\quad + P(\text{only life 1 alive BOY}) \times P(\text{life 1 dies during the year} | \text{only life 1 alive BOY}) \\ &\quad + P(\text{only life 2 alive BOY}) \times P(\text{life 2 dies during the year} | \text{only life 2 alive BOY}) \end{aligned}$$

4 DETAIL REPORTS

4.1 Overview

4.1.1 Producing the report

The AML can produce a detail report capturing the calculations for each call to the AML. A separate Excel workbook is created for each calculation for each model point.

This report is produced when the Produce AML audit file? variable (aml_audit_rpt_defn, on the Product Valuation input page or Rider Valuation input page under the Actuarial math library category) is Yes. Of course, this variable should be No for production runs.

4.1.2 Report names

The detail reports are located in the same directory as the output for the projection. Each Excel workbook is named as: [output file name]_[calculation type]_[model point ID].csv. The calculation type is one of the following:

- BasePlanCashValue
 - Basic cash values (Trad only)
 - AG 45 cash value for current premium basis (Term only)
 - AG 45 cash value for guaranteed premium basis (Term only)
- BasePlanStatRes
 - Nonforfeiture net premium
 - Statutory basic reserves
 - Statutory alternative minimum reserve for deficiency reserves
 - VM-20 net premium reserves
 - Statutory ROP pure endowment reserve (Term only)
 - Statutory ROP unusual cash value reserve (Term only)
 - Substandard basic reserve (Term only)
 - Substandard ROP pure endowment reserve (Term only)
 - Substandard ROP unusual cash value reserve (Term only)
- BasePlanTaxRes
 - Tax ROP pure endowment reserve (Term only)
 - Tax ROP unusual cash value reserve (Term only)
- RiderStatRes
 - Rider statutory reserves
 - Active and disabled life claim reserves
- RiderTaxRes
 - Rider tax reserves
 - Active and disabled life tax claim reserves

Long term care

The Long term care project includes these calculation types:

- StatRes
 - Statutory contract claim cost reserve
 - Statutory contract ROP reserve
- TaxRes
 - Tax contract claim cost reserve
 - Tax contract ROP reserve

4.1.3 Report structure

The detail report is produced in a comma separate value (csv) format so the file can be opened and displayed in Excel.

The first line of the report shows its location and name followed by rows and columns of information that captures the details in one of two main sections:

- Parameters detail
- Calculation detail

4.1.4 Per model basis

All values are per model basis of insurance in force which means unit for life insurance, and policy for long term care.

4.2 Parameters detail

The first section of the detail report shows the values passed to the AML. This section appears under the heading 'Parameters detail' that appears on the second line of the report.

4.2.1 Insurance data

Single value variables

Variables having single values are displayed as follows:

Life insurance data	
Mort rate treatment defn	Single
Valn defn	NY 147
Exp allow defn	Net Level
Claims timing defn	Continuous
Prem timing defn	Curtate
Maturity age	91
Prem term	5
Grade from pol yr	0
Grade to pol yr	5
XXX select grading	No
Max exp allow	1000
Initial NSP unitary	0
Initial NSP segmented	0

Value	Description
Mortality rate treatment defn	See the Joint life mortality section in the Life insurance valuations chapter.
Valn defn	See the Valuation definition section in the Life insurance valuations chapter.
Exp allow defn	See the Expense allowance definition section in the Life insurance valuations chapter.
Endow for death benefit at omega definition	If input endowment value is 0 at omega age, Yes means using death benefit for PV endowment calculation, No otherwise.
Claims timing defn	See the Function timing section in the Life insurance valuations chapter.
Prem timing defn	See the Function timing section in the Life insurance valuations chapter.
Maturity age	Issue age plus policy anniversary at maturity.
Prem term	Number of policy years when premiums are due.
Grade from pol yr	The specified expense allowance method applies through this policy anniversary.
Grade to pol yr	Net level terminal values apply beginning on this policy anniversary. The reserve or cash value premium between these anniversaries is a level percent calculated to match these grading terminal values.
XXX select grading	When the valuation definition is NY147, Yes means that the XXX mortality select multiple grades linearly from the multiple in year 10 to 1.0 in year 15. No otherwise.
Max exp allow	The expense allowance is capped at this amount per unit.
Initial NSP unitary	A non-negative double used to define initial unitary terminal value.
Initial NSP segmented	A non-negative double used to define initial segmented terminal value.

Value	Description
Initial level prem period (VM-20 NPR only)	When the valuation definition is VM20NPR, and the initial level premium period is greater than or equal to the premium term break for selecting the lapse rate, valuation net premiums for policy years after the shock lapse are limited and may result in two uniform percentages. The first percent applies to policy years prior to the shock lapse; the second applies following the shock lapse: i. Compute the actuarial present value of benefits for policy years following the shock lapse. ii. Compute the actuarial present value of valuation net premiums for policy years following the shock lapse. iii. If ii/i is greater than the shock lapse net premium limit, reduce the net valuation premiums in ii uniformly to produce a ratio of ii/i of the input limit value.
Prem term break for lapse rate (VM-20 NPR only)	See the description for the initial level prem period.
Shock lapse net premium limit (VM-20 NPR only)	See the description for the initial level prem period.

Variables varying by policy year

Variables having values by policy year are then displayed as follows:

Period	Prem payable	Seg end flag	Prem pattern gtd	Adj prem mult	Dth benefit	Dth benefit increm	Lapse benefit	Endow benefit	Claim costs	Int rates	Lapse rates	Init seg factors mult	Exp allow unitary	Exp allow segmented	ETI cash val
0	1	0	1200	0	1000	0	200	300	0	0.05	0.05	0	0	0	0
1	1	0	1200	0	1000	0	200	300	20	0.05	0.05	0	0	0	0
2	1	0	1200	0	990	0	200	300	30	0.05	0.05	0	0	0	0
3	1	0	1200	0	980	0	200	300	40	0.05	0.05	0	0	0	0
4	1	0	1200	0	970	0	200	300	50	0.05	0.05	0	0	0	0
5	1	0	1200	0	960	0	200	300	60	0.05	0.05	0	0	0	0

Value	Description
Period	Policy year
Prem payable	1 when premium paying; 0 otherwise
Seg end flag	1 when a new segment begins for the next policy year; 0 otherwise
Adj prem mult (VM-20 NPR only)	Expense allowance adjustment percent of premium for each policy year.
Prem pattern gtd	The guaranteed premium for each policy year. Net premiums are a level percent of this pattern of premiums.

Value	Description
Dth benefit	Death benefit for each policy year.
Dth benefit increm	When the valuation definition is ActuarialGuideline45, this incremental death benefit is used to calculate present value of death benefit.
Lapse benefit	Lapse benefit for each policy year.
Endow benefit	Endowment benefit at the end of the policy year.
Claim costs	Claim cost for each policy year.
Int rates	Valuation interest rate. This must be greater than zero when Claim timing or premium timing is Continuous.
Lapse rate	Lapse rate at the end of the policy year.
Init seg factors mult	When the valuation definition is NY147, this multiple applies to the XXX select input 1 values. The total select multiple is rounded to the nearest integer percent and capped at 100%. (This variable does not apply for Regulation XXX.)
Exp allow unitary	When the expense allowance definition is UserDefined, this is the expense allowance for unitary reserves.
Exp allow seg	When the expense allowance definition is UserDefined, this is the expense allowance for segmented reserves.
ETI cash val	Cash value at the end of the policy year. This value is used for Extended Term Insurance (ETI) calculations only.

Not all variables described above are needed for a specific calculation. For instance, Expense allowance unitary applies only if the Expense allowance definition is UserInput. You don't need to pass those parameters into AML if you don't need them for a specific model point. In that case, AML simply displays 0 in the detail report.

4.2.2 Life data

A section of life data is included for the number of lives defined. Joint life products will include two sections; one for life 1 and a second for life 2.

Single value variables

Issue age information is displayed as follows:

Life data	
Issue age	25

Variables varying by policy year

Variables varying by policy year are then displayed as follows:

Period	Mort base	Init seg select factor	Ren seg select factor	Mort improve mult
0	0.00052	1	1	0.932065
1	0.00055	1	1	0.932065
2	0.00058	1	1	0.922745
3	0.00063	1	1	0.913517
4	0.00068	1	1	0.904382
5	0.00072	1	1	0.895338

Value	Description
Mort base	Mortality rates are per dollar amounts, so a mortality rate of 0.001 is one death per thousand.
Init seg select factor	When the valuation definition is RegulationXXX or NY147, the base mortality rate is multiplied by this select factor during the first segment only.
Ren seg select factor	When the valuation definition is RegulationXXX or NY147, the base mortality rate is multiplied by this select factor after the first segment.
Mort improve mult	When mortality improvement are included, the select factor adjusted mortality rate is multiplied by this mortality improvement multiple. For NY147 Amendment 5, this applies in the first segment only.

4.2.3 Annuity data

Single value variables

Variables having single values are displayed as follows:

Annuity data	
Decrem rate treatment defn	Single
Maturity age	48

Value	Description
Decrement rate treatment defn	See the Joint life decrement section in the Annuity valuations chapter.
Maturity age	Issue age plus policy anniversary at maturity.

Variables varying by month

Variables having values by month are then displayed as follows:

Period	Benefit BOM	Benefit EOM	Int rates
0	0	0	0.02
1	0.569388	0	0.02
2	0	0	0.02
3	0	0	0.02
4	0	0	0.02
5	0	0	0.02
6	0	0	0.02
7	0.569388	0	0.02
8	0	0	0.02

Value	Description
Period	Month from the start of annuity
Benefit BOM	The annuity benefit, paid at the beginning of month
Benefit EOM	The annuity benefit, paid at the end of month
Int Rates	Annual valuation interest rate

4.2.4 Annuitant data

A section of annuitant data is included for the number of lives defined. Joint life products will include two sections; one for annuitant 1 and a second for annuitant 2.

Single value variables

Annuitant age information is displayed as follows:

Annuitant data	
Benefit Start Age	18

Variables varying by month

Variables having values by month are then displayed as follows:

Period	Decrem base	Decrem improve mult
0	0.999566	1
1	0.999566	1
2	0.998162	1
3	0.993715	1
4	0.981608	1
5	0.967736	1

Value	Description
Period	Month from the start of annuity
Decrement base	Decrement rates are per dollar amounts, so a decrement rate of 0.001 is one per thousand.

Value	Description
Decrement improve mult	When decrement improvements are included, the decrement rate is multiplied by this improvement multiple.

4.3 Basic calculation detail

The following topics describe the basic calculation details. These topics apply for both formulaic and VM-20 NPR calculations.

4.3.1 Segments

When the valuation definition is RegulationXXX, NY147, or Term, the segments calculation details are displayed as follows:

Segments			
Period	Prem payable	Seg end flag	Segment num
0	1	0	0
1	1	1	1
2	1	1	2
3	1	0	3
4	1	1	3
5	1	0	4

Value	Description
Premium payable	Input parameter for insurance data (described above)
Seg end flag	Input parameter for insurance data (described above)
Segment num	Number of segment for this policy year. It increases by 1 if there is a segment end last year.

4.3.2 Mortality rate select and improvement multiple

When the valuation definition is RegulationXXX, NY147, or Term, the mortality rate select and improvement multiple calculations are displayed as follows:

Mortality rate select and improvement multiple								
Period	Seg end flag	Segment num	Init seg factors mult	Init seg select factor	Ren seg select factor	Select mult unrounded	Select mult	Mort improve mult
1	0	1	1	1	1	1	1	0.932065
2	0	1	1	1	1	1	1	0.922745
3	0	1	1	1	1	1	1	0.913517
4	0	1	1	1	1	1	1	0.904382
5	0	1	1	1	1	1	1	0.895338

Value	Description
Segment end flag	Input parameter for insurance data (described above)
Segment num	Calculated in Segment section (described above)
Init seg factors mult	(Only for NY147) Input parameter for insurance data (described above)
Init seg select factor	Input parameter for life data (described above)
Ren seg select factor	Input parameter for life data (described above)
Select mult unrounded	(Only for NY147) Segment factor multiple * Segment select factor * Segment 1 multiple (first segment only)
Select mult	When the expense allowance is other than NY 147 Amendment 5, this is the select multiple rounded to the nearest integer percent and capped at 100%. For NY 147 Amendment 5, this select multiple is only capped at 100%
Mort improve mult	When mortality improvements are included, this is mortality improvement multiple. For NY 147 Amendment 5, the multiple is 1.0 after the first segment

For joint life products, the segment select factors for both life 1 and life 2 are used.

4.3.3 Life insurance decrements

Single life

For single life products, the decrements calculations are displayed as follows:

Period	Mort base	Select mult	Mort improve mult	Mort rate unrounded	Dth claim rate
1	0.00055	1	0.93207	0.000513	0.00051
2	0.00058	1	0.92275	0.000535	0.00054
3	0.00063	1	0.91352	0.000576	0.00058
4	0.00068	1	0.90438	0.000615	0.00061
5	0.00072	1	0.89534	0.000645	0.00064

Value	Description
Mort base	Input parameter for life data (described above)
Select mult	Calculated in Mortality rate select and improvement multiple section (described above)
Mort improve mult	Calculated in Mortality rate select and improvement multiple section (described above)
Mort rate unrounded	Mortality base * Select multiple * Mort improve mult

Dth claim rate	For NY 147 Amendment 5, this is the mortality rate unrounded value rounded to 5 decimals. For all other methods, this equals the Mort rate unrounded
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First to die

For first to die joint life products, the decrements calculations are displayed as follows:

Period	Mort base life 1	Select mult life 1	Mort improve mult life 1	Mort rate life 1 unrounded	Mort rate life 1	Mort base life 2	Select mult life 2	Mort improve mult life 2	Mort rate life 2 unrounded	Mort rate life 2	Dth claim rate
1	0.00091	1	0.932065	0.000848	0.00085	0.00172	1	0.932065	0.001603	0.0016	0.002449
2	0.00097	1	0.922745	0.000895	0.0009	0.00182	1	0.922745	0.001679	0.00168	0.002578
3	0.00102	1	0.913517	0.000932	0.00093	0.00192	1	0.913517	0.001754	0.00175	0.002678
4	0.0011	1	0.904382	0.000995	0.00099	0.00202	1	0.904382	0.001827	0.00183	0.002818
5	0.00114	1	0.895338	0.001021	0.00102	0.00203	1	0.895338	0.001818	0.00182	0.002838

Value	Description
Mort base life 1	Input parameter for life data (described above)
Select mult life 1	Calculated in Mortality rate select and improvement multiple section (described above)
Mort improve mult life 1	Calculated in Mortality rate select and improvement multiple section (described above)
Mort rate life 1 unrounded	Mortality base life 1 * Select multiple life 1 * Mort improve mult life 1
Mort rate life 1	For NY 147 Amendment 5, this is the mortality rate life 1 unrounded value rounded to 5 decimals. For all other methods, this equals the Mort rate life 1 unrounded
Mort base life 2	Input parameter for life data (described above)
Select mult life 2	Calculated in Mortality rate select and improvement multiple section (described above)
Mort improve mult life 2	Calculated in Mortality rate select and improvement multiple section (described above)
Mort rate life 2 unrounded	Mortality base life 2 * Select multiple life 2 * Mort improve mult life 2
Mort rate life 2	For NY 147 Amendment 5, this is the mortality rate life 2 unrounded value rounded to 5 decimals. For all other methods, this equals the Mort rate life 2 unrounded
Death claim rate	Mortality rate life 1 + Mortality rate life 2 – Mortality rate life 1 * Mortality rate life 2

Frasierized last survivor

For Frasierized last survivor joint life products, the decrements calculations are displayed as follows:

Period	Mort base life 1	Select mult life 1	Mort improve mult life 1	Mort rate life 1 unrounded	Mort rate life 1	Mort base life 2	Select mult life 2	Mort improve mult life 2	Mort rate life 2 unrounded	Mort rate life 2	Survivorship life 1	Survivorship life 2	Survivorship both	Survivorship life 1 only	Survivorship life 2 only	Dth claim rate both	Dth Last	Dth claim rate
1	0.0006	1	0.932065	0.000559	0.00056	0.00039	1	0.932065	0.000364	0.00036	1	1	1	0	0	0	0	0
2	0.00058	1	0.922745	0.000535	0.00054	0.00041	1	0.922745	0.000378	0.00038	0.99944	0.99964	0.99908	0.00036	0.00056	0	0.000001	0.000001
3	0.00058	1	0.913517	0.00053	0.00053	0.00044	1	0.913517	0.000402	0.0004	0.9989	0.99926	0.998161	0.000739	0.001099	0	0.000001	0.000001
4	0.00058	1	0.904382	0.000525	0.00052	0.00046	1	0.904382	0.000416	0.00042	0.998371	0.99886	0.997233	0.001138	0.001627	0	0.000001	0.000001
5	0.00059	1	0.895338	0.000528	0.00053	0.0005	1	0.895338	0.000448	0.00045	0.997852	0.998441	0.996296	0.001556	0.002145	0	0.000002	0.000002

Value	Description
Mort base life 1	Input parameter for life data (described above)
Select mult life 1	Calculated in Mortality rate select and improvement multiple section (described above)
Mort improve mult life 1	Calculated in Mortality rate select and improvement multiple section (described above)
Mort rate life 1 unrounded	Mortality base life 1_t * Select multiple life 1_t * Mort improve mult life 1_t
Mort rate life 1	For NY 147 Amendment 5, this is the mortality rate life 1 unrounded value rounded to 5 decimals. For all other methods, this equals the Mort rate life 1 unrounded
Mort base life 2	Input parameter for life data (described above)
Select mult life 2	Calculated in Mortality rate select and improvement multiple section (described above)
Mort improve mult life 2	Calculated in Mortality rate select and improvement multiple section (described above)
Mort rate life 2 unrounded	Mortality base life 2_t * Select multiple life 2_t * Mort improve mult life 2_t
Mort rate life 2	For NY 147 Amendment 5, this is the mortality rate life 2 unrounded value rounded to 5 decimals. For all other methods, this equals the Mort rate life 2 unrounded
Survivorship life 1	Survivorship life 1_{t-1} * (1 – Mortality rate life 1_{t-1})
Survivorship life 2	Survivorship life 2_{t-1} * (1 – Mortality rate life 2_{t-1})
Survivorship both	Survivorship life 1_t * Survivorship life 2_t
Survivorship life 1 only	Survivorship life 1_t * (1 - Survivorship life 2_t)
Survivorship life 2 only	Survivorship life 2_t * (1 - Survivorship life 1_t)
Dth claim rate both	Mortality rate life 1_t * Mortality rate life 2_t

Value	Description
Dth Last	$\text{Survivorship both}_t * \text{Death claim rate both}_t + \text{Survivorship life 1 only}_t * \text{Mortality rate life 1}_t + \text{Survivorship life 2 only}_t * \text{Mortality rate life 2}_t$
Dth claim rate	$\text{Death Last}_t / (\text{Survivorship both}_t + \text{Survivorship life 1 only}_t + \text{Survivorship life 2 only}_t)$

4.3.4 Death Claim timing

When the Claim timing definition is Continuous, the death claim timing calculations are displayed as follows:

Death claim timing			
Period	Int rates	$\ln(1 + \text{Int rates})$	Death claim timing
1	0.05	0.04879	1.024797
2	0.05	0.04879	1.024797
3	0.05	0.04879	1.024797
4	0.05	0.04879	1.024797
5	0.05	0.04879	1.024797

Value	Description
Int rates	Input parameter for insurance data (described above)
$\ln(1 + \text{Int rates})$	natural log of $(1 + \text{Int rates}_t)$
Death claim timing	$\text{Int rates}_t / \ln(1 + \text{Int rates}_t)$

4.3.5 Premium timing

When the premium timing definition is Continuous, the premium timing calculations are displayed as follows:

Premium timing								
Period	Int rates	Dth claim rate	Lapse rate	One year endow	Eff rate of disc	Force of int	Force of disc	Prem timing
1	0.05	0.026620	0.05	0.880677	0.047619	0.048790	0.048790	0.939555
2	0.05	0.029130	0.05	0.878406	0.047619	0.048790	0.048790	0.938401
3	0.05	0.031790	0.05	0.876000	0.047619	0.048790	0.048790	0.937177
4	0.05	0.034650	0.05	0.873412	0.047619	0.048790	0.048790	0.935862
5	0.05	0.037810	0.05	0.870553	0.047619	0.048790	0.048790	0.934409

Value	Description
Inter rates	Input parameter for insurance data (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
One year endow	$(1 - \text{Dth claim rate}_t) * (1 - \text{Lapse rate}_t) / (1 + \text{Int rates}_t)$

Value	Description
Eff rate of disc	$\text{Int rates}_t / (1 + \text{Int rates}_t)$
Force of int	$\ln(1 + \text{Int rates}_t)$
Force of disc	$\ln(1 + \text{Int rates}_t)$
Prem timing	$[\text{Force of int}_t - \text{Eff rate of disc}_t - \text{One year endow}_t * (\text{Force of int}_t - \text{Int rates}_t)] / (\text{Force of int}_t * \text{Force of disc}_t)$

4.3.6 Claim Cost timing

When the Claim timing definition is Continuous, the claim cost timing calculations are displayed as follows:

Claim cost timing								
Period	Int rates	Dth claim rate	Lapse rate	One year endow	Eff rate of disc	Force of int	Force of disc	Claim cost timing
1	0.05	0.0266	0.05	0.8807	0.0476	0.0488	0.0488	0.9396
2	0.05	0.0291	0.05	0.8784	0.0476	0.0488	0.0488	0.9384
3	0.05	0.0318	0.05	0.8760	0.0476	0.0488	0.0488	0.9372
4	0.05	0.0347	0.05	0.8734	0.0476	0.0488	0.0488	0.9359
5	0.05	0.0378	0.05	0.8706	0.0476	0.0488	0.0488	0.9344

Value	Description
Int rates	Input parameter for insurance data (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
One year endow	$(1 - \text{Dth claim rate}_t) * (1 - \text{Lapse rate}_t) / (1 + \text{Int rates}_t)$
Eff rate of disc	$\text{Int rates}_t / (1 + \text{Int rates}_t)$
Force of int	$\ln(1 + \text{Int rates}_t)$
Force of disc	$\ln(1 + \text{Int rates}_t)$
Claim cost timing	$(\text{Force of int} - \text{Eff rate of disc} - \text{One year endow} * (\text{Force of int} - \text{Int rates}_t)) / (\text{Force of int} * \text{Force of disc})$

4.3.7 One year cost of insurance

The one year cost of insurance calculation details are displayed as follows:

One year cost of insurance													
Period	Dth claim rate	Lapse rate	Int rates	Death claims timing	Prem timing	Claim cost timing	Death ben Cx	Lapse ben Cx	Claim cost surv factor	Death ben	Lapse ben	Claim cost	One year cost of insurance
1	0.0005	0	0.035	1.0174	1	0.982751	0.000491	0	0.982503	1000	0	0	0.491497
2	0.00054	0	0.035	1.0174	1	0.982731	0.000531	0	0.982464	1000	0	0	0.530817
3	0.00057	0	0.035	1.0174	1	0.982716	0.00056	0	0.982435	1000	0	0	0.560307
4	0.00061	0	0.035	1.0174	1	0.982697	0.0006	0	0.982395	1000	0	0	0.599627
5	0.00065	0	0.035	1.0174	1	0.982677	0.000639	0	0.982356	1000	0	0	0.638947

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Dth claim timing	Calculated in Death claim timing section (described above)
Prem timing	Calculated in Premium timing section (described above)
Claim cost timing	Calculated in Claim cost timing section (described above)
Death ben Cx	$[Dth\ claim\ timing_t * Dth\ claim\ rate_t / (1 + Int\ rates_t)] / Prem\ timing_t$
Lapse ben Cx	$[Dth\ claim\ timing_t * Lapse\ rate_t / (1 + Int\ rates_t)] / Prem\ timing_t$
Claim cost surv factor	$[Dth\ claim\ timing_t * (1 - Dth\ claim\ rate_t) * (1 - Lapse\ rate_t) / (1 + Int\ rates_t)] / Prem\ timing_t$
Death ben	Input parameter for insurance data (described above)
Lapse ben	Input parameter for insurance data (described above)
Claim cost	Input parameter for insurance data (described above)
One year cost of insurance	When claim timing definition is Continuous, equals $((Dth\ claim\ timing_t * Death\ ben_t * Dth\ claim\ rate_t + Dth\ claim\ timing_t * Lapse\ ben_t * Lapse\ rate_t) / (1 + Int\ rates_t) + Claim\ cost\ timing_t * Claim\ cost_t) / Prem\ timing_t$. When claim timing definition is Curtate, equals $(Death\ ben_t * Dth\ claim\ rate_t + Lapse\ ben_t * Lapse\ rate_t) / (1 + Int\ rates_t) / Prem\ timing_t + Claim\ cost\ surv\ factor_t * Claim\ cost_t$.

4.3.8 Annuity decrements

Single life

For single life annuities, the decrements calculations are displayed as follows:

Decrement			
Period	Decrem base	Decrem improve mult	Decrem rate
1	0.999566	1	0.475401
2	0.998162	1	0.408409
3	0.993715	1	0.34457
4	0.981608	1	0.28322
5	0.967736	1	0.24885

Value	Description
Period	Month from start of annuity
Decrement base	Input parameter for annuity data (described above)
Decrement improve mult	Input parameter for annuity data (described above)
Decrement rate	This equals the final monthly decrement rate

First to die

For first to die joint annuities, the decrements calculations are displayed as follows:

Decrement							
Period	Decrem base annuitant 1	Decrem improve mult annuitant 1	Decrem rate annuitant 1	Decrem base annuitant 2	Decrem improve mult annuitant 2	Decrem rate annuitant 2	Decrem rate
1	0.999566	1	0.475401	0.999566	1	0.475401	0.724796
2	0.998162	1	0.408409	0.998162	1	0.408409	0.65002
3	0.993715	1	0.34457	0.993715	1	0.34457	0.570412
4	0.981608	1	0.28322	0.981608	1	0.28322	0.486226
5	0.967736	1	0.24885	0.967736	1	0.24885	0.435774

Value	Description
Period	Month from start of annuity
Decrement base annuitant 1	Input parameter for annuitant data (described above)
Decrement improve mult annuitant 1	Input parameter for annuitant data (described above)

Value	Description
Decrement rate annuitant 1	$1 - (1 - \text{Decrem base annuitant 1} * \text{Decrem improve mult annuitant 1})^{1/12}$
Decrement base annuitant 2	Input parameter for annuitant data (described above)
Decrement improve mult annuitant 2	Input parameter for annuitant data (described above)
Decrement rate annuitant 2	$1 - (1 - \text{Decrem base annuitant 2} * \text{Decrem improve mult annuitant 2})^{1/12}$
Decrem rate	$\text{Decrem rate annuitant 1} + \text{Decrem rate annuitant 2} - \text{Decrem rate annuitant 1} * \text{Decrem rate annuitant 2}$

Frasierized last survivor

For Frasierized last survivor annuities, the decrements calculations are displayed as follows:

Decrement														
Period	Decrem base annuitant 1	Decrem improve mult annuitant 1	Decrem rate annuitant 1	Decrem base annuitant 2	Decrem improve mult annuitant 2	Decrem rate annuitant 2	Survivorship annuitant 1	Survivorship annuitant 2	Survivorship both	Survivorship annuitant 1 only	Survivorship annuitant 2 only	Decrem rate both	Decrem rate last	Decrem rate
1	0.999566	1	0.475401	0.999566	1	0.475401	1	1	1	0	0	0.226006	0.22601	0.226006
2	0.998162	1	0.408409	0.998162	1	0.408409	0.524599	0.524599	0.27520411	0.24939489	0.24939489	0.166798	0.24961	0.322501
3	0.993715	1	0.34457	0.993715	1	0.34457	0.31034805	0.31034805	0.09631591	0.21403214	0.21403214	0.118728	0.15893	0.303088
4	0.981608	1	0.28322	0.981608	1	0.28322	0.20341142	0.20341142	0.04137621	0.16203521	0.16203521	0.080214	0.0951	0.260235
5	0.967736	1	0.24885	0.967736	1	0.24885	0.14580124	0.14580124	0.021258	0.12454324	0.12454324	0.061926	0.0633	0.234152

Value	Description
Period	Month from start of annuity
Decrement base annuitant 1	Input parameter for annuitant data (described above)
Decrement improve mult annuitant 1	Input parameter for annuitant data (described above)
Decrem rate annuitant 1	$1 - (1 - (\text{Decrem base life } 1_t * \text{Decrem improve mult life } 1_t))^{1/12}$
Decrement base annuitant 2	Input parameter for annuitant data (described above)
Decrement improve mult annuitant 2	Input parameter for annuitant data (described above)
Decrem rate annuitant 2	$1 - (1 - (\text{Decrem base life } 2_t * \text{Decrem improve mult life } 2_t))^{1/12}$
Survivorship annuitant 1	$\text{Survivorship annuitant } 1_{t-1} * (1 - \text{Decrem rate annuitant } 1_{t-1})$
Survivorship annuitant 2	$\text{Survivorship annuitant } 2_{t-1} * (1 - \text{Decrem rate annuitant } 2_{t-1})$

Value	Description
Survivorship both	Survivorship annuitant 1 _t * Survivorship annuitant 2 _t
Survivorship annuitant 1 only	Survivorship annuitant 1 _t * (1 - Survivorship annuitant 2 _t)
Survivorship annuitant 2 only	Survivorship annuitant 2 _t * (1 - Survivorship annuitant 1 _t)
Decrem rate both	Decrem rate annuitant 1 _t * Decrem rate annuitant 2 _t
Decrem last	Survivorship both _t * Decrem rate both _t + Survivorship annuitant 1 only _t * Decrem rate annuitant 1 _t + Survivorship annuitant 2 only _t * Decrem rate annuitant 2 _t
Decrem rate	Decrem Last _t / (Survivorship both _t + Survivorship annuitant 1 only _t + Survivorship annuitant 2 only _t)

4.4 Net single and level premium calculation detail

The following topics describe the net single premium and net level premium calculation details for formulaic cash value and reserve methods.

4.4.1 Present value of benefits

The present value of benefits calculations details are displayed as follows:

Present value benefits																		
Period	Is the policy in force?	Policy matures?	Dth claim rate	Lapse rate	Int rates	Dth claims timing	Dth benefit	Dth benefit increm	PV dth benefit	Lapse benefit	PV lapse benefit	Endow benefit	PV endow benefit	Claim cost timing	Claim costs	PV claim costs	PV total	
0	1	0							311.65		70.23		2371.84			977.17	3730.89	
1	1	0	0.0266	0.05	0.05	1.0248	1000	0	324.37	200	68.67	300	2352.55	0.9396	20	1088.23	3833.82	
2	1	0	0.0291	0.05	0.05	1.0248	990	0	337.23	200	67.06	300	2336.68	0.9384	30	1206.82	3947.79	
3	1	0	0.0318	0.05	0.05	1.0248	980	0	350.25	200	65.41	300	2324.98	0.9372	40	1334.86	4075.50	
4	1	0	0.0347	0.05	0.05	1.0248	970	0	363.46	200	63.72	300	2318.47	0.9359	50	1474.75	4220.40	
5	1	0	0.0378	0.05	0.05	1.0248	960	0	376.81	200	61.98	300	2318.60	0.9344	60	1629.64	4387.03	

Value	Description
Is the policy in force?	1 if true; 0 otherwise
Policy matures?	1 in the maturity year; 0 otherwise
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Dth claim timing	Calculated in Death claim timing section (described above)

Value	Description
Dth benefit	Input parameter for insurance data (described above)
Dth benefit increment	Input parameter for insurance data (described above)
PV dth benefit	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Dth claim timing}_{t+1} * \text{Dth benefit}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$
Lapse benefit	Input parameter for insurance data (described above)
PV lapse benefit	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Dth claim timing}_{t+1} * \text{Lapse benefit}_{t+1} * \text{Lapse rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$
Endow benefit	Input parameter for insurance data (described above)
PV endow benefit	$PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Endow benefit}_t$
Claim cost timing	Calculated in Claim cost timing section (described above)
Claim costs	Input parameter for insurance data (described above)
PV claim costs	When the Claim timing definition is Continuous, $PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Claim cost timing}_{t+1} * \text{Claim cost}_{t+1}$ When the Claim timing definition is Curtate, $(PV_{t+1} + \text{Claim cost}_{t+1}) * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1})$
PV total benefit	PV dth benefit + PV lapse benefit + PV endow benefit + PV claim costs

Note that the lapse rate applies only for PBA VM-20 net premium reserves. It is zero for formulaic cash value and reserve methods.

4.4.2 Net level premium unitary

The unitary net level premium calculations details are displayed as follows:

Net level premium unitary											
Period	Prem payable	\$1 prem ratio	Dth claim rate	Lapse rate	Int rates	Prem timing	PV \$1 prem ratio	PV total benefits	Init terminal value	Net level prem at init period	Period prem
0	1	0	0	0.05	0.05	1	6.6894	3730.8912	0	557.7346	0
1	1	1	0.02662	0.05	0.05	0.939555	6.5289	3833.8216	0	557.7346	557.7346
2	1	1	0.02913	0.05	0.05	0.938401	6.3643	3947.7900	0	557.7346	557.7346
3	1	1	0.03179	0.05	0.05	0.937177	6.1954	4075.5005	0	557.7346	557.7346
4	1	1	0.03465	0.05	0.05	0.935862	6.0218	4220.3955	0	557.7346	557.7346
5	1	1	0.03781	0.05	0.05	0.934409	5.8438	4387.0336	0	557.7346	557.7346

Value	Description
Prem payable	Input parameter for insurance data (described above)
\$1 prem ratio	Input Premium pattern guaranteed _t / Input Premium pattern guaranteed ₁
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
PV \$1 prem ratio	$PV_{t+1} * (1 - Dth\ claim\ rate_{t+1}) * (1 - Lapse\ rate_{t+1}) / (1 + Int\ rates_{t+1}) + Prem\ timing_{t+1} * \$1\ prem\ ratio_{t+1}$
PV total benefits	Calculated in Present value of benefit section (described above)
Initial terminal value	Input parameter for insurance data (described above)
Net level prem at init period	$(PV\ total\ benefits_0 - Initial\ terminal\ value) / PV\ \$1\ prem\ ratio_0$
Period prem	Net level prem at init period _t * \$1 premium ratio _t

4.4.3 Net level premium segmented

When segments apply, the segmented net level premium calculation details are displayed as follows:

Net level premium segmented																			
										Prev PV \$1 prem ratio in segment	PV \$1 prem ratio in segment	PV total benefits	Endow benefit	PV net level prem	Prev PV future seg net level prem	PV future seg net level prem	Net level prem beg of segment	Prem ratio in segment	Period prem
Period	Segment num	Segme nt end	Prem payable	\$1 prem ratio	Dth claim rate	Lapse rate	Int rates	Prem timing											
0	0		1							18.12257	18.48354	202.6482		202.6482	162.6137	157.0346			
1	1	0	1		0.00051	0	0.035	0.982746	17.74931	18.12257	209.3287	0	207.3365	168.3961	162.6137	2.467795		1	2.467795
2	1	0	1		0.00054	0	0.035	0.982731	17.3635	17.74931	216.2226	0	212.1978	174.3912	168.3961	2.467795		1	2.467795
3	1	0	1		0.00058	0	0.035	0.982711	16.96448	17.3635	223.3298	0	217.2407	180.605	174.3912	2.467795		1	2.467795
4	1	0	1		0.00061	0	0.035	0.982697	16.55176	16.96448	230.6665	0	222.4699	187.0459	180.605	2.467795		1	2.467795
5	1	0	1		0.00064	0	0.035	0.982682	16.12515	16.55176	238.2412	0	227.8922	193.7262	187.0459	2.467795		1	2.467795

Value	Description
Segment num	Calculated in Segment section (described above)
Segment end	Calculated in Segment section
Prem payable	Input parameter for insurance data (described above)
\$1 prem ratio	Input Premium pattern guaranteed _t / Input Premium pattern guaranteed ₁
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)

Value	Description
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
Prev PV \$1 prem ratio in segment	If Segment end _{t+1} = 1, equals 0. Otherwise equals PV \$1 premium ratio in segment _{t+1}
PV \$1 prem ratio in segment	Prev PV \$1 premium ratio in segment _t * (1 - Dth claim rate _{t+1}) * (1 - Lapse rate _{t+1}) / (1 + Int rates _{t+1}) + Prem timing _{t+1} * Prem ratio in segment _{t+1} .
PV total benefits	Calculated in Present value of benefit section (described above)
Endow benefit	Input parameter for insurance data (described above)
PV net level premium	$PV_{t+1} * (1 - Dth \text{ claim rate}_{t+1}) * (1 - Lapse \text{ rate}_{t+1}) / (1 + Int \text{ rates}_{t+1}) + Prem \text{ timing}_{t+1} * Period \text{ prem}_{t+1}$.
Prev PV future seg net level premiums	If Segment end _{t+1} = 1, equals 0. Otherwise equals PV future seg net level premium _{t+1} .
PV future seg net level premiums	Prev PV future seg net level premium _t * (1 - Dth claim rate _{t+1}) * (1 - Lapse rate _{t+1}) / (1 + Int rates _{t+1}). If Segment end _{t+1} = 1, Previous PV = PV net level premium _{t+1} , otherwise, Previous PV = PV future net level premium _{t+1} .
Net level prem beg of segment	If Segment end _{t-1} = 1 or Period = 1, equals (PV total benefit _{t-1} – Endowment benefit _{t-1} - PV future seg net level premium _{t-1}) / PV \$1 premium ratio in segment _{t-1} Otherwise equals Net level prem beg of segment _{t-1}
Prem ratio in segment	\$1 prem ratio _t / \$1 prem ratio _s where s = the period when the segment begins
Period prem	Net level prem beg of segment _t * Prem ratio in segment _t

4.5 Expense allowance calculation detail

The following topics describe the expense allowance calculations.

4.5.1 Full preliminary term

When the expense allowance definition is full preliminary term, calculations are performed to determine, when applicable, unitary and segment FPT expense allowances and final expense allowances.

When unitary calculations apply, the FPT expense allowance unitary calculation details are displayed as follows:

Expense allowance unitary	
PV total benefits per 1	144.4163
Endow ben per 1	0
PV \$1 prem unitary per 1	51.92548
FPT prem unitary	2.781223
Cost of insurance per 1	0.68023
Prem timing per 1	0.977981
FPT allow unitary	2.054731

Value	Description
PV total benefits per 1	Calculated in Present value of benefit section (described above)
Endow ben per 1	Input parameter for insurance data (described above)
PV \$1 prem unitary per 1	Calculated in Net level premium unitary section (described above)
FPT premium unitary	PV total benefit per 1 / PV \$1 prem unitary per 1.
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
FPT allow unitary	$\max(0, (\text{FPT prem unitary} - \text{Cost of insurance per 1}) * \text{Premium timing per 1})$.

When segments apply, the FPT expense allowance segment calculation details are displayed as follows:

Expense allowance segment	
PV total benefits per 1	144.4163
Endow ben per 1	0
PV net level prem future seg per 1	107.6168
PV \$1 prem seg per 1	16.0838
FPT prem seg	2.287986
Cost of insurance per 1	0.68023
Prem timing per 1	0.977981
FPT allow seg	1.572354

Value	Description
PV total benefits per 1	Calculated in Present value of benefit section (described above)
Endow ben per 1	Input parameter for insurance data (described above)
PV net level prem future seg per 1	Calculated in Net level premium segment section (described above)
PV \$1 prem seg per 1	Calculated in Net level premium segment section (described above)

Value	Description
FPT prem seg	$(PV \text{ total benefit per 1} - \text{Endowment ben per 1} - PV \text{ net level prem future seg per 1}) / PV \$1 \text{ prem segmented per 1}$
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
FPT allow seg	$\max(0, (\text{FPT prem seg} - \text{Cost of insurance per 1}) * \text{Premium timing per 1})$;

The final expense allowance calculation is then displayed as follows:

Final expense allowance	
Max exp allow	2
Final expense allowance unitary	
FPT allow unitary	2.054731
Exp allow unitary	2
Final expense allowance segment	
FPT allow seg	1.572354
Exp allow seg	1.572354

Value	Description
Max exp allow	Input parameter for insurance data (described above)
FPT allow unitary	Calculated in the unitary expense allowance section (described above)
Exp allow unitary	$\min(\text{Max exp allow}, \text{FPT allow unitary})$;
FPT allow seg	Calculated in the segment expense allowance section (described above)
Exp allow seg	$\min(\text{Max exp allow}, \text{FPT allow seg})$;

4.5.2 1.5 year full preliminary term

When the expense allowance definition is 1.5 year full preliminary term, calculations are performed to determine a unitary expense allowance. Segments calculations do not apply.

The 1.5 year FPT expense allowance calculation detail is displayed as follows:

Expense allowance at issue

PV total benefits per 0	8352.88
PV total benefits per 1	8383.052
PV total benefits per 2	8422.55
PV \$1 prem unitary per 0	11.21848
PV \$1 prem unitary per 1	10.63744
PV \$1 prem unitary per 2	10.04223
Max exp allow	900
Exp allow	772.1386

Value	Description
PV total benefit per 0	Calculated in Present value of benefit section (described above)
PV total benefit per 1	Calculated in Present value of benefit section (described above)
PV total benefit per 2	Calculated in Present value of benefit section (described above)
PV \$1 prem unitary per 0	Calculated in Net level premium unitary section (described above)
PV \$1 prem unitary per 1	Calculated in Net level premium unitary section (described above)
PV \$1 prem unitary per 2	Calculated in Net level premium unitary section (described above)
Max exp allow	Input parameter for insurance data (described above)
Exp allow	$\max(0.0, \min([(PV \text{ total benefit per } 1 / PV \$1 \text{ prem unitary per } 1 + PV \text{ total benefit per } 2 / PV \$1 \text{ prem unitary per } 2) / 2 - PV \text{ total benefit per } 0 / PV \$1 \text{ prem unitary per } 0] * PV \$1 \text{ prem unitary per } 0), \text{Maximum expense allowance}))$.

4.5.3 2 year full preliminary term

When the expense allowance definition is 2 year full preliminary term, calculations are performed to determine a unitary expense allowance. Segments calculations do not apply.

The 2 year FPT expense allowance calculation detail is displayed as follows:

Expense allowance at issue

PV total benefits per 2	1700.347
Cost of insurance per 1	16.94154
Cost of insurance per 2	18.35077
Prem timing per 1	1
Prem timing per 2	1
PV \$1 prem per 2	7.564359
Dth claim rate per 1	0.017619
Int rate per 1	0.04
Lapse rate per 1	0
Max exp allow	800
Exp allow	402.8386

Value	Description
PV total benefits per 2	Calculated in Present value of benefit section (described above)
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Cost of insurance per 2	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
Prem timing per 2	Calculated in Premium timing section (described above)
PV \$1 prem per 2	Calculated in Net level premium section (described above)
Dth claim rate per 1	Calculated in Decrements section (described above)
Int rate per 1	Input parameter for insurance data (described above)
Lapse rate per 1	Input parameter for insurance data (described above)
Max exp allow	Input parameter for insurance data (described above)
Exp allow	$\max(0.0, \min((\text{PV total benefits per 2} / \text{PV \$1 prem per 2} - \text{Cost of insurance per 2}) * \text{Prem timing per 2} * (1 - \text{Dth claim rate per 1}) * (1 - \text{Lapse rate per 1}) / (1 + \text{Int rate per 1}) + (\text{PV total benefits per 2} / \text{PV \$1 prem per 2} - \text{Cost of insurance per 1}) * \text{Prem timing per 1}, \text{Max exp allow}))$.

There is no segmented expense allowance calculated for this method.

4.5.4 1980 CRVM

When the expense allowance definition is 1980 CRVM, calculations are performed to determine, when applicable, unitary and segment FPT expense allowances, a CRVM expense allowance cap, and final expense allowances.

When unitary calculations apply, the FPT expense allowance unitary calculation details are displayed as follows:

Expense allowance unitary	
PV total benefits per 1	144.4163
Endow ben per 1	0
PV \$1 prem unitary per 1	51.92548
FPT prem unitary	2.781223
Cost of insurance per 1	0.68023
Prem timing per 1	0.977981
FPT allow unitary	2.054731

Value	Description
PV total benefits per 1	Calculated in Present value of benefit section (described above)
Endow ben per 1	Input parameter for insurance data (described above)

Value	Description
PV \$1 prem unitary per 1	Calculated in Net level premium unitary section (described above)
FPT prem unitary	PV total benefit per 1 / PV \$1 prem unitary per 1.
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
FPT allow unitary	$\max(0, (\text{FPT prem unitary} - \text{Cost of insurance per 1}) * \text{Prem timing per 1})$.

When segments apply, the FPT expense allowance segment calculation details are displayed as follows:

Expense allowance segment	
PV total benefits per 1	144.4163
Endow ben per 1	0
PV net level prem future seg per 1	107.6168
PV \$1 prem seg per 1	16.0838
FPT prem seg	2.287986
Cost of insurance per 1	0.68023
Prem timing per 1	0.977981
FPT allow seg	1.572354

Value	Description
PV total benefits per 1	Calculated in Present value of benefit section (described above)
Endow ben per 1	Input parameter for insurance data (described above)
PV net level prem future seg per 1	Calculated in Net level premium segment section (described above)
PV \$1 prem seg per 1	Calculated in Net level premium segment section (described above)
FPT prem seg	$(\text{PV total benefit per 1} - \text{Endowment ben per 1} - \text{PV net level prem future seg per 1}) / \text{PV \$1 prem segmented per 1}$
Cost of insurance	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
FPT allow seg	$\max(0, (\text{FPT prem seg} - \text{Cost of insurance per 1}) * \text{Prem timing per 1})$;

Then, for the CRVM expense allowance cap, the average death benefit calculation details are first displayed as follows:

Average dth benefit	
Period	Dth benefits
2	1000
3	1000
4	1000
5	1000
6	1000
7	1000
8	1000
9	1000
10	1000
Avg dth benefit	1000

Value	Description
Death benefits	Input parameter for insurance data (described above)
Average dth benefit	Arithmetic average of death benefit from years 2 through 10

The whole life premium calculation details are then displayed as follows:

Whole life benefit						
Period	Dth claim rate	Lapse rate	Int rates	Claims timing	Avg dth benefits	PV dth benefits
1						144.4163
2	0.00072	0	0.045	1.022335	1000	150.2872
3	0.00078	0	0.045	1.022335	1000	156.3747
4	0.00082	0	0.045	1.022335	1000	162.7066
5	0.00086	0	0.045	1.022335	1000	169.2948

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Claims timing	Calculated in Death claim timing section (described above)
Avg dth benefits	Calculated in Average death benefit section (described above)
PV dth benefits	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Claims timing}_{t+1} * \text{Avg dth benefits}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$

Next, the 19-pay whole life premium calculation details are displayed as follows:

19 pay whole life net premium					
Period	Dth claim rate	Lapse rate	Int rates	Prem timing	PV prem
1					12.75931
2	0.00072	0	0.045	0.977961	12.32038
3	0.00078	0	0.045	0.977932	11.86211
4	0.00082	0	0.045	0.977913	11.38333
5	0.00086	0	0.045	0.977893	10.88304

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
PV prem	$PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Prem timing}_{t+1}$

Then, the ${}_{19}P_{x+1}$ premium and CRVM expense allowance calculation details are displayed as follows:

CRVM ${}_{19}P_{x+1}$	
PV dth benefits	144.4163
PV prem	12.75931
Net level prem	11.31851
Cost of insurance per 1	0.68023
Prem timing per 1	0.977981
CRVM allow	10.40403

Value	Description
PV dth benefits	Calculated in Whole life benefit section (described above)
PV prem	Calculated in 19 pay whole life net premium section (described above)
Net level prem	PV death benefits / PV prem
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
CRVM allow	$(\text{Net level prem} - \text{Cost of insurance per 1}) * \text{Prem timing per 1}$

Finally, the final expense allowance calculation details are displayed as follows:

Final expense allow	
Max exp allow	1000
Final expense allow unitary	
FPT allow unitary	2.054731
CRVM allow unitary	10.40403
Exp allow unitary	2.054731
Final expense allow segment	
FPT allow seg	1.572354
CRVM allow seg	10.40403
Exp allow seg	1.572354

Value	Description
Max exp allow	Input parameter for insurance data (described above)
FPT allow unitary	Calculated in the FPT expense allowance unitary section
CRVM allow unitary	Calculated in the ${}_{19}P_{x+1}$ premium and CRVM expense allowance section
Exp allow unitary	$\min(\text{Max exp allow}, \text{FPT allow unitary}, \max(0, \text{CRVM allow unitary}))$
FPT allow seg	Calculated in the FPT expense allowance seg section
CRVM allow seg	Same as CRVM allow unitary
Exp allow seg	$\min(\text{Max exp allow}, \text{FPT allow seg}, \max(0, \text{CRVM allow seg}))$

4.5.5 Pre 1980 CRVM

When the expense allowance definition is Pre 1980 CRVM, calculations are performed to determine, when applicable, unitary and segment FPT expense allowances, a Pre 1980 CRVM expense allowance cap, and final expense allowances.

When unitary calculations apply, the FPT expense allowance unitary calculation details are displayed as follows:

Expense allowance unitary	
PV total benefits per 1	144.4163
Endow ben per 1	0
PV \$1 prem unitary per 1	51.92548
FPT prem unitary	2.781223
Cost of insurance per 1	0.68023
Prem timing per 1	0.977981
FPT allow unitary	2.054731

Value	Description
PV total benefits per 1	Calculated in Present value of benefit section (described above)
Endow ben per 1	Input parameter for insurance data (described above)
PV \$1 prem unitary per 1	Calculated in Net level premium unitary section (described above)
FPT prem unitary	PV total benefit per 1 / PV \$1 prem unitary per 1.
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
FPT allow unitary	$\max(0, (\text{FPT prem unitary} - \text{Cost of insurance per 1}) * \text{Prem timing per 1})$.

When segments apply, the FPT expense allowance segment calculation details are displayed as follows:

Expense allowance segment	
PV total benefits per 1	144.4163
Endow ben per 1	0
PV net level prem future seg per 1	107.6168
PV \$1 prem seg per 1	16.0838
FPT prem seg	2.287986
Cost of insurance per 1	0.68023
Prem timing per 1	0.977981
FPT allow seg	1.572354

Value	Description
PV total benefits per 1	Calculated in Present value of benefit section (described above)
Endow ben per 1	Input parameter for insurance data (described above)
PV net level prem future seg per 1	Calculated in Net level premium segment section (described above)
PV \$1 prem seg per 1	Calculated in Net level premium segment section (described above)
FPT prem seg	$(\text{PV total benefit per 1} - \text{Endowment ben per 1} - \text{PV net level prem future seg per 1}) / \text{PV \$1 prem segmented per 1}$
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
FPT allow seg	$\max(0, (\text{FPT prem seg} - \text{Cost of insurance per 1}) * \text{Prem timing per 1})$;

Then, for the Pre 1980 CRVM expense allowance cap, the average death benefit calculation details are displayed as follows:

Average dth benefit							
Period	Dth benefit	Dth claim rate	Lapse rate	Claims timing	Int rates	PV dth benefit	PV \$1 dth benefit
1						144.4163	0.144416
2	1000	0.00072	0	1.022335	0.045	150.2872	0.150287
3	1000	0.00078	0	1.022335	0.045	156.3747	0.156375
4	1000	0.00082	0	1.022335	0.045	162.7066	0.162707
5	1000	0.00086	0	1.022335	0.045	169.2948	0.169295

PV dth benefits	144.4163
PV \$1 dth benefits	0.144416
Avg dth benefits	1000

Value	Description
Dth benefit	Input parameter for insurance data (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Claims timing	Calculated in Death claim timing section (described above)
Int rates	Input parameter for insurance data (described above)
PV dth benefit	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Claims timing}_{t+1} * \text{Dth benefit}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$
PV \$1 dth benefit	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Claims timing}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$
Avg dth benefits	PV dth benefit / PV \$1 dth benefit

The whole life premium calculation details are then displayed as follows:

Whole life benefit						
Period	Dth claim rate	Lapse rate	Int rates	Claims timing	Avg dth benefits	PV dth benefits
1						144.4163
2	0.00072	0	0.045	1.022335	1000	150.2872
3	0.00078	0	0.045	1.022335	1000	156.3747
4	0.00082	0	0.045	1.022335	1000	162.7066
5	0.00086	0	0.045	1.022335	1000	169.2948

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Claims timing	Calculated in Death claim timing section (described above)
Avg dth benefits	Calculated in Average death benefit section (described above)
PV dth benefits	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Claims timing}_{t+1} * \text{Avg dth benefits}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$

Next, the 19-pay whole life premium calculation details are displayed as follows:

19 pay whole life net premium					
Period	Dth claim rate	Lapse rate	Int rates	Prem timing	PV prem
1					12.75931
2	0.00072	0	0.045	0.977961	12.32038
3	0.00078	0	0.045	0.977932	11.86211
4	0.00082	0	0.045	0.977913	11.38333
5	0.00086	0	0.045	0.977893	10.88304

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
PV prem	$PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Prem timing}_{t+1}$

Then, the ${}_{19}P_{x+1}$ premium and CRVM expense allowance calculation details are displayed as follows:

CRVM ${}_{19}P_{x+1}$	
PV dth benefits	144.4163
PV prem	12.75931
Net level prem	11.31851
Cost of insurance per 1	0.68023
Prem timing per 1	0.977981
CRVM allow	10.40403

Value	Description
PV dth benefits	Calculated in Whole life benefit section (described above)
PV prem	Calculated in 19 pay whole life net premium section (described above)
Net level prem	PV death benefits / PV prem
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
CRVM allow	(Net level prem - Cost of insurance per 1) * Prem timing per 1

Finally, the final expense allowance calculation details are displayed as follows:

Final expense allow	
Max exp allow	1000
Final expense allow unitary	
FPT allow unitary	2.054731
CRVM allow unitary	10.40403
Exp allow unitary	2.054731
Final expense allow segment	
FPT allow seg	1.572354
CRVM allow seg	10.40403
Exp allow seg	1.572354

Value	Description
Max exp allow	Input parameter for insurance data (described above)
FPT allow unitary	Calculated in the FPT expense allowance unitary section
CRVM allow unitary	Calculated in the ${}_{19}P_{x+1}$ premium and CRVM expense allowance section
Exp allow unitary	$\min(\text{Max exp allow}, \text{FPT allow unitary}, \max(0, \text{CRVM allow unitary}))$
FPT allow seg	Calculated in the FPT expense allowance seg section
CRVM allow seg	Same as CRVM allow unitary
Exp allow seg	$\min(\text{Max exp allow}, \text{FPT allow seg}, \max(0, \text{CRVM allow seg}))$

4.5.6 NY 147 Amendment 5

For the NY 147 Amendment 5 expense allowance, separate unitary, based on a 1980 CRVM expense allowance method, and segment expense allowance, based on a 2 year full preliminary term method, are determined per the requirements of NY 147 Amendment 5.

For the unitary expense allowance, the FPT expense allowance calculation details are displayed as follows:

Expense allowance unitary	
PV total benefits per 1	173.4378
Endow ben per 1	0
PV \$1 prem unitary per 1	366.7217
FPT prem unitary	0.472941
Cost of insurance per 1	0.47011
Prem timing per 1	0.982765
FPT allow unitary	0.002783

Value	Description
PV total benefits per 1	Calculated in Present value of benefit section (described above)
Endow ben per 1	Input parameter for insurance data (described above)
PV \$1 prem unitary per 1	Calculated in Net level premium unitary section (described above)
FPT prem unitary	PV total benefit per 1 / PV \$1 prem unitary per 1.
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
FPT allow unitary	$\max(0, (\text{FPT prem unitary} - \text{Cost of insurance per 1}) * \text{Prem timing per 1})$.

Then, for the 1980 CRVM expense allowance cap, the average death benefit calculation details are displayed as follows:

Average dth benefit	
Period	Dth benefits
2	1000
3	1000
4	1000
5	1000
6	1000
7	1000
8	1000
9	1000
10	1000
Avg dth benefit	1000

Value	Description
Dth benefits	Input parameter for insurance data (described above)
Avg dth benefit	Arithmetic average of death benefit from years 2 through 10

The whole life premium calculation details are then displayed as follows:

Whole life benefit						
Period	Dth claim rate	Lapse rate	Int rates	Claims timing	Avg dth benefits	PV dth benefits
1						173.4378
2	0.0005	0	0.035	1.0174	1000	179.089
3	0.00052	0	0.035	1.0174	1000	184.9242
4	0.00055	0	0.035	1.0174	1000	190.942
5	0.00058	0	0.035	1.0174	1000	197.1492

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Claims timing	Calculated in Death claim timing section (described above)
Avg dth benefits	Calculated in Average death benefit section (described above)
PV dth benefits	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Claims timing}_{t+1} * \text{Avg dth benefits}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$

Next, the 19-pay whole life premium calculation details are displayed as follows:

19 pay whole life net premium					
Period	Dth claim rate	Lapse rate	Int rates	Prem timing	PV prem
1					13.86527
2	0.0005	0	0.035	0.982751	13.34008
3	0.00052	0	0.035	0.982741	12.7965
4	0.00055	0	0.035	0.982726	12.23398
5	0.00058	0	0.035	0.982711	11.65182

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
PV prem	$PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Prem timing}_{t+1}$

Then, the ${}_{19}P_{x+1}$ premium and CRVM expense allowance calculation details are displayed as follows:

CRVM 19Px+1	
PV dth benefits	173.43779
PV prem	13.865267
Net level prem	12.508795
Cost of insurance per 1	0.47011
Prem timing per 1	0.982765
CRVM allow	11.831201

Value	Description
PV dth benefits	Calculated in Whole life benefit section (described above)
PV prem	Calculated in 19 pay whole life net premium section (described above)
Net level prem	PV death benefits / PV prem
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
CRVM allow	(Net level prem - Cost of insurance per 1) * Prem timing per 1

When segments apply, the NY 147 Amendment 5, segment 2 year FPT expense allowance calculation details are then displayed as follows:

Expense allowance segment	
PV total benefits per 2	179.08895
Endow ben per 2	0
PV net level prem future seg per 2	171.55222
PV \$1 prem seg per 2	10.441672
Cost of insurance per 1	0.47011
Cost of insurance per 2	0.500124
Dth claim rate per 1	0.00047
Int rate per 1	0.035
Lapse rate per 1	0
Prem timing per 1	0.982765
Prem timing per 2	0.982751
FPT allow seg	0.457726

Value	Description
PV total benefits per 2	Calculated in Present value of benefit section (described above)
Endow per 2	Input parameter for insurance data (described above)
PV net level prem future seg per 2	Calculated in the Net level premium segmented section (described above)
PV \$1 prem seg per 2	Calculated in the Net level premium segmented section (described above); referred to in that section as PV \$1 prem ratio in segment.

Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Cost of insurance per 2	Calculated in One year cost of insurance section (described above)
PV \$1 prem per 2	Calculated in Net level premium section (described above)
Dth claim rate per 1	Calculated in Decrements section (described above)
Int rate per 1	Input parameter for insurance data (described above)
Lapse rate per 1	Input parameter for insurance data (described above)
Prem timing per 1	Calculated in Premium timing section (described above)
Prem timing per 2	Calculated in Premium timing section (described above)
FTP allow	$\max(0.0, ((\text{PV total benefits per 2} - \text{Endow ben per 2} - \text{PV net level prem future seg per 2}) / \text{PV \$1 prem seg per 2} - \text{Cost of insurance per 2}) * \text{Prem timing per 2} * (1 - \text{Dth claim rate per 1}) * (1 - \text{Lapse rate per 1}) / (1 + \text{Int rate per 1}) + ((\text{PV total benefits per 2} - \text{Endow ben per 2} - \text{PV net level prem future seg per 2}) / \text{PV \$1 prem seg per 2} - \text{Cost of insurance per 1}) * \text{Prem timing per 1})$

Finally, the final expense allowance calculation details are displayed as follows:

Final expense allow	
Max exp allow	1000
Final expense allow unitary	
FPT allow unitary	0.002783
CRVM allow unitary	11.831201
Exp allow unitary	0.002783
Final expense allow segment	
FPT allow seg	0.457726
Exp allow seg	0.457726

Value	Description
Max exp allow	Input parameter for insurance data (described above)
FPT allow unitary	Calculated in the FPT expense allowance unitary section
CRVM allow unitary	Calculated in the ${}_{19}P_{x+1}$ premium and CRVM expense allowance section
Exp allow unitary	$\min(\text{Max exp allow}, \text{FPT allow unitary}, \max(0, \text{CRVM allow unitary}))$
FPT allow seg	Calculated in the FPT expense allowance seg section
Exp allow seg	$\min(\text{Max exp allow}, \text{FPT allow seg})$

4.5.7 1980 SNFL minimum

When the expense allowance definition is 1980 SNFL minimum or the valuation definition is actuarial guideline 45, details for the 1980 SNFL calculations are displayed.

The present value of \$1 level premium for the plan of insurance calculation details are displayed as follows:

Present value \$1 level prem for plan of insurance					
Period	Dth claim rate	Lapse rate	Int rates	Prem timing	PV \$1 prem
0					22.71419
1	0.00028	0	0.04	1	22.58908
2	0.00031	0	0.04	1	22.45961
3	0.00036	0	0.04	1	22.32603
4	0.00039	0	0.04	1	22.18772
5	0.00045	0	0.04	1	22.04515

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
PV \$1 prem	$PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Prem timing}_{t+1}$

The average death benefit calculation details are then displayed as follows:

10 yr average dth benefit	
Period	Dth benefits
1	1000
2	1000
3	1000
4	1000
5	1000
6	1000
7	1000
8	1000
9	1000
10	1000
Avg dth benefit	1000

Value	Description
Dth benefits	Input parameter for insurance data (described above)
Avg dth benefit	Arithmetic average of death benefit from years 1 through 10

Finally, the expense allowance calculation details are displayed as follows:

Net level prem	
PV benefits	128.8884
PV \$1 level prem plan of ins	22.71419
Net level prem	5.674356
1% Avg dth benefit	10
1.25 * Min of NLP and 4% Avg dth benefit	7.092945
Max exp allow	1000
Exp allow	17.09295

Value	Description
PV benefits	Calculated in Present value benefit section (described above)
PV \$1 level prem plan of ins	Calculated in Present value \$1 level premium for plan of insurance section (described above)
Net level prem	PV benefits / PV \$1 level prem plan of ins
1% Avg dth benefit	0.01 * Average death benefit which was Calculated in 10 year average Death benefit section (described above)
1.25 * Min of NLP and 4% Avg dth benefit	1.25 * min(Net level prem, 0.04 * Avg dth benefit)
Max exp allow	Input parameter for insurance data (described above)
Exp allow	min(Max exp allow, max(0, 1% Avg dth benefit + 1.25 * Min of NLP and 4% Avg dth benefit))

4.5.8 Pre 1980 SNFL minimum

When the expense allowance definition is Pre 1980 SNFL minimum expense allowance calculations, details for the Pre 1980 SNFL minimum calculations are displayed.

The average death benefit calculation details are displayed as follows:

Average dth benefit							
Period	Dth benefit	Dth claim rate	Lapse rate	Claims timing	Int rates	PV dth benefit	PV \$1 dth benefit
0						152.4141	0.152414
1	1000	0.00108	0	1.022335	0.045	158.3396	0.15834
2	1000	0.001152	0	1.022335	0.045	164.4766	0.164477
3	1000	0.001224	0	1.022335	0.045	170.8358	0.170836
4	1000	0.001323	0	1.022335	0.045	177.4056	0.177406
5	1000	0.00135	0	1.022335	0.045	184.2574	0.184257

PV dth benefits	152.4141
PV \$1 dth benefits	0.152414
Avg dth benefits	1000

Value	Description
Dth benefit	Input parameter for insurance data (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Claims timing	Calculated in Claim timing section (described above)
Int rates	Input parameter for insurance data (described above)
PV dth benefit	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Claims timing}_{t+1} * \text{Dth benefit}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$
PV \$1 dth benefit	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Claims timing}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$
Avg dth benefits	PV dth benefit / PV \$1 dth benefit

The whole life premium calculation details are then displayed as follows:

Whole life plan								
Period	Dth claim rate	Lapse rate	Claims timing	Int rates	Avg dth benefit	PV dth benefit	Prem timing	PV \$1 prem
0						152.4141		19.25593
1	0.00108	0	1.022335	0.045	1000	158.3396	0.977787	19.12131
2	0.001152	0	1.022335	0.045	1000	164.4766	0.977752	18.98188
3	0.001224	0	1.022335	0.045	1000	170.8358	0.977717	18.83741
4	0.001323	0	1.022335	0.045	1000	177.4056	0.977669	18.68816
5	0.00135	0	1.022335	0.045	1000	184.2574	0.977655	18.53249

Value	Description
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Claims timing	Calculated in Death claim timing section (described above)
Int rates	Input parameter for insurance data (described above)
Avg dth benefits	Calculated in Average death benefit section (described above)
PV dth benefits	$[PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) + \text{Claims timing}_{t+1} * \text{Avg dth benefits}_{t+1} * \text{Dth claim rate}_{t+1}] / (1 + \text{Int rates}_{t+1})$
Prem timing	Calculated in Premium timing section (described above)
PV \$1 prem	$PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Prem timing}_{t+1}$

The plan of insurance adjusted premium calculation details are then displayed as follows:

Avg dth benefit	1000							
PV benefits whole life	152.4141							
PV \$1 prem whole life	19.25593							
PV benefits plan of ins	152.4141							
PV \$1 prem plan of ins	19.25593							
Plan of insurance adj prem calculation								
Condition	PV ben	Db mult	Avg db	Prem mult	Prem	PV \$1 prem	Subtraction	Adj Prem
Whole life < 40	152.4141	0.02	1000	0	0	19.25593	0.65	9.26662
Whole life >=40	152.4141	0.046	1000	0	0	19.25593	0	10.304051
Plan < Whole life < 40	152.4141	0.02	1000	0	0	19.25593	0.65	9.26662
Whole life < Plan < 40	152.4141	0.02	1000	0.25	9.26662	19.25593	0.4	9.26662
Plan < 40 < Whole life	152.4141	0.03	1000	0	0	19.25593	0.4	9.674097
Plan >= 40	152.4141	0.046	1000	0	0	19.25593	0	10.304051
Plan adj prem	9.26662							

Value	Description
Avg dth benefit	Calculated in Average death benefit section (described above)
PV benefits whole life	Calculated in Whole life plan section (described above)
PV \$1 prem whole life	Calculated in Whole life plan section (described above)
PV benefits plan of ins	Calculated in Present value benefit section (described above)
PV \$1 prem plan of ins	Calculated in Net level premium section (described above)
PV ben	For Whole life < 40 and Whole life >= 40, equals PV benefits whole life. Otherwise, equals PV benefits plan of ins.
Db mult	Expense allowance percent of death benefit as specified in the valuation law.
Avg db	Calculated in Average death benefit section (described above)
Prem mult	Expense allowance percent of premium as specified in the valuation law.
Prem	For Whole life < Plan < 40, equals Adjusted Premium for Whole life < 40. Otherwise, 0.
PV \$1 prem	For Whole life < 40 and Whole life >= 40, equals PV \$1 prem whole life. Otherwise, equals PV \$1 prem plan of ins.
Subtraction	Expense allowance PV death benefits adjustment as specified in the valuation law.
Adj Prem	$(PV\ ben + Db\ mult * Avg\ dbt) / (PV\ \$1\ prem - Subtraction)$

The formula for the plan of insurance adjusted premium is:

If Adjusted Premium for	Test order	Plan Adjusted Premium = Adjusted Premium for
(Plan $\geq$ 40) > 40	1	Plan $\geq$ 40
(Whole life $\geq$ 40) > 40	2	Plan < 40 < Whole life
(Whole life < Plan < 40) > (Whole life < 40)	3	Whole life < Plan < 40
None of above	4	Plan < Whole life < 40

Finally, the final expense allowance calculation details are displayed as follows:

Plan adj prem	9.26662
PV \$1 prem plan of ins	19.25593
PV benefits plan of ins	152.4141
Max exp allow	1000
Exp allow	26.0233

Value	Description
Plan adj prem	Calculated in Plan of insurance Adjusted Premium calculation section (described above)
PV \$1 prem plan of ins	Calculated in Net level premium section (described above)
PV benefits plan of ins	Calculated in Present value benefit section (described above)
Max exp allow	Input parameter for insurance data (described above)
Exp allow	$\min(\text{Max exp allow}, \max(0, \text{Plan adj prem} * \text{PV \$1 prem plan of ins} - \text{PV benefits plan of ins}))$

4.6 Valuation net premiums calculation detail

4.6.1 Modified premiums unitary

When unitary calculations apply, for valuation definitions other than VM20NPR, modified premiums are calculated.

When terminal values grade from the expense allowance result to the net level result between two specified policy anniversaries, premium grading calculation results are displayed as follows:

Modified premium unitary	
Grade from pol yr	2
Grade to pol yr	10

Present value grading prem							
Period	Prem payable	\$1 prem ratio	Dth claim rate	Int rates	Prem timing	PV \$1 prem ratio	PV \$1 graded prem
2	1	1.003289	0.00164	0.04	0.979845	7.349162	7.301128
3	1	1.006579	0.00161	0.04	0.979859	6.628042	6.584721
4	1	1.082237	0.00157	0.04	0.979879	5.799389	5.761485
5	1	1.092105	0.00152	0.04	0.979903	4.925888	4.893693

Value	Description
Grade from pol yr	Input parameter for insurance data (described above)
Grade to pol yr	Input parameter for insurance data (described above)
Prem payable	Input parameter for insurance data (described above)
\$1 prem ratio	Calculated in Net level premium section (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
PV \$1 prem ratio	Calculated in Net level premium section (described above)
PV \$1 graded prem	$PV_{t+1} * (1 - Dth\ claim\ rate_{t+1}) * (1 - Lapse\ rate_{t+1}) / (1 + Int\ rates_{t+1}) + Prem\ timing_{t+1} * Input\ Premium\ pattern\ guaranteed_{t+1} / Input\ Premium\ pattern\ guaranteed_{grade\ from+1}$

The unitary beta premium calculation details are then displayed as follows:

Beta premium	
Net level prem at issue	331.3861
Exp allow	9.670065
PV \$1 prem at issue	8.69575
Exp allow prem	1.112045
Beta premium	332.4982

Value	Description
Net level prem at issue	Calculated in Net level premium section (described above)
Exp allowance	Calculated in the appropriate expense allowance section (described above)
PV \$1 prem at issue	Calculated in Net level premium section (described above)
Exp allow prem	Expense allow / PV \$1 prem at issue
Beta premium	Net level prem at issue + Exp allow premium

Finally, the unitary modified premiums calculation details are displayed as follows:

Period	Net level prem	Cost of insurance	Exp allow	Exp allow prem	PV \$1 prem unitary at grade from	PV \$1 prem grading at grade from	Beta prem	Grading prem	Prem ratio	Compare prem	Compare prem prev	Exp allow applied	Exp allow avail	Prem timing	Modified prem
1	331.3861	1.671386					332.4982	0	1	332.4982	0	0	9.670065	0.97983	322.6291
2	332.4762	1.641337					332.4982	0	1.003289	333.5919	332.4982	9.670065	0	0.979845	333.5919
3	333.5663	1.611289	9.670065	1.112045	7.349162	7.301128	332.4982	334.6857	1.006579	334.6857	0	0	0	0.979859	334.6857
4	358.6383	1.571225	9.670065	1.112045	7.349162	7.301128	332.4982	334.6857	1.082237	359.8418	0	0	0	0.979879	359.8418
5	361.9086	1.521149	9.670065	1.112045	7.349162	7.301128	332.4982	334.6857	1.092105	363.123	0	0	0	0.979903	363.123

Value	Description
Net level prem	Calculated in Net level premium section (described above)
Cost of insurance	Calculated in One year cost of insurance section (described above)
Expense allow	Calculated in the appropriate expense allowance section (described above)
Exp allow prem	Calculated in Beta premium section (described above)
PV \$1 prem unitary at grade from	Calculated in Net level premium section (described above)
PV \$1 prem grading at grade from	Calculated in Present value grading premium section (described above)
Beta prem	Calculated in Beta premium section (described above)
Grading prem	$\text{Net level prem}_{\text{grade from}+1} + (\text{Exp allow}_t / \text{PV \$1 prem unitary}_0) * \text{PV \$1 prem unitary}_{\text{grade from}} / \text{PV \$1 prem grading}_{\text{grade from}}$
Prem ratio	Calculated in Net level premium section (described above)
Compare prem	$\text{Beta prem}_t * \text{Prem ratio}_t$
Compare prem prev	If period = 2 and period <= grade to, equals Compare prem ₁ . Otherwise, 0.
Exp allow applied	If period = 2, equals $\min(\text{Exp allow}_t, \text{Prem timing}_{t-1} * (\text{Compare prem prev}_t - \text{Modified prem}_{t-1}))$. Otherwise, 0.
Exp allow available	If period = 1, equals Exp allow _t . If period = 2, equals $\max(0, \text{Exp allow}_t - \text{Exp allow applied}_t) * (1 + \text{Int rates}_{t-1}) / (1 - \text{Dth claim rate}_{t-1})$. Otherwise, 0.
Prem timing	Calculated in Premium timing section (described above)
Modified prem	$\max(\text{Compare prem}_t - \text{Exp allow available}_t / \text{Prem timing}_t, \min(\text{Cost of insurance}_t, \text{Net level prem}_t))$

4.6.2 Modified premiums segments

When segments calculations apply, for valuation definitions other than VM20NPR and ActuarialGuideline45, modified premiums are calculated.

The segment beta premium calculation details are displayed as follows:

Beta premium	
PV total benefits per 0	167.95599
PV future seg net level prem per 0	159.99053
Exp allow per 0	0.457726
PV \$1 prem seg per 0	11.669796
Beta	0.721793

Value	Description
PV total benefits per 0	Calculated in Present value benefit section (described above)
PV future seg net level prem per 0	Calculated in Net level premium section (described above)
Exp allow per 0	Calculated in the appropriate expense allowance section (described above)
PV \$1 prem seg per 0	Calculated in Net level premium section (described above)
Beta	$(\text{PV total benefits per 0} - \text{PV future seg net level prem per 0} + \text{Exp allow per 0}) / \text{PV \$1 prem seg per 0}$

For expense allowance methods other than NY 147 Amendment 5, the segment modified premium calculation details are displayed as follows:

Period	Segment num	Net level prem	Cost of insurance per 1	Beta	Exp allow	Prem ratio	Prem timing	Modified prem
1	1	0.776531	0.491497	0.802752	0.311255		1	0.491497
2	1	0.776531		0.802752		1	1	0.802752
3	1	0.776531		0.802752		1	1	0.802752
4	1	0.776531		0.802752		1	1	0.802752
5	1	0.776531		0.802752		1	1	0.802752

Value	Description
Segment num	Calculated in Segment section (described above)
Net level prem	Calculated in Net level premium section (described above)
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Beta	Calculated in Beta premium section (described above)

Value	Description
Exp allow	Calculated in the appropriate expense allowance section (described above)
Prem ratio	Calculated in Net level premium section (described above)
Prem timing	Calculated in Premium timing section (described above)
Modified prem	If period = 1, equals $\max(\text{Beta}_1 - \text{Exp allow}_1 / \text{Prem timing}_1, \min(\text{Cost of insurance per } 1_1, \text{Net level prem}_1))$. Otherwise, equals $\text{Beta}_t * \text{Prem ratio}_t$. If Segment num _t > 1, equals Net level prem _t .

For the NY 147 Amendment 5 expense allowance method, the modified premium calculation details are displayed as follows:

Period	Segment num	Net level prem	Cost of insurance	Beta	Compare prem	Compare prem prev	Exp allow applied	Exp allow avail	Prem ratio	Prem timing	Modified prem
1	1	0.68257	0.47011	0.721793	0.721793	0	0	0.457726	1	0.982765	0.47011
2	1	0.68257	0.500124	0.721793	0.721793	0.721793	0.247346	0.217845	1	0.982751	0.500124
3	1	0.68257	0.520134	0.721793	0.721793	0	0	0	1	0.982741	0.721793
4	1	0.68257	0.55015	0.721793	0.721793	0	0	0	1	0.982726	0.721793
5	1	0.68257	0.580167	0.721793	0.721793	0	0	0	1	0.982711	0.721793

Value	Description
Segment num	Calculated in Segment section (described above)
Net level prem	Calculated in Net level premium section (described above)
Cost of insurance per 1	Calculated in One year cost of insurance section (described above)
Beta	Calculated in Beta premium section (described above)
Compare prem	$\text{Beta}_t * \text{Prem ratio}_t$
Compare prem prev	If period = 2, equals Compare prem ₁ . Otherwise, 0.
Exp allow applied	If period = 2, equals $\min(\text{Exp allow}, \text{Prem timing}_{t-1} * (\text{Compare prem prev}_t - \text{Modified prem}_{t-1}))$. Otherwise, 0.
Exp allow available	If period = 1, equals Exp allow. If period = 2, equals $\max(0, \text{Exp allow} - \text{Exp allow applied}_t) * (1 + \text{Int rates}_{t-1}) / (1 - \text{Dth claim rate}_{t-1})$. Otherwise, 0.
Prem ratio	Calculated in Net level premium section (described above)
Prem timing	Calculated in Premium timing section (described above)
Modified prem	If period <= 2, equals $\max(\text{Beta}_t - \text{Exp allow available}_t / \text{Prem timing}_t, \min(\text{Cost of insurance per } 1_t, \text{Net level prem}_t))$. Otherwise, equals $\text{Beta}_t * \text{Prem ratio}_t$. If Segment num _t > 1, equals Net level prem _t .

4.6.3 VM-20 valuation net premiums

The VM20 valuation net premiums calculation details includes details for the calculations of a preliminary valuation net premium and shock lapse premium limitation adjustment ratios.

The preliminary valuation net premiums calculation details are displayed as follows:

Preliminary valuation net premiums													
Period	Prem payable	\$1 prem ratio	Adj prem mult	Dth claim rate	Lapse rates	Int rates	Prem timing	PV \$1 prem ratio	PV total benefits	Initial NSP unitary	PV expense allowance	Net level prem at init period	Period prem
0	1	0	0	0	0	0.06	1	8.182582	11.921157	0	2.5	1.762421	0
1	1	1	0	0.000083	0.06	0.06	1	9.22793	13.356169	0	0	1.762421	0
2	1	1	0.9	0.000274	0.06	0.06	1	9.39364	14.774131	0	0	1.762421	1.586179
3	1	1	0.9	0.00051	0.06	0.06	1	9.582822	16.125842	0	0	1.762421	1.586179
4	1	1	0.9	0.000804	0.06	0.06	1	9.799145	17.343164	0	0	1.762421	1.586179
5	1	1	0.9	0.001163	0.06	0.06	1	10.04689	18.341361	0	0	1.762421	1.586179

Value	Description
Prem payable	Input parameter for insurance data (described above)
\$1 prem ratio	Input Premium pattern guaranteed _t / Input Premium pattern guaranteed ₁
Adj prem mult	Input parameter for insurance data (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
PV \$1 prem ratio	$PV_{t+1} * (1 - Dth\ claim\ rate_{t+1}) * (1 - Lapse\ rate_{t+1}) / (1 + Int\ rates_{t+1}) + Prem\ timing_{t+1} * \$1\ prem\ ratio_{t+1} * Adj\ prem\ mult_{t+1}$
PV total benefits	Calculated in Present value of benefit section (described above)
Initial terminal value	Input parameter for insurance data (described above)
PV expense allowance	Input Expense allowance unitary _{t+1} for life insurance data (described above)
Net level prem at initial period	$(PV\ total\ benefits_0 - Initial\ terminal\ value + PV\ expense\ allowance_0) / PV\ \$1\ prem\ ratio_0$
Period prem	$Net\ level\ prem\ at\ init\ period_t * \$1\ prem\ ratio_t * Adj\ prem\ mult_t$

If input initial level premium period is greater than or equal to input premium term break for lapse rate and less than input premium term (i.e., there are more than one level premium period), shock lapse premium limitation apply. This ensures that the uniform percent of respective adjusted gross premiums after the shock lapse does not exceed the user specified shock lapse net premium limit, resulting in an offsetting adjustment to the uniform percent of respective adjusted gross premiums before the shock lapse.

Present values benefits and premiums post shock lapse calculation details are displayed as follows:

Shock lapse premium limitation												
Present value benefits and premiums post shock lapse												
Period	Prem payable	Dth claim rate	Lapse rate	Int rates	Claims timing	Dth benefit	PV dth benefit	Endow benefit	PV endow benefit	PV total benefits	Prem timing	PV net prem
0	1						11.236008		0	11.236008		13.394556
1	1	0.000083	0.06	0.06	1	0	12.67144	0	0	12.67144	1	15.105748
2	1	0.000274	0.06	0.06	1	0	14.292982	0	0	14.292982	1	17.038805
3	1	0.00051	0.06	0.06	1	0	16.125842	0	0	16.125842	1	19.223776
4	1	0.000804	0.06	0.06	1	1000	17.343164	0	0	17.343164	1	2.984752
5	1	0.001163	0.06	0.06	1	1000	18.341361	0	0	18.341361	1	2.984752

Value	Description
Prem payable	Input parameter for insurance data (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Claims timing	Calculated in Claim timing section (described above)
Dth benefit	0 during initial level premium period. After that, input death benefit for life insurance data (described above)
PV dth benefit	$[PV_{t+1} * (1 - Dth\ claim\ rate_{t+1}) * (1 - Lapse\ rate_{t+1}) + Claims\ timing_{t+1} * Dth\ benefit_{t+1} * Dth\ claim\ rate_{t+1}] / (1 + Int\ rates_{t+1})$
Endow benefit	0 during initial level premium period. After that, input endowment benefit for life insurance data (described above)
PV endow benefit	$(PV_{t+1} + Endow\ benefit_{t+1}) * (1 - Dth\ claim\ rate_{t+1}) * (1 - Lapse\ rate_{t+1}) / (1 + Int\ rates_{t+1})$
PV total benefits	$PV\ dth\ benefit_t + PV\ endow\ benefit_t$
Prem timing	Calculated in Premium timing section (described above)
Net prem	0 during initial level premium period. After that, period premium calculated in preliminary valuation net premiums section (described above)
PV net prem	$PV_{t+1} * (1 - Dth\ claim\ rate_{t+1}) * (1 - Lapse\ rate_{t+1}) / (1 + Int\ rates_{t+1}) + Prem\ timing_{t+1} * Net\ premium_{t+1}$

The post shock lapse adjusted ratio calculation details are displayed as follows:

Post shock lapse adjusted ratio	
PV total benefits post shock lapse	11.23601
PV premium post shock lapse	13.39456
Post shock lapse ratio	1.19211
Max post shock lapse ratio	1.35
Valuation net premium adj factor following shock lapse	1

Value	Description
PV total benefits post shock lapse	PV total benefit ₀ calculated in Present value benefits and premiums post shock lapse section (described above)
PV premium post shock lapse	PV net premium ₀ calculated in Present value benefits and premiums post shock lapse section (described above)
Post shock lapse ratio	PV premium post shock lapse / PV total benefits post shock lapse
Max post shock lapse ratio	Input parameter for insurance data (described above)
Valuation net premium adj factor following shock lapse	Min(1, Max post shock lapse ratio / Post shock lapse ratio)

The pre shock lapse adjusted ratio calculation details are displayed as follows:

Pre shock lapse adjusted ratio	
PV total benefits	11.92116
PV expense allowances	2.5
PV adjusted valuation premiums post shock lapse	13.39456
PV unadjusted premiums before shock lapse	1.026601
Valuation net premium ratio before shock lapse	1

Value	Description
PV total benefits	PV total benefit ₀ calculated in Present value benefits section (described above)
PV expense allowances	PV expense allowance ₀ calculated in Preliminary valuation net premiums section (described above)
PV adjusted valuation premiums post shock lapse	PV premium post shock lapse * Valuation net premium adjustment factor following shock lapse calculated in Post shock lapse adjusted ratio section (described above)
PV unadjusted premiums before shock lapse	PV total benefits + PV expense allowances - PV net premium ₀ calculated in Present value benefits and premiums post shock lapse section (described above)

Value	Description
Valuation net premium ratio before shock lapse	(PV total benefits + PV expense allowances - PV adjusted valuation premiums post shock lapse) / PV unadjusted premiums before shock lapse

4.6.4 Present value of premiums

For all valuation definitions other than VM20NPR, the present value of premiums calculation details are displayed as follows:

The number of entries = Maturity age - Issue age + 1.

Present value of premium									
Period	Prem payable	Dth claim rate	Lapse rate	Int rates	Prem timing	Net prem unitary	PV net prem unitary	Net prem segmented	PV net prem segmented
0	1						138.7687		138.768738
1	1	0.00068	0	0.045	0.977981	0.68023	144.4163	0.68023	144.416347
2	1	0.00072	0	0.045	0.977961	2.781223	148.1794	2.287986	148.683883
3	1	0.00078	0	0.045	0.977932	2.781223	152.1239	2.287986	153.155937
4	1	0.00082	0	0.045	0.977913	2.781223	156.2555	2.287986	157.839247
5	1	0.00086	0	0.045	0.977893	2.781223	160.5829	2.287986	162.743884

Value	Description
Prem payable	Input parameter for insurance data (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
Net prem unitary	If Expense allowance definition = Net Level, equals Net level premium unitary calculated in Net level premium section (described below). Otherwise, equals Modified premium unitary calculated in Modified premium section (described below).
PV net prem unitary	$PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Prem timing}_{t+1} * \text{Net prem unitary}_{t+1}$.
Net prem segmented	If Expense allowance definition = Net Level, equals Net level premium segments calculated in Net level premium section (described above). Otherwise, equals Modified premium segmented calculated in Modified premium section (described above).
PV net prem segmented	$PV_{t+1} * (1 - \text{Dth claim rate}_{t+1}) * (1 - \text{Lapse rate}_{t+1}) / (1 + \text{Int rates}_{t+1}) + \text{Prem timing}_{t+1} * \text{Net prem segmented}_{t+1}$.

When the valuation definition is VM20NPR, the present value of premiums calculations are displayed as follows:

Present value net premiums									
Period	Prem payable	Dth claim rate	Lapse rate	Int rates	Prem timing	Preliminary valuation net premium	Adjustment ratio	Period prem	PV valuation net premium
0	1								14.421157
1	1	0.000083	0.06	0.06	1	0	1	0	16.2635
2	1	0.000274	0.06	0.06	1	0.613687	1	0.613687	17.652492
3	1	0.00051	0.06	0.06	1	0.613687	1	0.613687	19.223776
4	1	0.000804	0.06	0.06	1	2.984752	1	2.984752	18.326824
5	1	0.001163	0.06	0.06	1	2.984752	1	2.984752	17.320777

Value	Description
Prem payable	Input parameter for insurance data (described above)
Dth claim rate	Calculated in Decrements section (described above)
Lapse rate	Input parameter for insurance data (described above)
Int rates	Input parameter for insurance data (described above)
Prem timing	Calculated in Premium timing section (described above)
Preliminary valuation net premium	Period prem calculated in Preliminary valuation net premiums section (described below)
Adjustment ratio	If shock lapse premium limitation does not apply, 1. Otherwise, valuation net premium ratio before shock lapse during initial level premium period, valuation net premium adjustment factor following shock lapse afterward (described below)
Period prem	Preliminary valuation net premium * Adjustment ratio
PV valuation net premium	$PV_{t+1} * (1 - Dth\ claim\ rate_{t+1}) * (1 - Lapse\ rate_{t+1}) / (1 + Int\ rates_{t+1}) + Prem\ timing_{t+1} * Period\ prem_{t+1}$

4.7 Terminal values calculation detail

4.7.1 Terminal values

The terminal reserve or cash value calculation results are displayed as follows:

Terminal values						
Period	PV benefits	PV net prem unitary	Terminal val unitary	Segment num	PV net prem segmented	Terminal val segmented
0	144.2758	144.2758	0	0	144.27582	0
1	149.1936	150.0544	-0.860819	1	149.19359	0
2	154.2784	156.0763	-1.797855	1	154.2635	0.014925
3	159.529	162.3536	-2.824554	1	159.54959	-0.020583
4	165.0126	168.8855	-3.872969	1	165.04969	-0.037131
5	170.7385	175.6824	-4.943949	1	170.7724	-0.03394

Value	Description
PV benefit	Calculated in Present value benefit section (described above)
PV net prem unitary	Calculated in Present value of premiums section (described above)
Terminal val unitary	$PV\ benefit_t - PV\ net\ premium\ unitary_t$
Segment num	Calculated in Segment section (described above)
PV net prem segmented	Calculated in Present value of premiums section (described above)
Terminal val segmented	$PV\ benefit_t - PV\ net\ premium\ segmented_t$

4.7.2 ETI values

When ETI values are calculated, those calculation results are displayed as follows:

ETI Values									
Period	Attained age	ETI cash val	PV dth benefit	PV \$1 endow benefit	Prelim yrs	Full yrs NSP	Part yr NSP	ETI yrs	ETI endow per unit
1	26	-13.3883	83.32294	0.076012	0	0	0	0	0
2	27	-1.28729	85.90771	0.079951	0	0	0	0	0
3	28	11.46064	88.64467	0.084093	7	10.35675	11.7669	7.782815	0
4	29	24.87866	91.53251	0.088448	15	23.73789	25.5879	15.61663	0
5	30	38.98194	94.56084	0.093029	21	37.9742	40.29426	21.43436	0

Value	Description
Attained age	Issue age + policy year
ETI cash val	Input parameter for insurance data (described above)
PV dth benefit	Calculated in Present value of benefit section (described above)
PV \$1 endow benefit	Calculated similarly as PV endowment benefit in Present value of benefit section (described above) with \$1 replacing input endowment benefit, i.e., $[PV_{t+1} + 1] * (1 - Dth\ claim\ rate_{t+1}) * (1 - Lapse\ rate_{t+1}) / (1 + Int\ rates_{t+1})$

Value	Description
Prelim yrs	If period $\geq$ Premium term, or input ETI cash value ≤ 0 or ETI cash value $\geq$ PV death benefit, Preliminary years = 0. If ETI cash value $<$ PV death benefit, in those years, at each period, AML will at first go over one year, then calculate PV death benefit using this formula: $PV \text{ death benefit}_t = (PV \text{ dth benefit}_{t+1} * (1 - Dth \text{ claim rate}_t) * (1 - Lapse \text{ rate}_t) + Claim \text{ timing}_t * Dth \text{ benefit}_t * Dth \text{ claim rate}_t) / (1 + Int \text{ rates}_t)$. AML then compares calculated PV dth benefit to see if it's larger than ETI cash value, if not, then go back for 2 years, and so on. Eventually, it will go back enough years, such that PV dth benefit just goes above ETI cash value. The number of years takes for this to happen is Prelim yrs.
Full years NSP	When Prelim yrs = 0, Full years NSP = 0. Otherwise, Full years NSP is the PV dth benefit calculated right before it goes above ETI cash value.
Part year NSP	When Prelim yrs = 0, Part year NSP = 0. Otherwise, Part year NSP is the PV dth benefit calculated right after it goes above ETI cash value.
ETI years	If period $\geq$ Premium term, or input ETI cash value ≤ 0, ETI years = 0. For those periods when ETI cash value $\geq$ PV dth benefit, ETI years = Maturity age - Issue age - t. For those periods when ETI cash value $<$ PV dth benefit, ETI years = Prelim yrs + (ETI cash value - Full years NSP) / (Part year NSP - Full years NSP)
ETI endow per unit	If period $\geq$ Premium term, or input ETI cash value ≤ 0 or ETI cash value $<$ PV death benefit, then ETI endowment per unit = 0. Otherwise, ETI endow per unit = (ETI cash value - PV dth benefit) / PV \$1 Endow benefit.

4.8 Annuity values calculation detail

4.8.1 Present value of benefits

The present value of annuity benefits calculations details are displayed as follows:

Present value benefits						
Period	Decrem rate	Int rates	Survivor and discount factor	Benefits BOM	Benefits EOM	PV benefits
0						0.803507
1	0.475401	0.001652	0.523734	0.569388	0	0.447021
2	0.408409	0.001652	0.590616	0	0	0.756873
3	0.34457	0.001652	0.65435	0	0	1.156679
4	0.28322	0.001652	0.715598	0	0	1.616381
5	0.24885	0.001652	0.749911	0	0	2.155429
6	0.21805	0.001652	0.780661	0	0	2.761032
7	0.17643	0.001652	0.822212	0.569388	0	2.665547
8	0.14914	0.001652	0.849457	0	0	3.137942

Value	Description
Period	Month since start of annuity
Decrement rate	Calculated in Annuity decrements section (described above)
Int rates	Input parameter for annuity data (described above) translated into monthly rates.
Survivor and discount factor	$[(1 - \text{Decrem rate}_t) / (1 + \text{Int rates}_t)]$
Benefits BOM	Input parameter for annuity data (described above)
Benefits EOM	Input parameter for annuity data (described above)
PV benefits	$[(PV_{t+1} + \text{Benefits EOM}_t) * \text{Survivor and discount factor}_{t+1} + \text{Benefits BOM}_{t+1}]$

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A01 BONDS

This book discusses bond assets and the bond submodels.

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1 INPUT ORGANIZATION

Input fields for bonds are located on the Economy and Investment input pages, or on model point records.

1.1 Economy input page

The Economy input page includes assumptions related to the general economic environment, which is independent of company and asset portfolio assumptions.

1.1.1 *Credit spreads*

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. See the Credit spread section in the Market value chapter, below.

1.1.2 *Default rates*

Specify bond default rates on the Economy input page under the Scenarios: Assumptions category. See the Defaults chapter, below.

1.2 Investment input page

The Investment input page includes assumptions related to projected asset portfolio and investment strategies.

1.2.1 *Sinking fund schedule*

Specify bond sinking fund schedules on the Investment input page under the Decrements: Sinking Fund Payments category. See the Sinking fund payments chapter, below.

1.2.2 *Call and put options*

Specify bond call and put option price schedules on the Investment input page under the Decrements: Calls/Puts/Prepayments category. Also specify option exercise assumptions under this category. See the Calls and Puts chapters, below.

1.2.3 *Expenses*

Investment expense rates vary by asset sub portfolio. See the Investment expenses section in the Cash flows chapter, below.

1.2.4 *Accounting*

Specify bond accounting rules on the Investment input page under the Balance Sheet category. These definitions apply for all bonds. See the Accounting values chapter, below.

1.2.5 *Purchase bond definitions*

Specify purchase bond definitions on the Investment input page under the Purchase Asset Definitions category. See the Purchase bond variables chapter, below.

1.3 Model points

Specify on the model point records the definition for each bond existing at the projection date. See the Model point records chapter, below.

Report groups

Assets typically do not use the report group feature that is used to group liability model points. Instead, asset sub portfolio reports automatically collect the individual assets assigned to each sub portfolio. This has the advantage that asset sub portfolio assignments can change dynamically as asset durations change through the projection.

However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

2 CASH FLOWS

2.1 Payment timing

2.1.1 Maturity

The bond matures for its remaining par value at the end of the maturity month specified on the model point record.

2.1.2 Payment Dates

Coupon payments occur either semi-annually, quarterly, annually, or monthly, at the end of the month related to the maturity date. For example, a bond maturing in April has semi-annual coupon payments at the end of April and October.

All payments occur at the end of the month.

2.2 Coupon payments

2.2.1 Initial coupon rate

The model point record contains the initial (at the projection date) coupon rate. Coupon rates are nominal based on the payment mode. For example, a 6.00% (the rate on the model point record) coupon semi-annual bond has an effective coupon rate of 6.09%.

The initial coupon rate is calculated for purchased assets.

2.2.2 Floating rate notes

Existing assets

The initial coupon rate applies until the next reset date. The rate resets on the payment date in or after that month. That is, the rate can only reset on a coupon payment date.

For fixed coupon bonds, specify the reset date simply as the maturity date or later, such as 9999/01/01.

For floating rate notes, also called adjustable rate bonds, the coupon rate is reset at the next reset date, and then periodically thereafter based on the reset period months. At each reset date, the coupon rate changes to be the scenario bond equivalent yield from a specified category id for a specified maturity years, times a multiple then plus an addition (nominal rate).

Each new coupon rate is constrained to be within the maximum rate (cap) and minimum rate (floor).

Purchase assets

For bonds purchased during the projection, the addition for rate resets is different from the addition that determines the initial rate.

Floating rate note examples

The following shows examples for how to specify some special types of bonds:

- **Reverse floater** – When coupon rates change in opposite proportion to the reference rate, the multiple is -1.00 (-100%), and the addition is the initial coupon rate plus the reference rate when the coupon rate was last set.
- **Super floater** – When coupon rates change in twice the proportion to the reference rate, the multiple is 2.0 (200%), and the addition is the initial coupon rate.

2.3 Sinking fund payments

Each sinking fund payment reduces the par value by a percent of the original par value. Sinking fund payments occur on a coupon payment date. If other dates are specified, the sinking fund reduction occurs at the next payment date.

Each payment is a percent of the original par value. For example, when the sinking fund payment is \$20 per \$100 original par, the value is 0.20 (20%).

Existing bond sinking fund schedules

Specify sinking fund schedules for existing bonds on the Investment input page under the Decrements: Sinking Fund Payments category. The schedules must be an external source varying by hidden variable lookup key `asset_id`, and by the date column.

Purchase bond sinking fund schedules

Specify sinking fund schedules for purchase bonds on the Investment input page under the Purchase Asset Definitions : Bonds category. The schedules must be an external source varying by hidden variable lookup key `purch_asset_id` of the `inv_strategy` submodel, and by the hidden lookup key `mths_since_issue` variable of the `inv_strategy` submodel.

2.4 Calls

2.4.1 Call price schedules

Bond call options allow the borrower to prepay the remaining par amount, subject to a call penalty. The call price is the par amount plus the prepayment penalty. The call price schedule defines the call prices over the time when the bond is callable. The bond is not callable before the first call date. Bonds with American call options are callable from the first call date until maturity. Bonds with European call options are callable only on the specified dates.

The values are the percent of the par value that must be paid to exercise the option. For example, when the call premium is 5%, the value is 1.05 (105%). The call premium, also called call penalty, is the call price less the par value.

Existing bond call price schedules

Specify call price schedules for existing bonds on the Investment input page under the Decrements: Calls/Puts/Prepayments category. The schedules must be an external source varying by hidden variable lookup key `asset_id` and by the date column.

Purchase bond call price schedules

Specify call price schedules for purchase bonds on the Investment input page under the Purchase Asset Definitions: Bonds category. The schedules must be an external source varying by hidden variable lookup key `purch_asset_id` of the `inv_strategy` submodel, and by the hidden `mths_since_issue` variable of the `inv_strategy` submodel.

2.4.2 Financial prepayment test

When a call option is available, a financial prepayment test determines if refinancing the bond at current interest rates would be favorable to the borrower. The comparison is against the borrower issuing a new bond to raise the proceeds for calling the current one. The refinancing rate is either:

- The bond's current bond equivalent yield to weighted average life plus the bond's option spread at issue. For existing bonds, the option spread at issue is estimated as of the projection date.
- The scenario bond equivalent yield for a specified scenario year, times a multiple, and plus a spread.

Exercising the call option makes financial sense when the interest savings exceed the option cost. The interest savings are the present value of scheduled coupon and par value payments discounted at the refinancing rate, less the par value. The option cost is the call premium plus the refinancing cost. The call premium is determined from the call price schedule, or through a yield maintenance provision (discussed below) calculation. The refinancing cost is the par value times the assumed refinancing cost percent.

2.4.3 Call option exercise behavior

When the financial prepayment test shows that exercising the call option would be favorable to the borrower, a portion of the bond may be called. Partial asset decrements simulate the proportion of borrowers which would take advantage of the call option in the current situation.

The part of the bond called is equal to the par value times the call prepayment percent. The prepayment percent is linearly interpolated from the call option exercise assumption. The interpolation is based on the call coupon rate differential, which is the bond's coupon rate, net yield rate, or net yield rate at issue, less the refinancing rate.

Defining the call coupon rate differential using the bond's coupon rate is appropriate for most bonds sold at about par value. Defining it using the net yield rate can be used for zero-coupon bonds and original-issue discount bonds, as an approximation that is better than using the coupon rate.

For European call options, the call prepayment percent applies directly on the call option date. For American call options, the call prepayment percent is an effective annual rate; the monthly prepayment rate is:

$$1 - (1 - \text{annual rate})^{1/12}$$

Call option exercise percent

Specify the call option exercise percent on the Investment input page under the Decrements: Calls/Puts/Prepayments category. This must be an external source varying by hidden variable lookup key category_id (asset category), and by hidden variable lookup key int_rate_diff. The example below assumes that holders of AAA bonds will not exercise their call option until the interest rate differential exceeds 1%, reaching 40% exercise rate when the differential is 2% and 100% when the differential reaches 4% or more.

category_id	AAA
int_rate_diff	
0.00%	0.0000%
1.00%	0.0000%
2.00%	40.0000%
4.00%	100.0000%

2.4.4 Yield maintenance provisions

Yield maintenance provisions require the borrower to compensate the lender for any loss in yield on the call. These provisions are sometimes referred to as 'make whole' provisions. Available yield maintenance options are:

- **Yield maintenance** – The call premium is sufficient to compensate the lender for any loss in yield under the assumption that the call amount is reinvested in par bond equivalent risk free debt at the current risk free yield to the remaining maturity less a spread.
- **Cash flow maintenance** – The call amount is the present value of scheduled future cash flows on the bond, discounted at the current risk free yield to maturity less a spread, but not less than the current par value.

Specify the yield maintenance spread as a negative value for the yield maintenance addition variable on the model point record.

Yield maintenance calculations are strictly correct only for bonds purchased at par value. As the amount of discount or premium increases, the approximation becomes less accurate.

2.5 Puts

2.5.1 Put price schedules

Bond put options allow the investor to demand that the remaining par amount be repaid, minus a put penalty. The put price is the par amount minus the put penalty. The put price schedule defines the put prices over the time when the bond is puttable. The bond is not puttable before the first put date. Bonds with American put options are puttable from the first put date until maturity. Bonds with European put options are puttable only on the specified date.

The values are the percent of the par value that must be paid to exercise the option. For example, when the put penalty is 5%, the value is 0.95 (95%). The put penalty is the par value less the put price.

Existing bond put price schedules

Specify put price schedules for existing bonds on the Investment input page under the Decrements: Calls/Puts/Prepayments category. The schedules must be an external source varying by hidden variable lookup key `asset_id` and by the date column.

Purchase bond put price schedules

Specify put price schedules for purchase bonds on the Investment input page under the Purchase Asset Definitions: Bonds category. The schedules must be an external source varying by hidden variable lookup key `purch_asset_id` of the `inv_strategy` submodel, and by the hidden `mths_since_issue` variable of the `inv_strategy` submodel.

2.5.2 Put option exercise behavior

Exercising the put option makes financial sense when the interest savings exceed the option cost. The interest savings are the par value, less the present value of the scheduled principal and interest payments, discounted at the refinancing rate. The option cost is the put premium plus the refinancing cost. The put price is determined from the put price schedule. The refinancing cost is the par value times the assumed refinancing cost percentage.

When the financial prepayment test shows that exercising the put option would be favorable to the investor, a portion of the bond is put. Partial asset decrements simulate the proportion of bonds that would be put in the current situation, similar to the process described above for call options. The put coupon rate differential is the refinancing rate, less either the bond's coupon rate or its net yield rate.

Put option exercise percent

Specify the put option exercise percent on the Investment input page under the Decrements: Calls/Puts/Prepayments category, in the same way as for the call assumptions discussed above. This must be an external source varying by hidden variable lookup key `category_id` (asset category), and by hidden variable lookup key `int_rate_diff`.

2.6 Investment expenses

Investment expense rates vary by asset sub portfolio and are determined by asset sub portfolio. Specify this rate as part of the asset sub portfolio definitions on the Investment input page under the Asset sub portfolio category.

Asset cash flows are reduced by investment expenses. Investment income is reported net of investment expenses, and asset yield rates are net of investment expenses.

Investment expenses occur at the end of each month (prior to processing investment strategies), equal to the book value of each asset times its annual investment expense rate, divided by 12.

3 DEFAULTS

3.1 Deterministic defaults

Defaults are processed at the beginning of each month. Defaults thus reduce the amount of payments that would otherwise occur in that month and in the future. Partial defaults are a proxy for the gradual effect of declining quality grading as a portfolio of bond assets age.

Deterministic default rates vary by asset category, and can also vary by date. Specify deterministic default rates on the Economy input page under the Scenarios: Assumptions category. These definitions must be an external source varying by the hidden variable lookup key `quality_id`; they can also vary by the date column.

The monthly default rate is:

$$1 - (1 - \text{annual default rate})^{1/12}$$

3.2 Stochastic defaults

Default rates can be stochastic when the scenario file includes default rates for scenario yield curves in addition to the risk free yield curve. Typically the scenario file includes default rates for AA and B securities.

The stochastic default rate is an addition to the deterministic default rate.

Specify the stochastic default rate as a multiple of the default rates for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys `yld_curve_id` and `quality_id`.

In the following example, the stochastic default rate for A-rated securities is 80% of the scenario default rate for AA-rated securities plus 20% of the scenario default rate for B-rated securities.

quality_id		A
yld_curve_id		
Govt		0.0000%
AA		80.0000%
B		20.0000%

4 ACCOUNTING VALUES

4.1 Accrued interest

Accrued interest is added to the market and book values of bond assets. That is, projections calculate 'clean' market and book values which exclude accrued interest for bonds.

The accrued interest is the coupon payment, pro-rated for the fraction of the payment period since the last payment.

The initial book and market values (on asset model point records) exclude accrued interest. Accrued interest is added to the initial market value to establish the full market value.

4.2 Statutory accounting

Specify bond accounting rules on the Investment input page under the Balance Sheet category. These definitions apply for all bonds.

The cost basis for U.S. bonds is typically amortized cost. The statutory book value is typically the cost basis. Other options provide flexibility to use asset accounting methods that apply in various countries.

4.3 Tax accounting

The statutory book value definition applies for tax calculations.

4.3.1 *Municipal bonds*

When the bond is specified to have a taxable percent of less than 1.0, the bond is treated as a municipal bond and a portion of the interest payment, positive growth in cost basis, and change in accrued interest is excluded from taxable income.

The treatment for excluding the positive cost basis growth assumes the municipal bond was purchased with an original issue discount from the primary market.

4.4 GAAP accounting

GAAP accounting treatment varies according to the sales classification of each asset:

- For GAAP income, trading assets are valued at market value, while other classifications are valued at the statutory cost basis.
- For the GAAP balance sheet, trading and available for sale assets are valued at market value, while other classifications are valued at the statutory cost basis.

4.5 Amortized yield

When the cost basis definition is Amortized Cost, the asset yield rate is the rate that amortizes the cost basis to par value by a defined date, using scheduled payments.

For fixed rate bonds, the cost basis value can be amortized to par value by the following dates:

- maturity date
- first call option date
- first date when the bond is callable at par value

- first put option date
- first date when the bond is puttable at par value, or
- to whichever of the above options results in the lowest yield rate (yield to worst option).

For options that amortize the cost basis to par value before the maturity date, the asset yield is the coupon rate after the cost basis equals the par value.

For floating rate notes, the asset yield rate is also the discount rate which amortizes the starting cost basis value to the par value by the date described above. The asset yield is calculated under the assumption that the payment rate that is in effect at the calculation date remains constant to maturity. Asset yields for floating rate notes are recalculated whenever the coupon rate is reset. This 'constant coupon method' is described in Fabozzi's 'Handbook of Mortgage-Backed Securities', 3rd edition, 1992, Probus Publishing Co., in the Accounting for Adjustable-Rate Securities section on pages 1154-1157 (second method).

Purchase bonds created during the projection are purchased at a premium or a discount so that the net yield is the market yield at the time of purchase. You can specify a positive or negative coupon rate differential to result in investment asset bonds purchased at discount or premium, respectively.

5 MARKET VALUE

5.1 Discounted cash flows

Bond market values are approximated by discounting the scheduled cash flows, then subtracting the accrued interest at the market valuation date. The discount rate is the current scenario risk adjusted spot rate consistent with the timing of each cash flow, plus the option spread, and the market spread (discussed below).

The market value is increased by the accrued interest. For a semi-annual bond, the accrued interest is:

$$\text{par value} * \left(\left(1 + \text{coupon rate} / 2 \right)^{2 * \text{years_since_last_payment}} - 1 \right)$$

5.2 Credit spread

5.2.1 Fixed credit spreads

The credit spread (also called risk spread) addition to the spot rates adjusts for the assumed credit risk associated with the bond.

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. This must be an external source varying by hidden variable lookup keys `quality_id` and `rate_term`; and can also vary by the date column. Credit spreads typically vary by date when spreads that are consistent with current market values are different from those assumed to apply for normal market conditions going forward.

5.2.2 Stochastic credit spreads

The credit spread can be stochastic when the scenario file includes credit yield curves in addition to the risk free yield curve. Typically the scenario file includes yield curves for AA and B securities.

The stochastic credit spread is an addition to the deterministic credit spread.

Specify the stochastic credit spread as a multiple of the credit spreads for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup key `yld_curve_id` and hidden variable lookup key `quality_id`.

In the following example, the stochastic credit spread for A-rated securities is 80% of the spread for AA-rated securities plus 20% of the spread for B-rated securities.

quality_id	A
yld_curve_id	
Govt	0.0000%
AA	80.0000%
B	20.0000%

5.3 Option spread

The option spread addition to the spot rates adjusts the bond's market value for its call and put options, priced consistently with the current scenario yield curve. The following process is used to calculate the American option spread:

1. The base price of the bond is calculated as the present value of scheduled coupon and par value payments, discounted using the current scenario spot rates, plus the bond's credit spread.
2. The call option price of the bond is calculated using Black's modification to the Black-Scholes formula. Over the period from first call date to maturity, Black's approximation formula is used to estimate the European call option prices at the end of each calendar month, where the strike price at the end of each calendar month is the bond's call price plus accrued interest. The call option price of the bond is the largest of these European call option prices.
3. The put option price of the bond is calculated using the same method as described above for the call option price.

The option spread is calculated so that the present value of scheduled coupon and par value payments equals the base price, less the call option price, plus the put option price. The discount rate is the current scenario spot rate for each cash flow, plus the credit spread and the option spread.

This calculation is sensitive to small changes in the bond's credit spread. If option spreads appear to be inappropriate, review the credit spread assumptions.

5.4 Market spread

The market spread is the additional spread required for the calculated market value to match the input market value. This spread is not calculated when the input market value is zero.

The following options are available for using initial the market spread:

- Ignore the market spread by treating it as zero during the projection. Appropriately small market spreads can help validate that the projection assumptions are consistent with the assumptions used to calculate the input market values.
- The market spread is held level at the initial rate. This treatment interprets the market spread as an adjustment to the credit spread assumptions. For example, you may have assumed one set of credit spread assumptions for all A-rated bonds, but the particular bond in question is an A-rated transportation sector bond, which has somewhat different spreads than for average A-rated bonds.
- The initial market spread is graded linearly to zero by the maturity date. This treatment interprets the market spread as a result of actual market values which fluctuate around analytical market value calculations. The current differential will eventually be corrected by market forces.

5.5 The bond pricing problem

The following explains the above approach for calculating bond market values.

5.5.1 Traditional methods

Well-established techniques are available for calculating bond market values. However, these methods are designed to price assets as of the current date. They are accurate, but slow to calculate.

Insurance company asset/liability analysis presents a different problem from the asset pricing problems that traditional pricing methods are designed to solve. Insurance projections demand that asset market values must be determined not only at the present time, but also along each of the scenario paths being processed. The long calculation times of standard asset pricing methods make it impractical to project results under a large number of scenario paths.

The traditional bond pricing techniques involve projecting asset cash flows over either a large number of scenarios or interest rate lattice paths. These classical techniques fail for insurance asset/liability analysis because there is not enough time to calculate enough scenarios or lattice nodes to produce reasonable asset market values. Applying the methods using a small number of paths destroys the reliability of the results, no matter how accurate the calculations for the paths processed.

One could simplify the asset pricing method by precalculating and storing results at different points in the life of each bond at a range of different interest rates. Future market values would then simply interpolate between the stored results. For example, a one-factor interest rate model simplifies the market value calculation to depend only on the risk free short-term interest rates. A two-factor interest rate model not only bases the calculation on a longer-term rate, but also greatly expands the number of calculations and storage requirements. Thus this solution has the same problems as the direct pricing methods.

5.5.2 Approximating market values

The bond market value approximation method described above uses an alternative method that is practical for the substantial number of market value calculations made during insurance company projections using multiple economic scenarios.

The solution for the US Library bond submodel uses a closed form direct calculation of the option component of the bond asset market value. The formula is based on Black's modification to the Black-Scholes formula. This is a well known solution for the calculation of the market value of European-type stock options that pay no dividends. This technique is adapted to apply to bonds with American-type call and put options.

This option pricing method for bonds is not intended for pricing assets for trading purposes. That is a different problem from the need for pricing assets repeatedly, at different times in the projection and under different economic conditions. We are trying to obtain most of the accuracy of the traditional methods while only going through a fraction of the work they involve.

5.5.3 Comparative test

The following two tests demonstrate the value of the approximation method. In each test, we compared the asset market values for a portfolio against the quoted market values, and also against the results of a sophisticated traditional market pricing system:

The first test used 21 callable U.S. treasury bonds, issued from 1974 through 1984, whose prices were quoted as of February 15, 1994. These bonds were issued with 25 and 30 years to maturity, and are callable during the last 5 years before their maturity date. They were used as a test that would not be distorted by any assumptions about the credit risk of the bonds.

The second test used 49 callable U.S. corporate bonds whose prices were quoted as of February 15, 1994. The bonds had credit ratings that varied from AAA to BB and belonged to five different sectors industrial, financial, utility, telephone, and oil. Maturity dates varied from 1999 to 2032, and terms to the first call option varied from 0 to 9 years. Credit spreads by sector and credit rating were derived from the traditional market pricing system. These credit spreads were based on all bonds tracked by the traditional system, rather than the 49 bonds in the test.

The results of the tests are presented in terms of the errors in the market yield rates for the bonds in the test portfolios. The table below shows the average error in the market yield rate, as well as the standard deviation:

Market Value Method	Treasury Test		Corporates Test	
	Avg Error	Std Deviation	Avg Error	Std Deviation
Traditional pricing	0.00%	0.03%	- 0.03%	0.24%
Approximation method	0.01%	0.03%	- 0.06%	0.28%

These are remarkable results, showing asset market values calculated with only a small error even though the approximation technique requires only a fraction of the calculations of the traditional method.

6 MODEL POINT RECORDS

The model point record layout for bond assets is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	up to 10 characters (such as CUSIP)
D	use_mp	Yes, No
E	category_id_mp	up to 20 characters (must match a bond asset category)
F	port_defn_mp	Investment, Hedging, Collateral Pool
G	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
H	issue_date_mp	yyyy/mm/dd or mm/dd/yyyy
I	maturity_date_mp	yyyy/mm/dd or mm/dd/yyyy
J	planned_sale_date_mp	yyyy/mm/dd or mm/dd/yyyy
K	init_par_val_mp	par value at projection date
L	init_cost_basis_mp	cost basis at projection date
M	init_bk_val_mp	book value at projection date
N	init_mkt_val_mp	market value at projection date**
O	pmt_mode_mp	Monthly, Quarterly, Semi Annual, Annual
P	init_pmt_rate_mp	0.00 (0%) to 1.00 (100%); nominal rate using payment mode
Q	par_val_at_issue_mp	par value at issue
R	taxable_pct_mp	0.00 (0%) to 1.00 (100%)
S	call_opt_first_date_mp	yyyy/mm/dd or mm/dd/yyyy
T	call_price_defn_mp	Price Schedule, Yield Maintenance, Cash Flow Maintenance
U	call_defn_mp	Not Applicable, American, European
V	yld_maint_addn_mp	-1.00 (100%) to 0.00; nominal rate using payment mode
W	put_opt_first_date_mp	yyyy/mm/dd or mm/dd/yyyy
X	put_defn_mp	Not Applicable, American, European
Y	sinking_fund_defn_mp	No, Yes
Z	adj_reset_category_id_mp	category id for rate resets (up to 20 characters); default is Govt
AA	adj_reset_scen_yr_mp	coupon reset scenario maturity years
AB	adj_reset_scen_mult_mp	default = 1.0000
AC	adj_reset_renewal_addn_mp	0.00 (0%) to 1.00 (100%); nominal rate using payment mode
AD	adj_next_reset_date_mp	yyyy/mm/dd or mm/dd/yyyy
AE	adj_reset_mths_mp	1 ... 999
AF	adj_max_incr_per_reset_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AG	adj_max_decr_per_reset_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AH	adj_min_rate_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AI	adj_max_rate_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AJ	avr_contribn_pct_mp	0.00 to 1.00 (100%)

Excel Column	Variable	Value
AK	avr_max_pct_mp	0.00 to 1.00 (100%)
AL	avr_obj_pct_mp	0.00 to 1.00 (100%)
AM	rbc_c1_pct_mp	0.00 to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

**The initial market value is used to calculate the market spread, which is discussed above in the Market Value chapter. The market spread is automatically zero when this variable is left at the default value of 0.00.

7 PURCHASE BONDS

Each bond purchased by the investment strategy creates a new bond model point. These bonds are created to be consistent with the scenario rates that apply at the time of purchase.

7.1 Fixed rate bonds

Each fixed rate bond purchased during the projection is created using the following process:

1. Determine the basic characteristics using the purchase asset definition. These include the asset category, years to maturity, coupon mode, sinking fund schedule, call type, call price schedule, put type, and put price schedule.
2. Calculate the weighted average life according to the scheduled par value payments per unit of par.
3. Calculate the credit spread for the asset category, based on the weighted average life.
4. Calculate the coupon rate as the refinancing rate for the asset category, minus the specified coupon rate differential.
5. Calculate the market value per \$1 par value. This calculation includes the calculation of a coupon rate and option spread consistent with the call and put options.
6. Set the book value, cost basis, and market value equal to the amount to invest.
7. Calculate the par value as the market value divided by the market value per \$1 par value.

Bonds are purchased at a premium or a discount so that the net yield is the market yield at the time of purchase. You can specify a positive or negative coupon rate differential to result in bonds purchased at discount or premium, respectively.

7.2 Floating rate notes

The process for floating rate notes is the same as for fixed rate bonds, except that the adjustable coupon rate is determined according to the definition involving the scenario year bond equivalent yield times a multiple and plus the initial spread, then resetting based on the renewal spread.

7.3 Purchase bond variables

Specify purchase bonds on the Investment input page, under the Purchase Asset Definitions category. These are variables in the inv_strategy submodel.

The variables must be linked to an external source using the hidden variable lookup key purch_asset_id.

Variable	Value
<u>purch_bond_grp_id</u>	model point report group*
<u>purch_bond_category_id</u>	up to 20 characters (must match a bond asset category)
<u>purch_bond_port_defn</u>	Investment
<u>purch_bond_sale_class</u>	Available for Sale, Trading, Held to Maturity, Not Assigned
<u>purch_bond_yrs_to_maturity</u>	0.25 or greater
<u>purch_bond_taxable_pct</u>	0.00 (0%) to 1.00 (100%)
<u>purch_bond_rate_defn</u>	Refinancing Rate, Fixed Rate, Floating Rate
<u>purch_bond_pmt_mode</u>	Monthly, Quarterly, Semi-Annual, Annual
<u>purch_bond_rate_diff</u>	-1.00 (-100%) to 1.00 (100%)
<u>purch_bond_adj_category_id</u>	category id for rate reset (up to 20 characters); default is Govt
<u>purch_bond_adj_scen_yr</u>	coupon reset scenario maturity years
<u>purch_bond_adj_scen_mult</u>	default = 1.0000
<u>purch_bond_adj_addn_init</u>	0.00 to 1.00 (100%)
<u>purch_bond_adj_addn_renewal</u>	0.00 to 1.00 (100%)
<u>purch_bond_adj_reset_mths</u>	1 ... 999
<u>purch_bond_adj_max_above_init</u>	0.00 to 1.00 (100%)
<u>purch_bond_adj_max_below_init</u>	0.00 to 1.00 (100%)
<u>purch_bond_adj_max_reset_incr</u>	0.00 to 1.00 (100%)
<u>purch_bond_adj_max_reset_decr</u>	0.00 (0%) to 1.00 (100%)
<u>purch_bond_call_price_defn</u>	Price Schedule, Yield Maintenance, Cash Flow Maintenance, Related to Coupon
<u>purch_bond_call_protect_yrs</u>	1 ... 999
<u>purch_bond_call_defn</u>	Not Applicable, American, European
<u>purch_bond_call_price_pct</u>	1.00 (100%) to 2.00 (200%); varies by additional lookup key <u>mths_since_issue</u>
<u>purch_bond_yld_maint_addn</u>	-1.00 (100%) to 0.00 (0%)
<u>purch_bond_call_prem_mult</u>	-1.00 (100%) to 2.00 (200%)
<u>purch_bond_call_prem_addn</u>	-1.00 (100%) to 1.00 (100%)
<u>purch_bond_first_par_call_yr</u>	0 ... 999
<u>purch_bond_put_price_defn</u>	Price Schedule, Related to Coupon
<u>purch_bond_put_protect_yrs</u>	0 ... 999
<u>purch_bond_put_defn</u>	Not Applicable, American, European
<u>purch_bond_put_price_pct</u>	1.00 (100%) to 2.00 (200%); varies by additional lookup key <u>mths_since_issue</u>
<u>purch_bond_put_prem_mult</u>	-1.00 (100%) to 2.00 (200%)
<u>purch_bond_put_prem_addn</u>	-1.00 (100%) to 1.00 (100%)
<u>purch_bond_first_par_put_yr</u>	0 ... 999
<u>purch_bond_sinking_fund_defn</u>	No, Yes
<u>purch_bond_sinking_fund_prin_pmt_pct</u>	0.00 (0%) to 1.00 (100%); varies by additional lookup key <u>mths_since_issue</u>
<u>purch_bond_avr_contribn_pct</u>	0.00 (0%) to 1.00 (100%)
<u>purch_bond_avr_max_pct</u>	0.00 (0%) to 1.00 (100%)
<u>purch_bond_avr_obj_pct</u>	0.00 (0%) to 1.00 (100%)
<u>purch_bond_rbc_c1_pct</u>	0.00 (0%) to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic sub portfolios as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

8 SUBMODELS STRUCTURE

8.1 Submodels

The `bond_asset` model class contains the formulas and variables for bond model point processing. The `bond_cf_asset` model class contains the formulas and variables for bond cash flow calculations. The variable values are not specified by the user, but are instead set during initialization processing.

The following describes the model tree:

- `bond` – a data driven array submodel used to determine bond experience results. The number of elements in the array is initially the number of existing bond model points. The array size increases during the projection as new bonds are purchased.
- `bond_cf` -- a standalone instance of the `bond_cf_asset` model class used to determine experience cash flows
- `bond_pv`, `bond_is`, `bond_ym` – standalone instances of the `bond_cf_asset` model class. These models are clones of the `bond_cf` submodel, and are used for the cash flow projections required for market value (pv), market value shifted (pv), interest savings present value (is), net yield calculations (pv) and yield maintenance (ym).

8.2 Present value calculations

The following are external functions for calculating present values:

- `call_opt_price_calc` – When a bond has a call option, this external function returns the call option price by executing a loop over all future option expiry dates. Call option prices for each expiry date are determined using the Black-Scholes option value function.
- `put_option_price_calc` – When a bond has a put option, this external function returns the put option price by executing a loop over all future option expiry dates. Put option prices for each expiry date are determined using the Black-Scholes option value function.
- `weighted_avg_life_calc` – returns the weighted average life of future principal payments
- `pvcalc` – returns the discounted present value of future cash flows. This function handles both discounting at spot rates and a single yield rate. This function may also fill arrays of cash flows for use in the call and put option price algorithms.
- `pvcalc_simple` – returns the discounted present value of future cash flows using a closed form solution to determine the present value of future cash flows. This function can only be used if when certain criteria are met.
- `solve_for_spread_newton_raphson` – yield rates or spreads can usually be solved for quickly using the Newton-Raphson method. During each solve iteration, the present value of the cash flow stream is determined and is compared to the target value. When the difference is small, the solving process is terminated and the yield rate or spread is returned.
- `solve_for_spread_inverse_interp` – when the Newton-Raphson method does not converge quickly enough, this function uses a combination of inverse interpolation and bisection to solve for the yield rate or spread. During each solve iteration, the present value of the cash flow stream is determined and compared to the target value. When the difference is small, the solving process is terminated and the yield rate or spread is returned.

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A02 MORTGAGES

This book discusses mortgage assets and the mtg submodels. Refer to the Securitized assets book for instructions on using Intex to project Mortgage Backed Securities (MBS) using existing mortgage model point records.

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1 INPUT ORGANIZATION

Input fields for mortgages are located on the Economy and Investment input pages, or on model point records.

1.1 Economy input page

The Economy input page includes assumptions related to the general economic environment, which is independent of company and asset portfolio assumptions.

1.1.1 Credit spreads

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. See the Credit spread section in the Market value chapter, below.

1.1.2 Default rates

Specify bond default rates on the Economy input page under the Scenarios: Assumptions category. See the Defaults chapter, below.

1.2 Investment input page

The Investment input page includes assumptions related to projected asset portfolio and investment strategies.

1.2.1 Prepayment options

Specify mortgage prepayment penalty schedules on the Investment input page under the Decrements: Calls/Puts/Prepayments category. Also specify prepayment option exercise assumptions under this category. See the Prepayments chapter, below.

1.2.2 Expenses

Investment expense rates vary by asset sub portfolio. See the Investment expenses section in the Cash flows chapter, below.

1.2.3 Accounting

Specify mortgage accounting rules on the Investment input page under the Balance Sheet category. These definitions apply for all mortgages. See the Accounting values chapter, below.

1.2.4 Purchase mortgage definitions

Specify purchase mortgage definitions on the Investment input page under the Purchase Asset Definitions category. See the Purchase mortgage variables chapter, below.

1.3 Model points

Specify on the model point records the definition for each mortgage existing at the projection date. See the Model point records chapter, below.

Report groups

Assets typically do not use the report group feature that is used to group liability model points. Instead, asset sub portfolio reports automatically collect the individual assets assigned to each group. This has the advantage that asset sub portfolio assignments can change dynamically as asset durations change through the projection.

However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

2 CASH FLOWS

2.1 Payment timing

2.1.1 *Maturity*

The mortgage matures for its remaining par value at the end of the maturity month specified on the model point record.

2.1.2 *Payment dates*

Payments occur either monthly, semi-annually, quarterly, or annually, at the end of the month related to the maturity date. For example, a mortgage maturing in April has quarterly payments at the end of January, April, July, and October.

All payments occur at the end of the month.

2.2 Interest payments

2.2.1 *Initial interest rate*

The model point record contains the initial (at the projection date) payment rate. Interest rates are nominal based on the payment mode. For example, a 6.00% (the rate on the model point record) monthly mortgage has an effective interest rate of 6.17%.

The interest paid by borrowers is reduced by a servicing fee percent charged by the company servicing the mortgage. Specify the servicing fee on the model point record. The servicing fee is a nominal rate; that is, the servicing fee for each payment is this annual rate divided by the number of payments per year.

The mortgage interest rate is used to calculate amortization payments, interest savings present values for prepayments, and adjustable rates. The net interest rate is used to calculate asset cash flows, net yields, and market values. To base market values on the mortgage interest rate, replace the servicing fee with an appropriate increase in the investment expense rate.

2.2.2 *Adjustable rate mortgages*

Existing assets

The initial interest rate applies until the next reset date. The rate resets on the payment date in or after that month. That is, the rate can only reset on a payment date. For fixed rate mortgages, the reset date is simply specified as the maturity date or later, such as the 9999/01/01.

For adjustable rate mortgages, the payment rate is reset at the next reset date, and then periodically thereafter based on the reset period months. At each reset date, the payment rate changes to be the scenario bond equivalent yield for a specified category id for a specified maturity years, times a multiple then plus an addition (nominal rate).

Each new payment rate is constrained to be within the maximum rate (cap) and minimum rate (floor).

Purchase assets

For mortgages purchased during the projection, the addition for rate resets is different from the addition that determines the initial rate.

2.2.3 *Principal payments*

Once the mortgage begins amortizing, mortgage payments include principal and interest amounts determined so that the mortgage is paid off at the maturity date.

2.3 *Balloon payments*

2.3.1 *Single balloon payment*

When the balloon definition on the model point record is 'Yes On Single Date' for existing mortgages or specified as 'Yes On Single Period' for purchase mortgages, the remaining principal is paid a balloon payment at the balloon payment date.

The balloon payment can only occur on a payment date. If another date is specified, the balloon payment occurs at the next payment date.

2.3.2 *Balloon schedule*

When the balloon definition is 'Yes Per Schedule', a schedule of balloon payment is specified.

Each balloon payment reduces the principal value by a percent of the original principal value. Balloon payments occur on a payment date. If other dates are specified, the balloon payment reduction occurs at the next payment date.

Each payment is a percent of the original principal value. For example, when the balloon payment is \$20 per \$100 original principal, the value is 0.20 (20%).

Existing mortgage balloon schedules

Specify balloon schedules for existing mortgages on the Investment input page under the Decrements: Balloon Payments category. The schedules must be an external source varying by hidden variable lookup key `asset_id`, and by the date column.

Purchase mortgage balloon schedules

Specify balloon schedules for purchase mortgages on the Investment input page under the Purchase Asset Definitions : Mortgages category. The schedules must be an external source varying by hidden variable lookup key `purch_asset_id` of the `inv_strategy` submodel, and by the hidden lookup key `mths_since_issue` variable of the `inv_strategy` submodel.

2.4 *Prepayments*

2.4.1 *Prepayment provision*

Mortgage prepayment provisions allow the borrower to pay off the remaining principal amount, in addition to a prepayment penalty. The projection includes prepayments varying based on the duration of the mortgage and economic conditions. The prepayment rate can be determined using either the prepayment model described below, or the prepayment models available from Andrew Davidson & Co, Inc. (discussed separately below).

Projecting the percent of the mortgage that is prepaid reduces the size of the remaining mortgage asset. The mortgage payments are reduced by the same percent as the outstanding principal value. That is, mortgage prepayments do not advance the time when the remaining principal is paid. Instead, they reduce the portion of the mortgage that continues. Partial prepayments of an individual mortgage are ignored, as this effect is typically insignificant compared to prepayments of the mortgage in full.

2.4.2 *Prepayment penalty*

Mortgage prepayments may be subject to a prepayment penalty. The prepayment amount is the remaining principle plus the prepayment penalty. The prepayment penalty schedule defines the prepayment percentages over the time when the mortgage is prepayable.

The values are the percent of the remaining principle that must be paid as a penalty to prepay the mortgage.

Existing mortgage prepayment penalty schedules

Specify prepayment penalty schedules for existing mortgages on the Investment input page under the Decrements: Calls/Puts/Prepayments category. The schedules must be an external source varying by hidden variable lookup key `asset_id` and by the date column.

Purchase mortgage prepayment penalty schedules

Specify prepayment penalty schedules for purchase mortgages on the Investment input page under the Purchase Asset Definitions: Mortgages category. The schedules must be an external source varying by hidden variable lookup key `purch_asset_id` of the `inv_strategy` submodel, and by the hidden `mths_since_issue` variable of the `inv_strategy` submodel.

Purchase mortgage prepayment penalty percent of payment rate

For purchase mortgages, the prepayment penalty may be specified to be a percentage of the payment rate at issue of the mortgage, grading to 0% over some specified months from issue. This feature allows for prepayment penalty sensitivity to the level of the payment rate.

Specify provision for prepayment penalty as a percent of the payment rate on the Investment input page under the Purchase Asset Definitions: Mortgages category.

2.4.3 *Yield maintenance*

Yield maintenance provisions require the borrower to compensate the lender for any loss in yield on the prepayment. These provisions are sometimes referred to as 'make whole' provisions. Available yield maintenance options are:

- **Yield maintenance** – The prepayment penalty is sufficient to compensate the lender for any loss in yield under the assumption that the prepayment is reinvested in par bond equivalent risk free debt at the current risk free yield to the remaining maturity less a spread.
- **Cash flow maintenance** – The prepayment penalty is the present value of scheduled future cash flows on the mortgage, discounted at the current risk free yield to maturity less a spread, but not less than the current principal.

Yield maintenance calculations are strictly correct only for mortgages purchased at par value. As the amount of discount or premium increases, the approximation becomes less accurate.

2.4.4 *Refinancing Financial test*

When prepayments are permitted, a financial test determines if refinancing the mortgage at current interest rates would be favorable to the borrower. The comparison is against the borrower obtaining a new mortgage to raise the proceeds for paying off the current one. The refinancing rate is either:

- The scenario bond equivalent yield, plus the credit spread, for the current mortgage's weighted average life (WAL), plus the (last known) prepayment option spread for the mortgage.
- The scenario bond equivalent yield for a specified scenario year, times a multiple, and plus a spread.

Refinancing the mortgage makes financial sense when the interest savings exceed the option cost. The interest savings are the present value of scheduled principal and interest payments discounted at the refinancing rate, less the principal value. The option cost is the prepayment amount plus the refinancing cost. The prepayment penalty is determined from the prepayment penalty schedule, or through a yield

maintenance provision (discussed below) calculation. The refinancing cost is the principal value times the assumed refinancing cost percent.

2.4.5 Financial prepayment behavior

When the financial prepayment test shows that refinancing the mortgage would be favorable to the borrower, a portion of the mortgage may be refinanced. Partial asset decrements simulate the proportion of borrowers which would take advantage of the refinancing option in the current situation.

The part of the mortgage refinanced is equal to the principal value times the financial prepayment percent. The financial prepayment percent is linearly interpolated from the prepayment option exercise assumption. The interpolation is based on the interest rate differential, which is the current interest rate less the refinancing rate.

Mortgage prepayment rates are often defined in terms of a multiple of the Public Security Association Standard Prepayment Model. Prepaying mortgages at '100% of PSA' means that prepayment rates grade linearly from 0.2% per annum in the first month following issue to 6% per annum in the 30th month following issue and remaining constant thereafter.

The prepayment percent is an effective annual rate. The monthly prepayment rate is:

$$1 - (1 - \text{annual rate})^{1/12}$$

Prepayment option exercise percent

Specify the prepayment option exercise percent on the Investment input page under the Decrements: Calls/Puts/Prepayments category. This must be an external source varying by hidden variable lookup key category_id (asset category), and by hidden variable lookup key int_rate_diff. The example below assumes that mortgage holders will not exercise their prepayment option until the interest rate differential exceeds 1%, reaching 40% exercise rate when the differential is 2% and 100% when the differential reaches 4% or more.

category_id	FNMA
int_rate_diff	
0.00%	0.0000%
1.00%	0.0000%
2.00%	40.0000%
4.00%	100.0000%

2.4.6 Nonfinancial prepayments

Mortgage prepayment assumptions also include a nonfinancial component, which is not subject to any financial tests. This rate can be adjusted for special conditions:

- Higher interest rates can reduce the nonfinancial prepayment activity as the interest differential increases between the mortgage refinancing rate and the current mortgage interest rate.
- A seasonality adjustment can skew the percent by calendar month.

The nonfinancial prepayment percent that applies in a particular month is:

$$\text{Prepayment nonfinancial percent} \\ - \text{Prepayment nonfinancial adjustments} \times \text{seasonality adjustment but } \geq 0$$

Nonfinancial prepayment adjustments

Specify the nonfinancial prepayment adjustments percent on the Investment input page under the Decrements: Calls/Puts/Prepayments category. This must be an external source varying by hidden variable lookup key category_id (asset category), and by hidden variable lookup key nonfinancial_int_rate_diff.

Specify the nonfinancial seasonality adjustment factor on the Investment input page under the under the Decrements: Calls/Puts/Prepayments category. This must be an external source varying by hidden variable lookup key category_id (asset category), and by cal_mth column.

2.5 Andrew Davidson prepayment behavior

An alternative to the above prepayment model is to license prepayment models software from Andrew Davidson & Co., Inc. These prepayment models provide monthly prepayment rates using specific pool characteristics, interest rate history, interest rate forecast, home price index history, and home price index forecast.

Mortgage model points are linked to the appropriate Andrew Davidson prepayment models through their individual model point characteristics and asset category ID. Mortgage asset category ID information includes Andrew Davidson category identification, collateral definition, subprime definition, and adjustable rate provisions.

Fixed rate mortgage-backed securities (MBS) prepayments use the fixed rate MBS prepayment model which considers the loan age, remaining term, payment rate, balloon features, and history and forecasts for interest rates and home price indexes.

Adjustable rate MBS, hybrid MBS, and adjustable rate home equity loans asset-based securities (ABS) prepayments use the unified prepayment model which considers the loan age, remaining term, adjustable rate provisions, forecasts of adjusted payment rates, along with history and forecasts of the interest rates and the reference index rate.

Tuning for the prepayment forecast is handled internally in the model based on tuning parameters included in the Andrew Davidson data files.

The fixed home equity loans, auto, and manufactured housing ABS prepayment models take into account the major factors affecting their prepayments.

You must license these models from Andrew Davidson & Co., Inc. The ABS prepayment models can only be used with the Intex link, which is discussed in the Securitized assets book. Contact Andrew Davidson & Co. Inc. for more information.

2.6 Investment expenses

Investment expense rates vary by asset sub portfolio and are determined by asset sub portfolio. Specify this rate as part of the asset sub portfolio definitions on the Investment input page under the Asset sub portfolios category.

Asset cash flows are reduced by investment expenses. Investment income is reported net of investment expenses, and asset yield rates are net of investment expenses.

Investment expenses occur at the end of each month (prior to processing investment strategies), equal to the book value of each asset times its annual investment expense rate, divided by 12.

3 DEFAULTS

3.1 Deterministic defaults

Defaults are processed at the beginning of each month. Defaults thus reduce the amount of payments that would otherwise occur in that month and in the future. Proportional defaults are a proxy for the gradual effect of declining quality grading as a portfolio of mortgage assets age.

Deterministic default rates vary by asset category, and can also vary by date. Specify deterministic default rates on the Economy input page under the Scenarios: Assumptions category. These definitions must be an external source varying by the hidden variable lookup key `quality_id`; they can also vary by the date column.

The monthly default rate is:

$$1 - (1 - \text{annual default rate})^{1/12}$$

3.2 Stochastic defaults

Default rates can be stochastic when the scenario file includes default rates for scenario yield curves in addition to the risk free yield curve. Typically the scenario file includes default rates for AA and B securities.

The stochastic default rate is an addition to the deterministic default rate.

Specify the stochastic default rate as a multiple of the default rates for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys `yld_curve_id` and `quality_id`.

In the following example, the stochastic default rate for FNMA mortgages is 80% of the scenario default rate for AA securities plus 20% of the scenario default rate for B securities.

quality_id	FNMA
yld_curve_id	
Govt	0.0000%
AA	80.0000%
B	20.0000%

3.3 Refinance defaults

In addition to proportional defaults (discussed above), mortgages can include default events that refinance restructure the mortgage, or foreclose on the collateral property. Specify refinancing assumptions on the Investment input page, under the Decrements: Defaults/Refinance/Foreclosure/Restructuring category.

The refinancing decrement refinances the mortgage consistent with current interest rates. Refinancing is typically a preventative measure, and occurs before the mortgage is actually in default. Amounts to be refinanced are used to purchase a new mortgage, with terms defined using the same procedures as for creating mortgage investment assets.

The refinance amounts are collected from all mortgages, then invested in one 'refinance' purchase mortgage each month. If no refinance purchase asset is specified, the refinance amount is added to general defaults.

Accrued interest on refinance amounts is added to proportional defaults.

3.4 Restructure defaults

Mortgages for borrowers in worse financial difficulty can be restructured. Restructuring a mortgage typically involves some write down of the investment amount. The remaining amount is refinanced in a new mortgage. The interest and principal payments can be reduced for some period after the restructuring, eventually increasing to 100% of the interest payments for the restructured mortgage.

The restructure amounts, net of write downs, are similarly collected from all mortgages, then invested in one 'restructure' purchase mortgage each month. If no restructure purchase asset is specified, the amount is added to general defaults.

Accrued interest and write downs on restructured amounts are added to proportional defaults.

3.5 Foreclosure defaults

The most severe type of mortgage default is foreclosure. The investor receives the collateral property, which is eventually sold. The asset value suffers a write down at the time of foreclosure. The cash flows and value from the sale of the property can be represented by a new 'foreclosure' purchase mortgage that has a balloon payment date simulating the proceeds from the eventual sale of the property. The interest payments for the new mortgage can be reduced to zero just after the foreclosure, then increasing to perhaps 100% of the new mortgage asset interest rate after a few years. The principal payments can likewise be reduced to zero, then increasing as the economic situation for the foreclosed property improves.

The principal payments can be assumed to increase to more than 100% of the principal payments for the new mortgage asset. This will create a capital gain at the time of the balloon payments representing the sale of the property, assuming a partial recovery of the amount of the asset written down at the time of the foreclosure.

The foreclosure amounts, net of write downs, are collected and invested in one 'foreclosure' purchase mortgage. If no foreclosure purchase asset is specified, the amount is added to proportional defaults.

Accrued interest and write downs on foreclosure amounts are added to general defaults.

4 ACCOUNTING VALUES

4.1 Accrued interest

Accrued interest is added to the market value and statutory book value of mortgage assets. That is, projections calculate 'clean' market and book values (which exclude accrued interest) for mortgages.

The accrued interest is the interest payment, compounded for the fraction of the payment period since the last payment.

The initial book and market value (on asset model points) excludes accrued interest. Accrued interest is added to the initial market value to establish the full market value for market consistent projection calibrations.

4.2 Statutory accounting

Specify mortgage accounting rules on the Investment input page under the Balance Sheet category. These definitions apply for all mortgages.

The cost basis for U.S. mortgages is typically amortized cost. The statutory book value for U.S. mortgages is typically the cost basis. Other options provide flexibility to use asset accounting methods that apply in various countries.

4.3 Tax accounting

The statutory book value applies for tax calculations.

4.4 GAAP accounting

GAAP accounting treatment varies according to the sales classification of each asset:

- For GAAP income, trading assets are valued at market value, while other classifications are valued at the statutory cost basis.
- For the GAAP balance sheet, trading and available for sale assets are valued at market value, while other classifications are valued at the statutory cost basis.

4.5 Amortized yield

When the cost basis definition is Amortized Cost, the asset yield rate is the rate that amortizes the cost basis to par value by a defined date, using scheduled payments.

For fixed rate mortgages, the amortization is always to maturity.

For adjustable rate mortgages, the amortization is either to the maturity date or to the next rate reset date. The net yield is calculated under the assumption that the payment rate that is in effect at the calculation date remains constant to maturity. This constant interest method is described in Fabozzi's 'Handbook of Mortgage-Backed Securities', 3rd edition, 1992, Probus Publishing Co., in the Accounting for Adjustable-Rate Securities section on pages 1154-1157 (second method). Net yields for adjustable rate mortgages are recalculated whenever the interest rate is reset.

Defaults are decrements of a portion of the mortgage. They do not affect the amortized yield of the remaining asset.

5 MARKET VALUE

Mortgage market values are calculated by discounting the cash flows projected assuming that the current scenario rates continue unchanged, based on prepayment assumptions. The discount rate is the current scenario risk-adjusted spot rate consistent with the timing of each cash flow, plus the market spread.

The market value is increased by the accrued interest.

5.1 Credit spread

5.1.1 Fixed credit spreads

The credit spread (also called risk spread) addition to the spot rates adjusts for the assumed prepayment and credit risk associated with the mortgage.

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. This must be an external source varying by hidden variable lookup keys `quality_id` and `rate_term`; and can also vary by the date column. Credit spreads typically vary by date when spreads that are consistent with current market values are different from those assumed to apply for normal market conditions going forward.

5.1.2 Stochastic credit spreads

The risk spread can be stochastic when the scenario file includes credit yield curves in addition to the risk free yield curve. Typically the scenario file includes yield curves for AA and B securities.

The stochastic credit spread is an addition to the deterministic credit spread.

Specify the stochastic credit spread as a multiple of the credit spreads for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup key `yld_curve_id` and hidden variable lookup key `quality_id`.

In the following example, the stochastic credit spread for FNMA mortgage is 80% of the spread for AA-rated securities plus 20% of the spread for B-rated securities.

quality_id	FNMA
yld_curve_id	
Govt	0.0000%
AA	80.0000%
B	20.0000%

5.2 Market spread

The market spread is the additional spread required for the calculated market value to match the input market value. This spread is not calculated when the input market value is zero.

The following options are available for using initial the market spread:

- Ignore the market spread by treating it as zero during the projection. Appropriately small market spreads can help validate that the projection assumptions are consistent with the assumptions used to calculate the input market values.
- The market spread is held level at the initial rate. This treatment interprets the market spread as an adjustment to the credit spread assumptions.
- The initial market spread is graded linearly to zero by the maturity date. This treatment interprets the market spread as a result of actual market values which fluctuate around analytical market value calculations. The current differential will eventually be corrected by market forces.

6 MODEL POINT RECORDS

The model point record layout for mortgage assets is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	up to 10 characters (such as CUSIP for agency pass throughs)
D	use_mp	Yes, No
E	category_id_mp	up to 20 characters (must match a mortgage asset category)
F	port_defn_mp	Investment, Hedging, Collateral Pool
G	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
H	issue_date_mp	yyyy/mm/dd or mm/dd/yyyy
I	maturity_date_mp	yyyy/mm/dd or mm/dd/yyyy
J	amortzn_start_cy_mp	the mortgage is interest-only (not amortizing) through the maturity month in this calendar year (yyyy)
K	planned_sale_date_mp	yyyy/mm/dd or mm/dd/yyyy
L	init_par_val_mp	principal value at projection date
M	init_cost_basis_mp	cost basis at projection date
N	init_bk_val_mp	book value at projection date
O	init_mkt_val_mp	market value at projection date**
P	pmt_mode_mp	Monthly, Quarterly, Semi-Annual, Annual
Q	init_pmt_rate_mp	0.00 to 1.00 (100%); nominal rate using payment mode
R	servicing_fee_mp	0.00 to 1.00 (100%); nominal rate using payment mode
S	prin_val_at_issue_mp	principal value at issue
T	prepmt_opt_first_date_mp	yyyy/mm/dd or mm/dd/yyyy
U	prepmt_penalty_defn_mp	Price Schedule, Yield Maintenance, Cash Flow Maintenance
V	balloon_defn	No, Yes On Single Date, Yes Per Schedule
W	balloon_date_mp	yyyy/mm/dd or mm/dd/yyyy
X	yld_maint_addn_mp	-1.00 (100%) to 0.00 (0%); nominal rate using payment mode
Y	adj_reset_category_id_mp	category id for rate resets (up to 20 characters); default is Govt
Z	adj_reset_scen_yr_mp	coupon reset scenario maturity years
AA	adj_reset_scen_mult_mp	default = 1.0000
AB	adj_reset_renewal_addn_mp	0.00 to 1.00 (100%); nominal rate using payment mode

Excel Column	Variable	Value
AC	adj_next_reset_date_mp	yyyy/mm/dd or mm/dd/yyyy When using the Andrew Davidson prepayment model, this date must be the date of the first reset from issue, not from the projection date. Otherwise, the Andrew Davidson model will not select the appropriate loan type for the prepayment model forecast.
AD	adj_reset_mths_mp	1 ... 999
AE	adj_max_incr_per_reset_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AF	adj_max_decr_per_reset_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AG	adj_min_rate_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AH	adj_max_rate_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AI	avr_contribn_pct_mp	0.00 to 1.00 (100%)
AJ	avr_max_pct_mp	0.00 to 1.00 (100%)
AK	avr_obj_pct_mp	0.00 to 1.00 (100%)
AL	rbc_c1_pct_mp	0.00 to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

**The initial market value is used to calculate the market spread, which is discussed above in the Market Value chapter. The market spread is automatically zero when this variable is left at the default value of 0.00.

7 PURCHASE MORTGAGES

Each mortgage purchased by the investment strategy creates a new mortgage model point. These mortgages are created to be consistent with the scenario rates that apply at the time of purchase.

7.1 Fixed rate mortgages

Each mortgage investment asset purchased during the projection is created using the following process:

1. Determine the basic characteristics using the purchase asset definition. These include the asset category, years to maturity, payment mode, balloon year, and prepayment penalty schedule.
2. Calculate the weighted average life according to the scheduled principal payments per unit of principal.
3. Calculate the credit spread for the asset category, based on the weighted average life.
4. Calculate the interest rate as the refinancing rate specified for the asset category, minus the specified interest rate differential.
5. Calculate the market value per \$1 principal.
6. Set the book value and market value equal to the amount to invest.
7. Calculate the principal value as the market value divided by the market value per \$1 principal.

Mortgages are purchased at a premium or a discount so that the book yield is the market yield at the time of purchase. You can specify a positive or negative interest rate differential to result in mortgages purchased at discount or premium, respectively.

7.2 Adjustable rate mortgages

The process for adjustable rate mortgages is the same as for fixed rate mortgages, except that the adjustable interest rate is determined according to the definition involving the scenario year bond equivalent yield times a multiple and plus the initial spread, then resetting based on the renewal spread.

7.3 Purchase mortgage variables

Specify purchase mortgages on the Investment input page, under the Purchase Asset Definitions category. These are variables in the inv_strategy submodel.

The variables must be linked to an external source using the hidden variable lookup key purch_asset_id.

Variable	Value
purch_mtg_grp_id	model point report group*
purch_mtg_category_id	up to 20 characters (must match a bond asset category)
purch_mtg_port_defn	Investment
purch_mtg_sale_class	Available for Sale, Trading, Held to Maturity, Not Assigned
purch_mtg_yrs_to_maturity	0.25 or greater
purch_mtg_amortzn_yrs	years until amortization begins (the mortgage is interest-only until this time), 0 ... (less than yrs_to_maturity)
purch_mtg_rate_defn	Refinancing Rate, Fixed Rate, Floating Rate
purch_mtg_pmt_mode	Monthly, Quarterly, Semi-Annual, Annual
purch_mtg_servicing_fee	0.00 to 1.00 (100%)
purch_mtg_rate_diff	-1.00 (-100%) to 1.00 (100%)
purch_mtg_adj_category_id	category id for rate resets (up to 20 characters); default is Govt
purch_mtg_adj_scen_yr	coupon reset scenario maturity years
purch_mtg_adj_scen_mult	default = 1.0000
purch_mtg_adj_addn_init	0.00 to 1.00 (100%)
purch_mtg_adj_addn_renewal	0.00 to 1.00 (100%)
purch_mtg_adj_reset_mths	1 ... 999
purch_mtg_adj_max_above_init	0.00 to 1.00 (100%)
purch_mtg_adj_max_below_init	0.00 to 1.00 (100%)
purch_mtg_adj_max_reset_incr	0.00 to 1.00 (100%)
purch_mtg_adj_max_reset_decr	0.00 to 1.00 (100%)
purch_mtg_prepmt_penalty_defn	Penalty Percent, Yield Maintenance, Cash Flow Maintenance, Related to Payment Rate
purch_mtg_prepmt_protect_yrs	1 ... 999
purch_mtg_prepmt_penalty_pct	1.00 (100%) to 2.00 (200%); varies by additional lookup key mths_since_issue
purch_mtg_yld_maint_addn	-1.00 (100%) to 0.00
purch_mtg_prepmt_penalty_mult	-1.00 (100%) to 2.00 (200%)
purch_mtg_prepmt_penalty_addn	-1.00 (100%) to 1.00 (100%)
purch_mtg_first_par_prepmt_yr	0 ... 999
purch_mtg_balloon_defn	No, Yes On Single Period, Yes Per Schedule
purch_mtg_yrs_to_balloon	0 ... 999
purch_mtg_balloon_prin_pmt_pct	0.00 (0%) to 1.00 (100%); varies by additional lookup key mths_since_issue
purch_mtg_avr_contribn_pct	0.00 to 1.00 (100%)
purch_mtg_avr_max_pct	0.00 to 1.00 (100%)
purch_mtg_avr_obj_pct	0.00 to 1.00 (100%)
purch_mtg_rbc_c1_pct	0.00 to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

8 SUBMODELS STRUCTURE

8.1 Submodels

The `mtg_asset` model class contains the formulas and variables for mortgage model point processing. The `mtg_cf_asset` model class contains the formulas and variables for mortgage cash flow calculations. The variable values are not specified by the user but are instead set during initialization processing.

The following describes the model tree:

- `mtg` – a data driven array of the `mtg_asset` model class used to determine mortgage experience results. The number of elements in the array is initially the number of existing mortgage model points. The array size increases during the projection as new mortgages are purchased.
- `mtg_cf` – a standalone instance of the `mtg_cf_asset` model class used to determine mortgage experience cash flows
- `mtg_pv`, `mtg_is`, `mtg_ym` – standalone instances of the `mtg_cf_asset` model class. These models are clones of the `mtg_cf` submodel, and are used for the cash flow projections required for market value (pv), market value shifted (pv), interest savings present value (is), net yield calculations (pv) and yield maintenance (ym).

8.2 Present value calculations

The primary external functions for present value calculations include:

- `weighted_avg_life_calc` – returns the weighted average life of future principal payments.
- `pv_calc` – returns the discounted present value of future cash flows. This function handles both discounting at spot rates and a single yield rate.
- `pv_calc_simple` – returns the discounted present value of future cash flows using a closed form solution to determine the present value of future cash flows. This function can only be used if when certain criteria are met.
- `solve_for_sprd_newton_raphson` – yield rates or spreads can usually be solved for quickly using the Newton-Raphson method. During each solve iteration, the present value of the cash flow stream is determined and is compared to the target value. When the difference is small, the solving process is terminated and the yield rate or spread is returned.
- `solve_for_sprd_inverse_interp` – when the Newton-Raphson method does not converge quickly enough, this function uses a combination of inverse interpolation and bisection to solve for the yield rate or spread. During each solve iteration, the present value of the cash flow stream is determined and compared to the target value. When the difference is small, the solving process is terminated and the yield rate or spread is returned.

A03 SECURITIZED ASSETS

This book discusses securitized assets, also called collateralized obligations, and the sec submodels.

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1 INPUT ORGANIZATION

Input fields for securitized assets are on the Economy and Investment input pages, or on model point records.

1.1 Economy input page

The Economy input page includes assumptions related to the general economic environment, which is independent of company and asset portfolio assumptions.

1.1.1 Credit spreads

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. See the Credit spread section in the Market value chapter, below.

1.1.2 Default rates

Specify securitized asset default rates on the Economy input page under the Scenarios: Assumptions category. See the Defaults chapter, below.

1.2 Investment input page

The Investment input page includes assumptions related to projected asset portfolio and investment strategies.

1.2.1 Intex processing

Specify parameters for Intex processing on the Investment input page under the Asset Run Control: Intex category. See the Intex chapter, below.

1.2.2 PAC schedules

Specify PAC schedules on the Investment input page under the Securitized Asset Cash Flows category. See the Cash flows chapter, below.

1.2.3 Expenses

Investment expense rates vary by asset sub portfolio. See the Investment expenses section in the Cash flows chapter, below.

1.2.4 Accounting

Specify securitized asset accounting rules on the Investment input page under the Balance Sheet category. These definitions apply for all securitized assets. See the Accounting values chapter, below.

1.2.5 Purchase securitized asset definitions

Specify purchase securitized asset definitions on the Investment input page under the Purchase Asset Definitions category. See the Securitized asset purchases chapter, below.

1.3 Model points

Specify on the model point records the definition for each securitized asset existing at the projection date. See the Model point records chapter, below.

Report groups

Assets typically do not use the report group feature that is used to group liability model points. Instead, asset sub portfolio reports automatically collect the individual assets assigned to each group. This has the advantage that asset sub portfolio assignments can change dynamically as asset durations change through the projection.

However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

1.4 Underlying collateral

A unique aspect of securitized assets is that its underlying collateral characteristics and assumptions are also required as part of the setup for securitized assets. The underlying collateral is identified using a `cp_asset_id` variable where the quality of the underlying collateral is defined using an associated `cp_category_id` variable.

The definitions and cash flow processing of the underlying collateral for the securitized asset are very similar to the definitions and cash flow processing for the mortgage assets. Please see the 'Underlying collateral' chapter below.

2 UNDERLYING COLLATERAL

2.1 Payment timing

2.1.1 Maturity

The collateral matures for its remaining par value at the end of the maturity month specified on the model point record.

2.1.2 Payment dates

Payments occur either monthly, semi-annually, quarterly, or annually, at the end of the month related to the maturity date. For example, a collateral pool maturing in April has quarterly payments at the end of January, April, July, and October.

All payments occur at the end of the month.

2.2 Interest payments

2.2.1 Initial interest rate

The model point record contains the initial (at the projection date) payment rate. Interest rates are nominal based on the payment mode. For example, a 6.00% (the rate on the model point record) monthly collateral has an effective interest rate of 6.17%.

The interest paid by borrowers is reduced by a servicing fee percent charged by the company servicing the mortgage. Specify the servicing fee on the model point record. The servicing fee is a nominal rate; that is, the servicing fee for each payment is this annual rate divided by the number of payments per year.

The collateral interest rate is used to calculate amortization payments, interest savings present values for prepayments, and adjustable rates. The net interest rate is used to calculate the asset cash flows.

2.2.2 Adjustable rate collateral

Existing assets

The initial interest rate applies until the next reset date. The rate resets on the payment date in or after that month. That is, the rate can only reset on a payment date. For fixed rate collateral, the reset date is simply specified as the maturity date or later, such as the 9999/01/01.

For adjustable rate collateral, the payment rate is reset at the next reset date, and then periodically thereafter based on the reset period months. At each reset date, the payment rate changes to be the scenario bond equivalent yield for a specified category id for a specified maturity years, times a multiple then plus an addition (nominal rate).

Each new payment rate is constrained to be within the maximum rate (cap) and minimum rate (floor).

Purchase assets

For collateral of securitized assets purchased during the projection, the addition for rate resets is different from the addition that determines the initial rate.

2.3 Principal payments

2.3.1 Scheduled payments

The collateral may begin paying down principal immediately or contain a period where payments are interest only. Once the collateral begins amortizing, collateral principal payments are determined so that the collateral is paid off at the maturity date.

2.3.2 Balloon payments

The remaining principal is a balloon payment at the balloon payment date specified on the model point record. When there is no balloon feature, specify the balloon date as the maturity date or later, such as 9999/01/01.

The balloon payment can only occur on a payment date. If another date is, the balloon payment occurs at the next payment date.

2.4 Prepayments

2.4.1 Prepayment provision

Collateral prepayment provisions allow the borrower to pay off the remaining principal amount, in addition to a prepayment penalty. The projection includes prepayments varying based on the duration of the collateral and economic conditions. The prepayment rate can be determined using either the prepayment model described below, or the prepayment models available from Andrew Davidson & Co, Inc. (discussed separately below).

Projecting the percent of the collateral that is prepaid reduces the size of the remaining collateral. The collateral payments are reduced by the same percent as the outstanding principal value. That is, collateral prepayments do not advance the time when the remaining principal is paid. Instead, they reduce the portion of the collateral that continues. Partial prepayments of an individual collateral are ignored, as this effect is typically insignificant compared to prepayments of the collateral in full.

2.4.2 Prepayment penalty

Collateral prepayments may be subject to a prepayment penalty. The prepayment amount is the remaining principle plus the prepayment penalty. The prepayment penalty schedule defines the prepayment percentages over the time when the collateral is prepayable.

The values are the percent of the remaining principle that must be paid as a penalty to prepay the collateral.

Existing securitized asset collateral prepayment penalty schedules

Specify prepayment penalty schedules for existing securitized asset collateral on the Investment input page under the Decrements: Calls/Puts/Prepayments category. The schedules must be an external source varying by hidden variable lookup key `asset_id` and by the date column. Since the input variable is linked to a variable in the underlying collateral submodel, the `asset_id` specified in the assumptions and used in the lookup is the `asset_id` for the underlying collateral, not the securitized asset.

Purchase securitized asset collateral prepayment penalty schedules

Specify prepayment penalty schedules for purchase securitized asset collateral on the Investment input page under the Purchase Asset Definitions: Securitized Assets category. The schedules must be an external source varying by hidden variable lookup key `purch_sec_cp_asset_id` of the `inv_strategy` submodel, and by the hidden `mths_since_issue` variable of the `inv_strategy` submodel.

Purchase securitized asset collateral prepayment penalty percent of payment rate

For purchase securitized asset collateral, the prepayment penalty may be specified to be a percentage of the payment rate at issue of the collateral, grading to 0% over some specified months from issue. This feature allows for prepayment penalty sensitivity to the level of the payment rate.

Specify provision for prepayment penalty as a percent of the payment rate on the Investment input page under the Purchase Asset Definitions: Securitized Assets category.

2.4.3 Yield maintenance

Yield maintenance provisions require the borrower to compensate the lender for any loss in yield on the prepayment. These provisions are sometimes referred to as 'make whole' provisions. Available yield maintenance options are:

- **Yield maintenance** – The prepayment penalty is sufficient to compensate the lender for any loss in yield under the assumption that the prepayment is reinvested in par bond equivalent risk free debt at the current risk free yield to the remaining maturity less a spread.
- **Cash flow maintenance** – The prepayment penalty is the present value of scheduled future cash flows on the collateral, discounted at the current risk free yield to maturity less a spread, but not less than the current principal.

2.4.4 Refinancing financial test

When prepayments are permitted, a financial test determines if refinancing the collateral at current interest rates would be favorable to the borrower. The comparison is against the borrower obtaining a new collateral to raise the proceeds for paying off the current one. The refinancing rate is either:

- The scenario bond equivalent yield, plus the credit spread, for the current collateral's weighted average life (WAL).
- The scenario bond equivalent yield for a specified scenario year, times a multiple, and plus a spread.

Refinancing the collateral makes financial sense when the interest savings exceed the prepayment cost. The interest savings are the present value of scheduled principal and interest payments discounted at the refinancing rate, less the principal value. The prepayment cost is the prepayment amount plus the refinancing cost. The prepayment penalty is determined from the prepayment penalty schedule, or through a yield maintenance provision calculation. The refinancing cost is the principal value times the assumed refinancing cost percent.

2.4.5 Financial prepayment behavior

When the financial prepayment test shows that refinancing the collateral would be favorable to the borrower, a portion of the collateral may be refinanced. Partial asset decrements simulate the proportion of borrowers which would take advantage of the refinancing option in the current situation.

The part of the collateral refinanced is equal to the principal value times the financial prepayment percent. The financial prepayment percent is linearly interpolated from the prepayment option exercise assumption. The interpolation is based on the interest rate differential, which is the current interest rate less the refinancing rate.

Collateral prepayment rates are often defined in terms of a multiple of the Public Security Association Standard Prepayment Model. Prepaying collateral at '100% of PSA' means that prepayment rates grade

linearly from 0.2% per annum in the first month following issue to 6% per annum in the 30th month following issue and remaining constant thereafter.

The prepayment percent is an effective annual rate. The monthly prepayment rate is:

$$1 - (1 - \text{annual rate})^{1/12}$$

Prepayment option exercise percent

Specify the prepayment option exercise percent on the Investment input page under the Decrements: Calls/Puts/Prepayments category. This must be an external source varying by hidden variable lookup key category_id (asset category), and by hidden variable lookup key int_rate_diff. The example below assumes that collateral holders will not exercise their prepayment option until the interest rate differential exceeds 1%, reaching 40% exercise rate when the differential is 2% and 100% when the differential reaches 4% or more.

category_id	FNMA
int_rate_diff	
0.00%	0.0000%
1.00%	0.0000%
2.00%	40.0000%
4.00%	100.0000%

2.4.6 Nonfinancial prepayments

Collateral prepayment assumptions also include a nonfinancial component, which is not subject to any financial tests. This rate can be adjusted for special conditions:

- Higher interest rates can reduce the nonfinancial prepayment activity as the interest differential increases between the collateral refinancing rate and the current collateral interest rate.
- A seasonality adjustment can skew the percent by calendar month.

The nonfinancial prepayment percent that applies in a particular month is:

$$\text{Prepayment nonfinancial percent} \\ - \text{Prepayment nonfinancial adjustments} \times \text{seasonality adjustment but} \geq 0$$

Nonfinancial prepayment adjustments

Specify the nonfinancial prepayment adjustments percent on the Investment input page under the Decrements: Calls/Puts/Prepayments category. This must be an external source varying by hidden variable lookup key category_id (asset category), and by hidden variable lookup key nonfinancial_int_rate_diff.

Specify the nonfinancial seasonality adjustment factor on the Investment input page under the under the Decrements: Calls/Puts/Prepayments category. This must be an external source varying by hidden variable lookup key category_id (asset category), and by cal_mth column.

2.5 Andrew Davidson prepayment behavior

An alternative to the above prepayment model is to license prepayment models software from Andrew Davidson & Co., Inc. These prepayment models provide monthly prepayment rates using specific collateral characteristics, interest rate history, interest rate forecast, home price index history, and home price index forecast.

Securitized asset collateral are linked to the appropriate Andrew Davidson prepayment models through their individual model point characteristics and asset category ID. Collateral asset category ID information includes

Andrew Davidson category identification, collateral definition, subprime definition, and adjustable rate provisions.

Fixed rate collateral use the fixed rate MBS prepayment model which considers the loan age, remaining term, payment rate, balloon features, and history and forecasts for interest rates and home price indexes.

Adjustable rate collateral, hybrid MBS, and adjustable rate home equity loans asset-based securities (ABS) prepayments use the unified prepayment model which considers the loan age, remaining term, adjustable rate provisions, forecasts of adjusted payment rates, along with history and forecasts of the interest rates and the reference index rate.

Tuning for the prepayment forecast is handled internally in the model based on tuning parameters included in the Andrew Davidson data files.

The fixed home equity loans, auto, and manufactured housing ABS prepayment models take into account the major factors affecting their prepayments.

You must license these models from Andrew Davidson & Co., Inc. The ABS prepayment models can only be used with the Intex link, which is discussed below. Contact Andrew Davidson & Co. Inc. for more information.

3 CASH FLOWS

3.1 Third party modeling

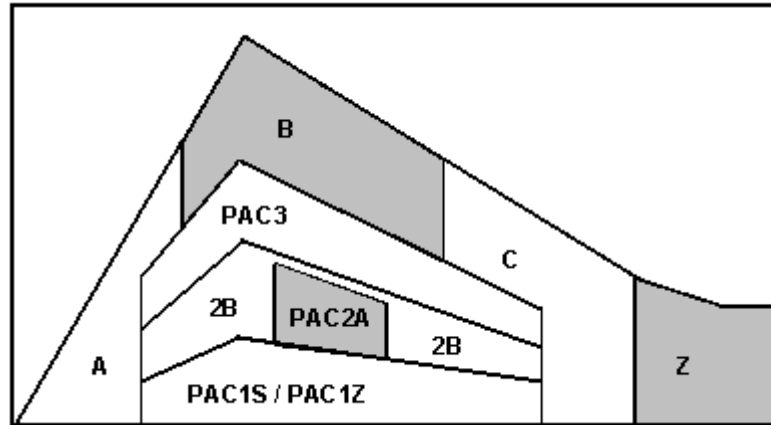
Existing securitized assets are typically projected using specialized tools available for handling these complex assets. This avoids the work involved in mapping the entire deal's tranche structure into the nine-tranche structure described in the Tranche structure section below. There are two methods available:

- Intex – See the Intex chapter below for information about the databases and subroutines for structured securities which are available from Intex Solutions, Inc.
- EPA – See the Externally projected assets book for information about EPA files created by an asset pricing system available from external vendors.

3.2 Tranche structure

The actual structure of each deal is complex. Projecting the results for a particular asset, however, does not require each individual tranche to be projected. Instead, various tranches in the deal structure can be grouped together. It is important that the cash flows for the other tranches are projected in their correct relation to the particular tranche owned by the asset portfolio. The entire deal structure must be projected together, since the presence of the other tranches affects the cash flows of the tranche owned.

The securitized asset deal structure is grouped into nine modeled tranches. This grouping does not change the dynamics of the tranche owned. The following diagram gives a summary of the tranches modeled. The tranches that are typically modeled as the owned tranche are shaded.



3.2.1 Modeled support tranches

The modeled A, B, and C tranches represent support or sequential tranches. These tranches pay sequentially.

A support tranche in the asset portfolio is modeled as tranche B. Assign the other tranches this way:

- Support tranches that pay before this tranche are grouped into tranche A.
- Support tranches that pay after this tranche are grouped into tranche C.
- All PAC/TAC tranches are grouped into tranche PAC1. Splitting the PAC/TAC tranches into different priority levels would not affect the projection results.

3.2.2 Modeled PAC/TAC tranches

The PAC and TAC tranches represent planned amortization classes and targeted amortization classes with different priorities over principal payments from the collateral. The modeled PAC1 tranche has first priority over principal payments until its scheduled payments are fulfilled. The modeled PAC3 tranche has lowest priority among the PAC/TAC tranches.

A PAC/TAC tranche in the asset portfolio is modeled as tranche PAC2A. Assign the other tranches this way:

- All higher payment priority PAC/TAC tranches are grouped into tranche PAC1.
- All lower payment priority PAC/TAC tranches are grouped into tranche PAC3.
- All equal payment priority PAC/TAC tranches are grouped into tranche PAC2B.
- All support tranches are grouped into tranche A, since it does not matter what the relative priorities are between these separate tranches.

3.2.3 Modeled Z tranches

When a Z (accrual) tranche is in the asset portfolio, all the support tranches are grouped into tranche A, and all the PAC/TAC tranches are grouped into tranche PAC1.

When the Z tranche is a PAC/TAC tranche, it is modeled as tranche PAC2A. Other tranches are assigned following the rules above for PAC/TAC tranches.

3.2.4 Owned tranche payments

The owned tranche may be specified as receiving interest only, principal only, or both interest and principal.

3.3 Payment timing

3.3.1 Payment dates

Payments occur either monthly, semi-annually, quarterly, or annually, at the same frequency as the underlying collateral.

All payments occur at the end of the month.

3.4 Interest payments

3.4.1 Tranche interest rate

The tranche interest rates are nominal based on the payment mode. For example, a 6.00% (the rate on the model point record for an existing securitized asset) monthly tranche has an effective interest rate of 6.17%.

Interest is payable on the tranche payment dates.

3.4.2 Adjustable rate tranche

Existing assets

When the tranche interest rate multiple is not zero, the tranche interest rate is adjustable. The initial interest rate applies until the next payment date. The rate resets on each subsequent payment date.

At each reset date, the payment rate changes to be the scenario bond equivalent yield for a specified category id for a specified maturity years, times a multiple then plus an addition (nominal rate).

Each new payment rate is constrained to be within the maximum rate (cap) and minimum rate (floor).

Purchase assets

The tranche interest rate is specified as either being a Fixed Rate or Floating Rate using the `purch_sec_tranche_rate_defn` variable.

If the tranche rate is a floating rate, the payment rate changes at each payment date to be the scenario bond equivalent yield for a specified category id for a specified maturity years, times a multiple then plus an addition (nominal rate).

Each new payment rate is constrained to be within the maximum rate (cap) and minimum rate (floor) which are determined when the asset is purchased using the maximum increase/decrease rates from the initial rate.

3.5 Principal payments

3.5.1 Distributable principal

Securitized asset principal payments are based on the amount of principal paid on the underlying collateral. The scheduled principal payments and cumulative prepayments of the collateral are available to be distributed to the tranches of the securitized asset.

The principal distributions from the collateral into the securitized asset occur at the collateral payment frequency. Principal prepayments received between payment date are aggregated and included with the scheduled principal payments at the next payment date.

3.5.2 Allocation of principal among tranches

The principal payments from the collateral are allocated among the modeled tranches as follows:

- To PAC1 and PAC1Z until the principal balances meet the scheduled balances.
- To PAC2A and PAC2B until the principal balances meet the scheduled balances.
- To PAC3 until the principal balance meets the scheduled balance.

- When Z has jumped in priority ahead of the support tranches, to Z until the principal balance is reduced to zero.
- To A, B and C, in order, until the principal balances are reduced to zero.
- When Z has not jumped, and the prepayment priority before jump is 'before PAC2', to Z until the principal balance is reduced to zero.
- To PAC3 until the principal balance is reduced to zero.
- To PAC2A and PAC2B until the principal balances are reduced to zero. Additional principal payments roughly follow the PAC schedules.
- When Z has not jumped and the prepayment priority before jump is 'after PAC2', to Z until the principal balance is reduced to zero.
- To PAC1 and PAC1Z until the principal balances are reduced to zero. Additional principal payments roughly follow the PAC schedules.

3.6 Z tranche types

3.6.1 Z tranche priority order

The following table summarizes the order in which principal is allocated to the nine modeled tranches. The three columns correspond to the three payment states of the Z tranche.

Z before jump: prepayment priority before jump is Before PAC2	Z before jump: prepayment priority before jump is After PAC2	Z after jump
Scheduled PAC1	Scheduled PAC1	Scheduled PAC1
Scheduled PAC2	Scheduled PAC2	Scheduled PAC2
Scheduled PAC3	Scheduled PAC3	Scheduled PAC3
A	A	Z
B	B	A
C	C	B
Z	Additional PAC3	C
Additional PAC3	Additional PAC2	Additional PAC3
Additional PAC2	Z	Additional PAC2
Additional PAC1	Additional PAC1	Additional PAC1

3.6.2 Interest on accrual (Z) tranches

Before any principal payments have been made to the Z tranche, the interest on the Z tranche accrues to increase the principal balance of the Z tranche, and to pay additional principal amounts on the tranches currently being paid down. Once a principal payment has been made on the Z tranche, the further accrual of interest depends on the Z tranche definition.

For an accrual-only Z tranche, the accrual of interest stops upon the first principal payment that exceeds the current month's interest. Even when principal payments are zero in some future months (due to requirements to pay PAC/TAC schedule amounts) further interest does not accrue. Instead, it is paid to the owner of the Z tranche.

For a jump or jump/stick Z tranche, the payment priority of the Z tranche moves ahead of other support tranches when the prepayment speed on the collateral pool exceeds a specified percent of PSA threshold. When the prepayment speed falls below this specified percent of PSA threshold, the payment priority of a jump Z tranche reverts to its original position (after all other support tranches), while the payment priority of a jump/stick Z tranche never reverts back to its original position.

3.6.3 *Interest accruals on Z PAC/TAC tranches*

In addition to PAC1Z, modeled tranches PAC2A, PAC2B and PAC3 can also be Z tranches. The following applies when the named tranches are Z PAC/TAC tranches.

For PAC1Z, PAC2A and PAC3, the accrual of interest stops upon the first principal payment that exceeds the current month's interest. Even if principal payments are zero in some future months (due to requirements to pay PAC/TAC schedule amounts), further interest does not accrue. Instead, it is paid to the owner of each of these tranches.

Interest accruals on tranche PAC2B stop only when the PAC2A balance is reduced to zero (and a principal payment is made that exceeds the current month's interest). PAC2B is treated this way because part of the schedule for PAC2A may include tranches that are still accruing interest.

3.7 PAC principal payments

Each principal payment is the difference between specified PAC balances. For existing securitized assets, the PAC balance are specified as the full amount of the PAC balance remaining at each date after the projection date. For purchased securitized assets, the PAC balances are specified as a percentage of the initial collateral amount for each month since issue.

Existing securitized asset PAC schedules

Specify PAC schedules for existing securitized assets on the Investment input page under the Securitized Asset Cash Flows category. The schedules must be an external source varying by hidden variable lookup key `asset_id` of the sec submodel, and by the date column of the sec submodel.

Purchase securitized asset PAC schedules

Specify PAC schedules for purchase securitized assets on the Investment input page under the Purchase Asset Definitions : Securitized Asset category. The schedules must be an external source varying by hidden variable lookup key `purch_asset_id` of the inv_strategy submodel, and by the hidden lookup key `mths_since_issue` variable of the inv_strategy submodel.

3.8 Investment expenses

Investment expense rates vary by asset sub portfolio and are determined by asset sub portfolio. Specify this rate as part of the asset sub portfolio definitions on the Investment input page under the Asset sub portfolio category.

Asset cash flows are reduced by investment expenses. Investment income is reported net of investment expenses, and asset yield rates are net of investment expenses.

Investment expenses occur at the end of each month (prior to processing investment strategies), equal to the book value of each asset times its annual investment expense rate, divided by 12.

4 DEFAULTS

4.1 Deterministic defaults

Defaults are processed at the beginning of each month. Defaults thus reduce the amount of payments that would otherwise occur in that month and in the future. Partial defaults are a proxy for the gradual effect of declining quality grading as a portfolio of securitized assets age.

Deterministic default rates vary by asset category, and can also vary by date. Specify deterministic default rates on the Economy input page under the Scenarios: Assumptions category. These definitions must be an external source varying by the hidden variable lookup key `quality_id`; they can also vary by the date column.

The monthly default rate is:

$$1 - (1 - \text{annual default rate})^{1/12}$$

4.2 Stochastic defaults

Default rates can be stochastic when the scenario file includes default rates for scenario yield curves in addition to the risk free yield curve. Typically the scenario file includes default rates for AA and B securities.

The stochastic default rate is an addition to the deterministic default rate.

Specify the stochastic default rate as a multiple of the default rates for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys `yld_curve_id` and `quality_id`.

In the following example, the stochastic default rate for A-rated securities is 80% of the scenario default rate for AA-rated securities plus 20% of the scenario default rate for B-rated securities.

quality_id		A
yld_curve_id		
Govt		0.0000%
AA		80.0000%
B		20.0000%

5 ACCOUNTING VALUES

5.1 Accrued interest

Accrued interest is added to the market and book values of securitized assets. That is, projections calculate 'clean' market and book values (which exclude accrued interest) for securitized assets.

The accrued interest is the interest payment, compounded for the fraction of the payment period since the last payment.

Note that the initial book and market value (on existing asset model points) excludes accrued interest. However, accrued interest is calculated and added to the initial market value to establish the full market value for market consistent projection calibrations. Accrued interest is added to the market value and statutory book value of securitized assets.

5.2 Statutory accounting

The cost basis for U.S. securitized assets is typically amortized cost. The statutory book value is typically the cost basis. Other options provide flexibility to use asset accounting methods that apply in various other countries.

The expected future payments for a tranche can be volatile in changing interest rate environments, due to the sensitivity of prepayments to the level of interest rates. The amortized cost is updated each month based on the amortized yield and the principal and interest payments received during the month. See the Amortized Yield section for the frequency for recalculating the amortized yield.

The statutory book value applies for tax calculations.

5.3 GAAP accounting

GAAP accounting treatment varies according to the sales classification of each asset:

- For GAAP income, trading assets are valued at market value, while other classifications are valued at the statutory cost basis.
- For the GAAP balance sheet, trading and available for sale assets are valued at market value, while other classifications are valued at the statutory cost basis.

5.4 Amortized yield

When the cost basis definition is amortized cost, the asset yield rate is the rate that amortizes the cost basis to par value by the maturity date, using scheduled payments.

The amortized yield for securitized asset tranches is calculated according to the prospective yield method. This is described on pages 1163-1165 of Fabozzi's 'Handbook of Mortgage-Backed Securities', 3rd edition, 1992, Probus Publishing Co. The net yield rate at the model start date is the rate that discounts the projected cash flows to the starting cost basis value. The calculation is based on projected principal and interest payments consistent with the starting scenario rates.

The amortized yield is recalculated periodically based on the specification for the securitized asset net yield. The amortized yield is recalculated by projecting principal and interest payments consistent with the scenario rates at the time of the recalculation, then solving for the rate which discounts the projected cash flows to the current cost basis value.

Defaults are decrements of a portion of the securitized asset tranche. They do not affect the amortized yield of the remaining asset.

6 MARKET VALUE

Securitized asset market values are approximated assuming that the current scenario rates continue unchanged, based on prepayment assumptions for the collateral, by discounting the expected future cash flows, then subtracting the accrued interest at the market valuation date. The discount rate is the current scenario risk adjusted spot rate consistent with the timing of each cash flow, plus the market spread.

The market value is increased by the accrued interest.

6.1 Credit spread

6.1.1 Fixed credit spreads

The credit spread (also called risk spread) addition to the spot rates adjusts for the assumed credit risk associated with the securitized asset.

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. This must be an external source varying by hidden variable lookup keys `quality_id` and `rate_term`; and can also vary by the date column. Credit spreads typically vary by date when spreads that are consistent with current market values are different from those assumed to apply for normal market conditions going forward.

6.1.2 Stochastic credit spreads

The credit spread can be stochastic when the scenario file includes credit yield curves in addition to the risk free yield curve. Typically the scenario file includes yield curves for AA and B securities.

The stochastic credit spread is an addition to the deterministic credit spread.

Specify the stochastic credit spread as a multiple of the credit spreads for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys `yld_curve_id` and `quality_id`.

In the following example, the stochastic credit spread for A-rated securities is 80% of the spread for AA-rated securities plus 20% of the spread for B-rated securities.

quality_id		A
yld_curve_id		
Govt		0.0000%
AA		80.0000%
B		20.0000%

6.2 Market spread

The market spread is the additional spread required for the calculated market value to match the input market value. This spread is not calculated when the input market value is zero.

The following options are available for using initial the market spread:

- Ignore the market spread by treating it as zero during the projection. Appropriately small market spreads can help validate that the projection assumptions are consistent with the assumptions used to calculate the input market values.
- The market spread is held level at the initial rate. This treatment interprets the market spread as a correction of the credit spread assumptions. This interprets that the difference is due to the nature of the securitized asset, rather than due to current market conditions being different from ultimate expectations.
- The initial market spread is graded linearly to zero by the maturity date. This treatment interprets the market spread as a result of actual market values which fluctuate around analytical market value calculations. The current differential will eventually be corrected by market forces.

7 MODEL POINT RECORDS

The model point record layout for securitized assets is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	unique identifier for the securitized asset. Up to 10 characters (such as CUSIP)
D	use_mp	Yes, No
E	category_id_mp	category identifier for the securitized asset. Up to 20 characters
F	cp_asset_id	identifier for the collateral. Up to 10 characters.

Excel Column	Variable	Value
G	cp_category_id	category identifier for the collateral. Up to 20 characters
H	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
I	planned_sale_date_mp	yyyy/mm/dd or mm/dd/yyyy
J	init_tranche_par_val_mp	par value at projection date
K	init_tranche_cost_basis_mp	cost basis at projection date
L	init_tranche_bk_val_mp	book value at projection date
M	init_tranche_mkt_val_mp	market value at projection date**
N	tranche_owned_mp	A, B, C, Z, PAC2A
O	io_po_defn_mp	Both, IO (interest only), PO (principal only)
P	tranche_rate_category_id	category id for rate resets (up to 20 characters); default is Govt
Q	tranche_rate_scen_yr_mp	tranche rate scenario maturity years
R	tranche_rate_scen_mult_mp	default = 1.00000
S	tranche_rate_addn_mp	addition = 0.00 (0%) to 1.00 (100%)
T	tranche_min_rate_mp	0.00 to 1.00 (100%)
U	tranche_max_rate_mp	0.00 to 1.00 (100%)
V	init_tranche_a_bal_mp	tranche A balance at projection date
W	init_tranche_b_bal_mp	tranche B balance at projection date
X	init_tranche_c_bal_mp	tranche C balance at projection date
Y	init_pac1_std_bal_mp	PAC 1S balance at projection date
Z	init_pac1z_bal_mp	PAC 1Z balance at projection date
AA	init_pac1z_z_rate_mp	PAC 1Z nominal interest rate, 0.00 to 1.00 (100%)
AB	init_pac2a_bal_mp	PAC 2A balance at projection date
AC	init_pac2b_bal_mp	PAC 2B balance at projection date
AD	pac2_accrual_defn_mp	Standard (interest is paid before principle payments begin), Z PAC (interest accumulates to be paid with tranche payments)
AE	init_pac2a_z_rate_mp	PAC 2Z nominal interest rate, 0.00 to 1.00 (100%)
AF	init_pac3_bal_mp	PAC 3 balance at projection date
AG	init_pac3_z_rate_mp	PAC 3Z nominal interest rate, 0.00 to 1.00 (100%)
AH	init_tranche_z_bal_mp	Tranche Z balance at projection date
AI	tranche_z_defn_mp	Accrual Only (pays after tranche C), Jump (paid before sequential tranches when the prepayment speed > Z tranche jump speed), Jump/Stick (paid before sequential tranches if the prepayment speed has ever exceeded the jump speed)
AJ	tranche_z_accrual_rate_mp	tranche Z nominal interest rate, 0.00 to 1.00 (100%)
AK	tranche_z_jump_speed_mp	percent PSA speed, 200 specifies 200%
AL	tranche_z_prepmt_priority_bef_jump_mp	Before PAC2 (if the sequential tranches have been paid, then prepayments above the planned payments for tranche PAC2 are paid to tranche Z rather than to tranche PAC2), After PAC2 (pays after PAC 2)
AM	avr_contribn_pct_mp	0.00 to 1.00 (100%)
AN	avr_max_pct_mp	0.00 to 1.00 (100%)
AO	avr_obj_pct_mp	0.00 to 1.00 (100%)
AP	rbc_c1_pct_mp	0.00 to 1.00 (100%)
AQ	cp_issue_date_mp	yyyy/mm/dd or mm/dd/yyyy
AR	cp_maturity_date_mp	yyyy/mm/dd or mm/dd/yyyy
AS	cp_init_prin_val_mp	collateral principal value at projection date
AT	cp_amortzn_start_cy_mp	the collateral is interest-only (not amortizing) through the maturity month in this calendar year (yyyy)
AU	cp_pmt_mode_mp	Monthly, Quarterly, Semi-Annual, Annual
AV	cp_init_pmt_rate_mp	0.00 to 1.00 (100%); nominal rate using payment mode

Excel Column	Variable	Value
AW	cp_servicing_fee_mp	0.00 to 1.00 (100%); nominal rate using payment mode
AX	cp_prin_val_at_issue_mp	collateral principal value at issue
AY	cp_prepmt_opt_first_date_mp	yyyy/mm/dd or mm/dd/yyyy
AZ	cp_prepmt_penalty_defn_mp	Not Applicable, Price Schedule, Yield Maintenance, Cash Flow Maintenance
BA	cp_balloon_date_mp	yyyy/mm/dd or mm/dd/yyyy
BB	cp_yld_maint_addn_mp	-1.00 (100%) to 0.00 (0%); nominal rate using payment mode
BC	cp_adj_reset_category_id_mp	category id for collateral rate resets (up to 20 characters); default is Govt
BD	cp_adj_reset_scen_yr_mp	coupon reset scenario maturity years
BE	cp_adj_reset_scen_mult_mp	default = 1.0000
BF	cp_adj_reset_renewal_addn_mp	0.00 to 1.00 (100%); nominal rate using payment mode
BG	cp_adj_next_reset_date_mp	yyyy/mm/dd or mm/dd/yyyy When using the Andrew Davidson prepayment model, this date must be the date of the first reset from issue, not from the projection date. Otherwise, the Andrew Davidson model will not select the appropriate loan type for the prepayment model forecast.
BH	cp_adj_reset_mths_mp	1 ... 999
BI	cp_adj_max_incr_per_reset_mp	0.00 to 1.00 (100%); nominal rate using payment mode
BJ	cp_adj_max_decr_per_reset_mp	0.00 to 1.00 (100%); nominal rate using payment mode
BK	cp_adj_min_rate_mp	0.00 to 1.00 (100%); nominal rate using payment mode
BL	cp_adj_max_rate_mp	0.00 to 1.00 (100%); nominal rate using payment mode
BM	cp_init_prepmt_cumul_mp	0; this variable is only used for nested projections, where the write-state-of-the-world function updates the cumulative prepayments for the next time step

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

**The initial market value is used to calculate the market spread, which is discussed above in the Market Value chapter. The market spread is automatically zero when this variable is left at the default value of 0.00.

8 SECURITIZED ASSET PURCHASES

8.1 Purchase process

Each securitized asset purchased by the investment strategy creates a new securitized asset model point. These securitized assets are created to be consistent with the scenario rates that apply at that time.

Defining securitized asset purchases involves specifying the rules for creating the collateral, as well as the rules for dividing the pool into tranches. The size of the tranche is determined by the amount to be invested. Since that amount changes each time the asset is purchased, securitized asset purchases define only the relative size of each tranche as a percent of the total collateral, which is always million dollars. When the amount of the investment asset tranche is calculated, the size of the other tranches is then known.

Each securitized asset purchased during the projection is created using the following process:

1. Determine the characteristics of the collateral using the purchase asset definition for the collateral. These include an asset id, category, years to maturity, number of years before amortization begins, payment mode, servicing fee, years to balloon payment, prepayment penalty provisions, and adjustable rate characteristics. The collateral interest rate is the current refinancing rate for the collateral category, less the specified interest rate differential.
2. Calculate the tranche interest rate, which is a fixed or floating rate. The initial rate is the scenario bond equivalent yield for a specified category id, scenario year, times a multiple, and plus a spread.
3. Calculate the interest rate for each of the modeled Z tranches. The interest rate for each is the collateral interest rate less a spread specified for each type of Z tranche.
4. Project the cash flows per one unit of principal for the owned tranche, based on the projected cash flows for the collateral, assuming a \$1 million dollar collateral, allocated among the modeled tranches according to the rules defined in the purchase asset definition.
5. Calculate the tranche's market value per \$1 of principal.
6. Set the tranche's book value and market value to the amount to invest in this tranche.
7. Calculate the tranche's principal as the market value divided by the market value per \$1 of principal. Calculate the principal of the other modeled tranches, based on the specified percent of the collateral allocated to each modeled tranche.
8. The interest rates are not defined for other tranches (except for the Z tranches), because the performance of the tranche being purchased is not affected by the interest rate on these tranches.

8.2 Purchase securitized asset variables

Specify purchase securitized assets on the Investment input page, under the Purchase Asset Definitions category. These are variables in the `inv_strategy` submodel.

The variables must be linked to an external source using the hidden variable lookup key `purch_asset_id`.

Variable	Value
<code>purch_sec_grp_id</code>	model point report group*
<code>purch_sec_category_id</code>	category identifier for securitized asset. Up to 20 characters
<code>purch_sec_cp_asset_id</code>	identifier for the collateral. Up to 10 characters.
<code>purch_sec_cp_category_id</code>	category identifier for the collateral. Up to 20 characters
<code>purch_sec_sale_class</code>	Available for Sale, Trading, Held to Maturity, Not Assigned
<code>purch_sec_tranche_owned</code>	Tranche A, Tranche B, Tranche C, Tranche Z, PAC2A
<code>purch_sec_io_po_defn</code>	Both, IO (interest only), PO (principal only)
<code>purch_sec_tranche_rate_defn</code>	Fixed (rate calculated at purchase), Float (calculated at the beginning of each payment period)
<code>purch_sec_tranche_rate_category_id</code>	category id for rate resets (up to 20 characters); default is Govt
<code>purch_sec_tranche_rate_scen_yr</code>	tranche rate scenario maturity years
<code>purch_sec_tranche_rate_scen_mult</code>	default = 1.0000
<code>purch_sec_tranche_rate_addn</code>	addition = 0.00 to 1.00 (100%)
<code>purch_sec_tranche_rate_max_decr_from_init</code>	0.00 to 1.00 (100%)
<code>purch_sec_tranche_rate_max_incr_from_init</code>	0.00 to 1.00 (100%)
<code>purch_sec_tranche_a_pct</code>	percent of principal in tranche A, 0.00 to 1.00 (100%)
<code>purch_sec_tranche_b_pct</code>	percent of principal in tranche B
<code>purch_sec_tranche_c_pct</code>	percent of principal in tranche C
<code>purch_sec_tranche_z_pct</code>	percent of principal in tranche Z

Variable	Value
<code>purch_sec_tranche_z_defn</code>	Accrual Only (pays after tranche C), Jump (paid before sequential tranches when the prepayment speed > Z tranche jump speed), Jump/Stick (paid before sequential tranches if the prepayment speed has ever exceeded the jump speed)
<code>purch_sec_tranche_z_jump_speed</code>	percent PSA speed, 200 specifies 200%
<code>purch_sec_tranche_z_prepmt_priority_bef_jump</code>	Before PAC2 (if the sequential tranches have been paid, then prepayments above the planned payments for tranche PAC2 are paid to tranche Z rather than to tranche PAC2), After PAC2 (pays after PAC 2)
<code>purch_sec_tranche_z_addn</code>	addition to collateral pool interest rate = 0.00 to 1.00 (100%)
<code>purch_sec_pac1_std_sch</code>	0.00 to 1.00 (100%); varies by additional lookup key <code>mths_since_issue</code>
<code>purch_sec_pac1z_sch</code>	0.00 to 1.00 (100%); varies by additional lookup key <code>mths_since_issue</code>
<code>purch_sec_pac1z_accrual_addn</code>	addition to collateral pool interest rate = 0.00 to 1.00 (100%)
<code>purch_sec_pac2_accrual_defn</code>	Standard, Z PAC
<code>purch_sec_pac2a_sch</code>	0.00 to 1.00 (100%); varies by additional lookup key <code>mths_since_issue</code>
<code>purch_sec_pac2b_sch</code>	0.00 to 1.00 (100%); varies by additional lookup key <code>mths_since_issue</code>
<code>purch_sec_pac2_z_accrual_addn</code>	addition to collateral pool interest rate = 0.00 to 1.00 (100%)
<code>purch_sec_pac3_sch</code>	0.00 to 1.00 (100%); varies by additional lookup key <code>mths_since_issue</code>
<code>purch_sec_pac3_z_accrual_addn</code>	addition to collateral pool interest rate = 0.00 to 1.00 (100%)
<code>purch_sec_avr_contribn_pct</code>	0.00 to 1.00 (100%)
<code>purch_sec_avr_max_pct</code>	0.00 to 1.00 (100%)
<code>purch_sec_avr_obj_pct</code>	0.00 to 1.00 (100%)
<code>purch_sec_rbc_c1_pct</code>	0.00 to 1.00 (100%)
<code>purch_sec_cp_yrs_to_maturity</code>	0.25 or greater
<code>purch_sec_cp_amortzn_yrs</code>	years until amortization begins (the mortgage is interest-only until this time), 0 ... (less than <code>yrs_to_maturity</code>)
<code>purch_sec_cp_rate_defn</code>	Refinancing Rate, Fixed Rate, Floating Rate
<code>purch_sec_cp_rate_diff</code>	Monthly, Quarterly, Semi-Annual, Annual
<code>purch_sec_cp_pmt_mode</code>	0.00 to 1.00 (100%)
<code>purch_sec_cp_servicing_fee</code>	-1.00 (-100%) to 1.00 (100%)
<code>purch_sec_cp_adj_category_id</code>	category id for collateral rate resets (up to 20 characters); default is Govt
<code>purch_sec_cp_adj_scen_yr</code>	coupon reset scenario maturity years
<code>purch_sec_cp_adj_scen_mult</code>	default = 1.0000
<code>purch_sec_cp_adj_addn_init</code>	0.00 to 1.00 (100%)
<code>purch_sec_cp_adj_addn_renewal</code>	0.00 to 1.00 (100%)
<code>purch_sec_cp_adj_reset_mths</code>	1 ... 999
<code>purch_sec_cp_adj_max_above_init</code>	0.00 to 1.00 (100%)
<code>purch_sec_cp_adj_max_below_init</code>	0.00 to 1.00 (100%)
<code>purch_sec_cp_adj_max_reset_incr</code>	0.00 to 1.00 (100%)
<code>purch_sec_cp_adj_max_reset_dec_r</code>	0.00 to 1.00 (100%)
<code>purch_sec_cp_prepmt_penalty_defn</code>	Not Applicable, Penalty Percent, Yield Maintenance, Cash Flow Maintenance, Related to Payment Rate
<code>purch_sec_cp_prepmt_protect_yrs</code>	1 ... 999
<code>purch_sec_cp_prepmt_penalty_pct</code>	1.00 (100%) to 2.00 (200%); MUST vary by lookup keys <code>purch_sec_cp_asset_id</code> and <code>mths_since_issue</code>
<code>purch_sec_cp_yld_maint_addn</code>	-1.00 (100%) to 0.00

Variable	Value
purch_sec_cp_prepmt_penalty_mult	-1.00 (100%) to 2.00 (200%)
purch_sec_cp_prepmt_penalty_adj	-1.00 (100%) to 1.00 (100%)
purch_sec_cp_first_par_prepmt_yr	0 ... 999
purch_sec_cp_yrs_to_balloon	0 ... 999

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

9 INTEx

9.1 Intex Solutions, Inc.

Intex Solutions, Inc. creates and maintains databases of information for structured securities such as collateralized mortgage obligations (CMO), asset backed securities (ABS), and commercial mortgage backed securities (CMBS). Intex has databases for agency pass through pool mortgage backed securities (MBS), and makes available subroutines that allow for third-party software to access the databases. When the subroutines are integrated, clients can utilize the Intex databases when running the third party software.

Instead of using the normal projection calculations for existing securitized assets, projections can use a link to the databases and subroutines provided by Intex Solutions. For more information on the asset class coverage provided by the Intex databases, contact Intex Solutions, Inc.

Intex is not available for purchase assets. Use the normal features provided for purchasing securitized assets during the projection.

9.2 Initiating the Intex link

Follow these steps to activate the Intex link:

1. Set the Include Intex calculations? option to Yes.
2. Specify the Collateralized obligation dll: reference to the Intex DLL. For 64 bit processing, specify the Collateralized obligation dll (64bit) reference to the Intex 64bit DLL.
3. Specify the CUSIP deal translation path. The path, which is provided with the file, is required by the Intex subroutines so that they can access the deal model (CDI) data and the deal update (CDU) data.

The Intex subroutines require the path to the deal model data (CDI), which is derived from the data storage scheme. The path to the deal model data is the cmo_cdi subdirectory of the Intex root directory. You must follow the Intex data storage scheme in order for the Intex subroutines to parse the deals successfully.

9.3 Existing assets data

The Intex link is applicable only for mortgage and securitized assets. Specify MBS assets using the mortgage asset type. Specify CMO, ABS, and CMBS assets using the securitized asset type. Assets to be projected using the Intex link require only a few the existing asset record fields.

9.3.1 MBS assets

For MBS assets, use the following variables on mortgage model point records:

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	up to 10 characters (such as CUSIP for agency pass throughs)
D	use_mp	Yes, No
E	category_id_mp	up to 20 characters (must match a mortgage asset category)
F	port_defn_mp	Investment, Collateral Pool
G	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
K	planned_sale_date_mp	yyyy/mm/dd or mm/dd/yyyy
L	init_par_val_mp	principal value at projection date
M	init_cost_basis_mp	cost basis at projection date
N	init_bk_val_mp	book value at projection date
O	init_mkt_val_mp	market value at projection date**
AK	avr_contribn_pct_mp	0.00 to 1.00 (100%)
AL	avr_max_pct_mp	0.00 to 1.00 (100%)
AM	avr_obj_pct_mp	0.00 to 1.00 (100%)
AN	rbc_c1_pct_mp	0.00 to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

**The initial market value is used to calculate the market spread, which is discussed above in the Market Value chapter. The market spread is automatically zero when this variable is left at the default value of 0.00.

9.3.2 CMO, ABS, and CMBS assets

For CMO, ABS, and CMBS assets, use the following variables on securitized asset model point records:

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	up to 10 characters (Must be the CUSIP)
D	use_mp	Yes, No
E	category_id_mp	category identifier for the securitized asset. Up to 20 characters
F	cp_asset_id	identifier for the collateral. Up to 10 characters. Typically the deal name.
G	cp_category_id	category identifier for the collateral. Up to 20 characters
H	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
I	planned_sale_date_mp	yyyy/mm/dd or mm/dd/yyyy
J	init_tranche_par_val_mp	par value at projection date
K	init_tranche_cost_basis_mp	cost basis at projection date
L	init_tranche_bk_val_mp	book value at projection date
M	init_tranche_mkt_val_mp	market value at projection date**
AM	avr_contribn_pct_mp	0.00 to 1.00 (100%)
AN	avr_max_pct_mp	0.00 to 1.00 (100%)
AO	avr_obj_pct_mp	0.00 to 1.00 (100%)
AP	rbc_c1_pct_mp	0.00 to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

**The initial market value is used to calculate the market spread, which is discussed above in the Market Value chapter. The market spread is automatically zero when this variable is left at the default value of 0.00.

9.4 Projections using Intex

9.4.1 Pre-projection process

Complex securitized assets require large amounts of memory to project accurately. A relatively simple CMO can require 2 Mb of RAM, and a larger multi-pool CMO can require upwards of 200 Mb. Memory requirements during the projection are reduced by pre-projecting assets linked to Intex. These results are stored in submodel class 'intex' arrays, which are used during the normal projection.

The Intex pre-projection is processed during the initialization step of the existing asset model points. A separate pre-projection is processed for each iteration. Each asset which meet certain criteria (see below) is pre-projected using an 'intex_tgt' column to project the asset through the required projection periods. Cash flows for the owned tranche are retrieved using the Intex cash flow library routines. Results for the cash flows, income statement, and balance sheet amounts are stored in the submodel class 'intex' arrays. The asset values are then reset.

Results from the Intex pre-projection process are not stored on disk. Therefore, every run must perform the Intex pre-projection process, even when nothing has changed between projections.

9.4.2 Assets excluded from pre-projection process

Not all mortgages or securitized assets are pre-projected using Intex. An asset is excluded from the pre-projection process when any of the following occur:

- The asset identification is not found in the Intex CUSIP-to-deal translation table.
- The deal cannot be parsed by the Intex subroutines.
- A deal information (CDI) file cannot be found for the asset.
- A projection date deal update (CDU) file cannot be found for the asset.
- The name (as translated from the CUSIP) cannot be found in the deal.
- The accounting definition for cost basis is not Amortized Cost.
- The accounting definition for book value is not Cost Basis.
- A collateral pool asset has not been specified.
- Both Par and Book value for the asset are zero.

Warning messages are written to the Run log when an asset is excluded from the Intex pre-projection process. Each asset not pre-projected using Intex is projected using the normal asset projection process.

10 SUBMODELS STRUCTURE

10.1 Submodels

The sec_asset model class contains the formulas and variables for securitized asset model point processing.

The sec_cf_asset model class contains the formulas and variables for securitized asset cash flow calculations. The variable values are not entered by the user but are instead set during initialization processing.

The sec_cp_asset model class contains the formulas and variables for securitized asset collateral cash flow calculations. This model class processes the scheduled payments and prepayments from the collateral using much of the same functionality as is used to process cash flows for mortgage assets.

The following describes the model tree:

- **sec** – A data driven array submodel used to determine securitized asset experience results. The number of elements in the array is initially the number of existing securitized asset models. The array size increases as new securitized assets are purchased.
- **sec_cf** – A standalone instance of the **sec_cf_asset** model class, used to determine securitized asset experience cash flows.
- **sec_pv** – A standalone instance of the **sec_cf_asset** model class. This model is a clone of the **sec_cf** model, which is used for the cash flow projections required for market value (pv), market value shifted (pv), and net yield calculations (pv).
- **sec_cp** – As standalone instance of the **sec_cp_asset** model class, used to determine securitized asset collateral cash flows. This submodel is a rebase submodel. Using a rebase submodel allows for this same submodel to be used for the experience collateral cash flows, and be rebased as required to determine projected cash flows for determining collateral prepayment interest savings present values and cash flow maintenance present values, as well as collateral cash flows required for the securitized asset present value calculations.

10.2 Present value calculations

Within the **sec_cf_asset** model class, the primary external functions for present value calculations include:

- **weighted_avg_life_calc** – The weighted average life of future principal payments
- **pv_calc** – The discounted present value of future cash flows. This function handles both discounting at spot rates and a single yield rate. This function also fills arrays of cash flows for use in other present value calculations which use the same cash flows.
- **solve_for_sprd_newton_raphson** – Yield rates or spreads can usually be solved for quickly using the Newton-Raphson method. During each solve iteration, the present value of the cash flow stream is determined and is compared to the target value. When the difference is small, the solving process is terminated and the yield rate or spread is returned.
- **solve_for_sprd_inverse_interp** – When the Newton_Raphson method does not converge quickly enough, this function uses a combination of inverse interpolation and bisection to solve for the yield rate or spread. During each solve iteration, the present value of the cash flow stream is determined and compared to the target value. When the difference is small, the solving process is terminated and the yield rate or spread is returned.

For the **sec_cp_asset** model classe, present value calculations utilize the following formulas:

- **int_savings_pv_calc** – Determines the discounted present value of future scheduled collateral cash flow for use in determining the refinancing financial test. The amount may be determined using either a simple closed form solution or through a projection of future cash flows.
- **yld_maint_cost_pct** – Determines the yield maintenance prepayment cost as a multiple of the outstanding payoff amount. The amount may be determined using either as simple closed form solution or through a projection of future cash flows.

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A04 INTEREST RATE DERIVATIVES

This book discusses interest rate derivative (IRD) assets, and the ird submodel.

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1 INPUT ORGANIZATION

1.1 Economy input page

The Economy input page includes assumptions related to the general economic environment, which is independent of company and asset portfolio assumptions.

1.1.1 *Credit spreads*

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. See the Credit spread section in the Market value chapter, below.

1.1.2 *Interest rate volatilities*

Specify the interest rate volatilities on the Economy input page under the Scenarios: Assumptions category. See the Interest rate volatilities section in the Market value chapter, below.

1.1.3 *Default rates*

Specify IRD default rates on the Economy input page under the Scenarios: Assumptions category. See the Defaults chapter, below.

1.2 Investment input page

The Investment input page includes assumptions related to projected asset portfolio and investment strategies.

1.2.1 *Expenses*

Investment expense rates vary by asset sub portfolio. See the Investment expenses section in the Cash flows chapter, below.

1.2.2 *Purchase IRD definitions*

Specify purchase IRD definitions on the Investment input page under the Purchase Asset Definitions category. See the Interest rate derivative purchases variables chapter, below.

1.3 Model points

Specify on the model point records the definition for each IRD existing at the projection date. See the Model point records chapter, below.

Report groups

Assets typically do not use the report group feature that is used to group liability model points. Instead, asset sub portfolio reports automatically collect the individual assets assigned to each group. This has the advantage that asset sub portfolio assignments can change dynamically as asset durations change through the projection.

2 CASH FLOWS

2.1 Scheduled payments

On each payment date, the interest payment is the interest rate differential, determined as of the beginning of the payment period, then times the notional amount.

2.2 Interest rates definition

2.2.1 Interest rate differential

The interest rate differential is 'rate 1' minus 'rate 2', subject to conditions noted below. The rates can be floating or fixed interest rates.

2.2.2 Floating rate

Specify a floating interest rate as the nominal semi-annual interest rate from a specified category for a specified number of maturity years. The nominal semi-annual rate is adjusted by the IRD rate multiple, then increased by the nominal semi-annual IRD spread. This final result is converted to be nominal for the rate payment frequency.

2.2.3 Floating rate reset timing

Floating interest rates are reset at the beginning of each payment period. For example, a floating interest rate with semi-annual payments resets at the beginning and middle of each year.

The initial interest rate reset for existing contracts is defined by the first payment date (on the model point record). For new IRD assets purchased during the projection, specify the first interest rate reset directly in the Months to first reset variable.

2.2.4 Fixed rate

For existing IRD assets: Specify a fixed interest rate by using a rate multiple of zero, and the spread equal to the fixed rate. The spread is a nominal semi-annual rate.

For IRD assets purchased during the projection: Specify the rate definition as Fixed. The rate determined at the time of purchase remains fixed.

2.3 Swaps

2.3.1 General

The contract is a swap when the payment definition is Always, meaning that the payment can be positive or negative. The payment is (rate 1 – rate 2) times the notional amount.

Swap payments can have a different payment frequency for each rate. When only one rate is currently at a payment date, the other rate is considered to be zero.

- For existing swaps that have payments at a different frequency, the first payment date specified on the model point record refers to the rate that makes more frequent payments.
- For newly purchased swaps that have payments at a different frequency, specify the Months to first reset as the number of months until the more frequent rate is reset. When the more frequent rate is a fixed rate, then this is the number of months until the fixed rate begins to apply.

2.3.2 *Forward starting swaps*

A forward starting swap is defined when the first interest rate reset occurs after the projection date for existing swaps, or after the purchase month for swaps purchased during the projection. The payments are zero prior to the scheduled payment period.

2.4 Caps and floors

The contract is a cap or a floor when the payment occurs only when rate 1 is higher than rate 2. That is, there is a payment only when the interest rate difference is positive. Both rates must have the same payment frequency.

- Cap – the difference is between a floating rate (rate 1) and a fixed rate (rate 2).
- Floor – the difference is between a fixed rate (rate 1) and a floating rate (rate 2).
- Hybrid – both rate 1 and rate 2 are floating rates.

2.5 Investment expenses

Investment expense rates vary by asset sub portfolio. Specify this rate as part of the asset sub portfolio definitions on the Investment input page under the Asset sub portfolios category.

Asset cash flows are reduced by investment expenses. Investment income is reported net of investment expenses. Asset yield rates are net of investment expenses.

Investment expenses occur at the end of each month (prior to processing investment strategies), equal to the book value of each asset times its annual investment expense rate, divided by 12.

3 DEFAULTS

3.1 Deterministic defaults

Defaults are processed at the beginning of each month. Defaults thus reduce the amount of payments that would otherwise occur in that month and in the future. Partial defaults are a proxy for the gradual effect of declining quality grading as a portfolio of IRDs age.

Deterministic default rates vary by asset category, and can also vary by date. Specify deterministic default rates on the Economy input page under the Scenarios: Assumptions category. These definitions must be an external source varying by the hidden variable lookup key `quality_id`; they can also vary by the date column.

The monthly default rate is:

$$1 - (1 - \text{annual default rate})^{1/12}$$

3.2 Stochastic defaults

Default rates can be stochastic when the scenario file includes default rates for scenario yield curves in addition to the risk free yield curve. Typically the scenario file includes default rates for AA and B securities.

The stochastic default rate is an addition to the deterministic default rate.

Specify the stochastic default rate as a multiple of the default rates for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys `yld_curve_id` and `quality_id`.

In the following example, the stochastic default rate for A securities is 80% of the scenario default rate for AA securities plus 20% of the scenario default rate for B securities.

quality_id	A
yld_curve_id	
Govt	0.0000%
AA	80.0000%
B	20.0000%

4 ACCOUNTING VALUES

4.1 Accrued interest

Accrued interest is added to the market value and statutory book value of IRD assets. The accrued interest is the interest payment calculated at the beginning of the payment period, compounded for the fraction of the payment period since the last payment. When a swap with different payment modes is defined, interest is accrued from the last payment date for each rate separately.

4.2 Statutory accounting

The cost basis for IRD assets held by U.S. insurers is typically straight-line amortization, which linearly grades the cost basis to zero at maturity. The statutory book value is typically the cost basis. Other options provide flexibility to use asset accounting methods that apply in various other countries.

4.3 Tax accounting

The statutory book value applies for tax calculations.

4.4 GAAP accounting

GAAP accounting treatment varies according to the sales classification of each asset:

- For GAAP income, trading assets are valued at market value, while other classifications are valued at the statutory cost basis.
- For the GAAP balance sheet, trading and available for sale assets are valued at market value, while other classifications are valued at the statutory cost basis.

5 MARKET VALUE

Different market valuation techniques apply based on the type of contract being valued.

5.1 Swaps

Swaps are valued as a portfolio of forward rate derivatives. The market value is the sum of the expected payments discounted at appropriate spot rates, where expected payments are either known or based on forward rates.

5.1.1 Estimated payments

Estimated payments are determined the same way as scheduled payments, except that when a payment is unknown a forward rate is used in place of the interest rate. Three possibilities are:

- Both payments are known – When rate 1 and rate 2 are known, the payment is calculated normally.

- Both payments are unknown – When both rate 1 and rate 2 are unknown, the payment is estimated by using appropriate forward rates determined for the time to the cash flow.
- One payment is unknown – Both an interest rate and a forward rate are used in estimating the payment. This case only applies to swap contracts where a different payment mode is specified for each rate. For example, suppose rate 1 makes monthly floating interest payments and rate 2 makes annual floating interest payments. The estimated payment uses a forward rate for rate 1, and using the known interest rate for rate 2.

5.1.2 Forward starting swaps

Market values for forward starting swaps are slightly different than for swaps currently making payments. Prior to the scheduled payment period, interest payments are zero. The market value is calculated using forward rates and will be non zero. Estimated payments are calculated as explained above.

5.1.3 Newly purchased swaps

For swaps purchased during the projection, the market value is zero when the swap is created, since the contract represents a hedge where neither party has an advantage. The market value fluctuates at future dates.

The swap is 'priced' at fair value at the time of purchase. The swap is priced by solving for a spread added to a particular rate such that the initial market value is zero. There are three main cases:

- Fixed versus floating swap – The spread (can be positive or negative) is added to the fixed rate.
- Floating versus floating swap and market value is negative – A positive spread is added to rate 1.
- Floating versus floating swap and market value is positive – A positive spread is added to rate 2.

5.2 Caps and floors

Caps are valued as a portfolio of European put options on 1/f-year zero coupon bonds, and floors are valued as a portfolio of European call options on 1/f-zero year coupon bonds, where f is the payment frequency. The market value is the sum of the market values of the European put and call options associated with each of the payments. The Black-Scholes formula is used to evaluate the market values of each of these European put and call options.

For example, a semi-annual cap payment 5.5 years from now is based on an interest rate equivalent to a European put option to sell, 5 years from now, a certain number of half-year zero coupon bonds that yield rate 1, at a strike price of a zero coupon bond that yields the fixed rate, rate 2.

5.3 Positive difference between floating rates

Hybrid contracts (when both rates are floating rates) are valued as the average of a cap and a floor. The cap pays the positive difference between rate 1, less a fixed rate determined by the forward rate underlying rate 2. The floor pays the positive difference between a fixed rate determined by the forward rate underlying rate 1 less the floating rate, rate 2.

5.4 Credit spread

5.4.1 Fixed credit spreads

The credit spread, also called risk spread, addition to the spot rates adjusts for the assumed credit risk associated with the contract.

Specify the credit spread on the Economy input page under the Scenarios: Assumptions category. This must be an external source varying by hidden variable lookup keys quality_id and rate term; and can also vary by

the date column. Credit spreads typically vary by date when spreads that are consistent with current market values are different from those assumed to apply for normal market conditions going forward.

5.4.2 Stochastic credit spreads

The credit spread can be stochastic when the scenario file includes credit yield curves in addition to the risk free yield curve. Typically the scenario file includes default rates for AA and B securities.

The stochastic credit spread is an addition to the deterministic credit spread.

Specify the stochastic credit spread as a multiple of the credit spreads for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys yld_curve_id and quality_id.

In the following example, the stochastic credit spread for A securities is 80% of the spread for AA securities plus 20% of the spread for B securities.

quality_id	A
yld_curve_id	
Govt	0.0000%
AA	80.0000%
B	20.0000%

5.5 Interest rate volatility

5.5.1 Deterministic volatility

Specify the assumed deterministic volatility on the Economy input page under the Scenarios: Assumptions category. This must be an external source varying by hidden variable lookup keys quality_id, moneyness, and rate_term; and can also vary by the date column. Interest rate volatility typically vary by date when volatilities that are consistent with current market values are different from those assumed to apply for normal market conditions going forward.

5.5.2 Stochastic volatility

The interest rate volatility can be stochastic when the scenario file includes credit yield curves in addition to the risk free yield curve. Typically the scenario file includes yield curves for AA and B securities.

The stochastic volatility is an addition to the deterministic volatility.

Specify the stochastic volatility as a multiple of the volatilities for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys yld_curve_id and quality_id.

In the following example, the stochastic volatility for A-rated securities is 80% of the volatility spread for AA-rated securities plus 20% of the volatility spread for B-rated securities.

quality_id	A
yld_curve_id	
Govt	0.0000%
AA	80.0000%
B	20.0000%

5.6 Market spreads

The market spot and volatility spreads are the additional spreads required for the calculated market value to match the specified initial (at the projection date) market value. The spread is not calculated when the initial market value is zero.

When the calculated market value is less than the initial market value for caps and floors, a market volatility spread is calculated. When the calculated market value is higher, a market spot spread is calculated.

For contract types other than caps and floors, a market spot spread is calculated.

The following options are available for using the initial market spread:

- Ignore the market spread by treating it as zero during the projection. Appropriately small market spreads can help validate that the projection assumptions are consistent with the assumptions used to calculate the input market values.
- The market spread is held level at the initial rate. This treatment interprets the market spread as a correction of the projection assumptions.
- The initial market spread is graded linearly to zero by the maturity date. This treatment interprets the market spread as a result of actual market values which fluctuate around analytical market value calculations. The current differential will eventually be corrected by market forces.

6 MODEL POINT RECORDS

The model point record layout for IRD assets is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	up to 10 characters (such as CUSIP)
D	use_mp	Yes, No
E	hdg_grp_code_mp	up to 20 characters, Leave as DEFAULT when the port_defn_mp is Investment When the port_defn_mp is Hedging, must be specified as [hedge group id][greek type][key rate name], where greek type can be 'Rho' or 'RhoConvexity', and the key rate name can be 'Parallel' or one of the key rate names defined in the hedge strategy input.
F	category_id_mp	up to 20 characters (must match an interest rate derivative asset category)
G	port_defn_mp	Investment, Hedging
H	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
I	maturity_date_mp	yyyy/mm/dd or mm/dd/yyyy
J	planned_sale_date_mp	yyyy/mm/dd or mm/dd/yyyy
K	init_notional_amt_mp	notional amount at projection date
L	init_cost_basis_mp	cost basis at projection date
M	init_bk_val_mp	book value at projection date
N	init_mkt_val_mp	market value at projection date**
O	rate_1_pmt_mode_mp	Monthly, Quarterly, Semi-Annual, Annual
P	rate_2_pmt_mode_mp	Monthly, Quarterly, Semi-Annual, Annual
Q	pmt_defn_mp	Always, If Positive (if rate 1 > rate 2)
R	pmt_first_date_mp	yyyy/mm/dd (default = 1900/01/01)
S	rate_1_category_id_mp	category id for rate 1 (up to 20 characters); default is Govt
T	rate_1_scen_yr_mp	rate 1 scenario maturity years
U	rate_1_scen_mult_mp	default = 1.0000
V	rate_1_addn_mp	addition = 0.00 to 1.00 (100%)
W	rate_2_category_id_mp	category id for rate 2 (up to 20 characters); default is Govt

Excel Column	Variable	Value
X	rate_2_scen_yr_mp	rate 2 scenario maturity years
Y	rate_2_scen_mult_mp	default = 1.0000
Z	rate_2_addn_mp	addition = 0.00 (0%) to 1.00 (100%)
AA	avr_contribn_pct_mp	0.00 to 1.00 (100%)
AB	avr_max_pct_mp	0.00 to 1.00 (100%)
AC	avr_obj_pct_mp	0.00 to 1.00 (100%)
AD	rbc_c1_pct_mp	0.00 to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

**The initial market value is used to calculate the market spread, which is discussed above in the Market Values section. The market spread is automatically zero if this field is left at the default value of 0.00.

7 INTEREST RATE DERIVATIVE PURCHASES

7.1 Payment rates

Each IRD asset purchased during the model projection is created consistently with the scenario rates that apply at that time. The initial values for rate 1 and rate 2 are calculated when the asset is purchased. When the payment definition is Fixed, the rate continues unchanged. When the payment definition is Float, the rate is recalculated on each payment date, based on the scenario rates in effect at that time.

7.2 Purchase IRD variables

Specify purchase IRDs on the Investment input page, under the Purchase Asset Definitions category. These are variables in the inv_strategy submodel.

The variables must be linked to an external source defining each purchase IRD using the hidden variable lookup key purch_asset_id.

Variable	Value
purch_ird_grp_id	model point report group*
purch_ird_category_id	up to 20 characters (must match an interest rate derivative asset category)
purch_ird_port_defn	Investment, Hedging
purch_ird_sale_class	Available for Sale, Trading, Held to Maturity, Not Assigned
purch_ird_pmt_defn	Always, If Positive (if rate 1 > rate 2)
purch_ird_rate_1_pmt_mode	Monthly, Quarterly, Semi-Annual, Annual
purch_ird_rate_2_pmt_mode	Monthly, Quarterly, Semi-Annual, Annual
purch_ird_mths_to_first_reset	0.25 or greater
purch_ird_yrs_to_maturity	0.25 or greater
purch_ird_rate_1_defn	Fixed Rate (at purchase), Floating Rate (calculated at each payment date)
purch_ird_rate_1_category_id	category id for rate 1 (up to 20 characters); default is Govt
purch_ird_rate_1_scen_yr	rate 1 scenario maturity years
purch_ird_rate_1_scen_mult	default = 1.0000
purch_ird_rate_1_addn	addition = 0.00 to 1.00 (100%)
purch_ird_rate_2_defn	Fixed Rate (at purchase), Floating Rate (calculated at each payment date)
purch_ird_rate_2_category_id	category id for rate 2 (up to 20 characters); default is Govt

Variable	Value
purch_ird_rate_2_scen_yr	rate 2 scenario maturity years
purch_ird_rate_2_scen_mult	default = 1.0000
purch_ird_rate_2_addn	addition = 0.00 to 1.00 (100%)
purch_ird_avr_contribn_pct	0.00 to 1.00 (100%)
purch_ird_avr_max_pct	0.00 to 1.00 (100%)
purch_ird_avr_obj_pct	0.00 to 1.00 (100%)
purch_ird_rbc_c1_pct	0.00 to 1.00 (100%)

*The report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

8 SUBMODELS STRUCTURE

8.1 Submodels

The ird_asset model class contains the formulas and variables for IRD model point processing. The following describes the model tree:

- ird – a data driven array submodel used to determine IRD experience results. The number of elements in the array is initially the number of existing IRD model points. The array size increases during the projection as new IRDs are purchased.

8.2 Present value calculations

The primary functions for present value calculations include:

- mkt_val_calc – calculates the market value of an index option using variations of the Black-Scholes option pricing formula
- spot_sprd_adj – calculates a market spread to the spot rate curve. This function is not called when a volatility spread is calculated.
- volatility_sprd_adj – calculates a market spread to the volatility curve. This function is not called when a spot spread is calculated.

A05 REAL ESTATE

This book discusses real estate (property) assets and the re submodel.

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1 INPUT ORGANIZATION

1.1 Scenario index

Real estate assets are projected based on a real estate index contained in the scenario file. Each real estate asset data record specifies the asset index which the real estate asset is linked to.

1.2 Economy input page

The Economy input page includes assumptions related to the general economic environment, which is independent of company and asset portfolio assumptions.

1.2.1 Default rates

Specify real estate default rates on the Economy input page under the Scenarios: Assumptions category. See the Defaults chapter, below.

1.3 Investment input page

The Investment input page includes assumptions related to projected asset portfolio and investment strategies.

1.3.1 Expenses

Investment expense rates vary by asset sub portfolio. See the Investment expenses section in the Cash flows chapter, below.

1.3.2 Purchase real estate definitions

Specify purchase real estate definitions on the Investment input page under the Purchase Asset Definitions category. See the Real estate purchases chapter, below.

1.4 Model points

Specify on the model point records the definition for each bond existing at the projection date. See the Model point records chapter, below.

Report groups

Assets typically do not use the report group feature that is used to group liability model points. Instead, asset sub portfolio reports automatically collect the individual assets assigned to each group. This has the advantage that asset sub portfolio assignments can change dynamically as asset durations change through the projection.

However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

2 CASH FLOWS

2.1 Rent payments

On each payment date, the rent payment is based on an initial rental rate:

- For existing assets, the initial rental rate is the annualized rent (on the equity model point record), divided by the initial market value in the existing equity record.
- For real estate purchased by the investment strategy, the initial rental rate is determined at the time of purchase.

The rental income can vary over time by applying rental income growth rate to the rental income base amount. The current rental income growth rate is the scenario growth rate for the index specified on the model point record, times the market value growth scenario multiple, then plus an addition.

2.2 Investment expenses

Investment expense rates vary by asset sub portfolio and are determined by asset sub portfolio. Specify this rate as part of the asset sub portfolio definitions on the Investment input page under the Asset sub portfolios category.

Asset cash flows are reduced by investment expenses. Investment income is reported net of investment expenses, and asset yield rates are net of investment expenses.

Investment expenses occur at the end of each month (prior to processing investment strategies), equal to the book value of each asset times its annual investment expense rate, divided by 12.

3 DEFAULTS

3.1 Deterministic defaults

Defaults are processed at the beginning of each month. Defaults thus reduce the amount of payments that would otherwise occur in that month and in the future. Partial defaults are a proxy for the gradual effect of declining quality grading as a portfolio of real estate assets age.

Deterministic default rates vary by asset category, and can also vary by date. Specify deterministic default rates on the Economy input page under the Scenarios: Assumptions category. These definitions must be an external source varying by the hidden variable lookup key `quality_id`; they can also vary by the date column.

The monthly default rate is:

$$1 - (1 - \text{annual default rate})^{1/12}$$

3.2 Stochastic defaults

Default rates can be stochastic when the scenario file includes default rates for scenario yield curves in addition to the risk free yield curve. Typically the scenario file includes default rates for AA and B securities.

The stochastic default rate is an addition to the deterministic default rate.

Specify the stochastic default rate as a multiple of the default rates for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys `yld_curve_id` and `quality_id`.

In the following example, the stochastic default rate for A-rated securities is 80% of the scenario default rate for AA-rated securities plus 20% of the scenario default rate for B-rated securities.

quality_id		A
yld_curve_id		
Govt		0.0000%
AA		80.0000%
B		20.0000%

4 ACCOUNTING VALUES

4.1 Accrued rent

Accrued rent is added to the market value and statutory book value of real estate assets. The accrued rent is the next projected rental income payment, times the fraction of the payment period since the last payment.

4.2 Statutory accounting

The cost basis for real estate is typically straight-line depreciation. The statutory book value for real estate held by U.S. insurers is typically the cost basis. Other options provide flexibility to use asset accounting methods that apply in various other countries.

4.3 Tax accounting

The statutory book value applies for tax calculations.

4.4 GAAP accounting

GAAP accounting treatment varies according to the sales classification of each asset:

- For GAAP income, trading assets are valued at market value, while other classifications are valued at the statutory cost basis.
- For the GAAP balance sheet, trading and available for sale assets are valued at market value, while other classifications are valued at the statutory cost basis.

5 MARKET VALUE

Real estate market values change according to the scenario market value index rate. The change in market value from the previous month, times a multiple and then plus an addition, applies to the previous market value.

The initial market value must be specified on the model point record. For real estate purchased by the investment strategy, the initial market value is the book value at purchase.

6 MODEL POINT RECORDS

The model point record layout for real estate assets is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	up to 10 characters
D	use_mp	Yes, No
E	asset_index_mp	(must match a scenario equity index)
F	category_id_mp	up to 20 characters (must match a real estate asset category)
G	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
H	next_pmt_date_mp	yyyy/mm/dd (default = 1900/01/01)
I	planned_sale_date_mp	yyyy/mm/dd (default = 9999/01/01)
J	init_cost_basis_mp	cost basis at projection date
K	init_bk_val_mp	book value at projection date

Excel Column	Variable	Value
L	init_mkt_val_mp	market value at projection date
M	pmt_mode_mp	Monthly, Quarterly, Semi-Annual, Annual
N	mkt_growth_scen_mult_mp	default = 1.0000
O	mkt_growth_addn_mp	addition = 0.00 (0%) to 1.00 (100%)
P	depreciation_yrs_mp	1 ... 999 (required when the cost basis is Straight-Line Depreciation)
Q	init_rent_annualzd_mp	default = 0.00**
R	rental_growth_mp	Yes (rental income is a constant percent of market value), No (rental income is adjusted only for partial defaults and sales)
S	avr_contribn_pct_mp	0.00 to 1.00 (100%)
T	avr_max_pct_mp	0.00 to 1.00 (100%)
U	avr_obj_pct_mp	0.00 to 1.00 (100%)
V	rbc_c1_pct_mp	0.00 to 1.00 (100%)

*The report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

**The rental income rate is the initial annualized rent divided by the initial market value.

7 REAL ESTATE PURCHASES

Specify purchase real estate on the Investment input page, under the Purchase Asset Definitions category. These are variables in the inv_strategy submodel.

The variables must be linked to an external source defining each purchase real estate using the hidden variable lookup key purch_asset_id.

Variable	Value
purch_re_grp_id	model point report group*
purch_re_asset_index	(must match a scenario real estate index)
purch_re_category_id	up to 20 characters (must match a real estate asset category)
purch_re_sale_class	Available for Sale, Trading, Held to Maturity, Not Assigned
purch_re_depreciation_yrs	0 ... 999 (required when the cost basis is Straight-Line Depreciation)
purch_re_pmt_mode	Monthly, Quarterly, Semi-Annual, Annual
purch_re_rent_rate_scen_mult	default = 1.0000
purch_re_rent_rate_scen_addn	addition = 0.00 to 1.00 (100%)
purch_re_mkt_growth_scen_mult	default = 1.0000
purch_re_mkt_growth_addn	addition = 0.00 to 1.00 (100%)
purch_re_avr_contribn_pct	0.00 to 1.00 (100%)
purch_re_avr_max_pct	0.00 to 1.00 (100%)
purch_re_avr_obj_pct	0.00 to 1.00 (100%)
purch_re_rbc_c1_pct	0.00 to 1.00 (100%)

* The report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

8 SUBMODELS STRUCTURE

The re_asset model class contains the formulas and variables for real estate (property) model point processing. The following describes the model tree:

- re – A data driven array submodel used to determine real estate experience results. The number of elements in the array is initially the number of existing real estate model points. The array size increases as real estate assets are purchased.

A06 EQUITIES

This book discusses equity assets and the eqt submodel.

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1 INPUT ORGANIZATION

1.1 Scenario index

Equities are projected based on an equity index contained in the scenario file. Each equity asset data record specifies the asset index which the stock holding is linked to. The index name specified for the equity must exactly match an index name contained in the scenario file.

1.2 Investment input page

The Investment input page includes assumptions related to projected asset portfolio and investment strategies.

1.2.1 Expenses

Investment expense rates vary by asset sub portfolio. See the Investment expenses section in the Cash flows chapter, below.

1.2.2 Purchase equity definitions

Specify purchase equity definitions on the Investment input page under the Purchase Asset Definitions category. See the Equity purchases chapter, below.

1.3 Model points

Specify on the model point records the definition for each bond existing at the projection date. See the Model point records chapter, below.

Report groups

Assets typically do not use the report group feature that is used to group liability model points. Instead, asset sub portfolio reports automatically collect the individual assets assigned to each group. This has the advantage that asset sub portfolio assignments can change dynamically as asset durations change through the projection.

However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

2 CASH FLOWS

2.1 Dividend payments

Dividends are paid monthly. The dividend payment is the market value at the end of the month, times the dividend yield rate, divided by 12.

The dividend yield rate is based on the scenario income rate index contained in the scenario file, determined as a nominal monthly rate. Each equity model point record gives the name of the equity scenario index to which this asset's performance is linked.

- For existing equities, the dividend yield rate maintains the relationship between the initial dividend yield (on the model point record) and the initial dividend yield that is based on initial dividend yield in the scenario rates file. The current dividend yield is the product of (1) an initial dividend ratio and (2) the current scenario dividend yield times a multiple and plus an addition. The initial dividend ratio is the ratio of (1) the input dividend yield to (2) the initial scenario dividend yield times a multiple and plus an addition (the same multiple and addition that is used to calculate the current dividend yield). To specify a fixed dividend yield, set both the dividend yield scenario multiple and addition to zero so that the initial dividend yield applies throughout the projection.

- For equities purchased by the investment strategy, the dividend yield rate is the scenario income return times a multiple and then plus an addition.

2.2 Investment expenses

Investment expense rates vary by asset sub portfolio and are determined by asset sub portfolio. Specify this rate as part of the asset sub portfolio definitions on the Investment input page under the Asset sub portfolios category.

Asset cash flows are reduced by investment expenses. Investment income is reported net of investment expenses, and asset yield rates are net of investment expenses.

Investment expenses occur at the end of each month (prior to processing investment strategies), equal to the book value of each asset times its annual investment expense rate, divided by 12.

3 ACCOUNTING VALUES

3.1 Accrued dividends

Since dividends are paid monthly, there are no accrued dividends.

3.2 Statutory accounting

The cost basis for equities is actual cost, which is the purchase price of the equity. The statutory book value for equities in U.S. projections is typically market value. Other options provide flexibility to use asset accounting methods that apply in various other countries.

3.3 Tax accounting

The statutory book value applies for tax calculations.

3.4 GAAP accounting

GAAP accounting treatment varies according to the sales classification of each asset:

- For GAAP income, trading assets are valued at market value, while other classifications are valued at the statutory cost basis.
- For the GAAP balance sheet, trading and available for sale assets are valued at market value, while other classifications are valued at the statutory cost basis.

4 MARKET VALUE

Equity market values change according to the scenario market value index rate. The change in market value from the previous month, times a multiple and then plus an addition, applies to the previous market value.

The initial market value must be specified on the model point record. For equities purchased by the investment strategy, the initial market value is the book value at purchase.

5 MODEL POINT RECORDS

The model point record layout for equity assets is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	up to 10 characters

Excel Column	Variable	Value
D	use_mp	Yes, No
E	asset_index_mp	(must match a scenario equity index)
F	category_id_mp	up to 20 characters (must match an equity asset category)
G	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
H	planned_sale_date_mp	yyyy/mm (default = 9999/12)
I	init_cost_basis_mp	cost basis at projection date
J	init_bk_val_mp	book value at projection date
K	init_mkt_val_mp	market value at projection date
L	mk_growth_scen_mult_mp	default = 1.0000
M	mk_growth_addn_mp	addition = 0.00 to 1.00 (100%)
N	init_div_yld_mp	0.00 (0%) to 1.00 (100%)
O	div_yld_scen_mult_mp	default = 1.0000
P	div_yld_sprd_mp	addition = 0.00 to 1.00 (100%)
Q	avr_contribn_pct_mp	0.00 to 1.00 (100%)
R	avr_max_pct_mp	0.00 to 1.00 (100%)
S	avr_obj_pct_mp	0.00 to 1.00 (100%)
T	rbc_c1_pct_mp	0.00 to 1.00 (100%)

* The report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

6 EQUITY PURCHASES

Specify purchase equity definition on the Investment input page, under the Purchase Asset Definitions category. These are variables in the inv_strategy submodel.

The variables must be linked to an external source defining each purchase equity using the hidden variable lookup key purch_asset_id.

Variable	Value
purch_eqt_grp_id	model point report group*
purch_eqt_asset_index	(must match a scenario equity index)
purch_eqt_category_id	up to 20 characters (must match an equity asset category)
purch_eqt_sale_class	Available for Sale, Trading, Held to Maturity, Not Assigned
purch_eqt_pmt_mode	Monthly
purch_eqt_div_yld_scen_mult	default = 1.0000
purch_eqt_div_yld_addn	addition = 0.00 to 1.00 (100%)
purch_eqt_mkt_growth_scen_mult	default = 1.0000
purch_eqt_mkt_growth_addn	addition = 0.00 to 1.00 (100%)
purch_eqt_avr_contribn_pct	0.00 to 1.00 (100%)
purch_eqt_avr_max_pct	0.00 to 1.00 (100%)
purch_eqt_avr_obj_pct	0.00 to 1.00 (100%)
purch_eqt_rbc_c1_pct	0.00 to 1.00 (100%)

* The report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

7 SUBMODELS STRUCTURE

The eqt_asset model class contains the formulas and variables for equity model point processing. The following describes the model tree:

- eqt – a data driven array submodel used to determine equity experience results. The number of elements in the array is initially the number of existing equity model points. The array size increases as new equities are purchased.

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A07 EQUITY INDEX OPTIONS

This book discusses equity index option (EIO) assets and the eio submodel.

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1 INPUT ORGANIZATION

1.1 Scenario index

Equity index options are projected based on an equity index contained in the scenario file. Each equity index option asset data record specifies the asset index which the option is linked to.

1.2 Investment input page

The Investment input page includes assumptions related to projected asset portfolio and investment strategies.

1.2.1 Expenses

Investment expense rates vary by asset sub portfolio. See the Investment expenses section in the Cash flows chapter, below.

1.2.2 Purchase equity index option definitions

Specify purchase equity index option definitions on the Investment input page under the Purchase Asset Definitions category. See the Equity index option purchases chapter, below.

1.3 Model point records

Variables located on data records vary between each individual asset.

The data record for each equity existing as of the projection date is included in the Existing Model Points data records. For example, that data record contains the notional amount, maturity date, and option type definitions for each individual equity index option.

2 CASH FLOWS

2.1 Index option payments

2.1.1 Call or Put options

At expiry, an index call option pays an amount that is based on the positive difference between an index value at expiry and a specified strike index value. An index put option pays an amount that is based on the positive difference between a specified strike index value and an index value at expiry. The index value can be the index value at expiry, a moving average index value, or the highest attained index value where index values are sampled at discrete points in time. The index value growth over time is based on the scenario growth rates for the scenario index specified on the model point record, times a multiple then plus a spread.

2.1.2 Futures contracts

At expiry, an index futures contract pays an amount equal to the difference between the index value at expiry and a specified strike index value. The index value is always the index value at expiry. The index value growth over time is based on the scenario growth rates for the scenario index specified on the model point record, times a multiple then plus a spread.

2.2 Investment expenses

Investment expense rates vary by asset sub portfolio and are determined by asset sub portfolio. Specify this rate as part of the asset sub portfolio definitions on the Investment input page under the Asset sub portfolios category.

Asset cash flows are reduced by investment expenses. Investment income is reported net of investment expenses, and asset yield rates are net of investment expenses.

Investment expenses occur at the end of each month (prior to processing investment strategies), equal to the book value of each asset times its annual investment expense rate, divided by 12.

3 ACCOUNTING VALUES

3.1 Statutory accounting

The cost basis for equity options held by U.S. insurers is typically market value. The statutory book value is typically the cost basis. Other options provide flexibility to use asset accounting methods that apply in various other countries.

3.2 Tax accounting

The statutory book value applies for tax calculations.

3.3 GAAP accounting

GAAP accounting treatment varies according to the sales classification of each asset:

- For GAAP income, trading assets are valued at market value, while other classifications are valued at the statutory cost basis.
- For the GAAP balance sheet, trading and available for sale assets are valued at market value, while other classifications are valued at the statutory cost basis.

4 MARKET VALUE

Different market valuation techniques apply based on the type of index option being valued:

- A vanilla call index option pays the positive difference between the index value at expiry less a strike price.
- An Asian call index option pays the positive difference between a moving average index value at expiry (such as a 24-month moving average) less a strike price.
- A highest gain call index option pays the positive difference between the maximum index value at discrete points in time at expiry (such as the maximum index value at the last five expiry anniversary dates) less a strike price.
- A futures contract pays the difference between the index value at expiry and a strike price.

4.1 Vanilla options

Vanilla index options are valued using the Black-Scholes formula.

4.2 Asian options

Asian index options are valued using a moment determination approach. The approach is a modification by Boyle of an approximation that appears in Levy's paper 'Pricing European Average Rate Currency Options', Journal of International Money and Finance, 11:474-92 (1992).

The moment method is based on the following assumptions:

- The sum of lognormally distributed variates is approximately lognormal.
- Simple expressions can be derived for the first two moments of the distribution of the arithmetic mean.
- A lognormally distributed random variable can be created which also has the same first two moments of the arithmetic mean.
- A Black-Scholes type formula values these options.

4.3 Highest gain option

Highest gain index options are valued using a closed form method developed by Boyle. A closed form solution exists for the continuously monitored case. Discrete monitored cases can be valued by making approximations from the continuously monitored case. A second order approximation is used, which usually but not always overestimates the true value.

4.4 Futures contracts

Futures contracts are valued as the index units owned times the future value of the index at expiry determined using the difference between the risk free spot rate for the term to expiry less the dividend yield.

4.5 Market spread

The market spot and volatility spreads are the additional spreads required for the calculated market value to match the input market value. If the calculated market value is less than the input market value and there is no maximum payout index value, a market volatility spread is calculated. Otherwise a market spot spread is calculated. This spread is not calculated when the input market value is zero.

These market spreads should be small for individual index options. Large market spreads are usually indicative of incorrect asset category assumptions.

The following options are available for using the initial market spread:

- Ignore the market spread by treating it as zero during the projection. Appropriately small market spreads can help validate that the projection assumptions are consistent with the assumptions used to calculate the input market values.
- The market spread is held level at the initial rate. This treatment interprets the market spread as a correction of the projection assumptions.
- The initial market spread is graded linearly to zero by the maturity date. This treatment interprets the market spread as a result of actual market values which fluctuate around analytical market value calculations. The current differential will eventually be corrected by market forces.

5 MODEL POINT RECORDS

The model point record layout for equity index option assets is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id_mp	segment ID
C	asset_id_mp	up to 10 characters
D	use_mp	Yes, No
E	hdg_grp_code_mp	up to 20 characters Leave as DEFAULT when the port_defn_mp is Investment When the port_defn_mp is Hedging and is used for FIA index credit hedging, must match the hedge group id When the port_defn_mp is Hedging and is used for IUL index crediting hedging, must be specified as [hedge_grp_id][Bucket_xx] where xx is the bucket number this option is linked to When the port_defn_mp is Hedging and is used for VA greek hedging, must be specified as [hedge group id][greek type][index name][index term name], where greek type can be 'Delta', 'Gamma' or 'Vega', the index name can be 'Parallel' or one of the index names defined in the hedge strategy input, and for Vega hedges, the index term can be 'AllTerms' or one of the index terms defined in the hedge strategy input.
F	asset_index_mp	(must match a scenario equity index)
G	category_id_mp	up to 20 characters (must match an equity index option asset category code)
H	port_defn_mp	Investment, Hedging
I	sale_class_mp	Available for Sale, Trading, Held to Maturity, Not Assigned
J	maturity_date_mp	yyyy/mm/dd (default = 1900/01/01)
K	planned_sale_date_mp	yyyy/mm/dd (default = 9999/01/01)
L	init_cost_basis_mp	cost basis at projection date
M	init_bk_val_mp	book value at projection date
N	init_mkt_val_mp	market value at projection date
O	index_units_owned_mp	default = 0.00000
P	init_index_val_mp	default = 0.00
Q	opt_strike_price_mp	default = 0.00
R	index_val_at_issue_mp	default = 0.00
S	payout_index_max_mp	default = 0.00
T	opt_defn_mp	Call (right to purchase), Put (right to sell), Futures
U	mkt_growth_scen_mult_mp	default = 1.0000
V	mkt_growth_addn_mp	addition = 0.00 (0%) to 1.00 (100%)
W	div_yld_scen_mult_mp	default = 1.0000
X	div_yld_sprd_mp	addition = 0.00 (0%) to 1.00 (100%)
Y	lookback_defn_mp	None (final index value is price at maturity), Asian (average of the sampling date values), Highest Gain (greatest of the sampling date values)
Z	lookback_sampling_interval_mp	Annual, Semiannual, Quarterly, Monthly
AA	lookback_yrs_mp	1 ... 999
AB	avr_contribn_pct_mp	0.00 (0%) to 1.00 (100%)
AC	avr_max_pct_mp	0.00 (0%) to 1.00 (100%)

Excel Column	Variable	Value
AD	avr_obj_pct_mp	0.00 (0%) to 1.00 (100%)
AE	rbc_c1_pct_mp	0.00 (0%) to 1.00 (100%)

* The report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

6 EQUITY INDEX OPTION PURCHASES

6.1 Liability hedging

Equity index options should only be purchased by liability hedging strategies, such as those required to hedge fixed indexed annuities, where the amount of each purchase is based on a predetermined notional amount. Investment strategies purchase asset amounts so that the initial market value is the amount to invest; since equity index options are highly leveraged, such purchase may produce unexpected results.

6.2 Purchase EIO variables

Specify purchase equity on the Investment input page, under the Purchase Asset Definitions category. These are variables in the inv_strategy submodel.

The variables must be linked to an external source using the hidden variable lookup key purch_asset_id.

Variable	Value
purch_eio_grp_id	model point report group*
purch_eio_asset_index	(must match a scenario equity index)
purch_eio_category_id	up to 20 characters (must match an equity index option asset category)
purch_eio_port_defn	Investment, Hedging
purch_eio_sale_class	Available for Sale, Trading, Held to Maturity, Not Assigned
purch_eio_opt_defn	Call, Put, Futures
purch_eio_opt_strike_price	default = 1.00000
purch_eio_payout_index_max	default = 1.0000
purch_eio_yrs_to_maturity	1 ...
purch_eio_index_scen_mult	default = 1.0000
purch_eio_index_scen_addn	addition = 0.00 to 1.00 (100%)
purch_eio_lookback_defn	None (final index value is price at maturity), Asian (average of the sampling date values), Highest Gain (greatest of the sampling date values)
purch_eio_lookback_sampling_interval	Annual, Semiannual, Quarterly, Monthly
purch_eio_lookback_yrs	1 ... 999
purch_eio_div_yld_scen_mult	default = 1.0000
purch_eio_div_yld_addn	addition = 0.00 to 1.00 (100%)
purch_eio_avr_contribn_pct	0.00 to 1.00 (100%)
purch_eio_avr_max_pct	0.00 to 1.00 (100%)
purch_eio_avr_obj_pct	0.00 to 1.00 (100%)
purch_eio_rbc_c1_pct	0.00 to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

7 SUBMODELS STRUCTURE

7.1 Submodels

The `eio_asset` model class contains the formulas and variables for index option model point processing. The following describes the model tree:

- `eio` – a data driven array submodel used to determine index option experience results. The number of elements in the array is initially the number of equity index option model points. The array size increases as new equity index options are purchased.

7.2 Present value calculations

The following are external functions for calculating present values:

- `mkt_val_calc` – calculates the market value of an index option using variations of the Black-Scholes option pricing formula.
- `spot_sprd_adj` – calculates a market spread to the spot rate curve. This function is not called when a volatility spread is calculated.
- `volatility_sprd_adj` – calculates a market spread to the volatility curve. This function is not called when a spot spread is calculated.

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A08 EXTERNALLY PROJECTED ASSETS

This book discusses externally projected assets (EPA) and the epa submodel. It also includes the file names and format for external vendor systems providing externally projected assets for use in asset/liability projections.

Externally projected assets can be used only for bonds, mortgages, securitized assets, interest rate derivatives, and equities.

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1 INPUT ORGANIZATION

1.1 Economy input page

The Economy input page includes assumptions related to the general economic environment, which is independent of company and asset portfolio assumptions.

1.1.1 Default rates

Specify default rates for the EPA assets on the Economy input page under the Scenarios: Assumptions category. See the Defaults chapter, below.

1.2 Model points

Specify on the model point records the definition for each bond existing at the projection date. See the Model point records chapter, below.

Report groups

Assets typically do not use the report group feature that is used to group liability model points. Instead, asset sub portfolio reports automatically collect the individual assets assigned to each group. This has the advantage that asset sub portfolio assignments can change dynamically as asset durations change through the projection.

However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

1.3 EPA files

Projection results required for the projection of externally projected assets are contained in EPA files. The EPA file may contain the results from a single asset or from a portfolio of assets. When the EPA file contains the results from a portfolio of assets, the EPA file is still treated as a single asset during the projection for processing of defaults, investment expenses, and sales. Thus, care should be used when creating EPA files and then in how sales strategies are specified in projections which use EPA files.

The number of items required is limited to a key set of cash flow, income statement, and balance sheet items. These amounts are assumed to be projected on a 'held to maturity' basis. That is, no impacts from purchases or sales should be included in the results of an EPA file.

Additional calculation results required for the projection are determined from those key items and other projection assumptions, such as default and investment expense assumptions.

1.4 Existing assets only

Externally projected assets are used only for existing assets. There is no capability to purchase an EPA.

2 CASH FLOWS

2.1 Payment timing

All externally projected payments are assumed occur at the end of the month.

2.2 Interest payments

Interest payments for each projection month must be specified in the EPA file. The notional amount (par value) at the end of each projection month is also specified.

2.3 Scheduled principal payments

Scheduled principal payments for each projection month must be specified in the EPA file.

2.4 Option payments

Principal payments resulting from calls, puts, and prepayments for each projection month must be specified in the EPA file. The amount of accrued interest repaid from a call, put, or prepayment must also be specified.

2.5 Investment expenses

Investment expenses are not included in the EPA file. Instead, investment expense rates vary by asset sub portfolio and are determined by asset sub portfolio. Specify this rate as part of the asset sub portfolio definitions on the Investment input page under the Asset sub portfolios category.

Asset cash flows are reduced by investment expenses. Investment income is reported net of investment expenses, and asset yield rates are net of investment expenses.

Investment expenses occur at the end of each month (prior to processing investment strategies), equal to the book value of each asset times its annual investment expense rate, divided by 12.

3 DEFAULTS

3.1 Deterministic defaults

Defaults are not included in the EPA file. Defaults are processed at the beginning of each month. Defaults thus reduce the amount of payments that would otherwise occur in that month and in the future. Partial defaults are a proxy for the gradual effect of declining quality grading as a portfolio of assets included in the EPA file age.

Deterministic default rates vary by asset category, and can also vary by date. Specify deterministic default rates on the Economy input page under the Scenarios: Assumptions category. These definitions must be an external source varying by the hidden variable lookup key `quality_id`; they can also vary by the date column.

The monthly default rate is:

$$1 - (1 - \text{annual default rate})^{1/12}$$

3.2 Stochastic defaults

Default rates can be stochastic when the scenario file includes default rates for scenario yield curves in addition to the risk free yield curve. Typically the scenario file includes default rates for AA and B securities.

The stochastic default rate is an addition to the deterministic default rate.

Specify the stochastic default rate as a multiple of the default rates for each of the yield curves in the scenario file. This must be an external source varying by hidden variable lookup keys `yld_curve_id` and `quality_id`.

In the following example, the stochastic default rate for A-rated securities is 80% of the scenario default rate for AA-rated securities plus 20% of the scenario default rate for B-rated securities.

quality_id		A
yld_curve_id		
Govt		0.0000%
AA		80.0000%
B		20.0000%

4 ACCOUNTING VALUES

4.1 Accrued interest

Accrued interest at the end of each projection month is specified in the EPA file.

4.2 Statutory accounting

EPA files are accounted for using U.S. accounting methods for fixed income assets.

The cost basis and book value at the projection date and end of each projection month must be specified in the EPA file. Realized capital gains resulting from option payments for a projection month should also be specified in the EPA file.

4.3 Tax accounting

The statutory book value applies for tax calculations.

4.3.1 *Municipal bonds*

When the EPA includes municipal bonds, specify the amount of tax exempt income in the EPA file.

The treatment for excluding the cost basis growth assumes the municipal bond was purchased with an original issue discount from the primary market.

Important Note: In order to include the tax exempt income in EPA processing, the EPA schedule format must be used.

4.4 GAAP accounting

GAAP accounting treatment varies according to the sales classification of each externally projected asset:

- For GAAP income, trading assets are valued at market value, while other classifications are valued at the statutory cost basis.
- For the GAAP balance sheet, trading and available for sale assets are valued at market value, while other classifications are valued at the statutory cost basis.

5 MARKET VALUE

The market value at the projection date and the end of each projection month must be specified in the EPA file.

The years to maturity, weighted average life, and effective duration at the projection date and end of each projection months should also be specified so that the externally projected asset is assigned to the appropriate asset sub portfolio. See the Asset sub portfolios chapter of the Asset processing book.

6 MODEL POINT RECORDS

The model point record layout for externally projected assets is described in the following table. The 'Excel column' refers to the column ID when the related comma-separated values file is opened in Excel.

The following shows the order of data fields for externally projected assets:

Excel Column	Variable	Value
A	group	model point report group*
B	seg_id	segment ID
C	asset_id	up to 10 characters
D	use_mp	Yes, No
E	epa_file_defn	Schedule File, TAS EPA File
F	file_name	EPA file path and name, up to 300 characters
G	asset_defn	Bond, Mortgage, Securitized Asset, Interest Rate Derivative, Equity
H	category_id	up to 20 characters
I	sale_class	Available for Sale, Trading, Held to Maturity, Not Assigned
J	planned_sale_date	yyyy/mm (default = 9999/12)
K	pct_to_use_mp	percent included, such as 1.00 (100%)
L	avr_contribn_pct	0.00 to 1.00 (100%)
M	avr_max_pct	0.00 to 1.00 (100%)
N	avr_obj_pct	0.00 to 1.00 (100%)
O	rbc_c1_pct	0.00 to 1.00 (100%)

*The model point report group feature is typically not used for assets since the asset sub portfolio reporting feature provides dynamic groups as asset duration changes through the projection. However, report groups can be used to review calculation details not included in the asset sub portfolio reports.

7 EPA FILE FORMATS

7.1 EPA schedule format

When the EPA files use the schedule structure, the EPA file names may be any valid file name, likely with a prefix of 'EPA' and an extension of '.csv'. A single EPA file should contain results for all iterations to be projected. Fields in the EPA file are comma delimited within records. Fields which are not used may be excluded.

7.1.1 Record format

Each EPA field contains a header record followed by blocks of monthly data organized within scenario loops, with records for the projection date and each projection month. The record layout of the data in an EPA file is presented below.

An (X) in the asset type column indicates the field is applicable for an EPA file of that asset type.

Bnd = Bond, Mtg= Mortgage, Sec = Securitized Asset, and Ird = Interest Rate Derivative

Record	Field	Description/Value	Bnd	Mtg	Sec	Ird	Eq
1		File Header Information					
	1	sched_val	X	X	X	X	X
	2	Period	X	X	X	X	X
	3	int_pmt	X	X	X	X	X
	4	sched_prin_pmt	X	X	X		
	5	opt_pmt	X	X			

Record	Field	Description/Value	Bnd	Mtg	Sec	Ird	Eq
	6	notional_amt	X	X	X	X	X
	7	cost_basis	X	X	X	X	X
	8	bk_val	X	X	X	X	X
	9	mkt_val	X	X	X	X	X
	10	accr_int	X	X	X	X	
	11	opt_inc	X	X			
	12	realzd_cap_gain_opt_pmt	X	X			
	13	tax_exempt_inc	X				
	14	yrs_to_maturity	X	X			
	15	weighted_avg_life	X	X	X		
	16	eff_dur	X	X	X	X	
2+		Projection task Loop/Projection Data					
	1	Schedule value (1 to ...)	X	X	X	X	X
	2	Projection period (0 to ...)	X	X	X	X	X
	3	Interest payment	X	X	X	X	X
	4	Scheduled principal payment	X	X	X		
	5	Option payment	X	X			
	6	Notional amount	X	X	X	X	X
	7	Cost basis	X	X	X	X	X
	8	Book value	X	X	X	X	X
	9	Market value	X	X	X	X	X
	10	Accrued interest	X	X	X	X	
	11	Option income	X	X			
	12	Realized capital gains on option payments	X	X			
	13	Tax exempt income	X				
	14	Years to maturity (0.00 to ...)	X	X			
	15	Weighted average life (0.00 to ...)	X	X	X		
	16	Effective duration (0 to ...)	X	X	X	X	

7.1.2 Data descriptions

The record format section provides a summary record layout for the EPA cash flows data file. This section describes the data that is provided in an EPA cash flow data file.

Record 1: File header information

This record contains the names of the items included in the EPA file. These names are included so that the schedule lookup can be flexible and will not fail if a column is not included.

The names of the columns **must** match the names specified in the record format layout.

Record 2+: Projection task loop/Projection data

Each EPA projection task loop/projection data record may contain 15 fields. When a field is not applicable to a particular asset type or would only contain all zeros, it may be excluded from the file or set to zero.

Each field is referenced here by a number and letter. The letters are useful when viewing EPA cash flow data in a spreadsheet.

In some cases formulas are provided. The formulas presented are consistent with asset cash flows occurring at the end of the month.

1.A	Schedule value: The projection task loop number for the data, starting at 1.
2.B	Projection period: The projection period for the data, starting at 0 and ending at the final projection period. This value will start at 0 for each projection task loop block of data.
3.C	Interest payment: The interest payment for the projection month.
4.D	Scheduled principal payments: Sinking fund and balloon payments for bonds and mortgages, and includes all principal payments on securitized assets. These payments are treated as return of principal and, as such, do not generate any capital gains or losses.
5.E	Option payments: The call, put, and prepayment amounts including call, put, and prepayment premium amounts. This can be calculated as: Call Amount = Par Value After Scheduled Principal Payments * (1 + Call Premium) * Call Factor
6.F	Notional amount: Par value for bonds, mortgages, and securitized assets as of the end of the projection month.
7.G	Cost basis: The cost basis as of the end of the projection month. This field is used to calculate unrealized and realized capital gains and losses on the assets in the EPA file when the projection defaults or sells these assets.
8.H	Book value: The book value as of the end of the projection month. The book value does not include accrued interest. For bonds and mortgages, this value may be calculated as: Book Value at end of projection month = (Book Value After Default - Scheduled Principal Payments + Amortization of Discount) * (1 - Call Factor) where Scheduled Principal Payments include sinking fund and balloon payments.
9.I	Market value: The market value as of the end of the projection month. Market value excludes accrued interest. This field is used to determine capital gains and losses on the assets in the EPA file when the projection sells these assets.
10.J	Accrued interest: The accrued interest as of the end of the projection month.
11.K	Option income: The accrued interest on the assets in the EPA file that have been called, put, or prepaid during the projection month.

12.L	Realized capital gain on option payments: The realized capital gains on calls, puts, and prepayments, including call, put, and prepayment premiums, on the assets in the EPA file during the projection month. This can be calculated as: Realized Capital Gain On Calls and Prepayments $= [\text{Par Value After Scheduled Principal Payments} * (1 + \text{Call Premium}) - \text{Book Value After Scheduled Payments}] * \text{Call Factor}$
13.M	Tax exempt income: The interest payment, change in accrued interest and original issue discount cost basis growth from municipal bonds that is excluded from taxable income.
14.N	Years to maturity: The average years to maturity of the assets in the EPA file as of the end of the projection month. This should be calculated as the sum over all assets of the market value of the individual asset times the years to maturity of the individual asset divided by the total market value of the assets in the EPA file. Years to maturity is used to estimate IMR/AVR for the assets in the EPA file. In general this is a very crude estimate.
15.O	Weighted average life: The average weighted average life (WAL) of the assets in the EPA file as of the end of the projection month. This should be calculated as the sum over all assets of the market value of the individual asset times the weighted average life of the individual asset divided by the total market value of the assets in the EPA file.
16.P	Effective duration: The average effective duration of the assets in the EPA file as of the end of the projection month. This should be calculated as the sum over all assets of the market value of the individual asset times the effective duration of the individual asset divided by the total market value of the assets in the EPA file. When the effective duration is not readily available, modified duration can be provided here instead.

7.2 TAS format

In order to reduce the burden on long time vendor partners, EPA modeling supports EPA files developed for TAS projections.

When the EPA files use the TAS format EPA structure the externally projected assets (EPA) data names are of the form EPAuuuuu.???, where:

- 'uuuuu' is a five character name that you assign to the EPA file.
- ??? is either the projected task loop number (for example, 001 for loop number 1) or ALL.

The EPA file name can be any five alphanumeric characters with no embedded blanks.

The file extension is either the number of the scenario used for the asset/liability model run or ALL. There must be an externally projected assets file for each scenario used in the model. Scenario-independent EPA files can be used in TAS, and have an extension of 'ALL'. If a requested EPA file with the proper numeric scenario extension cannot be located at run time, the projection looks for a corresponding EPA file with the extension ALL.

Fields in the EPA file are space delimited within records.

Each field must be represented in the file, either by user data, a default value, or zero.

7.2.1 Record format

Each EPA field contains four header records followed by end of month data, one record per month. The record layout of the data in an EPA file is presented below.

An (X) in the asset type column indicates the field is applicable for an EPA file of that asset type. An (N/A) indicates the field is not used in the projection. All fields included in the structure MUST be included in the EPA cash flow data file with the appropriate default value. Asset types are translated as follows:

Bnd = Bond, Mtg= Mortgage, Sec = Securitized Asset, and Ird = Interest Rate Derivative

Record	Field	Description	Bnd	Mtg	Sec	Ird	Eqt
1	1	Version numbers (2.00 1.00 8.00 0.00)	X	X	X	X	X
2	1	200-character description of assets	X	X	X	X	X
3		Model information for the file					
	1	Asset type code: 0 = Bonds 1 = Mortgages 2 = Securitized Asset 4 = Equity 5 = Interest Rate Derivative	X	X	X	X	X
	2	Name of scenario file for model, 5 characters	X	X	X	X	X
	3	Projection year (4 digits)	X	X	X	X	X
	4	Projection month (1 to 12)	X	X	X	X	X
	5	Quality grade for included assets, 6 characters	X	X	X	X	X
	6	Has investment expense been calculated? 1 = Yes 2 = No (default)	N/A	N/A	N/A	N/A	N/A
	7	Have defaults been calculated? 1 = Yes 2 = No (default)	N/A	N/A	N/A	N/A	N/A
	8	Has risk based capital been calculated? 1 = Yes 2 = No (default)	N/A	N/A	N/A	N/A	N/A
	9	Sales class 1 = not assigned 2 = available for sale 3 = trading 4 = held to maturity default selections: 1 = for interest rate derivatives 2 = for bonds, mortgages, and securitized assets	X	X	X	X	X
	10	Assets held at market? 1 = No (default) 2 = Yes	N/A	N/A	N/A	N/A	N/A

4		Projection date information					
	1	Accrued interest	X	X	X	X	
	2	Book value	X	X	X	X	X
	3	Initial book yield, such as 0.05 for 5%	N/A	N/A	N/A	N/A	N/A
	4	Market value	X	X	X	X	X
	5	Effective duration (0.00 to ...)	X	X	X		
	6	Notional amount	X	X	X	X	X
	7	Accrued dividend	N/A	N/A	N/A	N/A	N/A
	8	Actual cost	N/A	N/A	N/A	N/A	N/A
	9	Weighted average life (0 to ...)	X	X	X		
	10	Years to maturity (0 to ...)	X	X	X	X	
	11	Risk based capital C1 risk component	N/A	N/A	N/A	N/A	N/A
5+		End of month, one record per month					
	1	Accrued dividend	N/A	N/A	N/A	N/A	N/A
	2	Nominal monthly book yield or 0 (such as 0.05 for 5%)	N/A	N/A	N/A	N/A	N/A
	3	Accrued interest	X	X	X	X	
	4	Cost basis	X	X	X	X	X
	5	Amortization of discount	N/A	N/A	N/A	N/A	N/A
	6	Book value	X	X	X	X	X
	7	Notional amount (par value for bonds, mortgages, and securitized assets)	X	X	X	X	X
	8	Call/prepayment income	X	X			
	9	Call/prepayment amount	X	X			
	10	CMO book value appreciation	N/A	N/A	N/A	N/A	N/A
	11	Depreciation of real estate	N/A	N/A	N/A	N/A	N/A
	12	Increase in accrued dividend	N/A	N/A	N/A	N/A	N/A
	13	Interest payment	X	X	X	X	X
	14	Market value	X	X	X	X	X
	15	Effective duration	X	X	X		
	16	Realized capital gain on calls and prepayments	X	X			
	17	Realized capital loss on defaults	N/A	N/A	N/A	N/A	N/A
	18	Scheduled principal payments	X	X	X		

	19	Unrealized capital gain due to appreciation	N/A	N/A	N/A	N/A	N/A
	20	Unrealized capital gain on calls and prepayments	N/A	N/A	N/A	N/A	N/A
	21	Unrealized capital loss on defaults	N/A	N/A	N/A	N/A	N/A
	22	Investment expense	N/A	N/A	N/A	N/A	N/A
	23	Default amount on book value	N/A	N/A	N/A	N/A	N/A
	24	Risk based capital C1 risk component	N/A	N/A	N/A	N/A	N/A
	25	Years to maturity	X	X	X	X	
	26	Average time in years from payment date to end of month	N/A	N/A	N/A	N/A	N/A
	27	Weighted average life	X	X	X		

7.2.2 Data descriptions

The record format section provides a summary record layout for a TAS EPA cash flows data file.

Record 1: Version number

This record contains the version number for which the EPA file was created. It should be 2.00 1.00 8.00 0.00.

Record 2: Description

The second record of the EPA file contains a file description, which must be limited to 200 characters or less.

Record 3: Projection Parameters

These data provide information relating to the projection of the EPA file. The data can be overridden as the EPA existing asset records are entered in the existing asset data file. If you do this, be careful that you don't compromise the integrity of your model.

This record contains 10 required fields. None of the character fields can be left blank. Each field is reference here by a number and a letter. The letters are useful when viewing the EPA cash flow data in a spreadsheet.

1.A	Asset type: This one-digit code identifies the type of asset whose cash flows are recorded in the EPA Cash Flows Data file. Valid code values include: 0 for <i>Bonds</i> , 1 for <i>Mortgages</i> , 2 for <i>Securitized Assets</i> , 4 for <i>Equities</i> , and 5 for <i>Interest Rate Derivatives</i> . Asset type affects certain calculations such as whether investment expenses apply to book or market values.
2.B	Scenario name: The scenario file used to generate the EPA cash flows data file
3.C	Projection year: The projection date year (yyyy).
4.D	Projection month: The projection date month (mm).
5.E	Quality grade: The quality grade for the EPA which maps to assumptions for defaults. If you do not know what category will be used in the model, enter XXXXXX as a default category name.
6.F	Has investment expense been calculated? Not used, set to 2.

7.G	Have defaults been calculated? Not used, set to 2.
8.H	Has risk based capital been calculated? Not used, set to 2.
9.I	Sales classification: This one-digit code identifies the sales classification for the EPA assets. Valid codes are: 1 for Not Assigned, 2 for Available For Sale, 3 for Trading, and 4 for Held To Maturity. The default value for bonds, mortgages, and securitized assets is Available For Sale; and the default value for all other asset types is Not Assigned.
10.J	Asset held at market: Not used, set to 1.

Record 4: Projection date data

This record contains 11 required fields of information at the projection date. Each field is referenced here by a number and a letter. The letters are useful when viewing the EPA cash flow data in a spreadsheet.

1.A	Accrued interest: The accrued interest as of the projection date. This field is used to estimate the net asset yield and is also used to calculate accrued income during the first month of the asset projection.
2.B	Book value: The book value (excluding accrued interest) at the projection date.
3.C	Initial book yield: Not used.
4.D	Market value: The market value (excluding accrued interest) at the projection date.
5.E	Effective duration: The average effective duration of the assets in the EPA file as of the projection date. This should be calculated as the sum over all assets of the market value of the individual asset times the effective duration of the individual asset divided by the total market value of the assets in the EPA file. If the effective duration is not readily available, modified duration can be provided here instead. This information is used only for model start date reporting.
6.F	Notional amount: The outstanding par value for bonds and securitized assets, and principal value for mortgages at the projection date.
7.G	Accrued dividend: Not used.
8.H	Cost basis: The cost basis at the projection date. This field is used to calculate unrealized capital gains at the projection date.
9.I	Weighted average life: The average weighted average life (WAL) of the assets in the EPA file as of the projection date. This should be calculated as the sum over all assets of the market value of the individual asset times the weighted average life of the individual asset divided by the total market value of the assets in the EPA file. This information is used only for model start date reporting.
10.J	Years to maturity: The average years to maturity of the assets at the projection date. This should be calculated as the sum over all assets of the market value of the individual asset times the years to maturity of the individual asset, divided by the total market value of the assets in the EPA file.
11.K	Risk based capital C1 risk component – Not used.

Record 5+: Monthly cash flow data

EPA assets are projected monthly. The fifth record in each EPA cash flows data file contains cash flows occurring during the first projection month and statement values as of the end of the first projection month. The sixth record contains information for the second projection month, etc.

Each EPA cash flow record contains 27 required fields. When a field is not used in the application or not applicable to a particular asset type, set the field value to zero. Each field is reference here by a number and letter. The letters are useful when viewing EPA cash flow data in a spreadsheet.

In some cases formulas are provided. The formulas presented are consistent with asset cash flows occurring at the end of the month.

1.A	Accrued dividend: Not used.
2.B	Nominal monthly book yield: Not used.
3.C	Accrued interest: The accrued interest as of the end of the projection month.
4.D	Cost basis: The cost basis as of the end of the projection month. This field is used to calculate unrealized and realized capital gains and losses on the assets in your EPA file when the projection defaults or sells these assets.
5.E	Amortization of discount: Not used.
6.F	Book value: The book value as of the end of the projection month. The book value does not include accrued interest. For bonds and mortgages, this value may be calculated as: Book Value at end of projection month $= (\text{Book Value After Default} - \text{Scheduled Principal Payments} + \text{Amortization of Discount}) * (1 - \text{Call Factor})$ where Scheduled Principal Payments include sinking fund and balloon payments.
7.G	Notional amount: The par value for bonds, mortgages, and securitized assets as of the end of the projection month.
8.H	Call and prepayment income: This field contains the accrued interest on the assets in your EPA file that have been called, put, or prepaid during the projection month.
9.I	Calls and prepayments: The call, put, and prepayment amounts including call, put, and prepayment premium amounts. This can be calculated as: Call Amount $= \text{Par Value After Scheduled Principal Payments} * (1 + \text{Call Premium}) * \text{Call Factor}$
10.J	CMO book value appreciation: Not used.
11.K	Depreciation of real estate: Not used.
12.L	Increase In accrued dividend: Not used.
13.M	Interest payment: The gross coupon/interest payments during the projection month. This value should not be adjusted for investment expense.
14.N	Market value: The market value as of the end of the projection month. Market value excludes accrued interest. This field is used to determine capital gains and losses on the assets in your EPA file when the projection sells these assets.
15.O	Effective duration: This field contains the average effective duration of the assets in the EPA file as of the end of the projection month. This should be calculated as the sum over all assets of the market value of the individual asset times the effective duration of the individual asset divided by the total market value of the assets in the EPA file. If the effective duration is not readily available, modified duration can be provided here instead.

16.P	Realized capital gain on calls and prepayments: The realized capital gains on calls, puts, and prepayments, including call, put, and prepayment premiums, on the assets in the EPA file during the projection month. This can be calculated as: Realized Capital Gain On Calls and Prepayments = [Par Value After Scheduled Principal Payments * (1 + Call Premium) - Book Value After Scheduled Payments] * Call Factor
17.Q	Realized capital loss on defaults: Not used.
18.R	Scheduled principal payments: Sinking fund and balloon payments for bonds and mortgages, including all principal payments on securitized assets. These payments are treated as return of principal and, as such, do not generate capital gains or losses.
19.S	Unrealized capital gain due to appreciation: Not used.
20.T	Unrealized capital gain on calls and prepayments: Not used.
21.U	Unrealized capital loss on defaults: Not used.
22.V	Investment expense: Not used.
23.W	Default amount on book value: Not used.
24.X	Risk based capital C1 risk component: Not used.
25.Y	Years to maturity: The average years to maturity of the assets in the EPA file as of the end of the projection month. This should be calculated as the sum over all assets of the market value of the individual asset times the years to maturity of the individual asset divided by the total market value of the assets in the EPA file. Years to maturity is used to estimate IMR/AVR for the assets in the EPA file. In general this is a very crude estimate.
26.Z	Average time in years from payment date to end of month: Not used.
27.AA	Weighted average life: The average weighted average life (WAL) of the assets in the EPA file as of the end of the projection month. This should be calculated as the sum over all assets of the market value of the individual asset times the weighted average life of the individual asset divided by the total market value of the assets in the EPA file.

8 SUBMODELS STRUCTURE

The `epa_asset` model class contains the formulas and variables for externally projected asset model point processing. The following describes the model tree:

- `epa` – A data driven array submodel used to determine externally projected asset experience results. The number of elements in the array is fixed at the number of EPA model points.